Introduction to Vector and Banach Lattices

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Introduction

Functional analysis originates from studies of function spaces. There are many spaces of functions that are important in analysis and in applications, including various spaces of continuous functions, spaces of measurable functions, L_p-spaces, etc. A Banach space is often viewed as an abstraction of a function space. However, spaces of real-valued functions have a structure which is not reflected in the concept of a Banach space: order structure. Given two functions f and g in a vector space of real-valued functions, we define that $f \leqslant g$ if $f(t) \leqslant g(t)$ for all t (or for almost all t in spaces of measurable functions). This makes our vector space into a (partially) ordered set. This order structure is certainly important. First, it is often important to know when one function is dominated by another. Second, in many applications one has to deal with positive functions. Third, many facts about function spaces involve increasing or decreasing sequences of functions. However, the formalism of Banach spaces does not provide a mechanism to study order structures.

This is where Banach lattices come to rescue. A Banach lattice is, essentially, a Banach space which is also equipped with a "nice" partial order structure. This concept of a Banach lattice allows one to describe and study order properties of function spaces in abstract terms. The category of Banach lattices is sufficiently large, comprising many of the classical spaces that appear in analysis, including the space $C(K)$ of real-valued continuous functions on a compact Hausdorff topological space; the space $L_p(\mu)$ of real-valued functions on a measure space whose p-th power is μ-integrable; as well as Orlicz spaces, Lorentz spaces, and various combinations and variants of these spaces. On the other hand, the category of Banach lattices is smaller and more "tame" then the category of all Banach spaces; it excludes some pathological Banach spaces.

This book focuses on two kinds of spaces, two categories: vector lattices and Banach lattices. A vector lattice is a vector space which is also a partially ordered space such that the two structures are compatible in a certain natural way and, in addition, the order is a lattice one, meaning that every two elements have a supremum and an infimum. A Banach lattice is a vector lattice which is also a Banach space and, again, the two structures are compatible in a natural way. One may view the concept of a

Banach lattice as a combination of three structures: vector space, complete metric, and lattice order. Table 1 on page 1 illustrates other mathematical categories arising from various combinations of these structures.

In some sense, vector lattices are to Banach lattices as vector spaces are to Banach spaces. However, there is an important difference. While vector space by themselves are very simple and completely understood, vector lattices have surprisingly rich structures. Thus, a considerable part of this book deals with vector lattices. Even though a vector lattice need not be a topological space, it enjoys topological-like structures such as order convergence, order completeness, and order continuity. Furthermore, there are important function spaces that are vector lattices but not Banach lattices: for example, the space of all real-valued continuous function on a Hausdorff topological space, or the space of all measurable functions on a measure space.

My initial plan was to focus on vector lattices in the first part of the book, and then shift the focus to Banach lattices at a later point, just as one studies vector spaces before moving on to Banach spaces. But this turned out to be impossible, primarily because most interesting examples of vector lattices happen to be Banach lattices. So we rather follow a "concept-by-concept" format. The first part of the book deals with vector lattice concepts; every vector lattice concept is first introduced for vector lattices, and then we discuss how this concept manifests itself in Banach lattices. The second part deals with concepts that pertain exclusively to Banach lattices. The third part is a collection of more special topics; most of the chapters in this part are independent of each other (although they all rely on the first two parts).

This book makes heavy use of representations. All classical vector and Banach lattices are spaces of functions. We will see throughout this book that one can often realize a given vector lattice as a function space. In particular, many vector lattices may "locally" be represented as $C(K)$ spaces for some compact space K. On the other hand, many vector lattices may be represented as sublattices of $L_1(\mu)$ for some measure μ. Thus, we may often view vector lattices as function spaces and use various function space techniques to work with them. Furthermore, since vector lattices often sit "between" $C(K)$ and $L_1(\mu)$ spaces, one may use methods of topology and measure theory to study them.

This book is based on previous monographs [AA02, AB06, LT79, LZ71, MN91, Sch74, Vul67, Wnuk99, Zaa83, Zaa97]. We refer the reader to these books for further information on the subject. With few exceptions, the notes are self-contained. We assume the reader is familiar with the basics of Real and Functional Analysis. Throughout the notes, our prime "testing" examples are $C(K)$ and $L_p(\mu)$ spaces. The reader is assumed to be familiar with topological and measure spaces, continuous functions, Lebesgue integration and general measure theory. We also assume that the reader is familiar with basics of Banach space theory. For the convenience of the reader, we provide a quick overview of these topics (to the extent needed for this book) in Appendix A.

Starred sections, as well as material typeset on grey background, delve deeper into the concepts and may be skipped on the first reading. Most of the "Exercises" in this book are meant as "propositions whose proof is straightforward and is left as an exercise".

Most of our notation and terminology are standard; however, there are a few notable exceptions. In modern literature, the terms "vector lattice" and "Riesz space" are equivalent; we use the former. When discussing Banach lattices, all "topological" adjectives relate to norm properties unless specified otherwise. For example, "bounded" means "norm bounded", "convergent" means "norm convergent", "completion" means "norm completion", etc. We use the term "order completeness" rather than "Dedekind completeness". Historically, there have been several non-equivalent definitions of order convergence; we use the one where the dominating net is allowed to have a different index set than the original net (see the discussion about various definitions of order convergence in Section 21.1). We write $\mathcal{L}_{\mathrm{ob}}(X,Y)$, $\mathcal{L}_{\mathrm{oc}}(X,Y)$, and $\mathcal{L}_{\sigma\text{-oc}}(X,Y)$ for the spaces of all order bounded, order continuous, and σ-order continuous linear operators between vector lattices X and Y, respectively; we write $X^{\sim}_{\mathrm{oc}}$ and $X^{\sim}_{\sigma\text{-oc}}$ for the order continuous and σ-order continuous dual of X, respectively. We write "order continuous Banach lattice" instead of "a Banach lattice with an order continuous norm".

		product	**partial order**	**lattice order**
	vector space	algebra	ordered vector space	vector lattice (Riesz space)
topological space	topological vector space	topological algebra	ordered topological vector space	locally solid vector lattice
metric space	normed space	normed algebra	ordered normed space	normed lattice
complete metric space	Banach space	Banach algebra	ordered Banach space	Banach lattice

Table 1: This table illustrates the place of vector and Banach lattices in Functional Analysis.

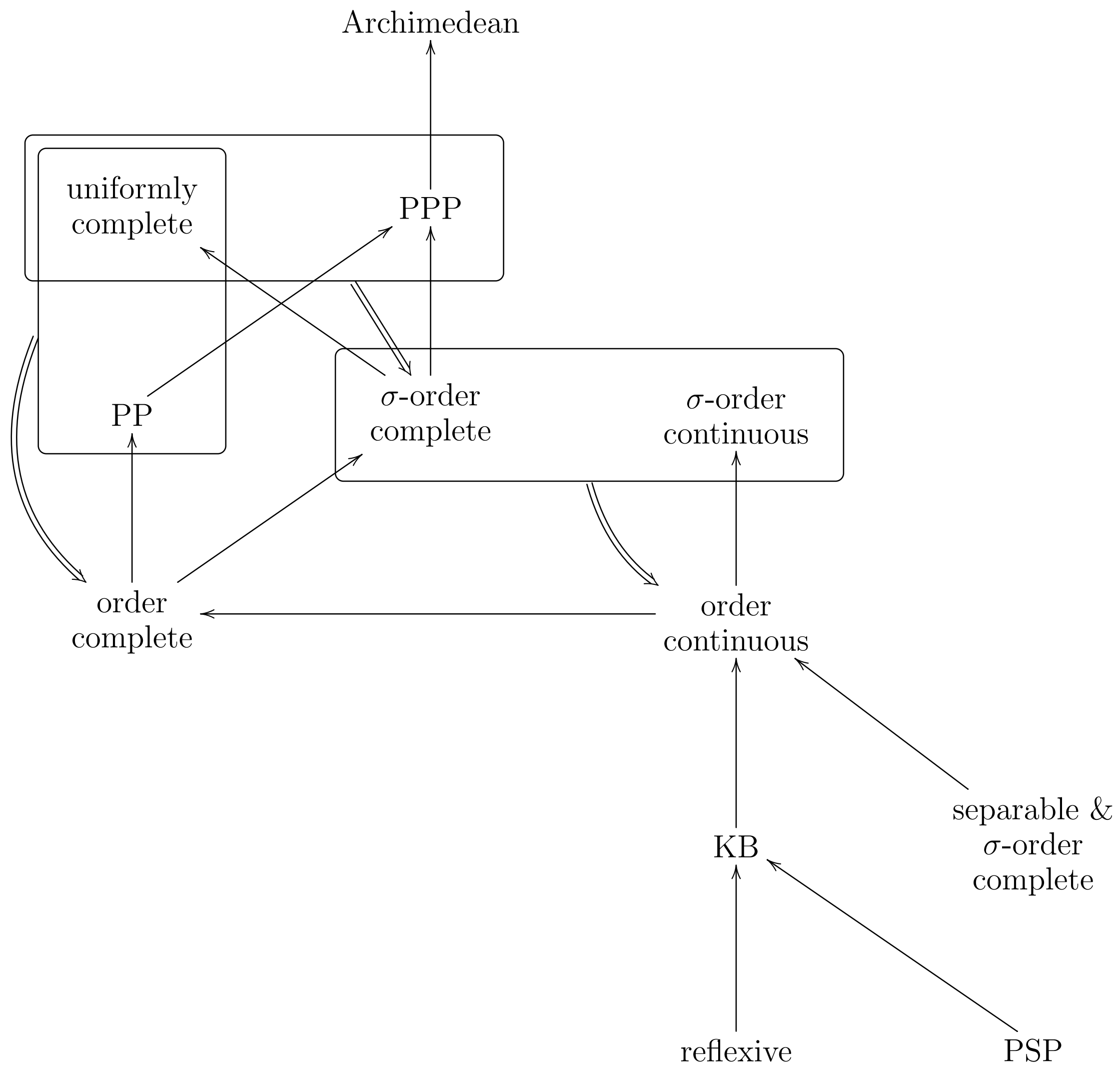

Table 2: Hierarchy of properties of vector lattices (on the left side) and Banach lattices (on the right side). Frames and double arrows indicate when two weaker properties together imply a stronger one.

Chapter 1

Basic definitions and examples

1.1 Ordered sets, nets, and lattices

In this section we briefly review concepts and notation related to ordered sets. A relation "$\leqslant$" on a set $\mathcal{S}$ is called a ***(partial) order*** if

(i) $\forall x \in \mathcal{S} \quad x \leqslant x$ (reflexivity);

(ii) $\forall x, y \in \mathcal{S} \quad x \leqslant y$ and $y \leqslant x$ imply $x = y$ (antisymmetry);

(iii) $\forall x, y, z \in \mathcal{S} \quad x \leqslant y$ and $y \leqslant z$ imply $x \leqslant z$ (transitivity).

In this case, we say that $\mathcal{S}$ is ***(partially) ordered***. A relation which is reflexive and transitive is called a ***pre-order***. We write $x \geqslant y$ if $y \leqslant x$, $x < y$ if $x \leqslant y$ and $x \neq y$, and $x > y$ when $y < x$.

A set Λ is ***directed*** if it is pre-ordered and every two elements have a common majorant, i.e.,

$$\forall \alpha, \beta \in \Lambda \quad \exists \gamma \in \Lambda \text{ such that } \alpha \leqslant \gamma \text{ and } \beta \leqslant \gamma.$$

A function from a directed set Λ to any set A is referred to as a ***net*** in A over Λ. In this case, instead of writing $x \colon \Lambda \to A$ we write $(x_\alpha)_{\alpha \in \Lambda}$ or just (x_α) if the index set is clear from the context or not important. Note that every directed set may be viewed as a net over itself via the identity map. A ***sequence*** is a net over $\mathbb{N}$. Let $(x_\alpha)_{\alpha \in \Lambda}$ be a net in A. Fix $\alpha_0 \in \Lambda$; put $\Lambda_0 = \{\alpha \in \Lambda \ : \ \alpha \geqslant \alpha_0\}$. The restriction of x to Λ_0 is called a ***tail*** of (x_α); it is sometimes denoted by $(x_\alpha)_{\alpha \geqslant \alpha_0}$. We refer the reader to Appendix A.1 for a further discussion about the definition of a net.

For the rest of this section, let $\mathcal{S}$ be an ordered set. Let A be a subset of $\mathcal{S}$. Clearly, A is still an ordered set under the induced order. If $x \in \mathcal{S}$ satisfies $a \leqslant x$ for all $a \in A$, we say that x is an ***upper bound*** of A and write $A \leqslant x$. We say that A is ***bounded above*** if it has an upper bound. Suppose that $x \in A \subseteq \mathcal{S}$; we say that x is the ***greatest element*** of A if $A \leqslant x$; we say that x is a ***maximal element*** of A if no $a \in A$ satisfies $x < a$. It is easy to see that A has at most one greatest element;

furthermore, if x is the greatest element of A then it is the unique maximal element of A. Similarly, one defines ***lower bounds***, ***least*** elements, and ***minimal*** elements. We will denote by $A^{\uparrow}$ and $A^{\downarrow}$ the sets of all upper and lower bounds of A, respectively, i.e.,

$$A^{\uparrow} = \bigl\{x \in \mathcal{S} \,:\, A \leqslant x\bigr\} \qquad \text{and} \qquad A^{\downarrow} = \bigl\{x \in \mathcal{S} \,:\, x \leqslant A\bigr\}.$$

If $A^{\uparrow}$ has a least element, this element is called the ***supremum*** or the ***least upper bound*** of A and is denoted by $\sup A$. Similarly, the ***infimum*** $\inf A$ or ***greatest lower bound*** of A is defined to be the greatest element of $A^{\downarrow}$ (if it exists). Again, it is easy to see that if x is the greatest element of A then $\sup A = x$. Note that, in general, the concepts of maximal elements, greatest elements, and supremum are different.

1.1.1 Exercise. Let $\mathcal{C}$ be a collection of subsets of $\mathcal{S}$, such that $\inf A$ exists for each set in $\mathcal{C}$. Then $\inf \bigcup \mathcal{C} = \inf_{A \in \mathcal{C}} \inf A$.

We say that A is ***order bounded*** if it is bounded both above and below. It is easy to see that A is order bounded if and only if it is contained in an ***order interval***, where by an order interval we mean a set of the form $[a, b] = \{x \in \mathcal{S} \,:\, a \leqslant x \leqslant b\}$ for some $a, b \in \mathcal{S}$ with $a \leqslant b$.

Suppose that (x_n) is a sequence in $\mathcal{S}$. We write $\sup x_n$ and $\inf x_n$ for the supremum and the infimum of the set $\{x_n \,:\, n \in \mathbb{N}\}$, respectively. We will write $x_n \uparrow$ or $x_n \downarrow$ to indicate that (x_n) is increasing or decreasing, respectively. For example, $a \leqslant x_n \uparrow \leqslant b$ means that (x_n) is an increasing sequence in $[a, b]$. The notation $x_n \uparrow x$ means that (x_n) is increasing and $x = \sup x_n$. Similar notations will be used for nets. Hence, $a \geqslant x_\alpha \downarrow z$ means that (x_α) is a decreasing net, $a \geqslant x_\alpha$ for every α, and $z = \inf x_\alpha$.

For $x, y \in \mathcal{S}$ we write $x \vee y = \sup\{x, y\}$ and $x \wedge y = \inf\{x, y\}$. It can be easily verified that $\vee$ and $\wedge$ are associative and commutative binary operations (whenever they are defined), so that expressions like $x_1 \vee \dots \vee x_n$ or $\bigwedge_{n=1}^{\infty} x_n$ are unambiguous.

1.1.2 Exercise. Suppose that $x_1 \leqslant y_1$ and $x_2 \leqslant y_2$. Then $x_1 \vee x_2 \leqslant y_1 \vee y_2$ and $x_1 \wedge x_2 \leqslant y_1 \wedge y_2$ whenever they exist.

An ordered set $\mathcal{S}$ is a ***lattice*** if $x \vee y$ and $x \wedge y$ exist for every $x, y \in \mathcal{S}$; the order on $\mathcal{S}$ is then said to be a ***lattice order***. It follows immediately that every finite subset of $\mathcal{S}$ has supremum and infimum.

1.1.3 Example. Let $\Lambda = \mathbb{N} \times \mathbb{N}$ ordered component-wise, i.e., $(m_1, n_1) \leqslant (m_2, n_2)$ whenever $m_1 \leqslant m_2$ and $n_1 \leqslant n_2$. It is easy to see that Λ is a lattice with coordinate-wise lattice operations, i.e.,

$$(m_1, n_1) \vee (m_2, n_2) = \bigl(m_1 \vee m_2, n_1 \vee n_2\bigr)$$

and

$$(m_1, n_1) \wedge (m_2, n_2) = \bigl(m_1 \wedge m_2, n_1 \wedge n_2\bigr).$$

Here $m \vee n = \max\{m, n\}$ and $m \wedge n = \min\{m, n\}$ in $\mathbb{N}$. In particular, Λ is a directed set. Nets indexed by Λ are sometimes called ***double sequences***. For example, $x_{n,m} = \frac{n}{m+n^2}$ is a double-sequence in $\mathbb{R}$.

Let $A \subseteq \Lambda$ given by $A = \bigl\{(1,1), (1,2), (1,3), (2,1), (2,2), (3,1)\bigr\}$. Then $(1,1)$ is the least and the only minimal element of A. Note that A has no greatest element; however, $(1,3)$, $(2,2)$, and $(3,1)$ are all maximal in A. $A^{\downarrow} = \bigl\{(1,1)\bigr\}$ and $A^{\uparrow} = \bigl\{(m,n) : m, n \geqslant 3\bigr\}$, so that $\inf A = (1,1)$ while $\sup A = (3,3)$. Note also that A is order bounded as $A \subseteq \bigl[(1,1), (3,3)\bigr]$.

1.1.4 Example. Let Ω be an arbitrary set. Then $\mathcal{P}(\Omega)$, the power set of Ω, is a lattice under the inclusion order, i.e., $A \leqslant B$ whenever $A \subseteq B$. In this case $A \vee B = A \cup B$ and $A \wedge B = A \cap B$ for any $A, B \in \mathcal{P}(\Omega)$. Note that Ω is the greatest element of $\mathcal{P}(\Omega)$, while $\varnothing$ is the least element. For every subset $\mathcal{G}$ of $\mathcal{P}(\Omega)$, $\sup \mathcal{G} = \bigcup \mathcal{G}$ while $\inf \mathcal{G} = \bigcap \mathcal{G}$.

1.1.5 Exercise. Let A be a subset of a lattice L and $A \leqslant x$ for some $x \in L$. Then $x = \sup A$ iff $A \leqslant y \leqslant x$ implies $y = x$.

1.1.6. Suppose that L is a lattice and $A \subseteq L$. We define $A^{\vee}$ as the set of the suprema of all the finite subsets of A and $A^{\wedge}$ as the set of the infima of all the finite subsets of A:

$$\begin{aligned} A^{\vee} &= \bigl\{x_1 \vee \dots \vee x_n : n \in \mathbb{N},\ x_1, \dots, x_n \in A\bigr\} \quad \text{and} \\ A^{\wedge} &= \bigl\{x_1 \wedge \dots \wedge x_n : n \in \mathbb{N},\ x_1, \dots, x_n \in A\bigr\}. \end{aligned}$$

Clearly, $A \subseteq A^{\vee}$ and $A \subseteq A^{\wedge}$. It is easy to see that $A^{\vee}$ and $A^{\wedge}$ are closed under $\vee$ and $\wedge$ operations, respectively. In particular, $A^{\vee}$ is a directed set with respect to the order induced from L. Hence, $A^{\vee}$ can be viewed as an increasing net in L indexed by itself. The set $A^{\wedge}$ becomes directed if we equip it with the order which is reverse to the one induced from L, i.e., for $a, b \in A^{\wedge}$ put $a \preceq b$ if $a \geqslant b$. Then the identity map from $(A^{\wedge}, \preceq)$ to $(L, \leqslant)$ makes $A^{\wedge}$ into a decreasing net.

It is easy to see that $\sup A$ exists iff $\sup A^{\vee}$ exists, and in this case $\sup A = \sup A^{\vee}$. Similarly, $\inf A$ exists iff $\inf A^{\wedge}$ exists, and in this case $\inf A = \inf A^{\wedge}$. This approach allows one to reduce the supremum (or infimum) of a set to the supremum of an increasing net (infimum of a decreasing net, respectively).

1.1.7 Exercise. Let $\varphi\colon L \to M$ be a bijection between two lattices. TFAE

(i) $\varphi(a \vee b) = \varphi(a) \vee \varphi(b)$ and $\varphi(a \wedge b) = \varphi(a) \wedge \varphi(b)$ for all $a, b \in L$;

(ii) φ preserves suprema and infima of arbitrary sets;

(iii) $a \leqslant b$ iff $\varphi(a) \leqslant \varphi(b)$.

The preceding exercise essentially asserts that a bijection between two lattices that preserves the order structure also preserves the lattice structure. This should not be really surprising because the lattice structure is determined by the order structure. We will call such a φ an ***order isomorphism***.

1.2 Ordered vector spaces and cones

A vector space X is said to be an ***ordered vector space*** if it is an ordered set and

(i) $\forall x, y, z \in X \quad x \leqslant y$ implies $x + z \leqslant y + z$, and

(ii) $\forall x, y \in X$ and $\lambda \in \mathbb{R}_+ \quad x \leqslant y$ implies $\lambda x \leqslant \lambda y$.

The axioms (i) and (ii) guarantee compatibility between the vector space structure of X and the order structure of X. Such an order is called a ***linear order***[1].

1.2.1 Exercise. In an ordered vector space,

(i) If $x_1 \leqslant y_1$ and $x_2 \leqslant y_2$ then $x_1 + x_2 \leqslant y_1 + y_2$;

(ii) If $x, y \geqslant 0$ and $x + y = 0$ then $x = y = 0$.

Suppose that X is an ordered vector space. An element x of X satisfying $x \geqslant 0$ is referred to as ***positive***. There is a slight collision of notations here as the zero vector will also be considered as a positive vector. We will write X_+ for the set of all positive elements in X. In particular, $\mathbb{R}_+ = [0, \infty)$. A vector x is said to be ***negative*** if $-x$ is positive.

Generally, for a given vector space X there may be many orders making it into an ordered vector space. We will see that every such order corresponds to a cone in X.

1.2.2 Definition. A non-empty subset C of a vector space X is said to be a ***cone*** if

(i) $\forall x, y \in C \quad x + y \in C$,

(ii) $\forall x \in C$ and $\lambda \in \mathbb{R}_+ \quad \lambda x \in C$, and

(iii) $\forall x \in X \quad x \in C$ and $-x \in C$ iff $x = 0$.

[1] In some texts, a linear order is defined as a partial order in which every two elements are comparable; we will call such an order ***total***.

In other words, $C = C + C = \lambda C$ for every $\lambda \geqslant 0$ and $-C \cap C = \{0\}$. Note that (ii) yields $0 \in C$. A set which only satisfies (i) and (ii) is called a ***wedge***. For example, the closed upper half-plane in $\mathbb{R}^2$ is a wedge but not a cone.

1.2.3 Exercise. If X is an ordered vector space then X_+ is a cone in X. In view of this, X_+ is usually referred to as the ***positive cone*** of X.

Now, suppose that X is a vector space and C a cone in X. For $x, y \in X$ put $x \leqslant y$ if $y - x \in C$. Then "$\leqslant$" is an order relation making X into an ordered vector space with $X_+ = C$ (check this). Thus, for an arbitrary vector space X there is a one-to-one correspondence between cones in X and linear orders on X. From this point of view, the theory of ordered vector spaces is nothing but the theory of cones. In particular, many properties of ordered vector spaces can be characterized in terms of their cones.

Here is one of such properties. Let C be a cone in a vector space X. We write $C - C$ for the set $\{x - y \ : \ x, y \in C\}$. It is easy to see that $C - C = \operatorname{span} C$, the subspace of X generated by C. We say that C is ***generating*** if $C - C = X$, i.e., every vector in X can be written as the difference of two vectors which are positive in the order induced by C.

1.2.4 Example. Let $X = \mathbb{R}^2$. Then the cone $\{(x, y) \ : \ x, y \geqslant 0\}$ is generating, while the cone $\{(x, 0) \ : \ x \geqslant 0\}$ is not. Describe the orders corresponding to these two cones.

A vector in an ordered vector space is said to be ***regular*** if it can be written as a difference of two positive vectors. In particular, every positive vector is regular; X_+ is generating iff every vector is regular.

1.2.5 Exercise. For an element z in an ordered vector space, TFAE:

(i) z is regular, i.e., $z = x - y$ for some positive x and y;
(ii) $z \leqslant x$ for some positive x;
(iii) $-x \leqslant z \leqslant x$ for some positive x;
(iv) $\pm z \leqslant x$ for some positive x.

1.2.6 Exercise. The set $X_+ - X_+$ of all regular elements in an ordered vector space X is a vector subspace.

1.2.7 Exercise. An ordered vector space X is directed iff X_+ is generating.

1.2.8 Exercise. Let C be a cone in a vector space X. Then $-C$ is also a cone. The map $x \mapsto -x$ in as order isomorphism from the order generated on X by C to the order generated on X by $-C$. The cone $-C$ is the positive cone for the "reverse" order $\geqslant$.

Let X be an ordered vector space. We say that X is ***Archimedean*** if it has the ***Archimedean Property***: if $x \in X$ and $u \in X_+$ satisfy $nx \leqslant u$ for every $n \in \mathbb{N}$ then $x \leqslant 0$.

Note that the assumption that u is positive is redundant. Indeed, suppose that X has the Archimedean Property, $x, u \in X$ and $nx \leqslant u$ for every n. In particular, $x \leqslant u$. Put $v = u - x$; then $v \geqslant 0$. For every n, we have $(n+1)x \leqslant u$, so that $nx \leqslant v$, and, therefore, $x \leqslant 0$.

1.2.9 Exercise. Suppose that X is an ordered vector space. Then X is Archimedean iff $\frac{1}{n}u \downarrow 0$ for every $u \in X_+$.

1.2.10 Example. *A non-Archimedean ordered vector space.* Put $X = \mathbb{R}^2$, equipped with ***lexicographic*** (sometimes called *dictionary*) order, that is, $(x_1, y_1) \leqslant (x_2, y_2)$ if $x_1 < x_2$ or $x_1 = x_2$ and $y_1 \leqslant y_2$. Then X is an ordered vector space with generating cone. However, if we take $x = (0,1)$ and $u = (1,0)$ then $nx \leqslant u$ for every n but $x > 0$.

Intuitively, the Archimedean Property means that the scalar multiplication $(\alpha, x) \mapsto \alpha x$ viewed as a map from $\mathbb{R} \times X \to X$ is continuous on α; we will make this idea more formal later. Most ordered vector spaces that we consider in these notes will be Archimedean. Moreover, we will see that the Archimedean property is a very weak property; it is implied by almost every other property we will study. It is easy to see that the Archimedean property is inherited by subspaces.

Note that in the lexicographic order on $\mathbb{R}^2$, every non-zero vector is either positive or negative. Even though it looks very nice at the first glance, this means that the lattice structure is trivial: you do not get anything "new" by taking the sup or the inf of two elements.

1.2.11 Exercise. A subspace of an Archimedean ordered vector space is Archimedean.

1.2.12 Exercise. If X is an Archimedean ordered vector space with $\dim X > 1$ then there exists a vector in X which is neither positive nor negative.

The following proposition essentially says that there is no "gap" between A and $A^{\uparrow}$:

1.2.13 Proposition. *Let A be a non-empty bounded above set in an Archimedean ordered vector space. Then* $\inf(A^{\uparrow} - A) = 0$.

Proof. Let $B = A^{\uparrow} - A = \{x - a \ : \ a \in A, x \geqslant A\}$. Clearly, $0 \leqslant B$. Suppose that $z \leqslant B$. Fix $x \in A^{\uparrow}$, let $a \in A$, then $z \leqslant x - a$ implies $a \leqslant x - z$. Since this is true for all $a \in A$, we have $x - z \in A^{\uparrow}$. Iterating, we conclude that $x - nz \in A^{\uparrow}$ for every $n \in \mathbb{N}$. Fix any $a \in A$; we have $a \leqslant x - nz$ and, therefore, $nz \leqslant x - a$ for all $n \in \mathbb{N}$. It follows that $z \leqslant 0$. $\square$

1.3 Vector lattices

A ***vector lattice*** is an ordered vector space which is also a lattice[2]. In the literature, vector lattices are often also referred to as ***Riesz spaces***. A note on notation: when writing expressions involving addition and lattice operations, we always assume that the lattice operations have higher priority than addition. Hence, $x + y \wedge z$ means $x + (y \wedge z)$, not $(x + y) \wedge z$ (these are not the same!). Throughout the rest of this chapter, X will always stand for a vector lattice. For each $x \in X$ we define the ***positive part*** of x, the ***negative part*** of x, and the ***modulus*** of x, respectively, as follows

$$x^+ = x \vee 0, \quad x^- = (-x) \vee 0, \quad \text{and} \quad |x| = x \vee (-x).$$

The maps that take x into $|x|$, x^+, x^-, as well as the binary operations $x \vee y$ and $x \wedge y$ are called ***lattice operations*** of X.

1.3.1 Example. Let Ω be an arbitrary set; consider the vector space $\mathbb{R}^\Omega$ of all real-valued functions on Ω. This set will usually be ordered pointwise, i.e., $f \leqslant g$ if $f(t) \leqslant g(t)$ for every $t \in \Omega$. It is easy to see that this is a vector lattice with

$$(f \vee g)(t) = \max\bigl\{f(t), g(t)\bigr\} = f(t) \vee g(t) \text{ and}$$
$$(f \wedge g)(t) = \min\bigl\{f(t), g(t)\bigr\} = f(t) \wedge g(t) \text{ for every } t.$$

It follows that for $f \in \mathbb{R}^\Omega$ we have $|f|(t) = \bigl|f(t)\bigr|$, $f^+(t) = f(t)^+$, and $f^-(t) = f(t)^-$ for all $t \in \Omega$. That is, lattice operations can be computed pointwise. It can be easily shown that this vector lattice is Archimedean (again, verify this pointwise).

When working with $\mathbb{R}^\Omega$ or other spaces of functions, we will write $\mathbb{1}_A$ for the characteristic function of $A \subseteq \Omega$ and $\mathbb{1} = \mathbb{1}_\Omega$ for the "identically one" function. For $f \in \mathbb{R}^\Omega$ and $a \in \mathbb{R}$, we write $\{f < a\}$ for $\bigl\{t \in \Omega \,:\, f(t) < a\bigr\}$.

1.3.2 Example. Consider the special case of the preceding example when Ω is finite; then we can identify $\mathbb{R}^\Omega$ with $\mathbb{R}^n$ for some $n \in \mathbb{N}$. The order is defined coordinate-wise, i.e., if $x, y \in \mathbb{R}^n$ then $x \leqslant y$ iff $x_i \leqslant y_i$ for every $i = 1, \dots, n$. This is a lattice order with

$$x \vee y = \bigl(x_1 \vee y_1, \dots, x_n \vee y_n\bigr) \text{ and } x \wedge y = \bigl(x_1 \wedge y_1, \dots, x_n \wedge y_n\bigr).$$

and for $x = (x_1, \dots, x_n)$ we have

$$|x| = \bigl(|x_1|, \dots, |x_n|\bigr), \quad x^+ = \bigl(x_1^+, \dots, x_n^+\bigr), \quad x^- = \bigl(x_1^-, \dots, x_n^-\bigr).$$

[2] After a certain point in the exposition, we will focus on Archimedean vector lattices.

Unless specified otherwise, $\mathbb{R}^n$ will always be assumed to be equipped with the coordinate-wise order. (Occasionally, we will consider $\mathbb{R}^n$ with the lexicographic order, but in those cases we will clearly specify this.)

Let $x, y \in \mathbb{R}^2$ given by $x = (1, 1)$ and $y = (2, 3)$. Then the order interval $[x, y]$ is the closed rectangle with vertices $(1, 1)$, $(1, 3)$, $(2, 1)$, $(2, 3)$.

1.3.3 Exercise. For all $x, y, z \in X$ we have the following identities:

(i) $-(x \wedge y) = (-x) \vee (-y)$ and $-(x \vee y) = (-x) \wedge (-y)$;

(ii) $x + y \wedge z = (x + y) \wedge (x + z)$ and $x + y \vee z = (x + y) \vee (x + z)$;

(iii) $(x - y)^+ = x \vee y - y = x - x \wedge y$;

(iv) $x + y = x \wedge y + x \vee y$;

(v) $\lambda(x \wedge y) = (\lambda x) \wedge (\lambda y)$ and $\lambda(x \vee y) = (\lambda x) \vee (\lambda y)$ for every $\lambda \in \mathbb{R}_+$;

(vi) $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$ and $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$.

Identities (i), (ii), (v), and (vi) may be viewed as distributive laws. In particular, (vi) says that every vector lattice is a distributive lattice. We must warn the reader that addition cannot be interchanged with $\vee$ or sup in (ii), that is, in general it is not true that $(x + y) \wedge z = x \wedge z + y \wedge z$ or $(x + y) \vee z = x \wedge z + y \vee z$; the lack of these distributive laws distinguishes a vector lattice from an algebra.

At this stage, the reader is expected to deduce Exercise 1.3.3 from the axioms of a vector lattice. Alternatively, one can deduce (ii) and (v) from Exercise 1.1.7 and the easy observation that the maps $y \mapsto x + y$ and $x \mapsto \lambda x$ are order isomorphisms, while (i) follows from Exercises 1.1.7 and (1.2.8). We will later prove in Theorems 4.3.3 that any equality or inequality which is valid for real numbers remains valid in every vector lattice; this will make proving such identities much easier.

Some of the distributive laws in the preceding exercise remain valid for infinite sets:

1.3.4 Exercise. For every $x \in X$, and $A \subseteq X$, we have

(i) $\sup A = -\inf(-A)$ and $\inf A = -\sup(-A)$;

(ii) $\lambda \sup A = \sup(\lambda A)$ and $\lambda \inf A = \inf(\lambda A)$ for every $\lambda \in \mathbb{R}_+$;

(iii) $x + \sup A = \sup\{x + a \,:\, a \in A\}$ and $x + \inf A = \inf\{x + a \,:\, a \in A\}$;

(iv) $x \wedge \sup A = \sup\{x \wedge a \,:\, a \in A\}$ and $x \vee \inf A = \inf\{x \vee a \,:\, a \in A\}$.

In each of these identities, the existence of $\sup A$ or $\inf A$ in the left hand side implies the existence of the sup or inf, in the right hand side.

Iterating identities in the preceding exercise, we get distributive laws for sets:

1.3.5 Exercise. (i) If $A, B \subseteq X$ such that $\inf A$ and $\inf B$ exist then $\inf(A + B) = (\inf A) + (\inf B)$, $\inf(A \vee B) = (\inf A) \vee (\inf B)$, and $\inf(A \cup B) = \inf(A \wedge B) = (\inf A) \wedge (\inf B)$.

(ii) If $a_\alpha \downarrow x$ and $b_\alpha \downarrow y$ then $a_\alpha + b_\alpha \downarrow x + y$.

Similar statements are valid for suprema. (The exercise remains valid for ordered vector spaces.)

1.3.6 Exercise. For every $x \in X$ the following are true:

(i) x^+, x^-, and $|x|$ are positive;
(ii) $|\lambda x| = |\lambda||x|$ for all $\lambda \in \mathbb{R}$;
(iii) $x = x^+ - x^-$ and $|x| = x^+ + x^- = x^+ \vee x^-$;
(iv) $x^+ \wedge x^- = 0$;

These properties uniquely determine x^+ and x^-:

1.3.7 Exercise. If $x = u - v$ for some $u, v \in X$ such that $u \wedge v = 0$, then $u = x^+$ and $v = x^-$.

Together with (i) and (iii) in Exercise 1.3.3, the following exercise shows that the lattice operations can all be expressed via each other.

1.3.8 Exercise. Prove the following identities:

(i) $x^- = (-x)^+$ and $x^+ = (-x)^-$;
(ii) $x^+ = \frac{1}{2}\bigl(|x| + x\bigr)$ and $x^- = \frac{1}{2}\bigl(|x| - x\bigr)$;
(iii) $x \vee y = \frac{1}{2}(x + y) + \frac{1}{2}|x - y|$ and $x \wedge y = \frac{1}{2}(x + y) - \frac{1}{2}|x - y|$.

1.3.9. The fact that all the lattice operations can be expressed via each other has numerous applications. For example, if X is also a topological vector space, then the continuity of any of the lattice operations immediately implies the continuity of all the lattice operations. Moreover, using Exercise 1.3.3(iii), we immediately get the following:

1.3.10 Proposition. *Suppose that, in addition to being a vector lattice, X is a topological vector space such that the map $x \mapsto x^+$ is continuous. Then $\vee$ and $\wedge$ viewed as maps from $X \times X$ to X are jointly continuous.*

In the following exercise, we collect several useful facts:

1.3.11 Exercise. For every x, y, a, b in a vector lattice,

(i) $x \leqslant y$ implies $x^+ \leqslant y^+$ and $x^- \geqslant y^-$;
(ii) The ***triangle inequality***: $\bigl||x| - |y|\bigr| \leqslant |x \pm y| \leqslant |x| + |y|$;
(iii) $(x + y)^+ \leqslant x^+ + y^+$;
(iv) If $x, a, b \geqslant 0$ then $x \wedge (a + b) \leqslant x \wedge a + x \wedge b$.

Note that every order interval $[a, b]$ is contained in a symmetric order interval $[-u, u]$ for some $u \geqslant 0$; just put $u = |a| \vee |b|$. Note that $x \in [-u, u]$ iff $|x| \leqslant u$. It also follows that a set A is order bounded iff it is contained in a symmetric order interval, i.e., there exists $u > 0$ such that $|x| \leqslant u$ whenever $x \in A$.

1.3.12. All the identities and inequalities proved in this section remain valid in ordered vector spaces as long as the expressions involved are defined. For example, $x + y \vee z = (x+y) \vee (x+z)$ as long as $y \vee z$ exists; we still have the triangle inequality $|x+y| \leqslant |x|+|y|$ as long as $|x|$, $|y|$, and $|x + y|$ exist, where $|x| = x \vee (-x)$. It follows that if X is an ordered vector space such that $|x|$ exists for every x (defined as $x \vee (-x)$) then X is a vector lattice.

1.3.13 Exercise. Let Z be an ordered vector space such that $|x| \coloneqq x \vee (-x)$ exists for every $x \in Z$. Then Z is a vector lattice with lattice operations given by Exercise 1.3.8(iii).

1.3.14 Exercise. If X is a vector lattice then X_+ is generating. More precisely, an ordered vector space Z is a vector lattice iff Z_+ is generating and Z_+ is itself a lattice.

A (linear) subspace Z of a vector lattice X is said to be a ***linear sublattice*** or simply a ***sublattice*** of X if it is closed under any lattice operation (and, therefore, under every lattice operations). That is, a subset Z of X is a sublattice iff it is closed under all linear and lattice operations. It can be easily verified that a subspace Z of X is a sublattice iff it is itself a vector lattice under the order induced from X.

1.3.15 Exercise. Which one-dimensional subspaces of $\mathbb{R}^2$ are sublattices?

1.3.16 Exercise. A vector lattice X is Archimedean iff the following is satisfied: if for some $x, u \geqslant 0$ we have $nx \leqslant u$ for all n then $x = 0$.

1.3.17. Instead of defining a vector lattice as an ordered vector space which is a lattice, one can define it as a set equipped with operations of addition, scalar multiplication, and $\vee$, satisfying certain axioms; this will be discussed in Exercise 22.3.1. One can also do it through the modulus operation as in the following exercise, which is based on [MW74]:

1.3.18 Exercise. Let X be a vector space and $m\colon X \to X$ such that

(i) $m\big(m(x)\big) = m(x) = m(-x)$ for every $x \in X$;
(ii) $C \coloneqq \operatorname{Range} m$ is a generating cone in X;
(iii) $m(x + y) \leqslant m(x) + m(y)$ for all $x, y \in X$ with respect to the order induced on X by C.

Then X is a vector lattice under this order, with $m(x) = |x|$.

1.3.1 Normed lattices

We refer the reader to Section A.4 for a brief review of terminology and basic facts about normed and Banach spaces. A ***normed lattice*** is a vector lattice which is also a normed space such that the following two conditions relating the vector lattice structure and the norm are satisfied:

(i) $x \leqslant y$ implies $\|x\| \leqslant \|y\|$ for any two positive vectors x and y;
(ii) $\big\| |x| \big\| = \|x\|$ for every vector x.

A norm satisfying these conditions is called a ***lattice norm***. It can be easily verified that these two conditions can be combined into a single condition, which is often more practical to verify: $|x| \leqslant |y|$ implies $\|x\| \leqslant \|y\|$ for any two vectors x and y. If, in addition, the space is a Banach space (i.e., the norm is complete), then it is said to be a ***Banach lattice***.

1.3.19 Proposition. *The lattice operations $|x|$, x^+, and x^- on a normed lattice are (norm) continuous. The operations $x \vee y$ and $x \wedge y$ are jointly continuous.*

Proof. Suppose that X is a normed lattice. It follows from $\big| |x| - |y| \big| \leqslant |x - y|$ that $\big\| |x| - |y| \big\| \leqslant \|x - y\|$ for all $x, y \in X$. Hence, the map $x \mapsto |x|$ is norm continuous. It follows now from Proposition 1.3.8 and 1.3.9 that all the lattice operations on X are norm continuous, with $\vee$ and $\wedge$ being jointly norm continuous. $\square$

1.3.20. The preceding proposition implies that the positive cone X_+ of a normed lattice X is closed because X_+ is the kernel of x^-. It follows that we may pass to the limit in an inequality. That is, if $x_n \leqslant y_n$ for all n, $x_n \to x$, and $y_n \to y$, then $x \leqslant y$.

1.3.21. It also follows that order intervals are closed because $[a, b] = (a+X_+)\cap(b-X_+)$. It is also easy to see that order intervals are bounded: for every $x \in [a, b]$ we have $|x| \leqslant |a| \vee |b|$, so that $\|x\| \leqslant \big\| |a| \vee |b| \big\|$. It follows that order bounded sets are (norm) bounded.

When dealing with norm or Banach lattices, we will often use the following conventions: we will drop the "norm" modifier before adjectives like "bounded", "continuous", "closed", "convergent", "complete", etc. For example, we say that 1.3.21 asserts that every order bounded set in a normed lattice is bounded (meaning norm bounded). By a continuous operator between normed spaces we will mean a norm continuous operator, as opposed to order continuous or uniformly continuous (to be defined later). The notation $x_n \to x$ will always mean norm convergence.

1.3.22 Exercise. Every normed lattice is Archimedean.

1.3.23. If X a normed lattice then its lattice structure extends to the norm completion $\overline{X}$, making the latter into a Banach lattice. This may be proved directly by extending the order and the lattice operations from X to $\overline{X}$ "by continuity", but this approach requires tedious computations. We will later see a "shortcut" in Exercise 6.7.11: X^{**} is a Banach lattice and $\overline{X}$ is a sublattice of it, hence a Banach lattice on its own.

1.4 Examples of vector and Banach lattices

Most of the examples below are classical Banach spaces; see their definitions and further details in Appendix A.4.2.

1.4.1 Example. $C(\Omega)$ *spaces.* Let Ω be a topological space. *All topological spaces in this book will be assumed Hausdorff unless specified otherwise.* The space $C(\Omega)$ of all real-valued continuous functions on Ω is a sublattice of $\mathbb{R}^\Omega$ because the point-wise maximum of two continuous functions is again a continuous function. It follows that $C(\Omega)$ is itself an Archimedean vector lattice under point-wise lattice operations, so that, for example, $|f|(t) = \bigl|f(t)\bigr|$ for every $t \in \Omega$.

Note that $C(\Omega)$ does not have a "natural" norm. However, the vector subspace $C_b(\Omega)$ consisting of all bounded functions in $C(\Omega)$ is a Banach space under the sup-norm. Being a sublattice of $C(\Omega)$, $C_b(\Omega)$ is a vector lattice. It is easy to see that the sup-norm is a lattice norm; hence $C_b(\Omega)$ is a Banach lattice.

Let K be a compact space. *Again, all compact spaces in this book will be assumed Hausdorff unless specified otherwise.* In this case, $C(K) = C_b(K)$, hence $C(K)$ is a Banach lattice under the point-wise order and lattice operations. Spaces of this type will be referred to as $C(K)$-spaces; they will play a major role in our exposition. Our primary tool when dealing with $C(K)$ spaces will be Urysohn's Lemma A.2.2.

Let Ω be a locally compact topological space. The space $C_0(\Omega)$ of all functions in $C(\Omega)$ vanishing at infinity is a Banach lattice. For example, $C_0(\mathbb{R}) = \bigl\{f \in C(\mathbb{R}) : \lim\limits_{|t|\to\infty} f(t) = 0\bigr\}$. Clearly, $C_0(\Omega)$ is a closed sublattice of $C_b(\Omega)$, hence a Banach lattice itself. Note that if K is compact then $C_0(K) = C(K)$.

1.4.2 Example. Even though the supremum and infimum in $C(K)$ are computed pointwise for pairs, and, therefore, for finite sets, this is not true for infinite sets. Consider the following sequence in $C[0,1]$: suppose that f_n equals one at the origin, vanishes on $\bigl[\frac{1}{n}, 1\bigr]$ and is linear on $\bigl[0, \frac{1}{n}\bigr]$. Then $f_n \downarrow 0$ in $C[0,1]$. Note that $f_n \downarrow \mathbb{1}_{\{0\}}$ in $\mathbb{R}^{[0,1]}$.

Put $g_n = \mathbb{1} - f_n$. Then $g_n \uparrow \mathbb{1}$ in $C[0,1]$ even though $g_n(0) = 0$ for all n.

Similarly, consider the following sequence in $C[0,1]$: $h_n(t) = \sqrt[n]{t}$. Then $h_n \uparrow \mathbb{1}$ in $C[0,1]$, even though h_n converges pointwise to $\mathbb{1}_{(0,1]}$, and, therefore, $h_n \uparrow \mathbb{1}_{(0,1]}$ in $\mathbb{R}^{[0,1]}$.

1.4.3 Example. $L_p(\mu)$ *spaces.* Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. See Appendix A.3.1 for the definition of a measure space. In particular, while we do not generally assume measures to be finite or even σ-finite, we do assume that measures have no atoms of infinite measure.

Consider the space $L_0(\mu)$ of all real-valued measurable functions on Ω. As usual, we identify functions that are equal almost everywhere (***a.e.***), so that, formally speaking, $L_0(\mu)$ consists of equivalence classes of functions. For $f, g \in L_0(\mu)$, we write $f \leqslant g$ if $f(t) \leqslant g(t)$ a.e. It is an easy exercise that under this order $L_0(\mu)$ is an Archimedean vector lattice with lattice operations defined a.e. pointwise, i.e., $(f \vee g)(t) = f(t) \vee g(t)$, $(f \wedge g)(t) = f(t) \wedge g(t)$, and $|f|(t) = \bigl|f(t)\bigr|$ a.e.

Let $0 < p < \infty$. As usual, $L_p(\mu)$ stands for the linear subspace of $L_0(\mu)$ consisting of all $f \in L_0(\mu)$ such that $\int |f|^p \, d\mu$ is finite. Observe that $L_p(\mu)$ is a sublattice of $L_0(\mu)$ because if $f \in L_p(\mu)$ then $|f| \in L_p(\mu)$. Hence, $L_p(\mu)$ is itself a vector lattice. For $p \geqslant 1$, the space $L_p(\mu)$ is a Banach space under the norm $\|f\|_p = \bigl(\int |f|^p \, d\mu\bigr)^{\frac{1}{p}}$. Similarly, the space $L_\infty(\mu)$ of all essentially bounded functions in $L_0(\mu)$ equipped with the norm $\|f\|_\infty = \operatorname{ess\,sup}_{t \in \Omega} |f(t)|$ is a Banach lattice.

In most sequence and function spaces, elements have natural ***supports***. For $f \in \mathbb{R}^\Omega$ we write

$$\operatorname{supp} f = \{t \in \Omega \ : \ f(t) \neq 0\} = \{f \neq 0\}.$$

In particular, for $x = (x_i)$ in $\mathbb{R}^{\mathbb{N}}$, we write

$$\operatorname{supp} x = \{i \ : \ x_i \neq 0\}.$$

In a similar fashion one can define $\operatorname{supp} f$ for $f \in L_0(\mu)$ and, in particular, for any $f \in L_p(\mu)$, but one should keep in mind that in this case $\operatorname{supp} f$ is defined up to a set of measure zero. More precisely, f is an equivalence class, and if f_1 and f_2 are two representatives then the sets $\{f_1 \neq 0\}$ and $\{f_2 \neq 0\}$ may be different, but they differ by a set of zero measure. To avoid this ambiguity, one can identify $\operatorname{supp} f$ with the (equivalence class of the) characteristic function $\mathbb{1}_{\operatorname{supp} f}$ in $L_0(\mu)$.

1.4.4 Exercise. Show that if $0 < p < \infty$ and $f \in L_p(\mu)$ for some measure μ then $\operatorname{supp} f$ is σ-finite.

1.4.5. The preceding exercise often allows one to reduce arbitrary measures to σ-finite ones. For example, suppose that $0 < p < \infty$, (Ω, μ) is an arbitrary measure space, and

(f_n) is a sequence in $L_p(\Omega,\mu)$. Let $A=\bigcup_{n=1}^{\infty}\operatorname{supp} f_n$. By the exercise, the restriction of μ to A is σ-finite. We can consider the f_n's as elements of $L_p(A,\mu)$.

1.4.6 Example. Let Ω be an arbitrary set and $0<p\leqslant\infty$. Recall that $\ell_p(\Omega)=L_p(\Omega,\mu)$, where μ is the counting measure on Ω. It follows immediately that $\ell_p(\Omega)$ is a vector lattice under point-wise order and lattice operations; it is a sublattice of $\mathbb{R}^\Omega$. Furthermore, $\ell_p(\Omega)$ is a Banach lattice when $1\leqslant p\leqslant\infty$. In particular, $\ell_p=\ell_p(\mathbb{N})$ and $\ell_p^n=\ell_p\bigl(\{1,\dots,n\}\bigr)$ are Banach lattices.

1.4.7 Example. The following spaces are sublattices of $\mathbb{R}^{\mathbb{N}}$ and, therefore, are vector lattices under coordinate-wise order and lattice operations:

- the space c of all convergent sequences;
- the space c_0 of all sequences consisting converging to zero;
- the space c_{00} of all sequences that are eventually zero: $(x_1,\dots,x_n,0,0,\dots)$.

Furthermore, being closed sublattices of ℓ_∞, the spaces c and c_0 are Banach lattices under the sup-norm. The space c_{00} has no "canonical" norm; however, it is a dense sublattice of c_0 and ℓ_p for $1\leqslant p<\infty$. Thus, viewed as a subspace of ℓ_p or c_0 (or, equivalently, of ℓ_∞), c_{00} is a normed lattice. We will denote it by $\bigl(c_{00},\lVert\cdot\rVert_p\bigr)$.

Throughout the book, we will write e_n for the n-th unit vector in $\mathbb{R}^{\mathbb{N}}$, i.e., $e_n=(0,\dots,0,1,0,\dots)$, with the 1 in position n.

1.4.8 Example. Let X be the space of all affine functions (i.e., of the form $f(t)=at+b$) on $[-1,1]$, ordered pointwise. Note that while X is a linear subspace of $C[-1,1]$, it is not a sublattice. Indeed, let $f(t)=t$ and $g(t)=-t$. Then the supremum of f and g in $C[-1,1]$ is the function $|t|$, while the supremum of f and g in X (i.e., the least affine function on $[-1,1]$ which dominates both f and g) is $\mathbb{1}$. So the lattice operations on X are different from those on $C[-1,1]$.

Still, X is a vector lattice in its own right. For any $a,b\in\mathbb{R}$ there exists a unique $f\in X$ such that $f(-1)=a$ and $f(1)=b$; we denote this function by $f_{a,b}$. Moreover, if $c,d\in\mathbb{R}$ then $f_{a,b}\leqslant f_{c,d}$ iff $a\leqslant c$ and $b\leqslant d$. Hence, X can be identified with $\mathbb{R}^2$ with the coordinate-wise order. In particular, X is a vector lattice with

$$f_{a,b}\vee f_{c,d}=f_{a\vee c,b\vee d}\quad\text{and}\quad f_{a,b}\wedge f_{c,d}=f_{a\wedge c,b\wedge d}.$$

1.4.9 Exercise. Let X be the space of all affine functions on $\mathbb{R}$ with pointwise order. Show that X is an ordered vector space, but not a vector lattice. Note that the only positive elements are the positive constant functions; hence, the positive cone is not generating.

1.4.10 Exercise. Let $C^1[0,1]$ be the space of all continuously differentiable functions on $[0,1]$, with pointwise order. Show that it is an Archimedean ordered vector space. Show that not every element of $C^1[0,1]$ has a modulus; hence $C^1[0,1]$ is not a vector lattice. Observe, however, that every element of $C^1[0,1]$ is dominated by a positive constant function, hence the positive cone of $C^1[0,1]$ is generating.

1.4.11 Exercise. Consider $\mathbb{R}^2$ with lexicographic order as in Example 1.2.10. Verify that it is a vector lattice even though, as we showed in Example 1.2.10, it fails the Archimedean property. Show that it admits no lattice norm.

1.4.12 Example. *Order bounded sets depend on the ambient space.* The sequence (e_n) is order bounded in ℓ_∞ as it is contained in $[0,\mathbb{1}]$. However, the same sequence is not order bounded when viewed as a sequence in c_0, which is a closed sublattice of ℓ_∞. Nevertheless, it is easy to see that a set which is order bounded in a sublattice remains order bounded in the "big" space.

1.4.13. We observed in 1.3.21 that every order bounded set in a normed lattice X is bounded. Is the converse true? Clearly, it suffices to check whether the unit ball B_X is order bounded. This is clearly true in $C(K)$ and $L_\infty(\mu)$ spaces because $B_X=[-\mathbb{1},\mathbb{1}]$ there. However, it fails in c_0 or ℓ_p when $1\leqslant p<\infty$: the standard unit sequence (e_n) is contained in B_X but is not order bounded. The unit ball of $L_p[0,1]$ when $1\leqslant p<\infty$ is not order bounded either: the set of all the functions of the form $\lambda\mathbb{1}_A$ in the ball already fails to be order bounded. *Spoiler alert:* we will show in Section 8.1 that $C(K)$ spaces are the only Banach lattices (up to a lattice isomorphism) where the unit ball is order bounded and, correspondingly, where norm bounded and order bounded sets agree.

Note also that a convergent sequence in a Banach lattice need not be order bounded; take, for example, the sequence $x_n=\frac{1}{n}e_n$ in ℓ_1.

1.4.14. *The supremum/infimum of a set depends on the ambient space.* Let X be a vector lattice, Y a sublattice of X, and A a subset of Y. We write $\sup_Y A$ for the supremum of A in Y and $\sup A$ for the supremum of A in X. Several situations are possible:

(i) $\sup A$ and $\sup_Y A$ both exist but are different: see Example 1.4.2. In this case, $\sup_Y A\geqslant\sup A$;

(ii) $\sup A$ exists but $\sup_Y A$ doesn't: Put $X=\ell_\infty$, $Y=c_0$, and $A=\{e_n : n\in\mathbb{N}\}$.

(iii) $\sup_Y A$ exists but $\sup A$ doesn't: Put $X=C[0,1]$, and Y the sublattice of X consisting of the functions that are constant on $[\frac{1}{2},1]$. Let f_n be constant 1 on

$[0, \frac{1}{2} - \frac{1}{n}]$, constant 0 on $[\frac{1}{2}, 1]$, and linear in between. Then $\sup_Y f_n = \mathbb{1}$, but $\sup f_n$ does not exist.

(iv) If $\sup A$ exists and belongs to Y then $\sup_Y A$ exists and equals $\sup A$. Same for infima. In particular, if $\inf A = 0$ then $\inf_Y A = 0$.

1.4.15 Exercise. We know that an order interval in a Banach lattice is always closed and bounded. What about compactness? Verify whether order intervals are compact in ℓ_p when $1 \leqslant p < \infty$ but not when $p = \infty$. (We will discuss this issue further in 6.4.9 and in Theorem 10.8.2.)

1.4.16 Exercise. Even though $\mathbb{R}^{\mathbb{N}}$ is a vector lattice by Example 1.3.1, it admits no lattice norm.

1.4.17 Exercise. Let (Ω, ρ) be a metric space. A function $f\colon \Omega \to \mathbb{R}$ is said to be Lipschitz if there exists a constant $C > 0$ such that $\bigl|f(s) - f(t)\bigr| \leqslant C\rho(s,t)$ for all $s, t \in \Omega$. The least value of such C is called the ***Lipschitz constant*** of f. The space of all Lipschitz functions on (Ω, ρ) is denoted $\mathrm{Lip}(\Omega, \rho)$. Show that $\mathrm{Lip}(\Omega, \rho)$ is a vector lattice under pointwise order.

Fix $t_0 \in \Omega$ and consider the set Y of all functions in $\mathrm{Lip}(\Omega, \rho)$ that vanish at t_0. This is a sublattice of $\mathrm{Lip}(\Omega, \rho)$. The Lipschitz constant is a norm on Y. Show that, in general, it need not be a lattice norm.

1.4.18. Suppose that Y is a closed sublattice of a Banach lattice X, and (x_n) a norm null sequence in Y. If (x_n) is order bounded in X, does this imply that (x_n) is order bounded in Y? We will show in Example 18.8.7 that the answer is "no".

Throughout the book, we will see more examples of vector and Banach lattices, including spaces of measures in Sections 8.2 and 8.3 and Orlicz spaces in Chapter 18.

1.5 Disjoint vectors, order units, and Riesz decomposition

In this section, we collect several concepts that will be useful for us in the future.

1.5.1 Disjoint vectors

Given two vectors x and y in a vector lattice X, we say that x and y are ***disjoint*** and write $x \perp y$ if $|x| \wedge |y| = 0$.

1.5.1 Example. For two functions f, g in the space $\mathbb{R}^\Omega$ of all functions from Ω to $\mathbb{R}$, we have $f \perp g$ iff their supports are disjoint sets. The same is true in $C(\Omega)$ and in "sequence" spaces $\mathbb{R}^N$, ℓ_p, c_0, etc. In $L_p(\mu)$ for $0 \leqslant p \leqslant \infty$, two functions are disjoint iff the intersection of their support has zero measure.

1.5.2. For every vector x we have $x^+ \perp x^-$. Let X be an Archimedean vector lattice of dimension greater than one. By Exercise 1.2.12, X contains a non-zero vector which is neither positive nor negative. Hence X contains two non-zero disjoint vectors.

1.5.3 Exercise. If $x \perp y$ then

(i) $(\alpha x) \perp (\beta y)$ for any scalars α and β;
(ii) $|x + y| = |x| + |y| = |x| \vee |y|$;
(iii) If $|z| \leqslant |y|$ then $x \perp z$;
(iv) If $x \perp y$ and $x \perp z$ then $x \perp (y \vee z)$ and $x \perp (y + z)$.

1.5.4 Exercise. Given any x and y, the vectors $x - x \wedge y$ and $y - x \wedge y$ are positive and disjoint.

1.5.5. Given any vectors x and y and a positive scalar λ, it follows from $(x - \lambda y)^+ \perp (x - \lambda y)^-$ and $(x - \lambda y)^- = -(\lambda y - x)^+ = -\lambda(y - \frac{1}{\lambda}x)^+$ that $(x - \lambda y)^+ \perp (y - \frac{1}{\lambda}x)^+$.

A subset of X (or a sequence in X) is said to be disjoint if its members are pairwise disjoint. It is easy to see that Exercise 1.5.3 extends to finite sets. Moreover, we have the following.

1.5.6 Exercise. Let $x_1, \dots, x_n$ be a disjoint collection of vectors in a vector lattice and $\alpha_1, \dots, \alpha_n$ be scalars. Then

$$\Bigl|\sum_{i=1}^{n} \alpha_i x_i\Bigr| = \sum_{i=1}^{n} |\alpha_i||x_i| = \bigvee_{i=1}^{n} |\alpha_i||x_i|.$$

1.5.7 Exercise. Every disjoint collection of non-zero vectors is linearly independent.

1.5.8 Exercise. If a disjoint sequence in a normed lattice converges in norm then the limit is zero.

There are important connections between disjoint sequences and (Schauder) bases. For motivation, note that the canonical basis of ℓ_p for $1 \leqslant p < \infty$ or c_0 is a disjoint sequence. See Appendix A.4.1 for an overview of bases and basic sequences.

1.5.9 Proposition. *Every disjoint sequence (e_i) of non-zero vectors in a Banach lattice is a 1-unconditional basic sequence.*

Proof. We will use Theorem A.4.28. Suppose that $|\alpha_i| \leqslant |\beta_i|$ as $i = 1, \dots, n$. By Exercise 1.5.6,

$$\Bigl|\sum_{i=1}^{n} \alpha_i e_i\Bigr| = \bigvee_{i=1}^{n} |\alpha_i||e_i| \leqslant \bigvee_{i=1}^{n} |\beta_i||e_i| = \Bigl|\sum_{i=1}^{n} \beta_i e_i\Bigr|.$$

It follows that $\bigl\|\sum_{i=1}^{n} \alpha_i e_i\bigr\| \leqslant \bigl\|\sum_{i=1}^{n} \beta_i e_i\bigr\|$. □

Somewhat conversely, every 1-unconditional basis in a Banach space X yields a natural Banach lattice structure on X:

1.5.10 Exercise. Let X be a Banach space with a 1-unconditional basis (e_i). Define an order on X as follows: put $\sum_{i=1}^\infty \alpha_i e_i \leqslant \sum_{i=1}^\infty \beta_i e_i$ whenever $\alpha_i \leqslant \beta_i$ for every i. Show that under this order X is a Banach lattice with coordinate-wise lattice operations.

1.5.11. In particular, if a Banach lattice has a basis and the order is compatible with the basis (in the sense that $\sum_{i=1}^\infty \alpha_i x_i \leqslant \sum_{i=1}^\infty \beta_i x_i$ whenever $\alpha_i \leqslant \beta_i$ for every i) then the basis has to be positive, disjoint, and 1-unconditional. Such spaces are sometimes called ***Banach sequence spaces***. There are really just Banach spaces with a 1-unconditional basis.

1.5.2 Order units.

Let e be a positive vector in a vector lattice X. We say that e is a

- ***strong order unit*** if $\forall x \in X \;\; \exists \lambda \in \mathbb{R}_+ \;\; |x| \leqslant \lambda e$.
- ***weak order unit*** if $\forall x \in X \;\; x \perp e$ implies $x = 0$;

For brevity, we will often shorten these terms to *strong unit* and *weak unit.*

1.5.12 Proposition. *Every strong unit is also a weak unit.*

Proof. Suppose that $e \in X_+$ is a strong unit, and $x \perp e$. Then $|x| \leqslant \lambda e$ for some $\lambda \in \mathbb{R}_+$. Replacing λ with $\lambda \vee 1$ we may assume that $\lambda \geqslant 1$. Now $x \perp e$ implies

$$0 = |x| \wedge e \geqslant |x| \wedge \tfrac{1}{\lambda}|x| = \tfrac{1}{\lambda}|x|,$$

hence $x = 0$. □

As suggested by the name, having a strong unit is a very strong and restrictive property. We will see in Chapter 4 that vector lattices with strong units are in a certain sense close to $C(K)$ spaces.

1.5.13 Exercise. (i) Let K be a compact space. Show that $\mathbb{1}$ is a strong unit in $C(K)$. Show that a function $f \in C(K)_+$ is a strong unit iff $f(t) > 0$ for every $t \in K$;

(ii) Let $f \in C[0,1]$ be given by $f(t) = t$. Then f is a weak unit in $C[0,1]$ without being a strong order unit.

(iii) Let Ω be a topological space. Show that $\mathbb{1}$ is a strong unit in $C_b(\Omega)$. Is $\mathbb{1}$ a strong unit in $C(\Omega)$? Is it a weak unit?

(iv) Show that $\mathbb{1}$ is a strong unit in $L_\infty(\mu)$.
(v) Does $C_0(\mathbb{R})$ have a weak unit? A strong unit?
(vi) Show that c_{00} contains no weak units.
(vii) Show that $L_p[0,1]$ has no strong units when $0 \leqslant p < \infty$.
(viii) Show that c_0 and ℓ_p with $0 < p < \infty$ have no strong units.
(ix) Let $0 \leqslant f \in L_p(\mu)$ where $0 \leqslant p < \infty$. Show that f is a weak unit iff $f(t) \neq 0$ a.e., that is, $\operatorname{supp} f$ has full measure.
(x) Let $0 < p < \infty$. Show that $L_p(\mu)$ admits a weak unit iff μ is σ-finite. In particular, $\ell_p(\Omega)$ admits a weak unit iff Ω is countable.
(xi) Find a function f in $C[0,1]$ which is a weak unit in $C[0,1]$ but not in $L_p[0,1]$ ($0 \leqslant p < \infty$).

1.5.3 Riesz Decomposition Theorem

1.5.14 Theorem (Riesz Decomposition Theorem)**.** *Suppose that $u \leqslant z_1 + z_2$ for some positive u, z_1, and z_2. Then there exist positive x_1 and x_2 such that $u = x_1 + x_2$, $x_1 \leqslant z_1$, and $x_2 \leqslant z_2$.*

Proof. Put $x_1 = u \wedge z_1$ and $x_2 = u - x_1$. Then $x_1 \in [0, z_1]$ and $u = x_1 + x_2$. It follows from $x_1 \leqslant u$ that $x_2 \geqslant 0$. We only need to show that $x_2 \leqslant z_2$. By Exercise 1.3.3(iii),

$$x_2 = u - u \wedge z_1 = (u - z_1)^+ \leqslant z_2^+ = z_2.$$

$\square$

1.5.15 Exercise. Prove the analogous statement for $u \leqslant z_1 + \dots + z_n$. (Of course, use induction!)

The preceding exercise leads to another interesting property. Suppose that $u + v = z_1 + \dots + z_n$ for some positive u, v, $z_1, \dots, z_n$. By the exercise, we find $x_1, \dots, x_n$ such that $u = x_1 + \dots + x_n$ and $0 \leqslant x_i \leqslant z_i$ as $i = 1, \dots, n$. Put $y_i = z_i - x_i$; then $0 \leqslant y_i \leqslant z_i$, $z_i = x_i + y_i$, and $v = y_1 + \dots + y_n$. Thus, one can split each z_i into the sum of two positive vectors x_i and y_i such that $u = x_1 + \dots + x_n$ and $v = y_1 + \dots + y_n$. Using induction on the number of terms in the left hand side, one obtains the following extension of Riesz Decomposition Theorem.

1.5.16 Theorem. *Suppose that $\sum_{i=1}^m u_i = \sum_{j=1}^n z_j$ for some positive $u_1, \dots u_m$ and $z_1, \dots, z_n$. Then there exists a double array (x_{ij}) in X_+ with $i = 1, \dots, m$ and $j = 1, \dots, n$ such that*

$$u_i = \sum_{j=1}^n x_{ij} \text{ for every } i = 1, \dots, m, \text{ and } z_j = \sum_{i=1}^m x_{ij} \text{ for every } j = 1, \dots, n.$$

1.5.17 Exercise. Suppose that $\sum_{i=1}^k u_i \leqslant x + y$ for some positive $u_1, \dots, u_k, x, y$. Then there exist positive $v_1, \dots, v_k, w_1, \dots, w_k$ such that $\sum_{i=1}^k v_i \leqslant x$, $\sum_{i=1}^k w_i \leqslant y$, and $u_i = v_i + w_i$ for every $i = 1, \dots, k$.

1.5.18 Exercise. Let $|x| = u + v$ for some x and some positive vectors u and v in a vector lattice. Then there exist y and z such that $x = y + z$, $|y| = u$, and $|z| = v$.

The statement of this exercise may be easily extend to the case of more than two summands by induction. Combining it with Theorem 1.5.14 results in the following generalization of Riesz Decomposition Theorem.

1.5.19 Theorem. *Suppose that $|x| \leqslant v_1 + \dots + v_n$ for some positive $v_1, \dots, v_n$. Then $x = x_1 + \dots + x_n$ for some $x_1, \dots, x_n$ satisfying $|x_i| \leqslant |x| \wedge v_i$ as $i = 1, \dots, n$.*

Proof. Exercise 1.5.15 yields positive $u_1, \dots, u_n$ such that $|x| = u_1 + \dots + u_n$ and $u_i \leqslant v_i$ as $i = 1, \dots, n$. Applying the n-version of Exercise 1.5.18 to this identity finishes the proof. □

1.5.20 Exercise. For any $x, y \geqslant 0$, one has $[0, x] + [0, y] = [0, x + y]$.

1.6 Positive operators and lattice homomorphisms

Given two vector spaces X and Y, we write $\mathcal{L}(X, Y)$ for the vector space of all the linear operators from X to Y. Suppose that X and Y are ordered vector spaces such that X_+ is generating. A linear operator $T\colon X \to Y$ is said to be ***positive*** if it maps positive vectors to positive vectors. In this case we write $T \geqslant 0$. Again, with a slight abuse of notation, the zero operator is viewed as positive. It is easy to see that $T \geqslant 0$ iff $T(X_+) \subseteq Y_+$ iff $Tx \leqslant Ty$ whenever $x \leqslant y$ in X. Clearly, the sum of two positive operators is positive.

If $T \geqslant 0$ and $-T \geqslant 0$ then $T = 0$ because T vanishes on X_+; it now follows from $X = X_+ - X_+$ that T vanishes on X. Therefore, the set of all positive operators $\mathcal{L}(X, Y)_+$ is a cone in $\mathcal{L}(X, Y)$, making the latter into an ordered vector space. In particular, for two operators $S, T\colon X \to Y$ we write $T \leqslant S$ if $S - T \geqslant 0$ or, equivalently, $Tx \leqslant Sx$ whenever $x \in X_+$. An operator is ***regular*** if it can be written as the difference of two positive operators. In particular, Exercise 1.2.5 applies to regular operators. Observe also that the product (the composition) of positive operators is again positive.

1.6.1 Example. Let $n, m \in \mathbb{N}$ and suppose that $\mathbb{R}^n$ and $\mathbb{R}^m$ are ordered coordinate-wise as in Example 1.3.2. It is easy to see that an $m \times n$ matrix viewed as a linear operator from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a positive operator iff all its entries are non-negative. Of course, every matrix is a regular operator.

1.6.2 Example. Let X and Y be among ℓ_p with $1 \leqslant p < \infty$ or c_0 and T a bounded operator from X to Y. For every $i, j \in \mathbb{N}$, define $t_{ij} = (Te_j)_i$, where (e_j) is the standard unit basis of X. Then the infinite matrix $(t_{ij})_{i,j=1}^{\infty}$ completely determines T: for any $x \in X$, the i-th entry of Tx is

$$(Tx)_i = \Big(T\sum_{j=1}^{\infty} x_j e_j\Big)_i = \sum_{j=1}^{\infty} x_j (Te_j)_i = \sum_{j=1}^{\infty} t_{ij} x_j.$$

Furthermore, $T \geqslant 0$ iff $t_{ij} \geqslant 0$ for every $i, j \in \mathbb{N}$. In particular, for two operators $S, T\colon X \to Y$ we have $S \leqslant T$ iff $s_{ij} \leqslant t_{ij}$ for all $i, j \in \mathbb{N}$.

The preceding paragraph remains valid if ℓ_p or c_0 is replaced with any Banach sequence space as in 1.5.11. In particular, every basis projection is a positive operator.

Let X and Y be vector lattices. We have observed earlier that $\mathcal{L}(X, Y)$ is an ordered vector space. However, we will see later that it need not be a vector lattice.

1.6.3 Exercise. Suppose that $T\colon X \to Y$ is a positive operator between vector lattices. Then for every x and y in X one has

(i) $T(x \vee y) \geqslant (Tx) \vee (Ty)$,
(ii) $T(x \wedge y) \leqslant (Tx) \wedge (Ty)$, and
(iii) $|Tx| \leqslant T|x|$.

1.6.4 Example. Let $X = L_1(\mu)$ and $T\colon X \to \mathbb{R}$ given by $Tf = \int f$. Then T is a positive operator (functional). The inequality $|Tf| \leqslant T|f|$ corresponds to the standard fact of integration theory that $\left|\int f\right| \leqslant \int |f|$.

Suppose that $T \geqslant 0$ and $x \leqslant y$. It is easy to see that $T[x, y] \subseteq [Tx, Ty]$. We say that T is ***interval preserving*** if $T \geqslant 0$ and $T[x, y] = [Tx, Ty]$ whenever $x \leqslant y$ or, equivalently, if $T \geqslant 0$ and $T[0, a] = [0, Ta]$ for all $a \geqslant 0$.

1.6.5 Example. Let $X = L_p[-1, 1]$ and $T\colon X \to X$ is the "projection onto the right half": $Tf = f \cdot \mathbb{1}_{[0,1]}$. Then T is interval preserving.

1.6.6 Exercise. Let $T\colon \mathbb{R} \to \mathbb{R}^2$ given by $Tx = (x, x)$. Show that $T \geqslant 0$ but T is not interval preserving.

A linear operator $T\colon X \to Y$ is said to be a ***lattice homomorphism*** if T preserves the lattice operations, i.e., $T(x \vee y) = (Tx) \vee (Ty)$, $|Tx| = T|x|$, etc, for all $x, y \in X$.

Actually, it suffices that T preserves any of the lattice operations because then it preserves all of them by 1.3.9. A lattice homomorphism is positive because if $x \geqslant 0$ then $Tx = T|x| = |Tx| \geqslant 0$.

From the categorical perspective, lattice homomorphisms are the "natural" morphisms in the category of vector lattices: they are the maps between vector lattices that preserve all vector lattice operations (both linear and lattice). Similarly, positive operators are the morphisms in the category of ordered vector spaces: they preserve linear operations and order.

1.6.7 Exercise. Let $X = C(\Omega)$ for some topological space Ω, let $t \in \Omega$, and let $\delta_t\colon X \to \mathbb{R}$ be the evaluation functional of t, i.e., $\delta_t(f) = f(t)$. Show that δ_t is a lattice homomorphism.

1.6.8 Exercise. The range of a lattice homomorphism is a sublattice. The preimage of a sublattice under a lattice homomorphism is a sublattice.

Note that every sublattice is the range of a lattice homomorphism: just take the inclusion map.

1.6.9 Exercise. Let $S, T\colon X \to Y$ be two lattice homomorphisms and $Z = \{x \in X : Tx = Sx\}$. That is, $Z = \ker(S - T)$. Then Z a sublattice.

The following exercise provides a convenient criterion for an operator to be a lattice homomorphism: it asserts that a positive operator is a lattice homomorphism iff it preserves disjointness.

1.6.10 Exercise. A positive linear operator $T\colon X \to Y$ is a lattice homomorphism iff $a \perp b$ implies $Ta \perp Tb$ for all $a, b \in X$ (equivalently, for all $a, b \in X_+$).

1.6.11 Exercise. Every surjective lattice homomorphism is interval preserving.

If, in addition to being a lattice homomorphism, T is also bijection, we say that T is a ***lattice isomorphism***.

1.6.12 Exercise. Let $X = L_p[-1, 1]$ and $T\colon X \to X$ via $(Tf)(t) = f(-t)$. Show that T is a lattice isomorphism.

Two vector lattices X and Y are said to be lattice isomorphic if there exists a lattice isomorphism from X to Y. The following theorem guarantees that its inverse is again a lattice isomorphism.

1.6.13 Theorem. *Let $T\colon X \to Y$ be a linear bijection between vector lattices. TFAE.*

(i) *T is a lattice isomorphism;*
(ii) *T preserves suprema and infima of arbitrary sets;*
(iii) *$x \leqslant y$ iff $Tx \leqslant Ty$ for all $x, y \in X$;*
(iv) *$T \geqslant 0$ and $T^{-1} \geqslant 0$;*
(v) *T is interval preserving.*

Proof. The equivalence (i)⇔(ii)⇔(iii) follows from Exercise 1.1.7. The equivalence (iii)⇔(iv) is trivial. (i)⇒(v) by Exercise 1.6.11

(v)⇒(iv) We only need to show that $T^{-1} \geqslant 0$. Suppose $Tx \geqslant 0$ for some $x \in X$. Then $0 \leqslant Tx = |Tx| \leqslant T|x|$ since $T \geqslant 0$. Hence, $Tx \in \big[0, T|x|\big] = T\big[0, |x|\big]$. Since T is one-to-one, we have $x \in \big[0, |x|\big]$, so that $x \geqslant 0$. □

1.6.14 Example. Theorem 1.6.13 fails without the "bijection" assumption. For example, the natural inclusion map from $C[0,1]$ to $L_\infty[0,1]$ is a lattice homomorphism but fails to be interval preserving. On the other hand, the map $T\colon L_1[0,1] \to \mathbb{R}$ given by $Tf = \int f$ is clearly interval preserving, but fails to be a lattice homomorphism: put $f = \mathbb{1}_{[0,\frac{1}{2}]} - \mathbb{1}_{[\frac{1}{2},1]}$, then $Tf = 0$ but $T|f| = T\mathbb{1} = 1 \neq |Tf|$.

1.6.15. Suppose that $T\colon X \to Y$ is an injective lattice homomorphism. By Exercise 1.6.8, $\operatorname{Range} T$ is a sublattice of Y, hence a vector lattice itself. Viewed as an operator from X to $\operatorname{Range} T$, T is a lattice isomorphism between vector lattices, hence X is lattice isomorphic to a sublattice of Y. We say that T is a ***lattice isomorphic embedding*** of X into Y.

1.6.16 Exercise. Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be a linear bijection. Show that the following are equivalent.

(i) T is a lattice isomorphism;
(ii) $T \geqslant 0$ and the matrix of T has exactly one non-zero entry in every row and in every column;
(iii) T is a weighted permutation, i.e., there exists a permutation (a bijection) σ of $\{1, \dots, n\}$ and positive reals $\lambda_1, \dots, \lambda_n$ such that

$$T(x_1, \dots, x_n) = \big(\lambda_1 x_{\sigma(1)}, \dots, \lambda_n x_{\sigma(n)}\big)$$

for every $(x_1, \dots, x_n)$ in $\mathbb{R}^n$.

If X and Y are normed lattices, by a ***lattice isometry*** from X onto Y we mean a lattice isomorphism which is also a norm isometry, i.e., $\|Tx\| = \|x\|$ for all $x \in X$. In this case, we say that X and Y are ***lattice isometric***.

1.6.17. Consider $L_p(\mu) = L_p(\Omega, \mathcal{F}, \mu)$ with $1 \leqslant p < \infty$. Every weak unit u in $L_p(\mu)_+$ may be "converted" into $\mathbb{1}$ in the following sense. By Exercise 1.5.13(ix), $u(t) > 0$ a.e. Consider the measure $\nu = u^p \cdot \mu$ defined by $\nu(A) = \int_A u^p \, d\mu$ for a $A \in \mathcal{F}$. This measure is clearly finite. Define an operator $T\colon L_p(\nu) \to L_p(\mu)$ by $Tf = fu$. Observe that

$$\|Tf\|^p_{L_p(\mu)} = \int |Tf|^p \, d\mu = \int |f|^p u^p \, d\mu = \int |f|^p \, d\nu = \|f\|^p_{L_p(\nu)}.$$

for every $f \in L_p(\nu)$. It follows that T is a lattice isometry. Note that $u = T\mathbb{1}$. Note also that if, at the beginning of this process, we scale u so that $\|u\|_{L_p(\mu)} = 1$ then the resulting measure ν is a probability measure.

1.6.18. Consider $L_\infty(\mu) = L_\infty(\Omega, \mathcal{F}, \mu)$ where μ is σ-finite. Let (Ω_n) be a disjoint sequence of sets of finite measure in $\mathcal{F}$ such that $\Omega = \bigcup_{n=1}^\infty \Omega_n$. Define a new measure ν on $\mathcal{F}$ via

$$\nu(A) = \sum_{n=1}^\infty \frac{\mu(A \cap \Omega_n)}{2^n \mu(\Omega_n)}.$$

It is easy to see that ν is a probability measure. For a measurable function f, its essential suprema with respect to μ and to ν are the same. It follows that $L_\infty(\nu) = L_\infty(\mu)$ and the identity map is a lattice isometry.

1.6.19 Proposition. *Let $L_p(\mu) = L_p(\Omega, \mathcal{F}, \mu)$, where $1 \leqslant p \leqslant \infty$ and μ is a σ-finite measure. Then $L_p(\mu)$ is lattice isometric to $L_p(\nu)$ for some probability measure ν on $\mathcal{F}$.*

Proof. If $p = \infty$, the result follows immediately from 1.6.18. If $p < \infty$ then $L_p(\mu)$ admits a weak unit by Exercise 1.5.13(x) and the result follows from 1.6.17. $\square$

A normed lattice combines two structures: a normed space and a vector lattice. Hence, when dealing with operators between normed lattices, we have to distinguish between lattice isomorphisms (maps that preserve the vector lattice structure) and norm isomorphisms (maps that preserve the normed space structure). We will sometimes call the former "vector lattice isomorphism" to more clearly distinguish them from the latter.

Further, one can ask whether one of these two structures determines the other? If two normed lattices are isomorphic as normed spaces (norm isomorphic), do they have to be isomorphic as vector lattices (lattice isomorphic)? What about the converse? These are difficult questions, and we will see several results of this kind through the book. Here we present a short preview of some of these results.

Consider $\bigl(c_{00}, \|\cdot\|_p\bigr)$ and $\bigl(c_{00}, \|\cdot\|_q\bigr)$ where $p, q \in [1, \infty]$ with $p \neq q$. These two spaces are the same as vector lattices, but the norms are not equivalent. This shows that there exist normed lattices that are vector lattice isomorphic but not norm isomorphic. Hence, a vector lattice isomorphism T need not be a norm isomorphism.

In Banach lattices, the situation is dramatically different. We will show in Theorem 6.1.9 that every positive operator (in particular, every lattice homomorphism) between Banach lattices is bounded. It follows by Banach's Theorem A.4.4 that every vector lattice isomorphism between Banach lattices is automatically a norm isomorphism. So we will just use the term "lattice isomorphism" in such situations.

We now consider the converse question: to what extent does the norm structure determine the lattice structure? It is known that $L_2[0,1]$ and ℓ_2 are isomorphic (even isometric) as Banach spaces. However, they are not lattice isomorphic. Indeed, let a be a standard basis vector in ℓ_2, then $[0,a]$ consists of scalar multiples of a, i.e., a is an ***atom***. Since atoms are defined in vector lattice terms, every lattice isomorphism must send atoms to atoms. However, it is easy to see that $L_2[0,1]$ lacks any atoms. It follows that the two spaces cannot be lattice isomorphic.

However, there are several situations when the norm does tell us something about the lattice structure. We will show in Theorem 10.8.8 that if a Banach lattice is norm isomorphic to ℓ_1 then it is lattice isomorphic to ℓ_1. We will see in Corollary 10.7.2 that if a Banach lattice is a Hilbert space (under the same norm) then it is lattice isometric to $L_2(\mu)$ for some measure μ.

Obviously, every Banach lattice is a Banach space. Is the converse true: does every Banach space admit an order that makes it into a Banach lattice? The answer is "no"; we will provide counterexamples in Example 5.4.14 and Example 10.5.7.

Chapter 2

Completenesses and convergences

2.1 Order completeness

Let A and B be two subsets of an ordered set. We write $A \leqslant B$ if $a \leqslant b$ for all $a \in A$ and $b \in B$.

2.1.1 Proposition. *Let $\mathcal{S}$ be an ordered set. TFAE:*

(i) *For every two non-empty subsets A and B of $\mathcal{S}$, if $A \leqslant B$ then there exists $x \in \mathcal{S}$ such that $A \leqslant x \leqslant B$;*

(ii) *Every non-empty bounded above subset of $\mathcal{S}$ has a supremum;*

(iii) *Every non-empty bounded below subset of $\mathcal{S}$ has an infimum.*

Proof. (i)⇒(ii) Suppose that $A \subseteq \mathcal{S}$ is non-empty and bounded above, so that $A^{\uparrow}$ is nonempty. Then there exists $x \in \mathcal{S}$ such that $A \leqslant x \leqslant A^{\uparrow}$. Clearly, $x = \sup A$.

(ii)⇒(i) Suppose A and B are two non-empty subsets of $\mathcal{S}$ such that $A \leqslant B$. It follows that A is bounded above, hence $x = \sup A$ exists. Then $x \in A^{\uparrow} \supseteq B$, so that $A \leqslant x \leqslant B$.

The proof of (i)⇔(iii) is similar. □

If an ordered set $\mathcal{S}$ satisfies any of the equivalent conditions in Proposition 2.1.1, we say that $\mathcal{S}$ is ***order complete***. In literature, this property is often referred to as *Dedekind completeness* or *conditional order completeness*. Note that order completeness does not necessarily imply that $\mathcal{S}$ is a lattice. However, an ordered set $\mathcal{S}$ is an order complete lattice iff it satisfies the equivalent conditions of Proposition 2.1.1 and every pair of elements is order bounded.

In the categories of lattices or vector lattices, order completeness has nice equivalent characterizations. Recall that a net (x_α) in an ordered set $\mathcal{S}$ is bounded above if the set $\{x_\alpha\}$ is bounded above; that is, there exists an $s \in \mathcal{S}$ such that $x_\alpha \leqslant s$ for all α. Bounded below nets are defined similarly.

2.1.2 Proposition. *Let L be a lattice. Then TFAE:*

(i) *L is order complete;*

(ii) *Every increasing bounded above net in L has a supremum;*

(iii) *Every decreasing bounded below net in L has an infimum.*

Proof. The implications (i)⇒(ii) and (i)⇒(iii) follow immediately from Proposition 2.1.1.

To prove (ii)⇒(i), let A be a non-empty subset of L such that A is bounded above by some $u \in L$. In view of 1.1.6, $A^\vee$ is an increasing net, and $A^\vee \leqslant u$. By assumption, $\sup A^\vee$ exists. It is easy to see that $\sup A^\vee = \sup A$. Hence, L is order complete by Proposition 2.1.1(ii).

The proof of the implication (iii)⇒(i) is similar. □

In the case of a vector lattice, it suffices to consider sets (or nets) in X_+:

2.1.3 Exercise. Let X be a vector lattice. TFAE:

(i) X is order complete;

(ii) Every bounded above non-empty subset of X_+ has a supremum;

(iii) Every positive increasing order bounded net has a supremum.

It is often desirable to consider a "countable" or "sequential" variant of order completeness. A vector lattice X is said to be ***σ-order complete*** if it satisfies the equivalent conditions of Exercise 2.1.4:

2.1.4 Exercise. Let X be a vector lattice. TFAE:

(i) every bounded above countable set has a supremum;

(ii) every bounded below countable set has an infimum;

(iii) every increasing order bounded sequence has a supremum;

(iv) every decreasing order bounded sequence has an infimum;

(v) every positive increasing order bounded sequence has a supremum.

Condition (v) may be abbreviated as follows: $0 \leqslant x_n\uparrow \leqslant u \;\Rightarrow\; \sup x_n$ exists.

The following example shows that conditions (iii) and (iv) in Exercise 2.1.4 are not equivalent for general lattices.

2.1.5 Example. Let Ω be an uncountable set and let $\mathcal{S}$ be the set of all subsets A of Ω such that either A is at most countable or its complement is finite; let $\mathcal{S}$ be ordered by inclusion. It is easy to see that $\mathcal{S}$ is a lattice, and every increasing sequence of sets in $\mathcal{S}$ has its union (which is also its supremum) in $\mathcal{S}$. However, it is easy to find a decreasing sequence in $\mathcal{S}$ which has no infimum (even though it is trivially bounded below by the empty set).

We will show in Theorems 5.6.26 and 5.6.27 that, under some mild assumptions (which are satisfied when X is a Banach lattice), X is order complete iff every *disjoint* order bounded set has a supremum; similarly, X is σ-order complete iff every *disjoint* order bounded sequence has a supremum;

It is easy to see that order completeness implies σ-order completeness.

2.1.6 Proposition. *Every σ-order complete (in particular, every order complete) vector lattice is Archimedean.*

Proof. Suppose that $nx \leqslant u$ for all $n \in \mathbb{N}$. This means that the set $\{nx : n \in \mathbb{N}\}$ is bounded above by u. Then σ-order completeness implies that $s = \sup_n nx$ exists. For every $n \in \mathbb{N}$ we have $(n+1)x \leqslant s$, so that $nx \leqslant s - x$. It follows that $s \leqslant s - x$, hence $x \leqslant 0$. □

We will show later in Theorem 3.2.22 that every Archimedean vector lattice admits a unique *order completion*.

2.1.1 Examples of order complete vector lattices

2.1.7 Exercise. Show that

(i) $\mathbb{R}^\Omega$ is order complete for every set Ω;
(ii) ℓ_p with $0 < p \leqslant \infty$ and c_0 are order complete.

2.1.8 Example. *c is not σ-order complete, hence not order complete.* Let $x_n = (\underbrace{1,0,1,0,\dots,1,0,}_{2n}0,0,\dots)$. Then $x_n \uparrow\leqslant \mathbb{1}$, but $\sup x_n$ does not exist. To see this, suppose that $x = \sup x_n$ in c. Then every odd coordinate of x is equal to 1, so that $\lim x \geqslant 1$. It follows that some even coordinate of x must be positive. Replacing this coordinate with 0 we obtain a vector which is strictly less than x yet still dominates (x_n).

Note that, when viewed as a sequence in c_0, (x_n) is not order bounded, while when viewed as a sequence in ℓ_∞, it has a supremum there.

2.1.9. *Spaces c and c_0 are isomorphic as Banach spaces, but not as Banach lattices:* The map $T\colon c \to c_0$ given by

$$T\colon (x_1, x_2, x_3, \dots) \mapsto \bigl(x_\infty, x_1 - x_\infty, x_2 - x_\infty, \dots\bigr), \text{ where } x_\infty = \lim_{n\to\infty} x_n$$

is a surjective Banach space isomorphism. However, T is not a lattice isomorphism; it is not even positive. From the preceding exercises, we conclude that c and c_0 are not lattice isomorphic because a lattice isomorphism preserves order completeness.

2.1.10 Exercise. Let Ω be an uncountable set. Let X be the subspace of $\mathbb{R}^\Omega$ defined by

$$X = \{\lambda\mathbb{1} + f \;:\; \lambda \in \mathbb{R},\ \operatorname{supp} f \text{ is countable } \}.$$

That is, X consists of all countable (including finite) perturbations of constant functions. Show that X is a σ-order complete vector lattice which fails to be order complete.

2.1.11. Let X be a normed lattice. What is the relationship between its norm completeness and order completeness? Generally, none. For example, c is norm complete but not order complete. On the other hand, let X be ℓ_1 viewed as a normed sublattice of c_0, i.e., $X = (\ell_1, \|\cdot\|_\infty)$. As a vector lattice, X is just ℓ_1, hence is order complete. However, X is not complete as a normed space; its norm completion is c_0.

2.1.2 Order complete L_p-spaces.

Throughout this section, μ is an arbitrary measure and $0 \leqslant p \leqslant \infty$. We will show that $L_p(\mu)$ spaces are σ-order complete and "almost always" order complete. Recall that the pointwise supremum of a sequence of measurable functions is measurable; see Theorem A.3.3.

2.1.12 Proposition. *Let (f_n) be a sequence in $L_p(\mu)$. If (f_n) is bounded above then* $\sup f_n$ *exists in $L_p(\mu)$ and agrees (a.e.) with the pointwise supremum. If (f_n) is bounded below then* $\inf f_n$ *exists in $L_p(\mu)$ and agrees with the pointwise infimum.*

Proof. Let $L_p(\mu) = L_p(\Omega, \mathcal{F}, \mu)$; let $u \in L_p(\mu)$ such that $f_n \leqslant u$ for every n; let h be the pointwise supremum of (f_n). Let $A_n = \{f_n > u\}$, then $\mu(A_n) = 0$ for every n, so that the set $\Omega_0 = \Omega \setminus \left(\bigcup_{n=1}^\infty A_n\right)$ has full measure and $f_n(t) \leqslant u(t)$ for all $n \in \mathbb{N}$ and all $t \in \Omega_0$. It follows that $h(t) \leqslant u(t)$ for all $t \in \Omega_0$; hence $h(t) \leqslant u(t)$ a.e. It now follows from $f_n(t) \leqslant h(t) \leqslant u(t)$ a.e. that $h \in L_p(\mu)$ and $f_n \leqslant h \leqslant u$ for all n. Since u was an arbitrary upper bound of (f_n), we conclude that $h = \sup f_n$ in $L_p(\mu)$. The proof for infimum is similar. $\square$

2.1.13 Corollary. *$L_p(\mu)$ is σ-order complete.*

2.1.14 Proposition. *Let (f_n) be an increasing sequence in $L_p(\mu)$ and $f \in L_p(\mu)$. Then $f_n \uparrow f \;\Leftrightarrow f_n \xrightarrow{\text{a.e.}} f$. Same for decreasing sequences.*

Proof. For each n, the set $A_n = \{f_n > f_{n+1}\}$ has zero measure. Then the set $\Omega_0 = \Omega \setminus \left(\bigcup_{n=1}^\infty A_n\right)$ has full measure and $(f_n(t))$ is an increasing sequence for every $t \in \Omega_0$, hence a.e.

If $f_n \uparrow f$ then $\sup_n f_n(t) = f(t)$ a.e. by Proposition 2.1.12, so that $f(t) = \lim_n f_n(t)$ a.e. Conversely, if $f(t) = \lim_n f_n(t)$ a.e. then $f(t) = \sup_n f_n(t)$ a.e., hence $f = \sup f_n$ by Proposition 2.1.12. □

2.1.15 Example. Propositions 2.1.12 and 2.1.14 fail for uncountable nets! More precisely, the concepts of pointwise supremum of an uncountable set and of a.e.-limits of a net are not well defined in $L_p(\mu)$.

For every $t \in [0,1]$, let $f_t = \mathbb{1}_{\{t\}}$, the characteristic function of the singleton $\{t\}$. Then the pointwise supremum of this family $\{f_t : 0 \leqslant t \leqslant 1\}$ is $\mathbb{1}$. On the other hand, $f_t = 0$ a.e. for every t, so that, viewed as a family in $L_p[0,1]$, this family consists just of the zero function, hence its supremum in $L_p[0,1]$ is zero!

Furthermore, for every finite subset α of $[0,1]$, consider its characteristic function $\mathbb{1}_\alpha$. We may view $(\mathbb{1}_\alpha)$ as a net indexed by all finite subsets of $[0,1]$ ordered by inclusion. Clearly $\mathbb{1}_\alpha(t) \uparrow 1$ for every t, so the pointwise supremum and the pointwise limit of the net $(\mathbb{1}_\alpha)$ is $\mathbb{1}$. However, for every α we have $\mathbb{1}_\alpha = 0$ in $L_p[0,1]$, hence $\sup \mathbb{1}_\alpha$ in $L_p[0,1]$ is zero.

2.1.16 Theorem. *$L_p(\mu)$ is order complete provided that μ is σ-finite or $0 < p < \infty$.*

Proof. As in Exercise 2.1.3, it suffices to show that if D is a subset of $L_p(\mu)_+$ and $D \leqslant u$ for some $u \in L_p(\mu)$ then $\sup D$ exists. Replacing D with $D^\vee$, we may assume WLOG that D is closed under taking finite sups. For every $f \in D$, it follows from $0 \leqslant f \leqslant u$ that $\operatorname{supp} f \subseteq \operatorname{supp} u$. Hence, restricting all the functions involved to $\operatorname{supp} u$, we may assume that $\operatorname{supp} u = \Omega$, i.e., $u(t) > 0$ a.e. We may also assume WLOG that μ is σ-finite: if $0 < p < \infty$ then this follows from Exercise 1.4.4 (cf 1.4.5); if $p = 0$ or $p = \infty$ then this is a part of the assumption.

Special case: $u \in L_1(\mu)$. Then the set $A := \bigl\{\int f : f \in D\bigr\}$ is bounded above by $\int u$ in $\mathbb{R}$, hence $\lambda := \sup A$ exists. Find a sequence (f_n) in D such that $\int f_n \uparrow \lambda$. Replacing f_n with $f_1 \vee \dots \vee f_n$ if necessary, we may assume that $f_n \uparrow$. Proposition 2.1.14 guarantees that $h = \sup f_n$ exists in $L_p(\mu)$ and $f_n \xrightarrow{\text{a.e.}} h$. Monotone Convergence Theorem A.3.12 yields $\int h = \lim_n \int f_n = \lambda$.

We claim that $h = \sup D$. Clearly, $f_n \uparrow h$, so it suffices to show that $D \leqslant h$. Let $f \in D$. Since $f \vee f_n \in D$ for every n, we have

$$\lambda \geqslant \int f \vee f_n \geqslant \int f_n \ \uparrow\ \lambda, \quad \text{so that} \quad \int f \vee f_n \to \lambda.$$

On the other hand, $f \vee h = f \vee \sup f_n = \sup(f \vee f_n)$. Proposition 2.1.14 yields $(f \vee f_n) \xrightarrow{\text{a.e.}} (f \vee h)$, so that $\int f \vee f_n \to \int f \vee h$ by the Monotone Convergence

Theorem. This yields $\int f \vee h = \lambda = \int h$, so that $\int (f \vee h - h) = 0$. It follows that $f \vee h - h = 0$, so that $f \leqslant h$. Hence $D \leqslant h$.

General case. As μ is σ-finite, we can write $\Omega = \bigcup_{n=1}^{\infty} \Omega_n$ for some sets $\Omega_1 \subseteq \Omega_2 \subseteq \dots$ of finite measure. For each $0 \leqslant f \in L_p(\mu)$ and each $m \in \mathbb{N}$, put $\Phi_m(f) = (f \cdot \mathbb{1}_{\Omega_m}) \wedge m\mathbb{1}$. Let $D_m = \Phi_m(D)$ and $u_m = \Phi_m(u)$; then $D_m \leqslant u_m$. By *Special case*, $h_m = \sup D_m$ exists. Moreover, $h_m \uparrow \leqslant u$. By Proposition 2.1.13, $h := \sup h_m$ exists. We claim that $h = \sup D$. Indeed, for every $f \in D$ we have $\Phi_m(f) \in D_m \leqslant h_m \leqslant h$ for every m. Observe that $\Phi_m(f) \uparrow$ and $\Phi_m(f) \xrightarrow{\text{a.e.}} f$; it follows from Proposition 2.1.14 that $\Phi_m(f) \uparrow f$. Combining this with $\Phi_m(f) \leqslant h$, we get $f \leqslant h$. Hence, $D \leqslant h$. On the other hand, if $D \leqslant g$ for some $g \in L_p(\mu)$ then $D_m \leqslant g$ for every m, hence $h_m \leqslant g$, so that $h \leqslant g$.

$\square$

2.1.17 Example. *$L_0(\mu)$ and $L_\infty(\mu)$ need not be order complete when μ is not σ-finite.* Let Ω be an uncountable set. Declare a subset $A \subseteq \Omega$ measurable if either A or its complement A^C is countable. It is easy to see that such sets form a σ-algebra. Let μ be a counting measure, i.e., $\mu(A)$ is the cardinality of A if A is finite and $\mu(A) = \infty$ otherwise. Clearly, μ is not σ-finite. Fix $U \subset \Omega$ such that both U and U^C are uncountable. Consider the following collection of characteristic functions in $L_\infty(\mu)$:

$$\big\{\mathbb{1}_A \,:\, A \subset U \text{ and } A \text{ is countable}\big\}.$$

It is easy to see that this set has no supremum in $L_\infty(\mu)$ or in $L_0(\mu)$ even though it is bounded above by $\mathbb{1}$.

2.1.18 Exercise. Let $D \subseteq L_p(\mu)$ such that μ is σ-finite or $0 < p < \infty$. If D is bounded above then $\sup D = \sup x_n$ for some increasing sequence (x_n) in $D^\vee$.

2.1.19 Remark. We say that X satisfies the ***countable sup property*** if for every $D \subseteq X$ for which $\sup D$ exists there exists a countable subset C of D such that $\sup D = \sup C$. It follows from Exercise 2.1.18 that $L_p(\mu)$ spaces enjoy the countable sup property provided that μ is σ-finite or $0 < p < \infty$. Indeed, let (x_k) be the sequence in the exercise, then each x_k is the supremum of a finite subset of D. Let C be the union of all these finite subsets, then C is a countable subset of D and $\sup C = \sup x_k = \sup D$.

It is well known that for metrizable topological spaces, including normed spaces, many topological properties may be expressed using sequences rather than nets. Somewhat similarly, for a vector lattices with a countably sup property, certain properties may be expressed using sequences rather than nets.

2.1.20 Example. *$L_0(\mu)$ and $L_\infty(\mu)$ may fail the countable sup property when μ is not σ-finite.* Let Ω be an uncountable set, μ the counting measure on all the subsets of Ω, and D the set of all characteristic functions of finite subsets of Ω. Then $\sup D = \mathbb{1}$, but there is no sequence in D whose supremum is $\mathbb{1}$.

2.1.21 Exercise. $\mathbb{R}^{\mathbb{N}}$ has the countable sup property.

2.1.22. Here is an alternative proof of the *General case* in the proof of Theorem 2.1.16. The *Special case* of the proof shows that $L_1(\mu)$ is order complete. Observe that order completeness is an order property of X_+, hence it is preserved by order isomorphisms. When $0 < p < \infty$, $L_p(\mu)_+$ is order isomorphic to $L_1(\mu)_+$ via $\Phi(f) = f^p$; it follows that $L_p(\mu)$ is order complete. For $p = \infty$, observe that by Proposition 1.6.19, we may assume WLOG that μ is finite. Then $L_\infty(\mu) \subseteq L_1(\mu)$. It follows from the *Special case* that $h := \sup D$ exists in $L_1(\mu)$. It is now easy to verify that h is still equals $\sup D$ in $L_\infty(\mu)$. Finally, suppose that $p = 0$. Since $u(t) > 0$ a.e., the map $\Psi\colon L_0(\mu) \to L_0(\mu)$ given by $\Psi(f) = \frac{1}{u} f$ is a lattice isomorphism. Note that $\Psi(u) = \mathbb{1} \in L_\infty(\mu)$ and $\Psi(D) \subseteq [0, \mathbb{1}]$. By the preceding argument, $v := \sup \Psi(D)$ exists in $L_\infty(\mu)$. It is now easy to verify that $\Psi^{-1} v = uv$ is the supremum of D in $L_0(\mu)$.

2.1.3 Order complete $C(K)$ spaces.

2.1.23 Example. $C[a, b]$ *is not σ-order complete.* Let f_n in $C[-1, 1]$ be defined as follows:

$$f_n(t) = \begin{cases} 0 & \text{if } t \leqslant 0, \text{ and} \\ \sqrt[n]{t} & \text{if } t > 0. \end{cases}$$

Then $f_n \uparrow\leqslant \mathbb{1}$, but is an easy exercise that $\sup f_n$ does not exist. Hence, $C[-1, 1]$ is not σ-order complete. Similarly, one can show that $C[a, b]$ is not σ-order complete whenever $a < b$ in $\mathbb{R}$. Of course, this space is also not order complete.

Are $C(K)$ spaces ever order or σ-order complete? If the answer is “yes”, can we characterize such spaces in terms of topological properties of K? The first question has an easy affirmative answer: we know that $\mathbb{R}^n$ is order complete for every n, and we may view $\mathbb{R}^n$ as $C(K)$ where K is a finite set equipped with the discrete topology. We are going to show that $C(K)$ is order complete whenever K is “sufficiently discrete” or, rather, “sufficiently disconnected”.

A subset of a topological space Ω is ***clopen*** if it is simultaneously closed and open. Note that A is clopen iff its characteristic function $\mathbb{1}_A$ is continuous. Recall that Ω is disconnected if it can be written as a union of two non-empty disjoint open (and, therefore, clopen) sets. Otherwise, Ω is connected. That is, Ω is connected iff the only clopen sets in Ω are $\varnothing$ and Ω itself; Ω is disconnected iff there exist proper non-empty clopen sets. We will consider a hierarchy of conditions that require that a topological space has “sufficiently many” clopen sets, hence is “highly disconnected”. They have rather fancy names and may look somewhat pathological at the first glance. But we will see they arise naturally in many situations. As before, we restrict our attention to compact (Hausdorff) spaces.

Let K be a compact space. We say that K is ***totally disconnected*** if every point has a base of clopen neighbourhoods. K is said to be ***extremally disconnected*** if the closure of every open set is open (hence, clopen). The latter condition has a sequential variant: K is ***basically disconnected*** if the closure of every open F_σ set is open; recall that a set is F_σ if it is a union of a sequence of closed sets. Open F_σ sets come up naturally as supports of continuous functions:

2.1.24 Exercise. For a subset U of a compact topological space K, TFAE:

(i) U is an open F_σ set;

(ii) $U = \bigcup_{n=1}^{\infty} V_n$ for some sequence (V_n) of open sets such that $\overline{V_n} \subseteq V_{n+1}$ for every n;

(iii) $U = \{f \neq 0\}$ for some $f \in C(K)$ (i.e., U is a ***cozero set***).

2.1.25 Exercise. For a compact space, we have the following implications:

$$\text{extremally disconnected} \quad \Rightarrow \quad \text{basically disconnected} \quad \Rightarrow \quad \text{totally disconnected.}$$

Extremally and basically disconnected compact spaces are sometimes called ***Stonean*** and ***quasi-Stonean***, respectively. One may think of basically disconnected as σ-extremally disconnected.

2.1.26 Exercise. A compact space K is totally disconnected iff every connected subset of K is a singleton.

2.1.27 Exercise. Let Δ be the Cantor ternary set. Show that Δ is totally disconnected but not basically disconnected.

Recall that for every set Γ there is a compact space $\beta\Gamma$, called the ***Stone-Čech compactification*** of Γ, such that Γ is a dense subset of $\beta\Gamma$ and every function from Γ to any compact space K extends uniquely to a continuous function from $\beta\Gamma$ to K. In particular, every real-valued bounded function on Γ extends uniquely to a continuous real-valued function on $\beta\Gamma$. This space is unique up to a homeomorphism fixing Γ.

2.1.28 Exercise. Show that $\beta\Gamma$ is extremally disconnected.

2.1.29 Theorem. *Given a compact space K.*

(i) *$C(K)$ is order complete iff K is extremally disconnected;*

(ii) *$C(K)$ is σ-order complete iff K is basically disconnected.*

Proof. We will only prove (ii); the other statement is similar and is left as an exercise. Suppose that the closure of every open F_σ subset of K is open. Let $f_n \uparrow\leqslant h$ in $C(K)$; we need to show that $\sup f_n$ exists. Let g be the point-wise supremum, i.e., $g(t) = \sup f_n(t)$ for every t. Clearly, $f_n(t) \leqslant g(t) \leqslant h(t)$ for all n and t. However, g

need not be continuous. For each $t \in K$ let $\mathcal{N}_t$ be the set of all open neighbourhoods of t and put

$$f(t) = \inf_{V \in \mathcal{N}_t} \sup_{s \in V} g(s).$$

It is clear that $g(t) \leqslant f(t) \leqslant h(t)$. It is left to show that f is continuous; then $f = \sup f_n$.

Fix $a \in \mathbb{R}$. We want to show that the sets $\{f < a\}$ and $\{f > a\}$ are open. Suppose $f(t) < a$ for some t. Then there exist $V \in \mathcal{N}_t$ such that $\sup_{s \in V} g(s) < a$. Take any $r \in V$, then $V \in \mathcal{N}_r$; it follows that $f(r) \leqslant \sup_{s \in V} g(s) < a$. It follows that $V \subseteq \{f < a\}$. Therefore, the set $\{f < a\}$ is open and, therefore, $\{f \geqslant a\}$ is closed.

Observe that

$$\{g > a\} = \bigcup_{n=1}^{\infty} \{f_n > a\} = \bigcup_{n,m=1}^{\infty} \{f_n \geqslant a + \tfrac{1}{m}\}.$$

The first equality implies that $\{g > a\}$ is open; the second equality yields that it is an F_σ set. Therefore, by assumption, the set $\overline{\{g > a\}}$ is clopen. Clearly, $\{g > a\} \subseteq \{g \geqslant a\} \subseteq \{f \geqslant a\}$. Since $\{f \geqslant a\}$ is closed, we have $\overline{\{g > a\}} \subseteq \{f \geqslant a\}$.

Fix any $\varepsilon > 0$; if $f(t) \geqslant a$ then for every $V \in \mathcal{N}_t$ we have $\sup_{s \in V} g(s) \geqslant a$, so that there exists $s \in V$ with $g(s) > a - \varepsilon$. This yields $\{f \geqslant a\} \subseteq \overline{\{g > a - \varepsilon\}}$. Combining the inclusions, we get

$$\overline{\{g > a\}} \subseteq \{f \geqslant a\} \subseteq \overline{\{g > a - \varepsilon\}}.$$

It follows that

$$\overline{\{g > a + \tfrac{1}{m}\}} \subseteq \{f \geqslant a + \tfrac{1}{m}\} \subseteq \overline{\{g > a + \tfrac{1}{m+1}\}}$$

for every m, so that

$$\{f > a\} = \bigcup_{m=1}^{\infty} \{f \geqslant a + \tfrac{1}{m}\} = \bigcup_{m=1}^{\infty} \overline{\{g > a + \tfrac{1}{m}\}},$$

a union of clopen sets. Therefore, $\{f > a\}$ is open.

Suppose now that $C(K)$ is σ-order complete and let U be an open subset of K with $U = \bigcup_{n=1}^{\infty} F_n$, where each F_n is closed. We need to show that $\overline{U}$ is clopen or, equivalently, that $\mathbb{1}_{\overline{U}}$ is continuous.

For every n, By Urysohn's Lemma A.2.2, there exists $f_n \in C(K)$ such that $0 \leqslant f_n \leqslant \mathbb{1}$, f_n equals 1 on F_n and vanishes outside of U. Let $f = \sup f_n$ in $C(K)$. Clearly, $0 \leqslant f \leqslant \mathbb{1}$. We claim that $f = \mathbb{1}_{\overline{U}}$; this will finish the proof.

Let $t \in U$. Then $t \in F_n$ for some n, so that $1 \geqslant f(t) \geqslant f_n(t) = 1$. It follows that f equals 1 on U and, therefore, on $\overline{U}$. On the other hand, let $t \notin \overline{U}$. By Urysohn's lemma, we can find $g \in C(K)$ such that $0 \leqslant g \leqslant \mathbb{1}$, g equals 1 on $\overline{U}$ and $g(t) = 0$. It follows that $f_n \leqslant g$ for every n, so that $f \leqslant g$ and, therefore, $f(t) \leqslant g(t) = 0$. Hence, $f(t) = 0$. □

2.2 Principal ideals and uniform completeness

Let X be a vector lattice and $e \in X_+$. We define the ***principal ideal of*** e as follows:

$$I_e = \bigl\{x \in X \,:\, \lvert x\rvert \leqslant \lambda e \text{ for some } \lambda \in \mathbb{R}_+\bigr\}.$$

That is, $x \in I_e$ iff $\lvert x\rvert$ is dominated by a positive scalar multiple of e. We summarize some basic properties of principal ideals in the following exercise.

2.2.1 Exercise. (i) $x \in I_e$ iff $\lvert x\rvert \leqslant ne$ for some $n \in \mathbb{N}$.

(ii) I_e is a sublattice of X.

(iii) $I_e = \bigcup_{n=1}^{\infty}[-ne, ne] = \operatorname{span}[-e, e]$.

(iv) e is a strong unit iff $I_e = X$.

(v) e is a strong unit for I_e.

(vi) If $x \in I_e$ and $\lvert y\rvert \leqslant \lvert x\rvert$ then $y \in I_e$.

(vii) Let $x_1, \dots, x_n \in X$. Put $e = \bigvee_{i=1}^{n}\lvert x_i\rvert$. Then $x_1, \dots, x_n \in I_e$.

(viii) Let (x_n) be a sequence in a Banach lattice X. Then (x_n) is contained in a principal ideal.

2.2.2. Even when X is a Banach lattice, I_e is rarely closed in X. However, one can equip I_e with a norm of its own. For $x \in X$, we define

$$\lVert x\rVert_e = \inf\bigl\{\lambda \geqslant 0 \,:\, \lvert x\rvert \leqslant \lambda e\bigr\}.$$

That is, $\lVert\cdot\rVert_e$ is the Minkowski functional of $[-e, e]$. Clearly, $\lVert x\rVert_e < \infty$ iff $x \in I_e$.

2.2.3 Exercise. Let X be an Archimedean vector lattice and $e \in X_+$.

(i) $\lvert x\rvert \leqslant \lVert x\rVert_e e$ for every $x \in I_e$.

(ii) $\bigl(I_e, \lVert\cdot\rVert_e\bigr)$ is a normed lattice.

(iii) The closed unit ball of $\lVert\cdot\rVert_e$ is the order interval $[-e, e]$.

(iv) If $e \leqslant u$ then $\lVert\cdot\rVert_e \geqslant \lVert\cdot\rVert_u$.

(v) If X is a normed lattice then the inclusion of $\bigl(I_e, \lVert\cdot\rVert_e\bigr)$ into X is continuous.

2.2.4 Example. (i) Let $X = C(K)$ and $e = \mathbb{1}$. Then $I_e = X$ and $\lVert\cdot\rVert_e$ is the usual supremum norm on $C(K)$.

(ii) Let $X = C[0,1]$ and $e \in X_+$ is given by $e(t) = t$. Clearly, if $f \in I_e$ then $f(0) = 0$. The converse is false: the function $\sqrt{t}$ is not in I_e, even though it vanishes at 0. More generally, for $e \in C(K)_+$ and $f \in I_e$, we always have $\operatorname{supp} f \subseteq \operatorname{supp} e$.

(iii) Let $X = L_p(\mu)$, where μ is a finite measure and $0 \leqslant p \leqslant \infty$, and $e = \mathbb{1}$. Then $I_e = L_\infty(\mu)$ and $\|\cdot\|_e = \|\cdot\|_\infty$. This example shows that I_e need not be norm closed.

(iv) Let $X = L_p(\mu)$ for some measure μ and $0 \leqslant p \leqslant \infty$. Let $e = \mathbb{1}_A$ where A is a measurable set with $\mu(A) < \infty$. Then $I_e = L_\infty(A, \mu)$ and $\|f\|_e = \|f \cdot \mathbb{1}_A\|_\infty$.

Let X be an Archimedean vector lattice and $e \in X_+$. Exercise 2.2.3 says that $\left(I_e, \|\cdot\|_e\right)$ is a normed lattice. “Spoiler alert”: it will be proved in Chapter 4 that the completion of $\left(I_e, \|\cdot\|_e\right)$ is lattice isometric to $C(K)$ for some compact K. We will see that quite often $\left(I_e, \|\cdot\|_e\right)$ is already complete, in which case $\left(I_e, \|\cdot\|_e\right)$ itself is lattice isometric to $C(K)$.

In view of the preceding paragraph, the following exercise is not surprising because these identities clearly hold in $C(K)$-spaces:

2.2.5 Exercise. Let X be a vector lattice, $x, y \in X$, and $e \in X_+$. Then:

(i) If $x, y \geqslant 0$ then $\|x \vee y\|_e = \|x\|_e \vee \|y\|_e$;
(ii) If $x \perp y$ then $\|x + y\|_e = \|x\|_e \vee \|y\|_e$.

The following exercise yields concrete descriptions of principal ideals in $C(K)$, and $L_p(\mu)$, and shows that they are rarely closed.

2.2.6 Exercise. Let K be a compact space.

(i) Let $u \in C(K)_+$. Put $\Omega = \{u \neq 0\}$. Then $g \in I_u$ iff there exists $h \in C_b(\Omega)$ such that $g(t) = u(t)h(t)$ for every $t \in \Omega$ and g vanishes on Ω^C. The map that sends g to h is a lattice isomorphism between I_u and $C_b(\Omega)$.
(ii) If $u \in C(K)_+$ then $\sqrt{u} \in \overline{I_u}$, where $\overline{I_u}$ is the (norm) closure of I_u in $C(K)$.
(iii) Let $X = C[0,1]$ and $u(t) = t$. Show that $\sqrt{u}$ is in $\overline{I_u}$ but not in I_u, hence I_u is not closed.
(iv) For $u \in C(K)$, I_u is closed iff the set $\{u \neq 0\}$ is clopen.

2.2.7 Exercise. Let $X = L_p(\mu)$ for some measure space $(\Omega, \mathcal{F}, \mu)$ and $0 \leqslant p \leqslant \infty$. Let $u \in L_p(\mu)_+$. Show that I_u is lattice isomorphic to $L_\infty(A, \mu)$, where $A = \{u \neq 0\}$ up to a set of measure zero.

2.2.8. *Non-Archimedean case.* Let X be a vector lattice. If we do not assume that X is Archimedean, $\|\cdot\|_e$ is only a seminorm. In fact, X is Archimedean iff $\|\cdot\|_e$ is a norm for every $e \in X_+$. Indeed, suppose that X is not Archimedean. Then we can find $0 < x < e$ such that $nx \leqslant e$ for all $n \in \mathbb{N}$. It follows that $\|x\|_e \leqslant \frac{1}{n}$ for all n, hence $\|x\|_e = 0$.

An Archimedean vector lattice X is ***uniformly complete*** if $\left(I_e, \|\cdot\|_e\right)$ is norm complete for every positive e in X. We are going to prove that the class of uniformly complete vector lattices is very large: it includes σ-order complete vector lattices and Banach lattices. Recall that by Proposition 2.1.6 and Exercise 1.3.22, every σ-order complete vector lattice and every normed lattice is Archimedean.

2.2.9 Theorem. *Every σ-order complete vector lattice and every Banach lattice is uniformly complete.*

Proof. Suppose that $e \in X_+$ and (x_n) is a Cauchy sequence in $\bigl(I_e, \|\cdot\|_e\bigr)$. We want to show that it converges in $\|\cdot\|_e$. There is a subsequence (x_{n_k}) such that $\|x_{n_k} - x_n\|_e \leqslant \frac{1}{k}$ whenever $n \geqslant n_k$. Passing to this subsequence and applying Exercise 2.2.3(i), we may assume that

$$|x_k - x_n| \leqslant \tfrac{1}{k}e \tag{2.1}$$

whenever $n \geqslant k$.

Case: X is σ-order complete. Fix k. It follows from (2.1) that

$$x_k - \tfrac{1}{k}e \leqslant x_n \leqslant x_k + \tfrac{1}{k}e.$$

for all $n \geqslant k$. Put $a_k \coloneqq \inf\limits_{n \geqslant k} x_n$ and $b_k \coloneqq \sup\limits_{n \geqslant k} x_n$; these exist because X is σ-order complete. Note that

$$x_k - \tfrac{1}{k}e \leqslant a_k \leqslant b_k \leqslant x_k + \tfrac{1}{k}e \tag{2.2}$$

for every k. Moreover, $a_k \uparrow$, $b_k \downarrow$, and $a_k \leqslant b_m$ for all k, m. It follows that $a_k \uparrow a$ and $b_k \downarrow b$ for some $a, b \in X$, and $a \leqslant b$. For each k, it follows from (2.2) that

$$0 \leqslant b - a \leqslant b_k - a_k \leqslant \bigl(x_k + \tfrac{1}{k}e\bigr) - \bigl(x_k - \tfrac{1}{k}e\bigr) = \tfrac{2}{k}e.$$

Therefore, $a = b$. It follows that $x_k - \frac{1}{k}e \leqslant a \leqslant x_k + \frac{1}{k}e$ for every k, so that $|x_k - a| \leqslant \frac{1}{k}e$, which yields $\|x_k - a\|_e \leqslant \frac{1}{k}$.

Case: X is a Banach lattice. Then (2.1) yields $\|x_k - x_n\| \leqslant \frac{1}{k}\|e\|$ whenever $n \geqslant k$. It follows that (x_n) is Cauchy in X, hence $x_n \xrightarrow{\|\cdot\|} a$ for some $a \in X$. Since lattice operations are norm continuous, it follows from (2.1) that $|x_k - a| \leqslant \frac{1}{k}e$. It now follows easily $a \in I_e$ and that $x_n \xrightarrow{\|\cdot\|_e} a$. $\square$

2.2.10 Corollary. *Let X be a Banach lattice with a strong unit e. Then $\|\cdot\|_e$ is equivalent to the original norm of X.*

Proof. The identity operator $I\colon \bigl(X, \|\cdot\|_e\bigr) \to \bigl(X, \|\cdot\|\bigr)$ is continuous by Exercise 2.2.3(v), hence is an isomorphism by Banach's Theorem A.4.4. $\square$

This yields some useful equivalent characterizations of strong units in Banach lattices:

2.2.11 Exercise. Suppose that X is a Banach lattice and $e \in X_+$. TFAE:

(i) e is a strong unit;

(ii) $e \in \operatorname{Int} X_+$;

(iii) $\lambda B_X \subseteq [-e, e]$ for some $\lambda > 0$.

In particular, X has a strong unit iff X_+ has non-empty interior iff B_X is order bounded.

2.2.12. It is easy to see that the positive cone of $L_p[0,1]$ with $1 \leqslant p < \infty$ has empty interior because every positive function may be perturbed on a set of arbitrary small measure so that the resulting function is no longer positive. Similarly, c_0 and ℓ_p with $1 \leqslant p < \infty$ has empty interior. In view of Exercise 2.2.11, this implies that these spaces have no strong units, which we already observed in Exercise 1.5.13.

2.2.13 Example. *Uniform completeness does not imply order or σ-order completeness:* $C[0,1]$ is uniformly complete but not σ-order complete.

2.2.14 Example. Corollary 2.2.10 and Exercise 2.2.11 may fail for normed lattices. Let X be $C[0,1]$ equipped with the integral norm $\|f\| = \int_0^1 \bigl|f(t)\bigr|dt$ instead of the usual supremum norm, and put $e = \mathbb{1}$. Then e is a strong unit, and $\|\cdot\|_e$ is the supremum norm on $C[0,1]$, which is not equivalent to the integral norm. Furthermore, e is not an interior point of X_+. This example also shows that uniform completeness need not imply norm completeness.

2.2.15 Example. *A normed lattice need not be uniformly complete.* Let X be the subset of ℓ_∞ consisting of all eventually constant sequences. That is, X is the linear span of all standard unit vectors e_n and $\mathbb{1}$. Clearly, X is a sublattice of ℓ_∞, hence $\bigl(X, \|\cdot\|_\infty\bigr)$ is a normed lattice, with $\mathbb{1}$ being a strong unit. Note that X is not closed in ℓ_∞, hence is not norm complete. Since $X = I_{\mathbb{1}}$, it follows that X is not uniformly complete.

Uniform completeness is preserved under a surjective lattice homomorphism:

2.2.16 Proposition. *Let X and Y be two Archimedean vector lattices and $T\colon X \to Y$ a surjective lattice homomorphism. If X is uniformly complete then so is Y.*

Proof. Let $u \in Y_+$, it suffices to show that $\bigl(I_u, \|\cdot\|_u\bigr)$ is complete.

Find $e \in X$ such that $u = Te$. Then $u = |u| = T|e|$, hence, WLOG, $e \geqslant 0$. For every $x \in I_e$ we have $|x| \leqslant \|x\|_e e$, so that $|Tx| \leqslant \|x\|_e u$, hence $\|Tx\|_u \leqslant \|x\|_e$.

We claim that for every $y \in I_u$ there exists $x \in I_e$ such that $Tx = y$ and $\|x\|_e \leqslant \|y\|_u$. Indeed, let $y \in I_u$. Let $\lambda = \|y\|_u$, then $-\lambda u \leqslant y \leqslant \lambda u$. Take $z \in X$ such that $Tz = y$. "Truncate" z above and below: put $x = (z \wedge \lambda e) \vee (-\lambda e)$. Then $x \in I_e$, moreover, $-\lambda e \leqslant x \leqslant \lambda e$ implies $\|x\|_e \leqslant \lambda$, and

$$Tx = (Tz \wedge \lambda Te) \vee (-\lambda Te) = (y \wedge \lambda u) \vee (-\lambda u) = y$$

This proves the claim.

It follows that the operator from $I_e/\ker T$ to I_u defined by $x+\ker T \mapsto Tx$ is a surjective isometry. Therefore, I_u is isometric to $I_e/\ker T$, hence is complete. $\square$

2.3 Uniform convergence

Let X be an Archimedean vector lattice. For a net (x_α) in X, we say that the net converges ***relatively uniformly*** or just ***uniformly*** to some $x \in X$ and write $x_\alpha \xrightarrow{\mathrm{u}} x$ if there exists $e > 0$ such that $\|x_\alpha - x\|_e \to 0$. In this case, we say that (x_α) converges to x ***uniformly relative to*** e. We say that e is a ***regulator*** for the convergence. Note that this implies that $\|x_\alpha - x\|_e$ is finite for all sufficiently large α's, i.e., $x - x_\alpha \in I_e$.

It follows from the definition of $\|\cdot\|_e$ and from Exercise 2.2.3(i) that $x_\alpha \xrightarrow{\mathrm{u}} x$ iff

$$\exists e \in X_+ \quad \forall \varepsilon > 0 \quad \exists \alpha_0 \quad \forall \alpha \geqslant \alpha_0 \quad |x_\alpha - x| \leqslant \varepsilon e. \tag{2.3}$$

Since $\big(I_e, \|\cdot\|_e\big)$ is a normed space for every e, uniform convergence is sequential in nature:

2.3.1 Exercise. (i) Uniform limits are unique.

(ii) If $x_\alpha \xrightarrow{\mathrm{u}} x$ then there exists an increasing sequence of indices (α_n) such that $x_{\alpha_n} \xrightarrow{\mathrm{u}} x$.

(iii) If $x_n \xrightarrow{\mathrm{u}} x$ and (x_{n_k}) is a subsequence of (x_n) then $x_{n_k} \xrightarrow{\mathrm{u}} x$.

(iv) If $x_\alpha \xrightarrow{\mathrm{u}} x$ in X and $\lambda_\alpha \to \lambda$ in $\mathbb{R}$ then $\lambda_\alpha x_\alpha \xrightarrow{\mathrm{u}} \lambda x$.

(v) If $x_\alpha \xrightarrow{\mathrm{u}} x$ and $y_\alpha \xrightarrow{\mathrm{u}} y$ then $x_\alpha + y_\alpha \xrightarrow{\mathrm{u}} x+y$, $x_\alpha \wedge y_\alpha \xrightarrow{\mathrm{u}} x \wedge y$, and $x_\alpha \vee y_\alpha \xrightarrow{\mathrm{u}} x \vee y$;

(vi) If $x_\alpha \geqslant 0$ for all α and $x_\alpha \xrightarrow{\mathrm{u}} x$ then $x \geqslant 0$;

(vii) Every uniformly convergent net has an order bounded tail. Every uniformly convergent sequence is order bounded.

(ii) says that uniform convergence is sequential in nature. It follows from (iv) and (v) that linear and lattice operations are uniformly continuous.

2.3.2 Exercise. If e is a strong unit in X then $x_\alpha \xrightarrow{\mathrm{u}} x$ iff $\|x_\alpha - x\|_e \to 0$.

2.3.3 Example. If $X = C(K)$ for some compact space K and $e = \mathbb{1}$, then $\|\cdot\|_e$ is the usual sup-norm of $C(K)$. In particular, $x_\alpha \xrightarrow{\mathrm{u}} x$ iff $x_\alpha \xrightarrow{\|\cdot\|} x$ (this justifies the term "uniform convergence"). Similarly, on $L_\infty(\mu)$, uniform convergence agrees with norm convergence.

We will show later that in $L_p(\mu)$ with $p < \infty$, a sequence converges uniformly iff it is order bounded and converges almost everywhere; see Exercise 2.5.19.

A subset A of X is ***uniformly closed*** if for every net (x_α) in A, $x_\alpha \xrightarrow{\mathrm{u}} x$ implies $x \in A$. In view of Exercise 2.3.1(ii), nets may be replaced with sequences in this definition. By Exercise 2.3.1(vi), X_+ is uniformly closed. It also follows that order intervals are uniformly closed because $[a,b] = (a + X_+) \cap (b - X_+)$.

2.3.4 Exercise. A subset A of X is uniformly closed iff $A \cap I_e$ is closed in $\bigl(I_e, \|\cdot\|_e\bigr)$ for every $e \in X_+$.

In case of a non-Archimedean vector lattice, one may still define uniform convergence using (2.3). However, one must keep in mind that in this case uniform limits need not be unique and $\|\cdot\|_e$ is just a seminorm.

Suppose that X is a normed lattice. It is easy to see that uniform convergence implies norm convergence. Indeed, suppose that $x_\alpha \xrightarrow{\mathrm{u}} x$. Then there exists $e \in X_+$ such that $\|x_\alpha - x\|_e \to 0$. It follows from Exercise 2.2.3(i) that $\|x_\alpha - x\| \leqslant \|x_\alpha - x\|_e \cdot \|e\| \to 0$, where $\|\cdot\|$ stands for the "native" norm of X. This argument shows that uniform convergence is a very strong convergence. We will consider several natural convergences throughout this book, and uniform convergence will be the strongest among them.

2.3.5 Example. *A sequence which converges in norm but not uniformly.* Let $X = \ell_1$ and $x_n = \frac{1}{n} e_n$. Clearly, $x_n \xrightarrow{\|\cdot\|} 0$. However, (x_n) is not order bounded and, therefore, (x_n) cannot be uniformly convergent in view of Exercise 2.3.1(vii). Indeed, if there were a vector $u \in X_+$ such that $|x_n| \leqslant u$ for every n, it would imply that the n-th component of u is greater than $\frac{1}{n}$, which would contradict u being in ℓ_1.

Even though, in general, norm convergence does not imply uniform convergence, if a sequence in a Banach lattice converges in norm "sufficiently fast" then it converges uniformly:

2.3.6 Proposition. *Let (x_n) be a sequence in a Banach lattice. If the series $\sum_{n=1}^\infty n\|x_n\|$ converges then $x_n \xrightarrow{\mathrm{u}} 0$.*

Proof. Put $e = \sum_{n=1}^\infty n|x_n|$; the series converges because it converges absolutely. For every n, we have $|nx_n| \leqslant e$, so that $\|x_n\|_e \leqslant \frac{1}{n}$ and, therefore, $x_n \xrightarrow{\mathrm{u}} 0$. □

This proposition immediately yields the following result, which will be at the heart of many results in Banach lattice theory.

2.3.7 Theorem. *Every (norm) convergent sequence in a Banach lattice has a subsequence which converges uniformly to the same limit.*

2.3.8 Example. We observed in Example 2.3.5 that the sequence $x_n = \frac{1}{n}e_n$ in ℓ_1 converges in norm but not uniformly. Nevertheless, (x_n) has lots of uniformly convergent subsequences. For example, let $n_k = 2^k$. Put $e = \sum_{k=1}^{\infty} \frac{k}{2^k} e_{2^k}$; this series converges in ℓ_1. Then $0 \leqslant x_{n_k} \leqslant \frac{1}{k}e$ for every k. It follows that $x_{n_k} \xrightarrow{\mathrm{u}} 0$.

2.3.9 Corollary. *In a normed lattice, every norm closed set is uniformly closed. In a Banach lattice, norm closed and uniformly closed sets agree.*

Proof. Let A is a norm closed set in a normed lattice. Suppose that (x_α) is a net in A, and $x_\alpha \xrightarrow{\mathrm{u}} x$. It follows that $x_n \xrightarrow{\|\cdot\|} x$, hence $x \in A$ and, therefore, A is uniformly closed. Conversely, suppose that A is a uniformly closed set in a Banach lattice, (x_n) is a sequence in A, and $x_n \xrightarrow{\|\cdot\|} x$. Then a subsequence of (x_n) converges to x uniformly, hence $x \in A$ and, therefore, A is norm closed. □

Let us pause for a moment. It follows immediately from Theorem 2.3.7 that a sequence in a Banach space is norm convergent iff every subsequence has a uniformly convergent subsequence. In view of Proposition A.2.1, doesn't this imply that norm convergent and uniformly convergences have to agree on sequences? Furthermore, since every topology is determined by its closed sets, and we know that uniformly closed sets in a Banach lattice agree with norm closed sets, one would expect that norm convergence and uniform convergence must agree on all nets. Yet Example 2.3.5 provides a norm convergent sequence in a Banach lattice that is not uniformly convergent! The explanation to this mystery is that uniform convergence is generally not topological, i.e., there may be no topology on X which induces uniform convergence. In particuar, Example 2.3.5 shows that uniform convergence on ℓ_1 is not topological.

2.3.10 Example. *Uniform convergence on $L_1[0,1]$ is not topological.* Let $X = L_1[0,1]$. If uniform convergence on $L_1[0,1]$ were topological, it would have to agree with the norm convergence by Corollary 2.3.9. However, the typewriter sequence (x_n) in Example A.3.8 is norm null but not uniformly null. Here is another way to see that uniform convergence on $L_1[0,1]$ is not topological: while (x_n) is not uniformly null, every subsequence of (x_n) has a further subsequence which converges to zero uniformly by Theorem 2.3.7; now apply Proposition A.2.1.

Even though uniform convergence is not topological, it fits into the framework of *convergence structures*, which will be discussed in Section 21.5. At this point, we would only mention that many topological concepts may be extended to convergence structures. We have already discussed uniformly closed sets. One can define a ***uniformly continuous*** function in the natural way: $f(x_\alpha) \xrightarrow{\mathrm{u}} f(x)$ whenever $x_\alpha \xrightarrow{\mathrm{u}} x$. Replacing nets with sequences yields the definition of a sequentially uniformly continuous function. A sequence (x_n) in an Archimedean vector lattice X is said to be ***uniformly Cauchy*** if there exists $e > 0$ such that $\|x_n - x_m\|_e \to 0$ as $n, m \to \infty$. Equivalently, for every $\varepsilon > 0$ there exists n_0 such that for all $n, m \geqslant n_0$ one has $|x_n - x_m| < \varepsilon e$. (Note that (x_n) need not be in I_e.) One can define uniformly Cauchy nets in a similar fashion.

2.3.11 Proposition. *An Archimedean vector lattice X is uniformly complete iff every uniformly Cauchy sequence (or net) is uniformly convergent.*

Proof. Suppose that X is uniformly complete and (x_α) is a uniformly Cauchy net. Then there exists $e > 0$ such that $\|x_\alpha - x_\beta\|_e \to 0$. In particular, there exists α_0 such that $|x_\alpha - x_{\alpha_0}| < e$ for all $\alpha \geqslant \alpha_0$. It follows from $|x_\alpha| \leqslant |x_{\alpha_0}| + |x_\alpha - x_{\alpha_0}| \leqslant |x_{\alpha_0}| + e =: u$ that $x_\alpha \in I_u$. Also, $\|x_\alpha - x_\beta\|_u \leqslant \|x_\alpha - x_\beta\|_e \to 0$. Passing to a tail, we may assume that (x_α) is contained in I_u and is $\|\cdot\|_u$-Cauchy. Since $\big(I_u, \|\cdot\|_u\big)$ is complete, it follows that (x_α) converges in $\big(I_u, \|\cdot\|_u\big)$, hence uniformly converges in X.

Conversely, suppose that every uniformly Cauchy sequence is uniformly convergent. Let $e \in X_+$; we want to show that $\big(I_e, \|\cdot\|_e\big)$ is complete. Let (x_n) be a $\|\cdot\|_e$-Cauchy sequence in I_e. Then (x_n) is uniformly Cauchy in X, hence converges uniformly to some $x \in X$. Fix $\varepsilon > 0$. Since (x_n) is $\|\cdot\|_e$-Cauchy, there exists m such that $\|x_n - x_m\|_e < \varepsilon$ for every $n \geqslant m$. It follows that $x_n \in [x_m - \varepsilon e, x_m + \varepsilon e]$. Since order intervals are uniformly closed, passing to the uniform limit on n, we conclude that $x \in [x_m - \varepsilon e, x_m + \varepsilon e]$. Therefore, $x \in I_e$ and $\|x - x_m\|_e \leqslant \varepsilon$, hence $x_n \to x$ in $\big(I_e, \|\cdot\|_e\big)$. □

The fact that uniform convergence is not topological leads to certain issues that do not arise for topological convergences. We mention two of them. The first one deals with the uniform closure of a set: let A be a subset of X; the set of the uniform limits of all uniformly convergent nets in A need not be uniformly closed. Second, uniform convergence depends on the ambient space. Let Y be a sublattice of X and let (x_n) be a sequence in Y. It is easy to see that if $x_n \xrightarrow{\mathrm{u}} 0$ in Y then $x_n \xrightarrow{\mathrm{u}} 0$ in X. The converse is false in general:

2.3.12 Example. *Uniform convergence depends on the ambient space.* We observed in Example 2.3.5 that the sequence $x_n = \frac{1}{n}e_n$ in ℓ_1 fails to converge uniformly. Recall that ℓ_1 is a sublattice of ℓ_∞. The same sequence converges to zero uniformly in ℓ_∞ because $0 \leqslant x_n \leqslant \frac{1}{n}\mathbb{1}$ for every n.

We will show that uniform convergence does not depend on the ambient space in the category of uniformly complete vector lattices.

2.3.13 Proposition. *Let Y be a uniformly closed sublattice of a uniformly complete vector lattice X and (x_n) a sequence in Y. Then $x_n \xrightarrow{\mathrm{u}} 0$ in Y iff $x_n \xrightarrow{\mathrm{u}} 0$ in X.*

Proof. The forward implication is trivial. Suppose $x_n \xrightarrow{\mathrm{u}} 0$ in X. Find $e \in X_+$ such that $\|x_n\|_e \to 0$. For every k there exists n_k such that $|x_n| \leqslant \frac{1}{k^3}e$ for all $n \geqslant n_k$. WLOG, (n_k) is an increasing sequence. For every k, put $v_k = \bigvee_{n=n_k}^{n_{k+1}-1}|x_n|$. Then $v_k \leqslant \frac{1}{k^3}e$ and, therefore, $\|v_k\|_e \leqslant \frac{1}{k^3}$. Since X is uniformly complete, $\big(I_e, \|\cdot\|_e\big)$ is a Banach lattice. The series $\sum_{k=1}^{\infty}\|kv_k\|_e$ converges in $\mathbb{R}$, it follows that the series $\sum_{k=1}^{\infty} kv_k$ converges in $\big(I_e, \|\cdot\|_e\big)$ to some $w \in I_e$. Since Y is uniformly closed, we have $w \in Y$. It is left to show that (x_n) converges to zero uniformly in Y relative to w. Let $k \in \mathbb{N}$. Take any $n \geqslant n_k$. Find $m \geqslant k$ such that $n_m \leqslant n < n_{m+1}$. Then $|x_n| \leqslant v_m \leqslant \frac{1}{m}w \leqslant \frac{1}{k}w$. □

Recall that a subspace of a Banach space is closed iff it is complete.

2.3.14 Corollary. *Let Y be a uniformly closed sublattice of a uniformly complete vector lattice X. Then Y is uniformly complete.*

Proof. Let (x_n) be a uniformly Cauchy sequence in Y. Clearly, it remains uniformly Cauchy when viewed as a sequence in X. Hence $x_n \xrightarrow{\mathrm{u}} x$ for some $x \in X$ by Proposition 2.3.11. Since Y is uniformly closed, we have $x \in Y$. We now have $x_n \xrightarrow{\mathrm{u}} x$ in Y by Proposition 2.3.13. □

2.3.15 Exercise. Every principal ideal in c_{00} is finite dimensional; every subspace of c_{00} is uniformly closed.

The following exercise shows that the converse of Corollary 2.3.14 fails:

2.3.16 Example. *A uniformly complete sublattice of a uniformly complete vector lattice need not be uniformly closed.* Consider c_{00} as a sublattice of c_0. Being a Banach lattice, c_0 is uniformly complete by Theorem 2.2.9. Since every principal ideal I_e in c_{00} is finite-dimensional, it is complete in any norm, in particular, in $\|\cdot\|_e$; this shows that c_{00} is uniformly complete. However, for every element x of c_0 there is a sequence in c_{00} that converges to x in norm, which, by Theorem 2.3.7 has a subsequence that converges to x uniformly. This shows that c_{00} is not uniformly closed in c_0 (rather, it is ***uniformly dense***).

Since Banach lattices are uniformly complete by Theorem 2.2.9, combining Proposition 2.3.13 with Corollary 2.3.9, we get the following.

2.3.17 Theorem. *Let Y be a norm closed sublattice of a Banach lattice X and (x_n) a sequence in Y. Then $x_n \xrightarrow{\mathrm{u}} 0$ in Y iff $x_n \xrightarrow{\mathrm{u}} 0$ in X.*

The preceding theorem fails for nets. Consider the double sequence $x_{n,m} = \frac{1}{n}e_m$ in ℓ_∞. It follows from $x_{n,m} \leqslant \frac{1}{n}\mathbb{1}$ that $x_{n,m} \xrightarrow{\mathrm{u}} 0$ in ℓ_∞. However, viewed as a net in c_0, it's tails are not order bounded, hence it fails to converge uniformly to zero.

Proposition 2.3.6 may be improved:

2.3.18 Exercise. Let (x_n) be a sequence in a Banach lattice. If the series $\sum_{n=1}^{\infty}\|x_n\|$ converges then $x_n \xrightarrow{\mathrm{u}} 0$.

The following is a variant of Monotone Convergence Lemma for uniform convergence.

2.3.19 Lemma. *If (x_α) is a monotone net in a Banach lattice and $x_\alpha \to x$ then $x_\alpha \xrightarrow{\mathrm{u}} x$.*

Proof. WLOG, $x_\alpha \downarrow$ and $x = 0$, i.e., $x_\alpha \to 0$; we need to show that $x_\alpha \xrightarrow{\mathrm{u}} 0$. Note that $x_\alpha \geqslant 0$ for every α: suppose that $x_\beta^- \neq 0$ for some β; since $x_\alpha^- \geqslant x_\beta^-$ for every $\alpha \geqslant \beta$, we have $\|x_\alpha\| \geqslant \|x_\alpha^-\| \geqslant \|x_\beta^-\| > 0$, which contradicts $x_\alpha \to 0$. For each n, find α_n such that $\|x_{\alpha_n}\| < \frac{1}{2^n}$. It follows that the series $e := \sum_{n=1}^\infty n x_{\alpha_n}$ converges. Let $n \in \mathbb{N}$. For every $\alpha \geqslant \alpha_n$, we have $0 \leqslant x_\alpha \leqslant x_{\alpha_n} \leqslant \frac{1}{n} e$, so that $\|x_\alpha\|_e \leqslant \frac{1}{n}$. □

Some of the results in this section may fail when a Banach lattice is replaced with a normed lattice. Let X be as in Example 2.2.14, since X agrees, as a vector lattice, with $C[0,1]$, then, by Corollary 2.3.9, its uniformly closed subsets are the ones that are closed under $\|\cdot\|_{\sup}$, which need not be closed under $\|\cdot\|_1$, so they need not be norm closed in X. Thus, norm closed and uniformly closed sets need not agree. Furthermore, Lemma 2.3.19 also fails in X, just consider the sequence (f_n) as in Example 1.4.2. Even though $f_n \to 0$ in norm (that is, in $\|\cdot\|_1$-norm), it fails to converge to zero uniformly. Moreover, no subsequence of (f_n) converges to zero uniformly. This means that Theorem 2.3.7 fails in normed lattices. Here is another example of this:

2.3.20 Example. *A norm convergent sequence in a normed lattice need not have a uniformly convergent subsequence.* Let $X = \big(c_{00}, \|\cdot\|_\infty\big)$ and put $x_n = \frac{1}{n} e_n$. Then (x_n) is norm null, yet every subsequence of (x_n) is not order bounded, hence not uniformly null.

2.4 Order convergence

Recall that for a net (x_α) in a vector lattice X, $x_\alpha \uparrow x$ means $x_\alpha\uparrow$ and $x = \sup x_\alpha$, while $x_\alpha \downarrow x$ means $x_\alpha\downarrow$ and $x = \inf x_\alpha$. We will now extend the concept of order convergence to non-monotone nets. Our definition of order convergence is of an ε-δ type.

We say that a net $(x_\alpha)_{\alpha\in\Lambda}$ in a vector lattice X ***converges in order*** to $x \in X$ and write $x_\alpha \xrightarrow{\mathrm{o}} x$ if there exists a net $(u_\gamma)_{\gamma\in\Gamma}$ in X such that $u_\gamma \downarrow 0$ and for every $\gamma \in \Gamma$, the order interval $[x - u_\gamma, x + u_\gamma]$ contains a tail of (x_α), that is, there exists $\alpha_0 \in \Lambda$ such that $|x_\alpha - x| \leqslant u_\gamma$ whenever $\alpha \geqslant \alpha_0$. We will refer to the net (u_γ), which "witnesses" the order convergence of (x_α), as a ***dominating*** net. Note that a dominating net is not unique and we allow a dominating net to have a different index sets than the original net.

We will show later that, similarly to uniform convergence, order convergence is not topological, but defines a convergence structure. Hence, one can still define various topologically sounding concepts for order convergence. For example, a subset A of X is said to be ***order closed*** if the limit of every order convergent net in A belongs to A; a function $f\colon X \to Y$ between two vector lattices is said to be ***order continuous*** if $x_\alpha \xrightarrow{\mathrm{o}} x$ in X implies $f(x_\alpha) \xrightarrow{\mathrm{o}} f(x)$ in Y, etc. We say that a net (x_α) is ***order null*** if $x_\alpha \xrightarrow{\mathrm{o}} 0$.[1]

2.4.1 Exercise. (i) $x_\alpha \xrightarrow{\mathrm{o}} x \quad\Leftrightarrow\quad x_\alpha - x \xrightarrow{\mathrm{o}} 0 \quad\Leftrightarrow\quad |x_\alpha - x| \xrightarrow{\mathrm{o}} 0$.

(ii) If $x_\alpha \xrightarrow{\mathrm{o}} 0$ and $|y_\alpha| \leqslant |x_\alpha|$ for every α then $y_\alpha \xrightarrow{\mathrm{o}} 0$.

(iii) In $\mathbb{R}$, order convergence agrees with the usual convergence.

(iv) Every order convergent net has an order bounded tail.

(v) Every order convergent sequence is order bounded.

There is a sequential version of order convergence: we say that a net (x_α) ***σ-order converges*** to x and write $x_\alpha \xrightarrow{\sigma o} x$ if it satisfies the definition of order convergence with a dominating net being a sequence. That is, there exists a sequence (u_n) such that $u_n \downarrow 0$ and for every $n \in \mathbb{N}$ the order interval $[-u_n, u_n]$ contains a tail of the net. It is easy to see that Exercise 2.4.1 remains valid for σ-order convergence. We get a "hierarchy" of convergences:

2.4.2 Exercise. $x_\alpha \xrightarrow{\sigma o} x$ implies $x_\alpha \xrightarrow{\mathrm{o}} x$. If X is Archimedean then $x_\alpha \xrightarrow{\mathrm{u}} x$ implies $x_\alpha \xrightarrow{\sigma o} x$.

While we defined σ-order convergence for nets, we will primarily apply it to sequences. In this case, the definition of σ-order convergence may be somewhat simplified:

2.4.3 Exercise. For a sequence (x_n), $x_n \xrightarrow{\sigma o} x$ iff there exists a sequence (u_n) such that $u_n \downarrow 0$ and and $|x_n - x| \leqslant u_n$ for every n.

2.4.4 Example. Consider the unit vector sequence (e_n) in ℓ_∞. Then $e_n \xrightarrow{\sigma o} 0$. To see this, take

$$u_n = (\overbrace{0, \dots, 0}^{n-1}, 1, 1, \dots),$$

then $0 \leqslant e_n \leqslant u_n \downarrow 0$. In particular, $e_n \xrightarrow{\mathrm{o}} 0$.

2.4.5 Example. Consider the following net in $\mathbb{R}$ indexed by two consecutive copies on $\mathbb{N}$:

$$(1, 2, 3, 4, \dots, 0, 0, 0, 0 \dots).$$

[1]Here is a word of warning. There are several non-equivalent definitions of order convergent nets in literature. Up to a certain point in the theory of vector lattices, it makes little difference which definition one uses. For example, these definitions lead to the same concepts of order continuous linear functionals. In fact, order continuous linear functionals (and several other concepts) may be defined using only order bounded monotone nets, and all definitions of order convergence agree on such nets, so, up to a certain point, one may avoid order convergence altogether. However, many of the more advanced results in this book are only valid provided that one uses the definition of order convergence given at the beginning of this section. See Section 21.1 for further details on various definitions of order convergence.

This net has a tail of zeroes, hence it converges to zero in order (and in any other reasonable convergence). Yet, the net fails to be bounded.

2.4.6 Example. *A sequence which is order convergent but not σ-order convergent.* The following example shows that when dealing with an order convergent sequence, one still has to consider a dominating net and not just a sequence in the definition of order convergence.

Let X be the space of functions from $\mathbb{R}$ to $\mathbb{R}$ of form $c\mathbb{1} + f$ where f is a function with finite support. It is easy to see that X is a vector lattice under point-wise operations. For each $n \in \mathbb{N}$, let x_n be the characteristic function of the singleton $\{n\}$.

We claim that $x_n \xrightarrow{o} 0$. Let Λ be the set of all finite subsets of $\mathbb{R}$ directed by inclusion. For every $\alpha \in \Lambda$, let z_α be the characteristic function of α. Clearly, $(z_\alpha)_{\alpha\in\Lambda}$ is a net, and $z_\alpha \uparrow \mathbb{1}$. Put $y_\alpha = \mathbb{1} - z_\alpha$, then $y_\alpha \downarrow 0$. We claim that (y_α) dominates (x_n). Indeed, for each $\alpha \in \Lambda$ there exists $m \in \mathbb{N}$ such that for all $n \geqslant m$ we have $n \notin \alpha$ and, therefore, $x_n \leqslant y_\alpha$. This proves that $x_n \xrightarrow{o} 0$. Actually, we will prove in Proposition 3.2.33 that every disjoint order bounded sequence in an Archimedean vector lattice is order null.

On the other hand, (x_n) fails to σ-order converge to zero. Indeed, suppose that (u_n) is a sequence in X such that $u_n \downarrow 0$ and $x_n \leqslant u_n$ for every n. It follows that $x_k \leqslant u_n$ for all $k \geqslant n$, so that u_n is greater than or equal to $\mathbb{1}$ except on a finite set, say, F_n. Take any $t \notin \bigcup_{n=1}^{\infty} F_n$, then $u_n(t) \geqslant 1$ for every n. This contradicts $u_n \downarrow 0$.

Thus, even though $x_n \xrightarrow{o} 0$, one cannot choose a dominating net to be a sequence.

Next, we will show that vector and lattice operations are order continuous and σ-order continuous:

2.4.7 Proposition. *Suppose that $x_\alpha \xrightarrow{o} x$, $y_\alpha \xrightarrow{o} y$, and $\lambda \in \mathbb{R}$. Then*

(i) $\lambda x_\alpha \xrightarrow{o} \lambda x$;

(ii) $x_\alpha + y_\alpha \xrightarrow{o} x + y$;

(iii) $|x_\alpha| \xrightarrow{o} |x|$, $x_\alpha^+ \xrightarrow{o} x^+$, *and* $x_\alpha^- \xrightarrow{o} x^-$;

(iv) $x_\alpha \wedge y_\alpha \xrightarrow{o} x \wedge y$ *and* $x_\alpha \vee y_\alpha \xrightarrow{o} x \vee y$.

Same for σ-order convergence.

Proof. For simplicity, we will prove the statements for σ-order convergent sequences first. Suppose that $x_n \xrightarrow{\sigma o} x$ and $y_n \xrightarrow{\sigma o} y$. Then $|x_n - x| \leqslant a_n \downarrow 0$ and $|y_n - y| \leqslant b_n \downarrow 0$ for some sequences (a_n) and (b_n). Then

$$\big|(x_n + y_n) - (x + y)\big| \leqslant |x_n - x| + |y_n - y| \leqslant a_n + b_n \downarrow 0$$

by Exercise 1.3.5(ii), hence $x_n + y_n$ σ-order converges to $x + y$. Furthermore, for every $\lambda \in \mathbb{R}$, we have

$$\big|\lambda x_n - \lambda x\big| = |\lambda| \cdot |x_n - x| \leqslant |\lambda| a_n \downarrow 0,$$

hence $\lambda x_n \xrightarrow{\sigma o} \lambda x$. Finally,

$$\big||x_n| - |x|\big| \leqslant |x_n - x| \leqslant a_n \downarrow 0,$$

so that $|x_n| \xrightarrow{\sigma o} |x|$, hence the modulus is σ-order continuous. Since by 1.3.9 all the lattice operations can be expressed via the modulus operation, addition, and scalar multiplication, they all are σ-order continuous.

The proof for order (and σ-order) convergent nets goes along the same lines. Find dominating nets $(a_\lambda)_{\lambda\in\Lambda}$ and $(b_\delta)_{\delta\in\Delta}$ for $x_\alpha \xrightarrow{o} x$ and $y_\alpha \xrightarrow{o} 0$, respectively. Observe that the dominating nets need not have the same index set! But this can be fixed by the following trick. Define a new index set $\Gamma = \Lambda \times \Delta$ ordered coordinate-wise. Clearly, Γ is a directed set. Now define two new nets indexed by Γ via $\tilde{a}_{(\lambda,\delta)} := a_\lambda$ and $\tilde{b}_{(\lambda,\delta)} := b_\delta$. It is easy to see that, WLOG, we may replace $(a_\lambda)_{\lambda\in\Lambda}$ and $(b_\delta)_{\delta\in\Delta}$ with $(\tilde{a}_\gamma)_{\gamma\in\Gamma}$ and $(\tilde{b}_\gamma)_{\gamma\in\Gamma}$. So, relabelling, we may assume that our dominating nets are $(a_\gamma)_{\gamma\in\Gamma}$ and $(b_\gamma)_{\gamma\in\Gamma}$. By Exercise 1.3.5(ii), $a_\gamma + b_\gamma \downarrow 0$.

For every $\gamma \in \Gamma$, we find

$$\begin{aligned} &\alpha_1 \in \Lambda \quad \text{such that} \quad |x_\alpha - x| \leqslant a_\gamma \text{ for all } \alpha \geqslant \alpha_1, \text{ and} \\ &\alpha_2 \in \Lambda \quad \text{such that} \quad |y_\alpha - y| \leqslant b_\gamma \text{ for all } \alpha \geqslant \alpha_2. \end{aligned}$$

Find $\alpha_0 \in \Lambda$ such that $\alpha_0 \geqslant \alpha_1, \alpha_2$. Then for every $\alpha \geqslant \alpha_0$, we have

$$\Big|(x_\alpha + y_\alpha) - (x+y)\Big| \leqslant |x_\alpha - x| + |y_\alpha - y| \leqslant a_\gamma + b_\gamma.$$

It follows that $x_\alpha + y_\alpha \xrightarrow{o} 0$. The other statements are proved in a similar way. □

2.4.8 Exercise. If $x_\alpha \geqslant a$ for every α and $x_\alpha \xrightarrow{o} x$ then $x \geqslant a$. Same for σ-order convergence.

It follows from the preceding exercise that X_+ is order closed. Furthermore, an order interval $[a, b]$ is order closed.

2.4.9 Exercise. Order limits and σ-order limits are unique.

2.4.10 Exercise. If $x_\alpha \xrightarrow{o} x$ then every subnet of (x_α) converges in order to x. In particular, if $x_n \xrightarrow{o} x$ then every subsequence of (x_n) converges in order to x. Same for σ-order convergence.

2.4.11 Exercise. For monotone nets, the new definition of order convergence extends the old one:

- for an increasing net (x_α), $x_\alpha \xrightarrow{o} x$ iff $x_\alpha \uparrow x$;
- for a decreasing net (x_α), $x_\alpha \xrightarrow{o} x$ iff $x_\alpha \downarrow x$.

Same for σ-order convergent sequences.

In Archimedean vector lattices, scalar multiplication is jointly order continuous:

2.4.12 Exercise. Let X be an Archimedean ordered vector space. If $\lambda_\alpha \to \lambda$ in $\mathbb{R}$ and $x_\alpha \xrightarrow{\text{o}} x$ in X then $\lambda_\alpha x_\alpha \xrightarrow{\text{o}} \lambda x$. Same for σ-order convergence.

2.4.13. The previous exercise actually characterizes the Archimedean property. That is, a vector lattice X is Archimedean if scalar multiplication is order or σ-order continuous. Indeed, suppose that $0 \leqslant y \leqslant \frac{1}{n}x$ for all n. Then $\frac{1}{n}x \xrightarrow{\text{o}} 0$, hence $y = 0$.

2.4.14. *Order convergence depends on the ambient space.* We observed in Example 2.4.4 that the standard unit sequence (e_n) in ℓ_∞ is σ-order null and, therefore, order null. However, viewed as a sequence in c_0, it is not even order bounded, hence is not order null. Thus, order convergence depends on the ambient space.

Here is another example that order convergence depends on the ambient space, even for monotone sequences. Recall the sequence (f_n) in Example 1.4.2. We have $f_n \downarrow 0$ and, therefore, $f_n \xrightarrow{\sigma o} 0$ and $f_n \xrightarrow{\text{o}} 0$ in $C[0,1]$. However, if we view (f_n) as a sequence in $\mathbb{R}^{[0,1]}$, then $f_n \downarrow \mathbb{1}_{\{0\}}$, so that $f_n \xrightarrow{\sigma o} \mathbb{1}_{\{0\}}$ and $f_n \xrightarrow{\text{o}} \mathbb{1}_{\{0\}}$ in $\mathbb{R}^{[0,1]}$.

Next, we discuss several equivalent definitions of order convergence. The following exercise asserts that dominating nets in the definition of order convergence may be replaced with pairs of nets "dominating on both sides", or with sets.

2.4.15 Exercise. Given a net (x_α) in X. TFAE:

(i) $x_\alpha \xrightarrow{\text{o}} x$;

(ii) There exist two nets $(a_\gamma)_{\gamma\in\Gamma}$ and $(b_\gamma)_{\gamma\in\Gamma}$ such that $a_\gamma \uparrow x$, $b_\gamma \downarrow x$, and for every $\gamma \in \Gamma$ the order interval $[a_\gamma, b_\gamma]$ contains a tail of (x_α);

(iii) There exist non-empty sets A and B with $\sup A = x = \inf B$ and for every $a \in A$ and $b \in B$, the order interval $[a, b]$ contains a tail of (x_α);

(iv) There exist a non-empty set C with $\inf C = 0$, such that for every $c \in C$ there exists α_0 such that $|x_\alpha - x| \leqslant c$ whenever $\alpha \geqslant \alpha_0$;

(v) There is a family $\mathcal{F}$ of order intervals, such that $\bigcap \mathcal{F} = \{x\}$, and each interval in $\mathcal{F}$ contains a tail of (x_α).

Exercise 2.4.15 motivates the following definitions. Consider a net (x_α) in X. We say that a net (x_α) in X is ***eventually bounded below*** if it has a tail that is bounded below. That is, there exists a and an index α_0 such that $x_\alpha \geqslant a$ for all $\alpha \geqslant \alpha_0$. We say that a is an ***eventual lower bound*** for (x_α). In a similar fashion, one can define eventual upper bounds. We can also consider a "symmetrized" version: an element $c \in X_+$ is said to ***eventually dominate*** (x_α) if a tail of the net is contained in $[-c, c]$; we say that c is an ***eventual dominant*** of the net.

Condition (iii) in Exercise 2.4.15 says that $x_\alpha \xrightarrow{\text{o}} x$ iff there are sets A and B consisting of eventual lower (respectively, upper) bounds such that $\sup A = x = \inf B$. Condition (iv) means that there exists a set C consisting of eventual dominants of the net $(x_\alpha - x)$ with $\inf C = 0$. We may, actually, consider the set of *all* eventual upper/lower bounds or dominants:

2.4.16 Exercise. For a net (x_α) in X, TFAE:

(i) $x_\alpha \xrightarrow{\mathrm{o}} 0$;
(ii) The sets A and B of all eventual lower (respectively, upper) bounds of (x_α) are both non-empty and satisfy $\sup A = 0 = \inf B$;
(iii) The set of all eventual dominants of (x_α) is non-empty and has infimum zero.

In the special case when $X = \mathbb{R}$, Exercise 2.4.16 essentially says that $x_\alpha \to 0$ iff $\liminf x_\alpha = \limsup x_\alpha = 0$ iff $\liminf|x_\alpha| = 0$.

Exercises 2.4.15 and 2.4.16 allow one to extend the concept of order convergence beyond vector lattices, to ordered sets. Note, however, that while statements (iii) and (v) in Exercise 2.4.15, as well as Exercise 2.4.16(ii) remain equivalent in ordered sets, they are no longer equivalent to Exercise 2.4.15(ii); the latter condition is stronger. All these conditions are equivalent when the ordered set is a lattice.

2.4.17 Exercise. Suppose that X has countable sup property and $(x_\alpha)_{\alpha\in\Lambda}$ is an order bounded net in X. Then $x_\alpha \xrightarrow{\mathrm{o}} x$ iff there exists a net $(u_\alpha)_{\alpha\in\Lambda}$ such that $u_\alpha \downarrow 0$ and $|x_\alpha - x| \leqslant u_\alpha$ for every α. In particular, order convergence and σ-order convergence agree for sequences in X.

2.4.18. Exercise 2.4.2 fails for non-Archimedean spaces. Let $X = \mathbb{R}^2$ with the lexicographic order, put $x_n = (\frac{1}{n}, 0)$ and $e = (1,0)$. Then $x_n \downarrow$ and $\|x_n\|_e \to 0$, so that $x_n \xrightarrow{\mathrm{u}} 0$. However, $\inf x_n$ does not exist, so that (x_n) is not order convergent. In fact, it is easy to see that uniform convergence implies order convergence iff X is Archimedean.

It is also easy to see that if $x_n \xrightarrow{\mathrm{u}} 0$ then we can find a sequence of positive reals (λ_n) such that $\lambda_n \uparrow \infty$ but $\lambda_n x_n \xrightarrow{\mathrm{u}} 0$; we say that uniform convergence is ***stable***.

2.4.19 Example. *σ-order convergence need not be stable.* Let $X = \ell_\infty$ and $x_n = (0,\dots,0,1,1,\dots)$, with n zeros at the beginning. Then $x_n \downarrow 0$, but if $\lambda_n \uparrow \infty$ then the sequence $(\lambda_n x_n)$ is not even order bounded, so it cannot be order null.

2.4.20 Exercise. σ-order convergence of sequences in an Archimedean vector lattice is stable iff it agrees with uniform convergence.

2.4.1 Order convergence and order completeness

In order complete spaces, there is a simple formula for the order limit of a net. Furthermore, one may choose a dominating net over the same index set as the original net.

2.4.21 Proposition. *For an order bounded net (x_α) in an order complete vector lattice, TFAE:*

(i) $x_\alpha \xrightarrow{\mathrm{o}} x$;
(ii) $\inf\limits_\alpha \sup\limits_{\beta\geqslant\alpha} |x_\beta - x| = 0$;
(iii) $x = \inf\limits_\alpha \sup\limits_{\beta\geqslant\alpha} x_\beta = \sup\limits_\alpha \inf\limits_{\beta\geqslant\alpha} x_\beta$.

(Note that the infima and the suprema exist because (x_α) is order bounded and X is order complete.)

Proof. Replacing x_α with $x_\alpha - x$, we may assume WLOG that $x = 0$. For every α, put $v_\alpha = \sup\limits_{\beta\geqslant\alpha}|x_\beta|$. It is easy to see that $v_\alpha \downarrow$ and $|x_\alpha| \leqslant v_\alpha$ for every α. The implication (ii)⇒(i) is immediate because (v_α) is a dominating net for (x_α) and $v_\alpha \downarrow 0$. To prove (i)⇒(ii), suppose that $x_\alpha \xrightarrow{\mathrm{o}} 0$; then there exists a net (u_γ) such that $u_\gamma \downarrow 0$ and for every γ there exists α_0 such that $|x_\alpha| \leqslant u_\gamma$ whenever $\alpha \geqslant \alpha_0$. It follows that $v_{\alpha_0} \leqslant u_\gamma$. Since $v_\alpha\downarrow$, it follows from $u_\gamma \downarrow 0$ that $v_\alpha \downarrow 0$.

(ii)⇒(iii) It is easy to see that $\inf\limits_{\beta\geqslant\alpha_1} x_\beta \leqslant \sup\limits_{\beta\geqslant\alpha_2} x_\beta$ for all α_1 and α_2, so that $\sup\limits_\alpha \inf\limits_{\beta\geqslant\alpha} x_\beta \leqslant \inf\limits_\alpha \sup\limits_{\beta\geqslant\alpha} x_\beta$. It follows that

$$0 = -\inf_\alpha \sup_{\beta\geqslant\alpha}|x_\beta| \leqslant \sup_\alpha \inf_{\beta\geqslant\alpha} x_\beta \leqslant \inf_\alpha \sup_{\beta\geqslant\alpha} x_\beta \leqslant \inf_\alpha \sup_{\beta\geqslant\alpha}|x_\beta| = 0.$$

(iii)⇒(i) Put $a_\alpha = \inf\limits_{\beta\geqslant\alpha} x_\beta$ and $b_\alpha = \sup\limits_{\beta\geqslant\alpha} x_\beta$. It is easy to see that $a_\alpha \leqslant x_\alpha \leqslant b_\alpha$. By assumption, $a_\alpha \uparrow 0$ and $b_\alpha \downarrow 0$. It follows that $x_\alpha \xrightarrow{\mathrm{o}} 0$ by Exercise 2.4.15(ii). □

2.4.22. The assumption that the net is order bounded is not restrictive because every order convergent net has an order bounded tail, so one may usually assume that the net is order bounded by passing to a tail. One can now take $u_\alpha = \sup\limits_{\beta\geqslant\alpha}|x_\beta - x|$ as a dominating net in the definition of order convergence.

2.4.23 Corollary. *Suppose that X is order complete and $(x_\alpha)_{\alpha\in\Lambda}$ is order bounded. Then $x_\alpha \xrightarrow{\mathrm{o}} x$ iff there exists a net $(u_\alpha)_{\alpha\in\Lambda}$ such that $u_\alpha \downarrow 0$ and $|x_\alpha - x| \leqslant u_\alpha$ for every α.*

The simple formula in Proposition 2.4.21(iii) is a convenient tool for computing order limits. Moreover, it provides a pair of nets $a_\alpha = \inf\limits_{\beta\geqslant\alpha} x_\beta$ and $b_\alpha = \sup\limits_{\beta\geqslant\alpha} x_\beta$ for the alternative definition of order convergence in Exercise 2.4.15(ii): $a_\alpha\uparrow$, $b_\alpha\downarrow$, $a_\alpha \leqslant x_\alpha \leqslant b_\alpha$ for every α and $x \xrightarrow{\mathrm{o}} x$ iff $a_\alpha \uparrow x$ and $b_\alpha \downarrow x$. The two nets are indexed by the same index set as (x_α).

Proposition 2.4.21 applies to σ-order convergence of sequences. In this case, the proof still works when the space is just σ-order complete instead of order complete. Also, order boundedness assumption is now redundant in (i) because every order convergent sequence is automatically order bounded. Thus, for an order convergent sequence in a σ-order complete vector lattice, one may always find a dominating sequence rather than a net in the definition of order convergence, hence order convergence and σ-order convergence agree:

2.4.24 Proposition. *For a sequence (x_n) in a σ-order complete vector lattice, TFAE:*

(i) $x_n \xrightarrow{\mathrm{o}} x$;

(ii) $x_n \xrightarrow{\sigma\mathrm{o}} x$;

(iii) (x_n) *is order bounded and* $\inf\limits_n \sup\limits_{k\geqslant n}|x_k - x| = 0$;

(iv) (x_n) *is order bounded and* $x = \inf\limits_n \sup\limits_{k\geqslant n} x_k = \sup\limits_n \inf\limits_{k\geqslant n} x_k$.

2.4.25. Similarly to the net case, if $x_n \xrightarrow{\mathrm{o}} x$ then we can define a dominating *sequence* in the definition of order convergence via $u_n = \sup_{k\geqslant n}|x_k - x|$. We may also define two "control" sequences

$$a_n = \inf_{k\geqslant n} x_k \text{ and } b_n = \sup_{k\geqslant n} x_k,$$

then $a_n \leqslant x_n \leqslant b_n$ for every n, $a_n \uparrow x$, and $b_n \downarrow x$.

2.4.26 Exercise. Every order bounded disjoint sequence in a σ-order complete vector lattice is σ-order null.

We will show later in Proposition 3.2.33 that every order bounded disjoint sequence in an Archimedean vector lattice is order null.

We can now characterize order convergent sequences in $L_p(\mu)$ spaces:

2.4.27 Proposition. *Let (f_n) be a sequence in $L_p(\mu)$, $0 \leqslant p \leqslant \infty$, and $f \in L_p(\mu)$. Then $f_n \xrightarrow{\mathrm{o}} f$ iff $f_n \xrightarrow{\sigma\mathrm{o}} 0$ iff (f_n) is order bounded and $f_n \to f$ a.e.*

Proof. Recall that $L_p(\mu)$ is σ-order complete by Corollary 2.1.13. By Proposition 2.4.24, we only need to show that $f_n \xrightarrow{\text{a.e.}} f$ iff $\inf_n v_n = 0$ where $v_n = \sup_{k\geqslant n}|f_n - f|$. It is easy to see that $f_n \xrightarrow{\text{a.e.}} f$ iff $v_n \xrightarrow{\text{a.e.}} 0$. Since $v_n\downarrow$, the latter is equivalent to $v_n \downarrow 0$ by Proposition 2.1.14. □

When $p = 0$, the order boundedness assumption is redundant:

2.4.28 Exercise. Given (f_n) and f in $L_0(\mu)$. If (f_n) as a.e. convergent then it is order bounded. Hence, $f_n \xrightarrow{\mathrm{o}} f$ iff $f_n \xrightarrow{\text{a.e.}} f$.

Since the a.e. convergence in $L_0(\mu)$ need not be topological (see Example A.3.8), it follows that order convergence on a vector lattice need not be topological. One can consider the so called ***order topology*** on a vector lattice: the topology whose closed sets are the order closed sets. However, the convergence generated by this topology need not agree with order convergence. Order topology may be quite poor: for example, we will show in Chapter 21 that on $C[0,1]$ is is not even Hausdorff or linear. On the other hand, it may be nice: we will see in Exercise 2.5.15 that on $L_p(\mu)$ spaces with $1 \leqslant p < \infty$, order topology agrees with the norm topology.

We showed in this section that order convergence "behaves better" in order complete spaces. We will later expose another important connection between these two concepts: we will prove in Theorem 6.10.9 that an Archimedean vector lattice is order complete iff every order Cauchy net is order convergent.

2.4.2 * Order convergence in $C(K)$ spaces

This section is based on [vdW18, BT22]. Throughout this subsection, K stands for a compact space. While lattice operations in $C(K)$ are point-wise for finite sets of vectors, this fails for infinite sets; see, e.g., Example 1.4.2. The following lemma provides a way of finding infima (and, therefore, suprema) of infinite sets.

2.4.29 Lemma. *For $G \subseteq C(K)_+$, $\inf G = 0$ iff for every non-empty open set U and every $\varepsilon > 0$ there exists $t \in U$ and $g \in G$ with $g(t) < \varepsilon$.*

Proof. Suppose that $\inf G \neq 0$. Then there exists $f \in C(K)$ with $0 < f \leqslant G$. We can then find an open non-empty set U and $\varepsilon > 0$ such that f is greater that ε on U. It follows that every $g \in G$ greater that ε on U.

Suppose now that there exists an open non-empty U and $\varepsilon > 0$ such that every $g \in G$ is greater than or equal to ε on U. By Urysohn's Lemma A.2.2, we find a non-zero $f \in C(K)_+$ such that $f \leqslant \varepsilon\mathbb{1}$ and f vanishes outside of U. Then $0 < f \leqslant G$. □

Next, we are going to characterize order convergence in $C(K)$. Since $f_\alpha \xrightarrow{o} f$ iff $|f_\alpha - f| \xrightarrow{o} 0$, it suffices to characterize order convergence of positive nets to zero. Also, every order convergent net has an order bounded tail, so we may assume WLOG that the net is order bounded.

2.4.30 Theorem. *For an order bounded net (f_α) in $C(K)_+$, $f_\alpha \xrightarrow{o} 0$ iff for every non-empty open set U and every $\varepsilon > 0$ there exists an open non-empty $V \subseteq U$ and an index α_0 such that f_α is less than ε on V whenever $\alpha \geqslant \alpha_0$.*

Proof. Suppose $f_\alpha \xrightarrow{o} 0$. By Exercise 2.4.16, there exists a non-empty set G of eventual dominants of (f_α) with $\inf G = 0$. Fix a non-empty open U and $\varepsilon > 0$. Let t and g be as in Lemma 2.4.29. Put $V = U \cap \{g < \varepsilon\}$; this set is open and non-empty as $t \in V$. Since g dominates a tail of (f_α), there exists α_0 such that $f_\alpha \leqslant g$ for all $\alpha \geqslant \alpha_0$. In particular, $f_\alpha(t) \leqslant g(t) < \varepsilon$ for all $t \in V$.

To prove the converse, we will again use Exercise 2.4.16. Suppose that the condition in the proposition is satisfied. Since (f_α) is order bounded, we may assume WLOG that $f_\alpha \leqslant \mathbb{1}$ for every α. Fix an open non-empty set U and $\varepsilon > 0$. Let V and α_0 be as in the assumption. Fix any $t \in V$. By Urysohn's Lemma A.2.2, find $h \in C(K)_+$ such that $h(t) = 0$ and h equals 1 outside of V. Put $g = h \vee \varepsilon\mathbb{1}$. Then $g(t) = \varepsilon$. We claim that $f_\alpha \leqslant g$ for every $\alpha \geqslant \alpha_0$. Indeed, if $s \in V$ then $f_\alpha(s) < \varepsilon \leqslant g(s)$, and if $s \notin V$ then $f_\alpha(s) \leqslant 1 = h(s) \leqslant g(s)$.

Repeat this process for every pair (U, ε), where U is open and non-empty and $\varepsilon > 0$; let G be the set of the resulting functions g. Each such g dominates a tail of (f_α). Lemma 2.4.29 yields $\inf G = 0$; this completes the proof. □

Next, we explore relationship between order convergence and pointwise convergence on "large" subsets. Recall that a subset A of K is A ***meagre*** if it is a union of a sequence of nowhere dense sets; A is ***co-meager*** if its complement is meagre; see A.2.5.

2.4.31 Lemma. *For $G \subseteq C(K)_+$, TFAE:*

(i) $\inf G = 0$*;*

(ii) *There exists a dense set D such that* $\inf_{g\in G} g(t) = 0$ *for every $t \in D$;*

(iii) *There exists a co-meagre set D such that* $\inf_{g\in G} g(t) = 0$ *for every $t \in D$;*

Proof. (ii)⇒(i) by Lemma 2.4.29. (iii)⇒(ii) because every co-meagre set is dense by Baire Category Theorem as in Corollary A.2.7. To prove (i)⇒(iii), assume that $\inf G = 0$. For $n \in \mathbb{N}$, put $W_n = \bigcup_{g\in G}\{g < \frac{1}{n}\}$. Clearly, W_n is open. For every non-empty open set U, Lemma 2.4.29 yields a point $t \in U$ and $g \in G$ with $g(t) < \frac{1}{n}$; hence $t \in W_n$. This means that W_n is dense. Put $D \coloneqq \bigcap_{n=1}^\infty W_n$, then D is co-meagre as in A.2.5. Let $t \in D$, then for every $n \in \mathbb{N}$ we have $t \in W_n$, hence $\inf_{g\in G} g(t) \leqslant \frac{1}{n}$. It follows that $\inf_{g\in G} g(t) = 0$. □

2.4.32 Theorem. *[[BT22, vdW18]] If $f_\alpha \xrightarrow{\text{o}} f$ in $C(K)$ then f_α converges to f pointwise on a co-meagre set. The converse is true for order bounded sequences.*

Proof. WLOG, $f = 0$. Suppose that $f_\alpha \xrightarrow{\text{o}} 0$. Let (g_γ) be a dominating net for this convergence. Put $G = \{g_\gamma\}$. Then $\inf G = \inf g_\gamma = 0$. By Lemma 2.4.31, there exists a co-meagre set D such that for every $t \in D$ we have $0 = \inf_{g\in G} g(t) = \inf_\gamma g_\gamma(t)$ and, therefore, $\lim_\alpha f_\alpha(t) = 0$.

Suppose that an order bounded sequence (f_n) converges to zero on a co-meagre set D. We will use Theorem 2.4.30 to show that $f_n \xrightarrow{\text{o}} 0$. WLOG, $f_n \geqslant 0$ for all n. Fix an open non-empty set U and $\varepsilon > 0$. For each m, put $W_m = \bigcup_{n\geqslant m}\{f_n > \varepsilon\}$. Clearly, W_m is open. If $t \in \bigcap_m W_m$ then $\limsup_n f_n(t) \geqslant \varepsilon$, hence $t \notin D$. It follows that $\bigcap_m W_m$ is contained in D^C, hence is meagre. Since W_m is open, ∂W_m is nowhere dense for every m; we conclude that $\bigcup_m \partial W_m$ is meagre. It follows from

$$\bigcap_m \overline{W_m} \subset \Bigl(\bigcap_m W_m\Bigr) \cup \Bigl(\bigcup_m \partial W_m\Bigr)$$

that $\bigcap_m \overline{W_m}$ is meagre. We conclude that its complement $\bigcup_m \overline{W_m}^C$ is co-meagre, hence dense. It follows that it meets U and, therefore, $U \cap \overline{W_m}^C \neq \varnothing$ for some m. Denote this intersection by V. For every $t \in V$, it follows from $t \notin W_m$ that $f_n(t) \leqslant \varepsilon$ for all $n \geqslant m$. □

2.4.33 Example. The following example shows that one cannot replace "co-meagre" with "dense" in Theorem 2.4.32. Let (f_n) be the (standard) Schauder basis in $C[0,1]$. Let D be the set of all ***dyadic points*** in $[0,1]$, i.e., the set of all the points of the form $\frac{k}{2^n}$ where $k, n \in \mathbb{N}$ with $k \leqslant 2^n$. Then D is dense in $[0,1]$ and $\big(f_n(t)\big)$ is eventually zero for every $t \in D$, so that (f_n) converges to zero pointwise on D. Yet, it is easy to see that $\sup_{n \geqslant m} f_n = \mathbb{1}$ for every m, hence (f_n) does not converge to zero in order.

2.4.34 Example. We will construct an order bounded net that converges to zero pointwise at every point, yet fails to converge in order. This shows that the converse implication in Theorem 2.4.32 generally fails for nets. Let $X = C[0,1]$; let Λ be the set of all finite subsets of $[0,1]$ containing 0 and 1, ordered by inclusion. For each $\alpha \in \Lambda$, $\alpha = \{t_1, \dots, t_n\}$ with $0 = t_0 < t_1 < \dots < t_n = 1$, we define f_α as follows: we put $f_\alpha(t_i) = 0$ for every i, we put f_α to be 1 at the midpoints, i.e., $f_\alpha\big(\frac{t_{i-1}+t_i}{2}\big) = 1$ for all $i = 1, \dots, n$, and define f_α linearly in between. It is easy to see that (f_α) converges to zero pointwise on $[0,1]$. On the other hand, we claim that the supremum of every tail of this net is $\mathbb{1}$, hence the nets fails to converge to zero in order. Indeed, fix $\alpha \in \Lambda$, and suppose that $h \geqslant f_\beta$ for all $\beta \geqslant \alpha$. Let $s \notin \alpha$. Then one can find $\beta \supseteq \alpha$ such that s is a midpoint for β. Hence, $h(s) \geqslant f_\beta(s) = 1$. Thus, $h(s) \geqslant 1$ for all $s \notin \alpha$. Since α is a finite set and h is continuous, it follows that $h \geqslant \mathbb{1}$.

2.5 Convergences: norm vs order vs uniform

We start with an easy but very useful exercise:

2.5.1 Exercise. Let (x_n) be a monotone sequence in a normed lattice. If it has a convergence subsequence then the entire sequence is convergent.

2.5.2 Exercise. A normed lattice X is norm complete iff every increasing Cauchy sequence in X_+ is norm convergent.

We already know that uniform convergence implies both order convergence and norm convergence. What is the relationship between order convergence and norm convergence in X? In general, neither implies the other:

- As per Examples 2.4.4, the unit vector sequence (e_n) in ℓ_∞ converges to zero in order but not in norm;
- As per Example 2.3.5, the sequence $\frac{1}{n}e_n$ in ℓ_1 converges to zero in norm, but is not order bounded, hence does not converge in order.

The following two results show that norm convergence "often" implies order convergence. First, norm convergence implies order convergence for monotone nets:

2.5.3 Lemma (Monotone Convergence Lemma)**.** *If (x_α) is an increasing net in a normed lattice then $x_\alpha \xrightarrow{\|\cdot\|} x$ implies $x_\alpha \uparrow x$. Similarly, $x_\alpha\downarrow$ and $x_\alpha \xrightarrow{\|\cdot\|} x$ imply $x_\alpha \downarrow x$.*

Proof. Suppose $x_\alpha\uparrow$ and $x_\alpha \xrightarrow{\|\cdot\|} x$. Fix any α_0, then $x_\alpha \geqslant x_{\alpha_0}$ for every $\alpha \geqslant \alpha_0$. Passing to the limit (see 1.3.20), we get $x \geqslant x_{\alpha_0}$. Since α_0 is arbitrary, it follows that x is an upper bound of the net. Suppose that y is also an upper bound of (x_α). Again, passing to the limit, we get $y \geqslant x$. Hence, $x = \sup x_\alpha$. □

Recall that, according to Theorem 2.3.7, every norm convergent sequence in a Banach lattice has a uniformly convergent subsequence. Since uniformly convergence implies σ-order convergence, this immediately yields the following:

2.5.4 Theorem. *Every norm convergent sequence in a Banach lattice has a subsequence which σ-order converges (and, therefore, order converges) to the same limit.*

2.5.5 Exercise. In a Banach lattice, every order closed set is norm closed.

2.5.6 Exercise. (i) Does every norm convergent sequence in a normed lattice have an order null subsequence? An order bounded subsequence?

(ii) Does every norm convergent net in a Banach lattice have an order convergent subnet? An order bounded subnet?

2.5.7 Exercise. Find an example of a Banach lattice where the unit ball fails to be order closed.

2.5.8. Suppose that Y is a closed sublattice of a Banach lattice X and (x_n) is a norm null sequence in Y. If $x_n \xrightarrow{o} 0$ in X, does this imply that $x_n \xrightarrow{o} 0$ in Y? We will show in Example 18.8.7 that the answer is "no".

2.5.1 Order continuous Banach lattices

A normed lattice X is said to be ***order continuous*** if $x_\alpha \xrightarrow{o} x$ implies $x_\alpha \to x$ for every net (x_α) and every x in X. Order continuous Banach lattices play a major role the theory of Banach lattices.[2] In this section, we present only a few relatively easy

[2]In literature, such spaces are sometimes called *Banach lattices with an order continuous norm.*

(but useful) observations; we will study order continuous Banach lattices in more detail in Chapter 10. We start by listing several equivalent definitions of order continuity.

2.5.9 Proposition. *Suppose that X is a normed lattice. Then the following are equivalent.*

(i) *X is order continuous;*

(ii) *$x_\alpha \xrightarrow{o} x$ implies $\|x_\alpha\| \to \|x\|$;*

(iii) *$x_\alpha \downarrow 0$ implies $x_\alpha \to 0$;*

(iv) *$0 \leqslant x_\alpha \uparrow x$ implies $x_\alpha \to x$.*

Proof. The implications (i)⇒(ii)⇒(iii) and (i)⇒(iv) are trivial.

(iii)⇒(i) Suppose that $x_\alpha \xrightarrow{o} x$. Let $u_\gamma \downarrow 0$ be a dominating net for the net $|x_\alpha - x|$ as in the definition of order convergence. It follows from $\|x_\alpha - x\| \leqslant \|u_\gamma\| \to 0$ that $\|x_\alpha - x\| \to 0$.

(iv)⇒(iii) Assume that $x_\alpha \downarrow 0$. Fix any α_0. For all $\alpha \geqslant \alpha_0$ we have $0 \leqslant (x_{\alpha_0} - x_\alpha) \uparrow x_{\alpha_0}$, so that $x_{\alpha_0} - x_\alpha \to x_{\alpha_0}$ and, therefore, $x_\alpha \to 0$. $\square$

Note that (ii) means that the norm is an order continuous function, while (iv) and (iii) provide a way to define order continuity without using order convergence, and only using monotone nets.

There is also a sequential variant of order continuity: a normed lattice is ***σ-order continuous*** if $x_n \xrightarrow{\sigma o} x$ implies $x_n \to x$ for every sequence (x_n) and every x in X. The following is the sequential version of Proposition 2.5.9:

2.5.10 Proposition. *Suppose that X is a normed lattice. Then the following are equivalent.*

(i) *X is σ-order continuous;*

(ii) *$x_n \xrightarrow{\sigma o} x$ implies $\|x_n\| \to \|x\|$;*

(iii) *$x_n \downarrow 0$ implies $x_n \to 0$;*

(iv) *$0 \leqslant x_n \uparrow x$ implies $x_n \to x$.*

Since $x_n \xrightarrow{\sigma o} x$ implies $x_n \xrightarrow{o} x$, order continuity implies σ-order continuity. Example 2.4.4 shows that ℓ_∞ fails to be σ-order continuous (and, therefore, fails to be order continuous).

2.5.11 Example. *$C[0,1]$ is not σ-order continuous.* Let $x_n \in C[0,1]$ be given by $x_n(t) = t^n$. Then $x_n \downarrow 0$, but $\|x_n\| = 1$ for all n, hence the sequence (x_n) fails (iii) in Proposition 2.5.9.

2.5.12 Exercise. Show that $L_\infty[0,1]$ is not σ-order continuous.

2.5.13 Proposition. *$L_p(\mu)$ is order continuous when $1 \leqslant p < \infty$.*

Proof. We first observe that $L_p(\mu)$ is σ-order continuous. This is, essentially, Lebesgue Dominated Convergence Theorem A.3.13. Indeed, suppose that $f_n \downarrow 0$. By Proposition 2.1.14, we have $f_n \xrightarrow{\text{a.e.}} 0$. Hence $f_n^p \xrightarrow{\text{a.e.}} 0$. By Lebesgue Dominated Convergence Theorem, $\int f_n^p \to 0$, hence $\|f_n\| \to 0$.

For order continuity, suppose that $f_\alpha \downarrow 0$. By Exercise 2.1.18, we can find a sequence of indices (α_n) such that $f_{\alpha_n} \downarrow 0$. As above, we have $\|f_{\alpha_n}\| \to 0$. It now follows from $f_\alpha \downarrow$ that $\|f_\alpha\| \to 0$. □

In particular, ℓ_p spaces are order continuous when $1 \leqslant p < \infty$. More generally, all Banach sequence spaces in the sense of 1.5.11 are order continuous:

2.5.14 Proposition. *Let X be a Banach lattice whose order is given by a basis. Then X is order continuous.*

Proof. Suppose that $x_\alpha \downarrow 0$; we will prove that $x_\alpha \to 0$. Let $\varepsilon > 0$. Fix any index α_0. By the definition of a basis, $P_n x_{\alpha_0} \to x_{\alpha_0}$ as $n \to \infty$, where P_n is the n-th basis projection. Find n_0 so that $\|x_{\alpha_0} - P_n x_{\alpha_0}\| < \varepsilon$. Since $x_\alpha \downarrow 0$, it converges to zero coordinate-wise. It follows that $P_{n_0} x_\alpha \to 0$ as $\alpha \to \infty$. Therefore, there exists $\alpha_1 \geqslant \alpha_0$ such that $\|P_{n_0} x_{\alpha_1}\| < \varepsilon$. It follows that

$$\|x_\alpha\| \leqslant \|P_{n_0} x_\alpha\| + \|(I - P_{n_0}) x_\alpha\| \leqslant \|P_{n_0} x_{\alpha_1}\| + \|(I - P_{n_0}) x_{\alpha_0}\| < 2\varepsilon.$$

whenever $\alpha \geqslant \alpha_1$. □

It follows, in particular, that c_0 is order continuous.

Recall that in a general Banach lattice, every order closed sets are norm closed by Exercise 2.5.5. It is easy to see that in an order continuous Banach lattice the converse is also true:

2.5.15 Exercise. In an order continuous Banach lattice, order closed and norm closed sets agree.

It follows that order topology in order continuous Banach lattices agrees with the norm topology. In particular, it is Hausdorff and linear. This is one of the reasons why order continuous Banach lattices have nice vector lattice properties.

2.5.2 * Order vs uniform convergences

We will now use machinery of order continuity to explicitly characterize order and uniformly convergence sequences in some spaces.

2.5.16 Corollary. *Norm, order, and uniform convergences agree for sequences in c_0.*

Proof. WLOG, it suffices to consider convergence to zero. Uniform convergence always implies order convergence. Since c_0 is order continuous, order convergence in it implies norm convergence. Suppose $x_n \to 0$. Then $x_n \to 0$ in ℓ_∞. Exercise 2.3.2 yields that $x_n \xrightarrow{\mathrm{u}} 0$ in ℓ_∞. It follows from Theorem 2.3.17 that $x_n \xrightarrow{\mathrm{u}} 0$ in c_0. ☐

Recall that in a general vector lattice, uniform convergence implies order convergence. In the order continuous case, these convergences agree and, moreover, this property characterizes order continuous Banach lattices:

2.5.17 Theorem. *A Banach lattice X is order continuous iff uniform convergence of nets in X agrees with order convergence.*

Proof. Suppose that X is order continuous. We already know that uniform convergence implies order convergence. Suppose that $x_\alpha \xrightarrow{\mathrm{o}} x$, we will show that $x_\alpha \xrightarrow{\mathrm{u}} x$. Let (u_γ) be a dominating net as in the definition of order convergence. Since $u_\gamma \downarrow 0$ and X is order continuous, we have $u_\gamma \to 0$. Lemma 2.3.19 yields $u_\gamma \xrightarrow{\mathrm{u}} 0$. Then there exists $e \in X_+$ such that for every $\varepsilon > 0$ there exists γ_0 such that $\|u_{\gamma_0}\|_e < \varepsilon$. There exists α_0 such that for all $\alpha \geqslant \alpha_0$ one has $|x_\alpha - x| \leqslant u_{\gamma_0}$, hence $\|x_\alpha - x\|_e < \varepsilon$. Therefore, $\|x_\alpha - x\|_e \to 0$, so that $x_\alpha \xrightarrow{\mathrm{u}} x$.

If uniform convergence and order convergence agree on nets in X then $x_\alpha \xrightarrow{\mathrm{o}} x$ implies $x_\alpha \xrightarrow{\mathrm{u}} x$, which, in turn, implies $x_\alpha \to x$, so that X is order continuous. ☐

2.5.18 Exercise. A Banach lattice X is σ-order continuous iff every σ-order convergent sequence converges uniformly (cf. Exercise 2.4.20).

Curiously, we will show later in Corollary 10.1.5 that uniform convergence agrees with order convergence on sequences in X iff X is order continuous.

Recall that in Proposition 2.4.27 and Exercise 2.4.28, we characterized order convergence for sequences in $L_p(\mu)$, $0 \leqslant p \leqslant \infty$. By Theorem 2.5.17, uniform and order convergences agree on $L_p(\mu)$ when $1 \leqslant p < \infty$. For sequences, they agree even when $p < 1$:

2.5.19 Exercise. For sequences in $L_p(\mu)$ with $0 \leqslant p < \infty$, uniform convergence agrees with order convergence (when $p = 0$ we assume, in addition, that μ is σ-finite).

2.5.20 Example. Corollary 2.5.16 asserts that norm, order, and uniform convergences agree for sequences in c_0. What about nets? By Theorem 2.5.17, uniform and order convergences agree for nets in c_0. However, a norm convergent net need not be uniformly convergent. Indeed, let $\Lambda = \mathbb{N}^2$ indexed lexicographically, and define $x_{(n,m)} = \frac{1}{n}e_m$. That is,

$$(e_1, e_2, e_3, \dots, \tfrac{1}{2}e_1, \tfrac{1}{2}e_2, \tfrac{1}{2}e_3, \dots \tfrac{1}{3}e_1, \tfrac{1}{3}e_2, \tfrac{1}{3}e_3, \dots).$$

Then $x_n \to 0$, but no tail of (x_n) is order bounded.

2.5.21 Exercise. Show that a net in $\mathbb{R}^{\mathbb{N}}$ converges uniformly iff it converges in order iff it has an order bounded tail and converges coordinate-wise. The order boundedness condition may be dropped for sequences (but not for nets).

2.5.3 Order continuity and order completeness

2.5.22 Theorem. *Every order continuous Banach lattice is order complete.*

Proof. We present a direct proof here, but we will present a much easier proof in 6.10.7.

Let X be an order continuous Banach lattice and suppose that $0 \leqslant x_\gamma \uparrow\leqslant u$ for some net $(x_\gamma)_{\gamma\in\Gamma}$ and some u. Let $A = \{x_\gamma\}$. Since every Banach lattice is Archimedean by Exercise 1.3.22, we conclude from Proposition 1.2.13 that $\inf(A^\uparrow - A) = 0$.

Clearly, with any $a, b \in A^\uparrow$ we have $a \wedge b \in A^\uparrow$. Hence, we may view $A^\uparrow$ as a decreasing net, $A^\uparrow = (y_\lambda)_{\lambda\in\Lambda}$. For each $\lambda \in \Lambda$ and $\gamma \in \Gamma$, put $z_{\lambda,\gamma} = y_\lambda - x_\gamma$. This defines a net indexed by $\Lambda \times \Gamma$. This net is, clearly, decreasing and it follows from $\inf(A^\uparrow - A) = 0$ that $z_{\lambda,\gamma} \downarrow 0$. Order continuity yields $z_{\lambda,\gamma} \to 0$. Fix $\varepsilon > 0$. Then there exist λ_0 and γ_0 such that $\|z_{\lambda,\gamma}\| < \varepsilon$ whenever $\lambda \geqslant \lambda_0$ and $\gamma \geqslant \gamma_0$. Hence, for any $\alpha, \beta \geqslant \gamma_0$ we have

$$\|x_\alpha - x_\beta\| \leqslant \|y_{\lambda_0} - x_\alpha\| + \|y_{\lambda_0} - x_\beta\| \leqslant 2\varepsilon.$$

It follows that the net (x_γ) is Cauchy. Hence, $x_\gamma \to x$ for some x. Finally, $x = \sup x_\gamma$ by the Monotone Convergence Lemma 2.5.3. □

2.5.23 Exercise. Every closed sublattice of an order continuous Banach lattice is order continuous.

We know that order continuity implies order completeness, and each of the two properties implies its σ version. The following theorem asserts that, surprisingly, the two "σ-" properties together yield the two non-σ ones; see the diagram on page 2. It also provides a convenient criterion for order continuity.

2.5.24 Theorem (Nakano). *For a Banach lattice X, the following are equivalent.*

(i) *X is order continuous;*

(ii) *X is σ-order complete and σ-order continuous;*

(iii) *every increasing order bounded sequence converges.*

Proof. The implications (i)$\Rightarrow$(ii)$\Rightarrow$(iii) are straightforward. Suppose that X satisfies (iii), and assume that $0 \leqslant x_\alpha \uparrow x$. We need to show that $x_\alpha \to x$. Actually, it suffices to simply show that (x_α) converges, as in this case it has to converge to x by the Monotone Convergence Lemma 2.5.3. Suppose that (x_α) is not convergent, hence not Cauchy. Then we can find some $\varepsilon > 0$ and $\alpha_1 < \alpha_2 < \alpha_3 < \dots$ such that $\|x_{\alpha_{i+1}} - x_{\alpha_i}\| > \varepsilon$ for every i. But the sequence (x_{α_i}) is increasing and bounded above by x, hence converges; a contradiction. $\square$

Note that ℓ_∞ is order complete but not order continuous, hence, not σ-order continuous. We will further study the relationship between the properties in Nakano's Theorem in Section sec:s-o-cont

2.5.25 Proposition. *Let X be an order continuous Banach lattice, and (x_n) a disjoint order bounded sequence in X, Then $x_n \to 0$.*

Proof. While this proposition may be deduced from Exercise 2.4.26, we present here a short direct proof. Suppose that $|x_n| \leqslant u$ for all n. For every m, we have $v_m \coloneqq \sum_{n=1}^m |x_n| = \bigvee_{n=1}^m |x_n| \leqslant u$. Since X is order complete, $v \coloneqq \sup v_m$ exists. Since X is order continuous, it follows from $v_m \uparrow v$ that $v_m \to v$. Therefore, the series $\sum_{n=1}^\infty |x_n|$ converges. It follows that $|x_n| \to 0$ and, therefore, $x_n \to 0$. $\square$

In fact, we will show later, in Meyer-Nieberg Theorem 10.1.2, that this property characterizes order continuous Banach lattices.

2.5.26 Exercise. Show using only tools of Measure Theory that every order bounded disjoint sequence in $L_p[0,1]$ with $1 \leqslant p < \infty$ is norm null.

2.5.27 Corollary. *A disjoint order bounded collection in an order continuous Banach lattice is at most countable.*

Proof. Let Λ be a disjoint order bounded collection in an order continuous Banach lattice. WLOG, the elements of Λ are positive and non-zero. Suppose that $\Lambda \subseteq [0,u]$ for some u. For each n, put $\Lambda_n = \{x \in \Lambda \;:\; \|x\| \geqslant \frac{1}{n}\}$. By Proposition 2.5.25, Λ_n is finite for every n, hence Λ is at most countable. $\square$

2.5.28 Example. The last two results fail without the order continuity assumption. For example, the standard unit sequence in ℓ_∞ is order bounded and disjoint, yet not norm null. The space $\ell_\infty(\Omega)$, where Ω is an uncountable set, clearly contains an uncountable disjoint collection.

2.5.29 Example. *A Banach lattice that is σ-order continuous but neither order continuous nor σ-order complete.* Let Ω be an uncountable set, let $X = \ell_\infty(\Omega)$, the space of all bounded real-valued functions on Ω with $\|f\| = \sup_{\omega\in\Omega}|f(\omega)|$. For $\omega \in \Omega$, we write δ_ω for $\mathbb{1}_{\{\omega\}}$.

Clearly, X is a Banach lattice. Let Y be the closed sublattice generated by the characteristic functions of finite sets and $\mathbb{1}$. It is easy to see that Y consists of functions of the form $\lambda_0\mathbb{1} + \sum_{i=1}^{\infty} \lambda_i\delta_{\omega_i}$ where $\lambda_i \to 0$ in $\mathbb{R}$ and (ω_i) is a sequence of distinct points in Ω.

Fix a sequence (ω_i) of distinct points in Ω and put $x_n = \sum_{i=1}^{n} \delta_{\omega_i}$. Then $x_n \uparrow\leqslant \mathbb{1}$, yet (x_n) has no supremum in Y. It follows that Y is not σ-order complete and, therefore, not order continuous by Nakano's Theorem 2.5.24.

There is also a direct way to see that Y is not order continuous: let Λ be the collection of all finite subsets of Ω; it is a directed set when ordered by inclusion. The net $(\mathbb{1}_A)_{A\in\Lambda}$ is an increasing net in Y with supremum $\mathbb{1}$; yet it does not converge to $\mathbb{1}$ in norm.

Finally, one can show that Y is σ-order continuous by proving that if $x_n \downarrow 0$ in Y then $x_n \to 0$; we leave details to the reader.

2.5.30 Example. Most of the results in this section fail when Banach lattices are replaced with norm lattices. Let X be the norm dense sublattice of $L_1[0,1]$ consisting of all the simple functions. Observe that X is order continuous because if $x_\alpha \downarrow 0$ in X then $x_\alpha \downarrow 0$ in $L_1[0,1]$, hence $x_\alpha \xrightarrow{\|\cdot\|} 0$. On the other hand, it is easy to see that X fails to be σ-order complete.

Chapter 3

Sublattices and ideals

In this chapter we will consider various types of subspaces of a vector lattice. By a ***subspace*** we always mean a linear subspace. Throughout this chapter, X will always stand for a vector lattice. Suppose that Y is a subspace of X. There are several ways in which Y may "respect" the vector lattice structure of X. This leads to several important types of subspaces. A subspace Y of X is a ***lattice subspace*** if it is a lattice under the order induced from X; however, lattice operations in Y may yield different results from those in X. For a subset A of Y, we write $\sup_Y A$ and $\inf_Y A$ for the supremum and the infimum of A in Y, respectively. We say that Y is a ***sublattice*** of X if it is closed under (finite) lattice operations of X. We have observed before that it suffices that Y is closed under any one lattice operation, e.g., $x \vee y$ is in Y whenever $x, y \in Y$. In particular, for every finite set $A \subseteq Y$ we have $\sup_Y A = \sup_X A$. If this property remains valid for every infinite subset A of Y for which $\sup_Y A$ exists, we say that Y is a ***regular*** sublattice. A sublattice Y of X is an ***ideal*** if $0 \leqslant x \leqslant y$ and $y \in Y$ imply $x \in Y$. A ***band*** is an order closed ideal. A band B is a ***projection band*** if every $x \in X$ admits a unique decomposition $x = y + z$ where y is in B and z is disjoint to every vector in B. These classes satisfy the following relations:

$$\begin{aligned}
&\text{lattice subspaces} \supseteq \text{sublattices} \supseteq \text{regular sublattices} \\
&\qquad \supseteq \text{ideals} \supseteq \text{bands} \supseteq \text{projection bands}
\end{aligned}$$

Each of these classes is transitive in the following sense: if Y is a sublattice of X and Z is a sublattice of Y then Z is a sublattice of X.

3.0.1 Exercise. Let $Z \subseteq Y \subseteq X$, where Y is a sublattice of a vector space X and Z a subset of Y.

(i) If Z is a sublattice of X then Z is a sublattice of Y;
(ii) If Z is a regular sublattice of X then Z is a regular sublattice of Y;
(iii) If Z is an ideal of X then Z is an ideal of Y.

It can be easily verified that the intersection of any family of sublattices of X is again a sublattice. Therefore, every set $A \subseteq X$, we can consider the intersection of all sublattices of X containing A; this is the smallest sublattice of X containing A. We will call it the ***sublattice generated by*** A and denote it by $\mathrm{Lat}(A)$. Similarly, the classes of ideals and bands are closed under intersection, so we define $I(A)$ to be the ideal generated by A and $B(A)$ to be the band generated by A. Clearly, $\mathrm{Lat}(A) \supseteq I(A) \supseteq B(A)$.

3.1 Sublattices

A subspace Y of a vector lattice X is a ***sublattice*** if it is closed under the lattice operations. Since all the lattice operations can be expressed via each other, it suffices to verify that Y is closed under any one lattice operation. For example, it suffices to check that $|x|$ is in Y whenever $x \in Y$. Every sublattice is itself a vector lattice; in particular, it is a lattice subspace.

Recall that by Exercise 1.6.8, the range of a lattice homomorphism is a sublattice.

3.1.1 Exercise. Suppose that Y is a lattice subspace of a vector lattice X. Then Y is a sublattice iff the lattice operations in Y agree with those inherited from X iff the formal inclusion map is a lattice homomorphism.

3.1.2 Exercise. Suppose that Y is a lattice subspace of a vector lattice X. Then Y is a sublattice iff $x \wedge y = 0$ in Y implies $x \wedge y = 0$ in X for any $x, y \in Y$.

3.1.3 Example. If $0 \leqslant p \leqslant q \leqslant +\infty$ then $L_q[0,1]$ is a sublattice of $L_p[0,1]$ (even an ideal). Also, the set of all simple functions in $L_p[0,1]$ is a sublattice.

3.1.4 Example. The set of all even functions in $C[-1,1]$ is a sublattice of $C[-1,1]$.

3.1.5 Example. The space of all affine functions on $[0,1]$ introduced in Example 1.4.8 is a lattice subspace of $C[0,1]$, but it is not a sublattice!

3.1.6. Suppose that Y is a sublattice of a normed lattice X. Since lattice operations are norm continuous (by Proposition 1.3.19), the closure $\overline{Y}$ is again a sublattice of X.

3.1.7 Exercise. Is the sum of two sublattices a sublattice?

3.1.1 Sublattice generated by a set

Suppose that X is a vector lattice and $A \subseteq X$. As before, we write $\mathrm{Lat}(A)$ for the sublattice of X generated by A. Clearly, $\mathrm{Lat}(A)$ is the intersection of all the sublattices of X containing A. We say that X is ***finitely generated*** if $X = \mathrm{Lat}(A)$ for some finite subset A. Recall that $\operatorname{span} A$ is the set of all linear combinations of elements of A.

3.1.8 Exercise. $\mathrm{Lat}(A) = \mathrm{Lat}(\operatorname{span} A)$.

3.1.9 Exercise. A function $f \in C[0,1]$ is ***piece-wise affine*** if there exist $0 = t_0 < t_1 < \cdots < t_n = 1$ such that f is linear on $[t_{k-1}, t_k]$ for every $k = 1, \dots, n$. Let Y be the set of all piece-wise affine functions in $C[0,1]$. Then Y is a sublattice of $C[0,1]$. Furthermore, Y is the sublattice of $C[0,1]$ generated by just two functions: $\mathbb{1}$ and $u(t) = t$.

By a ***lattice-linear combination*** of vectors $x_1, \dots, x_n$, we consider any expressions obtained from these vectors using vector and lattice operations. For example, $x_1 + (3x_2 - x_3)^+$ is a lattice-linear combination of x_1, x_2, and x_3. Note that every lattice-linear combination only involves finitely many vectors.

3.1.10 Exercise. $\mathrm{Lat}(A)$ is the set of all lattice-linear combinations of elements of A.

We will show next that every vector in $\mathrm{Lat}(A)$ may be represented by a "canonical" lattice-linear combination of elements of $\operatorname{span} A$. However, we need some preliminaries first. Recall that a subset W of X is a *wedge* if it is closed under addition and non-negative scalar multiplication; $W^\vee$ and $W^\wedge$ are defined as in 1.1.6.

3.1.11 Exercise. If W is a wedge then so are $W^\vee$ and $W^\wedge$.

3.1.12 Proposition. *Let Y be a linear subspace of X. Then $Y^{\vee\wedge} = Y^{\wedge\vee} = \mathrm{Lat}(Y)$.*

Proof. By Exercise 3.1.11, $Y^\vee$, $Y^\wedge$, $Y^{\vee\wedge}$, and $Y^{\wedge\vee}$ are wedges. We claim that $Y^{\vee\wedge} = Y^{\wedge\vee}$. Let $x, y \in Y^\vee$. Then $x = \bigvee_{i=1}^m x_i$ and $y = \bigvee_{j=1}^n y_j$ for some $x_1, \dots, x_m$, $y_1, \dots, y_n$ in Y. By Exercise 1.3.5,

$$x \wedge y = \bigvee_{\substack{i=1,\dots,m \\ j=1,\dots,n}} (x_i \wedge y_j)$$

It follows that $Y^{\vee\wedge} \subseteq Y^{\wedge\vee}$. Similarly, $Y^{\wedge\vee} \subseteq Y^{\vee\wedge}$, so that $Y^{\vee\wedge} = Y^{\wedge\vee}$. Clearly, this set is closed under finite sups and infs. Being a wedge, it is closed under addition and non-negative scalar multiplication. Furthermore, distributive laws yield $-(Y^{\vee\wedge}) = (-Y)^{\wedge\vee} = Y^{\wedge\vee} = Y^{\vee\wedge}$. Thus, $Y^{\vee\wedge}$ is closed under negation and is, therefore, a sublattice of X. Clearly, every sublattice of X which contains Y also contains $Y^{\vee\wedge}$. $\square$

It follows that every element of $\mathrm{Lat}(Y)$ may be written as a finite supremum of finite infima of elements of Y or as a finite infimum of finite suprema of elements of Y. We will next deduce two more "canonical" representations of elements of $\mathrm{Lat}(Y)$ in terms of lattice-linear combinations of elements of Y. We will use the following lemma, which is of independent interest.

3.1.13 Exercise. Suppose that W is a wedge in X such that W is closed under the $\vee$ operation. Then $W - W$ is a sublattice of X.

3.1.14 Example. Let Ω be an open convex subset of $\mathbb{R}^n$ for some n. It follows from Exercise 3.1.13 that the set of differences of real-valued convex functions on Ω is a sublattice of $C(\Omega)$.

3.1.15 Proposition. *If W is a wedge (in particular, a subspace), then* $\mathrm{Lat}(W) = W^\vee - W^\vee = W^\wedge - W^\wedge$.

Proof. By Exercise 3.1.11, $W^\vee$ is a wedge. Clearly, $W^\vee$ is closed under $\vee$. It follows from Exercise 3.1.13 that $W^\vee - W^\vee$ is a sublattice. It follows from $W \subseteq W^\vee - W^\vee$ that $\mathrm{Lat}(W) \subseteq W^\vee - W^\vee$. On the other hand, every element of $W^\vee - W^\vee$ is a lattice-linear expression of elements of W, hence $W^\vee - W^\vee \subseteq \mathrm{Lat}(W)$.

Observe that $-W$ is again a wedge. Clearly, $-W \subseteq \mathrm{Lat}(W)$, so that $\mathrm{Lat}(-W) \subseteq \mathrm{Lat}(W)$; by symmetry, we have $\mathrm{Lat}(-W) = \mathrm{Lat}(W)$. It follows that $\mathrm{Lat}(W) = (-W)^\vee - (-W)^\vee = -W^\wedge + W^\wedge$. □

Hence, if Y is a linear subspace of a vector lattice X, then every element of $\mathrm{Lat}(Y)$ can be written in the form $\bigvee_{i=1}^m x_i - \bigvee_{j=1}^n y_j$ for some $x_1, \dots, x_m, y_1, \dots, y_n$ in Y.

Let A be a subset of a vector lattice X. How "large" can $\mathrm{Lat}(A)$ be compared to A? If A is finite then $\operatorname{span} A$ is necessarily finite dimensional, but $\mathrm{Lat}(A)$ need not be finite-dimensional; Exercise 3.1.9 provides an example of an infinite-dimensional sublattice generated by just two vectors.

3.1.16 Exercise. Let X be a normed lattice. If Y is a separable subspace of X then $\mathrm{Lat}(Y)$ is separable. If A is a countable subset of X then the closed sublattice of X generated by A is separable.

Some of the preceding techniques apply even when X is only an ordered vector space. In this case, lattice operations may still be defined for some elements of X. The following proposition shows that under a certain condition, a subspace of X still "generates a sublattice" in some rather strong sense:

3.1.17 Proposition. *Let Y be a subspace of an ordered vector space X such that $\bigvee_{i=1}^n y_i$ exists in X for every finite collection $y_1, \dots, y_n$ in Y. Then for elements of $Y^\vee - Y^\vee$, lattice operations are defined in X, and $Y^\vee - Y^\vee$ is a vector lattice under these operations.*

Furthermore, Y is a generating set in $Y^\vee - Y^\vee$.

Proof. The proof is a combination of ideas of Exercises 3.1.11 and 3.1.13. Observe that $Y^\vee$ is a wedge by Exercise 1.3.5(i). Let $a \in Y^\vee - Y^\vee$, with $a = x - y$ for some $x, y \in Y^\vee$. Then $x \vee y - y \in Y^\vee - Y^\vee$. It follows from $x \vee y - y = (x - y)^+ = a^+$ that a^+ is defined in X and belongs to $Y^\vee - Y^\vee$. We conclude that for elements of $Y^\vee - Y^\vee$, lattice operations are defined in X. It is easy to see that they also yield lattice operations in $Y^\vee - Y^\vee$, making $Y^\vee - Y^\vee$ into a vector lattice. Finally, by Proposition 3.1.15, Y is generating in $Y^\vee - Y^\vee$. □

3.1.2 Sublattice generated by a disjoint sequence

3.1.18 Exercise. Let X be a vector lattice and A be a disjoint set in X_+. Then $\operatorname{span} A$ is a sublattice of X and $\operatorname{Lat}(A) = \operatorname{span} A$.

3.1.19 Exercise. Consider a linear combination $\sum_{i=1}^n \lambda_i x_i$ of distinct vectors in A. By Exercise 1.5.6,

$$\Bigl|\sum_{i=1}^n \lambda_i x_i\Bigr| = \sum_{i=1}^n |\lambda_i| x_i \in \operatorname{span} A.$$

It follows that $\operatorname{span} A$ is a sublattice. Finally, $\operatorname{Lat}(A) = \operatorname{Lat}(\operatorname{span} A) = \operatorname{span} A$.

We will see in Corollary 5.4.12 that every finite-dimensional sublattice can be constructed this way.

Suppose now that X is a Banach lattice and (x_n) is a disjoint sequence in X_+. By 3.1.6, its closed linear span $[x_n]$ is a sublattice of X. It follows from Proposition 1.5.9 that (x_n) is a 1-unconditional basis of $[x_n]$. So $[x_n]$ is a Banach sequence space in the sense of 1.5.11, i.e., it is a Banach lattice whose order is given by a basis.

When talking about a sequence of vectors in a Banach sequence space, it is sometimes convenient to use subscripts and superscripts. For example, $x_i^{(n)}$ stands for the i-th coordinate of the n-th term of the sequence.

3.1.20 Theorem. *Let X be a Banach sequence space. Closed sublattices of X are exactly the closed spans of (finite or infinite) disjoint positive sequences.*

Proof. Let Y be a closed sublattice of X. We will construct a disjoint sequence $\bigl(x^{(n)}\bigr)$ such that $Y = \bigl[x^{(n)}\bigr]$. Here is the idea: if we knew the $x^{(n)}$'s, then every vector in Y restricted to $\operatorname{supp} x^{(n)}$ would be just a scalar multiple of $x^{(n)}$. Note also that multiplying every $x^{(n)}$ by a positive scalar does not affect $\bigl[x^{(n)}\bigr]$.

Define Λ to be the support of Y, i.e., $\Lambda := \{i \in \mathbb{N} : \exists y \in Y \ y_i \neq 0\}$. For $i, j \in \Lambda$, we write $i \sim j$ if $y_i = 0$ iff $y_j = 0$ for every $y \in Y$. Clearly, this is an equivalence relation on Λ. The equivalence classes of this relation will be the supports of our $x^{(n)}$'s.

Claim: if $i, j \in \Lambda$ and $i \not\sim j$ then there exists $y \in Y$ such that $y_i = 1$ and $y_j = 0$. Indeed, if there exists $y \in Y$ such that $y_i \neq 0$ and $y_j = 0$ then we are done by scaling

y. Otherwise, we can find $u \in Y$ such that $u_i = 0$ and $u_j \neq 0$. Since $i \in \Lambda$, we can also find $v \in Y$ such that $v_i \neq 0$. Now we can choose y to be an appropriate linear combination of u and v. This proves the claim. Note also that replacing y with $|y|$, we may assume that $y \in Y_+$.

Fix an equivalence class A of "$\sim$" and any $n \in A$. Let P_A be the natural basis projection onto A, that is, $P_A x$ is the sequence that agrees with x on A and vanishes outside of A. It follows from the definition of "$\sim$" that $P_A x$ and $P_A y$ are proportional for any $x, y \in Y$. That is, $P_A(Y)$ is one-dimensional. Enumerating $\Lambda \setminus A$ and applying the *Claim* to every $j \in \Lambda \setminus A$, we produce a sequence $(y^{(m)})$ in Y_+ such that $y_n^{(m)} = 1$ for all m, and for every $j \notin A$ there exists m with $y_j^{(m)} = 0$. Replacing $y^{(m)}$ with $y^{(1)} \wedge \dots \wedge y^{(m)}$, we may assume that $y^{(m)} \downarrow$. Since X is order complete and $y^{(m)} \geqslant 0$, we have $y^{(m)} \downarrow x$ for some $x \in X_+$. Since X is order continuous by Theorem 2.5.14, $y^{(m)}$ converges to x in norm, hence $x \in Y$. Also, it follows from $y^{(m)} \downarrow x$ that $x_n = 1$ and $x_i = 0$ for all $i \notin A$. By the definition of "$\sim$", $x_k \neq 0$ for all $k \in A$. This yields $\operatorname{supp} x = A$. It follows that $P_A(Y) = \operatorname{span}\{x\}$.

Let $A_1, A_2, \dots$ be the equivalence classes of "$\sim$". For each n, fix $x^{(n)} \in Y_+$ such that $P_{A_n}(Y) = \operatorname{span} x^{(n)}$. Clearly, $(x^{(n)})$ is a disjoint sequence in Y. Take any $z \in Y$, it suffices to show that $z \in [x^{(n)}]$. WLOG, $z \geqslant 0$ as, otherwise, we can consider z^+ and z^-. For each m, put $z^{(m)} = P_{A_1} z + \dots + P_{A_m} z$. Then $z^{(m)} \in \operatorname{span} x^{(n)}$. It follows from $z^{(m)} \uparrow z$ that that $z^{(m)}$ converge to z in norm; hence $z \in [x^{(n)}]$. $\square$

3.1.21 Corollary. *A subspace of $\mathbb{R}^n$ is a sublattice iff it is spanned by a disjoint positive sequence.*

3.1.22 Corollary. *Every closed sublattice of ℓ_p with $1 \leqslant p < \infty$ is lattice isometric to ℓ_p or to ℓ_p^n for some n. Every closed sublattice of c_0 is lattice isometrically isomorphic to c_0 or to ℓ_∞^n for some n.*

3.1.23 Example. The corollary may fail for non-closed sublattices: ℓ_1 is a sublattice of c_0, yet ℓ_1 is not lattice isomorphic to c_0 or ℓ_∞^n for some n.

3.1.24 Exercise. Show that c is a closed sublattice of ℓ_∞, but it is not the closed span of any disjoint positive sequence. Why does not this contradict Theorem 3.1.20?

3.1.3 Sublattices of $L_p(\mu)$

3.1.25 Theorem. *Suppose that $X = L_p(\Omega, \mathcal{F}, \mu)$ for some $1 \leqslant p < \infty$ and a finite measure μ. Norm closed sublattices of X containing $\mathbb{1}$ are of the form $L_p(\Omega, \mathcal{G}, \mu_{|\mathcal{G}})$ where $\mathcal{G}$ is a sub-σ-algebra of $\mathcal{F}$.*

Proof. Let $Y = L_p(\Omega, \mathcal{G}, \mu_{|\mathcal{G}})$ for some sub-σ-algebra $\mathcal{G}$ of $\mathcal{F}$. Clearly, Y is a closed subspace of X (with the same norm) and $\mathbb{1} \in Y$. Since Y is an L_p-space, it is itself a Banach lattice. Since the norm on Y agrees with that of X, Y is a norm closed subspace of X. Finally, since lattice operations in Y are a.e.-pointwise, they agree with the lattice operations in X, so that Y is a sublattice of X.

Conversely, suppose that Y is a closed sublattice of X with $\mathbb{1} \in Y$. Put $\mathcal{G} = \big\{A \in \mathcal{F} : \mathbb{1}_A \in Y\big\}$. We claim that $\mathcal{G}$ is a σ-algebra. It follows from $0, \mathbb{1} \in Y$ that $\varnothing$ and Ω are in $\mathcal{G}$. If $A, B \in \mathcal{G}$ then $\mathbb{1}_{\Omega \setminus A} = \mathbb{1} - \mathbb{1}_A \in Y$ and $\mathbb{1}_{A \cup B} = \mathbb{1}_A \vee \mathbb{1}_B \in Y$, so that $\Omega \setminus A$ and $A \cup B$ are in $\mathcal{G}$. Thus, $\mathcal{G}$ is a subalgebra of $\mathcal{F}$. Suppose that $A_1 \subseteq A_2 \subseteq \dots$ is a sequence in $\mathcal{G}$. Put $A = \bigcup_{n=1}^{\infty} A_n$. It suffices to show that $A \in \mathcal{G}$; see Proposition A.3.1. Note that $A \in \mathcal{F}$ because $\mathcal{F}$ is a σ-algebra. Note that $\mathbb{1}_{A_n}\uparrow$ and $\mathbb{1}_{A_n} \to \mathbb{1}_A$ pointwise. It follows that $\mathbb{1}_{A_n} \uparrow \mathbb{1}_A$ by Proposition 2.1.14 and, therefore, $\mathbb{1}_{A_n} \xrightarrow{\|\cdot\|} \mathbb{1}_A$ since $L_p(\mu)$ is order continuous. Since Y is norm closed, this yields $\mathbb{1}_A \in Y$, so that $A \in \mathcal{G}$.

We have $\mathbb{1}_A \in Y$ for every $A \in \mathcal{G}$. It follows that Y contains all $\mathcal{G}$-simple functions. Since $\mathcal{G}$-simple functions are norm dense in $L_p(\Omega, \mathcal{G}, \mu_{|\mathcal{G}})$, this yields $L_p(\Omega, \mathcal{G}, \mu_{|\mathcal{G}}) \subseteq Y$.

To prove the opposite inclusion, let $f \in Y$. Put $A = \{f > 0\}$. The sequence $(nf^+ \wedge \mathbb{1})$ is increasing and converges to $\mathbb{1}_A$ pointwise, hence $(nf^+ \wedge \mathbb{1}) \uparrow \mathbb{1}_A$ and, therefore, $(nf^+ \wedge \mathbb{1}) \xrightarrow{\|\cdot\|} \mathbb{1}_A$, which yields $\mathbb{1}_A \in Y$, hence $A \in \mathcal{G}$. Similarly, $\{f > \lambda\} = \{f - \lambda\mathbb{1} > 0\} \in \mathcal{G}$ for every $\lambda \in \mathbb{R}$. Therefore, f is $\mathcal{G}$-measurable, so that $f \in L_p(\Omega, \mathcal{G}, \mu_{|\mathcal{G}})$. □

3.1.26 Exercise. Given a sublattice Y in $L_p[-1,1]$ ($1 \leqslant p < \infty$), find $\mathcal{G}$ as in Theorem 3.1.25:

(i) Y consists of all the functions that are a.e. constant on the intervals $[-1,0]$ and $[0,1]$.

(ii) Y consists of a.e. even functions.

Theorem 3.1.25 may be extended to general closed sublattices of $L_p(\mu)$; see Exercise 5.1.30.

3.1.4 Sublattices of $C(K)$

We will now present the lattice version of the celebrated Stone-Weierstrass Theorem. We will then deduce the classical Stone-Weierstrass Theorem for subalgebras from it. Throughout this subsection, K stands for a compact space.

3.1.27 Lemma. *Let Y be a sublattice of $C(K)$ and $f \in C(K)$. Then $f \in \overline{Y}$ iff for every finite set F in K there exists h in Y that agrees with f on F. Actually, it suffices to consider sets F consisting of just two points.*

Proof. Let $f \in \overline{Y}$ and $F = \{t_1, \dots, t_n\}$. Define $T\colon C(K) \to \mathbb{R}^n$ via $Th = \bigl(h(t_1), \dots, h(t_n)\bigr)$. Clearly, $TY \subseteq T\overline{Y} \subseteq \overline{TY}$. Since T acts into a finite-dimensional space, $TY = \overline{TY}$, hence $T\overline{Y} = TY$. Therefore, there exists $h \in Y$ with $Tf = Th$. This means that f and h agree on F.

Suppose now that for any $s \neq t$ in K there exists $h \in Y$ such that $h(s) = f(s)$ and $h(t) = f(t)$. Fix $\varepsilon > 0$. For every s and t in K find $f_{s,t} \in Y$ so that $f_{s,t}(s) = f(s)$ and $f_{s,t}(t) = f(t)$. Put

$$\begin{aligned} U_{s,t} &= \bigl\{z \in K \,:\, f_{s,t}(z) < f(z) + \varepsilon\bigr\}, \text{ and} \\ V_{s,t} &= \bigl\{z \in K \,:\, f_{s,t}(z) > f(z) - \varepsilon\bigr\}. \end{aligned}$$

Clearly, each of these two sets is open and contains both s and t. It follows that for each $t \in K$ the collection of sets $\{U_{s,t} \,:\, s \in K\}$ is an open cover of K. Hence, there is a finite subcover, $K = \bigcup_{i=1}^n U_{s_i,t}$. Put $f_t = \bigwedge_{i=1}^n f_{s_i,t}$. Then $f_t \in Y$, $f_t(t) = f(t)$, and $f_t \leqslant f + \varepsilon\mathbb{1}$. Put $V_t = \bigcap_{i=1}^n V_{s_i,t}$. Then V_t is open, $t \in V_t$, and $f_t(z) \geqslant f(z) - \varepsilon$ for all $z \in V_t$. Again, the collection $\{V_t \,:\, t \in K\}$ is an open cover of K, hence there is a finite subcover $K = \bigcup_{j=1}^m V_{t_j}$. Put $g = \bigvee_{j=1}^m f_{t_j}$, then $g \in Y$ and $f - \varepsilon\mathbb{1} \leqslant g \leqslant f + \varepsilon\mathbb{1}$, so that $\lVert f - g\rVert \leqslant \varepsilon$. Since ε was chosen arbitrarily, we conclude that $f \in \overline{Y}$. □

A subset Y of $C(K)$ is said to ***separate points*** of K if for any s and t in K with $s \neq t$ there is $f \in Y$ such that $f(s) \neq f(t)$.

3.1.28 Theorem (Stone-Weierstrass Theorem for sublattices)**.** *Suppose that Y is a sublattice of $C(K)$ such that $\mathbb{1} \in Y$ and Y separates points of K. Then Y is dense in $C(K)$.*

Proof. Let $f \in C(K)$; fix $s \neq t$ in K. There exists $g \in Y$ such that $g(s) \neq g(t)$. It is easy to see that there exist two real numbers λ and μ such that $h \coloneqq \lambda g + \mu\mathbb{1}$ agrees with f at s and at t. Since $h \in Y$, it now follows from Lemma 3.1.27 that $f \in \overline{Y}$. □

3.1.29 Example. Let Y be the sublattice of $C[0,1]$ generated by $\mathbb{1}$ and $h(t) = t$. Since these two functions already separate the points of $[0,1]$, it follows from Theorem 3.1.28 that Y is dense in $C[0,1]$. Alternatively, by Exercise 3.1.9, Y consists of all piecewise affine functions in $C[0,1]$; it is easy to see directly that Y is dense in $C[0,1]$.

Note also that even though h alone already separates the points of $[0,1]$, the sublattice of $C[0,1]$ generated by h only is just the linear span of h and fails to be dense in $C[0,1]$. Hence the theorem fails without the assumption $\mathbb{1} \in Y$.

3.1.30 Lemma. *Every closed subalgebra of $C(K)$ is a sublattice.*

Proof. Let A be a closed subalgebra of $C(K)$; let $f \in A$. It suffices to show that $|f| \in A$. WLOG, $\operatorname{Range} f \subseteq [-1,1]$. Fix any $\varepsilon > 0$. It is a calculus exercise that there exists a polynomial $p(t)$ such that $p(0) = 0$ and $\bigl|p(x) - |x|\bigr| < \varepsilon$ for all $x \in [-1,1]$. It follows that $\bigl|p(f(t)) - |f(t)|\bigr| < \varepsilon$ for all $t \in K$. This means that $\bigl\|p \circ f - |f|\bigr\| < \varepsilon$. Note that $p(0) = 0$ implies that p has no constant term, so that $p \circ f \in A$. It follows that $|f| \in A$ because A is closed. $\square$

The lemma may fail when the subalgebra is not closed. For example, the subalgebra of all polynomials in $C[0,1]$ is not a sublattice.

3.1.31 Theorem (Stone-Weierstrass Theorem for subalgebras)**.** *Suppose that A is a subalgebra of $C(K)$. If A contains $\mathbb{1}$ and separates points then A is dense in $C(K)$.*

Proof. Since addition and multiplication are continuous on $C(K)$, the closure $\overline{A}$ of A is again a subalgebra of $C(K)$. By Lemma 3.1.30, $\overline{A}$ is a sublattice and, therefore, $\overline{A} = C(K)$ by Theorem 3.1.28. $\square$

Next, we consider sublattices which do not separate points. The idea is that if our sublattice does not distinguish certain points, we "merge" them together.

3.1.32 Theorem. *Let Y be a closed sublattice of $C(K)$ containing $\mathbb{1}$. Then Y is a subalgebra of $C(K)$ and is lattice and algebraically isometric to $C(Q)$ for some compact space Q.*

Proof. For $s, t \in K$, write $s \sim t$ if $f(s) = f(t)$ for all $f \in Y$. Clearly, "$\sim$" is an equivalence relation. Put $Q = K/\sim$. For $s \in K$ we write $\tilde{s}$ for the equivalence class of s. Induce a topology on Q as follows: a set $A \subseteq Q$ is open whenever the set $\{s \in K : \tilde{s} \in A\}$ is open in K. This implies that the map $s \in K \mapsto \tilde{s} \in Q$ is continuous.

For $f \in Y$, we define $\tilde{f} \colon Q \to \mathbb{R}$ via $\tilde{f}(\tilde{s}) \coloneqq f(s)$. It is easy to see that $\tilde{f}$ is well-defined, continuous and $\max \tilde{f} = \max f$. If $\tilde{s} \neq \tilde{t}$ for some $s, t \in K$, then $f(s) \neq f(t)$ for some $f \in Y$; it follows that $\tilde{f}(\tilde{s}) \neq \tilde{f}(\tilde{t})$. Therefore, $\tilde{s}$ and $\tilde{t}$ are separated by a real-valued continuous function on Q. This implies that Q is Hausdorff. Being the image of K under the continuous map $s \mapsto \tilde{s}$, Q is compact.

Define $T \colon Y \to C(Q)$ via $Tf = \tilde{f}$. Then T is an isometry. Also, it follows from $|Tf| = T|f|$ for every $f \in Y$ that T is a lattice homomorphism. By Lemma 1.6.8, $\operatorname{Range} T$ is a sublattice of $C(Q)$. Clearly, $T\mathbb{1} = \mathbb{1}_Q$. It follows from the definition of Q that $\operatorname{Range} T$ separates points of Q. Thus, Theorem 3.1.28 yields that $\operatorname{Range} T$ is dense in $C(Q)$. Since T is an isometry, $\operatorname{Range} T$ is closed, hence $\operatorname{Range} T = C(Q)$.

Let $f, g \in Y$. Since $\operatorname{Range} T = C(Q)$, it follows that $\tilde{f}\tilde{g} = \tilde{h}$ for some $h \in Y$. It is easy to see that $fg = h$, hence Y is a subalgebra of $C(K)$ and T is an algebra homomorphism. □

It is clear from the proof that Y consists exactly of all the functions in $C(K)$ that are constant on certain sets, namely, on the equivalence classes of $\sim$. Combining this theorem with Lemma 3.1.30, we immediately get the following:

3.1.33 Corollary. *Let A be a closed subset of $C(K)$ and $\mathbb{1} \in A$. Then A is a sublattice iff it is a subalgebra.*

This fails for non-closed subsets. The set of all piece-wise affine functions on $[0, 1]$ is a dense sublattice of $C[0, 1]$, but not a subalgebra.

Next, we will characterize closed sublattices of $C(K)$ that are not necessarily unital. We start with some motivation and examples. It can be easily verified that the set of all functions f in $C[0, 1]$ satisfying $f(1) = 2f(0)$ forms a closed sublattice. More generally, consider $X = C(K)$, fix $s \neq t$ in K and a non-negative scalar α, and put $Y = \bigl\{f \in C(K) : f(s) = \alpha f(t)\bigr\}$; then Y is a closed sublattice of $C(K)$. Note that $Y = \ker \mu$ where $\mu \in C(K)^*$ is defined by $\mu = \delta_s - \alpha\delta_t$; we write δ_s for the point-evaluation functional $\delta_s(f) = f(s)$. In case $\alpha = 0$, we just have $\mu = \delta_s$; in this case, t is irrelevant.

In the preceding paragraph, Y was determined by a single constraint $f(s) = \alpha f(t)$. We may generalize this construction to an arbitrary family of constraints as follows. Consider a collection of triples (s, t, α), where s and t are two distinct points in K and $\alpha \geqslant 0$, and let Y be the set of all functions f in $C(K)$ such that $f(s) = \alpha f(t)$ for every triple (s, t, α) in the collection. There is an alternative way to describe Y: let $\mathcal{M}$ be a family of functionals in $C(K)^*$ of the form $\mu = \delta_s - \alpha\delta_t$, where the triple (s, t, α) in the collection; then $Y = \bigcap_{\mu \in \mathcal{M}} \ker \mu = \mathcal{M}_\perp$. It is easy to see that Y is a closed sublattice of $C(K)$. We are going to show that every closed sublattice of $C(K)$ is of this form. Again, we first characterize the closure of a sublattice; this will easily yield a characterization of closed sublattices.

3.1.34 Proposition. *Let Y be a sublattice of $C(K)$. Let $\mathcal{M}$ be the set of all functionals $\mu \in C(K)^*$ of the form $\mu = \delta_s - \alpha\delta_t$, where $s \neq t$ in K and $\alpha \geqslant 0$, such that μ vanishes on Y. Then $\overline{Y} = \mathcal{M}_\perp$.*

Proof. For every $\mu \in \mathcal{M}$, we have $Y \subseteq \ker \mu$, hence $\overline{Y} \subseteq \ker \mu$; it follows that $\overline{Y} \subseteq \mathcal{M}_\perp$. Suppose that $f \in \mathcal{M}_\perp$; we will use Proposition 3.1.27 to show that $f \in \overline{Y}$. Fix $s \neq t$ in K. Put $T : Y \to \mathbb{R}^2$ via $Th = \bigl(h(s), h(t)\bigr)$. If $\operatorname{Range} T$ is all of $\mathbb{R}^2$ then we can find $h \in Y$ such that $\bigl(h(s), h(t)\bigr) = Th = \bigl(f(s), f(t)\bigr)$. If $\operatorname{Range} T = \{0\}$ then δ_s and δ_t vanish on Y, hence belong to $\mathcal{M}$; it follows that they vanish on f and we can take $h = 0$. Finally, suppose that T has rank 1. Then, possibly after interchanging s and t, we find $\alpha \in \mathbb{R}$ such that $h(s) = \alpha h(t)$ for all $h \in Y$. It also follows that $g(t) \neq 0$ for some $g \in Y$. Since $|g| \in Y$, we have $|g(s)| = \alpha|g(t)|$, which yields $\alpha \geqslant 0$. Let $\mu := \delta_s - \alpha\delta_t$. Then $\mu \in \mathcal{M}$, hence $f \in \ker \mu$. Scaling g, if necessary, we may assume that $g(t) = f(t)$. It now follows that $g(s) = \alpha g(t) = \alpha f(t) = f(s)$. □

We now immediately get the following:

3.1.35 Theorem. *Closed sublattices of $C(K)$ are the sets of the form $\mathcal{M}_\perp$, where $\mathcal{M}$ is as in Proposition 3.1.34.*

3.1.36 Corollary. *Every closed sublattice of $C(K)$ of co-dimension one is of the form $\{f \in C(K) : f(s) = \alpha f(t)\}$ for some $s \neq t$ in K and $\alpha \geqslant 0$.*

We know that every closed subspace of a Banach space may be expressed as the intersection of a family of hyperplanes. We now have a somewhat analogous result:

3.1.37 Corollary. *Every closed sublattice of $C(K)$ is the intersection of a family of closed sublattices of co-dimension 1.*

In Proposition 3.1.34, $\mathcal{M}$ consists of all the functionals in $Y^\perp$ of the form $\delta_s - \alpha\delta_t$. We can actually replace it with a much larger set:

3.1.38 Corollary. *Let Y be a sublattice of X, put $\mathcal{M}' = Y^\perp \cap \operatorname{span}\{\delta_s : s \in K\}$. Then $\overline{Y} = \mathcal{M}'_\perp$.*

Proof. It follows from $Y \subseteq \mathcal{M}'_\perp$ that $\overline{Y} \subset \mathcal{M}'_\perp$. Let $\mathcal{M}$ be as in Proposition 3.1.34. It follows from $\mathcal{M} \subseteq \mathcal{M}'$ that $\mathcal{M}'_\perp \subseteq \mathcal{M}_\perp = \overline{Y}$. ☐

3.1.39. We are now going to further explore the structure of a closed sublattice or, more generally, of the closure of a sublattice in $C(K)$. Let Y be a sublattice of $C(K)$, then $\overline{Y} = \mathcal{M}_\perp$ as in Proposition 3.1.34. Let $\ker Y = \{t \in K : \delta_t \in \mathcal{M}\}$. That is, $t \in \ker Y$ iff $f(t) = 0$ for all $t \in Y$. Also, $\ker Y = \bigcap_{f\in Y} \ker f$. For any $s, t \in (\ker Y)^C$, we set $s \sim t$ if there exists $\alpha \neq 0$ such that $f(s) = \alpha f(t)$ for all $f \in Y$. Note that α has to be positive. Indeed, since $t \notin \ker Y$, there exist $f, g \in Y$ with $f(s) \neq 0$ and $g(t) \neq 0$. Put $h = |f| \vee |g|$; then $h \in Y$. It follows from $h(s) = \alpha h(t)$ and $h(s), h(t) > 0$ that $\alpha > 0$. It also follows that α is uniquely determined by s and t. One can easily verify that $s \not\sim t$ iff there exists $f \in Y$ such that $f(s) = 1$ and $f(t) = 0$.

It is clear that $s \sim t$ iff δ_s and δ_t are proportional on Y and, therefore, on $\overline{Y}$. Clearly, this is an equivalence relation on $(\ker Y)^C$; hence yields a partition of $(\ker Y)^C$ into equivalence classes $\{A_\gamma\}$. Let A_γ be such an equivalence class. Fix $f_\gamma \in Y_+$ which does not vanish at some point (and, therefore, at every point) of A_γ. Furthermore, let $g \in Y$ and fix $t \in A_\gamma$. For every $s \in A_\gamma$, it follows from $s \sim t$ that $\frac{g(s)}{f_\gamma(s)} = \frac{g(t)}{f_\gamma(t)}$. Therefore, every function in Y is proportional to f_γ on A_γ. It now follows that for $g \in C(K)$, we have $g \in \overline{Y}$ (or $g \in Y$ if Y is closed) iff g vanishes on $\ker Y$ and is proportional to f_γ on A_γ for every γ. This description is somewhat similar to the situation in Theorem 3.1.20.

We may view $\mathbb{R}^n$ as $C(K)$ where $K = \{1, \dots, n\}$. In this case, 3.1.39 yields a characterization of sublattices of $\mathbb{R}^n$ that agrees with Corollary 3.1.21.

3.2 Regular, order dense, and majorizing sublattices

3.2.1 Regular sublattices

Let Y be a sublattice of a vector lattice X. The definition of a sublattice guarantees that the inclusion map of Y into X preserves the suprema and the infima of finite sets.

This may fail for infinite sets though. For $A \subseteq Y$, we write $\sup_Y A$ for the supremum of A in Y, while $\sup A$ stands for the supremum of A in X; same for infima. We say that Y is a ***regular sublattice*** of X if for every subset A of Y for which $\sup_Y A$ exists, $\sup A$ also exists and equals $\sup_Y A$. The following exercise presents several useful equivalent characterizations of regularity.

3.2.1 Exercise. For a sublattice Y of X, TFAE:

(i) Y is regular;
(ii) For any $A \subseteq Y$, if $\inf_Y A$ exists then $\inf A$ exists and $\inf A = \inf_Y A$;
(iii) for every net (y_α) in Y, if $y_\alpha \downarrow 0$ in Y then $y_\alpha \downarrow 0$ in X;
(iv) for every net (y_α) in Y, if $y_\alpha \xrightarrow{o} y$ in Y then $y_\alpha \xrightarrow{o} y$ in X;
(v) for every net (y_α) in Y, if $y_\alpha \xrightarrow{o} 0$ in Y then $y_\alpha \xrightarrow{o} 0$ in X;
(vi) The inclusion of Y into X is order continuous.

We observed in 2.4.14 that the order convergences in a sublattice may not agree with order convergence in the whole space. The preceding exercise shows that a sublattice Y is regular iff order convergence "passes up" from Y to X.

3.2.2 Example. *A non-regular sublattice.* Let f_n be the function in $C[0,1]$ such that $f_n(0) = 1$, f_n vanishes on $[\frac{1}{n}, 1]$ and f_n is linear on $[0, \frac{1}{n}]$. Then $f_n \downarrow 0$ in $C[0,1]$. However, viewed as a sequence in $\mathbb{R}^{[0,1]}$, $f_n \downarrow \mathbb{1}_{\{0\}}$. Hence, $C[0,1]$ is not regular in $\mathbb{R}^{[0,1]}$.

3.2.3 Exercise. Let Y be a sublattice of a normed lattice X. If Y is order continuous (in the norm inherited from X) then Y is regular in X.

We will now characterize two situations when order convergence passes down to a regular sublattice.

3.2.4 Exercise. Suppose that Y is a regular order complete sublattice of X, (y_α) a net in Y which is order bounded in Y. If $y_\alpha \xrightarrow{o} x$ in X then $x \in Y$ and $y_\alpha \xrightarrow{o} x$ in Y.

3.2.5 Theorem. *Let Y be a regular sublattice of an Archimedean vector lattice X and (y_α) a net in Y which is order bounded in Y. If $y_\alpha \xrightarrow{o} 0$ in X then $y_\alpha \xrightarrow{o} 0$ in Y.*

Proof. WLOG, $y_\alpha \geqslant 0$ for all α; otherwise consider the net $\big(|y_\alpha|\big)$. We will use Exercise 2.4.16. Let A and B be the sets of all eventual dominants of the net in Y and in X, respectively. Then A and B are non-empty subsets of X_+ and $\inf_X B = 0$. We need to prove that $\inf_Y A = 0$.

Suppose not. Then there exists $w \in Y$ with $0 < w \leqslant A$. Since $\inf_X B = 0$, there exists $b_0 \in B$ such that $b_0 \ngeqslant w$ and, therefore, $(w - b_0)^+ > 0$. Since $b_0 \in B$, there exists α_0 such that for every $\alpha \geqslant \alpha_0$ we have $y_\alpha \leqslant b_0$ and, therefore, for every $a \in A$ we have $a - y_\alpha \geqslant w - b_0$, so that $(a - y_\alpha)^+ \geqslant (w - b_0)^+$.

Put $C = \{(a - y_\alpha)^+ : a \in A, \alpha \geqslant \alpha_0\}$. By the preceding, $\inf_X C = 0$ is impossible. Since Y is regular, $\inf_Y C = 0$ is also impossible. Therefore, there exists $v \in Y$ with $0 < v \leqslant C$.

Fix $a \in A$. There exists $\alpha_1 \geqslant \alpha_0$ such that for every $\alpha \geqslant \alpha_1$ we have $y_\alpha \leqslant a$. It follows that $a - y_\alpha = (a - y_\alpha)^+ \geqslant v$, and, therefore, $y_\alpha \leqslant a - v$ for all $\alpha \geqslant \alpha_1$. This yields $a - v \in A$. Iterating this process, we conclude that $a - nv \in A$ for every $n \in \mathbb{N}$. Then $a - nv \geqslant 0$. The Archimedean property asserts that $v = 0$, which is a contradiction. □

3.2.2 Order dense and majorizing sublattices

Let Y be a sublattice of a vector lattice X. We say that Y is

- ***order dense*** in X if $\forall\, 0 < x \in X \quad \exists\, 0 < y \in Y \quad y \leqslant x$;
- ***majorizing*** in X if $\forall\, 0 \leqslant x \in X \quad \exists\, y \in Y \quad y \geqslant x$.

3.2.6 Exercise. (i) c_{00} is order dense in c_0 and in ℓ_p, $(0 < p \leqslant \infty)$.

(ii) ℓ_p is order dense in c_0, $(0 < p < \infty)$.

(iii) c is order dense and majorizing in ℓ_∞.

(iv) c_0 is order dense but not majorizing in ℓ_∞.

(v) $L_\infty(\mu)$ is order dense in $L_p(\mu)$, $(0 \leqslant p \leqslant \infty)$.

(vi) Constant functions are majorizing in $L_\infty(\mu)$ and in $C(K)$ (but not order dense unless the space is one-dimensional).

(vii) Piece-wise affine functions are order dense and majorizing in $C[0,1]$.

(viii) Let V be the set of all simple functions in $L_0(\mu)$, that is, $f \in V$ iff f is a.e. equal to a function with a finite range. Then V is an order dense sublattice in $L_p(\mu)$ for every $0 \leqslant p \leqslant \infty$. Furthermore, V is majorizing in $L_\infty(\mu)$.

3.2.7 Exercise. Let Ω be an arbitrary set and Y a sublattice of $\mathbb{R}^\Omega$. Then Y is order dense in $\mathbb{R}^\Omega$ iff it contains the characteristic function of every singleton.

3.2.8 Exercise. The assumption that $0 \leqslant x$ in the definition of a majorizing sublattice is redundant.

3.2.9 Exercise. Show that if Y is an order dense sublattice of X and Z is an order dense sublattice of Y then Z is order dense in X. Same for “majorizing”.

3.2.10 Exercise. Let Y be a majorizing sublattice of X and $A \subseteq Y$. Then A is order bounded in X iff it is order bounded in Y.

3.2.11 Exercise. Let Y be an order dense sublattice of X and $e \in Y_+$. Then e is a weak unit in Y iff it is a weak unit in X.

3.2.12 Exercise. If Y is a dense sublattice of a $C(K)$ space then Y is order dense in $C(K)$. The converse is false even if we assume, in addition, that Y is an ideal or contains $\mathbb{1}$.

One can say that Y is order dense in X iff every order interval $[0, x]$ in X with $x > 0$ meets Y. Here is a word of caution: this is not equivalent to saying that every order interval $[x, y]$ in X with $x < y$ meets Y. The later condition is, actually so strong it is rarely satisfied. For example, even though c is order dense and majorizing in ℓ_∞, it fails this condition, — just take

$$x = (0, 1, 0, 1, 0, 1, 0, 1, \dots) \text{ and } y = (1, 1, 0, 1, 0, 1, 0, 1, \dots).$$

3.2.13 Proposition. *Every order dense sublattice is regular.*

Proof. Let Y be an order dense sublattice of X; let (y_α) be a net in Y such that $y_\alpha \downarrow 0$ in Y. We need to show that $y_\alpha \xrightarrow{o} 0$ in X. Suppose, for the sake of contradiction, that there $x \in X$ such that $0 < x \leqslant y_\alpha$ for all α. Since Y is order dense, there exists $y \in Y$ such that $0 < y \leqslant x$. It follows that $0 < y \leqslant y_\alpha$ for every α, which contradicts $y_\alpha \downarrow 0$ in Y. □

Not every regular sublattice is order dense: it is easy to see that the x-axis in $\mathbb{R}^2$ is regular but not order dense.

3.2.14 Example. *A (norm) dense non-regular sublattice of a Banach lattice.* Let $X = c_0$ and Y be the set of all vectors of the form $\left(t_1, t_2, \dots, t_n, \frac{t_1}{n+1}, \frac{t_1}{n+2}, \dots\right)$. It is easy to see that Y is a sublattice of X. It is dense because every element of c_{00} can be approximated by elements on Y. Consider the sequence (y_n) in Y where $y_n = \left(1, 0, \dots, 0, \frac{1}{n+1}, \frac{1}{n+2}, \dots\right)$ Then $y_n \downarrow e_1$ in c_0, but $y_n \downarrow 0$ in Y. Hence, Y is not regular.

The rest of the section will critically use the following characterization of order density in Archimedean vector lattices. For this reason, *for the rest of the section, we assume that X is Archimedean.*

3.2.15 Proposition. *Y is order dense iff $x = \sup Y \cap [0, x]$ for every $x \in X_+$.*

Proof. Suppose Y is order dense. Let $x \in X_+$; put $D = Y \cap [0, x]$. Clearly, $D \leqslant x$. Suppose that $\sup D = x$ fails. Then $D \leqslant z < x$ for some $z \in X$, as in by Exercise 1.1.5. By assumption, there exists $y \in Y$ such that $0 < y \leqslant x - z$. It follows from $D \leqslant z$ that

$$y + D \leqslant y + z \leqslant x. \tag{3.1}$$

Since $0 \in D$, it follows that $y \leqslant x$ so that $y \in D$. Applying (3.1) again, we get $2y \leqslant x$, so that $2y \in D$. Iterating, we get $ny \leqslant x$ for every n, which contradicts the Archimedean property of X.

The converse implication is trivial. □

Note that the set $Y \cap [0,x]$ in the preceding proposition may be viewed as an increasing net. So the proposition says that Y is order dense iff every element of X_+ may be expressed as the order limit of an increasing net in Y. Recall that a subset of a topological space is dense iff every point of the space is the limit of a net in the subset. Does this apply to order density? We will show that the answer is "yes" for regular sublattices and "no" in general.

3.2.16 Theorem. *A regular sublattice Y in X is order dense iff every element of X is the order limit of a net in Y.*

Proof. Suppose Y is order dense; we need to show that every $x \in X$ is an order limit of a net in Y. If x is positive, this follows immediately from Proposition 3.2.15 because the set $Y \cap [0,x]$ is directed upward and, therefore, may be viewed as an increasing net. If we do not assume that x is positive, we find nets (a_α) and (b_β) in Y such that $a_\alpha \xrightarrow{o} x^+$ and $b_\beta \xrightarrow{o} x^-$. As in the proof of Proposition 2.4.7, we may assume that the two nets have the same index set. Then $a_\alpha - b_\alpha \xrightarrow{o} x^+ - x^- = x$.

To prove the converse, fix $0 < x \in X$ and find a net (y_α) in Y such that $y_\alpha \xrightarrow{o} x$ in X. Replacing y_α with y_α^+, we may assume that $y_\alpha \geqslant 0$ for every α. Let (u_γ) be a dominating net in X for $y_\alpha \xrightarrow{o} x$. Since $u_\gamma \downarrow 0$, there exists γ_0 such that $u_{\gamma_0} \not\geqslant x$. Put $u = u_{\gamma_0}$, then $(x-u)^+ > 0$. Furthermore, there exists α_0 such that $|x - y_\alpha| \leqslant u$ for all $\alpha \geqslant \alpha_0$. It follows that $y_\alpha \geqslant x - u$ and, therefore, $y_\alpha \geqslant (x-u)^+$.

We claim that there exists $y \in Y$ such that $0 < y \leqslant y_\alpha$ whenever $\alpha \geqslant \alpha_0$. Suppose not. Then $\inf_{\alpha \geqslant \alpha_0} y_\alpha = 0$ in Y and, therefore, in X (be regularity of Y). This contradicts $y_\alpha \geqslant (x-u)^+ > 0$ for all $\alpha \geqslant \alpha_0$.

It follows from the claim and the fact that $y_\alpha \xrightarrow{o} x$ that $x \geqslant y$. □

That is, a sublattice is order dense iff it is regular and dense with respect to order convergence.

3.2.17 Corollary. *Every norm dense regular sublattice of a Banach lattice is order dense.*

Proof. Let Y be a regular norm dense sublattice of a Banach lattice X. Every $x \in X$ is a norm limit of a sequence in Y. Passing to a subsequence as in Theorem 2.5.4, we get a sequence in Y that converges to x in order. Now apply Theorem 3.2.16. □

3.2.18 Corollary. *Let Y be a norm dense sublattice of a normed lattice X. If Y is order continuous then it is order dense in X.*

Proof. By Exercise 3.2.3, Y is regular in X. If X is a Banach lattice then Y is order dense by Corollary 3.2.17. If X is only a normed space, the previous argument shows that Y is order dense in the norm completion of X and, therefore, in X itself. $\square$

3.2.19 Example. *An non-regular sublattice which is dense with respect to order convergence but not order dense.* Let X be the set of real-valued functions on $[0,1]$ of the form $f = g + h$ where g is continuous and h vanishes except at finitely many points. Being a sublattice of $\mathbb{R}^{[0,1]}$, X is a vector lattice. Let $Y = C[0,1]$. Clearly, Y is a sublattice of X.

To see that Y is not regular, let (f_n) be a decreasing sequence in Y_+ such that $f_n(\frac{1}{2}) = 1$ for every n and $f_n(t) \to 0$ for every $t \neq \frac{1}{2}$. Then $f_n \downarrow 0$ in Y but $f_n \downarrow \mathbb{1}_{\{\frac{1}{2}\}}$ in X.

Since Y is not regular, it cannot be order dense. This may also be seen directly: there is no $y \in Y$ with $0 < y \leqslant \mathbb{1}_{\{\frac{1}{2}\}}$. On the other hand, it is easy to see that for every $x \in X$ there is a sequence (y_n) in Y such that $y_n \xrightarrow{\mathrm{o}} x$.

The concepts of Y being order dense or majorizing are somewhat dual to each other: one says that each positive non-zero element of X dominates a positive non-zero element of Y, while the other says that each positive non-zero element of X is dominated by an element of Y. So one may think of order dense sublattices as "minorizing". Note that Y is both order dense *and* majorizing iff every positive non-zero element of X is contained in an interval whose endpoints are positive non-zero vectors in Y. In some sense, this means that every element of x can be approximated both from above and from below by elements of Y:

3.2.20 Proposition. *TFAE:*

(i) *Y is both order dense and majorizing in X;*
(ii) *For every $x \in X$ one has $x = \inf\{y \in Y : y \geqslant x\}$;*
(iii) *For every $x \in X$ one has $x = \sup\{y \in Y : y \leqslant x\}$.*

Proof. It is straightforward (replacing x with $-x$) that (ii)$\Leftrightarrow$(iii). To show that (i)$\Rightarrow$(ii), put $A = \{y \in Y : y \geqslant x\}$. Since Y is majorizing, there exists $y_0 \in Y$ such that $x \leqslant y_0$. In particular, A is non-empty and $[x, y_0] \cap Y \subseteq A$. By Proposition 3.2.15,

$$\inf[x,y_0] \cap Y = \inf\bigl(y_0 - [0, y_0 - x] \cap Y\bigr) = y_0 - \sup[0, y_0 - x] \cap Y = y_0 - (y_0 - x) = x;$$

hence $\inf A = x$.

It is left to deduce (i) from the other two statements. It follows immediately from (ii) that Y is majorizing. Fix $x \in X_+$. By (iii), $x = \sup B$ where $B = \{y \in Y : y \leqslant x\}$. On the other hand, for every $y \in B$ we have $y \leqslant y^+ \in [0,x] \cap Y$. It follows that $x = \sup[0,x] \cap Y$. Now Proposition 3.2.15 completes the proof. $\square$

Combining Proposition 3.2.13 and Exercise 3.2.1, we conclude that if Y is an order dense sublattice than order converges passes up from Y to X. In the case of an order dense and majorizing sublattice, the situation is even better: order convergence passes both ways:

3.2.21 Theorem. *[AS05] Suppose that Y is order dense and majorizing. Then $x_\alpha \xrightarrow{\text{o}} x$ in Y iff $x_\alpha \xrightarrow{\text{o}} x$ in X for any net (x_α) and any vector x in Y.*

Proof. WLOG, $x = 0$. Being order dense, Y is regular, hence the forward implication is immediate from Proposition 3.2.13. The converse follows immediately from Theorem 3.2.5 because if $x_\alpha \xrightarrow{\text{o}} x$ in X then it has a tail that is order bounded in X and, therefore, in Y.

Here is another proof of the converse implication. Suppose that $x_\alpha \xrightarrow{\text{o}} 0$ in X. Let (u_γ) be a dominating net in X for this convergence. Put

$$A = \bigl\{y \in Y\ :\ y \geqslant u_\gamma \text{ for some } \gamma\bigr\}.$$

Clearly, $A \subseteq Y_+$; A is non-empty because Y is majorizing. We claim that $\inf A = 0$ in X and, therefore, in Y. Indeed, if $X_+ \ni z \leqslant A$ then for every γ we have $z \leqslant \{y \in Y\ :\ y \geqslant u_\gamma\}$, so that $z \leqslant u_\gamma$ by Proposition 3.2.20. Hence, $z = 0$.

Since A is directed downwards, we may view A as a decreasing net in Y. It is easy to see that this net dominates (x_α), so that $x_\alpha \xrightarrow{\text{o}} 0$ in Y. □

3.2.3 Order completion

Just as every normed space can be completed to a Banach space, every Archimedean vector lattice X can be enlarged to an order complete vector lattice.

3.2.22 Theorem. *Every Archimedean vector lattice X is lattice isomorphic to an order dense and majorizing sublattice of some order complete vector lattice.*

Such a vector lattice is called an ***order completion*** or ***Dedekind completion*** of X. We will show later in Theorem 6.10.1 that it is unique in the sense that every two order completions of X are lattice isomorphic; so one can speak about the order completion of X. The order completion of X will be denoted by X^δ.

The proof of the theorem is similar to the classical construction of $\mathbb{R}$ as the set of all Dedekind cuts in $\mathbb{Q}$. We will actually, prove a more general fact that every Archimedean ordered vector space with a generating cone embeds "densely" into an complete vector lattice:

3.2.23 Theorem. *Let X be an Archimedean ordered vector space with X^+ generating. Then there exists an order complete vector lattice X^δ and a linear operator $T\colon X \to X^\delta$ such that T is linear, $x \geqslant 0$ iff $Tx \geqslant 0$. Put $Y = \operatorname{Range} T$; then $a = \sup\{b \in Y\ :\ b \leqslant a\}$ for every $a \in X^\delta$.*

We present the proof as a sequence of exercises. As before, for a set A in X we write $A^{\uparrow}$ and $A^{\downarrow}$ for the sets of all upper and lower bounds of A, respectively. We put $\varnothing^{\uparrow} = \varnothing^{\downarrow} = X$. For a vector $x \in X$ we write $x^{\uparrow} = \{x\}^{\uparrow}$ and $x^{\downarrow} = \{x\}^{\downarrow}$. In particular, $X_+ = 0^{\uparrow}$.

3.2.24 Exercise. Let A and B be two subsets of X and $x \in X$. Then

(i) if $A \subseteq B$ then $A^{\uparrow} \supseteq B^{\uparrow}$ and $A^{\downarrow} \supseteq B^{\downarrow}$, hence $A^{\uparrow\downarrow} \subseteq B^{\uparrow\downarrow}$;
(ii) $A \subseteq A^{\uparrow\downarrow}$ and $A \subseteq A^{\downarrow\uparrow}$;
(iii) $(x + A)^{\uparrow\downarrow} = x + A^{\uparrow\downarrow}$;
(iv) $x^{\uparrow\downarrow} = x^{\downarrow}$;
(v) $A^{\uparrow\downarrow\uparrow} = A^{\uparrow}$ and $A^{\downarrow\uparrow\downarrow} = A^{\downarrow}$;
(vi) If $x = \sup A$ then $A^{\uparrow} = x^{\uparrow}$, hence $A^{\uparrow\downarrow} = x^{\downarrow}$;
(vii) $(A + B^{\uparrow\downarrow})^{\uparrow\downarrow} = (A + B)^{\uparrow\downarrow}$.

If A is a non-empty proper subset of X, we say that A is a *down-cone* if $A = A^{\uparrow\downarrow}$. Let X^{δ} be the set of all down-cones in X. We order X^{δ} by inclusion. We will use lower-case letters a, b and c for elements of X^{δ}, while symbols x, y, and z will stand for elements of X.

3.2.25 Exercise. (i) For every $A \subseteq X$, if A is bounded below than $A^{\downarrow}$ is a down-cone; for every $x \in X$, the set $x^{\downarrow}$ is a down-cone.
(ii) For a down-cone a, we have $x \in a$ iff $x^{\downarrow} \subseteq a$;
(iii) Every down-cone is convex and bounded above.
(iv) If the intersection of a family of down-cones is non-empty then is again a down-cone.
(v) X^{δ} is an order complete lattice (here we use that X_+ is generating).

Define addition on X^{δ} via $a \oplus b := \big\{x + y \,:\, x \in a,\ y \in b\big\}^{\uparrow\downarrow}$. We use "$\oplus$" rather than "$+$" because we normally utilize "$+$" for the set-theoretic (Minkowski) sum $a + b = \{x + y : x \in a, y \in b\}$. Thus, $a \oplus b = (a + b)^{\uparrow\downarrow}$.

3.2.26 Exercise. $(X, \oplus)$ is an abelian group, with $\nvdash := 0^{\downarrow} = -X_+$ being the neutral element, and the negation $\ominus a$ of a is given by $\{-x \,:\, x \in a^{\uparrow}\} = -(a^{\uparrow})$. (Here we use that X is Archimedean).

The following exercise (in linear algebra) essentially says that, when verifying that something is a vector space, it suffices to verify the axioms that involve scalar multiplication for positive scalars only.

3.2.27 Exercise. Let $(E, +)$ be an abelian group with positive scalar multiplication such that $r(x + y) = rx + ry$, $r(sx) = (rs)x$, $(r + s)x = rx + sx$ and $1 \cdot x = x$ for all $x, y \in E$ and positive real r and s. Then extending the scalar multiplication to $E \times \mathbb{R}$ via $(-r)x = -(rx)$ for $r > 0$ and $0 \cdot x = 0$ turns E into a vector space.

Define scalar multiplication on X^{δ} as follows. For $a \in X^{\delta}$ and $r > 0$, put $r \odot a := ra = \{rx \,:\, x \in a\}$, $(-r) \odot a := \ominus(ra)$, and $0 \odot a := \nvdash$.

3.2.28 Exercise. X^{δ} is an order complete vector lattice.

Define $T\colon X \to X^{\delta}$ via $Tx = x^{\downarrow}$. Let $Y = \operatorname{Range} T$.

3.2.29 Exercise. (i) T is one-to-one, with the inverse map $T^{-1}\colon Y \to X$ given by $x = \sup x^{\downarrow}$.

(ii) $T\colon X \to Y$ is an order isomorphism: $x \geqslant y$ iff $Tx \geqslant Ty$.
(iii) T is linear.
(iv) $a = \sup\{b \in Y \,:\, b \leqslant a\}$ for every $a \in X^\delta$.

This completes the proof of Theorem 3.2.23.

In order to complete the proof of Theorems 3.2.22, suppose, in addition, that X is a vector lattice. We already know that, viewed as a map from X to Y, T is a linear and order isomorphism. It follows that Y is a vector lattice and T is a lattice isomorphism. Finally, Y is order dense and majorizing in X^δ by Proposition 3.2.20.

3.2.30 Exercise. ℓ_∞ is an order completion of c.

3.2.31 Exercise. Let X be the space of eventually constant sequences; then ℓ_∞ is an order completion of X.

The following result is an immediate consequence of Theorem 3.2.21.

3.2.32 Corollary. *For any net (x_α) and a vector x in X we have $x_\alpha \xrightarrow{\mathrm{o}} x$ in X iff $x_\alpha \xrightarrow{\mathrm{o}} x$ in X^δ.*

Here is in interesting application of this corollary. Recall that, according to Exercise 2.4.26, every order bounded disjoint sequence in a σ-order complete vector lattice is σ-order null.

3.2.33 Proposition. *Every disjoint order bounded sequence in X is order null.*

Proof. Let (x_n) be an order bounded disjoint sequence in X. Then it is still disjoint and order bounded in X^δ. By Exercise 2.4.26, (x_n) is order null in X^δ. It follows from Corollary 3.2.32 that (x_n) is order null in X. □

3.2.34 Example. An order bounded disjoint sequence need not be σ-order null. Let Y be as in Example 2.5.29. Choose any distinct sequence (ω_n) in Ω. Then the sequence (δ_{ω_n}) is disjoint and order bounded by $\mathbb{1}$ in Y, yet it is not σ-order null in Y.

3.2.35 Exercise. Let X be a normed lattice. For $u \in X^\delta$, put $|||u||| = \inf\bigl\{\|x\| \,:\, |u| \leqslant x \in X\bigr\}$. Then $|||\cdot|||$ is an extension of the original norm on X to a lattice norm on X^δ. If X is norm complete then so is X^δ. (Interestingly, this extension need not be unique: there may be other lattice norms on X^δ extending $\|\cdot\|$; see [Sol66].)

3.3 Ideals

A sublattice J in a vector lattice X is said to be an ***order ideal*** if $x \in J$ and $0 \leqslant y \leqslant x$ imply $y \in J$. When there is no risk to confuse it with algebraic ideals, we will just use the term ***ideal***. In case of a normed lattice, we are especially interested in (norm) closed ideals.

3.3.1 Exercise. (i) If J is an ideal and $|x| \in J$ then $x \in J$;

(ii) A subspace J is an ideal iff $x \in J$ and $|y| \leqslant |x|$ imply $y \in J$.

3.3.2 Example. Here we list a few examples of ideals:

(i) ℓ_p is an ideal in ℓ_q whenever $p \leqslant q$.

(ii) ℓ_p is an ideal in c_0 for every $0 < p < \infty$; c_0 is a closed ideal in ℓ_∞.

(iii) c is not an ideal in ℓ_∞.

(iv) For any measure μ and any $0 \leqslant p \leqslant \infty$, the space $L_p(\mu)$ is an ideal in $L_0(\mu)$. Furthermore, if μ is finite and $0 \leqslant p \leqslant q \leqslant \infty$ then $L_q(\mu)$ is an ideal in $L_p(\mu)$.

(v) Let $X = L_p(\mu)$ for some measure μ and A a measurable set. The set $L_p(A)$ consisting of all the functions in $L_p(\mu)$ whose support is (a.e.) contained in A is an ideal in $L_p(\mu)$. It is closed when $1 \leqslant p \leqslant \infty$.

(vi) The set of all the functions in $\mathbb{R}^\Omega$ or in $C(K)$ that vanish at some fixed point is a closed ideal.

There are several motivations for the name "ideal". First, if J is an ideal then $x \in J_+$ and $y \in X_+$ imply $x \wedge y \in J$. Second, we will see that in $C(K)$ spaces, the closed order ideals are exactly the closed algebraic ideals. Third, we will show that ideals are precisely the kernels of lattice homomorphisms.

3.3.3 Exercise. In this exercise, we collect some easy properties of ideals.

(i) If J is an ideal in X and Y a sublattice in X then $J \cap Y$ is an ideal in Y.

(ii) The sum of an ideal and a sublattice is a sublattice (cf Exercise 3.1.7).

(iii) An interval preserving operator maps ideals to ideals. In particular, the range of such an operator is an ideal.

(iv) The pre-image of an ideal under a lattice homomorphism is again an ideal.

(v) An ideal is a regular sublattice.

(vi) The norm closure of an ideal in a normed lattice is again an ideal.

Next, we consider some examples of ideals. Recall that for a set $A \subseteq X$, we write $I(A)$ for the ideal generated by A. It is easy to see that if $A = \{a\}$ for some $a \in X_+$, then $I(A) = I_a$, the principal ideal of a; see Exercise 2.2.1.

3.3.4 Exercise. If $\dim X > 1$ then X contains a proper non-trivial ideal.

3.3.5 Exercise. If T is a positive operator between vector lattices and $Ta = 0$ for some $a \geqslant 0$ then T vanishes on I_a.

3.3.6 Exercise. Let A be a subset of a vector lattice X, then

$$I(A)=\Bigl\{x\in X\ :\ |x|\leqslant\sum_{i=1}^{n}\lambda_i|a_i|,\ n\in\mathbb{N},\ \lambda_1,\dots\lambda_n\in\mathbb{R}_+,\ a_1,\dots,a_n\in A\Bigr\}.$$

3.3.7. Let Y be a sublattice of X. Exercise 3.3.6 implies that $x\in I(Y)$ iff $|x|\leqslant a$ for some $a\in Y_+$. It follows that Y is majorizing in $I(Y)$. In particular, Y is majorizing in X iff $I(Y)=X$.

3.3.8 Exercise. The ideals in $\mathbb{R}^n$ are the sets of the form $\operatorname{span}\{e_i\ :\ i\in A\}$ for some subset A of $\{1,\dots,n\}$.

Let K be a compact space. Recall that, in addition to being a Banach lattice, $C(K)$ is also an algebra. Recall also that a subspace J of an algebra $\mathcal{A}$ is said to be an ***algebraic ideal*** if $x\in J$ and $a\in\mathcal{A}$ imply ax and xa are in J.

3.3.9 Exercise. Every order ideal in $C(K)$ is also an algebraic ideal. This remains valid in $C_0(\Omega)$, where Ω is a locally compact space.

It is well known that the closed algebraic ideals in $C(K)$ are exactly the sets of the form

$$J_F=\{f\in C(K)\ :\ f\text{ vanishes on }F\},$$

where F is a closed subset of K. It is easy to see that J_F is also an order ideal.

3.3.10 Proposition. *Let A be a subset of $C(K)$. Then the closed order ideal generated by A, $\overline{I(A)}$, equals J_F, where $F=\bigcap\limits_{f\in A}\ker f$.*

Proof. Clearly, F is closed, hence J_F is a closed order ideal. It follows from $A\subseteq J_F$ that $\overline{I(A)}\subseteq J_F$. To show the opposite inclusion, take any $f\in J_F$ and show that $f\in\overline{I(A)}$. Since f^+ and f^- are in J_F, we may assume without loss of generality that $f\geqslant 0$.

Take an arbitrary $\varepsilon>0$ and put $G=\{f\geqslant\varepsilon\}:=\{t\in K\ :\ f(t)\geqslant\varepsilon\}$. Then G is closed, hence compact since K is compact. Also, $G\cap F=\varnothing$. For each $s\in G$ we have $s\notin F$, hence there exists $g_s\in A$ such that $g_s(s)\neq 0$. Replacing g_s with $|g_s|$, if necessary, we may assume that $g_s\in I(A)$, $g_s\geqslant 0$, and $g_s(s)>0$. The sets $\{g_s>0\}$ as s runs through G form an open cover of G. Hence, $G\subseteq\bigcup_{i=1}^n\{g_{s_i}>0\}$ for some $s_1,\dots,s_n\in G$. Put $g=g_{s_1}\vee\dots\vee g_{s_n}$. Then $0\leqslant g\in I(A)$, and g is positive at every point of G. Since G is compact, g attains its minimum on G, let $\delta=\min_G g$. Note that $\delta>0$. It follows that there exists a positive real number λ such that $\lambda g\geqslant f$ on G. Let $h=f\wedge(\lambda g)$. Then $0\leqslant h\leqslant\lambda g$, so that $h\in I(A)$. On the other hand, h agrees with f on G, so that $\|f-h\|\leqslant\varepsilon$. It follows that $f\in\overline{I(A)}$. $\square$

Applying the preceding proposition with A being a closed order ideal immediately yields the following:

3.3.11 Theorem. *Closed order ideal in $C(K)$ are the sets of the form J_F for some closed subset F of K.*

That is, closed order ideals and closed algebraic ideals in $C(K)$ are the same.

3.3.12 Exercise. The map $F \mapsto J_F$ is a one-to-one correspondence between the closed sets in K and the closed ideals in $C(K)$, with converse map is given by $F = \bigcap_{f \in J} \ker f$.

3.3.13 Remark. While the J_F notation is convenient and efficient, it is also somewhat awkward, because it emphasizes the set where functions vanish rather than the set on which they are supported. So, occasionally, it is more convenient to use an alternative notation: for an open set U, we put

$$C_U = \bigl\{f \in C(K) \,:\, \operatorname{supp} f \subseteq U\bigr\}.$$

Clearly, $C_U = J_{U^C}$, so the C_U notation and the J_F notation are dual to each other. In this notation, Exercise 3.3.10 says that for any ideal J in $C(K)$, we have $\overline{J} = C_U$ where $U = \bigcup_{f \in J} \operatorname{supp} f$.

In a similar fashion, Section 5.1 that every closed ideal in $L_p(\mu)$ consists of all the functions supported on a certain set.

3.3.14 Exercise. Let J_1 and J_2 be two ideals in a vector lattice X.

(i) $J_1 \perp J_2$ iff $J_1 \cap J_2 = \{0\}$.
(ii) If J_1 and J_2 are order dense then so is $J_1 \cap J_2$.
(iii) $J_1 + J_2$ is again an ideal in X;
(iv) $(J_1 + J_2)_+ = (J_1)_+ + (J_2)_+$. That is, for every positive $y \in J_1 + J_2$ one can write $y = y_1 + y_2$ with $0 \leqslant y_1 \in J_1$ and $0 \leqslant y_2 \in J_2$;
(v) For every $y \in J_1 + J_2$ one can write $y = y_1 + y_2$ with $y_i \in J_i$ and $|y_i| \leqslant |y|$ as $i = 1, 2$.

Recall that the sum of two closed subspaces in a Banach space need not be closed.

3.3.15 Theorem. *If J_1 and J_2 are two closed ideals in a Banach lattice X then $J_1 + J_2$ is again a closed ideal in X.*

Proof. By Exercise 3.3.14, $J_1 + J_2$ is an ideal in X. Suppose that (x_n) is a sequence in $J_1 + J_2$ and $x_n \to x$ for some $x \in X$; we need to show that $x \in J_1 + J_2$. Passing to a subsequence, we may assume WLOG that $\|x_n - x_{n-1}\| \leqslant \frac{1}{2^n}$ for all $n \geqslant 1$ (we take $x_0 = 0$). Note that $x = \sum_{n=1}^{\infty}(x_n - x_{n-1})$. By Exercise 3.3.14, for every n one can find $u_n \in J_1$ and $v_n \in J_2$ such that $x_n - x_{n-1} = u_n + v_n$ and $|u_n|, |v_n| \leqslant |x_n - x_{n-1}|$, so that $\|u_n\|, \|v_n\| \leqslant \frac{1}{2^n}$. Therefore, the series $u \coloneqq \sum_{n=1}^{\infty} u_n$ and $v \coloneqq \sum_{n=1}^{\infty} v_n$ converge. Since J_1 and J_2 are closed, we have $u \in J_1$ and $v \in J_2$, hence $x = u + v \in J_1 + J_2$. □

3.3.16 Exercise. If J_1 and J_2 are two ideals in a Banach lattice X then $\overline{J_1 + J_2} = \overline{J_1} + \overline{J_2}$.

Recall that if Y is a subspace in a normed space X and $A \subseteq Y$ then $\overline{A}^Y = \overline{A}^X \cap Y$, where $\overline{A}^Y$ and $\overline{A}^X$ stand for the closures of A in Y and in X, respectively.

3.3.17 Exercise. Let Y is a dense sublattice of a normed lattice X and J an ideal in Y. Let J_0 be the ideal generated by J in X. Then $\overline{J}^X = \overline{J_0}^X$, this set is an ideal in X, and $\overline{J}^Y = \overline{J_0}^X \cap Y$. (The statement may fail without the density assumption.)

3.3.18 Proposition. *Let J be an ideal in a vector lattice X. If X is order complete then so is J. Conversely, if every principal ideal in X is order complete then so is X. Same for σ-order completeness.*

Proof. Suppose that X is order complete and J is an ideal in X. Let $0 \leqslant x_\alpha \uparrow \leqslant u$ in J. Since X is order complete, $x \coloneqq \sup x_\alpha$ exists in X. Clearly, $0 \leqslant x \leqslant u$, so that $x \in J$. By Exercise 3.3.3(v), $x_\alpha \uparrow x$ in J.

Suppose that every principal ideal in X is order complete. Let $0 \leqslant x_\alpha \uparrow \leqslant u$ in X. Then we also have $0 \leqslant x_\alpha \uparrow \leqslant u$ in I_u. By assumption, $x \coloneqq \sup x_\alpha$ exists in I_u and, therefore, in X by Exercise 3.3.3(v).

The proof for σ-order completeness is similar. □

Let μ be a σ-finite measure. We proved in Theorem 2.1.16 that $L_p(\mu)$ is order complete for every $0 \leqslant p \leqslant +\infty$. We see now that it was sufficient to establish order completeness of $L_0(\mu)$; order completeness of $L_p(\mu)$ for all $p > 0$ follows automatically by Proposition 3.3.18 because $L_p(\mu)$ is an ideal in $L_0(\mu)$.

3.3.19 Proposition. *Let Y be an order dense sublattice in an Archimedean vector lattice X. If Y is order complete then Y is an ideal in X.*

Proof. Suppose that $0 \leqslant x \leqslant y$ for some $x \in X$ and $y \in Y$; it suffices to show that $x \in Y$. By Proposition 3.2.15, $x = \sup [0, x] \cap Y$. Since Y is order complete and $[0, x] \cap Y \leqslant y$ in Y, the set $[0, x] \cap Y$ has supremum in Y. Since Y is regular by Proposition 3.2.13, this supremum has to be x, so that $x \in Y$. □

The concept of an ideal is closely related to that of a *solid sets*. A subset A of X is ***solid*** if $x \in A$ and $|y| \leqslant |x|$ imply $y \in A$.

3.3.20 Exercise. (i) The order interval $[-u,u]$ is solid for every $u\in X_+$;

(ii) Ideals are precisely the solid subspaces.

(iii) A norm on a vector lattice is a lattice norm iff its unit ball is solid;

Let $A\subseteq X$. We write $\mathrm{Sol}(A)$ for the ***solid hull*** of A, i.e., the intersection of all solid sets containing A.

3.3.21 Exercise. Let $A\subseteq X$.

(i) $\mathrm{Sol}(A)$ is the least solid set containing A;

(ii) $\mathrm{Sol}(A)=\bigl\{y\in X\ :\ \lvert y\rvert\leqslant\lvert x\rvert \text{ for some } x\in A\bigr\}=\bigcup_{x\in A}\bigl[-\lvert x\rvert,\lvert x\rvert\bigr]$;

(iii) If $u\in X_+$ then $\mathrm{Sol}\bigl(\{u\}\bigr)=[-u,u]$;

3.3.22 Exercise. A solid set is order closed iff it is closed under taking sups of increasing positive nets.

What is the relationship between solid sets and convex sets?

3.3.23 Example. *"(Absolutely) convex" and "solid" do not imply each other.* Let $X=\mathbb{R}^2$. The straight line segment connecting $(1,-1)$ and $(-1,1)$ is absolutely convex but not solid. On the other hand, let $u=(1,2)$ and $v=(2,1)$, then the set $[-u,u]\cup[-v,v]$ is solid but not convex.

Recall that the ***convex hull*** of a set A in a vector space is the intersection of all convex sets containing A; it is the least convex set containing A. We denote it by $\mathrm{conv}(A)$. It is easy to see that $\mathrm{conv}(A)$ is precisely the set of all convex combinations of elements of A:

$$\mathrm{conv}(A)=\Bigl\{\sum_{i=1}^{n}\lambda_i a_i\ :\ a_1,\dots,a_n\in A,\ \lambda_1,\dots,\lambda_n\in\mathbb{R}_+,\ \text{and } \lambda_1+\dots+\lambda_n=1\Bigr\}.$$

3.3.24 Proposition. *[Nakano] The convex hull of a solid set is solid.*

Proof. Suppose that $\lvert x\rvert\leqslant\lvert y\rvert$ and $y\in\mathrm{conv}(A)$ for some solid set A; we need to show that $x\in\mathrm{conv}(A)$. We can write $y=\sum_{i=1}^n\lambda_i a_i$, where $a_1,\dots,a_n\in A$, $\lambda_1,\dots,\lambda_n\in(0,1]$, and $\sum_{i=1}^n\lambda_i=1$. Then $\lvert x\rvert\leqslant\sum_{i=1}^n\lvert\lambda_i a_i\rvert$. By Riesz Decomposition Theorem 1.5.19, we may write $x=\sum_{i=1}^n x_i$ such that $\lvert x_i\rvert\leqslant\lvert\lambda_i a_i\rvert$ for every $i=1,\dots,n$. It follows from $\bigl\lvert\frac{1}{\lambda_i}x_i\bigr\rvert\leqslant\lvert a_i\rvert$ that $\frac{1}{\lambda_i}x_i\in A$ and, therefore, $x\in\mathrm{conv}(A)$. □

3.3.25 Example. *The solid hull of a convex set need not be convex; the solid hull of a subspace need not be a subspace.* Let $X=\mathbb{R}^{\mathbb{R}}$, the set of all functions from $\mathbb{R}$ to $\mathbb{R}$. Let $f(t)=t$ and let Y be the subspace of X spanned by f and $\mathbb{1}$. It is easy to see that Y consists of all affine functions in X. Clearly, Y is convex. We are going to show that $\mathrm{Sol}(Y)$ is not convex; in particular, $\mathrm{Sol}(Y)$ is not a subspace. Assume for the sake of contradiction that $\mathrm{Sol}(Y)$ is convex. Clearly, $\lvert f\rvert$ and $\lvert f-\mathbb{1}\rvert$ are in $\mathrm{Sol}(Y)$, so that $g:=\frac{1}{2}\lvert f\rvert+\frac{1}{2}\lvert f-\mathbb{1}\rvert$ is in $\mathrm{Sol}(Y)$. Note that g is strictly positive and $\lim_{t\to\pm\infty}g(t)=\infty$. Since every function in Y is either constant or has a root, $g\leqslant\lvert h\rvert$ is impossible for any $h\in Y$; hence $g\notin\mathrm{Sol}(Y)$, which is a contradiction.

However, if we only consider “one-sided” solid hull then convexity is preserved; cf Exercise 3.3.21(ii):

3.3.26 Exercise. Let A be a convex set in X_+. The set $\bigcup_{a\in A}[0,a]$ is convex.

3.3.27 Proposition. *Let Y be a subspace of $\mathbb{R}^n$ for some $n\in\mathbb{N}$. Then* $\mathrm{Sol}(Y)=I(Y)$.

Proof. Clearly, $\mathrm{Sol}(Y)\subseteq I(Y)$. Let A be the set of all indices i such that Y contains a vector, say, x_i, whose i-th component is non-zero. Let $J_A=\mathrm{span}\{e_i\,:\,i\in A\}$. Since $Y\subseteq J_A$ and J_A is an ideal, we have $I(Y)\subseteq J_A$. There is a linear combination x of $x_1,\dots,x_i$ with $\operatorname{supp}x=A$. It follows that $J_A=I_x$. Furthermore, $x\in Y$ yields $I_x\subseteq\mathrm{Sol}(Y)$. Combining the inclusions, we get $\mathrm{Sol}(Y)\subseteq I(Y)\subseteq J_A=I_x\subseteq\mathrm{Sol}(Y)$, which completes the proof. □

3.3.28 Exercise. Does the interior A of a solid set in a Banach lattice have to be solid?

3.4 Quotients and direct sums

There is a natural relationship between ideals, quotients, and lattice homomorphisms. We start with the following easy observation.

3.4.1 Exercise. If T is a lattice homomorphism between vector lattices then $\ker T$ is an ideal.

We are going to show that the quotient of a vector lattice over an ideal is again a vector lattice. Recall that we showed in Section 1.2 that every cone C in a vector space induces a linear order via $x\geqslant y$ whenever $x-y\in C$.

3.4.2 Theorem. *Let X be a vector lattice and J an ideal in X; let $Q\colon X/J$ be the quotient map. Then $Q(X_+)$ is a cone in X/J, the corresponding order makes X/J into a vector lattice, and Q is a lattice homomorphism.*

Proof. For $x\in X$, we write $\tilde{x}$ for the equivalence class of x in X/J. That is, $\tilde{x}=Qx$.

Put $C=Q(X_+)$. It is clear that C is closed under addition and positive scalar multiplication. To show that $C\cap(-C)=\{0\}$, let $\varphi\in C\cap(-C)$. Then $\varphi=Qx$ and $-\varphi=Qy$ for some $x,y\in X_+$. Therefore, $0=Qx+Qy=Q(x+y)$, so that $x+y\in J$. It follows from $0\leqslant x\leqslant x+y$ that $x\in J$ and, therefore, $\varphi=0$. We conclude that C is a cone. It follows from $C=Q(X_+)$ that Q is a positive operator.

We claim that if $\varphi\geqslant\tilde{x}$ then $\varphi=\tilde{y}$ for some $y\geqslant x$. Indeed, it follows from $\varphi-\tilde{x}\in Q(X_+)$ that $\varphi-Qx=Qz$ for some $z\in X_+$. Put $y=x+z$, then $y\geqslant x$ and $Qy=Qx+Qz=\varphi$.

Finally, we show that $(Qx)\vee(Qy)=Q(x\vee y)$ for any $x,y\in X$; this will imply that X/J is a vector lattice and Q is a lattice homomorphism. It follows from $x,y\leqslant x\vee y$ that $Qx,Qy\leqslant Q(x\vee y)$. Suppose that $Qx,Qy\leqslant\varphi$; it is left to show that $Q(x\vee y)\leqslant\varphi$.

Using the preceding paragraph, we find $a, b \in \varphi$ such that $x \leqslant a$ and $y \leqslant b$. Since $|a \vee b - a| \leqslant |b - a|$ and $b - a \in J$, we have $a \vee b - a \in J$ and, therefore, $a \vee b \in \varphi$. It follows from $x \vee y \leqslant a \vee b$ that $Q(x \vee y) \leqslant Q(a \vee b) = \varphi$. □

3.4.3 Corollary. *In a vector lattice, ideals are exactly the kernels of lattice homomorphisms.*

Next, we will discuss some *lifting* properties of vector lattices. That is, given a collection of elements in X/J, which satisfies a certain property, when can one pick a representative for each member of the collection in such a way that the resulting subset of X has the same property?

3.4.4 Exercise. Let X be a vector lattice and J an ideal in X.

(i) Lifting positivity: for every positive $\varphi \in X/J$ there exists a positive $x \in X$ with $\tilde{x} = \varphi$;

(ii) Lifting inequalities: if $\varphi \leqslant \psi$ in X/J then there exist $x \in \varphi$ and $y \in \psi$ with $x \leqslant y$;

(iii) If $x \leqslant z$ in X and $\tilde{x} \leqslant \varphi \leqslant \tilde{z}$ in X/J then there exists $y \in \varphi$ such that $x \leqslant y \leqslant z$.

(iv) Let $\varphi \in X/J$ and $0 \leqslant y \in |\varphi|$. Then there exists $x \in \varphi$ such that $|x| \leqslant y$.

(v) Lifting increasing sequences: Given an increasing positive sequence (φ_n) in X/J, one can choose $x_n \in \varphi_n$ for every n in such a way that (x_n) is an increasing positive sequence in X.

3.4.5 Exercise. Let $T\colon X \to Y$ be a linear operator between two vector lattices and J an ideal in X with $J \subseteq \ker T$. Consider the operator $\widetilde{T}\colon X/J \to Y$ defined via $\widetilde{T}\tilde{x} = Tx$.

(i) If $T \geqslant 0$ then $\widetilde{T} \geqslant 0$;

(ii) If T is a lattice homomorphism then so is $\widetilde{T}$.

In particular, if $T\colon X \to Y$ is a surjective lattice homomorphism then, applying the preceding exercise with $J = \ker T$, we conclude that $\widetilde{T}\colon X/J \to Y$ is a lattice isomorphism. It follows that Y is lattice isomorphic to X/J. This way, we may identify T with the quotient map from X onto X/J. Thus, every surjective lattice homomorphism may be viewed as a quotient map. It now follows from Exercise 3.4.4(iii) that every surjective lattice homomorphism is interval preserving.

3.4.6 Theorem. *The quotient of a normed lattice over a norm closed ideal is a normed lattice. The quotient of a Banach lattice over a normed closed ideal is a Banach lattice.*

Proof. Let X be a normed lattice and $J \subseteq X$ a closed ideal. As we already know that X/J is a vector lattice, it is left to prove that the quotient norm is a lattice norm.

Suppose that $|\varphi| \leqslant |\psi|$ in X/J. Fix $\varepsilon > 0$ and find $y \in \psi$ such that $\|y\| \leqslant \|\psi\| + \varepsilon$. Note $|y| \in |\psi|$. By Exercise 3.4.4(iii), there exists $h \in |\varphi|$ such that $0 \leqslant h \leqslant |y|$. By Exercise 3.4.4(iv), there exists $x \in \varphi$ such that $|x| \leqslant h$. Therefore, $\|\varphi\| \leqslant \|x\| \leqslant \|y\| \leqslant \|\psi\| + \varepsilon$. Since ε is arbitrary, we have $\|\varphi\| \leqslant \|\psi\|$.

If X is a Banach lattice then X/J is complete, hence also is a Banach lattice. $\square$

In this book, we are mostly interested in Archimedean vector lattices. However, the quotient of a vector lattice over an ideal may be non-Archimedean even when the original lattice is Archimedean (an example is coming)! When are quotients Archimedean? The following proposition, due to Luxemburg and Moore, answers this question.

3.4.7 Proposition. *Let X be a vector lattice (not necessarily Archimedean) and $J \subseteq X$ an ideal. Then X/J is Archimedean iff J is uniformly closed.*

Proof. Suppose that X/J is Archimedean. Let (x_n) be a sequence in J such that $x_n \xrightarrow{\mathrm{u}} x$ for some $x \in X$. We need to show that $x \in J$. There exists $e \in X_+$ such that $\|x_n - x\|_e \to 0$. Passing to a subsequence, if necessary, we may assume that $\|x_n - x\|_e < \frac{1}{n}$, so that $|x_n - x| < \frac{1}{n}e$. Since the quotient map is a lattice homomorphism by Theorem 3.4.2 and $\tilde{x}_n = 0$, we have $|\tilde{x}| < \frac{1}{n}\tilde{e}$. Since X/J is Archimedean, this yields $\tilde{x} = 0$, hence $x \in J$.

Conversely, suppose that J is uniformly closed. Let $x, e \in X$ such that $0 \leqslant \tilde{x} \leqslant \frac{1}{n}\tilde{e}$ for every n; we need to show that $\tilde{x} = 0$. WLOG, replacing e with $|e|$, we may assume that $e \geqslant 0$. For every n, it follows from $0 \leqslant \tilde{x} \leqslant \frac{1}{n}\tilde{e}$ that there exists $x_n \in J$ such that $0 \leqslant x - x_n \leqslant \frac{1}{n}e$. Therefore, $\|x - x_n\|_e \to 0$. It follows that $x \in J$, hence $\tilde{x} = 0$. $\square$

3.4.8 Example. *A non-Archimedean quotient.* $C[0,1]$ is clearly is Archimedean. Let

$$J = \bigl\{f \in C[0,1] \,:\, f \text{ vanishes on } [0,\delta] \text{ for some } \delta > 0\bigr\}.$$

It can be easily verified that J is an ideal in $C(K)$. The quotient vector lattice $C[0,1]/J$ consists of “germs” of continuous functions. Let for $f, g \in C[0,1]$. Their equivalence classes $\tilde{f}$ and $\tilde{g}$ coincide iff f and g agree on $[0,\delta]$ for some $\delta > 0$. Likewise, $\tilde{f} \leqslant \tilde{g}$ if there exists $\delta > 0$ such that $f(t) \leqslant g(t)$ for all $t \in [0,\delta]$.

We claim that the quotient vector lattice $C[0,1]/J$ is not Archimedean. Indeed, take $f, g \in C[0,1]$ where $f(t) = t$ and $g(t) = \sqrt{t}$ for all t. It is easy to see that for every $n \in \mathbb{N}$ there is $\delta > 0$ such that $nf(t) \leqslant g(t)$ on $[0,\delta]$, hence $n\tilde{f} \leqslant \tilde{g}$.

We will show that J is not uniformly closed. For each $n \in \mathbb{N}$, define $f_n \in C[0,1]$ as follows: f_n vanishes on $[0,\frac{1}{n}]$, $f_n(t) = t$ when $t \geqslant \frac{2}{n}$, and f_n is linear on $[\frac{1}{n},\frac{2}{n}]$. It is easy to see that $f_n \in J$ for all n and $f_n \xrightarrow{\|\cdot\|} f$ and, therefore, $f_n \xrightarrow{\mathrm{u}} f$, where $f(t) = t$ for all t. Yet $f \notin J$.

Finally, we will discuss lifting disjointness.

3.4.9 Exercise. Let J be an ideal in a vector lattice X. Let $x, y \in X_+$ such that $\tilde{x} \perp \tilde{y}$ in X/J. Then there exist $x_0 \in \tilde{x}$ and $y_0 \in \tilde{y}$ such that $0 \leqslant x_0 \leqslant x$, $0 \leqslant y_0 \leqslant y$, and $x_0 \perp y_0$.

If X is a Banach lattice and $\varepsilon > 0$ then, in addition, we can ensure that $\|x_0\| \leqslant (1+\varepsilon)\|\tilde{x}\|$ and $\|y_0\| \leqslant (1+\varepsilon)\|\tilde{y}\|$.

3.4.10 Theorem ([dPW15]). (i) *Let J be an ideal in a vector lattice X, and suppose that $\varphi_1, \dots, \varphi_m$ are disjoint in X/J. One can choose $x_i \in \varphi_i$, $i = 1, \dots, m$ so that $x_1, \dots, x_m$ are disjoint.*

(ii) *If J is a closed ideal in a Banach lattice X then the same statement is true for infinite sequences.*

Proof. We will prove (ii) first. One may assume WLOG that $\sum_{i=1}^{\infty}\|\varphi_i\| < \infty$. Indeed, otherwise, replace each φ_i with φ_i/α_i where $\alpha_i = 2^i\|\varphi_i\|$; then replace the resulting x_i's with $\alpha_i x_i$.

Assume first that each φ_n is positive. For each n, put $\theta_n = \sum_{i=n+1}^{\infty} \varphi_i$. Note that $\varphi_k \perp \theta_n$ for each $k = 1, \dots, n$ because $0 = \varphi_k \wedge \sum_{i=n+1}^{m} \varphi_i \to \varphi_k \wedge \theta_n$ as $m \to \infty$.

The proof is inductive. By Exercise 3.4.9, one can find $0 \leqslant x_1 \in \varphi_1$ and $0 \leqslant z_1 \in \theta_1$ such that $x_1 \perp z_1$. Suppose that we have already constructed disjoint positive $x_1, \dots, x_n, z_n$ such that $x_i \in \varphi_i$ as $i = 1, \dots, n$ and $z_n \in \theta_n$. Since $\varphi_{n+1}, \theta_{n+1} \leqslant \theta_n$, by Exercise 3.4.4(iii) we can find $u, v \in [0, z_n]$ such that $u \in \varphi_{n+1}$ and $v \in \theta_{n+1}$. Applying Exercise 3.4.9 to u and v we find $x_{n+1} \in \varphi_{n+1}$ and $z_{n+1} \in \theta_{n+1}$ such that $x_{n+1} \perp z_{n+1}$, $x_{n+1} \leqslant u$ and $z_{n+1} \leqslant v$. In particular, $x_{n+1}, z_{n+1} \in [0, z_n]$ which ensures that the set $x_1, \dots, x_n, x_{n+1}, z_{n+1}$ is disjoint. Now proceed inductively.

Without the assumption that each φ_i is positive, we use the preceding proof to find a disjoint positive sequence (y_i) in X which "lifts" the sequence $\big(|\varphi_i|\big)$, and then, using Exercise 3.4.4(iv) find $x_i \in \varphi_i$ with $|x_i| \leqslant y_i$ for each i.

The proof of (i) goes along the same lines except that we define $\theta_n = \sum_{i=n+1}^{m} \varphi_i$; in this case this is a finite sum and we do not have to worry about its convergence. □

Let J be a closed ideal in a Banach lattice X. It follows from the definition of the quotient norm that for every positive φ in X/J and every $\varepsilon > 0$ we can find a representative x of φ X such that $\|x\| \leqslant \|\varphi\| + \varepsilon$. The following exercise shows that, in addition, we can chose x to be dominated by any specified representative. It follows that the x_i's in Theorem 3.4.10(ii) may be chosen so that their norms are arbitrarily close to that of φ_i's.

3.4.11 Exercise. Let J be a closed ideal in a Banach lattice X, $Q\colon X \to X/J$ the corresponding quotient map, $u \in X_+$, and $\varepsilon > 0$. There exists $v \in [0, u]$ with $Qv = Qu$ and $\|v\| \leqslant \|Qu\| + \varepsilon$.

Let $(X_\alpha)_{\alpha\in\Lambda}$ be a family of vector lattices. Let X be the Cartesian product of the family. We can equip X with a coordinate-wise order, i.e., $(x_\alpha) \leqslant (y_\alpha)$ if $x_\alpha \leqslant y_\alpha$ for every α. It can be easily verified that X is again a vector lattice, with coordinate-wise lattice operations. We say that X is the ***direct product*** of the family $(X_\alpha)_{\alpha\in\Lambda}$ and write $X = \prod_{\alpha\in\Lambda} X_\alpha$.

Fix some α_0. It follows from the definition that the canonical embeddings of X_{α_0} into $\prod_{\alpha\in\Lambda} X_\alpha$ is a lattice homomorphisms, and X_{α_0} may be viewed as an ideal in $\prod_{\alpha\in\Lambda} X_\alpha$. Similarly, the map $\pi_{\alpha_0}\colon X \to X_{\alpha_0}$ which maps (x_α) to x_{α_0} and the canonical projection $P_{\alpha_0}\colon X \to X$, which takes (x_α) into the family (y_α) where $y_\alpha = x_\alpha$ when $\alpha = \alpha_0$ and $y_\alpha = 0$ otherwise, are lattice homomorphisms.

For many purposes, $\prod_{\alpha\in\Lambda} X_\alpha$ may be "too large". One would often consider a sublattice of $\prod_{\alpha\in\Lambda} X_\alpha$ consisting of all families (x_α) with only finitely many non-zero coordinates, or with only countably many non-zero coordinates, or with coordinates being summable in some way. Such sublattices are often called ***direct sums*** and denoted $\bigoplus_{\alpha\in\Lambda} X_\alpha$, though one has to specify the summability criterion in each case. In particular, the classical direct ℓ_p-sum of Banach spaces $\bigoplus_p X_\alpha$ as in A.4.32 is a special case of the preceding construction when every X_α is a Banach lattice. For a finite family of vector lattices, all definitions of direct sums or products agree (as vector lattices), so we can freely use notations $X \oplus Y$ or $\bigoplus_{k=1}^n X_k$.

3.4.12 Exercise. Let $X = \prod_{\alpha\in\Lambda} X_\alpha$.

(i) X is Archimedean iff each X_α is Archimedean.

(ii) X is order complete iff each X_α is order complete.

Chapter 4

$C(K)$ representations

The reader may have noticed that all the examples of vector lattice that we have seen happen to be spaces of functions. We will show in this chapter that every Archimedean vector lattice may "locally" be represented as a $C(K)$ space. This allows one to treat elements of a vector lattice as continuous functions. In particular, it allows one to define functions of elements of a vector lattice (function calculus).

We will use the $C(K)$-Representation Theorem to deduce Yudin's theorem, which, essentially, says that every identity or inequality that involves only linear and lattice operations and which is valid in $\mathbb{R}$ remains valid in every vector lattice.

More generally, $C(K)$ representation theory allows one to reduce various questions about vector lattices to questions about $C(K)$ spaces, and then use topology to solve these questions. This technique leads to many deep and interesting results in the theory of vector and Banach lattices. While many of these results may be proved directly from the axioms (and this is the approach used in some of the literature on the subject), we find the representation approach easier and more intuitive. It shows that $C(K)$ spaces play a very special role in the theory. It also explains deep connections between vector lattices and topology.

4.1 $C(K)$-Representation Theorem

In this section, we present the most important fact of this chapter: an Archimedean vector lattice X with a strong unit may be represented as a sublattice $C(K)$ space. Here is the idea. Suppose that X equals $C(K)$; how can one "recover" K from the vector lattice structure of $C(K)$? For every $t \in K$, the map $f \mapsto f(t)$ defines a continuous linear functional on $C(K)$, denoted δ_t; it is called a ***point evaluation*** functional or a ***Dirac measure*** at t.

4.1.1 Proposition. *Real valued lattice homomorphisms on $C(K)$ are exactly the non-negative scalar multiples of point evaluation functionals.*

Proof. It is easy to see that every point evaluation functional is a lattice homomorphism; cf. Exercise 1.6.7. Let $\xi\colon C(K) \to \mathbb{R}$ be a non-zero lattice homomorphism. It follows from

$$\left|\xi(f)\right| = \xi\bigl(|f|\bigr) \leqslant \xi\bigl(\|f\|\mathbb{1}\bigr) = \xi(\mathbb{1})\|f\|$$

that ξ is norm bounded. Then $\ker\xi$ is a closed ideal, hence $\ker\xi = J_F$ for some non-empty closed set F by Theorem 3.3.11. Fix any $t \in F$, then $\mathbb{1} \notin J_{\{t\}} \supseteq J_F$. Since $\ker\xi$ has co-dimension one, we have $\ker\xi = J_{\{t\}}$. On the other hand, $\ker\delta_t = J_{\{t\}}$. It follows that ξ is a scalar multiple of δ_t. Since both functionals are positive, so is the scalar. □

In particular, there is a one-to-correspondence between K and the set of all real valued lattice homomorphism on $C(K)$ which map $\mathbb{1}$ to 1. It follows also that the sum of two lattice homomorphisms need not be a lattice homomorphism!

Now let X be a vector lattice with a strong unit e; take K to be the set of all real valued lattice homomorphism φ on X with $\varphi(e) = 1$. We can now view an element x of X as a function on K via $\varphi \mapsto \varphi(x)$. To make this idea into a formal proof, we need some technical facts about lattice homomorphisms; in particular, that there is a one-to-one correspondence between real valued lattice homomorphism on X and maximal ideals in X.

4.1.2. Let J_0 be an ideal in a vector lattice X and $x_0 \notin J_0$. Let $\mathcal{G}$ be the collection all ideals in X containing J_0 but not containing x_0. This collection is not empty because $J_0 \in \mathcal{G}$. If $\mathcal{C}$ is a chain in $\mathcal{G}$ then $\bigcup\mathcal{C}$ is clearly also in $\mathcal{G}$. By Zorn's Lemma, $\mathcal{G}$ has a maximal element.

In particular, given $x_0, x_1 \in X$ such that $x_0 \notin I_{x_1}$, we apply the preceding argument with $J_0 = I_{x_1}$ and conclude that there is an ideal J which contains x_1 but not x_0 and which is maximal with respect to this condition.

4.1.3. If X has a strong unit e then an ideal is proper iff it does not contain e. It follows from the preceding paragraph that for every $x \in X$ which is not a strong unit, there exists a maximal proper ideal M containing x. By "maximal proper" we mean that $M \neq X$ and $M \subseteq J$ implies $J = M$ or $J = X$ for every ideal J.

4.1.4 Exercise. If M is a maximal proper ideal in a vector lattice X then $\dim X/M = 1$.

4.1.5 Exercise. A proper ideal in a vector lattice is maximal iff it is the kernel of a non-zero real-valued lattice homomorphism.

4.1.6 Lemma. *Let X be a vector lattice with a strong unit e. Let $x_0 \in X_+$ with $x_0 \nleqslant e$. Then there exists a non-zero lattice homomorphism $\varphi\colon X \to \mathbb{R}$ such that $\varphi(x_0) \geqslant \varphi(e)$.*

Note that this lemma is trivial for $C(K)$-spaces. Indeed, in the case of a $C(K)$ space, the assumption $x_0 \nleqslant e$ means that there exists $t_0 \in K$ such that $x_0(t_0) > e(t_0)$. Now take $\varphi = \delta_{t_0}$.

Proof. It follows from $x_0 \nleqslant e$ that $(x_0 - e)^+ \neq 0$, so that the principal ideal of $(x_0 - e)^-$ is proper. Let M be a maximal proper ideal containing $(x_0 - e)^-$. Then $\dim X/M = 1$, so we may identify X/M with $\mathbb{R}$. Let $q\colon X \to X/M$ be the quotient map. Then q is a lattice homomorphism, and it follows from $q\bigl((x_0 - e)^-\bigr) = 0$ that

$$q(x_0) = q(x_0 - e) + q(e) = q\bigl((x_0 - e)^+\bigr) + q(e) \geqslant q(e).$$

□

We now recall a few facts from Section 2.2. Let e be a positive vector in an Archimedean vector lattice X. Then the principal ideal I_e is a normed lattice under the norm

$$\|x\|_e = \inf\bigl\{\lambda \geqslant 0 \,:\, |x| \leqslant \lambda e\bigr\}.$$

In particular, if e is a strong unit then $\|\cdot\|_e$ is a lattice norm on X. For every $x \in I_e$, we have $|x| \leqslant \|x\|_e e$. X is uniformly complete if $\bigl(I_e, \|\cdot\|_e\bigr)$ is complete for every $e \in X_+$. Every σ-order complete vector lattice as well as every Banach lattice is uniformly complete.

4.1.7 Theorem ($C(K)$-Representation Theorem, Kakutani-Krein)**.** *Let X be an Archimedean vector lattice with a strong unit e. Then the completion of $\bigl(X, \|\cdot\|_e\bigr)$ is lattice isometric to $C(K)$ for some compact space K, with e corresponding to $\mathbb{1}$.*

Proof. Let K be the set of all lattice homomorphisms $\varphi\colon X \to \mathbb{R}$ satisfying $\varphi(e) = 1$. We view K as a subset of the set $\mathbb{R}^X$ of all real-valued functions on X, which may also be identified with the Cartesian product $\prod_{x\in X}\mathbb{R}$. We equip this product with the product topology, which is just the topology of point-wise convergence; see Appendix A.2. This induces a topology on K. Since the product topology on $\mathbb{R}^X$ is Hausdorff, K is Hausdorff as well.

Let $\varphi \in K$. For every $x \in X$, it follows from $|x| \leqslant \|x\|_e e$ that $\bigl|\varphi(x)\bigr| = \varphi\bigl(|x|\bigr) \leqslant \|x\|_e$, so that $\varphi(x) \in \bigl[-\|x\|_e, \|x\|_e\bigr]$. Thus, not only do we have $K \subseteq \prod_{x\in X}\mathbb{R}$ but actually $K \subseteq \prod_{x\in X}\big[-\|x\|_e, \|x\|_e\big]$. By Tychonoff's Product Theorem A.2.8, the latter product is compact. Note that K is closed in this product space: if (φ_α) is a net in K and $\varphi_\alpha \to \varphi$ point-wise in $\prod_{x\in X}\big[-\|x\|_e, \|x\|_e\big]$ then $\varphi(e) = \lim_\alpha \varphi_\alpha(e) = 1$; it is also easy to see that φ is a lattice homomorphism, thus, $\varphi \in K$. Being a closed subset of a compact set, K is itself compact.

We show next that K is non-empty; moreover, K is "sufficiently rich". It will follow from this *Claim:* for every $x \in X_+$ with $\|x\|_e > 1$ there exists $\varphi \in K$ with $\varphi(x) \geqslant 1$. Indeed, $\|x\|_e > 1$ implies $x \nleqslant e$. Now apply the lemma preceding the theorem and scale the resulting φ so that $\varphi(e) = 1$.

Fix $x \in X$. The map $\varphi \in K \mapsto \varphi(x)$ is continuous because $\varphi_\alpha \to \varphi$ in K implies $\varphi_\alpha(x) \to \varphi(x)$. Denote this map by Tx. This yields an operator $T\colon X \to C(K)$ with $(Tx)(\varphi) = \varphi(x)$. We will show that T is a surjective lattice isometry with $Te = \mathbb{1}$; this will complete the proof.

It is easy to see that T is linear. Observe that $(Te)(\varphi) = \varphi(e) = 1$ for every $\varphi \in K$, so that $Te = \mathbb{1}$. We will now verify that T is a lattice homomorphism. Indeed, for every $x \in X$ and $\varphi \in K$, we have

$$\big(T|x|\big)(\varphi) = \varphi\big(|x|\big) = \big|\varphi(x)\big| = \big|(Tx)(\varphi)\big| = |Tx|(\varphi),$$

where the last identity follows from the fact that lattice operations in $C(K)$ are computed point-wise. It follows that $|Tx| = T|x|$, hence T is a lattice homomorphism.

We show next that T is an isometry. Let $x \in X$. On one hand, $|x| \leqslant \|x\|_e e$ and $Te = \mathbb{1}$ imply $|Tx| = T|x| \leqslant \|x\|_e\mathbb{1}$, so that $\|Tx\| \leqslant \|x\|_e$. To prove the other inequality, suppose first that $\|x\|_e > 1$. Then the *Claim* above yields a $\varphi \in K$ with $\varphi\big(|x|\big) \geqslant 1$; it now follows from the definition of the norm in $C(K)$ that $\|Tx\| \geqslant \big|(Tx)(\varphi)\big| = \varphi\big(|x|\big) \geqslant 1$. Suppose now that $\|x\|_e = 1$. For every $\lambda > 1$, we have $\|\lambda x\|_e > 1$, hence $\lambda\|Tx\| = \big\|T(\lambda x)\big\| \geqslant 1$. Passing to the limit as $\lambda \searrow 1$, we get $\|Tx\| \geqslant 1$. It follows that $\|Tx\| = \|x\|_e$ for every $x \in X$.

Since T is a lattice homomorphism, $\operatorname{Range} T$ is a sublattice of $C(K)$. Furthermore, $\mathbb{1} = Te \in \operatorname{Range} T$. We can now use Stone-Weierstrass Theorem 3.1.28 to claim that $\operatorname{Range} T$ is (norm) dense in $C(K)$; we only need to verify that $\operatorname{Range} T$ separates points. This is easy: if $\varphi \neq \psi$ in K then $\varphi(x) \neq \psi(x)$ for some $x \in X$ and, therefore, $(Tx)(\varphi) \neq (Tx)(\psi)$.

We have proved that T is a lattice isometric embedding of $(X, \|\cdot\|_e)$ onto a dense sublattice $C(K)$. It follows that T extends to a (surjective) isometry $\widetilde{T}$ from the completion of $(X, \|\cdot\|_e)$ to $C(K)$. Recall that the completion of a normed lattice is a Banach lattice; see 1.3.23. Continuity of lattice operations ensures that $\widetilde{T}$ is still a lattice homomorphism. $\square$

Next, we present several corollaries (which are, actually, equivalent reformulations) of the $C(K)$-representation theorem. Under the assumptions of the theorem, X is obviously dense in the completion of $\bigl(X, \lVert\cdot\rVert_e\bigr)$. This yields the following:

4.1.8 Theorem. *Every Archimedean vector lattice with a strong unit e is lattice isomorphic to a dense sublattice of $C(K)$, with e corresponding to $\mathbb{1}$.*

Exercise 3.2.12 implies that the resulting sublattice is not only norm dense but also order dense.

4.1.9 Theorem. *Let X be a Archimedean vector lattice and $e \in X_+$. Then $(I_e, \lVert\cdot\rVert_e)$ is lattice isometric to a dense sublattice of $C(K)$ for some compact space K, with e corresponding to $\mathbb{1}$. If, in addition, X is uniformly complete then the embedding is onto.*

Theorem 4.1.9 essentially says a uniformly complete Archimedean vector lattice locally behaves like a $C(K)$-space.

4.1.10 Theorem. *Let X be a vector lattice with a strong unit e. If X is σ-order complete or is a Banach lattice then $(X, \lVert\cdot\rVert_e)$ is lattice isometric to $C(K)$ for some compact space K, e corresponding to $\mathbb{1}$. If X is order complete then K is extremally disconnected.*

Proof. X is Archimedean by Proposition 2.1.6 or Exercise 1.3.22, and uniformly complete by Theorem 2.2.9. It follows, in particular, that $(X, \lVert\cdot\rVert_e) = (I_e, \lVert\cdot\rVert_e)$ is norm complete. Theorem 4.1.7 yields that $(X, \lVert\cdot\rVert_e)$ is lattice isometric to $C(K)$ for some compact space K, with e corresponding to $\mathbb{1}$. If X is order complete then $C(K)$ or order complete, hence K is extremally disconnected by Theorem 2.1.29. □

Let $\bigl(X, \lVert\cdot\rVert\bigr)$ be a Banach lattice and $e \in X$ a strong unit. By Corollary 2.2.10, the norms $\lVert\cdot\rVert$ and $\lVert\cdot\rVert_e$ are equivalent, hence the identity map from $\bigl(X, \lVert\cdot\rVert\bigr)$ to $\bigl(X, \lVert\cdot\rVert_e\bigr)$ is a lattice and norm isomorphism. Thus, we have the following:

4.1.11 Corollary. *Every Banach lattice with a strong unit is lattice and norm isomorphic to $C(K)$ for some compact space K.*

4.1.12 Example. Let $X = L_\infty(\mu)$; take $e = \mathbb{1}$. Then e is a strong unit and $\lVert\cdot\rVert_e$ is exactly the usual norm $\lVert\cdot\rVert_\infty$ on $L_\infty(\mu)$. Theorem 4.1.10 says that $L_\infty(\mu)$ is lattice isometric to some $C(K)$ space, with the constant one function of $L_\infty(\mu)$ going into the constant one function of $C(K)$.

4.2 * Prime ideals and totally ordered vector lattices

Throughout this section, X will stand for a vector lattice. We observed in Exercise 4.1.4 that the quotient of X over a maximal proper ideal is one-dimensional, hence may be identified with $\mathbb{R}$. In particular, this quotient is totally ordered. In this section, we will extend this idea by replacing a maximal proper ideal with a more general concept of a *prime* ideal. While we generally shun away from non-Archimedean vector lattices in this book, this section is exceptional as non-Archimedean vector lattices play a major role in it.

We say that X is ***totally ordered*** if every two elements are comparable, i.e., for every $x, y \in X$ either $x \leqslant y$ or $y \leqslant x$. An example of a totally ordered vector lattice is provided by $\mathbb{R}^n$ equipped with the lexicographic order as in Exercise 1.2.10; we will denote this vector lattice by ${}^n\mathbb{R}$ in order to distinguish this space from the "usual" $\mathbb{R}^n$. Note that ${}^n\mathbb{R}$ is clearly non-Archimedean.

4.2.1 Exercise. The only Archimedean totally ordered vector lattice is $\mathbb{R}$.

An ideal J in a vector lattice X is said to be ***prime*** if $J = J_1 \cap J_2$ implies that either $J = J_1$ or $J = J_2$ for any two ideals J_1 and J_2. Clearly, every maximal proper ideal is prime.

4.2.2 Example. There are exactly three distinct ideals in ${}^2\mathbb{R}$, namely, $\{0\}$, the y-axis, and the whole space; all the three are prime.

The following observation guarantees that there are "sufficiently many" prime ideals in every vector lattice. We saw in 4.1.2 that given any ideal J_0 in x and $x_0 \notin J_0$, one can find an ideal J such that $J_0 \subseteq J$ and $x_0 \notin J$ and, furthermore, J is maximal with respect to this property. We claim that J is prime. Suppose not, then we can find two ideals J_1 and J_2 in X such that $J = J_1 \cap J_2$ yet $J \subsetneq J_1$ and $J \subsetneq J_2$. By maximality of J, we have $x_0 \in J_1$ and $x_0 \in J_2$, so that $x_0 \in J_1 \cap J_2 = J$; a contradiction.

4.2.3 Corollary. *For every non-zero vector $x \in X$ there exists a prime ideal J such that $x \notin J$.*

4.2.4 Proposition. *Let J be a prime ideal and $a \notin J$. Then either $a^+ \in J$ or $a^- \in J$.*

Proof. Let J_1 be the ideal generated by J and a^+; let J_2 be the ideal generated by J and a^-. Clearly, $J \subseteq J_1 \cap J_2$. We claim that $J = J_1 \cap J_2$; it would then follow that either $J = J_1$, in which case $a^+ \in J$, or $J = J_2$, in which case $a^- \in J$.

To prove the claim, take an arbitrary $x \in J_1 \cap J_2$. Then

$$\begin{array}{ll} x \in J_1 \text{ implies} & |x| \leqslant y + \alpha a^+ \text{ for some } y \in J_+ \text{ and } \alpha \in \mathbb{R}_+, \text{ and} \\ x \in J_2 \text{ implies} & |x| \leqslant z + \beta a^- \text{ for some } z \in J_+ \text{ and } \beta \in \mathbb{R}_+. \end{array}$$

Put $\gamma = \max\{\alpha, \beta\}$ and $u = y \vee z$. Then $|x| \leqslant u + \gamma a^+$ and $|x| \leqslant u + \gamma a^-$. It follows that

$$|x| \leqslant (u + \gamma a^+) \wedge (u + \gamma a^-) = u + \gamma(a^+ \wedge a^-) = u.$$

It now follows from $u \in J$ that $x \in J$. □

4.2.5 Theorem. *Let J be an ideal in X. TFAE:*

(i) *J is prime;*
(ii) *X/J is totally ordered;*
(iii) *If $a \wedge b \in J$ then $a \in J$ or $b \in J$;*
(iv) *If $a \wedge b = 0$ then $a \in J$ or $b \in J$.*

Proof. (i)⇒(ii) Let $x, y \in J$. Then either $(x-y)^+$ is in J or $(x-y)^-$ is in J. Say, $(x-y)^- \in J$. It follows that $(x-y)+J = (x-y)^+ + J$, so that $(x-y)+J$ is positive in X/J, which yields $x + J \geqslant y + J$.

(ii)⇒(iii) Suppose that $a \wedge b \in J$. Since the quotient map is a lattice homomorphism, we have $(a+J) \wedge (b+J) = 0$ in X/J. Since X/J is totally ordered, either $a + J = 0$, in which case $a \in J$, or $b + J = 0$, in which case $b \in J$.

(iii)⇒(iv) is trivial.

(iv)⇒(i) Suppose that $J = J_1 \cap J_2$ for two ideals J_1 and J_2, yet neither $J = J_1$ nor $J = J_2$. Find $x_1 \in J_1 \setminus J_2$ and $x_2 \in J_2 \setminus J_1$. Replacing x_1 and x_2 with their moduli, if necessary, we may assume that $x_1, x_2 \geqslant 0$. It follows from $(x_1 - x_1 \wedge x_2) \wedge (x_2 - x_1 \wedge x_2) = 0$ that $x_1 - x_1 \wedge x_2 \in J$ or $x_2 - x_1 \wedge x_2 \in J$. Say, $x_1 - x_1 \wedge x_2$ is in J and, therefore, in J_2. Note that $0 \leqslant x_1 \wedge x_2 \leqslant x_2$, so that $x_1 \wedge x_2 \in J_2$. This yields that $x_1 \in J_2$; a contradiction. □

4.2.6 Exercise. Let K be a compact space.

(i) Let $t_0 \in K$. The set $J_{\{t_0\}}$ of all functions in $C(K)$ that vanish at t_0 is a prime ideal.
(ii) Let J be a proper prime ideal in $C(K)$. Then there is exactly one point t_0 of K such that $f(t_0) = 0$ for all $f \in J$.

For further information on prime ideals, we refer the reader to Chapter 5 in [LZ71].

4.2.7 Theorem (Totally Ordered Representation Theorem)**.** *Every vector lattice embeds lattice isomorphically into a direct product of totally ordered vector lattices.*

Proof. Let $\mathcal{P}$ be the collection of all proper prime ideals in X. For every $J \in \mathcal{P}$, the quotient vector lattice X/J is totally ordered by Theorem 4.2.5(ii). Let $Q_J\colon X \to X/J$ be the quotient map. Being a quotient map, Q_J is a lattice homomorphism by Theorem 3.4.2.

Consider the map Q from X to the direct product $\prod_{J\in\mathcal{P}} X/J$ defined by $(Qx)_J = Q_J x$ for $x \in X$. Q is a lattice homomorphism as $\bigl(Q|x|\bigr)_J = Q_J|x| = |Q_J x| = \bigl|(Qx)_J\bigr|$. It is left to show that Q is one-to-one. Suppose that $x \in X$ with $x \neq 0$. By Corollary 4.2.3, there exists $J \in \mathcal{P}$ such that $x \notin J$. It follows that $Q_J x \neq 0$ and, therefore, $Qx \neq 0$. $\square$

The preceding theorem suggests that totally ordered vector lattices play an important role in the theory. In the rest of this section, we discuss some properties of such spaces. We will see that ${}^n\mathbb{R}$ spaces play a special role here.

Let X be a vector lattice, and $a, b \in X_+$. We write $a \ll b$ if $na \leqslant b$ for every $n \in \mathbb{N}$. In an Archimedean vector lattice, this may only happen when $a = 0$.

4.2.8 Exercise. Let $e_1, \dots, e_n$ be the standard unit vectors in ${}^n\mathbb{R}$. Verify that $e_1 \gg e_2 \gg \cdots \gg e_n$.

4.2.9 Exercise. Let X be a vector lattice and $x_1, \dots, x_n \in X$ such that $x_1 \gg x_2 \gg \cdots \gg x_n > 0$. Then the linear map $T\colon {}^n\mathbb{R} \to X$ defined by $Te_i = x_i$ is a lattice isomorphic embedding.

Recall that I_a stands for the principal ideal of a.

4.2.10 Exercise. Let X be a totally ordered vector lattice and $a, b \in X_+ \setminus \{0\}$. Then there is a dichotomy: either $a \ll b$ or $I_b \subseteq I_a$.

4.2.11 Lemma. *Let X be a totally ordered vector lattice and $a, b \in X_+ \setminus \{0\}$. If $I_a = I_b$ then there exists a unique $\lambda > 0$ such that $|b - \lambda a| \ll a$.*

Proof. For every positive real λ we have $\lambda a \leqslant b$ or $\lambda a \geqslant b$. Let

$$A = \bigl\{\lambda > 0 \,:\, \lambda a \leqslant b\bigr\} \quad \text{and} \quad B = \bigl\{\lambda > 0 \,:\, \lambda a \geqslant b\bigr\}.$$

Clearly $A \cup B = (0, \infty)$. It follows from $I_a = I_b$ that both A and B are non-empty. It is also clear that if $\lambda \in A$ and $0 < \mu < \lambda$ then $\mu \in A$; similarly, if $\lambda \in B$ and $\mu > \lambda$ then $\mu \in B$. It follows that A and B are intervals with $\sup A = \inf B$; denote this number by λ_0. For every sufficiently large $n \in \mathbb{N}$, we have $\lambda_0 - \frac{1}{n} \in A$ and $\lambda_0 + \frac{1}{n} \in B$. This yields $-\frac{1}{n}a \leqslant b - \lambda_0 a \leqslant \frac{1}{n}a$. It follows that $n|b - \lambda_0 a| \leqslant a$. Since $n \in \mathbb{N}$ is arbitrary, we get $|b - \lambda a| \ll a$. $\square$

It is easy to see that e_1 is a strong unit in ${}^n\mathbb{R}$.

4.2.12 Exercise. Let X be a totally ordered vector lattice with a strong unit e. If $\dim X > 1$ then there exists $x \in X$ such that $0 < x \ll e$.

Recall that a vector lattice X is *finitely generated* if there exist vectors $x_1, \dots, x_m$ in X such that the sublattice generated by $x_1, \dots, x_m$ is all of X.

4.2.13 Proposition. *Every finitely generated totally ordered vector lattice is lattice isomorphic to* ${}^n\mathbb{R}$ *for some* $n \in \mathbb{N}$.

Proof. Let X be a totally ordered vector lattice generated by some non-zero vectors $x_1, \dots, x_m$. Replacing x_i with $-x_i$ if necessary, we may assume that $x_i > 0$ for every i. Relabelling, we may assume that $x_1 > x_i$ whenever $i > 1$.

We may assume WLOG that $x_1 \gg x_i$ for each $i > 1$. Indeed, if $x_1 \gg x_i$ fails then $I_{x_1} = I_{x_i}$ by Exercise 4.2.10. We then use Lemma 4.2.11 to find λ_i such that $|x_i - \lambda_i x_1| \ll x_1$. Replacing x_i with either $x_i - \lambda_i x_1$ or $-(x_i - \lambda_i x_1)$, whichever is positive, we may assume that $x_i \ll x_1$. Note that this operation does not affect the fact that $x_1, \dots, x_m$ generate X. If any of the resulting vectors is zero, drop it from the list and relabel. Now apply the same procedure to $\{x_2, \dots, x_m\}$. Proceeding inductively, we arrive to some vectors $x_1, \dots, x_n$ which still generate X and satisfy $x_1 \gg x_2 \gg \dots \gg x_n > 0$. Now use Exercise 4.2.9. □

4.2.14 Example. Recall that c_{00} stands for the space of all sequences of real numbers that are eventually zero; it is an ideal in the vector lattice $\mathbb{R}^{\mathbb{N}}$ of all real sequences (with the usual coordinate-wise order). We consider the quotient $\mathbb{R}^{\mathbb{N}}/c_{00}$; here we identify sequences that are eventually equal, i.e., one can be obtained from the other by perturbing finitely many terms. Note that $\mathbb{R}^{\mathbb{N}}/c_{00}$ fails to be Archimedean. Indeed, let $x = (1, \frac{1}{2}, \frac{1}{3}, \dots)$. Then $nx \leqslant \mathbb{1}$ for every $n \in \mathbb{N}$ because the terms of nx are eventually less than one. That is, $x \ll \mathbb{1}$. Yet $x > 0$.

4.2.15 Proposition. $\mathbb{R}^{\mathbb{N}}/c_{00}$ *contains a lattice isomorphic copy of* ${}^n\mathbb{R}$ *for every* $n \in \mathbb{N}$.

Proof. We use the fact that the real function $f(t) = t^k$ asymptotically dominates every polynomial of degree less than k.

Let $Q\colon \mathbb{R}^{\mathbb{N}} \to \mathbb{R}^{\mathbb{N}}/c_{00}$ be the quotient map. Fix $n \in \mathbb{N}$. Let $x_1, \dots, x_n \in \mathbb{R}^{\mathbb{N}}$ defined by $x_k = (1^k, 2^k, 3^k, \dots)$. It is easy to see that $0 < Qx_1 \ll \dots \ll Qx_n$ in $\mathbb{R}^{\mathbb{N}}/c_{00}$. Now apply Exercise 4.2.9. □

We finish this section with a few comments on prime *subspaces*. The following definition is motivated by Theorem 4.2.5: a subspace Y of X is a ***prime subspace*** if $x \wedge y \in Y$ implies $x \in Y$ or $y \in Y$.

4.2.16 Exercise. (i) A subspace Y is prime iff $x \wedge y = 0$ implies $x \in Y$ or $y \in Y$;

(ii) Every prime subspace is a sublattice;

(iii) Every subspace that contains a prime subspace is itself prime.

(iv) Every subspace that contains a prime sublattice is itself a sublattice.

4.2.17 Theorem. *A vector lattice X admits a sublattice of codimension n for every natural number $n \leqslant \dim X$.*

Proof. It suffices to prove the statement for $n = 1$; we can then proceed by induction. By Corollary 4.2.3, X admits a proper prime ideal, say, J. Take any subspace Y of X of codimension 1 such that $J \subseteq Y$. Then Y is a sublattice by the preceding exercise. $\square$

It was shown in [AL90a] that one can even find a majorizing sublattice of arbitrary finite codimension.

4.3 Lattice-linear function calculus

In this section, we present an easy "generic" way of proving various identities and inequalities like the ones in Exercises 1.3.3 and 1.3.11 in any vector lattice. To illustrate the method, let's present a new proof of the triangle inequality $|x + y| \leqslant |x| + |y|$ for any x and y in an Archimedean vector lattice X (we will extend the method to the non-Archimedean case later in this section). Put $e = |x| \vee |y|$. Clearly, $x, y \in I_e$. By Theorem 4.1.7, we may view I_e as a sublattice of some $C(K)$ space, so we may view x and y as functions on K. For every $t \in K$ we have $\bigl|x(t) + y(t)\bigr| \leqslant \bigl|x(t)\bigr| + \bigl|y(t)\bigr|$; this is the usual triangle inequality in $\mathbb{R}$. Since lattice operations in $C(K)$ are point-wise, $|x + y| \leqslant |x| + |y|$ is satisfied in $C(K)$, hence in I_e, hence in X.

4.3.1 Exercise. Use this technique to prove that the following statements are true in every Archimedean vector lattice X.

(i) $(x - y)^+ \wedge (y - z)^+ \wedge (z - x)^+ = 0$.

(ii) If $|a| \leqslant |b|$ and $a \perp c$ then $|a| \leqslant |b + c|$.

(iii) If $a, b, c \in X_+$ then $0 \leqslant a - (a - b - c)^+ \leqslant c + a \wedge b$.

This method effectively yields that any identity or inequality which involves only linear and lattice operations and finitely many variables and is valid for real numbers is automatically satisfied in every Archimedean vector lattice. In order to state (and prove) this formally, we need to introduce some notation.

Similarly to *linear expessions* in vector spaces, there is a natural concept of a *lattice-linear expression* in a vector lattice. Fix formal variables $t_1, \dots, t_n$; by a (formal)

lattice-linear expression of $t_1,\dots,t_n$ we mean an expression which is constructed out of $t_1,\dots,t_n$ using only linear operations (addition and scalar multiplication) and lattice operations and following the usual rules of syntax in vector lattices. For example, $2t_1^+ + t_2\wedge(t_3-5t_1)$ and $t_2\wedge(t_3-5t_1)+2t_1^+$ are two (different) lattice-linear expression of t_1, t_2, and t_3. For an interested reader, we will provide a more formal definition of a lattice-linear expression in Section 22.3.

Consider a lattice-linear expression of $t_1,\dots,t_n$; denote it by $\Phi[t_1,\dots,t_n]$. We use square brackets to underscore that $t_1,\dots,t_n$ are formal variables and we do not (yet) view $\Phi[t_1,\dots,t_n]$ as a function. However, if we replace $t_1,\dots,t_n$ in Φ with some $x_1\dots,x_n$ in a vector lattice X, the resulting expression yields an element of X, which we will denote by $\Phi(x_1,\dots,x_n)$. We say that this vector is a ***lattice-linear combination*** of $x_1,\dots,x_n$. Furthermore, this yields a function from X^n to X, for every vector lattice X. In particular, if we interpret $t_1,\dots,t_n$ as real numbers, we get a function from $\mathbb{R}^n$ to $\mathbb{R}$. With some abuse of notation, we denote this function by $\Phi(t_1,\dots,t_n)$. We call such a function a ***lattice-linear function***. In other words, a function from $\mathbb{R}^n$ to $\mathbb{R}$ is lattice-linear iff it is a composition of linear and lattice operations.

Note that given $x_1,\dots,x_n$ in a vector lattice X, the set of all lattice-linear combinations of $x_1,\dots,x_n$ is precisely the sublattice of X generated by $x_1,\dots,x_n$.

Let L_n stand for the set of all lattice-linear functions of n variables. Clearly, L_n is a sublattice of $C(\mathbb{R}^n)$ and of the vector lattice $\mathbb{R}^{\mathbb{R}^n}$ of all the real-valued functions of n variables. In particular, L_n itself is a vector lattice. For every $i=1,\dots,n$, let $\delta_i\colon\mathbb{R}^n\to\mathbb{R}$ be the i-th coordinate functions defined by $\delta_i(t_1,\dots,t_n)=t_i$. It is easy to see that L_n is exactly the sublattice of $C(\mathbb{R}^n)$ (and, therefore, of $\mathbb{R}^{\mathbb{R}^n}$) generated by $\delta_1,\dots,\delta_n$.

4.3.2 Example. Consider the formal lattice-linear expression $\Phi[t_1,t_2,t_3]=2t_1+t_2\wedge t_3$. The function from $\mathbb{R}^3$ to $\mathbb{R}$ given by $\Phi(t_1,t_2,t_3)=2t_1+t_2\wedge t_3$ is a lattice-linear function. This function is precisely $2\delta_1+\delta_2\wedge\delta_3$. Given three vectors x_1, x_2, and x_3 in *any* vector lattice X, the expression $\Phi(x_1,x_2,x_3)=2x_1+x_2\wedge x_3$ defines an element of X. So Φ induces a function from X^3 to X.

Let $\Psi[t_1,t_2,t_3]=(2t_1+t_2)\wedge(2t_1+t_3)$. Φ and Ψ differ as formal lattice-linear expressions. However, it can be easily deduced from the distributive law that $\Phi(x_1,x_2,x_3)=\Psi(x_1,x_2,x_3)$ for any x_1,x_2,x_3 in X. In particular, Φ and Ψ yield the same lattice-linear function.

The following theorem asserts that if two lattice-linear expressions agree as functions from $\mathbb{R}^n$ to $\mathbb{R}$ then they also agree when viewed as functions from X^n to X.

4.3.3 Theorem (Yudin)**.** *Let $\Phi[t_1,\dots,t_n]$ and $\Psi[t_1,\dots,t_n]$ be two lattice-linear expressions such that $\Phi(t_1,\dots,t_n)=\Psi(t_1,\dots,t_n)$ for all $t_1,\dots,t_n\in\mathbb{R}$. Then $\Phi(x_1,\dots,x_n)=\Psi(x_1,\dots,x_n)$ for any vectors $x_1,\dots,x_n$ in an arbitrary vector lattice X.*

Proof. Replacing Φ with $\Phi-\Psi$ we may assume WLOG that $\Psi=0$. We say that Φ vanishes on a vector lattice X if $\Phi(x_1,\dots,x_n)=0$ for all $x_1,\dots,x_n$ in X. Thus, it is given that Φ vanishes on $\mathbb{R}$, and we need to show that Φ vanishes on every vector lattice X.

Suppose that X is Archimedean. Fix $x_1,\dots,x_n\in X$. Take any $e\in X_+$ satisfying $x_1,\dots,x_n\in I_e$; for example, we can take $e=\lvert x_1\rvert\vee\dots\vee\lvert x_n\rvert$. Since I_e is a sublattice, we have $\Phi(x_1,\dots,x_n)\in I_e$. By $C(K)$ Representation Theorem 4.1.9, there exists a lattice isomorphic embedding T from I_e to $C(K)$ for some compact space K. Since T is a lattice homomorphism, we have

$$T\Phi(x_1,\dots,x_n)=\Phi(Tx_1,\dots,Tx_n).$$

View this as a function in $C(K)$. For every $t\in K$, we have

$$\big(T\Phi(x_1,\dots,x_n)\big)(t)=\Phi\big((Tx_1)(t),\dots,(Tx_n)(t)\big)=0$$

because $(Tx_1)(t),\dots,(Tx_n)(t)$ are real numbers. It follows that $T\Phi(x_1,\dots,x_n)=0$ and, therefore, $\Phi(x_1,\dots,x_n)=0$.

In the general case, when we do not assume that X is Archimedean, we use the Totally Ordered Representation Theorem 4.2.7 instead of the $C(K)$ Representation Theorem. By Theorem 4.2.7, it suffices to prove that Φ vanishes on every totally ordered vector lattice. So we may assume that X is totally ordered. Since $\Phi(x_1,\dots,x_n)$ belongs to the sublattice generated by $x_1,\dots,x_n$, we may assume WLOG that X is finitely generated. By Proposition 4.2.13, we reduce it to the case $X={}^n\mathbb{R}$ for some $n\in\mathbb{N}$.

By Proposition 4.2.7, it suffices to consider the case when X is a sublattice of $\mathbb{R}^{\mathbb{N}}/c_{00}$. Since Φ vanishes on $\mathbb{R}$ and since the lattice operations on $\mathbb{R}^{\mathbb{N}}$ are coordinate-wise, it is clear that Φ vanishes on $\mathbb{R}^{\mathbb{N}}$. It follows that it vanishes on $\mathbb{R}^{\mathbb{N}}/c_{00}$, because the quotient map is a lattice homomorphism. In particular, Φ vanishes on X.

□

4.3.4. Yudin's theorem remains valid for inequalities because $\Phi\leqslant\Psi$ is equivalent to $(\Phi-\Psi)^+=0$.

Fix vectors $x_1,\dots,x_n$ in a vector lattice X. Let $f\in L_n$, i.e., f is a lattice-linear real function of n variables. Let $\Phi[t_1,\dots,t_n]$ be a lattice-linear expression which yields f. That is, $f(t_1,\dots,t_n)=\Phi(t_1,\dots,t_n)$ for all $t_1,\dots,t_n\in\mathbb{R}$. Note that Φ need not be unique; see Example 4.3.2. We can now consider the element $\Phi(x_1,\dots,x_n)$ in X.

Thanks to Yudin's Theorem, this element does not depend on the choice of a particular lattice-linear expression Φ representing f. We denote it by $f(x_1,\dots,x_n)$. This defines a map $T\colon L_n \to X$, where $Tf = f(x_1,\dots,x_n)$ in X. This map may be viewed as ***lattice-linear function calculus***. Note that T depends on the choice of $x_1,\dots,x_n$.

4.3.5 Exercise. Show that $\bigl|x\wedge z - y\wedge z\bigr| \leqslant |x-y|$ for any three vectors x, y, and z in a vector lattice.

4.3.6 Exercise. Let X be a vector lattice and $x_1,\dots,x_n \in X$. The map T that sends $f \in L_n$ to $f(x_1,\dots,x_n)$ is the unique lattice homomorphism from L_n to X satisfying $T\delta_i = x_i$ as $i=1,\dots,n$.

4.4 Positively homogeneous function calculus

In the preceding section we showed that one can define lattice-linear function calculus on every vector lattice. Can one extend it beyond lattice-linear functions? Say, could we make sense of expressions like $x^2+y^3+\sin z$ or $\sqrt{x^2+y^2}$ in every vector lattice? We will show that the answer it "yes", as long as we restrict ourselves to continuous *positively homogeneous* functions and uniformly complete Archimedean vector lattices.

Throughout this section, X will be a uniformly complete Archimedean vector lattice and $x_1,\dots,x_n \in X$. As before, choose any $e \in X_+$ such that $x_1,\dots,x_n \in I_e$; e.g., one may take $e = |x_1|\vee\dots\vee|x_n|$. By Theorem 4.1.9, we may identify I_e with a $C(K)$ space, hence we may view $x_1,\dots,x_n$ as functions in $C(K)$. So now expressions like $x_1^2+x_2^3+\sin x_3$ are defined in $C(K)$ and, therefore, in I_e and, hence, in X. Here we use the assumption that X is uniformly complete; otherwise, $x_1^2+x_2^3+\sin x_3$ may be in $C(K)$ but not in I_e.

More generally, given any continuous function $f\colon \mathbb{R}^n \to \mathbb{R}$, the composition $f(x_1,\dots,x_n)$ given by

$$f(x_1,\dots,x_n)(t) = f\bigl(x_1(t),\dots,x_n(t)\bigr)$$

is in $C(K)$ and, therefore, may be viewed as an element of X. Unfortunately, there is a problem with this approach: the resulting vector is not well-defined: it depends on the choice of e. However, we will show in this section that this procedure is well-defined when f is *positively homogeneous.*

A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is ***positively homogeneous*** if $f(\lambda t_1,\dots,\lambda t_n) = \lambda f(t_1,\dots,t_n)$ for any real $t_1,\dots,t_n$ and positive real λ. We write H_n for the set of all continuous positively homogeneous functions of n variables.

4.4.1 Example. For every positive real p, the function $f(t_1,\dots,t_n) = \Bigl(\sum_{i=1}^n |t_i|^p\Bigr)^{\frac{1}{p}}$ is positively homogeneous. In the case $p=\infty$, the function $u(t_1,\dots,t_n) = |t_1|\vee\dots\vee|t_n|$

is also positively homogeneous. This particular function will play a special role in our exposition.

4.4.2 Example. The geometric mean function is positively homogeneous. More generally, if we fix positive real numbers $\theta_1, \dots, \theta_n$ such that $\sum_{k=1}^n \theta_k = 1$ then the function $f(t_1, \dots, t_n) = |t_1|^{\theta_1} \dots |t_n|^{\theta_n}$ is positively homogeneous.

It is easy to see that H_n is a sublattice of $C(\mathbb{R}^n)$. Every lattice-linear function is positively homogeneous, hence L_n is a sublattice of H_n. Let B_∞^n and S_∞^n be the closed unit ball and the unit sphere of ℓ_∞^n respectively, that is,

$$B_\infty^n = \bigl\{x \in \mathbb{R}^n \,:\, \bigvee_{i=1}^n |x_i| \leqslant 1\bigr\} \text{ and } S_\infty^n = \bigl\{x \in \mathbb{R}^n \,:\, \bigvee_{i=1}^n |x_i| = 1\bigr\}.$$

If $f \in H_n$ then its restriction to S_∞^n belongs to $C(S_\infty^n)$. We denote the restriction $f_{|S_\infty^n}$. Conversely, given $g \in C(S_\infty^n)$, we can extend it to a $f \in H_n$ by putting $f(x) = \|x\|_\infty g\bigl(\frac{x}{\|x\|_\infty}\bigr)$ for every non-zero $x \in \mathbb{R}^n$ and $f(0) = 0$. It follows that the restriction map $R\colon H_n \to C(S_\infty^n)$ defined by $R\colon f \mapsto f_{|S_\infty^n}$ is a (surjective) lattice isomorphism. Hence, H_n may be identified with $C(S_\infty^n)$. Under this identification, the constant one function $\mathbb{1}$ in $C(S_\infty^n)$ corresponds to the function $u(t_1, \dots, t_n) = |t_1| \vee \dots \vee |t_n|$ in H_n (it is, actually, in L_n). In particular, u is a strong unit in H_n. In a similar fashion, H_n may be identified with a sublattice of $C(B_\infty^n)$ by using the restriction to B_∞^n.

This allows us to "pull" the norm from $C(S_\infty^n)$ or $C(B_\infty^n)$ to H_n. That is, for $f \in H_n$, we define

$$\|f\| = \bigl\|f_{|S_\infty^n}\bigr\|_{C(S_\infty^n)} = \max\bigl\{f(t_1, \dots, t_n) \,:\, |t_1| \vee \dots \vee |t_n| = 1\bigr\}.$$

Note that $\|f\| = \|f\|_u$. With respect to this norm, R is a lattice isometry from H_n onto $C(S_\infty^n)$. It is easy to see that $\|f\| = \bigl\|f_{|B_\infty^n}\bigr\|_{C(B_\infty^n)}$.

It the special case when $n = 2$, S_∞^2 is a square, and H_2 is identified with the space $C(S_\infty^2)$ of all continuous functions on the square. In this case, L_2 corresponds to all piece-wise affine functions on the square. In particular, it is dense in $C(S_\infty^2)$, hence in H_2.

More generally, we claim that L_n is dense in H_n. It suffices to show that $R(L_n)$ is dense in $C(S_\infty^n)$. We will use Stone-Weierstrass Theorem 3.1.28. For every $i = 1, \dots, n$, consider the coordinate function $\delta_i(t_1, \dots, t_n) = t_i$; then $\delta_i \in L_n$. It follows that $R(L_n)$ separates the points of S_∞^n. We also have $\mathbb{1} = Ru \in R(L_n)$. Therefore, L_n is dense in H_n. Moreover, since L_n contains a strong unit of H_n, it is majorizing in H_n.

We now show that lattice-linear calculus from the preceding section extends to H_n. This extension is referred to as a ***positively homogeneous (continuous) function calculus***.

4.4.3 Theorem. *Let X be a uniformly complete Archimedean vector lattice and $x_1,\dots,x_n \in X$. There exists a unique lattice homomorphism $V\colon H_n \to X$ such that $Vf = f(x_1,\dots,x_n)$ for every lattice-linear function f.*

Naturally, for $f \in H_n$, we write $Vf = f(x_1,\dots,x_n)$.

Proof. Let $T\colon L_n \to X$ be the lattice-linear function calculus from Exercise 4.3.6 given by $Tf = f(x_1,\dots,x_n)$. Put $e = \lvert x_1\rvert \vee \dots \vee \lvert x_n\rvert$. By Theorem 4.1.9, $\bigl(I_e, \lVert\cdot\rVert_e\bigr)$ is lattice isometric to a $C(K)$ space; we identify I_e with $C(K)$, with e becoming $\mathbb{1}$. In particular, $x_1,\dots,x_n$ may be viewed as elements of $C(K)$ satisfying $\lvert x_i\rvert \leqslant \mathbb{1}$. For $f \in L_n$, since Tf is in I_e, we may view it as an element of $C(K)$, and

$$(Tf)(t) = f(x_1,\dots,x_n)(t) = f\bigl(x_1(t),\dots,x_n(t)\bigr)$$

for all $t \in K$. It follows that

$$\begin{aligned}\lVert Tf\rVert = \sup_{t\in K}\bigl\lvert (Tf)(t)\bigr\rvert = \sup_{t\in K}\Bigl\lvert f\bigl(x_1(t),\dots,x_n(t)\bigr)\Bigr\rvert \\ \leqslant \sup_{u_1,\dots,u_n\in[-1,1]}\bigl\lvert f(u_1,\dots,u_n)\bigr\rvert = \bigl\lVert f_{|B^n_\infty}\bigr\rVert_{C(B^n_\infty)} = \lVert f\rVert.\end{aligned}$$

Thus, $T\colon L_n \to I_e$ is norm bounded. Therefore, it has a unique extension to a bounded operator V from H_n to I_e. Since T is a lattice homomorphism, so is V, because lattice operations are norm continuous.

To show uniqueness, let $W\colon H_n \to X$ be a lattice homomorphism which agrees with T on L_n. As before, put $u(t_1,\dots,t_n) = \lvert t_1\rvert \vee \dots \vee \lvert t_n\rvert$, then $u \in L_n$ and $Wu = Tu = e$. As we observed earlier, u is a strong unit in H_n; it follows that $\operatorname{Range} W$ is contained in I_e, so we may view W as an operator from H_n to I_e. For $f \in H_n$, it follows from $\lVert f\rVert = \lVert f\rVert_u$ that $\lvert f\rvert \leqslant \lVert f\rVert u$, so that $\lvert Wf\rvert = W\lvert f\rvert \leqslant \lVert f\rVert Wu$ and, therefore, $\lVert Wf\rVert \leqslant \lVert f\rVert \lVert Wu\rVert$. It follows that W is bounded. Since W agrees with V on L_n and L_n is dense in H_n, we have $W = V$. □

4.4.4 Remark. It follows from the proof that $\operatorname{Range} V$ is contained in the ideal generated by $x_1,\dots,x_n$. This ideal coincides with I_e where $e = \lvert x_1\rvert \vee \dots \vee \lvert x_n\rvert$. Viewed as a map from H_n to I_e, V is a contraction.

4.4.5 Example. For every positive real p, the expression $\Bigl(\sum_{i=1}^n \lvert x_i\rvert^p\Bigr)^{\frac{1}{p}}$ is well defined for any vectors $x_1,\dots,x_n$ in every uniformly complete Archimedean vector lattice; in particular, in every Banach lattice.

4.4.6 Exercise. *Positively Homogeneous Function Calculus is preserved under lattice homomorphisms.* Let $T\colon X \to Y$ be a lattice homomorphism between two uniformly complete Archimedean vector lattices. Then $Tf(x_1,\dots,x_n) = f\bigl(Tx_1,\dots,Tx_n\bigr)$ for any $x_1,\dots,x_n \in X$ and any positively homogeneous function f.

The next corollary shows that positively homogeneous function calculus is preserved under a quotient map.

4.4.7 Corollary. *Let X be a uniformly complete Archimedean vector lattice, $J \subseteq X$ a uniformly closed ideal, and $q\colon X \to X/J$ the quotient map. Then*

$$q\big(f(x_1,\dots,x_n)\big) = f\big(q(x_1),\dots,q(x_n)\big)$$

for every $x_1,\dots,x_n \in X$ and every $f \in H_n$.

Proof. Note that X/J is Archimedean by Proposition 3.4.7 and uniformly complete by Proposition 2.2.16. Since q is a lattice homomorphism, the result follows from Exercise 4.4.6. □

4.4.8. *Stability of Positively Homogeneous Function Calculus.* First, we show that it does not depend on the ambient space. Let Y be a sublattice of X such that Y is uniformly complete; let $x_1,\dots,x_n \in Y$. Formally speaking, we then have two function calculi: one from H_n to X and the other one from H_n to Y. However, they agree by the uniqueness clause in Theorem 4.4.3 because the inclusion map of Y into X is a lattice homomorphism.

Second, we show that Positively Homogeneous Function Calculus is stable under adding "fake" variables. Consider the following example. Fix x_1, x_2, x_3 in X. Consider the expression $\big(\lvert x_1\rvert^2 + \lvert x_2\rvert^2\big)^{\frac{1}{2}}$. We may view this expression as $f(x_1,x_2)$ for $f \in H_2$ given by $f(t_1,t_2) = \big(\lvert t_1\rvert^2 + \lvert t_2\rvert^2\big)^{\frac{1}{2}}$ with respect to the function calculus defined by x_1 and x_2. Alternatively, we could view the same expression as $g(x_1,x_2,x_3)$ for $g \in H_3$ defined by $g(t_1,t_2,t_3) = \big(\lvert t_1\rvert^2 + \lvert t_2\rvert^2\big)^{\frac{1}{2}}$ with respect to the function calculus defined by x_1, x_2, and x_3. Will we get the same result?

The answer is "yes". Formally, we have two function calculi here: $V_{x_1,x_2}\colon H_2 \to X$ and $V_{x_1,x_2,x_3}\colon H_3 \to X$. It is easy to see that H_2 may be identified with a sublattice of H_3. The restriction of V_{x_1,x_2,x_3} to H_2 is a lattice homomorphism, and it clearly agrees the lattice-linear function calculus on L_2. Hence, by uniqueness, this restriction agrees with V_{x_1,x_2}. In the example, this translates into

$$f(x_1,x_2) = V_{x_1,x_2} f = (V_{x_1,x_2,x_3|H_2}) f = V_{x_1,x_2,x_3} g = g(x_1,x_2,x_3).$$

4.4.9. *Positively Homogeneous Function Calculus depends continuously on the function.* Using notation of Theorem 4.4.3 and Remark 4.4.4, we observe that $V\colon H_n \to I_e$ is a positive operator between Banach lattices, hence is continuous. It follows that (f_m) is a sequence in H_n which converges to f in norm (i.e., uniformly on S^n_∞ or, equivalently, on B^n_∞), then $f_m(x_1,\dots,x_n)$ converges to $f(x_1,\dots,x_n)$ in $\lVert\cdot\rVert_e$ as $k \to \infty$. In particular, if X is a Banach lattice then $f_m(x_1,\dots,x_n)$ converges to $f(x_1,\dots,x_n)$ in the norm of X.

4.4.10. *Positively Homogeneous Function Calculus depends continuously on the vectors.* Fix $x_1,\dots,x_n \in X$ and $f \in H_n$. Suppose that each x_k $(1 \leqslant k \leqslant n)$ is the uniform limit of some sequence $(x_k^{(i)})$ in X. We claim that

$$f\big(x_1^{(i)},\dots,x_n^{(i)}\big) \xrightarrow{\mathrm{u}} f(x_1,\dots,x_n).$$

By taking the sup of all the regulators, we can find a single $e \in X_+$ such that $\|x_k^{(i)} - x_k\|_e \to 0$ as $i \to \infty$ for every $k = 1, \dots, n$. In particular, each of the sequences $(x_k^{(i)})$ is bounded in $\|\cdot\|_e$. Again, scaling e and passing to subsequences, if necessary, we may assume that $|x_k| \leqslant e$ and $|x_k^{(i)}| \leqslant e$ for all k, i. Represent I_e as $C(K)$ for some compact K such that e corresponds to $\mathbb{1}$, then x_k and $x_k^{(i)}$ are in $[-\mathbb{1}, \mathbb{1}]$ for all k, i.

We may view f as a continuous function on B_∞^n. Since the ball is compact, f is uniformly continuous. Fix $\varepsilon > 0$. There exists $\delta > 0$ such that

$$\bigl|f(s_1, \dots, s_n) - f(t_1, \dots, t_n)\bigr| < \varepsilon \qquad \text{whenever } |s_k| \leqslant 1,\ |t_k| \leqslant 1, \text{and } |s_k - t_k| < \delta \text{ as } k = 1, \dots, n.$$

Choose i_0 sufficiently large so that $\|x_k^{(i)} - x_k\|_e < \delta$ for every k and every $i \geqslant i_0$. It follows that $\bigl|x_k^{(i)}(t) - x_k(t)\bigr| < \delta$ for all $t \in K$ and $i \geqslant i_0$. Therefore,

$$\Bigl|f\bigl(x_1^{(i)}(t), \dots, x_n^{(i)}(t)\bigr) - f\bigl(x_1(t), \dots, x_n(t)\bigr)\Bigr| < \varepsilon$$

for all $t \in K$ and $i \geqslant i_0$. This means that

$$\bigl|f(x_1^{(i)}, \dots, x_n^{(i)}) - f(x_1, \dots, x_n)\bigr| < \varepsilon e.$$

This proves the claim.

In the special case when X is a Banach lattice, we may replace uniform convergence with norm convergence in the preceding result because uniform convergence implies norm convergence and, on the other hand, every norm convergent sequence has a uniformly convergent subsequence by Theorem 2.3.7.

In this section, we defined Positively Homogeneous Function Calculus in uniformly complete vector lattices. While uniform completeness is a very "mild" assumption, one can naturally ask whether it is necessary. It is easy to see that one cannot extend Positively Homogeneous Function Calculus to arbitrary vector lattices: for example, let X be the vector lattice of all piece-wise affine functions on $[0, 1]$ and let $f \in X$ be defined by $f(t) = t$. It is clear that $\bigl(\mathbb{1}^2 + f^2\bigr)^{\frac{1}{2}}$ does not exist in X. Nevertheless, there are vector lattices which admit Positively Homogeneous Function Calculus, yet fail to be uniformly complete; see [LT20] for details.

4.5 Disjointification Lemma

In this section, we deduce from $C(K)$-Representation Theorem a very useful technique of converting a sequence into a disjoint sequence. We start with a motivation. Given a sequence (A_n) of subsets of some set Ω, one can "disjointify" it by replacing A_n with $B_n = A_n \setminus \bigcup_{k<n} A_k$. Note that some of the terms in the resulting sequence may be empty. In terms of characteristic functions, this is equivalent to

$$\mathbb{1}_{B_n} = \Bigl(\mathbb{1}_{A_n} - \bigvee_{k<n} \mathbb{1}_{A_k}\Bigr)^+ = \Bigl(\mathbb{1}_{A_n} - \sum_{k<n} \mathbb{1}_{A_k}\Bigr)^+.$$

Suppose now that $X = C(K)$, and let (f_n) be a sequence in X_+. By analogy, we could consider

$$u_n = \Big(f_n - \sum_{k<n} f_k\Big)^+.$$

However, it is easy to see that this sequence need not be disjoint. We could somewhat improve the construction by putting

$$u_n = \Big(f_n - \lambda_n \sum_{k<n} f_k\Big)^+.$$

If λ_n is "sufficiently large" then the resulting sequence is "almost disjoint" in the sense that the "error terms" $u_n \wedge u_k$ are small (where $k \neq n$). Furthermore, we may "kill" these error terms by subtracting a constant function:

$$u_n = \Big(f_n - \lambda_n \sum_{k<n} f_k - \mu_n \mathbb{1}\Big)^+.$$

Choosing the μ_n's is somewhat of a balancing act: they have to be sufficiently large to kill the error terms, but if they are too large then all u_n's may equal zero.

4.5.1 Lemma (Disjointification Lemma)**.** *Let X be an Archimedean vector lattice, (x_n) a sequence in X_+, and $e \in X_+$ such that $\frac{x_n}{2^n} \leqslant e$ for every n. Put*

$$u_n = \Big(x_n - 4^n \sum_{k=1}^{n-1} x_k - 2^{-n} e\Big)^+.$$

Then (u_n) is disjoint.

Proof. Note that both (x_n) and (u_n) are contained in I_e. Using $C(K)$-representation Theorem 4.1.9, we may represent I_e as a sublattice of $C(K)$ for some compact K, with e corresponding to $\mathbb{1}$. We need to show that $u_n \perp u_k$ whenever $k < n$. Suppose $u_n(t) > 0$ for some $t \in K$. It follows that

$$x_n(t) - 4^n \sum_{i=1}^{n-1} x_i(t) - 2^{-n} > 0.$$

This yields $x_k(t) < 4^{-n} x_n(t) \leqslant 4^{-n} \cdot 2^n = 2^{-n} < 2^{-k}$, and, therefore, $x_k(t) - 2^{-k} < 0$. It follows that $u_k(t) = 0$. □

4.5.2 Remark. (i) If (x_n) is a bounded sequence in a Banach lattice, one can take $e = \sum_{n=1}^{\infty} \frac{x_n}{2^n}$.

(ii) In general, the lemma does not guarantee that the resulting sequence is non-zero. This lemma is usually used in conjunction with an argument that (u_n) has a non-zero subsequence.

(iii) A direct proof of the lemma which does not use a $C(K)$-representation may be found in Lemma 4.35 of [AB06].

Chapter 5

Bands

In this chapter, we continue our study of various types of subspaces of vector lattices. Recall that a ***band*** is an order closed ideal. There is a critical distinction between ideals and bands. While ideals are defined by axioms involving finite sets, the definition of a band involves convergence of infinite nets. As a consequence, while ideals are closer to Algebra, bands truly belong to Analysis.

5.1 Bands and disjoint complements

Since every ideal is a solid set, Exercise 3.3.22 tells us that it suffices to only check increasing positive nets in the definition of a band. That is, an ideal J is a band iff for every net (x_α) in J, if $0 \leqslant x_\alpha \uparrow x$ then $x \in J$.

It follows from the definition of a band that the intersection of any collection of bands is again a band. In particular, the intersection of all bands containing a given set A clearly the least band containing A; we denote this band by $B(A)$ and call it the ***band generated by*** A. While Exercise 3.3.6 provides an easy way to verify whether a given vector is in $I(A)$, it is more difficult with $B(A)$. Since $B(A) = B\big(I(A)\big)$, we may assume that A is an ideal.

5.1.1 Proposition. *Let J be an ideal in X and $B(J)$ the band generated by J. The following are equivalent.*

(i) $x \in B(J)_+$;
(ii) $0 \leqslant x_\alpha \uparrow x$ *for some net* (x_α) *in* J;
(iii) $x = \sup D$ *for some non-empty* $D \subseteq J_+$;
(iv) $x = \sup J \cap [0, x]$.

Proof. (iv)⇒(iii) because we can take $D = J \cap [0, x]$. (iii)⇒(ii) because $\sup D = \sup D^\vee$ and $D^\vee$ can be viewed as an increasing net, see 1.1.6. It is straightforward that (ii)⇒(iv). Hence (ii)⇔(iii)⇔(iv). (ii)⇒(i) because (x_α) is a net in $B(J)$ and $B(J)$ is order closed. It remains to show that (i)⇒(iv). Let

$$B = \Big\{x \in X \;:\; |x| = \sup J \cap \big[0, |x|\big]\Big\}.$$

That is, $x \in B$ whenever $|x|$ satisfies (iv) and, therefore, (ii) and (iii). We need to show that $B(J)_+ \subseteq B$. Since $J \subseteq B$; it suffices to show that B is a band.

It is straightforward that B is closed under scalar multiplication. Let $x, y \in B$. By (ii), there exist nets (x_α) and (y_α) in J_+ such that $x_\alpha \uparrow |x|$ and $y_\alpha \uparrow |y|$. Then

$$(x_\alpha + y_\alpha) \wedge |x+y| \;\uparrow\; \big(|x| + |y|\big) \wedge |x+y| \;=\; |x+y|.$$

Since $\big((x_\alpha + y_\alpha) \wedge |x+y|\big)$ is a net in J_+, it follows that $x + y \in B$. Therefore, B is a subspace.

To verify that B is an ideal, assume that $x \in B$ and $|y| \leqslant |x|$. Find a net (x_α) in J_+ with $x_\alpha \uparrow |x|$. Then $\big(x_\alpha \wedge |y|\big)$ is a net in J_+ such that $x_\alpha \wedge |y| \uparrow |y|$; it follows that $y \in B$.

To check that B is a band, suppose that $x_\alpha \uparrow x$ for a net (x_α) in B_+. Then

$$x = \sup_\alpha x_\alpha = \sup_\alpha \sup J \cap [0, x_\alpha] = \sup J \cap \bigcup_\alpha [0, x_\alpha].$$

Now (iii) yields that $x \in B$. □

5.1.2 Exercise. Every ideal J is order dense in $B(J)$. An ideal J in an Archimedean vector lattice X is order dense in X iff $B(J) = X$.

A ***principal band*** is a band generated by a single vector; we write B_a instead of $B(\{a\})$. Clearly, $I_a \subseteq B_a$, where I_a is the principal ideal of a, and B_a is the band generated by I_a. It is interesting to see the difference between I_a and B_a. Given $x \in X_+$, it is easy to see that $x \in I_a$ iff $x \leqslant na$ or, equivalently, $x \wedge na = x$, for all sufficiently large $n \in \mathbb{N}$. We will now show that $x \in B_a$ whenever $x \wedge na \uparrow x$.

5.1.3 Proposition. *Let $a, x \in X_+$. Then*

$$x \in B_a \quad\Leftrightarrow\quad x = \sup\{x \wedge na \;:\; n \in \mathbb{N}\}.$$

Proof. If $x \wedge na \uparrow x$ then $x \in B_a$ because $x \wedge na \in B_a$ for every n and B_a is order closed.

Conversely, suppose that $x \in B_a$. Let $D = \{x \wedge na \;:\; n \in \mathbb{N}\}$. Clearly, $D \leqslant x$. Suppose that $D \leqslant z$, it suffices to show that $x \leqslant z$. By Proposition 5.1.1(iv) we have $x = \sup I_a \cap [0, x]$. If $y \in I_a \cap [0, x]$ then $y \leqslant na$ for some n, and $y \leqslant x$, so that $y \leqslant x \wedge na \leqslant z$. Hence $I_a \cap [0, x] \leqslant z$, so that $x \leqslant z$. □

5.1.1 Disjoint complements

For $A \subseteq X$, we define

$$A^d \coloneqq \bigl\{x \in X \,:\, x \perp a \text{ for all } a \in A\bigr\}.$$

The set A^d is termed as the ***disjoint complement*** of A.

5.1.4 Exercise. If $f \in L_p(\mu)$ then $\{f\}^d$ consists of all the functions in $L_p(\mu)$ supported on the complement of $\operatorname{supp} f$. Furthermore, for $A \subseteq L_p(\mu)$ we have

$$A^d = \bigl\{g \in L_p(\mu) \,:\, fg = 0 \text{ a.e. for all } f \in A\bigr\}.$$

5.1.5 Exercise. For every $A, B \subseteq X$ and $a \in X$ we have

(i) if $A \subseteq B$ then $B^d \subseteq A^d$;

(ii) $(A \cup B)^d = A^d \cap B^d$

(iii) $A \cap A^d = \{0\}$

(iv) $\{a\}^d = \{x \in X \,:\, x \perp a\}$

(v) $A^d = \bigcap_{a \in A} \{a\}^d$

(vi) $A \subseteq A^{dd}$

(vii) A vector $e \in X_+$ is a weak order unit iff $\{e\}^d = \{0\}$.

5.1.6 Proposition. *A^d is a band for any $A \subseteq X$.*

Proof. It suffices to show that $\{a\}^d$ is a band for every $a \in X$, the result will then follow from Exercise 5.1.5(v). Suppose that $a \in X$. Since $\{a\}^d = \{|a|\}^d$, we may assume without loss of generality that $a \geqslant 0$. It follows from Exercise 1.5.3 that $\{a\}^d$ is an ideal. Finally, to show that $\{a\}^d$ is order closed, assume that (x_α) is a net in $\{a\}^d$ such that $x_\alpha \xrightarrow{o} x$ for some x. Since lattice operations are order continuous, we have $0 = |x_\alpha| \wedge a \xrightarrow{o} |x| \wedge a$, hence $x \in \{a\}^d$. ☐

Throughout the rest of this subsection, let X be an *Archimedean* vector lattice. The next theorem provides a convenient characterization of bands.

5.1.7 Theorem. *A subset B of X is a band iff $B = B^{dd}$.*

Proof. If $B = B^{dd}$ then B is a band by Proposition 5.1.6. Suppose now that B is a band. Clearly, $B \subseteq B^{dd}$ and B^{dd} is a band. To finish the proof, it suffices to show that B is order dense in B^{dd}; it would then follow from Exercise 5.1.2 that B^{dd} equals the band generated by B, which is B itself. Let $0 < x \in B^{dd}$. Then $x \notin B^d$, so that $x \wedge v > 0$ for some $v \in B$. Since B is an ideal, we have $x \wedge v \in B$. ☐

5.1.8 Corollary. *A subset B of X is a band iff it is of form A^d for some set A.*

5.1.9 Corollary. *For any set A in X, A^{dd} is the band generated by A.*

Proof. Clearly, $A \subseteq A^{dd}$. By Proposition 5.1.6, A^{dd} is a band. Let B be a band such that $A \subseteq B$. Then $A^d \supseteq B^d$ and $A^{dd} \subseteq B^{dd} = B$. □

5.1.10 Exercise. An ideal J in X is order dense iff $J^d = \{0\}$.

5.1.11 Exercise. Let J be an ideal in X. Then $J + J^d$ is an order dense ideal in X (but need not equal X).

5.1.12 Exercise. Is $C_0(\mathbb{R})$ a sublattice of $C(\mathbb{R})$? An ideal? A band?

Let $a \in X$. Corollary 5.1.9 yields that $\{a\}^{dd}$ equals B_a, the principal band generated by a.

5.1.13 Exercise. Suppose that $e \in X_+$. Then TFAE

(i) e is a weak unit;
(ii) $B_e = X$;
(iii) $x \wedge ne \uparrow x$ for every positive x.

5.1.14 Exercise. Let $e \in X_+$. Then e is a weak unit in B_e.

5.1.15 Exercise. For a set A in a normed lattice, $A^d = (\overline{A})^d$.

5.1.16 Exercise. In a normed lattice, every band is norm closed.

5.1.17 Exercise. In an order continuous Banach lattice, bands and norm closed ideals agree.

5.1.18 Corollary. *Every separable Banach lattice has a weak unit.*

Proof. Let (x_n) be a dense sequence in X. Find $e \in X_+$ such that (x_n) is in I_e as in Exercise 2.2.1(viii). It follows that (x_n) is contained in $\overline{I_e}$. Since $I_e \subseteq B_e$ and B_e is closed, we have $B_e = \overline{I_e} = X$. It follows that e is a weak unit by Exercise 5.1.13. □

5.1.19 Example. *A non-separable order continuous Banach lattice with a weak unit.* Consider the measure λ on $\{0,1\}$ with $\lambda(\{0\}) = \lambda(\{1\}) = \frac{1}{2}$. Fix an uncountable set I. Let $(\Omega, \mathcal{F}, \mu)$ be the product of I copies of λ. That is, $\Omega = \{0,1\}^I$, $\mathcal{F}$ is the σ-algebra generated by the sets of the form $A_i = \{\omega \in \Omega : \omega_i = 1\}$ for $i \in I$, and μ is obtained by prescribing $\mu(A_i) = \mu(A_i^C) = \frac{1}{2}$ for every $i \in I$, then extending it "independently" to the sets generated by the A_i's via

$$\mu\bigl(A_{i_1} \cap \dots \cap A_{i_k} \cap A_{i_{k+1}}^C \cap \dots A_{i_n}^C\bigr) = \frac{1}{2^n}$$

for any $k \leqslant n$ and any distinct $i_1, \dots, i_n$ in I, and then, finitely, extending μ to all of $\mathcal{F}$.

Let $1 \leqslant p < \infty$. Since μ is a probability measure, $\mathbb{1}$ is a weak unit in $L_p(\mu)$. Yet, we claim that $L_p(\mu)$ is non-separable. Observe that $\mu(A_i \triangle A_j) = \frac{1}{2}$ whenever $i \neq j$. It follows that $\|\mathbb{1}_{A_i} - \mathbb{1}_{A_j}\|_{L_p} = (\frac{1}{2})^{\frac{1}{p}}$. Hence, $\{\mathbb{1}_{A_i} : i \in I\}$ is an uncountable family in $L_p(\mu)$ which cannot be approximated by a countable set.

5.1.2 Bands in $C(K)$ and L_p spaces

Let K be a compact space. According to Theorem 3.3.11, closed order ideals in $C(K)$ are the sets of the form J_F for some closed subset F of K. This yields a one-to-one correspondence between the closed subsets of K and the closed ideals in $C(K)$. By Exercise 5.1.16, every band in $C(K)$ is a closed ideal. It leads to a natural question: which closed subsets of K correspond to bands? That is, we want to characterize those closed subsets F in K for which J_F is not just a closed ideal but a band.

5.1.20 Example. Not every J_F is a band. Consider $C[0,1]$ and let $F = \{0\}$. Then J_F is not order closed, hence is not a band. There is a sequence (g_n) in J_F such that $g_n \uparrow \mathbb{1}$; see, e.g., Example 1.4.2. Yet $\mathbb{1} \notin J_F$. Note that $J_F^d = \{0\}$ and $J_F^{dd} = C[0,1]$.

5.1.21 Example. Again, let $K = [0,1]$. Then $J^d_{[0,\frac{1}{2}]} = J_{[\frac{1}{2},1]}$ and $J^{dd}_{[0,\frac{1}{2}]} = J_{[0,\frac{1}{2}]}$, so that $J_{[0,\frac{1}{2}]}$ is a band by Theorem 5.1.7.

5.1.22 Lemma. *If F is a closed subset of K then $J_F^d = J_{\overline{K \setminus F}}$.*

Proof. If $f \in J_{\overline{K \setminus F}}$ then f vanishes on $K \setminus F$, hence $f \in J_F^d$. On the other hand, suppose that $f \in J_F^d$, and let $t \in K \setminus F$. By Urysohn's Lemma, there exists $g \in C(K)$ such that $g(t) = 1$ and g vanishes on F. Then $g \in J_F$, so that $f \perp g$, hence $f(t) = 0$. Since $t \in K \setminus F$ was chosen arbitrarily, it follows that f vanishes on $K \setminus F$ and, therefore, on $\overline{K \setminus F}$. Thus, $f \in J_{\overline{K \setminus F}}$. □

5.1.23 Theorem. *A subset B in $C(K)$ is a band iff $B = J_{\overline{U}}$ for some open set U in K.*

Proof. Suppose that U is an open subset of K. Let $F = K \setminus U$. Lemma 5.1.22 yields $J_{\overline{U}} = J_F^d$, which is a band by Proposition 5.1.6.

Conversely, suppose that B is a band in $C(K)$. Then B is an ideal, and it is closed by Exercise 5.1.16. Theorem 3.3.11 yields that $B = J_F$ for some closed F. By Lemma 5.1.22 we have $J_F^d = J_{\overline{K \setminus F}}$. It follows from Theorem 5.1.7 and Lemma 5.1.22 that

$$B = B^{dd} = J_F^{dd} = J^d_{\overline{K \setminus F}} = J_{\overline{U}}.$$

where $U = K \setminus (\overline{K \setminus F}) = \operatorname{Int} F$. □

Let now $X = C(\Omega)$ for some Haursdorff topological space Ω; we do not assume that Ω is compact. It is easy to see that sets of the form J_F where F is a closed subset of Ω are still ideals in X. However, the converse is no longer true. For example, the set of all functions $f \in C(\mathbb{R})$ such that $\lim_{t\to\infty} f(t) = 0$ is an ideal, but it is not of this form. We refer the reader to [KV19] for characterizations of ideals and bands in $C(\Omega)$, $C_b(\Omega)$, and $C_0(\Omega)$ spaces.

5.1.24 Example. *The sum of two bands need not be a band.* Let $X = C[0,1]$, $B_1 = J_{[0,\frac{1}{2}]}$, and $B_2 = J_{[\frac{1}{2},1]}$. Then B_1 and B_2 are bands, but $B_1 + B_2 = J_{\{\frac{1}{2}\}}$ is not band.

We now move to L_p-spaces. Recall that $L_p(\mu)$ is an order continuous Banach lattice when $1 \leqslant p < \infty$, so Exercise 5.1.17 yields the following:

5.1.25 Proposition. *If $1 \leqslant p < \infty$ then bands and closed ideal in $L_p(\mu)$ agree.*

Let $X = L_p(\Omega, \mathcal{F}, \mu)$. For every $A \in \mathcal{F}$, we write $L_p(A)$ for the subspace of X consisting of all the functions f in X whose support is contained in A (recall that $\operatorname{supp} f$ is defined up to a set of zero measure). It is easy to see that $L_p(A)$ can be identified with $L_p(A, \mathcal{F}_A, \mu_A)$ where $\mathcal{F}_A = \{B \in \mathcal{F} : B \subseteq A\}$, and μ_A is the restriction of μ to $\mathcal{F}_A$.

5.1.26 Theorem. *Let $X = L_p(\Omega, \mathcal{F}, \mu)$ where μ is a σ-finite measure and $1 \leqslant p \leqslant \infty$. Then the bands in X are exactly the sets of the form $L_p(A)$ for some $A \in \mathcal{F}$.*

Proof. Let $A \in \mathcal{F}$. It is easy to see that $L_p(A) = L_p(\Omega \setminus A)^d$. Then Proposition 5.1.6 yields that $L_p(A)$ is a band. To complete the proof, it suffices to show that every band B equals $L_p(A)$ for some $A \in \mathcal{F}$.

The difficulty here is in constructing this A. A "naive" approach would be to take A to be the union of the supports of all the functions in B. However, the supports of functions in $L_p(\mu)$ are defined a.e. Since B need not be countable, the "a.e. errors" may accumulate and the union of the supports of all the functions in B may be way off the target.

We first assume that μ is a finite measure. Let D be the set of all characteristic functions in B. Then $D \leqslant \mathbb{1}$ and D is closed under finite sups, i.e., $D = D^\vee$. Since $L_p(\mu)$ is order complete by Theorem 2.1.16, $h \coloneqq \sup D$ exists. It follows from Exercise 2.1.18 that h is the supremum of an increasing sequence in D, i.e., $\mathbb{1}_{A_n} \uparrow h$ for some sequence (A_n) in $\mathcal{F}$. It follows that $h = \mathbb{1}_A$ for some $A \in \mathcal{F}$ by Proposition 2.1.14. Since B is order closed, we have $\mathbb{1}_A \in B$.

We claim that $B = L_p(A)$. For every measurable subset C of A, it follows from $0 \leqslant \mathbb{1}_C \leqslant \mathbb{1}_A$ that $\mathbb{1}_C \in B$. Hence, all the simple functions in $L_p(A)$ are in B. Since simple functions are dense in $L_p(A)$, it follows that $L_p(A) \subseteq B$.

On the other hand, suppose that $f \in B$; we will show that $f \in L_p(A)$. WLOG, $f \geqslant 0$; otherwise consider $|f|$. For every $n \in \mathbb{N}$, it follows from $\frac{1}{n}\mathbb{1}_{\{f>\frac{1}{n}\}} \leqslant f$ that $\mathbb{1}_{\{f>\frac{1}{n}\}} \in B$. Since the union of these sets for all $n \in \mathbb{N}$ is $\operatorname{supp} f$, we have $\mathbb{1}_{\{f>\frac{1}{n}\}} \uparrow \mathbb{1}_{\operatorname{supp} f}$ and, $\mathbb{1}_{\operatorname{supp} f} \in B$. It follows that $\mathbb{1}_{\operatorname{supp} f} \in D$ and, therefore, $\mathbb{1}_{\operatorname{supp} f} \leqslant \mathbb{1}_A$, so that $\operatorname{supp} f \subseteq A$ (up to a set of measure zero) and, therefore, $f \in L_p(A)$.

The case of a general σ-finite measure can be reduced to the case of a finite measure as follows. As in Proposition 1.6.19, there is a lattice isometry from $L_p(\mu)$ onto $L_p(\nu)$ where ν is a finite measure, and this map preserves sets of the form $L_p(A)$. Being a lattice isomorphism, it preserves bands. So we can now apply the first part of the proof to $L_p(\mu)$. $\square$

5.1.27 Exercise. Things work somewhat differently when $p = \infty$. For the following sets J in $X = L_\infty(\mu)$, verify that J is a closed ideal in X, yet it is not a band, and it is not of the form $L_p(A)$.

(i) $X = \ell_\infty$ and $J = c_0$;

(ii) $X = L_\infty[0,1]$ and J is the closure of

$$J_0 = \big\{ f \in X \ :\ \exists \varepsilon > 0 \quad f \text{ vanishes on } [0,\varepsilon] \big\}.$$

We observed in Exercise 4.1.12 that $L_\infty(\mu)$ is lattice isometric to some $C(K)$ space. Hence, we can use the characterizations of closed ideals and bands in $C(K)$ spaces in Theorems 3.3.11 and 5.1.23 to characterize closed ideals and bands in $L_\infty(\mu)$ spaces.

5.1.28 Exercise. Let $X = L_p(\mu)$ where μ is an arbitrary measure and $1 \leqslant p < \infty$. For every $f \in X$, the principal band B_f is of the form $L_p(A)$ where $A = \operatorname{supp} f$.

5.1.29 Remark. When dealing with sequences in $L_p(\mu)$ spaces with $1 \leqslant p < \infty$, one may often assume WLOG that the space has a weak unit and, furthermore, that the measure is finite. Indeed, let (f_n) be a sequence in $L_p(\mu)$ for some measure space $(\Omega, \mathcal{F}, \mu)$. By Exercise 2.2.1(viii), (f_n) is contained in the principal band B_u for some $u \geqslant 0$. By Exercise 5.1.28, we may identify B_u as $L_p(A)$, where $A = \operatorname{supp} u$. The restriction of μ to A is σ-finite by Exercise 1.4.4. Hence $L_p(A)$ has a weak unit by Exercise 1.5.13(x). Furthermore, it follows from Proposition 1.6.19 that $L_p(A)$ is lattice isometric to $L_p(\nu)$ for some finite measure ν. So we may view (f_n) as a sequence in $L_p(\nu)$.

5.1.30 Exercise. Let $1 \leqslant p < \infty$ and $(\Omega, \mathcal{F}, \mu)$ a σ-finite measure space. Characterize all closed sublattices of $L_p(\mu)$ (cf Theorem 3.1.25).

In [VV17], the authors prove the following characterization of bands in $L_p(\mu)$-spaces where $p < \infty$ and μ is not σ-finite. Consider all possible decompositions $\mu = \mu_1 + \mu_2$ into two measures such that μ_1 and μ_2 are ***locally disjoint*** in the sense that for every $A \in \mathcal{F}$ with $\mu(A)$ finite one can write $A = A_1 \cup A_2$ for two disjoint measurable sets A_1 and A_2 with $\mu_1(A_2) = \mu_2(A_1) = 0$. Such a decomposition of μ induces a decomposition of $L_p(\mu)$ into two disjoint bands $L_p(\mu) = L_p(\mu_1) \oplus L_p(\mu_2)$. Conversely, every band in $L_p(\mu)$ is of this form.

5.2 Projection bands

Suppose again that $X = L_p(\Omega, \mathcal{F}, \mu)$ where μ is a σ-finite measure and $1 \leqslant p \leqslant \infty$, and let B be a band in X. By the results of the preceding section, we know that $B = L_p(A)$ for some $A \in \mathcal{F}$ and $B^d = L_p(\Omega \setminus A)$. Note that for every $f \in X$ we have $f = g + h$ where $g = f \cdot \mathbb{1}_A \in L_p(A) = B$ and $h = f \cdot \mathbb{1}_{\Omega\setminus A} \in L_p(\Omega \setminus A) = B^d$. Hence, $X = B \oplus B^d$; the sum is direct because clearly $B \cap B^d = \{0\}$. Also, the map $P\colon X \to X$ defined by $Pf = f \cdot \mathbb{1}_A$ is a positive projection onto B, while $I - P$ is a positive projection onto B^d. These facts motivate the following definition.

Throughout this section, let X will be an Archimedean vector lattice. A band B in X is said to be a ***projection band*** if $X = B \oplus B^d$. That is, for every $x \in X$ there exist unique $y \in B$ and $z \in B^d$ such that $x = y + z$. Define $Px = y$, then $P\colon X \to X$ is a linear projection. It is called the ***band projection*** or the ***order projection*** associated with B.

5.2.1. Let B be a band. Since $B = B^{dd}$, B is a projective band iff B^d is a projective band. In particular, let $a \in X$; it follows from $B_a = \{a\}^{dd}$ that B_a is a projection band iff $\{a\}^d$ is a projection band.

5.2.2 Example. The discussion at the beginning of this section shows that every band in $L_p(\mu)$ where $1 \leqslant p < \infty$ and μ is a σ-finite measure is a projection band.

5.2.3 Example. Let $X = C[0,1]$ and put $B = J_{[0,\frac{1}{2}]}$. Then B is a band and $B^d = J_{[\frac{1}{2},1]}$ by Theorem 5.1.23 and Lemma 5.1.22. However, $B + B^d = J_{\{\frac{1}{2}\}} \neq X$, hence B is not a projection band.

5.2.4 Proposition. *Let K be a compact space. Projection bands in $C(K)$ are of the form J_A, where A is a clopen set. The corresponding band projection is given by $f \mapsto f \cdot \mathbb{1}_{A^C}$.*

Proof. Suppose that A is a clopen set. Then $K = A \cup A^C$ is a disconnection of K. It follows that $C(K) = J_A \oplus J_{A^C}$. It is easy to see that $J_{A^C} = J_A^d$ and $J_A = J_{A^C}^d$, hence both J_A and J_{A^C} are projection bands.

Conversely, suppose that B is a projection band in $C(K)$. Then Theorem 5.1.23 yields that $B = J_{\overline{U}}$ for some open set U in K. Furthermore, $B^d = J_{\overline{K\setminus\overline{U}}}$ by Lemma 5.1.22.

Note that the boundary ∂U is contained in both $\overline{U}$ and $\overline{K \setminus \overline{U}}$. It follows from $X = B \oplus B^d$ that every function in X vanishes on ∂U for every $f \in X$. In particular, this applies to $\mathbb{1}$, hence $\partial U = \varnothing$. It follows from $\overline{U} = U \cup \partial U = U$ that U is closed, hence clopen.

The map $f \mapsto f \cdot \mathbb{1}_{A^C}$ is the band projection for J_A because $f = f \cdot \mathbb{1}_A + f \cdot \mathbb{1}_{A^C}$, $f \cdot \mathbb{1}_{A^C} \in J_A$, and $f \cdot \mathbb{1}_A \in J_{A^C} = J_A^d$. $\square$

5.2.5 Corollary. *K is connected iff the only projection bands in $C(K)$ are $\{0\}$ and $C(K)$.*

The preceding corollary may be extended to the case when $X = C(\Omega)$ where Ω is a topological space. Recall that for $f \in C(\Omega)$, we write $\operatorname{supp} f = \{f \neq 0\}$; for $A \subseteq C(\Omega)$, put $\operatorname{supp} A = \bigcup_{f \in A} \operatorname{supp} f$. Clearly, $\operatorname{supp} A$ is an open set.

5.2.6 Exercise. Let X be a sublattice of $C(\Omega)$ such that $\operatorname{supp} X$ is connected. Then X has no projection bands other than $\{0\}$ and X.

5.2.7 Proposition. *Suppose that $X = J_1 \oplus J_2$ where J_1 and J_2 are two ideals. Then J_1 and J_2 are projection bands and $J_2 = J_1^d$.*

Proof. If $x \in J_1$ and $y \in J_2$ then $\lvert x\rvert \wedge \lvert y\rvert \in J_1 \cap J_2 = \{0\}$. It follows that $J_1 \perp J_2$. In particular, $J_2 \subseteq J_1^d$. To show that $J_2 = J_1^d$, take any $x \in J_1^d$. Since $X = J_1 \oplus J_2$, we have $x = x_1 + x_2$ for some $x_1 \in J_1$ and $x_2 \in J_2$. It follows that $x_1 \perp x_2$, so that $\lvert x\rvert = \lvert x_1\rvert + \lvert x_2\rvert$, so that $\lvert x_1\rvert \leqslant \lvert x\rvert$. Since $x \in J_1^d$ and J_1^d is a band by Proposition 5.1.6, hence an ideal, we conclude that $x_1 \in J_1^d$. Since $x_1 \in J_1$, it follows that $x_1 = 0$, so that $x = x_2 \in J_2$.

This proves $J_2 = J_1^d$. By Proposition 5.1.6 that J_2 is a band. Similarly, J_1 is a band. It now follows from $X = J_1 \oplus J_2$ that J_1 and J_2 are projection bands. $\square$

This provides a convenient criterion for an ideal to be a projection band:

5.2.8 Corollary. *An ideal J is a projection band iff $J + J^d = X$.*

5.2.9 Exercise. Let A and B be two subsets of X such that $A \perp B$ and $A + B = X$. Then $I(A)$ and $I(B)$ are complementary projection bands.

5.2.10 Exercise. Every band projection is a lattice homomorphism. In particular, it is positive.

The next result provides an easy and useful intrinsic characterization of band projections.

5.2.11 Theorem. *Let $P\colon X \to X$ be a linear operator. TFAE*

(i) *P is a band projection;*

(ii) *$P^2 = P$ and $0 \leqslant P \leqslant I$;*

(iii) $\operatorname{Range} P \perp \operatorname{Range}(I-P)$.

Proof. Put $Q = I - P$. (i)⇒(ii) is trivial.

(ii)⇒(iii) Let $x, y \in X_+$; put $z = Px \wedge Qy$. It follows from $P, Q \geqslant 0$ that $z \geqslant 0$. It follows from $z \leqslant Qy$ that $Pz = PQy = 0$. Similarly, $Qz = 0$. Hence, $z = Pz + Qz = 0$. Thus, $Px \perp Qy = 0$ for all $x, y \in X_+$. Suppose now that $x \in X$ and $y \in X$. By the previous argument, $\{Px^+, Px^-\} \perp \{Qy^+, Qy^-\}$, so that $Px \perp Qy$.

(iii)⇒(i) Let $J_1 = I\big(\operatorname{Range} P\big)$ and $J_2 = I\big(\operatorname{Range} Q\big)$. It follows from $\operatorname{Range} P \perp \operatorname{Range}(I-P)$ that $J_1 \perp J_2$, so that $J_1 \cap J_2 = \{0\}$. Also, $J_1 + J_2 \supseteq \operatorname{Range} P + \operatorname{Range} Q = X$, so that $X = J_1 \oplus J_2$. By Proposition 5.2.7, J_1 and J_2 are bands and, therefore, projection bands with $J_1^d = J_2$ and $J_2^d = J_1$. For each $x \in X$, we have $x = Px + Qx$, with $Px \in J_1$ and $Qx \in J_2$. It follows that P and Q are the band projections associated with J_1 and J_2, respectively. □

5.2.12 Example. Let $X = \mathbb{R}^2$ and P the orthogonal projection onto the line $y = x$. Since the line is not a band, P is not a band projection. Observe that $P^2 = P$ and $0 \leqslant P$. However, $P \leqslant I$ fails because, e.g., $Pe_1 \nleqslant e_1$.

5.2.13 Exercise. Let B_1 and B_2 be two projection bands in X, with band projections P_1 and P_2, respectively. Then $B_1 \subseteq B_2$ iff $P_1 \leqslant P_2$ iff $P_1 P_2 = P_1$.

While the intersection of two bands is again a band, we saw in Exercise 5.2.2 that the sum of two bands need not be a band. For projection bands, the situation is much better.

5.2.14 Theorem. *Let B_1 and B_2 be two projection bands in X, with band projections P_1 and P_2, respectively. Then $B_1 \cap B_2$ and $B_1 + B_2$ are also projection bands, with projections $P_1 P_2$ and $P_1 + P_2 - P_1 P_2$. In particular, $P_1 P_2 = P_2 P_1$.*

Proof. Since $0 \leqslant P_1 \leqslant I$ and B_2 is a band, we conclude that $\operatorname{Range} P_1 P_2 = P_1(B_2) \subseteq B_2$. Combining this with $\operatorname{Range} P_1 P_2 \subseteq \operatorname{Range} P_1 = B_1$ we conclude that $\operatorname{Range} P_1 P_2 \subseteq B_1 \cap B_2$. On the other hand, for every $x \in B_1 \cap B_2$, we have $P_1 P_2 x = P_1 x = x$; it follows that $P_1 P_2$ is a projection onto $B_1 \cap B_2$. Since clearly $0 \leqslant P_1 P_2 \leqslant I$, we conclude that $P_1 P_2$ is a band projection, and its range $B_1 \cap B_2$ is the corresponding projection band. It also follows from $B_1 \cap B_2 = B_2 \cap B_1$ that $P_1 P_2 = P_2 P_1$.

Clearly, $I - P_1$ and $I - P_2$ are the band projections for B_1^d and B_2^d. It follows from the preceding paragraph that $B_1^d \cap B_2^d$ is a projection band, and the corresponding band projection is given by $(I - P_1)(I - P_2) = I - P_1 - P_2 + P_1 P_2$. Hence, $P_1 + P_2 - P_1 P_2$

is a band projection for $(B_1^d \cap B_2^d)^d$. It is left to show that $(B_1^d \cap B_2^d)^d = B_1 + B_2$. It follows from $B_1^d \supseteq B_1^d \cap B_2^d$ that $B_1 = B_1^{dd} \subseteq (B_1^d \cap B_2^d)^d$. Similarly, $B_2 \subseteq (B_1^d \cap B_2^d)^d$, so that $B_1 + B_2 \subseteq (B_1^d \cap B_2^d)^d$. On the other hand, if $x \in (B_1^d \cap B_2^d)^d$ then $x = (P_1 + P_2 - P_1P_2)x = P_1(x - P_2x) + P_2x \in B_1 + B_2$. □

We will show in Section 7.3 that all bands, all projection bands, and all band projections on X form *Boolean algebras.*

Recall that we first defined bands in terms of order limits, and only later studied bands using the language of disjoint complements. For projection bands, we do it in the opposite order. We defined them using disjoint complements. The following most useful theorem may be viewed as an equivalent definition of projection bands in the language of order limits.

5.2.15 Theorem. *A band B in X is a projection band iff* $\sup B \cap [0, x]$ *exists for every $x \in X_+$. In this case,* $\sup B \cap [0, x] = Px$*, where P is the band projection associated with B.*

Proof. Suppose that B is a projection band, and $x \in X_+$. Then $x = y + z$ for some $y \in B_+$ and $z \in B_+^d$. We need to show that $y = \sup B \cap [0, x]$. Since, clearly, $y \in B \cap [0, x]$, it suffices to show that $B \cap [0, x] \leqslant y$. Take any $h \in B \cap [0, x]$. Then $0 \leqslant h \leqslant x = y + z$, so that by Riesz Decomposition Theorem 1.5.14 we have $h = u + v$ for some $0 \leqslant u \leqslant y$ and $0 \leqslant v \leqslant z$. Then $u \in B$ and $v \in B^d$. Since $h \in B$, we have $v = 0$, so that $h = u \leqslant y$.

Conversely, suppose that $\sup B \cap [0, x]$ exists for every $x \in X_+$. Take $x \in X_+$, and put $y = \sup B \cap [0, x]$ and $z = x - y$. Then, clearly, $x = y + z$ and $y, z \in [0, x]$. Since B is a band, we have $y \in B$. It is left to show that $z \in B^d$. Let $u \in B_+$, we will show that $z \perp u$. It follows from $0 \leqslant z \wedge u \leqslant u$ that $z \wedge u \in B_+$. Then

$$0 \leqslant z \wedge u + y \leqslant z + y = x,$$

so that $z \wedge u + y \in B \cap [0, x]$. It follows that $z \wedge u + y \leqslant y$, and, therefore, $z \wedge u = 0$. □

Next, we consider the case when B is a principal band. The following analogue of Theorem 5.2.15 will be used extensively.

5.2.16 Exercise. Let $a \in X_+$. Then the band B_a is a projection band iff $\sup_n x \wedge na$ exists for all $x \in X_+$. In this case, $\sup_n x \wedge na = P_a x$, where P_a is the band projection corresponding to B_a.

5.2.17 Exercise. Let J be an ideal in X and $B(J)$ the band generated by it.

(i) For $x \in X_+$, the sets $[0,x] \cap J$ and $[0,x] \cap B(J)$ have the same upper bounds;
(ii) $B(J)$ is a projection band iff $\sup[0,x] \cap J$ exists for all $x \in X_+$.

5.2.18 Exercise. Let $a \in X_+$ and P a band projection on X. Then $P = P_a$ iff $Pa = a$ and P vanishes on $\{a\}^d$.

5.2.19 Exercise. Let $a \in X_+$ and P a projection band on X. Then $P(B_a) = B_{Pa}$.

5.2.20 Exercise. Let Y be a sublattice in an Archimedean vector lattice X. In this exercise, we explore relationships between bands in Y and bands in X. For $A \subset Y$, we write $B^Y(A)$ for the band generated by A in Y.

(i) Let Y be a sublattice of X and B a band in Y. Then $B = Y \cap B^{dd}$, where B^{dd} is computed in X.
(ii) If Y is a regular sublattice of X and B is a band in X then $B \cap Y$ is a band in Y.
(iii) If Y is order dense then the map $B \mapsto B \cap Y$ is a bijection between the bands of X and the bands of Y; the converse map is given by $B \mapsto B^{dd}$, where B^{dd} is computed in X.
(iv) Let B is a projection band in X; let $P\colon X \to X$ be its band projection. If Y is an ideal in X then $B \cap Y$ is a projection band in Y with band projection $P_{|Y}$.
(v) A band B is a projection band iff $B \cap I_a$ is a projection band in I_a for every $a \in X_+$.
(vi) Let Y and J be ideals in X. Then $B^Y(Y \cap J) = Y \cap B(J)$.
(vii) If Y is an ideal in X and $a \in Y_+$ then $B_a^Y = B_a^X \cap Y$, where B_a^Y and B_a^X are the bands generated by a in Y and in X, respectively. If B_a^X is a projection band in X then B_a^Y is a projection band in Y.
(viii) Let Y be a sublattice of X and $a \in Y$. Does it follow that $B_a^Y \subseteq B_a^X$?
(ix) If Y is regular in X, $a \in Y$ and the principal band projection P_a^X and P_a^Y exists in X and in Y, respectively. Then P_a^Y is the restriction of P_a^X to Y.

5.2.21 Example. *If B is a projection band in X and $Y \subseteq X$ is a sublattice, $Y \cap B$ need not be a projection band in Y.* Let

$$X = C\big([0,1]\cup[2,3]\big), \quad Y = \big\{f \in X : f(1) = f(2)\big\}, \quad \text{and} \quad B = \big\{f \in X : f \text{ vanishes on } [2,3]\big\}.$$

5.2.22 Exercise. Suppose that X is uniformly complete and $e \in X_+$. Then I_e is uniformly closed iff it is a band iff it is a projection band.

A vector lattice X is said to satisfy the ***Projection Property*** or ***PP*** if every band in X is a projection band. We say that X has the ***Principal Projection Property*** or ***PPP*** if every principal band in X is a projection band. Clearly, PP implies PPP. The following is an immediate corollary of Theorem 5.2.15 and Exercise 5.2.16.

5.2.23 Corollary. *If X is order complete then X has PP. If X is σ-order complete then X has PPP.*

5.2.24 Example. We observed earlier that if μ is σ-finite and $1 \leqslant p < \infty$ then $L_p(\mu)$ has PP and, therefore, PPP. What happens if $p = \infty$ or μ is not σ-finite? It follows from Proposition 2.1.13 and Corollary 5.2.23 that $L_p(\mu)$ has PPP for every $0 \leqslant p \leqslant \infty$ and any measure μ. If $p = \infty$ or μ is not σ-finite but not both, then $L_p(\mu)$ is order complete by Theorem 2.1.16, hence has PP.

Intuitively, PPP means that one can "cut" a vector into a sum of disjoint "pieces". We will show next that every positive decomposition of a vector may be modified into a positive *disjoint* decomposition.

5.2.25 Lemma. *Let X be a vector lattice with PPP and $x, y \in X_+$. Then there exist* disjoint $u, v \in X_+$ *such that $u \leqslant 2x$, $v \leqslant 2y$, and $u + v = x + y$.*

Proof. Let $a = x+y$ and P be the band projection on X for $(x-y)^+$. Put $u = Pa$ and $v = a - u = (I - P)a$. Then $a = u + v$ and $u \perp v$. It follows from $a = 2x + (y - x) \leqslant 2x+(x-y)^-$ that $u \leqslant 2Px \leqslant 2x$. Similarly, using $I-P$ instead of P, we get $v \leqslant 2y$. □

5.2.26 Proposition. *Let X be a vector lattice with PPP, and $x_1, \dots, x_n \in X_+$. There exists disjoint $y_1, \dots, y_n$ such that $0 \leqslant y_i \leqslant x_i$ for every i and $\sum_{i=1}^n y_i = \bigvee_{i=1}^n x_i$.*

Proof. The proof in by induction. Suppose first that $n = 2$. Let P be the band projection for $(x_1 - x_2)^+$. Put $y_1 = Px_1$ and $y_2 = (I - P)x_2$. Then $0 \leqslant y_1 \leqslant x_1$, $0 \leqslant y_2 \leqslant x_2$, $y_1 \perp y_2$, and

$$y_1 + y_2 = x_2 + P(x_1 - x_2) = x_2 + (x_1 - x_2)^+ = x_1 \vee x_2$$

by Exercise 1.3.3(iii).

Suppose that the statement is true for n; we will now prove it for $n + 1$. Let $x_1, \dots, x_{n+1} \in X_+$. By the induction hypothesis, we can find disjoint $y'_1, \dots, y'_n$ such that $0 \leqslant y'_i \leqslant x_i$ for every i and $\sum_{i=1}^n y'_i = \bigvee_{i=1}^n x_i =: u$ Again, by the induction hypothesis, we find disjoint z and y_{n+1} such that $0 \leqslant z \leqslant u$, $0 \leqslant y_{n+1} \leqslant x_{n+1}$, and $z + y_{n+1} = u \vee x_{n+1} = \bigvee_{i=1}^{n+1} x_i$. Riesz Decomposition Theorem 1.5.15 yields $y_1, \dots, y_n$ such that $0 \leqslant y_i \leqslant y'_i$ for all $i = 1, \dots, n$ and $y_1 + \dots, y_n = z$. It is now easy to see that the collection $y_1, \dots, y_{n+1}$ does the job. □

Corollary 5.2.23 suggests that PP and PPP and relatively weak properties (yet, $C[0, 1]$ fails them). The following exercise shows that the Archimedean Property is even weaker.

5.2.27 Exercise. Every vector lattice satisfying PP or PPP is Archimedean.

5.3 Infinite band decompositions

Let $(X_\alpha)_{\alpha\in\Lambda}$ be a family of vector lattices; we write $X = \prod_{\alpha\in\Lambda} X_\alpha$ for the direct product of the family. We have observed earlier that each X_α may be viewed as an ideal in X. In fact, it is easy to see that X_α is a projection band in X, with $X_\alpha^d = \{x \in X \,:\, x_\alpha = 0\}$.

Now let X be a vector lattice, and consider the collection of all disjoint sets of positive vectors in X. Order this collection by inclusion. By Zorn's lemma, this collection has a maximal element. Thus, every vector lattice admits a maximal disjoint set of positive vectors.

Note that we consider maximality with respect to inclusion, not refinement. For example, the set $\{\mathbb{1}_{[0,\frac{1}{2}]}, \mathbb{1}_{[\frac{1}{2},1]}\}$ in $L_1[0,1]$ is maximal because it is not properly contained in any disjoint set, even though it obviously admits further refinements. It is easy to see that a disjoint collection Λ in X_+ is maximal iff $\Lambda^d = \{0\}$. As before, for $a \in X_+$, we write B_a for the principal band generated by a and P_a for the corresponding band projection. Recall that a is a weak unit in B_a.

5.3.1 Proposition. *Let X be a vector lattice with PPP and Λ a maximal family of disjoint positive elements of X. For every $x \in X_+$ we have $x = \sup_{a\in\Lambda} P_a x$.*

Proof. Clearly, $x \geqslant P_a x$ for every a. Suppose that $x \geqslant y \geqslant P_a x$ for every $a \in \Lambda$. Then $P_a x \geqslant P_a y \geqslant P_a x$, hence $P_a(x-y) = 0$. Therefore, $x - y$ is disjoint with every $a \in \Lambda$. It follows from the maximality of Λ that $x - y = 0$. Hence, $x = \sup_{a\in\Lambda} P_a x$. □

We know that a vector lattice need not have a weak unit; see, e.g., Exercise 1.5.13(x). However, the following construction often allows one reduce a problem about general vector lattices to those with a weak unit.

5.3.2 Proposition. *Let X be a vector lattice with PPP and Λ a maximal disjoint family in X_+. The map $T\colon X \to \prod_{a\in\Lambda} B_a$ defined by $Tx = (P_a x)_{a\in\Lambda}$ is a lattice homomorphic embedding whose range is order dense.*

Proof. Note that we consider $\prod_{a\in\Lambda} B_a$ as a formal direct product and *not* as a subset of X.

Since each P_a is a lattice homomorphism by Exercise 5.2.10, and since lattice and algebraic operations in $\prod_{a\in\Lambda} B_a$ are computed component-wise, T is a (linear) lattice homomorphism.

To show that T is one-to-one, let $x \in X$ with $Tx = 0$; we need to show that $x = 0$. Since T is a lattice homomorphism, replacing x with $|x|$ we may assume WLOG that

$x \geqslant 0$. $Tx = 0$ means $P_a x = 0$ for every $a \in \Lambda$, hence $x \perp \Lambda$. This contradicts the maximality of Λ.

It is left to show that $\operatorname{Range} T$ is order dense. Fix $v = (v_a)_{a\in\Lambda}$ in $\prod_{a\in\Lambda} B_a$ such that $v > 0$. Then $v_b > 0$ for some $b \in \Lambda$. Note that $v_b \in B_b$. Let $w = Tv_b$. Write $w = (w_a)_{a\in\Lambda}$. Then $w_b = P_b v_b = v_b$, while $w_a = P_a v_b = 0$ whenever $a \neq b$. It follows that $0 < w \leqslant v$. □

5.3.3 Corollary. *Every Archimedean vector lattice embeds lattice isomorphically as an order dense sublattice into a direct product of a family of order complete vector lattices with weak units.*

Proof. Let X be an Archimedean vector lattice. Since X embeds lattice isomorphically into its order completion, we may assume WLOG that X is order complete. In particular, X has PPP and every projection band in X is order complete. Now apply the preceding proposition. □

5.3.4 Example. This embedding need not be surjective. For example, let $X = c_0$ and let Λ be the set of all standard unit vectors. Note that X already has a weak unit. The corollary yields the natural embedding of c_0 into $\mathbb{R}^{\mathbb{N}}$. Actually, we may replace c_0 with any sequence space.

If X is order continuous, we get a somewhat better result. The following lemma essentially says that every order continuous Banach lattice X may viewed as a direct sum of principal bands (hence each of these bands has a weak unit). Recall that by Corollary 2.5.27 a disjoint order bounded collection in an order continuous Banach lattice is at most countable.

5.3.5 Lemma (Decomposition Lemma)**.** *Let X be an order continuous Banach lattice and Λ a maximal family of disjoint positive elements of X. For every $x \in X$, the collection $(P_z x)_{z\in\Lambda}$ has at most countably many non-zero terms. Let $(P_{z_i}x)$ be any enumeration of those terms, then $x = \sum_{i=1}^{\infty} P_{z_i}x$.*

Proof. First assume that $x \geqslant 0$. By Proposition 5.3.1, $x = \sup_{z\in\Lambda} P_z x$. The collection $(P_z x)_{z\in\Lambda}$ is disjoint and order bounded, hence it has at most countably many terms. Let $(P_{z_i}x)$ be any enumeration of those terms. Put $u_m = \sum_{i=1}^{m} P_{z_i}x = \bigvee_{i=1}^{m} P_{z_i}x$. Then $\sup_m u_m = \sup_{z\in\Lambda} P_z x = x$, hence $u_m \uparrow x$. Order continuity yields $u_m \to x$, so that $x = \sum_{i=1}^{\infty} P_{z_i}x$.

For a general $x \in X$, it follows from $|P_z x| = P_z|x|$ that the collection $(P_z x)_{z\in\Lambda}$ has at most countably many non-zero terms. Furthermore, $x = x^+ - x^- = \sum_{i=1}^{\infty} P_{z_i}x^+ - \sum_{i=1}^{\infty} P_{z_i}x^- = \sum_{i=1}^{\infty} P_{z_i}x$. □

5.4 Discrete vector lattices

Throughout this section, let X be an Archimedean vector lattice. A positive non-zero vector $a \in X$ is said to be an ***atom*** or a ***discrete element*** if $0 \leqslant x \leqslant a$ implies that x is a scalar multiple of a.

5.4.1 Exercise. Let $0 < a \in X$. TFAE:

(i) a is an atom;
(ii) I_a is one-dimensional (hence, $I_a = \operatorname{span}\{a\}$);
(iii) $[0, a]$ contains no two non-zero disjoint vectors.

5.4.2 Exercise. Any two atoms are either disjoint or are scalar multiples of each other.

5.4.3 Exercise. If Y is an order dense sublattice of X then X and Y have the same atoms.

5.4.4 Exercise. (i) In $\mathbb{R}^n$, $\mathbb{R}^{\mathbb{N}}$, c_0, and ℓ_p with $0 < p \leqslant \infty$, every standard unit vector e_i is an atom. Moreover, a vector is an atom iff it is a positive scalar multiple of e_i for some $i \in \mathbb{N}$. The same remains true in Banach sequence spaces (i.e., Banach lattices where the order is given by an unconditional basis).

(ii) $C[0,1]$ and $L_p[0,1]$ $(0 \leqslant p \leqslant \infty)$ have no atoms (i.e., these spaces are ***non-atomic***).

5.4.5. Recall that in Measure Theory, a measurable set A is said to be an ***atom of a measure*** μ if $\mu(A) > 0$ and no measurable subset B of A satisfies $0 < \mu(B) < \mu(A)$. It is easy to see that atoms in the vector lattice $L_p(\mu)$ are precisely the functions of the form $\lambda \mathbb{1}_A$, where $\lambda > 0$ and A is an atom in $\mathcal{F}$. In particular, $L_p(\mu)$ is non-atomic iff μ is non-atomic. It also follows that for $1 \leqslant p \leqslant \infty$, the extreme points of $B_{L_p(\mu)}$ are of the form $\pm x$, where x is an atom of norm one.

5.4.6 Exercise. Let X and Y be two Archimedean vector lattices. The atoms in $X \oplus Y$ are precisely the vectors of the form $a \oplus 0$ and $0 \oplus b$, where a is an atom in X and b is an atom in Y.

5.4.7 Exercise. Suppose that X has PPP and $0 < e \in X$ is not an atom. Then $e = u + v$ for some disjoint $u, v > 0$.

Let A be a set of atoms in X. Consider two atoms in A equivalent if they are scalar multiples of each other, and pick one representative from each equivalence class; denote this set by A_0. Then A_0 is a disjoint set and each atom in A is a scalar multiple of an element of A_0. We say that A_0 is a ***maximal disjoint collection of atoms*** in A. If A is the set of *all* atoms in X then we say that A_0 is a maximal collection of atoms in X. For example, if X is a Banach lattice then the set of all atoms of norm one is a maximal disjoint collection of atoms.

5.4.8 Example. Let $X = \mathbb{R}^{\mathbb{N}}$, c_0, c, or ℓ_p $(0 < p \leqslant \infty)$; let A be the set of all atoms in X. The set A_0 of all the standard unit vectors forms a maximal disjoint collection of atoms in X. Note that $I(A) = I(A_0) = c_{00} = \operatorname{span} A = \operatorname{span} A_0$ and $A^d = A_0^d = \{0\}$, so that $B(A) = B(A_0) = A^{dd} = X$.

5.4.9 Lemma. *Let A be a collection of atoms in X. Then $I(A) = \operatorname{span} A$. If A is finite then this set is a projection band.*

Proof. WLOG, A is disjoint; otherwise replace A with a maximal disjoint collection of atoms in A; this will not affect $\operatorname{span} A$, $I(A)$, or $B(A)$. In particular, the elements of A are linearly independent. It suffices to show that $\operatorname{span} A$ is an ideal. Let $x \in \operatorname{span} A$. We have $x = \lambda_1 a_1 + \dots + \lambda_n a_n$ for some scalars $\lambda_1, \dots, \lambda_n$ and some distinct atoms $a_1, \dots, a_n \in A$. Since the atoms are disjoint, we have $|x| = |\lambda_1| a_1 + \dots + |\lambda_n| a_n$ by Proposition 1.5.6. Hence $|x| \in \operatorname{span} A$, so that $\operatorname{span} A$ is a sublattice. It is left to show that if $x \in \operatorname{span} A$ and $y \in X$ such that $0 \leqslant y \leqslant x$ then $y \in \operatorname{span} A$. Indeed, we can again write $x = \lambda_1 a_1 + \dots + \lambda_n a_n$ with $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ and $a_1, \dots, a_n \in A$. It follows from $x = |x|$ that $\lambda_i \geqslant 0$ for all i. By Riesz Decomposition Theorem, we can decompose $y = y_1 + \dots + y_n$ for some $0 \leqslant y_i \leqslant \lambda_i a_i$ for every i. It follows that y_i is itself a scalar multiple of a_i, hence $y \in \operatorname{span} A$.

Let a be an atom in X. We claim that I_a is a projection band. Let $x \in X_+$. For every $n \in \mathbb{N}$, $x \wedge na \in I_a$, hence $x \wedge na = \lambda_n a$ for some $\lambda_n \geqslant 0$. Clearly, the sequence (λ_n) is increasing; it is bounded because $\lambda_n a \leqslant x$ for all n and X is Archimedean. Let $\lambda = \lim \lambda_n$, then $x \wedge na = \lambda_n a \uparrow \lambda a$. If $x \in B_a$ then Proposition 5.1.3 yields $x = \sup_n x \wedge na = \lambda a \in I_a$, hence $B_a = I_a$ and, therefore I_a is a band. By the preceding computation, $\sup_n (x \wedge na)$ exists for every $x \in X_+$; it follows that I_a is a projection band by Exercise 5.2.16.

Suppose now that $A = \{a_1, \dots, a_n\}$ is finite. Then $I(A) = I_{a_1} + \dots + I_{a_n}$ is a projection band by Theorem 5.2.14. $\square$

Recall that X is assumed to be an Archimedean vector lattice. We say that X is ***atomic*** or ***discrete*** if X equals the band generated by all the atoms in X. Example 5.4.8 shows that $\mathbb{R}^{\mathbb{N}}$, c_0, c, and ℓ_p $(0 < p \leqslant \infty)$ are discrete. Clearly, $\mathbb{R}^n$ is discrete for every n. The following theorem provides several convenient equivalent characterizations of atomic spaces.

5.4.10 Theorem. *TFAE*

(i) *X is discrete;*

(ii) *For every $0 < x \in X$ there is an atom a such that $a \leqslant x$;*

(iii) *For every $x \in X$, if x is disjoint to every atom then $x = 0$.*

Proof. Let A be the set of all atoms in X.

(i)⇒(ii) Suppose that $X = B(A)$. Since $B(A) = B\bigl(I(A)\bigr)$, it follows from Exercise 5.1.2 that $I(A)$ is order dense in X. Since $I(A) = \operatorname{span} A$ by Lemma 5.4.9, this implies (ii).

(ii)⇒(iii) If $x \neq 0$ then there is an atom a such that $a \leqslant |x|$, so that $x \not\perp a$.

(iii)⇒(i) It follows from (iii) that $A^d = \{0\}$, so that $A^{dd} = X$. But A^{dd} is the band generated by A by Theorem 5.1.7. □

We are now ready to completely characterize finite-dimensional Archimedean vector lattices.

5.4.11 Theorem. *[Yudin] TFAE:*

(i) *X is finite-dimensional;*

(ii) *X contains no disjoint (infinite) sequence of non-zero vectors;*

(iii) *X is lattice isomorphic to $\mathbb{R}^n$ for some n.*

Proof. (iii)⇒(i) is trivial. (i)⇒(ii) is trivial as well, because every disjoint sequence is linearly independent. It is left to prove (ii)⇒(iii).

Suppose that X contains no disjoint sequence of non-zero vectors. We will show first that X is discrete. Suppose not; then there exists $u > 0$ such that $[0, u]$ contains no atoms by Theorem 5.4.10(ii). It follows from Exercise 5.4.1 that there exist two non-zero disjoint vectors in $[0, u]$, denote them by x_1 and u_1. Since $u_1 \in [0, u]$, it is not an atom. Applying Exercise 5.4.1 again, we find two non-zero disjoint vectors in $[0, u_1]$; denote them by x_2 and u_2. It follows from $x_1 \perp u_1$ that the set $\{x_1, x_2, u_2\}$ is disjoint. Proceeding inductively, we obtain a disjoint sequence (x_n); a contradiction.

Let A be a maximal disjoint collection of atoms in X. By assumption, A is finite, $A = \{a_1, \dots, a_n\}$, and $X = B(A) = I(A) = \operatorname{span} A$ by Lemma 5.4.9. Define a linear operator $T \colon \mathbb{R}^n \to X$ via $Te_i = a_i$. Clearly, T is a bijection. It follows from Exercise 1.5.6 that T is a lattice homomorphism, hence a lattice isomorphism. □

5.4.12 Corollary. *Every finite dimensional sublattice of an Archimedean vector lattice is spanned by a disjoint finite set of positive vectors.*

Since every disjoint non-zero sequence in a Banach lattice is unconditional by Proposition 1.5.9, Theorem 5.4.11 immediately yields the following:

5.4.13 Corollary. *Every infinite-dimensional Banach lattice contains an unconditional basic sequence.*

5.4.14 Example. The preceding corollary helps us answer the question whether every Banach space admits an order that makes it into a Banach lattice. It is known that a hereditary indecomposable Banach space admits no unconditional basic sequences, hence it cannot be made into a Banach lattice.

5.4.15 Exercise. Every finite dimensional sublattice of an Archimedean vector lattice is regular.

5.4.16 Example. Let $X = \ell_\infty$. We have already observed that X is discrete and the set $\{e_n\}$ of all standard unit vectors is a maximal disjoint collection of atoms, so that $X = B(\{e_n\})$. Note also that $I(\{e_n\}) = c_{00}$, so that $\overline{I(\{e_n\})} = c_0$. This gives an example of a closed ideal which is not a band. Even though the sequence (e_n) is not a Schauder basis for X (since ℓ_∞ is non-separable, it has no Schauder basis), this sequence still acts like a basis in order sense: for every positive sequence $x = (\lambda_n)$ in ℓ_∞, we have $x = \bigvee_{n=1}^\infty \lambda_n e_n$.

5.4.17. The preceding example can be generalized to any discrete Archimedean vector lattice X. Let a be an atom in X. It follows from Lemma 5.4.9 that $B_a = I_a = \operatorname{span} a$, and this is a (principal) projection band. Let P_a be the corresponding band projection. Fix $x \in X$. Then $P_a x$ is a scalar multiple of a, hence $P_a x = \lambda a$ for some real number λ, which depends on a and on x; we will denote it by $\lambda_a(x)$. Thus, $P_a x = \lambda_a(x)a$ for every atom a and $x \in X$. It is easy to see that for a fixed a, the map $x \mapsto \lambda_a(x)$ is a positive linear functional on X; it is often called the ***coordinate functional*** of a. On the other hand, fix a maximal disjoint collection A of atoms in X; for every $x \in X$, the map $a \in A \mapsto \lambda_a(x) \in \mathbb{R}$ may be viewed as the "coordinate expansion" of x with respect to A, with $\lambda_a(x)$ being the "coefficient at a" in this expansion. The following lemma asserts that x is determined by its expansion.

5.4.18 Lemma. *Suppose that X is discrete, A is a maximal disjoint collection of atoms in X, and $x \in X_+$. Then $x = \bigvee\limits_{a\in A} \lambda_a(x)a$.*

Proof. Clearly, $x \geqslant P_a x = \lambda_a(x)a$ for each $a \in A$. Suppose that there exists z such that $x > z$ and $z \geqslant \lambda_a(x)a$ for every $a \in A$. It follows from $x - z > 0$ that there exists $a \in A$ and a positive real μ such that $x - z \geqslant \mu a$ by Theorem 5.4.10(ii). Then $x \geqslant \mu a + z \geqslant \mu a + \lambda_a(x)a$, so that $\lambda_a(x)a = P_a x \geqslant \mu a + \lambda_a(x)a$, a contradiction. $\square$

Thus, every $x \in X$ can be associated with a function in $\mathbb{R}^A$. This yields a map $T\colon X \to \mathbb{R}^A$. One can verify that this map is a lattice isomorphism of X onto a sublattice of $\mathbb{R}^A$. For every $a \in A$, we have $\mathbb{1}_{\{a\}} = Ta \in \operatorname{Range} T$. Example 5.4.16 shows that T need not be onto. It follows that discrete Archimedean vector lattices are precisely the order dense sublattices of $\mathbb{R}^A$ for some A:

5.4.19 Proposition. *X is discrete iff it is lattice isomorphic to an order dense sublattice of $\mathbb{R}^A$ for some set A. Moreover, X is order complete iff this sublattice is an ideal in $\mathbb{R}^A$.*

Proof. Recall that by Exercise 3.2.7, a sublattice of $\mathbb{R}^A$ is order dense iff it contains the characteristic functions of all singletons. Suppose that X is discrete; let $Y = \operatorname{Range} T$ where T is as before. By the paragraph before the theorem, Y is an order dense sublattice of $\mathbb{R}^A$ and T is a lattice isomorphism between X and Y.

Conversely, let A be any set and Y an order dense sublattice of $\mathbb{R}^A$. Then $\mathbb{1}_{\{a\}} \in Y$ for every $a \in A$. It can be easily verified that $\mathbb{1}_{\{a\}}$ is an atom in Y. It follows that Y is discrete by Theorem 5.4.10.

Let's now prove the "moreover" clause. It suffices to prove that an order dense sublattice Y of $\mathbb{R}^A$ is an ideal iff it is order complete. If Y is an ideal then it is order complete because $\mathbb{R}^A$ is order complete; see Proposition 3.3.18. Conversely, if Y is order complete then it is an ideal by Exercise 3.3.19. □

5.4.20 Exercise. Every positive vector in a discrete order continuous Banach lattice may be written as a supremum of a finite or countable collection of atoms.

We will study discrete order continuous Banach latices in more detail in Section 10.8. Also, we will show in Theorem 8.6.1 that $L_p(\Omega,\mu)$ with $1 \leqslant p < \infty$ is discrete iff it is lattice isometric to $\ell_p(\Gamma)$ for some Γ.

5.4.21 Exercise. If μ is a σ-finite measure then $L_\infty(\mu)$ is discrete iff it is lattice isometric to ℓ_∞.

Next, we investigate discrete $C(K)$ spaces.

5.4.22 Exercise. Atoms in $C(K)$ are exactly the functions of the form $\lambda\mathbb{1}_{\{t\}}$, where t is an isolated point of K and $\lambda > 0$.

5.4.23 Proposition. *$C(K)$ is discrete iff isolated points are dense in K.*

Proof. By Theorem 5.4.10, $C(K)$ is discrete iff every $0 < f \in C(K)$ dominates an atom. By Exercise 5.4.22, this means that there exists an isolated point in $\operatorname{supp} f$.

The latter is clearly satisfied if isolated points are dense. To see the converse, note that by Urysohn's Lemma A.2.2, every non-empty open set contains the support of some $f > 0$. $\square$

We will now show that a discrete order complete $C(K)$ spaces is just $\ell_\infty(\Gamma)$ for an appropriate set Γ. Recall that $\beta\Gamma$ stands for the ***Stone-Čech compactification*** of Gamma, see 35.

5.4.24 Proposition. *Let K be a compact space. Then $C(K)$ is discrete and order complete iff K is homeomorphic to $\beta\Gamma$ for some set Γ; in this case $C(K)$ is lattice isometric to $\ell_\infty(\Gamma)$ and Γ is the set of isolated points in K.*

Proof. We prove first that for every set Γ, $\ell_\infty(\Gamma)$ is lattice isometric to $C(\beta\Gamma)$. Given $f \in C(\beta\Gamma)$, f is bounded, hence the restriction of f to Γ belongs to $\ell_\infty(\Gamma)$. Define this restriction Tf. This defines $T\colon C(\beta\Gamma) \to \ell_\infty(\Gamma)$. It is clear that T is linear and positive. The definition of $\beta\Gamma$ yields that T is a bijection. Clearly, T is a contraction. Since Γ is dense in $\beta\Gamma$, $f \geqslant 0$ iff $Tf \geqslant 0$, so that T is a lattice isomorphism. Also, if $Tf \leqslant \mathbb{1}$ in $\ell_\infty(\Gamma)$ then $f \leqslant \mathbb{1}$ in $C(\beta\Gamma)$, so that T is an isometry.

Now suppose that $C(K)$ is discrete and order complete. By Theorem 2.1.29, K is extremally disconnected. Let Γ be the set of all isolated points in K. By Proposition 5.4.23, Γ is dense in K.

It is left to show that every bounded function $h\colon \Gamma \to \mathbb{R}$ extends to K. WLOG, $0 \leqslant h(t) \leqslant 1$ for every $t \in \Gamma$ as, otherwise, we can find $M \in \mathbb{R}$ such that $-M \leqslant h(t) \leqslant M$ for all $t \in \Gamma$ and replace h with $h' = \frac{1}{2M}\big(h + M\big)$

Let f be the supremum in $C(K)$ of $\big\{h(s)\mathbb{1}_{\{s\}} \ :\ s \in \Gamma\big\}$; the supremum exists because this set is bounded above by $\mathbb{1}$ and $C(K)$ is order complete; it also follows that $f \leqslant \mathbb{1}$. We claim that f extends h. Let $t \in \Gamma$. Using distributivity, we have

$$\mathbb{1}_{\{t\}} \wedge f = \sup\big\{\mathbb{1}_{\{t\}} \wedge h(s)\mathbb{1}_{\{s\}} \ :\ s \in \Gamma\big\} = h(t)\mathbb{1}_{\{t\}}.$$

It follows that $f(t) = \min\big\{1, f(t)\big\} = (\mathbb{1}_{\{t\}} \wedge f)(t) = h(t)\mathbb{1}_{\{t\}}(t) = h(t)$. $\square$

5.4.25 Example. The preceding two propositions fail without the order completeness assumption. Indeed, the space c is clearly discrete; the standard unit vectors e_i form a maximal collection of atoms. Yet c is not an ideal in $\mathbb{R}^{\mathbb{N}}$. Furthermore, c may be identified with $C(K)$ where K is a one-point compactification of $\mathbb{N}$, that is, $K = \mathbb{N} \cup \{\infty\}$. Yet is not lattice isometric to an $\ell_\infty(\Gamma)$.

As before, let X be an (Archimedean) vector lattice. We write X_a for the band generated by the atoms of X; we call it the ***atomic*** or ***discrete part*** of X. Its disjoint complement X_a^d is called the ***non-atomic*** part of X. These names are justified by the following:

5.4.26 Exercise. (i) X_a is discrete;

(ii) For $x \in X_+$, $x \in X_a$ iff x is the supremum of a family of atoms in X;

(iii) X_a^d is non-atomic.

Note that most of the results in this section are not valid for non-Archimedean vector lattices. In particular, many of the results fail when X is $\mathbb{R}^2$ equipped with the lexicographic order.

5.5 * Quasi-interior points

Let e be a positive vector in a vector lattice X. Recall that B_e is the order closure of I_e. Recall that for $x \in X_+$, we have $x \in B_e$ iff $x \wedge ne \uparrow x$ by Proposition 5.1.3. Clearly, $x \in I_e$ iff $x \wedge ne = x$ for all sufficiently large n. Recall also that e is a strong unit iff $I_e = X$ and a weak unit iff $B_e = X$.

Suppose now that X is a Banach lattice and $e \in X_+$. By Exercise 5.1.16, B_e is (norm) closed, hence $I_e \subseteq \overline{I_e} \subseteq B_e$. In general, both inclusions may be proper (an example is forthcoming).

5.5.1 Exercise. Let X be a Banach lattice and $e, x \in X_+$. Then $x \in \overline{I_e}$ iff $x \wedge ne \to x$.

Combining the definition of an ideal, Exercise 5.5.1, and Proposition 5.1.3, we get the following: suppose that $e, x \in X_+$. Then

- $x \in I_e$ if $x = x \wedge ne$ for all sufficiently large n;
- $x \in \overline{I_e}$ if $x = \lim_n (x \wedge ne)$;
- $x \in B_e$ if $x = \sup_n (x \wedge ne)$.

We say that e is a ***quasi-interior point*** or a ***topological order unit*** in X if I_e is dense in X, i.e., $\overline{I_e} = X$. This notion is intermediate between being a weak unit and a strong unit:

$$\text{strong unit, } I_e = X \quad \Rightarrow \quad \text{quasi-interior point, } \overline{I_e} = X \quad \Rightarrow \quad \text{weak unit, } B_e = X.$$

5.5.2 Example. *A quasi-interior point which is not a strong unit.* Let $X = L_p[0,1]$ with $1 \leqslant p < \infty$, put $e = \mathbb{1}$. Then $I_e = L_\infty[0,1]$, so that I_e is a dense but proper subset of X. Thus, e is a quasi-interior point but not a strong unit.

5.5.3 Example. *A weak unit which is not a quasi-interior point.* Let $X = C[0,1]$; let $e \in C[0,1]$ be given by $e(t) = t$. Clearly, no non-zero element of $C[0,1]$ is disjoint to e; hence e is a weak unit. In particular, $B_e = C[0,1]$. However, $\overline{I_e}$ is the set of all functions in $C[0,1]$ which vanish at 0 by, e.g., Theorem 3.3.11. Hence, $\overline{I_e} \neq B_e$ and e is not quasi-interior.

Exercise 5.5.1 yields the following.

5.5.4 Proposition. *Let e be a positive vector in a Banach lattice X. Then e is quasi-interior iff $x \wedge ne \to x$ for every $x \in X_+$.*

5.5.5 Exercise. Every separable Banach lattice admits a quasi-interior point.

Finally, observe that in order continuous spaces, the notions of a weak unit and of a quasi-interior point coincide. The idea is that in this case, the order closure is the same as the norm closure.

5.5.6 Proposition. *Let X be order continuous and $e \in X_+$. Then e is quasi-interior iff it is a weak unit.*

Proof. We only need to prove the converse. Let e be a weak unit and $x \in X_+$. Then $x \wedge ne \uparrow x$ and, therefore, $x \wedge ne \to x$. Hence, e is quasi-interior by Proposition 5.5.4. $\square$

We will show in Corollary 14.1.19 that all quasi-interior points in X are "equivalent" in a certain way.

5.6 * Ideals and bands in dense sublattices of $C(K)$

We know from $C(K)$ Representation Theorem 4.1.8 that an Archimedean vector lattice X with a strong unit may be identified with a dense sublattice of $C(K)$ for some K. In this section, we will relate vector lattice properties of X with that of $C(K)$ and, further, with topological properties of K. As a byproduct, we will prove that for uniformly complete vector lattices (including Banach lattices), PP is equivalent to order completeness, while PPP is equivalent to σ-order completeness. Many of the results in this section are due to Veksler and Geiler [VG72].

Throughout this section, K will stand for a compact space, and X for a dense sublattice of $C(K)$, equipped with the norm of $C(K)$. Recall that X is automatically order dense, hence regular in $C(K)$; see Exercise 3.2.12. We start with a useful generalization of Urysohn's Lemma (which goes back to Veksler and Geller):

5.6.1 Lemma (Urysohn's Lemma for dense sublattices)**.** *Let X be a dense sublattice of $C(K)$, $f \in X_+$, and F and G two disjoint closed subsets of K. Then there exists $g \in X$ such that $0 \leqslant g \leqslant f$, g agrees with f on F and vanishes on G.*

Proof. By the original Urysohn's Lemma A.2.2, we can find $h \in C(K)$ such that h equals 3 on F and vanishes on G. By density of X, we can find $u \in X$ such that $\|h-u\| < 1$. Note that u is greater than 2 on F and less than 1 on G. In a similar way, find $v \in X$ such that v is less that 1 on F and greater than 2 on G. Put $w = (u-v)^+$, then $w \in X$, w is greater than 1 on F and vanishes on G. Now put $g = \big(\|f\|w\big) \wedge f$; it is easy to see that g satisfies the requirements. □

5.6.2 Corollary. *Let X be a dense sublattice of $C(K)$, $f \in X$, and F and G two disjoint closed subsets of K. Then there exists $g \in X$ such that $|g| \leqslant |f|$, g agrees with f on F and vanishes on G.*

Proof. Apply Lemma 5.6.1 to f^+ and f^-. □

5.6.3 Corollary. *Let X be a dense sublattice of $C(K)$ such that $\mathbb{1} \in X$. If $F \subseteq U$ such that F is closed and U is open in K, then there exists $f \in X$ such that f equals 1 on F and vanishes outside of U. In particular, X contains the characteristic function of every clopen set.*

5.6.4 Remark. The extra assumption in the corollary that $\mathbb{1} \in X$ is, actually, very mild for the following reason. Let X be a dense sublattice of $C(K)$, and fix $0 < \varepsilon < 1$. We can find $e \in X$ with $\|e-\mathbb{1}\| < \varepsilon$, i.e., $(1-\varepsilon)\mathbb{1} \leqslant e \leqslant (1+\varepsilon)\mathbb{1}$. Viewed as an operator $M_e\colon C(K) \to C(K)$, multiplication by e is a lattice isomorphism, $\|M_e - I\| \leqslant \varepsilon$, and the image of X under M_e^{-1} is a dense sublattice of $C(K)$ containing $M_e^{-1}e = \mathbb{1}$.

Recall that there is a one-to-one correspondence between closed subsets F of K and closed ideals J in $C(K)$ given by $F \mapsto J_F$ and $J \mapsto \bigcap_{f\in J} \ker f$; see Exercise 3.3.12. We will show that this correspondence extends to closed ideals in a dense sublattice X of $C(K)$; by "closed" we mean that the ideals are closed in X with respect to the norm induced from $C(K)$. For a closed set F in K, we write $J_F^X = J_F \cap X = \big\{f \in X : f \text{ vanishes on } F\big\}$. It is easy to see that it is a closed ideal in X. We can recover F from J_F^X:

5.6.5 Lemma. *For a closed set F in K, $F = \bigcap_{f\in J_F^X} \ker f$.*

Proof. It is clear that $F \subseteq \bigcap_{f\in J_F^X} \ker f$. Suppose that $t \notin F$. Since X is dense, we find $f \in X$ such that $f(t) \neq 0$. By Corollary 5.6.2, we may assume that f vanishes on F. It follows that $f \in J_F \cap X = J_F^X$, so that $f(t) = 0$. Therefore $t \notin \bigcap_{f\in J_F^X} \ker f$. □

5.6.6 Proposition. *Closed ideals in X are the sets of the form J_F^X, where F is a closed subset of K. Furthermore, J_F^X is dense in J_F.*

Proof. It is clear that J_F^X is a closed ideal of X whenever F is closed.

Let J be an ideal in X. By Exercise 3.3.17, $\overline{J}^{C(K)}$ is an ideal in $C(K)$. Clearly, this is the smallest closed ideal of $C(K)$ containing J. It follows from Proposition 3.3.10 that $\overline{J}^{C(K)} = J_F$ where $F = \bigcap_{f\in J} \ker f$. If, in addition, J is closed in X then $J = \overline{J}^{C(K)} \cap X = J_F \cap X = J_F^X$.

Applying the preceding argument with $J = J_F^X$ and using Lemma 5.6.5, we conclude that $\overline{J_F^X}^{C(K)} = J_F$, hence J_F^X is dense in J_F. $\square$

5.6.7. Thus, there are one-to-one correspondences between closed subsets of K, closed ideals in $C(K)$, and closed ideals in X given by $F \leftrightarrow J_F \leftrightarrow J_F^X$.

5.6.8 Exercise. Let F and G be two closed subsets of K. Then $F \cap G = \varnothing$ iff $J_F^X + J_G^X = X$.

As in Remark 3.3.13, it is occasionally convenient to use the complementary notation for ideals: for an open set U in K, write $C_U^X := \bigl\{f \in X \,:\, \operatorname{supp} f \subseteq U\bigr\}$. It follows immediately that $C_U^X = C_U \cap X$ and $C_U^X = J_{U^C}^X$. It follows from Proposition 5.6.6 that C_U^X is dense in C_U and that closed ideals in X are precisely the sets of the form C_U^X for some open U.

For an ideal J in X, we define the ***support*** of J, denoted $\operatorname{supp} J$, to be the union of the supports of all the functions in J. The following exercise extends Exercise 3.3.10 and Remark 3.3.13:

5.6.9 Proposition. *Let J be an ideal in X. Its closure $\overline{J}$ in X equals $C_{\operatorname{supp} J}^X$.*

Proof. Let J_0 be the ideal of $C(K)$ generated by J. Note that $\operatorname{supp} J_0 = \operatorname{supp} J =: U$. Combining Exercise 3.3.17 with Remark 3.3.13, we get $\overline{J}^X = \overline{J_0}^{C(K)} \cap X = C_U \cap X = C_U^X$. $\square$

We write $\overline{\operatorname{supp}}\, f$ for the closure of $\operatorname{supp} f$. Given an open set U in K, put $D_U^X = \bigl\{f \in X \,:\, \overline{\operatorname{supp}}\, f \subseteq U\bigr\}$. It is clear that $D_U^X \subseteq C_U^X$. It follows from

$$\overline{\operatorname{supp}}(f+g) \subseteq (\overline{\operatorname{supp}}\, f) \cup (\overline{\operatorname{supp}}\, g) \tag{5.1}$$

that D_U^X is itself an ideal.

5.6.10 Exercise. $\overline{D_U^X} = C_U^X$ for every open set U.

The following exercise shows that D_U^X and C_U^X are, respectively, the least and the greatest ideals in X whose support is U.

5.6.11 Proposition. *Let J be an ideal in X; put $U = \operatorname{supp} J$. Then $D_U^X \subseteq J \subseteq C_U^X$.*

Proof. It follows from Proposition 5.6.9 that $J \subseteq C_U^X$. Let $f \in D_U^X$. We know that $\overline{\operatorname{supp}}\, f \subseteq \bigcup_{g\in J} \operatorname{supp} g$. This represents an open cover of a compact set. It follows that there are $g_1, \dots, g_n \in J$ such that $\overline{\operatorname{supp}}\, f \subseteq \bigcup_{i=1}^n \operatorname{supp} g_i = \operatorname{supp} g$, where $g = |g_1| \vee \dots \vee |g_n|$. Then $g \in J$ and g is positive on $\overline{\operatorname{supp}}\, f$. Since the latter set is compact, g is bounded below by some positive constant on it. It follows that $|f| \leqslant \lambda g$ for some $\lambda > 0$ and, therefore, $f \in J$. □

Exercise 5.6.8 yields that if U and V are open with $U \cup V = K$ then $C_U^X + C_V^X = X$. It may be natural to conjecture that $C_U^X + C_V^X = C_{U\cup V}^X$ for any two open U and V. In general, this is false:

5.6.12 Example. *$C_U^X + C_V^X$ need not equal $C_{U\cup V}^X$.* Let $K = [-1, 1]$, $U = [-1, 0)$, $V = (0, 1]$, and X be the space of all $f \in C(K)$ such that f is even on some interval centred at zero:

$$X = \big\{ f \in C(K) \,:\, \exists \varepsilon > 0 \ \forall t \in (0, \varepsilon) \ f(-t) = f(t) \big\}.$$

Then $f \in C_U^X$ iff f vanishes on $[-\varepsilon, 1]$ for some $\varepsilon > 0$ and, similarly, $f \in C_V^X$ iff f vanishes on $[-1, \varepsilon]$ for some $\varepsilon > 0$. It follows that $f \in C_U^X + C_V^X$ iff f vanishes on $[-\varepsilon, \varepsilon]$ for some $\varepsilon > 0$. Let $f(x) = |x|$; then $f \in C_{U\cup V}^X$ but $f \notin C_U^X + C_V^X$.

5.6.13 Proposition. *$D_U^X + D_V^X = D_{U\cup V}^X$ for any two open sets U and V.*

Proof. $D_U^X + D_V^X \subseteq D_{U\cup V}^X$ follows from (5.1). Let $h \in D_{U\cup V}^X$, i.e., $\overline{\operatorname{supp}}\, h \subseteq U \cup V$. Find an open W such that $\overline{\operatorname{supp}}\, h \subseteq W \subseteq \overline{W} \subseteq U \cup V$. Put $U' = U \cap W$ and $V' = V \cap W$. Let

$$F \coloneqq \overline{\operatorname{supp}}\, h \setminus V' \quad \text{and} \quad G \coloneqq \overline{\operatorname{supp}}\, h \setminus U'.$$

Using Urysohn's Lemma for dense sublattices 5.6.1, we find $f \in X$ such that $0 \leqslant f \leqslant h$, f agrees with h on F and vanishes on G. Put $g \coloneqq h - f$, then g agrees with h on G and vanishes on F.

Observe that $\operatorname{supp} f \subseteq U'$ because if $t \in (\operatorname{supp} f) \setminus U'$ then $t \in G$, so that $f(t) = 0$; a contradiction. It follows that $\overline{\operatorname{supp}}\, f \subseteq U$, so that $f \in D_U^X$. Similarly, $g \in D_V^X$. It follows that $h \in D_U^X + D_V^X$. □

In the special case when $X = C(K)$, we get the following:

5.6.14 Corollary. *$C_U + C_V = C_{U\cup V}$ for every open sets U and V.*

Proof. $C_U + C_V \subseteq C_{U\cup V}$ is obvious. On the other hand, by Exercise 5.6.10, Proposition 5.6.13, and Exercise 3.3.16, we get

$$C_{U\cup V} = \overline{D_{U\cup V}} = \overline{D_U + D_V} = \overline{D_U} + \overline{D_V} = C_U + C_V.$$

□

5.6.15 Theorem. *For every strictly positive f and every open cover $U_1,\dots,U_n$ of K there exist $f_1,\dots,f_n\in X_+$ such that $f=\sum_{i=1}^n f_i$ and $\overline{\operatorname{supp}} f_i\subseteq U_i$ for every $i=1,\dots,n$.*

Proof. Applying Proposition 5.6.13 with $U=U_1$ and $V=\bigcup_{i=1}^n U_i$, we get $X=D^X_{U_1}+D^X_V$. Iterating, we get $X=D^X_{U_1}+\dots+D^X_{U_n}$. The result now follows immediately. □

5.6.16 Corollary. *If $U_1,\dots,U_n$ form an open cover of K then there exist positive $f_1,\dots,f_n\in C(K)$ such that* $\operatorname{supp} f_i\subseteq U_i$ *for every i and $f_1+\dots+f_n=\mathbb{1}$.*

Such an n-tuple of functions $f_1,\dots,f_n$ is called a ***partition of unity*** subordinate to the open cover $U_1,\dots,U_n$.

Next, we consider bands. Theorem 5.1.23 and Proposition 5.2.4 assert that a band in $C(K)$ is a set of the form J_F where F is the closure of an open set; it is a projection band iff F is clopen. We will now extend these characterizations to bands in X. *All disjoint complements throughout the rest of this section are computed in X.*

5.6.17 Exercise. Let X be a dense sublattice of $C(K)$.

(i) For $f\in X$, $\{f\}^d=J^X_F$, where $F=\overline{(\ker f)^C}$;
(ii) For $A\subseteq X$, $A^d=J^X_F$, where $F=\overline{\left(\bigcap_{f\in A}\ker f\right)^C}$;
(iii) For a closed set F, $(J^X_F)^d=J^X_{\overline{F^C}}=J^X_{(\operatorname{Int} F)^C}$;
(iv) For an ideal J in X, $J^d=J^X_{\overline{\operatorname{supp} J}}$.

5.6.18 Proposition. *Let F be a closed subset of K. Then J^X_F is a band in X iff F is the closure of an open set; J^X_F is a projection band iff F is clopen.*

Proof. The first claim is proved analogously to Theorem 5.1.23. Suppose now J^X_F is a projection band. Since it is a band, $F=\overline{U}$ for some open U. Then $(J^X_F)^d=J^X_{U^C}$. Since X is dense in $C(K)$, it contains a strictly positive function, say, e. We can decompose $e=f+g$, where $f\in J^X_F$ and $g\in J^X_{U^C}$. This implies that e vanishes on ∂F, hence $\partial F=\varnothing$ and, therefore, F is clopen.

If F is clopen, by Exercises 5.6.17 and 5.6.8, we have $X=J^X_F+J^X_{F^C}=J^X_F+(J^X_F)^d$. Hence, J^X_F is a projection band. □

5.6.19 Exercise. Let U be an open subset of K, $A\subseteq X$ such that each function in A is supported on U, and $f=\sup A$. Then $\operatorname{supp} f\subseteq\overline{U}$.

5.6.1 PP/PPP in $C(K)$ and its dense sublattices

The reader is advised to review Section 2.1.3 before preceding. We defined three degrees of disconnectedness of a compact set there: total, basic, and extremal, and showed that $C(K)$ is order complete iff it is extremally disconnected; $C(K)$ is σ-order complete iff K is basically disconnected. We now bring dense sublattices and PP/PPP into the picture.

5.6.20 Theorem. *K is totally disconnected iff $C(K)$ has a dense sublattice with PPP.*

Proof. Suppose that K is totally disconnected. Let X be the linear span of characteristic functions of clopen sets or, equivalently, the set of all functions in $C(K)$ that take only finitely many values. Clearly, X is a sublattice of $C(K)$ and $\mathbb{1} \in X$. If $s, t \in K$ with $s \neq t$ then there is a clopen set U such that $s \in U$ and $t \in U^C$, hence $\mathbb{1}_U$ separates s and t. Thus, X is dense in $C(K)$ by Stone-Weierstrass Theorem 3.1.28. It is left to show that X has PPP. Let $f \in X$. It is clear from the definition of X that $F := \ker f$ is clopen. By Exercise 5.6.17 $\{f\}^d = J^X_{F^C}$ and, therefore the band generated by f in X is $\{f\}^{dd} = J^X_F$, which is projective by Proposition 5.6.18.

To prove the converse, suppose that X is a dense sublattice of $C(K)$ with PPP. As in Remark 5.6.4, we may assume that $\mathbb{1} \in X$. Let $t \in V$ for some open set V; we need to find a clopen set G such that $t \in G \subseteq V$. By Urysohn's Lemma for dense sublattices 5.6.1, there exists $f \in X_+$ such that $f(t) = 0$ and $f(t) = 1$ whenever $t \notin V$. Put $F = \ker f$; then $F \subseteq V$ and $t \in \operatorname{Int} F$. By Exercise 5.6.17, the band generated by f is $\{f\}^{dd} = J^X_G$, where $G = \overline{K \setminus \overline{K \setminus F}} = \overline{\operatorname{Int} F}$. By assumption, this band is a projection band. Proposition 5.6.18 asserts that G is clopen. Note that $t \in G \subseteq F \subseteq V$. $\square$

A vector lattice is said to be ***disjointly order complete*** if every disjoint set that is bounded above has supremum. Clearly, every order complete vector lattice is disjointly order complete. Similarly, a vector lattice is said to be ***disjointly σ-order complete*** if every disjoint sequence that is bounded above has supremum; every σ-order complete vector lattice is disjointly σ-order complete. Recall that an open subset U of K is F_σ iff $U^C = \ker f$ for some $f \in C(K)$; see Exercise 2.1.24(iii).

5.6.21 Theorem. *TFAE:*

(i) *K is basically disconnected;*

(ii) *$C(K)$ has PPP;*

(iii) *Every dense sublattice of $C(K)$ has PPP;*

(iv) *$C(K)$ has a dense sublattice that is disjointly σ-order complete.*

Proof. We will prove that (i)⇒(iii)⇒(ii)⇒(i) and (i)⇔(iv)

(i)⇒(iii) Suppose that K is basically disconnected; let X be a dense sublattice of $C(K)$ and $f \in X$. The band generated by f in X is $\{f\}^{dd}$. By Exercise 5.6.17, $\{f\}^d = J^X_{\overline{F^C}}$, where $F = \ker f$. Observe that F^C is an F_σ-set, hence $\overline{F^C}$ is clopen. Hence $\{f\}^d$ is a projection band by Proposition 5.6.18. It follows that $X = \{f\}^d \oplus \{f\}^{dd}$ and, therefore, $\{f\}^{dd}$ is a projection band.

(iii)⇒(ii) trivially.

(ii)⇒(i) Let U be an open F_σ set. Find $f \in C(K)$ such that $U^C = \ker f$. Then $\{f\}^d = J_{\overline{U}}$ by Exercise 5.6.17. Since $C(K)$ has PPP, this set is a projection band, hence $\overline{U}$ is clopen by Proposition 5.2.4 and Exercise 3.3.12.

(i)⇒(iv) If K is basically disconnected then $C(K)$ is σ-order complete, hence $C(K)$ itself is disjointly σ-order complete.

(iv)⇒(i) Suppose that there exists a dense disjointly σ-order complete sublattice X of $C(K)$. By Remark 5.6.4, we may assume that $\mathbb{1} \in X$. Let U be an open F_σ set. By Exercise 2.1.24, there exists a sequence (V_n) of open sets such that $\overline{V_n} \subseteq V_{n+1}$ and $\bigcup_{n=1}^\infty V_n = U$. Put $V_0 = V_{-1} \coloneqq \varnothing$. For every $n \geqslant 1$, the closed set $\overline{V_n} \setminus V_{n-1}$ is contained in the open set $V_{n+1} \setminus \overline{V_{n-2}}$. By Corollary 5.6.3, we can find $f_n \in X$ such that f_n equals 1 on $\overline{V_n} \setminus V_{n-1}$ and vanishes outside of $V_{n+1} \setminus \overline{V_{n-2}}$. The sequence (f_n) need not be disjoint as both f_n and f_{n+1} may have support in $V_{n+1} \setminus \overline{V_n}$. However, each of the three sequences (f_{3n}), (f_{3n-1}), and (f_{3n-2}) is disjoint and bounded above by $\mathbb{1}$. By assumption,

$$f \coloneqq \bigl(\sup f_{3n}\bigr) \vee \bigl(\sup f_{3n-1}\bigr) \vee \bigl(\sup f_{3n-2}\bigr) = \sup_n f_n$$

exists in X. Clearly, $0 \leqslant f \leqslant \mathbb{1}$.

We claim that $f = \mathbb{1}_{\overline{U}}$ so that $\overline{U}$ is clopen; this would complete the proof. If $t \in U$ then we can find n such that $t \in V_n \setminus V_{n-1}$. It follows that $f_n(t) = 1$, hence $f(t) = 1$. Hence, f equals 1 on U and, therefore, on $\overline{U}$. On the other hand, since the support of each f_n is contained in U, f vanishes outside $\overline{U}$ by Exercise 5.6.19. □

As usual, for extremally disconnected K's we get the best of all conditions. We repeatedly use Proposition 5.6.18 in the proof.

5.6.22 Theorem. *TFAE:*

(i) *K is extremally disconnected;*

(ii) *$C(K)$ has PP;*

(iii) *Every dense sublattice of $C(K)$ has PP;*

(iv) *There exists a dense sublattice of $C(K)$ with PP;*

(v) *There exists a dense disjointly order complete sublattice of $C(K)$.*

Proof. (iii)⇒(ii)⇒(iv) is straightforward.

(i) implies (ii), (iii), and (v): Suppose that K is extremally disconnected. Then $C(K)$ is order complete and, therefore, has PP; this yields (ii) and (v). To prove (iii), let X be a dense sublattice of $C(K)$. Every band in X is of the form $J^X_{\overline{U}}$ for some open set U. The ideal $J_{\overline{U}}$ in $C(K)$ is a band, hence a projection band, which implies that $\overline{U}$ is clopen. It follows that $J^X_{\overline{U}}$ is a projection band in X.

(iv)⇒(i) Let X be a dense sublattice of $C(K)$ with PP and U an open subset of K. Then $J^X_{\overline{U}}$ is a band, hence a projection band, hence $\overline{U}$ is clopen.

(v)⇒(i) Let X be a dense disjointly order complete sublattice of $C(K)$. By Remark 5.6.4, we may assume that $\mathbb{1} \in X$. Let U be a non-empty open subset of K; we want to show that $\overline{U}$ is clopen. By Theorem 5.6.21(iv), K is basically and, therefore, totally disconnected, so that U contains a non-empty clopen subset. Let $\mathcal{G}$ be a maximal disjoint collection of clopen subsets of U; such collections exist by Zorn's Lemma. Note that $\bigcup\mathcal{G}$ is dense in U because otherwise we could find a non-empty clopen set in $U \setminus \big(\bigcup\mathcal{G}\big)$, which would contradict the maximality of $\mathcal{G}$. By Corollary 5.6.3, $\mathbb{1}_V \in X$ for every $V \in \mathcal{G}$. Since the collection $\{\mathbb{1}_V : V \in \mathcal{G}\}$ is disjoint, the assumption yields existence of $f \coloneqq \sup\{\mathbb{1}_V : V \in \mathcal{G}\}$. We claim that $f = \mathbb{1}_{\overline{U}}$; this would imply that $\overline{U}$ is clopen.

Clearly, $0 \leqslant f \leqslant \mathbb{1}$. Observe that f equals one on each V, hence on $\bigcup\mathcal{G}$. Since this union is dense in U, f equals 1 on U and, therefore, on $\overline{U}$. On the other hand, f vanishes outside $\overline{U}$ by Exercise 5.6.19. □

5.6.2 PP/PPP and disjoint order completeness

We will now apply preceding results to more general classes of vector lattices. Throughout this subsection, X stands for a vector lattice. The following result is essentially an application of Exercise 5.2.20.

5.6.23 Proposition. *A vector lattice X has PPP iff every principal ideal in it has PPP. Furthermore, X has PP iff every principal ideal has PP.*

Proof. It is easy to see that each of the four properties implies that X is Archimedean. Suppose that X has PP. Let $e \in X_+$ and let B be a band in I_e. Then $B = I_e \cap B^{dd}$ by Exercise 5.2.20(i), where B^{dd} is computed in X. Since B^{dd} is a projection band in X, B is a projection band in I_e by Exercise 5.2.20(iv). It follows that every principal ideal in X has PP. Using Exercise 5.2.20(vii), we can prove in a similar fashion that if X has PPP then every principal ideal in X has PPP.

Suppose that every principal ideal in X has PP and $e \in X_+$. Then $B \cap I_e$ is a band in I_e by Exercise 5.2.20(ii). By assumption, $B \cap I_e$ is a projection band in I_e. It follows from Exercise 5.2.20(v) that B is a projection band in X. Hence, X has PP.

Suppose that every principal ideal in X has PPP. In order to show that X has PPP, it suffices, by Exercise 5.2.16, to show that $\sup_n x \wedge na$ exists in X for any $a, x \in X_+$. Find $e \in X_+$ such that $a, x \in I_e$; e.g., take $e = a \vee x$. By assumption, $\sup_n x \wedge na$ exists in I_e and, therefore, in X.

□

5.6.24 Exercise. A vector lattice X is disjointly order complete iff so is every principal ideal in X. Same for disjoint σ-completeness.

Recall that every order complete vector lattice is disjointly order complete and has PP. Similarly, every σ-order complete vector lattice is disjointly σ-order complete and has PPP.

5.6.25 Theorem. *Let X be an Archimedean vector lattice. If X is disjointly order complete then it has PP. If X is disjointly σ-order complete then it has PPP.*

Proof. Suppose that X is disjointly order complete. By Proposition 5.6.23, it suffices to prove that every principal ideal I_e in X has PP. By the preceding exercise, I_e is disjointly order complete. By Theorem 4.1.9, I_e is lattice isomorphic to a dense sublattice of $C(K)$ for some compact space K. Now apply the equivalence (v)⇔(iii) in Theorem 5.6.22.

The proof of the second claim is similar; use the equivalence (iv)⇔(iii) in Theorem 5.6.21. □

5.6.26 Theorem (Veksler-Geller)**.** *Let X be a uniformly complete Archimedean vector lattice. TFAE:*

(i) *X is order complete;*

(ii) *X is disjointly order complete;*

(iii) *X has PP.*

Proof. Order completeness trivially implies disjoint order completeness, and the latter implies PP by the preceding theorem. Suppose X has PP. By Proposition 3.3.18, it suffices to prove that every principal ideal I_e in X is order complete. By Proposition 5.6.23, I_e has PP. Since X is uniformly complete, by Theorem 4.1.9, I_e is lattice isomorphic to $C(K)$ for some compact space K. Now Theorem 5.6.22 implies that K is extremally disconnected and, therefore, $C(K)$ is order complete. □

Analogously, one can prove the following:

5.6.27 Theorem. *Let X be a uniformly complete Archimedean vector lattice. TFAE:*

(i) *X is σ-order complete;*

(ii) *X is disjointly σ-order complete;*

(iii) *X has PPP.*

Recall that a σ-order complete vector lattice is Archimedean and uniformly complete. Recall also that every vector lattice with PPP is Archimedean. We now get the following:

5.6.28 Corollary. *A vector lattice is order complete iff it is uniformly complete and has PP. A vector lattice is σ-order complete iff it is uniformly complete and has PPP.*

Since Banach lattices are uniformly complete and Archimedean, we get the following:

5.6.29 Corollary. *A Banach lattice X is order complete iff it is disjointly order complete iff it has PP. Furthermore, X is σ-order complete iff it is disjointly σ-order complete iff it has PPP.*

Recall that for a set A in an Archimedean vector lattice, the band $B(A)$ generated by A equals A^{dd}; see Corollary 5.1.9. Recall also that by Theorem 5.2.15, a band B is a projection band iff $\sup[0,x]\cap B$ exists for every $x \in X_+$. We now provide two theorems due to Bilokopytov that characterize projection bands in terms of disjoint sets. The first theorem is stated for ideals that are generated by countable sets; it is easy to see that in a Banach lattice, such ideals are just principal ideals (use Exerise 2.2.1(viii)).

5.6.30 Theorem. *Let J be a countably generated ideal in an Archimedean vector lattice X, and $B = B(J)$. B is a projection band iff every disjoint sequence in J_+ that is order bounded in X is also order bounded in B.*

Proof. The forward implication is straightforward; we only prove the converse.

We reduce the theorem to the case when X has a strong unit. Indeed, suppose we know the theorem for this special case. By Exercise 5.2.20(v), it suffices to prove that $B \cap I_e$ is a principle band in I_e for every $e \in X_+$. Fix $e \in X_+$. Let $J_0 = J \cap I_e$ and $B_0 = B \cap I_e$. Then B_0 is the band generated by J_0 in I_e by Exercise 5.2.20(vi). If a disjoint sequence (x_n) in J_0^+ is bounded above by some $u \in J_0$, then, by the assumption, it is bounded above by some $b \in B$, hence by $u \wedge b \in B_0$. By the special case, B_0 is a projection band in I_e.

So we now assume that X has a strong unit e. We identify X with a dense sublattice of $C(K)$ for some compact K, with e corresponding to $\mathbb{1}$. Suppose that $J = I\big(\{g_n\}\big)$ for some sequence (g_n). By Exercise 5.6.17, $J^d = J^X_{\overline{U}}$, where $U = \operatorname{supp} J = \bigcup_{n=1}^\infty \operatorname{supp} g_n$. Since $\operatorname{supp} g_n$ is an F_σ set for every n by Exercise 2.1.24(ii), then so is U. It suffices to show that $\overline{U}$ is clopen; this would imply that J^d and, therefore, B are projection bands by Proposition 5.6.18.

Following the argument in the proof of (iv)⇒(i) in Theorem 5.6.21 on page 5.6.1, we construct a sequence (f_n) in X_+ such that $0 \leqslant f_n \leqslant \mathbb{1}$ and $\overline{\operatorname{supp} f_n} \subseteq U$ for each n, for every $t \in U$ there exists n such that $f_n(t) = 1$, and each of the three sequences (f_{3n}), (f_{3n-1}), and (f_{3n-2}) is disjoint. By Proposition 5.6.11 we have $f_n \in J$ for every n. Since each of the three subsequences is bounded by e, the assumption of the theorem implies that (f_n) is bounded above by some $h \in B$. It follows that h is greater than equal to 1 on U and, therefore, on $\overline{U}$. On the other hand, it follows from $h \in B = J^{dd}$ and Exercise 5.6.17 that h vanishes on $K \setminus \overline{U}$. Therefore, $\overline{U}$ is clopen. □

5.6.31 Theorem. *Let B be a band in an Archimedean vector lattice X. B is a projection band iff every subset of B_+ that is order bounded in X is also order bounded in B.*

Proof. As in the proof of Theorem 5.6.30, the forward implication is straightforward and we may assume that X contains a strong unit. Indeed, it suffices to prove that for every $e \in X_+$, the set $B \cap I_e$, which is a band in I_e, is a projection band in I_e. On the other hand, ever disjoint subset A of $(I_e \cap B)_+$ that is order bounded in I_e is order bounded in B (by the assumption) and, therefore, in $I_e \cap B$.

So we now assume that X has a strong unit e. Let A be a maximal disjoint collection in $[0, e] \cap B$. By assumption, $A \leqslant u$ for some $u \in B$. We claim that u is a weak unit in B. Indeed, otherwise, we could find $y \in B_+$ such that $y \perp u$ and, therefore, $y \perp A$. Scaling y, if necessary, we may also assume that $y \in [0, e]$. This would contradict maximality of A.

Since u is a weak unit in B, we have $B = B_u = B(I_u)$. Applying Theorem 5.6.30 to I_u, we conclude that B_u is a projection band. □

Chapter 6

Operators between vector lattices

6.1 Regular and order bounded operators

Throughout this chapter, X and Y will be two vector lattices. As in Section 1.6, we will write $\mathcal{L}(X,Y)$ for the vector space of all (linear) operators from X to Y, and $\mathcal{L}(X,Y)_+$ for the cone of all positive operators in $\mathcal{L}(X,Y)$. With this cone, $\mathcal{L}(X,Y)$ is an ordered vector space. However, we will see that $\mathcal{L}(X,Y)$ need not be a lattice. Moreover, $\mathcal{L}(X,Y)_+$ need not even be generating, i.e., it is possible that not every operator is the difference of two positive operators. Recall that an operator is regular if it is the difference of two positive operators. By Exercise 1.2.5, TFAE:

- T is regular;
- $T \leqslant S$ for some $S \geqslant 0$;
- $-S \leqslant T \leqslant S$ for some $S \geqslant 0$;
- $\pm T \leqslant S$ for some $S \geqslant 0$.

6.1.1 Exercise. For $T, S \in \mathcal{L}(X,Y)$ with $S \geqslant 0$ we have $\pm T \leqslant S$ iff $|Tx| \leqslant S|x|$ for every $x \in X$.

We write $\mathcal{L}_r(X,Y)$ for the set of regular operators from X to Y; hence $\mathcal{L}_r(X,Y) = \mathcal{L}(X,Y)_+ - \mathcal{L}(X,Y)_+$. We will see that for certain choices of X and Y the space $\mathcal{L}(X,Y)$ happens to be a lattice; it is clear that in this case every operator is regular, i.e., $\mathcal{L}_r(X,Y) = \mathcal{L}(X,Y)$.

An operator $T \in \mathcal{L}(X,Y)$ is said to be ***order bounded*** if it maps order bounded sets to order bounded sets. Since order bounded sets are exactly subsets of order intervals, T is order bounded iff the image under T of every order interval is contained in an order interval. We write $\mathcal{L}_{\mathrm{ob}}(X,Y)$ for the set of all order bounded operators in $\mathcal{L}(X,Y)$. When $X = Y$ we will write $\mathcal{L}(X)$, $\mathcal{L}(X)_+$, $\mathcal{L}_r(X)$, and $\mathcal{L}_{\mathrm{ob}}(X)$ instead of $\mathcal{L}(X,X)$, $\mathcal{L}(X,X)_+$, $\mathcal{L}_r(X,X)$, and $\mathcal{L}_{\mathrm{ob}}(X,X)$ respectively.

6.1.2 Exercise. An operator T is order bounded iff

$$\forall x \in X_+ \quad \exists y \in Y_+ \quad T[0,x] \subseteq [-y,y].$$

6.1.3 Exercise. $\mathcal{L}_r(X,Y)$ and $\mathcal{L}_{\mathrm{ob}}(X,Y)$ are (linear) subspaces of $\mathcal{L}(X,Y)$ and

$$\mathcal{L}(X,Y)_+ \subset \mathcal{L}_r(X,Y) \subseteq \mathcal{L}_{\mathrm{ob}}(X,Y) \subseteq \mathcal{L}(X,Y). \tag{6.1}$$

Later, we will present examples showing that each of these inclusions can be proper. On the other hand, we will prove the Riesz-Kantorovich Theorem which asserts that if Y is order complete then $\mathcal{L}_r(X,Y) = \mathcal{L}_{\mathrm{ob}}(X,Y)$, and this space is itself an order complete vector lattice.

Even when $\mathcal{L}(X,Y)$ fails to be a lattice, $S \vee T$ and $S \wedge T$ may still exist for some pairs S and T in $\mathcal{L}(X,Y)$. Here, as before, by $S \vee T$ and $S \wedge T$ we mean the least upper bound and, respectively, the greatest lower bound of S and T in $\mathcal{L}(X,Y)$. It is easy to see that $S \vee T$ exists iff $S \wedge T$ exists, and, in this case, the identities in Exercise 1.3.3 remain valid for S and T. Furthermore, we define $|T| = T \vee (-T)$, $T^+ = T \vee 0$ and $T^- = (-T) \vee 0$ if they exist. In this case, it can be easily verified that the properties described in Exercises 1.3.6, 1.3.8, and 1.3.11, and 1.3.7 remain valid.

6.1.4. In particular, T^+, T^-, and $|T|$ exist or fail to exist simultaneously. If they exist, they can be expressed via each other as in Exercise 1.3.8(ii).

6.1.5 Exercise. If $|T|$ exists then T is regular and $|Tx| \leqslant |T||x|$ for every x.

6.1.6 Example. Let $X = \mathbb{R}^N$ and $Y = \mathbb{R}^M$. Then $\mathcal{L}(X,Y)$ can be identified with the space of all $M \times N$ matrices. Suppose that $S, T \in \mathcal{L}(X,Y)$. Then we can write them in matrix form $S = (s_{ij})$ and $T = (t_{ij})$. It is easy to see that $S \leqslant T$ iff $s_{ij} \leqslant t_{ij}$ for all i and j. Furthermore, $S \wedge T$ and $S \vee T$ exist and are given by the matrices $(s_{ij} \wedge t_{ij})$ and $(s_{ij} \vee t_{ij})$, respectively. In other words, $\mathcal{L}(X,Y)$ is a lattice, with the lattice operations defined entry-wise. In particular, $|T|$ is given by $(|t_{ij}|)$, and every operator is regular. Hence, $\mathcal{L}_r(X,Y) = \mathcal{L}_{\mathrm{ob}}(X,Y) = \mathcal{L}(X,Y)$.

The following result asserts, in particular, that an operator is order bounded iff it is ***uniformly continuous***:

6.1.7 Theorem. *[TT20] Suppose that Y is Archimedean and $T \in \mathcal{L}(X,Y)$. TFAE:*

(i) *T is order bounded;*

(ii) *$x_\alpha \xrightarrow{\mathrm{u}} x$ implies $Tx_\alpha \xrightarrow{\mathrm{u}} Tx$ for every (x_α) and x in X;*

(iii) *$x_\alpha \xrightarrow{\mathrm{u}} x$ implies $Tx_\alpha \xrightarrow{\mathrm{o}} Tx$ for every (x_α) and x in X.*

Proof. (i)⇒(ii) Suppose that T is order bounded and $x_\alpha \xrightarrow{\mathrm{u}} x$; we need to show that $Tx_\alpha \xrightarrow{\mathrm{u}} Tx$. WLOG, $x=0$. There exists $e \in X_+$ such that for every $\varepsilon>0$ there exists α_0 such that $|x_\alpha| \leqslant \varepsilon e$ whenever $\alpha \geqslant \alpha_0$. Since T is order bounded, we can find $a \in Y_+$ such that $T[-e,e] \subseteq [-a,a]$. It follows that $|Tx_\alpha| \leqslant \varepsilon a$ for all $\alpha \geqslant \alpha_0$, hence $Tx_\alpha \xrightarrow{\mathrm{u}} 0$.

(ii)⇒(iii) is trivial because uniform convergence implies order convergence.

(iii)⇒(i) Take $e \in X_+$; we need to show that $T[0,e]$ is order bounded in Y. Let $\Lambda = \bigl\{(n,x) \,:\, n \in \mathbb{N},\ x \in [0,e]\bigr\}$, pre-ordered by the first component: $(n,x) \leqslant (m,y)$ whenever $n \leqslant m$. It is easy to see that Λ is a directed set. Consider the following net in X indexed by Λ: $v_{(n,x)} = \frac{1}{n}x$. This net converges uniformly to zero because $0 \leqslant v_{(n,x)} \leqslant \frac{1}{n}e$. By assumption, the net $(Tv_{(n,x)})$ is order null; in particular, it has an order bounded tail, i.e., there exist $n_0 \in \mathbb{N}$, $x_0 \in [0,e]$, and $u \in Y_+$ such that $|Tv_{(n,x)}| \leqslant u$ whenever $(n,x) \geqslant (n_0,x_0)$. Let $x \in [0,e]$; it follows from $(n_0+1,x) \geqslant (n_0,x_0)$ that $|Tv_{(n_0+1,x)}| \leqslant u$. Since $v_{(n_0+1,x)} = \frac{1}{n_0+1}x$, we get $|Tx| \leqslant (n_0+1)u$. □

6.1.8 Example. The theorem fails without the Archimedean assumption. Let $X=\mathbb{R}$, Y be $\mathbb{R}^2$ with the lexicographic order, and $T\colon X \to Y$ be given by $Tx=(x,0)$. Then T is order bounded, yet it fails to be u-to-o continuous.

Theorem 6.1.7 provides an alternative proof that if the quotient of a vector lattice X over an ideal J is Archimedean then J is uniformly closed, cf. Theorem 3.4.7. Indeed, in this case, the quotient map Q is an order bounded operator between Archimedean vector lattices, hence uniformly continuous. It follows that $J=\ker Q$ is uniformly closed.

In Section 13.4, we will consider ***sequentially uniformly continuous*** operators, that is, the operators that preserve uniform convergence of ***sequences***.

6.1.1 Order bounded operators on Banach lattices

The following theorem is one of the gems of the theory of Banach lattices.

6.1.9 Theorem. *Every order bounded (in particular, every positive or regular) operator from a Banach lattice to a normed lattice is bounded.*

Proof. Let $T \in \mathcal{L}_{\mathrm{ob}}(X,Y)$, where X is a Banach lattice and Y is a normed lattice. By Theorem 6.1.7, T is uniformly continuous. Suppose that $x_n \to 0$; let (x_{n_k}) be a subsequence. By Theorem 2.3.7, there exists a further subsequence $(x_{n_{k_i}})$ such that $x_{n_{k_i}} \xrightarrow{\mathrm{u}} 0$. This yields $Tx_{n_{k_i}} \xrightarrow{\mathrm{u}} 0$ and, therefore $Tx_{n_{k_i}} \to 0$. Thus, every subsequence of (Tx_n) has a further subsequence which converges in norm to zero; it follows that $Tx_n \to 0$; see Proposition A.2.1. □

Here is an alternative proof of the theorem. Again, observe that T is uniformly continuous. Recall that, by Corollary 2.3.9, every closed set in a normed lattice is uniformly closed; the converse is true for Banach lattices. If $F \subseteq Y$ is closed, then it is uniformly closed, hence $T^{-1}(F)$ is uniformly closed in X and, therefore, closed. It follows that T is continuous.

Let X and Y be two Banach lattices. Recall that $L(X,Y)$ stands for the space of all norm bounded operators from X to Y. Theorem 6.1.9 asserts that $\mathcal{L}_{\mathrm{ob}}(X,Y) \subseteq L(X,Y)$. This inclusion may be proper: in 6.4.1, we will see an example of an operator which is bounded but not order bounded. In Chapter 13, we will show that $\mathcal{L}_r(X,Y)$ and $\mathcal{L}_{\mathrm{ob}}(X,Y)$ are Banach spaces under appropriate norms.

Theorem 6.1.9 leads to numerous important consequences. Let $T\colon X \to Y$ be a surjective lattice isomorphism between Banach lattices. Then T is positive, so the theorem asserts that T is bounded. It follows that T is a norm isomorphism. We can summarize this as follows:

6.1.10 Corollary. *If two Banach lattices are lattice isomorphic (as vector lattices) then they are isomorphic (as normed spaces).*

This implies uniqueness of a lattice norm:

6.1.11 Corollary. *Any two complete lattice norms on a given vector lattice are equivalent.*

Proof. Suppose that $\|\cdot\|_1$ and $\|\cdot\|_2$ are complete lattice norms on a vector lattice X. The identity operator from $\bigl(X, \|\cdot\|_1\bigr)$ to $\bigl(X, \|\cdot\|_2\bigr)$ is a lattice isomorphism between two Banach lattices, hence it is a (norm) isomorphism. It follows that the norms are equivalent. $\square$

6.1.12 Example. The preceding corollary may fail for norms that are not complete. For example, let $1 \leqslant p \leqslant \infty$; let X be ℓ_1 viewed as a sublattice of ℓ_p, i.e., $X = \bigl(\ell_1, \|\cdot\|_{\ell_p}\bigr)$. This gives a continuum of pair-wise non-equivalent lattice norms on ℓ_1.

6.1.13 Exercise. Let X be a normed lattice and Y a Banach lattice. If Y is a sublattice of X then the inclusion map is norm continuous.

6.1.14 Exercise. If $T\colon X \to Y$ is a positive bounded operator between normed lattices then $\|T\| = \sup\limits_{x \in S_X^+} \|Tx\|$.

6.1.15 Exercise. Let $T, S \in L(X,Y)$ where X and Y are among ℓ_p with $1 \leqslant p < \infty$ or c_0. Recall that by Example 1.6.2, $T \leqslant S$ iff $t_{ij} \leqslant s_{ij}$ for all $i, j \in \mathbb{N}$. The modulus of T exists iff the infinite matrix $\bigl(|t_{ij}|\bigr)$ represents a linear operator; in this case, the resulting operator is norm bounded and is precisely $|T|$.

6.1.16 Example. We claim that $\mathcal{L}_r(\ell_1)=\mathcal{L}_{\rm ob}(\ell_1)=L(\ell_1)$. That is, a linear operator $T\colon \ell_1\to\ell_1$ is norm bounded iff it is order bounded iff it is regular. Indeed, we already know that

$$T\text{ is regular}\quad\Rightarrow\quad T\text{ is order bounded}\quad\Rightarrow\quad T\text{ is norm bounded.}$$

It is easy to see that an infinite matrix (t_{ij}) corresponds to a norm bounded operator on ℓ_1 iff $\sup_j\sum_{i=1}^\infty\lvert t_{ij}\rvert<\infty$. Therefore, if T is a norm bounded operator on ℓ_1 then the matrix $\bigl(\lvert t_{ij}\rvert\bigr)$ also corresponds to a bounded operator, hence $\lvert T\rvert$ exists by Exercise 6.1.15, and, therefore, T is regular. Note that, in this case, $\lVert T\rVert=\sup_j\sum_{i=1}^\infty\lvert t_{ij}\rvert=\bigl\lVert\lvert T\rvert\bigr\rVert$.

In fact, we will show in Theorem 13.3.5 that every norm bounded operator $T\colon L_1(\mu)\to L_1(\nu)$ is regular.

6.2 Extension Lemma

The following lemma is a valuable tool for constructing positive operators.

6.2.1 Extension Lemma. *Let X and Y be vector lattices, with Y Archimedean, and $\tau\colon X_+\to Y_+$ an additive map. Then τ extends uniquely to a linear operator $T\colon X\to Y$. Furthermore, T is positive and satisfies $Tx=\tau x^+-\tau x^-$ for all $x\in X$.*

Proof. It is easy to see that if τ extends to a linear operator $T\colon X\to Y$ then T is positive and satisfies $Tx=Tx^+-Tx^-=\tau x^+-\tau x^-$ for all $x\in X$;[1] the latter yields that the extension is unique.

Let $\tau\colon X_+\to Y_+$ be an additive map; extend τ to $T\colon X\to Y$ via $Tx=\tau y-\tau z$ whenever $x=y-z$ for some $y,z\geqslant 0$. This is well defined because if we also have $x=y'-z'$ for some $y',z'\geqslant 0$ then $y+z'=y'+z$, which yields $\tau y+\tau z'=\tau y'+\tau z$, so that $\tau y-\tau z=\tau y'-\tau z'$. Note also that $Tx=\tau x^+-\tau x^-$ for every $x\in X$.

It is easy to see that T is additive: given x_1 and x_2, we write $x_1=y_1-z_1$ and $x_2=y_2-z_2$ for some $y_1,y_2,z_1,z_2\in X_+$; it follows from $x_1+x_2=(y_1+y_2)-(z_1+z_2)$ that

$$T(x_1+x_2)=\tau(y_1+y_2)-\tau(z_1+z_2)=\tau y_1+\tau y_2-\tau z_1-\tau z_2=Tx_1+Tx_2.$$

Next, we observe that T is monotone: if $x\leqslant y$ then, using additivity and the fact that $y-x\geqslant 0$, we get

$$Ty=T\bigl(x+(y-x)\bigr)=Tx+T(y-x)=Tx+\tau(y-x)\geqslant Tx.$$

[1] Recall that Tx^+ stands for $T(x^+)$ as opposed to $(Tx)^+$.

We will now show that T is homogeneous. Let $x \in X$. If $n \in \mathbb{N}$ then $T(nx) = nTx$ by additivity. Also, $T(-nx) + T(nx) = T0 = 0$, so that $T(-nx) = -T(nx) = -nTx$. Hence, $T(nx) = nTx$ for every $n \in \mathbb{Z}$. Let now $n \in \mathbb{Z}$ and $m \in \mathbb{N}$, then

$$mT\big(\tfrac{n}{m}x\big) = T\big(m\tfrac{n}{m}x\big) = T(nx) = nTx,$$

so that $T\big(\frac{n}{m}x\big) = \frac{n}{m}Tx$. Thus, $T(qx) = qTx$ for every $q \in \mathbb{Q}$.

Now let $\lambda \in \mathbb{R}$ and $x \geqslant 0$. Take sequences (p_n) and (q_n) in $\mathbb{Q}$ so that $p_n \uparrow \lambda$ and $q_n \downarrow \lambda$. Then $p_n x \leqslant \lambda x \leqslant q_n x$ for every n, so that

$$p_n Tx = T(p_n x) \leqslant T(\lambda x) \leqslant T(q_n x) = q_n Tx.$$

Since Y is Archimedean, we have $p_n Tx \uparrow \lambda Tx$ and $q_n Tx \downarrow \lambda Tx$ by Exercise 2.4.12, hence $T(\lambda x) = \lambda Tx$.

Finaly, suppose that $x \in X$ and $\lambda \in \mathbb{R}$ are arbitrary. Take any $y, z \in X_+$ such that $x = y - z$. Then

$$\begin{aligned} T(\lambda x) = T\big(\lambda y + (-\lambda z)\big) = T(\lambda y) + T(-\lambda z) \\ = \lambda Ty + (-\lambda)Tz = \lambda(Ty - Tz) = \lambda Tx \end{aligned}$$

by the definition of T. $\square$

6.2.2 Remark. We didn't actually use lattice structures of X and Y in the proof. The lemma (and the proof) remain valid if X and Y are just ordered vector spaces with X_+ generating and Y Archimedean.

6.2.3 Corollary. *Suppose that Y is Archimedean and $\tau\colon X_+ \to Y_+$ an additive bijection. Then τ extends to a lattice isomorphism from X onto Y.*

Proof. By Extension Lemma 6.2.1, τ extends to a positive operator $T\colon X \to Y$ via $Tx = \tau(x^+) - \tau(x^-)$. It follows from $Y_+ = \operatorname{Range}\tau \subseteq \operatorname{Range} T$ that T is onto. To show that T is one-to-one, suppose that $Tx = 0$, then $\tau(x^+) = \tau(x^-)$. Since τ is one-to-one, we have $x^+ = x^-$, so that $x = 0$.

Hence, T is a bijection. It is now easy to see that $x \geqslant 0$ iff $Tx \geqslant 0$. It follows from Theorem 1.6.13 that T is a lattice isomorphism. $\square$

6.3 Riesz-Kantorovich Theorem

The following theorem is probably the most important and the most useful fact in the theory of vector lattices.

6.3.1 Theorem (Riesz-Kantorovich)**.** *Let X and Y be vector lattices with Y order complete. Then $\mathcal{L}_r(X,Y)=\mathcal{L}_{\mathrm{ob}}(X,Y)$, and this space is itself a vector lattice, with lattice operations given by the following formulas:*

$$\begin{aligned} T^+x &= \sup T[0,x], \quad T^-x=-\inf T[0,x], && (6.2)\\ |T|x &= \sup T[-x,x]=\sup\bigl\{|Tu|\ :\ -x\leqslant u\leqslant x\bigr\}, && (6.3)\\ (S\vee T)x &= \sup\bigl\{Su+Tv\ :\ u,v\geqslant 0,\ u+v=x\bigr\},\ \textit{and} && (6.4)\\ (S\wedge T)x &= \inf\bigl\{Su+Tv\ :\ u,v\geqslant 0,\ u+v=x\bigr\} && (6.5)\end{aligned}$$

for any $S,T\in\mathcal{L}_r(X,Y)$ and any $x\in X_+$.

Proof. We already know that $\mathcal{L}_r(X,Y)\subseteq\mathcal{L}_{\mathrm{ob}}(X,Y)$. Take $T\in\mathcal{L}_{\mathrm{ob}}(X,Y)$. We start by showing that T^+ exists and is given by (6.2). Observe that $T[0,x]$ is order bounded for every $x\in X_+$, hence $\tau x\coloneqq\sup T[0,x]$ exists because Y is order complete. Then $\tau\colon X_+\to Y_+$. If $x,y\in X_+$, by Riesz Decomposition Theorem we have $[0,x]+[0,y]=[0,x+y]$. It follows that $T[0,x]+T[0,y]=T[0,x+y]$, hence, $\tau x+\tau y=\tau(x+y)$ by Exercise 1.3.5(i).

By the Extension Lemma 6.2.1, τ extends to a positive operator $S\in\mathcal{L}(X,Y)$ (the lemma applies because being order complete, Y is Archimedean). For every $x\in X_+$ we have $Sx=\tau x\geqslant Tx$, hence $S\geqslant T$. In order to prove that $S=T^+$ it is left to show that $S\leqslant R$ whenever $R\geqslant T$ and $R\geqslant 0$ for some $R\in\mathcal{L}(X,Y)$. But, indeed, if R is such an operator then for each $x\in X_+$ and each $u\in[0,x]$ we have $Tu\leqslant Ru\leqslant Rx$, so that $Sx=\sup T[0,x]\leqslant Rx$, hence $S\leqslant R$. Thus, $S=T^+$.

It follows from 6.1.4 that $|T|$ and T^- also exist, with $|T|=-T+2T^+$ and $T^-=T^+-T$. Hence, for every $x\in X_+$ we have

$$\begin{aligned} T^-x=T^+x-Tx=\sup\bigl\{T(u-x)\ :\ 0\leqslant u\leqslant x\bigr\}&\\ =-\inf\bigl\{Tv\ :\ 0\leqslant v\leqslant x\bigr\}=-\inf T[0,x],&\end{aligned}$$

(here $v=x-u$), and

$$\begin{aligned} |T|x=-Tx+2T^+x=\sup\bigl\{T(-x+2u)\ :\ 0\leqslant u\leqslant x\bigr\}&\\ =\sup\bigl\{Tv\ :\ -x\leqslant v\leqslant x\bigr\}=\sup T[-x,x].&\end{aligned}$$

Furthermore,

$$\begin{aligned} |T|x=\sup\bigl\{Tv\ :\ -x\leqslant v\leqslant x\bigr\}=\sup\bigl\{(Tu)\vee(-Tu)\ :\ -x\leqslant u\leqslant x\bigr\}&\\ =\sup\bigl\{|Tu|\ :\ -x\leqslant u\leqslant x\bigr\}.&\end{aligned}$$

Since $T = T^+ - T^-$, we conclude that T is regular, hence $\mathcal{L}_r(X,Y) = \mathcal{L}_{\mathrm{ob}}(X,Y)$ and this space is a vector lattice.

Suppose that $S, T \in \mathcal{L}_r(X,Y)$ and $x \in X_+$. Since $\mathcal{L}_r(X,Y)$ is a vector lattice, we have $S \vee T = T + (S-T) \vee 0 = T + (S-T)^+$ by Exercise 1.3.3(iii). By (6.2) we have

$$\begin{aligned}(S \vee T)x &= Tx + \sup\bigl\{(S-T)u \,:\, 0 \leqslant u \leqslant x\bigr\} \\ &= \sup\bigl\{Su + T(x-u) \,:\, 0 \leqslant u \leqslant x\bigr\} = \sup\bigl\{Su + Tv \,:\, u, v \geqslant 0,\ u+v = x\bigr\}.\end{aligned}$$

The formula for $S \wedge T$ is proved in a similar fashion. □

Formulas (6.2)–(6.5) are often referred to as the ***Riesz-Kantorovich formulas***. The assumption that $x \geqslant 0$ is not very restrictive: once we know how to compute $|T|x$ for a positive x, we can now use $|T|x = |T|x^+ - |T|x^-$ for an arbitrary x. It is often convenient to state the formulas for $S \vee T$ and $S \wedge T$ as follows:

$$(S \vee T)x = \sup_{0 \leqslant u \leqslant x} Su + T(x-u) \quad \text{and} \quad (S \wedge T)x = \inf_{0 \leqslant u \leqslant x} Su + T(x-u).$$

The following result naturally complements Theorem 6.3.1 by characterizing infinite lattice operations in $\mathcal{L}_{\mathrm{ob}}(X,Y)$. Essentially, it says that suprema and infima of nets of operators are computed point-wise.

6.3.2 Theorem (Riesz-Kantorovich). *Let X and Y be vector lattices with Y order complete. Then $\mathcal{L}_{\mathrm{ob}}(X,Y)$ is order complete. Moreover, $T_\alpha \uparrow T$ in $\mathcal{L}_{\mathrm{ob}}(X,Y)$ iff $T_\alpha x \uparrow Tx$ for every $x \in X_+$.*

Proof. Claim: If $0 \leqslant T_\alpha \uparrow \leqslant S$ in $\mathcal{L}_{\mathrm{ob}}(X,Y)$ then there exists $T \coloneqq \sup T_\alpha$ in $\mathcal{L}_{\mathrm{ob}}(X,Y)$ and $T_\alpha x \uparrow Tx$ for every $x \in X_+$.

Indeed, for each $x \in X_+$ we have $T_\alpha x \uparrow \leqslant Sx$ in Y. Since Y is order complete, $\sup T_\alpha x =: \tau x$ exists. Then $\tau\colon X_+ \to Y_+$. We will now verify that τ is additive. Given $x, y \in X$, we have

$$\tau(x+y) = \sup_\alpha T_\alpha(x+y) \leqslant \sup_\alpha T_\alpha x + \sup_\alpha T_\alpha y = \tau x + \tau y.$$

On the other hand, for every α_1 and α_2 there exists an α such that $\alpha_1, \alpha_2 \leqslant \alpha$, so that

$$T_{\alpha_1} x + T_{\alpha_2} y \leqslant T_\alpha(x+y) \leqslant \tau(x+y).$$

Taking sup over α_1 and α_2, we get $\tau x + \tau y \leqslant \tau(x+y)$.

By the Extension Lemma 6.2.1, τ can be extended to a positive operator $T \in \mathcal{L}(X,Y)$. For every $x \in X_+$ and for every α we have $T_\alpha x \leqslant \tau x = Tx \leqslant Sx$, hence

$T_\alpha \leqslant T \leqslant S$. This remains true for every choice of an upper bound S of (T_α). It follows that $T = \sup T_\alpha$. This proves the claim.

The claim, together with Exercise 2.1.3(iii), implies that $\mathcal{L}_{\mathrm{ob}}(X, Y)$ is order complete. It also proves that if $0 \leqslant T_\alpha \uparrow T$ then $T_\alpha x \uparrow Tx$ for every $x \geqslant 0$.

To remove the assumption $0 \leqslant T_\alpha$ in the preceding sentence, suppose that $T_\alpha \uparrow T$ in $\mathcal{L}_{\mathrm{ob}}(X, Y)$ and $x \in X_+$. Fix an index α_0, and consider the tail $(T_\alpha - T_{\alpha_0})_{\alpha \geqslant \alpha_0}$. For this tail, we have $0 \leqslant (T_\alpha - T_{\alpha_0}) \uparrow (T - T_{\alpha_0})$, hence $(T_\alpha - T_{\alpha_0})x \uparrow (T - T_{\alpha_0})x$, and, therefore, $T_\alpha x \uparrow Tx$.

The converse implication is straightforward. □

In the rest of this section, we provide variants and refinements of Riesz-Kantorovich formulas for certain special cases.

6.3.3. By iterating (6.4) and (6.5), we obtain their variants for more than two operators:

$$\begin{aligned}
\Big(\bigvee_{i=1}^n T_i\Big)(x) &= \sup\Big\{\sum_{i=1}^n T_i x_i \;:\; x_1, \dots, x_n \geqslant 0,\; x_1 + \dots + x_n = x\Big\} \\
\Big(\bigwedge_{i=1}^n T_i\Big)(x) &= \inf\Big\{\sum_{i=1}^n T_i x_i \;:\; x_1, \dots, x_n \geqslant 0,\; x_1 + \dots + x_n = x\Big\}
\end{aligned}$$

for every $x \geqslant 0$.

6.3.4. The Riesz-Kantorovich formula for $|T|$ may be re-written to represent $|T|x$ as the supremum of an increasing net, namely,

$$|T|x = \sup\Big\{\sum_{i=1}^n |Tx_i| \;:\; n \in \mathbb{N}, x_1, \dots, x_n \geqslant 0, \sum_{i=1}^n x_i = x\Big\}$$

The set in the sup may be viewed as an increasing net indexed by finite partitions of x into sums of positive vectors; it is directed by Riesz Decomposition Theorem 1.5.16. To prove this identity, observe that

$$\sum_{i=1}^n |Tx_i| \leqslant \sum_{i=1}^n |T|x_i = |T| \sum_{i=1}^n x_i = |T|x.$$

On the other hand, using (6.4), we get

$$\begin{aligned}
|T|x = \big(T \vee (-T)\big)x &\leqslant \sup\big\{Tu - Tv \;:\; u, v \geqslant 0, u + v = x\big\} \\
&\leqslant \sup\big\{|Tu| + |Tv| \;:\; u, v \geqslant 0, u + v = x\big\} \\
&\leqslant \sup\Big\{\sum_{i=1}^n |Tx_i| \;:\; n \in \mathbb{N}, x_1, \dots, x_n \geqslant 0, \sum_{i=1}^n x_i = x\Big\}.
\end{aligned}$$

Analogously, we get:

6.3.5 Exercise.

$$\Bigl(\bigvee_{i=1}^{n} T_i\Bigr)x = \sup\Bigl\{\sum_{j=1}^{m}\bigvee_{i=1}^{n} T_i x_j \;:\; x_1,\dots,x_m \geqslant 0,\; \sum_{j=1}^{m} x_j = x\Bigr\}.$$

for every $x \geqslant 0$. The set in the right hand side may be viewed as an increasing net.

PPP allows one to partition vectors into disjoint sums; consequently, it suffices to only consider disjoint decompositions in the Riesz-Kantorovich formulas for $S\vee T$ and $S\wedge T$:

6.3.6 Theorem ([Abr71]). *If X has PPP, Y is order complete, $S,T \in \mathcal{L}_{\mathrm{ob}}(X,Y)$, and $x \in X_+$ then*

$$\begin{aligned} (S\vee T)x &= \sup\bigl\{Su+Tv \;:\; u\wedge v=0,\; u+v=x\bigr\} \text{ and} && (6.6)\\ (S\wedge T)x &= \inf\bigl\{Su+Tv \;:\; u\wedge v=0,\; u+v=x\bigr\}. && (6.7)\end{aligned}$$

Proof. We first prove (6.7) in the special case when $S\wedge T=0$. In this case, it follows from the Riesz-Kantorovich formulas that $\inf\bigl\{Sy+Tz \;:\; y,z\geqslant 0,\; y+z=x\bigr\}=0$. For any $y,z\geqslant 0$ with $y+z=x$, by Lemma 5.2.25 we can find disjoint positive u and v such that $u+v=x$, $u\leqslant 2y$ and $v\leqslant 2z$. It follows that $Su+Tv\leqslant 2(Sy+Tz)$. Therefore,

$$\inf\bigl\{Su+Tv \;:\; u\wedge v=0,\; u+v=x\bigr\}=0.$$

Now let S and T be arbitrary. Since the operators $S-S\wedge T$ and $T-S\wedge T$ are disjoint by Exercise 1.5.4, applying the first part of the proof, we get

$$\begin{aligned}0 &= \inf\bigl\{(S-S\wedge T)y+(T-S\wedge T)z \;:\; y\wedge z=0,\; y+z=x\bigr\}\\ &= \inf\bigl\{Sy+Tz \;:\; y\wedge z=0,\; y+z=x\bigr\}-(S\wedge T)x.\end{aligned}$$

This proves (6.7). To prove (6.6), apply (6.7) to $-S$ and $-T$. □

6.3.7. What happens if we drop the assumption that Y is order complete in Riesz-Kantorovich Theorem 6.3.1? An inspection of the proof reveals that this assumption was only used to conclude that the suprema in the Riesz-Kantorovich formulae exist, and that Y is Archimedean, so we could use Extension Lemma 6.2.1. Therefore, if we assume that Y is Archimedean and $T\colon X\to Y$ a linear operator such that $\sup T[0,x]$ exists for every $x\in X_+$ then T^+ exists and is given by (6.2). Similarly for the rest of Riesz-Kantorovich formulae.

It is also possible that $|T|$, T^+, and T^- may still exist, but not satisfy the Riesz-Kantorovich formula: in [Ell19], M.Elliott constructed an example of a regular operator $T\colon L_1[0,1]\to C(K)$, where K is a compact space, such that $|T|$ exists but is not given by the Riesz-Kantorovich formula.

Next, we show discuss how one may "restrict" or, rather, "project" an operator to an ideal.

6.3.8 Remark. Suppose now that $T \in \mathcal{L}(X,Y)_+$, X has PPP, and $u \in X$. Put $S = TP_u$, where P_u is the band projection for u. Then $0 \leqslant S \leqslant T$, S agrees with T on B_u (and, in particular, on I_u), and vanishes on $\{u\}^d$.

6.3.9 Theorem. *Suppose that $T \in \mathcal{L}(X,Y)_+$, Y is order complete, and J is an ideal in X. Then the formulas*

$$Sx = \sup\bigl\{Tv \,:\, v \in J \text{ and } 0 \leqslant v \leqslant x\bigr\} = \sup\bigl\{T(x \wedge v) \,:\, v \in J_+\bigr\}, \quad x \in X_+$$

define $S \in \mathcal{L}(X,Y)$ such that $0 \leqslant S \leqslant T$, S agrees with T on J, and vanishes on J^d.

Proof. It is easy to see that the two suprema are taken over the same sets, hence they are equal. They exist because Y is order complete. Thus, the formula defines a map from X_+ to Y_+. Once we show that it is additive, Extension Lemma 6.2.1 will guarantee that S extends to a positive linear operator from X to Y. Let $x, y \in X_+$ and $v \in J_+$. It follows from $(x+y) \wedge v \leqslant x \wedge v + y \wedge v$ that

$$T\bigl((x+y) \wedge v\bigr) \leqslant T(x \wedge v) + T(y \wedge v) \leqslant Sx + Sy.$$

Taking supremum over all $v \in J_+$, we get $S(x+y) \leqslant Sx + Sy$. On the other hand, it is easy to see that $x \wedge u + y \wedge v \leqslant (x+y) \wedge (u+v)$ for all $u, v \in J_+$. This implies

$$T(x \wedge u) + T(y \wedge v) \leqslant T\bigl((x+y) \wedge (u+v)\bigr) \leqslant S(x+y).$$

Consecutively takings suprema over u and v, we get $Sx + Sy \leqslant S(x+y)$.

Thus, we may view S as an element of $\mathcal{L}(X,Y)_+$. Let $x \in X_+$. For every $v \in J$ with $0 \leqslant v \leqslant x$ we clearly have $Tv \leqslant Tx$. It follows that $Sx \leqslant Tx$ and, therefore, $S \leqslant T$.

It is easy to see that S agrees with T on J. If $0 \leqslant x \in J^d$ then $x \wedge v = 0$ for all $v \in J_+$, hence $Sx = 0$. Now for every $x \in J^d$, we have $Sx = S(x^+) - S(x^-) = 0$. □

6.3.10 Remark. In the special case when J is a principal ideal, $J = I_u$ for some $u \in X_+$, it is easy to see that the formula for S takes a somewhat simpler form $Sx = \sup_n T(x \wedge nu)$. Therefore, in this case, it suffices to assume that Y is σ-order complete. Note that $Su = Tu$ and S vanishes on $\{u\}^d$.

Applying Theorem 6.3.9 to *every* operator in $\mathcal{L}(X,Y)_+$ leads to "restriction" operator on $\mathcal{L}_{\mathrm{ob}}(X,Y)$:

6.3.11 Theorem. *Let Y be order complete and J an ideal in X. The map $\mathcal{R}\colon \mathcal{L}(X,Y)_+ \to \mathcal{L}(X,Y)_+$ defined by*

$$(\mathcal{R}T)(x) = \sup\bigl\{Tv \,:\, v \in J \text{ and } 0 \leqslant v \leqslant x\bigr\}$$

for $x \in X_+$ and $T \in \mathcal{L}(X,Y)_+$ extends to a band projection $\mathcal{R}\colon \mathcal{L}_{\mathrm{ob}}(X,Y) \to \mathcal{L}_{\mathrm{ob}}(X,Y)$. For every positive operator T, $\mathcal{R}T$ is a component of T; it agrees with T on J and vanishes on J^d.

Proof. By Riesz-Kantorovich Theorem, $\mathcal{L}_{\mathrm{ob}}(X,Y)$ is a vector lattice. As in Theorem 6.3.9, $\mathcal{R}T$ is defined for every $T \in \mathcal{L}(X,Y)_+$, $\mathcal{R}T$ agrees with T on J and vanishes on J^d.

We claim that $\mathcal{R}$ is additive on $\mathcal{L}(X,Y)_+$; we can then extend it to a positive linear operator on $\mathcal{L}_{\mathrm{ob}}(X,Y)$ by Extension Lemma 6.2.1. Indeed, let $S,T \in \mathcal{L}(X,Y)_+$ and $x \in X_+$. Since the set $J \cap [0,x]$ is directed upward, we may view it as an increasing net, say, (v_α). In these terms, he definition of $\mathcal{R}$ means that $Sv_\alpha \xrightarrow{\mathrm{o}} (\mathcal{R}S)(x)$, $Tv_\alpha \xrightarrow{\mathrm{o}} (\mathcal{R}T)(x)$, and $(S+T)v_\alpha \xrightarrow{\mathrm{o}} \bigl(\mathcal{R}(S+T)\bigr)(x)$. On the other hand, $(S+T)v_\alpha = Sv_\alpha + Tv_\alpha \xrightarrow{\mathrm{o}} (\mathcal{R}S)(x) + (\mathcal{R}T)(x)$. It follows that $\mathcal{R}(S+T) = \mathcal{R}S + \mathcal{R}T$.

We may now view $\mathcal{R}$ as a positive linear operator on $\mathcal{L}_{\mathrm{ob}}(X,Y)$. Clearly, $0 \leqslant \mathcal{R} \leqslant \mathcal{I}$, where $\mathcal{I}$ is the identity operator on $\mathcal{L}_{\mathrm{ob}}(X,Y)$. We need to verify that $\mathcal{R}^2 = \mathcal{R}$; Theorem 5.2.11 will then imply that $\mathcal{R}$ is a band projection. Let $T \in \mathcal{L}(X,Y)_+$. Since $\mathcal{R}T$ agrees with T on J, we have

$$(\mathcal{R}^2T)(x) = \sup\bigl\{(\mathcal{R}T)v \,:\, v \in J \text{ and } v \leqslant x\bigr\} = \sup\bigl\{Tv \,:\, v \in J \text{ and } v \leqslant x\bigr\} = (\mathcal{R}T)(x)$$

for every $x \in X_+$. It follows that $\mathcal{R}^2T = \mathcal{R}T$ for every positive T, hence $\mathcal{R}^2 = \mathcal{R}$.

Let $T \geqslant 0$. Since $\mathcal{R}$ is a band projection, we have

$$0 = \mathcal{R}T \wedge (\mathcal{I} - \mathcal{R})T = \mathcal{R}T \wedge (T - \mathcal{R}T).$$

It follows that $\mathcal{R}T$ is a component of T. □

Let J is a projection band in X; let P be its band projection. It is easy to see that the operator $\mathcal{R}$ constructed in the preceding theorem agrees with the map $T \mapsto TP$ because $\mathcal{R}T$ and TP agree on J and on J^d, hence on X, for every T.

Suppose that J is just an ideal such that the band generated by it is a projection band (which is always the case when X has PP). Let P be the corresponding band projection. In this case, we still have $0 \leqslant TP \leqslant T$, and TP agrees with T on J, and vanishes on J^d for every $T \in \mathcal{L}_{\mathrm{ob}}(X,Y)$. One may conjecture that in this case we still have $\mathcal{R}T = TP$. The following example shows that this is false.

6.3.12 Example. Let $X = \ell_\infty$, $Y = \mathbb{R}$, and $J = I_u$ where $u = (1, \frac{1}{2}, \frac{1}{3}, \dots)$. Since $u \in c_0$, we clearly have $J \subseteq c_0$. On the other hand, u is a weak unit, hence $B_u = \ell_\infty$ and the corresponding band projection P is the identity operator on ℓ_∞. Hence, the map $f \mapsto f \circ P$ is the identity map on $\mathcal{L}_{\mathrm{ob}}(X,Y)$. (This map is actually the adjoint of P; we will discuss adjoints in Subsection 6.11.1.) Let $\mathcal{R}$ be as in the theorem; we claim that $\mathcal{R} \neq \mathcal{I}$. Indeed, let f be a Banach limit; let $x = \mathbb{1}$. Clearly, $f(\mathbb{1}) = 1$. For every n, we have $\mathbb{1} \wedge nu \in c_0$, hence $f(\mathbb{1} \wedge nu) = 0$; it follows that $(Rf)(\mathbb{1}) = 0$. Hence, $Rf \neq f$.

6.3.13 Exercise. Suppose that Y is order complete. Let J_1 and J_2 be two ideals in X, and $\mathcal{R}_1$ and $\mathcal{R}_2$ the corresponding restriction operators on $\mathcal{L}_{\mathrm{ob}}(X,Y)$. If $J_1 \perp J_2$ then the projection bands $\operatorname{Range}\mathcal{R}_1$ and $\operatorname{Range}\mathcal{R}_2$ are disjoint.

Next, we establish (under some mild assumptions) the "dual" versions of the Riesz-Kantorovich formulas. Given $S \leqslant T$ in $\mathcal{L}(X,Y)$ and $x \in X$, we write $[S,T]x := \bigl\{Rx : R \in [S,T]\bigr\}$.

6.3.14 Theorem. *Suppose that X has PPP or Y is σ-order complete. Then for every $T \in \mathcal{L}(X,Y)_+$ and $x,y \in X$,*

$$Tx^+ = \sup[0,T]x, \quad Tx^- = -\inf[0,T]x, \quad \textit{and} \quad T|x| = \sup[-T,T]x,$$

$$\begin{aligned} T(x \vee y) &= \sup\{Rx + Sy \,:\, R,S \geqslant 0,\ R+S=T\} \ \textit{and} \\ T(x \wedge y) &= \inf\{Rx + Sy \,:\, R,S \geqslant 0,\ R+S=T\}. \end{aligned}$$

Moreover, these sup*'s and* inf*'s are actually attained.*

Proof. Let $x \in X$. By Remarks 6.3.8 and 6.3.10 we can find S such that $0 \leqslant S \leqslant T$, $Sx^+ = Tx^+$, and $Sy = 0$ whenever $y \perp x^+$. In particular, $Sx^- = 0$, so that $Sx = Tx^+$. We claim that Sx is the greatest element of $[0,T]x$. Indeed, if $R \in [0,T]$ then $Rx \leqslant Rx^+ \leqslant Tx^+ = Sx$. This proves the first identity. The other identities follow easily (as in the proof of Theorem 6.3.1). □

6.3.15. Iterating the last two formulas in 6.3.14, we get

$$\begin{aligned} T\Bigl(\bigvee_{i=1}^n x_i\Bigr) &= \sup\Bigl\{\sum_{i=1}^n S_i x_i \,:\, S_1,\dots,S_n \geqslant 0, \sum_{i=1}^n S_i = T\Bigr\} \text{ and} \\ T\Bigl(\bigwedge_{i=1}^n x_i\Bigr) &= \inf\Bigl\{\sum_{i=1}^n S_i x_i \,:\, S_1,\dots,S_n \geqslant 0, \sum_{i=1}^n S_i = T\Bigr\}. \end{aligned}$$

for any $x_1,\dots,x_n \in X$. More over, the sup and the inf are actually attained.

6.3.16 Exercise. Let $X = L_p[0,1]$, $1 \leqslant p \leqslant \infty$, I the identity operator on X, and $T = \varphi \otimes \mathbb{1}$, where $\varphi(f) = \int f$. That is, $Tf = \bigl(\int_0^1 f(t)\,dt\bigr) \cdot \mathbb{1}$ for every $f \in X$. Show that $I \wedge T = 0$.

6.3.17 Exercise. Let X be a Banach lattice.

(i) Show that every norm bounded operator from X to a $C(K)$ space is order bounded.

(ii) Show that every norm bounded operator from X to a ℓ_∞ is regular.

6.4 Examples of non-regular and non-order-bounded operators

Recall that, in general, $\mathcal{L}_r(X,Y)\subseteq\mathcal{L}_{\rm ob}(X,Y)$. Riesz-Kantorovich Theorem guarantees that $\mathcal{L}_r(X,Y)=\mathcal{L}_{\rm ob}(X,Y)$ when Y is order complete, while Theorem 6.1.9 asserts that $\mathcal{L}_{\rm ob}(X,Y)\subseteq L(X,Y)$ when X is a Banach lattice and Y is a normed lattice. In this section we will present examples illustrating that these inclusions may be proper.

6.4.1 Example. *A non-order-bounded operator.* Let $T\colon C[0,1]\to c_0$ defined by $(Tf)_n=f(\frac{1}{n})-f(0)$. This sequence is indeed in c_0 because $f(\frac{1}{n})\to f(0)$ for every continuous function f. Note that T is norm bounded because $\bigl|(Tf)_n\bigr|\leqslant\bigl|f(\frac{1}{n})\bigr|+\bigl|f(0)\bigr|\leqslant 2\|f\|_\infty$ for every n, so that $\|T\|\leqslant 2$. We claim that $T[0,\mathbb{1}]$ is not order bounded. Indeed, for every n, one can find $f_n\in[0,\mathbb{1}]$ such that $f_n(\frac{1}{n})=1$ and f_n vanishes outside of $\bigl(\frac{1}{n+1},\frac{1}{n-1}\bigr)$. It follows that $Tf_n=e_n$. Therefore, $T[0,\mathbb{1}]$ contains the sequence (e_n), and it is easy to see that this sequence is not order bounded in c_0.

Similarly, one can show that the natural Banach space isomorphism from c onto c_0, described in 2.1.9, is not order bounded.

6.4.1 Krengel's example.

A non-order-bounded isometry $T\colon\ell_2\to\ell_2$. The example is based on the following observation. Consider the following 2×2 matrix: $A=\left[\begin{smallmatrix}\frac{1}{\sqrt2}&\frac{1}{\sqrt2}\\-\frac{1}{\sqrt2}&\frac{1}{\sqrt2}\end{smallmatrix}\right]$. Geometrically, it represents the rotation of $\mathbb{R}^2$ about the origin by $\frac{\pi}{4}$. In particular, A is an isometry on ℓ_2^2. On the other hand, $|A|=\left[\begin{smallmatrix}\frac{1}{\sqrt2}&\frac{1}{\sqrt2}\\\frac{1}{\sqrt2}&\frac{1}{\sqrt2}\end{smallmatrix}\right]$. Put $x=\left[\begin{smallmatrix}\frac{1}{\sqrt2}\\\frac{1}{\sqrt2}\end{smallmatrix}\right]$. Then $\|x\|_2=1$ and $\bigl\||A|x\bigr\|=\sqrt2$, so that $\bigl\||A|\bigr\|\geqslant\sqrt2$, hence $\bigl\||A|\bigr\|>\|A\|$. By "amplifying" A to higher dimensions as follows, we can now make $|A|$ have arbitrarily large norm.

Let $X_n=\ell_2^{2^n}$. For each $n=0,1,2,\dots$, define $H_n\colon X_n\to X_n$ as follows: $H_0=(1)$, $H_{n+1}=\left[\begin{smallmatrix}H_n&H_n\\-H_n&H_n\end{smallmatrix}\right]$. Note that all the entries of H_n are ±1's. Consider their normalized versions: let $T_n=2^{-\frac{n}{2}}H_n$. Note that $T_n=\frac{1}{\sqrt2}\left[\begin{smallmatrix}T_{n-1}&T_{n-1}\\-T_{n-1}&T_{n-1}\end{smallmatrix}\right]$ for all $n\geqslant1$. We will use induction to show that T_n is an isometry of X_n. The case $n=0$ is trivial. Suppose that T_n is an isometry for some n, and let $x\in X_{n+1}$. We can write $x=\left[\begin{smallmatrix}y\\z\end{smallmatrix}\right]$ for some $y,z\in X_n$. Then, using the parallelogram law, we get

$$\begin{aligned}\|T_{n+1}x\|^2&=\tfrac12\Bigl\|\left[\begin{smallmatrix}T_n&T_n\\-T_n&T_n\end{smallmatrix}\right]\left[\begin{smallmatrix}y\\z\end{smallmatrix}\right]\Bigr\|^2=\tfrac12\Bigl\|\left[\begin{smallmatrix}T_ny+T_nz\\-T_ny+T_nz\end{smallmatrix}\right]\Bigr\|^2\\&=\tfrac12\Bigl[\|T_ny+T_nz\|^2+\|-T_ny+T_nz\|^2\Bigr]\\&=\|T_ny\|^2+\|T_nz\|^2=\|y\|^2+\|z\|^2=\|x\|^2.\end{aligned}$$

Next, we consider the direct sum of all the T_n's. We have $\ell_2 = \bigoplus_{n=0}^{\infty} X_n$, where $(x_n) \in \ell_2$ is mapped into

$$\Big((x_1), (x_2, x_3), (x_4, x_5, x_6, x_7), (x_8, x_9, x_{10}, x_{11}, x_{12}, x_{13}, x_{14}, x_{15}), \dots\Big).$$

Let P_n be the canonical projection from ℓ_2 to X_n. Define $T\colon \ell_2 \to \ell_2$ via $T = \bigoplus_{n=0}^{\infty} T_n$. That is, T is represented by a block-diagonal infinite matrix with diagonal blocks T_0, T_1, T_2, etc. Then T is an isometry. Indeed, if $x \in \ell_2$ then

$$\|Tx\|^2 = \sum_{n=0}^{\infty} \|T_n P_n x\|^2 = \sum_{n=0}^{\infty} \|P_n x\|^2 = \|x\|^2.$$

Assume for the sake of contradiction that T is order bounded. By Riesz-Kantorovich Theorem, $|T|$ exists. By Exercise 6.1.15, the matrix of $|T|$ is $\big(|t_{ij}|\big)$, i.e.,

$$T = \begin{pmatrix} 1 & & & & & & & \\ & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & & & & & \\ & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & & & & & \\ & & & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \\ & & & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \\ & & & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \\ & & & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \\ & & & & & & & \ddots \end{pmatrix}, \quad |T| = \begin{pmatrix} 1 & & & & & & & \\ & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & & & & & \\ & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & & & & & \\ & & & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \\ & & & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \\ & & & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \\ & & & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \\ & & & & & & & \ddots \end{pmatrix}.$$

However, the latter matrix does not correspond to an operator on ℓ_2. Indeed, if $|T|$ were an operator, then, being positive, it would be bounded. Let u_n be the vector in X_n with all the coordinates being 1's. Then $|H_n|u_n = 2^n u_n$, hence $|T_n|u_n = 2^{\frac{n}{2}} u_n$ and, therefore, $\big\||T|\big\| \geqslant \big\||T_n|\big\| \geqslant 2^{\frac{n}{2}} \to \infty$ an $n \to \infty$; a contradiction.

6.4.2 Rademacher functions

For each $n \in \mathbb{N}$, the n-th Rademacher function r_n is defined to be the function in $L_\infty[0,1]$ such that $r_n(t) = 1$ whenever $\frac{2k-1}{2^n} \leqslant t < \frac{2k}{2^n}$ for some integer k and $r_n(t) = -1$ otherwise. The Rademacher sequence has many important applications in Functional Analysis.

There are equivalent ways to define and view Rademacher functions. For example, $r_n(t) = 1$ iff the n-th digit in the binary expansion of t equals 0. Also, $r_n(t) = \operatorname{sign} \sin 2^n \pi t$ for $t \in [0,1]$. Note that $|r_n| = \mathbb{1}$ for every n. From a probabilistic point of view, the sequence (r_n) may be viewed as a sequence of independent identically distributed random variables on $[0,1]$ such that $\mathbb{P}(r_n = 1) = \mathbb{P}(r_n = -1) = \frac{1}{2}$.

6.4.2. Consider a linear combination $\sum_{i=1}^n \alpha_i r_i$ for some $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. Split $[0,1]$ into 2^n equal subintervals. On each of these subintervals, every r_i is constant 1 or -1; as we run over all the intervals we get all possible choices of signs. Thus,

$$\Bigl\|\sum_{i=1}^n \alpha_i r_i\Bigr\|_{L_1} = \int_0^1 \Bigl|\sum_{i=1}^n \alpha_i r_i\Bigr| = \frac{1}{2^n} \sum_{\varepsilon_i=\pm 1} \Bigl|\sum_{i=1}^n \alpha_i \varepsilon_i\Bigr|,$$

where the first sum in the right hand side is taken over all possible choices of "signs" $\varepsilon_i = \pm 1$. This is the average of the expression $\bigl|\sum_{i=1}^n \varepsilon_i \alpha_i\bigr|$ over all choices of $\varepsilon_i = \pm 1$; we will abbreviate it by $\operatorname*{Ave}_{\varepsilon_i=\pm 1} \bigl|\sum_{i=1}^n \varepsilon_i \alpha_i\bigr|$.

The following fact is central to many applications of the Rademacher sequence; we refer the reader to [AK06, Car05] for its proof.

6.4.3 Theorem (Khinchin's Inequality)**.** *Suppose that $1 \leqslant p < \infty$. There exist positive constants A_p and B_p such that*

$$A_p\Bigl(\sum_{i=1}^n |\alpha_i|^2\Bigr)^{\frac{1}{2}} \leqslant \Bigl\|\sum_{i=1}^n \alpha_i r_i\Bigr\|_{L_p} \leqslant B_p\Bigl(\sum_{i=1}^n |\alpha_i|^2\Bigr)^{\frac{1}{2}}$$

for any scalars $\alpha_1, \dots, \alpha_n$.

6.4.4. (i) In case $p = 1$, Khinchin's inequality may be reformulated as:

$$\operatorname*{Ave}_{\varepsilon_i=\pm 1} \Bigl|\sum_{i=1}^n \varepsilon_i \alpha_i\Bigr| \sim \Bigl(\sum_{i=1}^n |\alpha_i|^2\Bigr)^{\frac{1}{2}}$$

for any scalars $\alpha_1, \dots, \alpha_n$. For a general p, a similar argument yields

$$\Bigl(\frac{1}{2^n} \sum_{\varepsilon_i=\pm 1} \Bigl|\sum_{i=1}^n \varepsilon_i x_i\Bigr|^p\Bigr)^{\frac{1}{p}} \sim \Bigl(\sum_{i=1}^n |\alpha_i|^2\Bigr)^{\frac{1}{2}},$$

(ii) Khinchin's Inequality means that the sequence (r_n) in $L_p[0,1]$ is a basic sequence equivalent to the unit vector basis of ℓ_2. Equivalently, the operator $T\colon \ell_2 \to L_p[0,1]$ defined by $Te_n = r_n$ is an isomorphic embedding. In particular, the closed linear span $[r_n]$ of the sequence (r_n) in $L_p[0,1]$ is isomorphic to ℓ_2.

(iii) Since T is norm continuous and $e_n \xrightarrow{\mathrm{w}} 0$ in ℓ_2, it follows that $r_n \xrightarrow{\mathrm{w}} 0$ in $L_p[0,1]$ $(1 \leqslant p < \infty)$.

(iv) The adjoint operator $T^*\colon L_q[0,1] \to \ell_2$, where q is the conjugate of p (hence $1 < q \leqslant \infty$), is given by $(T^*f)_n = \int f r_n$. In particular, this operator is norm bounded and surjective.

(v) If $1 < p < \infty$ then the subspace $[r_n]$ in $L_p[0,1]$ is complemented. Define $S\colon L_p[0,1] \to \ell_2$ via $(Sf)_n = \int f r_n$. This operator is norm bounded and surjective by (iv). Then the operator $TS\colon L_p[0,1] \to L_p[0,1]$ is a bounded projection onto $[r_n]$.

(vi) $\lim\limits_{n\to\infty} \int f r_n = 0$ for every $f \in L_1[0,1]$. That is, $r_n \xrightarrow{\mathrm{w}^*} 0$ in $L_\infty[0,1]$.

(vii) For $p = \infty$, the map $R\colon \ell_1 \to L_\infty[0,1]$ defined by $Re_n = r_n$ is an isometry. Hence, viewed as a sequence in $L_\infty[0,1]$, (r_n) is equivalent to the unit vector basis of ℓ_1, and $[r_n]$ is an isometric copy of ℓ_1 in $L_\infty[0,1]$.

6.4.5 Example. *The isomorphism $T\colon \ell_2 \to L_p[0,1]$ defined by $Te_n = r_n$ ($1 \leqslant p < \infty$) is not order bounded.* Indeed, suppose that T is order bounded. Then $|T|$ exists by Riesz-Kantorovich Theorem. Then $|T|e_n \geqslant |Te_n| = |r_n| = \mathbb{1}$. Let $x_m = \sum_{n=1}^m \frac{e_n}{n}$. The sequence (x_m) is order and norm bounded in ℓ_2, but

$$\big\||T|x_m\big\| = \Big\||T|\sum_{n=1}^m \frac{e_n}{n}\Big\| \geqslant \Big\|\Big(\sum_{n=1}^m \tfrac{1}{n}\Big)\mathbb{1}\Big\| = \sum_{n=1}^m \tfrac{1}{n} \to \infty,$$

a contradiction.

6.4.6 Example. *The operator $S\colon L_p[0,1] \to \ell_2$ defined by $(Sf)_n = \int f r_n$ ($1 < p \leqslant \infty$) is not order bounded.* Indeed, suppose it is. Then $S[-\mathbb{1},\mathbb{1}] \subseteq [-a,a]$ for some positive $a = (a_n) \in \ell_2$. Note that $r_n \in [-\mathbb{1},\mathbb{1}]$ for every n and $Sr_n = e_n$. It follows that $e_n \leqslant a$, hence $1 \leqslant a_n$. This contradicts $a \in \ell_2$.

6.4.7 Example. *A non-order-bounded operator $S\colon L_1[0,1] \to c_0$.* Recall that $\lim\limits_{n\to\infty} \int f r_n = 0$ for every $f \in L_1[0,1]$. Define $S\colon L_1[0,1] \to c_0$ via $(Sf)_k = \int f r_k$. Note that S is bounded because

$$\|Sf\| = \sup\big|(Sf)_k\big| = \sup\Big|\int f r_k\Big| \leqslant \int |f| = \|f\|.$$

However, as in Example 6.4.6, $Sr_n = e_n$ and, therefore, S is not order bounded.

There is a variation of this example due to Lozanovsky: consider $V\colon L_1[0,2\pi] \to c_0$ given by $(Vx)_k = \int_0^{2\pi} x(t) \sin kt\,dt$, where $k \in \mathbb{N}$. A similar argument shows that V is norm bounded but not order bounded.

6.4.8 Example. Let $R\colon \ell_1 \to L_\infty[0,1]$ be the isometry defined by $Re_n = r_n$. Then R is regular. Indeed, put $U\colon \ell_1 \to L_\infty[0,1]$ via $Ue_i = \mathbb{1}$. It is easy to see that U is norm bounded and $\pm R \leqslant U$.

6.4.9. It follows from Khinchin's Inequality that order intervals in $L_p(\mu)$ are, generally, not compact. Indeed, the Rademacher sequence (r_n) is contained in $[-\mathbb{1},\mathbb{1}]$, yet it follows from Khinchin's Inequality that is has no convergent subsequences.

6.4.3 Lotz' example

An order (and norm) bounded non-regular operator. Define $T\colon C[-1,1]\to C[-1,1]$ via
$$(Tf)(t)=\begin{cases} f\bigl(\sin\frac{1}{t}\bigr)-f\bigl(\sin(t+\frac{1}{t})\bigr) & \text{if } t\neq 0;\\ 0 & \text{if } t=0.\end{cases}$$
First, we show that Tf is continuous. This is trivial if $t\neq 0$. In order to show that Tf is continuous at zero, fix $\varepsilon>0$. Since $f\in C[-1,1]$, hence is uniformly continuous, there exists $\delta>0$ such that $|s-t|<\delta$ implies that $\bigl|f(s)-f(t)\bigr|<\varepsilon$. Let $0\neq t\in(-\delta,\delta)$. By the Mean Value Theorem, we have
$$\Bigl|\sin\tfrac{1}{t}-\sin(t+\tfrac{1}{t})\Bigr|\leqslant|t|<\delta,\quad\text{so that}\quad\Bigl|f\bigl(\sin\tfrac{1}{t}\bigr)-f\bigl(\sin(t+\tfrac{1}{t})\bigr)\Bigr|<\varepsilon,$$
Hence $\bigl|(Tf)(t)\bigr|<\varepsilon$.

Next, we will show that T is order bounded. Since every order interval in $C[-1,1]$ is contained in $[-\lambda\mathbb{1},\lambda\mathbb{1}]$ for some $\lambda>0$, it suffices to show that $T[-\mathbb{1},\mathbb{1}]$ is order bounded. If $f\in[-\mathbb{1},\mathbb{1}]$ then for every $t\neq 0$ we have
$$\bigl|(Tf)(t)\bigr|\leqslant\Bigl|f\bigl(\sin\tfrac{1}{t}\bigr)\Bigr|+\Bigl|f\bigl(\sin(t+\tfrac{1}{t})\bigr)\Bigr|\leqslant 2,$$
so that $Tf\in[-2\mathbb{1},2\mathbb{1}]$. Hence, T is order bounded. Moreover, this also shows that $\|T\|\leqslant 2$.

Finally, we show that T is not regular. Suppose that $T\leqslant S$ for some positive operator S. Let $0\leqslant f\in C[-1,1]$. We claim that $Sf(0)\geqslant\max f$. Pick any $\alpha\in[0,2\pi]$ and let $t_n=\frac{1}{\alpha+2\pi n}$. Then $t_n\to 0$. For every n find $g_n\in C[-1,1]$ so that $0\leqslant g_n\leqslant f$, $g_n(\sin\alpha)=f(\sin\alpha)$, and $g_n\bigl(\sin(\alpha+t_n)\bigr)=0$. Then
$$\begin{aligned} Sf(t_n)\geqslant Sg_n(t_n)\geqslant Tg_n(t_n)&=g_n\bigl(\sin\tfrac{1}{t_n}\bigr)-g_n\bigl(\sin(t_n+\tfrac{1}{t_n})\bigr)\\ &=g_n\bigl(\sin(\alpha+2\pi n)\bigr)-g_n\bigl(\sin(t_n+\alpha+2\pi n)\bigr)\\ &=g_n(\sin\alpha)-g_n\bigl(\sin(\alpha+t_n)\bigr)=f(\sin\alpha).\end{aligned}$$
Since Sf is continuous, it follows that $Sf(0)\geqslant f(\sin\alpha)$. Since α was chosen arbitrarily, it follows that $Sf(0)\geqslant\max f$.

Let $n\in\mathbb{N}$, and $f_1,\dots,f_n$ be disjoint positive functions in $C[-1,1]$ such that $\max f_i=1$ as $i=1,\dots,n$. Then $0\leqslant\sum_{i=1}^n f_i\leqslant\mathbb{1}$, so that $0\leqslant S\bigl(\sum_{i=1}^n f_i\bigr)\leqslant S\mathbb{1}$, so that $S\mathbb{1}(0)\geqslant\sum_{i=1}^n Sf_i(0)\geqslant n$. Since n was chosen arbitrarily, this is a contradiction.

6.5 Order continuous operators

Suppose that X and Y are two vector lattices and $T\in\mathcal{L}(X,Y)$. As before, we say that T is order continuous if $Tx_\alpha\xrightarrow{\text{o}}Tx$ whenever $x_\alpha\xrightarrow{\text{o}}x$; we say that T is σ-order continuous if $Tx_n \xrightarrow{\sigma o} Tx$ whenever $x_n \xrightarrow{\sigma o} x$. Since T is linear, we may always assume in these definitions that $x = 0$.

6.5.1 Exercise. Suppose that $T \geqslant 0$ then

(i) T is order continuous iff $Tx_\alpha \downarrow 0$ whenever $x_\alpha \downarrow 0$;
(ii) T is σ-order continuous iff $Tx_n \downarrow 0$ whenever $x_n \downarrow 0$;

6.5.2. An operator $T\colon X \to X$ is said to be ***central*** if $-\lambda I \leqslant T \leqslant \lambda I$ for some $\lambda > 0$. This class of operators includes band projections because every band projection P satisfies $0 \leqslant P \leqslant I$. Also, every multiplication operator on $C(K)$, i.e., an operator of the form $Tx = xy$, where y is a fixed element of $C(K)$, is central. The next statement asserts that all these operators are order continuous.

6.5.3. A sublattice of X is regular iff the inclusion map is order continuous; see Exercise 3.2.1. In particular, the canonical inclusion of an Archimedean vector lattice into its order completion is order continuous.

6.5.4 Exercise. Every central operator is order continuous and σ-order continuous.

Since every uniformly convergent net in an Archimedean vector lattice is order convergent, Theorem 6.1.7 immediately yields the following result.

6.5.5 Theorem ([AS05])**.** *If X and Y are Archimedean then every order continuous operator $T\colon X \to Y$ is order bounded.*

6.5.6 Example. The preceding theorem fails without the Archimedean assumption. Let X be $\mathbb{R}^2$ equipped with the lexicographic order, while $Y = \mathbb{R}^2$ with the usual order; let T be the formal identity operator. Since the y-axis is order bounded in X but not in Y, T is not order bounded. However, T is order continuous because if $x_\alpha \xrightarrow{o} 0$ in X then it has a tail which is contained in the y-axis which converges to zero in the usual topology, hence $x_\alpha \xrightarrow{o} 0$ in Y.

6.5.7 Example. *An order continuous operator need not be regular.* Consider Lotz' example from Section 6.4.3. We already know that T is order bounded but fails to be regular. We claim that T is order continuous.

For $f \in C[-1,1]$ and $t \neq 0$ we have $(Tf)(t) = f\bigl(g_1(t)\bigr) - f\bigl(g_2(t)\bigr)$, where $g_1(t) = \sin\frac{1}{t}$ and $g_2(t) = \sin(t + \frac{1}{t})$. Suppose that $f_\alpha \xrightarrow{o} 0$ in $C[-1,1]$; we need to show that $Tf_\alpha \xrightarrow{o} 0$. We will use Theorem 2.4.30. WLOG, (f_α) is order bounded. Since T is order bounded, (Tf_α) is order bounded. Fix an open non-empty subset U of $[-1,1]$ and $\varepsilon > 0$. Find an open interval $(a_0, b_0) \subseteq U$ such that $0 \notin (a_0, b_0)$ and g_1 is strictly monotone on (a_0, b_0). Then $g_1\bigl((a_0, b_0)\bigr)$ is an open interval and, therefore, by Theorem 2.4.30, there exists an open subinterval $(c_1, d_1) \subseteq g_1\bigl((a_0, b_0)\bigr)$ and an index α_1 such that $\bigl|f_\alpha(s)\bigr| < \frac{\varepsilon}{2}$ for all $s \in (c_1, d_1)$ and all $\alpha > \alpha_1$. The pri-image $g_1^{-1}\bigl(c_1, d_1)\bigr)$ is a subinterval of (a_0, b_0); denote it by (a_1, b_1). We conclude that $\bigl|f_\alpha\bigl(g_1(t)\bigr)\bigr| < \frac{\varepsilon}{2}$ for all $t \in (a_1, b_1)$ and all $\alpha \geqslant \alpha_1$. Iterating this argument, we find a subinterval (a_2, b_2) of (a_1, b_1) and an index α_2 such that $\bigl|f_\alpha\bigl(g_2(t)\bigr)\bigr| < \frac{\varepsilon}{2}$ for all

$t \in (a_2, b_2)$ and all $\alpha \geqslant \alpha_2$. It follows that for all $t \in (a_2, b_2)$ and all $\alpha > \alpha_1, \alpha_2$, we have $\big|(Tf_\alpha)(t)\big| \leqslant \big|f_\alpha\big(g_1(t)\big)\big| + \big|f_\alpha\big(g_2(t)\big)\big| < \varepsilon$. By Theorem 2.4.30, it follows that $Tf_\alpha \xrightarrow{\text{o}} 0$.

The converse of Theorem 6.5.5 is false even when $Y = \mathbb{R}$: we will show later that when $X = C[0,1]$ then norm bounded and order bounded (linear) functionals on X agree, hence X admits plenty of order bounded functionals, yet X admits no non-zero order continuous functionals.

6.5.8 Proposition. *A linear operator from an order continuous Banach lattice to an Archimedean vector lattice is order continuous iff it is order bounded.*

Proof. The forward implication follows from the preceding theorem. To prove the converse, let $T\colon X \to Y$ be an order bounded operator from an order continuous Banach lattices X to a vector lattice Y. Suppose that $x_\alpha \xrightarrow{\text{o}} x$ in X. By Theorem 2.5.17, we have $x_\alpha \xrightarrow{\text{u}} x$. Then $Tx_\alpha \xrightarrow{\text{u}} Tx$ by Theorem 6.1.7, hence $Tx_\alpha \xrightarrow{\text{o}} Tx$. ☐

6.5.9 Example. *A σ-order continuous operator which is not order bounded.* Recall that the operator $S\colon L_1[0,1] \to c_0$ in Example 6.4.7 given by $(Sf)_k = \int f r_k$, fails to be order bounded. Yet, it is σ-order continuous. Indeed, if $f_n \xrightarrow{\sigma\text{o}} 0$ in $L_1[0,1]$ then $f_n \xrightarrow{\|\cdot\|} 0$ because $L_1[0,1]$ is order continuous, hence $Sf_n \xrightarrow{\|\cdot\|} 0$ in c_0 because S is norm bounded. Since norm and order convergences agree for sequences in c_0 by Corollary 2.5.16, we conclude that $Sf_n \xrightarrow{\text{o}} 0$.

In view of Theorem 6.5.5, this example also shows that a σ-order continuous operator need not be order continuous.

For the next few results, we assume that Y is order complete.

6.5.10 Theorem. *Suppose that $T \in \mathcal{L}_{\text{ob}}(X,Y)$, where Y is order complete. Then the following are equivalent.*

(i) *T is order continuous;*
(ii) $\inf|Tx_\alpha| = 0$ *whenever $x_\alpha \downarrow 0$;*
(iii) *T^+ and T^- are order continuous;*
(iv) *$|T|$ is order continuous.*

Similar equivalences hold for σ-order continuity.

Proof. Note that, under the assumptions of the theorem, the operators T^+, T^-, and $|T|$ exist by Riesz-Kantorovich formulas.

Implications (i)⇒(ii) and (iii)⇒(iv) are trivial.

To verify that (iv)⇒(i), suppose that $|T|$ is order continuous and $x_\alpha \xrightarrow{\text{o}} 0$. Then $|x_\alpha| \xrightarrow{\text{o}} 0$, so that $|Tx_\alpha| \leqslant |T||x_\alpha| \xrightarrow{\text{o}} 0$, hence $Tx_\alpha \xrightarrow{\text{o}} 0$.

It is left to show that (ii)⇒(iii). Suppose that $x_\alpha \downarrow 0$. By Exercise 6.5.1 it suffices to show that $T^+x_\alpha \downarrow 0$. Since Y is order complete, $z \coloneqq \inf T^+x_\alpha$ exists. It is left to show that $z = 0$. Clearly, $z \geqslant 0$. Fix any α_0 and put $x = x_{\alpha_0}$. Take any $u \in [0, x]$. For every $\alpha \geqslant \alpha_0$ we have $x_\alpha \leqslant x$, so that

$$0 \leqslant u - u \wedge x_\alpha = u \wedge x - u \wedge x_\alpha \leqslant x - x_\alpha,$$

so that

$$Tu - T(u \wedge x_\alpha) = T(u - u \wedge x_\alpha) \leqslant T^+(u - u \wedge x_\alpha) \leqslant T^+(x - x_\alpha) = T^+x - T^+x_\alpha.$$

Therefore,

$$0 \leqslant z \leqslant T^+x_\alpha \leqslant T^+x + T(u \wedge x_\alpha) - Tu \leqslant T^+x + \bigl|T(u \wedge x_\alpha)\bigr| - Tu.$$

It follows from $x_\alpha \downarrow 0$ that $u \wedge x_\alpha \downarrow 0$. Hence, by assumption, $\inf_\alpha \bigl|T(u \wedge x_\alpha)\bigr| = 0$, so that $0 \leqslant z \leqslant T^+x - Tu$. By Riesz-Kantorovich formula (6.2),

$$z \leqslant T^+x - \sup\bigl\{Tu \,:\, 0 \leqslant u \leqslant x\bigr\} = 0.$$

Hence, $z = 0$.

The proof for σ-order continuity is similar. □

We will show in Corollary 6.5.17 that the equivalence (i)⇔(ii) remains valid without the assumption that Y is order complete.

Given vector lattices X and Y, we write $\mathcal{L}_{\mathrm{oc}}(X, Y)$ and $\mathcal{L}_{\sigma\text{-oc}}(X, Y)$ for the collections of all order continuous operators and of all σ-order continuous operators in $\mathcal{L}_{\mathrm{ob}}(X, Y)$, respectively.[2] Clearly, both $\mathcal{L}_{\mathrm{oc}}(X, Y)$ and $\mathcal{L}_{\sigma\text{-oc}}(X, Y)$ are subspaces of $\mathcal{L}_{\mathrm{ob}}(X, Y)$, and $\mathcal{L}_{\mathrm{oc}}(X, Y) \subseteq \mathcal{L}_{\sigma\text{-oc}}(X, Y)$.

Suppose that Y is order complete. Then $\mathcal{L}_{\mathrm{ob}}(X, Y)$ is an order complete vector lattice by Riesz-Kantorovich Theorem 6.3.2. It follows immediately from Theorem 6.5.10 that $\mathcal{L}_{\mathrm{oc}}(X, Y)$ and $\mathcal{L}_{\sigma\text{-oc}}(X, Y)$ are sublattices of $\mathcal{L}_{\mathrm{ob}}(X, Y)$. In fact, projection bands:

6.5.11 Theorem (Ogasawara)**.** *If Y is order complete then $\mathcal{L}_{\mathrm{oc}}(X, Y)$ and $\mathcal{L}_{\sigma\text{-oc}}(X, Y)$ are projection bands in $\mathcal{L}_{\mathrm{ob}}(X, Y)$.*

Proof. We already know that $\mathcal{L}_{\mathrm{oc}}(X, Y)$ and $\mathcal{L}_{\sigma\text{-oc}}(X, Y)$ are sublattices of $\mathcal{L}_{\mathrm{ob}}(X, Y)$. Since $\mathcal{L}_{\mathrm{ob}}(X, Y)$ is order complete by the Riesz-Kantorovich Theorem, every band in

[2] In the literature, the notations $\mathcal{L}_n(X, Y)$ and $\mathcal{L}_c(X, Y)$ are often used instead of $\mathcal{L}_{\mathrm{oc}}(X, Y)$ and $\mathcal{L}_{\sigma\text{-oc}}(X, Y)$.

it is a projection band by Corollary 5.2.23. Therefore, it suffices to prove that they are bands.

It is easy to see that $\mathcal{L}_{\mathrm{oc}}(X,Y)$ and $\mathcal{L}_{\sigma\text{-oc}}(X,Y)$ are ideals in $\mathcal{L}_{\mathrm{ob}}(X,Y)$. Indeed, suppose that $0 \leqslant S \leqslant T \in \mathcal{L}_{\sigma\text{-oc}}(X,Y)$, and $x_n \downarrow 0$. Then $Tx_n \downarrow 0$, hence $Sx_n \downarrow 0$. Hence $S \in \mathcal{L}_{\sigma\text{-oc}}(X,Y)$ by Exercise 6.5.1, so that $\mathcal{L}_{\sigma\text{-oc}}(X,Y)$ is an ideal.

To verify that $\mathcal{L}_{\sigma\text{-oc}}(X,Y)$ is order closed, suppose that $0 \leqslant T_\lambda \uparrow T$ for some net (T_λ) in $\mathcal{L}_{\sigma\text{-oc}}(X,Y)$ and some $T \in \mathcal{L}_{\mathrm{ob}}(X,Y)$. It suffices to show that $T \in \mathcal{L}_{\sigma\text{-oc}}(X,Y)$. Take a sequence (x_n) such that $x_n \downarrow 0$. For every λ and every n, we have

$$0 \leqslant Tx_n = Tx_1 - T(x_1 - x_n) \leqslant Tx_1 - T_\lambda(x_1 - x_n) = (T - T_\lambda)x_1 + T_\lambda x_n.$$

Since T_λ is order continuous, $T_\lambda x_n \downarrow 0$ (in n), so that $0 \leqslant \inf Tx_n \leqslant (T - T_\lambda)x_1$ (note that $\inf Tx_n$ exists because Y is order complete). As we now take inf over λ, we get $\inf_\lambda (T - T_\lambda)x_1 = 0$ by Riesz-Kantorovich Theorem 6.3.2. Hence, $\inf Tx_n = 0$.

The argument for $\mathcal{L}_{\mathrm{oc}}(X,Y)$ is similar. $\square$

Let $\mathcal{L}_{\mathrm{oc}}(X,Y)^d$ be the complimentary band of $\mathcal{L}_{\mathrm{oc}}(X,Y)$ in $\mathcal{L}_{\mathrm{ob}}(X,Y)$. Operators in $\mathcal{L}_{\mathrm{oc}}(X,Y)^d$ are sometimes called ***singular***. Suppose that $T \geqslant 0$; let S be the component of T in $\mathcal{L}_{\mathrm{oc}}(X,Y)$, i.e., the "order continuous part" of T. Similarly, let R be the component of T in $\mathcal{L}_{\sigma\text{-oc}}(X,Y)$, i.e., the "$\sigma$-order continuous part" of T. Then

$$Sx = \inf_{0 \leqslant x_\alpha \uparrow x} \sup Tx_\alpha \quad \text{and} \quad Rx = \inf_{0 \leqslant x_n \uparrow x} \sup Tx_n$$

for every $x \geqslant 0$; here the infs are over all positive increasing nets (or sequences) order converging to x. We refer the reader to [Zaa97] for a proof of this.

6.5.12. Again, suppose that Y is order complete and $T \in \mathcal{L}_{\mathrm{ob}}(X,Y)$. Put

$$N_T = \bigl\{x \in X \ :\ |T||x| = 0\bigr\}.$$

It is easy to see that N_T is an ideal. It is called the ***null ideal*** of T. The set $C_T \coloneqq N_T^d$ in X is referred to as the ***carrier*** of T.

Note that C_T is a band by Proposition 5.1.6. Also, if T is order continuous, then $|T|$ is also order continuous by Theorem 6.5.10, and it easily follows that in this case N_T is a band.

The following exercise shows that $C_T = \{0\}$ need not imply $T = 0$.

6.5.13 Exercise. Let $f\colon c \to \mathbb{R}$ be the limit functional, i.e., $f(x)$ is the limit of x. Show that $C_f = \{0\}$.

6.5.14 Theorem ([AB06]). *Suppose that Y is order complete and $T \in \mathcal{L}_{\mathrm{ob}}(X, Y)$. Let J_T be the principal ideal generated by T in $\mathcal{L}_{\mathrm{ob}}(X, Y)$. Then*

(i) *T is order continuous iff N_S is a band for every $S \in J_T$;*
(ii) *T is σ-order continuous iff N_S is a σ-closed ideal for every $S \in J_T$.*

We are now going to consider order continuous lattice homomorphisms.

6.5.15 Exercise. A lattice homomorphism $T\colon X \to Y$ is order continuous iff it preserves suprema of arbitrary sets. That is, for every non-empty subset A of X, if $\sup A$ exists in X then $\sup T(A)$ exists in Y and $\sup T(A) = T\big(\sup A\big)$. Same for infima.

6.5.16 Exercise. Every lattice isomorphism between vector lattices is order continuous.

We can now partially extend Theorem 6.5.10

6.5.17 Corollary. *For $T \in \mathcal{L}(X, Y)$, where Y is Archimedean, TFAE:*

(i) *T is order continuous;*
(ii) *T is order bounded and $\inf|Tx_\alpha| = 0$ whenever $x_\alpha \downarrow 0$.*

Proof. (i)⇒(ii) follows from Theorem 6.5.5.

(ii)⇒(i) Let $J\colon Y \hookrightarrow Y^\delta$ be the canonical inclusion map. Clearly, it is a lattice homomorphism. If $x_\alpha \downarrow 0$ then $\inf|Tx_\alpha| = 0$; it follows from Exercise 6.5.15 that $\inf|JTx_\alpha| = \inf J|Tx_\alpha| = 0$. Theorem 6.5.10 yields that JT is order continuous. If $x_\alpha \xrightarrow{\mathrm{o}} 0$ in X then $JTx_\alpha \xrightarrow{\mathrm{o}} 0$ in Y^δ and, therefore, $Tx_\alpha \xrightarrow{\mathrm{o}} 0$ in X by Corollary 3.2.32. □

Recall that if $T\colon X \to Y$ is a lattice homomorphism then $\ker T$ is an ideal in X and $\operatorname{Range} T$ is a sublattice in Y. Recall also the standard fact that a linear functional on a Banach space is continuous iff its kernel is norm closed. The following result is somewhat analogous:

6.5.18 Theorem. *Let $T\colon X \to Y$ be a lattice homomorphism and X is Archimedean. Then T is order continuous iff $\ker T$ is a band and $\operatorname{Range} T$ is regular.*

The theorem follows immediately from this proposition, which characterizes connections between several properties.

6.5.19 Proposition. *Let $T\colon X \to Y$ be a lattice homomorphism.*

(i) *If T is order continuous then $\ker T$ is a band;*
(ii) *If T is order continuous and X is Archimedean then $\operatorname{Range} T$ is regular;*
(iii) *If $\ker T$ is a band and T is onto then T is order continuous;*
(iv) *If $\ker T$ is a band and $\operatorname{Range} T$ is regular then T is order continuous.*

Proof. (i) Suppose that $0 \leqslant x_\alpha \uparrow x$ for some net (x_α) in $\ker T$. It follows from $Tx_\alpha \uparrow Tx$ that $Tx = 0$, hence $\ker T$ is a band.

(ii) Let $H = \operatorname{Range} T$ and B a non-empty subset of H_+ with $v = \sup_H B$; we need to show that $v = \sup_Y B$. Find $u \in X$ with $Tu = v$. Replacing u with u^+, we may assume that $u \geqslant 0$. Put $A = [0, u] \cap T^{-1}(B)$.

We claim that $T(A) = B$. Indeed, we clearly have $T(A) \subseteq B$. On the other hand, let $b \in B$. We can find $x \in X$ with $Tx = b$. Put $a = x^+ \wedge u$, then $Ta = b^+ \wedge v = b$. It follows, in particular, that A is non-empty.

Claim: $\sup A = u$. Indeed, suppose that $A \leqslant x \leqslant u$. It follows from $B = T(A) \leqslant Tx \leqslant Tu = v$ and $v = \sup B$ that $Tx = Tu = v$. Fix any $a \in A$. Then $a + u - x \geqslant a \geqslant 0$ and $T(a + u - x) = Ta \in B$; it follows that $a + u - x \in A$. Iterating, we get $a + n(u - x) \in A$ and, therefore, $n(u - x) \leqslant u - a$ for every $n \in \mathbb{N}$. Since X is Archimedean, this implies $x = u$ and, therefore, $u = \sup A$.

By Exercise 6.5.15, $v = Tu = \sup_Y T(A) = \sup_Y B$.

(iii) Since T is a lattice homomorphism, $\ker T$ is an ideal and the induced operator $\widetilde{T}\colon X/\ker T \to Y$ is a lattice isomorphism by Exercise 3.4.5, hence $\widetilde{T}$ and $(\widetilde{T})^{-1}$ are both positive by Theorem 1.6.13.

Let $0 \leqslant x_\alpha \uparrow x$; we need to show that $Tx_\alpha \uparrow Tx$. Clearly, $Tx_\alpha \leqslant Tx$ for all α. Suppose that $h \in Y$ with $Tx_\alpha \leqslant h \leqslant Tx$ for all α. Then $h = Ty$ for some $y \in X$. WLOG, $0 \leqslant y \leqslant x$; otherwise, replace y with $y^+ \wedge x$. It follows that $\widetilde{T}\tilde{x}_\alpha \leqslant \widetilde{T}\tilde{y}$ and, therefore, $\tilde{x}_\alpha \leqslant \tilde{y}$ for every α. Put $z_\alpha = x_\alpha \vee y - y$. Then $\tilde{z}_\alpha = \tilde{x}_\alpha \vee \tilde{y} - \tilde{y} = 0$, hence $z_\alpha \in \ker T$. It follows from $x_\alpha \uparrow x$ that $z_\alpha \uparrow x \vee y - y = x - y$. Since $\ker T$ is a band, we conclude that $x - y \in \ker T$, and, therefore, $h = Ty = Tx$.

(iv) Suppose that $x_\alpha \downarrow 0$ in X Let $H = \operatorname{Range} T$. Viewed as an operator from X to H, T is order continuous by (iii), hence $Tx_\alpha \downarrow 0$ in H and, therefore, in Y because H is regular. $\square$

Since a quotient map over an ideal is a lattice homomorphism by Theorem 3.4.2, and it is always surjective, we get the following:

6.5.20 Corollary. *Let J be an ideal in X. The canonical quotient map $Q\colon X \to X/I$ is order continuous iff J is a band.*

6.5.21 Example. Recall that c_0 is a closed ideal in ℓ_∞, but not a band. The quotient operator $Q\colon \ell_\infty \to \ell_\infty/c_0$ is a surjective lattice homomorphism. However, it is not order continuous. Indeed, let

$$x_n = (\overbrace{0, \dots, 0}^{n-1}, 1, 1, \dots).$$

Clearly $x_n \downarrow 0$ in ℓ_∞. On the other hand, $\mathbb{1} - x_n$ is in c_0 for every n, so that $Qx_n = Q\mathbb{1}$ and, therefore, (Qx_n) is a non-zero constant sequence in ℓ_∞/c_0, hence it does not converge to zero in order.

6.5.22 Example. The following example shows that the regularity assumption in (iv) cannot be dropped. Let $T\colon C[0,1] \to L_1[0,1]$ be the natural embedding. Clearly, T is a lattice homomorphism, and $\ker T = \{0\}$ is a band. On the other hand, T fails to be order continuous. This may be shown directly, but we will actually see in Exercise 6.6.14 that no operator from $C[0,1]$ to $L_1[0,1]$ is order continuous.

6.6 Order dual

Suppose that X is a vector lattice. The space $\mathcal{L}_{\mathrm{ob}}(X, \mathbb{R})$, that is, the space of all order bounded linear functionals on X, is called the ***order dual*** of X and is denoted by $X^\sim$. The elements of $X^\sim_+$ are called ***positive functionals***.

We will prove in Theorem 6.7.1 that the order dual of a Banach lattice coincides with its norm dual. In particular, this will determine the order duals of $C(K)$ and $L_p(\mu)$ (when $1 \leqslant p \leqslant \infty$). We will show in Exercise 6.7.2 that a vector lattice may have a trivial order dual.

Since $\mathbb{R}$ is an order complete vector lattice, the following are immediate corollaries of Riesz-Kantorovich Theorems 6.3.1 and 6.3.2.

6.6.1 Proposition. (i) $X^\sim = \mathcal{L}_r(X,\mathbb{R})$, *i.e., every order bounded linear functional is regular;*

(ii) $X^\sim$ *is itself an order complete vector lattice;*

(iii) $f_\alpha \uparrow f$ *in* $X^\sim$ *iff* $f_\alpha(x) \uparrow f(x)$ *for every* $x \in X_+$.

6.6.2. It also follows that all Riesz-Kantorovich formulas in Section 6.3 are valid in $X^\sim$. In particular, for any $f,g \in X^\sim$ and any $x \in X_+$, we have

$$\begin{aligned}
f^+(x) &= \sup\bigl\{f(u) \,:\, 0 \leqslant u \leqslant x\bigr\},\\
f^-(x) &= -\inf\bigl\{f(u) \,:\, 0 \leqslant u \leqslant x\bigr\},\\
\lvert f\rvert(x) &= \sup\bigl\{f(u) \,:\, -x \leqslant u \leqslant x\bigr\}\\
&= \sup\bigl\{\bigl\lvert f(u)\bigr\rvert \,:\, -x \leqslant u \leqslant x\bigr\},\\
(f\vee g)(x) &= \sup\bigl\{f(u)+g(v) \,:\, u,v \geqslant 0,\ u+v=x\bigr\},\ \text{and}\\
(f\wedge g)(x) &= \inf\bigl\{f(u)+g(v) \,:\, u,v \geqslant 0,\ u+v=x\bigr\}.
\end{aligned}$$

Furthermore, for every $f \in X^\sim_+$ and every $x,y \in X$, we have

$$\begin{aligned}
f(x^+) &= \max\bigl\{g(x) \,:\, 0 \leqslant g \leqslant f\bigr\},\\
f(x^-) &= -\min\bigl\{g(x) \,:\, 0 \leqslant g \leqslant f\bigr\},\\
f\bigl(\lvert x\rvert\bigr) &= \max\bigl\{g(x) \,:\, -f \leqslant g \leqslant f\bigr\},\\
f(x\vee y) &= \max\{g(x)+h(v) \,:\, g,h \geqslant 0,\ g+h=f\bigr\},\\
f(x\wedge y) &= \min\{g(x)+h(v) \,:\, g,h \geqslant 0,\ g+h=f\bigr\}.
\end{aligned}$$

6.6.3 Exercise. $X^\sim$ is Archimedean.

6.6.4 Exercise. Let $X = C[0,1]$, f and g functionals on X defined by $f(x) = \int x$ and $g(x) = x(0)$. Show that $f,g \in X^\sim_+$ and $f \perp g$.

6.6.5. Since $\mathbb{R}$ is order continuous, the generalized Riesz-Kantorovich formulas 6.3.3 and 6.3.15

remain valid for functionals. That is, if $f_1,\dots,f_n\in X^\sim$ and $x\geqslant 0$ then

$$\begin{aligned}\Bigl(\bigvee_{i=1}^n f_i\Bigr)(x) &= \sup\Bigl\{\sum_{i=1}^n f_i(x_i)\ :\ x_1,\dots,x_n\geqslant 0,\ \sum_{i=1}^n x_i=x\Bigr\}\ \text{and}\\ \Bigl(\bigwedge_{i=1}^n f_i\Bigr)(x) &= \inf\Bigl\{\sum_{i=1}^n f_i(x_i)\ :\ x_1,\dots,x_n\geqslant 0,\ \sum_{i=1}^n x_i=x\Bigr\}\end{aligned}$$

If $f\in X_+^\sim$ and $x_1,\dots,x_n\in X$ then

$$\begin{aligned}f\Bigl(\bigvee_{i=1}^n x_i\Bigr) &= \max\Bigl\{\sum_{i=1}^n f_i(x_i)\ :\ f_1,\dots,f_n\geqslant 0,\ \sum_{i=1}^n f_i=f\Bigr\}\ \text{and}\\ f\Bigl(\bigwedge_{i=1}^n x_i\Bigr) &= \min\Bigl\{\sum_{i=1}^n f_i(x_i)\ :\ f_1,\dots,f_n\geqslant 0,\ \sum_{i=1}^n f_i=f\Bigr\}.\end{aligned}$$

We have the following "disjoint" variant of the Riesz-Kantorovich Theorem for functionals, cf. Proposition 6.3.6.

6.6.6 Proposition. *Let $f,g\in X^\sim$ such that $f\wedge g=0$, and $z\in X_+$. Then*

$$(f\vee g)(z)=\sup\bigl\{f(x)+g(y)\ :\ x\wedge y=0,\ x+y\leqslant z\bigr\}.$$

Proof. Clearly, if $x\wedge y=0$ and $x+y\leqslant z$ then $f(x)+g(y)\leqslant(f+g)(z)=(f\vee g)(z)$. Hence, $(f\vee g)(z)$ is greater than or equal to the sup, and we only need to prove the other inequality. Fix $\varepsilon>0$. By the Riesz-Kantorovich formula for $f\wedge g$ in 6.6.2, find positive u and v such that $u+v=z$ and $f(v)+g(u)<\varepsilon$. Put $x=u-u\wedge v$ and $y=v-u\wedge v$. Then $x\wedge y=0$ by Exercise 1.5.4, and $x+y\leqslant u+v=z$. Note also that $f(u\wedge v)+g(u\wedge v)\leqslant f(v)+g(u)<\varepsilon$. It follows that

$$\begin{aligned}(f\vee g)(z)=(f+g)(z)=f(u)+g(u)+f(v)+g(v)&\leqslant f(u)+g(v)+\varepsilon\\ &\leqslant f(u-u\wedge v)+g(v-u\wedge v)+2\varepsilon=f(x)+g(y)+2\varepsilon.\end{aligned}$$

□

Note that $x+y\leqslant z$ cannot be replaced with $x+y=z$ in the proposition because there may be no non-trivial disjoint x and y satisfying $x+y=z$.

6.6.7 Proposition. *Let X be a vector lattice, $f,g\in X_+^\sim$ with $f\perp g$, $u,v\in X_+$, and $\varepsilon>0$. Then there exist $x\in[0,u]$ and $y\in[0,v]$ such that $x\perp y$, $f(x)>f(u)-\varepsilon$, and $g(y)>g(v)-\varepsilon$.*

Proof. Using the Riesz-Kantorovich formulas,

$$\begin{aligned}&\text{since } (f\wedge g)(u)=0,\quad \exists u_1,u_2\geqslant 0 &&u=u_1+u_2 \text{ and } f(u_1)+g(u_2)<\varepsilon;\\ &\text{since } (f\wedge g)(v)=0,\quad \exists v_1,v_2\geqslant 0 &&v=v_1+v_2 \text{ and } f(v_1)+g(v_2)<\varepsilon.\end{aligned}$$

Put $x = u_2 - u_2 \wedge v_1$ and $y = v_1 - v_1 \wedge u_2$. It is clear that $0 \leqslant x \leqslant u$ and $0 \leqslant y \leqslant v$. Furthermore, $x \perp y$ by Exercise 1.5.4. Also,

$$f(x) = f(u_2) - f(u_2 \wedge v_1) \geqslant f(u - u_1) - f(v_1) = f(u) - f(u_1) - f(v_1) \geqslant f(u) - 2\varepsilon.$$

Similarly, $g(y) \geqslant g(v) - 2\varepsilon$. □

Next, we discuss restriction maps.

6.6.8 Exercise. Let J be an ideal in X. Then $\big|f_{|J}\big| = |f|_{|J}$ for every $f \in X^\sim$. That is, the restriction map, viewed as an operator from $X^\sim$ to $J^\sim$, is a lattice homomorphism.

While X may have no projection bands, $X^\sim$ is order complete and, therefore, has lots of projection bands. We will now show that every ideal in X induces a projection band in $X^\sim$. In the preceding exercise, the restriction map acts from $X^\sim$ to $J^\sim$. We now apply Theorems 6.3.9 and 6.3.11 together with Remark 6.3.10 to show that the restriction map may be viewed as a band projection on $X^\sim$:

6.6.9 Theorem. *Let J be an ideal in X. There exists a band projection R on $X^\sim$ such that Rf agrees with f on J and vanishes on J^d for every $f \in X^\sim$. Furthermore,*

$$(Rf)(x) = \sup\big\{f(v) \,:\, v \in J \text{ and } v \leqslant x\big\} = \sup\big\{f(x \wedge v) \,:\, v \in J_+\big\}$$

for $x \in X_+$ and $f \in X_+^\sim$. In the special case when $J = I_u$ for some $u \in X_+$, we have $(Rf)(x) = \sup_n f(x \wedge nu)$ for $f \in X_+^\sim$ and $x \in X_+$.

The following example provides some intuition for the restriction map.

6.6.10 Example. Let $X = L_p(\mu)$ for some $1 \leqslant p < \infty$ and a σ-finite measure μ; we identify $X^\sim = X^*$ with $L_q(\mu)$ where $q = p^*$. Let $u \in X_+$ and put $J = I_u$; let $0 \in f \in L_q(\mu)$, and let R be as in the preceding theorem. We claim that as a function in $L_q(\mu)$, Rf is the restriction of f to $\operatorname{supp} u$.

Indeed, let g be restriction of f to $\operatorname{supp} u$. It is easy to see that g agrees with f on I_u and vanishes on I_u^d. It follows that g agrees with Rf on I_u and on I_u^d, hence on $I_u + I_u^d$. Since X is order continuous, $I_u + I_u^d$ is dense in X, hence $g = Rf$.

Let $A = \operatorname{supp} g$. Note that the null ideal N_g of g consists of all the functions in $L_p(\mu)$ that vanish on A (up to a set of measure zero), and X/N_g may be identified with $L_p(A)$, with the quotient map $q\colon X \to X/N_g$ being just the restriction to A. In particular, it follows from $A \subseteq \operatorname{supp} u$ that $q(u)$ is a weak unit in X/N_g.

Let $u \in X_+$. Recall that u is a weak unit in B_u. If X has PPP then $B_u = \{u\}^{dd}$ is a projection band, hence B_u may be identified with $X/\{u\}^d$. Motivated by the previous example, we will show that even if X fails PPP, we may still view u as a weak unit in an appropriate quotient space.

6.6.11 Proposition. *Let $u \in X_+$, $f \in X_+^\sim$ and $h = Rf \in X^\sim$ be the restriction of f to I_u in the sense of Theorem 6.6.9. Then $Q(u)$ is a weak unit in X/N_h, where $Q\colon X \to N_h$ is the quotient map.*

Proof. Suppose $Q(x) \perp Q(u)$ for some $x \in X_+$; it suffices to show that $Q(x) = 0$. Observe that $Q(x) \perp nQ(u)$ for every $n \in \mathbb{N}$; it follows that $Q(x \wedge nu) = Q(x) \wedge nQ(u) = 0$, hence $h(x \wedge nu) = 0$. By Theorem 6.6.9, $0 = h(x \wedge nu) = \sup_m f(x \wedge nu \wedge mu) = f(x \wedge nu)$, and, therefore, $h(x) = \sup_n f(x \wedge nu) = 0$. It follows that $Q(x) = 0$. □

6.6.1 Order continuous dual

We write $X^\sim_{\mathrm{oc}}$ and $X^\sim_{\sigma\text{-oc}}$ for the spaces of all order continuous functionals and of all σ-order continuous functionals in $X^\sim$, respectively[3]. The following is a special case of Ogasawara's Theorem 6.5.11.

6.6.12 Theorem. *$X^\sim_{\mathrm{oc}}$ and $X^\sim_{\sigma\text{-oc}}$ are projection bands in $X^\sim$.*

It is easy to see that if X is an order continuous Banach lattice then every functional in X^* is order continuous and σ-order continuous. On the other hand, the following example shows that $X^\sim_{\mathrm{oc}}$ and $X^\sim_{\sigma\text{-oc}}$ may be trivial.

6.6.13 Theorem. *Let $X = C[0,1]$. Then $X^\sim_{\mathrm{oc}} = X^\sim_{\sigma\text{-oc}} = \{0\}$.*

Proof. Suppose that f is a σ-order continuous functional on X; we will show that $f = 0$. WLOG, $f > 0$; otherwise replace f with $|f|$. If $f(\mathbb{1}) = 0$ then $f = 0$ and we are done. Suppose $f(\mathbb{1}) > 0$. WLOG, scaling f if necessary, we may assume that $f(\mathbb{1}) > 1$.

Claim: for every $s \in [0,1]$ and every $\varepsilon > 0$ there exists $x \in X$ with $0 \leqslant x \leqslant \mathbb{1}$, $x(s) = 1$, and $f(x) < \varepsilon$. Indeed, for every n, consider the "peak" function which vanishes outside of $[s - \frac{1}{n}, s + \frac{1}{n}]$, equals one at s and is linear in between. Let y_n be the restriction of this function to $[0,1]$. Then $y_n \downarrow 0$, so that $f(y_n) \downarrow 0$, hence $f(y_{n_0}) < \varepsilon$ for a sufficiently large n_0. Take $x = y_{n_0}$. This proves the claim.

Let (t_n) be a dense sequence in $[0,1]$, e.g., an enumeration of the rationals in $[0,1]$. For each n, let x_n be as in the Claim with $s = t_n$ and $\varepsilon = 2^{-n}$. Put $z_n = \bigvee_{i=1}^n x_i$. Then $z_n \leqslant \sum_{i=1}^n x_i$ implies $f(z_n) \leqslant \sum_{i=1}^n f(x_i) < 1$ for every n. On the other hand, $z_n \uparrow \mathbb{1}$ because if $y \geqslant z_n$ for every n then $y(t_n) \geqslant z_n(t_n) = 1$, hence $y \geqslant \mathbb{1}$. It follows that $f(z_n) \uparrow f(\mathbb{1})$, so that $f(\mathbb{1}) \leqslant 1$; a contradiction. □

6.6.14 Exercise. If X is an order continuous Banach lattice then $\mathcal{L}_{\mathrm{oc}}\big(C[0,1], X\big)$ and $\mathcal{L}_{\sigma\text{-oc}}\big(C[0,1], X\big)$ are trivial.

6.6.15 Exercise. Let $X = \ell_\infty$ and $f \in X^*_+$ be a Banach limit. Show that that f is not order continuous. Similarly, if $g \in X^*_+$ is the limit along a free ultrafilter, then g is not order continuous. Observe also that g is a lattice homomorphism, while f is not.

[3] In literature, $X^\sim_{\mathrm{oc}}$ and $X^\sim_{\sigma\text{-oc}}$ are often denoted by $X^\sim_n$ and $X^\sim_c$, respectively.

We observed in Theorem 6.6.13 that $X^\sim_{\mathrm{oc}}$ and $X^\sim_{\sigma\text{-oc}}$ may be trivial. Actually, it will be seen in Exercise 6.7.2 in the next section that even the entire order dual of a vector lattice may be trivial. We will later see some results that are valid for vector lattices for which $X^\sim$, $X^\sim_{\mathrm{oc}}$, or $X^\sim_{\sigma\text{-oc}}$ is "sufficiently large". Here is a way to define what "sufficiently large" means. Let X be a vector lattice; let Z be an ideal in $X^\sim$ (in application, Z will often be either $X^\sim$ or $X^\sim_{\mathrm{oc}}$). We say that Z ***separates points of*** X if for every $x, y \in X$ with $x \neq y$ there exists $f \in Z$ such that $f(x) \neq f(y)$.

6.6.16 Exercise. Let Z be an ideal in $X^\sim$. TFAE:

(i) Z separates points of X;

(ii) For every $x > 0$ there exists $f \in Z_+$ such that $f(x) > 0$;

(iii) For every $x \in X$, $x \geqslant 0$ iff $f(x) \geqslant 0$ for every $f \in Z_+$.

The following theorem says that two order continuous functionals are disjoint iff their carriers are disjoint.

6.6.17 Theorem (Nakano)**.** *Let X be an Archimedean vector lattice and f and g order continuous functionals on X. Then the following are equivalent.*

(i) $f \perp g$

(ii) $C_f \subseteq N_g$

(iii) $C_g \subseteq N_f$

(iv) $C_f \perp C_g$

Proof. We may assume without loss of generality that f and g are positive. Note that N_f is a band by 6.5.12.

(i)⇒(ii) Suppose that $f \wedge g = 0$ and $0 \leqslant x \in C_f$. By a Riesz-Kantorovich formula 6.6.2, we have

$$0 = (f \wedge g)(x) = \inf\bigl\{ f(y) + g(x-y) \ :\ y \in [0,x]\bigr\}.$$

Fix $\varepsilon > 0$, then for every $n \in \mathbb{N}$ one can find $y_n \in [0,x]$ such that $f(y_n)+g(x-y_n) < \frac{\varepsilon}{2^n}$. Put $z_n = \bigwedge_{i=1}^n y_i$. Clearly, $z_n \downarrow\geqslant 0$. We claim that $z_n \downarrow 0$. Indeed, suppose that there exists $z \geqslant 0$ such that $z \leqslant z_n$ for all n. Then $0 \leqslant f(z) \leqslant f(z_n) \leqslant f(y_n) < \frac{\varepsilon}{2^n}$, so that $f(z) = 0$. It follows that $z \in N_f$. But $z \leqslant z_1 = y_1 \leqslant x$ implies $z \in C_f$. Since $N_f \cap C_f = \{0\}$ by Exercise 5.1.5(iii), we conclude that that $z = 0$. Further,

$$\begin{aligned} 0 \leqslant g(x - z_n) &= g\Bigl(x - \bigwedge_{i=1}^n y_i\Bigr) = g\Bigl(\bigvee_{i=1}^n (x-y_i)\Bigr) \\ &\leqslant g\Bigl(\sum_{i=1}^n (x-y_i)\Bigr) = \sum_{i=1}^n g(x-y_i) < \sum_{i=1}^n \frac{\varepsilon}{2^i} < \varepsilon. \end{aligned}$$

It follows that $g(x-z_n)<\varepsilon$. Since g is order continuous, $g(x-z_n)\uparrow g(x)$, so that $g(x)\leqslant\varepsilon$. It follows that $g(x)=0$, hence $x\in N_g$.

(ii)⇒(iii) Since $C_f\subseteq N_g$, we have $C_g=N_g^d\subseteq C_f^d=N_f^{dd}=N_f$ by Theorem 5.1.7.

(iii)⇒(i) Suppose $C_g\subseteq N_f$. If $x=y+z$ where $y\in N_g$ and $z\in C_g$ then

$$0\leqslant(f\wedge g)(x)=(f\wedge g)(y)+(f\wedge g)(z)\leqslant g(y)+f(z)=0,$$

so that $f\wedge g$ vanishes on $N_g\oplus C_g$. Next, observe that $(N_g\oplus C_g)^d=\{0\}$. Indeed, if $x\perp(N_g\oplus C_g)$ then $x\in N_g^d$ and $x\in C_g^d=N_g$, so that $x=0$ by Exercise 5.1.5(iii). It follows that $(N_g\oplus C_g)^{dd}=X$, hence, by Corollary 5.1.9, X is the band generated by $N_g\oplus C_g$. Since $f\wedge g$ is order continuous by Ogasawara's Theorem 6.6.12 and vanishes on $N_g\oplus C_g$, it vanishes on X by Proposition 5.1.1(ii).

(iii)⇔(iv) If $C_g\subseteq N_f$ then $C_f\perp C_g$ because $N_f\perp C_f$. Conversely, suppose that $C_f\perp C_g$. Then $C_g\subseteq C_f^d=N_f^{dd}=N_f$ by Theorem 5.1.7. □

6.6.18. *Second order dual.* Again, let X be a vector lattice. Put $X^{\sim\sim}=(X^\sim)^\sim$. For each $x\in X$, the map $\hat{x}\colon X^\sim\to\mathbb{R}$ defined by $\hat{x}(f)=f(x)$ is linear. If $x\geqslant0$ then $\hat{x}\geqslant0$. Furthermore,

$$\hat{x}(f)=f(x)=f(x^+)-f(x^-)=\widehat{x^+}(f)-\widehat{x^-}(f)$$

for every $f\in X^\sim$. Hence, $\hat{x}=\widehat{x^+}-\widehat{x^-}$. Therefore, $\hat{x}$ is regular, hence $\hat{x}\in X^{\sim\sim}$. Moreover, the map $\hat{\ }\colon X\to X^{\sim\sim}$ is a lattice homomorphism. Indeed, let $x\in X$, then for every $f\in X_+^\sim$ it follows from Riesz-Kantorovich formulas 6.6.2 that

$$(\hat{x})^+(f)=\sup\{\hat{x}(g)\ :\ 0\leqslant g\leqslant f\}=\sup\{g(x)\ :\ 0\leqslant g\leqslant f\}=f(x^+)=\widehat{x^+}(f).$$

It follows that $(\hat{x})^+=\widehat{x^+}$, so that $\hat{\ }$ preserves the lattice operations.

Note that this map need not be one-to-one, as $X^\sim$ could be trivial. However, if $X^\sim$ separates the points of X, then $\hat{\ }$ is one-to-one (and, therefore, a lattice isomorphism) by Exercise 6.6.16. In this case, we may view X as a sublattice of $X^{\sim\sim}$.

6.6.19 Proposition. *$\hat{x}$ is order continuous for every $x\in X$.*

Proof. Since $\hat{x}=\widehat{x^+}-\widehat{x^-}$, we may assume without loss of generality that $x\geqslant0$. Suppose that $f_\alpha\downarrow0$ in $X^\sim$. Then $f(x_\alpha)\downarrow0$ by Riesz-Kantorovich Theorem 6.6.1(iii), so that $\hat{x}(f_\alpha)\downarrow0$; hence $\hat{x}$ is order continuous by Exercise 6.5.1. □

Suppose that $X^\sim$ separates points of X. By 6.6.18, the map $x\in X\mapsto\hat{x}\in X^{\sim\sim}$ is a lattice isomorphic embedding, so that, with some abuse of notation, we may view X as a sublattice of $X^{\sim\sim}$. Moreover, Proposition 6.6.19 asserts that $X\subseteq(X^\sim)^\sim_{\mathrm{oc}}$.

Could this inclusion be an equality? Generally no, because $(X^\sim)^\sim_{\mathrm{oc}}$ is a band in $X^{\sim\sim}$ by Theorem 6.6.12, while X need not be a band there. For example, if $X = c_0$ then $X^\sim = \ell_1$ and $X^{\sim\sim} = \ell_\infty$, but c_0 is not a band in ℓ_∞ because $c_0^{dd} = \ell_\infty$. However, in this case, $(X^\sim)^\sim_{\mathrm{oc}} = \ell_\infty$ as well.

6.6.20 Proposition. *If $X^\sim$ separates points of X then $(X^\sim)^\sim_{\mathrm{oc}}$ equals X^{dd}, the band generated by X in $X^{\sim\sim}$.*

Proof. Note that both $X^\sim$ and $X^{\sim\sim}$ are order complete. Since $X \subseteq (X^\sim)^\sim_{\mathrm{oc}}$ and $(X^\sim)^\sim_{\mathrm{oc}}$ is a band in $X^{\sim\sim}$, we have $X^{dd} \subseteq (X^\sim)^\sim_{\mathrm{oc}}$. To show the equality, it suffices to prove that $(X^\sim)^\sim_{\mathrm{oc}} \perp X^d$ or, equivalently, $(X^\sim)^\sim_{\mathrm{oc}} \cap X^d = \{0\}$. Let $\xi \in (X^\sim)^\sim_{\mathrm{oc}} \cap X^d$. Since ξ is order continuous, N_ξ is a band.

Fix $x \in X_+$. Then $\xi \perp \hat{x}$. Since both ξ and $\hat{x}$ are order continuous, we have $C_\xi \subseteq N_{\hat{x}}$ by Nakano's Theorem 6.6.17. Therefore, for every positive $f \in C_\xi$ we have $f(x) = \hat{x}(f) = 0$. Since this is true for every $x \in X_+$, we conclude that $f = 0$. It follows that $C_\xi = 0$, so that $N_\xi = C_\xi^d = X^\sim$, hence $\xi = 0$. □

6.7 Positive functionals on a Banach lattice

We write X^* for the norm dual of a normed space X.

6.7.1 Theorem. *If X is a Banach lattice then $X^\sim = X^*$.*

Proof. Theorem 6.1.9 yields

$$X^\sim = \mathcal{L}_{\mathrm{ob}}(X, \mathbb{R}) \subseteq L(X, \mathbb{R}) = X^*.$$

On the other hand, suppose that $f \in X^*$. Fix $x \geqslant 0$, then for every $u \in [0, x]$ we have $\lvert f(u) \rvert \leqslant \lVert f \rVert \lVert u \rVert \leqslant \lVert f \rVert \lVert x \rVert$, so that $f[0, x] \subseteq \bigl[-\lVert f \rVert \lVert x \rVert, \lVert f \rVert \lVert x \rVert\bigr]$. Hence, f is order bounded. □

Let $X = L_p(\mu)$ where $1 \leqslant p < \infty$ and μ is a σ-finite measure. It is well known that X^* is isometric (as a Banach space) to $L_q(\mu)$, where q is the conjugate of p. It is easy to see that this isometry is also a lattice isomorphism, i.e., positive functionals on $L_p(\mu)$ correspond to a.e. positive functions in $L_q(\mu)$. Thus, $L_q(\mu)$ is still the dual space of $L_p(\mu)$ in the category of Banach lattices. Similarly, that the classical isometry between c_0^* and ℓ_1 is a lattice isomorphism.

6.7.2 Exercise. Show that the order dual of $L_p[0, 1]$ with $0 \leqslant p < 1$ is trivial.

If X is a normed lattice but not a Banach lattice, the situation is slightly more subtle.

6.7.3 Proposition. *Let X be a normed lattice. Then X^* is a Banach lattice and an ideal in $X^\sim$.*

Proof. As in the proof of Theorem 6.7.1, we have $X^* \subseteq X^\sim$. Suppose $f \in X^*$. For every $x \in X$, it follows from Riesz-Kantorovich formulas that

$$\bigl||f|(x)\bigr| \leqslant |f|\bigl(|x|\bigr) = \sup\bigl\{\bigl|f(y)\bigr| \;:\; |y| \leqslant |x|\bigr\} \leqslant \|f\|\|x\|$$

because $\bigl|f(y)\bigr| \leqslant \|f\|\|y\| \leqslant \|f\|\|x\|$ whenever $|y| \leqslant |x|$. It follows that $|f| \in X^*$ with $\bigl\||f|\bigr\| \leqslant \|f\|$ and, therefore, X^* is closed under lattice operations; hence a sublattice of $X^\sim$.

For every $x \in X$, we have $\bigl|f(x)\bigr| \leqslant |f|\bigl(|x|\bigr) \leqslant \bigl\||f|\bigr\|\|x\|$, so that $\|f\| = \bigl\||f|\bigr\|$.

Suppose that $0 \leqslant f \leqslant g$ for some $f \in X^\sim$ and $g \in X^*$. For every $x \in X$, we have

$$\bigl|f(x)\bigr| \leqslant f\bigl(|x|\bigr) \leqslant g\bigl(|x|\bigr) \leqslant \|g\|\|x\|.$$

It follows that $f \in X^*$ and $\|f\| \leqslant \|g\|$. Therefore, X^* is an ideal in $X^\sim$ and the norm on X^* is a lattice norm. □

6.7.4 Example. *A normed lattice for which $X^* \subsetneq X^\sim$.* Let $X = c_{00}$ equipped with $\|\cdot\|_{\ell_1}$. For $x \in c_{00}$, put $f(x) = \sum_{n=1}^\infty nx_n$ (note that this sum has only finitely many non-zero terms). Observe that f is a positive linear functional on X, yet f is not bounded.

6.7.5. Let X be a normed space. It follows from Hahn-Banach Theorem that X^* separates the points of X. Exercise 6.6.16 now immediately yields that for every $x \in X$,

- if $f(x) \geqslant 0$ for all $f \in X^*$ then $x \geqslant 0$;
- if $x > 0$ then there exists $f \in X^*_+$ such that $f(x) > 0$.

The latter statements may be somewhat refined. By Hahn-Banach Theorem, for every $x \in X$ one can find a functional $f \in S_{X^*}$ which attains its norm on x, i.e., $f(x) = \|x\|$. Conversely, given $f \in X^*$, it follows from the definition of $\|f\|$ that for every $\varepsilon > 0$ one can find $x \in S_X$ such that $f(x) \geqslant \|f\| - \varepsilon$; i.e., f "almost" attains its norm on x. We observe next that if one of x or f is positive, then the other one can also be chosen to be positive.

6.7.6 Exercise. Let X be a normed lattice.

(i) If $x \in X_+$ then there exists $f \in S^+_{X^*}$ such that $f(x) = \|x\|$;
(ii) For every $f \in X^*_+$ and $\varepsilon > 0$ there exists $x \in S^+_X$ such that $f(x) \geqslant \|f\| - \varepsilon$.

The next proposition shows that if two functionals are disjoint then they "almost" attain their norms on disjoint vectors; cf. Theorem 6.6.17.

6.7.7 Proposition. *Let X be a normed lattice, $f,g \in X^*_+$ with $f \perp g$, and $\varepsilon > 0$. There exist* disjoint *x and y in B_X^+ such that $f(x) \geqslant \|f\| - \varepsilon$ and $g(y) \geqslant \|g\| - \varepsilon$.*

Proof. Using Exercise 6.7.6, find $u, v \in S_X^+$ such that $f(u) \geqslant \|f\| - \varepsilon$ and $g(v) \geqslant \|g\| - \varepsilon$. Find x and y as in Proposition 6.6.7; it is clear that they satisfy the required condition. □

Recall that if J is a proper closed ideal of X then Hahn-Banach theorem yields that there exists a non-zero functional $f \in X^*$ such that f vanishes on J. It follows from the Riesz-Kantorovich formulas that $|f|$ also vanishes on J. Thus, we get the following.

6.7.8 Proposition. *If J is a proper closed ideal in a normed lattice X then there exists a positive non-zero functional on X which vanishes on J.*

6.7.9 Exercise. Let $T\colon X \to Y$ be a positive bounded operator between normed lattices. The adjoint T^* of T is positive.

6.7.10 Proposition. *Let X be a normed lattice. The canonical embedding of X into X^{**} is a lattice isometry.*

Proof. Recall that the embedding is given by $j(x) = \hat{x}$ where $\hat{x}(f) = f(x)$ and is always an isometry. In case when X is a Banach lattice, Theorem 6.7.1 guarantees that $X^* = X^\sim$, while 6.6.18 asserts that the map $x \in X \mapsto \hat{x} \in X^\sim$ is a one-to-one lattice homomorphism. Suppose that X is a normed lattice. It follows from Proposition 6.7.3 that X^{**} is a Banach lattice. It is easy to see that $x \geqslant 0$ iff $j(x) \geqslant 0$, hence j is a lattice isomorphic embedding; see 1.6.15. □

6.7.11 Exercise. Use Proposition 6.7.10 to show that if X is a normed lattice, its norm completion $\overline{X}$ is a Banach lattice. More precisely, there is a unique Banach lattice order on $\overline{X}$ which agrees with the original order on X.

6.7.12 Corollary. *If X is a normed lattice then X^* is order complete. In particular, every reflexive Banach lattice is order complete.*

Proof. If X is a Banach lattice then $X^* = X^\sim$ by Theorem 6.7.1, and $X^\sim$ is always order complete by Proposition 6.6.1(ii). If X is only a normed lattice then $X^* = \overline{X}^*$ where $\overline{X}$ is the completion of X, and $\overline{X}$ is a Banach lattice. □

Since $L_p(\mu)$ is reflexive when $1 < p < \infty$, this gives an easy proof that this space is order complete; this is much easier than our proof of Theorem 2.1.16.

6.7.13 Example. We now identify order continuous duals of some classical spaces. If X is an order continuous Banach lattice then it is easy to see that every functional in X^* is order continuous and, therefore, $X^\sim_{\mathrm{oc}} = X^*$. This settles the question for c_0, ℓ_p, and $L_p(\mu)$ spaces when $1 \leqslant p < \infty$. We observed in Theorem 6.6.13 that the order continuous dual of $C[0,1]$ is trivial. We will show in Example 10.4.12 that the order continuous dual of $L_\infty(\mu)$ is $L_1(\mu)$ when μ is σ-finite; in particular, $(\ell_\infty)^\sim_{\mathrm{oc}} = \ell_1$. Thus, $\big(L_p(\mu)\big)^\sim_{\mathrm{oc}} = L_{p^*}(\mu)$ for all $1 \leqslant p \leqslant \infty$. For a reader interested in Orlicz spaces, will show in Theorem 18.9.10 that this duality extends to Orlicz spaces: the order continuous dual of the Orlicz space L_φ is the Orlicz space L_ψ, where ψ is the conjugate function for φ.

The following is an immediate corollary of Proposition 6.6.20.

6.7.14 Proposition. *Let X be a Banach lattice. The band X^{dd} generated by X in X^{**} coincides with $(X^*)^\sim_{\mathrm{oc}}$.*

6.7.15 Exercise. Let $X = \ell_1$ and let $\varphi \in \ell_\infty^*$ be a Banach limit. Show that $\varphi \in X^d$ in X^{**}, so that $(X^*)^\sim_{\mathrm{oc}} \subsetneq X^{**}$.

6.7.16 Example. *The order dual and bidual of c.* We observed in 2.1.9 that c and c_0 are isomorphic as Banach spaces but not as vector lattices. It follows that the dual c^* is Banach space isomorphic to $c_0^* = \ell_1$. We will show that c^* is actually lattice isometric to ℓ_1.

Let $X = \mathbb{R} \oplus_1 \ell_1$. That is, X consists of pairs (r,f) with $r \in \mathbb{R}$ and $f \in \ell_1$ with $\big\|(r,f)\big\| = |r| + \sum_{i=1}^\infty |f_i|$. We will write $r \oplus f$ instead of (r,f). It is clear that X is lattice isometric to ℓ_1. Define $T\colon X \to c^*$ via

$$\big(T(r\oplus f)\big)(x) = rx_\infty + \sum_{i=1}^\infty f_i x_i, \text{ where } f \in \ell_1,\ r \in \mathbb{R},\ x \in c, \text{ and } x_\infty = \lim_{i\to\infty} x_i.$$

Clearly, T is linear and one-to-one. To show that T is onto, fix $\varphi \in c^*$. Let $f_i = \varphi(e_i)$ as $i \in \mathbb{N}$. Note that $f = (f_i)$ is in ℓ_1 because for every $m \in \mathbb{N}$ we have

$$\sum_{i=1}^m |f_i| = \sum_{i=1}^m \big|\varphi(e_i)\big| \leqslant \sum_{i=1}^m |\varphi|(e_i) \leqslant |\varphi|(\mathbb{1}).$$

In particular, the series $\sum_{i=1}^\infty \varphi(e_i)$ converges. Put $r = \varphi(\mathbb{1}) - \sum_{i=1}^\infty \varphi(e_i)$. Let $x \in c$; put $x_\infty = \lim_i x_i$. Then

$$\varphi(x) = \varphi\Big(x_\infty \mathbb{1} + \sum_{i=1}^\infty (x_i - x_\infty)e_i\Big) = rx_\infty + \sum_{i=1}^\infty f_i x_i = \big(T(f\oplus r)\big)(x),$$

so that $T(r\oplus f) = \varphi$.

It is easy to see that $T \geqslant 0$. To show that T is a lattice isomorphism, suppose that $T(r\oplus f)$ is positive and show that $r \geqslant 0$ and $f \geqslant 0$. We have $f_i = \big(T(r\oplus f)\big)(e_i) \geqslant 0$ for every i, hence $f \geqslant 0$. Suppose that $r < 0$. Since $f \in \ell_1$, find n so that $\sum_{i=n}^{\infty}|f_i| < |r|$. Put $x_n = (\overbrace{0, \dots, 0}^{n \text{ times}}, 1, 1, \dots)$ Then $\bigl(T(r \oplus f)\bigr)(x) = \sum_{i=n}^{\infty} f_i + r < 0$, which is a contradiction. Hence, T is a lattice isomorphism.

It is left to show that T is an isometry:

$$\bigl\|T(r \oplus f)\bigr\| = \Bigl\|\bigl|T(r \oplus f)\bigr|\Bigr\| = \Bigl\|T|r \oplus f|\Bigr\| = \Bigl(T\bigl(|r| \oplus |f|\bigr)\Bigr)(\mathbb{1}) = |r| + \sum_{i=1}^{\infty}|f_i| = \|r \oplus f\|.$$

Since c^* is lattice isometric to ℓ_1, we know that c^{**} is lattice isometric to ℓ_∞. However, the canonical embedding of c into c^{**} given by Proposition 6.7.10 is not the natural inclusion of c into ℓ_∞. It follows from the definition of T above that for $x \in c$, we may identify $\hat{x} \in c^{**}$ with the vector $\hat{x} = (x_\infty, x_1, x_2, \dots)$ in ℓ_∞.

It is interesting that ℓ_∞ is also the order completion of c, with the canonical embedding being the inclusion map; see Exercise 3.2.30. Thus, ℓ_∞ is both the double dual and the order completion of c; however, the canonical embeddings are different!

6.8 Weak convergence in Banach lattices

6.8.1 Exercise. Let X be a normed lattice.

(i) X_+ is weakly closed;

(ii) X^*_+ is weak*-closed.

(iii) Order intervals in X are weakly closed; order intervals in X^* are weak* closed.

It is a standard fact that norm convergence implies weak convergence. The following theorem shows that, for monotone nets, the two convergences agree.

6.8.2 Theorem (Dini). *Given a monotone net (x_α) in a normed lattice X, if $x_\alpha \xrightarrow{\mathrm{w}} x$ then $x_\alpha \xrightarrow{\|\cdot\|} x$.*

Proof. WLOG, $x_\alpha\uparrow$. For every positive functional $f \in X^*$ we have $f(x_\alpha)\uparrow$ in $\mathbb{R}$ and $f(x_\alpha) \to f(x)$. It follows that $f(x_\alpha) \leqslant f(x)$ for every α, hence $x_\alpha \leqslant x$.

Let $A = \operatorname{conv}\{x_\alpha\}$, the convex hull of (x_α). Since the weak closure of a convex set agrees with its norm closure (by Proposition A.4.11), we have $x \in \overline{A}^w = \overline{A}$. Fix $\varepsilon > 0$. There exists $y \in A$ such that $\|x-y\| < \varepsilon$. By the definition of A, $y = \lambda_1 x_{\alpha_1} + \dots + \lambda_n x_{\alpha_n}$ for some indices $\alpha_1, \dots, \alpha_n$ and positive real numbers $\lambda_1, \dots, \lambda_n$ with $\sum_{k=1}^n \lambda_k = 1$. Let α_0 be such that $\alpha_0 \geqslant \alpha_k$ as $k = 1, \dots, n$. It follows from $x_\alpha\uparrow$ that $x_{\alpha_0} \geqslant x_{\alpha_k}$ as $k = 1, \dots, n$. Therefore,

$$y = \lambda_1 x_{\alpha_1} + \dots + \lambda_n x_{\alpha_n} \leqslant \lambda_1 x_{\alpha_0} + \dots + \lambda_n x_{\alpha_0} = x_{\alpha_0}.$$

For all $\alpha \geqslant \alpha_0$, it follows from $y \leqslant x_{\alpha_0} \leqslant x_\alpha \leqslant x$ that $\|x - x_\alpha\| \leqslant \|x - y\| < \varepsilon$. Hence, $x_\alpha \xrightarrow{\|\cdot\|} x$. $\square$

6.8.3 Remark. For sequences, the preceding theorem may also be deduced from the classical theorem of Dini that asserts that for a monotone sequence in $C(K)$, point-wise convergence implies uniform convergence. Indeed, suppose (x_n) is an increasing sequence in a Banach lattice X such that $x_n \xrightarrow{\mathrm{w}} x$. Let $\hat{x}$ be the image of x in X^{**} under the canonical embedding of X into X^{**}. In particular, $\hat{x}$ is weak*-continuous. Similarly, $\hat{x}_n$ is weak*-continuous for every n. It follows from $x_n \xrightarrow{\mathrm{w}} x$ that $\hat{x}_n \xrightarrow{\mathrm{w}^*} \hat{x}$. Let $K = X_+^* \cap B_{X^*}$ with the weak* topology, then K is a compact space by Alaoglu's Theorem. We can view $\hat{x}$ as an elements $C(K)$ and $(\hat{x}_n)$ as a monotone sequence in $C(K)$. It follows from $\hat{x}_n \xrightarrow{\mathrm{w}^*} \hat{x}$ that $(\hat{x}_n)$ converges to $\hat{x}$ point-wise on K. By the classical Dini's theorem, $(\hat{x}_n)$ converges to $\hat{x}$ uniformly in $C(K)$. It follows that $\hat{x}_n \xrightarrow{\|\cdot\|} \hat{x}$ in X^{**} and, therefore, $x_n \xrightarrow{\|\cdot\|} x$ in X.

6.8.4 Example. *Dini's Theorem fails for weak*-convergence.* Let $X = \ell_\infty = \ell_1^*$ and $x_n = (1, \dots, 1, 0, \dots)$. Then $x_n \uparrow$, (x_n) converges weak* to $\mathbb{1}$, but fails to converge weakly or in norm.

We will see another case when weak convergence agrees with norm convergence in Theorem 10.1.11.

6.8.5 Corollary. *A Banach lattice X is order continuous iff every functional in X^* is order continuous.*

Proof. The forward implication is straightforward. To prove the converse, suppose that $x_\alpha \downarrow 0$. By assumption, we have $f(x_\alpha) \to 0$ for every $f \in X^*$. It follows that $x_\alpha \xrightarrow{\mathrm{w}} 0$ and, therefore, $x_\alpha \to 0$ by Dini's Theorem. □

6.8.6. *Lattice operations need not be weakly continuous.* Indeed, it was mentioned in 6.4.4(iii) that the Rademacher sequence (r_n) in L_p with $1 \leqslant p < \infty$ is weakly null, yet $|r_n| = \mathbb{1}$ for each n, so the sequence $\big(|r_n|\big)$ fails to be weakly null. For $1 < p \leqslant \infty$, the same example shows that lattice operations need not be weak*-continuous (when $p = \infty$, we have $r_n \xrightarrow{\mathrm{w}^*} 0$ in L_∞ by 6.4.4(vi)). In Exercise 8.4.7 and Theorem 10.8.3, we will show that under certain assumptions on the space, lattice operations may be weakly sequentially continuous.

6.8.7 Exercise. Lattice operations in a Banach lattice X are weakly sequentially continuous iff $x_n \xrightarrow{\mathrm{w}} 0$ implies $|x_n| \xrightarrow{\mathrm{w}} 0$.

A disjoint sequence need not be weakly null: just consider the standard unit vector basis of ℓ_1. However, the following exercise shows that disjoint sequences are often weakly null.

6.8.8 Exercise. Let (x_n) be a disjoint sequence in a Banach lattice X. Each of the following conditions implies that (x_n) is weakly null.

(i) (x_n) is order bounded;
(ii) (x_n) is weakly convergent;
(iii) (x_n) is norm bounded and X is reflexive.

What is the relationship between weak and order convergences? Combining Dini's Theorem 6.8.2 with Monotone Convergence Lemma 2.5.3, we conclude that for monotone nets weak convergence implies uniform and, therefore, order convergence. On the other hand, in order continuous Banach lattices, order convergence implies norm convergence and, therefore, weak convergence. However, the following example shows that, in general, order convergence and weak convergence do not imply each other.

6.8.9 Example. Let $x_n = (\overbrace{0,\dots,0}^{n},1,1,\dots)$ in ℓ_∞. Then $x_n \xrightarrow{\text{o}} 0$ but $\varphi(x_n) = 1$ for all n for any Banach limit φ; it follows that (x_n) fails to be weakly null.

On the other hand, the Rademacher sequence (r_n) is weakly null in $L_p[0,1]$ with $1 \leqslant p < \infty$, but is fails to converge in order.

We will see later that the relationship between weak compactness and the order structure on a Banach lattice is deep and leads to many exciting results.

6.9 Extending operators

In this section, we consider several results of the same type: given a sublattice Y of a vector lattice X and a positive operator T from Y to another vector lattice Z, when can we extend T to a positive operator $\widetilde{T}$ on the whole space:

$$\begin{tikzcd} X \arrow[r, dotted, "\widetilde{T}"] & Z \\ Y \arrow[u, hook] \arrow[ur, "T"'] & \end{tikzcd}$$

6.9.1 Hahn-Banach extensions

We start by observing that the classical Hahn-Banach Theorem A.4.1 for extending real-valued functionals on vector spaces remains valid (with essentially the same proof) if functionals are replaced with linear operators acting into an order complete vector lattice. A function $p\colon E \to X$, where E is a (real) vector space and X is a vector lattice, is called ***sublinear*** if

(i) $p(x+y) \leqslant p(x) + p(y)$ for all $x, y \in E$, and
(ii) $p(\lambda x) = \lambda p(x)$ for all $x \in E$ and $0 \leqslant \lambda \in \mathbb{R}$.

6.9.1 Theorem (Hahn-Banach Theorem for vector lattices)**.** *Let F be a subspace of a vector space E, X an order complete vector lattice, $p\colon E\to X$ a sublinear map, and $T\colon F\to X$ a linear operator such that $Tx\leqslant p(x)$ for all $x\in F$. Then T extends to a linear operator $\widetilde{T}\colon E\to X$ such that $\widetilde{T}x\leqslant p(x)$ for all $x\in E$.*

Proof. Take $a\notin F$. Let G be the subspace of E generated by F and a, that is, $G=F+\operatorname{span}\{a\}=\bigl\{x+\lambda a\ :\ x\in F,\lambda\in\mathbb{R}\bigr\}$. Pick any $z\in X$; let S be the extension of T to G such that $Sa=z$, that is, $S(x+\lambda a)=Tx+\lambda z$ for all $x\in F$ and $\lambda\in\mathbb{R}$. We claim that one can choose z so that S is dominated by p on G, that is, $S(x+\lambda a)\leqslant p(x+\lambda a)$ or, equivalently, $Tx+\lambda z\leqslant p(x+\lambda a)$ for all $x\in F$ and $\lambda\in\mathbb{R}$. Separately considering cases when $\lambda\geqslant 0$ and $\lambda<0$ and scaling by $\pm\lambda$, this is equivalent to $Tx+z\leqslant p(x+a)$ and $Tx-z\leqslant p(x-a)$ or, equivalently,

$$Tx-p(x-a)\leqslant z\leqslant p(x+a)-Tx \tag{6.8}$$

for all $x\in F$. Let

$$A=\bigl\{Tx-p(x-a)\ :\ x\in F\bigr\}\text{ and }B=\bigl\{p(x+a)-Tx\ :\ x\in F\bigr\}.$$

For every $x,y\in F$, we have

$$Tx+Ty=T(x+y)\leqslant p(x+y)\leqslant p(x-a)+p(y+a),$$

so that $Tx-p(x-a)\leqslant p(y+a)-Ty$. It follows that $A\leqslant B$. Since X is order complete, there exists z such that $A\leqslant z\leqslant B$, which yields (6.8).

Thus, we can always extend T to a proper superspace G of F such that the extension is dominated by p. Consider the set of all such extensions, ordered in the natural way: one extension is greater than another if it is an extension of the other. By Zorn's lemma, there must be a maximal extension. It is easy to see that it has to be an extension of T to all of E. □

We note that extensions constructed in this section need not be unique.

6.9.2 Theorem (Hahn-Banach Theorem for operators)**.** *Let Y be a sublattice of a vector lattice X, Z an order complete vector lattice, $S\in\mathcal{L}(Y,Z)_+$ and $T\in\mathcal{L}(X,Z)_+$ such that $0\leqslant S\leqslant T_{|Y}$. Then S extends to a linear operator $\widetilde{S}\colon X\to Z$ such that $0\leqslant\widetilde{S}\leqslant T$.*

Proof. For $x\in X$, put $p(x)=Tx^+$; it is sublinear by Exercise 1.3.11(iii). For each $x\in Y$, we have $Sx\leqslant Sx^+\leqslant Tx^+=p(x)$. By Hahn-Banach Theorem 6.9.1, S extends to a linear operator $\widetilde{S}\colon X\to Z$ such that $\widetilde{S}x\leqslant p(x)$ for all $x\in X$. Fix $x\in X_+$. Then $\widetilde{S}x\leqslant p(x)=Tx$ and $-\widetilde{S}x=\widetilde{S}(-x)\leqslant p(-x)=0$, so that $\widetilde{S}x\geqslant 0$. It follows that $0\leqslant\widetilde{S}\leqslant T$. □

6.9.3 Theorem (Hahn-Banach Theorem for positive functionals)**.** *Let Y be a sublattice of a normed lattice X and $f\in Y^*_+$. Then f extends to a positive functional $\tilde{f}$ on X with $\|\tilde{f}\|=\|f\|$.*

Proof. WLOG, $\|f\| = 1$. For $x \in X$, put $p(x) = \|x^+\|$. Then p is sublinear and $f(x) \leqslant f(x^+) \leqslant \|x^+\| = p(x)$ for every $x \in Y$. By the usual Hahn-Banach Theorem, f extends to a linear functional $\tilde{f}\colon X \to \mathbb{R}$ such that $\tilde{f}(x) \leqslant p(x)$ for all $x \in X$. For every $x \in X_+$, we have $-\tilde{f}(x) = \tilde{f}(-x) \leqslant p(-x) = 0$, so that $\tilde{f}(x) \geqslant 0$; it follows that $\tilde{f} \geqslant 0$. Since $\tilde{f}$ is an extension of f, we have $\|\tilde{f}\| \geqslant 1$. On the other hand, for every $x \in X$ we have $\bigl|\tilde{f}(x)\bigr| \leqslant \tilde{f}\bigl(|x|\bigr) \leqslant p\bigl(|x|\bigr) = \|x\|$, so that $\|\tilde{f}\| \leqslant 1$. □

Besides extending a single functional, one may simultaneously extend a monotone sequence of functionals preserving their order:

6.9.4 Exercise. Let Y be a sublattice of a normed lattice X; (f_n) a decreasing positive sequence in the unit ball of Y^*. There exists a decreasing positive sequence $(\tilde{f}_n)$ in the unit ball of X^* such that $\tilde{f}_n$ extends f_n for every n.

6.9.5 Theorem (Kantorovich)**.** *Every positive operator $T\colon Y \to Z$ from a majorizing sublattice Y of a vector lattice X to an order complete vector lattice Z admits an extension to positive operator $\widetilde{T}\colon X \to Z$.*

Proof. Fix $x \in X$ and consider the set $A = \{y \in Y \,:\, x \leqslant y\}$. Since Y is majorizing, this set is non-empty. It is bounded below in Y because there exists $v \in Y$ such that $-x \leqslant v$, so that $-v \leqslant A$. It follows that $T(A)$ is a non-empty bounded below subset of Z, hence $\inf T(A)$ exists. Denote it $p(x)$, that is, $p(x) = \inf\bigl\{Ty \,:\, x \leqslant y \in Y\bigr\}$. This defines a function from X to Z. It is easy to see that it is sublinear; it agrees with T on Y. By Theorem 6.9.1, T extends to $\widetilde{T}\colon X \to Z$. If $x \leqslant 0$ in X, it follows from the definition of p that $p(x) \leqslant 0$, hence $\widetilde{T}(x) \leqslant 0$. This yields $\widetilde{T} \geqslant 0$. □

6.9.6. Note that the proof does not use the lattice structure of X. Hence, Kantorovich's Theorem generalized as follows: Every positive operator $T\colon Y \to Z$ from a majorizing subspace Y of an ordered vector space X to an order complete vector lattice Z admits an extension to positive operator $\widetilde{T}\colon X \to Z$.

We will now show that Kantorovich's Theorem remains valid for lattice homomorphisms:

6.9.7 Theorem (Lipecki, Luxemburg, Schep)**.** *Every lattice homomorphism $T\colon Y \to Z$ from a majorizing sublattice Y of a vector lattice X to an order complete vector lattice Z admits an extension to lattice homomorphism $\widetilde{T}\colon X \to Z$.*

Proof. The following proof is based on [Lip85] and heavily relies on Section 3.1.1. Let $a \in X \setminus Y$ and let Y' be the sublattice of X generated by Y and a; we will show first that T extends to a lattice homomorphism on Y'. As in the proof of Kantorovich's Theorem 6.9.5, we define a map $p\colon X \to Z$ via $p(x) = \inf\bigl\{Ty \;:\; x \leqslant y \in Y\bigr\}$. It is easy to see that p is positively homogeneous and agrees with T on Y. It follows from distributive laws that

$$p(x_1 \vee x_2) = p(x_1) \vee p(x_2) \text{ for all } x_1, x_2 \in X, \tag{6.9}$$

and, furthermore, that

$$p(x + y) = p(x) + Ty \text{ whenever } x \in X \text{ and } y \in Y. \tag{6.10}$$

It is easy to see that $Y' = \mathrm{Lat}(Y + \mathbb{R}_+ a)$. Since the set $Y + \mathbb{R}_+ a$ is a wedge, Proposition 3.1.15 yields $Y' = W - W$ where $W = (Y + \mathbb{R}_+ a)^{\vee}$. By Exercise 3.1.11, W is a wedge.

We claim that p is additive on W. Indeed, let $u, v \in W$. We can write $u = \bigvee_{i=1}^{m}(u_i + s_i a)$ and $v = \bigvee_{j=1}^{n}(v_i + t_i a)$ for some $u_1, \dots, u_m, v_1, \dots, v_n$ in Y and $s_1, \dots, s_m, t_1, \dots, t_n$ in $\mathbb{R}_+$. By Exercise 1.3.5, we have

$$u + v = \bigvee_{\substack{i=1,\dots,m\\ j=1,\dots,n}} \bigl(u_i + v_j + (s_i + t_j)a\bigr).$$

It follows from (6.9) that

$$p(u + v) = \bigvee_{\substack{i=1,\dots,m\\ j=1,\dots,n}} p\bigl(u_i + v_j + (s_i + t_j)a\bigr)$$

For every i and j, (6.10) and positive homogeneity of p yield

$$\begin{aligned} p\bigl(u_i + v_j + (s_i + t_j)a\bigr) &= T(u_i + v_j) + p\bigl((s_i + t_j)a\bigr) \\ &= Tu_i + Tv_j + (s_i + t_j)p(a) = p(u_i + s_i a) + p(v_j + t_j a), \end{aligned}$$

Using Exercise 1.3.5 again, we get

$$p(u + v) = \bigvee_{\substack{i=1,\dots,m\\ j=1,\dots,n}} p(u_i + s_i a) + p(v_j + t_j a) = \bigvee_{i=1}^{m} p(u_i + s_i a) + \bigvee_{j=1}^{n} p(v_j + t_j a) = p(u) + p(v).$$

This proves the claim.

Recall that $Y' = W - W$. Define $S\colon Y' \to Z$ via $S(u - v) = p(u) - p(v)$ where $u, v \in W$. This is well-defined: if $u - v = u' - v'$ for some $u', v' \in W$ then $u + v' = u' + v$ yields $p(u) + p(v') = p(u') + p(v)$ and, therefore, $p(u) - p(v) = p(u') - p(v')$. It is easy to see that S is linear and agrees with p on W and, therefore, it agrees with T on Y. It is left to show that S is a lattice homomorphism. Let $x \in Y'$. Then $x = u - v$ for some $u, v \in W$. By Exercise 1.3.3(iii), we have $(u - v)^+ = u \vee v - v$. It follows that

$$S(x^+) = p(u \vee v) - p(v) = p(u) \vee p(v) - p(v) = \bigl(p(u) - p(v)\bigr)^+ = \Bigl(S(u - v)\Bigr)^+ = (Sx)^+,$$

and, therefore, S is a lattice homomorphism.

To complete the proof, we use a standard Zorn's Lemma extension argument. Let $\mathcal{G}$ be the set of all pairs (U,S) where U is a sublattice of X containing Y and $S\colon U\to Z$ is a lattice homomorphism extending T. Order $\mathcal{G}$ as follows: $(U,S)\leqslant(V,R)$ if $U\subseteq V$ and $S=R_{|U}$. It can be easily verified that $\mathcal{G}$ satisfies the assumptions of Zorn's Lemma. Therefore, it contains a maximal element; denote it by (U,S). Finally, observe that $U=X$ because otherwise we could use the first part of the proof find a further extension, which would contradict maximality of (U,S). □

In particular, the theorem allows one to extend lattice homomorphisms from X to X^δ. Note that injectivity is preserved under such an extension:

6.9.8 Exercise. Let $T\colon X\to Z$ be a lattice homomorphism between two vector lattices, and Y an order dense sublattice of X. If the restriction of T to Y is injective then T is injective on X.

6.9.2 Extending order continuous operators

It is well known that every continuous operator from a normed space E to a Banach space F extends uniquely to the norm completion of E or, more generally, to any normed space that contains E as a dense subspace. In particular, one can extend a continuous operator defined on a norm dense subspace to an operator defined on the whole space. In this section, we will prove an analogous statement: every order continuous positive operator acting on X into an order complete vector lattice extends uniquely to an order continuous positive operator on X^δ. First, we observe that if such an extension exists, it has to be given by a certain simple formula.

6.9.9 Exercise. Let $T\colon X\to Z$ be an order continuous positive operator between two Archimedean vector lattices, and Y an order dense sublattice of X. Then $Tx=\sup T\bigl([0,x]\cap Y\bigr)$ for every $x\in X$.

6.9.10 Lemma. *Let Y be an order dense sublattice of an Archimedean vector lattice X, and $T\colon Y\to Z$ an order continuous positive operator from Y to an order complete vector lattice Z. If the set $T\bigl([0,x]\cap Y\bigr)$ is order bounded in Z for every $x\in X_+$ then T has a unique order continuous extension $\widetilde{T}\colon X\to Z$ given by $\widetilde{T}x=\sup T\bigl([0,x]\cap Y\bigr)$ for all $x\in X_+$.*

Proof. Let $x\in X_+$. Put $\widetilde{T}x\coloneqq\sup T\bigl([0,x]\cap Y\bigr)$; the supremum exists because Z is order complete.

We claim that if $y_\alpha\uparrow x$ then $Ty_\alpha\uparrow\widetilde{T}x$ for every net (y_α) in Y_+. Indeed, $\sup Ty_\alpha$ exists because $Ty_\alpha\leqslant Ty_0$. Since $y_\alpha\in[0,x]\cap Y$ for every α we have $\sup Ty_\alpha\leqslant\widetilde{T}x$. On the other hand, take any $y\in[0,x]\cap Y$. It follows from $y_\alpha\uparrow x$ that $(y_\alpha\wedge y)\uparrow(x\wedge y)=y$ in X. By Theorem 3.2.5, we have $(y_\alpha\wedge y)\uparrow y$ in Y. Since T is order continuous, we have

$Ty = \sup T(y_\alpha \wedge y) \leqslant \sup Ty_\alpha$. Taking sup over all y in $[0,x]\cap Y$, we get $\widetilde{T}x \leqslant \sup Ty_\alpha$. This proves the claim.

It is now easy to see that $\widetilde{T}$ is additive on X_+. Indeed, let $x_1, x_2 \in X_+$. Take two nets (y_α) and (z_β) in Y_+ such that $y_\alpha \uparrow x_1$ and $z_\beta \uparrow x_2$. Then $(y_\alpha + z_\beta)_{(\alpha,\beta)}$ is an increasing net and $y_\alpha + z_\beta \uparrow x_1 + x_2$. Hence $T(y_\alpha + z_\beta) \uparrow \widetilde{T}(x_1 + x_2)$. On the other hand, $T(y_\alpha + z_\beta) = Ty_\alpha + Tz_\beta \uparrow \widetilde{T}x_1 + \widetilde{T}x_2$. Therefore, $\widetilde{T}(x_1 + x_2) = \widetilde{T}x_1 + \widetilde{T}x_2$.

By the Extension Lemma 6.2.1, we extend $\widetilde{T}$ to a positive operator $\widetilde{T}\colon X \to Z$ which is also an extension of T. It is left to show that $\widetilde{T}$ is order continuous on X. Suppose that $0 \leqslant x_\alpha \uparrow x$ in X; we need to show that $\widetilde{T}x_\alpha \uparrow \widetilde{T}x$ in Z. Clearly, $\widetilde{T}x_\alpha \uparrow\leqslant \widetilde{T}x$. Let

$$A = \bigl\{y \in Y_+ \,:\, y \leqslant x_\alpha \text{ for some } \alpha\bigr\} = \bigcup_\alpha [0, x_\alpha] \cap Y.$$

Note that

$$x = \sup_\alpha x_\alpha = \sup_\alpha \sup [0, x_\alpha] \cap Y = \sup A.$$

Since we may view A as an increasing net, it follows that $\widetilde{T}x = \sup T(A)$. On the other hand, for every $y \in A$ there exists α such that $y \leqslant x_\alpha$, so that $Ty \leqslant \widetilde{T}x_\alpha \leqslant \sup_\alpha \widetilde{T}x_\alpha$. Taking sup over $y \in A$, we get $\sup T(A) \leqslant \sup_\alpha \widetilde{T}x_\alpha$.

Uniqueness of the extension follows from Exercise 6.9.9. □

6.9.11 Theorem (Veksler). *Let Y be an order dense majorizing sublattice of an Archimedean vector lattice X, and T an order continuous positive operator from Y to an order complete vector lattice Z. Then T has a unique order continuous extension $\widetilde{T}\colon X \to Z$, which is positive and satisfies*

$$\widetilde{T}x = \sup T\bigl([0,x] \cap Y\bigr) \textit{ for all } x \in X_+. \tag{6.11}$$

Proof. Let $x \in X_+$. Since Y is majorizing, there exists $y_0 \in Y_+$ such that $x \leqslant y_0$. It follows that $\sup T\bigl([0,x]\cap Y\bigr) \leqslant Ty_0$ in Z. Now apply Lemma 6.9.10. □

Veksler's Theorem implies that there exists a one-to-one correspondence between order continuous operators on Y and on X:

6.9.12 Corollary. *Let Y be an order dense majorizing sublattice of an Archimedean vector lattice X, and Z an order complete vector lattice. For $S \in \mathcal{L}_{\mathrm{oc}}(X,Z)$, let $\Psi(S)$ be the restriction of S to Y. Then Ψ is a lattice isomorphism between $\mathcal{L}_{\mathrm{oc}}(X,Z)$ and $\mathcal{L}_{\mathrm{oc}}(Y,Z)$.*

Proof. Note that by Theorem 3.2.21, order limits in Y and in X agree. It follows that $\Psi(S) \in \mathcal{L}_{\mathrm{oc}}(Y,Z)$ whenever $S \in \mathcal{L}_{\mathrm{oc}}(X,Z)$. For $T \in \mathcal{L}_{\mathrm{oc}}(Y,Z)_+$, let $\widetilde{T}$ be as in Theorem 6.9.11; put $\phi(T) = \widetilde{T}$. Then $\phi\colon \mathcal{L}_{\mathrm{oc}}(Y,Z)_+ \to \mathcal{L}_{\mathrm{oc}}(X,Z)_+$. We will show that φ is an additive bijection. It is straightforward that φ is one-to-one. To show that φ is onto, let $S \in \mathcal{L}_{\mathrm{oc}}(X,Z)_+$. Then $\Psi(S) \in \mathcal{L}_{\mathrm{oc}}(Y,Z)_+$ and $S = \widetilde{\Psi(S)} = \varphi\bigl(\Psi(S)\bigr)$ by the uniqueness clause in the theorem.

Finally, if $S, T \in \mathcal{L}_{\mathrm{oc}}(Y,Z)_+$ then the operator $\widetilde{S}+\widetilde{T}$ agrees with $S+T$ on Y, hence, by uniqueness of extension, $\widetilde{S}+\widetilde{T}=\widetilde{S+T}$, so that $\varphi(S)+\varphi(T)=\varphi(S+T)$.

It now follows from Corollary 6.2.3 that φ extends to a lattice isomorphism Φ from $\mathcal{L}_{\mathrm{oc}}(Y,Z)$ onto $\mathcal{L}_{\mathrm{oc}}(X,Z)$. The restriction of $\Phi(T)$ to Y agrees with T for every $T \in \mathcal{L}_{\mathrm{oc}}(Y,Z)$ because

$$\big[\Phi(T)\big](x)=\big[\varphi(T^+)-\varphi(T^-)\big](x)=T^+x-T^-x=Tx$$

for every $x \in X$. It follows that $\Phi=\Psi^{-1}$, hence Ψ is a lattice isomorphism. □

6.9.13 Exercise. If the operator T in Veksler's Theorem 6.9.11 is a lattice homomorphism, then $\widetilde{T}$ is also a lattice homomorphism. It follows that every order continuous lattice homomorphism $T\colon X \to Y$ between two Archimedean vector lattices extends uniquely to an order continuous lattice homomorphism $T^\delta\colon X^\delta \to Y^\delta$.

6.10 Order completion and order completeness

In this section, we present several important applications of Veksler's Extension Theorem 6.9.11. We suggest that the reader reviews Section 3.2.1 before proceeding. Throughout this section, we assume that X is an Archimedean vector lattice. We already know from Theorem 3.2.22 that X has an order completion; the inclusion map of X into its order completion is order continuous by Proposition 3.2.13.

6.10.1 Theorem. *Order completion is unique. That is, if Y and Z are two order completions of X then there is a lattice isomorphism $S\colon Y \to Z$ which preserves X.*

Proof. Let $R\colon X \to Z$ be the inclusion map. Clearly, R is positive and order continuous. Since X is order dense and majorizing in Y, Veksler's Theorem 6.9.11 guarantees that R extends to a positive order continuous operator $S\colon Y \to Z$. Similarly, the inclusion map of X into Y extends to a positive order continuous operator $T\colon Z \to Y$. The composition $TS\colon Y \to Y$ is a positive order continuous operator which fixes X. By the uniqueness clause in Veksler's Theorem, TS is the identity of Y. Similarly, ST is the identity of Z. It follows that $T=S^{-1}$. Since both S and S^{-1} are positive, S is a lattice isomorphism. □

The order completion of X is denoted by X^δ. Recall that by Corollary 3.2.32, $x_\alpha \xrightarrow{\mathrm{o}} x$ in X iff $x_\alpha \xrightarrow{\mathrm{o}} x$ in X^δ. Since X is order dense and majorizing in X^δ, the following is a special case of Veksler's Theorem 6.9.11.

6.10.2 Corollary. *Let $T\colon X \to Z$ be an order continuous positive operator from X to an order complete vector lattice Z. Then T uniquely extends to an order continuous positive operator from X^δ to Z.*

6.10.3. *X^δ is the least order complete vector lattice containing X as an order dense sublattice.* Indeed, suppose that X is an order dense sublattice of an order complete vector lattice Z. The ideal $I(X)$ generated by X in Z is order complete by Proposition 3.3.18. It is easy to see that X is order dense and majorizing in $I(X)$, hence $I(X)$ may be identified with X^δ.

6.10.4. *X^δ is the greatest vector lattice that contains X as an order dense and majorizing sublattice.* Suppose that X is an order dense and majorizing sublattice of an Archimedean vector lattice Z. We claim that the inclusion map of X into X^δ extends to an order continuous lattice isomorphic embedding from Z to X^δ. Thus, we may identify Z with an (order dense and majorizing) sublattice of X^δ. Moreover, if Z is order complete, this map is onto.

Indeed, via the diagram, $X \hookrightarrow Z \hookrightarrow Z^\delta$, we may view X is an order dense majorizing sublattice of Z^δ. Hence, Z^δ is an order completion of X. Theorem 6.10.1 guarantees that there exists a lattice isomorphism $S\colon Z^\delta \to X^\delta$ which preserves X. The restriction of S to Z yields the required map.

A net $(x_\alpha)_{\alpha\in\Lambda}$ in X is said to be ***order Cauchy*** if the net $(x_\alpha - x_\beta)_{(\alpha,\beta)}$ is order null. The latter net is indexed by Λ^2, ordered component-wise. Clearly, every order convergent net is order Cauchy.

6.10.5 Proposition. *Every monotone order bounded net in X is order Cauchy.*

Proof. Suppose that $x_\alpha \uparrow\leqslant u$ in X. Then $z := \sup x_\alpha$ exists in X^δ. It follows that the double net $(x_\alpha - x_\beta)_{(\alpha,\beta)}$ is order null in X^δ and, therefore, in X. The case of a decreasing net may be handled in a similar way. □

6.10.6 Exercise. For $n \in \mathbb{N}$, let $x_n = (\overbrace{0,1,\dots,0,1}^{2n},0,0,\dots)$ in c. Show that (x_n) is order Cauchy but not order convergent.

6.10.7. As an application, we present a simple and intuitive proof of Theorem 2.5.22 that every order continuous Banach lattice is order complete. Let X be an order continuous Banach lattice and $x_\alpha \downarrow\geqslant 0$. By Proposition 6.10.5, (x_α) is order Cauchy. Then the double net $(x_\alpha - x_\beta)_{(\alpha,\beta)}$ is order null, hence norm null. It follows that (x_α) is norm Cauchy and, therefore, norm convergent. By Monotone Convergence Lemma 2.5.3, $\inf x_\alpha$ exists.

6.10.8 Exercise. Every order Cauchy net has an order bounded tail.

6.10.9 Theorem. *X is order complete iff every order Cauchy net is order convergent.*

Proof. Suppose that X is order complete; let (x_α) be an order Cauchy net in X. We want to show that it is order convergent. By the preceding exercise, (x_α) has an order bounded tail; we may assume WLOG that (x_α) is order bounded. Put

$$x = \sup a_\alpha \text{ where } a_\alpha = \inf_{\beta\geqslant\alpha} x_\beta \text{ and } y = \inf b_\alpha \text{ where } b_\alpha = \sup_{\beta\geqslant\alpha} x_\beta.$$

By Proposition 2.4.21, it suffices to show that $x = y$. As in the proof of Proposition 2.4.21, we have $a_\alpha \leqslant x \leqslant y \leqslant b_\alpha$ for every α. Let $(v_{\alpha,\beta})$ be a net such that $v_{\alpha,\beta} \downarrow 0$ and $|x_\alpha - x_\beta| \leqslant v_{\alpha,\beta}$. Fix a pair (α_0, β_0). Let α be such that $\alpha \geqslant \alpha_0$ and $\alpha \geqslant \beta_0$. For every β with $\beta \geqslant \alpha$, we have $(\alpha, \beta) \geqslant (\alpha_0, \beta_0)$, so that $|x_\alpha - x_\beta| \leqslant v_{\alpha_0,\beta_0}$. It follows that $x_\beta \in [x_\alpha - v_{\alpha_0,\beta_0}, x_\alpha + v_{\alpha_0,\beta_0}]$, which yields $a_\alpha, b_\alpha \in [x_\alpha - v_{\alpha_0,\beta_0}, x_\alpha + v_{\alpha_0,\beta_0}]$ and, therefore, $0 \leqslant b_\alpha - a_\alpha \leqslant 2v_{\alpha_0,\beta_0}$, and hence $0 \leqslant y - x \leqslant 2v_{\alpha_0,\beta_0}$. If follows that $x - y = 0$.

Conversely, suppose that every order Cauchy net in X is order convergent. Let $0 \leqslant x_\alpha\uparrow\leqslant u$. By Proposition 6.10.5, (x_α) is order Cauchy, hence order convergent; it follows that $\sup x_\alpha$ exists. □

The following exercise is a sequential version of the preceding theorem. Recall that a sequence is σ-order convergent if the dominating net in the definition of order convergence may be chosen to be a sequence.

6.10.10 Exercise. TFAE:

(i) X is σ-order complete;
(ii) Every order Cauchy sequence is order convergent;
(iii) Every order Cauchy sequence is σ-order convergent.

6.10.1 * Regular sublattices and order and uniform completions

Recall that for a linear subspace E of a Banach space F, the completion of E may be identified with its closure in F. We will now prove a vector lattice analogue of this statement: if Y is a regular sublattice of an order complete vector lattice X then Y^δ is also a regular sublattice of X. Actually, we prove a more general fact. Recall that Y is regular in X iff the inclusion map is order continuous.

6.10.11 Theorem. *Let Y be a regular sublattice of X. Then Y^δ is a regular sublattice of X^δ. More precisely, the inclusion map $Y \hookrightarrow X$ extends to an order continuous lattice isomorphic embedding from Y^δ into X^δ.*

Proof. Since X is regular in X^δ, it follows that Y is regular in X^δ. Therefore, replacing X with X^δ, we may assume that X is order complete. The inclusion map $Y \hookrightarrow X$ is order continuous and, therefore, by Corollary 6.10.2, it extends to an order continuous positive operator $S\colon Y^\delta \to X$. It is left to show that S is a lattice isomorphism.

To show that S is a lattice homomorphism, let $a \in Y^\delta$. By Proposition 3.2.20, there is a net (y_α) in Y such that $y_\alpha \uparrow a$ in Y^δ. It follows that $y_\alpha^+ \uparrow a^+$ in Y^δ. Since S is order continuous and fixes Y, we have $y_\alpha^+ = Sy_\alpha^+ \uparrow Sa^+$ and $y_\alpha = Sy_\alpha \uparrow Sa$ in X. The latter yields $y_\alpha^+ \uparrow (Sa)^+$. Hence, $(Sa)^+ = Sa^+$.

In order to show that S is one-to-one, suppose that $Sa = 0$ for some non-zero $a \in Y^\delta$. Since Y is order dense in Y^δ, there exists $y \in Y$ with $0 < y \leqslant |a|$. It follows that $y = Sy \leqslant S|a| = |Sa| = 0$, which is a contradiction.

□

This theorem yields a short and elegant proof of Theorem 3.2.5 that order convergence passes down to regular sublattices for order bounded nets. Indeed, let Y be a regular sublattice of X, (y_α) a net in Y such that (y_α) is order bounded in Y and $y_\alpha \xrightarrow{\mathrm{o}} 0$ in X. Since the order convergencies in X and X^δ agree, we have $y_\alpha \xrightarrow{\mathrm{o}} 0$ in X^δ. By Theorem 6.10.11 we may view Y^δ as a regular sublattice of X^δ. Exercise 3.2.4 guarantees that $y_\alpha \xrightarrow{\mathrm{o}} 0$ in Y^δ and, therefore, in Y.

6.10.12 Exercise. If Y is an order dense sublattice of X then Y^δ is an ideal in X^δ.

(Relatively) uniform completion. Let Y be a sublattice of a uniformly complete vector lattice X. Let (x_n) and x be in Y. When we say that $x_n \xrightarrow{\mathrm{u}} x$ in Y, we implicitly imply that the regulator for this convergence is also in Y. It is clear that if $x_n \xrightarrow{\mathrm{u}} x$ in Y then $x_n \xrightarrow{\mathrm{u}} x$ in X, but the converse may be false; see Example 2.3.12. However, it is easy to see that if, in addition, Y is majorizing in X, then $x_n \xrightarrow{\mathrm{u}} x$ in X implies $x_n \xrightarrow{\mathrm{u}} x$ in Y. We observed in Corollary 2.3.14 and Example 2.3.16 that if Y is uniformly closed in X then it is itself uniformly complete, but the converse is, generally, false.

6.10.13 Exercise. Let Y be a majorizing sublattice of a uniformly complete Archimedean vector lattice X. Then Y is uniformly closed in X iff it is uniformly complete.

6.10.14 Lemma. *[EG:FUCVL] Let Y be a sublattice of an Archimedean vector lattice X. Suppose that $x_n \xrightarrow{\mathrm{u}} 0$ in X for some (x_n) in Y. If (x_n) is uniformly Cauchy in Y then $x_n \xrightarrow{\mathrm{u}} 0$ in Y.*

Proof. By assumption, there exists $v \in Y_+$ such that for every $\varepsilon > 0$ there exists m_0 such that for all $n \geqslant m \geqslant m_0$, we have

$$x_m - x_n \in [-\varepsilon v, \varepsilon v]_Y \subseteq [-\varepsilon v, \varepsilon v]_X. \tag{6.12}$$

By the Archimedean property, order intervals in X are uniformly closed. Since $x_n \xrightarrow{\mathrm{u}} 0$ in X, passing to the limit on n in (6.12) yields $x_m \in [-\varepsilon v, \varepsilon v]_X$, hence $x_m \in [-\varepsilon v, \varepsilon v]_Y$. We conclude that $x_n \xrightarrow{\mathrm{u}} 0$ in Y. □

Recall that being order complete, X^δ is also uniformly complete by Theorem 2.2.9.

6.10.15 Proposition. *Let X be an Archimedean vector lattice. The intersection of any family of uniformly complete sublattices of X is again uniformly complete.*

Proof. Replacing X with X^δ, we may assume WLOG that X is uniformly complete. Let $\mathcal{A}$ be a family of uniformly complete sublattices of X, and let $Z = \bigcap \mathcal{A}$. Let (x_n) be a uniformly Cauchy sequence in Z; in particular, there is a regulator in Z. Let $Y \in \mathcal{A}$. It follows from $Z \subseteq Y$ that (x_n) is still uniformly Cauchy in Y. Since Y is uniformly complete, $x_n \xrightarrow{\mathrm{u}} x$ in Y for some $x \in Y$ (and with a regulator in Y). It follows that $x_n \xrightarrow{\mathrm{u}} x$ in X. Since uniform limits are unique, we conclude that x does not depend on Y. Then x belongs to every member of $\mathcal{A}$, hence to Z. Since $x_n \xrightarrow{\mathrm{u}} x$ in Y, the preceding lemma yields $x_n \xrightarrow{\mathrm{u}} x$ in Z. □

Let X be an Archimedean vector lattice. Let Z be the intersection of the family of all uniformly complete sublattices of X^δ that contain X. We claim that Z is uniformly complete. This follows immediately from Proposition 6.10.15. Alternatively, we may observe that sublattices in the family contain X and, therefore, are majorizing in X^δ, hence uniformly closed in X^δ by Exercise 6.10.13; this implies that Z is uniformly closed and, therefore, uniformly complete. We call Z the ***relatively uniform*** or just ***uniform completion*** of X and denote it X^{ru} (we use X^{ru} rather than X^u to distinguish it from the universal completion).

It follows immediately from $X \subseteq X^{ru} \subseteq X^\delta$ that X is order dense and majorizing in X^{ru}.

We claim that X^{ru} is the least uniformly complete vector lattice that contains X as an order dense sublattice. Indeed, let Y be a uniformly complete vector lattice which contains X as an order dense sublattice. By Theorem 6.10.11, we may view X^δ as a sublattice of Y^δ (even as an ideal by Exercise 6.10.12). Since Y and X^δ are two uniformly complete sublattices of Y^δ, so is $Y \cap X^\delta$. It now follows from the definition of X^{ru} that $X^{ru} \subseteq Y \cap X^\delta \subseteq Y$.

We show next that X^{ru} is a "universal" object in the category of uniformly complete vector lattices in the following sense: We use the easy fact that every order bounded operator is uniformly continuous (cf. Theorem 6.1.7).

6.10.16 Proposition. *Every lattice homomorphism T from X to any uniformly complete vector lattice Y extends uniquely to a lattice homomorphism $\widetilde{T}\colon X^{ru}\to Y$.*

$$\begin{tikzcd} X \arrow[r, hook] \arrow[dr, "T"'] & X^{ru} \arrow[d, "\widetilde{T}"] \\ & Y \end{tikzcd}$$

Moreover, if T is order continuous then so is $\widetilde{T}$.

Proof. Let $j\colon Y\hookrightarrow Y^\delta$ be the canonical inclusion. Clearly, $jT\colon X\to Y^\delta$ is a lattice homomorphism.

$$\begin{tikzcd} X \arrow[r, hook] \arrow[dr, "T"'] & X^{ru} \arrow[r, hook] & X^\delta \arrow[d, "S"'] \\ & Y \arrow[r, hook, "j"] & Y^\delta \end{tikzcd}$$

By Theorem 6.9.7, jT extends to a lattice homomorphism $S\colon X^\delta\to Y^\delta$. It suffices to show that $S(X^{ru})\subseteq Y$ as we can then take the restriction of S to X^{ru} as $\widetilde{T}$.

Since S is a lattice homomorphism, $S^{-1}(Y)$ is a sublattice of X^δ. Clearly, $X\subseteq S^{-1}(Y)$. Being order bounded, S is uniformly continuous; since Y is uniformly closed in Y^δ, we conclude that $S^{-1}(Y)$ is uniformly closed in X^δ. It follows that $X^{ru}\subseteq S^{-1}(Y)$. This yields $S(X^{ru})\subseteq Y$.

To show uniqueness, suppose that $\widetilde{T}$ and $\widehat{T}$ are two lattice homomorphism extensions of T to X^{ru}, acting into Y. Then $\ker(\widetilde{T}-\widehat{T})$ contains X, is a sublattice by Exercise 1.6.9. Since $\widetilde{T}-\widehat{T}$ is order bounded, it is uniformly continuous, hence $\ker(\widetilde{T}-\widehat{T})$ is uniformly closed in X^{ru} and, therefore, uniformly complete. It follows that $\ker(\widetilde{T}-\widehat{T})=X^{ru}$, so that $\widetilde{T}=\widehat{T}$.

Suppose now that, in addition, T is order continuous. Since Y is regular in Y^δ, j is order continuous, hence so is jT. By Exercise 6.9.13, there is an order continuous lattice homomorphism $S\colon X^\delta\to Y^\delta$ extending jT. We now proceed as above. Note that X^{ru} and Y are order dense and, majorizing in X^δ and Y^δ, respectively. It follows that the restriction $\widetilde{T}$ of S to X^{ru} is still order continuous. □

6.11 * Lattice homomorphisms

Many important operators appearing in Analysis happen to be lattice homomorphisms. For example, ***weighted composition operators*** on $C(K)$ and on $L_p(\mu)$ spaces are lattice homomorphisms as long as the weight function is positive. More precisely, let $T\colon C(K)\to C(K)$ be given by $Tf(t)=w(t)f\bigl(\varphi(t)\bigr)$, where $\varphi\colon K\to K$ and $w\colon K\to\mathbb{R}$ are continuous functions; it is easy to see that T is a lattice homomorphism provided that $w \geqslant 0$. Similarly, let (Ω, μ) be a measure space, $0 \leqslant p \leqslant \infty$, and $T\colon L_p(\mu) \to L_p(\mu)$ given by $Tf(t) = w(t)f\bigl(\varphi(t)\bigr)$ where $w \in L_\infty(\mu)$ and $\varphi\colon \Omega \to \Omega$ is measurable; then T is a lattice homomorphism as long as $w \geqslant 0$. For a finite measure μ and $p > q$, the natural embedding of $L_p(\mu)$ into $L_q(\mu)$ is a lattice homomorphism.

Let X and Y be two vector lattices. We should warn the reader that the set of all lattice homomorphisms from X to Y lacks "nice" structures: it is not closed under negation (as a lattice homomorphism has to be positive); it is also need not be closed under addition. Indeed, take $X = Y = \mathbb{R}^2$ and $T = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$, then I and T are lattice homomorphisms, while $I + T$ is not, as $(I+T)\lvert e_1 - e_2\rvert \neq \bigl\lvert (I+T)(e_1 - e_2)\bigr\rvert$.

Recall that if $T\colon X \to Y$ is a lattice homomorphism then $\ker T$ is an ideal, $\operatorname{Range} T$ is a sublattice, and the pre-image of a sublattice under T is again a sublattice.

We start with the special case $Y = \mathbb{R}$. The following simple fact is most useful:

6.11.1 Exercise. For $f \in X^\sim$, f is a lattice homomorphism iff it is an atom in $X^\sim$.

6.11.2 Exercise. Every two lattice homomorphisms in $X^\sim$ are either proportional or disjoint.

6.11.3 Exercise. Let X be a Banach function space, i.e., a Banach lattice whose order is generated by an unconditional basis (e_i). Then $T\colon X \to X$ is a lattice isomorphism iff T is a strictly positive weighted permutation, that is, there exists a bijection $\sigma\colon \mathbb{N} \to \mathbb{N}$ and a sequence of positive reals (w_i) such that $Te_i = w_i e_{\sigma(i)}$ for every i.

Recall that every band projection is a lattice homomorphism by Exercise 5.2.10. In particular, if a is an atom in a vector lattice X them I_a is a projection band by Lemma 5.4.9, hence the corresponding band projection is a rank one lattice homomorphism; it follows that the resulting coordinate functional (as in 5.4.17) is a lattice homomorphism from X to $\mathbb{R}$. In particular, if X is a Banach sequence space generated by an unconditional basis (e_i) then every coordinate functional e_i^* is a lattice homomorphism and an atom in X^*.

Since the dual of $L_p[0,1]$ when $1 \leqslant p < \infty$ is non-atomic, it follows from Exercise 6.11.1 that there are no non-trivial lattice homomorphisms from $L_p[0,1]$ to $\mathbb{R}$!

6.11.4 Exercise. There are no non-trivial lattice homomorphisms from $L_p[0,1]$ to $C(K)$ or to $L_q(\mu)$, where K is a compact space, μ is a measure, and $1 \leqslant p < q \leqslant \infty$.

Interestingly, there are plenty of real-valued lattice homomorphisms on $C(K)$, where K is a compact space. We proved in Proposition 4.1.1 that real-valued lattice homomorphisms on $C(K)$ are exactly the non-negative scalar multiples of point evaluation functionals. Recall that for $t \in K$, we write δ_t for the point evaluation function on $C(K)$.

6.11.5 Exercise. Let X be an Archimedean vector lattice with a strong unit. Show that lattice homomorphisms in $X^\sim$ separate points of X. That is, if $\varphi(x)=0$ for every lattice homomorphism in $\varphi\in X^\sim$ then $x=0$. Furthermore, if $\varphi(x)\leqslant\varphi(y)$ for every lattice homomorphism in $\varphi\in X^\sim$ then $x\leqslant y$.

Let K and L be two compact spaces. An operator $T\colon C(K)\to C(L)$ is said to be a ***composition operator*** if it is of the form $Tf=f\circ\varphi$ for some continuous function $\varphi\colon L\to K$. More generally, we say that T is a ***weighted composition operator*** if $Tf=w\cdot(f\circ\varphi)$, i.e., $(Tf)(t)=w(t)f\bigl(\varphi(t)\bigr)$ for some function w in $C(L)$ and $\varphi\colon L\to K$, such that φ is continuous on the set $\{w\neq 0\}$. We say that w is a ***weight function***.

6.11.6 Theorem. *Let K and L be compact spaces and $T\colon C(K)\to C(L)$. Then T a lattice homomorphism iff T is a weighted composition operator with a positive weight. In this case, the weight function equals $T\mathbb{1}_K$.*

Proof. It is easy to see that if T is a weighted composition operator with a positive weight function w then T is a lattice homomorphism and $w=T\mathbb{1}$. Suppose that T a lattice homomorphism. Being a positive operator, T is (norm) continuous, so that its adjoint $T^*\colon C(L)^*\to C(K)^*$ is defined. For every $t\in L$, $T^*\delta_t=\delta_t\circ T$ is a lattice homomorphism on $C(K)$. Hence there exists $s\in K$ (which need not be unique) and a non-negative scalar λ such that $T^*\delta_t=\lambda\delta_s$. Put $w(t)=\lambda$ and $\varphi(t)=s$. This defines functions $w\colon L\to\mathbb{R}_+$ and $\varphi\colon L\to K$, and $T^*\delta_t=w(t)\delta_{\varphi(t)}$. For every $f\in C(K)$ and $t\in L$ we have

$$(Tf)(t)=\delta_t(Tf)=(T^*\delta_t)(f)=w(t)\delta_{\varphi(t)}(f)=w(t)f\bigl(\varphi(t)\bigr),$$

so that $Tf=w\cdot(f\circ\varphi)$.

Note that $w=T\mathbb{1}_K$; in particular, $w\in C(L)_+$. Suppose that $t\in L$ such that $w(t)\neq 0$; we will show that φ is continuous at t. Let $t_\alpha\to t$ and $w(t)\neq 0$. Take any $f\in C(K)$. Since Tf is continuous, we have $(Tf)(t_\alpha)\to(Tf)(t)$ so that $w(t_\alpha)f\bigl(\varphi(t_\alpha)\bigr)\to w(t)f\bigl(\varphi(t)\bigr)$. Since w is continuous, it follows that $f\bigl(\varphi(t_\alpha)\bigr)\to f\bigl(\varphi(t)\bigr)$. Since this is true for all $f\in C(K)$, we conclude that $\varphi(t_\alpha)\to\varphi(t)$. □

6.11.7 Corollary. *Let K and L be compact spaces and $T\colon C(K)\to C(L)$. Then T a lattice homomorphism with $T\mathbb{1}_K=\mathbb{1}_L$ iff T is a composition operator.*

6.11.8 Exercise. Let K and L be two compact spaces. Then $C(K)$ and $C(L)$ are lattice isomorphic iff K and L are homeomorphic.

6.11.9 Theorem (Abramovich)**.** *A positive surjective isometry between two normed lattices is a lattice isomorphism.*

Proof. Let $T\colon X \to Y$ be such an isometry. By Theorem 1.6.13, it suffices to show that $Tx \geqslant 0$ implies $x \geqslant 0$. Suppose not; suppose that there exists $x \in X$ such that $Tx \geqslant 0$ yet $x^- \neq 0$. WLOG, $\|x\| = \|Tx\| = 1$. Put $y_1 = Tx^+$ and $y_2 = Tx^-$. Note that $y_1, y_2 \geqslant 0$, $y_1 - y_2 = Tx \geqslant 0$, and $\|y_1 \pm y_2\| = \|x^+ \pm x^-\| = \|x\| = 1$.

Since $x^+ \perp x^-$, we have $\|x^+ + nx^-\| \geqslant \|x^+ + x^-\| = 1$ for every n. We claim that actually $\|x^+ + nx^-\| = 1$. It will then follow that $\|\frac{1}{n}x^+ + x^-\| = \frac{1}{n}$ and, after passing to the limit, we get $\|x^-\| = 0$, which is a contradiction.

We already know that the claim is true for $n = 1$. Suppose we have already proved it for n. Note that

$$-\bigl(y_1 + ny_2\bigr) \leqslant y_1 - (n+1)y_2 \leqslant y_1 + ny_2.$$

Indeed, the second inequality is obvious, while the first one follows from the fact that the difference equals $y_1 + (y_1 - y_2) \geqslant 0$. The double inequality is equivalent to $\bigl|y_1 - (n+1)y_2\bigr| \leqslant |y_1 + ny_2|$. Therefore,

$$\begin{aligned}\bigl\|x^+ + (n-1)x^-\bigr\| = \bigl\|x^+ - (n-1)x^-\bigr\| = \bigl\|y_1 - (n+1)y_2\bigr\| \\ \leqslant \|y_1 + ny_2\| = \|x^+ + nx^-\| \leqslant 1,\end{aligned}$$

□

There are two important comments to the preceding theorem. First, the result fails without positivity of the operator. Indeed, ℓ_2 is isometrically isomorphic to $L_2[0,1]$, yet the two are not lattice isomorphic. Second, the proof actually shows more: if T is a positive isometry from X *into* Y (not necessarily surjective), then $x \geqslant 0$ iff $Tx \geqslant 0$.

6.11.1 Adjoint operators

In the rest of this section, we will show that there is a duality between lattice homomorphisms and interval preserving operators. Let $T\colon X \to Y$ be an order bounded operator between vector lattices. For $g \in Y^\sim$, we define $T^\sim g = g \circ T$. Then $T^\sim$ is a linear operator from $Y^\sim$ to $X^\sim$; we call it the ***order adjoint*** of T. It is easy to see that if T is positive then so is $T^\sim$. In the case when T is a bounded operator between Banach lattices, $T^\sim$ equals the usual adjoint T^*. In case of normed lattices, T^* is the restriction of $T^\sim$ to Y^* (cf Proposition 6.7.3).

6.11.10 Theorem. *If $T\colon X \to Y$ is an order bounded operator between vector lattices then $T^\sim$ is order bounded and order continuous.*

Proof. We will prove the theorem under the additional assumption that X is Archimedean and $T \geqslant 0$; see Theorem 1.73 in [AB06] for a proof that avoids these assumptions. By Theorem 6.5.5, it suffices to show that T is order continuous. Let $g_\alpha \downarrow 0$ in $Y^\sim$. By Proposition 6.6.1(iii), this is equivalent to $g_\alpha(y) \downarrow 0$ for every $y \in Y_+$. Then $(T^\sim g_\alpha)(x) = g_\alpha(Tx) \downarrow 0$ for every $x \in X_+$, hence $T^\sim g_\alpha \downarrow 0$. □

Recall that a positive operator T is interval preserving if $[0, Tx] = T[0, x]$ for every $x \geqslant 0$.

6.11.11 Theorem (Kim-Andô)**.** *Let $T\colon X \to Y$ be a linear operator between vector lattices, such that $X^\sim$ separates the points of X. Then T is a lattice homomorphism iff $T^\sim$ is interval preserving.*

Proof. Suppose that T is a lattice homomorphism (in particular, $T \geqslant 0$); let $f \in [0, T^\sim g]$ for some $g \in Y^\sim$. We need to show that $f = T^\sim h$ for some $h \in [0, g]$. Recall that $\operatorname{Range} T$ is a sublattice of Y (see Exercise 1.6.8). For every $y \in \operatorname{Range} T$, we have $y = Tx$ for some $x \in X$; define $h_0(y) = f(x)$. Observe that $h_0(y)$ is well-defined: if we also have $y = Tx'$ for some $x' \in X$ then

$$\bigl|f(x) - f(x')\bigr| \leqslant f\bigl(|x - x'|\bigr) \leqslant (T^\sim g)\bigl(|x - x'|\bigr) = g\bigl(T|x - x'|\bigr) = g\bigl(|Tx - Tx'|\bigr) = 0$$

because T is a lattice homomorphism. It is easy to see that h_0 is a linear functional on $\operatorname{Range} T$.

Let $0 \leqslant y \in \operatorname{Range} T$. Find $x \in X$ with $y = Tx$. Since $y = |Tx| = T|x|$, we may assume WLOG that $x \geqslant 0$. It follows from $0 \leqslant f(x) \leqslant (T^\sim g)(x) = g(y)$ that $0 \leqslant h_0(y) \leqslant g(y)$, so that $0 \leqslant h_0 \leqslant g$ on $\operatorname{Range} T$. We now use Theorem 6.9.2 to extend h_0 to $h \in Y^\sim$ so that $0 \leqslant h \leqslant g$. For every $x \in X$, we have $f(x) = h_0(Tx) = h(Tx) = (T^\sim h)(x)$, so that $f = T^\sim h$.

Conversely, suppose that $T^\sim$ is interval preserving. Fix $x \in X$; it suffices to show that $(Tx)^+ = Tx^+$. Let $f \in X^\sim_+$. By Riesz-Kantorovich formulas 6.6.2, we have

$$\begin{aligned} f\bigl((Tx)^+\bigr) &= \max\Bigl\{g(Tx) \,:\, 0 \leqslant g \leqslant f\Bigr\} = \max\Bigl\{(T^\sim g)(x) \,:\, 0 \leqslant g \leqslant f\Bigr\} \\ &= \max\Bigl\{h(x) \,:\, 0 \leqslant h \leqslant T^\sim f\Bigr\} = (T^\sim f)(x^+) = f(Tx^+). \end{aligned}$$

Since $X^\sim$ separates the points of X, it follows that $(Tx)^+ = Tx^+$. □

For Banach lattices, a partial converse of Theorem 6.11.11 is true. We say that a positive operator $T\colon X \to Y$ between two Banach lattices is ***densely interval preserving*** if $T[0, x]$ is dense in $[0, Tx]$ for every $x \in X_+$. Clearly, every interval preserving operator is densely interval preserving.

6.11.12 Theorem (Lotz)**.** *A positive operator $T\colon X \to Y$ between two Banach lattices is densely interval preserving iff T^* is a lattice homomorphism.*

Proof. Suppose that T is densely interval preserving. Fix $g \in Y^*$; it suffices to show that $(T^*g)^+ = T^*g^+$. Let $x \in X_+$. By Riesz-Kantorovich formulas 6.6.2,

$$(T^*g)^+(x) = \sup(T^*g)[0,x] = \sup g\bigl(T[0,x]\bigr).$$

By assumption, $T[0,x]$ is dense in $[0,Tx]$. Since g is continuous, $g\bigl(T[0,x]\bigr)$ is dense in $g\bigl([0,Tx]\bigr)$, hence the two sets have the same supremum in $\mathbb{R}$:

$$\sup g\bigl(T[0,x]\bigr) = \sup g\bigl([0,Tx]\bigr) = g^+(Tx) = (T^*g^+)(x).$$

Therefore, $(T^*g)^+ = T^*g^+$.

Suppose now that T fails to be densely interval preserving but T^* is a lattice homomorphism. It follows that there exists $x \in X_+$ and $z \in [0,Tx]$ such that $z \notin \overline{T[0,x]}$. Since the latter set is convex, it can be strictly separated from z with a hyperplane, that is, there exists $g \in Y^*$ such that $g(z) > 1$ while $g(y) \leqslant 1$ for all $y \in T[0,x]$; see Theorem A.4.3. It follows from $0 \leqslant z \leqslant Tx$ that

$$g^+(z) \leqslant g^+(Tx) = (T^*g^+)(x) = (T^*g)^+(x) = \sup g\bigl(T[0,x]\bigr) \leqslant 1 < g(z) \leqslant g^+(z),$$

a contradiction. □

We can now refine Kim-Andô Theorem 6.11.11 for Banach spaces.

6.11.13 Corollary. *For a positive operator $T\colon X \to Y$ between two Banach lattices, TFAE:*

(i) *T is a lattice homomorphism;*
(ii) *T^* is interval preserving;*
(iii) *T^* is densely interval preserving.*

Proof. (i)⇔(ii) by Theorem 6.11.11. (ii)⇒(iii) is obvious. (iii)⇒(i), observe that if T^* is ensely interval preserving then T^{**} is a lattice homomorphism by Theorem 6.11.12. It follows that T is a lattice homomorphism. □

Recall (see A.4) that if E is a closed subspace of a Banach space F and $Q\colon E \to E/F$ is the quotient map then Q^* is an isometric embedding.

6.11.14 Exercise. Let J be a closed ideal in a Banach lattice X, let $Q\colon X \to X/J$ be the quotient map. Then Q^* is a lattice isometric embedding. In particular, $(X/J)^*$ is lattice isometric to a closed sublattice of X^*.

Chapter 7

* Vector lattices and Boolean algebras

Theory of vector lattices is closely related to that of Boolean algebras. Many Boolean algebras arising naturally in the setting of vector lattices. In particular, projection bands in a given vector lattice form a Boolean algebra.

Let e be a strong unit in an Archimedean vector lattice X. If we can "split" e into the sum of disjoint "pieces", these "pieces" are called *components* of e. We will show that all the components of e form a Boolean algebra. Furthermore, consider the $C(K)$ representation of X as in Theorem 4.1.9. Then the Boolean algebra of components of e is isomorphic (as a Boolean algebra) to the collection of all clopen subsets of K.

This latter observation leads us to Freudenthal Spectral Theorem, which essentially says that if X has PPP then every vector in X may be approximated by "step functions": linear combinations of components of e. In particular, PPP means that every vector has plenty of components.

7.1 Boolean algebras

In this section, we provide a quick introduction to Boolean algebras. Recall that an ordered set is a lattice if $x \wedge y$ and $x \vee y$ exist for every x and y. A lattice is called ***distributive*** if these two operations satisfy the distributive laws as in Exercise 1.3.3(vi). A ***Boolean algebra*** is a distributive lattice with least and greatest elements, which are usually denoted by $\nvdash$ and $\nVdash$ respectively, and an extra operation $x \mapsto \bar{x}$ satisfying $x \wedge \bar{x} = \nvdash$ and $x \vee \bar{x} = \mathbb{1}$ for every x. We call $\bar{x}$ the ***complement*** of x.

7.1.1 Exercise. Let x and y be elements of a Boolean algebra.

(i) The complement of x is unique. That is, if $x \wedge y = \nvdash$ and $x \vee y = \mathbb{1}$ then $y = \bar{x}$;

(ii) $\bar{\bar{x}} = x$;

(iii) $\overline{x \vee y} = \bar{x} \wedge \bar{y}$ and $\overline{x \wedge y} = \bar{x} \vee \bar{y}$.
(iv) $x \wedge y = \mathbb{0}$ iff $x \leqslant \bar{y}$.

7.1.2 Example. Let Ω be an arbitrary set. Then the power set $\mathcal{P}(\Omega)$ is a Boolean algebra under the operations $A \leqslant B$ if $A \subseteq B$, $A \vee B = A \cup B$, $A \wedge B = A \cap B$, and $\bar{A} = \Omega \setminus A$. In this case $\mathbb{0} = \varnothing$ and $\mathbb{1} = \Omega$.

7.1.3 Example. In measure theory, one deals with algebras and σ-algebras of subsets of a given set Ω. It is easy to see that every such algebra (and, in particular, every σ-algebra) is a Boolean algebra. One may view it as a Boolean subalgebra of $\mathcal{P}(\Omega)$.

7.1.4 Example. Let Ω be a topological space, and suppose that Ω consists of finitely many connected components $\Omega_1, \dots, \Omega_n$. Recall that every connected component of Ω is *clopen*, i.e., simultaneously closed and open. It is easy to see that the set of arbitrary unions of the components forms a Boolean algebra, which is a Boolean subalgebra of $\mathcal{P}(\Omega)$. It consists of all clopen subsets of Ω.

7.1.5 Example. More generally, let Ω be a topological space. Let $\mathrm{Clop}(\Omega)$ be the collection of all clopen subsets of Ω. In particular, it contains $\varnothing$, Ω, and every connected component of Ω. Then $\mathrm{Clop}(\Omega)$ is a Boolean subalgebra of $\mathcal{P}(\Omega)$. In particular, it is itself a Boolean algebra.

The last example may seem somewhat exotic, but it is important because the classical representation theorem for Boolean algebras due to M. Stone asserts that every Boolean algebra can be represented as the algebra of all clopen sets in some compact space. In particular, every Boolean algebra can be represented as an algebra of sets. This theorem (together with its variants) is at the heart of many important facts of Functional Analysis. Its power is that it starts with a Boolean algebra, which is an object of Set Theory, and produces a compact space, which is an object of Topology. $C(K)$-Representation Theorem 4.1.7 as well as Gelfand's theorem on representations of commutative C^*-algebras are "relatives" of Stone's theorem. (Actually, one can deduce the $C(K)$-Representation Theorem from Stone's Theorem and vice versa.)

Before we present the theorem, we need to introduce some terminology. A map φ between two Boolean algebras is said to be a ***Boolean algebra homomorphism*** if it preserved the Boolean operations, i.e., $\varphi(x \vee y) = \varphi(x) \vee \varphi(y)$, $\varphi(x \wedge y) = \varphi(x) \wedge \varphi(y)$, $\varphi(\mathbb{0}) = \mathbb{0}$, and $\varphi(\mathbb{1}) = \mathbb{1}$. It follows easily that $\varphi(\bar{x}) = \overline{\varphi(x)}$. If, in addition, φ is a bijection, it is said to be a ***Boolean algebra isomorphism***, or just an ***isomorphism*** when it causes no confusion. It is easy to see that in this case φ^{-1} is also a Boolean algebra homomorphism.

7.1.6 Example. Let Ω be a topological space. Let $\mathcal{B}$ be set of all characteristic functions of clopen sets. Under the order induced from $C(\Omega)$, $\mathcal{B}$ is a Boolean algebra. The map that takes $A \in \mathrm{Clop}(\Omega)$ into $\mathbb{1}_A$ is a Boolean lattice isomorphism between $\mathrm{Clop}(K)$ and $\mathcal{B}$.

Recall that in 2.1.3 we introduced the concept of a totally disconnected compact space.

7.1.7 Theorem (Stone). *Every Boolean algebra is isomorphic to* $\mathrm{Clop}(K)$ *for some totally disconnected compact space K. This K is unique up to a homeomorphism.*

Before we prove the theorem, we need some additional machinery. Let $\mathcal{B}$ be a Boolean algebra. A non-empty subset F of $\mathcal{B}$ is said to be a ***filter*** if it does not contain $\not\mathbb{1}$, is closed under $\wedge$ operation, and with every a it contains all the elements of $\mathcal{B}$ that are greater than a. For example, for every non-zero $a \in \mathcal{B}$, the set $F_a = \{b \in \mathcal{B} \,:\, b \geqslant a\}$ is a filter.

By Zorn's Lemma, the set of all filters on $\mathcal{B}$, ordered by inclusion, has maximal elements (naturally, they are called ***maximal filters***) and every filter is contained in a maximal filter. In particular, for every non-zero $a \in \mathcal{B}$ it follows from $a \in F_a$ that a is contained in some maximal filter.

7.1.8 Exercise. A filter F in a Boolean algebra $\mathcal{B}$ is maximal iff for every $a \in \mathcal{B}$ either $a \in F$ or $\bar{a} \in F$.

7.1.9 Exercise. If p is a maximal filter in a Boolean algebra $\mathcal{B}$ and $a, b \in \mathcal{B}$ then $a \vee b \in p$ iff a or b is in p.

The difficulty in proving Stone's Theorem is how one constructs the set K from $\mathcal{B}$. Let's work our way backwards: suppose that $\mathcal{B}$ is already a Boolean algebra of subsets of some set K. Then, for each $p \in K$, the set $\{a \in \mathcal{B} \,:\, p \in a\}$ is a maximal filter in $\mathcal{B}$. So one can identify K with a set of maximal filters in $\mathcal{B}$. Motivated by this observation, for an arbitrary Boolean algebra $\mathcal{B}$ we will define K to be the set of all maximal filters in $\mathcal{B}$.

Proof of Stone's Theorem. Let K be the set of all maximal filters in $\mathcal{B}$. For each $a \in \mathcal{B}$, define $V_a \coloneqq \{p \in K \,:\, a \in p\}$. That is, $p \in V_a$ iff $a \in p$. We claim that the map $V\colon a \in \mathcal{B} \mapsto V_a \in \mathcal{P}(K)$ is a Boolean algebra homomorphism. Indeed, $V_{\not\mathbb{1}} = \varnothing$ and $V_{\mathbb{1}} = K$. One can verify that $V_{a\wedge b} = V_a \cap V_b$ and $V_{a\vee b} = V_a \cup V_b$ (use Exercise 7.1.9 for the latter).

Next, we will show that the map V is one-to-one. First, observe that for each non-zero $a \in \mathcal{B}$ there exists $p \in K$ such that $a \in p$, so that $p \in V_a$, hence $V_a \neq \varnothing$. Therefore, $V_a = \varnothing$ iff $a = \not\mathbb{1}$. Now suppose that $V_a = V_b$. Then $\varnothing = V_a \cap V_b^C = V_{a\wedge\bar{b}}$, so that $a \wedge \bar{b} = \not\mathbb{1}$ and, therefore, $a \leqslant b$ by Exercise 7.1.1(iv). Similarly, $b \leqslant a$, so that $a = b$.

Next, we will define a topology on K. For a fixed $p \in K$, the collection $\{V_a \,:\, a \in p\}$ is closed under finite intersections and each member of this collection contains p. We take this collection as a base of neighborhoods of p in K. That is, we declare a set U to be a neighborhood of p if $V_a \subseteq U$ for some $a \in p$. This defines a topology on K. This topology is Hausdorff: if $p \neq q$ in K then there exists a such that $a \in p$ but $a \notin q$ (or vice versa), then $p \in V_a$ while $q \in V_{\bar{a}}$ and $V_a \cap V_{\bar{a}} = \varnothing$.

Note that each V_a is clopen. Indeed, by definition, V_a is a neighborhood of every point in it, hence is open. But $V_a^C = V_{\bar{a}}$ is open as well.

We will now show that K is compact. We need to show that every open cover has a finite subcover. It is easy to see that it suffices to consider covers by base neighborhoods, so we need to prove that every cover of K by sets of the form V_a has a finite subcover. Passing to complements and using $V_a^C = V_{\bar{a}}$, it suffices to show that every collection $\{V_a : a \in A\}$ for any $A \subseteq \mathcal{B}$ has the finite intersection property, i.e., if $\bigcap_{a\in I} V_a$ is non-empty for every finite $I \subseteq A$ then $\bigcap_{a\in A} V_a$ is non-empty. Since $\bigcap_{a\in I} V_a = V_{\inf I}$, the assumption means exactly that $\not\Vdash \notin A^\wedge$. Put

$$F = \bigl\{b \in \mathcal{B} : b \geqslant a \text{ for some } a \in A^\wedge\bigr\}.$$

It is easy to see that F is a filter and $A \subseteq F$; it follows that there exists a maximal filter p such that $F \subseteq p$. For every $a \in A$ we have $a \in p$, so that $p \in V_a$, hence $\bigcap_{a\in A} V_a$ is non-empty.

It is left to show that $\operatorname{Range} V = \operatorname{Clop}(K)$. Suppose that C is a clopen set in K. Since C is open, for every $p \in C$ there exists a base neighborhood U of p such that $U \subseteq C$. It follows that there is a cover of C by base neighborhoods. Since C is closed, it is compact, hence we can find a finite subcover $V_{a_1}, \dots, V_{a_n}$. This yields $C = V_{a_1} \cup \dots \cup V_{a_n} = V_{a_1 \vee \dots \vee a_n} \in \operatorname{Range} V$.

To prove uniqueness, let Q be a totally disconnected compact space such that $\mathcal{B}$ is isomorphic to $\operatorname{Clop}(Q)$. WLOG, $\mathcal{B} = \operatorname{Clop}(Q)$. For every $q \in Q$, the set $\{a \in \operatorname{Clop}(Q) : q \in a\}$ is a maximal filter, hence defines an element of K; denote it $\varphi(q)$. Thus, $\varphi\colon Q \to K$ and $p = \varphi(q)$ iff $q \in a$ for all $a \in p$ iff $q \in \bigcap p$. Next, we will show that φ is onto. Let $p \in K$. Being a filter in $\operatorname{Clop}(Q)$, p has a finite intersection property, so, by compactness of Q, $\bigcap p$ is nonempty. For every $q \in \bigcap p$ we have $\varphi(q) = p$ and, therefore, φ is onto. It is also one-to-one: if $q_1 \neq q_2$ then by total disconnectedness of Q there exists $a \in \operatorname{Clop}(Q)$ such that $q_1 \in a$ while $q_2 \notin a$, hence $a \in \varphi(q_1)$ but $a \notin \varphi(q_2)$ and, therefore, $\varphi(q_1) \neq \varphi(q_2)$. Thus, φ is a bijection. Furthermore, for every $q \in Q$ and $a \in \operatorname{Clop}(Q)$, we have $q \in a$ iff $a \in \varphi(q)$ iff $\varphi(q) \in V_a$, hence φ maps a onto V_a. Thus, for every $q \in Q$, φ maps a base of neighborhoods of q to a base of neighborhoods of $\varphi(q)$; this yields that φ is a homeomorphism. $\square$

7.1.10 Remark. A maximal filter in a Boolean algebra is also called an ***ultrafilter***. It can be easily verified that F is a maximal filter iff its complement in $\mathcal{P}(\mathcal{B})$ is a ***maximal ideal***. Furthermore, maximal ideals in $\mathcal{B}$ are exactly the kernels of Boolean algebra homomorphisms from $\mathcal{B}$ to the Boolean algebra $\{\not\Vdash, \mathbb{1}\}$; such a homomorphism is obviously determined by its kernel. Therefore, one can equivalently define K as the set of all maximal ideals in $\mathcal{B}$ or the set of all Boolean algebra homomorphisms from $\mathcal{B}$ to $\{\not\Vdash, \mathbb{1}\}$.

We will call the space K constructed in the theorem the ***Stone space*** of $\mathcal{B}$.

Since every set in $\mathcal{B}$ is order bounded by $\not\Vdash$ and $\mathbb{1}$, it follows that $\mathcal{B}$ is order complete (in the sense of Section 2.1) iff every non-empty set in it has a supremum. We now characterize order completeness of $\mathcal{B}$ in terms of its Stone space. This result should be compared to Theorem 2.1.29, which asserts that $C(K)$ is order complete iff K is extremally disconnected.

7.1.11 Proposition. *Let $\mathcal{B}$ be a Boolean algebra and K its Stone space. Then $\mathcal{B}$ is order complete iff K is extremally disconnected.*

Proof. By Stone's Theorem, we may assume that $\mathcal{B} = \operatorname{Clop}(K)$. Suppose that K is extremally disconnected. Let $\mathcal{A}$ be a non-empty subset of $\operatorname{Clop}(K)$. Let U be the union of $\mathcal{A}$. Then U is open and, therefore, $\overline{U}$ is clopen; it is clear that $\overline{U}$ is the supremum of $\mathcal{A}$ in $\mathrm{Clop}(K)$. Therefore, $\mathrm{Clop}(K)$ is order complete.

Conversely, suppose that $\mathrm{Clop}(K)$ is order complete, and let O be an open subset of K. Since K is totally disconnected (by Stone's Theorem), for every $t \in O$ we find a clopen neighborhood V_t with $V_t \subseteq O$. Let V be the supremum of the family $\{V_t : t \in O\}$ in $\mathrm{Clop}(K)$. Then V is clopen; it is left to show that $V = \overline{O}$. It follows from $t \in V_t \subseteq V$ for all $t \in O$ that $\overline{O} \subseteq V$. Suppose now that $s \notin \overline{O}$. Find a clopen neighborhood W of s such that $\overline{O} \cap W = \varnothing$. It follows that $V_t \subseteq W^C$ for every $t \in O$, so that $V \subseteq W^C$ and, therefore, $s \notin V$. □

7.1.12. The following construction comes up naturally in various parts of analysis. Let $\varphi\colon \mathcal{B} \to Y$ be a map from a Boolean algebra $\mathcal{B}$ to a vector space Y. Suppose that φ is disjointly additive, i.e., $\varphi(a \vee b) = \varphi(a) + \varphi(b)$ whenever $a \wedge b = 0$. Let X be the vector space of all formal linear combinations of elements of $\mathcal{B}$, H the subset of X given by $H = \{a \vee b - a - b : a \wedge b = 0\}$, and V the subspace of X spanned by H. We extend φ to a linear operator $T_0\colon X \to Y$ be linearity, i.e., $T_0\left(\sum_{i=1}^n r_i a_i\right) = \sum_{i=1}^n r_i \varphi(a_i)$. Then T_0 vanishes on H and, therefore, on V. It follows that T_0 induces a linear operator $T\colon X/V \to Y$.

Here is a typical application of the preceding construction. Let Ω be a set, and $\mathcal{F}$ an algebra of subsets of Ω, i.e., a Boolean subalgebra of $\mathcal{P}(\Omega)$. Let $\mathrm{s}(\mathcal{F})$ be the vector space of all simple functions, i.e., functions of the form $\sum_{i=1}^n r_i \mathbb{1}_{A_i}$, where $r_1, \dots, r_n \in \mathbb{R}$ and $A_1, \dots, A_n \in \mathcal{F}$. Let μ be a finitely-additive function on $\mathcal{F}$, i.e., $\mu\colon \mathcal{F} \to \mathbb{R}$ such that $\mu(A \cup B) = \mu(A) + \mu(B)$ whenever $A \cap B = \varnothing$. For $f \in \mathrm{s}(\mathcal{F})$ given by $f = \sum_{i=1}^n r_i \mathbb{1}_{A_i}$, one defines $\int f\,d\mu = \sum_{i=1}^n r_i \mu(A_i)$. However, one has to verify that this is well defined, i.e., does not depend on the choice of a representation of f. To see this, we apply the preceding paragraph with $\mathcal{B} = \mathcal{F}$, $Y = \mathbb{R}$, and $\varphi = \mu$. We observe that two expressions $\sum_{i=1}^n r_i \mathbb{1}_{A_i}$ and $\sum_{j=1}^m s_j \mathbb{1}_{B_j}$ represent the same function in $\mathrm{s}(\mathcal{F})$ precisely when one may be obtained from the other by a finite number of "splittings" and "mergers", where we replace $\mathbb{1}_{A \cup B}$ with $\mathbb{1}_A + \mathbb{1}_B$ for some $A, B \in \mathcal{F}$ with $A \cap B = \varnothing$. This means that $\mathrm{s}(\mathcal{F})$ equals X/V from the preceding paragraph. Then $\int f\,d\mu$ is Tf, and is well defined.

7.2 Boolean algebras of components

Let X be a vector lattice and $e \in X_+$. Throughout this section, we will restrict our attention to I_e, so the reader may assume WLOG that e is a strong unit and, therefore, $X = I_e$. A vector $x \in [0, e]$ is said to be a ***component*** of e if $x \perp (e - x)$. Let $\mathcal{C}_e$ be the set of all components of e.

7.2.1 Lemma. *A vector $x \in [0,e]$ is a component of e iff $2x \wedge e = x$.*

Proof. Adding x to both sides as in Exercise 1.3.3(ii), we conclude that $x \wedge (e-x) = 0$ iff $2x \wedge e = x$. □

7.2.2 Example. Let $X = L_p(\Omega)$ for a finite measure μ and $0 \leqslant p \leqslant \infty$. The components of $\mathbb{1}$ are exactly the functions of the form $\mathbb{1}_A$ where A is a measurable set. More generally, if μ is an arbitrary measure and $e \in L_p(\mu)_+$ then the components of e are exactly the functions of the form $e \cdot \mathbb{1}_A$ for some measurable set A.

7.2.3 Example. Let $X = C(\Omega)$. The components of $\mathbb{1}$ are exactly the characteristic functions of clopen sets. By Example 7.1.6, $\mathcal{C}_{\mathbb{1}}$ is Boolean algebra isomorphic to $\mathrm{Clop}(\Omega)$. In particular, the only components of $\mathbb{1}$ in $C[0,1]$ are 0 and $\mathbb{1}$.

7.2.4 Proposition. *Suppose that X is a vector lattice and $e \in X_+$. Then $\mathcal{C}_e$ is a Boolean algebra with $\not{\mathbb{1}} = 0$, $\mathbb{1} = e$, and $\bar{x} = e - x$.*

Proof. First, we use Lemma 7.2.1 to check that $\mathcal{C}_e$ is closed under lattice operations. Indeed, for $x, y \in \mathcal{C}_e$,

$$\begin{aligned}\big[2(x \wedge y)\big] \wedge e &= (2x) \wedge (2y) \wedge e = (2x \wedge e) \wedge (2y \wedge e) = x \wedge y, \text{ and} \\ \big[2(x \vee y)\big] \wedge e &= (2x \wedge e) \vee (2y \wedge e) = x \vee y,\end{aligned}$$

so that $x \wedge y$ and $x \vee y$ are in $\mathcal{C}_e$. Since X is a distributive lattice, so is $\mathcal{C}_e$. Clearly, 0 and e are the least and the greatest elements of $\mathcal{C}_e$, respectively. For $x \in \mathcal{C}_e$, put $\bar{x} = e - x$. Then $\bar{x} \in \mathcal{C}_e$ and $x \perp \bar{x}$, so that $x \wedge \bar{x} = 0$. Also, $x \vee \bar{x} = x + \bar{x} = x + (e-x) = e$. □

In the following, we will be interested in the linear span of $\mathcal{C}_e$ in X. Clearly, it is a subspace of I_e. Elements of $\operatorname{span} \mathcal{C}_e$ can be written in the form $\sum_{i=1}^n \lambda_i x_i$, where $\lambda_1, \dots, \lambda_n$ are scalars and $x_1, \dots, x_n$ are elements of $\mathcal{C}_e$. We call such a representation ***disjoint*** if $x_1, \dots, x_n$ are disjoint.

We are now going to investigate an important relationship between the $C(K)$-Representation Theorem 4.1.9 and Stone Theorem 7.1.7. By $C(K)$-Representation Theorem, there exists a lattice isometry $T \colon I_e \to C(K)$ for some compact K, such that $Te = \mathbb{1}$ and $\operatorname{Range} T$ is dense. We will consider three cases: when X is just Archimedean, when X is uniformly complete, and when X has PPP. Recall that by Corollary 5.6.28, X σ-order complete iff it is uniformly complete and has PPP.

Suppose that X is uniformly complete. Then the situation is crystal clear. In this case, the lattice isometry T is onto by Theorem 4.1.9. It is easy to see that the restriction of T to $\mathcal{C}_e$ is a Boolean algebra isomorphism of $\mathcal{C}_e$ onto $\mathcal{C}_{\mathbb{1}}$. Hence, $\mathcal{C}_e$ is isomorphic to $\mathrm{Clop}(K)$.

We will show next that even though without uniform completeness assumption $T\colon I_e \to C(K)$ need not be surjective, it still yields a Boolean algebra isomorphism between $\mathcal{C}_e$ and $\mathcal{C}_{\mathbb{1}}$ (and, therefore, between $\mathcal{C}_e$ and $\mathrm{Clop}(K)$.)

7.2.5 Proposition. *Let X be an Archimedean vector lattice, $e \in X_+$, and $T\colon I_e \to C(K)$ as in $C(K)$-Representation Theorem 4.1.9. Then the restriction of T to $\mathcal{C}_e$ is a Boolean algebra isomorphism of $\mathcal{C}_e$ onto $\mathcal{C}_{\mathbb{1}}$. Hence $\mathcal{C}_e$ is isomorphic to* $\mathrm{Clop}(K)$. *Furthermore,* $\operatorname{span}\mathcal{C}_e$ *is a sublattice of I_e and the restriction of T it is a lattice isomorphism between* $\operatorname{span}\mathcal{C}_e$ *and* $\operatorname{span}\mathcal{C}_{\mathbb{1}}$.

Proof. It is easy to see that the restriction of T to $\mathcal{C}_e$ is an injective Boolean algebra homomorphism from $\mathcal{C}_e$ to $\mathcal{C}_{\mathbb{1}}$; it remains to prove it is onto. Take an element of $\mathcal{C}_{\mathbb{1}}$; it has to be of the form $\mathbb{1}_A$ for some $A \in \mathrm{Clop}(K)$. Since $\operatorname{Range} T$ is dense in $C(K)$, by Urysohn's Lemma for dense sublattices 5.6.2, we can find $x \in I_e$ such that $Tx = \mathbb{1}_A$. Let $y = x^+ \wedge e$, then $0 \leqslant y \leqslant e$ and $Ty = (Tx)^+ \wedge Te = \mathbb{1}_A$. Since T is injective, it follows from $T\bigl(y \wedge (e-y)\bigr) = \mathbb{1}_A \wedge (\mathbb{1} - \mathbb{1}_A) = 0$ that $y \perp (e-y)$ and, therefore, $y \in \mathcal{C}_e$. This proves that the restriction of T to $\mathcal{C}_e$ is a Boolean algebra isomorphism of $\mathcal{C}_e$ onto $\mathcal{C}_{\mathbb{1}}$.

It is easy to see that T is a bijection between $\operatorname{span}\mathcal{C}_e$ and $\operatorname{span}\mathcal{C}_{\mathbb{1}}$. For every $x \in \operatorname{span}\mathcal{C}_e$, we have $Tx \in \operatorname{span}\mathcal{C}_{\mathbb{1}}$, so that $T\lvert x\rvert = \lvert Tx\rvert \in \operatorname{span}\mathcal{C}_{\mathbb{1}}$ because $\operatorname{span}\mathcal{C}_{\mathbb{1}}$ is a sublattice of $C(K)$. It follows that $\lvert x\rvert \in \operatorname{span}\mathcal{C}_e$, hence $\operatorname{span}\mathcal{C}_e$ is a sublattice. Since T is a lattice isomorphism on I_e, its restriction to $\operatorname{span}\mathcal{C}_e$ remains a lattice isomorphism. □

7.2.6 Exercise. Let $x, y \in \mathcal{C}_e$. If $x \leqslant y$ then $y - x \in \mathcal{C}_e$ and $x \perp (y-x)$. Furthermore, if $x_1 \leqslant \ldots \leqslant x_n$ in $\mathcal{C}_e$ then $x_1, x_2 - x_1, \ldots, x_n - x_{n-1}$ are disjoint.

7.2.7 Corollary. *Let X be an Archimedean vector lattice and $e \in X_+$. Then every element in* $\operatorname{span}\mathcal{C}_e$ *admits a disjoint representation. Furthermore, if $x = \sum_{i=1}^n \lambda_i x_i$ is a disjoint decomposition of x then $\lVert x\rVert_e = \max_{i \leqslant n}\lvert\lambda_i\rvert$.*

Proof. Let T be as in the preceding propositions. It is easy to see that $\operatorname{span}\mathcal{C}_{\mathbb{1}}$ is a sublattice of $C(K)$ consisting of simple functions with respect to the algebra $\mathrm{Clop}(K)$. In particular, every element of $\operatorname{span}\mathcal{C}_{\mathbb{1}}$ admits a disjoint decomposition; hence the same is true for $\mathcal{C}_e$. Let $x = \sum_{i=1}^n \lambda_i x_i$ be a disjoint decomposition of $x \in \operatorname{span}\mathcal{C}_e$. Then $Tx = \sum_{i=1}^n \lambda_i T x_i$ is a disjoint decomposition of Tx in $\operatorname{span}\mathcal{C}_{\mathbb{1}}$. Since T is an isometry and $Tx_i \in \mathcal{C}_{\mathbb{1}}$ for every i, we have

$$\|x\|_e = \Big\|\sum_{i=1}^{n} \lambda_i T x_i\Big\|_{C(K)} = \max_{i\leqslant n} |\lambda_i|.$$

□

7.2.8 Exercise. Let X and Y be two Archimedean vector lattices, $e \in X_+$ and $u \in Y_+$. Suppose that $\varphi\colon \mathcal{C}_e \to \mathcal{C}_u$ be such that if $x_1 \perp x_2$ in $\mathcal{C}_e$ then $\varphi(x_1) \perp \varphi(x_2)$ and $\varphi(x_1 + x_2) = \varphi(x_1) + \varphi(x_2)$. Then

$$T\Big(\sum_{i=1}^{n} \lambda_i x_i\Big) = \sum_{i=1}^{n} \lambda_i \varphi(x_i) \tag{7.1}$$

well-defines a lattice homomorphism $T\colon \operatorname{span}\mathcal{C}_e \to \operatorname{span}\mathcal{C}_u$.

7.2.9 Exercise. Let X and Y be two Archimedean vector lattices, $e \in X_+$ and $u \in Y_+$. Let $\varphi\colon \mathcal{C}_e \to \mathcal{C}_u$ be an injective Boolean algebra homomorphism. Let T be as in (7.1). Then T is a well-defined lattice isometric embedding from $\big(\operatorname{span}\mathcal{C}_e, \|\cdot\|_e\big)$ to $\big(\operatorname{span}\mathcal{C}_u, \|\cdot\|_u\big)$.

7.2.10 Exercise. It is easy to see that φ satisfies the assumptions of Exercise 7.2.8. Hence T is well-defined and is a lattice homomorphism. It follows from Corollary 7.2.7 that $\|Tx\| = \|x\|_e$ for every $x \in \operatorname{span}\mathcal{C}_e$.

Proposition 7.2.5 provides a Boolean algebra isomorphism between $\mathcal{C}_e$ and $\mathcal{C}_{\mathbb{1}}$ and a lattice isometry between $\operatorname{span} C_e$ and $\operatorname{span} C_{\mathbb{1}}$. Note, however, that $\mathcal{C}_e$ may be rather poor.

7.2.11 Example. Let $X = C[0,1]$ and $e = \mathbb{1}$. Applying Proposition 7.2.5, one gets $I_e = X$, $K = [0,1]$, $\mathcal{C}_e = \{0, \mathbb{1}\}$, and $\operatorname{span}\mathcal{C}_e$ consists only of constant functions. In particular, one recovers little information about I_e from $\mathcal{C}_e$ or $\operatorname{span}\mathcal{C}_e$. The same example also shows that K need not be the Stone space of $\mathcal{C}_e$ (even though X is uniformly complete).

7.2.12 Example. Let $X = L_p(\mu)$ for some finite measure μ; take $e = \mathbb{1}$. Then $I_e = L_\infty(\mu)$, $\|\cdot\|_e = \|\cdot\|_\infty$, $\mathcal{C}_e$ consists of characteristic functions of measurable sets, and $\operatorname{span}\mathcal{C}_e$ consists of all simple functions. It is a standard fact of measure theory that simple functions are dense in $L_\infty(\mu)$.

The difference between these two examples lies in the fact that $L_p(\mu)$ has PPP while $C(K)$ fails it. We will show next that abundance of band projection guarantees that $\mathcal{C}_e$ is sufficiently rich so that $\operatorname{span}\mathcal{C}_e$ is dense in I_e.

7.2.13 Proposition. *Let X be a vector lattice with PPP; let $e \in X_+$. Then K in $C(K)$-Representation Theorem 4.1.9 coincides with the Stone space for the Boolean algebra $\mathcal{C}_e$.*

Proof. Let $T\colon I_e \to C(K)$ be as in $C(K)$-Representation Theorem 4.1.9. Let $Y = \operatorname{Range} T$. By Proposition 7.2.5, $\mathcal{C}_e$ is isomorphic (as a Boolean algebra) to $\mathcal{C}_{\mathbb{1}}$ and, therefore, to $\operatorname{Clop}(K)$.

Since X has PPP, so does I_e by 5.6.23; and so does Y because T is a lattice isomorphism between I_e and Y. Theorem 5.6.20 now yields that K is totally disconnected; this completes the proof. □

7.2.14 Theorem (Freudenthal). *Let X be a vector lattice with PPP and $e \in X_+$. Then $\operatorname{span} \mathcal{C}_e$ is dense in $\big(I_e, \|\cdot\|_e\big)$. Moreover, for every $x \in I_e$ and every $\varepsilon > 0$ there exists $y \in \operatorname{span} \mathcal{C}_e$ such that $y \leqslant x \leqslant y + \varepsilon e$.*

Proof. Let $T\colon X \to C(K)$ be as in $C(K)$-Representation Theorem 4.1.9. By Proposition 7.2.13, K is totally disconnected. It follows that for every $s, t \in K$ with $s \neq t$ there exists a clopen set U such that $s \in U$ and $t \notin U$ and, therefore, $\mathbb{1}_U$ separates s and t. By Stone-Weierstrass Theorem 3.1.28, $\operatorname{span} \mathcal{C}_{\mathbb{1}}$ is dense in $C(K)$. Using Proposition 7.2.5 and the fact that T is an isometry, we conclude that $\operatorname{span} \mathcal{C}_e$ is dense in $\big(I_e, \|\cdot\|_e\big)$.

In particular, for every $x \in I_e$ and every $\varepsilon > 0$ there exists $z \in \operatorname{span} \mathcal{C}_e$ such that $\|x - z\|_e < \frac{\varepsilon}{2}$. It follows that $z - \frac{\varepsilon}{2} e \leqslant x \leqslant z + \frac{\varepsilon}{2} e$. Taking $y = z - \frac{\varepsilon}{2} e$ completes the proof. □

Let C be a convex set in a vector space; let $x \in C$. Recall that x is an ***extreme point*** of C if $x = \lambda a + \mu b$ for some $a, b \in C$ and some $\lambda, \mu > 0$ with $\lambda + \mu = 1$ implies $a = b = x$.

7.2.15 Proposition. *Let X be an Archimedean vector lattice and $e \in X_+$. Then $\mathcal{C}_e$ is the set of all extreme points of $[0, e]$.*

Proof. Suppose that x is an extreme point of $[0, e]$. Let $y = x \wedge (e - x)$; we need to show that $y = 0$. It follows from $0 \leqslant y \leqslant x \leqslant e$ that $0 \leqslant x - y \leqslant e$. It follows from $0 \leqslant y \leqslant e - x$ that $0 \leqslant x + y \leqslant e$. Since we have $x = \frac{1}{2}(x + y) + \frac{1}{2}(x - y)$, we conclude that $x + y = x - y = x$ and, therefore, $y = 0$.

To prove the converse, let $x \in \mathcal{C}_e$, and suppose that $x = \lambda a + \mu b$ for some $a, b \in [0, e]$ and some $\lambda, \mu > 0$ with $\lambda + \mu = 1$. Using $C(K)$-Representation Theorem 4.1.9, we may represent x, a, and b as continuous functions on some $C(K)$ space, with $e = \mathbb{1}$. By Proposition 7.2.5, $x = \mathbb{1}_U$ for some clopen set U in K. It follows easily from $\mathbb{1}_U = \lambda a + \mu b$ and $a, b \in [0, \mathbb{1}]$ that $a = b = \mathbb{1}_U = x$. □

7.3 Boolean algebras of projection bands

Throughout this section, X will be an Archimedean vector lattice. We start with two motivational examples.

7.3.1 Example. Let $X = C(K)$. By Proposition 5.2.4, there is a one-to-one correspondence between projection bands in $C(K)$ and clopen subsets of K. In particular, this correspondence preserves inclusion order. Since $\mathrm{Clop}(K)$ is a Boolean algebra, it follows that projection bands in $C(K)$ also form a Boolean algebra, isomorphic to $\mathrm{Clop}(K)$. It is easy to see from this isomorphism that its Boolean operations are given by $B_1 \wedge B_2 = B_1 \cap B_2$, $B_1 \vee B_2 = B_1 + B_2$, and the complement of B is B^d. Note also that this Boolean algebra is also isomorphic to $\mathcal{C}_{\mathbb{1}}$.

7.3.2 Example. Let $X = L_p(\Omega, \mathcal{F}, \mu)$ with μ σ-finite and $1 \leqslant p \leqslant \infty$. Since X is order complete, every band is a projection band, and by Theorem 5.1.26 there is a one-to-one correspondence between bands and $\mathcal{F}$. It allows us to put a structure of Boolean algebra on the set of all bands in X, with the same Boolean operations as in Example 7.3.1. If $e \in X_+$ is a weak unit (equivalently, $e(t) > 0$ a.e.), then the map $A \in \mathcal{F} \to e \cdot \mathbb{1}_A$ is a Boolean algebra isomorphism between $\mathcal{F}$ and $\mathcal{C}_e$.

Let $\mathcal{B}(X)$ stand for the set of all projection bands in X, while $\mathcal{P}(X)$ stands for the set of all band projections in X. The two are partially ordered sets: $\mathcal{B}(X)$ is ordered by inclusion, while $\mathcal{P}(X)$ inherits its order from $\mathcal{L}(X)$.

7.3.3 Theorem. *The set $\mathcal{B}(X)$ of all projection bands in X forms a Boolean algebra with*

$$\not\mathbb{1} = \{0\}, \quad \mathbb{1} = X, \quad B_1 \wedge B_2 = B_1 \cap B_2, \quad B_1 \vee B_2 = B_1 + B_2, \quad \textit{and} \quad \overline{B} = B^d.$$

The set $\mathcal{P}(X)$ of all band projections on X forms a Boolean algebra with

$$\not\mathbb{1} = 0, \quad \mathbb{1} = I, \quad \overline{P} = I - P, \quad P_1 \wedge P_2 = P_1 P_2, \quad \textit{and} \quad P_1 \vee P_2 = P_1 + P_2 - P_1 P_2.$$

The map $\Phi\colon \mathcal{B}(X) \to \mathcal{P}(X)$, where $\Phi(B) = P_B$, is a Boolean algebra isomorphism.

Proof. Clearly, Φ is a bijection. It follows from Exercise 5.2.13 that Φ is an isomorphism of $\mathcal{B}(X)$ and $\mathcal{P}(X)$ as partially ordered sets. It is clear that $\{0\}$, X, 0, and I are the least and the greatest elements in $\mathcal{B}(X)$ and $\mathcal{P}(X)$, respectively.

Let $B_1, B_2 \in \mathcal{B}(X)$; it follows from Theorem 5.2.14 that $B_1 \cap B_2$ and $B_1 + B_2$ are in $\mathcal{B}(X)$; it is now easy to see that $B_1 \cap B_2 = B_1 \wedge B_2$ and $B_1 + B_2 = B_1 \vee B_2$, hence $\mathcal{B}(X)$ is a lattice. For every $B \in \mathcal{B}(X)$, we clearly have $B \wedge \overline{B} = B \cap B^d = \{0\}$ and $B \vee \overline{B} = B + B^d = X$, hence the map $B \mapsto B^d$ is a complement operation in $\mathcal{B}(X)$.

Since Φ is an order isomorphism, it follows by Theorem 5.2.14 that $\mathcal{P}(X)$ is also a lattice with $P_1 \wedge P_2 = P_1P_2$ and $P_1 \vee P_2 = P_1 + P_2 - P_1P_2$, and with complement $\overline{P} = I - P$.

A direct computation verifies that $\mathcal{P}(X)$ is distributive, hence a Boolean algebra. Since Φ is an order isomorphism, we conclude that $\mathcal{B}(X)$ is also a Boolean algebra. $\square$

7.3.4 Remark. All bands in X also form a Boolean algebra. While this may be shown directly, we will easily deduce it from a "big" theorem in Theorem 7.4.19.

7.3.5 Exercise. Verify that the operations "$\vee$" and "$\wedge$" in $\mathcal{P}(X)$ agree with those in $\mathcal{L}(X)$.

Next, we are going to explore connections between projection bands and components.

7.3.6 Exercise. If $e \in X_+$ and P is a band projection then Pe is a component of e.

7.3.7 Proposition. *Let $e \in X_+$. The map $P \mapsto Pe$ is a Boolean algebra homomorphism from $\mathcal{P}(X)$ to $\mathcal{C}_e$.*

Proof. Clearly, the map $P \mapsto Pe$ preserves order. Let's show that it preserves infima. Let $P_1, P_2 \in \mathcal{P}$; put $P = P_1 \wedge P_2 = P_1P_2$. Let B_1, B_2, and B bet the corresponding bands. Put $x_1 = P_1e$ and $x_2 = P_2e$; we need to show that $x_1 \wedge x_2 = Pe$. On one hand, it follows from $Pe = P_1P_2e = P_1x_2 \leqslant x_2$ and $Pe \leqslant x_1$ that $Pe \leqslant x_1 \wedge x_2$. On the other hand, we have $x_1 \wedge x_2 \in B_1 \cap B_2 = B$; applying Theorem 5.2.15, we get $Pe = \sup\{x \in B \,:\, x \leqslant e\} \geqslant x_1 \wedge x_2$.

Next, observe that the map $P \mapsto Pe$ preserves complements: $\overline{P}e = (I - P)e = e - Pe = \overline{Pe}$. It now follows from Exercise 7.1.1(iii) that this map preserves suprema. $\square$

For the rest of the section, we focus on the case when X has PPP.

7.3.8 Exercise. Let X be a vector lattice with PPP, $e, x \in X_+$. TFAE:

(i) $x \in \mathcal{C}_e$;
(ii) $x = Pe$ for some band projection P;
(iii) $x = P_ae$ for some $a \in X$;
(iv) $x = P_xe$.

7.3.9 Exercise. Let X be a vector lattice with PPP, $e \in X_+$. If $x \in \mathcal{C}_e$ and $P \in \mathcal{P}(X)$ then $Px \in \mathcal{C}_e$.

7.3.10 Exercise. Let X have PPP and $e \in X_+$. Then the map $P \mapsto Pe$ from $\mathcal{P}(X)$ to $\mathcal{C}_e$ is onto.

7.3.11 Proposition. *Let X be a vector lattice with PPP and $e \in X_+$ a weak unit. Then the map $P \mapsto Pe$ is a Boolean algebra isomorphism from $\mathcal{P}(X)$ onto $\mathcal{C}_e$.*

Proof. By Proposition 7.3.7 and Exercise 7.3.10, this map is a surjective Boolean algebra homomorphism. It is left to show that it is one-to-one. By Exercise 5.2.19, if $P \in \mathcal{P}(X)$ then $\operatorname{Range} P = P(X) = P(B_e) = B_{Pe}$. It follows that if $P_1 e = P_2 e$ for some $P_1, P_2 \in \mathcal{P}(X)$ then $\operatorname{Range} P_1 = \operatorname{Range} P_2$, hence $P_1 = P_2$. $\square$

7.3.12 Corollary. *If X has PPP and e is a weak unit then the Boolean algebras $\mathcal{B}(X)$ of projection bands in X, $\mathcal{P}(X)$ of band projections on X, and $\mathcal{C}_e$ of components of e are isomorphic.*

7.3.13 Exercise. If X has PPP and admits a weak unit then every band projection is principal.

7.3.14. Let X be an Archimedean vector lattice and $e \in X_+$. We have repeatedly used $C(K)$-Representation Theorem 4.1.9 to identify I_e with a dense sublattice of $C(K)$ for some compact space, with e corresponding to $\mathbb{1}$. By Proposition 7.2.5, we may identify $\mathcal{C}_e$ with $\mathcal{C}_{\mathbb{1}}$. That is, we identify components of e and characteristic functions of clopen sets in K. Furthermore, we may identify $\operatorname{span} \mathcal{C}_e$ with $\operatorname{Clop}(K)$-simple functions in $C(K)$.

We now add band projections to this picture. Let $P \in \mathcal{P}(X)$. It follows from $0 \leqslant P \leqslant I$ that $P(I_e) \subseteq I_e$ and the restriction $P_{|I_e}$ is bounded with respect to $\|\cdot\|_e$. It follows that $P_{|I_e}$ extends to a bounded operator on $C(K)$; denote it $\widetilde{P}$. By continuity, $\widetilde{P}$ still satisfies $0 \leqslant \widetilde{P} \leqslant I$ and $\widetilde{P}^2 = \widetilde{P}$, hence $\widetilde{P}$ is again a band projection. Therefore, by Proposition 5.2.4, there exists a clopen set U in K such that $\widetilde{P} f = f \cdot \mathbb{1}_U$ for all $f \in C(K)$. Thus, every band projections in X induces an operator of the form $f \mapsto f \cdot \mathbb{1}_U$ in $C(K)$ for some $U \in \operatorname{Clop}(K)$. This leads to a map from $\mathcal{P}(X)$ to $\operatorname{Clop}(K)$.

Suppose that, in addition, X has PPP and e is a weak unit. Then by Proposition 7.2.13, K is the Stone space of $\mathcal{C}_e$; in particular, it is totally disconnected. Proposition 7.3.11 says that the map from $\mathcal{P}(X)$ to $\operatorname{Clop}(K)$ described above is a Boolean algebra isomorphism. So we may identify band projections on X with operators of the form $f \mapsto f \cdot \mathbb{1}_U$ with $U \in \operatorname{Clop}(K)$. By Exercise 7.3.13, every projection band in $\mathcal{P}(X)$ is principal. Let $a \in X$, then the principal band P_a in corresponds to the operator $f \mapsto f \cdot \mathbb{1}_U$ in $C(K)$ for some $U \in \operatorname{Clop}(K)$. What is the relationship between a and U? Put $x = P_a e$. Then $x \in \mathcal{C}_e$ by Exercise 7.3.6, hence $x = P_x e$ by Exercise 7.3.8. It follows that $P_a = P_x$ by Theorem 7.3.11. Being a component of e, x corresponds to $\mathbb{1}_U$ for a clopen set U in K, and P_x (and, therefore, P_a) agrees with the operator $f \mapsto f \cdot \mathbb{1}_U$. We say that U is the ***generalized support*** of a.

7.3.15 Exercise. Suppose that X has PPP, $e \in X_+$ is a weak unit, and $C(K)$ be as above.

(i) If $a \in I_e$ then the generalized support of a is $\overline{\{a \neq 0\}}$, when we view a as an element of $C(K)$;

(ii) If $a \in X$ then the generalized support of a is $\overline{\{|a| \wedge e \neq 0\}}$, where we view $|a| \wedge e$ as an element of $C(K)$.

7.3.16 Exercise. Let X be an Archimedean vector lattice, let $a \in X_+$ such that P_a exists. Let $\widetilde{P}_a$ stand for the principal projection band for a in X^δ. Then $\widetilde{P}_a$ agrees with P_a on X.

7.3.17 Example. The following example demonstrates a typical application of techniques developed in this section. Say, we would like to prove that the identity

$$P_{(x-y)^+}x + (I - P_{(x-y)^+})y = x \vee y$$

holds in every Archimedean vector lattice X, provided that $P_{(x-y)^+}$ exists. Replacing X with X^δ, and using Exercise 7.3.16, we may assume WLOG that X is order complete; in particular, it is uniformly complete and has PPP. Fix some $e \in X_+$ such that $x, y \in I_e$; say, $e = |x| \vee |y|$. Replacing X with I_e and using Exercise 5.2.20, we may assume WLOG that e is a strong unit in X. Realize X as a $C(K)$ space for some compact K. It now suffices to prove that

$$x \cdot \mathbb{1}_U + y \cdot (\mathbb{1} - \mathbb{1}_U) = x \vee y, \tag{7.2}$$

in $C(K)$, where $U = \overline{\{(x-y)^+ > 0\}}$. Let $t \in K$. If $(x-y)^+(t) > 0$ then $x(t) > y(t)$ and $t \in U$; a direct verification shows that (7.2) holds true at t. Since the functions involved are continuous, it follows that it holds true for all $t \in U$. If $t \notin U$ then $(x-y)^+(t) = 0$, so that $x(t) \leqslant y(t)$; it follows again that (7.2) holds true at t. Therefore, (7.2) holds true for all $t \in K$.

7.4 Universal completion

Throughout this section, X wll stand for an Archimedean vector lattice. We start with a couple of motivational examples.

7.4.1 Example. Let $X = \ell_p$ with $0 < p \leqslant \infty$ or c_0. Then X is contained in the vector lattice $\mathbb{R}^{\mathbb{N}}$ of all real sequences as an order dense sublattice. The same works when X is any sequence space. While $\mathbb{R}^{\mathbb{N}}$ has no natural norm, it may still be useful to have this huge (but not too huge as X is order dense in it) "ambient" space. In many proofs, we first construct some real sequence as a member of $\mathbb{R}^{\mathbb{N}}$ and only later show that it actually belongs to X.

7.4.2 Example. Let $X = L_p(\Omega, \mathcal{F}, \mu)$ for some measure space $(\Omega, \mathcal{F}, \mu)$ and $0 \leqslant p \leqslant \infty$. Then X is an order dense sublattice of the vector lattice $L_0(\mu) = L_0(\Omega, \mathcal{F}, \mu)$ of all (equivalence classes of) measurable functions. Again, while $L_0(\mu)$ lacks a norm, it is often useful in studies of $L_p(\mu)$ to have access to functions that are measurable but not necessarily integrable. One may think of $L_0(\mu)$ as a "huge" ambient space. In this construction, $L_p(\mu)$ may be replaced with Orlicz spaces, Lorentz spaces, or many other spaces of measurable functions.

This examples are special cases of a more general phenomenon: every vector lattice X embeds as an order dense sublattice into some "huge" vector lattice, which may be viewed as a completion of X. We now pass to formal definitions.

A vector lattice X is said to be ***laterally complete*** if every disjoint set in X_+ has a supremum. We say that X is ***universally complete*** if it is both order complete and laterally complete. In this section, we will focus on universally complete spaces; we refer the reader to [AB03] for a thorough study of laterally complete spaces. We will later provide an example of a vector lattice that is laterally complete but not universally complete; see Example 7.6.4.

7.4.3 Exercise. A vector lattice X is universally complete iff it is order complete and every disjoint set in X_+ has an upper bound.

It is easy to see that $\mathbb{R}^{\mathbb{N}}$ and, more generally, $\mathbb{R}^A$ for every set A, is universally complete.

7.4.4 Exercise. If μ is a σ-finite measure than $L_0(\mu)$ is universally complete.

7.4.5 Exercise. No infinite-dimensional Banach lattice is universally complete.

7.4.6 Exercise. Every universally complete vector lattice has a weak unit.

If X embeds as an order dense sublattice into a universally complete vector lattice Y, we say that Y is a ***universal completion*** of X. We write $Y = X^u$.

7.4.7 Theorem (Maeda–Ogasawara)**.** *Every Archimedean vector lattice X admits a universal completion, which is unique up to a lattice isomorphism preserving X.*

The rest of this section will be devoted to proving this theorem: existence of a universal completion will be proved in Theorem 7.4.17, while its uniqueness will be established on page 220. Along the way, we will also discover a few useful facts about universal completions.

7.4.1 Existence and $C^\infty(K)$ representation

We will prove the existence of a universal completion by explicitly constructing it as a certain function space. In many applications, it is sufficient to know that X^u exists, while its specific construction is not important, just as a specific construction of real numbers from rational numbers is not important for most applications. However, there are situations where having a functional representation of X^u is helpful. It means, in particular, that every Archimedean vector lattice may be represented as a function space (though, the functions take values in $\overline{\mathbb{R}} = [-\infty, \infty]$).

Here is a brief overview of the construction. First, we observe that X embeds into its order completion X^δ, hence we may assume WLOG that X is order complete. Second, by embedding X into a product of vector lattices with weak units as in Corollary 5.3.3, we reduce the theorem to the case when X has a weak unit, say e. By $C(K)$-representation Theorem, we represent the principal ideal I_e as $C(K)$ for some extremally disconnected compact space K. The last (and the most difficult) part is that we can identify X with an order dense sublattice of the universally complete vector lattice $C^\infty(K)$ consisting of continuous functions $f\colon K \to \overline{\mathbb{R}}$ that are finite "almost everywhere".

We start by introducing $C^\infty(K)$ spaces and investigating their properties. Let K be an extremally disconnected compact space. We write $C^\infty(K)$ for the space of continuous functions from K to $\overline{\mathbb{R}}$ such that $f^{-1}\big(\{-\infty, \infty\}\big)$ is nowhere dense. Since this set is closed, the condition is equivalent to f being finite on an open dense set. We equip $C^\infty(K)$ with the point-wise order.

There is a simple but very useful observation that allows to reduce extended real functions to functions with values in $[-1, 1]$. We write $C(K, \overline{\mathbb{R}})$ for the set of all continuous functions from K to $\overline{\mathbb{R}}$, ordered point-wise. Note that $\overline{\mathbb{R}}$ is topologically and order isomorphic to $[-1, 1]$: let $\varphi\colon [-1, 1] \to \overline{\mathbb{R}}$ be any strictly increasing continuous bijection. For example, $\varphi(t) = \tan \frac{\pi t}{2}$ extended to the endpoints in the natural way, would do the job. Then φ is a homeomorphism and an order isomorphism. Then the composition operator $T\colon f \mapsto f \circ \varphi$ is a bijection between $C(K, \overline{\mathbb{R}})$ and $C\big(K, [-1, 1]\big)$. Clearly, T is an order isomorphism in the sense that $f \leqslant g$ iff $Tf \leqslant Tg$, hence $C(K, \overline{\mathbb{R}})$ and $C\big(K, [-1, 1]\big)$ are isomorphic as ordered spaces.

Since we assumed that K is extremally disconnected, $C(K)$ is order complete by Theorem 2.1.29. It may be easily deduced that $C\big(K, [-1, 1]\big)$ is order complete (as a partially ordered set), hence so is $C(K, \overline{\mathbb{R}})$.

The restriction of T to functions with finite values only yields an order isomorphism between $C(K)$ and $C\big(K, (-1, 1)\big)$. Furthermore, the restriction of T to $C^\infty(K)$ yields an order isomorphism between $C^\infty(K)$ and the set of all continuous functions from K to $[-1,1]$ for which the set $\{f=\pm 1\}$ is nowhere dense; we (tentatively) denote this set by $C_1^\infty(K)$.

For a general compact space K, the set $C^\infty(K)$ need not have a good vector space structure. While scalar multiplication does not cause any problems, how do we define $f+g$ at a point t where $f(t)=\infty$ and $g(t)=-\infty$? For example, let $K=[-1,1]$ and put

$$f(t)=\begin{cases}-\frac{1}{t} & \text{if } t<0,\\ \frac{1}{t^2} & \text{if } t>0,\\ \infty & \text{if } t=0\end{cases}$$

and $g(t)=f(-t)$. Then $f,g\in C^\infty[-1,1]$. However,

$$\lim_{t\to 0^-}(f-g)(t)=-\infty,\quad\text{while}\quad \lim_{t\to 0^+}(f-g)(t)=\infty,$$

so that there is no way to define $f-g$ at zero so that the resulting function is in $C^\infty[-1,1]$! Fortunately, the assumption that K is extremally disconnected saves the day. Here is the critical lemma:

7.4.8 Lemma. *Suppose that K is extremally disconnected and $f\colon U\to\mathbb{R}$ is a continuous function on an open dense subset U of K. Then f extends uniquely to a function in $C^\infty(K)$.*

Proof. By the preceding discussion, we may replace $C^\infty(K)$ with $C_1^\infty(K)$. So suppose that $f\colon U\to(-1,1)$. Let

$$G=\Big\{g\in C\big(K,[-1,1]\big)\;:\;\forall t\in U\quad g(t)\geqslant f(t)\Big\}.$$

Since $C(K)$ is order complete by Theorem 2.1.29 and $G\geqslant -\mathbb{1}$, $h:=\inf G$ exists in $C(K)$. Clearly, h takes values in $[-1,1]$, hence $h\in C\big(K,[-1,1]\big)$.

We will now show that h extends f. Fix $t\in U$. Since K is totally disconnected, we can find a clopen set V such that $t\in V\subseteq U$. Put $g=f\cdot\mathbb{1}_V+\mathbb{1}_{K\setminus V}$; then $g\in G$ and, therefore, $h\leqslant g$; it follows that $h(t)\leqslant g(t)=f(t)$. On the other hand, for every $g\in G$ we have $g\geqslant f\cdot\mathbb{1}_V$, so that $h\geqslant f\cdot\mathbb{1}_V$ and, therefore, $h(t)\geqslant f(t)$.

This proves that h extends f. Since U is open and dense, the extension is unique and the set $\{h=\pm 1\}$ is nowhere dense. $\square$

Suppose that K is extremally disconnected and $f,g\in C^\infty(K)$. Let U be the set on which both f and g are finite, i.e., U is the complement of $f^{-1}\big(\{-\infty,\infty\}\big)\cup g^{-1}\big(\{-\infty,\infty\}\big)$. Clearly, U is open and dense, and $f+g$ is defined, finite, and continuous on U. By the lemma, it extends uniquely to a function in $C^\infty(K)$. Naturally, we denote this function by $f+g$. Equivalently, we define $f+g$ to be the unique function h in $C^\infty(K)$ such that h agrees with $f+g$ on an open dense set where f and g are both finite. It is now a straightforward verification that $C^\infty(K)$ is a vector lattice.

Again, suppose that K is extremally disconnected compact space.

7.4.9 Exercise. $C(K)$ is an order dense ideal in $C^\infty(K)$.

7.4.10 Exercise. $C^\infty(K)$ is order complete.

Proof. Recall that $C^\infty(K)$ is order isomorphic to $C_1^\infty(K)$. Being a solid subset of an order complete vector lattice $C(K)$, $C_1^\infty(K)$ is order complete, hence so is $C^\infty(K)$. □

We proved in Proposition 5.2.4 that projection bands in $C(K)$ correspond to clopen sets in K. We now extend this characterization to $C^\infty(K)$:

7.4.11 Proposition. *Bands in $C^\infty(K)$ are projection bands; they are precisely the sets of the form $\{f \in C^\infty(K) : \operatorname{supp} f \subseteq U\}$, where U is a clopen set in K. The corresponding band projection is given by $P_U f = f \cdot \mathbb{1}_U$.*

Proof. Since $C^\infty(K)$ is order complete, every band in it is a projection band.

Let U be a clopen subset of K. Then $\mathbb{1}_U$ is continuous. Let $B_U = \{f \in C^\infty(K) : \operatorname{supp} f \subseteq U\}$. It is easy to see that one may identify B_U with $C^\infty(U)$. Clearly, $C^\infty(K) = B_U \oplus B_{K\setminus U}$. It follows from Corollary 5.2.8 that B_U is a projection band. It is easy to see that the band projection P_U is given by $P_U f = f \cdot \mathbb{1}_U$.

Conversely, let B be a band in $C^\infty(K)$. Since $C(K)$ is order dense in $C^\infty(K)$, $B \cap C(K)$ is a band in $C(K)$ by Exercise 5.2.20(iii). By Proposition 5.2.4, $B \cap C(K)$ consists of all functions whose support is contained in some clopen set U. If $f \in B$ then $f \wedge \mathbb{1} \in B \cap C(K)$, hence $\operatorname{supp}(f \wedge \mathbb{1}) \subseteq U$ and, therefore, $f \in B_U$. This yields $B \subseteq B_U$. To prove the converse inclusion, observe that if $f \in B^d$ then $f \perp B \cap C(K)$, hence $f \perp J_U$. It follows that $f \perp \mathbb{1}_U$, so that $\operatorname{supp} f \cap U = \varnothing$; it follows that $f \in B_U^d$. We conclude that $B^d \subseteq B_U^d$ and, therefore, $B \supseteq B_U$. □

We will need the following technical lemma which says that a subset of $C^\infty(K)$ is order bounded iff it is “almost locally bounded”:

7.4.12 Lemma. *Let A be a non-empty subset of $C^\infty(K)_+$ for some extremally disconnected K. A is bounded above iff for every non-empty clopen subset U of K there exists a non-empty clopen $V \subseteq U$ and a real $r \geqslant 0$ such that $f(t) \leqslant r$ for all $f \in A$ and $t \in V$.*

Proof. Suppose that A is order bounded, so that $A \leqslant h$ for some $h \in C^\infty(K)$. Let U be a non-empty clopen subset of K. Since the set $\{h = \pm\infty\}$ is nowhere dense, there exists $s \in U$ such that $h(s)$ is finite. Since h is continuous and K is totally disconnected, we can then find a clopen neighborhood V of x such that h is bounded on V for by some positive real r. Replacing V with $V \cap U$, we may assume that $V \subseteq U$. For every $f \in A$ and $t \in V$ we now have $f(t) \leqslant h(t) \leqslant r$.

Suppose now that A satisfies the condition; we need to show that it is order bounded. Since $C(K, \overline{\mathbb{R}})$ is order complete, $h := \sup A$ in $C(K, \overline{\mathbb{R}})$ exists. It suffices to show that $h \in C^\infty(K)$. Suppose not. Then there exists a non-empty clopen set U such that h equals infinity on U. By assumption, we can find a non-empty clopen $V \subseteq U$ and $r > 0$ such that $f(t) \leqslant r$ for all $f \in A$ and $t \in V$. Put

$$g(t) = \begin{cases} h(t) & \text{when } t \notin V \\ r & \text{when } t \in V. \end{cases}$$

Note that g is continuous because V is clopen. Then $f \leqslant g$ for all $f \in A$, hence $h \leqslant g$, which contradicts h being infinite on V. □

7.4.13 Theorem. *If K is extremally disconnected then $C^\infty(K)$ is a universally complete vector lattice.*

Proof. We have already shown that $C^\infty(K)$ is an order complete vector lattice. Let D be a disjoint subset of $C^\infty(K)_+$; we need to show that D is bounded above. We will use the preceding lemma. Fix a non-empty clopen set U. If all functions in D vanish on U then we take $V = U$ and $r = 0$. Suppose now that there exists $s \in U$ and $f \in D$ such that $f(s) > 0$. It follows that there exists $t \in U$ such that $0 < f(t) < \infty$. We can then find a clopen set V such that $t \in V \subseteq U$ and f positive and bounded on V. It also follows that all the other functions in D vanish on V, so the condition in the lemma is satisfied. □

Recall that if X is an order complete vector lattice and $e \in X_+$ then the $C(K)$-Representation Theorem 4.1.9 guarantees that there is a lattice isomorphism from I_e onto $C(K)$ for some extremally disconnected compact space K.

7.4.14 Theorem. *Let X be an order complete vector lattice with a weak unit e, let $T\colon I_e \to C(K)$ be its $C(K)$-representation. Then T extends to a lattice isomorphic embedding $S\colon X \to C^\infty(K)$.*

Proof. Since X is order complete, T is onto by Theorem 4.1.9; since I_e is order complete by Proposition 3.3.18, so is $C(K)$; it follows that K is extremally disconnected by

Theorem 2.1.29. Let $x \in X^+$. Then $x = \sup_n (x \wedge ne)$ in X by Exercise 5.1.13. It would be natural to define Sx as $\sup_n T(x \wedge ne)$ in $C^\infty(K)$. Note that $T(x \wedge ne)$ is defined as $x \wedge ne \in I_e$. But why does the supremum exists?

Since $C^\infty(K)$ is order complete, it suffices to show that the sequence $\bigl(T(x \wedge ne)\bigr)$ is order bounded in $C^\infty(K)$. We will use Lemma 7.4.12 to establish this. Let U be a clopen subset of K. Being order complete, X has PPP, so, by 7.3.14, there is a band projection P on X which corresponds to U in the sense that $TPz = (Tz) \cdot \mathbb{1}_U$ for every $z \in I_e$. In particular, $TPe = \mathbb{1}_U$. By the Archimedean Property, there exits $m \in \mathbb{N}$ such that $mPe \nleqslant Px$. Put $h \coloneqq (mPe - Px)^+$; then $h > 0$. Let Q be the band projection for h; let V be the corresponding set in $\mathrm{Clop}(K)$. It follows from $h > 0$ that $Q > 0$ and, therefore, V is non-empty. It follows from $h \leqslant mPe$ that $Q \leqslant P$ and, therefore, $V \subseteq U$. We have

$$Q(me - x) = QP(me - x) = Q(mPe - Px) = h;$$

the last identity follows from the fact that $P_{a^+}a = a^+$ for every a in a vector lattice. This yields $Qx \leqslant mQe$, hence $Q(x \wedge ne) \leqslant mQe$ for every n. Applying T to both sides, we get $T(x \wedge ne) \cdot \mathbb{1}_V \leqslant m\mathbb{1}_V$. This means that assumption in Lemma 7.4.12 is satisfied, and, therefore, the sequence $\bigl(T(x \wedge ne)\bigr)$ is order bounded in $C^\infty(K)$.

We put $\tau(x) = \sup_n T(x \wedge ne)$ in $C^\infty(K)$. We will use Extension Lemma 6.2.1 to show that it extends to a lattice homomorphism from X to $C^\infty(K)$; we just need to verify that τ is additive. Indeed, let $x, y \in X_+$. It follows from $(x+y)\wedge ne \leqslant x\wedge ne+y\wedge ne$ that $\tau(x+y) \leqslant \tau(x)+\tau(y)$. On the other hand, $(x+y)\wedge(n+m)e \geqslant x\wedge ne+y\wedge me$ yields $\tau(x+y) \geqslant \tau(x)+\tau(y)$.

Let $S\colon X \to C^\infty(K)$ be the extension of τ as in Extension Lemma 6.2.1. In particular, $Sx = \tau(x) = Tx$ for every $x \in X_+$; it follows that S extends T.

Suppose $x \wedge y = 0$ in X. Then $(x \wedge ne) \wedge (y \wedge ne) = 0$ and, hence, $T(x \wedge ne) \wedge T(y \wedge ne) = 0$ as T is a lattice homomorphism. Continuity of lattice operations yields $Sx \perp Sy$. It now follows from Exercise 1.6.10 that S is a lattice homomorphism.

It is left to show that S is one-to-one. For every $x \in X_+$ and $n \in \mathbb{N}$, we have

$$T(x \wedge ne) = S(x \wedge ne) = (Sx) \wedge n\mathbb{1}.$$

Hence, if $Sx = Sy$ for some $x, y \in X_+$, then $T(x \wedge ne) = T(y \wedge ne)$ and, since T is one-to-one, $x \wedge ne = y \wedge ne$ for every $n \in \mathbb{N}$; it follows that $x = y$. Finally, $Sx = 0$ for some $x \in X$ then $Sx^+ = Sx^-$ and, therefore, $x^+ = x^- = 0$, so that $x = 0$. □

7.4.15 Theorem. *Every Archimedean vector lattice embeds as an order dense sublattice into $C^\infty(K)$ for some extremally disconnected compact space K.*

Proof. Let X be an Archimedean vector lattice. By Corollary 5.3.3, X embeds as an order dense sublattice into a direct product $Y=\prod_{\alpha\in\Lambda}X_\alpha$ of vector lattices where each X_α has a weak units, say e_α. It is easy to see that $e=(e_\alpha)_{\alpha\in\Lambda}$ is a weak unit in Y.

Y embeds as an order dense sublattice into its order completion Y^δ; e is still a weak unit in Y^δ; see Exercise 3.2.11. By the preceding theorem, Y^δ embeds as an order dense sublattice into $C^\infty(K)$ for some extremally disconnected compact K. □

7.4.16 Remark. If X has a weak unit e, then we can skip the first part of the proof. In this case, e corresponds to $\mathbb{1}$ in $C^\infty(K)$.

Combining the preceding theorem with Theorem 7.4.13, we get the existence part of Maeda-Ogasawara Theorem 7.4.7:

7.4.17 Theorem. *Every Archimedean vector lattice X admits a universal completion, which is lattice isomorphic to $C^\infty(K)$ for some extremally disconnected compact topological space K. If X has a weak unit e, the representation may be chosen so that e becomes $\mathbb{1}$.*

We showed in 7.3.14 that $C(K)$ representations "respect" band projections. We will now extend this to $C^\infty(K)$-representations as follows. The following proposition provides a characterization of bands, projection bands, and band projections in X: bands in X correspond to clopen sets in K, and band projections correspond to multiplications by characteristic functions.

7.4.18 Theorem. *Let X be an order dense sublattice of $C^\infty(K)$, where K is extremally disconnected (as in Theorem 7.4.15). A subset B of X is a band iff $B=B_U$ where $B_U=\{f\in X\ :\ \operatorname{supp} f\subseteq U\}$ for some clopen set U in K. In this case, $U=\overline{\bigcup_{f\in B}\operatorname{supp} f}$. Furthermore, B is a projection band iff $f\cdot\mathbb{1}_U\in X$ for every $f\in X$; in this case, the map $f\mapsto f\cdot\mathbb{1}_U$ is the corresponding projection band on X and $B_U^d=B_{K\setminus U}$.*

Proof. A subset B of X is a band in X iff $B=B_0\cap X$ for some band B_0 in $C^\infty(K)$; this follows from Exercise 5.2.20(iii). It now follows from Proposition 7.4.11 that $B=B_U$ for some clopen set U is a K.

It is clear that if $f\in B_U$ then $\operatorname{supp} f\subseteq U$ and, therefore, $\overline{\bigcup_{f\in B}\operatorname{supp} f}\subseteq U$. For the sake of contradiction, assume that the two sets are different. Then $V\coloneqq U\setminus\overline{\bigcup_{f\in B}\operatorname{supp} f}$ is a non-empty clopen set. Since X is order dense, there exists $g\in X$ with $0<g\leqslant\mathbb{1}_V$. But then $\operatorname{supp} g\subseteq U$, so that $g\in B_U$, which is a contradiction.

Suppose now that $f\cdot\mathbb{1}_U\in X$ for every $f\in X$. Then $f\cdot\mathbb{1}_{K\setminus U}=f-f\cdot\mathbb{1}_U$ is also in X, hence in $B_{K\setminus U}$. It follows that $X=B_U\oplus B_{K\setminus U}$, hence B_U is a projection band.

In this case, it is clear that the map $f \mapsto f \cdot \mathbb{1}_U$ is the corresponding projection band on X.

Conversely, suppose that B_U is a projection band. Clearly, $B_{K \setminus U} \subseteq B_U^d$. To prove the opposite inclusion, suppose that there exists $g \in B_U^d$ but $g \notin B_{K \setminus U}$. WLOG, $g \geqslant 0$. It follows that $g \wedge \mathbb{1}_U > 0$ in $C^\infty(K)$. By order density of X, there exists $h \in X$ with $0 < h \leqslant g \wedge \mathbb{1}_U$. It follows from $h \leqslant g$ that $h \in B_U^d$; while $h \leqslant \mathbb{1}_U$ yields that $h \in B_U$; a contradiction. This proves that $B_U^d = B_{K \setminus U}$. It follows that for every $f \in X$, we can write $f = g + h$ for some $g \in B_U$ and $h \in B_{K \setminus U}$; we conclude that $g = f \cdot \mathbb{1}_U \in X$. $\square$

7.4.19 Theorem. *Let X be an Archimedean vector lattice. The set $\mathcal{B}(X)$ of bands in X, ordered by inclusion, forms a complete Boolean algebra with Boolean operations $B_1 \wedge B_2 = B_1 \cap B_2$, $B_1 \vee B_2 = (B_1 \cup B_2)^{dd}$, and $\overline{B} = B^d$. Furthermore, if X is represented as an order dense ideal in $C^\infty(K)$ as in Theorem 7.4.15, then $\mathcal{B}(X)$ is isomorphic (as a Boolean algebra) to* $\mathrm{Clop}(K)$.

Proof. We represent X as an order dense ideal in $C^\infty(K)$ as in Theorem 7.4.15. The map that sends $U \in \mathrm{Clop}(K)$ to B_U in Theorem 7.4.18 is a bijection, it preserves order both ways, sends $\varnothing$ to $\{0\}$, K to X, and complements to complements. It follows that $\mathcal{B}(X)$ is a Boolean algebra, isomorphic to $\mathrm{Clop}(K)$. Since $\mathrm{Clop}(K)$ is order complete by Proposition 7.1.11, so is $\mathcal{B}(X)$. The identities $B_1 \wedge B_2 = B_1 \cap B_2$ and $B_1 \vee B_2 = (B_1 \cup B_2)^{dd}$ are straightforward. $\square$

It follows that K is the Stone space for the Boolean algebra $\mathcal{B}(X)$ as in Stone Theorem 7.1.7. In particular, K is uniquely determined by X. We call K the ***Stone space*** of X. Thus, every Archimedean vector space can be represented as an order dense sublattice in $C^\infty(K)$, where K is the Stone space of X or, equivalently, the Stone space of $\mathcal{B}(X)$.

7.4.20 Exercise. Let X be an order dense sublattice of $C^\infty(K)$, where K is extremally disconnected (as in Theorem 7.4.15) and $a \in X_+$. Let $V = \overline{\operatorname{supp} a}$. Then the principal band B_a generated by a in X equals $\{f \in X : \operatorname{supp} f \subseteq V\}$. If, in addition, B_a is a projection band then the corresponding band projection is given by $P_a f = f \cdot \mathbb{1}_U$ for every $f \in X$.

Recall that by "function calculus", we mean the continuous positively homogeneous function calculus constructed in Section 4.4. We will show that, under the assumptions of Theorem 7.4.17, the function calculus in X is computed "almost everywhere pointwise".

7.4.21 Proposition. *Let X be a uniformly complete Archimedean vector lattice; let $T\colon X \to C^\infty(K)$ be the lattice isomorphic embedding as in Theorem 7.4.17. Let $x_1,\dots,x_n \in X$ and f a continuous positively homogeneous function, and $y = f(x_1,\dots,x_n)$. Then Ty is the unique continuous extension to K of the function*

$$t \mapsto f\bigl(x_1(t),\dots,x_n(t)\bigr) \tag{7.3}$$

for all $t \in K$ for which $x_1(t),\dots,x_n(t)$ are all finite.

Proof. First, we will show that the function calculus in $C^\infty(K)$ itself is given by (7.3). Being universally complete, $C^\infty(K)$ is order complete and, therefore, uniformly complete. Hence, it has a function calculus. Let $x_1,\dots,x_n \in C^\infty(K)$. Let U be the subset of K where all the x_i's are finite; clearly, U is an open and dense. Let $f \in H_n$, i.e., $f\colon \mathbb{R}^n \to \mathbb{R}$ is a continuous positively homogeneous function. For every $t \in U$, the function $t \mapsto f\bigl(x_1(t),\dots,x_n(t)\bigr)$ is well-defined and continuous. By Lemma 7.4.8, this function extends uniquely to an element of $C^\infty(K)$; let's denote it by Vf. It is easy to see that $V\colon H_n \to C^\infty(K)$ is a lattice homomorphism. It follows from Theorem 4.4.3 that V is the function calculus, because $V\delta_i = x_i$ as $i = 1,\dots,n$.

Suppose now that $x_1,\dots,x_n \in X$. Let $W\colon H_n \to X$ be the function calculus for X and $V\colon H_n \to C^\infty(K)$ be the function calculus for $C^\infty(K)$, both corresponding to $x_1,\dots,x_n$. By uniqueness of function calculus in Theorem 4.4.3, we conclude that $V = TW$. This means that if $f \in H_n$ and $y = f(x_1,\dots,x_n)$ then $Ty = TWf = Vf$, which implies the required conclusion. □

7.4.2 Uniqueness

Throughout this subsection, X will be an Archimedean vector lattice. In the previous subsection, we proved that X admits a universal completion. In this subsection, we prove its uniqueness and investigate its properties. Here is the idea. Let Y be some universal completion of X and $y \in Y_+$. Since X is order dense in Y, we have $y = \sup[0,y] \cap X$ by Proposition 3.2.15, where the supremum is computed in Y, but the set $[0,y] \cap X$ is a subset of X. Thus, every element of Y_+ may be represented as the supremum of a subset of X which is bounded in Y. This kind of sets will be called *dominable*, and they will play a major role in our study. While being dominable really just means being bounded in a universal completion, using this as a definition is not practical until we prove uniqueness of universal completion. Fortunately, Fremlin discovered that a dominable set may be defined intrinsically in terms of X only, and we will adopt this definition:

7.4.22 Definition. A subset A of X_+ is said to be ***dominable*** if for every $u > 0$ there exists $0 < v \leqslant u$ and a real $r > 0$ such that $(ru - a)^+ \geqslant v$ for every $a \in A$.

Essentially, it says that a scalar multiple of u dominates A "on some subset"; it is somewhat motivated by Lemma 7.4.12.

7.4.23 Exercise. A subset A of X is dominable iff for every $u > 0$ there exists $0 < v \leqslant u$ and a real $r > 0$ such that $(a - ru)^+ \perp v$ for all $a \in A$.

Proof. Suppose A is dominable; let $u > 0$. Let v and r be as in Definition 7.4.22. Let $a \in A$. It follows from $(a - ru)^+ \perp (ru - a)^+ \geqslant v$ that $(a - ru)^+ \perp v$. □

To prove the converse, let $u > 0$. Find $0 < v \leqslant u$ and a real $r > 0$ such that $(a - ru)^+ \perp v$ for all $a \in A$. Then $\bigl((r+1)u - a\bigr)^+ \geqslant v$. for every $a \in A$. The latter can be easily verified using a $C(K)$ representation for a principal ideal containing u and a.

7.4.24 Exercise. A subset of a dominable set is dominable. Every order bounded set is dominable (this is, essentially, the Archimedean property).

7.4.25 Exercise. Let X be an order dense sublattice in a vector lattice Y and $A \subseteq X_+$. Then A is dominable in X iff it is dominable in Y.

In the case when X has PPP, the definition of a dominable set is even closer to the condition in Lemma 7.4.12 (and this is not an accident).

7.4.26 Proposition. *Suppose that X has PPP and $A \subseteq X_+$. If A is dominable then for every $0 < u \in X$ then there is a non-zero component w of u and a number $r > 0$ such that $P_w(A) \leqslant rw$.*

Proof. Fix $0 < u \in X$; let v and r be as in Definition 7.4.22. Put $w = P_v u$. Clearly, w is a component of u and $P_w = P_v$. Represent X as in Theorem 7.4.15. By Exercise 7.4.20, P_v is the restriction to X of the multiplication by $\mathbb{1}_V$ in $C^\infty(K)$, where $V = \overline{\{v \neq 0\}}$. Let $a \in A$. Applying P_v to $(ru - a)^+ \geqslant v$, we get

$$(rw - \mathbb{1}_V a)^+ \geqslant v. \tag{7.4}$$

Take $t \in K$. If $v(t) > 0$ then, evaluating (7.4) at t we get $\bigl(rw(t) - a(t)\bigr)^+ > 0$, so that $a(t) < rw(t)$. By continuity, we conclude that $a(t) \leqslant rw(t)$ for all $t \in V$. It follows that $P_w a = P_v a \leqslant rw$. □

We observed in Section 5.3 that every vector lattice X admits a maximal (with respect to inclusion) disjoint collection Λ of positive vectors; if, in addition, X has PPP then $x = \sup_{a \in \Lambda} P_a x$ for every $x \in X_+$. We will now use this to prove the following key fact.

7.4.27 Theorem. *Let $A \subseteq X_+$; let Y be a universal completion of X. Then A is dominable in X iff it is order bounded in Y.*

Proof. If A is order bounded in Y then it is dominable in Y by Exercise 7.4.24 and hence in X by Exercise 7.4.25.

Suppose that A is dominable in X; we need to show that it is order bounded in Y. Here is the idea: A is "locally bounded" in X; by gluing all these disjoint local bounds together in Y, we get a genuine upper bound. First, we show that WLOG we may assume that X has PPP. Indeed, as in 6.10.3, we know that Y contains X^δ, and by Exercise 7.4.25, A is dominable in X^δ. Hence, replacing X with X^δ, we may assume that X has PPP.

Claim: there exists a maximal disjoint collection $\{e_\alpha\}_{\alpha\in\Lambda}$ in X_+ and a collection $\{r_\alpha\}_{\alpha\in\Lambda}$ of positive reals such that $P_{e_\alpha}(A) \leqslant r_\alpha e_\alpha$ for every $\alpha \in \Lambda$.

Indeed, consider the family of all disjoint collections $\{e_\alpha\}_{\alpha\in\Lambda}$ in X_+ such that for every $\alpha \in \Lambda$ there exists a positive real r_α such that $P_{e_\alpha}(A) \leqslant r_\alpha e_\alpha$. Such a family is non-empty by Proposition 7.4.26. Hence, by Zorn's Lemma, it has a maximal element, say Λ. We need to show that Λ is also a maximal disjoint collection.

Suppose not. Then there exists $u > 0$ in X such that $u \perp \Lambda$. Let w be as in Proposition 7.4.26. Then $w \perp \Lambda$, and $P_w(A) \leqslant rw$, which contradicts the maximality of Λ. This proved the *Claim.*

Put $v = \sup_Y\{r_\alpha e_\alpha : \alpha \in \Lambda\}$. For every $a \in A$ and $\lambda \in \Lambda$, we have $P_{e_\alpha} a \leqslant r_\alpha e_\alpha \leqslant v$; it now follows from $a = \sup_Y P_{e_\alpha} a$ that $a \leqslant v$ and, therefore, v is an upper bound of A in Y. $\square$

We are now ready to prove that a universal completion of a vector lattice is unique:

Proof of the uniqeness part of Maeda-Ogasawara Theorem 7.4.7. Let X be an Archimedean vector lattice; let Y and Z be two universally complete vector lattices. Suppose that X is an order dense sublattice in both Y and Z. We need to show that there is a lattice isomorphism between Y and Z preserving X.

Again, using 6.10.3, we may assume WLOG that X is order complete. Hence X is an ideal in both Y and Z by Proposition 3.3.19.

Fix $y \in Y_+$. The set $A \coloneqq [0, y] \cap X$ is a subset of X; $y = \sup_Y A$ by Proposition 3.2.15 because X is order dense in Y. Since A is order bounded in Y, it is dominable in X and, therefore, is order bounded in Z, hence $\sup_Z A$ exists; denote it $\tau(y)$. This defines a map $\tau \colon Y_+ \to Z_+$.

Following the proof of Theorem 6.3.9, one can show that τ is additive. Extension Lemma 6.2.1 guarantees that τ extends to a positive operator $T \colon Y \to Z$. If $y_1 \perp y_2$

in Y_+ then the sets $[0,y_1]\cap X$ and $[0,y_2]\cap X$ are disjoint in X, hence $\tau(y_1)\perp\tau(y_2)$ in Z; it follows from Exercise 1.6.10 that T is a lattice homomorphism.

In order to complete the proof, it suffices to show that τ is a bijection. Define $\sigma\colon Z_+\to Y_+$ symmetrically: for $z\in Z_+$, we put $\sigma(z)=\sup_Y[0,z]\cap X$. Note that while we compute the set $[0,z]\cap X$ in Z, it is, actually, a subset of X. We will show that $\sigma=\tau^{-1}$. Again, let $y\in Y_+$ and put $A=[0,y]\cap X$ and $z\coloneqq\tau(y)=\sup_Z A$. It follows that $A\subseteq[0,z]\cap X$. On the other hand, let $x\in[0,z]\cap X$. Since X is order complete, $\sup_{a\in A}x\wedge a$ exists in X; since X is regular in both Y and Z, the value of this supremum is the same in X, Y, and Z. We then have

$$x=x\wedge z=x\wedge\sup_Z A=\sup_{a\in A}x\wedge a\leqslant y$$

in Y. It follows that $x\leqslant y$ and, therefore, $x\in A$. This yields $[0,z]\cap X=A$, hence $\sigma(z)=\sup_Y A=y$. □

We will write X^u for *the* universal completion of X.

7.4.3 Further properties of X^u

7.4.28 Exercise. X is order complete iff X is an ideal in X^u.

The following corollary says that if X is order dense in Y then $X\subseteq Y\subseteq X^u$ (up to a lattice isomorphism).

7.4.29 Corollary. *Let X be a sublattice of an Archimedean vector lattice Y. If X is order dense in Y then there exists a lattice isomorphic embedding from Y to X^u which preserves X.*

Proof. Since X is order dense in Y and Y is order dense in Y^u, X is order dense in Y^u. By uniqueness of universal completion, there is a lattice isomorphism $T\colon Y^u\to X^u$ which preserves X. The restriction of T to Y provides a required embedding. □

7.4.30. We know from 6.10.3 that X^δ is the least order complete vector lattice containing as an order dense sublattice X. The preceding corollary is a somewhat dual result: is says that X^u is the greatest vector lattice which contains X as an order dense sublattice (but generally not majorizing). In particular, for every order complete vector lattice Y that contains X as an order dense sublattice, we have $X^\delta\subseteq Y\subseteq X^u$ (up to a lattice isomorphism), and the both inclusions are ideals by Propositions 3.3.19. Since X is majorizing in X^δ, it also follows that X^δ is the ideal of X^u generated by X.

We can now clarify the role of $C^\infty(K)$ spaces. Clearly, if X is universally complete then $X=X^u$. Combining this with Theorem 7.4.17, we get the following:

7.4.31 Corollary. *Every universally complete vector lattice X is lattice isomorphic to $C^\infty(K)$ for some extremally disconnected compact space K.*

7.4.32 Exercise. Let $X = L_p(\mu)$ for some σ-finite measure μ and $0 \leqslant p \leqslant \infty$. Then $X^u = L_0(\mu)$.

There is a σ-variant of universal completeness: a vector lattice is ***σ-universally complete*** if it is σ-order complete and every disjoint sequence has a supremum. We refer the reader to [AB03] for a thorough review of σ-universally complete spaces.

We now deduce some useful properties of universal completions. For the rest of the section, X is an Archimedean vector lattice, with X^u represented as $C^\infty(K)$. Recall that X is order complete iff it is an ideal in $C^\infty(K)$. We are going to prove that, in this case, every positive element of X^u can be expressed as the supremum of a disjoint collection in X.

7.4.33 Lemma. *Suppose that X is order complete and $0 < u \in C^\infty(K)$. Then there exists a non-empty clopen subset Q of K with $u \cdot \mathbb{1}_Q \in X$.*

Proof. Find $t_0 \in K$ such that $0 < u(t_0) < \infty$. By continuity of u, we can find a clopen set V and $0 < a < b < \infty$ such that u is between a and b on V. Since X is order dense, there exist $x \in X$ such that $0 < x \leqslant u \cdot \mathbb{1}_V$. Find $t_1 \in K$ such that $x(t_1) > 0$. Again, by continuity of x, x dominates a scalar multiple of $\mathbb{1}_Q$ for some clopen subset Q of V. Since X is an ideal, $\mathbb{1}_Q \in X$. Since $u \cdot \mathbb{1}_Q \leqslant b\mathbb{1}_Q$, we conclude that $u \cdot \mathbb{1}_Q \in X$. □

7.4.34 Proposition. *Suppose that X is order complete and $0 < u \in C^\infty(K)$. Then there exists a family $\mathcal{D}$ of pairwise disjoint clopen sets in K such that $u = \sup\{u \cdot \mathbb{1}_Q : Q \in \mathcal{D}\}$ and $u \cdot \mathbb{1}_Q \in X$ for every $Q \in \mathcal{D}$.*

Proof. The proof is a standard combination of Lemma 7.4.33 and Zorn's Lemma. Let $\mathcal{G}$ be the collection whose members are disjoint families $\mathcal{D}$ of clopen subsets of K such that $u \cdot \mathbb{1}_Q \in X$ for every $A \in \mathcal{D}$. By Lemma 7.4.33, $\mathcal{G}$ is non-empty. We order $\mathcal{G}$ by inclusion. By Zorn's Lemma, $\mathcal{G}$ has a maximal element. Let $\mathcal{D}$ be such a maximal element. We claim that the open set $\bigcup \mathcal{D}$ is dense in K. Indeed, otherwise, the set $K_0 := K \setminus \overline{\bigcup \mathcal{D}}$ is clopen and disjoint with all the sets in $\mathcal{D}$; applying Lemma 7.4.33 to $u \cdot \mathbb{1}_{K_0}$ yields a non-empty clopen set Q with $u \cdot \mathbb{1}_Q \in X$, which contradicts the maximality of $\mathcal{D}$.

It follows from the universal completeness of $C^\infty(K)$ that $w := \sup\{u \cdot \mathbb{1}_Q : Q \in \mathcal{D}\}$ exists in $C^\infty(K)$. On one hand, we clearly have $u \geqslant w$. On the other hand, u is dominated by w on every set from $\mathcal{D}$, hence on $\bigcup \mathcal{D}$. It follows by continuity of u and w that $u \leqslant w$ and, therefore, $u = w$. □

The proposition immediately yields the following:

7.4.35 Theorem. *Suppose that X is order complete and $0 < u \in X^u$. Then $u = \sup A$ for some disjoint subset A of X.*

The following result is somewhat analogous to Exercise 6.9.13.

7.4.36 Theorem (Fremlin)**.** *If $T\colon X \to Y$ is an order continuous lattice homomorphism between vector lattices extends to an order continuous lattice homomorphism $T^u\colon X^u \to Y^u$ given by $T^u v = \sup\bigl(T[0,v] \cap X\bigr)$ for all $v \in X^\delta_+$.*

Proof. By Exercise 6.9.13, we may assume WLOG that X and Y are order complete. Let $u \in X^u_+$. Applying Theorem 7.4.35, we find a disjoint set A in X_+ such that $u = \sup A$ in X^u. Then $T(A)$ is a disjoint set in Y, hence $v \coloneqq \sup T(A)$ exists in Y^u.

For every $x \in X_+$ with $x \leqslant u$, we have $x = x \wedge u = \sup(x \wedge A)$. Exercise 6.5.15 yields $Tx = \sup T(x \wedge A) = \sup Tx \wedge T(A) \leqslant v$. We conclude that $T\bigl([0,u] \cap X\bigr) \leqslant v$. The result now follows from Lemma 6.9.10. □

7.5 Truncated vector lattices

Truncated vector lattices were introduced and investigated in [Ball14, BH22]. Throughout this subsection, X stands for an Archimedean vector lattice. If we fix $a \in X_+$, we can consider the (non-linear) map from X^+ to X^+ defined by $x \mapsto x \wedge a$. This motivates the following definition: a unary operation $x \mapsto x^*$ from X^+ to X^+ is called a ***truncation*** if it satisfies the following two axioms:

(i) $x^* \wedge y = x \wedge y^*$ for all x, y;
(ii) If $(nx)^* = nx$ for all $n \in \mathbb{N}$ then $x = 0$

By a ***truncated vector lattice*** we mean a vector lattice equipped with a truncation.

Clearly, for every fixed $a \in X_+$, the map $x \mapsto x \wedge a$ is a truncation. Such a truncation (and the resulting truncated vector lattice) is said to be ***unital***. Here is an example of a non-unital truncation. Let $X = L_1(\mathbb{R})$; for $x \in X_+$, define $x^* = x \wedge \mathbb{1}$. Note that while $\mathbb{1} \notin X$, we have $\mathbb{1} \in L_0(\mathbb{R}) = X^u$. This motivates the following representation theorem for truncated vector lattices; it shows that every truncated vector lattice "sits" inside a unital one.

7.5.1 Theorem. *Let X be a truncated vector lattice. There exists $e \in X^u_+$ such that $x^* = x \wedge e$ for all $x \in X^+$.*

Proof. Let $x \in X^+$. It follows from $x^* = x^* \wedge x^* = x \wedge x^{**}$ that $x^* \leqslant x, x^{**}$. Applying this to x^*, we get $x^{**} \leqslant x^*$, hence $x^{**} = x^*$, that is, the truncation operation is idempotent. Also, it is monotone because if $x \leqslant y$ in X_+ then it follows from $x^* \leqslant x \leqslant y$ that $x^* = x^* \wedge y = x \wedge y^* \leqslant y^*$.

Let $A = \{x^* : x \in X^+\}$. Clearly, $x \in A$ iff $x = x^*$. We now show next that A is dominable in X. Let $0 < x \in X$. Let $n \in \mathbb{N}$ be the least such that $(nx)^* \neq nx$. Put $v \coloneqq nx - (nx)^*$. Then $v > 0$. We claim that $v \leqslant x$. Indeed, if $n = 1$ then $v = x - x^* \leqslant x$. If $n > 1$ then it follows from $\big((n-1)x\big)^* = (n-1)x$, and $\big((n-1)x\big)^* \leqslant (nx)^*$ that $v \leqslant x + (n-1)x - \big((n-1)x\big)^* = x$. For every $y \in A$, we now have

$$v \leqslant nx - (nx)^* \wedge y = nx - (nx) \wedge y^* = nx - (nx) \wedge y = (nx - y)^+,$$

so that $0 < v \leqslant (nx - y)^+$. This proves that A is dominable.

It follows that A is order bounded in X^u. Since X^u is order complete, $e \coloneqq \sup A$ exists in X^u. We now claim that $x^* = x \wedge e$ for every $x \in X_+$. Indeed, it follows from $x^* \in A$ that $x^* \leqslant e$. Since $x^* \leqslant x$, we get $x^* \leqslant x \wedge e$. On the other hand,

$$x \wedge e = x \wedge \sup\{y^* : y \in X_+\} = \sup\{x \wedge y^* : y \in X_+\} = \sup\{x^* \wedge y : y \in X_+\} \leqslant x^*,$$

where the infinite sup's are evaluated in X^u. It follows that $x \wedge e = x^*$. □

Under the setup of Theorem 7.5.1, suppose that $e \notin X$ (i.e., X is not unital), and let Y be the subspace of X^u spanned by X and e. We claim that Y is a sublattice of X^u. It suffices to show that Y is closed under the $\wedge$ operation. Take two generic elements of Y: $x + \alpha e$ and $y + \beta e$, where $x, y \in X$ and $\alpha, \beta \in \mathbb{R}$. Using distributive laws in X^u, we get

$$(x + \alpha e) \wedge (y + \beta e) = y + \alpha e + (x - y) \wedge \big((\beta - \alpha)e\big).$$

If $\alpha = \beta$, this equals $y + (x-y)^+ + \alpha e$, which is clearly an element of Y. Otherwise, it equals $y + \alpha e + (\beta - \alpha)\big(\frac{x-y}{\beta-\alpha} \wedge e\big) = y + \alpha e + (\beta - \alpha)\big(\frac{x-y}{\beta-\alpha}\big)^*$, which is again an element of Y. Thus, we have just proved the following:

7.5.2 Corollary. *For every non-unital truncated vector lattice X there exists a vector lattice Y and $e \in Y_+$ such that X is a sublattice of Y of co-dimension 1, $e \notin X$, and $x^* = x \wedge e$ for all $x \in X_+$.*

Such a pair (Y, e) is called a ***unitization*** of X.

Let (Y, e) be a unitization of X. We will characterize in terms of $x \in X$ and $r \in \mathbb{R}$ when $x + re \geqslant 0$ in Y. Obviously, if $r = 0$ then $x + re \geqslant 0$ in Y iff $x \geqslant 0$ in X. Suppose that $r > 0$. Then $x + re \geqslant 0$ in Y iff

$$0 = (x + re) \wedge 0 = -x^- + (re) \wedge x^- = -x^- + r\big(e \wedge (\tfrac{x^-}{r})\big) = -x^- + r\big(\tfrac{x^-}{r}\big)^*.$$

Finally, if $r < 0$ then $x + re \geqslant 0$ is impossible because in this case we would have $e \leqslant -\frac{x}{r} \leqslant -\frac{x^+}{r}$, so that $e = \left(-\frac{x^+}{r}\right) \wedge e = \left(-\frac{x^+}{r}\right)^*$, hence $e \in X$, which is a contradiction.

It is clear that, as a vector space, Y may be identified with $X \oplus \mathbb{R}$ via $x + re \mapsto (x, r)$. The preceding paragraph says that if we now use this linear isomorphism to "transfer" the order structure from Y to $X \oplus \mathbb{R}$, the latter becomes vector lattice and a unitization of X. Let's summarize:

7.5.3 Corollary. *Let X be a non-unital truncated vector lattice. Define an order on $X \oplus \mathbb{R}$ as follows: $(x, r) \geqslant 0$ if $r > 0$ and $r\left(\frac{x^-}{r}\right)^* = x^-$ or $r = 0$ and $x \geqslant 0$. This order makes $X \oplus \mathbb{R}$ into a unitization of X under the natural embedding $x \mapsto (x, 0)$. Moreover, for every unitization (Y, e) of X, the map $x + re \mapsto (x, r)$ is a lattice isomorphism between Y and $X \oplus \mathbb{R}$ preserving X and sending the unit to the unit.*

This corollary yields uniqueness of unitizaton, as well as a way of construction the unitization, avoiding using X^u.

7.5.4 Exercise. Under the notation of Theorem 7.5.1,

(i) e is a weak unit in X^u iff $x^* = 0$ implies $x = 0$ for every $x \in X_+$;
(ii) $X \subseteq I_e$ in X^u iff $\forall x \in X_+ \, \exists \lambda > 0 \, (\lambda x)^* = \lambda x$.

The two conditions in the exercise are called ***weak unit truncation*** and ***strong unit truncation***, respectively. Suppose that X satisfies the strong unit truncation property. Then it also satisfies the weak one because $X \subseteq I_e$ implies e being a weak unit in X^u (as X is order dense in X^u). Furthermore, we may represent X^u as $C^\infty(K)$ so that e becomes $\mathbb{1}$ as in Theorem 7.4.15 and the remark after it. In this case, the strong truncation condition $X \subseteq I_e$ becomes $X \subseteq C(K)$.

7.5.5 Corollary. *Every strongly truncated vector lattice may be represented as an order dense sublattice of some $C(K)$ space, with $x^* = x \wedge \mathbb{1}$. If, in addition, X is unital then we may take $\mathbb{1}$ to be the unit; in this case, X is norm dense in $C(K)$.*

Proof. The first claim has already been proved. Suppose that X is unital, then $e \in X$. It follows that e is a strong unit in X. The result now follows from Krein-Kakutani Representation Theorem 4.1.7. $\square$

In the non-unital case, we have the following.

7.5.6 Theorem. *Let X be a non-unital strongly truncated vector lattice. Then X may be represented as a norm dense sublattice of $C_0(\Omega)$ for some locally compact Ω, with the truncation given by $x^* = x \wedge \mathbb{1}$.*

Proof. By the preceding argument, we can find a unitization (Y, e) of X such that e is a strong unit in Y. Note that X contains no strong units of Y. Indeed, if $u \in X_+$ is a strong unit of Y then $e \leqslant \lambda u$ for some $\lambda > 0$ and, therefore, $(\lambda u)^* = (\lambda u) \wedge e = e$ is in X, which is a contradiction.

By Krein-Kakutani Representation Theorem 4.1.7, we may represent the completion $\overline{Y}$ of $\big(Y, \|\cdot\|_e\big)$ as $C(K)$ for some compact Hausdorff K, with e corresponding to $\mathbb{1}$.

We claim that X is closed in Y. Indeed, suppose that some sequence (x_n) in X converges in $\|\cdot\|_e$ to $x + \alpha e$ for some $x \in X$ and $\alpha \in \mathbb{R}$. It suffices to show that $\alpha = 0$. Suppose not. WLOG, $\alpha = 1$. Then $\|x_n - (x + e)\| < \frac{1}{2}$ for all sufficiently large n. It follows that $-\frac{1}{2}e \leqslant x_n - x - e \leqslant \frac{1}{2}e$, and, therefore, $x_n - x \geqslant \frac{1}{2}e$, which implies that $x_n - x$ is a strong unit, which is a contradiction.

Since X is a closed sublattice of codimension 1 in Y, its closure $\overline{X}$ in $\overline{Y}$ is a closed sublattice of co-dimension 1 in $\overline{Y} = C(K)$.

We claim that all the functions in X have a common root. Suppose not, suppose that for every $t \in K$ there exists $x \in X$ such that $x(t) \neq 0$. By compactness of K, we can then find a finite collection $x_1, \dots, x_m$ in X such that for every $t \in K$ there is $i \leqslant m$ with $x_i(t) \neq 0$. Now take $x = \bigvee_{i=1}^m |x_i|$, then $x \in X$, x never vanishes on K and is, therefore, a strong unit; a contradiction. It follows that there exists $t_0 \in K$ such that all functions in X vanish at t_0. Hence, X and, therefore, $\overline{X}$ is contained in the ideal J_{t_0}. Since both $\overline{X}$ and J_{t_0} have codimension 1 in $C(K)$, we conclude that $\overline{X} = J_{t_0}$. It now follows that $\overline{X}$ may be identified with $C_0(\Omega)$ where $\Omega = K \setminus \{t_0\}$. $\square$

7.6 $S(\Omega)$ spaces

Results of this section are due to A.Wickstead. We start with a motivation. Let K be an extremally disconnected compact space. For every $f \in C^\infty(K)$, the set where f is finite is an open and dense subset of K, the restriction of f to this set is a continuous real-valued function, and f is uniquely determined by this restriction by Lemma 7.4.8.

Let Ω be a Hausdorff topological space. We write $S(\Omega)$ for the set of all real-valued continuous functions that are defined on open dense subsets of Ω; we identify two such functions if they agree on the intersection of their domains (note that this intersection is again an open dense set).

Formally speaking, members of $S(\Omega)$ are equivalence classes. Let a be an equivalence class. Let $U = \bigcup\{\operatorname{dom} f \ :\ f \in a\}$. Clearly, U is open and dense. Define $g\colon U \to \mathbb{R}$ as follows: for every $t \in U$, find $f \in a$ such that $t \in \operatorname{dom} f$ and put $g(t) = f(t)$. It is easy to see that g is well-defined; for every $f \in a$, g agrees with f on

$\operatorname{dom} f$. It follows that g is continuous and belongs to a. It is clear that every element of a is a restriction of g. Thus, every equivalence class a in $S(\Omega)$ has a "canonical" representative g with the greatest domain, which will be denoted $\operatorname{Dom} a$. This allows us to view $S(\Omega)$ as a space of functions rather than classes, when convenient.

Next, we make $S(\Omega)$ into a vector lattice by defining operations "almost everywhere". Let f and g be two continuous real-valued functions defined on two open dense subsets of Ω. Than $f+g$ is defined and continuous on the open dense set $(\operatorname{dom} f)\cap(\operatorname{dom} g)$. It is easy to see that this procedure "respects" equivalence classes: if $f'\sim f$ and $g'\sim g$ then $f'+g'\sim f+g$. Thus, this induces an addition operation on $S(\Omega)$. We can define scalar multiplication and order in a similar way; $f\leqslant g$ if $f(t)\leqslant g(t)$ for all $t\in(\operatorname{dom} f)\cap(\operatorname{dom} g)$. It is a straightforward verification that these operations make $S(\Omega)$ into a vector lattice. For example, $h=f\vee g$ if $h(t)=\max\bigl\{f(t),g(t)\bigr\}$ for every $t\in(\operatorname{dom} f)\cap(\operatorname{dom} g)$. It can be easily verified that $S(\Omega)$ is Archimedean.

7.6.1 Proposition. *If K is extremally disconnected then $C^\infty(K)$ is lattice isomorphic to $S(K)$; here $f\in C^\infty(K)$ is mapped to its restriction to the set on which it is finite. In particular, $S(K)$ is universally complete.*

Proof. It is obvious that this restriction map defines a lattice homomorphic injection from $C^\infty(K)$ to $S(K)$. It is surjective by Lemma 7.4.8, hence the map is a lattice isomorphism. Since $C^\infty(K)$ is universally complete, so is $S(K)$. $\square$

Theorem 7.4.17 immediately yield the following:

7.6.2 Corollary. *Every Archimedean vector lattice X admits a universal completion, which is lattice isomorphic to $S(K)$ for some extremally disconnected compact topological space K. If X has a weak unit e, the representation may be chosen so that e becomes $\mathbb{1}$.*

7.6.3 Exercise. Let Ω be a Hausdorff topological space. Then $S(\Omega)$ is laterally complete.

7.6.4 Example. *A space that is laterally complete but not uniformly complete.* Let $\Omega=[0,1]$. By the preceding exercise, $S(\Omega)$ is laterally complete. We will show that it is not order complete, hence nor universally complete. We put $g_n=\sum_{k=1}^{2^n-1}\frac{k}{2^n}\mathbb{1}_{(\frac{k}{2^n},\frac{k+1}{2^n})}$ and $f_m=\sum_{n=1}^{m}\frac{1}{2^n}g_n$. It is easy that $f_n\in S(\Omega)$ and $0\leqslant f_n\uparrow\leqslant\mathbb{1}$. Yet, we claim that $\sup f_n$ does not exist in $S(\Omega)$. Indeed, if $f=\sup f_n$ then $\operatorname{dom} f$, being an open set, must contain a dyadic point, because the set of all dyadic points is dense. It is then straightforward to verify that f has to be discontinuous at this point, which is a contradiction.

Chapter 8

AM- and AL-spaces

Up to this point, even though we considered both vector and Banach lattices, we have focused on vector lattice concepts. From now on, we will focus on Banach lattice concepts.

The norm on a normed lattice X is said to be an

- ***AL-norm*** if $\|x+y\| = \|x\| + \|y\|$ whenever $x \perp y$;
- ***AM-norm*** if $\|x+y\| = \max\big\{\|x\|, \|y\|\big\}$ whenever $x \perp y$.

We may assume WLOG that x and y are positive because $x \perp y$ implies $|x+y| = |x| + |y|$, so that $\|x+y\| = \big\||x|+|y|\big\|$. Moreover, once we assume that $x, y \geqslant 0$, then $x + y = x \vee y$. Thus, the definition of an AM-norm may be reformulated as follows: $\|x \vee y\| = \|x\| \vee \|y\|$ whenever $x \wedge y = 0$.

If, in addition, X is a Banach lattice, then we say that X is an ***AM-space*** or an ***AL-space***, respectively. It is clear that every closed sublattice of an AM-space is again an AM-space; same for AL.

8.0.1 Example. (i) $L_1(\mu)$ is an AL-space for every measure μ.

(ii) $C(K)$ is an AM-space for every compact space K. More generally, $C_b(\Omega)$ is an AM-space for every topological space Ω. It follows that every closed sublattice of $C_b(\Omega)$ is an AM-space. In particular, if Ω is locally compact then $C_0(\Omega)$ is an AM-space.

(iii) $L_\infty(\mu)$ and its closed sublattices are AM-spaces for every measure μ. In particular, c_0 is an AM-space as a closed sublattice of ℓ_∞.

In this chapter, we will prove that AM- and AL-spaces admit "nice" functional representations. Namely, every AM-space may be represented as a closed sublattice of a $C(K)$ space, while every AL-space is lattice isometric to an $L_1(\mu)$ space. The abbreviation "AL" stands for "abstract L_1" or "abstract linear", while "AM" stand for "abstract maximum".

8.0.2 Exercise. Let X be a normed lattice with an AL-norm. Then its completion is an AL-space. Same for AM.

8.1 Properties of AM- and AL-spaces

8.1.1. Let X be an Archimedean vector lattice with a strong unit e. Consider the norm $\|\cdot\|_e$ on X induced by e as in Section 2.2. This is an AM-norm by Exercise 2.2.5. In particular, if it is complete, then X is an AM-space. We call such a space an ***AM-space with a unit***. That is, an AM-space with a unit is a Banach lattice with a strong unit e such that the norm of the space agrees with $\|\cdot\|_e$.

8.1.2 Example. $L_\infty(\mu)$ is an AM-space with a unit for every measure μ.

$C(K)$-Representation Theorem 4.1.7 immediately yields the following:

8.1.3 Proposition. *Every AM-space with a unit is lattice isometric to $C(K)$ for some compact space K, with the unit corresponding to $\mathbb{1}$.*

Corollary 4.1.11 of $C(K)$-Representation Theorem now yields the following:

8.1.4 Proposition. *Every Banach lattice with a strong unit is lattice and norm isomorphic to an AM-space with a unit.*

8.1.5 Example. One has to distinguish a strong unit and an "AM unit". First, a Banach lattice with a strong unit need not be an AM-space with respect to its original norm. For example, consider $C(K)$ with the equivalent lattice norm $|||x||| = \|x\|_\infty + \|x\|_{L_1}$. This is not an AM-norm.

Second, if X is an AM-space and e is a strong unit in X, this does not imply that X is an AM-space with unit e, because the norm of X need not agree with $\|\cdot\|_e$, even thought the two norms must be equivalent. For example, let $X = C(K)$ and take $e(t) = 1 + t$; then e is a strong unit in X but $\|\cdot\|_e$ is different from the original norm of $C(K)$, so X is not an AM-space with unit e. However, X is an AM-space with unit $\mathbb{1}$.

Could it be that every AM-space with a strong unit is an AM-space with a unit, but possible with another unit? No, the following example shows that an AM-space with a strong unit need not even be lattice *isometric* to a $C(K)$ space. Let X be the closed sublattice of $C[0,1]$ consisting of all those functions x that satisfy $x(1) = 2x(0)$. Clearly, X is an AM-space and the function $e(t) = 1 + t$ is a strong unit in X. Yet X cannot be lattice isometric to a $C(K)$ space because B_X has no maximal element. In particular, this means that X is not an AM-space with a unit.

8.1.6 Exercise. Suppose that X is a Banach lattice and Y is a Banach lattice with a strong unit e. Then a linear operator $T\colon X \to Y$ is order bounded iff it is norm bounded, i.e., $\mathcal{L}_{\mathrm{ob}}(X,Y) = L(X,Y)$.

We will prove next that AL- and AM-spaces are dual to each other.

8.1.7 Theorem. (i) *X is an AM-space $\quad\Leftrightarrow\quad$ X^* is an AL-space;*
(ii) *X is an AL-space $\quad\Leftrightarrow\quad$ X^* is an AM-space.*

Proof. Suppose that X is an AM-space; we will show that X^* is an AL-space. Let $f,g\in X^*_+$ and $f\perp g$. Clearly, $\|f+g\|\leqslant\|f\|+\|g\|$. Fix $\varepsilon>0$; let x and y be as in Proposition 6.7.7. Then

$$\|x+y\|=\|x\vee y\|=\max\bigl\{\|x\|,\|y\|\bigr\}\leqslant 1.$$

It follows that

$$\|f\|+\|g\|\leqslant f(x)+g(y)+2\varepsilon\leqslant(f+g)(x+y)+2\varepsilon\leqslant\|f+g\|+2\varepsilon.$$

This yields $\|f+g\|=\|f\|+\|g\|$.

Next, suppose that X is an AL-space; we will show that X^* is an AM-space. Suppose that $f\perp g$ for some $f,g\in X^*_+$. It follows from $0\leqslant f,g\leqslant f+g$ that $\|f\|\vee\|g\|\leqslant\|f+g\|$; we only need to prove the other inequality. Fix $\varepsilon>0$. Find $x\in B_X$ such that $(f+g)(x)\geqslant\|f+g\|-\varepsilon$. WLOG, replacing x with $|x|$, we may assume WLOG that $x\geqslant 0$. Since $f\wedge g=0$, Riesz-Kantorovich formulas 6.6.2 imply that there exist $u,v\geqslant 0$ such that $u+v=0$ and $f(u)+g(v)<\varepsilon$. It follows that $f(u\wedge v)+g(u\wedge v)<\varepsilon$. Then

$$\begin{aligned}\|f+g\|&\leqslant(f+g)(x)+\varepsilon=f(u)+f(v)+g(u)+g(v)+\varepsilon\leqslant f(v)+g(u)+2\varepsilon\\&=f(v-u\wedge v)+g(u-u\wedge v)+f(u\wedge v)+g(u\wedge v)+2\varepsilon\\&\leqslant f(v-u\wedge v)+g(u-u\wedge v)+3\varepsilon\leqslant\max\bigl\{\|f\|,\|g\|\bigr\}\cdot\Bigl(\|v-u\wedge v\|+\|u-u\wedge v\|\Bigr)+3\varepsilon.\end{aligned}$$

Since the vectors $v-u\wedge v$ and $u-u\wedge v$ are disjoint, and

$$0\leqslant(v-u\wedge v)+(u-u\wedge v)=u+v-2(u\wedge v)\leqslant x,$$

and since X is an AL-space, we have

$$\|v-u\wedge v\|+\|u-u\wedge v\|=\bigl\|u+v-2(u\wedge v)\bigr\|\leqslant\|x\|\leqslant 1.$$

It follows that $\|f+g\|\leqslant\max\bigl\{\|f\|,\|g\|\bigr\}+3\varepsilon$. Since ε was arbitrary, we get $\|f+g\|\leqslant\max\bigl\{\|f\|,\|g\|\bigr\}$.

Suppose now that X^* is an AL-space. By the preceding argument, X^{**} is an AM-space. Since X is a closed sublattice of X^{**} by Proposition 6.7.10, it is an AM-space itself. In a similar fashion, one can prove that if X^* is an AM-space then X is an AL-space. $\square$

We will show that in the definition of AL- and AM-spaces, one may replace the assumption $x \perp y$ with $x, y \geqslant 0$. We start with AM-spaces.

8.1.8 Proposition. *A Banach lattice X is an AM-space iff $\|x \vee y\| = \max\{\|x\|, \|y\|\}$ for all $x, y \in X_+$.*

Proof. The converse implication is trivial. Suppose X is an AM-space. The inequality $\|x \vee y\| \geqslant \max\{\|x\|, \|y\|\}$ is straightforward; so we only need no prove that $\|x \vee y\| \leqslant \max\{\|x\|, \|y\|\}$ for all $x, y \in X_+$. Since X is a closed sublattice of X^{**}, it suffices to prove the required statement in X^{**}. By Theorem 8.1.7, X^* is an AL-space. Furthermore, being a dual space, X^* is order complete and, therefore, has PPP. Thus, after relabelling, it suffices to prove that if X is an AL-space with PPP then $\|f \vee g\| \leqslant \max\{\|f\|, \|g\|\}$ for all $f, g \in X^*_+$.

Fix $\varepsilon > 0$. Find $x \in S_X$ such that $\|f \vee g\| \leqslant (f \vee g)(x) + \varepsilon$. Replacing x with $|x|$, we may assume WLOG that $x \geqslant 0$. By Theorem 6.3.6, we can find positive disjoint u and v such that $u + v = x$ and $(f \vee g)(x) \leqslant f(u) + g(v) + \varepsilon$. It follows from $\|u\| + \|v\| = \|x\| = 1$ that

$$f(u) + g(v) \leqslant \|f\|\|u\| + \|g\|\|v\| \leqslant \max\{\|f\|, \|g\|\}.$$

This yields

$$\|f \vee g\| \leqslant (f \vee g)(x) + \varepsilon \leqslant f(u) + g(v) + 2\varepsilon \leqslant \max\{\|f\|, \|g\|\} + 2\varepsilon.$$

Since ε is arbitrary, this implies $\|f \vee g\| \leqslant \max\{\|f\|, \|g\|\}$. $\square$

8.1.9 Corollary. *If X is an AL-space then X^* is an AM-space with a unit e, and $e(x) = \|x\|$ for every $x \geqslant 0$.*

Proof. By Theorem 8.1.7, X^* is an AM-space. It follows from Proposition 8.1.8 that if $0 \leqslant f, g \in B_{X^*}$ then $f \vee g \in B^+_{X^*}$, so that the set $B^+_{X^*}$ is directed upwards. We may view it as an increasing net. By Alaoglu's Theorem, there is a subnet (f_α) which weak* converges to some $e \in B^+_{X^*}$. In particular, if $x \geqslant 0$ then $f_\alpha(x) \uparrow$ and $f_\alpha(x) \to e(x)$, so that $f_\alpha(x) \uparrow e(x)$.

We claim that $B_{X^*} = [-e, e]$. Indeed, it follows from $e \in B^+_{X^*}$ that $[-e, e] \subseteq B_{X^*}$. On the other hand, let $f \in B_{X^*}$. Then $|f| \in B^+_{X^*}$, so that $|f| \leqslant f_\alpha$ for some α and, therefore, $\bigl|f(x)\bigr| \leqslant |f|(x) \leqslant f_\alpha(x) \leqslant e(x)$. This yields $f \in [-e, e]$.

It follows from $B_{X^*} = [-e, e]$ that e is a strong unit in X^*. Furthermore, since $[-e, e]$ is exactly the unit ball of $\|\cdot\|_e$ by Exercise 2.2.3(iii), the norm of X^* agrees with $\|\cdot\|_e$ (because a norm is determined by its unit ball). Therefore, X^* is an AM-space with a unit e.

Finally, for every positive vector x in X we have $e(x) \leqslant \|x\|$ because $\|e\| \leqslant 1$. On the other hand, there exists $f \in S^+_{X^*}$ such that $\|x\| = f(x)$ by Exercise 6.7.6. It follows from $f \leqslant e$ that $\|x\| \leqslant e(x)$. Therefore, $e(x) = \|x\|$. □

8.1.10 Corollary. *X is an AM-space iff it is lattice isometric to a closed sublattice of $C(K)$ for some compact space K.*

Proof. Let X be an AM-space. By Proposition 6.7.10, X is lattice isometric to a closed sublattice of X^{**}. Combining Theorem 8.1.7 with Proposition 8.1.9, we conclude that X^{**} is an AM-space with a unit, hence is lattice isometric to $C(K)$ for some compact space K by Proposition 8.1.3.

For the converse, recall that $C(K)$ is an AM-space and every closed sublattice of an AM-space is again an AM-space, □

We now have a counterpart of Proposition 8.1.8 for AL-spaces:

8.1.11 Proposition. *A Banach lattice X is an AL-space iff $\|x+y\| = \|x\| + \|y\|$ for all $x, y \in X_+$.*

Proof. Clearly, if $\|x+y\| = \|x\| + \|y\|$ for all $x, y \geqslant 0$ then X is an AL-space. Let X be an AL-space and $x, y \in X_+$. Let e be as in Theorem 8.1.9. Then

$$\|x+y\| = e(x+y) = e(x) + e(y) = \|x\| + \|y\|.$$

□

8.1.12 Proposition. *Every AL-space is order continuous.*

Proof. Let X be an AL-space. By Nakano's Theorem 2.5.24, it suffices to prove that every increasing positive order bounded sequence in X converges in norm. Let $0 \leqslant x_n \uparrow\leqslant u$ in X. Put $x_0 = 0$. Let e be as in Corollary 8.1.9. It follows from $e(x_n) \uparrow\leqslant e(u)$ that the sequence $\big(e(x_n)\big)$ is Cauchy in $\mathbb{R}$. If $n \leqslant m$ then $\|x_m - x_n\| = e(x_m - x_n) = e(x_m) - e(x_n) \to 0$ as $n, m \to \infty$. It follows that (x_n) is Cauchy, hence convergent. □

8.1.13 Exercise. Let $T\colon X \to Y$ be a lattice homomorphism between two AL-spaces such that $\|Tx\| = \|x\|$ for every $x \geqslant 0$. Then T is an isometry.

Combining Corollary 8.1.10 with Theorem 3.1.35, we get the following:

8.1.14 Theorem (Kakutani)**.** *Every AM-space is lattice isometric to $\mathcal{M}_\perp$ for some compact space K and some $\mathcal{M}$.*

Fix a set Γ. We write $c_{00}(\Gamma)$ for the set of all functions from Γ to $\mathbb{R}$ with finite support. Clearly, $c_{00}(\Gamma)$ is a vector lattice; it is a sublattice of $\ell_p(\Gamma)$ for every $0 < p \leqslant \infty$. We write $c_0(\Gamma)$ for the closure of $c_{00}(\Gamma)$ in $\ell_\infty(\Gamma)$.

8.1.15 Exercise. For $f\colon \Gamma \to \mathbb{R}$, $f \in c_0(\Gamma)$ iff for every $\varepsilon > 0$ the set $\big\{|f| \geqslant \varepsilon\big\}$ is finite.

8.1.16 Theorem ([Bohn40])**.** *An AM-space is order continuous iff it is of the form $c_0(\Gamma)$ for some set Γ.*

Proof. Let Γ be a set. Being a closed sublattice of an AM-space $\ell_\infty(\Gamma)$, $c_0(\Gamma)$ is itself an AM-space. To show that $c_0(\Gamma)$ is order continuous, let $x_\alpha \downarrow 0$ and fix $\varepsilon > 0$. WLOG, passing to a tail, we may assume that (x_α) is order bounded, hence contained in $[0, u]$ for some u. Find a finite subset A of Γ such that $u(t) \leqslant \varepsilon$ for all $t \notin A$. Since A is finite, we can find α_0 such that $x_\alpha(t) \leqslant \varepsilon$ for all $\alpha \geqslant \alpha_0$ and all $t \in A$; it follows that $\|x_\alpha\| \leqslant \varepsilon$.

Suppose that X is an order continuous AM-space. Let Γ be the set of all atoms in X_+ of norm one. Let $Y = \overline{\operatorname{span}}\,\Gamma$. By Lemma 5.4.9, $Y = \overline{I(A)}$. Since X is order continuous, this implies that $Y = B(\Gamma)$.

We claim that $Y = X$ or, equivalently, that $Y^d = \{0\}$. Indeed, otherwise we can find $e \in Y^d$ with $\|e\| = 1$. Let $\mathcal{G}$ be the collection of all downward directed chains $\{z_\alpha\}$ in $[0, e]$ with $\|z_\alpha\| = 1$ for every α. The collection is non-empty as the chain $\{e\}$ is there. Order $\mathcal{G}$ by inclusion. For every chain of elements of $\mathcal{G}$ (so this would be a chain of chains), the union is an upper bound of the chain, hence $\mathcal{G}$ has a maximal element by Zorn's Lemma. Let $\{z_\alpha\}$ be such a maximal chain. We may view it as a net. Since X is order continuous, $x \coloneqq \inf z_\alpha$ exists and $\|x\| = \lim_\alpha \|z_\alpha\| = 1$, so that $x > 0$. Since $x \in Y^d$ and Y^d contains no atoms, by Exercise 5.4.7 we can write $x = u + v = u \vee v$ for some positive disjoint non-zero u and v. Since X is an AM-space, either $\|u\| = 1$ or $\|v\| = 1$, which contradicts the maximality of $\{z_\alpha\}$. This proves $X = Y = \overline{\operatorname{span}}\,\Gamma$.

Clearly, $\operatorname{span}\Gamma$ in X may be identified with $c_{00}(\Gamma)$. Since X is an AM-space, the norm of X on $\operatorname{span}\Gamma$ agrees with $\|\cdot\|_\infty$ on $c_{00}(\Gamma)$. It follows that $X = c_0(\Gamma)$. ☐

8.2 Space of finitely additive measures $\mathrm{ba}(\mathcal{F})$.

Throughout this section, let $\mathcal{F}$ be an algebra of subsets of a given set Ω (for this section, $\mathcal{F}$ need not be a σ-algebra). We write $\mathrm{ba}(\mathcal{F})$ for the set of all ***signed finitely additive bounded measures*** on $\mathcal{F}$, that is, all the functions $\mu\colon \mathcal{F} \to \mathbb{R}$ such that μ has bounded range and $\mu(A \cup B) = \mu(A) + \mu(B)$ whenever A and B are two disjoint sets in $\mathcal{F}$. The space $\mathrm{ba}(\mathcal{F})$ is equipped with a natural order: $\mu \leqslant \nu$ if $\mu(A) \leqslant \nu(A)$ for every $A \in \mathcal{F}$. Clearly, this is an ordered vector space. Positive elements in $\mathrm{ba}(\mathcal{F})$ are called ***finitely additive finite measures***[1] on $\mathcal{F}$.

We will show that $\mathrm{ba}(\mathcal{F})$ is a vector lattice; moreover, it is an AL-space under an appropriate norm. While this may be done directly (through some tedious computations), we will instead prove this showing that $\mathrm{ba}(\mathcal{F})$ is the dual of some normed lattice with an AM-norm, namely, the space of all simple functions.

We write $\mathrm{s}(\mathcal{F})$ for the space of all simple functions, that is, functions of the form $\sum_{i=1}^n r_i \mathbb{1}_{A_i}$ where $r_1, \dots, r_n \in \mathbb{R}$ and $A_1, \dots, A_n \in \mathcal{F}$. Each function in $\mathrm{s}(\mathcal{F})$ admits a

[1]The terminology is somewhat confusing. A *signed finitely additive bounded measure* need not be a *finitely additive finite measure*, and a *finitely additive finite measure* need not be a *measure*!

unique representation $\sum_{i=1}^n r_i \mathbb{1}_{A_i}$ where the sets $A_1,\dots,A_n$ are pairwise disjoint. It is easy to see that $\mathrm{s}(\mathcal{F})$ is a sublattice of $\mathbb{R}^\Omega$, hence is itself a vector lattice.

8.2.1 Exercise. $\mathrm{s}(\mathcal{F})$ has PPP.

Proof. Let $f\in \mathrm{s}(\mathcal{F})$; put $A=\operatorname{supp} f$. It is easy to see that $\{f\}^d$ consists of all the functions in $\mathrm{s}(\mathcal{F})$ that are supported on A^C. It follows that $\{f\}^{dd}$ consists of all the functions in $\mathrm{s}(\mathcal{F})$ that are supported on A. Since $g=g\cdot\mathbb{1}_A+g\cdot\mathbb{1}_{A^C}$ for every $g\in \mathrm{s}(\mathcal{F})$, we have $\mathrm{s}(\mathcal{F})=\{f\}^d\oplus\{f\}^{dd}=B_f\oplus B_f^d$. □

Given $\mu\in \mathrm{ba}(\mathcal{F})$ and $f\in \mathrm{s}(\mathcal{F})$ given by $\sum_{i=1}^n r_i\mathbb{1}_{A_i}$, we define $\varphi(f)=\int f\,d\mu=\sum_{i=1}^n r_i\mu(A_i)$. This is well-defined by 7.1.12 (one can also verify this via a direct computation). Note that $\mu(A)=\varphi(\mathbb{1}_A)$ for every $A\in\mathcal{F}$. Put $\varphi=T\mu$.

8.2.2 Exercise. The map T constructed above is a linear bijection between $\mathrm{ba}(\mathcal{F})$ and $\big(\mathrm{s}(\mathcal{F})\big)^\sim$, such that $\mu\geqslant 0$ iff $T\mu\geqslant 0$.

8.2.3 Theorem. $\mathrm{ba}(\mathcal{F})$ *is a vector lattice, with lattice operations given as follows:*

$$\begin{aligned}
|\mu|(A)&=\sup\big\{\mu(B)-\mu(A\setminus B)\;:\;B\subseteq A,\;B\in\mathcal{F}\big\},\\
\mu^+(A)&=\sup\big\{\mu(B)\;:\;B\subseteq A,\;B\in\mathcal{F}\big\},\\
(\mu\vee\nu)(A)&=\sup\big\{\mu(B)+\nu(A\setminus B)\;:\;B\subseteq A,\;B\in\mathcal{F}\big\},\\
(\mu\wedge\nu)(A)&=\inf\big\{\mu(B)+\nu(A\setminus B)\;:\;B\subseteq A,\;B\in\mathcal{F}\big\}.
\end{aligned}$$

Proof. It follows from Exercise 8.2.2 that $\mathrm{ba}(\mathcal{F})$ is a vector lattice. Since $\mathrm{s}(\mathcal{F})$ has PPP, we may use the "disjoint versions" of Riesz-Kantorovich formulas as in Theorem 6.3.6 to characterize lattice operations on $\mathrm{ba}(\mathcal{F})$. Let $\mu,\nu\in\mathrm{ba}(\mathcal{F})$; let φ and ψ be the corresponding elements of $\big(\mathrm{s}(\mathcal{F})\big)^\sim$, then

$$(\mu\vee\nu)(A)=(\varphi\vee\psi)(\mathbb{1}_A)=\sup\big\{\varphi(f)+\psi(g)\;:\;f\wedge g=0,f+g=\mathbb{1}_A\big\}.$$

It easily follows from $f\wedge g=0$ and $f+g=\mathbb{1}_A$ that f and g are characteristic functions. That is, there exists $B\subseteq A$ such that $f=\mathbb{1}_B$ and $g=\mathbb{1}_{A\setminus B}$. Then $\varphi(f)+\psi(g)=\mu(B)+\nu(A\setminus B)$. This yields the required formula for $\mu\vee\nu$. The other formulas may be proved in a similar fashion (or deduced from this one). □

Even though it is an abuse of notation, it is common to use the same symbol for the measure and the corresponding functional, i.e., we write $\mu(f)=\int f\,d\mu$. For $A\in\mathcal{F}$, $|\mu|(A)$ is called the ***variation*** of μ on A, and $|\mu|(\Omega)$ is called the ***total variation*** of μ.

We may view $\mathrm{s}(\mathcal{F})$ as a sublattice of $\ell_\infty(\Omega)$. Hence, we may equip $\mathrm{s}(\mathcal{F})$ with $\|\cdot\|_\infty$, which is clearly an AM-norm. Note that $\mathrm{s}(\mathcal{F})$ need not be an AM-space because it need not be complete.

Since the unit ball of $\mathrm{s}(\mathcal{F})$ is $[-\mathbb{1},\mathbb{1}]$, every order bounded functional on $\mathrm{s}(\mathcal{F})$ is automatically norm bounded. It follows that, as a vector lattice, $\mathrm{ba}(\mathcal{F})=\big(\mathrm{s}(\mathcal{F})\big)^\sim=\big(\mathrm{s}(\mathcal{F})\big)^*$. Thus, we may equip $\mathrm{ba}(\mathcal{F})$ with the lattice norm induced from $\big(\mathrm{s}(\mathcal{F})\big)^*$. Since the dual space of $\mathrm{s}(\mathcal{F})$ equals the dual of its completion, and the latter is an AM-space by Exercise 8.0.2, we conclude by Theorem 8.1.7 that $\mathrm{ba}(\mathcal{F})$ is an AL-space. We claim that the norm on $\mathrm{ba}(\mathcal{F})$ that we have just defined agrees with total variation. Indeed, since we already know that this norm is a lattice norm, for $\mu\in\mathrm{ba}(\mathcal{F})$, we have

$$\|\mu\|=\big\||\mu|\big\|=\sup\big\{|\mu|(f)\ :\ \|f\|_\infty\leqslant 1\big\}=|\mu|(\mathbb{1})=|\mu|(\Omega).$$

Let's summarize:

8.2.4 Theorem. $\mathrm{ba}(\mathcal{F})$ *is an AL-space under the total variation norm;* $\mathrm{ba}(\mathcal{F})$ *is the order and the norm dual of* $\mathrm{s}(\mathcal{F})$*.*

8.2.5 Exercise. For an increasing net (μ_α) in $\mathrm{ba}(\mathcal{F})$ we have $\mu_\alpha\uparrow\mu$ for some $\mu\in\mathrm{ba}(\mathcal{F})$ iff $\mu_\alpha(A)\to\mu(A)$ for every $A\in\mathcal{F}$.

Given two positive measures μ and ν in $\mathrm{ba}(\mathcal{F})$, we say that ν is ***absolutely continuous*** with respect to μ and write $\nu\ll\mu$ if for every $\varepsilon>0$ there exists $\delta>0$ such that $\nu(A)<\varepsilon$ whenever $\mu(A)<\delta$[2]. More generally, given arbitrary μ and ν in $\mathrm{ba}(\mathcal{F})$, we say that $\nu\ll\mu$ if $|\nu|\ll|\mu|$.

8.2.6 Exercise. Given μ and ν in $\mathrm{ba}(\mathcal{F})$, show that $\nu\ll\mu$ iff for every $\varepsilon>0$ there exists $\delta>0$ such that $\big|\nu(A)\big|<\varepsilon$ whenever $|\mu|(A)<\delta$.

8.2.7 Theorem. *Let* $\mu,\nu\in\mathrm{ba}(\mathcal{F})$*. Then* $\nu\ll\mu$ *iff* ν *is in the band generated by* μ *in* $\mathrm{ba}(\mathcal{F})$*. That is,* $B_\mu=\big\{\nu\in\mathrm{ba}(\mathcal{F})\ :\ \nu\ll\mu\big\}$.

Proof. WLOG, μ and ν are positive. Suppose that $\nu\in B_\mu$. Fix $\varepsilon>0$. Then $\nu\wedge n\mu\uparrow\nu$ by Proposition 5.1.3, so that $\nu-\nu\wedge n\mu\downarrow 0$. By Exercise 8.2.5, $(\nu-\nu\wedge n\mu)(\Omega)\downarrow 0$. Hence, for a sufficiently large n, we have $(\nu-\nu\wedge n\mu)(\Omega)<\varepsilon$. It follows that $(\nu-\nu\wedge n\mu)(A)<\varepsilon$ for every $A\in\mathcal{F}$, so that

$$\nu(A)<\varepsilon+(\nu\wedge n\mu)(A)\leqslant\varepsilon+n\mu(A)<2\varepsilon$$

whenever $\mu(A)<\frac{\varepsilon}{n}$.

[2]This notation is unrelated to the "$\ll$" notation we used for elements of non-Arhimedean vector lattices in Section 4.2.

Conversely, suppose that $\nu \ll \mu$. Since $\mathrm{ba}(\mathcal{F})$ is an AL-space, it is σ-order complete by Proposition 8.1.12 (or by direct verification), hence it has PPP by Corollary 5.2.23, so that $\mathrm{ba}(\mathcal{F}) = B_\mu \oplus B_\mu^d$. It follows that $\nu = \nu_1 + \nu_2$ with $\nu_1 \in B_\mu$ and $\nu_2 \perp \mu$. It suffices to show that $\nu_2 = 0$.

By the first part of the proof, we have $\nu_1 \ll \mu$ and, therefore, $\nu_2 \ll \mu$. Fix $\varepsilon > 0$; let $\delta > 0$ be such that $\nu_2(A) < \varepsilon$ whenever $\mu(A) < \delta$. WLOG, $\delta < \varepsilon$. It follows from $\nu_2 \perp \mu$ and Theorem 8.2.3 that there exists $A \in \mathcal{F}$ such that $\nu_2(A)+\mu(\Omega \setminus A) < \delta$. In particular, $\mu(\Omega \setminus A) < \delta$, which yields $\nu_2(\Omega \setminus A) < \varepsilon$. Now $\nu_2(\Omega) = \nu_2(A) + \nu_2(\Omega \setminus A) < \delta + \varepsilon < 2\varepsilon$, and, therefore $\nu_2(\Omega) = 0$. □

For $\mu, \nu \in \mathrm{ba}(\mathcal{F})$, we say that μ and ν are ***singular*** if $\mu \perp \nu$. The preceding theorem immediately yields the following result.

8.2.8 Theorem (Lebesgue Decomposition for $\mathrm{ba}(\mathcal{F})$)**.** *Given any $\mu, \nu \in \mathrm{ba}(\mathcal{F})$, one can decompose $\nu = \nu_1 + \nu_2$ with $\nu_1, \nu_2 \in \mathrm{ba}(\mathcal{F})$ such that $\nu_1 \ll \mu$ and ν_2 is singular to μ.*

8.3 Space of signed measures $\mathrm{ca}(\mathcal{F})$.

Throughout this section, $\mathcal{F}$ is a σ-algebra of subsets of some set Ω. Let $\mathrm{ca}(\mathcal{F})$ be the set of all signed measures in $\mathrm{ba}(\mathcal{F})$. That is,

$$\mathrm{ca}(\mathcal{F}) = \bigl\{\mu \in \mathrm{ba}(\mathcal{F}) \;:\; \mu \text{ is countably additive}\bigr\}.$$

Note that the positive cone of $\mathrm{ca}(\mathcal{F})$ consists of all finite measures on $\mathcal{F}$.

The following exercise provides a convenient equivalent characterization of countable additivity. For a sequence (A_n) in $\mathcal{F}$, we write $A_n \downarrow \varnothing$ if the sequence is nested decreasing and $\bigcap_{n=1}^\infty A_n = \varnothing$.

8.3.1 Exercise. Given a finitely additive function $\mu\colon \mathcal{F} \to \mathbb{R}$, μ is countably additive iff $\mu(A_n) \to 0$ whenever $A_n \downarrow \varnothing$.

The following proposition implies that the boundedness requirement in the definition of $\mathrm{ca}(\mathcal{F})$ is redundant; $\mathrm{ca}(\mathcal{F})$ is the set of all countably additive functions from $\mathcal{F} \to \mathbb{R}$.

8.3.2 Proposition. *Every countably additive function $\mu\colon \mathcal{F} \to \mathbb{R}$ is bounded.*

Proof. For $A \in \mathcal{F}$, put $\nu(A) = \sup\bigl\{\bigl|\mu(B)\bigr| \;:\; B \subseteq A,\ B \in \mathcal{F}\bigr\}$. It can be easily verified that ν is subadditive on disjoint sets. It suffices to show that $\nu(\Omega) < \infty$. Suppose, for the sake of contradiction, that $\nu(\Omega) = \infty$. We will construct a nested decreasing sequence (A_n) in $\mathcal{F}$ such that $\nu(A_n)=\infty$ and $\bigl|\mu(A_n)-\mu(A_{n+1})\bigr|\geqslant 1$ for every n. This will lead to a contradiction because Exercise 8.3.1 yields $\mu\bigl(\bigcap_{n=1}^\infty A_n\bigr)=\lim_n\mu(A_n)$, hence the sequence $\bigl(\mu(A_n)\bigr)$ converges.

Take $A_1=\Omega$. Suppose that $A_1,\dots,A_n$ have already been constructed. Since $\nu(A_n)=\infty$, we can find $B\subseteq A_n$ with $\bigl|\mu(B)\bigr|>\bigl|\mu(A_n)\bigr|+1$. If $\nu(B)=\infty$ then we take $A_{n+1}\coloneqq B$, and the required conditions are satisfies. Else, the subadditivity of ν implies $\nu(A_n\setminus B)=\infty$. We put $A_{n+1}\coloneqq A_n\setminus B$. In this case, $\bigl|\mu(A_n)-\mu(A_{n+1})\bigr|=\bigl|\mu(B)\bigr|\geqslant 1$. □

8.3.3 Theorem. $\mathrm{ca}(\mathcal{F})$ *is a band in* $\mathrm{ba}(\mathcal{F})$.

Proof. First, show that $\mathrm{ca}(\mathcal{F})$ is a sublattice of $\mathrm{ba}(\mathcal{F})$. It suffices to prove that for every $\mu\in\mathrm{ca}(\mathcal{F})$, we have $\mu^+\in\mathrm{ca}(\mathcal{F})$. We will use the formula for μ^+ from Theorem 8.2.3. Let (A_n) be a disjoint sequence in $\mathcal{F}$ and $A=\bigcup_{n=1}^\infty A_n$. We need to show that $\sum_{n=1}^\infty\mu^+(A_n)=\mu^+(A)$. For every m, we have

$$\sum_{n=1}^m\mu^+(A_n)=\mu^+\Bigl(\bigcup_{n=1}^m A_n\Bigr)\leqslant\mu^+(A).$$

Passing to the limit on m, we get $\sum_{n=1}^\infty\mu^+(A_n)\leqslant\mu^+(A)$. To show the converse inequality, observe that for every measurable B such that $B\subseteq A$ we have $B=\bigcup_{n=1}^\infty B\cap A_n$. It follows that

$$\mu(B)=\sum_{n=1}^\infty\mu(B\cap A_n)\leqslant\sum_{n=1}^\infty\mu^+(A_n),$$

so that $\mu^+(A)\leqslant\sum_{n=1}^\infty\mu^+(A_n)$.

Next, we prove that $\mathrm{ca}(\mathcal{F})$ is an ideal. It suffices to show that if $\mu\in\mathrm{ca}(\mathcal{F})$ and $0\leqslant\nu\leqslant\mu$ then $\nu\in\mathrm{ca}(\mathcal{F})$. But this follows immediately from Exercise 8.3.1.

To finish the proof, we need to show that if (μ_α) is a net in $\mathrm{ca}(\mathcal{F})$ and $0\leqslant\mu_\alpha\uparrow\mu$ for some $\mu\in\mathrm{ba}(\mathcal{F})$ then μ is countably additive. Again, we will use Exercise 8.3.1. Suppose that (A_n) is a nested decreasing sequence in $\mathcal{F}$ with $\bigcap_{n=1}^\infty A_n=\varnothing$. It suffices to prove that $\mu(A_n)\to 0$. Fix $\varepsilon>0$. By Exercise 8.2.5, $\mu_\alpha(\Omega)\uparrow\mu(\Omega)$. Find α such that $(\mu-\mu_\alpha)(\Omega)<\varepsilon$. Since μ_α is countably additive, there exists n_0 such that $\mu_\alpha(A_n)<\varepsilon$ for all $n\geqslant n_0$; then

$$\mu(A_n)=(\mu-\mu_\alpha)(A_n)+\mu_\alpha(A_n)\leqslant(\mu-\mu_\alpha)(\Omega)+\varepsilon<2\varepsilon.$$

Hence, $\mu(A_n)\to 0$. □

8.3.4 Exercise. Given $\mu\in\mathrm{ba}(\mathcal{F})$, show that μ is countably additive iff it is σ-order continuous (as a functional on $\mathrm{s}(\mathcal{F})$).

In view of this exercise, Theorem 8.3.3 may be viewed as a special case of Ogasawara's Theorem 6.6.12.

8.3.5 Corollary. $\mathrm{ca}(\mathcal{F})$ *is an AL-space.*

Proof. By Theorem 8.2.4, $\mathrm{ba}(\mathcal{F})$ is an AL-space. Being a band, $\mathrm{ca}(\mathcal{F})$ is a closed sublattice of $\mathrm{ba}(\mathcal{F})$ by Exercise 5.1.16. Clearly, the norm is still additive on positive measures. □

Combining Theorems 8.3.3 and 8.2.7, we get the following:

8.3.6 Corollary. *For $\mu, \nu \in \mathrm{ca}(\mathcal{F})$, $\nu \ll \mu$ iff $\nu \in B_\mu$.*

It also follows that the Lebesgue Decomposition Theorem 8.2.8 remains valid for countably additive measures.

8.3.7 Theorem (Lebesgue Decomposition for $\mathrm{ca}(\mathcal{F})$)**.** *Given any $\mu, \nu \in \mathrm{ca}(\mathcal{F})$, one can decompose $\nu = \nu_1 + \nu_2$ with $\nu_1, \nu_2 \in \mathrm{ca}(\mathcal{F})$ such that $\nu_1 \ll \mu$ and ν_2 is singular to μ.*

8.3.8 Exercise. Given two measures μ and ν on $\mathcal{F}$, $\mu \perp \nu$ iff there exists $B \in \mathcal{F}$ such that $\mu(B) = 0$ while $\nu(B^C) = 0$. That is, two measure are singular iff they are "disjointly supported".

8.3.9 Exercise. Let $\mu, \nu \in \mathrm{ca}(\mathcal{F})$. Then the suprema and infima in Theorem 8.2.3 are attained.

8.3.10 Proposition (Hahn Decomposition)**.** *Let $\mu \in \mathrm{ca}(\mathcal{F})$. Then there exists B such that $\mu(A) \geqslant 0$ for all $A \subseteq B$ and $\mu(A) \leqslant 0$ for all $A \subseteq B^C$.*

Proof. Since $\mu^+ \perp \mu^-$, by Exercise 8.3.8, there exists $B \in \mathcal{F}$ such that $\mu^-(B) = 0$ and $\mu^+(B^C) = 0$. It now follows from Theorem 8.2.3 that for every $A \subseteq B^C$ we have $\mu(A) \leqslant \mu^+(B^C) = 0$. The other case is similar. □

Recall that for positive elements x and e in a vector lattice, x is a *component* of e if $0 \leqslant x \leqslant e$ and $x \perp (e - x)$; cf Section 7.2.

8.3.11 Lemma. *Let $\mu, \nu \in \mathrm{ca}(\mathcal{F})_+$. Then ν is a component of μ iff there exists $B \in \mathcal{F}$ such that $\nu(A) = \mu(A \cap B)$ for all $A \in \mathcal{F}$.*

Proof. Suppose that ν is a component of μ; put $\lambda = \mu - \nu$. Then ν and λ are positive and $\nu \perp \lambda$. By Exercise 8.3.8, there is $B \in \mathcal{F}$ such that $\nu(B^C) = 0$ and $\lambda(B) = 0$. For every $A \in \mathcal{F}$, we have

$$\mu(A \cap B) = \nu(A \cap B) + \lambda(A \cap B) = \nu(A \cap B) = \nu(A \cap B) + \nu(A \cap B^C) = \nu(A).$$

To prove the converse, observe that $0 \leqslant \nu \leqslant \mu$. Let $\lambda = \mu - \nu$. Then $\nu(B^C) = 0 = \lambda(B)$. Hence $\nu \perp \lambda$ by Exercise 8.3.8. $\square$

8.3.12 Exercise. Let $0 \leqslant \mu, \nu \in \mathrm{ca}(\mathcal{F})$. Then $\nu \ll \mu$ iff $\nu(A) = 0$ whenever $\mu(A) = 0$.

We finish this section with the Radon-Nikodym Theorem, which provides another important characterization of absolute continuity. We start with the following auxiliary fact:

8.3.13 Proposition. *For a measure μ in $\mathrm{ca}(\mathcal{F})$, there exists a surjective lattice isometry $T\colon L_1(\mu) \to B_\mu$, where B_μ is the principal band of μ in $\mathrm{ca}(\mathcal{F})$, such that $(Tf)(A) = \int_A f\,d\mu$ for every $f \in L_1(\mu)_+$ and every $A \in \mathcal{F}$.*

Proof. For every $f \in L_1(\mu)$ and $A \in \mathcal{F}$, define $(Tf)(A) = \int_A f\,d\mu$. Clearly, Tf is finitely additive; it follows from $\bigl|(Tf)(A)\bigr| \leqslant \int_A |f| d\mu \leqslant \|f\|$ that Tf is bounded. Hence, $Tf \in \mathrm{ba}(\mathcal{F})$. It is easy to see that $T\colon L_1(\mu) \to \mathrm{ba}(\mathcal{F})$ is linear.

We claim that $Tf \in \mathrm{ca}(\mathcal{F})$. Suppose first that $f \geqslant 0$. If $A_n \downarrow \varnothing$ in $\mathcal{F}$, then $(Tf)(A_n) = \int f \cdot \mathbb{1}_{A_n} d\mu \to 0$ by Monotone Convergence Theorem A.3.12; it follows from Exercise 8.3.1 that Tf is a countably additive measure, hence $Tf \in \mathrm{ca}(\mathcal{F})$. For a general $f \in L_1(\mu)$, we have $Tf = Tf^+ - Tf^- \in \mathrm{ca}(\mathcal{F})$. Hence $T\colon L_1(\mu) \to \mathrm{ca}(\mathcal{F})$.

Suppose that $f, g \in L_1(\mu)_+$ with $f \perp g$. Let $B = \operatorname{supp} g$. Then $(Tf)(B) = 0$ and $(Tg)(B^C) = 0$; Exercise 8.3.8 yields $(Tf) \wedge (Tg) = 0$. It follows from Exercise 1.6.10 that T is a lattice homomorphism. If $f \geqslant 0$ then $\|Tf\| = (Tf)(\Omega) = \int f\,d\mu = \|f\|$; it follows from Exercise 8.1.13 that T is an isometry. In particular, $\operatorname{Range} T$ is closed.

We claim that $\operatorname{Range} T = B_\mu$, the band generated by μ in $\mathrm{ca}(\mathcal{F})$. For every $f \in L_1(\mu)$ we have $Tf \ll \mu$ by Exercise 8.3.12, hence $Tf \in B_\mu$ by Corollary 8.3.6. Hence, $\operatorname{Range} T \subseteq B_\mu$.

Fix $B \in \mathcal{F}$. For every $A \in \mathcal{F}$, we have $(T\mathbb{1}_B)(A) = \mu(A \cap B)$. Therefore, by Lemma 8.3.11, $\mathcal{C}_\mu = \bigl\{T\mathbb{1}_B \;:\; B \in \mathcal{F}\bigr\}$. It follows that $\mathcal{C}_\mu \subseteq \operatorname{Range} T$ and, therefore, $\operatorname{span} \mathcal{C}_e \subseteq \operatorname{Range} T$. By Freudenthal's Theorem 7.2.14, $\operatorname{span} \mathcal{C}_e$ is dense in the principal ideal I_μ in $\mathrm{ca}(\mathcal{F})$ in the norm induces by μ. Since this norm is stronger than the original norm of $\mathrm{ca}(\mathcal{F})$, we conclude that $\operatorname{Range} T$ contains the closure $\overline{I_\mu}$ of I_μ in $\mathrm{ca}(\mathcal{F})$. Since $\mathrm{ca}(\mathcal{F})$ is an AL-space by Corollary 8.3.5, it is order continuous by Proposition 8.1.12 and, therefore, $\overline{I_\mu} = B_\mu$ by Exercise 5.1.17. Thus, $B_\mu \subseteq \operatorname{Range} T$. This concludes the proof that $\operatorname{Range} T = B_\mu$. $\square$

8.3.14 Theorem (Radon-Nikodym)**.** *Suppose that $\mu, \nu \in \mathrm{ca}(\mathcal{F})_+$. Then $\nu \ll \mu$ iff there exists a (unique) $f \in L_1(\mu)_+$ such that $\nu(A) = \int_A f\,d\mu$ for every $A \in \mathcal{F}$.*

Proof. Let $T\colon L_1(\mu) \to B_\mu$ be as in Proposition 8.3.13. If $\nu \ll \mu$ then $\nu \in B_\mu$ by Corollary 8.3.6, hence $\nu = Tf$ for some $f \in L_1(\mu)$.

To prove the converse, observe that $\nu \ll \mu$ by Exercise 8.3.1. □

8.3.15 Remark. Suppose that $\mu, \nu \in \mathrm{ca}(\mathcal{F})_+$ with $\nu \ll \mu$. The function f in Radon-Nikodym Theorem is called the ***Radon-Nikodym derivative*** of ν with respect to μ; we write $f = \frac{d\nu}{d\mu}$. The theorem yields that $f = \frac{d\nu}{d\mu}$ iff $\nu = f\mu$. In particular, $\int g\,d\nu = \int gf\,d\mu$ for any $g \in L_1(\nu)$.

If we do not assume that ν is positive, we define $\frac{d\nu}{d\mu} = \frac{d\nu^+}{d\mu} - \frac{d\nu^-}{d\mu}$. Furthermore, Radon-Nikodym Theorem easily extends to σ-finite measure; in this case, however, we cannot assume that f is integrable, only that it is measurable.

Observe that for the operator T in Proposition 8.3.13, T^{-1} is precisely the Radon-Nikodym derivative, i.e., $T^{-1}\nu = \frac{d\nu}{d\mu}$.

Recall that if μ is a σ-finite measure then $L_1(\mu)^* = L_\infty(\mu)$. However, in general, $L_1(\mu)$ is not reflexive; $L_1(\mu)$ is a proper subspace of $L_\infty(\mu)$ (under the canonical embedding). The following proposition shows that $L_1(\mu)$ is reflexive in a certain weaker sense. We include a direct proof here; a shorter proof will be given later in Example 10.4.12.

8.3.16 Proposition. *$L_\infty(\mu)^\sim_{\mathrm{oc}} = L_1(\mu)$ for every σ-finite measure μ, with the duality given by $\langle f, g\rangle = \int fg\,d\mu$ for all $g \in L_\infty(\mu)$ and $f \in L_1(\mu)$.*

Proof. Let $L_\infty(\mu) = L_\infty(\Omega, \mathcal{F}, \mu)$. WLOG, by Proposition 1.6.19, μ is finite. Let $\varphi \in L_\infty(\mu)^\sim_{\mathrm{oc}}$. For every $A \in \mathcal{F}$, define $\nu(A) = \varphi(\mathbb{1}_A)$. This defines a finitely-additive function on $\mathcal{F}$. By Exercise 8.3.1, ν is countably additive because if $A_n \downarrow \varnothing$ then $\mathbb{1}_{A_n} \xrightarrow{o} 0$ and, therefore, $\nu(A_n) \to 0$. Hence, $\nu \in \mathrm{ca}(\mathcal{F})$. Furthermore, $\nu \ll \mu$ because $\mu(A) = 0$ implies $\mathbb{1}_A = 0$ in $L_\infty(\mu)$ and, therefore, $\nu(A) = 0$. Let $f = \frac{d\nu}{d\mu}$, then $f \in L_1(\mu)$. It can be easily verified that $\varphi(g) = \int fg\,d\mu$ for every $g \in L_\infty(\mu)$: first, check this for simple functions in $L_\infty(\mu)$, and then extend it to all of $L_\infty(\mu)$ by continuity.

Put $f = S\varphi$. Then $S\colon L_\infty(\mu)^\sim_{\mathrm{oc}} \to L_1(\mu)$. Clearly, S is a positive linear operator. To see that it is onto, fix $f \in L_1(\mu)_+$ and put $\varphi(g) = \int fg\,d\mu$ for every $g \in L_\infty(\mu)$, i.e., $\varphi = \hat{f}$. It suffices to show that φ is order continuous. It is easy to see that if $g_\alpha \downarrow 0$ in $L_\infty(\mu)$ then $fg_\alpha \downarrow 0$ in $L_1(\mu)$ and, therefore, $\varphi(g_\alpha) = \int fg_\alpha\,d\mu \downarrow 0$. □

8.4 The dual of $C(K)$

Throughout this section, K is a compact space. Let $\mathcal{B}$ be the ***Baire σ-algebra*** on K, that is, the smallest σ-algebra of subsets of K such that every function in $C(K)$ is measurable. A ***Baire measure*** on K is a measure on $\mathcal{B}$. Please see Appendix A.3.3 for more information about Baire measures.

We put $\mathcal{M}(K) = \mathrm{ca}(\mathcal{B})$, the space of all finite signed Baire measures on K. Every Baire measure μ defines a positive functional on $C(K)$ via $f \mapsto \int f\,d\mu$. The classical Riesz-Markov Theorem 8.4.1 asserts that the converse is also true:

8.4.1 Theorem (Riesz-Markov)**.** *Let K be a compact space and $0 \leqslant \xi \in C(K)^*$. There exists a finite Baire measure μ such that $\xi(f) = \int f\,d\mu$ for every $f \in C(K)$.*

While direct proofs of Riesz-Markov Theorem are available in the literature, e.g., in [Roy88], we provide here a proof based on Banach lattice techniques, as well as some basic topology. Recall that a subset A of K is A ***meagre*** if it is a union of a sequence of nowhere dense sets; see A.2.5. A set of the form $U \triangle A$, where U is open and A is meagre, is called ***almost open***. Almost open subsets of K form a σ-algebra, which contains the Borel σ-algebra.

Proof of Riesz-Markov Theorem. Since $C(K)$ is an AM-space with a unit, $C(K)^*$ is an AL-space by Theorem 8.1.7. It is easy to see that $\|\zeta\| = \zeta(\mathbb{1}_K)$ for every positive $\zeta \in C(K)^*$ because $\mathbb{1}_K$ is the maximal element in the unit ball of $C(K)$. Proposition 8.1.9 asserts that $C(K)^{**}$ is an AM-space with a unit e, such that $e(\zeta) = \|\zeta\|$ for every positive $\zeta \in C(K)^*$. It follows that the canonical injection $j\colon C(K) \to C(K)^{**}$ maps $\mathbb{1}_K$ to e. By Proposition 8.1.3, we may identify $C(K)^{**}$ with $C(L)$ for some compact L, such that e becomes $\mathbb{1}_L$. We may now view j as a map from $C(K)$ into $C(L)$. Since j is a lattice homomorphism by Proposition 6.7.10 and $j(\mathbb{1}_K) = \mathbb{1}_L$, by Theorem 6.11.7, we can find a continuous map $\varphi\colon L \to K$ such that $j(f) = f \circ \varphi$ for every $f \in C(K)$. Since $C(L)$ is a dual space, it is order complete by Corollary 6.7.12, hence L is extremally disconnected by Theorem 2.1.29. We will use this fact to build a required measure on L; we will then pull it down to K using φ.

For every open subset of L, its closure is clopen and its boundary is meagre; it follows that every almost open set in L can be written in the form $U \triangle A$, where U is clopen and A is meagre. Moreover, such a representation is unique: if $U \triangle A = V \triangle B$ for some clopen U and V and meagre A and B then $U \triangle V = A \triangle B$ and this set is open and meagre and, therefore, empty, so that $U = V$ and $A = B$. In a similar fashion, it may be verified that if $U \triangle A \subseteq V \triangle B$ then $U \subseteq V$. Let $\mathcal{F}$ be the σ-algebra of all almost open subsets of L.

Since $\xi \in C(K)^*$, we may consider its image $\hat{\xi}$ under the canonical embedding $C(K)^* \hookrightarrow C(K)^{***} = C(L)^*$. It follows from $\xi \geqslant 0$ that $\hat{\xi} \geqslant 0$. Furthermore, $\hat{\xi}$ is order continuous by Proposition 6.6.19. We define $\mu_0\colon \mathcal{F} \to \mathbb{R}_+$ via $\mu_0(U \triangle A) = \hat{\xi}(\mathbb{1}_U)$ whenever U is clopen and A is meagre. To show that μ_0 is countably additive, let (U_n) and (A_n) be sequences of clopen and meagre sets, respectively, such that the sequence $(U_n \triangle A_n)$ is nested increasing. It follows that the sequence (U_n) is nested increasing. Put $U = \overline{\bigcup_{n=1}^\infty U_n}$, then U is clopen and $\bigcup_{n=1}^\infty (U_n \triangle A_n) = U \triangle A$ for some meagre set A. It follows from $\mathbb{1}_{U_n} \uparrow \mathbb{1}_U$ that

$$\mu_0(U_n \triangle A_n) = \hat{\xi}(\mathbb{1}_{U_n}) \uparrow \hat{\xi}(\mathbb{1}_U) = \mu_0(U \triangle A) = \mu_0\Big(\bigcup_{n=1}^\infty (U_n \triangle A_n)\Big).$$

This shows that μ_0 is countably additive, hence a finite measure on $\mathcal{F}$.

The map $g \mapsto \int g\,d\mu_0$ is a continuous linear functional on $C(L)$; it agrees with $\hat{\xi}$ on the characteristic functions of clopen sets. The linear span of such functions is dense in $C(L)$, hence the two functionals agree on $C(L)$.

Since φ is continuous, $\varphi^{-1}(A)$ is Borel in L whenever A is a Borel subset of K; hence $\varphi^{-1}(A) \in \mathcal{F}$. Put $\mu(A) = \mu_0\big(\varphi^{-1}(A)\big)$. It is easy to see that μ is a finite Borel measure on K, hence its restriction to the Baire σ-algebra is a finite Baire measure. For every $f \in C(K)$, we have

$$\xi(f) = \hat{\xi}\big(j(f)\big) = \int (f \circ \varphi) d\mu_0 = \int f\, d\mu.$$

□

We can now identify the dual of $C(K)$.

8.4.2 Theorem (Riesz Representation). *Let K be a compact space. Then $C(K)^*$ is lattice isometric to $\mathcal{M}(K)$. For $0 \leqslant \mu \in \mathcal{M}(K)$, the corresponding functional on $C(K)$ is given by $\varphi(f) = \int f\, d\mu$.*

Proof. Note that $C(K)^*$ and $\mathcal{M}(K)$ are both AL-spaces: the former by Theorem 8.1.7 and the latter by Corollary 8.3.5.

Define $\tau\colon \mathcal{M}(K)_+ \to C(K)^*_+$ via $\tau(\mu)\colon f \mapsto \int f\, d\mu$. Clearly, τ is additive. By the Riesz-Markov Theorem, τ is a bijection. If $\mu \geqslant 0$ then, since $\tau(\mu)$ is a positive functional on $C(K)$ and $\mathbb{1}$ is the maximal element in the unit ball of $C(K)$, we have $\big\|\tau(\mu)\big\| = \big\langle \tau(\mu), \mathbb{1}\big\rangle = \mu(K) = \|\mu\|$. By Corollary 6.2.3, τ extends to a surjective lattice isomorphism $T\colon \mathcal{M}(K) \to C(K)^*$. It is an isometry by Exercise 8.1.13.

□

8.4.3. Recall that real-valued lattice homomorphisms on $C(K)$ are precisely the positive scalar multiples of point evaluation functionals δ_t by Proposition 4.1.1. On the other hand, By Exercise 6.11.1, real valued lattice homomorphisms on $C(K)$ correspond to atoms in $C(K)^*$. Thus, positive scalar multiples of point evaluation functionals are precisely the atoms in $\mathcal{M}(K)$ (of course, this may be also verified directly).

Consider the band in $\mathcal{M}(K)$ generated by all the atoms, that is, the discrete part of $\mathcal{M}(K)$. Its elements are called ***discrete*** signed measures.

8.4.4. Let $K = [0,1]$ and λ be the Lebesgue measure. Since $\lambda \perp \delta_t$ for every $t \in K$, the band B_{discr} of all discrete measures in $\mathcal{M}(K)$ (as in 8.4.3) is disjoint to the band B_λ of all the measures that are absolutely continuous with respect to λ. One may naturally conjecture that B_{discr} and B_λ form a disjoint decomposition of $\mathcal{M}(K)$. However, this is false as there are measure that are disjoint to both B_{discr} and B_λ, e.g., the Lebesgue-Stieltjes measure corresponding to the Cantor function; see Exercise 9 on p. 127 in [Hal50]. Thus, we can decompose $\mathcal{M}(K)$ into the sum of three pair-wise disjoint bands, $\mathcal{M}(K) = B_{\mathrm{discr}} \oplus B_\lambda \oplus B_{\mathrm{cs}}$; the elements of B_{cs} are sometimes called ***continuous singular*** signed measures. It follows that every measure μ in $\mathcal{M}(K)$ may be uniquely decomposed into $\mu = \mu_1 + \mu_2 + \mu_3$, so that μ_1 is discrete, $\mu_2 \ll \lambda$, and μ_3 is continuous singular.

8.4.5 Exercise. Show that every discrete element μ of $\mathcal{M}(K)$ is supported on a countable set. That is, there exists a countable subset C of K such that $\mu(A) = 0$ whenever $A \cap C = \varnothing$.

8.4.6 Exercise. For a sequence (f_n) in $C(K)$, $f_n \xrightarrow{\mathrm{w}} 0$ iff (f_n) is bounded and converges to zero pointwise.

Recall that in a general Banach lattice, lattice operations need not be weakly continuous; see 6.8.6.

8.4.7 Exercise. In an AM-space, lattice operations are weakly sequentially continuous.

8.4.8 Exercise. $\mathcal{M}[0,1]$ has no order continuous predual. That is, no order continuous Banach lattice X satisfies $X^* = \mathcal{M}[0,1]$.

8.5 Representations of AL-spaces

A Banach lattice is said to be an L_1-space if it is lattice isometric to $L_1(\mu)$ for some measure μ. The goal of this section is to prove that every AL-space is an L_1-space.

The next lemma says that in order to show that an AL-space is an L_1-space, it suffices to show that it is "locally" an L_1-space.

8.5.1 Lemma. *Let X be an AL-space. If every principal band in X is an L_1-space then X is an L_1-space.*

Proof. Let Λ be a maximal disjoint collection in X_+. By assumption, for every $z \in \Lambda$ there exists a measure space $(\Omega_z, \mathcal{F}_z, \mu_z)$ and a lattice isometry T_z from B_z onto $L_1(\Omega_z, \mathcal{F}_z, \mu_z)$. Put Ω to be the disjoint union of all Ω_z over $z \in \Lambda$. Let $\mathcal{F}$ be the σ-algebra on Ω generated by all $\mathcal{F}_z$ as $z \in \Lambda$, that is, $\mathcal{F}$ consists of all the unions of countable collections of sets in $\mathcal{F}_z$'s and their complements in Ω. Let μ be the measure on $\mathcal{F}$ such that μ agrees with μ_z on Ω_z.

By Decomposition Lemma 5.3.5, for any $x \in X$, the collection $(P_z x)_{z\in\Lambda}$ has at most countably many non-zero terms and $x = \sum_{i=1}^n P_{z_i} x = \sum_{i=1}^n x_i$ for any enumeration of the non-zero terms in $(P_z x)_{z\in\Lambda}$ It follows from

$$\|x\| = \lim_m \Bigl\|\sum_{i=1}^m x_i\Bigr\| = \lim_m \sum_{i=1}^m \|x_i\| = \sum_{i=1}^\infty \|x_i\|.$$

In particular, the sequence $\bigl(\|x_i\|\bigr)$ is summable. Let $f_i = T_{z_i} x_i$ in $L_1(\mu_{z_i})$. Let f be the function on Ω which agrees with f_i on Ω_{z_i} and is zero otherwise. Since the sequence $\bigl(\|x_i\|\bigr)$ is summable, we have $f \in L_1(\mu)$ and

$$\|f\| = \sum_{i=1}^\infty \|f_i\| = \sum_{i=1}^\infty \|x_i\| = \|x\|.$$

Put $f = Tx$. That is, $Tx = \sum_{z\in\Lambda} T_z P_z x$, where the sum has only countably many non-zero pair-wise disjoint terms and converges independently of how one orders the terms. This defines a linear isometry $T\colon X \to L_1(\mu)$.

It is left to show that T is onto. Let $f \in L_1(\mu)$. It is easy to see that the set of restrictions of f to Ω_z as $z \in \Lambda$ has at most countably many non-zero terms. Enumerate them: let f_i be the restriction of f to Ω_{z_i}. Then $\|f\| = \sum_{i=1}^{\infty}\|f_i\|$. For every i, put $x_i = T_{z_i}^{-1} f_i$, $x_i \in B_{z_i}$. Then $\sum_{i=1}^{\infty}\|x_i\|$ is summable, so that $x = \sum_{i=1}^{\infty} x_i$ converges in X. It is easy to see that $Tx = f$. □

8.5.2 Corollary. *The spaces* $\mathrm{ca}(\mathcal{F})$ *and* $\mathcal{M}(K)$ *introduced in Sections 8.3 and 8.4, respectively, are* L_1*-spaces.*

Proof. Every principal band in $\mathrm{ca}(\mathcal{F})$ is an L_1-space by Proposition 8.3.13. Lemma 8.5.1 yields that $\mathrm{ca}(\mathcal{F})$ is an L_1-space. By definition, $\mathcal{M}(K) = \mathrm{ca}(\mathcal{B})$, where $\mathcal{B}$ is the Baire σ-algebra on K; hence $\mathcal{M}(K)$ is an L_1-space. □

We can now easily deduce the following fact of Banach space theory:

8.5.3 Theorem (Arens-Kelly)**.** *The extreme points of the unit ball of* $C(K)^*$ *are the functionals of the form* $\pm\delta_t$*, where* $t \in K$*.*

Proof. By 8.4.3, the functionals of the form $\pm\delta_t$, where $t \in K$, are the atoms in $\mathcal{M}(K) = C(K)^*$. By Corollary 8.5.2, $\mathcal{M}(K) = L_1(\mu)$ for some measure μ. Now apply 5.4.5. □

8.5.4 Lemma. *Let* Y *be a closed sublattice of* $L_1(\mu)$*. Suppose that there exists* $u \in Y$ *such that* $u > 0$ *a.e. Then* Y *is an* L_1*-space.*

Proof. Suppose that Y is a closed sublattice of $L_1(\Omega, \mathcal{F}, \mu)$. As in 1.6.17, let $\nu = u\mu$. The map $T\colon L_1(\Omega, \mathcal{F}, \mu) \to L_1(\Omega, \mathcal{F}, \nu)$ defined by $Tg = u^{-1}g$ is a surjective lattice isometry. Observe that $T(Y)$ is a closed sublattice of $L_1(\Omega, \mathcal{F}, \nu)$ containing $\mathbb{1}$. The conclusion now follows from Theorem 3.1.25. □

8.5.5 Theorem (Kakutani's AL Representation Theorem)**.** *Every AL-space is lattice isometric to* $L_1(\mu)$ *for some measure* μ*.*

Proof. Let X be an AL-space. By Theorem 8.1.9, X^* is an AM-space with unit. By Corollary 8.1.3, X^* is lattice isometric to $C(K)$ for some compact K. Riesz Representation Theorem 8.4.2 yields that X^{**} is lattice isometric to $\mathcal{M}(K)$. Corollary 8.5.2 asserts that $\mathcal{M}(K)$ is an L_1-space. Therefore, we may assume that X is a closed sublattice of $L_1(\mu)$.

Let $x \in X_+$. By Lemma 8.5.1, it suffices to prove that the band B_x^X generated by x in X is an L_1-space. First, we claim that B_x^X is contained in the band $B_x^{L_1(\mu)}$ generated by x in $L_1(\mu)$. Indeed, suppose that $0 \leqslant y \in B_x^X$. Then $y \wedge nx \uparrow y$ in X. Since X is order continuous, it follows that $y \wedge nx \to y$ in norm in X and, therefore, in $L_1(\mu)$. Now Monotone Convergence Lemma 2.5.3 yields that $y \wedge nx \uparrow y$ in $L_1(\mu)$ and, therefore, $y \in B_x^{L_1(\mu)}$.

By Exercise 5.1.28, $B_x^{L_1(\mu)}$ consists of all the functions in $L_1(\mu)$ whose support is contained in some measurable set A. Moreover, $B_x^{L_1(\mu)}$ may be identified with $L_1(A, \mathcal{F}_{|A}, \mu_{|A})$. We can identify B_x^X with a closed sublattice of $L_1(A, \mathcal{F}_{|A}, \mu_{|A})$. Being a weak unit in $L_1(A, \mathcal{F}_{|A}, \mu_{|A})$, x is a.e. positive on A. It follows from Lemma 8.5.4 that B_x^X is an L_1-space. □

Note that the measure constructed in Theorem 8.5.5 need not be σ-finite. Indeed, if we apply the theorem to X where $X = L_1(\mu)$ for a non-σ-finite measure μ, then the theorem yields the same measure μ back.

8.5.6 Corollary. *Let X be an AL-space with a weak unit $e \geqslant 0$. Then X is lattice isometric to $L_1(\nu)$ for some* finite *measure ν such that e corresponds to $\mathbb{1}$.*

Proof. Let $L_1(\mu)$ be the representation of X given by the AL Representation Theorem 8.5.5 and let e corresponds to some $u \in L_1(\mu)$. The map $Tv = \frac{v}{u}$ is a lattice isometry from $L_1(\mu)$ onto $L_1(\nu)$, where $\nu = u\mu$, and $Tu = \mathbb{1}$; see 1.6.17. It follows from $\mathbb{1} \in L_1(\nu)$ that ν is finite. □

8.6 Classification of $L_p(\mu)$ spaces

We have proved in the previous section that AL-spaces are precisely those Banach lattices that are lattice isometric to $L_1(\mu)$ for some measure μ. We will later extend this in Section 10.7 to a characterization of Banach lattice that are lattice isometric to some $L_p(\mu)$ space for $1 \leqslant p < \infty$. In the current section, we will show that, up to a lattice isometry, μ may actually be chosen to be a combination of most familiar and easy measures. In particular, in the separable case, μ may be taken to be the sum of the counting measure on a finite or a countable set and the Lebesgue measure on $[0, 1]$.

Here is the outline. Since every $L_p(\mu)$ space is order continuous, it may be written as a disjoint direct sum of the band generated by the atoms, and its disjoint complement, which has to be non-atomic. Thus, the problem reduces to separately characterizing discrete and non-atomic L_p-spaces. We will show that every discrete L_p-space is lattice isometric to $\ell_p(\Gamma)$ for some Γ; in particular, in the separable case, we end up with either ℓ_p or ℓ_p^n for some n. On the other hand, we will show that every separable non-atomic L_p-space is lattice isometric to $L_p[0, 1]$ with the Lebesgue measure. Finally, we will look at the non-separable case.

8.6.1 Theorem. *A discrete $L_p(\mu)$ space $(1 \leqslant p < \infty)$ is lattice isometric to $\ell_p(\Lambda)$ for some set Λ.*

Proof. Let Λ be the set of all (positive) atoms of norm one in $L_p(\mu)$. For every $a \in \Lambda$, let P_a and λ_a be the coordinate projection and the coordinate functional of a, i.e., $P_a(x) = \lambda_a(x)a$ for every $x \in L_p(\mu)$, as in 5.4.17. By Theorem 5.4.10, Λ is a maximal disjoint family of positive elements. By Decomposition Lemma 5.3.5, for every $x \in L_p(\mu)$ there exists a sequence (a_i) in Λ such that $x = \sum_{i=1}^{\infty} \lambda_{a_i}(x)a_i$. It follows that

$$\|x\| = \lim_{n\to\infty} \Bigl\|\sum_{i=1}^{n} \lambda_{a_i}(x)a_i\Bigr\| = \lim_{n\to\infty} \Bigl(\sum_{i=1}^{n} |\lambda_{a_i}(x)|^p\Bigr)^{\frac{1}{p}} = \Bigl(\sum_{i=1}^{\infty} |\lambda_{a_i}(x)|^p\Bigr)^{\frac{1}{p}}.$$

Hence, the map that sends $x \in L_p(\mu)$ into the family $\bigl(\lambda_a(x)\bigr)_{a\in\Lambda}$ is a lattice isometry between $L_p(\mu)$ and $\ell_p(\Lambda)$. □

It is easy to see that if Λ is uncountable then $\ell_p(\Lambda)$ is non-separable. This immediately yields the following:

8.6.2 Corollary. *A separable discrete $L_p(\mu)$ space $(1 \leqslant p < \infty)$ is lattice isometric to ℓ_p or ℓ_p^n for some $n \in \mathbb{N}$.*

We now proceed to separable non-atomic case. We will make use of the concept of a *measure algebra*. Our approach is based on that in pp. 398-400 in [Roy88]. A Boolean algebra $\mathcal{B}$ is said to be a ***Boolean σ-algebra*** if every countable subset of $\mathcal{B}$ has a supremum. If, in addition, there is a function $\mu\colon \mathcal{B} \to \mathbb{R}_+$ satisfying $\mu(a) = 0$ iff $a = \nvdash$ and $\mu\bigl(\bigvee_{i=1}^{\infty}\bigr) = \sum_{i=1}^{\infty} \mu(a_i)$ for every disjoint sequence (a_i) in $\mathcal{B}$ ("disjoint" here means $a_i \wedge a_j = \nvdash$ whenever $i \neq j$), we say that $(\mathcal{B}, \mu)$ is a ***measure algebra***. We call such a μ a ***measure*** on $\mathcal{B}$. Two measure algebras $(\mathcal{B}_1, \mu_1)$ and $(\mathcal{B}_2, \mu_2)$ are said to be isomorphic if there is a Boolean algebra isomorphism Φ from $\mathcal{B}_1$ onto $\mathcal{B}_2$ that preserves the measure: $\mu_2\bigl(\Phi(a)\bigr) = \mu_1(a)$ for all $a \in \mathcal{B}_1$.

A measure algebra is a metric space under the metric $d(a, b) = \mu(a\triangle b)$, where $a\triangle b = (a \wedge \bar{b}) \vee (\bar{a} \wedge b)$, the ***symmetric difference*** of a and b. It is easy to verify that d is indeed a metric. A measure algebra is said to be ***separable*** when it is separable as a metric space.

A non-zero element a in a measure algebra $\mathcal{B}$ is said to be an ***atom*** if $b \leqslant a$ implies $b = 0$ or $b = a$. We say that $\mathcal{B}$ is non-atomic if it has no atoms.

Let $(\Omega, \mathcal{F}, \mu)$ be a finite measure space. In general, $(\mathcal{F}, \mu)$ need not be a measure algebra as some sets may have zero measure. However, we can make it into a measure algebra as follows. Let $\mathcal{N}$ be the collection of all sets of zero measure in $\mathcal{F}$. For $A, B \in \mathcal{F}$, declare that $A \sim B$ if $\mu(A\triangle B) = 0$. It can be easily verified that this defines an equivalence relation on $\mathcal{F}$. Let $\mathcal{F}/\mathcal{N}$ be the set of all equivalence classes. It is a Boolean σ-algebra under the operations induced from $\mathcal{F}$. We write $[A]$ for the equivalence class of A. It is easy to see that if $A \sim B$ then $\mu(A) = \mu(B)$, so that μ induces a function on $\mathcal{F}/\mathcal{N}$; we will denote this function by $\bar{\mu}$. Finally, it is a straightforward exercise that the pair $(\mathcal{F}/\mathcal{N}, \bar{\mu})$ is a measure algebra. We call it the measure algebra induced by μ.

There is an alternative way to view the preceding construction. Let $L_1(\mu) = L_1(\Omega, \mathcal{F}, \mu)$. It is clear that $A \sim B$ in $\mathcal{F}$ iff $\mathbb{1}_A = \mathbb{1}_B$ a.e., i.e., $\mathbb{1}_A$ and $\mathbb{1}_B$ are equal in $L_1(\mu)$. Thus, we may identify the Boolean algebra $\mathcal{F}/\mathcal{N}$ with the Boolean algebra of all characteristic functions in $L_1(\mu)$. The latter is precisely $\mathcal{C}_{\mathbb{1}}$, the Boolean algebra of the components of $\mathbb{1}$ in $L_1(\mu)$. Furthermore, $\bar{\mu}\big([A]\big) = \mu(A) = \|\mathbb{1}_A\|_{L_1}$. Thus, $\big(\mathcal{F}/\mathcal{N}, \bar{\mu}\big)$ is isometric (as a measure algebra) to $\big(\mathcal{C}_{\mathbb{1}}, \|\cdot\|_{L_1}\big)$.

The following two exercises essentially verify that our use of terminology is consistent.

8.6.3 Exercise. Let $1 \leqslant p < \infty$; let $(\Omega, \mathcal{F}, \mu)$ be a finite measure space. TFAE:

(i) $(\Omega, \mathcal{F}, \mu)$ is non-atomic as a measure space;
(ii) $\big(\mathcal{F}/\mathcal{N}, \bar{\mu}\big)$ is non-atomic as a measure algebra;
(iii) $L_p(\mu)$ is non-atomic as a vector lattice.

8.6.4 Exercise. Let $1 \leqslant p < \infty$; let $(\Omega, \mathcal{F}, \mu)$ be a finite measure space. Then its measure algebra $(\mathcal{F}/\mathcal{N}, \bar{\mu})$ is separable iff $L_p(\mu)$ is separable as a Banach space.

We are going to critically use the following important fact of Measure Theory; see, e.g., [Roy88, p. 399].

8.6.5 Theorem (Caratheodory)**.** *Every non-atomic separable measure algebra with $\mu(\mathbb{1}) = 1$ is isomorphic to the measure algebra induced by the Lebesgue measure on $[0, 1]$.*

We are now ready to characterize nonatomic separable $L_p(\mu)$-spaces.

8.6.6 Theorem. *Let $1 \leqslant p < \infty$. Every non-atomic separable $L_p(\mu)$ space is lattice isometric to $L_p[0, 1]$.*

Proof. Suppose that $L_p(\mu) = L_p(\Omega, \mathcal{F}, \mu)$ is separable and non-atomic. WLOG, μ is a probability measure. Indeed, since $L_p(\mu)$ is separable, it has a weak unit by Corollary 5.1.18 and, therefore, μ is σ-finite by Exercise 1.5.13(x). By Proposition 1.6.19, $L_p(\mu)$ is lattice isometric to L_p on some probability measure.

Since $L_p(\mu)$ is non-atomic and separable, so is its measure algebra $(\mathcal{F}/\mathcal{N}, \bar{\mu})$. By Caratheodory's theorem, this measure algebra is isomorphic to the measure algebra induced the Lebesgue measure λ on $[0,1]$. We may view it as a bijection Φ from the set of all (equivalence classes of) characteristic functions of measurable sets in $L_p(\mu)$ onto the set of all (equivalence classes of) characteristic functions of measurable sets in $L_p[0,1]$ which preserves the L_1-norm of the function, i.e., for every $A \in \mathcal{F}$ there is a Lebesgue measurable set B such that $\Phi(\mathbb{1}_A) = \mathbb{1}_B$ such that $\lambda(B) = \|\mathbb{1}_B\|_{L_1} = \|\mathbb{1}_A\|_{L_1} = \mu(A)$.

We can then extend Φ to the linear operator from the space of all simple functions in $L_p(\mu)$ to the space of all simple functions in $L_p(\mu)$ via $T\Bigl(\sum_{i=1}^n \lambda_i \mathbb{1}_{A_i}\Bigr) = \sum_{i=1}^n \lambda_i \Phi(\mathbb{1}_{A_i})$. By Exercise 7.2.9, T is well-defined and is a lattice isomorphism between the spaces of simple functions. Since every simple function admits a disjoint representation, it is easy to see that T is an isometry in the L_p-norm. Hence, T extends to a lattice isometry from $L_p(\mu)$ onto $L_p[0,1]$. □

8.6.7 Remark. Analyzing the proof of the theorem, we observe that in the case when μ is a probability measure, characteristic functions in $L_p(\mu)$ correspond to characteristic functions in $L_p[0,1]$ and $\mathbb{1}$ goes to $\mathbb{1}$.

8.6.8 Corollary. *Let $1 \leqslant p < \infty$. Every separable $L_p(\mu)$ space is lattice isometric to one of the following: ℓ_p, ℓ_p^n, $L_p[0,1]$, $\ell_p \oplus L_p[0,1]$, or $\ell_p^n \oplus L_p[0,1]$, where $n \in \mathbb{N}$ and the sums are disjoint band decompositions.*

Proof. As in the preceding proof, we may assume WLOG that the measure is finite. Let B be the band in $L_p(\mu)$ generated by all the atoms. Then $L_p(\mu) = B \oplus B^d$, and B^d is non-atomic. There exists a measurable set Ω_0 such that $B = L_p(\Omega_0, \mu)$ and $B^d = L_p(\Omega \setminus \Omega_0, \mu)$, B is discrete, and B^d is non-atomic. Applying the preceding theorems completes the proof. □

8.6.9 Remark. Again, it is clear from the preceding proof that in the case when μ is finite and $L_p[0,1]$ is among the summands, then the corresponding band projection sends constant functions in $L_p(\mu)$ to constant functions in $L_p[0,1]$.

8.6.10. *Non-separable case.* Suppose now that $L_p(\mu)$ is not separable, where $1 \leqslant p < \infty$ and μ is a measure. As before, we may decompose it into a disjoint direct sum of two L_p-spaces, one discrete and the other non-atomic. The discrete one is lattice isometric to $\ell_p(\Lambda)$ by Theorem 8.6.1, so it is left to characterize non-separable non-atomic $L_p(\mu)$-spaces. This may be done by replacing Caratheodory's Theorem in the preceding argument with Maharam's classification of general measure algebras. One can prove that every non-atomic $L_p(\mu)$ may be written as a disjoint direct sum of (finitely or infinitely many) spaces of the form $L_p\bigl([0,1]^{\mathfrak{a}}\bigr)$, where $\mathfrak{a}$ is a cardinal number, and $[0,1]^{\mathfrak{a}}$ is equipped with the product Lebesgue measure. We refer the reader to Chapter 5 in [Lac74] for details.

Here is another (easier) approach, which often allows one to reduce non-separable case to the separable one. We use the fact that a non-atomic measure has the following intermediate value property: if $0 < \alpha < \mu(A) < \infty$ then $\alpha = \mu(B)$ for some $B \subset A$; see, e.g., [Hal50, p. 74].

8.6.11 Proposition. *Let $1 \leqslant p < \infty$ and $L_p(\mu) = L_p(\Omega, \mathcal{F}, \mu)$ a non-atomic finite measure space. There exists a sub-σ-algebra $\mathcal{G}$ of $\mathcal{F}$ such that μ is non-atomic and separable on $\mathcal{G}$. In particular, the closed sublattice $L_p(\Omega, \mathcal{G}, \mu_{|\mathcal{G}})$ of $L_p(\Omega, \mathcal{F}, \mu)$ is non-atomic and separable.*

Proof. WLOG μ is a probability measure. Partition Ω into a disjoint union $\Omega = A_0 \cup A_1$ such that $\mu(A_0) = \mu(A_1) = \frac{1}{2}$. Further, partition each of A_0 and A_1 into two subsets of measure $\frac{1}{4}$. Proceeding inductively, we get a "dyadic tree" of sets in $\mathcal{F}$. Let $\mathcal{F}_0$ be the σ-algebra generated by these sets. It is straightforward to verify that $\mathcal{G}$ is as required. $\square$

Chapter 9

L_1-representations

In Chapter 4, we showed that every principal ideal in Archimedean vector lattice may be represented in a $C(K)$ space, so that vector lattices locally behave like spaces of continuous functions. This representation allowed us to reduce many problems about vector lattices to topology. In this chapter, we explore a somewhat dual approach. We will show that under some (rather mild) assumption, one can naturally embed a vector lattice into an $L_1(\mu)$ space for some measure μ. This allows one to reduce many problems about vector lattices to Measure Theory.

It shows that L_1-spaces play very special role in the theory of vector lattices. It also explains the deep connections between vector lattices and measure theory. This also "places" the theory of vector lattices somewhere "between" $C(K)$ and $L_1(\mu)$ spaces, between topology and measure theory.

We will also explore in this Chapter some properties of $L_1(\mu)$ spaces that will be useful in the context of L_1-representations of vector and Banach lattices.

9.1 Strictly positive functionals

Let X be a vector lattice. A functional f on X is said to be ***strictly positive*** if $f(x) > 0$ whenever $x > 0$. This clearly implies that $f \geqslant 0$.

9.1.1 Example. Let $X = L_1(\mu)$ for a measure μ. Put $f(x) = \int x$. Then f is strictly positive. Furthermore, if μ is finite then this also defines a strictly positive functional on $L_p(\mu)$ for all $1 \leqslant p \leqslant \infty$. Similarly, let $X = C[0,1]$ and, again, $f(x) = \int x$; it is still a strictly positive functional.

9.1.2 Exercise. There is no strictly positive functional on $\mathbb{R}^{\mathbb{N}}$.

9.1.3 Exercise. Let $X = \ell_\infty(\Omega)$, where Ω is an uncountable set. Then X admits no strictly positive functional.

We will see that the existence of a strictly positive functional leads to many useful consequences.

9.1.4 Exercise. If a vector lattice X admits a strictly positive functional then X is Archimedean.

In the proof of Theorem 2.1.16 that L_p spaces are order complete we used the integral (which is a strictly positive functional) to reduce order completeness to σ-order completeness. This argument may be extended as follows.

9.1.5 Proposition. *Let X be a vector lattice with a strictly positive functional. Then X is order complete iff it is σ-order complete.*

Proof. If X is order complete then it is trivially σ-order complete. Suppose now that X is σ-order complete and h is a strictly positive functional on X. Let $A \subseteq X$ be non-empty and bounded above. Let $B = A^{\uparrow}$, the set of all upper bounds of A. It suffices to prove that B has a minimal element.

The set $h(B)$ is non-empty and bounded below in $\mathbb{R}$, hence $\lambda = \inf h(B)$ exists. It follows that there exists (y_n) in B such that $h(y_n) \to \lambda$. Since X is σ-order complete, $y_0 := \inf y_n$ exists. We have $y_0 \leqslant y_n$ for every n, so that $h(y_0) \leqslant h(y_n)$; this yields $h(y_0) \leqslant \lambda$.

It is easy to see that $y_0 \in B$. We claim that $y_0 = \min B$. Suppose not, then there exists $y \in B$ such that $y_0 \wedge y < y_0$. Taking into account that $y_0 \wedge y \in B$ and h is strictly positive, we get $\lambda \leqslant h(y_0 \wedge y) < h(y_0) \leqslant \lambda$, a contradiction. □

9.1.6 Exercise. If a vector lattice X admits a strictly positive functional then X has the countable sup property. The converse is false in general.

9.1.7. The following process allows one to construct strictly positive functionals out of positive functionals. Let f be a positive functional on a vector lattice X. Consider the null ideal $N_f = \bigl\{x \in X \ :\ f\bigl(|x|\bigr) = 0\bigr\}$ as in 6.5.12. Let X/N_f be the quotient space. f induces a functional $\tilde{f}$ on X/N_f via $\tilde{f}(x + N_f) = f(x)$. Note that $\tilde{f}$ is strictly positive. Indeed, let $0 < \varphi \in X/N_f$. By Exercise 3.4.4(i), there exists $x \in X_+$ with $\varphi = \tilde{x}$. It follows from $\varphi \neq 0$ that $x \notin N_f$, so that $\tilde{f}(\varphi) = f(x) > 0$.

The following proposition shows that strictly positive functionals are related to weak units.

9.1.8 Proposition. *Suppose that $X^{\sim}$ separates points of X.*

(i) *Every weak unit of $X^{\sim}$ is a strictly positive functional;*

(ii) *Let $v \in X_+$. If $\hat{v}$ is strictly positive (as a functional on $X^\sim$) then v is a weak unit.*

Proof. (i) Let $f \in X^\sim$ be a weak unit but not strictly positive. Then the band B_f generated by f equals $X^\sim$ and there exists $x \in X$ such that $x > 0$ but $f(x) = 0$. Let $g \in X^\sim_+$. Then $g \wedge nf \uparrow g$ by Proposition 5.1.3. Note that $(g \wedge nf)(x) = 0$ for every n, so that $g(x) = \lim_n (g \wedge nf)(x) = 0$ by Proposition 6.6.1(iii). It follows that for every $g \in X^\sim$ we have $g(x) = g^+(x) - g^-(x) = 0$. This contradicts the assumption that $X^\sim$ separates points of X.

(ii) Suppose not. Then there exists $u > 0$ such that $u \perp v$. Since $X^\sim$ separates points of X, there exists a positive functional f such that $f(u) > 0$. By Lemma 6.3.10, there exists $g \in [0, f]$ such that $g(u) = f(u) \neq 0$ and $g(v) = 0$; a contradiction. □

9.1.9 Example. The converses of both statements in Proposition 9.1.8 are false.

(i) Let $X = C[0, 1]$. Let f and g be as in Exercise 6.6.4. Note that f is strictly positive; it is not a weak unit in $X^\sim$ as $f \perp g$.

(ii) Let $X = C[0, 1]$, $v \in X$ defined by $v(t) = t$, and $f \in X^\sim$ defined by $f(x) = x(0)$; a point evaluation functional. Then v is a weak unit in X, still $f(v) = 0$.

Recall that if X is a Banach lattice than $X^\sim = X^*$ and it separates points of X, so that Proposition 9.1.8 applies. For Banach lattices, we will refine it in Section 5.5.

9.1.10 Proposition. *Let v be a positive vector in a Banach lattice X. Then v is quasi-interior iff $\hat{v}$ is a strictly positive functional on X^*.*

Proof. Suppose that v is quasi-interior and $f \in X^*_+$. If $f(v) = 0$ then f vanishes on I_v. Since I_v is dense, it follows that $f = 0$. Conversely, suppose $\overline{I_v} \neq X$. Then, by Proposition 6.7.8, there exists a positive non-zero functional which vanishes at v. □

9.1.11 Proposition. *Every separable Banach lattice admits a strictly positive functional.*

Proof. Suppose that X is a separable Banach lattice. Let (x_n) be a dense sequence in S_X^+. Using Exercise 6.7.6, for each n we find $f_n \in S_{X^*}^+$ such that $f_n(x_n) = 1$. Put $f = \sum_{n=1}^\infty \frac{1}{2^n} f_n$. Let $x \in S_X^+$. Then $\|x - x_n\| < \frac{1}{2}$ for some n. It follows from

$$\bigl|f_n(x) - 1\bigr| = \bigl|f_n(x) - f_n(x_n)\bigr| \leqslant \|f_n\| \|x - x_n\| < \tfrac{1}{2}$$

that $f_n(x) > \frac{1}{2}$. Hence $f(x) \geqslant \frac{1}{2^n} f_n(x) > \frac{1}{2^{n+1}} > 0$. It follows that f is strictly positive. □

Recall from Corollary 5.1.18 that every separable Banach lattice has a weak unit. On the other hand, Example 5.1.19 shows that an order continuous Banach lattice with a weak unit need not be separable.

9.1.12 Theorem. *Every order continuous Banach lattice with a weak unit admits a strictly positive functional (which is order continuous by Corollary 6.8.5).*

Proof. Let X be an order continuous Banach lattice with a weak order unit e. Note that X is order complete by Theorem 2.5.22, and every functional on X is order continuous (Corollary 6.8.5). Let f be a non-zero positive functional on X. It follows from 6.5.12 that $X = N_f \oplus C_f$ is a band decomposition. Let P_f be the band projection for C_f. Put $e_f = P_f e$. Since e is a weak unit, we have $0 \neq f(e) = f\big(e_f + (I - P_f)e\big) = f(e_f)$ because $(I - P_f)e \in N_f$. Thus, $e_f \neq 0$ for every non-zero $f \in X^*_+$.

Let Λ be a maximal disjoint collection in X^*_+. Without loss of generality, $\|f\| = 1$ for every $f \in \Lambda$. For any distinct $f, g \in \Lambda$, $f \perp g$ yields $C_f \perp C_g$ by Theorem 6.6.17, so that $e_f \perp e_g$. It follows that $\{e_f : f \in \Lambda\}$ is a disjoint collection of non-zero vectors in $[0, e]$. Since X is order continuous, it follows that Λ is at most countable by Lemma 2.5.27. Let (f_n) be an enumeration of Λ. Put $f = \sum_{n=1}^{\infty} \frac{1}{2^n} f_n$.

It is left to show that f is strictly positive. Suppose not. Then $X = N_f \oplus C_f$ is a non-trivial band decomposition of X. Let g be any non-zero positive functional on N_f; extend it to all of X by setting it to zero on C_f. Then $C_f \subseteq N_g$ yields $g \perp f$ by Theorem 6.6.17. Since each $f_n \leqslant 2^n f$ for every n, it follows that $g \perp f_n$. This contradicts the maximality of Λ. $\square$

9.2 L_1-representations of vector lattices

Throughout this section, we fix a vector lattice X and a strictly positive functional h on X. For $x \in X$, define $\|x\|_h = h\big(|x|\big)$.

9.2.1 Exercise. $\|\cdot\|_h$ is an AL-norm on X.

It follows by Exercise 8.0.2 that the completion of X under this norm is an AL-space. We will denote it by $\widetilde{X}$. By AL Representation Theorem 8.5.5, $\widetilde{X}$ is lattice isometric to $L_1(\mu)$ for some measure μ. Thus, we have proved the following:

9.2.2 Theorem. *Every vector lattice with a strictly positive linear functional h embeds, as a dense sublattice, into $L_1(\mu)$ for some measure μ, such that $h(x) = \int x \, d\mu$ for every $x \in X_+$.*

We will call $L_1(\mu)$ an L_1***-representation space for*** X ***and*** h. This representation allows us to view X as a space of measurable functions, and we may now use the power of measure theory to study X.

9.2.3 Example. Let $X = C[0, 1]$. Let $h(x) = \int_0^1 x$. Clearly, h is a strictly positive functional on X. Then $\|\cdot\|_h$ is the L_1-norm; hence $\widetilde{X} = L_1[0, 1]$.

9.2.4 Example. Let $X = L_p(\mu)$, where μ is a finite measure and $1 \leqslant p \leqslant \infty$. Put $h(x) = \int x\,d\mu$. Then $\widetilde{X} = L_1(\mu)$.

9.2.5 Exercise. Let $L_1(\mu)$ be an L_1-representation for X and h. If (x_n) is a weakly null positive sequence in X then $\|x_n\|_{L_1(\mu)} \to 0$. Is this statement true without the positivity assumption?

The L_1-representation approach described above is dual to the $C(K)$-representation approach described in Chapter 4 in the following sense. Consider the principal ideal I_h generated by h in the order dual $X^\sim$. On I_h, we define a norm as in 2.2.2, that is, $\|f\|_h = \inf\bigl\{\lambda \geqslant 0 \;:\; |f| \leqslant \lambda h\bigr\}$. By Proposition 6.6.1(ii), $X^\sim$ is order complete and, therefore, Archimedean by Proposition 2.1.6 and uniformly complete by Theorem 2.2.9. It follows from the $C(K)$-Representation Theorem 4.1.9 that $\bigl(I_h, \|\cdot\|_h\bigr)$ is lattice isometric to $C(K)$ for some compact space K.

9.2.6 Proposition. *Let $\widetilde{X}$ be the L_1-representation for a vector lattice X and a strictly positive functional h. Then its dual $\widetilde{X}^*$ is lattice isometric to the principal ideal I_h in $X^\sim$. In particular, $L_1(\mu)^*$ is lattice isometric to $C(K)$, where μ and K are as above.*

Proof. Let $f \in I_h$. For every $x \in X$, we have

$$\bigl|f(x)\bigr| \leqslant |f|\bigl(|x|\bigr) \leqslant \|f\|_h h\bigl(|x|\bigr) = \|f\|_h \|x\|_h.$$

Since X is norm dense in $\widetilde{X}$, it follows that f extends to a functional $\tilde{f}$ on $\widetilde{X}$ with $\|\tilde{f}\| \leqslant \|f\|_h$. This defines a contraction $T\colon I_h \to \widetilde{X}^*$.

Conversely, let $g \in \widetilde{X}^*$ with $\|g\|_{\widetilde{X}^*} = 1$. Let f be its restriction to X. For every $x \in X_+$, we have $\bigl|f(x)\bigr| = \bigl|g(x)\bigr| \leqslant \|x\|_h = h(x)$. It follows that $|f| \leqslant h$, so that $f \in I_h$ with $\|f\|_h \leqslant 1$. Clearly, $g = Tf$. It follows that T is a surjective isometry. It is a lattice isomorphism by Theorem 1.6.13(iii). $\square$

In the next few pages we will prove several refinements of Theorem 9.2.2 that show that if X has various additional properties, e.g., order completeness, having a weak unit, or being an order continuous Banach lattice, then we can say more about the way X "sits" in $L_1(\mu)$. We will routinely identify $\widetilde{X}$ with $L_1(\mu)$.

Recall that every L_1-space is order continuous by Exercise 2.5.13 and, therefore, order complete. Recall also that a sublattice Y of a vector lattice X is order dense iff it is regular and dense with respect to order convergence; see Theorem 3.2.16. Also, every ideal in X is a regular sublattice.

9.2.7 Theorem. *Let $L_1(\mu)$ be an L_1-representation for a vector lattice X and a strictly positive functional h.*

(i) *X is order dense in $L_1(\mu)$ iff X is regular in $L_1(\mu)$ iff h is order continuous;*

(ii) *X is an ideal in $L_1(\mu)$ iff h is order continuous and X is order complete.*

Proof. As before, we can identify $L_1(\mu)$ with the completion $\widetilde{X}$ of $\bigl(X, \|\cdot\|_h\bigr)$.

(i) If X is order dense in $\widetilde{X}$ then it is regular by Proposition 3.2.13. Conversely, suppose that X is regular in $\widetilde{X}$. Since X is norm dense in $\widetilde{X}$, it is also dense with respect to order convergence by Theorem 2.5.4. It follows from Theorem 3.2.16 that X is order dense in $\widetilde{X}$.

Suppose that X is regular in $\widetilde{X}$. Let $x_\alpha \downarrow 0$ in X. Then $x_\alpha \downarrow 0$ in $\widetilde{X}$. Since $\widetilde{X}$ is order continuous, it follows that $h(x_\alpha) = \|x_\alpha\|_h \to 0$. Therefore, h is order continuous.

Finally, suppose that h is order continuous and show that X is regular. Suppose $x_\alpha \downarrow 0$ in X. It follows that $\|x_\alpha\|_h = h(x_\alpha) \downarrow 0$, so that $x_\alpha \downarrow 0$ in $\widetilde{X}$ by the Monotone Convergence Lemma 2.5.3.

(ii) If X is an ideal then it is regular, so that h is order continuous by (i). Being an ideal in an order complete space, X is itself order complete. Conversely, suppose that X is order complete and h is order continuous. Then X is order dense (hence regular) in $\widetilde{X}$ by (i) and is an ideal by Proposition 3.3.19.

□

Recall that by Proposition 9.1.5, the order completeness assumption in (ii) may be replaced with σ-order completeness.

9.2.8 Exercise. Let X be a Banach lattice with a quasi-interior point e. Then X^* is lattice (but not norm) isomorphic to an ideal in $L_1(\mu)$ for some measure μ and $\int f\,d\mu = f(e)$ for every $f \in X^*$.

9.2.9. *L_1-representations with a weak unit.* Suppose that, in addition to having a strictly positive order continuous functional h, X has a weak unit e. As before, let $\widetilde{X}$ be the completion of X with respect to $\|\cdot\|_h$. Since X is order dense in $\widetilde{X}$ by Theorem 9.2.7(i), it follows that e is a weak unit in $\widetilde{X}$ by Exercise 3.2.11 By Corollary 8.5.6, we can find a *finite* measure μ such that $\widetilde{X}$ is lattice isometric to $L_1(\mu)$ and e corresponds to $\mathbb{1}$. In this case, $\mu(\Omega) = \int \mathbb{1}\,d\mu = h(e)$. Therefore, if we scale e and h so that $h(e) = 1$ then the resulting measure is a probability measure.

If, in addition, X is order complete then, by Theorem 9.2.7, X is an ideal in $L_1(\mu)$. In this case, it follows from $\mathbb{1} \in X$ that X contains $L_\infty(\mu)$ as an ideal. It can be easily verified that the inclusion of $L_\infty(\mu)$ into X is a lattice isometry between $L_\infty(\mu)$ and $\bigl(I_e, \|\cdot\|_e\bigr)$. Since e is a weak unit, it follows that $L_\infty(\mu)$ is order dense in X. Let's summarize:

9.2.10 Theorem. *Let X be a vector lattice with an order continuous strictly positive functional h and a weak unit e. One can choose an L_1-representation $L_1(\mu)$ for X and h so that μ is finite and e corresponds to $\mathbb{1}$. If, in addition, X is order complete, then X contains $L_\infty(\mu)$ as an order dense ideal.*

9.2.11 Example. The preceding construction breaks down if we do not assume that h is order continuous. Take $X = C[0,1]$ and let h be the integral. While h is strictly positive, it is not order continuous. In this case, $\widetilde{X} = L_1[0,1]$. However, $C[0,1]$ fails to be order dense in $L_1[0,1]$, and a weak unit of $C[0,1]$ need not be a weak unit in $L_1[0,1]$; see Exercise 1.5.13(xi).

9.2.12 Example. The following example shows that, in general, order continuity of h cannot be omitted if one wants X to contain $L_\infty(\mu)$ and be order dense in $L_1(\mu)$.

Let $X = \ell_\infty$. Clearly, X is order complete and contains a weak (even strong) unit. Let $f, g \in X^*_+$, where f is the limit with respect to some free ultrafilter (see Exercise 6.6.15) and $g(x) = \sum_{n=1}^\infty \frac{x_n}{2^n}$. Put $h = f + g$. It follows from $h \geqslant g$ that h is strictly positive. As in Exercise 6.6.15, f and h are not order continuous, but f is a lattice homomorphism.

Let $\widetilde{X}$ be the L_1-representation for X and h. We claim that $\widetilde{X}$ may be identified with $L_1(\Omega, \mu)$ where $\Omega = \mathbb{N} \cup \{\infty\}$ and μ is defined by $\mu\bigl(\{n\}\bigr) = 2^{-n}$ for every $n \in \mathbb{N}$ and $\mu\bigl(\{\infty\}\bigr) = 1$. Indeed, since f is a lattice homomorphism, X is lattice isomorphic to (and will be identified with) the space of all bounded functions x on Ω such that $x(\infty)$ equals the limit of $\bigl(x(n)\bigr)_{n\in\mathbb{N}}$ along the ultrafilter. It is easy to see that $h\bigl(\lvert x\rvert\bigr) = \lVert x\rVert_{L_1}$ for every $x \in X$. It is left to show that X is norm dense in $L_1(\Omega, \mu)$; it will follow that the completion of X with respect to this norm is exactly $L_1(\Omega, \mu)$.

Let $z \in L_1(\Omega, \mu)$ be the characteristic function of $\{\infty\}$. Note that z is the limit of a sequence in X. Indeed, for every n, define $z^{(n)} \in L_1(\Omega, \mu)$ as follows: $z^{(n)}(k) = 0$ when $k < n$, $z^{(n)}(k) = 1$ when $k \geqslant n$, and $z^{(n)}(\infty) = 1$. Then $z^{(n)} \to z$, so that $z \in \overline{X}$.

Let $x \in L_1(\Omega, \mu)$. We need to show that $x \in \overline{X}$. By subtracting a scalar multiple of z, we may assume that $x(\infty) = 0$. It follows that $\lVert x\rVert = \sum_{k=1}^\infty \frac{\lvert x(k)\rvert}{2^k}$; in particular, the series converges. For each n, let $x^{(n)} \in X$ such that $x^{(n)}(k) = x(k)$ if $k \leqslant n$ and zero otherwise. Then $\lVert x^{(n)} - x\rVert = \sum_{k=n+1}^\infty \frac{\lvert x(k)\rvert}{2^k} \to 0$.

Note that $z \in L_\infty(\mu)$, yet $z \notin X$, hence $L_\infty(\mu)$ is not contained in X. Moreover, if $0 \leqslant y \leqslant z$ for some $y \in X$ then $y = 0$. It follows that X is not order dense in $\widetilde{X}$.

We say in Exercise 9.2.8 that, under some nice conditions, we may be able to find an L_1-representation for the dual of X. In the case when $X = L_p(\mu)$ with a finite measure μ and $1 \leqslant p < \infty$, we actually have X and X^* both sitting as ideals in $L_1(\mu)$, with the duality given by $\langle f, g\rangle = \int f g \, d\mu$, where $f \in X$ and $g \in X^*$. Can we do extend this to other vector or Banach lattices?

9.2.13 Theorem. *In the setting of Theorem 9.2.10, assuming that X is order complete, one can represent $X^\sim_{\mathrm{oc}}$ as the ideal in $L_1(\mu)$ consisting of all $v \in L_1(\mu)$ such that $uv \in L_1(\mu)$ for every $u \in X$. The duality is given by $\langle u, v\rangle = \int uv \, d\mu$ for all $u \in X$ and $v \in X^*$. Furthermore, $L_\infty(\mu) \subseteq X$.*

Proof. First, represent X as in Theorem 9.2.10 for some $L_1(\mu) = L_1(\Omega, \mathcal{F}, \mu)$, with μ a finite measure. Let $J\colon L_\infty(\mu) \hookrightarrow X$ be the inclusion map. Its order adjoint, given by $J^\sim\varphi = \varphi \circ J$, is a lattice homomorphism from $X^\sim$ to $L_\infty(\mu)^*$ by Exercise 6.6.8. Since $L_\infty(\mu)$ is an ideal in X, it is regular, so that J is order continuous. If $\varphi \in X^\sim$ is order continuous then so is $\varphi \circ J$, so that $J^\sim$ maps $X^\sim_{\mathrm{oc}}$ to $L_\infty(\mu)^\sim_{\mathrm{oc}}$. Since the latter two spaces are projection bands in $X\ \sim$ and $L_\infty(\mu)^*$ by Ogasawara's Theorem 6.6.12, the restriction of $J^\sim$ to $X^\sim_{\mathrm{oc}}$ is a lattice homomorphism from $X^\sim_{\mathrm{oc}}$ to $L_\infty(\mu)^\sim_{\mathrm{oc}}$.

By Proposition 8.3.16, the image of $L_1(\mu)$ under the canonical embedding $j\colon L_1(\mu) \to L_1(\mu)^{**} = L_\infty(\mu)^*$ is $L_\infty(\mu)^\sim_{\mathrm{oc}}$. Let $T = j^{-1}J^\sim$, viewed as an operator from $X^\sim_{\mathrm{oc}}$ to $L_1(\mu)$. Then T is a lattice homomorphism.

Let $0 \leqslant \varphi \in X^\sim_{\mathrm{oc}}$ and $v = T\varphi$. Then for every $g \in L_\infty(\mu)$, we have

$$\varphi(g) = \langle \varphi, Jg\rangle = \langle J^\sim\varphi, g\rangle = \langle jv, g\rangle = \langle v, g\rangle = \int gv\, d\mu.$$

Fix $u \in X_+$. The sequence $(u \wedge n\mathbb{1})$ is in $L_\infty(\mu)$ and converges to u a.e.. It follows that $u \wedge n\mathbb{1} \uparrow u$ in $L_1(\mu)$ and, therefore, in X. Furthermore, the product $(u \wedge n\mathbb{1})v$ converges to the function uv a.e., and $\int (u \wedge n\mathbb{1})v\, d\mu = \varphi(u \wedge n\mathbb{1}) \leqslant \varphi(u)$. By Fatou's Lemma A.3.10, uv is integrable. By order continuity of φ, and by Monotone Convergence Theorem A.3.12, we have

$$\varphi(u) = \lim_n \varphi(u \wedge n\mathbb{1}) = \lim_n \int (u \wedge n\mathbb{1})v\, d\mu = \int uv\, d\mu.$$

It follows that for every $u \in X$ and $\varphi \in X^\sim_{\mathrm{oc}}$, we have $\varphi(u) = \int uv\, d\mu$, where $v = T\varphi$. We also conclude from this that T is one-to-one.

If $v \in \operatorname{Range} T$ then $v = T\varphi$ for some $\varphi \in X^\sim_{\mathrm{oc}}$, so that $\int |uv| d\mu = |\varphi||u|$ for every $u \in X$; it follows that $uv \in L_1(\mu)$. We now claim that if $v \in L_1(\mu)$ such that $uv \in L_1(\mu)$ for every $u \in X$ then $v \in \operatorname{Range} T$. WLOG, $v \geqslant 0$. For $u \in X$, define $\varphi(u) = \int uv\, d\mu$. It suffices to show that φ is order continuous as we then clearly have $v = T\varphi$. Suppose that $u_\alpha \downarrow 0$ in X. Since X is an ideal in $L_1(\mu)$, we have $u_\alpha \downarrow 0$ in $L_1(\mu)$. It follows that $u_\alpha v \downarrow 0$ in $L_1(\mu)$ and, therefore, $\varphi(u_\alpha) = \int u_\alpha v\, d\mu \to 0$. This completes the proof of the claim. It follows from the claim that $\operatorname{Range} T$ is an ideal in $L_1(\mu)$. It clearly contains $\mathbb{1}$; it follows that it contains $L_\infty(\mu)$. □

9.3 L_1-representations of Banach lattices.

9.3.1. Suppose that X is a Banach lattice and h a strictly positive functional on X. It follows that $h \in X^*_+$. Let $\widetilde{X} = L_1(\mu)$ be an L_1-representation of X and h. In the following, we write $\|\cdot\|$ for the original norm of X to distinguish it from $\|\cdot\|_{L_1}$ and $\|\cdot\|_{L_\infty}$.

We write $\|h\|$ for $\|h\|_{X^*}$. The inclusion of X into $L_1(\mu)$ is a lattice isomorphism, but, generally, not a norm isomorphism because the restriction of $\|\cdot\|_{L_1}$-norm to X need not be equivalent to the original norm on X. We know that X is a dense subspace of $L_1(\mu)$, but it need not be closed in $L_1(\mu)$. For example, this is the case when $X = L_p(\mu)$ with $p > 1$. Nevertheless, being a positive linear operator between Banach spaces, the inclusion is continuous. Moreover, we can estimate its norm: $\|x\|_{L_1} = h\bigl(|x|\bigr) \leqslant \|h\|\|x\|$ for every $x \in X$.

If, in addition X is order complete and has a weak unit e, and h is order continuous, then Theorem 9.2.10 yields $L_\infty(\mu) \subseteq X \subseteq L_1(\mu)$, where both inclusions are ideals, and e corresponds to $\mathbb{1}$. Observe that in this case the inclusion $L_\infty(\mu) \hookrightarrow X$ is continuous as well, and every $x \in L_\infty(\mu)$ we have $|x| \leqslant \|x\|_{L_\infty} e$, which yields $\|x\| \leqslant \|x\|_{L_\infty}\|e\|$. We conclude that the norms of the inclusions of X into $L_1(\mu)$ and of $L_\infty(\mu)$ into X are bounded by $\|h\|$ and $\|e\|$, respectively.

Furthermore, in this case, $X^\sim_{\mathrm{oc}}$ is a projection band in X^* by Ogasawara's Theorem 6.6.12, and Theorem 9.2.13 tells us that $L_\infty(\mu) \subseteq X^\sim_{\mathrm{oc}} \subseteq L_1(\mu)$. We can easily estimate the norms of the two embeddings. Let $\varphi \in X^\sim_{\mathrm{oc}}$. For every $x \in L_\infty(\mu)$ we have $\bigl|\varphi(x)\bigr| \leqslant \|\varphi\|\|x\| \leqslant \|\varphi\|\|x\|_{L_\infty(\mu)}\|e\|$; it follows that $\|\varphi\|_{L_1(\mu)} \leqslant \|e\|\|\varphi\|$. On the other hand, if $\varphi \in L_\infty(\mu)_+$ with $\varphi \leqslant \mathbb{1}$ then $\varphi(x) \leqslant \int x\,d\mu = h(x) \leqslant \|h\|\|x\|$ for every $u \in X_+$; it follows that $\|\varphi\| \leqslant \|h\|\|\varphi\|_{L_\infty(\mu)}$.

L_1-representation theory works best when X is an order continuous Banach lattice with a weak unit. In this case, all the assumptions we discussed earlier are satisfied automatically. Indeed, by Theorem 9.1.12, X admits a strictly positive functional, which is automatically order continuous, and X is order complete. Furthermore, $X^\sim_{\mathrm{oc}} = X^*$. Hence, 9.2.9 and 9.3.1 together with Theorem 9.2.13 yield the following.

9.3.2 Theorem. *Let X be an order continuous Banach lattice with a weak unit. Then, up to lattice isomorphisms, we can identify X and X^* with norm and order dense ideals in $L_1(\mu)$ for some probability measure μ, such that $L_\infty(\mu) \subseteq X, X^* \subseteq L_1(\mu)$, with all embeddings continuous in the appropriate norms, and with the duality given by $\langle u, v\rangle = \int uv\,d\mu$ for all $u \in X$ and $v \in X^*$.*

A few remarks are due at this point:

9.3.3. If X is order continuous but does not have a weak unit, we can still use Theorem 9.3.2 "locally": for every $e \in X_+$, B_e is an order continuous Banach lattice with a weak unit, so Theorem 9.3.2 applies to it. In particular, if (x_n) be a sequence in X then, by Exercise 2.2.1(viii), (x_n) is contained in a principal band, hence we may view (x_n) as a sequence in the appropriate $L_1(\mu)$.

9.3.4. Can one iterate Theorem 9.3.2 and apply it to X^* in order to represent X^{**} as an ideal in $L_1(\mu)$? Generally, this would fail because X^* need not be order continuous. For example, let $X = L_1(\mu)$. Then $X^* = L_\infty(\mu)$ is an ideal in $L_1(\mu)$, but $X^{**} = L_\infty(\mu)^*$ generally is not contained in $L_1(\mu)$.

9.3.5. Suppose that X is a Banach lattice which is continuously embedded as an ideal into some $L_1(\mu)$. The example of $L_\infty[0,1]$ shows that X need not be order continuous. However, being an ideal in an order complete vector lattice, X is order complete. We will see in Exercise 10.3.2 that if, in addition, X is separable then it is order continuous.

9.3.6. Under the assumption of Theorem 9.3.2, since $L_\infty(\mu)$ is order dense in $L_1(\mu)$, it is also order dense in X. Since X is order continuous, $L_\infty(\mu)$ is norm dense in X. By a similar argument, $L_\infty(\mu)$ is order dense in X^*. Does $L_\infty(\mu)$ have to be norm dense in X^*? We will see in Example 16.8.7 that the answer is "no".

The following is an application of Theorem 9.3.2. Recall that, in general, lattice operations are not weakly sequentially continuous; see 6.8.6.

9.3.7 Proposition. *Let (x_n) be a disjoint sequence in an order continuous Banach lattice X. If $x_n \xrightarrow{\mathrm{w}} 0$ then $\lvert x_n\rvert \xrightarrow{\mathrm{w}} 0$.*

Proof. Suppose not. Then there exists $f \in X^*_+$ such that $f\bigl(\lvert x_n\rvert\bigr) \geqslant 1$ for all n. As in 9.3.3, we may assume that X has a weak unit. We may now assume that X and X^* are represented as ideas in $L_1(\mu)$ as in Theorem 9.3.2. In particular, we may view (x_n) as a disjoint sequence in $L_1(\mu)$ and f as positive function in $L_1(\mu)$ such that $\int \lvert x_n\rvert f \geqslant 1$ for every n. It is easy to see that there exists a function $g \in L_1(\mu)$ such that $\lvert g\rvert = f$ and $x_n g = \lvert x_n\rvert f$ for every n. Since X^* is an ideal in $L_1(\mu)$, we have $g \in X^*$ and $g(x_n) = \int x_n g = \int \lvert x_n\rvert f \geqslant 1$ for every n. This contradicts $x_n \xrightarrow{\mathrm{w}} 0$. □

9.3.8. *Almost contractive embeddings.* Let X be an order continuous Banach lattice with a strictly positive functional h and a weak unit e. As in 9.3.1, we have $L_\infty(\mu) \overset{A}{\hookrightarrow} X \overset{B}{\hookrightarrow} L_1(\mu)$ such that h corresponds to the integral, e corresponds to $\mathbb{1}$, both inclusions are norm and order dense ideals, $\mu(\Omega) = \int \mathbb{1} = h(e)$, $\lVert A\rVert \leqslant \lVert e\rVert$, and $\lVert B\rVert \leqslant \lVert h\rVert$. It is often convenient to scale e and h so that $\lVert e\rVert = \lVert h\rVert = 1$; in this case, both inclusions are contractions. On the other hand, it is also desirable to have $h(e) = 1$ because this guarantees that μ is a probability measure. Generally, it is impossible to have $\lVert e\rVert = \lVert h\rVert = 1$ and $h(e) = 1$ simultaneously. However, the following trick shows that this may be done "up to" any positive ε by modifying e and h appropriately.

First, scale e and h so that $\lVert e\rVert = \lVert h\rVert = 1$. Fix any $\varepsilon > 0$. Using Hahn-Banach Theorem, find $f \in X^*_+$ such that $\lVert f\rVert = 1 = f(e)$. Put $h' = \frac{f+\varepsilon h}{\lVert f+\varepsilon h\rVert}$. Clearly, $\lVert h'\rVert = 1$. Also, h' is strictly positive because it dominates a positive scalar multiple of h. Since X is order continuous, h' is order continuous. Observe that $h'(e) = \frac{1+\varepsilon h(e)}{\lVert f+\varepsilon h\rVert} \geqslant \frac{1}{1+\varepsilon}$. Put $e' = \frac{e}{h'(e)}$. Then

e' is a weak unit, $h'(e') = 1$ and $\|e'\| \leqslant 1 + \varepsilon$.

If we now apply the L_1-representation construction with e' and h', we get $L_\infty(\mu) \overset{A}{\hookrightarrow} X \overset{B}{\hookrightarrow} L_1(\mu)$ with μ being a probability measure, $\|A\| \leqslant 1 + \varepsilon$, and $\|B\| \leqslant 1$. That is, $\|x\|_{L_1} \leqslant \|x\| \leqslant (1+\varepsilon)\|x\|_{L_\infty}$ for every x in the appropriate space.

To proceed, we need the following general fact.

9.3.9 Theorem (Amemiya). *Suppose that X and Y are two Banach lattices such that Y is order continuous, Y is an ideal in X, and the embedding is continuous. Then the norm topologies of X and Y agree on order intervals of Y.*

Proof. Fix $u \in Y_+$. Since Y is an ideal in X, $[-u, u]_Y = [-u, u]_X$; so we will just write $[-u, u]$. Since the embedding of Y into X is continuous, $x_n \xrightarrow{\|\cdot\|_Y} x$ implies $x_n \xrightarrow{\|\cdot\|_X} x$ whenever (x_n) and x are in $[-u, u]$. To prove the converse, suppose that $x_n \xrightarrow{\|\cdot\|_X} x$ in $[-u, u]$. WLOG, $x = 0$. Passing to a subsequence, we have $x_{n_k} \xrightarrow{\text{o}} 0$ in X by Theorem 2.5.4. Being an ideal, Y is regular; since (x_n) is order bounded, we conclude that $x_n \xrightarrow{\text{o}} 0$ in Y by Theorem 3.2.5. Since Y is order continuous, we have $x_{n_k} \xrightarrow{\|\cdot\|_Y} 0$. Similarly, every subsequence of (x_n) has a further subsequence that converges to zero in Y; it follows that $x_n \xrightarrow{\|\cdot\|_Y} 0$. □

9.3.10 Example. Let μ be a finite measure and $1 \leqslant p \leqslant q$. Then $L_q(\mu)$ is an ideal in $L_p(\mu)$. For a sequence (f_n) in $L_q(\mu)$, if $f_n \to 0$ in $\|\cdot\|_{L_q}$ then $f_n \to 0$ in $\|\cdot\|_{L_p}$. Amemiya's Theorem tells us that the two convergences are equivalent when the sequence is order bounded in $L_q(\mu)$, in particular, when the sequence is order bounded by a constant function.

9.3.11 Corollary. *Let X be an order continuous Banach lattice which is also an ideal in $L_1(\mu)$ for some μ such that the embedding is continuous. Then the norm topologies of X and of $L_1(\mu)$ agree on order intervals of X.*

In particular, the space X in Theorem 9.3.2 satisfies the assumptions of this corollary, hence the norm topologies of X and of $L_1(\mu)$ agree on order intervals of X. We will later prove versions of the preceding results for weak topologies; see Theorem 10.1.6 and Corollary 10.1.7.

9.4 * Extending a regular operator

Let $\widetilde{X}$ be an L_1-representation of a Banach lattice X. Given a bounded operator T on X, does it extend to a bounded operator on $\widetilde{X}$? In general, there is no hope to extend all operators (or even functionals) from X to $\widetilde{X}$. For example, take $X = L_p[0,1]$ with

$1<p<\infty$ and take a functional $f\in L_p^*$. Then $f\in L_q[0,1]$ where q is the conjugate of p. Note that f extends to a continuous functional on $L_1[0,1]$ only when $f\in L_\infty[0,1]$.

However, instead of first constructing $\widetilde{X}$ and then extending operators from X to $\widetilde{X}$, it is possible for a given positive or regular operator T on X to construct a "custom-made" L_1-representation $\widetilde{X}$ so that T extends to a regular operator on $\widetilde{X}$. This is achieved by "adjusting" h and e to T. We follow the approach of [Wei82].

Suppose that T is a positive operator on a Banach lattice X with a strictly positive functional h_0. WLOG, assume that $\|T\|<1$. Define $h=\sum_{n=0}^\infty T^{*n}h_0$. The series converges because $\|T^*\|=\|T\|<1$. Note that $h\geqslant h_0$, hence h is strictly positive. Observe that $T^*h\leqslant h$. Let $L_1(\mu)$ be an L_1-representation for X and h. For every $x\in X$, we have

$$\|Tx\|_{L_1}=h\big(|Tx|\big)\leqslant h\big(T|x|\big)=\big(T^*h\big)\big(|x|\big)\leqslant h\big(|x|\big)=\|x\|_{L_1}.$$

It follows that T extends to $\widetilde{T}\colon L_1(\mu)\to L_1(\mu)$ with $\|\widetilde{T}\|\leqslant 1$. Since X is a norm dense sublattice in $L_1(\mu)$, $\widetilde{T}$ is positive.

Suppose now that, in addition, T and h_0 are order continuous. We claim that in this case h is also order continuous, so that X is order dense in $L_1(\mu)$ by Theorem 9.2.7(i). Indeed, order continuity of h_0 and T yields order continuity of $T^{*n}h_0$ for every n (because it is the composition of order continuous functions). By Ogasawara's Theorem 6.6.12, $X^\sim_{\mathrm{oc}}$ is a band in $X^\sim=X^*$, hence $X^\sim_{\mathrm{oc}}$ is closed; it follows that h is order continuous.

Suppose that, in addition, X is order complete and has a weak unit e_0. We "adjust" e_0 to T in the same fashion as we did it with h_0: put $e=\sum_{n=0}^\infty T^ne_0$. Then $e\geqslant e_0$, so that e is a weak unit. We also have $Te\leqslant e$. Let $L_1(\mu)$ be an L_1-representation for X, h, and e as in Theorem 9.2.10. Now X is an ideal in $L_1(\mu)$, and $L_\infty(\mu)$ is an ideal in X. For every $x\in L_\infty(\mu)$, we have $|x|\leqslant\|x\|_{L_\infty}e$; it follows that $|Tx|\leqslant\|x\|_{L_\infty}Te\leqslant\|x\|_{L_\infty}e$. This yields $Tx\in L_\infty(\mu)$, hence, T leaves $L_\infty(\mu)$ invariant and $\|Tx\|_{L_\infty}\leqslant\|x\|_{L_\infty}$. Therefore, $\big\|T\colon L_\infty(\mu)\to L_\infty(\mu)\big\|\leqslant 1$.

The method also works for regular operators provided that X is order complete.[1] However, $\|T\|$ has to be replaced with the ***regular norm*** of T.[2] Let T be a regular operator on X. Define $\|T\|_r=\big\||T|\big\|$. Suppose that $\|T\|_r<1$. Apply the preceding construction with T replaced with $|T|$. In particular, put $h=\sum_{n=0}^\infty|T|^{*n}h_0$ and $e=\sum_{n=0}^\infty|T|^ne_0$; so that $|T|e\leqslant e$ and $|T|^*h\leqslant h$. Note that if T is order continuous then

[1] We cannot just separately apply the preceding construction to T^+ and T^- as we need *the same* representation space for T^+ and T^-.

[2] The regular norm will be discussed in more detail in Section 13.2.

so is $|T|$ by Ogasawara's Theorem 6.5.11. Then

$$\|Tx\|_{L_1} = h\bigl(|Tx|\bigr) \leqslant h\bigl(|T||x|\bigr) = \bigl(|T|^*h\bigr)\bigl(|x|\bigr) \leqslant h\bigl(|x|\bigr) = \|x\|_{L_1} \tag{9.1}$$

for every $x \in X$. It follows that T extends to $\widetilde{T}$ on $L_1(\mu)$ with $\bigl\|\widetilde{T}\colon L_1(\mu) \to L_1(\mu)\bigr\| \leqslant 1$. Also, for every $x \in L_\infty(\mu)$ we have $|x| \leqslant \|x\|_{L_\infty}e$, so that $|Tx| \leqslant \|x\|_{L_\infty}|T|e \leqslant \|x\|_{L_\infty}e$. It follows that $Tx \in L_\infty(\mu)$, so that T leaves $L_\infty(\mu)$ invariant and $\bigl\|T\colon L_\infty(\mu) \to L_\infty(\mu)\bigr\| \leqslant 1$.

Finally, observe that $\widetilde{T}$ is regular. Indeed, similarly to (9.1), $\bigl\||T|x\bigr\|_{L_1} \leqslant \|x\|_{L_1}$ for every $x \in X$. It follows that $|T|$ also extends to $L_1(\mu)$ and, clearly, $\widetilde{T} \leqslant \widetilde{|T|}$. In particular, we get the following.

9.4.1 Theorem. *Let T be a regular order continuous operator on an order complete Banach lattice X with a weak unit and an order continuous strictly positive functional. Then there is an L_1-representation of T as in Theorem 9.2.10 such that T extends to a regular operator $\widetilde{T}$ on $L_1(\mu)$. Moreover, T leaves invariant $L_\infty(\mu)$ and the norms $\bigl\|\widetilde{T}\colon L_1(\mu) \to L_1(\mu)\bigr\|$ and $\bigl\|T\colon L_\infty(\mu) \to L_\infty(\mu)\bigr\|$ are less than or equal to $\|T\|_r$.*

Using a similar computation to that in (9.1), we can show that every operator S in the ideal generated by T can also be extended to a bounded regular operator on $L_1(\mu)$, and the restriction of S to $L_\infty(\mu)$ is bounded. Thus, we can extend not just a single operator but the entire principal ideal of operators. In particular, we can simultaneously extend a sequence of regular operators, because it is contained in such a principal ideal.

In case when X is order continuous, we can simultaneously control the norms of $\widetilde{T}$ and of the inclusions $L_\infty(\mu) \subseteq X \subseteq L_1(\mu)$:

9.4.2 Theorem ([Wei82])**.** *Let T be a regular operator on an order continuous Banach lattice X with a weak unit. Let $\varepsilon > 0$ and $a > 1$ such that $(1+\varepsilon)(1-a^{-1})^2 > 1$. Then there exists an L_1-representation $L_1(\mu)$ of X as in 9.3.8 such that, in addition, T extends to a regular operator $\widetilde{T}$ on $L_1(\mu)$, T leaves invariant $L_\infty(\mu)$, and the norms $\bigl\|\widetilde{T}\colon L_1(\mu) \to L_1(\mu)\bigr\|$ and $\bigl\|T\colon L_\infty(\mu) \to L_\infty(\mu)\bigr\|$ are less than or equal to $a\|T\|_r$.*

Proof. Fix $\delta > 0$ so that $(1+\varepsilon)(1-a^{-1})^2 > 1+\delta$. As in 9.3.8, find a weak unit e' and a strictly positive functional h' such that $\|h'\| = h'(e') = 1$ and $\|e'\| < 1+\delta$. WLOG, $\|T\|_r \leqslant 1$. We define $e_0 \coloneqq \sum_{n=0}^\infty a^{-n}|T|^n e'$ and $h_0 \coloneqq \sum_{n=0}^\infty a^{-n}|T|^{*n}h'$. It follows from $e_0 \geqslant e'$ and $h_0 \geqslant h'$ that e_0 is a weak unit, h_0 is a strictly positive functional, and $h_0(e_0) \geqslant 1$. It also follows that $|T|e_0 \leqslant ae_0$ and $|T|^*h_0 \leqslant ah_0$. Furthermore,

$$\|e_0\| \leqslant \sum_{n=0}^\infty a^{-n}\|e'\| \leqslant \frac{1+\delta}{1-a^{-1}} \qquad \text{and} \qquad \|h_0\| \leqslant \frac{1}{1-a^{-1}}.$$

Put $h = \frac{h_0}{\|h_0\|}$ and $e = \frac{e_0}{h(e_0)}$. Then $\|h\| = 1$ and $h(e) = 1$. Since h and e are positive scalar multiples of h_0 and e_0, we still have $|T|e \leqslant ae$ and $|T|^*h \leqslant ah$. It follows from $h_0(e_0) \geqslant 1$ that

$$\|e\| = \frac{\|e_0\|}{h(e_0)} \leqslant \|h_0\| \cdot \|e_0\| \leqslant \frac{1+\delta}{(1-a^{-1})^2} < 1+\varepsilon.$$

Let $L_1(\mu)$ be an L_1-representation for X, h, and e. As in 9.3.8, $h(e) = 1$ yields that μ is a probability measure, $\|h\| = 1$ yields that $\|x\|_{L_1} \leqslant \|x\|$ for every $x \in X$, and $\|e\| < 1+\varepsilon$ yields that $\|x\| \leqslant (1+\varepsilon)\|x\|_{L_\infty}$ for every $x \in L_\infty(\mu)$.

For every $x \in X$, we have

$$\|Tx\|_{L_1} = h\big(|Tx|\big) \leqslant h\big(|T||x|\big) = \big(|T|^*h\big)\big(|x|\big) \leqslant ah\big(|x|\big) = a\|x\|_{L_1}.$$

It follows that T extends to an operator $\widetilde{T}$ on $L_1(\mu)$ and $\|\widetilde{T}\| \leqslant a$.

For every $x \in L_\infty(\mu)$ we have $|x| \leqslant \|x\|_{L_\infty} e$, so that $|Tx| \leqslant \|x\|_{L_\infty}|T|e \leqslant \|x\|_{L_\infty} ae$. It follows that $\|Tx\|_{L_\infty} \leqslant a\|x\|_{L_\infty}$, so that $\big\|T\colon L_\infty(\mu) \to L_\infty(\mu)\big\| \leqslant a$. □

9.4.3 Example. In Theorem 9.4.2 one could, for example, take $\varepsilon = 1$ and $a = 4$.

9.5 * Representating with positive functionals

9.5.1. If we have just a positive but not strictly positive functional, we may still recover some of the preceding representation theory. Let h be a positive functional on a vector lattice X. Consider X/N_h, the quotient of X over the null ideal of h. Let $Q\colon X \to X/N_h$ be the quotient map. Recall that X/N_h is a vector lattice and Q is a lattice homomorphism by Theorem 3.4.2.

Since $N_h \subseteq \ker h$, h induces a functional $\tilde{h}$ on X/N_h via $\tilde{h}(\tilde{x}) = h(x)$, where $\tilde{x}$ is the equivalence class of $x \in X$. It is easy to see that $\tilde{h}$ is a strictly positive functional on X/N_h. So we can now represent X/N_h in some $L_1(\mu)$: there is a lattice isomorphic embedding $J\colon X/N_h \hookrightarrow L_1(\mu)$. Thus, the composition $JQ\colon X \to L_1(\mu)$ is a lattice homomorphism. This construction is of little value by itself because JQ is generally not one-to-one; it is usually used in conjunction with some additional assumptions.

Here is a typical example of this technique. Suppose now that X is a Banach lattice and $h \in X^*_+$ such that $\|h\| = 1$. Let $Q\colon X \to X/N_h$ and $J\colon X/N_h \to L_1(\mu)$ be as above. Since Q is a quotient map, we have $\|Q\| \leqslant 1$. Let $x \in X$; for every $y \in N_h$, we have $h(x) = h(x+y) \leqslant \|x+y\|$; taking infimum over $y \in N_h$ we get $\tilde{h}(\tilde{x}) = h(x) \leqslant \|\tilde{x}\|$. It follows that $\|\tilde{h}\| \leqslant 1$ and, therefore, $\|J\| = \|\tilde{h}\| \leqslant 1$. It also follows that $JQ\colon X \to L_1(\mu)$ is a lattice homomorphism and a contraction.

Suppose now that X is a Banach lattice and $x \in X_+$. Let $h \in X_+^*$ such that $h(x) = \|x\|$. Let Q and J be as above, so that $JQ\colon X \to L_1(\mu)$ is a lattice homomorphism and a contraction. In addition, $\|JQx\| = \tilde{h}(Qx) = h(x) = \|x\|$. So for any given x we can construct Q and J so that $\|JQx\| = \tilde{h}(Qx) = h(x) = \|x\|$.

We have seen many times that AM- and AL-spaces are the two "extreme" cases in the theory of Banach lattices. We now prove that every Banach lattice is, in some sense, a combination of these extremes. Namely, it may be identified with a closed sublattice of an AM-sum of AL-spaces.

9.5.2 Theorem. *For every Banach lattice X there exists a family $(\mu_\alpha)_{\alpha\in\Lambda}$ of probability measures such that X embeds lattice isometrically into $\Big(\bigoplus\limits_{\alpha\in\Lambda} L_1(\mu_\alpha)\Big)_\infty$.*

Proof. Let $0 \leqslant u \in S_X$. Find $f \in X_+^*$ such that $f(u) = \|f\| = 1$. Let $h \in X^*$ be the restriction of f to I_u in the sense of Theorem 6.6.9 and Proposition 6.6.11. It follows from $0 \leqslant h \leqslant f$ and $h(u) = f(u) = 1$ that $\|h\| = 1$. We now construct lattice homomorphisms $Q\colon X \to X/N_h$ and $J\colon X/N_h \to L_1(\mu_u)$ for some measure μ_u as in 9.5.1; we then have $\|JQ\| \leqslant 1 = \|JQu\|$. Furthermore, since Qu is a weak unit in X/N_h by Proposition 6.6.11, and $\tilde{h}(Qu) = h(u) = 1$, we can choose μ_u to be a probability measure as in 9.2.9. Denote JQ by T_u. Put $\Lambda = S_X \cap X^+$, and consider $T\colon X \to \Big(\bigoplus_{u\in\Lambda} L_1(\mu_u)\Big)_\infty$ as the sum of T_u's. Since each T_u is a lattice homomorphism and a contraction, so is T. For every $x \in S_X$, put $u = |x|$, then, since T is a lattice homomorphism, we get $\|Tx\| = \|Tu\| \geqslant \|T_u u\| = 1$; it follows that T is an isometry. □

9.5.3 Remark. The preceding argument partially extends to a vector lattice X whose order dual $X^\sim$ separate points as follows. Take $\Lambda = X_+ \setminus \{0\}$. For every $u \in \Lambda$, we find $f \in X_+^\sim$ such that $f(u) \neq 0$. Put $h \in X^\sim$ be the restriction of f to I_u as in Proposition 6.6.11. Then X/N_h has a weak unit $Q(u)$ and a strictly positive functional $\tilde{h}$, hence we can find a finite measure μ_u and a lattice isomorphic embedding $J\colon X/N_h \to L_1(\mu)$. Put $T_u = JQ$. Then $T_u u \neq 0$. Define $T\colon X \to \prod_{u\in\Lambda} L_1(\mu_u)$ as the product of T_u's. Then T is a lattice homomorphism. For every non-zero $x \in X_+$, put $u = |x|$, then $\big|T_u(x)\big| = T_u u \neq 0$, hence $Tx \neq 0$ and, therefore, T is one-to-one.

9.6 Asymptotically disjoint subsequences

Recall that Disjointification Lemma 4.5.1 provides a way to find an "almost disjoint" subsequence of a given sequence. In this section, we discuss a somewhat different approach to the same issue. This section is based on results by Kadeč and Pełczyński.

A sequence (x_n) in a Banach lattice X is ***asymptotically disjoint*** if it can be decomposed as follows: $x_n = d_x + r_n$, where (d_n) is a disjoint sequence and (r_n) is a norm null sequence. That is, there exists a disjoint sequence (d_n) such that $x_n - d_n \to 0$.

9.6.1 Exercise. (i) In the preceding definition, if x_n is positive for every n, then one can chose (d_n) and (r_n) to be positive as well.

(ii) If X has PPP then, in the preceding definition, one can chose (d_n) and (r_n) so that $d_n \perp r_n$ for every n.

(iii) (x_n) is asymptotically disjoint iff $\big(|x_n|\big)$ is asymptotically disjoint.

9.6.2 Example. The following example motivates much of what is coming later in this section. Let $X = L_1[0,1]$, and put $f_n = n\mathbb{1}_{[0,\frac{1}{n}]}$. Note that $\|f_n\| = 1$. Even though the sequence (f_n) is far from being disjoint, (f_n) has an asymptotically disjoint subsequence! Indeed, take $n_k = k$, so that $f_{n_k} = (k!)\mathbb{1}_{[0,\frac{1}{k!}]}$, and split f_{n_k} into a disjoint sum as follows: $f_{n_k} = d_k + r_k$, where let $r_k = f_{n_k} \cdot \mathbb{1}_{[0,\frac{1}{(k+1)!}]}$ and $d_k = f_{n_k} \cdot \mathbb{1}_{[\frac{1}{(k+1)!},\frac{1}{k!}]}$. Then (d_k) is disjoint and $\|r_k\| = \frac{1}{k+1} \to 0$. This example may be easily generalized to $L_p[0,1]$ where $1 < p < \infty$.

9.6.3 Theorem. *Every infinite-dimensional subspace of a Banach sequence space (as in 1.5.11) contains a normalized asymptotically disjoint sequence.*

Proof. The following proof is often referred to as *"gliding hump"*. Let Y be an infinite-dimensional subspace of a Banach sequence space X. For every n, let P_n be the n-th basis projection and $Z_n = \ker P_n$. That is, Z_n consists of those vectors x such that $x_1 = \dots = x_n = 0$. Since Z_n is a subspace of finite codimension, $Z_n \cap Y$ is non-trivial for every n.

Pick any $x^{(1)}$ in Y with $\|x^{(1)}\| = 1$. Since $P_n x^{(1)} \to x^{(1)}$, we can find n_1 such that $\|x^{(1)} - d^{(1)}\| < \frac{1}{2}$, where $d^{(1)} = P_{n_1}x^{(1)}$. Since $Z_{n_1} \cap Y$ is non-trivial, pick any $x^{(2)}$ in $Z_{n_1} \cap Y$ with $\|x^{(2)}\| = 1$. Find $n_2 > n_1$ such that $\|x^{(2)} - d^{(2)}\| < \frac{1}{3}$, where $d^{(2)} = P_{n_2}x^{(2)}$. Note that $d^{(1)} \perp d^{(2)}$ because they are supported on $\{1,\dots,n_1\}$ and $\{n_1+1,\dots,n_2\}$, respectively. Proceeding inductively, we produce sequences $(x^{(k)})$, $(d^{(k)})$ in X, and an increasing sequence (n_k) in $\mathbb{N}$ such that $x^{(k)} \in Z_{n_k} \cap Y$ with $\|x^{(k)}\| = 1$, $d^{(k)} = P_{n_k}x^{(k)}$, (d_k) is disjoint, and $\|x^{(k)} - d^{(k)}\| \leqslant \frac{1}{k+1}$. $\square$

A net (x_α) in a Banach lattice X is said to ***un-converge*** to x if $|x_\alpha - x| \wedge u \to 0$ for every $u \geqslant 0$. We write $x_\alpha \xrightarrow{\text{un}} x$. It is clear that $x_\alpha \xrightarrow{\text{un}} x$ iff $|x_\alpha - x| \xrightarrow{\text{un}} 0$, so we can reduce un-convergence to positive nets converging to zero. For a positive net (x_α), we have $x_\alpha \xrightarrow{\text{un}} 0$ iff $x_\alpha \wedge u \to 0$ for every $u \geqslant 0$. In this definition, "un" stands for "unbounded norm", because while all "truncated" nets $(x_\alpha \wedge u)$ have to be null, the original net (x_α) need not even be order or norm bounded. It is clear that if (x_α) is order bounded then $x_\alpha \xrightarrow{\mathrm{un}} x$ iff $x_\alpha \to x$. In this section, we will use un-convergence as a convenient tool; we will study this convergence in more detail in Section 15.3.

9.6.4 Theorem. *If $x_\alpha \xrightarrow{\mathrm{un}} 0$ then there exists an increasing sequence of indices (α_k) such that the sequence (x_{α_k}) is asymptotically disjoint.*

Proof. WLOG, $x_\alpha \geqslant 0$ for all α; otherwise, we replace (x_α) with $\big(|x_\alpha|\big)$ and use Exercise 9.6.1(iii).

Pick any α_1. Suppose that $\alpha_1, \dots, \alpha_{k-1}$ have been constructed. Note that $x_\alpha \wedge x_{\alpha_i} \to 0$ for every $i = 1, \dots, k-1$. Choose $\alpha_k > \alpha_{k-1}$ so that $\big\|x_{\alpha_k} \wedge x_{\alpha_i}\big\| \leqslant \frac{1}{2^{k+i}}$ for every $i = 1, \dots, k-1$. This produces an increasing sequence of indices (α_k) such that $\|z_{i,k}\| \leqslant \frac{1}{2^{k+i}}$ where $z_{i,k} = x_{\alpha_i} \wedge x_{\alpha_k}$, $1 \leqslant i < k$.

For every k, put $v_k = \sum_{i=1}^{k-1} z_{i,k} + \sum_{i=k+1}^{\infty} z_{k,i}$. It follows from

$$\sum_{i=1}^{k-1}\|z_{i,k}\| + \sum_{i=k+1}^{\infty}\|z_{k,i}\| < \sum_{i=1}^{\infty}\frac{1}{2^{k+i}} = \frac{1}{2^k}$$

that the series in the definition of v_k is convergent and, furthermore, that $\|v_k\| < \frac{1}{2^k}$. Put $d_k \coloneqq x_{n_k} - x_{n_k} \wedge v_k = (x_{\alpha_k} - v_k)^+$. It follows from $0 \leqslant x_{\alpha_k} - d_k \leqslant v_k$ that $\|x_{\alpha_k} - d_k\| \to 0$ as $k \to \infty$. It is left to show that the sequence (d_k) is disjoint. Let $k < m$. Then

$$\begin{aligned} d_k &= (x_{\alpha_k} - v_k)^+ \leqslant (x_{\alpha_k} - z_{k,m})^+ = x_{\alpha_k} - x_{\alpha_k} \wedge x_{\alpha_m}, \text{ and} \\ d_m &= (x_{\alpha_m} - v_m)^+ \leqslant (x_{\alpha_m} - z_{k,m})^+ = x_{\alpha_m} - x_{\alpha_k} \wedge x_{\alpha_m}. \end{aligned}$$

It follows that $d_k \perp d_m$. □

Recall that every order continuous Banach lattice with a weak unit can be continuously embedded as a dense ideal in $L_1(\mu)$ for a probability measure μ; with the weak unit corresponding to $\mathbb{1}$; see Theorem 9.3.2. The following proposition essentially says that un-convergence may be viewed as an abstraction of convergence in measure.

9.6.5 Proposition. *Let X be an order continuous Banach lattice, embedded continuously as an ideal in $L_1(\mu)$ for some finite measure μ, with $\mathbb{1} \in X$. Let (x_α) be a net in X. Then $x_\alpha \xrightarrow{\mathrm{un}} 0$ in X iff (x_α) converges to zero in measure in $L_1(\mu)$.*

Proof. WLOG, μ is a probability measure and $x_\alpha \geqslant 0$ for every α; otherwise, consider $|x_\alpha|$ instead. Suppose that $x_\alpha \xrightarrow{\mathrm{un}} 0$. Fix $\delta > 0$. By assumption, $\big\|x_\alpha \wedge \delta\mathbb{1}\big\|_X \to 0$ as $\alpha \to \infty$. By the continuity of the embedding, we have $\big\|x_\alpha \wedge \delta\mathbb{1}\big\|_{L_1} \to 0$. It is easy to see that $\delta \cdot \mathbb{1}_{\{x_\alpha \geqslant \delta\}} \leqslant x_\alpha \wedge \delta\mathbb{1}$. It follows that $\mu(x_\alpha \geqslant \delta) \leqslant \frac{1}{\delta}\big\|x_\alpha \wedge \delta\mathbb{1}\big\| \to 0$. Therefore, (x_α) converges to zero in measure.

Suppose that (x_α) converges to zero in measure and $0 \leqslant u \in X$. In particular, $u \in L_1(\mu)_+$. Fix $\varepsilon > 0$. Find $\delta > 0$ such that $\int_A u\,d\mu < \varepsilon$ whenever $\mu(A) < \delta$. Find α_0 such that $\mu\big(x_\alpha > \varepsilon\big) < \delta$ whenever $\alpha \geqslant \alpha_0$. In this case, put $A = \{x_\alpha > \varepsilon\}$, then $\mu(A) < \delta$ and, therefore,

$$\|x_\alpha \wedge u\|_{L_1} = \int (x_\alpha \wedge u)\,d\mu \leqslant \int_A u\,d\mu + \int_{A^C} x_\alpha\,d\mu \leqslant 2\varepsilon.$$

It follows that $\|x_\alpha \wedge u\|_{L_1} \to 0$. Since this sequence is contained in $[0,u]$, we have $\|x_\alpha \wedge u\|_X \to 0$ by Amemiya's Theorem 9.3.9 □

Combining Theorem 9.6.4 and 9.6.5, we get the following:

9.6.6 Theorem (Kadeč-Pełczyński)**.** *Let X be an order continuous Banach lattice, embedded continuously as an ideal in $L_1(\mu)$ for some finite measure μ, with $\mathbb{1} \in X$. Let (x_n) be a sequence in X. If (x_n) converges to zero in measure in $L_1(\mu)$ then (x_n) has an asymptotically disjoint subsequence in X.*

Recall that a sequence (x_n) in a Banach space is ***seminormalized*** if it is both bounded and bounded below in norm, i.e., there exist positive C_1 and C_2 such that $C_1 \leqslant \|x_n\| \leqslant C_2$ for all n.

9.6.7 Corollary. *Let X be an order continuous Banach lattice, embedded continuously as an ideal in $L_1(\mu)$ for some finite measure μ, with $\mathbb{1} \in X$. Let (x_n) be a bounded non-zero sequence in X_+. Then either (x_n) is seminormalized in $L_1(\mu)$ or has an asymptotically disjoint subsequence in X.*

Proof. Since the embedding of X into $L_1(\mu)$ is continuous, (x_n) is bounded in $L_1(\mu)$. Suppose (x_n) is not bounded below in norm in $L_1(\mu)$. Then there is a subsequence (x_{n_k}) such that $\|x_{n_k}\|_{L_1(\mu)} \to 0$. It follows that (x_{n_k}) converges to zero in measure. Now apply Theorem 9.6.6. □

It follows from Khinchin's Inequality 6.4.4(ii) that all L_p-norms as $1 \leqslant p < \infty$ are equivalent on the span of the Rademacher sequence. We now use Kadeč-Pełczyński's Theorem 9.6.6 to investigate subspaces with this property.

9.6.8 Exercise. Let X be an order continuous Banach lattice, embedded continuously as an ideal in $L_1(\mu)$ for some finite measure μ, with $\mathbb{1} \in X$. Let Y be a subspace of X. Then (i)⇔(ii)⇒(iii), where

(i) Y contains no sequence which, when viewed as a sequence in X, is normalized and asymptotically disjoint.

(ii) There exists $\delta > 0$ such that $\mu(x \geqslant \delta) \geqslant \delta$ for every $x \in Y_+$ with $\|x\|_X = 1$.

(iii) The norms $\|\cdot\|_X$ and $\|\cdot\|_{L_1(\mu)}$ are equivalent on Y.

Generally, (iii) need not imply (i) or (ii).

9.6.9 Exercise. Every seminormalized asymptotically disjoint sequence (x_n) in $L_p(\mu)$ has a subsequence equivalent to the unit vector basis of ℓ_p.

9.6.10 Proposition. *Let Y be a subspace of $L_p(\mu)$ where $1 < p < \infty$ and μ is a finite measure. TFAE:*

(i) *The norms $\|\cdot\|_{L_p(\mu)}$ and $\|\cdot\|_{L_1(\mu)}$ are equivalent on Y;*

(ii) *The norms $\|\cdot\|_{L_p(\mu)}$ and $\|\cdot\|_{L_q(\mu)}$ are equivalent on Y for every $q \in [1, p)$;*

(iii) *The norms $\|\cdot\|_{L_p(\mu)}$ and $\|\cdot\|_{L_q(\mu)}$ are equivalent on Y for some $q \in [1, p)$;*

(iv) *Y contains no sequence which, when viewed as a sequence in $L_p(\mu)$, is seminormalized and asymptotically disjoint;*

(v) *There exists $\delta > 0$ such that for $\mu(x \geqslant \delta) \geqslant \delta$ for every $x \in Y_+$ with $\|x\|_{L_p(\mu)} = 1$.*

Proof. WLOG, μ is a probability measure. Since $\|\cdot\|_{L_q(\mu)}$ is a monotone function of q, we get (i)⇒(ii)⇒(iii) trivially.

(iii)⇒(iv) Suppose that there exists a sequence (x_n) in Y which, when viewed as a sequence in $L_p(\mu)$, is seminormalized and almost disjoint. By (iii), (x_n) is also seminormalized in $L_q(\mu)$. Find a disjoint sequence (d_n) in $L_p(\mu)$ such that $\|x_n - d_n\|_{L_p(\mu)} \to 0$. Observe that (d_n) is in $L_q(\mu)$; it follows from $\|\cdot\|_{L_q(\mu)} \leqslant \|\cdot\|_{L_p(\mu)}$ that $\|x_n - d_n\|_{L_q(\mu)} \to 0$. Hence (x_n) is seminormalized and asymptotically disjoint in $L_q(\mu)$. By Exercise 9.6.9, (x_n) is equivalent to the unit vector bases of both ℓ_p and ℓ_q, hence the latter two bases are equivalent; this is a contradiction.

(iv)⇒(v)⇒(i) by Exercise 9.6.8. □

Here is another corollary of Kadeč-Pełczyński's Theorem 9.6.6:

9.6.11 Corollary. *Every positive weakly null sequence in an order continuous Banach lattice has an asymptotically disjoint subsequence.*

Proof. Let (x_n) be a weakly null positive sequence in an order continuous Banach lattice X. We may assume that X is continuously embedded as an ideal into $L_1(\mu)$ for some finite measure μ by Theorem 9.3.2 and by 9.3.3. Since the embedding is continuous, it is weak-to-weak continuous, so that $x_n \xrightarrow{\mathrm{w}} 0$ in $L_1(\mu)$. Since the integral is a functional on $L_1(\mu)$, we get $\|x_n\|_{L_1(\mu)} = \int x_n \to 0$. It follows that (x_n) converges to zero in measure. Now apply Kadeč-Pełczyński's Theorem 9.6.6. □

9.7 Almost order bounded and equi-integrable sets

A subset A of a normed lattice X is said to be ***almost order bounded*** if for every $\varepsilon > 0$ there exists $u > 0$ such that $A \subseteq [-u, u] + \varepsilon B_X$. That is, A is "approximately" contained in an order interval. Recall that $x = x \wedge u + (x - u)^+$ for every $x, u \geqslant 0$; this identity is extremely useful when dealing with almost order bounded sets.

9.7.1 Exercise. (i) $x \in [-u, u] + \varepsilon B_X$ iff $\big\|\big(|x| - u\big)^+\big\| \leqslant \varepsilon$;

(ii) $A \subseteq [-u, u] + \varepsilon B_X$ iff $\sup\limits_{x \in A}\big\|\big(|x| - u\big)^+\big\| \leqslant \varepsilon$;

(iii) A is almost order bounded iff for every $\varepsilon > 0$ there exists $u > 0$ such that $\big\|\big(|x| - u\big)^+\big\| \leqslant \varepsilon$ for every $x \in A$.

9.7.2 Exercise. Every almost order bounded set is norm bounded.

9.7.3 Exercise. Show that in Amemiya's Theorem 9.3.9, one can replace order intervals with almost order bounded sets.

For the rest of this section, let (Ω, Σ, μ) be a measure space. Let $0 \leqslant f \in L_1(\mu)$; let M be a positive real. As before, we write $\mu(f \geqslant M)$ for the measure of the set $\big\{\omega \in \Omega \,:\, f(\omega) \geqslant M\big\}$. It follows from $\|f\| \geqslant \int_{\{f \geqslant M\}} f \geqslant M\mu(f \geqslant M)$ that $\mu(f \geqslant M) \leqslant \frac{1}{M}\|f\|$; this is often referred to as ***Chebyshev's Inequality***. It is a standard fact of Measure Theory that for every $0 \leqslant f \in L_1(\mu)$ and every $\varepsilon > 0$ there exists $\delta > 0$ such that for every measurable set A with $\mu(A) < \delta$ we have $\int_A f\,d\mu < \varepsilon$ (this may be proved directly or deduced from the fact that $f\mu \ll \mu$ as in Remark 8.3.15.) A bounded subset F of $L_1(\mu)$ is said to be ***equi-integrable*** or ***uniformly integrable*** if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every measurable set A with $\mu(A) < \delta$ we have $\int_A |f|\,d\mu < \varepsilon$ for every $f \in F$.

9.7.4 Exercise. A bounded subset F of $L_1(\mu)$ fails to be equi-integrable iff there exists an $\varepsilon > 0$, a sequence of measurable sets (A_n) with $\mu(A_n) \to 0$, and a sequence (f_n) in F with $\int_{A_n} |f_n|\,d\mu > \varepsilon$ for every n.

9.7.5. A bounded sequence (f_n) in $L_1(\mu)$ is said to be equi-integrable if the set $\{f_n\}_n$ is equi-integrable. The preceding exercise implies that uniform integrability is a sequential property: F is uniformly integrable iff every sequence in it is uniformly integrable.

9.7.6. Note that F is equi-integrable iff $\big\{|f| \,:\, f \in F\big\}$ is equi-integrable iff $\operatorname{Sol} F$ is equi-integrable. This often allows us to assume WLOG that F consists of positive functions.

9.7.7 Example. Let μ be a finite measure and (f_n) a disjoint sequence in $L_1(\mu)$ with $\|f_n\| = 1$ for every n. It is easy to see that (f_n) is not equi-integrable.

9.7.8 Exercise. If F and G are equi-integrable then so is $F + G$.

9.7.9 Exercise. Let F be a bounded subset of $L_1(\mu)_+$. Then F is equi-integrable iff for every $\varepsilon > 0$ there exists $M > 0$ such that $\int_{\{f \geqslant M\}} f < \varepsilon$ for all $f \in F$. That is,

$$\lim_{M \to \infty} \sup_{f \in F} \int_{\{f \geqslant M\}} f = 0.$$

Next, we are going to prove that for bounded subsets of $L_1(\mu)$ where μ is finite, a set is equi-integrable iff it is almost order bounded.

9.7.10 Proposition. *Let F be a bounded set in $L_1(\mu)$. If F is almost order bounded then it is equi-integrable. The converse is true when μ is a finite measure.*

Proof. WLOG, $F \subset L_1(\mu)_+$. Suppose that F is almost order bounded. Fix $\varepsilon > 0$. Find $u \geqslant 0$ such that $F \subseteq [-u, u] + \varepsilon B_{L_1(\mu)}$. Find $\delta > 0$ such that $\int_A u < \varepsilon$ whenever $\mu(A) < \delta$. For every $f \in F$ we have $f = f \wedge u + (f - u)^+$, so that

$$\int_A f = \int_A f \wedge u + \int_A (f - u)^+ \leqslant \int_A u + \big\|(f - u)^+\big\| < 2\varepsilon.$$

Suppose now that μ is finite and F is equi-integrable. Fix $\varepsilon > 0$. Let M be as in Exercise 9.7.9. Put $u = M\mathbb{1}$. For every $f \in F$, we have $\big\|(f - u)^+\big\| \leqslant \int_{\{f \geqslant M\}} f < \varepsilon$. □

The assumption that the measure is finite in the last clause of the preceding proposition cannot be removed; see Example 9.8.7.

9.7.11 Exercise. If $f_n \to 0$ then (f_n) is equi-integrable.

We observed in 9.7.5 that equi-integrability is a sequential property: a set has it iff every countable subset has it. For a subset of $L_1(\mu)$, where μ is a finite measure, Proposition 9.7.10 tells us that almost order boundedness is equivalent to equi-integrability; hence almost order boundedness is also sequential. This remains valid for infinite measures, though, the proof is more involved:

9.7.12 Proposition. *Let A be a subset of $L_1(\mu)$. If every countable subset of A is almost order bounded then so is A itself.*

Proof. WLOG $A \subseteq X_+$; otherwise, replace A with $\big\{|a| : a \in A\big\}$. Suppose A is not almost order bounded. Then there exists $\varepsilon > 0$ such that for every $u \in X_+$ there exists $x \in A$ with $\big\|(x - u)^+\big\| > \varepsilon$. Applying this inductively, construct a sequence (x_n) in A such that $\Big\|\big(x_{n+1} - \bigvee_{i=1}^n x_i\big)^+\Big\| > \varepsilon$. It suffices to prove that the sequence (x_n) is not almost order bounded. Suppose it is. Then there exists $u \in X_+$ such that $\big\|(x_n - u)^+\big\| < \frac{\varepsilon}{2}$ for every n. Put $y_n = x_n \wedge u$ and $v_n = \big(y_{n+1} - \bigvee_{i=1}^n y_i\big)^+$. We are going to show that, on one hand,

$\lVert v_n\rVert \geqslant \frac{\varepsilon}{2}$, but, on the other hand, $u \geqslant \sum_{n=1}^{m} v_n$ and, therefore, $\lVert u\rVert \geqslant \frac{m\varepsilon}{2}$ for every m, which is a contradiction.

Applying the inequality $(a-c)^+ - (b-c)^+ \leqslant |a-b|$, where $a,b,c \geqslant 0$, we get

$$v_n \geqslant \Bigl(x_{n+1} \wedge u - \bigvee_{i=1}^{n} x_i\Bigr)^+ \geqslant \Bigl(x_{n+1} - \bigvee_{i=1}^{n} x_i\Bigr)^+ - \bigl|x_{n+1} - x_{n+1} \wedge u\bigr|$$
$$= \Bigl(x_{n+1} - \bigvee_{i=1}^{n} x_i\Bigr)^+ - (x_{n+1} - u)^+$$

for every n, so that

$$\varepsilon < \Bigl\lVert \bigl(x_{n+1} - \bigvee_{i=1}^{n} x_i\bigr)^+ \Bigr\rVert \leqslant \lVert v_n\rVert + \bigl\lVert (x_{n+1} - u)^+ \bigr\rVert.$$

It follows from $\bigl\lVert (x_{n+1} - u)^+ \bigr\rVert < \frac{\varepsilon}{2}$ that $\lVert v_n\rVert \geqslant \frac{\varepsilon}{2}$.

Applying inductively the identity $a + (b-a)^+ = a \vee b$, we get

$$\sum_{n=1}^{m} v_n \leqslant y_1 + (y_2 - y_1)^+ + (y_3 - y_1 \vee y_2)^+ + \dots + (y_{m+1} - y_1 \vee \dots \vee y_m)^+$$
$$= y_1 \vee \dots \vee y_{m+1} \leqslant u$$

for every m. □

Equi-intergability is really a property of distributions of functions rather than the functions themselves; see A.3.14. Given $f \in L_1(\mu)_+$, where μ is a probability measure, it follows from (A.3.14) that

$$\int_{\{f \geqslant M\}} f = M \cdot \mu(f \geqslant M) + \int_{M}^{\infty} \mu(f \geqslant t)\,dt. \tag{9.2}$$

It now follows from Exercise 9.7.9 that an equi-integrable set $F \subseteq L_1(\mu)_+$ remains equi-integrable if we replace each function in F with another function with the same distribution. Moreover, these new functions may even "live" in a different probability space.

Given $f \in L_1(\mu)_+$ and $g \in L_1(\nu)_+$, we say that f is ***dominated by g in distribution*** if $\mu(f \geqslant t) \leqslant \nu(g \geqslant t)$ for every t. By Proposition 9.7.10, every order bounded set is equi-integrable. In fact, "order bounded" may be replaced with "bounded in distribution":

9.7.13 Exercise. Let $F \subseteq L_1(\mu)$ for some probability measure μ, and there exists $g \in L_1(\nu)$ such that every $f \in F$ is bounded by g in distribution. Then F is equi-integrable.

In Kadeč-Pełczyński's Theorem 9.6.6, the sequence was assumed to be bounded and to converge in measure to zero. The following lemma, which is also due to Kadeč and Pełczyński, is valid for every bounded sequence. However, of course, the conclusion is weaker.

9.7.14 Lemma (Subsequence Splitting Lemma)**.** *Let (f_n) be a bounded sequence in*

$L_1(\mu)$, where μ is a finite measure. Then there exists a subsequence (f_{n_k}) such that $f_{n_k} = g_k + h_k$ where (g_k) is disjoint, (h_k) is equi-integrable, and $g_k \perp h_k$ for every k.

Proof. WLOG, μ is a probability measure. Arguing similarly to Exercise 9.6.1(i), we may assume that $f_n \geqslant 0$ for all n. It is clear that if we prove the lemma for (f_n), using the same subsequence and splitting along the same support sets will yield a splitting for any sequence (f'_n) satisfying $0 \leqslant f'_n \leqslant f_n$ for every n. Hence, WLOG, replacing each f_n with its ceiling $\lceil f_n \rceil$, we may assume that each f_n is integer valued (it follow from $\lceil f_n \rceil \leqslant f_n + \mathbb{1}$ that the sequence $\bigl(\lceil f_n \rceil\bigr)$ is still bounded). WLOG, $\|f_n\| \leqslant 1$ for every n.

We will now apply a standard diagonal process to ensure that for every k the sequence $\bigl(\mu(f_n \geqslant k)\bigr)_n$ converges in n, and does so "sufficiently fast". Since the sequence $\bigl(\mu(f_n \geqslant 1)\bigr)_n$ is bounded, there is a subsequence $(f_n^{(1)})$ of (f_n) such that $\mu(f_n^{(1)} \geqslant 1)$ converges to some α_1 and, moreover, such that $\mu(f_n^{(1)} \geqslant 1) \leqslant \alpha_1 + \frac{1}{2}$ for all n. Similarly, there is a subsequence $(f_n^{(2)})$ of $(f_n^{(1)})$ such that $\alpha_2 := \lim_n \mu(f_n^{(2)} \geqslant 2)$ exists and $\mu(f_n^{(2)} \geqslant 2) \leqslant \alpha_2 + \frac{1}{4}$ for all n. Proceeding inductively, for every $k \in \mathbb{N}$ we get $\alpha_k = \lim_n \mu(f_n^{(k)} \geqslant k)$ and $\mu(f_n^{(k)} \geqslant k) \leqslant \alpha_k + \frac{1}{2^k}$ for all n. The "diagonal" sequence $(f_n^{(n)})$ is a subsequence of (f_n); we relabel $f_n^{(n)}$ as f_n. We now have $\mu(f_n \geqslant k) \leqslant \alpha_k + \frac{1}{2^k}$ as $k = 1, \dots, n$.

For every $m \in \mathbb{N}$,

$$\sum_{k=1}^{m} \alpha_k = \sum_{k=1}^{m} \lim_{n\to\infty} \mu(f_n \geqslant k) = \lim_{n\to\infty} \sum_{k=1}^{m} \mu(f_n \geqslant k) \leqslant \lim_{n\to\infty} \int f_n \leqslant 1.$$

It follows that $\sum_{k=1}^{\infty} \alpha_k$ converges. Then, of course, $\sum_{k=1}^{\infty}(\alpha_k + \frac{1}{2^k})$ converges. It is now easy to find an integer-valued $g \in L_1[0,1]_+$ such that $\mu(g \geqslant k) = \alpha_k + \frac{1}{2^k}$ for all $k \in \mathbb{N}$.

We split each f_n into a disjoint sum $f_n = u_n + v_n$, where $u_n = f_n \cdot \mathbb{1}_{\{f_n \leqslant n\}}$ and $v_n = f_n \cdot \mathbb{1}_{\{f_n > n\}}$. We claim that the sequence (u_n) is bounded by g in distribution and, therefore, (u_n) is equi-integrable by Exercise 9.7.13. Indeed, let $k, n \in \mathbb{N}$. If $k \leqslant n$ then

$$\mu(u_n \geqslant k) \leqslant \mu(f_n \geqslant k) \leqslant \alpha_k + \frac{1}{2^k} = \mu(g \geqslant k),$$

while if $k > n$ then $\mu(u_n \geqslant k) = 0$. Hence, $\mu(u_n \geqslant k) \leqslant \mu(g \geqslant k)$ for all $n, k \in \mathbb{N}$. Since the functions are positive integer valued, we have $\mu(u_n \geqslant t) \leqslant \mu(g \geqslant t)$ for all $n \in \mathbb{N}$ and $t \in \mathbb{R}$.

Since, by Chebyshev's Inequality, $\mu(f_n > n) \leqslant \frac{1}{n} \to 0$, we conclude that (v_n) converges to zero in measure. By Kadeč-Pełczyński Theorem 9.6.6, after passing to a further subsequence of (f_n), we may split each v_n into a disjoint sum $d_n + g_n$ where (d_n) is a disjoint sequence and $g_n \to 0$. In particular, (g_n) is equi-integrable by Exercise 9.7.11. Put $h_n = u_n + g_n$, then (h_n) is equi-integrable and, therefore, $f_n = d_n + h_n$ is a required splitting. $\square$

9.8 Dunford-Pettis and Grothendieck Theorems

Recall that a subset A of a Banach space X is ***relatively weakly compact*** if it is contained in a weakly compact set. By Uniform Boundedness Principle, a relatively weakly compact set is bounded. Moreover, if X is reflexive then it follows from the Alaoglu-Bourbaki Theorem that a set is relatively weakly compact iff it is bounded.

Eberlein-Šmulian's Theorem A.4.20 asserts that a set is relatively weakly compact iff every sequence in it has a weakly convergent subsequence. It follows that relative weak compactness is a sequential property; see A.4.21.

In this section we present two classical theorems of Functional Analysis, — the Dunford-Pettis Theorem and Grothendieck's Theorem, — which provide several equivalent characterizations of relatively weakly compact subsets of L_1-spaces and of $C(K)^*$. At the first glance, these theorems seemingly only deal with very specific spaces. However, they lead to many important results about general Banach lattices via $C(K)$ and L_1-representations. They are also a source of motivation for many results in Banach lattice theory: while not all the characterizations they present remain equivalent in general Banach lattices, the conditions themselves turn out to be very interesting and prompt studies of classes of Banach lattices satisfying them. The proofs the two theorem are rather long and technical, and are based on Measure Theory machinery. We include them for convenience of the reader but an impatient reader may want to skip them at the first reading. Our proofs of these theorems are loosely based on those of [AK06].

We start with some motivations and preliminaries.

9.8.1 Exercise. Every relatively compact set in a normed lattice is almost order bounded.

On the other hand, the unit ball in $C[0,1]$ is order bounded, hence almost order bounded, yet not relatively compact (and not even relatively weakly compact).

Let A be a subset of a Banach space X and let (f_n) be a sequence in X^*. We say that (f_n) ***converges to zero uniformly on*** A if for every $\varepsilon > 0$ there exists n_0 such that $\lvert\langle x, f_n\rangle\rvert < \varepsilon$ whenever $x \in A$ and $n \geqslant n_0$[3].

9.8.2 Exercise. For a subset A in a Banach space X and a sequence (f_n) in X^*, TFAE:

(i) (f_n) converges to zero uniformly on A;

(ii) $\lim\limits_{n\to\infty} \sup\limits_{x\in A}\lvert\langle x, f_n\rangle\rvert = 0$;

(iii) $\langle x_n, f_n\rangle \to 0$ for every sequence (x_n) in A.

9.8.3. The exercise implies that uniform convergence is sequential: (f_n) converges to zero uniformly on A iff (f_n) converges to zero uniformly on every countable subset of A. Clearly, in the definition of uniform convergence, we may interchange X and X^*. For

[3]This should not be confused with (relative) uniform convergence on vector lattices in Section 2.3

readers familiar with the terminology of dual pairs we note that this definition naturally extends to the case when (X,Y) is a dual pair, $A \subseteq X$, and (f_n) is a sequence in Y.

9.8.4. Uniform convergence is given by a seminorm. Let A be a bounded subset of a Banach space X. For $f \in X^*$, we define $\rho_A(f) = \sup_{x\in A}|f(x)|$. Then ρ_A is a seminorm on X^* and (f_n) converges to zero uniformly on A iff $\rho_A(f_n) \to 0$. In a similar fashion, if A is a bounded subset of X^*, the resulting uniform convergence on X is given by a seminorm, namely, the restriction of ρ_A from X^{**} to X.

Throughout this section, (e_n) stands for the standard unit vector basis of ℓ_1. A basic sequence (x_n) is said to be ***complemented*** if its closed linear span $[x_n]$ is a complemented subspace.

9.8.5 Exercise. Let (x_n) be a bounded sequence in a Banach space X, $T\colon X \to \ell_1$ a bounded operator, and $\lambda < 1$ such that $\|Tx_n - e_n\| \leqslant \lambda$ for all n. Then (x_n) is a basic sequence, (x_n) is equivalent to (e_n), and $[x_n]$ is complemented in X.

9.8.6 Theorem (Dunford-Pettis)**.** *Let F be a bounded set in $L_1(\mu)$ where μ is a finite measure. TFAE:*

(i) *F is relatively weakly compact;*
(ii) *F is equi-integrable;*
(iii) *F is almost order bounded;*
(iv) *F contains no basic sequence equivalent to (e_n) in ℓ_1;*
(v) *F contains no complemented basic sequence equivalent to (e_n) in ℓ_1;*
(vi) *For every disjoint sequence (A_n) in Σ, $\int_{A_n}|f| \to 0$ as $n \to \infty$ uniformly on F;*
(vii) *Every disjoint bounded sequence in $L_\infty(\mu)_+$ converges to zero uniformly on F.*

All the conditions except (ii) *remain equivalent for infinite measures.*

Proof. We start with easier implications. We already know that (ii)⇔(iii) by Proposition 9.7.10. (i) implies (iv) because $\{e_n\}$ is not relatively weakly compact in ℓ_1. (iv)⇒(v) is trivial. (ii)⇒(vi) because $\mu(A_n) \to 0$.

(vi)⇒(vii) Let (g_n) be a disjoint bounded sequence in $L_\infty(\mu)$. WLOG, $|g_n| \leqslant \mathbb{1}$. Let $A_n = \operatorname{supp} g_n$, then (A_n) is a disjoint sequence in Σ. We have $\bigl|\langle g_n, f\rangle\bigr| \leqslant \int_{A_n}|f| \to 0$ uniformly on F.

(vii)⇒(vi) Observe first that (vii) remains valid for $|F|$. Indeed, let (g_n) be a disjoint bounded sequence in $L_\infty(\mu)_+$ and (f_n) a sequence in F. Note the $\bigl(g_n \cdot \mathbb{1}_{\{f_n>0\}}\bigr)$ is a disjoint bounded sequence in $L_\infty(\mu)_+$, so that, by assumption and by Exercise 9.8.2, we have $\bigl\langle g_n, f_n^+\bigr\rangle = \bigl\langle g_n \cdot \mathbb{1}_{\{f_n>0\}}, f_n\bigr\rangle \to 0$. Similarly, $\bigl\langle g_n, f_n^-\bigr\rangle \to 0$, hence $\bigl\langle g_n, |f_n|\bigr\rangle \to 0$. By Exercise 9.8.2, this means that (g_n) converges to zero uniformly on $|F|$. Now apply this with $g_n = \mathbb{1}_{A_n}$.

(ii)⇒(i) Suppose that F is equi-integrable. It suffices to show that the set $\bigl\{f^+, f^- : f \in F\bigr\}$ is relatively weakly compact because this would imply that F is relatively weakly compact by consecutively applying Eberlein-Šmulian Theorem A.4.20 to positive and negative parts. So we assume WLOG that $F \subseteq L_1(\mu)_+$.

Recall that a bounded set A in a Banach space X is relatively weakly compact iff $\overline{A}^{w^*} \subseteq X$ in X^{**}; see Proposition A.4.18. Let $\xi \in \overline{F}^{w^*}$ in $L_1(\mu)^{**}$. Then $\xi \geqslant 0$. For $A \in \Sigma$, put $\nu(A) = \langle \xi, \mathbb{1}_A \rangle$. It is easy to see that ν is a finitely-additive measure on Σ.

We claim that $\forall \varepsilon > 0\ \exists \delta > 0$ such that if $\mu(A) < \delta$ then $\nu(A) < \varepsilon$. Indeed, let $\varepsilon > 0$. Since F is equi-integrable, there exists $\delta > 0$ such that if $\mu(A) < \delta$ then we have $\langle f, \mathbb{1}_A \rangle = \int_A f < \varepsilon$, for all $f \in F$, where we view $\mathbb{1}_A$ as an element of $L_\infty(\mu)$. It follows from $\xi \in \overline{F}^{w^*}$ that $\nu(A) = \langle \xi, \mathbb{1}_A \rangle \leqslant \varepsilon$. This proves the claim. It follows that ν is countably additive by Exercise 8.3.1 and $\nu \ll \mu$. Radon-Nikodym Theorem 8.3.14 yields that $\nu = g \cdot \mu$ for some $g \in L_1(\mu)_+$. Note that $\langle \xi, \mathbb{1}_A \rangle = \nu(A) = \int_A g = \langle g, \mathbb{1}_A \rangle$ for every $A \in \Sigma$. It follows that ξ and g agree on simple functions. Since simple functions are dense in $L_\infty(\mu)$, we have $\xi = g$, so that $\xi \in L_1(\mu)$.

To complete the proof of the theorem, it is left to show that (v) and (vi) both imply (ii). Suppose that (ii) fails. Then there exists a sequence (f_n) in F and $c > 0$ such that $\int_{\{|f_n|>n\}} |f_n| > c$. WLOG (scaling F), we may assume that $c = 1$. Applying the Subsequence Splitting Lemma 9.7.14, we find a subsequence (f_{n_k}) and a splitting of $\operatorname{supp} f_n$ into the disjoint union of two measurable sets A_k and B_k such that the sequence (A_k) is disjoint and the sequence $(f_{n_k} \cdot \mathbb{1}_{B_k})$ is equi-integrable. It follows that

$$\int_{B_k \cap \{|f_{n_k}|>n_k\}} |f_{n_k}| \to 0 \quad \text{and, therefore,} \quad \int_{A_k} |f_{n_k}| \geqslant \int_{A_k \cap \{|f_{n_k}|>n_k\}} |f_{n_k}| > \frac{1}{2}$$

for all sufficiently large n. It follows that (vi) fails.

To prove that (v) fails, re-label (f_{n_k}) as (f_n) and put $\lambda_n = \int_{A_n} |f_n|$ for every n. Then, after passing to a tail, we have $\frac{1}{2} < \lambda_n \leqslant \|f_n\|$, so that the sequence (λ_n) is bounded above because F is bounded.

Fix $k \in \mathbb{N}$. Since $(f_n \cdot \mathbb{1}_{B_n})$ is equi-integrable, we can find $\delta > 0$ such that if $\mu(A) < \delta$ then $\int_A \bigl|f_m \cdot \mathbb{1}_{B_m}\bigr| < \frac{1}{2^{k+2}}$ for all m. Since the sequence (A_n) is disjoint, we have $\mu(A_n) \to 0$, so that there exists n_k such that $\mu(A_{n_k}) < \delta$ and, therefore, $\int_{A_{n_k}} \bigl|f_m \cdot \mathbb{1}_{B_m}\bigr| < \frac{1}{2^{k+2}}$. Iterating this, we get a subsequence (n_k). Hence, passing to this subsequence, we get $\int_{A_n} \bigl|f_m \cdot \mathbb{1}_{B_m}\bigr| < \frac{1}{2^{n+2}}$ for all n and m. We will prove that (f_n) is a complemented basic sequence equivalent to (e_n).

If $n \neq m$ then, since $\operatorname{supp} f_m = A_m \cup B_m$ and $A_n \cap A_m = \varnothing$, we have

$$\int_{A_n} |f_m| = \int_{A_n \cap B_m} |f_m| = \int_{A_n} |f_m \cdot \mathbb{1}_{B_m}| < \tfrac{1}{2^{n+2}}.$$

For each n, put $h_n = \operatorname{sign} f_n \cdot \mathbb{1}_{A_n}$, so that $f_n h_n = |f_n| \cdot \mathbb{1}_{A_n}$. Define an operator $T\colon L_1(\mu) \to \ell_1$ via $(Tf)_n = \int f h_n$ for $f \in L_1(\mu)$. That is, Tf is a vector in ℓ_1 whose n-th component is $\int f h_n$. Note that T is bounded:

$$\|Tf\|_{\ell_1} = \sum_{n=1}^{\infty} \Bigl| \int f h_n \Bigr| \leqslant \sum_{n=1}^{\infty} \int_{A_n} |f| \leqslant \|f\|.$$

Fix m and consider Tf_m. It's m'th component is $\int f_m h_m = \int_{A_m} |f_m| = \lambda_m$. It follows that

$$\|Tf_m - \lambda_m e_m\| = \Big\| \sum_{n\neq m} \Big(\int f_m h_n \Big) e_n \Big\| = \sum_{n\neq m} \Big| \int f_m h_n \Big|$$
$$\leqslant \sum_{n\neq m} \int_{A_n} |f_m| < \sum_{n=1}^{\infty} \frac{1}{2^{n+2}} = \frac{1}{4}.$$

It follows from $\lambda_m > \frac{1}{2}$ that $\big\|T\big(\frac{1}{\lambda_m} f_m\big) - e_m\big\| < \frac{1}{2}$. Lemma 9.8.5 now yields that $\big(\frac{1}{\lambda_n} f_n\big)$ and, therefore, (f_n), is a complemented basic sequence equivalent to (e_n).

It is left to prove that conditions (iii) through (vii) remain equivalent for infinite measures. We will show that this case reduces to the case of a finite measure. Observe that all these conditions are sequential in the sense that F satisfies the condition iff every countable subset of F satisfies it; see (A.4.21), (9.8.3), and Proposition 9.7.12. Therefore, one may assume WLOG that F is countable. As in Remark 5.1.29, one may assume that F is contained in a principal band which, after lattice isometry, may be identified with $L_1(\nu)$ for some *finite* measure ν. □

9.8.7 Example. Condition (ii) is *not* equivalent to the other conditions when μ is infinite. Consider $X = \ell_1$; every subset of ℓ_1 is equi-integrable. For another example consider $X = L_1(\mathbb{R})$, and let F be the set of characteristic functions of the intervals $[n, n+1]$ as $n \in \mathbb{Z}$. It is easy to see that F is equi-integrable, yet it fails the other conditions in the theorem.

9.8.8 Exercise. Let F be a subset of $L_1(\mu)$. TFAE:

(i) F is relatively weakly compact;

(ii) its solid hull $\mathrm{Sol}(F)$ is relatively weakly compact;

(iii) the set $|F| = \big\{|f| \,:\, f \in F\big\}$ is relatively weakly compact.

9.8.9 Exercise. Prove that order intervals in $L_p(\mu)$ with $1 \leqslant p < \infty$ are weakly compact. (They are, generally, not compact by 6.4.9.)

A subset A of a Banach lattice X is said to be ***confined*** if every disjoint sequence in $\mathrm{Sol}\,A$ converges in norm to zero. This means that A is "small" in the sense that it has little room for disjoint sequences. Note that the condition is effectively imposed on $\mathrm{Sol}(A)$ rather than on A itself. This is done to ensure that there are "sufficiently many" disjoint sequences to start with. For example, if $A = \{\mathbb{1}\}$ in $L_1(\mu)$, then A contains no disjoint sequences, while $\mathrm{Sol}\,A = [-\mathbb{1}, \mathbb{1}]$ contains plenty. In many applications, this condition is only applied to solid sets anyway. We can now extend the list in Dunford-Pettis Theorem:

9.8.10 Corollary. *A bounded subset F of $L_1(\mu)$ is relatively weakly compact iff F is confined.*

Proof. By Exercise 9.8.8, we may assume WLOG that F is solid. Suppose that F is relatively weakly compact but fails to be confined. Then there exists a disjoint sequence (f_n) in F such that (f_n) does not converge to zero. Passing to a subsequence, we may assume that (f_n) is seminormalized. Being a disjoint seminormalized sequence in $L_1(\mu)$, (f_n) is equivalent to (e_n) in ℓ_1, which contradicts (iv) in Dunford-Pettis Theorem 9.8.6.

Conversely, suppose that F is confined but fails to be relatively weakly compact. By Dunford-Pettis Theorem 9.8.6(vi), we can find a sequence (f_n) in F, a sequence (A_n) of disjoint measurable sets, and $\varepsilon > 0$ such that $\int_{A_n} |f_n| > \varepsilon$ for all n. Put $h_n = |f_n|\mathbb{1}_{A_n}$, then $h_n \in F$ because F is solid, the sequence (h_n) is disjoint, and $\|h_n\| > \varepsilon$ for all n; a contradiction. $\square$

Motivated by this result, confined sets are often called L-weakly compact.

9.8.11 Corollary. *$L_1(\mu)$ is weakly sequentially complete for every measure μ.*

Proof. Suppose that (f_n) is a weakly Cauchy sequence in $L_1(\mu)$; we need to show that it is weakly convergent. Let $F = \{f_n\}$. It suffices to show that F is relatively weakly compact because this would imply that (f_n) has a weakly convergent subsequence by Eberlein-Šmulian Theorem A.4.20. By part (iv) of Dunford-Pettis Theorem 9.8.6, it suffices to show that F contains no basic sequence equivalent to (e_n) in ℓ_1. Suppose it does. Then, passing to a subsequence, we conclude that a subsequence of (f_n) is equivalent to (e_n). This is a contradiction because (e_n) is not weakly Cauchy. $\square$

9.8.12 Exercise. Every bounded sequence in $L_1(\mu)$, where μ is finite, may, after passing to a subsequence, be split into a disjoint sum of a disjoint and a weakly convergent sequences.

We are now going to consider the special case $X = C(K)^*$, where K is a compact space. By the Riesz-Representation Theorem 8.4.2, $C(K)^*$ may be identified with $\mathcal{M}(K)$, which is the space of all finite signed Baire measures on K. Since every finite Baire measure on K extends uniquely to a regular Borel measure, we may view elements of $\mathcal{M}(K)$ as finite signed regular Borel measures; see Appendix A.3.3. By Corollary 8.5.2, $\mathcal{M}(K)$ is lattice isometric to $L_1(\mu)$ for some measure μ, so that the Dunford-Pettis Theorem applies to $C(K)^*$ (except condition (ii) because μ need not be finite).

Since every measure $\mu \in \mathcal{M}(K)_+$ is regular, for every Borel set B and every $\varepsilon > 0$ there exists a closed (and, therefore, compact) set $F \subseteq B$ with $\mu(B \setminus F) < \varepsilon$. Taking complements, this is equivalent to the existence of an open set $U \supseteq B$ with $\mu(U \setminus B) < \varepsilon$.

A subset $\mathcal{A}$ of $\mathcal{M}(K)$ is said to be ***uniformly regular*** if for every open set U and every $\varepsilon > 0$ there exists a compact set $F \subseteq U$ such that $|\nu|(U \setminus F) < \varepsilon$ for all $\nu \in \mathcal{A}$.

9.8.13 Theorem (Grothendieck)**.** *For a bounded subset $\mathcal{A}$ of $\mathcal{M}(K)$, TFAE:*

(i) *$\mathcal{A}$ is relatively weakly compact;*

(ii) *For every disjoint sequence (U_n) of open sets, $\lim\limits_{n\to\infty} \sup\limits_{\nu\in\mathcal{A}} |\nu|(U_n) = 0$;*

(iii) *Every disjoint bounded sequence in $C(K)_+$ converges to zero uniformly on $\mathcal{A}$;*

(iv) *$\mathcal{A}$ is uniformly regular.*

Proof. As we mentioned before, we may identify $\mathcal{M}(K)$ with an L_1-space, so that the Dunford-Pettis Theorem applies. Also, we assume WLOG that $\mathcal{A} \subseteq \mathcal{M}(K)_+$. Indeed, put $|\mathcal{A}| = \{|\nu| : \nu \in \mathcal{A}\}$. We claim that each of the conditions is satisfied for $\mathcal{A}$ iff it is satisfied for $|\mathcal{A}|$, so that WLOG we may replace $\mathcal{A}$ with $|\mathcal{A}|$. For (ii) and (iv), this is obvious. For (i), it follows from Exercise 9.8.8. For (iii), the argument is the same as in the proof of (vii)⇒(vi) in the Dunford-Pettis Theorem.

(i)⇒(ii) Apply the equivalence of (i) and (vi) in Dunford-Pettis Theorem 9.8.6.

(ii)⇒(iii) Let (x_n) be a disjoint positive bounded sequence in $C(K)$. WLOG, $\|x_n\| \leqslant 1$ for every n. Put $U_n = \{x_n \neq 0\}$. Then (U_n) is a disjoint sequence of open subsets of K. By (ii), the sequence $(\mathbb{1}_{U_n})$ converges to zero uniformly on $\mathcal{A}$. Since $x_n \leqslant \mathbb{1}_{U_n}$ for every n, we conclude that the same is true for (x_n).

(iii)⇒(iv) Suppose (iv) fails. Then there exists an open set U and $\delta > 0$ such that

$$\text{for every closed subset } F \text{ of } U \text{ there exists } \mu \in \mathcal{A} \text{ with } \mu(U \setminus F) > \delta. \tag{9.3}$$

We will construct sequences (G_n) of closed sets, (V_n) of open sets, and (μ_n) in $\mathcal{A}$ with $G_n \subseteq V_n \subseteq \overline{V_n} \subseteq U$ in such a way that $\mu_n(G_n) > \delta$ for every n and that the sequence $(\overline{V_n})$ is disjoint.

Applying (9.3) with $F = \varnothing$, find $\mu_1 \in \mathcal{A}$ with $\mu_1(U) > \delta$. Regularity of μ_1 implies the existence of a closed set $G_1 \subseteq U$ with $\mu_1(G_1) > \delta$. Find an open set V_1 with $G_1 \subseteq V_1 \subseteq \overline{V_1} \subseteq U$.

Suppose that G_k, V_k, and μ_k as $k = 1, \dots, n-1$ have been constructed. Applying (9.3) with $F = F_n = \overline{V_1} \cup \dots \cup \overline{V_{n-1}}$, we find $\mu_n \in \mathcal{A}$ with $\mu_n(U \setminus F_n) > \delta$. Since μ_n is regular, there exists a closed subset G_n of $U \setminus F_n$ with $\mu_n(G_n) > \delta$. Find an open V_n with $G_n \subseteq V_n \subseteq \overline{V_n} \subseteq U \setminus F_n$. It follows that $\overline{V_n} \cap \overline{V_k} = \varnothing$ as $k = 1, \dots, n-1$.

For each n, use Urysohn's Lemma to find $x_n \in C(K)$ with $0 \leqslant x_n \leqslant \mathbb{1}$ such that x_n equals one on G_n and vanishes outside of $\overline{V_n}$. It follows that (x_n) is disjoint and $\int x_n \, d\mu_n \geqslant \mu_n(G_n) > \delta$ for every n. By Exercise 9.8.2, this means that (x_n) fails (iii).

(iv)⇒(i) By Eberlein-Šmulian Theorem A.4.20, we may assume WLOG that $\mathcal{A}$ is countable. By Exercise 2.2.1(viii), $\mathcal{A}$ is contained in a principal ideal I_μ and, therefore, in B_μ for some $\mu \in \mathcal{M}(K)_+$ (in particular, μ is a finite measure). By Proposition 8.3.13 and Remark 8.3.15, B_μ is lattice isometric to $L_1(\mu)$ with $\nu \in B_\mu \mapsto \frac{d\nu}{d\mu} \in L_1(\mu)$. Let F be the image of $\mathcal{A}$ under this lattice isometry. It suffices to prove that F is relatively weakly compact in $L_1(\mu)$. By Dunford-Pettis Theorem 9.8.6, it suffices to prove that F is equi-integrable. Suppose not. Then there exists $\varepsilon > 0$ and measurable sets (A_n) such that for every $n \in \mathbb{N}$ we have $\mu(A_n) < \frac{1}{2^{n+1}}$ and $\sup_{f\in F} \int_{A_n} f \, d\mu > \varepsilon$. The latter inequality means that there exists $\nu_n \in \mathcal{A}$ such that $\nu_n(A_n) > \varepsilon$. Since μ is regular, we can find an open set $U_n \supseteq A_n$

such that $\mu(U_n) < \frac{1}{2^{n+1}}$. For each n, put $V_n = \bigcup_{k=n}^{\infty} U_k$. Then V_n is open, $V_{n+1} \subseteq V_n$, $\mu(V_n) < \frac{1}{2^n}$, and $\nu_n(V_n) > \varepsilon$ (because $A_n \subseteq U_n \subseteq V_n$).

By the uniform regularity of $\mathcal{A}$, for every n we can find a closed set $F_n \subseteq V_n$ such that $\nu(V_n \setminus F_n) < \frac{\varepsilon}{2^{n+1}}$ for all $\nu \in \mathcal{A}$. Put $F_0 = \bigcap_{n=1}^{\infty} F_n$. Clearly, F_0 is compact. It follows from $\mu(F_n) \leqslant \mu(V_n) \to 0$ that $\mu(F_0) = 0$. Since $\mathcal{A} \subseteq B_\mu$, for every measure ν is $\mathcal{A}$ we have $\nu \ll \mu$ by Theorem 8.2.7 and, therefore, $\nu(F_0) = 0$. Applying the uniform regularity of $\mathcal{A}$ to the complement of F_0, we find a closed set $G \subseteq F_0^C$ such that $\nu(F_0^C \setminus G) < \frac{\varepsilon}{2}$ for all $\nu \in \mathcal{A}$. Let $W = G^C$, then W is open, $F_0 \subseteq W$, and $\nu(W) \leqslant \nu(W \setminus F_0) + \nu(F_0) = \nu(F_0^C \setminus G) < \frac{\varepsilon}{2}$ for all $\nu \in \mathcal{A}$.

Since each F_n is compact and $\bigcap_{n=1}^{\infty} F_n \subseteq W$, we conclude that $\bigcap_{n=1}^{m} F_n \subseteq W$ for some m. It can be easily verified that

$$V_m \subseteq \Bigl(\bigcap_{n=1}^{m} F_n\Bigr) \cup \Bigl(\bigcup_{n=1}^{m} (V_m \setminus F_n)\Bigr) \subseteq W \cup \Bigl(\bigcup_{n=1}^{m} (V_n \setminus F_n)\Bigr).$$

It follows that for every $\nu \in \mathcal{A}$ we have

$$\nu(V_m) \leqslant \nu(W) + \sum_{n=1}^{m} \nu(V_n \setminus F_n) \leqslant \frac{\varepsilon}{2} + \sum_{n=1}^{m} \frac{\varepsilon}{2^{n+1}} < \varepsilon.$$

This contradicts $\nu_m(V_m) > \varepsilon$. $\square$

In view of Exercise 9.8.2(iii), condition (vi) in Dunford-Pettis Theorem 9.8.6 may be re-stated as follows: $\int_{A_n} |f_n| \to 0$ for every sequence in F and every disjoint sequence (A_n) of measurable sets. Similarly, condition (iii) in Grothendieck Theorem 9.8.13 means that $|\nu_n|(U_n) \to 0$ for every sequence (ν_n) in $\mathcal{A}$ and every disjoint sequence (U_n) of open sets.

Both the Dunford-Pettis and Grothendieck Theorems characterize relative weak compactness of a bounded set in an AL-space in terms of its "action" on disjoint bounded sequences in the dual space. While the Dunford-Pettis Theorem applies to arbitrary AL-spaces, Grothendieck's Theorem only applies to AL-spaces of the form $X = C(K)^*$. However, the advantage of Grothendieck's Theorem is that it suffices to consider disjoint sequences in $C(K)$, the pre-dual space of X, instead of X^*.

While some of the conditions that appear in the Dunford-Pettis Theorem may still be defined for arbitrary Banach lattices, they are generally not equivalent beyond $L_1(\mu)$ spaces. We will various relationships between these and other similar conditions in Chapter 11.

Chapter 10

Order continuous Banach lattices

In this chapter, we will focus on order continuous Banach lattices. In these spaces, the relationship between norm and order structures in these spaces is especially nice. Another reason why this class of spaces is important is that it admits several important equivalent characterizations.

Let's recall a few facts. A Banach lattice X is order continuous if $x_\alpha \xrightarrow{o} x$ implies $x_\alpha \to x$ or, equivalently, $x_\alpha \downarrow 0$ implies $x_\alpha \to 0$. Nakano's Theorem 2.5.24 asserts that X is order continuous iff it is both σ-order complete and σ-order continuous iff every increasing order bounded sequence converges. The latter condition my be rephrased symbolically as follows: $[0,u] \ni x_n \uparrow$ implies (x_n) is convergent.

Furthermore, if X is order continuous then

- X is order complete;
- Every functional in X^* is order continuous;
- Every closed sublattice of X is order continuous;
- A subset of X is closed iff it is order closed;
- Bands and closed ideals in X agree;
- "quasi-interior points = weak units".

If, in addition, X has a weak unit then X admits a strictly positive functional and, furthermore, a very nice AL-representation by Theorem 9.3.2.

Recall that c_0, ℓ_p and $L_p(\mu)$ are order continuous when $1 \leqslant p < \infty$. On the other hand, ℓ_∞ and L_∞ fail to be order continuous.

Table 10.1 illustrates some of the relationships between convergences in a Banach lattice.

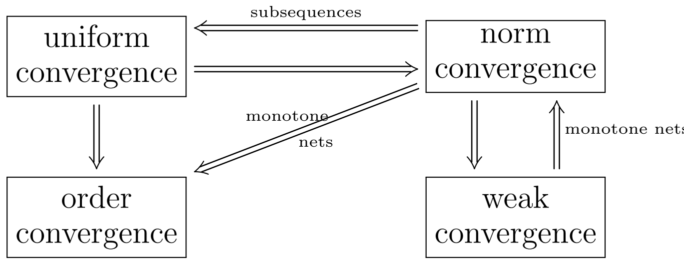

Table 10.1: Relationships between convergences in a Banach lattice

10.1 Characterizations of order continuity

The following result is an application of Grothendieck's theorem to general Banach lattices. Recall that every disjoint order bounded sequence in a Banach lattice is weakly null and every order interval $[a,b]$ is weakly closed; see Exercise 6.8.8(i) and 6.8.1(iii), respectively. Recall also that a solid set A is *confined* (or L-weakly compact) if every disjoint sequence in A is null.

10.1.1 Theorem. *Let e be a positive vector in a Banach lattice X. Then $[0,e]$ is weakly compact iff it is confined.*

Proof. By $C(K)$-representation theory, $\big(I_e, \|\cdot\|_e\big)$ is lattice isometric to $C(K)$ for some compact K, with e corresponding to $\mathbb{1}$. Let $T\colon I_e \to X$ be the inclusion map; it is continuous because $\|x\|_X \leqslant \|x\|_e \|e\|_X$ for every $x \in I_e$. Observe that $[0,e]$ is weakly compact (in X) iff $[-e,e]$ is weakly compact iff T is weakly compact because $[-e,e] = B_{I_e}$. By Gantmacher's Theorem A.4.23, this is equivalent to $T^*\colon X^* \to I_e^*$ being weakly compact, that is $T^*(B_{X^*})$ being relatively weakly compact in I_e^*. Since $I_e \cong C(K)$, Grothendieck's Theorem 9.8.13(iii) implies that this is equivalent to every disjoint sequence (x_n) in $[0,e]$ converge to zero uniformly on $T^*(B_{X^*})$. Since T^* is a restriction operator, this is equivalent to $\|x_n\|_X \to 0$. $\square$

Condition (ii) in the following theorem is most useful; it is due to Meyer-Nieberg. Recall that Nakano's Theorem 2.5.24 asserts that X is order continuous iff $0 \leqslant x_n \uparrow \leqslant u$ implies that (x_n) converges.

10.1.2 Theorem (Meyer-Nieberg)**.** *For a Banach lattice X, TFAE:*

(i) *X is order continuous;*
(ii) *Every order bounded disjoint sequence converges to zero;*
(iii) *Every order interval is weakly compact.*

Proof. (i)⇒(ii) Suppose that X is order continuous (and, therefore, order complete). Let (x_n) be a disjoint order bounded sequence. Then $v_m := \sum_{n=1}^m |x_n|$ is an increasing order bounded sequence. Hence it converges by Nakano's Theorem 2.5.24. Therefore, the series $\sum_{n=1}^{\infty}\lvert x_n\rvert$ converges, so that $x_n \to 0$.

(ii)$\Leftrightarrow$(iii) by Theorem 10.1.1.

(iii)$\Rightarrow$(i) We will use Nakano's Theorem 2.5.24. Let $0 \leqslant x_n \uparrow\leqslant u$. By assumption, $[0,u]$ is weakly compact. Eberlein-Šmulian's Theorem A.4.20 yields that (x_n) has a weakly convergent subsequence, say, (x_{n_k}). Since (x_{n_k}) is monotone, it is norm convergent by Dini's Theorem 6.8.2. Again, the monotonicity of (x_n) implies that (x_n) is convergent. $\square$

10.1.3 Exercise. Every reflexive Banach lattice is order continuous.

When $1 < p < \infty$ and μ is σ-finite, Exercise 10.1.3 provides an alternative proof that $L_p(\mu)$ is order continuous.

10.1.4 Corollary. *A Banach lattice X is order continuous iff $x_n \xrightarrow{\mathrm{o}} x$ implies $x_n \to x$ for every sequence (x_n) and x in X.*

Proof. The forward implication is trivial. To prove the converse, suppose that (x_n) is an order bounded disjoint sequence. Then (x_n) is order null by Proposition 3.2.34, hence norm null by assumption. Now apply Theorem 10.1.2. $\square$

This corollary should be compared with the definitions of order continuity and σ-order continuity; see Section 2.5.1, as well as with Nakano's Theorem 2.5.24. Note that the assumption in the corollary is stronger than σ-order continuity because not every order convergent sequence is σ-order convergent; see Example 2.5.29.

Combining the preceding corollary with Theorem 2.5.17 and the easy fact that uniform convergence implies order convergence and norm convergence, we get the following.

10.1.5 Corollary. *A Banach lattice X is order continuous iff order and uniform convergences agree on sequences.*

As an application of Theorem 10.1.2, we present a version of Amemiya's Theorem 9.3.9 for weak topologies.

10.1.6 Theorem (Amemiya)**.** *Suppose that X and Y are two order continuous Banach lattices such that Y is an ideal in X and the embedding is continuous. Then the weak topologies of X and Y agree on order intervals of Y.*

Proof. Let $j\colon Y \hookrightarrow X$ be the embedding. Write w_X and w_Y for the weak topologies of X and Y, respectively. Since j is continuous, it is weak-to-weak continuous, i.e., continuous as a map $j\colon (Y,w_Y) \to (X,w_X)$; see Proposition A.4.9. Fix $u \in Y_+$. Since Y is an ideal in X, the interval $[-u,u]$ in Y is also an interval in X. Consider topological spaces $K_1 = \bigl([-u,u],w_Y\bigr)$ and $K_2 = \bigl([-u,u],w_X\bigr)$. By Theorem 10.1.2(iii), K_1 and K_2 are compact. The restriction of j to $[-u,u]$ is a continuous bijection from K_1 onto K_2. It is now a standard exercise in topology that a continuous bijection between compact spaces is a homeomorphism. Therefore, w_Y agrees with w_X on $[-u,u]$. $\square$

This also yields an analogue of Corollary 9.3.11 for weak topologies.

10.1.7 Corollary. *Let X be an order continuous Banach lattice with a weak unit, and let $L_1(\mu)$ be an L_1-representation of X. On order intervals of X, the weak topologies of X and $L_1(\mu)$ agree.*

Recall that every Banach lattice X is a sublattice of X^{**}. We can now characterize order continuity of X in terms of how X sits in X^{**}.

10.1.8 Theorem. *For a Banach lattice X, TFAE:*

(i) *X is order continuous;*

(ii) *X is an ideal in X^{**};*

(iii) *X is regular in X^{**}.*

Proof. (i)⇒(ii) Suppose that X is order continuous. Let $j\colon X \to X^{**}$ be the canonical inclusion. Suppose $u \in X_+$; put $\hat{u} = j(u) \in X^{**}$. Let $\xi \in X^{**}$ with $|\xi| \leqslant \hat{u}$. We need to prove that $\xi \in j(X)$. Suppose not. Since $[-u,u]$ is weakly compact by Theorem 10.1.2(iii), and since the weak topology on X agrees with the restriction of the weak* topology of X^{**} to $j(X)$, we conclude that $j\bigl([-u,u]\bigr)$ is weak* compact in X^{**}. Clearly, this set is convex. By Theorem A.4.17, there exists $f \in X^*$ which separates ξ from $j\bigl([-u,u]\bigr)$. Scaling, if necessary, we may assume that $f(x) \leqslant 1$ for every $x \in [-u,u]$ but $\langle f,\xi\rangle > 1$. This leads to a contradiction because

$$\langle f,\xi\rangle \leqslant \bigl\langle |f|,|\xi|\bigr\rangle \leqslant |f|(u) = \sup\bigl\{f(x) \,:\, x \in [-u,u]\bigr\} \leqslant 1$$

by the Riesz-Kantorovich Formula.

(ii)⇒(iii) because every ideal is regular.

(iii)⇒(i) Suppose that X is regular in X^{**}. Let $x_\alpha \downarrow 0$ in X. By regularity, $\hat{x}_\alpha \downarrow 0$ in X^{**}. It follows that $\hat{x}_\alpha(f) \downarrow 0$ for every $f \in X^*_+$ by Proposition 6.6.1(iii). This yields $f(x_\alpha) \to 0$ for every $f \in X^*_+$ and, therefore, for every $f \in X^*$. Hence $x_\alpha \xrightarrow{\mathrm{w}} 0$. Since (x_α) is monotone, it follows from Dini's Theorem 6.8.2 that $x_\alpha \to 0$. Thus, X is order continuous. $\square$

10.1.9 Example. *$C[0,1]$ is not regular in $C[0,1]^{**}$.* Put $X = C[0,1]$. Since we already know that $C[0,1]$ is not order complete, hence not order continuous, Theorem 10.1.8 says that it is not regular in its dual. This may also be easily seen directly. Indeed, define x_n as follows: $x_n(0) = 1$, $x_n(\frac{1}{n}) = 0$, $x_n(1) = 0$, and linear in between. Then $x_n \downarrow 0$ in X. Let $f \in X^*$ be the point-evaluation at zero, that is, $f(x) = x(0)$. Then

$\hat{x}_n(f) = f(x_n) = 1$ for each n, so that $\hat{x}_n$ does not converge in order to zero by Proposition 6.6.1(iii). Since every order dense sublattice is regular by Proposition 3.2.13, X is not order dense in X^{**}.

The following result is due to Ando.

10.1.10 Theorem. *For a Banach lattice X, TFAE:*

(i) *X is order continuous;*
(ii) *Every closed ideal is a band;*
(iii) *Every closed ideal is a projection band;*
(iv) *Every closed ideal is the range of a positive projection.*

Proof. (i)⇒(ii) Since X is order continuous, every closed ideal is order closed and, therefore, is a band.

(ii)⇒(iii) Let J be a closed ideal in X; hence a band. Clearly, J^d is also a band; in particular, it is a closed ideal. By Theorem 3.3.15, $J + J^d$ is also a closed ideal, hence a band. It suffices to show that $J + J^d = X$. Being a band, $J + J^d$ equals $(J + J^d)^{dd}$. Since J and J^d are both contained in $J + J^d$, we have $(J + J^d)^d \subseteq J^d \cap J^{dd} = \{0\}$. It follows that $J + J^d = (J + J^d)^{dd} = \{0\}^d = X$[1].

(iii)⇒(iv) Trivial.

(iv)⇒(i) Let $u \in X_+$ and (x_n) a disjoint sequence in $[0, u]$. By Meyer-Nieberg's Theorem 10.1.2, it suffices to show that $x_n \to 0$. Fix $\varepsilon > 0$. Let J be the ideal generated by (x_n). By assumption, there exists a positive projection P on X with $\operatorname{Range} P = \overline{J}$. Note that $x_n = Px_n \leqslant Pu$ for every n, and $Pu \in \overline{J}$. Find $v \in J$ with $\|Pu - v\| < \varepsilon$. Then $|v| \leqslant \sum_{n=1}^{n_0} \lambda_n x_n$ for some n_0 and some non-negative $\lambda_1, \dots, \lambda_{n_0}$; see Exercise 3.3.6. Let $n > n_0$. Then $v \perp x_n$. Exercise 4.3.1(ii) yields $x_n \leqslant |Pu - v|$, so that $\|x_n\| \leqslant \|Pu - v\| < \varepsilon$. It follows that $x_n \to 0$. □

10.1.11 Theorem. *For a Banach lattice X, TFAE:*

(i) *X is order continuous;*
(ii) *Every positive order bounded weakly null net in X is norm null;*
(iii) *Every positive order bounded weakly null sequence in X is norm null.*

Proof. (i)⇒(ii) Suppose that X is order continuous, a net (x_α) is contained in $[0, u]$ for some $u \in X_+$, and $x_\alpha \xrightarrow{w} 0$. WLOG, u is a weak unit; otherwise, replace X with B_u. Let $L_1(\mu)$ be an L_1-representation for X and u. Since the embedding of X into $L_1(\mu)$ is continuous and therefore, weak-to-weak continuous, $x_\alpha \xrightarrow{w} 0$ in $L_1(\mu)$. It follows that

[1] Here is another way to conclude that $J + J^d$ equals X: it is order dense by Exercise 5.1.11; since it is a band, $J + J^d = X$ by Exercise 5.1.2.

$\|x_\alpha\|_{L_1(\mu)} = \int x_\alpha \to 0$. Since the norm topologies of X and $L_1(\mu)$ agree on $[0, u]$ by Corollary 9.3.11, we conclude that $\|x_\alpha\|_X \to 0$.

(ii)⇒(iii) is trivial.

(iii)⇒(i) Every disjoint order bounded sequence is weakly null by Exercise 6.8.8(i), hence $x_n \xrightarrow{\|\cdot\|} 0$. hence, X is order continuous by Meyer-Nieberg's Theorem 10.1.2. □

10.1.12 Corollary. *Suppose that (x_α) is a net in an order continuous Banach lattice X contained in an order interval $[a, b]$. If $x_\alpha \xrightarrow{\mathrm{w}} a$ then $x_\alpha \to a$.*

10.1.13 Example. The following example shows that the order boundedness assumption in Theorem 10.1.11 and Corollary 10.1.12 is necessary. Let $X = \ell_p$ with $1 < p < \infty$. The unit vector sequence (e_n) is positive but not order bounded. It is weakly null but not norm null.

10.1.14 Example. The following example shows that in Theorem 10.1.11 and Corollary 10.1.12, it is important that the limit is an endpoint. Let $X = L_2[0, 1]$, put $x_n = \mathbb{1} + r_n$, where (r_n) is the Rademacher sequence. Then (x_n) is positive and order bounded, and $x_n \xrightarrow{\mathrm{w}} \mathbb{1}$. However, (x_n) does not converge in norm.

There are many more equivalent characterizations of order continuity. In particular, we will show in Theorem 11.4.5 that X is order continuous iff every disjoint bounded sequence in X^* is weak*-null.

10.2 *Order continuous sublattices and vectors

10.2.1 Exercise. Let Y be a regular sublattice of an order continuous normed lattice X. Then Y is also order continuous.

10.2.2 Theorem. *Let Y be a norm dense sublattice of a normed lattice X. If Y is order continuous then so is X.*

Proof. Note that Y is order dense in X by Corollary 3.2.18.

Special case: X is a Banach lattice. We will use Meyer-Nieberg Theorem: let (x_n) be a disjoint sequence in $[0, u]$ in X; it suffices to show that $x_n \to 0$ in norm. Fix $\varepsilon > 0$. Find $v \in Y$ such that $\|u - v\| < \varepsilon$. Replacing v with v^+, we may assume WLOG that $v \geqslant 0$; indeed, $|u - v^+| \leqslant |u - v|$, so that $\|u - v^+\| \leqslant \|u - v\| < \varepsilon$. It follows from $|x_n - x_n \wedge v| = |x_n \wedge u - x_n \wedge v| \leqslant |u - v|$ that $\|x_n - x_n \wedge v\| < \varepsilon$. Since the sequence $(x_n \wedge v)$ is disjoint and order bounded X, we have $x_n \wedge v \xrightarrow{\mathrm{o}} 0$ in X by Proposition 3.2.33.

For every n, find $y_n \in Y$ with $\|x_n - y_n\| < \frac{1}{2^n}$. Again, WLOG, $y_n \geqslant 0$. Let $z_n = x_n \wedge v - y_n \wedge v$. It follows from $|z_n| \leqslant |x_n - y_n|$ that $\|z_n\| \leqslant \frac{1}{2^n}$. By Proposition 2.3.6, (z_n) is uniformly null and, therefore, order null in X. It follows that $y_n \wedge v \xrightarrow{\mathrm{o}} 0$ in X. Since $(y_n \wedge v)$ is order bounded in Y and Y is regular, it follows from Theorem 3.2.5 that $y_n \wedge v \xrightarrow{\mathrm{o}} 0$ in Y. Order continuity of Y yields $y_n \wedge v \xrightarrow{\|\cdot\|} 0$ in Y and, therefore, in X. It follows that $x_n \wedge v \xrightarrow{\|\cdot\|} 0$. Now $\|x_n - x_n \wedge v\| < \varepsilon$ implies that $\|x_n\| < \varepsilon$ for all sufficiently large n, hence $x_n \xrightarrow{\|\cdot\|} 0$.

General case. By the Special Case, the norm completion $\overline{X}$ of X is order continuous. Since Y is order dense in $\overline{X}$, we conclude that X is order dense in $\overline{X}$, hence X is regular in $\overline{X}$ by Proposition 3.2.13. Now Exercise 10.2.1 yields that X is order continuous. $\square$

Let X be a Banach lattice. A vector $u \in X_+$ is said to be ***order continuous*** if $u \geqslant x_\alpha \downarrow 0$ implies $x_\alpha \to 0$. We write

$$X_{\mathrm{oc}} = \big\{x \in X \ :\ |x| \text{ is order continuous } \big\}$$

and call X_{oc} the ***order continuous part***[2] of X. Of course, if X is a on order continuous Banach lattice, $X_{\mathrm{oc}} = X$.

The reader should recognize most of the conditions in the next theorem as "localized" versions of various characterizations of order continuous Banach lattices from preceding sections.

10.2.3 Theorem. *For a positive vector u in a Banach lattice X, TFAE:*

(i) *u is order continuous;*
(ii) *I_u is order continuous*[3]*;*
(iii) *$\overline{I_u}$ is order continuous;*
(iv) *$0 \leqslant x_n \uparrow\leqslant u$ implies that (x_n) is convergent;*
(v) *every disjoint sequence in $[0,u]$ is norm null;*
(vi) *$[0,u]$ is weakly compact;*
(vii) *every weakly null net in $[0,u]$ is norm null;*
(viii) *I_u (in X) is an ideal in X^{**};*
(ix) *I_u (in X) is regular in X^{**}.*

[2]In some of the literature, the order continuous part of X is denoted by X^a. We avoid this notation as it may be easily confused with the atomic part of X. Certain care should be exercised when this notation is used along our notation for the order continuous dual: $X^\sim_{\mathrm{oc}}$ stands for the order continuous dual of X, $(X_{\mathrm{oc}})^\sim$ stands for the order dual of the order continuous part of X, while $(X^\sim)_{\mathrm{oc}}$ stands for the order continuous part of the order dual of X.

[3]In this theorem, we consider I_u with the norm induced from X, not $\|\cdot\|_u$!

Proof. (i)⇒(ii) Suppose that $x_\alpha \downarrow 0$ in I_u. Passing to a tail, we may assume that $(x_\alpha) \subseteq [0, v]$ for some positive $v \in I_u$. It follows that $v \leqslant \lambda u$ for some $\lambda > 0$. Hence, $u \geqslant \frac{1}{\lambda} x_\alpha \downarrow 0$ in I_u and, therefore, in X, because I_u is an ideal. It follows that $\frac{1}{\lambda} x_\alpha \to 0$ and, hence, $x_\alpha \to 0$.

(ii)⇒(iii) by Theorem 10.2.2.

(iii)⇒(iv) follows by Nakano's Theorem 2.5.24(iii).

(iv)⇒(v) Let (d_k) be a disjoint sequence in $[0, u]$. Put $x_n = \sum_{k=1}^n d_k$. Then $0 \leqslant x_n \uparrow \leqslant u$. It follows that (x_n) is norm convergent and, therefore, $d_k \xrightarrow{\|\cdot\|} 0$.

(v)⇔(vi) by Theorem 10.1.1.

(vi)⇒(i)[4] Suppose that $u \geqslant x_\alpha \downarrow 0$. There exists a subnet (y_β) of (x_α) such that $y_\beta \xrightarrow{\mathrm{w}} y$ for some $y \in [0, u]$. By Dini's Theorem 6.8.2, $y_\beta \xrightarrow{\|\cdot\|} y$. It follows easily from monotonicity that $x_\alpha \xrightarrow{\|\cdot\|} y$. Now Monotone Convergence Lemma 2.5.3 yields $y = 0$.

We have now proved that (i) through (vi) are equivalent.

(iii)⇒(vii) Apply Theorem 10.1.11 to $\overline{I_u}$.

(vii)⇒(v) Every disjoint sequence in $[0, u]$ is weakly null by Exercise 6.8.8(i), hence norm null.

The proof of (vi)⇒(viii)⇒(ix)⇒(i) is similar to the proof of Theorem 10.1.8. □

10.2.4 Theorem. *X_{oc} is a closed order continuous ideal in X.*

Proof. Let $C = \{x \in X_+ \ :\ x \text{ is order continuous}\}$. By the preceding theorem, $x \in C$ iff $[0, x]$ is weakly compact ($x \in X_+$). Clearly, C is closed under positive scalar multiplication. If $x, y \in C$ then $[0, x + y] = [0, x] + [0, y]$ by Riesz Decomposition Theorem, so that weak continuity of addition implies that $[0, x + y]$ is weakly compact and, therefore, $x + y \in C$. It is also clear that if $x \in C$ and $0 \leqslant y \leqslant x$ then $y \in C$.

We claim that C is closed. Let $u \in \overline{C}$ and $0 \leqslant x_n \uparrow \leqslant u$. We will show that (x_n) is Cauchy and, therefore, convergent; it will imply by Theorem 10.2.3(iv) that $u \in C$. Fix $\varepsilon > 0$, and find $v \in C$ with $\|u - v\| < \varepsilon$. Note that for every $x \in [0, u]$ we have $|x - x \wedge v| = |x \wedge u - x \wedge v| \leqslant |u - v|$, so that $\|x - x \wedge v\| < \varepsilon$. Also, observe that $0 \leqslant x_n \wedge v \uparrow \leqslant v$. It follows that $(x_n \wedge v)$ converges. Fix n_0 such that $\|x_n \wedge v - x_m \wedge v\| < \varepsilon$ whenever $n, m \geqslant n_0$. It follows that

$$\|x_n - x_m\| \leqslant \|x_n - x_n \wedge v\| + \|x_n \wedge v - x_m \wedge v\| + \|x_m \wedge v - x_m\| < 3\varepsilon.$$

It follows that C is a closed cone in X_+. Since $x \in X_{\mathrm{oc}}$ iff $|x| \in C$, it follows easily that X_{oc} is a closed ideal. By Nakano's Theorem 2.5.24(iii), it is order continuous. □

10.2.5 Exercise. c_0 is the order continuous part of ℓ_∞.

[4] While this implication is redundant for the proof, we include it as a "shortcut".

10.3 σ-order complete Banach lattices

Let X be a Banach lattice. Recall that if X is order continuous then it is order complete by Theorem 2.5.22; furthermore, by Nakano's Theorem 2.5.24, X is order continuous iff it is σ-order complete and σ-order continuous. How big is the "gap" between order completeness and order continuity? The following theorem, due to Lozanovskii and Mekler, provides a partial answer to this question. Given two Banach lattices X and Y, we say that X ***contains a lattice copy of*** Y if X has a closed sublattice which is lattice isomorphic to Y (it follows that it is also norm isomorphic to Y).

10.3.1 Theorem. *Suppose that X is a σ-order complete Banach lattice. Then X is order continuous iff it contains no lattice copy of ℓ_∞.*

Proof. Suppose that X contains a lattice copy of ℓ_∞, i.e., there is a closed sublattice Y of X such that Y is lattice and norm isomorphic to ℓ_∞. Then Y and, therefore, ℓ_∞ is σ-order continuous by Exercise 2.5.23; a contradiction.

Suppose now that X fails to be order continuous. Then, applying Meyer-Nieberg's Theorem 10.1.2 and scaling, we can find $u \in X_+$ and a disjoint sequence (x_n) in $[0, u]$ such that $\|x_n\| \geqslant 1$ for every n. Let $0 \leqslant a \in \ell_\infty$, with $a = (\alpha_n)$. Then $\alpha_n x_n \leqslant \|a\|_\infty u$ for every n. Since X is σ-order complete, $\bigvee_{n=1}^\infty \alpha_n x_n$ exists in X. Denote it by $\tau(a)$. This defines a map $\tau\colon (\ell_\infty)_+ \to X_+$.

Observe that τ is additive. Take $a = (\alpha_n)$ and $b = (\beta_n)$ in $(\ell_\infty)_+$. Then $(\alpha_k + \beta_k)x_k = \alpha_k x_k + \beta_k x_k \leqslant \tau(a) + \tau(b)$ for each k. Taking supremum over k, we get $\tau(a + b) \leqslant \tau(a) + \tau(b)$. Conversely, for every i and j we have $\alpha_i x_i + \beta_j x_j \leqslant \tau(a + b)$. Consecutively taking sups over i and j, we get $\tau(a) + \tau(b) \leqslant \tau(a + b)$. Therefore, $\tau(a + b) = \tau(a) + \tau(b)$.

By Extension Lemma 6.2.1, τ extends to a positive operator $T\colon \ell_\infty \to X$. We claim that T is a lattice homomorphism. By Exercise 1.6.10, it suffices to prove that $a \wedge b = 0$ implies $Ta \wedge Tb = 0$. Indeed, using distributive laws and disjointness of (x_n), we get

$$Ta \wedge Tb = \tau(a) \wedge \tau(b) = \Bigl(\bigvee_{n=1}^\infty \alpha_n x_n\Bigr) \wedge \Bigl(\bigvee_{k=1}^\infty \beta_k x_k\Bigr) = \bigvee_{n=1}^\infty (\alpha_n \wedge \beta_n) x_n = 0.$$

Finally, we show that T is an isomorphism. Since $T \geqslant 0$, T is bounded. Let $a = (\alpha_i) \in \ell_\infty$. Then $|\alpha_k| \geqslant \frac{1}{2}\|a\|_\infty$ for some k. Then $|Ta| = T|a| = \bigvee_{i=1}^\infty |\alpha_i| x_i \geqslant |\alpha_k| x_k$, so that $\|Ta\| \geqslant |\alpha_k|\|x_k\| \geqslant \frac{1}{2}\|a\|_\infty$. Hence, T is bounded below, and, therefore, T is an isomorphism. $\square$

10.3.2 Exercise. For a separable Banach lattice X, TFAE:

(i) X is σ-order complete;

(ii) X is order complete;

(iii) X is order continuous.

10.4 KB-spaces

Throughout this section, X is a Banach lattice. Recall that by Nakano's theorem, X is order continuous iff every increasing order bounded sequence in X_+ is convergent. We now consider a somewhat stronger condition: X is said to be a ***KB-space***[5] if every increasing bounded (i.e., *norm* bounded) sequence in X_+ is convergent. That is, if $B_X^+ \ni x_n \uparrow$ then (x_n) is convergent.

10.4.1 Exercise. Prove that

(i) Every KB-space is order continuous.

(ii) Every closed sublattice of a KB-space is a KB-space.

(iii) Every reflexive Banach lattice is a KB-space.

(iv) Every L_1-space is a KB-space.

(v) No infinite-dimensional AM-space is a KB-space.

10.4.2 Exercise. Show that in the definition of a KB-space, sequences may be replaced with nets, and X may be replaced with X_+, i.e., X is a KB-space iff every increasing bounded net in X is convergent.

Similarly to order continuous spaces, KB-spaces admit several equivalent characterizations. Our next goal is to prove that X is a KB-space iff it contains no lattice copy of c_0. We say that a Banach space contains a copy of c_0 if it has a closed subspace isomorphic to c_0. A Banach lattice X contains a lattice copy of c_0 if it has a closed sublattice which is lattice and norm isomorphic to c_0. Equivalently, X_+ contains a disjoint sequence (x_n) which, as a basic sequence, is equivalent to the unit vector basis of c_0. In this case, we say that (x_n) is a ***lattice copy of the unit vector basis*** of c_0 in X. Recall that a sequence (x_n) in a Banach space is ***norm bounded below*** if $\inf\|x_n\| > 0$.

10.4.3 Lemma. *A positive sequence (x_n) in a Banach lattice X is a lattice copy of the unit vector basis of c_0 iff it is disjoint, norm bounded below, and its partial sums are bounded.*

[5] "KB" stands for Kantorovich-Banach.

Proof. The forward implication is trivial because the unit vector basis of c_0 satisfies the required conditions. To prove the converse, suppose that (x_n) is a disjoint sequence in X_+ and there exist positive real numbers δ and M such that

$$\|x_n\| \geqslant \delta \text{ and } \|x_1 + \dots + x_n\| \leqslant M \text{ for every } n.$$

Take any scalars $\lambda_1, \dots, \lambda_n$. Assume that $\max|\lambda_i|$ is attained at $i = i_0$. Since the sequence (x_n) is positive and disjoint, we have

$$\bigl|\lambda_1 x_1 + \dots + \lambda_n x_n\bigr| \geqslant |\lambda_{i_0}| x_{i_0},$$

so that

$$\bigl\|\lambda_1 x_1 + \dots + \lambda_n x_n\bigr\| \geqslant \delta \max|\lambda_i|.$$

On the other hand,

$$\bigl|\lambda_1 x_1 + \dots + \lambda_n x_n\bigr| = |\lambda_1| x_1 + \dots + |\lambda_n| x_n \leqslant \max|\lambda_i| \cdot (x_1 + \dots + x_n),$$

so that

$$\bigl\|\lambda_1 x_1 + \dots + \lambda_n x_n\bigr\| \leqslant M \max|\lambda_i|.$$

It follows that (x_n) is equivalent to the unit vector basis of c_0. □

10.4.4 Theorem. *X is a KB-space iff X contains no lattice copy of c_0.*

Proof. Suppose that X contains a lattice copy of c_0; let (x_n) be a lattice copy of the unit vector basis of c_0 in X_+. Put $u_n = x_1 + \dots + x_n$. Then (u_n) is an increasing norm bounded sequence in X_+, yet it diverges. It follows that X fails to be a KB-space.

Suppose that X contains no lattice copy of c_0 but fails to be a KB-space. By Meyer-Nieberg Theorem 10.1.2, X is order continuous. Indeed, suppose that (x_n) is a disjoint sequence in $[0, u]$ such that (x_n) does not converge to zero. Passing to a subsequence, we may assume that (x_n) is bounded below in norm. By Lemma 10.4.3, (x_n) is a lattice copy of the unit vector basis of c_0; this is a contradiction.

Since X is not a KB-space, there exists an increasing positive divergent sequence (x_n) in B_X. Passing to a subsequence, we may assume that the sequence $y_n = x_{n+1} - x_n$ is norm bounded below, i.e., there exists $\delta > 0$ such that $\|y_n\| > \delta$ for all n.

WLOG, X has a weak unit. Indeed, otherwise, replace X with a principal band B_e such that (x_n) is contained in B_e; e.g., put $e = \sum_{n=1}^{\infty} \frac{x_n}{2^n}$. Let $L_1(\mu)$ be an L_1-representation for X as in Theorem 9.3.2. WLOG, $x_n(t) \uparrow$ for every t; put $x(t) = \lim_n x_n(t)$. Since the inclusion of X into $L_1(\mu)$ is continuous, (x_n) is bounded in $L_1(\mu)$. Monotone Convergence Theorem implies that $x \in L_1(\mu)$ and $x_n \xrightarrow{\|\cdot\|_{L_1(\mu)}} x$. It follows that $y_n \xrightarrow{\|\cdot\|_{L_1(\mu)}} 0$ in $L_1(\mu)$. In particular, (y_n) converges to zero in measure. By Kadeč-Pełczyński's Theorem 9.6.6, there is a subsequence (y_{n_k}) and a disjoint sequence (d_k) in X such that $0 \leqslant d_k \leqslant y_{n_k}$ and $\|y_{n_k} - d_k\|_X \to 0$. Furthermore, we have $\|d_k\|_X > \frac{\delta}{2}$ for all sufficiently large k. Hence, passing to a tail, if necessary, we may assume that (d_k) is norm bounded below in X. Finally, observe that

$$d_1 + \cdots + d_k \leqslant y_{n_1} + \cdots + y_{n_k} \leqslant y_1 + \cdots + y_{n_k} = x_{n_k+1} - x_1,$$

so that $\|d_1 + \cdots + d_k\|_X \leqslant 2$. By Lemma 10.4.3, (d_k) is a lattice copy of the unit vector basis of c_0; this is a contradiction. □

Combining Theorem 10.4.4 with Lemma 10.4.3, we get the following.

10.4.5 Corollary. *A Banach lattice X is a KB-space iff for every disjoint bounded below sequence (x_n) we have $\|x_1 + \cdots + x_n\| \to \infty$.*

10.4.6 Example. Let $X = L_p(\mu)$ with $1 \leqslant p < \infty$. We already know from Exercise 10.4.1 that X is a KB-space. Here is another way to see this. Note that every seminormalized disjoint sequence in $X$is equivalent to the unit vector basis of ℓ_p, hence X has no lattice copy of the unit vector basis of c_0.

In the case $p = \infty$, if $L_\infty(\mu)$ is infinite-dimensional then every positive normalized disjoint sequence in it is a lattice copy of the unit vector basis of c_0, so that the space is not a KB-space.

Our next goal is two-fold. First, we will show that in Theorem 10.4.4 one may replace lattice copies of c_0 with arbitrary (isomorphic) copies. Second, we will show that a Banach lattice is a KB-space iff it is weakly sequentially complete. Recall that a Banach space is ***weakly sequentially complete*** if every weakly Cauchy sequence is weakly convergent. Note that weak sequential completeness is a Banach space property, yet it characterizes KB-spaces among all Banach lattices. Recall that $L_1(\mu)$ is weakly sequentially complete by Corollary 9.8.11, while c_0 fails to be weakly sequentially complete by Proposition A.4.24. Recall also that by Proposition A.4.25, X is weakly sequentially complete iff it is sequentially weak* closed in X^{**}.

10.4.7 Theorem. *For a Banach lattice X, TFAE:*

(i) *X is a KB-space;*
(ii) *X is weakly sequentially complete;*
(iii) *X contains no copies of c_0.*

Proof. (ii)⇒(iii) because c_0 fails to be weakly sequentially complete. (iii)⇒(i) by Theorem 10.4.4. It is left to prove (i)⇒(ii).

Suppose X is a KB-space (in particular, it is order continuous). Let (x_n) be a weakly Cauchy sequence in X; we need to prove that it is weakly convergent. Replacing X with a principal band, we may assume WLOG that X has a weak unit as in 9.3.3. By Theorem 9.1.12, X admits a strictly positive functional, say h. Let $L_1(\mu)$ be an L_1-representation for X and h as in Theorem 9.3.2. Since the inclusion of X into $L_1(\mu)$ is continuous, (x_n) is still weakly Cauchy as a sequence in $L_1(\mu)$. Since $L_1(\mu)$ is weakly sequentially complete, (x_n) converges weakly in $L_1(\mu)$ to some $x \in L_1(\mu)$. It is left to show that $x \in X$ and that $x_n \xrightarrow{\mathrm{w}} x$ in X.

Since $x_n \xrightarrow{\mathrm{w}} x$ in $L_1(\mu)$, there exists a sequence (y_n) such that each y_n is a convex combination of $x_1, \dots, x_n$ and $y_n \xrightarrow{\|\cdot\|_{L_1(\mu)}} x$; see Proposition A.4.12. It follows that $|y_n| \xrightarrow{\|\cdot\|_{L_1(\mu)}} |x|$ and, therefore, $|y_{n_k}| \xrightarrow{\mathrm{o}} |x|$ for some subsequence (y_{n_k}) by Theorem 2.5.4. Since $L_1(\mu)$ is order complete, by 2.4.25 there is an increasing sequence (z_k) in $L_1(\mu)$ such that $z_k \uparrow |x|$ and $z_k \leqslant |y_{n_k}|$ for every k. WLOG, $z_k \geqslant 0$; otherwise, replace z_k with z_k^+. Hence, $0 \leqslant z_k \leqslant |y_{n_k}|$. Since X is an ideal in $L_1(\mu)$ and $y_{n_k} \in X$, it follows that $z_k \in X$.

Being weakly Cauchy, (x_n) is bounded in X by Uniform Boundedness Principle. It follows that the sequences (y_n) and, therefore, (z_k) are bounded in X. Since X is a KB-space, we conclude that (z_k) converges to some z in X. It follows that $z_k \xrightarrow{\|\cdot\|_{L_1(\mu)}} z$. Since $z_k \uparrow |x|$ in $L_1(\mu)$, this yields $z = |x|$ by Lemma 2.5.3, so that $|x|$ and, therefore, x itself are in X.

We have proved that for a strictly positive functional h in X^* there exists $x \in X$ such that $x_n \xrightarrow{\mathrm{w}} x$ in $L_1(\mu)$ where $L_1(\mu)$ is an L_1-representation for X and h. By Proposition 9.2.6, $L_1(\mu)^*$ may be identified with the ideal I_h in X^*. Hence, we have $f(x_n) \to f(x)$ for every $f \in I_h$. Fix an arbitrary $g \in X_+^*$. Then $h + g$ is still strictly positive, so that, by a similar argument, one can find $y \in X$ such that $f(x_n) \to f(y)$ for all $f \in I_{h+g}$. In particular, $f(x_n) \to f(y)$ for every $f \in I_h$, hence $x_n \xrightarrow{\mathrm{w}} y$ in $L_1(\mu)$. It follows that $x = y$. Since $g \in I_{h+g}$, we have $g(x_n) \to g(x)$. Therefore, $f(x_n) \to f(x)$ for every $f \in X^*$, hence $x_n \xrightarrow{\mathrm{w}} x$ in X. □

10.4.8 Remark. The two preceding theorems imply, in particular, that if a Banach lattice X contains a subspace copy of c_0 then it also contains a lattice copy of c_0. That is, if X contains a closed subspace that is isomorphic to c_0 as a Banach space then X also contains a closed sublattice that is lattice isomorphic to c_0. This is a very special property of c_0. For example, ℓ_2 fails this property. Indeed, let $X = L_p[0, 1]$ for some $1 \leqslant p < \infty$ with $p \neq 2$. The closed subspace of $L_p[0, 1]$ spanned by the Rademacher functions is isomorphic (as a Banach space) to ℓ_2 by 6.4.4(ii). Yet, $L_p[0,1]$ contains no lattice copy of ℓ_2. Indeed, let Y be a closed sublattice of $L_p[0,1]$ which is lattice isomorphic to ℓ_2. Let (x_n) be the sequence in Y which corresponds to the unit vector basis of ℓ_2. Then (x_n) is disjoint in Y, hence in $L_p[0,1]$, so that (x_n) is equivalent (as a basic sequence) to the unit vector basis of ℓ_p. This yields that the unit vector bases of ℓ_2 and ℓ_p are equivalent, which is a contradiction.

Recall that $f_\alpha \uparrow f$ in X^* iff $f_\alpha(x) \uparrow f(x)$ for every $x \in X_+$; see Proposition 6.6.1(iii).

10.4.9 Exercise. Let X be a Banach lattice and (f_n) an increasing norm bounded sequence in X^*_+. Then $f_n \xrightarrow{w^*} g$ for some $g \in X^*_+$ and $f_n \uparrow g$.

Recall that X is order continuous iff X is an ideal in X^{**}.

10.4.10 Theorem. *X is a KB-space iff X is a band in X^{**}.*

(In this case, this band is projective because X^{**} is order complete.)

Proof. Suppose that X is a KB-space. Since X is order continuous, X is an ideal in X^{**} by Theorem 10.1.8. and $0 \leqslant \hat{x}_\alpha \uparrow \xi$ for a net (x_α) in X and a vector $\xi \in X^{**}$. By Proposition 5.1.1, it suffices to show that $\xi = \hat{x}$ for some $x \in X$. Clearly, $0 \leqslant x_\alpha \uparrow$ and $\|x_\alpha\| = \|\hat{x}_\alpha\| \leqslant \|\xi\|$. Hence, (x_α) is an increasing norm bounded net in X. Since X is a KB-space, $x_\alpha \to x$ for some $x \in X$. It follows that $\hat{x}_\alpha \to \hat{x}$, so that $\hat{x}_\alpha \uparrow \hat{x}$ by Monotone Convergence Lemma 2.5.3. Therefore, $\xi = \hat{x}$.

Conversely, suppose that X is a band in X^{**} and (x_n) is a norm bounded increasing sequence in X_+. It follows that $(\hat{x}_n)$ is an increasing norm bounded sequence in X^{**}_+. By Exercise 10.4.9, we have $\hat{x}_n \xrightarrow{w^*} \xi$ and $\hat{x}_n \uparrow \xi$ for some $\xi \in X^{**}_+$. Since X is a band in X^{**}, we have $\xi = \hat{x}$ for some $x \in X$. It follows from $\hat{x}_n \xrightarrow{w^*} \hat{x}$ that $x_n \xrightarrow{w} x$ in X. Since the sequence (x_n) is monotone, we have $x_n \to x$ by Dini's Theorem 6.8.2. ☐

10.4.11 Corollary. *X is a KB-space iff $(X^*)^\sim_{\mathrm{oc}} = X$ in X^{**}.*

Proof. By Proposition 6.7.14, $(X^*)^\sim_{\mathrm{oc}}$ is the band generated by X in X^{**}. Now apply Theorem 10.4.10. ☐

10.4.12 Example. Consider $L_1(\mu)$ for some measure μ. Since it is an AL-space, $L_1(\mu)^{**}$ is again an AL-space by Theorem 8.1.7. By AL-representation Theorem 8.5.5, $L_1(\mu)^{**}$ is lattice isometric to $L_1(\nu)$ for some measure ν. Since $L_1(\mu)$ is KB-space, it is an projection band in $L_1(\mu)^{**}$ by Theorem 10.4.10.

Suppose, in addition, that μ is σ-finite. Then $L_1(\mu) = L_\infty(\mu)$. By the preceding corollary, we conclude that $L_\infty(\mu)^\sim_{\mathrm{oc}} = L_1(\mu)$. This yields a short proof of Proposition 8.3.16.

10.5 Order continuity of X^*

Throughout this section, X is a Banach lattice. Recall that every KB-space is order continuous. Surprisingly, the two conditions are equivalent for dual spaces:

10.5.1 Proposition. *X^* is a KB-space iff it is order continuous.*

Proof. The forward implication is trivial. Suppose that X^* is order continuous. Let (f_n) be an increasing norm bounded sequence in X^*_+. By Exercise 10.4.9, $f_n \uparrow g$ for some $g \in X^*$. Order continuity now yields $f_n \to g$, hence X is a KB-space. □

Our next goal is to prove that X^* is order continuous (or, equivalently, a KB-space) iff X contains no lattice copy of ℓ_1.

10.5.2 Proposition. *Every lattice copy of ℓ_1 in a Banach lattice is the range of a positive projection (in particular, is complemented).*

Proof. Let $T\colon \ell_1 \to Y$ be a lattice isomorphism from ℓ_1 onto a closed sublattice Y of X. Put $v_n = Te_n$. Note that (v_n) is a disjoint sequence in X_+ and $Y = [v_n]$. Furthermore, $T^*\colon Y^* \to \ell_\infty$ is a surjective lattice isomorphism by, e.g., Theorem 1.6.13(iv). For each n, find $f_n \in Y^*$ such that

$$T^* f_n = (\overbrace{0, \dots, 0}^{n}, 1, 1, \dots).$$

It follows that (f_n) is a decreasing sequence in Y^* and

$$\langle f_n, v_m \rangle = \langle f_n, Te_m \rangle = \langle T^* f_n, e_m \rangle = \begin{cases} 1 & \text{if } m > n; \\ 0 & \text{otherwise.} \end{cases}$$

By Exercise 6.9.4, we extend each f_n to $\tilde{f}_n$ in X^*_+ such that $\tilde{f}_n \downarrow$.

Fix $x \in X$. For each m, we have

$$\sum_{n=1}^{m} \bigl|\langle \tilde{f}_n - \tilde{f}_{n+1}, x \rangle\bigr| \leqslant \sum_{n=1}^{m} \bigl\langle \tilde{f}_n - \tilde{f}_{n+1}, |x| \bigr\rangle = \bigl\langle \tilde{f}_1 - \tilde{f}_{m+1}, |x| \bigr\rangle \leqslant \langle \tilde{f}_1, |x| \rangle.$$

It follows that the series is summable. Since $(v_n) \sim (e_n)$, the series

$$Px \coloneqq \sum_{n=1}^{\infty} \langle \tilde{f}_n - \tilde{f}_{n+1}, x \rangle v_n$$

converges and defines a positive (and, therefore, continuous) operator on X. Clearly, $\operatorname{Range} P \subseteq Y$. Note that $Pv_m = v_m$ for every m; it follows that P is a projection onto Y. □

Recall that X is a KB-space iff it contains no copy of c_0 iff it contains no lattice copy of c_0. Recall also that X^* is a KB-space iff it is order continuous.

10.5.3 Theorem. *For a Banach lattice X, TFAE:*

(i) *X^* is not order continuous;*
(ii) *X^* contains a lattice copy of ℓ_∞;*
(iii) *X^* contains a copy of ℓ_∞;*
(iv) *X contains a lattice copy of ℓ_1;*
(v) *X contains a complemented copy of ℓ_1.*

Proof. (i)⇒(ii) Suppose that X^* is not order continuous. Being a dual space, X^* is order complete. It follows from Theorem 10.3.1 that X^* contains a lattice copy of ℓ_∞.

(ii)⇒(iii) is trivial.

(iii)⇒(i) If X^* contains a copy of ℓ_∞ then it contains a copy of c_0, hence it fails to be a KB-space. Thus, (i)⇔(ii)⇔(iii)

(iv)⇒(v) by Proposition 10.5.2.

(v)⇒(iii) X can be written as a Banach space direct sum $X = Y \oplus Z$ of two closed subspaces where Y is isomorphic to ℓ_1. It follows that $X^* = Y^* \oplus Z^*$ and Y^* is isomorphic to ℓ_∞.

(i)⇒(iv) Suppose that X^* is not order continuous, hence not a KB-space. Therefore, contains a lattice copy of c_0. By Lemma 10.4.3, there is a sequence (f_n) in X^*_+ and a constant $M > 0$ such that $\|f_n\| \geqslant 1$ and $\left\|\sum_{i=1}^n f_i\right\| \leqslant M$ for all n. Fix a sufficiently small positive real ε, say, $\varepsilon = \frac{1}{4}$. For each n, find $x_n \in B_X^+$ such that $f_n(x_n) \geqslant 1 - \varepsilon$.

We are going to disjointify (x_n). Fix k. It follows from

$$\sum_{i=1}^n f_i(x_k) = \Big(\sum_{i=1}^n f_i\Big)(x_k) \leqslant M\|x_k\|$$

that the series $\sum_{i=1}^\infty f_i(x_k)$ converges. In particular, $\lim_n f_n(x_k) = 0$.

We will now inductively construct a subsequence (x_{k_m}) of (x_k). Let $k_1 = 1$. Suppose that $k_1, \dots, k_{m-1}$ have already been constructed. Since

$$\lim_{n\to\infty} f_n(x_{k_1} + \dots + x_{k_{m-1}}) = 0,$$

there exists $k_m > k_{m-1}$ such that $f_{k_m}(x_{k_1} + \dots + x_{k_{m-1}}) < \frac{\varepsilon}{4^m}$. Proceeding inductively, we produce a subsequence of (x_k). Passing to this subsequence (in both (x_k) and (f_k)) and relabelling, we may assume WLOG that $f_n(x_1 + \dots + x_{n-1}) < \frac{\varepsilon}{4^n}$ for every $n > 1$.

Let $e = \sum_{n=1}^\infty \frac{x_n}{2^n}$. Then $\|e\| \leqslant 1$. Let u_n be as in Disjointification Lemma 4.5.1. Then (u_n) is disjoint, $0 \leqslant u_n \leqslant x_n$ for every n, and

$$f_n(u_n) \geqslant f_n(x_n) - 4^n \sum_{i=1}^{n-1} f_n(x_i) - 2^{-n} f_n(e) \geqslant (1-\varepsilon) - \varepsilon - 2^{-n}M \geqslant 1 - 3\varepsilon$$

for all sufficiently large n. Thus, passing to a tail, we may assume that $f_n(u_n) \geqslant 1-3\varepsilon$.

We claim that (u_n) is a lattice copy of the unit vector basis of ℓ_1. Let $\alpha_1,\dots,\alpha_n \in \mathbb{R}$. Then

$$\Bigl\|\sum_{i=1}^n \alpha_i u_i\Bigr\| \leqslant \sum_{i=1}^n |\alpha_i|\|u_i\| \leqslant \sum_{i=1}^n |\alpha_i|\|x_i\| \leqslant \sum_{i=1}^n |\alpha_i|.$$

On the other hand,

$$\begin{aligned}\Bigl\|\sum_{i=1}^n \alpha_i u_i\Bigr\| = \Bigl\|\sum_{i=1}^n |\alpha_i| u_i\Bigr\| &\geqslant \tfrac{1}{M}(f_1+\dots+f_n)\Bigl(\sum_{i=1}^n |\alpha_i| u_i\Bigr)\\ &\geqslant \tfrac{1}{M}\sum_{i=1}^n f_i\bigl(|\alpha_i| u_i\bigr) = \tfrac{1}{M}\sum_{i=1}^n |\alpha_i| f_i(u_i) \geqslant \frac{1-3\varepsilon}{M}\sum_{i=1}^n |\alpha_i|.\end{aligned}$$

It follows that (u_n) is equivalent to the unit vector basis of ℓ_1 and, therefore, $[u_n]$ is a lattice copy of ℓ_1 in X. □

We will show in Theorem 11.4.3 that X^* is order continuous iff every disjoint bounded sequence in X is weakly null.

Recall that a Banach space is reflexive iff its dual space is reflexive and that every closed subspace of a reflexive Banach space is again reflexive. In particular, no reflexive Banach space contains copies of c_0 or ℓ_1.

10.5.4 Theorem. *For a Banach lattice X, TFAE:*

(i) *X is reflexive;*
(ii) *X and X^* are both KB-spaces;*
(iii) *neither X nor X^* contains a lattice copy of c_0;*
(iv) *neither X nor X^* contains a lattice copy of ℓ_1;*
(v) *X contains no lattice copies of either c_0 or ℓ_1.*

Proof. (ii)⇔(iii) by Theorem 10.4.4. Combining Theorems 10.4.4 and 10.5.3, we get (ii)⇔(v). We trivially have (i)⇒(v).

(ii)⇒(i) Since X is a KB-space, Corollary 10.4.11 yields $(X^*)^\sim_{\mathrm{oc}} = X$ in X^{**}. On the other hand, since X^* is a KB-space, it is order continuous, hence every continuous functional on it is order continuous; it follows that $(X^*)^\sim_{\mathrm{oc}} = X^{**}$. Thus, $X^{**} = X$.

We have proved (i)⇔(ii)⇔(iii)⇔(v). Clearly, (i)⇒(iv) because if X is reflexive then X^* is also reflexive. It is left to show (iv)⇒(i). Suppose that neither X nor X^* contains a lattice copy of ℓ_1. Applying Theorem 10.5.3 separately to X and to X^*, we conclude that both X^* and X^{**} are KB-spaces. As we already know that (i)⇔(ii), this yields that X^* is reflexive. Therefore, X is reflexive. □

We observed in Remark 10.4.8 that a Banach lattice contains a copy of c_0, it also contains a lattice copy of c_0. The following two examples show that this fails if ℓ_1.

10.5.5 Example. Let $X = C[0,1]$. By Banach-Mazur Theorem A.4.31, $C[0,1]$ contains an isometric copy of every separable Banach space; in particular, it contains a copy of ℓ_1. However, $C[0,1]$ contains no lattice copy of ℓ_1 because $C[0,1]$ is an AM-space, hence every seminormalized disjoint sequence in it is equivalent to the unit vector basis of c_0.

Now let $X = L_\infty[0,1]$. By 6.4.4(vii), the Rademacher sequence spans a subspace copy of ℓ_1 in $L_\infty[0,1]$. However, as in the preceding example, $L_\infty[0,1]$ contains no lattice copies of ℓ_1.

In the preceding examples, X is an AM-space, hence X^* is an AL-space; in particular, it is order continuous. Thus, in Theorem 10.5.3(iv), one cannot replace a lattice copy of ℓ_1 with just a copy of ℓ_1. Somewhat surprisingly (and using some heavy machinery), one can replace lattice copies with non-lattice copies in Theorem 10.5.4(v):

10.5.6 Theorem. *A Banach lattice is reflexive iff it contains no copy of either c_0 or ℓ_1.*

Proof. The forward implication is obvious. Suppose that X contains no copy of c_0 or ℓ_1; we want to prove that X is reflexive. By Theorem A.4.15, it suffices to show that B_X is weakly compact. By Eberlein-Šmulian Theorem A.4.20, we only need to show that B_X is sequentially weakly compact. Let (x_n) be a sequence in B_X. Since X contains no copy of ℓ_1, Rosenthal's ℓ_1 Theorem A.4.33 guarantees that (x_n) has a subsequence (y_k) that is weakly Cauchy. Since X contains no copy of c_0, it is weakly sequentially complete by Theorem 10.4.7; it follows that (y_n) is weakly convergent. □

10.5.7 Example. We can now present another example of a Banach space that does not admit a Banach lattice structure. Recall that James space is not reflexive, yet it contains no copy of c_0 or ℓ_1; see, e.g., [FHH11, p. 205]. Hence, by Theorem 10.5.6, it cannot be made into a Banach lattice.

In this section, we have discussed spaces that contain a lattice copy of ℓ_1. We will show next that if X has ℓ_1 as a quotient, it also contain a lattice copy of ℓ_1. This is related to the following concept, which is of independent interest. A Banach lattice Z is ***projective*** if for every lattice homomorphism T from Z to a quotient of an arbitrary Banach lattice X over a closed ideal J there is a lifting of T to a lattice homomorphism $\widetilde{T}$ from Z to X with

$\|\widetilde{T}\| \leqslant (1+\varepsilon)\|T\|$:

$$\begin{tikzcd} & X \arrow[d, "Q"] \\ Z \arrow[ur, dotted, "\widetilde{T}"] \arrow[r, "T"'] & X/J \end{tikzcd}$$

Note that it follows from $T = Q\widetilde{T}$ and $\|Q\| = 1$ that $\|T\| \leqslant \|\widetilde{T}\|$. On the other hand, we cannot avoid the ε and just require that $\|\widetilde{T}\| = \|T\|$ because of the definition of the quotient norm.

10.5.8 Theorem ([dPW15]). *ℓ_1 is a projective Banach lattice.*

Proof. Let $T\colon \ell_1 \to X/J$ be a lattice homomorphism; fix $\varepsilon > 0$. For each i, put $\varphi_i = Te_i$. Clearly, $\varphi_i \geqslant 0$ for all i. By Theorem 3.4.10 and the remark after it, we can find a disjoint positive sequence (v_i) in X such that $Qv_i = \varphi_i$ and $\|v_i\| \leqslant (1+\varepsilon)\|\varphi_i\| \leqslant (1+\varepsilon)\|T\|$ We define $\widetilde{T}\colon \ell_1 \to X$ via $\widetilde{T}e_i = v_i$. It follows from

$$\Bigl\|\widetilde{T}\sum_{i=1}^n \alpha_i e_i\Bigr\| = \Bigl\|\sum_{i=1}^n \alpha_i v_i\Bigr\| \leqslant \sum_{i=1}^n |\alpha_i|\|v_i\| \leqslant (1+\varepsilon)\|T\| \sum_{i=1}^n |\alpha_i|$$

that $\widetilde{T}$ is bounded and $\|\widetilde{T}\| \leqslant (1+\varepsilon)\|T\|$. Since the sequence (v_i) is positive and disjoint, we have $\widetilde{T}|x| = \bigl|\widetilde{T}x\bigr|$ for every $x \in \ell_1$, so that $\widetilde{T}$ is a lattice homomorphism. □

10.5.9 Corollary. *If X/J contains a lattice copy of ℓ_1 for some some closed ideal J (in particular, if X/J is lattice isomorphic to ℓ_1 for some J) then X contains a lattice copy of ℓ_1.*

Proof. Let $T\colon \ell_1 \to X/J$ be a lattice isomorphic embedding. Let $\widetilde{T}\colon \ell_1 \to X$ be the lattice homomorphism as in Theorem 10.5.8. There exists $C > 0$ such that for every $x \in \ell_1$ we hace $C\|x\| \leqslant \|Tx\| = \|Q\widetilde{T}x\| \leqslant \|\widetilde{T}x\|$, so that $\widetilde{T}$ is an isomorphis embedding as well. □

10.6 Hierarchy of properties of Banach lattices

Combining several facts that have been proved throughout these notes, we get the following "hierarchy" of properties of vector and Banach lattices.

10.6.1 Theorem. *The following properties are listed in order of increasing strength. Here X is assumed to be a vector lattices in* (i)-(iv) *and a Banach lattice in* (v)-(vi).

(i) *X is Archimedean;*

(ii) *X has the Principal Projection Property (PPP);*

(iii) *X is σ-order complete;*

(iv) *X is order complete;*

(v) *X is order continuous;*

(vi) *X is a KB-space.*

Proof. (i)$\Leftarrow$(ii) Exercise 5.2.27.

(ii)$\Leftarrow$(iii) Corollary 5.2.23.

(iii)$\Leftarrow$(iv) Trivial.

(iv)$\Leftarrow$(v) Theorem 2.5.22.

(v)$\Leftarrow$(vi) by Nakano's Theorem 2.5.24. □

There are other important properties of Banach lattices.

10.6.2 Definition. Let X be a Banach lattice.

- X is ***monotonically bounded*** (sometimes called ***boundedly order bounded***) if every increasing norm bounded net is order bounded;
- X is ***monotonically complete*** or ***Levi*** if every increasing norm bounded net has a supremum;
- X has the ***Fatou Property*** if $0 \leqslant x_\alpha \uparrow x$ implies $\|x_\alpha\| \uparrow \|x\|$ for every net (x_α) in X.

10.6.3 Exercise. In the definitions of monotonically bounded and monotonically complete Banach lattices, it suffices to consider positive nets.

10.6.4 Exercise. Let X be a Banach lattice.

(i) X is monotonically complete iff it is monotonically bounded and order complete;
(ii) X is a KB-space iff it is monotonically complete and order continuous;
(iii) If X is order continuous then it has the Fatou property.

10.6.5 Example. ℓ_∞ is monotonically complete but not KB.

Exercise 3.3.22 implies the following:

10.6.6 Proposition. *X has the Fatou property iff B_X is order closed.*

10.6.7 Exercise. For a Banach lattice X, TFAE:

(i) X is monotonically complete and has the Fatou property;
(ii) If (x_α) is an increasing norm bounded net in X_+ then $x = \sup x_\alpha$ exists and $\|x_\alpha\| \to \|x\|$.

We note that in some of the literature, statement (ii) of Exercise 10.6.7 is used as the definition of the Fatou Property, see, e.g., [LT79]. Essentially, in one of the statements, the existence of $\sup x_\alpha$ is a part of the assumption, while in the other it is part of the conclusion. These statements are not equivalent and this causes some confusion. For example, it is easy to see that c_0 satisfies "our" Fatou property (because it is order continuous); however, it is easy to see that c_0 is not monotonically complete and, therefore, fails (ii) above.

10.6.8 Exercise. Let X be a Banach lattice. Every increasing net in $B^+_{X^*}$ is order bounded by some $f \in B^+_{X^*}$. It follows that X^* is monotonically bounded. Furthermore, X^* is monotonically complete and has the Fatou property.

10.6.9 Exercise. Let A be a subset of a Banach lattice X.

(i) If X is monotonically bounded then A is order bounded iff $A^\vee$ is norm bounded;

(ii) If X is monotonically bounded and has PPP (in particular, if X is monotonically complete) then A is order bounded iff the set

$$\Bigl\{\sum_{i=1}^n x_i \;:\; x_1,\dots,x_n \in \mathrm{Sol}(A),\ \text{disjoint}\Bigr\}$$

is norm bounded;

An order bounded subset of a Banach lattice clearly remains order bounded when viewed as a subset of X^{**}. However, the converse is false. For example, the set $\{e_n\}$ in c_0 fails to be order bounded even though it is order bounded in ℓ_∞.

10.6.10 Exercise. X is a KB-space iff X is order continuous and for every $A \subset X_+$, if A is order bounded in X^{**} then A is order bounded in X.

Note that the order continuity assumption in the exercise is critical. Indeed, $X := C[0,1]$ fails to be a KB-space, but both X and X^{**} are AM-spaces with units, hence order boundedness agrees with norm boundedness in both X and X^{**}.

The norm on a Banach lattice is said to be ***strictly monotone*** if $0 \leqslant x < y$ implies $\|x\| < \|y\|$.

10.6.11 Exercise. $L_p(\mu)$ is strictly monotone iff $p < \infty$. $C(K)$ is strictly monotone only when K is a singletone.

10.6.12 Exercise. Let $T\colon X \to Y$ be a positive isometric embedding between Banach lattices, with Y being strictly monotone. Then T is a lattice homomorphism.

10.7 * Abstract L$_p$-spaces

Let $1 \leqslant p < \infty$. A Banach lattice X is said to be an ***abstract L_p-space*** or an ***AL$_p$-space*** if $\|x+y\|^p = \|x\|^p + \|y\|^p$ whenever $x \perp y$. Clearly, every $L_p(\mu)$ space is an AL$_p$-space. In the special case when $p = 1$ we get the familiar AL-spaces. The following is a natural extension of Kakutani's AL-representation Theorem to AL$_p$-spaces. The proof we present here does not rely Kakutani's AL-representation Theorem. Thus, in the case $p = 1$, it may be viewed as an alternative proof of Kakutani's AL-representation theorem.

10.7.1 Theorem (Kakutani-Bohnenblust-Nakano). *Every AL$_p$-space with $1 \leqslant p < \infty$ is lattice isometric to $L_p(\mu)$ for some measure μ.*

Proof. Let X be an AL$_p$-space. Meyer-Nieberg Theorem 10.1.2 guarantees that X is order continuous because if (x_n) is a disjoint sequence in an order interval $[0,u]$ then $\sum_{i=1}^n \|x_i\|^p = \left\|\sum_{i=1}^n x_i\right\|^p \leqslant \|u\|^p$ implies $x_n \to 0$. In particular, X has PPP.

Similar to the proof of Lemma 8.5.1, we argue that it suffices to prove the theorem for a principal band in X. Thus, we may assume WLOG that X has a weak unit, say e. The set $\mathcal{C}_e$ of all components of e forms a Boolean algebra by Proposition 7.2.4. By Stone's Theorem 7.1.7, there is a Boolean algebra isomorphism φ between $\mathcal{C}_e$ and the algebra $\mathrm{Clop}(K)$ of all clopen set of some totally disconnected compact space K.

We now define a measure on $\mathrm{Clop}(K)$ as follows: if $A = \phi(x)$ for $x \in \mathcal{C}_e$, we put $\mu(A) = \|x\|^p$. Since X is an AL$_p$ space, μ is finitely additive. To show that μ is countably additive, suppose that $A = \bigcup_{i=1}^\infty A_i$ for some A and some disjoint sequence (A_n) in $\mathrm{Clop}(K)$. Being a closed subset of K, A is compact. Therefore, $A = \bigcup_{i=1}^\infty A_i$ is an open cover of a compact set. It follows that $A = \bigcup_{i=1}^n A_i$ for some n. Thus, every countable union in $\mathrm{Clop}(K)$ is actually finite, hence finite additivity of μ yields countable additivity.

Let $L_p(\mu) = L_p\bigl(K, \mathrm{Clop}(K), \mu\bigr)$. Let Y be the subspace of $L_p(\mu)$ consisting of simple functions. By Exercise 7.2.9, φ extends to a lattice isomorphism T from $\operatorname{span}\mathcal{C}_e$ onto Y via

$$T\Bigl(\sum_{i=1}^n \lambda_i x_i\Bigr) = \sum_{i=1}^n \lambda_i \mathbb{1}_{\varphi(x_i)}.$$

Let $x \in \operatorname{span}\mathcal{C}_e$; we can write $x = \sum_{i=1}^n \lambda_i x_i$ for some $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ and some *disjoint* $x_1, \dots, x_n$ in $\mathcal{C}_e$; this is possible by Corollary 7.2.7. It follows that

$$\|x\|^p = \sum_{i=1}^n |\lambda_i|^p \|x_i\|^p = \sum_{i=1}^n |\lambda_i|^p \mu\bigl(\varphi(x_i)\bigr) = \|Tx\|^p_{L_p(\mu)},$$

so that T is an isometry from $\bigl(\operatorname{span}\mathcal{C}_e, \|\cdot\|_X\bigr)$ onto Y.

Clearly, Y is norm dense in $L_p(\mu)$. Freudenthal's Theorem 7.2.14 guarantees that $\operatorname{span}\mathcal{C}_e$ is dense in I_e with respect to $\|\cdot\|_e$ and, therefore, to $\|\cdot\|_X$. Since X is order continuous, I_e is norm dense in $B_e = X$. It follows that T extends to a lattice isometry from X to $L_p(\mu)$. □

We already know that a Hilbert space may have different Banach lattice structures: e.g., $L_2[0,1]$ and ℓ_2 are isometric as Hilbert spaces, but not lattice isomorphic. Yet, we have the following:

10.7.2 Corollary. *Let X be a Banach lattice and a Hilbert space (under the same norm). Then X is lattice isometric to $L_2(\mu)$ for some measure μ.*

Proof. It suffices to show that X is an AL_2 space. Let $x \perp y$ in X. Then $|x+y| = |x-y|$ yields $\|x+y\| = \|x-y\|$. By the Parallelogram Law, we have

$$\|x\|^2 + \|y\|^2 = \tfrac{1}{2}\big(\|x+y\|^2 + \|x-y\|^2\big) = \|x+y\|^2.$$

□

10.8 * Discrete order continuous spaces

Let X be a Banach lattice. Theorem 10.1.2 asserts that X is order continuous iff every order interval is weakly compact. Furthermore, Theorem 10.2.3 refines it by characterizing those positive vectors u for which $[0,u]$ is weakly compact. In this section, we will show that strengthening weak compactness to norm compactness corresponds to adding discreteness or, equivalently, sequential weak continuity of lattice operations.

10.8.1 Proposition. *Let $X = L_p(\mu)$ for some finite measure μ and $1 \leqslant p < \infty$. If $[-\mathbb{1},\mathbb{1}]$ is compact then X is lattice isometric to ℓ_p or ℓ_p^n for some n.*

Proof. Suppose that $[-\mathbb{1},\mathbb{1}]$ is compact. Then $L_1(\mu)$ is separable because one can then find a dense countable subset in $[0,\mathbb{1}]$ and, therefore, in $\bigcup_{n=1}^\infty[-n\mathbb{1},n\mathbb{1}] = L_\infty(\mu)$, and the latter set is dense in $L_1(\mu)$. By Corollary 8.6.8, $L_p(\mu)$ is lattice isometric to one of ℓ_p, ℓ_p^n, $L_p[0,1]$, $\ell_p \oplus L_p[0,1]$, or $\ell_p^n \oplus L_p[0,1]$. Thus, it suffices to show that no band in $L_p(\mu)$ is isometric to $L_p[0,1]$. Indeed, suppose otherwise. Then by Remark 8.6.9, the interval $[-\mathbb{1},\mathbb{1}]$ in $L_p[0,1]$ is the image of $[-\mathbb{1},\mathbb{1}]$ in $L_p(\mu)$, hence compact. On the other hand, the Rademacher sequence (r_n) in $L_p[0,1]$ is contained in $[-\mathbb{1},\mathbb{1}]$ but has no convergent subsequences; a contradiction. □

Recall that lattice operations in a Banach lattice need not be weakly continuous; see 6.8.6.

10.8.2 Theorem. *Let u be a positive vector in a Banach lattice X. TFAE:*

(i) $u = \sum_{n=1}^\infty a_n$ *for a sequence of disjoint atoms* (a_n)*;*
(ii) $[0,u]$ *is compact;*
(iii) I_u *is order continuous and discrete;*
(iv) $\overline{I_u}$ *is order continuous and discrete;*
(v) $\overline{I_u}$ *is order continuous and* $x_n \xrightarrow{\mathrm{w}} 0$ *in* $\overline{I_u}$ *implies* $|x_n| \xrightarrow{\mathrm{w}} 0$.

Proof. (i)⇒(ii) Fix $\varepsilon > 0$. Find m such that $\big\|\sum_{n=m+1}^\infty\big\| < \varepsilon$. Write $u = v + w$ where $v = \sum_{n=1}^n a_n$ and $w = \sum_{n=m+1}^\infty a_n$. Let $x \in [0,u]$. By Riesz Decomposition Theorem 1.5.14, we have $x = y + z$ for some $y \in [0,v]$ and $z \in [0,w]$. It follows that $\|z\| \leqslant \|w\| < \varepsilon$ and, therefore, $[0,u] \subseteq [0,v] + \varepsilon B_X$. By Lemma 5.4.9, $I_v = \operatorname{span}\{a_1,\dots,a_m\}$. Hence, $[0,v]$ is an order bounded (and, therefore, norm bounded) closed subset of a finite-dimensional subspace of x, hence $[0,v]$ is compact. It follows that $[0,u]$ is compact by Corollary A.4.7.

(iii)⇔(iv) Since I_u is an ideal in $\overline{I_u}$, it is regular there. Corollary 3.2.17 implies that I_u is order dense in $\overline{I_u}$. It follows that I_u and $\overline{I_u}$ have the same atoms by Exercise 5.4.3. It now follows easily that I_u is discrete iff $\overline{I_u}$ is discrete. Furthermore, I_u is order continuous iff $\overline{I_u}$ is order continuous by Exercise 10.2.1 and Theorem 10.2.2.

(ii)⇒(iv) Suppose that $[0,u]$ is compact. In particular, it is weakly compact. Theorem 10.2.3 yields that $\overline{I_u}$ is order continuous. By Proposition 5.5.6, u is a weak unit in $\overline{I_u}$. Represent $\overline{I_u}$ as an ideal in $L_1(\mu)$ as in L_1-representation Theorem 9.3.2. Since the inclusion map is continuous and maps $[0,u]$ onto $[0,\mathbb{1}]$, we conclude that $[0,\mathbb{1}]$ is compact in $L_1(\mu)$, so that $L_1(\mu)$ is discrete by Proposition 10.8.1. Being an order dense ideal in $L_1(\mu)$, $\overline{I_u}$ is discrete as well.

(iv)⇒(i) by Decomposition Lemma 5.3.5. This proves (i)$\Leftrightarrow \cdots \Leftrightarrow$(iv).

(iv)⇒(v) WLOG, $\overline{I_u}$ is infinite-dimensional. Fix a strictly positive functional h_0 on $\overline{I_u}$; it exists by Theorem 9.1.12. Represent $\overline{I_u}$ as an ideal in $L_1(\mu)$ as in Theorem 9.3.2. Since $[0,u]$ is compact in $\overline{I_u}$ by (ii), we conclude that $[0,\mathbb{1}]$ is compact in $L_1(\mu)$; it follows that $L_1(\mu)$ is lattice isometric to ℓ_1 by Proposition 10.8.1. Suppose that $x_n \xrightarrow{\mathrm{w}} 0$ in $\overline{I_u}$. Since the inclusion map is continuous, hence weakly continuous, we have $x_n \xrightarrow{\mathrm{w}} 0$ in $L_1(\mu)$. By Schur's Theorem A.4.13, $x_n \to 0$ in norm. It follows that $h_0\big(|x_n|\big) = \int |x_n| d\mu = \|x_n\|_{L_1} \to 0$. This proves that $h\big(|x_n|\big) \to 0$ for every strictly positive functional h in $(\overline{I_u})^*$. Since every positive functional f on $\overline{I_u}$ is dominated by the strictly positive functional $f+h$, it follows that $f\big(|x_n|\big) \to 0$ and, therefore, $|x_n| \xrightarrow{\mathrm{w}} 0$.

(v)⇒(ii) Let (x_n) be a sequence in $[0,u]$. Since $\overline{I_e}$ is order continuous, $[0,u]$ is weakly compact. By Eberlein-Šmulian Theorem A.4.20, there is a subsequence (x_{n_k}) which converges weakly to some x in $[0,u]$. It follows that $x_{n_k} - x \xrightarrow{\mathrm{w}} 0$ and, therefore, $|x_{n_k} - x| \xrightarrow{\mathrm{w}} 0$. By Theorem 10.1.11, we have $|x_{n_k} - x| \to 0$ and, therefore, $x_{n_k} \to x$. It follows that $[0,u]$ is compact. □

The preceding theorem was local in nature. We now present its global counterpart (cf. Exercise 5.4.20). It shows that discrete order continuous Banach lattices is an important class of spaces.

10.8.3 Theorem. *For a Banach lattice X, TFAE:*

(i) *X is order continuous and discrete;*

(ii) *Every principal ideal in X is order continuous and discrete;*

(iii) *Order intervals in X are compact;*

(iv) *For every $u \in X_+$, $u = \sum_{n=1}^{\infty} a_n$ for some disjoint sequence (a_n) of atoms;*

(v) *Every order bounded weakly convergent sequence (or net) in X converges in norm.*

Proof. (i)⇒(ii) Let $u \in X_+$. It is easy to see that I_u is discrete. Being an ideal, I_u is regular, hence it is order continuous by Exercise 10.2.1.

(ii)⇔(iii)⇔(iv) by Theorem 10.8.2.

(ii) and (iii) imply (i). Indeed, if order intervals are compact, they are weakly compact, hence X is order continuous. If every principal ideal is discrete, it is easy to see that X is discrete. This yields (i)⇔(ii)⇔(iii)⇔(iv)

(iii)⇒(v) It is an easy fact that on a norm compact subset of a Banach space, the norm and the weal topologies agree.

(v)⇒(iii) Let $u \in X_+$; let (x_n) be a sequence in $[-u, u]$. Since $[-u, u]$ is weakly compact by Theorem 10.1.2, (x_n) has a subsequence that converges weakly. By assumption, this subsequence also converges in norm, hence $[-u, u]$ is compact. □

10.8.4 Example. Order continuity is critical in the preceding theorem. The space ℓ_∞ is discrete, yet the order interval $[-\mathbb{1}, \mathbb{1}]$ coincides with the unit ball, hence fails to be compact.

10.8.5 Corollary. *If X is a separable Banach lattice with compact order intervals than the order on X is given by an unconditional basis.*

Proof. By the preceding theorem, X is atomic. Let A be a maximal collection of normalized atoms in X_+. Since X is separable, A is at most countable. Hence, we can enumerate A into a sequence. Being disjoint, it is an unconditional basic sequence. It follows from (iv) that every vector in X belongs to the closed linear span of this sequence, hence it is a basis. □

10.8.6 Theorem. *An order continuous Banach lattice is discrete iff its lattice operations are weakly sequentially continuous.*

Proof. Suppose that X is discrete. Let $x_n \xrightarrow{\mathrm{w}} 0$ in X. Find $u \in X_+$ such that $(x_n) \subseteq I_u$. By Theorem 10.8.3(ii) combined with Theorem 10.8.2(v), we have $|x_n| \xrightarrow{\mathrm{w}} 0$.

To prove the converse, let $u \in X_+$. Being an ideal in an order continuous Banach lattice, $\overline{I_u}$ is order continuous by Exercise 10.2.1. By the implication (v)⇒(iii) of Theorem 10.8.2, $\overline{I_u}$ is discrete. The result now follows from Theorem 10.8.3(ii). □

10.8.7 Theorem (Nakamura). *A separable Banach lattice X is discrete and order continuous iff lattice operations on X^* are sequentially weak* continuous.*

Proof. Suppose that X is discrete and order continuous; let $f_n \xrightarrow{\mathrm{w}^*} 0$ in X^*. For every $x \in X_+$, the interval $[0,x]$ is compact by Theorem 10.8.3. It follows that (f_n) converges to zero uniformly on $[0,x]$. That is, for every $\varepsilon > 0$ there exists n_0 such that for all $n \geqslant n_0$ and all $y \in [0,x]$, we have $\lvert f_n(y)\rvert \leqslant \varepsilon$ and, therefore, by Riesz-Kantorovich formula for f_n^+, we have $\lvert f_n^+(x)\rvert \leqslant \varepsilon$. It follows that $f_n^+(x) \to 0$ and, therefore, $f_n^+ \xrightarrow{\mathrm{w}^*} 0$.

To prove the converse, we will use Gelfand-Phillips Theorem, which asserts that, given a subset A in a separable Banach space E, if every weak*-null sequence in E^* converges to zero uniformly on A then A is relatively compact; see, e.g., 5.86 on p. 286 in [FHH11].

Fix $x \in X_+$. Suppose that $f_n \xrightarrow{\mathrm{w}^*} 0$ in X^*. By assumption, we have $f_n^+ \xrightarrow{\mathrm{w}^*} 0$. Applying Riesz-Kantorovich formula for f_n^+ again, we conclude that f_n converges to zero uniformly on $[0,x]$. Gelfand-Phillips Theorem yields that $[0,x]$ is compact. □

A deep classical result of Lindenstrauss and Pelczynski states that every normalized unconditional basis of ℓ_1 is equivalent to the standard basis; see Theorem 2.b.9 of [LT77]. That is, ℓ_1 has a unique normalized unconditional basis, up to an equivalence. We now prove that ℓ_1 has a unique Banach lattice structure up to a lattice isomorphism.

10.8.8 Theorem ([AW75])**.** *If a Banach lattice X is isomorphic to ℓ_1 as a Banach space then X is lattice isomorphic to ℓ_1.*

Proof. Recall that being a KB-space is equivalent to weak sequentially complete by Theorem 10.4.7, and the latter is a Banach space property. Since ℓ_1 is a KB-space, so is X. In particular, X is order continuous. It follows that order intervals in X are weakly compact by Theorem 10.1.2.

Combining Schur's Theorem and Eberlein-Šmulyan Theorem, we conclude that every weakly compact set is compact in ℓ_1 and, therefore, in X. Hence, order intervals in X are compact. It now follows from Theorem 10.8.3 that X is discrete. Furthermore, Corollary 10.8.5 implies that X the order on X is given by a normalized unconditional basis.

By the aforementioned theorem of Lindenstrauss and Pelczynski, this basis is equivalent to the canonical basis of ℓ_1. This equivalence clearly yields a lattice isomorphism between X and ℓ_1. □

This result is in sharp contrast with the case of ℓ_2. Indeed, ℓ_2 and $L_2[0,1]$ are isomorphic (even isometric) as Banach spaces, yet they are not lattice isomorphism as one is discrete and the other is non-atomic.

We now turn to the other end of the scale: non-atomic Banach lattices. In general, if X is non-atomic, X^* need not be non-atomic. For example, while $C[0,1]$ is clearly non-atomic, every point evaluation in $C[0,1]^*$ is a lattice homomorphism and, therefore, is an atom in $C[0,1]^*$ as per Exercise 6.11.1. Even if we require that X is non-atomic and order complete, X^* need not be non-atomic. Indeed, consider $X = L_\infty[0,1]$. Clearly, $L_\infty[0,1]$ is non-atomic and order complete. However, being a unital AM-space, $L_\infty[0,1]$ is lattice isometric to $C(K)$ for some compact K. It follows that $L_\infty[0,1]^*$ is lattice isometric to $C(K)^*$. Again, every point evaluation functional on $C(K)$ is an atom in $C(K)^*$, which then corresponds to an atom in $L_\infty[0,1]^*$. Now for some good news:

10.8.9 Theorem (Abramovich-Lipecki)**.** *If a Banach lattice X is non-atomic and order continuous then X^* is non-atomic.*

Proof. Suppose that X^* contains an atom, say f. Then f is a lattice homomorphism by Exercise 6.11.1. Then $J := \ker f$ is a closed ideal in X of codimension 1. It follows that J is not dense. Since X is order continuous, J is not order dense. It follows that there exists $u \in X_+$ such that $[0,u] \cap J = \{0\}$ and, therefore, $J \cap I_u = \{0\}$. Since J is of codimension 1, it follows that $\dim I_u = 1$ and, therefore, u is an atom in X, which is a contradiction. $\square$

Going down is never a problem:

10.8.10 Exercise. If X^* is non-atomic then so is X.

Chapter 11

Weak compactness and disjoint sequences

Let A be a subset of a Banach lattice X. Recall that A is *almost order bounded* if for every $\varepsilon > 0$ there exists $u \geqslant 0$ such that $A \subseteq [-u, u] + \varepsilon B_X$; A is *confined* if every disjoint sequence in $\mathrm{Sol}(A)$ is norm null. Dunford-Pettis Theorem 9.8.6 asserts that for a bounded set A in $L_1(\mu)$, TFAE:

(i) A is relatively weakly compact;

(ii) A is almost order bounded;

(iii) A is confined.

It is easy to see that if A is confined or almost order bounded then so is $\mathrm{Sol}(A)$. It follows that for a bounded set A in $L_1(\mu)$, A is relatively weakly compact iff the same is true for $\mathrm{Sol}(A)$.

In this chapter, we will investigate relationships between these three conditions in general Banach lattices. We will show that (iii) always implies the other two, while (ii) implies the other two when X is order continuous. In fact X is order continuous precisely when almost order bounded sets are confined; this may be viewed as a generalization of Meyer-Nieberg Theorem 10.1.2. See the table on page 332 for more details.

We will also investigate spaces where (i) implies the other two properties, as well as when relative weak compactness passes to the solid hull.

Furthermore, we will consider more general versions of these questions where the native norm of X in (ii) and (iii) is replaced with an arbitrary lattice seminorm.

We have already observed before that there are connections between weak convergence, weak compactness, and disjoint sequences in a Banach lattice. In this chapter, we are going to go further in this direction. Throughout this chapter, X stands for a Banach lattice. Let's first summarize some facts that we already know:

- Every disjoint order bounded sequence is weakly null; Exercise 6.8.8(i);
- Dini's theorem: for monotone nets, weak convergence implies norm convergence; Theorem 6.8.2;
- If X is order continuous and $(x_\alpha) \subseteq [0,u]$ then $x_\alpha \xrightarrow{\mathrm{w}} 0$ implies $x_\alpha \to 0$; Proposition 10.1.11;
- If X is order continuous then every positive weakly null sequence has an asymptotically disjoint subsequence; Corollary 9.6.11;
- $[0,u]$ is weakly compact iff every disjoint sequence in it is norm null; Theorem 10.1.1;
- Meyer-Nieberg Theorem: X is order continuous iff every order bounded disjoint sequence is norm null; Theorem 10.1.2;
- X is a KB-space iff it is weakly sequentially complete; Theorem 10.4.7.

11.1 Seminorms and related terminology

We start by introducing some notation that unifies and simplifies exposition. Recall that a subset A of a vector lattice is solid if $x \in A$ and $|y| \leqslant |x|$ imply $y \in A$. Recall also that $\mathrm{Sol}(A)$ stands for the solid hull of A. A seminorm ρ on a vector lattice X is said to be a ***lattice seminorm*** if $\rho(x) \leqslant \rho(y)$ whenever $|x| \leqslant |y|$. For a seminorm ρ, we write B_ρ for the closed unit ball of ρ, i.e., $B_\rho = \big\{x : \rho(x) \leqslant 1\big\}$.

11.1.1 Exercise. A seminorm ρ on a vector lattice is a lattice seminorm iff B_ρ is solid.

Let X be a Banach space and A is a non-empty bounded subset of X (as before, "bounded" always means "norm bounded"unless specified otherwise). For $f \in X^*$, we define $\rho_A(f) = \sup_{x\in A}\big|f(x)\big|$. It is easy to see that ρ_A is a seminorm on X^*, and $\rho_A(f_\alpha) \to 0$ iff (f_α) converges to zero uniformly on A; see Exercises 9.8.2 and 9.8.4. Symmetrically, for a non-empty norm bounded subset B of X^*, we define a seminorm on X via $\rho_B(x) = \sup\big\{\big|f(x)\big| : f \in B\big\}$. Again, this seminorm corresponds to uniform convergence on B.

11.1.2 Example. The norm on X^* agrees with ρ_{B_X}, while the norm on X is $\rho_{B_{X^*}}$.

11.1.3 Exercise. Let X be a Banach lattice, A and B non-empty norm bounded subsets of X and X^*, respectively.

(i) If A is solid then ρ_A is a lattice seminorm;
(ii) If B is solid then ρ_B is a lattice seminorm.

Let X be a Banach lattice. For $x \in X_+$, we define a lattice seminorm ρ_x on X^* via $\rho_x(f) = |f|(x)$. It follows from the Riesz-Kantorovich formula that

$$\rho_x(f) = |f|(x) = \sup\bigl\{\bigl|f(v)\bigr| \,:\, v \in [-x, x]\bigr\} = \rho_{[-x,x]}(f),$$

hence $\rho_x = \rho_{[-x,x]}$. Similarly, for $f \in X_+^*$ $\rho_f(x) = f\bigl(|x|\bigr) = \rho_{[-f,f]}(x)$ defines a lattice seminorm on X.

Here is another way seminorms commonly arise. Let $T\colon E \to F$ be a linear operator from a vector space E to a normed space F. For $x \in E$, define $r_T(x) = \|Tx\|$. It is easy to see that r_T is a seminorm on E. We use r_T rather than ρ_T to avoid inconsistency with the previously introduced notation $\rho_f(x) = f\bigl(|x|\bigr)$ $(f \in X_+^*)$.

A seminorm ρ on a Banach space is said to be continuous if $x_n \to 0$ implies $\rho(x_n) \to 0$. We have defined seminorms of the form ρ_A and r_T. It is easy to see that such seminorms are continuous. In fact, every continuous seminorm is of this form:

11.1.4 Exercise. For a seminorm ρ on a Banach space X, TFAE:

(i) ρ is continuous;

(ii) $\rho = r_T$ for a bounded operator T from X to some Banach space Y;

(iii) $\rho = \rho_B$ for some bounded subset B of X^*.

11.1.5 Exercise. For a seminorm ρ on a Banach lattice X, TFAE:

(i) ρ is a continuous lattice seminorm;

(ii) $\rho = r_T$ for a lattice homomorphism T from X to some Banach lattice Y;

(iii) $\rho = \rho_B$ for some bounded solid subset B of X^*.

We will now describe two common ways of converting an arbitrary seminorm into a lattice seminorm.

11.1.6 Exercise. Let ρ be a seminorm on a vector lattice X. For $x \in X$, put

$$\tau(x) = \inf\Bigl\{\sum_{i=1}^n \rho(u_i) \,:\, u_1, \dots, u_n \geqslant 0, |x| \leqslant \sum_{i=1}^n u_i\Bigr\}.$$

Then τ is a lattice seminorm and $\tau(x) \leqslant \rho\bigl(|x|\bigr)$ for every $x \in X$. Equivalently, $\tau(x) \leqslant \rho(x)$ for all $x \geqslant 0$. Moreover, τ is the greatest lattice seminorm with this property.

11.1.7. Again, let ρ be a seminorm on a vector lattice X. Suppose that

$$\tau(x) \coloneqq \sup\Bigl\{\sum_{i=1}^n \rho(u_i) \,:\, u_1, \dots, u_n \geqslant 0, \sum_{i=1}^n u_i \leqslant |x|\Bigr\}. \tag{11.1}$$

is finite for every $x \in X$. By Riesz Decomposition Theorem, the condition $\sum_{i=1}^n u_i \leqslant |x|$ may be replaces with $\sum_{i=1}^n u_i = |x|$. It is immediate that $\tau(\lambda x) = |\lambda|\tau(x)$ and that $|x| \leqslant |y|$ implies $\tau(x) \leqslant \tau(y)$.

It is straightforward that if $x, y \geqslant 0$ then $\tau(x+y) \geqslant \tau(x) + \tau(y)$ Moreover, using Riesz Decomposition Theorem 1.5.17, one shows that $\tau(x+y) = \tau(x) + \tau(y)$ for all $x, y \geqslant 0$. It follows that

$$\tau(x+y) = \tau\big(|x+y|\big) \leqslant \tau\big(|x|\big) + \tau\big(|y|\big) = \tau(x) + \tau(y)$$

for any $x, y \in X$, so that τ is a lattice seminorm on X. We also have $\rho\big(|x|\big) \leqslant \tau(x)$ for every $x \in X$.

Suppose that, in addition, X is Archimedean. Since τ is additive on X_+, by Extension Lemma 6.2.1, there exists $f \in X_+^\sim$ such that τ and f agree on X^+ and, therefore, $\tau(x) = f\big(|x|\big)$ for every $x \in X$, so that $\tau = \rho_f$.

Furthermore, for every $x \in X_+$, it follows from $\rho(x) \leqslant \tau(x)$ that $\rho(x) \leqslant f(x)$. It follows from (11.1) that f is the least element of $X_+^\sim$ with this property. In other words, τ is the least seminorm of the form ρ_f dominating ρ.

The preceding construction is often applied in the special case when $\rho = r_T$ for some operator T from X to a Banach space E. In this case, (11.1) becomes

$$\tau(x) = \sup\Big\{\sum_{i=1}^n \|Tu_i\| \;:\; u_1, \dots, u_n \geqslant 0, \sum_{i=1}^n u_i \leqslant |x|\Big\}.$$

Almost order bounded sets. The following definition extends that of an almost order bounded set. Let A be a subset of a vector lattice X and ρ a seminorm on X. Motivated by Exercise 9.7.1, we say that A is ***ρ-almost order bounded*** if for every $\varepsilon > 0$ there exists $u \in X_+$ such that $\rho\Big[\big(|x| - u\big)^+\Big] < \varepsilon$ for all $x \in A$. If X is a Banach lattice and ρ is its norm then we get back the concept of almost order boundedness. If ρ is a lattice seminorm, the condition $\forall x \in A \quad \rho\Big[\big(|x| - u\big)^+\Big] < \varepsilon$ is equivalent to $A \subseteq [-u, u] + \varepsilon B_\rho$ where B_ρ is the unit ball of ρ as in Exercise 9.7.1.

11.1.8 Exercise. Suppose that A is a solid set and ρ is a seminorm on X. If A is ρ-almost order bounded then for every $\varepsilon > 0$ there exists $u \geqslant 0$ such that $A \subseteq [-u, u] + \varepsilon B_\rho$. Is the converse true?

11.1.9 Exercise. Let ρ be a lattice seminorm on X and $A \subseteq X$. Then A is ρ-almost order bounded iff $\mathrm{Sol}(A)$ is ρ-almost order bounded.

Clearly, $A \subseteq X$ is ρ_f-almost order bounded for some $f \in X^*$ if

$$\forall \varepsilon > 0 \quad \exists u \in X_+ \quad \forall x \in A \quad f\Big[\big(|x| - u\big)^+\Big] \leqslant \varepsilon.$$

Similarly, a subset B of X^* is ρ_x-almost order bounded for some $x \in X$ if

$$\forall \varepsilon > 0 \quad \exists g \in X^*_+ \quad \forall f \in A \quad \bigl(\lvert f\rvert - g\bigr)^+(x) < \varepsilon.$$

We say that a subset A of X is ***weakly almost order bounded*** if A is ρ_f-almost order bounded for every $f \in X^*_+$. For $B \subseteq X^*$; we say that B ***weak* almost order bounded*** if B is ρ_x-almost order bounded for every $x \in X_+$. Clearly, every almost order bounded subset of X is weakly almost order bounded and every weakly almost order bounded subset of X^* is weak* almost order bounded.

Confined sets. Recall that a set A is ***confined*** if every disjoint sequence in $\mathrm{Sol}(A)$ converges to zero (in norm). Theorem 10.1.1 says that an order interval $[0, u]$ is weakly compact iff it is confined, while Meyer-Nieberg Theorem says that X is order continuous iff every order interval is confined. More generally, if ρ is a seminorm, we say that A is ***ρ-confined*** if $\rho(x_n) \to 0$ for every disjoint sequence in $\mathrm{Sol}(A)$. We say that A is ***weakly confined*** if every disjoint sequence in $\mathrm{Sol}(A)$ is weakly null. Similarly, a subset B of X^* is ***weak* confined*** if every disjoint sequence in $\mathrm{Sol}(B)$ is weak* null. It follows from Exercise 6.8.8 that every order bounded set is weakly confined and every bounded subset of a reflexive Banach lattice is weakly confined. Even though lattice operations need not be weakly continuous, we have the following:

11.1.10 Exercise. (i) For $A \subseteq X$, A is weakly confined iff A is ρ_f-confined for every $f \in X^*_+$;

(ii) For $B \subseteq X^*$, B is weak*-confined iff B is ρ_x-confined for every $x \in X_+$.

Recall that Corollary 9.8.10 of Dunford-Pettis Theorem asserts that a bounded set A in $L_1(\mu)$ is confined iff it is relatively weakly compact. Because of this, confined sets are often called ***L-weakly compact*** in the literature. We find the term "L-weakly compact" somewhat confusing as the definition involves neither compactness nor weak topology explicitly. Furthermore, saying "weakly L-weakly compact" instead of "weakly confined" would make a terrible notation.

11.1.11 Exercise. A is confined iff every disjoint sequence in $\mathrm{Sol}(A)$ is norm convergent.

11.1.12 Exercise. It suffices to consider only *positive* disjoint sequences in $\mathrm{Sol}(A)$ in the definitions of confinement, weak confinement, and weak*-confinement, as well as in the definition of ρ-confinement provided that ρ is a lattice seminorm.

Given $A \subseteq X$ and $B \subseteq X^*$, we can consider conditions like A being ρ_B-almost order bounded or ρ_B-confined. Taking appropriate A and B allows us to express various concepts and results in this notation. The following exercise often allows one to assume that the sets are solid.

11.1.13 Exercise. Let A and B be non-empty bounded subsets of X and X^*, respectively.

(i) A is ρ_B-confined iff A is $\rho_{\mathrm{Sol}(B)}$-confined.
(ii) B is ρ_A-confined iff B is $\rho_{\mathrm{Sol}(A)}$-confined.

Even though we defined confinement using sequences, the following exercise suggests that it is really not a sequential property. We formulate the exercise for a solid set; for an arbitrary set A, just consider $\mathrm{Sol}(A)$.

11.1.14 Exercise. Let A be a solid subset of X and ρ a seminorm on X.

(i) A is ρ-confined iff for every disjoint subset D of A and $\varepsilon > 0$ only finitely many $x \in D$ satisfy $\rho(x) > \varepsilon$. That is, $D \setminus \varepsilon B_\rho$ is finite.
(ii) A is ρ-confined iff every (infinite) disjoint net in A is ρ-null.

Using terminology introduced in this section, Dunford-Pettis Theorem 9.8.6 asserts that a bounded set A in $X = L_1(\mu)$ is relatively weakly compact iff it is ρ_{B_X}-almost order bounded iff $B^+_{X^*}$ is ρ_A-confined. Grothendieck Theorem 9.8.13 says that in the special case when $X = C(K)^*$, one may replace X^* with $C(K)$ in the last clause, so $B^+_{X^*}$ becomes $[0, \mathbb{1}]$.

In this chapter, we will explore how relative weakly compact, almost order bounded, and confined sets are related in more general Banach lattices. Throughout the chapter, we will repeatedly use the identity $x = x \wedge u + (x - u)^+$ for all $x, u \in X$. Many of the results in this chapter will have dual versions with X and X^* interchanged. This redundancy may be avoided by considering dual pairs. This more general approach may be found, e.g., in [AB03].

11.2 Confined vs almost order bounded

Throughout this section, X will stand for a Banach lattice, A and B for non-empty bounded subsets of X and X^*, respectively.

11.2.1 Theorem. *Let ρ be a lattice seminorm on X. Every bounded ρ-confined subset A of X is ρ-almost order bounded.*

Proof. WLOG, A is solid. Suppose that A is ρ-confined, yet not ρ-almost order bounded. Then there exists $\varepsilon > 0$ such that for every $u \in X_+$ there exists $0 \leqslant x \in A$

with $\rho\big((x-u)^+\big)>\varepsilon$. We are going to use Disjointification Lemma 4.5.1. We start by inductively constructing a positive sequence in A as follows. Applying the assumption with $u=0$, find $0\leqslant x_1\in A$ such that $\rho(x_1)>\varepsilon$. Suppose that $x_1,\dots,x_{n-1}$ have been constructed for some $n>1$. By assumption, we can find a positive $x_n\in A$ such that $\rho(w_n)>\varepsilon$, where

$$w_n=\Big(x_n-4^n\sum_{k=1}^{n-1}x_k\Big)^+.$$

Since $x_n\in A$ for every n, the resulting sequence (x_n) is bounded, so that the series $e:=\sum_{n=1}^{\infty}\frac{x_n}{2^n}$ converges. For each n, put

$$v_n=\Big(x_n-4^n\sum_{k=1}^{n-1}x_k-\tfrac{1}{2^n}e\Big)^+.$$

Disjointification Lemma 4.5.1 asserts that the sequence (v_n) is disjoint. Since $0\leqslant v_n\leqslant x_n\in A$, we conclude that $v_n\in A$. Since A is ρ-confined, we have $\rho(v_n)\to 0$. It follows from $0\leqslant w_n-v_n\leqslant\frac{1}{2^n}e$ that $\rho(w_n-v_n)\leqslant\frac{1}{2^n}\rho(e)\to 0$, hence $\rho(w_n)\to 0$; a contradiction. □

11.2.2 Remark. Careful analysis of the proof yields that not only is A ρ-almost order bounded but the vector u in the definition of ρ-almost order boundedness may be chosen in the ideal $I(A)$.

11.2.3 Exercise. The assumption in Theorem 11.2.1 that ρ is a lattice seminorm may be replaced by ρ being a norm continuous sub-additive function.

Theorem 11.2.1 immediately yields the following:

11.2.4 Corollary. *Every bounded confined set is almost order bounded.*

Naturally, we want to know when the converse of Theorem 11.2.1 holds, i.e., when every ρ-almost order bounded set is ρ-confined. The following exercise somewhat simplifies the situation:

11.2.5 Exercise. Let ρ be a lattice seminorm on X. TFAE:

(i) Every ρ-almost order bounded set is ρ-confined;
(ii) Every order bounded set is ρ-confined.

We will show next that the converse of Corollary 11.2.4 is true precisely when X is order continuous. In other words, in Meyer-Nieberg Theorem 10.1.2, one may replace order bounded sets with almost order bounded ones.

11.2.6 Proposition. *X is order continuous iff every almost order bounded set is confined.*

Proof. If X is order continuous, than every order bounded set is confined by Meyer-Nieberg Theorem 10.1.2; now apply Exercise 11.2.5. If X is not order continuous then by Meyer-Nieberg Theorem 10.1.2 there is an order bounded (hence almost order bounded) set that fails to be confined. □

11.2.7 Example. The unit balls of ℓ_p and $L_p[0,1]$ with $1 \leqslant p < 0$ and of c_0 fail to be confined, hence fail to be almost order bounded by Proposition 11.2.6. The unit ball of ℓ_∞ is order bounded but not confined. This is in agreement with Proposition 11.2.6 as ℓ_∞ fails to be order continuous. Similarly, the unit ball of $C[0,1]$ fails to be confined.

Recall that $X_{\mathrm{oc}} = \bigl\{x \in X \ :\ \bigl[0, |x|\bigr] \text{ is confined}\bigr\}$; it is a closed order continuous ideal in X; see Section 10.2.

11.2.8 Exercise. A set A in X is confined iff it is contained in X_{oc} and is almost order bounded there.

11.2.9 Example. *A is confined in $X \overset{\not\Leftarrow}{\Rightarrow}$ A is confined in an AL-representation of X.* We know from Corollary 9.8.10 that a bounded set in $L_1(\mu)$ is confined iff it is relatively weakly confined. Suppose now that X is represented as an ideal in $L_1(\mu)$ as in Section 9.2 and $A \subseteq X$ is bounded. If A is confined in X, it is still confined in $L_1(\mu)$ because the inclusion $X \hookrightarrow L_1(\mu)$ is continuous. The converse is false. Indeed, let A be the unit ball of $X = L_p[0,1]$ with $1 < p < \infty$. Since X is reflexive, A is weakly compact in X. The inclusion map $X \hookrightarrow L_1[0,1]$ is continuous, hence weakly continuous. It follows that A is weakly compact in $L_1[0,1]$ and, therefore, confined there. Yet A is not confined in X.

We proved in Proposition 9.7.12 that in $L_1(\mu)$ spaces almost order boundedness is a sequential property. We now extend this fact to order continuous spaces.

11.2.10 Corollary. *Let A be a subset of an order continuous Banach lattice. If every countable subset of A is almost order bounded then so is A.*

Proof. It is easy to see that A is bounded. Suppose that it is not almost order bounded. Then $\mathrm{Sol}(A)$ is not almost order bounded. By Corollary 11.2.4, there exists a disjoint sequence (v_n) in $\mathrm{Sol}(A)$ such that (v_n) is not norm null. For each n, find $x_n \in A$ such that $|v_n| \leqslant |x_n|$. Put $C = \{x_n\}$. Then C is a countable subset of A and $v_n \in \mathrm{Sol}(C)$ for every n; hence C fails to be confined. By Proposition 11.2.6, we conclude that C is not almost order bounded; a contradiction. □

11.2.11 Example. *A set A in which every sequence is almost order bounded, yet A itself is not.* Let Ω be an uncountable set and let X be the closed sublattice of $\ell_\infty(\Omega)$ consisting of all the functions with countable support. Let $A = \{e_\omega : \omega \in \Omega\}$, the set of all characteristic functions of singletons. Every sequence (e_{ω_n}) in A is order bounded as $e_{\omega_n} \leqslant \mathbb{1}_C$ where $C = \{\omega_n\}$. Howevery it is easy to see that A itself is not almost order bounded.

We finish this section with another application of Theorem 11.2.1.

11.2.12 Corollary. *Let A be a bounded subset of X. For every $f \in X^*$, A is ρ_f-confined iff A is ρ_f-almost order bounded. It follows that A is weakly confined iff it is weakly almost order bounded.*

Proof. If A is ρ_f-confined then A is ρ_f-almost bounded by Theorem 11.2.1. Since every disjoint order bounded sequence in X is weakly null, we conclude that every order bounded set is ρ_f-confined. It follows from Exercise 11.2.5 that every ρ_f-almost order bounded is ρ_f-confined. □

The following is a dual version of the preceding corollary; the proof is analogous.

11.2.13 Corollary. *A bounded set in X^* is weak* confined iff it is weak* almost order bounded.*

11.3 Duality between disjoint sequences

Recall that if X is a vector lattice and $u \in X$ then Theorem 6.6.9 provides a band projection $R_u\colon X^\sim \to X^\sim$, such that for every $f \in X^\sim_+$ we have $0 \leqslant R_u f \leqslant f$, $R_u f$ agrees with f on I_u and vanishes on $\{u\}^d$; it is given by $(R_u f)(x) = \sup_n f(x \wedge nx)$ for every $x \in X_+$. One may think of R_u as a "restriction operator". By Exercise 6.3.13, if $u \perp v$ in X_+ then $(R_u f) \perp (R_v f)$.

11.3.1. Consider a positive sequence (x_n) in a Banach lattice X and a positive sequence (f_n) in X^*. Suppose that (x_n) is disjoint. For each n, put $g_n = R_{x_n} f_n$. Then $0 \leqslant g_n \leqslant f_n$ for every n, $g_n(x_n) = f_n(x_n)$, and (g_n) is a disjoint sequence. In particular, the sequence (x_n, g_n) is biorthogonal.

It is often desirable to "dualize" this construction. Again, let (x_n) and (f_n) be sequences in X_+ and X^*_+, respectively. Suppose now that (f_n) is disjoint. Can we "lower" x_n's to make them disjoint, without changing $f_n(x_n)$ too much?

Let X be a vector lattice. We proved in Proposition 6.6.7 that if $f, g \in X^\sim_+$ with $f \perp g$, $x, y \in X_+$, and $\varepsilon > 0$ then there exist $u \in [0, x]$ and $v \in [0, y]$ such that $u \perp v$,

$f(u) > f(x) - \varepsilon$, and $g(v) > g(y) - \varepsilon$. Clearly, by induction, we can extend this to a finite number of functional and vectors:

11.3.2 Proposition. *Let X be a vector lattice, $f_1, \dots, f_n$ disjoint functionals in $X_+^\sim$, $x_1, \dots, x_n$ in X_+, and $\varepsilon > 0$. There exist disjoint vectors $v_1, \dots, v_n$ such that $v_i \in [0, x_i]$ and $f_i(v_i) > f_i(x_i) - \varepsilon$ for every $i = 1, \dots, n$.*

11.3.3 Corollary. *Let X be a Banach lattice, $f_1, \dots, f_n$ disjoint functionals in X_+^*, and $\varepsilon > 0$. We can then find disjoint vectors $v_1, \dots, v_n \in B_X^+$ such that $f_i(v_i) \geqslant (1-\varepsilon)\|f_i\|$ for every $i = 1, \dots, n$.*

Proof. Fix $\delta > 0$. For each i, using the "positive" Hahn-Banach Theorem (cf. Exercise 6.7.6), we find $x_i \in B_X^+$ such that $f_i(x_i) \geqslant \|f\| - \delta$. By Proposition 11.3.2, we find disjoint $v_1, \dots, v_n$ such that $v_i \in [0, x_i]$ and $f(v_i) \geqslant f_i(x_i) - \delta \geqslant \|f_i\| - 2\delta$ for each i. By choosing δ sufficiently small, we have $f(v_i) \geqslant (1-\varepsilon)\|f_i\|$ for all i. □

The following result is analogue to Proposition 5.2.26 in the sense that it approximates, in a certain sense, the supremum of positive vectors with the supremum (or, equivalently, the sum) of disjoint positive vectors:

11.3.4 Theorem. *Let X be a vector lattice such that $X^\sim$ separates points. Suppose that $x_1, \dots, x_n \in X_+$, $f \in X_+^\sim$, and $\varepsilon > 0$. There exist disjoint vectors $v_1, \dots, v_n$ such that $v_i \in [0, x_i]$ for every i and $\langle f, \bigvee_{i=1}^n x_i \rangle \leqslant \langle f, \bigvee_{i=1}^n v_i \rangle + \varepsilon$.*

Proof. By 6.6.18, X embeds as a sublattice into its second order dual $X^{\sim\sim}$ via $x \mapsto \hat{x}$; $X^{\sim\sim}$ is order continuous by Proposition 6.6.1(ii), hence has PPP. Using Proposition 5.2.26, find disjoint $\xi_1, \dots, \xi_n$ in $X^{\sim\sim}$ such that $\sum_{i=1}^n \xi_i = \bigvee_{i=1}^n \hat{x}_i$ and $0 \leqslant \xi_i \leqslant \hat{x}_i$ for every i. By Proposition 11.3.2, we can find disjoint $f_1, \dots, f_n$ in $X^\sim$ such that $0 \leqslant f_i \leqslant f$ and $\xi_i(f_i) > \xi_i(f) - \frac{\varepsilon}{n}$ for every i. Applying Proposition 11.3.2 again, find disjoint $v_1, \dots, v_n$ in X such that $0 \leqslant v_i \leqslant x_i$ and $f_i(v_i) > f_i(x_i) - \frac{\varepsilon}{n}$ for every n. We now have

$$\begin{aligned}\Big\langle f, \bigvee_{i=1}^n x_i \Big\rangle &= \Big\langle \bigvee_{i=1}^n \hat{x}_i, f \Big\rangle = \Big\langle \sum_{i=1}^n \xi_i, f \Big\rangle = \sum_{i=1}^n \xi_i(f) < \sum_{i=1}^n \xi_i(f_i) + \varepsilon \\ &\leqslant \sum_{i=1}^n f_i(x_i) + \varepsilon < \sum_{i=1}^n f_i(v_i) + 2\varepsilon \leqslant \Big\langle f, \sum_{i=1}^n v_i \Big\rangle + 2\varepsilon = \Big\langle f, \bigvee_{i=1}^n v_i \Big\rangle + 2\varepsilon.\end{aligned}$$

□

We now show that under some minor additional assumption and after passing to a subsequence, Proposition 11.3.2 extends to infinite sequences.

11.3.5 Theorem (Groenewegen)**.** *Let X be a vector lattice, (f_n) a disjoint sequence in $X_+^\sim$, (x_n) is a sequence in X contained in some principal ideal I_e. Suppose that there exists $C > 0$ such that $1 < f_n(x_n) < C$ for every n. Then for every $\varepsilon > 0$ there exists a subsequence (x_{n_k}) and a disjoint sequence (v_k) such that $v_k \in [0, x_{n_k}]$ and $f_{n_k}(v_k) > 1 - \varepsilon$ for every k.*

Before we proceed to the proof, note that the assumption that (x_n) is contained in I_e is rather mild; every sequence in a Banach lattice is contained in a principal ideal. One may identify I_e with a dense sublattice of a $C(K)$ space, so the theorem may be reduced to $C(K)$ spaces, with x_n's becoming continuous functions and f_n's becoming measures. However, while providing a better intuition, this approach does not really yield any simplifications, so we instead present a direct proof based on [MN91].

Proof. Claim: There exists $k \in \mathbb{N}$, $w \in [0, x_k]$, and a subsequence (x_{n_j}) such that $k < n_1$ and $(R_w f_{n_j})(x_{n_j}) < \varepsilon$ for all $j \geqslant 1$.

To prove the claim, take $m \in \mathbb{N}$ such that $m\varepsilon > C$. Applying Proposition 11.3.2 to $x_1, \dots, x_m$, we find disjoint vectors $w_1, \dots, w_m$ such that $w_k \in [0, x_k]$ and $f_k(w_k) > 1$ for every $k = 1, \dots, m$. We claim that taking $w = w_k$ for some $k \leqslant m$ would do the job. Suppose not. Then for every $k = 1, \dots, m$ all but finitely many indices n after m satisfy $(R_{w_k} f_n)(x_n) \geqslant \varepsilon$. Hence, there exists $n > m$ such that $(R_{w_k} f_n)(x_n) \geqslant \varepsilon$ for all $k = 1, \dots, m$. Since $w_1, \dots, w_m$ are disjoint, the functionals $R_{w_1} f_n, \dots, R_{w_m} f_n$ are disjoint and dominated by f_n. It follows that $f_n \geqslant \sum_{k=1}^m R_{w_k} f_n$, so that $f_n(x_n) \geqslant \sum_{k=1}^m (R_{w_k} f_n)(x_n) \geqslant m\varepsilon > C$, which is a contradiction. This proves the *Claim*.

Next, we use a "small perturbation" to make w disjoint to each (x_{n_j}) as follows. We use the standard fact that if $a, b \in X_+$ and $m \in \mathbb{N}$ then $(a - mb)^+ \perp (b - \frac{1}{m} a)^+$; see 1.5.5. First, we replace w with $v = (w - \delta e)^+$, for some positive δ. It follows from $0 \leqslant w - v \leqslant \delta e$ that $0 \leqslant f_k(w) - f_k(v) \leqslant \delta f_k(e)$; we choose δ small enough that $f_k(v) > 1$. Next, we replace every x_{n_j} with $y_j = (x_{n_j} - m_j w)^+$ for some m_j. We claim that we can chose m_j sufficiently large that $v \perp y_j$. Indeed, take $m_j \geqslant \frac{\|x_{n_j}\|_e}{\delta}$. Then $x_{n_j} \leqslant \|x_{n_j}\|_e e \leqslant m_j \delta e$, so that $\frac{1}{m_j} x_{n_j} \leqslant \delta e$. It follows that

$$y_j \wedge v = (x_{n_j} - m_j w)^+ \wedge (w - \delta e)^+ \leqslant (x_{n_j} - m_j w)^+ \wedge (w - \tfrac{1}{m_j} x_{n_j})^+ = 0.$$

It follows from $y_j = x_{n_j} - x_{n_j} \wedge m_j w$ that $f_{n_j}(y_j) = f_{n_j}(x_{n_j}) - f_{n_j}(x_{n_j} \wedge m_j w)$. The definition of R_w yeilds $f_{n_j}(x_{n_j} \wedge m_j w) \leqslant (R_w f_{n_j})(x_{n_j}) < \varepsilon$. It follows that $f_{n_j}(y_j) > 1 - \varepsilon$.

Let us summarize. We found $k < n_1 < n_2 < \dots$, $v \in [0, x_k]$, and $y_j \in [0, x_{n_j}]$ for all j such that $v \perp y_j$ for every j, and $f_k(v) > 1$. Also, replacing ε with $\frac{\varepsilon}{2}$ in the preceding argument, we have $f_{n_j}(y_j) > 1 - \frac{\varepsilon}{2}$ for all j. Put $v = v_1$ and repeat the preceding argument with ε replaced with $\frac{\varepsilon}{4}$, (x_n) replaced with (y_j), and (f_n) replaced with (f_{n_j}). We then get $k_2 < j_1 < j_2 < \dots$, $v_2 \in [0, y_{k_2}]$, and $z_i \in [0, y_{j_i}]$ for all i such that $v_2 \perp z_i$ and $f_{n_{j_i}}(z_i) > 1 - \frac{3\varepsilon}{4}$ for every i. Iterating the process, we obtain a required sequence (v_k). $\square$

Let X be a Banach lattice, $A \subseteq X$ and $B \subseteq X^*$ such that A and B are both non-empty, bounded, and solid. The condition "A is ρ_B-confined" means that every disjoint sequence in A converges to zero uniformly on B. This condition means that both A and B are sufficiently small. In view of Exercise 9.8.2, this condition and its dual version can be restated as follows:

(i) A is ρ_B-confined iff $\langle x_n, f_n\rangle \to 0$ for every disjoint sequence (x_n) in A and every sequence (f_n) in B.

(ii) B is ρ_A-confined iff $\langle x_n, f_n\rangle \to 0$ for every sequence (x_n) in A and every disjoint sequence (f_n) in B.

Since $\bigl|\langle x_n, f_n\rangle\bigr| \leqslant \bigl\langle |x_n|, |f_n|\bigr\rangle$, it suffices to consider positive sequences. Note that the two conditions only differ by the placement of the word "disjoint". The following theorem shows that they are, in fact, equivalent!

11.3.6 Theorem (Burkinshaw-Dodds)**.** *Let X be a Banach lattice, $A \subseteq X$ and $B \subseteq X^*$ such that A and B are both non-empty, bounded, and solid. TFAE:*

(i) *B is ρ_A-confined;*

(ii) *A is ρ_B-confined;*

(iii) *A is ρ_B-almost order bounded and B is ρ_A-almost order bounded.*

Proof. (i)&(ii)$\Rightarrow$(iii) by Theorem 11.2.1.

(i)$\Rightarrow$(ii) Let (x_n) be a disjoint positive sequence in A and (f_n) a positive sequence in B. By 11.3.1, we find a disjoint sequence (g_n) such that $0 \leqslant g_n \leqslant f_n$ and $g_n(x_n) = f_n(x_n)$ for every n. In particular, $g_n \in B$. It follows from (i) that $\langle x_n, f_n\rangle = \langle x_n, g_n\rangle \to 0$.

(ii)$\Rightarrow$(i) The proof is similar, but we use Groenewegen's Theorem 11.3.5 instead of 11.3.1. Suppose that (i) fails. Then there exist a positive sequence (x_n) in A, a positive disjoint sequence (f_n) in B, and $\varepsilon > 0$ such that $\langle x_n, f_n\rangle > \varepsilon$ for all n. Since both A and B are bounded, so is the sequence $(f(x_n))$. By Groenewegen's Theorem 11.3.5, after passing to a subsequence, we find a disjoint sequence (v_n) such that $0 \leqslant v_n \leqslant x_n$ (hence $v_n \in A$) and $f_n(v_n) > \frac{\varepsilon}{2}$ for every n. This contradicts (ii).

(iii)$\Rightarrow$(ii) Let (x_n) be a disjoint sequence in A; we want to show that $\rho_B(x_n) \to 0$. WLOG, $x_n \geqslant 0$ for all n; otherwise, replace x_n with $|x_n|$. In the spacial case when A and B are order intervals, say, $A = [-u, u]$ and $B = [-g, g]$, it follows from the fact that every disjoint order bounded sequence is weakly null (see Exercise 6.8.8(i)) that $\rho_B(x_n) \leqslant g(x_n) \to 0$. We will now reduce the general case to this special one.

Fix $\varepsilon > 0$. Find $u \in X_+$ and $g \in X_+^*$ such that $\rho_B\bigl((x-u)^+\bigr) < \varepsilon$ and $\rho_A\bigl((f-g)^+\bigr) < \varepsilon$ for all positive $x \in A$ and $f \in B$. Note that $x_n = (x_n - u)^+ + x_n \wedge u$ and $\rho_B\bigl((x_n-u)^+\bigr) < \varepsilon$ for every n; it is left to control $\rho_B(x_n \wedge u)$. The sequence $(x_n \wedge u)$ is disjoint and order bounded, hence $g(x_n \wedge u) \to 0$. Find n_0 such that $g(x_n \wedge u) < \varepsilon$ whenever $n \geqslant n_0$. It follows that for every $f \in B$ we have

$$\begin{aligned}\Bigl|f(x_n \wedge u)\Bigr| \leqslant |f|(x_n \wedge u) &\leqslant \bigl(|f| - g\bigr)^+(x_n \wedge u) + \bigl(|f| \wedge g\bigr)(x_n \wedge u) \\ &\leqslant \bigl(|f| - g\bigr)^+(x_n) + g(x_n \wedge u) < 2\varepsilon.\end{aligned}$$

Hence, $\rho_B(x_n \wedge u) \leqslant 2\varepsilon$ and so $\rho_B(x_n) \leqslant \rho_B\bigl((x_n-u)^+\bigr) + \rho_B(x_n \wedge u) < 3\varepsilon$ for all $n \geqslant n_0$. □

This leads to the following most useful characterization of confined sets:

11.3.7 Exercise. For a bounded subset A of X, A is confined iff B_{X^*} is $\rho(A)$-confined. That is, every disjoint sequence in $\mathrm{Sol}(A)$ converges to zero iff every disjoint bounded sequence in X^* converges to zero uniformly on A. Symmetrically, for a bounded subset B of X^*, B is confined iff B_X is $\rho(B)$-confined.

Corollary 11.2.13 characterizes weak* almost order boundedness of a set of X^* in terms of disjoint sequences in this it. In the next theorem, we apply Theorem 11.3.6 in the special case when one of the two sets is an order interval to characterize sets B in X^* that are ρ_x-almost order bounded for just one $x \in X_+$ in terms of disjoint sequences in $[0,x]$: every disjoint sequence in $[0,x]$ converges to zero uniformly on B.

11.3.8 Theorem (Fremlin)**.** (i) *Let B be a bounded subset of X^* and $x \in X_+$. B is ρ_x-almost order bounded iff $[0,x]$ is ρ_B-confined.*

(ii) *Let A be a bounded subset of X and $f \in X_+^*$. A is ρ_f-almost order bounded iff $[0,f]$ is ρ_A-confined.*

Proof. (i) The proof is a chain of equivalences:
B is ρ_x-almost order bounded
$\Leftrightarrow$ $\mathrm{Sol}(B)$ is $\rho_{[-x,x]}$-almost order bounded (by Exercise 11.1.9)
$\Leftrightarrow$ $\mathrm{Sol}(B)$ is $\rho_{[-x,x]}$-almost order bounded and $[-x,x]$ is $\rho_{\mathrm{Sol}(B)}$-almost order bounded (because $[-x,x]$ is order bounded)
$\Leftrightarrow$ $[-x,x]$ is $\rho_{\mathrm{Sol}(B)}$-confined (by Theorem 11.3.6)
$\Leftrightarrow$ $[-x,x]$ is ρ_B-confined (by Exercise 11.1.13)
$\Leftrightarrow$ $[0,x]$ is ρ_B-confined.

The proof of (ii) is similar. □

Note that in the special case when $X = C(K)$ for some compact K and $x = \mathbb{1}$, Fremlin's Theorem is a special case of Grothendieck's Theorem 9.8.13. Indeed, it is easy to see that $\rho_x(f) = |f|(\mathbb{1}) = \|f\|$ for every $f \in C(K)^*$, hence $\rho_x = \|\cdot\|_{C(K)^*}$. Thus, B is ρ_x-almost order bounded means that B is almost order bounded and, therefore, relatively weakly compact by the Dunford-Pettis Theorem 9.8.6. On the other hand, $[0,x]$ is ρ_B-confined means that every disjoint bounded sequence in $C(K)$ converges to zero uniformly on B.

We deduced Fremlin's Theorem 11.3.8 from Burkinshaw-Dodds' Theorem 11.3.6, which, in turn, was deduced from Groenewegen's Theorem 11.3.5. One can instead deduce Fremlin's Theorem from Grothendieck's Theorem 9.8.13. We will show this deduction for part (i) of Theorem 11.3.8; part (ii) may be proved in an analogous way. As above, we may assume WLOG that B is solid.

Suppose B is ρ_x-almost order bounded. Let (x_n) be a disjoint sequence in $[0,x]$. Fix $\varepsilon > 0$. Since B is ρ_x-almost order bounded, there exists $g \in X^*_+$ such that $\big(|f| - g\big)^+(x) < \varepsilon$ for all $f \in B$. Since the sequence (x_n) is order bounded, it is weakly null and, therefore, $g(x_n) \to 0$. It follows that $g(x_n) < \varepsilon$ for all sufficiently large n and, therefore,

$$\big|f(x_n)\big| \leqslant |f|(x_n) = \big(|f| \wedge g\big)(x_n) + \big(|f| - g\big)^+(x_n) \leqslant g(x_n) + \big(|f| - g\big)^+(x) < 2\varepsilon.$$

Suppose now that $[0,x]$ is ρ_B-confined. Let $Y = \big(I_x, \|\cdot\|_x\big)$. Y is lattice isometric to $C(K)$ for some compact K with x corresponding to $\mathbb{1}$ by Theorem 4.1.10. Let $J\colon Y \to X$ be the natural embedding; J is bounded by Exercise 2.2.3(i). Then $J^*\colon X^* \to Y^*$ is bounded; in particular, $J^*(B)$ is a bounded subset of Y^*, where the latter may be identified with $C(K)^*$ and is, therefore, an AL-space. Note that J^* is the restriction operator, i.e., $J^*f = f_{|Y}$. Since Y is an ideal, J^* is a lattice homomorphism as in Exercise 6.6.8.

Our assumption means that every disjoint sequence in $[0,\mathbb{1}]$ converges to zero uniformly on $J^*(B)$. By Grothendieck's Theorem 9.8.13, $J^*(B)$ is relatively weakly compact (or, equivalently, almost order bounded) in Y^*. Let (f_n) be a positive disjoint sequence in B. By Theorem 11.2.1, it suffices to show that $\rho_x(f_n) \to 0$. Note that $\rho_x(f_n) = f_n(x) = J^*f_n(x) = \|J^*f_n\|_{Y^*}$ because $\|f\| = f(\mathbb{1})$ for every $f \in C(K)^*_+$. Since J^* is a lattice homomorphism, the sequence (J^*f_n) is disjoint in Y^*. It now follows from Proposition 11.2.6 that $\|J^*f_n\|_{Y^*} \to 0$. Alternatively, one may argue that otherwise (J^*f_n) would be equivalent to the unit vector basis of ℓ_1, which would contradict (iv) in Dunford-Pettis Theorem 9.8.6.

11.4 Properties of B_X and B_{X^*}

11.4.1 Proposition. *B_X is almost order bounded iff X has a strong unit.*

Proof. If X has a strong unit then it is lattice isomorphic to $C(K)$ for some compact K by Corollary 4.1.11; hence B_X is order bounded.

Suppose that B_X is almost order bounded. Then there exists $u > 0$ such that $B_X \subseteq [-u,u] + \frac{1}{2}B_X$. We claim that u is a strong unit. Indeed, let $x \in B_X$. Then there exists $a_1 \in [-u,u]$ with

$$x - a_1 \in \tfrac{1}{2}B_X \subseteq \tfrac{1}{2}[-u,u] + \tfrac{1}{4}B_X.$$

It follows that there exists $a_2 \in \frac{1}{2}[-u,u]$ with

$$x - a_1 - a_2 \in \tfrac{1}{4}B_X \subseteq \tfrac{1}{4}[-u,u] + \tfrac{1}{8}B_X.$$

Proceeding inductively, we produce a sequence (a_n) such that $a_n \in \frac{1}{2^{n-1}}[-u,u]$ and $x - \sum_{i=1}^n a_i \in \frac{1}{2^n}B_X$. It follows that $x = \sum_{i=1}^\infty a_i$, so that

$$|x| \leqslant \sum_{i=1}^\infty |a_i| \leqslant \sum_{i=1}^\infty \frac{1}{2^{i-1}}u = 2u.$$

Therefore, $B_X \subseteq [-2u, 2u]$. It follows that u is a strong unit. □

11.4.2 Exercise. Let $X = \ell_p$, $1 < p < \infty$. Prove directly that B_X fails to be almost order bounded.

The following two theorems are due to Dodds and Fremlin [DF80]

11.4.3 Theorem. *TFAE*

(i) B_X *is weakly almost order bounded;*
(ii) B_X *is weakly confined;*
(iii) X^* *is order continuous.*

Proof. (i)⇔(ii) by Corollary 11.2.12. To prove that (i)⇔(iii), observe that (i) means that B_X is ρ_f-almost order bounded for every $f \in X^*_+$. By Fremlin's Theorem 11.3.8(ii), the latter means that $[0, f]$ is confined. By Meyer-Nieberg Theorem 10.1.2, this is equivalent to X^* being order continuous. □

11.4.4 Corollary. *TFAE:*

(i) *Every norm bounded set in X is weakly almost order bounded;*
(ii) *Every norm bounded solid set in X is weakly confined;*
(iii) X^* *is order continuous.*

"Dualizing" the preceding proof and using Corollary 11.2.12 instead of Corollary 11.2.13, we prove the dual version of Theorem 11.4.3:

11.4.5 Theorem. *TFAE*

(i) B_{X^*} *is weak* almost order bounded;*
(ii) B_{X^*} *is weak*-confined;*

(iii) *X is order continuous.*

11.4.6 Exercise. A bounded subset A of X is confined iff B_{X^*} is ρ_A-confined.

11.4.7 Theorem. *For $x \in X_+$, $x \in X_{\mathrm{oc}}$ iff B_{X^*} is ρ_x-almost order bounded. Similarly, $f \in X^*_+$, $f \in (X^*)_{\mathrm{oc}}$ iff B_X is ρ_f-almost order bounded.*

Proof. By Fremlin's Theorem 11.3.8, B_{X^*} is ρ_x-almost order bounded iff $[0, x]$ is confined. The latter is equivalent to $x \in X_{\mathrm{oc}}$ by Theorem 10.2.3(v). The dual statement is proved in a similar fashion. $\square$

Recall that Eberlein-Šmulian Theorem A.4.20 asserts that relative weak compactness is a sequential property. Alaoglu's Theorem A.4.14 guarantees that B_{X^*} is weak* compact. However, weak* compactness, in general, is not sequential.

11.4.8 Theorem. *Let X be an order continuous Banach lattice with a weak unit. Then B_{X^*} is sequentially weak* compact.*

Proof. Let u be a weak unit in X. Then u is a quasi-interior point by Proposition 5.5.6; hence $\hat{u} \in X^{**}$ is a strictly positive functional on X^* by Theorem 9.1.10. Let $L_1(\mu)$ be an L_1-representation for X^* and $\hat{u}$, so that $X^* \hookrightarrow L_1(\mu)$. By Proposition 9.2.6, $L_1(\mu)^*$ is lattice isometric to the ideal $I_{\hat{u}}$ in X^{**} equipped with the norm $\|\cdot\|_{\hat{u}}$. Since X is order continuous, X is an ideal in X^{**} by Theorem 10.1.8, so that $I_{\hat{u}}$ is contained in X; $I_{\hat{u}} = I_u$.

We claim that the weak* topology on X^* and the weak topology on $L_1(\mu)$ agree on B_{X^*}. Indeed, let (f_α) be a net in B_{X^*}. Then $f_\alpha \xrightarrow{\mathrm{w}} 0$ in $L_1(\mu)$ iff $\langle f_\alpha, \xi \rangle \to 0$ for every $\xi \in L_1(\mu)^*$. By the first paragraph, this is equivalent to $\langle f_\alpha, x \rangle \to 0$ for every $x \in I_u$. The latter is equivalent to $\langle f_\alpha, x \rangle \to 0$ for every $x \in X$ because I_u is dense in X and (f_α) is in B_{X^*}. This proves the claim.

By Alaoglu's Theorem A.4.14, B_{X^*} is weak* compact. By the preceding paragraph, it is weakly compact as a subset of $L_1(\mu)$. Let (f_n) be a sequence in B_{X^*}. Applying Eberlein-Šmulian Theorem A.4.20 to $L_1(\mu)$, we conclude that there is a subsequence (f_{n_k}) and $f \in B_{X^*}$ such that $f_{n_k} \xrightarrow{\mathrm{w}} f$ in $L_1(\mu)$. It follows that $f_{n_k} \xrightarrow{\mathrm{w}^*} f$ in X^*. $\square$

11.5 Relative (weak) compactness

First, we quickly discuss relative *norm* compactness. It is the strongest of the properties we consider in this chapter.

The following example shows that compactness is poorly compatible with the lattice structure.

11.5.1 Example. Let $X = L_1[0,1]$ and (x_n) be sequence of characteristic functions of dyadic intervals: $\mathbb{1}_{[0,1]}$, $\mathbb{1}_{[0,\frac{1}{2}]}$, $\mathbb{1}_{[\frac{1}{2},1]}$, $\mathbb{1}_{[0,\frac{1}{4}]}$, $\mathbb{1}_{[\frac{1}{4},\frac{2}{4}]}$, ... Let $A = \{x_n\}$. Clearly, A is order bounded. Furthermore, A is relatively compact because $x_n \to 0$. Yet we claim that $A^\vee$ fails to be relatively compact. Indeed, it is easy to see that $A^\vee$ contains the positive parts of the Rademacher functions. It follows from $r_n^+ = \frac{1}{2}(\mathbb{1} + r_n)$ that $r_n^+ - r_m^+ = \frac{1}{2}(r_n - r_m)$; then Khinchin's Inequality 6.4.4(ii) implies that the sequence (r_n^+) has no convergent subsequences.

However, things work better for "finite-dimensional" sets.

11.5.2 Theorem. *Let X be a Banach lattice, Y a finite-dimensional subspace, and A a bounded subset of Y. Then $A^\vee$ is a convergent net. In particular, $A^\vee$ is relatively compact and* $\sup A$ *exists and equals* $\lim A^\vee$.

Proof. Find $u \in X_+$ such that $Y \subseteq I_u$. By Krein-Kakutani's representation Theorem, we can identify I_u with $C(K)$ for some compact K. Since Y is finite-dimensional, $\|\cdot\|_X$ and $\|\cdot\|_{C(K)}$ are equivalent on it; it follows that A is bounded in $\|\cdot\|_{C(K)}$ and, is, therefore, relatively compact in $C(K)$. By Arzelá–Ascoli Theorem A.4.30, A is equicontinuous. It is easy to verify that $A^\vee$ is still equicontinuous and bounded in $C(K)$, hence relatively compact there. Since the embedding of $C(K) = I_e$ into X is continuous, $A^\vee$ is relatively compact in X.

We may view $A^\vee$ as an increasing net in X. By relative compactness, it has a subnet that converges to some x. It follows from monotonicity that the entire net converges to $x \in X$. By Monotone Convergence Lemma 2.5.3, we conclude that $x = \sup A^\vee = \sup A$. □

11.5.3 Exercise. Every relatively compact solid set is confined.

The "solid" assumption in the preceding exercise can not be removed: every singleton is clearly compact but need not be confined.

We observed in Exercise 9.8.1 that every relatively compact set is almost order bounded. On the other hand, Theorem 10.8.3 and Corollary A.4.7 yield the following:

11.5.4 Proposition. *Every almost order bounded subset of X is relatively compact iff X is discrete and order continuous.*

The situation is more subtle for weak compactness. In this section, we study relationships between relative weak compactness, confinement, almost order boundedness, and weak almost order boundedness. We recommend that at this point the reader reviews properties of relatively weakly compact sets in Banach spaces in Appendix A.4.

Recall that the unit ball in a Banach space X is always weakly closed; it is weakly compact iff X is reflexive; see Proposition A.4.15. Recall also that every relatively weakly compact set is bounded; see Proposition A.4.18.

Dunford-Pettis Theorem 9.8.6 asserts that a set A in $L_1(\mu)$ is relatively weakly compact iff it is confined iff it is almost order bounded. We showed in Corollary 11.2.4 that every bounded confined set in a Banach lattice X is almost order bounded; the two classes of sets agree precisely when X is order continuous by Proposition 11.2.6. The following examples show that, in general, neither relative weak compactness nor almost order boundedness imply each other; and a weakly compact set need not be confined.

11.5.5 Example. The unit ball of $X = C[0,1]$ is order bounded; in particular, it is almost order bounded. Yet it is not relatively weakly compact because $C[0,1]$ is not reflexive. It also fails to be confined.

11.5.6 Example. Let X be $L_p[0,1]$ and ℓ_p when $1 < p < \infty$. Since X is reflexive, its unit ball is weakly compact. Yet, B_X is clearly not confined. Since X is order continuous, B_X is not almost order bounded by Proposition 11.2.6.

More generally, let X be a reflexive Banach lattice with no strong unit. Then B_X is weakly compact but fails to be confined or almost order bounded by Propositions 11.4.1 and 11.2.6.

11.5.7 Theorem. *Every bounded confined set is relatively weakly compact.*

Proof. Let A be a confined set in X. WLOG, A is solid. Fix $\varepsilon > 0$. Theorem 11.2.1 asserts that A is almost order bounded, so that we can find $u \geqslant 0$ such that $A \subseteq [-u, u] + \varepsilon B_X$. By Remark 11.2.2, we may also assume that $u \in I(A)$. It follows that $u \leqslant \lambda(a_1 + \dots + a_n)$ for some positive $a_1, \dots, a_n \in A$ and some $\lambda > 0$. Riesz Decomposition Theorem yields that

$$\tfrac{1}{\lambda}[-u, u] \subseteq [-a_1, a_1] + \dots + [-a_n, a_n].$$

For each $i = 1, \dots, n$, it follows from $a_i \in A$ that $[-a_i, a_i]$ is confined and, therefore, weakly compact by Theorem 10.1.1. Therefore, $[-u, u]$ is weakly compact. Since ε was arbitrary, it follows from Proposition A.4.19 that A is relatively weakly compact. $\square$

Recall that, according to Theorem 10.1.2, X is order continuous iff order intervals are weakly compact. Using a simple approximation argument, extend this to the following result, which may be viewed as an analogue of Proposition 11.5.4.

11.5.8 Theorem. *TFAE*

(i) *Every almost order bounded set in X is relatively weakly compact;*

(ii) *X is order continuous.*

Proof. Suppose (i); then, in particular, every order interval is weakly compact, hence X is order continuous.

Now suppose that X is order continuous and A is almost order bounded. For every $\varepsilon > 0$ there exists $u > 0$ such that $A \subseteq [-u, u] + \varepsilon B_X$. Since $[-u, u]$ is weakly compact, it follows that A is relatively weakly compact; see Proposition A.4.19. $\square$

11.5.9 Theorem. *Every relatively weakly compact set is weakly almost order bounded.*

Proof. It suffices to prove that every weakly compact set A is ρ_f-almost order bounded for every $f \in X_+^*$. Fix $f \in X_+^*$ and $\varepsilon > 0$.

Special case: f is strictly positive. Let $L_1(\mu)$ be an L_1-representation for X and f as in 9.3.1. In particular, X continuously embeds into $L_1(\mu)$ as a dense sublattice and $\|x\|_{L_1(\mu)} = f\big(|x|\big) = \rho_f(x)$ for every $x \in X$. Since the embedding is continuous, it is weak-to-weak continuous (see Proposition A.4.9); hence A is weakly compact in $L_1(\mu)$. By Dunford-Pettis Theorem 9.8.6, A is almost order bounded in $L_1(\mu)$, i.e., there exists $u \in L_1(\mu)_+$ such that $A \subseteq [-u, u] + \varepsilon B_{L_1(\mu)}$. Since X is a dense sublattice of $L_1(\mu)$, there exists $v \in X_+$ such that $\|v - u\|_{L_1(\mu)} < \varepsilon$. It follows that $A \subseteq [-v, v] + 2\varepsilon B_{L_1(\mu)}$. Therefore,

$$\rho_f\Big[\big(|x| - v\big)^+\Big] = \Big\|\big(|x| - v\big)^+\Big\|_{L_1(\mu)} \leqslant 2\varepsilon$$

for every $x \in A$. Hence, A is ρ_f-almost order bounded.

General case. As in 9.1.7, consider the null ideal $N_f = \big\{x \in X \,:\, f\big(|x|\big) = 0\big\}$ and the strictly positive functional $\tilde{f}$ induced on X/N_f by f via $\tilde{f}(\tilde{x}) = f(x)$. It is easy to see that N_f is a closed ideal. Therefore, the quotient space X/N_f is a Banach lattice and the quotient map $Q\colon X \to X/N_f$ is a lattice homomorphism; see Theorems 3.4.2 and 3.4.6. Since Q is continuous, it is weakly continuous, hence $Q(A)$ is weakly compact in X/N_f. By the Special Case, we conclude that $Q(A)$ is $\rho_{\tilde{f}}$-almost order bounded. It follows that there exists $0 < \varphi \in X/N_f$ such that $\tilde{f}\Big[\big(|Qx| - \varphi\big)^+\Big] < \varepsilon$ for every $x \in A$. Find $u \in X$ with $Qu = \varphi$. We may assume that $u \geqslant 0$ by Exercise 3.4.4(i). Since Q is a lattice homomorphism, we have

$$\tilde{f}\Big[\big(|Qx| - \varphi\big)^+\Big] = \tilde{f}\Big[\big(|Qx| - Qu\big)^+\Big] = \tilde{f}\Big[Q\big(|x| - u\big)^+\Big] = f\Big[\big(|x| - u\big)^+\Big].$$

It follows that $f\Big[\big(|x| - u\big)^+\Big] < \varepsilon$, and, therefore, A is ρ_f-almost order bounded. $\square$

Combining Theorems 11.5.9 and 11.5.8, we immediately conclude that for a subset

A of an order continuous Banach lattice,

$$\begin{aligned}&A \text{ is almost order bounded}\\ &\qquad\qquad\Rightarrow A \text{ is relatively weakly compact}\\ &\qquad\qquad\qquad\qquad\Rightarrow A \text{ is weakly almost order bounded.}\end{aligned}$$

11.6 Passing to the solid hull

As before, let A be a subset of a Banach lattice X. Recall that if A is ρ-almost order bounded for some lattice seminorm ρ then so is $\mathrm{Sol}(A)$. It follows that if A is (weakly) almost order bounded then so is $\mathrm{Sol}(A)$. On the other hand, the concepts of relative weak compactness and of being confined generally do not survive passing to the solid hull:

11.6.1 Example. Let $X = C[0,1]$ and $A = \{\mathbb{1}\}$. Being a singleton, A is compact and confined. However, $\mathrm{Sol}(A) = B_X$ is neither relatively weakly compact nor confined.

11.6.2 Exercise. X is order continuous iff every relatively compact set is confined.

11.6.3 Proposition. *If A is a relatively weakly compact then A is weakly confined.*

Proof. By Theorem 11.5.9, A is weakly almost order bounded. Being relatively weakly compact, A is bounded. It follows that $\mathrm{Sol}(A)$ is bounded and weakly almost order bounded. The result now follows from Corollary 11.2.12. $\square$

Recall that for every relatively weakly compact subset A of $L_1(\mu)$, its solid hull $\mathrm{Sol}(A)$ is also relatively weakly compact by Exercise 9.8.8.

We will now characterize spaces where relative weak compactness of A implies that of $\mathrm{Sol}(A)$. Surprisingly, the answer is different if we additionally require that $A \subseteq X_+$. Recall that A is relatively weakly compact iff it is bounded and $\overline{A}^{w^*}$ in X^{**} is contained in X; see Proposition A.4.18. We will also use the following.

11.6.4 Lemma. *Let A be a relatively weakly compact set in X. If $f_\alpha \downarrow 0$ in X^* then (f_α) converges to zero uniformly on* $\mathrm{Sol}(A)$.

Proof. Theorem 11.5.9 yields that A and, therefore, $\mathrm{Sol}(A)$, is weakly almost order bounded. Let $\varepsilon > 0$. Fix some index α_0. There exists $u > 0$ such that $f_{\alpha_0}\big[\big(|x|-u\big)^+\big] < \varepsilon$ for every $x \in \mathrm{Sol}(A)$. It follows from $f_\alpha \downarrow 0$ that $f_\alpha(u) \downarrow 0$ by Proposition 6.6.1(iii). Hence, there exists $\alpha_1 \geqslant \alpha_0$ such that $f_{\alpha_1}(u) < \varepsilon$. It follows that

$$\big|f_\alpha(x)\big| \leqslant f_\alpha\big(|x|\big) = f_\alpha\big(|x| \wedge u\big) + f_\alpha\big[\big(|x|-u\big)^+\big] \leqslant f_{\alpha_1}(u) + f_{\alpha_0}\big[\big(|x|-u\big)^+\big] < 2\varepsilon$$

whenever $\alpha \geqslant \alpha_1$ and $x \in \mathrm{Sol}(A)$. $\square$

11.6.5 Theorem (Wickstead)**.** *TFAE*

(i) *For every $A \subseteq X_+$, if A is relatively weakly compact then so is* $\mathrm{Sol}(A)$*;*

(ii) *X is order continuous.*

Proof. (i)⇒(ii) Let $u \in X_+$. Applying (i) to $A = \{u\}$, we conclude that $[-u, u]$ is relatively weakly compact, hence weakly compact. Therefore, order intervals in X are weakly compact, so that X is order continuous by Theorem 10.1.2.

(ii)⇒(i) Suppose that X is order continuous and $A \subseteq X_+$ is relatively weakly compact. It follows that A is bounded and, therefore, $\mathrm{Sol}(A)$ is bounded. It suffices to show that the weak* closure of $\mathrm{Sol}(A)$, viewed as a subset of X^{**}, is contained in X; see Proposition A.4.18. Let $\xi \in \overline{\mathrm{Sol}(A)}^{w^*}$ in X^{**}. Then there exists a net (x_α) in $\mathrm{Sol}(A)$ such that $\hat{x}_\alpha \xrightarrow{\mathrm{w}^*} \xi$ in X^{**}. For each α, there exists $y_\alpha \in A$ with $\lvert x_\alpha \rvert \leqslant y_\alpha$. Since A is relatively weakly compact, we may assume (passing to a subnet) that $y_\alpha \xrightarrow{\mathrm{w}} y$ for some $y \in X$ and, therefore, $\hat{y}_\alpha \xrightarrow{\mathrm{w}^*} \hat{y}$ in X^{**}. Since $-y_\alpha \leqslant x_\alpha \leqslant y_\alpha$ for every α, we have $-\hat{y} \leqslant \xi \leqslant \hat{y}$. Note that X is an ideal in X^{**} by Theorem 10.1.8; this yields $\xi \in X$. □

11.6.6 Corollary. *Suppose that X is discrete and order continuous, and $A \subseteq X$. If A is relatively weakly compact then so is* $\mathrm{Sol}(A)$.

Proof. Let $\lvert A \rvert = \bigl\{\lvert x \rvert \ : \ x \in A\bigr\}$. Since $\mathrm{Sol}(A) = \mathrm{Sol}\bigl(\lvert A \rvert\bigr)$, by Theorem 11.6.5 it suffices to show that $\lvert A \rvert$ is relatively weakly compact. We will use Eberlein–Šmulian Theorem A.4.20. Every sequence in $\lvert A \rvert$ is of the form $\bigl(\lvert x_n \rvert\bigr)$, where $x_n \in A$ for every n. There is a subsequence (x_{n_k}) such that $x_{n_k} \xrightarrow{\mathrm{w}} x$ for some $x \in X$. Since lattice operations in X are sequentially weakly continuous by Theorem 10.8.3, we have $\lvert x_{n_k} \rvert \to \lvert x \rvert$. □

11.6.7 Theorem (Abramovich)**.** *Let X be a KB-space and $A \subseteq X$. If A is relatively weakly compact then so is* $\mathrm{Sol}(A)$.

Proof. Again, it suffices to show that if $\xi \in \overline{\mathrm{Sol}(A)}^{w^*}$ in X^{**} then $\xi \in X$. We claim that ξ is order continuous. Indeed, suppose that $f_\alpha \downarrow 0$ in X^*. Lemma 11.6.4 yields that (f_α) converges to zero uniformly on $\mathrm{Sol}(A)$. Let $\varepsilon > 0$. There exists α_0 such that $\bigl\lvert f_\alpha(x) \bigr\rvert \leqslant \varepsilon$ whenever $\alpha \geqslant \alpha_0$ and $x \in \mathrm{Sol}(A)$. Fix $\alpha \geqslant \alpha_0$. Since $\xi \in \overline{\mathrm{Sol}(A)}^{w^*}$, there is a net (x_γ) in $\mathrm{Sol}(A)$ such that $\hat{x}_\gamma \xrightarrow{\mathrm{w}^*} \xi$. It follows from $\hat{x}_\gamma(f_\alpha) \to \xi(f_\alpha)$ as $\gamma \to \infty$ and $\bigl\lvert \hat{x}_\gamma(f_\alpha) \bigr\rvert = \bigl\lvert f_\alpha(x_\gamma) \bigr\rvert \leqslant \varepsilon$ that $\bigl\lvert \xi(f_\alpha) \bigr\rvert \leqslant \varepsilon$. This yields $\xi(f_\alpha) \to 0$. Thus, $\xi \in (X^*)^\sim_{\mathrm{oc}}$. But $(X^*)^\sim_{\mathrm{oc}} = X$ by Corollary 10.4.11. □

11.6.8 Example. We construct an order continuous Banach lattice X which is not a KB-space, and a relatively weakly compact set A in X such that $\mathrm{Sol}(A)$ is not relatively weakly compact. Let $X = \big(\bigoplus_{n=1}^{\infty} L_1[0,1]\big)_{c_0}$. Note that X is order continuous, but fails to be a KB-space because it contains a copy of c_0. Let (r_n) be the Rademacher sequence in $L_1[0,1]$. Put $x_n \in X$ such that

$$x_n = (\overbrace{r_n, \dots, r_n}^{n}, 0, \dots).$$

We claim that $x_n \xrightarrow{\mathrm{w}} 0$. It suffices to show that $f(x_n) \to 0$ for every $f \in X^*_+$. Fix $\varepsilon > 0$. Note that $X^* = \big(\bigoplus_{n=1}^{\infty} L_\infty[0,1]\big)_{\ell_1}$, so we can write $f = (f_k)$ where $0 \leqslant f_k \in L_\infty[0,1]$ for every k and $\sum_{k=1}^{\infty} \|f_k\|_{L_\infty} < \infty$. It follows that the series $h = \sum_{k=1}^{\infty} f_k$ converges in $L_\infty[0,1]$. Choose m so that $\big\|h - \sum_{k=1}^{m} f_k\big\|_{L_\infty} < \varepsilon$. It follows from $r_n \xrightarrow{\mathrm{w}} 0$ in $L_1[0,1]$ that there exists $n_0 \geqslant m$ such that $\big|\langle h, r_n\rangle\big| < \varepsilon$ for every $n \geqslant n_0$. It follows from

$$\langle f, x_n\rangle = \sum_{k=1}^{n} \langle f_k, r_n\rangle = \Big\langle \sum_{k=1}^{n} f_k, r_n\Big\rangle$$

that

$$\big|\langle f, x_n\rangle\big| \leqslant \Big\|h - \sum_{k=1}^{n} f_k\Big\|_{L_\infty} + \big|\langle h, r_n\rangle\big| \leqslant \Big\|h - \sum_{k=1}^{m} f_k\Big\|_{L_\infty} + \varepsilon < 2\varepsilon.$$

Thus, $x_n \xrightarrow{\mathrm{w}} 0$. Put $A = \{x_n\}$. Hence, A is relatively weakly compact by Eberlein-Šmulian Theorem A.4.20. However, $|x_n| = (\mathbb{1}, \dots, \mathbb{1}, 0, \dots)$, so that the sequence $\big(|x_n|\big)$ is equivalent to the unit vector basis of c_0. In particular, it has no weakly convergent subsequences. It follows that $\mathrm{Sol}(A)$ is not relatively weakly compact.

11.7 Positive Schur Property

In general Banach spaces, weak convergence does not imply norm convergence. However, Schur's Theorem asserts that for *sequences* in ℓ_1, weak convergence agrees with norm convergence. This fact leads to the following definition: a Banach space has the ***Schur Property*** if every weakly convergent sequence is norm convergent (to the same limit). WLOG, we may assume that the limit is zero. Schur's Theorem asserts that ℓ_1 enjoys this property. However, this property is very restrictive. Among "classical" Banach spaces, only the spaces $\ell_1(\Gamma)$ enjoy Schur property. In this section, we consider a Banach lattice variant of this property: a Banach lattice has the ***Positive Schur Property (PSP)*** if every positive weakly null sequence is norm null. It happens that these are exactly the spaces where relatively weakly compact sets agree with almost order bounded sets.

11.7.1 Exercise. A Banach lattice with the PSP is a KB-space.

Positive Schur Property has several equivalent reformulations. The following says that in the definition of the PSP, one may replace positive sequences with positive disjoint or just disjoint sequences.

11.7.2 Theorem. *TFAE:*

(i) *X has the PSP;*

(ii) *every disjoint weakly null sequence in X is norm null;*

(iii) *every positive disjoint weakly null sequence in X is norm null.*

Proof. (i)⇒(ii) Suppose X has the PSP and let (x_n) be a disjoint weakly null sequence. Note that X is a KB-space; in particular, it is order continuous. Proposition 9.3.7 yields $|x_n| \xrightarrow{\mathrm{w}} 0$. It follows from the PSP that $|x_n| \to 0$ and, therefore, $x_n \to 0$.

(ii)⇒(iii) trivially.

(iii)⇒(i) Suppose that (iii) is satisfied but there exists a weakly null positive sequence (x_n) which is not norm null. Passing to a subsequence and scaling, we may assume that $\|x_n\| \geqslant 1$ for all n. Note that (iii) implies that X has no lattice copy of c_0 because the unit vector sequence (e_n) in c_0 fails (iii). It follows that X is a KB-space. In particular, X is order continuous. Applying Corollary 9.6.11 and passing to a subsequence, we find a disjoint sequence (d_n) such that $0 \leqslant d_n \leqslant x_n$ for every n and $\|x_n - d_n\| \to 0$. The latter implies that (d_n) is not norm null. On the other hand, it follows from $0 \leqslant d_n \leqslant x_n \xrightarrow{\mathrm{w}} 0$ that $d_n \xrightarrow{\mathrm{w}} 0$. Now (iii) yields $d_n \to 0$; a contradiction. □

11.7.3 Theorem. *X has the PSP iff every seminormalized disjoint sequence has a subsequence equivalent to the unit vector basis of ℓ_1.*

Proof. Suppose that X has the PSP; let (x_n) be a seminormalized disjoint sequence. In particular, there exists $C > 0$ such that $\|x_n\| \leqslant C$ for every n. Being seminormalized, the sequence $\big(|x_n|\big)$ is not norm null and, therefore, not weakly null by Theorem 11.7.2(ii). Therefore, there exists $f \in X^*_+$ with $\|f\| = 1$ and $\varepsilon > 0$ such that, after passing to a subsequence, we have $f\big(|x_n|\big) \geqslant \varepsilon$ for all n. Fix $\alpha_1, \dots, \alpha_m$. On one hand, the triangle inequality yields $\Big\|\sum_{n=1}^m \alpha_n x_n\Big\| \leqslant C \sum_{n=1}^m |\alpha_n|$. On the other hand, using disjointness, we have

$$\Big\|\sum_{n=1}^m \alpha_n x_n\Big\| = \Big\|\sum_{n=1}^m |\alpha_n||x_n|\Big\| \geqslant f\Big(\sum_{n=1}^m |\alpha_n||x_n|\Big) \geqslant \varepsilon \sum_{n=1}^m |\alpha_n|.$$

It follows that (x_n) is equivalent to the unit vector basis of ℓ_1.

To prove the converse, suppose that every seminormalized disjoint sequence has a subsequence equivalent to the unit vector basis of ℓ_1, and let (x_n) be a disjoint positive weakly null sequence. It suffices to show that $x_n \to 0$. Suppose not. WLOG, $\|x_n\| \geqslant 1$ for all n. Since (x_n) is weakly null, it is norm bounded, hence seminormalized. Passing to a subsequence, we may assume that (x_n) is equivalent to the unit vector basis of ℓ_1. This contradicts (x_n) being weakly null (because the the unit vector basis of ℓ_1 is not weakly null). $\square$

11.7.4 Exercise. (i) $L_1(\mu)$ has the PSP for every μ;

(ii) No reflexive infinite-dimensional Banach lattice has the PSP.

The following is a dual version of the Theorem 11.7.3.

11.7.5 Theorem ([FTT09]). *If every seminormalized disjoint sequence in X has a subsequence equivalent to the unit vector basis of c_0 then X^* has PSP.*

Proof. We will use Theorem 11.7.3. Let (f_n) be a seminormalized disjoint sequence in X^*. WLOG, $f_n \geqslant 0$ and $c < \|f_n\| < C$ for every n, where c and C are two constant. For every n, pick $x_n \in X$ with $\|x_n\| = 1$ and $f_n(x_n) > c$. Using Groenewegen's Theorem 11.3.5 and passing to a subsequence, we find a disjoint sequence (v_n) such that $0 \leqslant v_n \leqslant x_n$ and $f_n(v_n) > \frac{c}{2}$ for every n. In particular, $1 \geqslant \|v_n\| \geqslant \frac{c}{2C}$, hence (v_n) is seminormalized. Passing to a further subsequence, we may assume that (v_n) is C'-equivalent to the unit vector basis of c_0 for some constant C'. Fix scalars $\alpha_1, \dots, \alpha_n$. On one hand, $\left\|\sum_{n=1}^m \alpha_n f_n\right\| \leqslant C \sum_{n=1}^m |\alpha_n|$. On the other hand, it follows from $\left\|\sum_{i=1}^m v_i\right\| \leqslant C'$ that

$$\Bigl\|\sum_{n=1}^m \alpha_n f_n\Bigr\| = \Bigl\|\sum_{n=1}^m |\alpha_n| f_n\Bigr\| \geqslant \Bigl(\sum_{n=1}^m |\alpha_n| f_n\Bigr)\Bigl(\frac{1}{C'}\sum_{i=1}^m v_i\Bigr)$$
$$\geqslant \frac{1}{C'}\sum_{n=1}^m |\alpha_n| f_n(v_n) \geqslant \frac{c}{2C'}\sum_{n=1}^m |\alpha_n|.$$

It follows that (f_n) is equivalent to the unit vector basis of ℓ_1. $\square$

Proposition 10.5.2 asserts that every lattice copy of ℓ_1 is complemented. We now use this to prove the following result.

11.7.6 Theorem ([FHSTT14]). *If X has PSP then every disjoint sequence in X has a subsequence whose closed span is complemented.*

Proof. Let (x_n) be a disjoint sequence in X. WLOG, (x_n) is normalized. By Theorem 11.7.3, we may assume that (x_n) is equivalent to the unit vector basis (e_n) of ℓ_1.

Suppose that $x_{n_k}^- \to 0$ for some subsequence (n_k). Passing to a subsequence and using the Principle of Small Perturbations A.4.29, we may assume that (x_n) is equivalent to (x_n^+) and $[x_n]$ is complemented whenever $[x_n^+]$ is complemented. However, $[x_n^+]$ is complemented by Proposition 10.5.2. The argument for the case when $x_{n_k}^+ \to 0$ for some subsequence (n_k) is similar.

Suppose now that neither (x_n^+) nor (x_n^-) has a null subsequence. In particular, there exist $C > 0$ such that $C^{-1} \leqslant \|x_n^\pm\| \leqslant C$ for all n. Passing to subsequences of (x_n) twice, we may assume that both (x_n^+) and (x_n^-) are c-equivalent to the unit vector basis (e_n) of ℓ_1 for some $c > 0$. "Merge" them into a single sequence (z_n) as follows: put $z_{2k-1} = x_k^+$ and $z_{2k} = x_k^-$. We claim that (z_n) is still equivalent to the unit vector basis of ℓ_1. Indeed, consider a linear combination $z := \sum_{n=1}^{2m} \alpha_n z_n$. Then $\|z\| \leqslant C \sum_{n=1}^{2m} |\alpha_n|$. On the other hand, since (z_n) is disjoint, we have $\bigl\|\sum_{n=1}^{m} \alpha_{2n-1} x_n^+\bigr\| \leqslant \|z\|$ and $\bigl\|\sum_{n=1}^{m} \alpha_{2n} x_n^-\bigr\| \leqslant \|z\|$, so that

$$\sum_{n=1}^{2m} |\alpha_n| = \sum_{n=1}^{m} |\alpha_{2n-1}| + \sum_{n=1}^{m} |\alpha_{2n}| \leqslant c\Bigl(\Bigl\|\sum_{n=1}^{m} \alpha_{2n-1} x_n^+\Bigr\| + \Bigl\|\sum_{n=1}^{m} \alpha_{2n} x_n^-\Bigr\|\Bigr) \leqslant 2c\|z\|.$$

This proves the claim.

By Proposition 10.5.2, there exists a projection P on X with Range $P = [z_n]$. Since (z_n) and (x_n) are both equivalent to (e_n), the map

$$R\Bigl(\sum_{n=1}^{\infty} \alpha_n z_n\Bigr) = \sum_{n=1}^{\infty} \frac{\alpha_{2n-1} - \alpha_{2n}}{2} x_n$$

defines a bounded operator $R\colon [z_n] \to [x_n]$. It is easy to check that RP is a projection of X onto $[x_n]$. ☐

Dunford-Pettis Theorem 9.8.6 asserts that a set in $L_1(\mu)$ is relatively weakly compact iff it is almost order bounded. We now show that the spaces in which these two properties agree are exactly the spaces with PSP. Recall that we proved in Theorem 11.5.8 that every almost order bounded set is relatively weakly compact iff X is order continuous.

11.7.7 Theorem. *TFAE:*

(i) *A subset of X is relatively weakly compact iff it is almost order bounded.*
(ii) *X has the PSP.*

Proof. (i)⇒(ii) By Theorem 11.5.8, X is order continuous. Let (x_n) be a positive disjoint weakly null sequence. The set $\{x_n\}$ is relatively weakly compact, hence almost order bounded. Fix $\varepsilon > 0$. Find $u > 0$ such that $\bigl\|(x_n - u)^+\bigr\| < \varepsilon$ for all n. Meyer-Nieberg Theorem 10.1.2 yields $x_n \wedge u \to 0$, hence $\|x_n \wedge u\| < \varepsilon$ for all sufficiently large n. It follows from $x_n = x_n \wedge u + (x_n - u)^+$ that $\|x_n\| < 2\varepsilon$. Thus, $x_n \to 0$, so that X has the PSP by Theorem 11.7.2.

(ii)⇒(i) Suppose X has the PSP. Then it is a KB-space by Exercise 11.7.1. In particular, it is order continuous, so that every almost order bounded set in X is relatively weakly compact by Theorem 11.5.8. Conversely, suppose that A is relatively weakly compact. By Theorem 11.6.7, $\mathrm{Sol}(A)$ is relatively weakly compact. In particular, $\mathrm{Sol}(A)$ is bounded. By Corollary 11.2.4, it suffices to show that A is confined. Let (x_n) be a disjoint sequence in $\mathrm{Sol}(A)$; we want to show that $x_n \to 0$. Suppose not. WLOG, replacing x_n with $|x_n|$, we may assume that $x_n \geqslant 0$. Passing to a subsequence, we can find $\delta > 0$ such that $\|x_n\| > \delta$ for every n. Since $\mathrm{Sol}(A)$ is relatively weakly compact, applying Eberlein-Šmulian's Theorem A.4.20 and passing to a further subsequence, we may assume that $x_n \xrightarrow{\mathrm{w}} x$ for some x. Since (x_n) is disjoint, $x = 0$ by Exercise 6.8.8(ii), hence $x_n \xrightarrow{\mathrm{w}} 0$. The PSP yields $x_n \to 0$; a contradiction. $\square$

11.7.8 Corollary. *X has the PSP iff every relatively weakly compact set is confined.*

Proof. Suppose that X has the PSP. In particular, X is KB. Let $A \subseteq X$ be relatively weakly compact. Then $\mathrm{Sol}(A)$ is relatively weakly compact by Abramovich's Theorem 11.6.7, $\mathrm{Sol}(A)$ is almost order bounded by Theorem 11.7.7 and, therefore, confined by Theorem 11.2.13. It follows that A is confined.

Conversely suppose that every relatively weakly compact set is confined. Let (x_n) be a disjoint weakly null sequence in X. Put $A = \{x_n\}$. It is easy to see that A is relatively weakly compact, hence confined. It follows that $x_n \to 0$. $\square$

Table 11.7 summarizes several results on relationships between relative (weak) compactness, (weak) almost order boundedness, and (weak) confinement for a *solid* set.

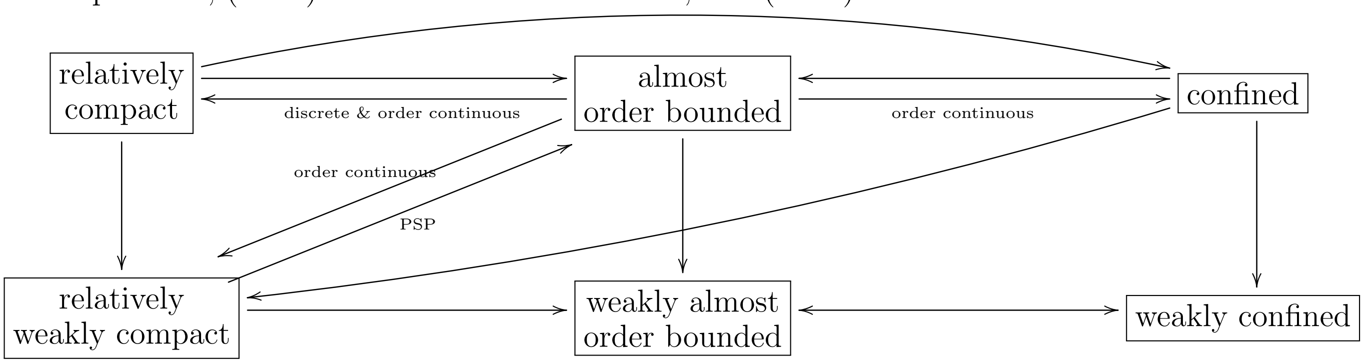

Table 11.1: Relationships between relative (weak) compactness, (weak) almost order boundedness, and (weak) confinement for a solid set.

The following examples show that all the implications in the diagram are sharp. The unit ball of $C[0,1]$ is order bounded, hence almost and weakly almost order bounded; yet it fails to be relatively weakly compact by Theorem A.4.15. Let X be a reflexive Banach lattice with no strong units, then B_X is weakly compact, it is weakly almost order bounded by Theorem 11.4.3 because X^* is reflexive hence order continuous; yet it fails to be almost order bounded by Proposition 11.4.1. The unit ball of ℓ_∞ is order bounded, hence almost order bounded, yet not confined. The unit ball of c_0 is weakly confined but not confined.

11.8 L- and M-weakly compact operators

The following fact has many applications, but its proof follows along the lines of that of Theorem 10.1.1, so we leave it as an exercise.

11.8.1 Exercise. Let $T\colon E \to X$ be a linear operator from a uniformly complete vector lattice E to a Banach space X; let $e \in E_+$ such that $T[0,e]$ is norm bounded. Then $T[0,e]$ is relatively weakly compact iff $Tx_n \to 0$ for every disjoint sequence in $[0,e]$.

Recall that for a bounded operator $T\colon X \to Y$ between Banach spaces, we define a seminorm $r_T(x) = \lVert Tx \rVert$; this seminorm is continuous but need not be a lattice even if X and Y are Banach lattices; see Exercises 11.1.4 and 11.1.5.

Throughout the rest of the section, E will stand for a Banach lattice and X for a Banach space. Exercise 11.8.1 immediately yields the following:

11.8.2 Corollary. *Let $T \in L(E,X)$ and $u \in E_+$. Then $T[0,u]$ is relatively weakly compact iff $[0,u]$ is r_T-confined.*

Let $T \in L(E,X)$. We say that T is ***order weakly compact*** if it maps order intervals to relatively weakly compact sets.

11.8.3 Exercise. Let $T \in L(E,X)$. T is order weakly compact iff $[0,u]$ is r_T-confined for every $u \in X_+$.

The theory of L- and M-weakly compact operators that we present below goes back to Meyer-Nieberg. Given $T \in L(X,E)$, we say that T is ***L-weakly compact*** if TB_X is confined (that is, if every disjoint sequence in $\mathrm{Sol}(TB_X)$ is norm null). Given $T \in L(E,X)$, we say that T is ***M-weakly compact*** if B_E is r_T-confined (that is, for every disjoint bounded sequence (x_n) in E we have $Tx_n \to 0$).

Recall that Theorem 11.2.1 and Exercise 11.2.3 assert that if ρ is a norm continuous seminorm on a Banach lattice then every ρ-confined set is ρ-almost order bounded. This immediately yields the following:

11.8.4 Proposition. (i) *If $T\colon X \to E$ is L-weakly compact then TB_X is almost order bounded;*

(ii) *If $T\colon E \to X$ is M-weakly compact then B_E is r_T-almost order bounded.*

The classes of L- and M-weakly compact operators are in duality to each other:

11.8.5 Theorem. (i) *An operator $T \in L(E,X)$ is M-weakly compact iff T^* is L-weakly compact;*

(ii) *An operator $T \in L(X,E)$ is L-weakly compact iff T^* is M-weakly compact.*

Proof. (i) Let $T \in L(E,X)$. For every $x \in E$, we have

$$r_T(x) = \lVert Tx \rVert = \sup_{f \in X^*} \bigl| f(Tx) \bigr| = \sup_{f \in X^*} \bigl| (T^*f)(Tx) \bigr| = \rho_A(x),$$

where $A = T^*B_{X^*}$. Using Exercise 11.1.13 and Burkinshaw-Dodds Theorem 11.3.6, we get a chain of equivalences:

$$\begin{aligned}
T \text{ is M-weakly compact} \quad &\Leftrightarrow \quad B_E \text{ is } r_T\text{-confined} \\
&\Leftrightarrow \quad B_E \text{ is } \rho_A\text{-confined} \\
&\Leftrightarrow \quad B_E \text{ is } \rho_{\mathrm{Sol}(A)}\text{-confined} \\
&\Leftrightarrow \quad \mathrm{Sol}(A) \text{ is } \rho_{B_E}\text{-confined} \\
&\Leftrightarrow \quad \mathrm{Sol}(T^*B_{X^*}) \text{ is confined} \\
&\Leftrightarrow \quad T^* \text{ is L-weakly compact}
\end{aligned}$$

The proof of (ii) is analogous. □

The following result explains the names of L- and M-weakly compact operators:

11.8.6 Proposition. (i) *An operator $T \in L(X, E)$, where E is an AL-space, is L-weakly compact if it is weakly compact;*

(ii) *An operator $T \in L(E, X)$, where E is an AM-space, is M-weakly compact if it is weakly compact;*

Proof. Suppose that E is an AL-space and $T \in L(X, E)$. By definition, T is L-weakly compact iff $\mathrm{Sol}(TB_X)$ is confined. Since E is an AL-space, by Corollary 9.8.10 this is equivalent to TB_X being relatively weakly compact, hence to T being weakly compact.

The second part is proved by duality. Suppose now that E is an AM-space. Then E^* is an AL-space; T is M-weakly compact iff T^* is L-weakly compact iff T^* is weakly compact (by the first part of the proof) iff T is weakly compact (by Gantmacher Theorem A.4.23). □

11.8.7 Proposition. *L- and M-weakly compact operators are weakly compact.*

Proof. If $T\colon X \to E$ is L-weakly compact then $\mathrm{Sol}(TB_X)$ is confined, hence relatively weakly compact by Theorem 11.5.7.

If $T\colon E \to X$ is M-weakly compact then T^* is L-weakly compact, hence weakly compact. It follows that T is weakly compact. □

11.8.8 Exercise. Let $1 < p < \infty$. Since ℓ_p is reflexive, the identity operator I is weakly compact. However, it is easy to see that it is neither L- nor M-weakly compact.

11.8.9 Exercise. Let $T\colon \ell_1 \to \ell_\infty$ be the rank-one operator given by $Tx = \bigl(\sum_{i=1}^\infty x_i\bigr)\mathbb{1}$. Then T is compact, yet fails to be L- or M-weakly compact.

11.8.10 Proposition. *The classes of L- and M-weakly compact operators are closed subspaces.*

Proof. It is straightforward that M-weakly compact operators form a linear subspace of $L(E,X)$. To show that it is closed, we will prove that every operator S in the closure of the set of M-weakly compact operators is again M-weakly compact. Let (x_n) be a disjoint sequence in B_E. Fix $\varepsilon > 0$. Find an M-weakly compact operator T with $\|S-T\| < \varepsilon$. Then $Tx_n \to 0$, hence we find n_0 such that $\|Tx_n\| < \varepsilon$ for all $n \geqslant n_0$; it follows that

$$\|Sx_n\| \leqslant \|(S-T)x_n\| + \|Tx_n\| < 2\varepsilon.$$

It follows that M-weakly compact operators in $L(E^*, X^*)$ is a closed subspace. Since the map $T \mapsto T^*$ from $L(X,E)$ to $L(E^*, X^*)$ is a linear isometry, it follows from Theorem 11.8.5 that L-weakly compact operators form a closed subspace of $L(X,E)$. □

It is easy to see from the definition that if we multiply an M-weakly compact operator by a continuous operator on the left, we again get an M-weakly compact operator. Similarly, the class of L-weakly compact operators is closed under right multiplication.

11.8.11 Example. *L- and M-weakly compact operators do not form ideals.* Let $1 \leqslant q < p < \infty$; let $T\colon L_p[0,1] \to L_q[0,1]$ be the canonical inclusion. We claim that T is M-weakly compact. Let (f_n) be a disjoint sequence in $L_p[0,1]$ such that $\|f_n\|_p \leqslant 1$ for every n. It is easy to see that $\|f_n\|_q \to 0$. Indeed, let $A_n = \operatorname{supp} f_n$ and $r > 0$ such that $\frac{1}{q} = \frac{1}{p} + \frac{1}{r}$; it follows from Hölder's inequality that $\|f_n\|_q \leqslant \|f_n\|_p \|\mathbb{1}_{A_n}\|_r \leqslant \mu(A_n)^{\frac{1}{r}} \to 0$. This proves that T is M-weakly compact

Let $S\colon \ell_1 \to L_p[0,1]$ be defined by $Se_n = r_n$, the n-th Rademacher function. It is bounded: for $\alpha_1, \dots, \alpha_n$ we have

$$\Bigl\|S\Bigl(\sum_{i=1}^n \alpha_i e_i\Bigr)\Bigr\|_p = \Bigl\|\sum_{i=1}^n \alpha_i r_i\Bigr\|_p \leqslant \sum_{i=1}^n |\alpha_i| = \Bigl\|\sum_{i=1}^n \alpha_i e_i\Bigr\|_{\ell_1}.$$

Moreover, S is regular: it is easy to see that $\pm S \leqslant R$ where $R\colon \ell_1 \to L_p[0,1]$ is the rank-one positive operator defined by $R\bigl(\sum_{i=1}^\infty \alpha_i e_i\bigr) = \bigl(\sum_{i=1}^\infty \alpha_i\bigr)\mathbb{1}$.

Finally, observe that $\|TSe_n\|_q = \|r_n\|_q = 1$, so that TS is not M-weakly compact.

Dualizing this construction, we see that T^* is L-weakly compact, S^* is bounded and regular, yet S^*T^* is not L-weakly compact.

Finally, we consider the case when both E and X are Banach lattices.

11.8.12 Proposition. *If X is an order continuous Banach lattice and $T \in \mathcal{L}_{\mathrm{ob}}(E,X)$ is M-weakly compact then T is L-weakly compact.*

Proof. We claim that TB_E is almost order bounded. Indeed, by Proposition 11.8.4, B_E is r_T-almost order bounded. Fix $\varepsilon > 0$. By Exercise 11.1.8, we find $u \in E_+$ such that $B_E \subseteq [-u,u] + \varepsilon B_{r_T}$. Let $x \in B_E$, then we can write $x = y + z$ where $|y| \leqslant u$ and $\|Tz\| < \varepsilon$. Since X is order continuous, $|T|$ exists and $|Ty| \leqslant |T|u$. Therefore,

$$Tx = Ty + Tz \in \bigl[-|T|u, |T|u\bigr] + \varepsilon B_X.$$

This proves that TB_E is almost order bounded, hence so is $\mathrm{Sol}(TB_E)$. Proposition 11.2.6 now guarantees that $\mathrm{Sol}(TB_E)$ is confined, so that T is L-weakly compact. □

Combining this proposition with Theorem 11.8.5, we immediately get the following:

11.8.13 Theorem (Dodds-Fremlin). *Let $T \in \mathcal{L}_{\mathrm{ob}}(E,X)$ where X and E^* are order continuous. Then TFAE:*

(i) *T is L-weakly compact;*
(ii) *T is M-weakly compact;*
(iii) *T^* is L-weakly compact;*
(iv) *T^* is M-weakly compact;*

11.8.14 Theorem (Dodds-Fremlin). *Let $T \in \mathcal{L}_{\mathrm{ob}}(E,X)$ where X and E^* are order continuous. T is M-weakly compact iff $\langle f_n, Tx_n\rangle \to 0$ for any disjoint norm bounded sequences (x_n) in E and (f_n) in X^*.*

Proof. Suppose that T is M-weakly compact; let (x_n) and (f_n) be disjoint bounded sequences in E and X^*, respectively. It follows that $\bigl|\langle f_n, Tx_n\rangle\bigr| \leqslant \|f_n\|\|Tx_n\| \to 0$.

Suppose now that $\langle f_n, Tx_n\rangle \to 0$ for any disjoint bounded sequences (x_n) in E and (f_n) in X^*, but T fails to me M-weakly compact. We can then find a disjoint bounded sequence (x_n) such that $\|Tx_n\| > 2$ for every n. Since E^* is order continuous, $|x_n| \xrightarrow{\mathrm{w}} 0$ by Corollary 11.4.4. Since X is order continuous, $|T|$ exists. Being continuous, it is weak-to-weak continuous, so $|T||x_n| \xrightarrow{\mathrm{w}} 0$. It then follows from $|Tx_n| \leqslant |T||x_n|$ that $|Tx_n| \xrightarrow{\mathrm{w}} 0$. Since X is order continuous, Corollary 9.6.11 asserts that every positive weakly null sequence in it has an asymptotically disjoint subsequence. Hence, passing to a subsequence, we may assume that there exists a disjoint sequence (d_n) such that $0 \leqslant d_n \leqslant |Tx_n|$ and $|Tx_n| - d_n \to 0$. Passing to a tail, we may assume that $\|d_n\| > 1$ for every n.

For every n, find $h_n \in B^+_{X^*}$ such that $h_n(d_n) > 1$. As in 11.3.1, we can find g_n such that $0 \leqslant g_n \leqslant h_n$ and $g_n(d_n) > 1$ for every n so that the sequence (g_n) is disjoint. It follows that $g_n\big(|Tx_n|\big) \geqslant g(d_n) > 1$ for every n. Using Riesz-Kantorovich formula, we find $f_n \in X^*$ such that $|f_n| \leqslant g_n$ and $f_n(Tx_n) > 1$ for every n. Note that the sequence (f_n) is still disjoint and bounded. So by assumption, we have $f_n(Tx_n) \to 0$; a contradiction. □

11.8.15 Remark. By splitting into positive and negative parts, the two sequences in the preceding theorem may be assumed to be positive.

Let X be a Banach space. It is easy to see that every weak*-null sequence (f_n) in X^* determines a bounded linear operator $T\colon X \to c_0$ via $Tx = \big(f_n(x)\big)_{n=1}^\infty$. It is a known fact that $f_n \xrightarrow{\text{w}} 0$ iff T is weakly compact; see, e.g., Theorem 5.26 in [AB06]. It is also easy to see that if $f_n \to 0$ then T is compact, because we can then approximate T in the operator norm by operators of finite rank.

11.8.16 Theorem ([Wnuk92])**.** *For a Banach lattice E, TFAE:*

(i) *E^* has PSP;*

(ii) *Every positive weakly compact operator from E to c_0 is M-weakly compact.*

Proof. Suppose that E^* has PSP; let $T\colon E \to c_0$ be a positive weakly compact operator. Let (f_n) be a weakly null positive sequence in E^* that defines T (as above). Since E^* has PSP, we have $f_n \to 0$. It follows that T is compact.

Since E^* has PSP, it is order continuous by Exercise 11.7.1. Hence, if (x_n) is a disjoint sequence in B_E then $x_n \xrightarrow{\text{w}} 0$ by Proposition 11.4.3, and then the compactness of T yields $Tx_n \to 0$. We conclude that T is M-weakly compact.

We will now prove the converse. Let (f_n) be a disjoint positive weakly null sequence in E^*. The corresponding operator $T\colon E \to c_0$ is positive and weakly compact, hence M-weakly compact. By Theorem 11.8.5(i), T^* is L-weakly compact. Note that $f_n = T^*e^*_n$ for every n, where (e^*_n) is the standard unit basis of ℓ_1. Thus, (f_n) is a disjoint sequence in a confined set $T^*B_{E^*}$, so that $f_n \to 0$. □

Chapter 12

Ordered Banach spaces

This chapter is incomplete.

12.1 Regular elements

Recall that an ***ordered normed space*** is a normed space endowed with a linear order, such that the positive cone X_+ is closed. If, in addition, the norm is complete, we call it an ***ordered Banach space***. The norm on an ordered normed space is said to be ***monotone*** if $0 \leqslant x \leqslant y$ implies $\|x\| \leqslant \|y\|$. This property can be characterized geometrically: for every $x \in S_X \cap X_+$ the set $x + X_+$ does not meet the interior of B_X.

12.1.1 Example. Let X and Y be two Banach lattices. The space of bounded operators $L(X,Y)$ is an ordered Banach space with a monotone norm.

12.1.2 Example. Let X be a Banach lattice and (E_n) is a filtration on X of positive contractive projections. Then the space of bounded martingales $M\big(X,(E_n)\big)$ is an ordered Banach space with a monotone norm.

12.1.3 Exercise. Every ordered normed space satisfies the Archimedean property.

Suppose that X is an ordered normed space. Let $X_r = X_+ - X_+$, the set of all ***regular*** elements of X (cf. Exercise 1.2.5). It is easy to see that X_r is a linear subspace of X. For $z \in X_r$ we define

$$\|z\|_r = \inf\big\{\|x\| \,:\, x \in X_+ \text{ and } -x \leqslant z \leqslant x\big\}.$$

Clearly, if $z \vee (-z) =: |z|$ exists (in particular, when X is a vector lattice) then $\|z\|_r = \big\||z|\big\|$.

12.1.4 Theorem. *Suppose that X is an ordered normed space with a monotone norm. Then $\|\cdot\|_r$ is a norm on X_r. For every $z \in X_r$, we have $\|z\| \leqslant 2\|z\|_r$. Moreover, if X is complete then $\big(X_r, \|\cdot\|_r\big)$ is complete.*

Proof. It is easy to see that $\|\cdot\|_r$ is positively homogeneous. To verify the triangle inequality, take $u, v \in X_r$, take any $\varepsilon > 0$, and find $x, y \in X_+$ such that $-x \leqslant u \leqslant x$ and $-y \leqslant v \leqslant y$, $\|x\| \leqslant \|u\|_r + \varepsilon$, and $\|y\| \leqslant \|v\|_r + \varepsilon$. It follows that $-(x+y) \leqslant u+v \leqslant x+y$, so that

$$\|u+v\|_r \leqslant \|x+y\| \leqslant \|x\| + \|y\| \leqslant \|u\|_r + \|v\|_r + 2\varepsilon.$$

It follows that $\|u+v\|_r \leqslant \|u\|_r + \|v\|_r$.

Fix $z \in X_r$. For every $x \in X_+$ satisfying $-x \leqslant z \leqslant x$, we have $0 \leqslant x \pm z \leqslant 2x$. Monotonicity of $\|\cdot\|$ implies $\|z \pm x\| \leqslant 2\|x\|$. It follows from $z = \frac{x+z}{2} - \frac{x-z}{2}$ that

$$\|z\| \leqslant \frac{\|x+z\|}{2} + \frac{\|x-z\|}{2} \leqslant 2\|x\|.$$

Taking the infimum over all such x, we get $\|z\| \leqslant 2\|z\|_r$. In particular, if $\|z\|_r = 0$ then $\|z\| = 0$ and, therefore, $z = 0$.

Now suppose that $\big(X, \|\cdot\|\big)$ is complete and show that $\big(X_r, \|\cdot\|_r\big)$ is complete. Let (z_n) be a $\|\cdot\|_r$-Cauchy sequence in X_r. Note that since $\|\cdot\| \leqslant 2\|\cdot\|_r$, the sequence is also Cauchy in the original norm, hence $z_n \xrightarrow{\|\cdot\|} z$ for some $z \in X$. It suffices to show that $z \in X_r$ and some subsequence of (z_n) converges to z in $\|\cdot\|_r$ because, in this case, the entire sequence would still converge to z in $\|\cdot\|_r$.

WLOG, passing to a subsequence, we may assume that for every n and every $k \geqslant n$ we have $\|z_n - z_k\|_r \leqslant \frac{1}{3^n}$. For each n, $z_{n+1} - z_n$ is regular, so we can find $x_n \in X_+$ such that $-x_n \leqslant z_{n+1} - z_n \leqslant x_n$ and $\|x_n\| \leqslant \frac{1}{2^n}$. Fix m. It follows from $z_n \xrightarrow{\|\cdot\|} z$ that $z - z_m = \sum_{n=m}^{\infty}(z_{n+1} - z_n)$, where the series converges in $\|\cdot\|$. Note that

$$\sum_{n=m}^{k}(z_{n+1} - z_n) \leqslant \sum_{n=m}^{k} x_n$$

for every $k > m$ and $\sum_{n=m}^{\infty} x_n$ converges in $\|\cdot\|$. Since X_+ is closed, $z - z_m \leqslant \sum_{n=m}^{\infty} x_n$. Similarly, $-(z - z_m) \leqslant \sum_{n=m}^{\infty} x_n$. It follows that

$$\|z - z_m\|_r \leqslant \Big\|\sum_{n=m}^{\infty} x_n\Big\| \leqslant \sum_{n=m}^{\infty}\|x_n\| \leqslant \tfrac{1}{2^{m-1}} \to 0.$$

□

12.1.5 Example. The following example shows that the estimate $\|z\| \leqslant 2\|z\|_r$ is sharp. Let $X = \mathbb{R}^2$ with X_+ being the closed first quadrant. Let B_X be the absolute convex hull of $(0,1)$, $(1,0)$, and $(1,1)$. It is easy to see that X_+ is closed and the norm is monotone. Let $z = (-1,1)$ and $x = (1,a)$. Clearly, $\|z\| = 2$. Note that $-x \leqslant z \leqslant x$, so that $\|z\|_r \leqslant \|x\| = 1$. It is easy to see that, actually, $\|z\|_r = 1$.

12.2 Duality of cones

To be completed.

If X is a Banach lattice and $x \in X$ then $x = x^+ - x^-$ and $\|x^\pm\| \leqslant \|x\|$. The following result can be viewed as an extension of this fact to ordered Banach spaces.

12.2.1 Theorem (Krein-Šmulian)**.** *Let X be an ordered Banach space such that X_+ is generating. There exists a constant $M > 0$ such that for every $x \in X$ there exist $a, b \geqslant 0$ such that $x = a - b$ and $\|a\|, \|b\| \leqslant M\|x\|$.*

Proof. For every $t \in \mathbb{R}_+$, define

$$E_t = \big\{x \in X \;:\; x = a - b \text{ for some positive } a \text{ and } b \text{ such that } \|a\|, \|b\| \leqslant t\big\}.$$

It can be easily verified that E_t is symmetric and convex and that $E_s \subseteq E_t$ whenever $s \leqslant t$. Since X_+ is generating, every vector in X is contained in some E_t, so that $X = \bigcup_{t\in\mathbb{R}_+} E_t = \bigcup_{n\in\mathbb{N}} E_n$. By Baire Category Theorem, there exists n such that $\overline{E_n}$ has a non-empty interior, hence it contains a ball. Since $\overline{E_n}$ is symmetric and convex, it contains a ball centred at the origin, say rB_X. It follows that $B_X \subseteq \overline{E_\lambda}$, where $\lambda = \frac{n}{r}$.

Fix $x \in S_X$. There exists $x_1 \in E_\lambda$ such that $\|x - x_1\| < \frac{1}{2}$. Since $2(x - x_1) \in B_X$, we can find $x_2 \in E_\lambda$ such that $\big\|2(x - x_1) - x_2\big\| < \frac{1}{2}$, so that $\big\|x - x_1 - \frac{x_2}{2}\big\| < \frac{1}{4}$. Proceeding inductively, we construct a sequence (x_k) in E_λ such that

$$\Big\|x - x_1 - \frac{x_2}{2} - \dots - \frac{x_k}{2^{k-1}}\Big\| < \frac{1}{2^k}.$$

It follows that $x = \sum_{k=1}^\infty \frac{x_k}{2^{k-1}}$.

Since $x_k \in E_\lambda$ for every k, we can find positive a_k and b_k such that $x_k = a_k - b_k$ and $\|a_k\|, \|b_k\| \leqslant \lambda$. It follows that the series $a \coloneqq \sum_{k=1}^\infty \frac{a_k}{2^{k-1}}$ and $b \coloneqq \sum_{k=1}^\infty \frac{b_k}{2^{k-1}}$ converge and $\|a\|, \|b\| \leqslant 2\lambda$. Since the cone is closed, a and b are positive. Finally,

$$x = \sum_{k=1}^\infty \frac{x_k}{2^{k-1}} = \sum_{k=1}^\infty \frac{a_k}{2^{k-1}} - \sum_{k=1}^\infty \frac{b_k}{2^{k-1}} = a - b.$$

$\square$

12.2.2 Corollary. *Let X be an ordered Banach space such that X_+ is generating. There exists a constant $M > 0$ such that for every $x \in X$ there exist $a \in X_+$ such that $x \in [-a, a]$ and $\|a\| \leqslant M\|x\|$.*

The following theorem appeared in [dJM14]. We show that one can prove it using the same method as Theorem 12.2.1.

12.2.3 Theorem. *Let $T\colon C \to X$ be an additive continuous map, where X is a Banach space and C is a closed cone (wedge) in a Banach space Y. TFAE*

(i) *T is surjective;*

(ii) *there exists $K > 0$ such that for every $x \in X$ there exists $c \in C$ such that $x = Tc$ and $\|c\| \leqslant K\|x\|$;*

(iii) *T is an open (i.e., maps open sets to open sets).*

Proof. Since T is additive and continuous, one can show that T is positively homogeneous.

(i)⇒(ii) For every $n \in \mathbb{N}$, define

$$E_n = \bigl\{x \in X \;:\; \exists a, b \in C \; \|a\|, \|b\| \leqslant n,\; Ta = x,\; \text{and } Tb = -x\bigr\}.$$

It is easy to see that E_n is convex and symmetric for every n and $\bigcup_{n=1}^{\infty} E_n = X$. By Baire Category Theorem, there exists k such that $\overline{E_k}$ contains a non-empty interior. Since E_k is convex and symmetric, it contains a ball centred at the origin. WLOG, (enlarging k, if necessary,) we assume that $B_X \subseteq \overline{E_k}$.

Fix $x \in S_X$. Find $x_1 \in E_k$ such that $\|x - x_1\| \leqslant \frac{1}{2}$. Since $2(x - x_1) \in B_X$, find $x_2 \in E_k$ such that $\bigl\|2(x - x_1) - x_2\bigr\| \leqslant \frac{1}{2}$, that is, $\bigl\|x - x_1 - \frac{x_2}{2}\bigr\| \leqslant \frac{1}{4}$. Proceeding inductively, find a sequence (x_n) in E_k such that

$$\Bigl\|x - x_1 - \frac{x_2}{2} - \dots - \frac{x_n}{2^{n-1}}\Bigr\| \leqslant \frac{1}{2^n}.$$

It follows that $x = \sum_{n=1}^{\infty} \frac{x_n}{2^{n-1}}$.

For every n, $x_n \in E_k$ implies that we can find some $a_n \in C$ such that $Ta_n = x_n$ and $\|a_n\| \leqslant k$. Let $a = \sum_{n=1}^{\infty} \frac{a_n}{2^{n-1}}$. Clearly, $\|a\| \leqslant 2k$. Since C is closed, $a \in C$. Since T is additive and continuous, we have $Ta = x$.

(ii)⇒(iii) Let $c \in C$; let V be an open neighborhood of c in V. It suffices to show that $T(V)$ is a neighborhood of $T(c)$. Find $\lambda > 0$ such that $V \supseteq (c + \lambda B_Y) \cap C$. By (ii), $T(C \cap KB_Y) \supseteq B_X$, so that $T(C \cap \lambda B_Y) \supseteq \frac{\lambda}{K} B_X$. It follows from $c + \lambda B_Y \supseteq c + C \cap \lambda B_y$ that $T(V) \supseteq T(c + C \cap \lambda B_y) \supseteq T(c) + \frac{\lambda}{K} B_X$.

(iii)⇒(i) Clearly, $0 = T(0) \in \operatorname{Range} T$. By (iii), 0 is an interior point of $\operatorname{Range} T$. It now follows from positive homogenuity that T is onto. □

Let X be an ordered Banach space. We say that the norm of X is ***monotone*** if $0 \leqslant x \leqslant y$ implies $\|x\| \leqslant \|y\|$. Clearly, the norm of every normed lattice is monotone. One should realize that this is an isometric property, — it may be lost under an equivalent norm. This motivates the following weaker notion: the positive cone X_+ in an ordered Banach spaces is ***normal*** if there is a positive constant K such that

$0 \leqslant x \leqslant y$ implies $\|x\| \leqslant K\|y\|$. It is easy to see that this property in invariant under an equivalent renorming.

A subset A in an ordered Banach space X is ***full*** if $x, y \in A$ imply $[x, y] \subseteq A$. The following theorem suggests that normality is an isomorphic version of monotonicity.

12.2.4 Theorem. *For an ordered Banach space X, TFAE:*

(i) *X_+ is normal;*
(ii) *X admits an equivalent monotone norm;*
(iii) *B_X is full.*

The following two theorems show that the concepts of normal and generating cones are dual to each other.

12.2.5 Theorem (Andô)**.** *If the cone of an ordered Banach space is generating then the dual cone is normal.*

12.2.6 Theorem. *If a cone of an ordered Banach space is normal then its dual cone (wedge) is generating.*

Chapter 13

Regular operators

This chapter is incomplete!

13.1 Order bounded operators

An operator T between ordered vector spaces is said to be ***order bounded*** if for every $a \in X_+$ there exists $b \in Y_+$ such that $T[-a,a] \subseteq [-b,b]$. Note that every regular (and, in particular, every positive) operator is order bounded.

13.1.1 Exercise. An operator between ordered vector spaces is order bounded iff it takes order intervals into order intervals.

13.1.2 Theorem. *Let $T\colon X \to Y$ be an order bounded operator from an ordered Banach space X to an ordered normed space Y. There exists a constant $C > 0$ such that for every $a \in X_+$ there exists $b \in Y_+$ with $T[-a,a] \subseteq [-b,b]$ and $\lVert b\rVert \leqslant C\lVert a\rVert$.*

Proof. Suppose there is no such constant. Then for every $n \in \mathbb{N}$, we can find $a_n \in X_+$ such that if $b \in Y_+$ and $T[-a_n,a_n] \subseteq [-b,b]$ then $\lVert b\rVert \geqslant n^3\lVert a_n\rVert$. WLOG, scaling a_n, we may assume that $\lVert a_n\rVert = \frac{1}{n^2}$. It follows that the series $a \coloneqq \sum_{n=1}^{\infty} a_n$ converges. Since T is order bounded, there exists $b \in Y_+$ such that $T[-a,a] \subseteq [-b,b]$. It follows that $T[-a_n,a_n] \subseteq [-b,b]$ for every n, so that $\lVert b\rVert \geqslant n^3\lVert a_n\rVert = n$; a contradiction. $\square$

We write $\lVert T\rVert_{\mathrm{ob}}$ for the infimum of the constants C in the theorem.

Recall that we proved in Theorem 6.1.9 that every order bounded operator between Banach lattices is bounded. We will now prove several extensions of this result.

13.1.3 Corollary. *Let $T\colon X \to Y$ be an operator from a Banach lattice X to a normed lattice Y. If T is order bounded then T is bounded and $\lVert T\rVert \leqslant \lVert T\rVert_{\mathrm{ob}}$.*

Proof. Let $x \in X$. Take any $C > \|T\|_{\mathrm{ob}}$. There exists $b \in Y_+$ such that $T\big[-|x|, |x|\big] \subseteq [-b, b]$ and $\|b\| \leqslant C\big\||x|\big\| = C\|x\|$. It follows that $Tx \in [-b, b]$, so that $\|Tx\| \leqslant \|b\| \leqslant C\|x\|$. Therefore, $\|Tx\| \leqslant \|T\|_{\mathrm{ob}}\|x\|$. ☐

13.1.4 Theorem. *Let $T\colon X \to Y$ be an order bounded operator between two ordered Banach spaces with X_+ generating. Then T is bounded.*

Proof. We will use Closed Graph Theorem. Let $x_n \to 0$ in X and $Tx_n \to y$ in Y; it suffices to show that $y = 0$. By Corollary 12.2.2, for every n there exist positive a_n such that $-a_n \leqslant x_n \leqslant a_n$ and $\|a_n\| \leqslant M\|x_n\|$. WLOG, passing to a subsequence, we may assume that $\sum_{n=1}^{\infty} n\|a_n\|$ converges. It follows that the series $a = \sum_{n=1}^{\infty} na_n$ converges in X and $a \geqslant 0$. Since T is order bounded, there exists $b \in Y_+$ such that $T[-a, a] \subseteq [-b, b]$. Fix $k \in \mathbb{N}$. For every $n \geqslant k$ we have $-a \leqslant kx_n \leqslant a$, so that $-b \leqslant kTx_n \leqslant b$. It follows that $-b \leqslant ky \leqslant b$. It follows from the Archimedean Property 12.1.3 that $y = 0$. ☐

13.1.5 Theorem. *Let $T\colon X \to Y$ be an order bounded operator from an ordered Banach space X with a generating cone to an ordered normed space Y with a normal cone. Then T is norm bounded. Moreover, $\|T\| \leqslant K\|T\|_{\mathrm{ob}}$, where K only depends on X and Y.*

Proof. Let M be as in Theorem 12.2.1, C be as in Theorem 13.1.2. Fix $x \in X$ with $\|x\| \leqslant 1$. By Theorem 12.2.1, we find $x_1, x_2 \geqslant 0$ such that $x = x_1 - x_2$ and $\|x_1\|, \|x_2\| \leqslant M$. Put $a = x_1 + x_2$, then $x \in [-a, a]$ and $\|a\| \leqslant 2M$. By Theorem 13.1.2, $T[-a, a] \subseteq [-b, b]$ for some $b \in Y$ with $\|b\| \leqslant C\|b\| \leqslant 2CM$. It follows that $\|Tx\| \leqslant k\|b\| \leqslant 2CMk$, where k is the normality constant of Y_+. ☐

13.1.6 Theorem. *Let X and Y be two ordered Banach spaces such that X_+ is generating and Y_+ is normal. Then $\mathcal{L}_{\mathrm{ob}}(X, Y)$ endowed with $\|\cdot\|_{\mathrm{ob}}$ is an ordered Banach space.*

Proof. It is easy to see that $\|\cdot\|_{\mathrm{ob}}$ is a norm on $\mathcal{L}_{\mathrm{ob}}(X, Y)$. It is left to check that it is complete. Suppose that (T_n) is a $\|\cdot\|_{\mathrm{ob}}$-Cauchy sequence. By Theorem 13.1.5, T_n's are bounded and the sequence is Cauchy in the operator norm. Since $L(X, Y)$ is complete, $\|T_n - T\| \to 0$ for some $T \in L(X, Y)$. We will show that $\|T_n - T\|_{\mathrm{ob}} \to 0$.

Since (T_n) is a $\|\cdot\|_{\mathrm{ob}}$-Cauchy, we may assume (after passing to a subsequence) that $\|T_{n+1} - T_n\|_{\mathrm{ob}} < \frac{1}{2^n}$ for every n. Fix $\varepsilon > 0$. There exists m such that $\sum_{n=m}^{\infty}\|T_{n+1} - T_n\|_{\mathrm{ob}} < \varepsilon$. Fix $a \in B_X \cap X_+$. For every $n \geqslant m$ there exists $c_n \in Y_+$ such that $(T_{n+1} - T_n)[-a, a] \subseteq [-c_n, c_n]$ and $\|c_n\| < 2\|T_{n+1} - T_n\|_{\mathrm{ob}}$. It follows that $c \coloneqq \sum_{n=m}^{\infty} c_n$

converges and $\|c\| \leqslant \sum_{n=m}^{\infty}\|c_n\| < 2\varepsilon$. Fix $x \in [-a,a]$. For every $n \geqslant m$, we have $-c_n \leqslant (T_{n+1} - T_n)x \leqslant c_n$. It follows that

$$-\sum_{n=m}^{k-1} c_n \leqslant \sum_{n=m}^{k-1}(T_{n+1} - T_n)x \leqslant \sum_{n=m}^{k-1} c_n$$

for every $k \geqslant m$. The sum in the middle equals $T_k x - T_m x$. This yields $-c \leqslant T_k x - T_m x \leqslant c$. Taking the limit as $k \to \infty$, we have $-c \leqslant Tx - T_m x \leqslant c$. It follows that $(T - T_m)[-a,a] \subseteq [-c,c]$. Since a was arbitrary and $\|c\| < 2\varepsilon$, it follows that $\|T - T_m\|_{\mathrm{ob}} < 2\varepsilon$.

Finally, observe that the cone of positive operators in $\mathcal{L}_{\mathrm{ob}}(X,Y)$ is closed in $\|\cdot\|_{\mathrm{ob}}$. Indeed, if $\|T_n - T\|_{\mathrm{ob}} \to 0$ in $\mathcal{L}_{\mathrm{ob}}(X,Y)$ and $T_n \geqslant 0$ for every n then $\|T_n - T\| \to 0$, hence $T \geqslant 0$. □

13.1.7 Corollary. *Let X and Y be two Banach lattices. Then $\mathcal{L}_{\mathrm{ob}}(X,Y)$ endowed with $\|\cdot\|_{\mathrm{ob}}$ is an ordered Banach space.*

13.2 Regular operators between Banach lattices

To be completed.

Let X and Y be Banach lattices. Recall that an operator from X to Y is ***regular*** if it is the difference of two positive operators. That is, the set $L_r(X,Y)$ of all regular operators from X to Y is exactly the set of all regular elements in $L(X,Y)$. As in Section 12.1, the regular norm of a regular operator T is defined via

$$\|T\|_r = \inf\bigl\{\|R\| \,:\, 0 \leqslant R\colon X \to Y \text{ and } -R \leqslant T \leqslant R\bigr\},$$

and if $|T|$ exists then $\|T\|_r = \bigl\||T|\bigr\|$.

13.2.1 Exercise. $\|T\| \leqslant \|T\|_r$ for every $T \in L_r(X,Y)$.

Theorem 12.1.4 combined with Riesz-Kantorovich Theorem 6.3.2 yield the following:

13.2.2 Theorem. *$L_r(X,Y)$ is a Banach space with respect to $\|\cdot\|_r$. Moreover, if Y is order complete then $L_r(X,Y)$ is an order complete Banach lattice*

13.2.3 Exercise. Let $T\colon X \to Y$ be a regular operator between Banach lattices and $Z \subseteq X$ a closed sublattice. Then the restriction $T_{|Z}$ is regular and $\bigl\|T_{|Z}\bigr\|_r \leqslant \|T\|_r$.

Let $T\colon X \to Y$ be an operator between Banach lattices such that $|T|$ exists. Clearly, T is regular. It follows that the adjoint operator $T^*\colon Y^* \to X^*$ is regular. Since X^* is order complete, T^* has a modulus by the Riesz-Kantorovich Theorem. We are going to discuss the relationship between $|T^*|$ and $|T|^*$.

13.2.4 Proposition. *If $|T|$ exists then $|T^*| \leqslant |T|^*$.*

Proof. Fix $g \in Y_+^*$. For every $x \in X^+$, we have

$$\langle \pm T^* g, x\rangle = \pm\langle T^* g, x\rangle \leqslant \bigl|\langle T^* g, x\rangle\bigr| = \bigl|\langle g, Tx\rangle\bigr| \leqslant \bigl\langle g, |T|x\bigr\rangle = \bigl\langle |T|^* g, x\bigr\rangle.$$

It follows that $\pm T^* g \leqslant |T|^* g$, hence $\pm T^* \leqslant |T|^*$ and, therefore, $|T^*| \leqslant |T|^*$. $\square$

13.2.5 Proposition. *If $|T|$ exists and $|T^*|$ is an adjoint operator then $|T^*| = |T|^*$.*

Proof. We already know that $|T^*| \leqslant |T|^*$. Suppose that $|T^*| = S^*$ for some $S\colon X \to Y$; we will show that $S = |T|$. It follows from $\pm T^* \leqslant S^*$ that $\pm T^{**} \leqslant S^{**}$, and, therefore, $\pm T \leqslant S$, hence $|T| \leqslant S$. On the other hand, fix any $x \in X_+$; for every $g \in Y_+^*$, we have

$$\langle Sx, g\rangle = \langle x, S^* g\rangle = \bigl\langle x, |T^*| g\bigr\rangle \leqslant \bigl\langle x, |T|^* g\bigr\rangle = \bigl\langle |T|x, g\bigr\rangle.$$

It follows that $S \leqslant |T|$. $\square$

13.2.6 Example. *An operator T such that $|T^*| \neq |T|^*$.* As usual, let c be the space of all convergent sequences. Define $T\colon c \to c$ via

$$T(x_1, x_2, x_3, \dots) = (x_1 - x_2, x_2 - x_3, x_3 - x_4, \dots).$$

We claim that $|T|$ exists and $|T| = S$ where

$$S(x_1, x_2, x_3, \dots) = (x_1 + x_2, x_2 + x_3, x_3 + x_4, \dots).$$

Indeed, we have $\pm Tx \leqslant Sx$ for every $x \in c_+$, so that $\pm T \leqslant S$. On the other hand, suppose that $\pm T \leqslant R$ for some $R\colon c \to c$. It suffices to show that $R \geqslant S$. Note that $\pm Tx \leqslant Rx$ and, therefore, $|Tx| \leqslant Rx$ for every $x \in c_+$. It follows, in particular, that $R \geqslant 0$. Fix $x = (x_1, x_2, x_3, \dots)$ in c_+. Then

$$Rx \geqslant R(x_1, 0, \dots) \geqslant T(x_1, 0, \dots) = (x_1, 0, \dots).$$

For every $n > 1$, we have

$$\begin{aligned} Rx \geqslant R(0, \dots, 0, \overbrace{x_n}^{n}, 0, \dots) &\geqslant \bigl|T(0, \dots, 0, \overbrace{x_n}^{n}, 0, \dots)\bigr| \\ &= \bigl|(0, \dots, 0, \overbrace{-x_n}^{n-1}, \overbrace{x_n}^{n}, \dots)\bigr| = (0, \dots, 0, \overbrace{x_n}^{n-1}, \overbrace{x_n}^{n}, \dots). \end{aligned}$$

It follows that for every $n \geqslant 1$ we have

$$\begin{aligned} Rx &\geqslant R(0,\dots,0,\overbrace{x_n}^{n},0,\dots) + R(0,\dots,0,\overbrace{x_{n+1}}^{n+1},0,\dots) \\ &\geqslant (0,\dots,0,\overbrace{x_n}^{n-1},\overbrace{x_n+x_{n+1}}^{n},\overbrace{x_{n+1}}^{n+1},0,\dots) \geqslant (0,\dots,0,\overbrace{x_n+x_{n+1}}^{n},0,\dots) =: y^{(n)}, \end{aligned}$$

It is easy to see that the sequence $(y^{(n)})$ has supremum in c, which is exactly Sx. It follows that $Rx \geqslant Sx$ and, therefore, $R \geqslant S$. This proves the claim that $|T| = S$.

Let $\varphi \in c_+^*$ be defined by $\varphi(x) = \lim_n x_n$; let $\mathbb{1} = (1,1,\dots) \in c$. Observe that

$$\bigl\langle |T|^*\varphi, \mathbb{1}\bigr\rangle = \bigl\langle \varphi, |T|\mathbb{1}\bigr\rangle = \varphi(2,2,\dots) = 2.$$

It follows that $|T|^*\varphi \neq 0$. On the other hand, since c^* is a dual space, it is order complete and, therefore, Riesz-Kantorovich formula yields

$$|T^*|\varphi = \sup\bigl\{T^*\psi \;:\; |\psi| \leqslant \varphi\bigr\}.$$

However, for every ψ with $|\psi| \leqslant \varphi$ and every $x \in c_+$, we have

$$\begin{aligned} \bigl|(T^*\psi)(x)\bigr| = \bigl|\psi(Tx)\bigr| &\leqslant \varphi\bigl(|Tx|\bigr) \\ &= \varphi\Bigl(|x_1 - x_2|, |x_2 - x_3|, |x_3 - x_4|, \dots\Bigr) = 0. \end{aligned}$$

It follows that $T^*\psi = 0$ and, therefore, $|T^*|\varphi = 0$. Hence $|T^*| \neq |T|^*$.

13.2.7 Theorem (Krengel-Synnatzschke)**.** *Let $T\colon X \to Y$ be an order bounded operator between Banach lattices with Y order complete. Then $|T^*|$ and $|T|^*$ agree on $Y^\sim_{\mathrm{oc}}$.*

Proof. The proof is based on Riesz-Kantorovich theorem. Recall that $|T^*| \leqslant |T|^*$ by Proposition 13.2.4. Thus, we only need to prove that $|T|^*g \leqslant |T^*|g$ for every $0 \leqslant g \in Y^\sim_{\mathrm{oc}}$. First, we claim that

$$\bigl\langle g, |Tx|\bigr\rangle \leqslant \bigl\langle |T^*|g, x\bigr\rangle \quad \text{for every } x \in X_+. \tag{13.1}$$

Indeed, by the "dual" Riesz-Kantorovich Theorem we have $g\bigl(|Tx|\bigr) = h(Tx)$ for some $h \in [-g, g]$, so that

$$\bigl\langle g, |Tx|\bigr\rangle = \bigl|\langle h, Tx\rangle\bigr| = \bigl|\langle T^*h, x\rangle\bigr| \leqslant \bigl\langle |T^*||h|, x\bigr\rangle \leqslant \bigl\langle |T^*|g, x\bigr\rangle$$

Fix $x \in X^+$. By the "net" Riesz-Kantorovich Theorem 6.3.4, we can write $|T|x$ as the supremum of the increasing net:

$$|T|x = \sup\Bigl\{\sum_{i=1}^n |Tx_i| \;:\; x_1,\dots,x_n \geqslant 0, \sum_{i=1}^n x_i = x\Bigr\}$$

Since g is order continuous, we have

$$\big\langle |T|^*g, x\big\rangle = \big\langle g, |T|x\big\rangle = \sup\Big\{\sum_{i=1}^n g\big(|Tx_i|\big) \;:\; x_1,\dots,x_n \geqslant 0, \sum_{i=1}^n x_i = x\Big\}.$$

By (13.1),

$$\big\langle |T|^*g, x\big\rangle \leqslant \sup\Big\{\sum_{i=1}^n \big(|T^*|g\big)x_i \;:\; x_1,\dots,x_n \geqslant 0, \sum_{i=1}^n x_i = x\Big\} = \big\langle |T^*|g, x\big\rangle,$$

which yields $|T|^*g \leqslant |T^*|g$. $\square$

13.2.8 Corollary. *Suppose that $T\colon X \to Y$ is order bounded operator between Banach lattices and Y is order continuous. Then $|T^*| = |T|^*$. That is, the map $T \mapsto T^*$ is a lattice homomorphism.*

13.2.9 Exercise. If T is regular then so is T^*. In this case, $\|T^*\|_r \leqslant \|T\|_r$.

Recall that if Y is order complete then $L_r(X,Y)$ is a Banach lattice; in particular, $|T|$ exists for every $T \in L_r(X,Y)$. Recall also that every dual Banach lattice is order complete by Corollary 6.7.12 and has the Fatou Property by Exercise 10.6.8. As usual, we write $j_X\colon X \to X^{**}$ for the canonical embedding; $\hat{x} = j_X(x)$ for $x \in X$. We will write j instead of j_X when there space involved is clear. Recall that j is a lattice isometry.

13.2.10 Theorem. *Suppose that Y is order complete and has the Fatou Property. If T is regular then $\|T^*\|_r = \|T\|_r$.*

Proof. By Exercise 13.2.9, T^* is regular and $\|T^*\|_r \leqslant \|T\|_r$. Similarly, $\|T^{**}\|_r \leqslant \|T^*\|_r$. Thus, it suffices to show that $\|T\|_r \leqslant \|T^{**}\|_r$.

Fix $\varepsilon > 0$. Find $x \in B_x$ such that $\big\||T|\big\| \leqslant \big\||T|x\big\| + \varepsilon$. WLOG, $x \geqslant 0$; otherwise, replace x with $|x|$. By Riesz-Kantorovich Formulas, $|T|x = \sup A$, where $A = \big\{|Ty| \in X \;:\; -x \leqslant y \leqslant x\big\}$. It follows that $|T|x = \sup A^\vee$. Since $A^\vee$ can be viewed as an increasing net in X_+, by the Fatou Property we have $\big\||T|x\big\| = \sup\limits_{a\in A^\vee}\|a\|$.

On the other hand, applying Riesz-Kantorovich Formulas to T^{**}, we get

$$|T^{**}|\hat{x} = \sup_{-\hat{x}\leqslant\xi\leqslant\hat{x}} |T^{**}\xi| \geqslant \sup_{-x\leqslant y\leqslant x} |T^{**}\hat{y}| = \sup_{-x\leqslant y\leqslant x} |j(Ty)| = \sup B,$$

where $B = \big\{\big|j(Ty)\big| \in X \;:\; -x \leqslant y \leqslant x\big\}$. Again, $\sup B = \sup B^\vee$; since Y^{**} has the Fatou Property, one has $\big\||T^{**}|\hat{x}\big\| = \sup\limits_{b\in B^\vee}\|b\|$. Note that $B = j(A)$. Since j is a lattice isometry, $B^\vee = j(A^\vee)$, so that $\big\||T^{**}|\hat{x}\big\| = \big\||T|x\big\|$. Therefore, $\big\||T^{**}|\hat{x}\big\| \geqslant \big\||T|\big\| - \varepsilon$, hence $\big\||T^{**}|\big\| \geqslant \big\||T|\big\| - \varepsilon$, and, since ε is arbitrary, $\big\||T^{**}|\big\| \geqslant \big\||T|\big\|$. Therefore, $\|T^{**}\|_r \geqslant \|T\|_r$. $\square$

13.2.11 Corollary. *If Y is a dual Banach lattice and T is regular then so is T^* and $\|T^*\|_r = \|T\|_r$.*

It is easy to see that $j_Y T = T^{**} j_X$ in $L(X, Y^{**})$. We will see that the relationship between T and jT (where $j = j_Y$) is somewhat analogous to the relationship between T and T^*. Since j is an isometry, we have $\|jT\| = \|T\|$. We will see that this generally fails for the regular norm. Clearly, if $T \geqslant 0$ then $jT \geqslant 0$. Since Y^{**} is order complete, $L_r(X, Y^{**})$ is a Banach lattice. In particular, $|jT|$ exists for every $T \in L_r(X, Y)$.

13.2.12 Exercise. Let $T \in L_r(X, Y)$. Then $\|jT\|_r \leqslant \|T\|_r$. If $|T|$ exists then $|jT| \leqslant j|T|$.

13.3 Pre-regular operators

Throughout this section, let $T\colon X \to Y$ be a bounded operator between two Banach lattices. We are going to study the relationship between regularity of T and of T^*. Operators whose adjoint is regular are sometimes called ***pre-regular***. This class of operators admits several surprising equivalent characterizations.

The following theorem shows that T is pre-regular iff it is regular when viewed as an operator from X to Y^{**}:

13.3.1 Theorem. *Let $T\colon X \to Y$ be a bounded operator between two Banach lattices. Let $j\colon Y \to Y^{**}$ be the canonical embedding. TFAE*

(i) *T^* is regular (i.e., T is pre-regular);*
(ii) *T^{**} is regular;*
(iii) *$jT\colon X \to Y^{**}$ is regular;*
(iv) *$jT\colon X \to Y^{**}$ is order bounded.*

In this case, the three operators have the same regular norms.

Proof. By Exercise 13.2.9, if T^* is regular then so is T^{**}, and $\|T^{**}\|_r \leqslant \|T^*\|_r$. Suppose T^{**} is regular. It is easy to see that jT is the restriction of T^{**} to X, hence jT is regular by Exercise 13.2.3 and $\|jT\|_r \leqslant \|T^{**}\|_r$. Since Y^{**} is order complete, Riesz-Kantorovich Theorem 6.3.1 yields that jT is regular iff it is order bounded.

Suppose that jT is regular. Then so is $(jT)^*$. Note that T^* is the restriction of $(jT)^*$ to Y^*, hence T^* is regular and $\|T^*\|_r \leqslant \big\|(jT)^*\big\|_r$. Being a dual Banach lattice, Y^{**} is order complete and has the Fatou Property, so that $\big\|(jT)^*\big\|_r = \|jT\|_r$ by Theorem 13.2.10. $\square$

13.3.2 Corollary. *Every order bounded operator is pre-regular.*

Proof. Suppose that $T\colon X \to Y$ is an order bounded operator between two Banach lattices. Then $jT\colon X \to Y^{**}$ is again order bounded. Therefore, T is pre-regular by Theorem 13.3.1. $\square$

Thus, we get the following:

$$\text{regular} \quad \Rightarrow \quad \text{order bounded} \quad \Rightarrow \quad \text{pre-regular}$$

13.3.3 Theorem. *Let $T\colon X \to Y$ be a bounded operator between Banach lattices.*

(i) *If Y has a strong unit then T is order bounded;*

(ii) *If Y has a strong unit and is order complete then T is regular;*

(iii) *If X is an AL-space then T is pre-regular;*

Proof. (i) is Exercise 8.1.6.

(ii) now follows from the Riesz-Kantorovich-Theorem 6.3.2.

(iii) Consider $T^*\colon Y^* \to X^*$. Note that Y^* is an AM space with a unit by Theorem 8.1.9. Furthermore, being a dual space, Y^* is order complete by Corollary 6.7.12. Thus, T^* satisfies the assumptions of (ii), so that T^* is regular. $\square$

13.3.4 Lemma. *Let $T\colon X \to Y$ be a norm bounded operator from a Banach lattice X to a unital order complete AM-space Y. Then T is regular and $\|T\|_r = \|T\|$.*

Proof. T is order bounded by Exercise 8.1.6. Hence, $\mathcal{L}_{\rm ob}(X,Y) = L(X,Y)$. By Riesz-Kantorovich Theorem 6.3.1, $\mathcal{L}_r(X,Y) = L(X,Y)$ and this space is a vector lattice. In particular, T is regular, $|T|$ exists, and $\|T\|_r = \big\||T|\big\|$. Observe that

$$|T|x = \sup\bigl\{|Ty| \;:\; |y| \leqslant x\bigr\}$$

for every $x \in X_+$. Note that $|y| \leqslant x$ yields $|Ty| \leqslant \|Ty\|e \leqslant \|T\|\|x\|e$, where e is the unit of Y. It follows that $\big\||T|x\big\| \leqslant \|T\|\|x\|$, and, therefore, $\big\||T|\big\| \leqslant \|T\|$ as in Exercise 6.1.14. It follows that $\|T\|_r = \big\||T|\big\| = \|T\|$. $\square$

13.3.5 Theorem. *Let $T\colon X \to Y$ be a bounded operator from an AL-space X to a KB-space Y. Then T is regular and $\|T\|_r = \|T\|$.*

Proof. Combining Theorem 13.3.3(iii) with Theorem 13.3.1, we conclude that $jT\colon X \to Y^{**}$ is regular. Let $P\colon Y^{**} \to Y$ is the band projection coming from Theorem 10.4.10. Clearly, $Pj = I$, so that $T = PjT$. Since P is positive, we conclude that T is regular.

Theorem 13.2.10 yields $\|T\|_r = \|T^*\|_r$. The preceding lemma now gives $\|T^*\|_r = \|T^*\| = \|T\|$. $\square$

13.3.6 Example. *Pre-regular operators that are not order bounded*, and, therefore, not regular. In Examples 6.4.1 and 6.4.7, we constructed continuous operators $T\colon C[0,1]\to c_0$ and $S\colon L_1\to c_0$, respectively, such that T and S are not order bounded. However, the adjoint operators $S^*\colon \ell_1\to L_\infty$ and $T^*\colon \ell_1\to C[0,1]^*$ are regular by Theorem 13.3.3(ii) and by Theorem 13.3.5, respectively. It follows that S and T are pre-regular.

13.3.7 Proposition. *Let X and Y be Banach lattices. The space $L_{\mathrm{pr}}(X,Y)$ of all pre-regular operators from X to Y is a Banach space under the norm $\lVert T\rVert_{\mathrm{pr}}=\lVert T^*\rVert_r$.*

Proof. Clearly, $L_{\mathrm{pr}}(X,Y)$ is a vector space. The "adjoint" map $T\in L_{\mathrm{pr}}(X,Y)\mapsto T^*\in L_r(Y^*,X^*)$ is an isometry, hence $L_{\mathrm{pr}}(X,Y)$ is a normed space. It is left to show that $\lVert\cdot\rVert_{\mathrm{pr}}$ is complete. Let (T_n) be a Cauchy sequence in $L_{\mathrm{pr}}(X,Y)$. By Theorem 13.3.1, the sequence (jT_n) is Cauchy in $L_r(X,Y^{**})$. Since the latter is complete, we have $\bigl\lVert jT_n-R\bigr\rVert_r\to 0$ for some regular operator $R\colon X\to Y^{**}$. It follows from $\lVert\cdot\rVert\leqslant\lVert\cdot\rVert_r$ that $\bigl\lVert jT_n-R\bigr\rVert\to 0$. Since $L(X,Y)$ is a closed subspace of $L(X,Y^{**})$ or, more precisely, the map $T\to jT$ is an isometry from $L(X,Y)$ into $L(X,Y^{**})$ (with respect to the operator norms), it follows that $R\in L(X,Y)$ or, more precisely, $R=jT$ for some $T\in L(X,Y)$. It follows that

$$\lVert T_n-T\rVert_{\mathrm{pr}}=\lVert jT_n-jT\rVert_r=\bigl\lVert jT_n-R\bigr\rVert_r\to 0.$$

$\square$

13.4 Sequentially uniformly continuous operators

We proved in Theorem 6.1.7 that an operator between Archimedean vector lattices is order bounded iff it is uniformly continuous, i.e., if it preserves uniform convergence of nets. We will now consider the sequential version of this concept: an operator is said to be ***sequentially uniformly continuous*** if it maps uniformly null sequences to uniformly null sequences. We will see that the sequential analogues of the conditions (ii) and (iii) in Theorem 6.1.7 are also equivalent, even though they do not imply order boundedness.

13.4.1 Proposition. *Let $T\colon X\to Y$ be a linear operator between Archimedean vector lattices. TFAE:*

(i) *T is sequentially uniformly continuous, i.e., $x_n\xrightarrow{\mathrm{u}}0$ implies $Tx_n\xrightarrow{\mathrm{u}}0$ for all sequences (x_n) in X;*

(ii) $x_n \xrightarrow{\mathrm{u}} 0$ *implies* $Tx_n \xrightarrow{\mathrm{o}} 0$ *for all sequences* (x_n) *in* X;

(iii) $x_n \xrightarrow{\mathrm{u}} 0$ *implies* (Tx_n) *is order bounded for all sequences* (x_n) *in* X.

Proof. (i)⇒(ii)⇒(iii) is trivial. Suppose that (iii) holds; let (x_n) be a net in X such that $x_n \xrightarrow{\mathrm{u}} 0$ in X. It is easy to see that there is a sequence (λ_n) in $\mathbb{R}_+$ such that $\lambda_n \uparrow \infty$ and $\lambda_n x_n \xrightarrow{\mathrm{u}} 0$. By assumption, the sequence $T(\lambda_n x_n)$ is order bounded. Let $a \in Y_+$ such that $T(\lambda_n x_n) \in [-a, a]$ for every n. It follows that $|Tx_n| \leqslant \frac{1}{\lambda_n} a$, so that $Tx_n \xrightarrow{\mathrm{u}} 0$. □

13.4.2 Exercise. Every sequentially uniformly continuous operator from a Banach lattice to a normed lattice is bounded.

An operator between Banach lattices is sequentially uniformly continuous iff it is pre-regular!

13.4.3 Theorem. *Let* $T\colon X \to Y$ *be an operator between Banach lattices. TFAE:*

(i) T *is pre-regular;*

(ii) T *is sequentially uniformly continuous;*

(iii) $x_n \xrightarrow{\mathrm{u}} 0$ *implies that the sequence* $\bigl(\bigvee_{k=1}^n |Tx_k|\bigr)_n$ *is norm bounded for all sequences* (x_n) *in* X;

(iv) *There exists a constant* $M > 0$ *such that*
$$\Bigl\|\bigvee_{k=1}^n |Tx_k|\Bigr\| \leqslant M \Bigl\|\bigvee_{k=1}^n |x_k|\Bigr\|$$
for all $x_1, \dots, x_n \in X$.

Moreover, if Y *is monotonically bounded then every pre-regular operator is order bounded.*

Proof. (i)⇒(ii) Suppose T is pre-regular. Then $jT\colon X \to Y^{**}$ is order bounded by Theorem 13.3.1. Theorem 6.1.7 yields that is is uniformly continuous, hence sequentially uniformly continuous. Since $j(Y)$ is a closed sublattice in Y^{**}, uniforms convergences in Y and Y^{**} agree by Theorem 2.3.17. It now follows that if $x_n \xrightarrow{\mathrm{u}} 0$ in X then $jTx_n \xrightarrow{\mathrm{u}} 0$ in Y^{**} and, therefore, $Tx_n \xrightarrow{\mathrm{u}} 0$ in Y.

(ii)⇒(iii) follows immediately from Proposition 13.4.1.

(iii)⇒(iv) Suppose that (iv) fails. Then for every m we can find vectors $x_1^{(m)}, \dots, x_{n_m}^{(m)}$ in X such that $\bigl\|\bigvee_{k=1}^{n_m} |x_k^{(m)}|\bigr\| < \frac{1}{2^m}$ yet $\bigl\|\bigvee_{k=1}^{n_m} |Tx_k^{(m)}|\bigr\| > m$. Let (x_k) be the sequence in X obtained by concatenating all of these finite sequences. We claim that $x_k \xrightarrow{\mathrm{u}} 0$. Indeed, for each m, let $v_m = \bigvee_{k=1}^{n_m} |x_k^{(m)}|$. Then $\|v_m\| < \frac{1}{2m}$, so that $v_m \xrightarrow{\mathrm{u}} 0$ by Proposition 2.3.6. It now follows that $x_k \xrightarrow{\mathrm{u}} 0$. On the other hand, it is easy to see that the sequence $\bigl(\bigvee_{k=1}^n |Tx_k|\bigr)_n$ fails to be norm bounded.

Before we prove the last implication, we need the following *Claim:* if Y is monotonically bounded then (iv) implies that T is order bounded. To prove this claim, let $u \in X_+$; it suffices to show that $T[0,u]$ is order bounded. For any $x_1,\dots,x_n \in [0,u]$ one has

$$\Big\|\bigvee_{i=1}^n |Tx_i|\Big\| = M\Big\|\bigvee_{i=1}^n |x_i|\Big\| \leqslant M\|u\|.$$

The set $\Big\{\bigvee_{i=1}^n |Tx_i| \;:\; x_1,\dots,x_n \in [0,u]\Big\}$ can be viewed as an increasing net indexed by finite subsets of $[0,u]$ directed by inclusion. Since this net is norm bounded, it is order bounded from above by some $y \in Y_+$. In particular, $|Tx| \leqslant y$ for every $x \in [0,u]$.

(iv)$\Rightarrow$(i) Suppose T satisfies (iv). Since $j(Y)$ is a closed sublattice of Y^{**}, we conclude that jT satisfies (iv) as well. Since Y^{**} is monotonically bounded by Exercise 10.6.8, the Claim yields that jT is order bounded, hence T is pre-regular by Theorem 13.3.1.

Once we know that (i)$\Leftrightarrow$(iv); the "moreover" clause follows immediately from the Claim. $\square$

The condition (iv) will come naturally in Chapter 20, where we will discuss multinormed spaces. In the language of multinormed spaces, this condition says that T is ∞-multibounded with respect to the canonical multinorms on X and Y. Moreover, using techniques of multinorms, we will show in Theorem 20.6.3 that the sups in (iv) may be replaced with sums.

Proposition 13.4.1 and Theorem 13.4.3 can be extended with essentially the same proofs to an operator $T\colon Z \to Y$, where Z is a subspace of X (we need Z to be closed in Theorem 13.4.3), provided that we extend our terminology appropriately. The sequence (x_n) should now come from Z, but the notation $x_n \xrightarrow{\mathrm{u}} 0$ is interpreted in X. By T being order bounded we now mean that $T(A)$ is order bounded in Y whenever A is a subset of Z which is order bounded in X. Finally, we need to extend the definition of a pre-regular operator to this more general setting. We say that a bounded operator $T\colon Z \to Y$ is pre-regular if $jT\colon Z \to Y^{**}$ is order bounded. For operators defined on the entire space, this new definition agrees with the old one by Theorem 13.3.1.

We know that uniformly continuous operators between Archimedean vector lattices are precisely the order bounded ones. As we now know that sequentially uniformly continuous operators between Banach lattices are precisely the pre-regular ones. We conclude that sequentially uniformly continuous operators need not be order bounded by Example 13.3.6. Here is another example.

13.4.4 Example. *Another pre-regular operator which fails to be order bounded.* Let $T\colon c \to c_0$, defined by

$$T\colon (a_1, a_2, \dots) \mapsto (a_\infty, a_\infty - a_1, a_\infty - a_2, \dots), \text{ where } a_\infty = \lim_n a_n.$$

Note that

$$T\colon (0, \dots, 0, 1, 1, \dots) \mapsto (1, \dots, 1, 0, 0, \dots),$$

so that $T[0, \mathbb{1}]$ is not order bounded in c_0, hence T is not order bounded. On the other hand, suppose that $x_n \xrightarrow{\mathrm{u}} 0$ in c. Then $x_n \xrightarrow{\|\cdot\|} 0$ in c. Since T is norm bounded, it follows that $Tx_n \xrightarrow{\|\cdot\|} 0$ in c_0. Since norm convergence and uniform convergence agree for sequences in c_0 by Corollary 2.5.16, we have $Tx_n \xrightarrow{\mathrm{u}} 0$ in c_0. Thus, T is sequentially uniformly continuous, hence pre-regular.

13.5 Finite-rank operators

We will show in this section that the situation is much better for finite rank operators. As before, X and Y are two Banach lattices. We write $F(X,Y)$ for the set of all operators of finite rank in $L(X,Y)$. Since every operator of rank one is regular, we have $F(X,Y) \subseteq L_r(X,Y)$. In this section, we will heavily use Riesz-Kantorovich formulae in $L_r(X,Y)$. Recall that even though $L_r(X,Y)$ is, generally, not a vector lattice, Riesz-Kantorovich formulae remain valid in it as long as the suprema involved in the formulae exist; see 6.3.7. We show next that $|T|$ is defined for every finite rank operator T:

13.5.1 Lemma. *Let $T_1, \dots, T_n \in F(X,Y)$. Then $\bigvee_{i=1}^n T_i$ exists and is compact. In particular, $|T|$ exists and is compact for every $T \in F(X,Y)$.*

Proof. By Riesz-Kantorovich Theorem 6.3.3 and by 6.3.7, it suffices to show that for every $x \in X_+$, the following set has supremum:

$$A_x := \Bigl\{\sum_{i=1}^n T_i x_i \,:\, x_1, \dots, x_n \geqslant 0,\ x_1 + \dots + x_n = x\Bigr\}.$$

It is easy to see that A_x is contained in the finite-dimensional subspace Y generated by the ranges of $T_1, \dots, T_n$. It is also easy to see that A_x is norm bounded by $\Bigl(\sum_{i=1}^n \|T_i\|\Bigr)\|x\|$. It follows from Theorem 11.5.2 that $\sup A_x$ exists. We conclude that $S := \bigvee_{i=1}^n T_i$ exists.

To prove that S is compact, put $C = \bigcup_{x \in B_X^+} A_x$. Then C is bounded by $\sum_{i=1}^n \|T_i\|$ and is contained in Y. So by Theorem 11.5.2, $\overline{C^\vee}$ is compact. For every $x \in B_X^+$, using Theorem 11.5.2 once again, we get $Sx = \sup A_x = \lim A_x^\vee \in \overline{C^\vee}$. It follows that $S(B_X) \subseteq \overline{C^\vee} - \overline{C^\vee}$. Since the latter set is compact, we conclude that S is compact. ☐

A word of warning: the operators $|T|$ and $\bigvee_{i=1}^n T_i$, constructed in the lemma, need not be finite rank. We will show next that not only $|T|$ and $\bigvee_{i=1}^n T_i$, but arbitrary lattice-linear combinations of finite rank operators are defined, and form a vector lattice in $L(X,Y)$. Let $G(X,Y) = F(X,Y)^\vee - F(X,Y)^\vee$, the set of all differences of finite suprema of finite-rank operators. It is clear that every operator in $G(X,Y)$ is regular and compact. Applying Proposition 3.1.17 to $F(X,Y)$ and $L(X,Y)$, we get the following:

13.5.2 Proposition. *The set $G(X,Y)$ is a normed lattice under the regular norm. Its lattice operations agree with that in $L(X,Y)$. Furthermore, $F(X,Y)$ is a generating set in $G(X,Y)$.*

13.5.3. The closure of $G(X,Y)$ in $L_r(X,Y)$ may be identified with the norm completion of $G(X,Y)$; it follows that it is a Banach lattice. It is clear that if $L_r(X,Y)$ is a vector lattice then $G(X,Y)$, as well as its closure, are sublattices of $L_r(X,Y)$.

As before, $j\colon Y \hookrightarrow Y^{**}$ is the canonical embedding. Let $J\colon L_r(X,Y) \to L_r(X,Y^{**})$ be the map that sends T to the composition jT. While J preserves the operator norm (in particular, it is one-to-one), it need not preserve the regular norm. Furthermore, while J is positive, it need not be a lattice homomorphism. We will show next that the restriction of J to $G(X,Y)$ (and, in particular, to $F(X,Y)$) is much better behaved.

13.5.4 Proposition. *The restriction of J to $G(X,Y)$ is a lattice isometric embedding.*

Proof. Let $T_1,\dots,T_n \in F(X,Y)$. We claim that $J\Big(\bigvee_{i=1}^n T_i\Big) = \bigvee_{i=1}^n JT_i$. By a Riesz-Kantorovich formula 6.3.5, we can write $\Big(\bigvee_{i=1}^n T_i\Big)x$ as $\sup A$, where

$$A = \Big\{\sum_{j=1}^m \bigvee_{i=1}^n T_i x_j \;:\; x_1,\dots,x_m \geqslant 0,\ \sum_{j=1}^m x_j = x\Big\}.$$

As noted in Exercise 6.3.5, we may view A as an increasing net. To show that it converges, we introduce an auxiliary set:

$$C = \Big\{\sum_{j=1}^m T_{i_j} x_j \;:\; m \in \mathbb{N},\ x_1,\dots,x_m \geqslant 0,\ \sum_{j=1}^m x_j = x,\ 0 \leqslant i_1,\dots,i_m \leqslant n\Big\}.$$

Clearly, C is contained in the finite-dimensional subspace generated by the ranges of $T_1,\dots,T_n$. Observe that C is order bounded and, therefore, norm bounded because for a generic element of C we have

$$\Big|\sum_{j=1}^m T_{i_j} x_j\Big| \leqslant \sum_{j=1}^m |T_{i_j}| x_j \leqslant \sum_{j=1}^m \Big(\bigvee_{i=1}^n |T_i|\Big) x_j \leqslant \Big(\bigvee_{i=1}^n |T_i|\Big) x.$$

Hence, by Theorem 11.5.2, $C^\vee$ is relatively compact.

We claim that $A \subseteq C^\vee$. Indeed, a generic element of A is of the form $\sum_{j=1}^m \bigvee_{i=1}^n T_i x_j$ for some $m \in \mathbb{N}$ and $x_1, \dots, x_m \geqslant 0$ with $\sum_{j=1}^m x_j = x$. For each $j = 1, \dots, m$, put $B_j = \{T_1 x_j, \dots, T_m x_j\}$. Using Exercise 1.3.5(i), we have

$$\sum_{j=1}^m \bigvee_{i=1}^n T_i x_j = \sum_{j=1}^m \sup B_j = \sup(B_1 + \dots + B_m) \in C^\vee.$$

It follows that A is relatively compact. Since it is an increasing net, we conclude that it converges in norm. Since j_Y is a (continuous) lattice homomorphism, using Monotone Convergence Lemma, we get

$$J\Big(\bigvee_{i=1}^n T_i\Big)x = j_Y(\lim A) = \lim j_Y(A) = \sup j_Y(A) = \Big(\bigvee_{i=1}^n JT_i\Big)x.$$

This proves that $J\Big(\bigvee_{i=1}^n T_i\Big) = \bigvee_{i=1}^n JT_i$.

Let $R \in G(X,Y)$. We can then write $R = S - T$ for some $S, T \in F(X,Y)^\vee$. It follows from the preceding paragraph that $J(S \vee T) = (JS) \vee (JT)$. This yields

$$J(R^+) = J(S \vee T - T) = J(S) \vee J(T) - J(T) = \big(J(S-T)\big)^+ = (JR)^+$$

We conclude that J is a lattice homomorphism on $G(X,Y)$.

Finally, J is an isometry on $G(X,Y)$ because

$$\|JR\|_r = \big\||JR|\big\| = \big\|J|R|\big\| = \big\||R|\big\| = \|R\|_r.$$

for every $R \in G(X,Y)$. $\square$

13.5.5 Proposition. *The adjoint map is a lattice isometry from $G(X,Y)$ to $G(Y^*,X^*)$. That is, $|T^*| = |T|^*$ and $\|T^*\|_r = \|T\|_r$ for every $T \in G(X,Y)$.*

Proof. Let $j_X \colon X \hookrightarrow X^{**}$ and $j_Y \colon Y \hookrightarrow Y^{**}$ be the canonical embeddings. It is easy to see that $j_Y T = T^{**} j_X$ for every $T \in L(X,Y)$.

Let $T \in G(X,Y)$. By Proposition 13.2.4, we have $|T^*| \leqslant |T|^*$. It suffices to prove that $|T|^*$ is the least majorant of $\pm T^*$. Suppose that $\pm T^* \leqslant V$ for some $V \in L(Y^*,X^*)$. Then $\pm j_Y T = \pm T^{**} j_X \leqslant V^* j_X$. By Proposition 13.5.4, $|T|^{**} j_X = j_Y |T| = \big|j_Y T\big| \leqslant V^* j_X$. It follows that for all $y^* \in Y_+^*$ and $x \in X_+$ we have $\big\langle |T|^{**} j_X x, y^*\big\rangle \leqslant \big\langle V^* j_X x, y^*\big\rangle$, which yields $\big\langle x, |T|^* y^*\big\rangle \leqslant \big\langle x, V y^*\big\rangle$, so that $|T|^* y^* \leqslant V y^*$, and, therefore, $|T|^* \leqslant V$. This completes the proof that $|T^*| = |T|^*$. It now follows that $\|T^*\|_r = \big\||T^*|\big\| = \big\||T|^*\big\| = \big\||T|\big\| = \|T\|_r$. Since the adjoint map is a lattice homomorphism, it follows from the definition of $G(X,Y)$ that $T^* \in G(Y^*,X^*)$ whenever $T \in G(X,Y)$. $\square$

Since $F(X,Y) \subseteq G(X,Y)$, we immediately get the following:

13.5.6 Corollary. *Let $T \in F(X,Y)$. Then $|T^*| = |T|^*$, $|jT| = j|T|$, and $\|T\|_r = \|jT\|_r = \|T^*\|_r$.*

The following proposition may be viewed as a dual version of Proposition 13.5.4. Note that the operator T in the proposition may be viewed as a "canonical" extension of S.

13.5.7 Proposition. *Let $S \in F(X,Y)$ and $T \in F(X^{**},Y)$ given by $Sx = \sum_{i=1}^n \langle x_i^*, x\rangle y_i$ and $Tx^{**} = \sum_{i=1}^n \langle x_i^*, x^{**}\rangle y_i$ for some $y_1,\dots,y_n \in Y$ and $x_1^*,\dots,x_n^*$ in X^*. Then $\|S\|_r = \|T\|_r$.*

Proof. Observe that $S^{**} = j_Y T$. It follow from Corollary 13.5.6 that $\|S\|_r = \|S^{**}\|_r = \|j_Y T\|_r = \|T\|_r$. □

We already now that $G(X,Y)$ consists of compact operators. We will show in Corollaries 19.7.6 and 19.7.7 that $F(X,Y)$ is uniformly dense in $G(X,Y)$ and, therefore, that $F(X,Y)$ and $G(X,Y)$ have the same closures in $L_r(X,Y)$. In particular, this means that the closure of $F(X,Y)$ in the regular norm is a Banach lattice, and that the adjoint map, as well as the map $T \mapsto j_Y T$, are lattice isometries on this space.

Chapter 14

Central operators and orthomorphisms

In this chapter, we consider several related classes of operators on vector lattices. Let X be a vector lattice, and $T \in \mathcal{L}(X)$, a linear operator from X to itself. We say that T is ***central*** if $\pm T \leqslant \lambda I$ for some $\lambda > 0$, ***band preserving*** if $T(B) \subseteq B$ for every band B, an ***orthomorphism*** if it is band preserving and order continuous, and ***disjointness preserving*** if $x \perp y$ implies $(Tx) \perp (Ty)$. These classes are nested:

$$\text{central} \quad \subseteq \quad \text{orthomorphisms} \quad \subseteq \quad \text{band preserving} \quad \subseteq \quad \text{disjointness preserving.}$$

Here we summarize the most important results that will be proved in this chapter. On Banach lattices, the first three classes coincide. The sets $\mathcal{Z}(X)$ and $\mathrm{Orth}(X)$ of all central operators and of all orthomorphisms on X, respectively, are vector lattices, with lattice operations computed point-wise, e.g., $|T|x = |Tx|$ for $x \in X_+$. If X is uniformly complete then $\mathcal{Z}(X)$ is an AM-space with unit I. For example, if $X = L_p(\mu)$ then $\mathcal{Z}(X) = \mathrm{Orth}(X)$ consists of all the multiplication operators and may be identifies with $L_\infty(\mu)$.

Weighted composition operators on function spaces are examples of disjointness preserving operators. Positive disjointness preserving operators are precisely the lattice homomorphisms. For every order bounded disjointness preserving operator T, the modulus $|T|$ exists, is a lattice homomorphism, and is computed point-wise.

Our study of central operators and orthomorphisms will be based on their representations as multiplication operators. A central operator T may be represented as a "local multiplication operator": for every $e \in X_+$, we identify I_e with a $C(K)$ space, then the restriction of T to I_e corresponds to some multiplication operator on I_e. If T is an orthomorphism, we identify the universal completion of X with a $C^\infty(K)$ space, then T is the restriction to X of a multiplication operator on $C^\infty(K)$. Unfortunately, there is no simple representation of disjointness preserving operators (though, in many particular cases they may be represented as weighted composition operators), so we have to use more algebraic proofs for properties of disjointness preserving operators.

Our exposition is not optimal: we start with central operators, then we proceed to orthomorphisms, and then to disjointness preserving operators. As a result, we sometimes have to prove the same fact two or three times (sometimes using different techniques). It would have been more efficient (and shorter) to present these classes in the opposite order. However, we believe that our approach is more intuitive as we start with the easiest object and proceed to more complicated ones.

14.1 Central operators

Throughout this chapter, let X be an Archimedean vector lattice and $T\colon X \to X$ a linear operator. We say that T is ***central*** if there exists a positive real λ such that $\pm T \leqslant \lambda I$. We are going to show that central operators may be interpreted as multiplication operators.

The set of central operators on X is called the ***center*** of X and is denoted by $\mathcal{Z}(X)$. It follows immediately from the definition that every central operator is regular. Furthermore, every central operator is order continuous and σ-order continuous; see Exercise 6.5.4.

14.1.1 Exercise. For $\lambda > 0$, TFAE:

(i) $\pm T \leqslant \lambda I$;
(ii) $|Tx| \leqslant \lambda x$ for every $x \geqslant 0$;
(iii) $|Tx| \leqslant \lambda |x|$ for every x;
(iv) $0 \leqslant \frac{1}{2}I + \frac{1}{2\lambda}T \leqslant I$.

14.1.2 Exercise. The central operators on $\mathbb{R}^n$ are exactly the diagonal matrices.

Let $X = C(K)$ for some compact space K. For each $f \in C(K)$, consider the operator M_f on X given by $M_f g = fg$. Such operators are called ***multiplication operators***.

14.1.3 Proposition. *For a linear operator T on $C(K)$, TFAE:*

(i) *T is central;*
(ii) *T is a multiplication operator;*
(iii) *T "preserves zeros": $g(t) = 0$ implies $(Tg)(t) = 0$.*

Proof. (ii)⇒(i)⇒(iii) is straightforward.

(iii)⇒(ii) Put $f = T\mathbb{1}$; we claim that $T = M_f$. Let $g \in C(K)$ and $t \in K$. Put $\alpha = g(t)$. It follows from $(g - \alpha\mathbb{1})(t) = 0$ and $\bigl|T(g - \alpha\mathbb{1})\bigr| \leqslant \lambda|g - \alpha\mathbb{1}|$ that $\bigl[T(g - \alpha\mathbb{1})\bigr](t) = 0$, so that $(Tg)(t) = \alpha(T\mathbb{1})(t) = g(t)f(t)$. □

The elegant argument used in the preceding proof will be repeatedly used throughout this chapter.

14.1.4 Corollary. *Let $X = C(K)$ for some compact K. Then $\mathcal{Z}(X)$ is lattice isomorphic to $C(K)$ via $f \in C(K) \mapsto M_f \in \mathcal{Z}(X)$ and $T \in \mathcal{Z}(X) \mapsto T\mathbb{1} \in C(K)$.*

14.1.5 Remark. Central operators on a vector lattice X may be viewed as "local multiplication operators". Let's make this precise. Let T be a central operator on X. Then $\pm T \leqslant \lambda I$ for some $\lambda > 0$. Let $e \in X_+$. Clearly, T leaves I_e invariant and the restriction of T to I_e is central. In particular, T is bounded with respect to $\|\cdot\|_e$ and, therefore, extends to an operator on the completion of $\bigl(I_e, \|\cdot\|_e\bigr)$, which is lattice isometric to a $C(K)$ space by Theorem 4.1.9. Therefore, T induces a central operator on $C(K)$, which is a multiplication operator by Proposition 14.1.3. That is, $Tx = fx$ for every $x \in I_e$, where $f = Te$ and fx is interpreted as a product in $C(K)$.

It follows that central operators share many properties of multiplication operators.

14.1.6 Lemma. *Let T be a central operator. Then $\bigl|T(x + y)\bigr| = |Tx| + |Ty|$ for all $x, y \in X_+$.*

Proof. Let $e = x \vee y$. In the notation of the preceding remark, we have

$$\bigl|T(x + y)\bigr| = \bigl|f(x + y)\bigr| = |fx| + |fy| = \bigl|Tx\bigr| + \bigl|Ty\bigr|.$$

□

14.1.7 Exercise. If $S, T \in \mathcal{L}(X, Y)$ satisfy $Sx = |Tx|$ for all $x \in X_+$ then $S = |T|$.

14.1.8 Exercise. If $T \in \mathcal{L}(X, Y)$ satisfies $\bigl|T(x + y)\bigr| = |Tx| + |Ty|$ for all $x, y \in X_+$ then $|T|$ exists and $|T|x = |Tx|$ for all $x \in X_+$.

Recall that if X is order complete then $\mathcal{L}_r(X)$ is an order complete vector lattice by Riesz-Kantorovich Theorems 6.3.1 and 6.3.2. In this case, it follows immediately from the definition of a central operator that $\mathcal{Z}(X)$ is the principal ideal generated by I in $\mathcal{L}_r(X)$. In particular, $\mathcal{Z}(X)$ is itself a vector lattice and I is a strong unit in $\mathcal{Z}(X)$. It turns out that completeness of X is not necessary! The following result is due to Bigard-Keimel and Conrad-Diem.

14.1.9 Theorem. *$\mathcal{Z}(X)$ is an Archimedean vector lattice with*

$$|T|x = \bigl|Tx\bigr|,\ (T \vee S)x = (Tx) \vee (Sx),\ \text{and } (T \wedge S)x = (Tx) \wedge (Sx)$$

for all $x \geqslant 0$ and $S, T \in \mathcal{Z}(X)$. The identity operator is a strong unit in $\mathcal{Z}(X)$.

Proof. Let $T \in \mathcal{Z}(X)$. Combining Lemma 14.1.6 with Exercise 14.1.8, we conclude that $|T|$ exists and is given by $|T|x = |Tx|$ for every $x \in X_+$. Since T is central, we have $\pm T \leqslant \lambda I$ for some $\lambda > 0$; it follows that $|T| \leqslant \lambda I$, hence $|T|$ is also central. The rest follows easily from Exercise 1.3.13. The Archimedean property is straightforward. □

Clearly, every scalar multiple of I is in the center. The following example shows that there may be nothing else in the center (in which case, we say that the center is trivial).

14.1.10 Exercise. Let X be the sublattice of $C[0, 1]$ consisting of all piece-wise linear functions. The center of X is trivial.

There are examples of Banach lattices with trivial centres; see [Wick88].

14.1.11 Exercise. If $T\colon X \to X$ is central then its order adjoint $T^{\sim}\colon X^{\sim} \to X^{\sim}$ is central. The converse is true provided that $X^{\sim}$ separates the points of X.

14.1.12 Exercise. Let $X = L_p(\mu)$ for a finite measure μ and $1 \leqslant p \leqslant \infty$. Show that $\mathcal{Z}(X)$ consists of all multiplication operators M_f with $f \in L_\infty(\mu)$. The map $f \mapsto M_f$ is a lattice isomorphism from $L_\infty(\mu)$ onto $\mathcal{Z}(X)$.

14.1.1 Center as a Banach lattice

We already know that $\mathcal{Z}(X)$ is a vector lattice with strong unit I. As in Section 2.2, we can make $\mathcal{Z}(X)$ into a normed lattice with respect to the "uniform" norm generated by I:

$$\|T\|_\infty = \inf\bigl\{\lambda > 0 \ :\ \pm T \leqslant \lambda I\bigr\}$$

(we write $\|T\|_\infty$ rather than $\|T\|_I$). From now on, we always equip $\mathcal{Z}(X)$ with this norm. Recall that every Banach lattice is uniformly complete, i.e., $\bigl(I_e, \|\cdot\|\bigr)$ is complete for every $e \in X_+$.

14.1.13 Proposition. *If X is uniformly complete then $\mathcal{Z}(X)$ is a Banach lattice. Moreover, it is an AM-space with unit I.*

Proof. By 8.1.1, we only need to prove that $\mathcal{Z}(X)$ is complete. The proof is straightforward. By Exercise 2.5.2, it suffices to show that every increasing Cauchy sequence in $\mathcal{Z}(X)$ is convergent. Let (T_n) be such a sequence. Fix $\varepsilon > 0$. There exists n_0 such that $0 \leqslant T_m - T_n \leqslant \varepsilon I$ whenever $m \geqslant n \geqslant n_0$. In particular, for every $u \in X_+$ we have $0 \leqslant T_m u - T_n u \leqslant \varepsilon u$; it follows that $(T_n u)$ is Cauchy with respect to $\|\cdot\|_u$ and, therefore, converges uniformly to some vector. It follows that for every $x \in X$, the sequence $(T_n x)$ converges uniformly to some vector; denote it Tx. It follows immediately that T is a linear operator.

Let $u \in X_+$. Clearly, $Tu = \sup T_n u$. For every $\varepsilon > 0$, let n_0 be as before, and we again have $0 \leqslant T_m u - T_n u \leqslant \varepsilon u$. Taking supremum over m, we get $0 \leqslant Tu - T_n u \leqslant \varepsilon u$, so that $0 \leqslant T - T_n \leqslant \varepsilon I$. It follows that $T \in \mathcal{Z}(X)$ and $\|T - T_n\|_\infty \leqslant \varepsilon$. □

14.1.14 Exercise. The preceding result fails without the uniform completeness assumption. Let X be the space of all piece-wise polynomial functions in $C[0,1]$. Show that $\mathcal{Z}(X) \simeq X$ and, therefore, fails to be complete.

14.1.15 Corollary. *If X is uniformly complete then $\mathcal{Z}(X)$ is lattice isometric to $C(K)$ for some compact K.*

Let K be as in the preceding corollary. We say that K is the ***carrier*** of X. It is unique up to a homeomorphism by Exercise 6.11.8.

14.1.16 Exercise. The centre of $C(K)$ is lattice isometric to $C(K)$. The carrier of $C(K)$ is K.

Every Banach lattice is uniformly complete, so the preceding results apply. In this case, more can be said:

14.1.17 Theorem. *If X be a Banach lattice then the norm $\|\cdot\|_\infty$ on $\mathcal{Z}(X)$ agrees with the operator norm.*

Proof. Being a Banach lattice, X is uniformly complete, so that $\mathcal{Z}(X)$ is a Banach lattice. Let $T \in \mathcal{Z}(X)$. Suppose first that $\|T\|_\infty < 1$. Then $\pm T \leqslant I$. For every $x \in X$, we have $|Tx| \leqslant |x|$, so that $\|Tx\| \leqslant \|x\|$. It follows that $\|T\| \leqslant 1$. Therefore, $\|T\| \leqslant \|T\|_\infty$.

Conversely, suppose that $\|T\| < 1$; it suffices to prove that $\|T\|_\infty \leqslant 1$. Assume first that $T \geqslant 0$. Let $x \in X_+$. Put $e = \sum_{k=0}^\infty T^k x$; the series converges because $\|T^k x\| \leqslant \|T\|^k \|x\|$ and $\|T\| < 1$. Observe that $0 \leqslant x \leqslant e$. As in Remark 14.1.5, we represent I_e as a $C(K)$ space, with e becoming $\mathbb{1}$, and T corresponds to a multiplication operator M_f on I_e with $f = T\mathbb{1} = Te = \sum_{k=1}^\infty T^k x \leqslant e = \mathbb{1}$. This yields $Tx = fx \leqslant x$. Since $x \in X_+$ was arbitrary, it follows that $T \leqslant I$ on X, hence $\|T\|_\infty \leqslant 1$.

We now move on to the general case. For every $x \in X$, it follows from $\bigl||T|x\bigr| \leqslant |T||x| = \bigl|T|x|\bigr|$ that $\bigl\||T|x\bigr\| \leqslant \bigl\|T|x|\bigr\| \leqslant \|T\|\|x\|$, so that $\bigl\||T|\bigr\| \leqslant \|T\| < 1$. By the special case, we conclude that $\bigl\||T|\bigr\|_\infty \leqslant 1$. However, $\|\cdot\|_\infty$ is a lattice norm, so that $\bigl\||T|\bigr\|_\infty = \|T\|_\infty$. □

Let X be a Banach lattice and $e \in X_+$. For every $T \in \mathcal{Z}(X)$, the restriction of T to I_e yields a map from $\mathcal{Z}(X)$ to $\mathcal{Z}(I_e)$, and the latter space may in turn be identified with I_e itself, so we get a map from $\mathcal{Z}(X)$ to I_e. The following proposition asserts that if e is a quasi-interior point then this map is a surjective lattice isometry.

14.1.18 Proposition. *Let e be a quasi-interior point in a Banach lattice X. Then $\bigl(I_e, \|\cdot\|_e\bigr)$ is lattice isometric to $\mathcal{Z}(X)$.*

Proof. If $T \in \mathcal{Z}(X)$ with $\pm T \leqslant \lambda I$, the same remains true for the restriction of T to I_e. Conversely, if $T \in \mathcal{Z}(I_e)$ with $\pm T \leqslant \lambda I$ on I_e then T is bounded with respect to the usual sup-norm of $C(K)$. Since I_e is dense, T extends uniquely to an operator $\widetilde{T}$ on $C(K)$. It is easy to see that $\pm\widetilde{T} \leqslant \lambda I$ on $C(K)$. It follows that the restriction map is a lattice isometry from $\mathcal{Z}(X)$ onto $\mathcal{Z}(I_e)$.

Since X is uniformly complete, $\bigl(I_e, \|\cdot\|_e\bigr)$ may be identified with some $C(K)$. Now by Exercise 14.1.16, we get $\mathcal{Z}(I_e) \cong \mathcal{Z}\bigl(C(K)\bigr) \cong C(K) \cong I_e$.

□

14.1.19 Corollary. *Let e_1 and e_2 be two quasi-interior points in a Banach lattice X. Then $\bigl(I_{e_1}, \|\cdot\|_{e_1}\bigr)$ and $\bigl(I_{e_2}, \|\cdot\|_{e_2}\bigr)$ are lattice isometric.*

Hence, all the quasi-interior points in a Banach lattice are "equivalent".

14.1.20 Example. Let $X = L_p(\mu)$ for a finite measure μ and $1 \leqslant p \leqslant \infty$. We showed in Example 14.1.12 that $\mathcal{Z}(X) \cong L_\infty(\mu)$. Here is another way to see this. Let $e = \mathbb{1}$. Clearly, it is a quasi-interior point in X and $\bigl(I_e, \|\cdot\|_e\bigr) = L_\infty(\mu)$. It now follows from Proposition 14.1.18 that $\mathcal{Z}(X) \cong L_\infty(\mu)$.

14.1.21 Exercise. Let $1 \leqslant p \leqslant \infty$. For $w \in \ell_\infty$, let $M_w\colon \ell_p \to \ell_p$ be the multiplication operator: $M_w x = (w_i x_i)$. Show that the map $w \mapsto M_w$ is a lattice isometry from ℓ_∞ onto $\mathcal{Z}(\ell_p)$.

14.1.22 Exercise. Let X be a Banach lattice with a quasi-interior point e, and $0 \leqslant v \leqslant e$. Then there exists $T \in L(X)$ such that $0 \leqslant T \leqslant I$ and $Te = v$.

14.1.2 Central vs ideal preserving

Let X be an Archimedean vector lattice and $T \in \mathcal{L}(X)$. We say that T is ***ideal preserving*** if it leaves every ideal invariant.

14.1.23 Exercise. T is ideal preserving iff for every $e \in X_+$ there exists $\lambda \geqslant 0$ such that $|Te| \leqslant \lambda e$.

14.1.24 Exercise. A central operator leaves every ideal invariant.

The next example shows that the converse is false in general. We will show in Corollary 14.2.34 that the converse is true on Banach lattices.

14.1.25 Example (Wickstead)**.** *An order bounded ideal preserving non-central operator.* Let c_{00} be the space of eventually zero sequences; let $T\colon c_{00} \to c_{00}$ be given by $Te_n = ne_n$. Suppose that $0 \leqslant x \in c_{00}$; let n be the index of the last non-zero component of x. It is easy to see that $0 \leqslant Tx \leqslant nx$. It follows that T is order bounded and leaves every ideal invariant. Yet, it is easy to see that T is not central.

We observed in Remark 14.1.5 that every central operator T is a "local multiplication operator". That is, for every $e \in X_+$ and a representation of I_e as a dense ideal in a $C(K)$ space, with e corresponding to $\mathbb{1}$, there exists $f \in C(K)$ such that T agrees with M_f on I_e (clearly, $f = Te$).

14.1.26 Proposition. *T is a local multiplication operator iff it is ideal preserving.*

Proof. Suppose that T is a local multiplication operator. Let e, K, and f be as in the preceding paragraph. Then $|f| \leqslant \lambda\mathbb{1}$ for some $\lambda > 0$; it follows that $|Te| = |fe| \leqslant \lambda e$. It follows that T is ideal preserving.

The proof of the converse is similar to that of Proposition 14.1.3. Suppose that T is ideal preserving; let $e \in X_+$, and represent I_e as a dense ideal in a $C(K)$ space with $e = \mathbb{1}$. Put $f = Te$. Let $g \in I_e$; fix $t \in K$. Observe that the set $J = \bigl\{h \in I_e : h(t) = 0\bigr\}$ is an ideal in X, hence $T(J) \subseteq J$. Put $\alpha \coloneqq g(t)$. Then $g - \alpha\mathbb{1} \in J$, hence $T(g - \alpha\mathbb{1}) \in J$. It follows that $(Tg)(t) = \alpha(T\mathbb{1})(t) = g(t)f(t)$. We conclude that $Tg = fg$. □

14.1.27 Corollary. *Every ideal preserving operator is order bounded.*

Proof. Suppose that $T \in \mathcal{L}(X)$ is ideal preserving and $e \in X_+$. We represent I_e and an order dense ideal in a $C(K)$ space with $e = \mathbb{1}$. By the preceding proposition, T agrees with M_f on I_e for some $f \in C(K)$. Find $\lambda > 0$ such that $|f| \leqslant \lambda\mathbb{1}$. It follows that for every $x \in [0, e]$, we have $|Tx| = |fx| \leqslant \lambda e$, hence $T[0, e] \subseteq [-\lambda e, \lambda e]$. □

14.2 Orthomorphisms

14.2.1 Band preserving operators and orthomorphisms

Throughout this section, X stands for an Archimedean vector lattice and $T\colon X \to X$ is a linear operator. We say that T is ***band preserving*** if $T(B) \subseteq B$ for every band B in X.

14.2.1 Exercise. TFAE

(i) T is band preserving;
(ii) $x \perp y$ implies $Tx \perp y$ for all $x, y \in X$;
(iii) $Tx \in B_x$ for every x.

14.2.2 Exercise. A positive band preserving operator is a lattice homomorphism.

We say that T is an ***orthomorphism*** if is band preserving and order bounded. We write $\mathrm{Orth}(X)$ for the set of all orthomorphisms on X. It is clear that $\mathrm{Orth}(X)$ is a subalgebra of $\mathcal{L}(X)$. Observe that

14.2.3 Exercise. Verify that

$$\text{central} \quad \Rightarrow \quad \text{ideal preserving} \quad \Rightarrow \quad \text{orthomorhism} \quad \Rightarrow \quad \text{band preserving.}$$

In particular, $\mathcal{Z}(X) \subseteq \mathrm{Orth}(X)$. We will show in Theorem 14.2.33 that on Banach lattices, these four classes of operators coincide. Example 14.1.25 together with following two examples show that in general these classes are different.

14.2.4 Example. Consider $T\colon \mathbb{R}^{\mathbb{N}} \to \mathbb{R}^{\mathbb{N}}$ given by

$$T\colon (x_1, x_2, x_3, x_4, \dots) \mapsto (x_1, 2x_2, 3x_3.4x_4, \dots).$$

It is easy to see that T is an orthomorphism. It is also not ideal preserving as it does not preserve the ideal of all bounded sequences.

14.2.5 Example. Let X be the sublattice of $\mathbb{R}^{[0,1)}$ consisting of left-continuous piecewise affine functions. That is, $f \in X$ iff there exist $0 = t_0 < t_1 < \cdots < t_n = 1$ such that f is of the form $m_i t + b_i$ on $[t_i, t_{i+1})$. Let $T\colon X \to X$, be the differentiation operator, i.e., Tf equals b_i on $[t_i, t_{i+1})$. Then T is band preserving. It is easy to see that T is not order bounded, hence not an orthomorphism.

14.2.6 Exercise. Let $T \in \mathcal{L}(X)$ be band preserving and B be a band in X.

(i) The restriction of T to B is band preserving;

(ii) The induced operator $\widetilde{T}\colon X/B \to X/B$ is well defined and is band preserving.

We will see that many properties of central operators may be extended to orthomorphisms. In the process, one often has to replace ideals with bands, strong units with weak units, and Kakutani-Krein's $C(K)$ representations with Maeda-Ogasawara's $C^\infty(K)$ representations. Most importantly, we will show that orthomorphisms form a vector lattice and correspond to multiplication operators on X^u, when the latter is represented as $C^\infty(K)$

14.2.2 Orthomorphisms are multiplication operators

Throughout this section, X is an Archimedean vector lattice. We fix a representation of the universal completion X^u as $C^\infty(K)$, where K is an extremally disconnected compact space (the Stone space of X) as in Maeda-Ogasawara's Theorem 7.4.17. Thus, we view X as an order dense ideal in $C^\infty(K)$. If, in addition, X has a weak unit, the representation may be chosen in such a way that the weak unit becomes $\mathbb{1}$. Given $w \in C^\infty(K)$, we consider the *multiplication operator* M_w given by $f \mapsto wf$. There is a minor subtlety here: formally speaking, wf is defined "almost everywhere", but it extends in a unique way to a function in $C^\infty(K)$ by Lemma 7.4.8. It is straightforward to verify that the resulting operator $M_w\colon C^\infty(K) \to C^\infty(K)$ is linear. With a minor abuse of notation, we say that an operator $T\colon X \to X$ is a ***multiplication operator*** if there exists $w \in C^\infty(K)$ such that $Tf = wf$ for every $f \in X$. That is, X is invariant under M_w and T is the restriction of M_w to X. Clearly, if $\mathbb{1} \in X$ then $w = T\mathbb{1}$; it particular, $w \in X$.

Observe that every multiplication operator is an orthomorphism:

14.2.7 Exercise. Let $T\colon X \to X$ be a multiplication operator. Then T is an orthomorphism.

We will eventually prove the converse of the preceding exercise. This requires a few preliminary facts. The following lemma essentially says that if T preserves zeros almost everywhere then it is a multiplication operator, cf. Proposition 14.1.3.

14.2.8 Lemma. *Suppose that $\mathbb{1} \in X$, $A \subseteq K$ such that $K \setminus A$ is nowhere dense, and $T \in \mathcal{L}(X)$. Suppose that for every $f \in X_+$ and $t \in A$, if $f(t) = 0$ then $(Tf)(t) = 0$. Then T is a multiplication operator.*

Proof. First, observe that the condition "$f \in X_+$" may be strengthened to "$f \in X$" by separately considering f^+ and f^-. Put $w = T\mathbb{1}$. Fix $f \in X$. Let $t \in A$ such that $\alpha := f(t)$ is finite. Then $(f - \alpha\mathbb{1})(t) = 0$, so that $\big[T(f - \alpha\mathbb{1})\big](t) = 0$; it follows that

$(Tf)(t) = \alpha(T\mathbb{1})(t) = f(t)w(t)$. Hence, Tf agrees with wf on the set of all $t \in A$ such that $f(t)$ is finite. Since the complement of this set is nowhere dense, $Tf = wf$ in $C^\infty(K)$ by Lemma 7.4.8. $\square$

The following is a variant of Urysohn's lemma adapted to our setup.

14.2.9 Exercise. Let $f \in X_+$ and U an open non-empty subset of K. There exists $g \in X$ and a non-empty open subset V of U such that $0 \leqslant g \leqslant f$, g agrees with f on V, and g vanishes outside U.

14.2.10 Lemma. *Let $T\colon X \to X$ be a band preserving operator, and $f \in X$. If f vanishes on an open set U then so does Tf.*

Proof. Replacing U with $\overline{U}$, we may assume that U is clopen. It suffices to show that the set B of all the functions in X that vanish on U is a band in X. Suppose that $f_\alpha \to f$ for some (f_α) in B and $f \in X$. Since X is order dense, it is regular in $C^\infty(K)$; it follows that $f_\alpha \to f$ in $C^\infty(K)$. Since the restriction map $Pf = f \cdot \mathbb{1}_{|U^C}$ is a band projection on $C^\infty(K)$, it is order continuous; it follows that $f_\alpha = Pf_\alpha \to Pf$, so that $f = Pf$ and, therefore, $f \in B$. $\square$

The following exercise allows one to "glue together" continuous functions:

14.2.11 Exercise. Let $\mathcal{U}$ be a collection of clopen subsets of K such that $\bigcup\mathcal{U}$ is dense in K. For every $U \in \mathcal{U}$, let w_U be a function in $C^\infty(U)$. Suppose that w_U and w_V agree on $U \cap V$ for any $U, V \in \mathcal{U}$. Then there exists $w \in C^\infty(K)$ such that $w_u = w_{|U}$ for every $U \in \mathcal{U}$.

The next exercise shows that every weak unit in $C^\infty(K)$ may be "converted" to $\mathbb{1}$.

14.2.12 Exercise. Let $e \in C^\infty(K)_+$ be a weak unit. Then so is $1/e$, and the multiplication operator $M_e\colon C^\infty(K) \to C^\infty(K)$ is a lattice isomorphism with $M_e^{-1} = M_{1/e}$.

For every two sets A and B in a topological space, we have $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$, but the converse inclusion may fail. However, we have the following:

14.2.13 Exercise. Let U and V be two open subsets of K. Then $\overline{U \cap V} = \overline{U} \cap \overline{V}$.

We are now ready to prove the representation theorem for orthomorphisms.

14.2.14 Theorem. *Let X be an order dense sublattice of $C^\infty(K)$ and $T \in \mathcal{L}(X)$. Then T is an orthomorphism iff it is a multiplication operator, that is, there exists (a unique) $w \in C^\infty(K)$ such that $Tf = wf$ for every $f \in X$.*

Proof. Every multiplication operator is an orthomorphism by Exercise 14.2.7. To show uniqueness, suppose that $w, v \in C^\infty(K)$ such that M_w and M_v agree on X. Put $u = w - v$, then M_u vanishes on X. Suppose that $u \neq 0$. Then there exists an open set U such that $u \neq 0$ on U. Find $f \in X$ with $0 < f \leqslant \mathbb{1}_{\overline{U}}$, then $M_u f = uf \neq 0$; a contradiction.

Suppose that $T \in \mathrm{Orth}(X)$; we need to show that T is a multiplication operator. Assume first that $\mathbb{1} \in X$. Since T is order bounded, there exists $u \in X_+$ such that $T[-\mathbb{1}, \mathbb{1}] \subseteq [-u, u]$; here we consider the intervals in X rather than in $C^\infty(K)$. Let $A = \{u < \infty\}$; then $K \setminus A$ is nowhere dense. Let $f \in X_+$ and $t \in A$ such that $f(t) = 0$. By Lemma 14.2.8, it suffices to show that $(Tf)(t) = 0$.

Since Tf is continuous, it suffices to show that for every open neighborhood U of t and every $\varepsilon > 0$ there exists $s \in U$ with $\bigl|(Tf)(s)\bigr| \leqslant \varepsilon$. Fix such U and ε. Since $u(t)$ is finite, we may assume by lessening U in necessary that u is bounded by some constant M on U. Since $f(t) = 0$, by lessening U even further we may assume that $f \leqslant \frac{\varepsilon}{M}$ on U. Let g and V be as in Exercise 14.2.9. Since g agrees with f on V, Tg agrees with Tf on V by Lemma 14.2.10. Furthermore, g vanishes outside U, while on U it satisfies $0 \leqslant g \leqslant f \leqslant \frac{\varepsilon}{M}$; it follows that $0 \leqslant g \leqslant \frac{\varepsilon}{M}\mathbb{1}$. By the definition of u, we get $\bigl|Tg\bigr| \leqslant \frac{\varepsilon}{M}u$. Hence, for every $s \in V$, we have $\bigl|(Tf)(s)\bigr| = \bigl|(Tg)(s)\bigr| \leqslant \frac{\varepsilon}{M}u(s) \leqslant \varepsilon$. This completes the proof under the extra assumption that $\mathbb{1} \in X$.

We are now going to get rid of this assumption. First, we will show that we may replace it with the assumption that X has a weak unit. Indeed, let e be a weak unit in X. Then e is a weak unit in $C^\infty(K)$. In view of Exercise 14.2.12, M_e and $M_{1/e}$ are lattice isomorphisms. Put $Y = M_{1/e}(X)$ and $S \in \mathcal{L}(Y)$ defined by $S = M_{1/e}TM_{e_{|Y}}$. It is easy to see that Y is an order dense sublattice of $C^\infty(K)$, $\mathbb{1} \in Y$, and $S \in \mathrm{Orth}(Y)$. Applying the preceding argument to S and Y, we find $w \in C^\infty(K)$ such that $Sg = wg$ for all $g \in Y$. It follows that $Tf = M_e S M_{1/e} f = ew(1/e)f = wf$ for every $f \in X$.

Finally, suppose that X has no weak units. We are going to use Exercise 14.2.11. Consider the following collection of clopen sets: $\mathcal{U} = \bigl\{\overline{\operatorname{supp} f} \ : \ f \in X\bigr\}$. For every $f \in X$, the band generated by f in X consists of those functions in X that are supported on U, where $U = \overline{\operatorname{supp} f}$; see Exercise 7.4.20. We denote this band by B_U. Restricting elements of B_U to U yields an order dense sublattice of $C^\infty(U)$. Clearly, f is a weak unit in B_U. By the preceding argument, there exists w_U in $C^\infty(U)$ such that Tg agrees with $w_U g$ on U for every $g \in B_U$. Since B_U is invariant under T, we have $Tg \in B_U$, hence Tg vanishes outside U.

Suppose that $U, W \in \mathcal{U}$ with $W \subseteq U$. For every $h \in B_W$, we have $h \in B_U$, hence Th agrees with $w_U h$ on U and vanishes outside U. It follows from $Th \in B_W$ that Th agrees with $w_U h$ on W and vanishes outside W. By uniqueness of multiplication operators, it follows that the restriction w_U to W agrees with w_W.

Suppose now that $U, V \in \mathcal{U}$. Then $U = \overline{\operatorname{supp} f}$ and $V = \overline{\operatorname{supp} g}$ for some $f, g \in X$. Put $h = |f| \wedge |g|$. Then $\operatorname{supp} h = (\operatorname{supp} f) \cap (\operatorname{supp} g)$, so that $W \coloneqq \overline{\operatorname{supp} h} = U \cap V$ by Exercise 14.2.13; it follows that $W \in \mathcal{U}$. By the preceding paragraph, we conclude that both w_U and w_V agree on W as each of them agrees with w_W there.

By Exercise 14.2.11, there exists $w \in C^\infty(K)$ such that $w_U = w_{|U}$ for each $U \in \mathcal{U}$. Let $f \in X$; put $U = \overline{\operatorname{supp} f}$. Then Tf agrees with $w_U f$ and, therefore, on wf on U, and both Tf and wf vanish outside of U; hence $Tf = wf$. □

14.2.15 Remark. If, in addition, $\mathbb{1} \in X$, then $w = T\mathbb{1}$ implies that w is actually in X, not just in $C^\infty(K)$.

14.2.3 Properties of orthomorphisms

We will now use Theorem 14.2.14 to deduce several important properties of orthomorphisms. The following result is due originally to to A. Bigart, K. Keimel, P.F. Conrad, and J.E. Diem.

14.2.16 Theorem. $\operatorname{Orth}(X)$ *is a vector lattice with point-wise lattice operations:*

$$|T|x = \bigl|Tx\bigr|,\ (T \vee S)x = (Tx) \vee (Sx),\ \text{and } (T \wedge S)x = (Tx) \wedge (Sx)$$

for all $x \geqslant 0$ and $S, T \in \operatorname{Orth}(X)$. Also, $|Tx| = |T||x|$ for all $x \in X$.

Proof. Clearly, $\operatorname{Orth}(X)$ is a subspace of $\mathcal{L}(X)$. Fix $T \in \operatorname{Orth}(X)$. Represent X as an order dense sublattice of $C^\infty(K)$; let w be as in Theorem 14.2.14. For every $f \in X_+$, we have $M_{|w|}f = |w|f = |wf| = |Tf|$. It follows that X is invariant under $M_{|w|}$. Let S be the restriction of $M_{|w|}$ to X. By Theorem 14.2.14, $S \in \operatorname{Orth}(X)$. If $f \in X_+$, we have $Sf = |Tf|$ as above; it follows from Exercise 14.1.7 that $S = |T|$. Clearly, $|Tf| = |wf| = |w||f| = |T||f|$ for all $f \in X$. Thus, for every $T \in \operatorname{Orth}(X)$, $|T|$ exists and is contained in $\operatorname{Orth}(X)$. The rest now follows from Exercise 1.3.13. □

For $T \in \operatorname{Orth}(X)$, let $w \in C^\infty(K)$ be as in Theorem 14.2.14; put $\Phi(T) = w$.

14.2.17 Proposition. *The map $\Phi\colon \operatorname{Orth}(X) \to C^\infty(K)$ constructed above is a lattice isomorphic embedding and an algebra homomorphism.* $\operatorname{Range}\Phi$ *consists of those $w \in C^\infty(K)$ for which $M_w(X) \subseteq X$.*

Proof. It is easy to see that Φ is linear and one-to-one. It follows from Theorem 14.2.14 that $w \in \operatorname{Range}\Phi$ iff $M_w(X) \subseteq X$. Φ is a lattice homomorphism because if $w = \Phi(T)$ then, by Theorem 14.2.16, we have $|T|x = |Tx| = |wx| = |w|x$ for every $x \in X_+$, so that $\Phi\bigl(|T|\bigr) = |w|$. It is easy to see that composition of orthomorphisms corresponds to the composition of multiplication operators on $C^\infty(K)$, which corresponds to product of multipliers. $\square$

14.2.18 Corollary. *Any two orthomorphisms on X commute.*

14.2.19 Corollary. *For $S, T \in \operatorname{Orth}(X)$, $S \perp T$ iff $ST = 0$.*

In the special case when $X = C^\infty(K)$, for every $w \in C^\infty(K)$ the operator M_w leaves X invariant, hence Φ is onto.

14.2.20 Corollary. *The map that takes $w \in C^\infty(K)$ to M_w is a lattice isomorphism from $C^\infty(K)$ onto $\operatorname{Orth}\bigl(C^\infty(K)\bigr)$.*

As $C^\infty(K) = X^u$, the preceding corollary yields $\operatorname{Orth}(X^u) \simeq X^u$. Recall that X^δ may be identified with the ideal generated by X in X^u. We will now show that every orthomorphism on X extends uniquely to an orthomorphism on X^δ and, further, on X^u.

14.2.21 Proposition. *An operator $T \in \mathcal{L}(X)$ is an orthomorhism iff it is the restriction to X of a (unique) orthomorphism on X^u (or on X^δ) that leaves X invariant.*

Proof. Clearly, for every orthomorphism on (or on X^δ) that leaves X invariant its restriction to X is in $\operatorname{Orth}(X)$.

Let $X^u = C^\infty(K)$ and Φ be as before. Let $T \in \operatorname{Orth}(X)$; put $w = \Phi(T)$. Then M_w is an orthomorphism on X^u and $T = M_{w|X}$. Hence, T is a restriction of an orthomorphism on X^u. We claim that M_w also leaves X^δ invariant. Indeed, let $g \in X^\delta_+$. Then there exists $f \in X_+$ such that $g \leqslant f$. It follows from $|M_w g| \leqslant |M_w f| = |Tf| \in X$ that $M_w g \in X^\delta$. Hence, $M_w(X^\delta) \subseteq X^\delta$. Therefore, the restriction of M_w to X^δ is an orthomorphism on X^δ extending T.

Uniqueness of extension follows from the uniqueness clause in Theorem 14.2.14. $\square$

We get the following chain of lattice isomorphic embeddings:

$$\operatorname{Orth}(X) \hookrightarrow \operatorname{Orth}(X^\delta) \hookrightarrow \operatorname{Orth}(X^u) \simeq X^u.$$

14.2.22 Corollary. *If X has a weak unit then $\operatorname{Orth}(X)$ embeds lattice isomorphically into X (via Φ).*

Proof. Let $e \in X$ be a weak unit. We can choose the representation $C^\infty(K)$ of X^u in such a way that e becomes $\mathbb{1}$. Let $T \in \operatorname{Orth}(X)$, then $T = M_{w|X}$, where $w = \Phi(T)$. Clearly, $w = w\mathbb{1} = Te \in X$. It follows that $\operatorname{Range}\Phi \subseteq X$. $\square$

14.2.23 Exercise. If an orthomorphism T vanishes on a weak unit then $T = 0$.

14.2.24 Proposition. *Every orthomorphism is order continuous.*

Proof. Let $T \in \mathrm{Orth}(X)$. We may assume that $T \geqslant 0$; otherwise, consider T^+ and T^- (they exist and are orthomorphisms by Theorem 14.2.16). Suppose that for the sake of contradiction that $x_\alpha \downarrow 0$ but there exists $y > 0$ such that $y \leqslant Tx_\alpha$ for every α. As before, we can represent X^u as $C^\infty(K)$ and T as M_w for some $w \in C^\infty(K)_+$. Since X is order dense in $C^\infty(K)$, we have $x_\alpha \downarrow 0$ in $C^\infty(K)$.

Since $y > 0$, there exists a non-empty open set U in K and $\delta > 0$ such that $y > \delta$ on U. Passing to a smaller open set if necessary, we may assume that w is bounded by some M on U. Hence, for every $t \in U$ we have $\delta < y(t) \leqslant (Tx_\alpha)(t) = w(t)x_\alpha(t) \leqslant Mx_\alpha(t)$. It follows that $x_\alpha \geqslant \frac{\delta}{M}\mathbb{1}_{\overline{U}}$. This contradicts $x_\alpha \downarrow 0$. ☐

14.2.25 Exercise. Let $T \in \mathrm{Orth}(X)$. Then $\ker T = (\operatorname{Range} T)^d$. In particular, $\ker T$ is a band.

14.2.26 Exercise ([Huij95])**.** Let $T \in \mathrm{Orth}(X)$. TFAE:

(i) T is bijective;
(ii) T is surjective;
(iii) T is injective and $\operatorname{Range} T$ is a band.

Recall that if X is order complete then $\mathcal{L}_{\mathrm{ob}}(X)$ is a vector lattice. Clearly, in this case, the centre $\mathcal{Z}(X)$ is the principal ideal of I in $\mathcal{L}_{\mathrm{ob}}(X)$.

14.2.27 Theorem. *If X is order complete then* $\mathrm{Orth}(X)$ *is the principal band B_I of I in $\mathcal{L}_{\mathrm{ob}}(X)$.*

Proof. Let $0 \leqslant T \in \mathcal{L}_{\mathrm{ob}}(X)$. By Proposition 5.1.3, $T \in B_I$ iff $T \wedge nI \uparrow T$ in $\mathcal{L}_{\mathrm{ob}}(X)$. By Riesz-Kantorovich Theorem 6.3.2, the latter is equivalent to $(T \wedge nI)x \uparrow Tx$ for every $x \geqslant 0$. It follows from $(T \wedge nI)x \leqslant nx$ that $(T \wedge nI)x \in B_x$ and, therefore, $Tx \in B_x$. By Exercise 14.2.1(iii), we conclude that $T \in \mathrm{Orth}(X)$.

Conversely, suppose that $T \in \mathrm{Orth}(X)$. For every $x \in X_+$, it follows from $Tx \in B_x$ that $Tx \wedge nx \uparrow Tx$. By Theorem 14.2.16, we get $(T \wedge nI)x \uparrow Tx$, hence $T \in B_I$.

For a general $T \in \mathcal{L}_{\mathrm{ob}}(X)$, consider T^+ and T^-. ☐

14.2.28 Exercise ([Huij95])**.** Let $T \in \mathrm{Orth}(X)$. If T is injective then T is a weak unit in $\mathrm{Orth}(X)$. The converse is true if X is order complete.

14.2.29 Exercise. If T is an invertible orthomorphism then T^{-1} is again an orthomorphism.

The analogue of this fact for band preserving operators was proved in [AK02].

14.2.4 Orthomorphisms vs central operators

From the previous section, we know that the map $w \to M_w$ is a lattice and algebra isomorphism between $C^\infty(K)$ and $\mathrm{Orth}\bigl(C^\infty(K)\bigr)$. Clearly, $\mathbb{1}$ corresponds to I. Restricting this map yields a lattice isomorphism between the principal ideals generated by $\mathbb{1}$ in $C^\infty(K)$ and by I in $\mathrm{Orth}\bigl(C^\infty(K)\bigr)$, respectively:

14.2.30 Proposition. *The center of $C^\infty(K)$ is lattice and algebra isomorphic to $C(K)$.*

Again, throughout this section, X is an Archimedean vector lattice and K is its Stone space; we identify X^u with $C^\infty(K)$ and $\Phi\colon \mathrm{Orth}(X) \to C^\infty(K)$ is as in the preceding section. Let $T \in \mathrm{Orth}(X)$; put $w = \Phi(T)$. It is easy to see that T is central iff w is bounded, hence belongs to $C(K)$. Therefore, the restriction of Φ to $\mathcal{Z}(X)$ is a lattice isomorphic embedding of $\mathcal{Z}(X)$ into $C(K)$.

If, in addition, X is order complete then this embedding is onto, i.e., $\mathcal{Z}(X) \simeq C(K)$. Indeed, in this case X is an ideal in $C^\infty(K)$, so that if $w \in C(K)$ then for every $f \in X$ we have $\lvert M_w f\rvert = \lvert wf\rvert \leqslant \lVert w\rVert \lvert f\rvert \in X$. It follows that M_w leaves X invariant. Hence, the restriction of M_w to X is an orthomorphism on X. Moreover, it follows from $\lvert M_w f\rvert = \lvert wf\rvert \leqslant \lVert w\rVert \lvert f\rvert$ that the restriction of M_w to X is central. Hence, we get the following:

14.2.31 Proposition. *If X is order complete then the map that sends an operator in $\mathcal{Z}(X^u)$ to its restriction to X is a lattice isomorphism and an algebraic homomorphism from $\mathcal{Z}(X^u)$ onto $\mathcal{Z}(X)$.*

14.2.32 Corollary. *Suppose that X is order complete; let K be the Stone space of X. Then $\mathcal{Z}(X) \cong C(K)$. That is, the Stone space of X equals the carrier of X.*

Proof. Combining Propositions 14.2.30 and 14.2.31, we get

$$\mathcal{Z}(X) \simeq \mathcal{Z}(X^u) = \mathcal{Z}\bigl(C^\infty(K)\bigr) \simeq C(K).$$

It is clear from the proofs of those propositions that the resulting lattice isomorphism between $\mathcal{Z}(X)$ and $C(K)$ is the restriction of Φ to $\mathcal{Z}(X)$. Let $T \in \mathcal{Z}(X)$ and $w = \Phi(T)$. Then $\lvert T\rvert \leqslant \lambda I$ iff $\lvert w\rvert \leqslant \lambda\mathbb{1}$. It follows that $\lVert T\rVert_\infty$ in $\mathcal{Z}(X)$ agrees with $\lVert w\rVert_{\sup}$ in $C(K)$, hence the map above is a lattice isometry. $\square$

Note that the preceding corollary may fail without order completeness, as the carrier of a vector lattice may be trivial, while its Stone space is not.

Surprisingly, on Banach spaces, the classes of band preserving operators, orthomorphisms, and central operators coincide! The following theorem is due to Y.A.Abramovich, A.I.Veksler, and A.V.Koldunov. The proof we present is due to A.Wickstead.

14.2.33 Theorem. *Let X be a Banach lattice and $T\colon X\to X$ a linear operator. Then T is band preserving iff it is central.*

Proof. Consider first the special case when T is positive and, therefore, is order and norm bounded; in particular, it as an orthomorphism. WLOG, $\lVert T\rVert<1$. Fix any $x\in X_+$ and put $e=\sum_{k=0}^\infty T^kx$. The series converges because $\lVert T^kx\rVert\leqslant\lVert T\rVert^k\lVert x\rVert$. Clearly, $Te=\sum_{k=1}^\infty T^kx\leqslant e$. Represent the universal completion B_e^u of the principal band B_e as $C^\infty(K)$ with e becoming $\mathbb{1}$. T leaves B_e invariant and the restriction of T to B_e is an orthomorphism. Theorem 14.2.14 yields that $Tf=wf$ for all $f\in B_e$, where $w=T\mathbb{1}\leqslant\mathbb{1}$. It follows that $Tx=wx\leqslant x$. Since $x\in X_+$ is arbitrary, it follows that $T\leqslant I$ on X, hence T is central.

We now move on to the general case. Suppose that T is band preserving but not central. Then there exists a sequence (x_n) such that $\lvert Tx_n\rvert\nleqslant\lvert x_n\rvert$ for every n. Find $e\in X_+$ such that (x_n) is contained in I_e; then the restriction of T to B_e is not central. Replacing X with B_e, we may assume WLOG that X has a weak unit. Represent X^u as $C^\infty(K)$ so that $\mathbb{1}\in X$. Note that T cannot be an orthomorphism: in this case, T^+ and T^- exist and are orthomorphisms by Theorem 14.2.16, it would then follow from the *special case* that T is central.

By Lemma 14.2.8, there exists $f\in X_+$ and $t\in K$ such that $f(t)=0$ but $(Tf)(t)\neq 1$. WLOG, scaling T if necessary, we may assume that $(Tf)(t)>1$. While $f(t)=0$ means $t\notin\operatorname{supp}f$, we claim that t belongs to the boundary of $\operatorname{supp}f$. Indeed, otherwise, f vanishes on some neighborhood U of t; Lemma 14.2.10 would imply that Tf vanishes on U, which contradicts $(Tf)(t)\neq 0$.

Fix a neighborhood U of t on which $Tf>1$. Since $t\in\partial(\operatorname{supp}f)$, there exists a sequence (t_n) in $U\cap\operatorname{supp}f$ such that $r_n\coloneqq f(t_n)\to 0$. WLOG, (r_n) is strictly decreasing and $r_n<\frac{1}{(n+1)^3}$. Find a disjoint sequence (I_n) of open intervals in $\mathbb{R}_+$ such that $r_n\in I_n$; put $U_n=f^{-1}(I_n)\cap U$. Then (U_n) is a disjoint sequence of non-empty open sets in $U\cap\operatorname{supp}f$ and f is less than $r_{n-1}=\frac{1}{n^3}$ on U_n. For every n, by Exercise 14.2.9, we can find a non-empty open $V_n\subseteq U_n$ and $v_n\in X_+$ such that v_n agrees with f on V_n and vanishes outside U_n. It follows that the sequence (v_n) is disjoint. By Lemma 14.2.10, Tv_n agrees with Tf on V_n. It follows that $n^3v_n=n^3f\leqslant 1<Tf=Tv_n$ on V_n. Therefore, $(Tv_n-n^3v_n)^+>0$.

Since T is band preserving, $Tv_n\in B_{v_n}$, hence $Tv_n-n^3v_n\in B_{v_n}$ for every n. It follows that the sequence $(Tv_n-n^3v_n)$ is disjoint. Moreover, $(Tv_n-n^3v_n)^+\perp(Tv_m-n^3v_m)^-$ for all n,m. Let B be the band in X generated by $\bigl\{(Tv_m-n^3v_m)^-\,:\,m\in\mathbb{N}\bigr\}$. Then $(Tv_n-n^3v_n)^+\perp B$ for every n. Observe that $v_n\notin B$ for every n: otherwise, we would have $(Tv_n-n^3v_n)^+\in B$, which would yield $(Tv_n-n^3v_n)^+=0$ and this would be a contradiction.

Since B is norm closed in X and B is invariant under T, X/B is again a Banach lattice, and T induces an operator $\widetilde{T}\colon X/B \to X/B$, which is band preserving by Exercise 14.2.6. Since $v_n \notin B$, its image $\tilde{v}_n$ under the quotient map satisfies $\tilde{v}_n > 0$. It follows from $(Tv_n - n^3 v_n)^- \in B$ that $\widetilde{T}\tilde{v}_n \geqslant n^3\tilde{v}_n$ in X/B.

For each n, put $a_n = \frac{\tilde{v}_n}{n^2\|\tilde{v}_n\|}$ in X/B, and put $a = \sum_{n=1}^{\infty} a_n$. Since (a_n) is disjoint, we have $(a - a_n) \perp a_n$ for every n. Then $\widetilde{T}(a-a_n) \perp \widetilde{T}a_n$ and, therefore, $|\widetilde{T}a| \geqslant |\widetilde{T}a_n|$. It follows that

$$\|\widetilde{T}a\| \geqslant \|\widetilde{T}a_n\| = \frac{\|\widetilde{T}\tilde{v}_n\|}{n^2\|\tilde{v}_n\|} \geqslant \frac{n^3\|\tilde{v}_n\|}{n^2\|\tilde{v}_n\|} = n$$

for every n, which is a contradiction. □

Exercise 14.2.3 now immediately yields:

14.2.34 Corollary. *On a Banach lattice, the following classes of operators coincide: central, orthomorphisms, ideal preserving, closed ideal preserving, and band preserving.*

The following example extends Exercise 14.1.10:

14.2.35 Example. Let X be the vector lattice of all piece-wise affine functions in $C[0,1]$. We claim that $\mathrm{Orth}(X)$ is trivial in the sense that if $T \in \mathrm{Orth}(X)$ then T is a scalar multiple of I.

Indeed, since $\mathrm{Orth}(X)$ is a vector lattice, we may assume WLOG that $T \geqslant 0$; otherwise, consider T^+ and T^-. View X as a linear subspace of $C[0,1]$. Being order bounded, T is also norm bounded, hence it extends to an operator $\widetilde{T}\colon C[0,1] \to C[0,1]$.

Suppose that $f \perp g$ for $0 \leqslant f, g \in C[0,1]$. Find increasing sequences (f_n) and (g_n) in X_+ such that $f_n \to f$ and $g_n \to g$ (in norm). Then $f_n \perp g_n$, hence $Tf_n \wedge g_n = 0$ and, by continuity of lattice operations, we get $\widetilde{T}f \wedge g$. Therefore, $\widetilde{T}$ is an orthomorphism.

It follows from Theorem 14.2.33 that $\widetilde{T}$ and, therefore, T itself is central. By Exercise 14.1.10, T is a scalar multiple of I.

The following extends Exercise 14.1.22. It asserts that if X is order complete then there are "sufficiently many" central operators (and, therefore, orthomorphisms).

14.2.36 Exercise. Let X be order complete and $e \in X_+$. If $0 \leqslant w \leqslant e$ then there exists $T \in \mathcal{L}(X)$ such that $0 \leqslant T \leqslant I$ and $Te = w$. Hence, for every $0 \leqslant w \in I_e$ there exists $T \in \mathcal{Z}(X)_+$ with $Te = w$.

14.2.37 Exercise. In the preceding exercise, can one replace I_e with B_e and $\mathrm{Orth}(X)$ with $\mathcal{Z}(X)$?

It is easy to see that if T is a central operator on a Banach lattice X, then $\pm T \leqslant \lambda I$ for some $\lambda > 0$, then $\pm T^* \leqslant \lambda I$ on X^*, hence T^* is also central. Applying the same argument to T^*, we get the converse (going through T^{**}). In view of Theorem 14.2.33, it follows that T is an orthomorphism on X iff T^* is an orthomorphism on X^*. In Theorem 14.3.12 we will extend this result to vector lattices.

14.2.5 Orthomorphisms vs lattice homomorphisms

We know that every positive orthomorphism is a lattice homomorphism (see Exercise 14.2.2). The converse is false: the shift operator $T\colon \ell_p \to \ell_p$ defined by $Te_i = e_{i+1}$ is a lattice homomorphism but not an orthomorphism. We now study more closely the relationship between the two classes of operators.

14.2.38 Exercise. For $T \geqslant 0$, T is an orthomorphism iff $I + T$ is a lattice homomorphism.

14.2.39 Theorem (Kutateladze)**.** *Let $0 \leqslant T\colon X \to Y$ where Y is an order complete vector lattice. Then T is a lattice homomorphism iff every $S \in [0,T]$ is of the form $S = RT$ for some $R \in \mathrm{Orth}(Y)$ (or, equivalently, $0 \leqslant R \leqslant I$).*

Proof. Suppose that T is a lattice homomorphism and $0 \leqslant S \leqslant T$. It follows that $\mathrm{Range}\,T$ is a sublattice of Y and that $\ker T \subseteq \ker S$: if $Tx = 0$ then $\lvert Sx\rvert \leqslant S\lvert x\rvert \leqslant T\lvert x\rvert = \lvert Tx\rvert = 0$. Therefore, we can (well) define $R_0\colon \mathrm{Range}\,T \to Y$ via $R_0(Tx) = Sx$.

Let $z \in (\mathrm{Range}\,T)_+$. Then there exists $x \in X$ such that $z = Tx$. Since $z = \lvert z\rvert = T\lvert x\rvert$, replacing x with $\lvert x\rvert$, we may assume that $x \geqslant 0$. It follows from $R_0 z = Sx$ that $0 \leqslant R_0 z \leqslant z$. By Hahn-Banach Theorem 6.9.2, R_0 extends to an operator $R\colon Y \to Y$ with $0 \leqslant R \leqslant I$. In particular, R is an orthomorphism and $S = RT$.

Suppose now that if $0 \leqslant S \leqslant T$ then $S = RT$ for some $R \in \mathrm{Orth}(Y)$. Suppose $x \wedge y = 0$; it suffices to show that $Tx \wedge Ty = 0$. By Theorem 6.3.9, there exists $S \in [0,T]$ such that S agrees with T on I_x and $Sy = 0$. Let $R \in \mathrm{Orth}(Y)$ be such that $S = RT$. For every $z \in I_x$, we have $Tz = Sz = RTz$. Hence R agrees with I on $T(I_x)$. Then $T(I_x) \subseteq \ker(I - R)$. Since this kernel is an ideal by Exercise 14.2.25, I and R agree on the ideal generated by $T(I_x)$. It follows that

$$Tx \wedge Ty = R(Tx \wedge Ty) = RTx \wedge RTy = Tx \wedge Sy = 0$$

as $Sy = 0$. $\square$

14.3 f-algebras

A vector lattice X is said to be a ***vector lattice algebra*** or a ***Riesz algebra*** if it is also an algebra and the product of two positive vectors is positive. A vector lattice algebra X is said to be an f-algebra if $x \perp y$ implies $x \perp (yz)$ and $x \perp (zy)$ for any $x, y, z \geqslant 0$.

14.3.1 Example. The following spaces are f-algebras under point-wise multiplication: $C(K)$, $C_0(\Omega)$, $C_b(\Omega)$, $C^\infty(K)$, $L_\infty(\mu)$, $L_0(\mu)$, $\ell_\infty(\Gamma)$, $\mathbb{R}^\Omega$, and ℓ_1. $L_1(\mathbb{R})$ with the convolution product is a vector lattice algebra but not an f-algebra.

14.3.2 Exercise. If X is a vector lattice then $\mathrm{Orth}(X)$ is an f-algebra (under the composition product).

Let X be an f-algebra and $v \in X$. Consider the map $L_v \colon X \to X$ defined by $L_v x = vx$; the operator of multiplication by v on the left.

14.3.3 Exercise. Let X be an f-algebra and $v \in X$. Then L_v is an orthomorphism.

14.3.4 Proposition. *Let X be an f-algebra. and $x, y \in X$. Then*

(i) *if $x \perp y$ then $xy = 0$;*

(ii) $|xy| = |x||y|$

(iii) $x^2 \geqslant 0$*;*

Proof. (i) If $x \perp y$ for $x, y \in X_+$ then $0 = x \wedge y = (xy) \wedge y = (xy) \wedge (xy) = xy$, so $xy = 0$. More generally, if $x \perp y$ for $x, y \in X$ then

$$xy = (x^+ - x^-)(y^+ - y^-) = x^+y^+ - x^+y^- - x^-y^+ + x^+y^+ = 0.$$

(ii) The vectors x^+y^+, x^+y^-, x^-y^+, and x^+y^+ are disjoint, so that

$$|xy| = x^+y^+ + x^+y^- + x^-y^+ + x^+y^+ = (x^+ + x^-)(y^+ + y^-) = |x||y|.$$

(iii) It follows from (i) that $x^2 = (x^+ - x^-)^2 = (x^+)^2 + (x^-)^2 \geqslant 0$. □

14.3.5 Theorem (Amemiya)**.** *Every f-algebra is commutative.*

Proof. Fix $v \in X$ and define $S, T \colon X \to X$ via $Sx = vx$ and $Tx = xv$. As in Exercise 14.3.3, S and T are orthomorphisms. Since $Sv = Tv$, we conclude that S and T agree on B_v by Exercise 14.2.23. By Proposition 14.3.4(i), both S and T vanish on $\{v\}^d = B_v^d$. Hence, S and T agree on $B_v + B_v^d$. The latter set is an order dense ideal in X by Exercise 5.1.11. Since both S and T are order continuous by Proposition 14.2.24, we conclude that $S = T$. □

14.3.6 Exercise. In our definition of an f-algebra, we assumed that X is an algebra, i.e., the multiplication operation is distributive and associative. Show that it suffices to assume that it is distributive; associativity follows automatically. (Hint: it can be proved analogously to the proof of commutativity above.)

14.3.7 Exercise. Let e be a multiplicative unit in X. Then $e \geqslant 0$ and e is a weak unit.

14.3.8 Theorem (Zaanen)**.** *Let X be an f-algebra. The map from X to* $\mathrm{Orth}(X)$ *given by $v \mapsto L_v$ is a lattice and algebra homomorphism. If, in addition, X has a multiplicative unit then this map is a bijection, hence X and* $\mathrm{Orth}(X)$ *are isomorphic as f-algebras.*

Proof. It is clear that this map is an algebra homomorphism. To see that it is a lattice homomorphism, fix $v \in X$. For every $x \in X_+$, we have $|L_v|x = |L_v x| = |vx| = |v|x = L_{|v|}x$ by Theorem 14.2.16 and Proposition 14.3.4(ii). It follows that $|L_v| = L_{|v|}$.

Suppose now that e is a multiplicative unit; hence $e \geqslant 0$ and e is a weak unit. The map is one-to-one because it $L_v = 0$ then $v = L_v e = 0$. To show that it is onto, let $T \in \mathrm{Orth}(X)$. Put $S = T - L_{Te}$. Then S is an orthomorphism. Since $Se = 0$, it follows from Exercise 14.2.23 that $S = 0$ and, therefore, $T = L_{Te}$. □

Finally, we show that unital f-algebras are just f-subalgebras of $C^\infty(K)$ containing $\mathbb{1}$.

14.3.9 Theorem. *Every f-algebra with a multiplicative unit e is lattice and algebraically isomorphic to an f-subalgebra of $C^\infty(K)$, where K is the Stone space of X, with e mapped to $\mathbb{1}$.*

Proof. Combine Theorem 14.3.8 with Proposition 14.2.17. Note that the unit e is mapped into the identity operator I in $\mathrm{Orth}(X)$, which then corresponds to the multiplication operator by $\mathbb{1}$; hence e goes to $\mathbb{1}$. □

The assumption that X is a multiplicative unit is critical in the preceding results. Indeed, consider any vector lattice X and equip it with the trivial product. This makes X into an f-algebra, yet Theorem 14.3.9 clearly fails. However, this assumption may be relaxed:

14.3.10 Exercise. Let X be an f-algebra such that for every non-zero x there exists y such that $xy \neq 0$. Then X is lattice and algebraically isomorphic to an f-subalgebra of $C^\infty(K)$.

Proof. By Theorem 14.3.8, the map $T\colon X \to \mathrm{Orth}(X)$ given by $Tv = L_v$ is a lattice and an algebra homomorphism. The assumption guarantees that T is one-to-one. The composition of T with the map Φ from Proposition 14.2.17 the required embedding of X into $C^\infty(K)$. □

Let X be an f-algebra with a multiplicative unit e, and $x \in X_+$. Since e is a weak unit, we have $x \wedge ne \uparrow x$. In fact, this convergence is uniform:

14.3.11 Proposition (de Pagter)**.** *Let X be an f-algebra with a multiplicative unit e, and $x \in X_+$. Then $x - x \wedge ne \leqslant \frac{1}{n}x^2$ for all n.*

Proof. by Theorem 14.3.9, we may assume that $X = C^\infty(K)$ and $e = \mathbb{1}$. It remains to verify the required inequality pointwise (almost everywhere), so it suffices to prove that it is valid in $\mathbb{R}$, which is an easy exercise. □

This yields an application of f-algebras to orthomorphisms. We refer the reader to 6.6.18 and 6.11.1 for the background.

14.3.12 Theorem (Wickstead)**.** *Let X be a vector lattice and $T \in \mathcal{L}_{\mathrm{ob}}(X)$. If T is an orthomorphism then its order adjoint $T^\sim$ is also an orthomorphism. The converse is true provided that $X^\sim$ separates points of X.*

Proof. Suppose that $T \in \mathrm{Orth}(X)$ and let $n \in \mathbb{N}$. Since $\mathrm{Orth}(X)$ is an f-algebra with multiplicative unit I, it follows from Proposition 14.3.11 that $T - T \wedge nI \leqslant T^2$ and, therefore. $T^\sim - T^\sim \wedge nI \leqslant (T^2)^\sim$. It follows that $T^\sim \wedge nI \uparrow T^\sim$ and, therefore, $T^\sim$ belongs to the principal band of I in $\mathcal{L}_{\mathrm{ob}}(X^\sim)$. Since $X^\sim$ is order complete, $T^\sim$ is an orthomorphism by Theorem 14.2.27.

Suppose now that $T^\sim \in \mathrm{Orth}(X^\sim)$. Then $T^{\sim\sim}$ is an orthomorphism on $X^{\sim\sim}$. Since $X^\sim$ separates points of X, X is a sublattice of $X^{\sim\sim}$ by 6.6.18. It is easy to see that T agrees with restriction of $T^{\sim\sim}$ to X. It now follows from Exercise 14.2.1(ii) that T is an orthomorphism. □

14.4 Disjointness preserving operators

Throughout this section, we assume that X and Y are Archimedean vector lattices and $T\colon X \to Y$ a linear operator. We say that T is ***disjointness preserving*** if $x \perp y$ implies $Tx \perp Ty$. By Exercise 1.6.10, a positive operator is disjointness preserving iff it is a lattice homomorphism. We will see that disjointness preserving operators may indeed be viewed as a generalization of lattice homomorphisms.

The class of disjointness preserving operators is quite broad: it includes band preserving operators and, therefore, orthomorphisms and central operators (by Exercise 14.2.1(ii)), and many more operators. We will see that this class is surprisingly "well behaved" (especially, when we also require the operator to be order bounded) and has several equivalent characterizations. However, unlike central operator or orthomorphisms, disjointness preserving operators (even order bounded ones) lack a good algebraic structure. This class is not even closed under addition, as the sum of two lattice homomorphisms need not be a lattice homomorphism; see Section 6.11.

14.4.1 Example. In Section 6.11, we considered weighted composition operators between $C(K)$ spaces or between $L_p(\mu)$ spaces i.e., operators of the form $Tf = w\cdot(f\circ\varphi)$. We observed in Section 6.11 that if the weight w is positive that T is a lattice homomorphism. Without the positivity assumption on w, T is clearly disjointness preserving. Moreover, it follows from Exercise 14.1.7 that $|T|$ exists and is given by $|T|f = |w|\cdot(f\circ\varphi)$. In particular, $|T|$ is a lattice homomorphism and $|T|f = |Tf|$ whenever $f \geqslant 0$.

14.4.2 Proposition. *TFAE:*

(i) *T is disjointness preserving;*
(ii) *$x\wedge y = 0$ implies $Tx\perp Ty$ for all x, y;*
(iii) *$\bigl|Tx\bigr| = \bigl|T|x|\bigr|$ for all x.*

Proof. (i)⇒(ii) is obvious.

(ii)⇒(iii) It follows from $x^+\wedge x^- = 0$ that $Tx^+\perp Tx^-$ and, therefore, $|Tx| = \bigl|Tx^+ - Tx^-\bigr| = \bigl|Tx^+ + Tx^-\bigr| = \bigl|T|x|\bigr|$.

(iii)⇒(i) Recall that $x\perp y$ iff $|x-y| = |x+y|$. If $x\perp y$ then $\bigl|Tx+Ty\bigr| = \bigl|T|x+y|\bigr| = \bigl|T|x-y|\bigr| = \bigl|Tx-Ty\bigr|$. Hence $Tx\perp Ty$. ☐

Example 14.2.5 shows that a disjointness preserving operator need not be order bounded. We will first focus on order bounded disjointness preserving operators. The situation is somewhat parallel to that of band preserving operators: first, the addition requirements of order boundedness yields a much "nicer" class of operators and, second, this requirement is not very restrictive as it is automatically satisfied for bounded operators on Banach lattices. We will prove that T is disjointness preserving and order bounded iff $|T|$ exists, is a lattice homomorphism, and satisfies $|Tx| = |T|x$ for all $x\geqslant 0$. Our exposition of this is based on [DM82].

14.4.3 Lemma. *If $\varphi\in X^\sim$ is disjointness preserving then φ or $-\varphi$ is a lattice homomorphism.*

Proof. Let $x \wedge y = 0$. Using Riesz-Kantorovich Formulae for $|\varphi|(x)$ and $\big|\varphi\big|(y)$ and distributive laws, we get

$$|\varphi|(x) \wedge |\varphi|(y) = \sup\Big\{\big|\varphi(u)\big| \wedge \big|\varphi(v)\big| \ : \ |u| \leqslant x,\ |v| \leqslant y\Big\} = 0$$

We conclude that $|\varphi|$ is a lattice homomorphism, hence an atom in $X^\sim$ by Exercise 6.11.1. It follows from $\varphi \in I_{|\varphi|}$ that φ is a scalar multiple of $|\varphi|$. It is now clear that $\varphi = |\varphi|$ or $\varphi = -|\varphi|$. □

14.4.4 Theorem. *TFAE:*

(i) *T is disjointness preserving and order bounded;*
(ii) *$|x| \leqslant |y|$ implies $|Tx| \leqslant |Ty|$ for all x, y.*
(iii) *$|T|$ exists and is a lattice homomorphism;*
(iv) *T^+ and T^- exist, are lattice homomorphisms, and $T^\pm x = (Tx)^\pm$ for all $x \geqslant 0$;*

Moreover, in this case, $|Tx| = |T||x|$ for all $x \in X$.

Proof. (i)⇒(ii) Suppose $|x| \leqslant |y|$. Find $u \geqslant 0$ such that $T\big[-|y|, |y|\big] \subseteq [-u, u]$. Clearly, $T(I_{|y|}) \subseteq I_u$. Fix a lattice homomorphism $\varphi \in I_u^\sim$ and define a linear functional ψ on $I_{|y|}$ via $\psi = \varphi \circ T$. Then ψ is order bounded and disjointness preserving. By Lemma 14.4.3, ψ or $-\psi$ is a lattice homomorphism and, therefore, equals $|\psi|$. This yields

$$\big|\psi(x)\big| = \big||\psi|(x)\big| = |\psi|\big(|x|\big) \leqslant |\psi|\big(|y|\big) = \big|\psi(y)\big|.$$

It follows from $\varphi\big(|Tx|\big) = \big|\varphi(Tx)\big| = |\psi(x)|$ that $\varphi\big(|Tx|\big) \leqslant \varphi\big(|Ty|\big)$ for every lattice homomorphism φ on I_u. We conclude that $|Tx| \leqslant |Ty|$ by Exercise 6.11.5.

(ii)⇒(iii) For every $x \in X_+$, we have $|Tx| = \sup\big\{|Tz| \ : \ z \in [-x, x]\big\}$. It follows from 6.3.7 that $|T|$ exists and $|T|x = |Tx|$ for every $x \geqslant 0$. Finally, $|T|$ is a lattice homomorphism because if $x \wedge y = 0$ then $|T|x \wedge |T|y = |Tx| \wedge |Ty| = 0$.

(iii)⇒(i) If $x \perp y$ then $|Tx| \wedge |Ty| \leqslant |T||x| \wedge |T||y| = |T|\big(|x| \wedge |y|\big) = 0$.

We have proved that (i)⇔(ii)⇔(iii) and, in addition, $|Tx| = |T|x$ for every $x \geqslant 0$. We will now deduce (iv). It is clear that $T^+ = \frac{1}{2}\big(T + |T|\big)$. It follows that $T^+x = \frac{1}{2}\big(Tx + |Tx|\big) = (Tx)^+$ for all $x \geqslant 0$. If $x \wedge y = 0$ then $0 \leqslant T^+x \wedge T^+y \leqslant |T|x \wedge |T|y = 0$, hence T^+ is a lattice homomorphism. The proof for T^- is similar.

(iv)⇒(ii) Clearly, $|T|$ exists. Let $x \in X$. It follows from

$$|T^+x| = T^+|x| = \big(T|x|\big)^+ \qquad \text{and} \qquad |T^-x| = T^-|x| = \big(T|x|\big)^-$$

that $T^+x \perp T^-x$. Then

$$|Tx| = \big|T^+x - T^-x\big| = |T^+x| + |T^-x| = T^+|x| + T^-|x| = |T||x|.$$

Hence, if $|x| \leqslant |y|$ then $|T||x| \leqslant |T||y|$ and, therefore, $|Tx| \leqslant |Ty|$. □

14.4.5 Remark. We proved in Theorems 14.1.9 and 14.2.16 that every central operator and every orthomorphism has a modulus, which is computed point-wise. We used representations to prove this. Alternatively, this can now be deduced from Theorem 14.4.4 without use of representations. Though, our proof of Theorem 14.4.4 invokes Exercise 6.11.5, so in order to avoid representations altogether, one has to use a proof of this exercise that does not use representations.

14.4.6 Exercise. TFAE:

(i) T is disjointness preserving and order bounded;

(ii) There exists a lattice homomorphism $S\colon X \to Y$ such that $|Tx| = Sx$ for all $x \geqslant 0$;

(iii) There exists a linear operator $R\colon X \to Y$ such that $|Tx| = R|x|$ for every $x \in X$.

In this case, $S = R = |T|$.

14.4.7 Exercise. Suppose that there exists a linear operator S such that $|Tx| = Sx$ for every $x \geqslant 0$. Does this imply that T is disjointness preserving?

14.4.8 Exercise. Let $T\colon X \to Y$ be order bounded. T is disjointness preserving iff $|x| = |y|$ implies $|Tx| = |Ty|$.

14.4.9 Exercise. Let $T\colon X \to Y$ be disjointness preserving and order bounded.

(i) $(\operatorname{Range} T^+) \perp (\operatorname{Range} T^-)$;

(ii) If T is onto then so is $|T|$.

(iii) $\ker T$ is an ideal and $\ker T = \ker|T|$;

(iv) $T^{-1}(J) = |T|^{-1}(J)$ for every ideal J in Y.

We will show that if an order bounded disjointess preserving operator is invertible, the inverse is again disjointness preserving.

14.4.10 Proposition. *Let $T\colon X \to Y$ be disjointness preserving, order bounded, and injective. Then $Tx \perp Ty$ implies $x \perp y$.*

Proof. By the preceding results, $|T|$ exists, and is an injective lattice homomorphism. It follows that $\operatorname{Range}|T|$ is a sublattice of Y and $|T|$ is a lattice isomorphism when viewed as an operator $|T|\colon X \to \operatorname{Range}|T|$. Therefore, $|T|^{-1}$ is a lattice homomorphism on $\operatorname{Range}|T|$. Then

$$\begin{aligned} Tx \perp Ty \quad &\Rightarrow \quad |Tx| \wedge |Ty| = 0 \quad \Rightarrow \quad |T||x| \wedge |T||y| = 0 \\ &\Rightarrow \quad |T|^{-1}\Big(|T||x| \wedge |T||y|\Big) = 0 \quad \Rightarrow \quad |x| \wedge |y| = 0 \quad \Rightarrow \quad x \perp y. \end{aligned}$$

□

14.4.11 Theorem. *Let $T\colon X \to Y$ be disjointness preserving, order bounded, and bijective. Then $\lvert T\rvert$ is a lattice isomorphism (and, therefore, X and Y are lattice isomorphic). Furthermore, T^{-1} is disjointness preserving and order bounded, and $\lvert T^{-1}\rvert = \lvert T\rvert^{-1}$.*

Proof. It follows from Exercise 14.4.9 that $\lvert T\rvert$ is a bijection, hence a lattice isomorphism. By the preceding proposition, T^{-1} is disjointness preserving. Fix $y \in Y_+$; find $x \in X$ with $Tx = y$. Then $y = \lvert Tx\rvert = \lvert T\rvert\lvert x\rvert$, so that $\lvert T^{-1}y\rvert = \lvert x\rvert = \lvert T\rvert^{-1}y$. Hence $\lvert T^{-1}\rvert = \lvert T\rvert^{-1}$ by Exercise 14.1.7. In particular, T^{-1} is regular and, therefore, order bounded. □

We are now going to characterize when disjointness preserving operators are order bounded. The next fact is based on results and ideas of many people, including Yu.A. Abramovich, A.I. Veksler, B. de Pagter, C.B. Huijsmans, P.McPolin, A. Wickstead, A.K. Kitover, and A.V. Koldunov.

14.4.12 Theorem. *Suppose that $T\colon X \to Y$ is disjoitness preserving. TFAE:*

(i) *T is regular;*
(ii) *T is order bounded;*
(iii) *T is sequentially uniformly continuous;*
(iv) *$0 \leqslant x_n \xrightarrow{\mathrm{u}} 0$ implies $\inf\lvert Tx_n\rvert = 0$.*

Proof. (i)⇒(ii) is trivial; (ii)⇒(i) follows from Theorem 14.4.4.

(ii)⇒(iii) because every order bounded operator is uniformly continuous; see Theorem 6.1.7.

(iii)⇒(iv) If $0 \leqslant x_n \xrightarrow{\mathrm{u}} 0$ then $Tx_n \xrightarrow{\mathrm{u}} 0$, hence there exists $e \in Y_+$ such that $\lvert Tx_n\rvert \leqslant r_n e$, where (r_n) is a sequence in $\mathbb{R}_+$ with $r_n \to 0$. It follows that $\inf\lvert Tx_n\rvert = 0$.

(iv)⇒(ii) Suppose T fails to be order bounded. Then there exists $u \geqslant 0$ such that $T[0,u] \not\subseteq \bigl[-\lvert Tu\rvert, \lvert Tu\rvert\bigr]$. It follows that there exists $x_0 \in [0,u]$ such that $\lvert Tx_0\rvert \nleqslant \lvert Tu\rvert$.

Define $x_n = \bigl\lvert x_{n-1} - \frac{u}{2^n}\bigr\rvert$ for $n \geqslant 1$. We claim that that $0 \leqslant x_n \leqslant \frac{u}{2^n}$ and $\lvert Tx_n\rvert \geqslant \lvert Tx_0\rvert - (1 - \frac{1}{2^n})\lvert Tu\rvert$ for all $n \geqslant 0$. Both inequalities can be verified by induction. Indeed they are trivially true for $n = 0$. Let $n \geqslant 1$. If $0 \leqslant x_{n-1} \leqslant \frac{u}{2^{n-1}}$ that $-\frac{u}{2^n} \leqslant x_{n-1} - \frac{u}{2^n} \leqslant \frac{u}{2^n}$ and, therefore, $x_n \leqslant \frac{u}{2^n}$. If $\lvert Tx_{n-1}\rvert \geqslant \lvert Tx_0\rvert - (1 - \frac{1}{2^{n-1}})\lvert Tu\rvert$ then, using Proposition 14.4.2(iii), we get

$$\begin{aligned}\lvert Tx_n\rvert = \bigl\lvert Tx_{n-1} - \tfrac{Tu}{2^n}\bigr\rvert &\geqslant \lvert Tx_{n-1}\rvert - \bigl\lvert \tfrac{Tu}{2^n}\bigr\rvert \\ &\geqslant \lvert Tx_0\rvert - (1 - \tfrac{1}{2^{n-1}})\lvert Tu\rvert - \tfrac{1}{2^n}\lvert Tu\rvert = \lvert Tx_0\rvert - (1 - \tfrac{1}{2^n})\lvert Tu\rvert.\end{aligned}$$

It follows that $\lvert Tx_n\rvert \geqslant \lvert Tx_0\rvert - \lvert Tu\rvert$ and, therefore, $\lvert Tx_n\rvert \geqslant \bigl(\lvert Tx_0\rvert - \lvert Tu\rvert\bigr)^+$ for all n, hence $\inf\lvert Tx_n\rvert \geqslant \bigl(\lvert Tx_0\rvert - \lvert Tu\rvert\bigr)^+ > 0$. □

14.4.13 Theorem. *Every continuous disjointness preserving operator between normed lattices is order bounded.*

Proof. Let $T\colon X \to Y$ be such an operator. If $0 \leqslant x_n \xrightarrow{\mathrm{u}} 0$ in X then $x_n \xrightarrow{\|\cdot\|} 0$, so that $Tx_n \xrightarrow{\|\cdot\|} 0$ and, therefore, $\inf|Tx_n| = 0$. By Theorem 14.4.12, T is order bounded. □

Combining the preceding theorem with Theorem 14.4.11, we get:

14.4.14 Corollary. *Suppose that X and Y are normed lattices, and there exists a continuous disjointness preserving invertible operator from X to Y. Then X and Y are lattice isomorphic.*

14.4.1 Without order boundedness assumption

In this subsection, we consider disjointness preserving operators that are not assumed to be order bounded. We will see that some results still work without order boundedness, but proofs require considerably more effort.

We proved in Proposition 14.4.10 that if T is a disjointness preserving order bounded injection that it preserves disjointness "both ways". We will show next that in some situation the order boundedness assumption in this result may be dropped.

For a linear operator $T\colon X \to Y$ between two vector lattices, A_T is the ***ideal of order boundedness*** of T, that is,

$$A_T = \bigl\{a \in X \;:\; T\bigl[-|a|, |a|\bigr] \text{ is order bounded.}\bigr\}.$$

14.4.15 Exercise. Show that A_T is an ideal. Every atom is contained in A_T.

14.4.16 Proposition ([dP84]). *Let $T\colon X \to Y$ be a disjointness preserving operator from a uniformly complete vector lattice X to a normed lattice Y. Then A_T is order dense.*

Proof. It suffices to show that A_T^d is trivial. Suppose not.

Claim: for every $0 < e \in A_T^d$, the set $T[0, e]$ fails to be norm bounded. Indeed, fix $m \in \mathbb{N}$; we will find $v \in [0, e]$ with $\|Tv\| \geqslant m$. Observe that the restriction of T to I_e is still disjointness preserving; since $e \in A_T^d$, the restriction fails to be order bounded. By Theorem 14.4.12, we find a sequence (x_n) in I_e such that $0 \leqslant x_n \xrightarrow{\mathrm{u}} 0$ in I_e but $\inf|Tx_n| > 0$. That is, $\|x_n\|_e \to 0$ but there exists $0 < y \in Y$ such that $y \leqslant |Tx_n|$ for all n. Find n sufficiently large so that $\|x_n\|_e \leqslant \frac{\|y\|}{m}$. Put $v = \frac{x_n}{\|x_n\|_e}$. It follows that $\|v\|_e = 1$, hence $v \leqslant e$. It also follows that $|Tv| = \frac{1}{\|x_n\|_e}|Tx_n| \geqslant \frac{y}{\|x_n\|_e}$ and, therefore, $\|Tv\| \geqslant \frac{\|y\|}{\|x_n\|_e} \geqslant m$. This proves the claim.

Fix $0 < e \in A_T^d$. Since A_T^d is non-atomic, we can find a disjoint sequence (u_m) such that $0 < u_m \leqslant e$ for every m. WLOG (by scaling), we may also assume that $\|u_m\|_e \to 0$. Using the Claim, for every m we find $v_m \in [0, u_m]$ such that $\|Tv_m\| \geqslant m$. Observe that (v_m) is a disjoint norm-null sequence. Since X is uniformly complete, $\bigl(I_e, \|\cdot\|_e\bigr)$ is an AM-space, so that the series $v \coloneqq \sum_{m=1}^\infty v_m$ converges in I_e. For every m, it follows from $v_m \perp (v-v_m)$ that $Tv_m \perp T(v-v_m)$ and, therefore, $|Tv| = |Tv_m|+\bigl|T(v-v_m)\bigr| \geqslant |Tv_m|$. This yields $\|Tv\| \geqslant \|Tv_m\| \geqslant m$ for all $m \in \mathbb{N}$, which is a contradiction. $\square$

14.4.17 Theorem ([Huij94]). *Let $T\colon X \to Y$ be an injective disjointness preserving operator from a uniformly complete vector lattice X to a normed lattice Y. Then $Tx_1 \perp Tx_2$ implies $x_1 \perp x_2$.*

Proof. Since $|Tx| = \bigl|T|x|\bigr|$ for every x, we may assume WLOG that x_1 and x_2 are positive. Suppose, for the sake of contradiction that $Tx_1 \perp Tx_2$ but $x_1 \wedge x_2 > 0$. Since A_T is order dense, we can find w such that $0 < w \leqslant x_1 \wedge x_2$. The restriction of T to I_w is order bounded, hence $|x| \leqslant |y|$ implies $|Tx| \leqslant |Ty|$ for all $x, y \in I_w$ by Theorem 14.4.4(ii).

By the Archimedean Property, there exists $\lambda \geqslant 1$ such that $\lambda w \nleqslant x_1 + x_2$. Put $y \coloneqq (\lambda w - x_1 - x_2)^+$, $u_1 = (\lambda w) \wedge x_1$ and $u_2 = (\lambda w) \wedge u_2$. Then $y, u_1, u_2 \in I_w$ and $y > 0$. It follows from $y \leqslant \lambda w$ that $y \wedge x_1 \leqslant u_1$. It can be easily verified (either directly or using representations of x_1, x_2, and w as continuous functions) that $y \wedge x_1 > 0$ and $(y \wedge x_1) \perp (x_1 - u_1)$. The latter yields $T(y \wedge x_1) \perp T(x_1 - u_1)$. It follows from $y \wedge x_1 \leqslant u_1$ that $\bigl|T(y \wedge x_1)\bigr| \leqslant |Tu_1| \leqslant |Tx_1| + |T(x_1 - u_1)|$. Since $T(y \wedge x_1) \perp T(x_1 - u_1)$ we conclude that $\bigl|T(y \wedge x_1)\bigr| \leqslant |Tx_1|$. Similarly, $\bigl|T(y \wedge x_2)\bigr| \leqslant |Tx_2|$. This implies $T(y \wedge x_1) \perp T(y \wedge x_2)$ On the other hand, it follows from $y \leqslant \lambda w \leqslant \lambda x_2$ that $y \wedge x_1 \leqslant \lambda(y \wedge x_2)$, so that $\bigl|T(y \wedge x_1)\bigr| \leqslant \lambda\bigl|T(y \wedge x_2)\bigr|$. This yields $T(y \wedge x_1) = 0$, which is a contradiction because $y \wedge x_1 > 0$ and T is injective. $\square$

14.4.18 Corollary. *If $T\colon X \to Y$ is a disjointness preserving invertible operator from a uniformly complete vector lattice X to a normed lattice Y then T^{-1} is again disjointness preserving.*

14.4.19 Example. *A disjointness preserving invertible operator whose inverse is not disjointness preserving,* [AA02]. As in Example 14.2.5, let X be space of all left-continuous piecewise affine functions on $[0, 1)$; let Y be the space of all left-continuous piecewise constant functions on $[0, 2)$. Let $f \in X$, then there exist $0 = t_0 < t_1 < \cdots < t_n = 1$ such that f is of the form $m_i t + b_i$ on $[t_i, t_{i+1})$. Define $Tf \in Y$ as follows: Tf equals m_i on $[t_i, t_{i+1})$ and b_i on $1 + [t_i, t_{i+1})$. T is onto because every partition of $[0, 2)$ into subintervals may be refined so that it is formed by a partition of $[0, 1)$ and its translate to $[1,2)$. It is straightforward that T is disjointness preserving. However, T^{-1} is not disjointness preserving: $\mathbb{1}_{[0,1)} \perp \mathbb{1}_{[1,2)}$ in Y, yet their preimages are not disjoint in X.

14.4.20 Theorem ([Huij94, Kol95])**.** *Let $T\colon X \to Y$ be disjointness preserving operator from a uniformly complete vector lattice X to a normed lattice Y. If T is invertible then T is order bounded.*

Proof. By Theorem 14.4.4(ii), it suffices to show that if $|x_1| \leqslant |x_2|$ in X then $|Tx_1| \leqslant |Tx_2|$. Suppose not, that is, suppose $d > 0$, where $d = \big(|Tx_1| - |Tx_2|\big)^+$. Since $|Tx_1| = \big|T|x_1|\big|$ and $|Tx_2| = \big|T|x_2|\big|$ by Proposition 14.4.2, we may assume that $0 \leqslant x_1 \leqslant x_2$.

We claim that $d \perp T\big((y-x_2)^+\big)$ for every $0 \leqslant y \in A_T$. Indeed, put $u = y \wedge x_1$ and $v = y \wedge x_2$. It follows from $x_2 - v = (x_2 - y)^+$ that $(x_2 - v) \perp (y - x_2)^+$. Furthermore, $x_1 - u = (x_1 - y)^+ \leqslant (x_2 - y)^+$ hence $(x_1 - u) \perp (y - x_2)^+$. Since T is disjointness preserving, we have

$$T(x_1 - u) \perp T\big((y-x_2)^+\big) \quad \text{and} \quad T(x_2 - v) \perp T\big((y-x_2)^+\big). \tag{14.1}$$

It follows from $0 \leqslant u \leqslant v \in A_T$ that $|Tu| \leqslant |Tv|$ by Theorem 14.4.4(ii), so that $\big(|Tu| - |Tv|\big)^+ = 0$. Using that $|a^+ - b^+| \leqslant |a - b|$ and the triangle inequality, we get

$$\begin{aligned} d = \Big|\big(|Tx_1| - |Tx_2|\big)^+ - \big(|Tu| - |Tv|\big)^+\Big| &\leqslant \Big|\big(|Tx_1| - |Tx_2|\big) - \big(|Tu| - |Tv|\big)\Big| \\ = \Big|\big(|Tx_1| - |Tu|\big) - \big(|Tx_2| - |Tv|\big)\Big| &\leqslant \big||Tx_1| - |Tu|\big| + \big||Tx_2| - |Tv|\big| \\ &\leqslant |Tx_1 - Tu| + |Tx_2 - Tv|. \end{aligned}$$

It now follows from (14.1) that $d \perp T\big((y-x_2)^+\big)$. This proves the claim.

Since T is onto, $d = Tx = |Tx| = \big|T|x|\big|$ for some $x \in X$. Put $a = |x|$; hence $a \geqslant 0$ and $d = |Ta|$. It follows from the claim that for every $0 \leqslant y \in A_T$, we have $Ta \perp T\big((y-x_2)^+\big)$. It follows from Theorem 14.4.17 that $a \perp (y-x_2)^+$. Put $b = a + x_2$. Since A_T is order dense by Theorem 14.4.16, we have $b = \sup\big\{y \,:\, 0 \leqslant y \in A_T, y \leqslant b\big\}$ by Proposition 3.2.15. It follows that

$$a = a \wedge a = a \wedge (b - x_2)^+ = \sup\big\{a \wedge (y - x_2)^+ \,:\, 0 \leqslant y \in A_T, y \leqslant b\big\} = 0.$$

This yields $d = |Ta| = 0$. $\square$

Applying Theorem 14.4.11, we conclude that, under the assumptions of the preceding theorem, T^{-1} is disjointness preserving and order bounded and $|T|$ is a lattice isomorphism.

14.4.21 Corollary. *An invertible disjointness preserving operator between Banach lattices is norm bounded.*

14.4.2 Disjointness preserving operators between $C(K)$ spaces

Let K and L be two compact spaces and $T\colon C(K) \to C(L)$ a linear operator. Recall that T is continuous iff it is order bouned. As in Section 6.11, T said to be a ***weighted composition operator*** if $Tf = w \cdot (f \circ \varphi)$, i.e., $(Tf)(t) = w(t)f\bigl(\varphi(t)\bigr)$ for some function w in $C(L)$ and $\varphi\colon L \to K$, such that φ is continuous on the set $\{w \neq 0\}$.

14.4.22 Theorem. *Every continuous disjointness preserving operator $T\colon C(K) \to C(L)$ is a weighted composition operator.*

Proof. By Theorem 14.4.4, $|T|$ exists, satisfies $|Tf| = |T|f$ for all $f \geqslant 0$, and is a lattice homomorphism. By Theorem 6.11.6, $|T|$ is a weighted composition operator, $|T|f = w \cdot (f \circ \varphi)$ for some $w \in C(L)_+$ and $\varphi\colon L \to K$, such that φ is continuous on the set $\{w \neq 0\}$.

Let $v = T\mathbb{1}$. Then $v \in C(L)$ and $|v| = |T\mathbb{1}| = |T|\mathbb{1} = w$. Let $f \in C(K)_+$, and fix $t \in K$. It follows from $\bigl|(Tf)(t)\bigr| = w(t)f\bigl(\varphi(t)\bigr)$ that $(Tf)(t) = \pm v(t)f\bigl(\varphi(t)\bigr)$. Furthermore, $\bigl|Tf + v\bigr| = \bigl|T(f + \mathbb{1})\bigr| = |T|(f + \mathbb{1}) = |Tf| + |v|$ implies that $(Tf)(t)$ and $v(t)$ have the same sign (or zero). This yields $(Tf)(t) = v(t)f\bigl(\varphi(t)\bigr)$. □

Combining this with Corollary 14.4.21, we get the following:

14.4.23 Corollary ([Jar90])**.** *If $T\colon C(K) \to C(L)$ is disjointness preserving and invertible then both T and T^{-1} are continuous weighted composition operators.*

14.4.3 Representations of disjointness preserving operators

We observed earlier that central operators and orthomorphisms may be represented as multiplication operators. Given Example 14.4.1, one may expect that order bounded disjointness preserving operators could be represented as weighted composition operators in a similar fashion. In this section, we will investigate this conjecture. We start with some notations and preliminary results.

For the rest of the section, let $T\colon X \to Y$ be an order bounded disjointness preserving operator between two (Archimedean) vector lattices. As before, we view X and Y as order dense sublattices of their universal completions realized as $C^\infty(K)$ and $C^\infty(L)$, respectively, where K and L are extremally disconnected compact spaces, as in Maeda-Ogasawara Theorem. Recall that by Theorem 14.4.4, $|T|$ exists, is a lattice homomorphism, and satisfies $|Tx| = |T|x$ for all $x \in X_+$. As before, for $w \in C^\infty(L)$, we write M_w for the multiplication operator by w on $C^\infty(L)$.

14.4.24 Theorem (Polar decomposition,[AAK92])**.** *Let $T\colon X \to Y$ be order bounded and disjointness preserving. Then $T = M_w|T|$ for some $w \in C(L)$ with $|w| = \mathbb{1}$.*

Proof. Observe that $\lvert Tx+Ty\rvert=\lvert Tx\rvert+\lvert Ty\rvert$ for all $x,y\in X_+$. It follows that, when viewed as functions, Ty and Ty have the same sign at every point of L at which they are both finite. Let

$$A_+=\bigl\{t\in L:\exists x\in X_+\ (Tx)(t)\in(0,\infty)\bigr\}\text{ and }A_-=\bigl\{t\in L:\exists x\in X_+\ (Tx)(t)\in(-\infty,0)\bigr\}.$$

From the preceding argument, we see that A_+ and A_- are disjoint open sets. Since K is extremally disconnected, their closures are clopen and, therefore, still disjoint by Exercise 14.2.13. We define w to be -1 on $\overline{A_-}$, and 1 everywhere else. It is clear that $w\in C(L)$. Let $x\in X_+$; put $y=Tx$. We claim that $y=w\lvert y\rvert$. Indeed, let $t\in L$ such that $y(t)$ is finite. If $t\in A_+$ then $y(t)>0$ and $w(t)=1$, if $t\in A_-$ then $y(t)<0$ and $w(t)=-1$, and if $t\notin A_+\cup A_-$ then $y(t)=0$. In each of the three cases, we get $y(t)=w(t)\bigl\lvert y(t)\bigr\rvert$. Since the set on which y is infinite is nowhere dense, we conclude that $y=w\lvert y\rvert$. It follows from $\lvert y\rvert=\lvert T\rvert x$ that $T=M_w\lvert T\rvert$. $\square$

Thus, if we can represent $\lvert T\rvert$ as a weighted composition operator then we also get a representation for T. This reduces the representation problem to lattice homomorphisms.

In the preceding theorem, we view M_w as an operator on $C^\infty(L)$. However, if Y is order complete, then it is an ideal in $C^\infty(L)$; it follows from $\lvert wy\rvert\leqslant\lvert y\rvert$ that M_w leaves Y invariant and induces a central operator on Y, say, V. Moreover, since lattice operations on central operators are computed pointwise, we have $\lvert V\rvert y=\lvert Vy\rvert=\lvert wy\rvert=y$ for every $y\in Y_+$; it follows that $\lvert V\rvert=I$. Thus, we get the following:

14.4.25 Corollary. *Let $T\colon X\to Y$ be disjointness preserving and order bounded, and Y order complete. Then $T=V\lvert T\rvert$, where $V\colon Y\to Y$ satisfies $\lvert V\rvert=I$ (in particular, V is central).*

Next we prove a special case of a lemma due to Hart. The reader may review Section 7.2 at this point. Recall that the components of $\mathbb{1}$ in $C(K)$ are the characteristic functions of clopen sets. These are precisely the algebraic idempotents of $C(K)$. We will denote this set $\mathcal{C}^K_{\mathbb{1}}$. Note that $\operatorname{span}\mathcal{C}^K_{\mathbb{1}}$ is a dense sublattice of $C(K)$. For $u\in C^\infty(K)$, we write $\mathbb{1}_u$ for the characteristic function of $\overline{\{u\neq 0\}}$, the closure of the support of u. It is easy to see that if $\lvert u\rvert\leqslant\lvert v\rvert$ then $\mathbb{1}_u\leqslant\mathbb{1}_v$ and if $u\perp v$ then $\mathbb{1}_u\perp\mathbb{1}_v$. The notation $x=y\sqcup z$ means $x=y+z$ and $y\perp z$. It is easy to see that in this case $\mathbb{1}_x=\mathbb{1}_y\sqcup\mathbb{1}_z$.

Recall that if X is order complete then it is an ideal in $C^\infty(K)$. Let $x\in X$ and $f\in C(K)$. It follows from $\lvert fx\rvert\leqslant\lVert f\rVert\lvert x\rvert$ that $fx\in X$. The following observation my serve as a motivation for Hart's lemma: if T is a weighted composition operator,

$T = M_w C_\varphi$, with $\varphi\colon L \to K$ continuous, and f and x are as above, then it is easy to see that $T(fx) = (C_\varphi f)(Tx)$.

14.4.26 Lemma (Hart)**.** *Suppose that X and Y are order complete. Let $T\colon X \to Y$ be an order bounded disjointness preserving operator. Then there exists a lattice and algebraic homomorphism $S\colon C(K) \to C(L)$ such that $T(fx) = (Sf)(Tx)$ for every $f \in C(K)$ and $x \in X$.*

Proof. To reduce clutter, we write Tfx instead of $T(fx)$. We write $\mathcal{B}(X)$ and $\mathcal{B}(Y)$ for the sets of all bands in X in Y respectively; since the spaces are order complete, their bands are projection bands. We define a map $\gamma\colon \mathcal{B}(X) \to \mathcal{B}(Y)$ via $\gamma(G) = B(TG)$, the band generated by $T(G)$ in Y. That is, $\gamma(G) = (TG)^{dd}$. Let $G, H \in \mathcal{B}(X)$ with $G \perp H$. Since T is disjointness preserving, we have $TG \perp TH$ and, therefore, $\gamma(G) \perp \gamma(H)$. It follows from Theorem 5.2.14 that $G+H$ and $\gamma(G)+\gamma(H)$ are bands. We claim that $\gamma(G+H) = \gamma(G)+\gamma(H)$, i.e., $B\big(T(G+H)\big) = B(TG)+B(TH)$. On one hand, it follows from $TG, TH \subseteq T(G+H)$ that $B(TG), B(TH) \subseteq B\big(T(G+H)\big)$, hence $B(TG)+B(TH) \subseteq B\big(T(G+H)\big)$. On the other hand, $B(TG)+B(TH)$ is a band containing both TG and TH and, hence $T(G+H)$ and, therefore, $B\big(T(G+H)\big)$. This proves the claim.

By Theorem 7.4.19, $\mathcal{B}(X)$ is isomorphic (as a Boolean algebra) to $\mathrm{Clop}(K)$ and, therefore, to $\mathcal{C}^K_{\mathbb{1}}$. Namely, $e \in \mathcal{C}^K_{\mathbb{1}}$ corresponds to the band $eX = \{ex : x \in X\}$. Hence, γ induces a map from $\mathcal{C}^K_{\mathbb{1}}$ to $\mathcal{C}^L_{\mathbb{1}}$; denote it by S. Note also that $S\mathbb{1}$ corresponds to $B(TX)$, hence is the characteristic function of closure of the support of $\operatorname{Range} T$. It follows from the preceding paragraph that if $e_1 \perp e_2$ in $\mathcal{C}^K_{\mathbb{1}}$ then $(Se_1) \perp (Se_2)$ and $S(e_1+e_2) = Se_1 + Se_2$.

Let $e \in \mathcal{C}^K_{\mathbb{1}}$ and $x \in X$; it follows from $x = (ex) \sqcup \big((\mathbb{1}-e)x\big)$ that $Tx = Tex \sqcup T(\mathbb{1}-e)x$. Since $Tex \in T(eX) \subseteq \gamma(eX)$ and $T(\mathbb{1}-e)x \in \gamma\big((\mathbb{1}-e)X\big)$, we conclude that Tex is the projection of Tx onto the band $\gamma(eX)$, hence $Tex = (Tx)\cdot\mathbb{1}_U$, where U is the (clopen) support of $\gamma(eX)$. That is, $Tex = (Se)(Tx)$.

By Exercise 7.2.8, S extends to a lattice homomorphism from $\operatorname{span}\mathcal{C}^K_{\mathbb{1}}$ to $\operatorname{span}\mathcal{C}^L_{\mathbb{1}}$ via $S\Big(\sum_{i=1}^n \alpha_i e_i\Big) = \sum_{i=1}^n \alpha_i Se_i$. Observe that for every $e_1, e_2 \in \mathcal{C}^K_{\mathbb{1}}$ we have $S(e_1e_2) = S(e_1 \wedge e_2) = Se_1 \wedge Se_2 = (Se_1)(Se_2)$. It can now be easily verified that S is an algebraic homomorphism. If $0 \leqslant f \in \operatorname{span}\mathcal{C}^K_{\mathbb{1}}$, $f = \sum_{i=1}^n \alpha_i e_i$, and $x \in X_+$, we have

$$Tfx = \sum_{i=1}^n \alpha_i Te_i x = \sum_{i=1}^n \alpha_i (Se_i)(Tx) = (Sf)(Tx).$$

Since S is clearly a a contraction with respect to the sup-norm, it extends to a lattice and algebraic homomorphism from $C(K)$ to $C(L)$.

Fix $f \in C(K)$ and $x \in X$; we will show that $Tfx = (Sf)(Tx)$. Find (f_n) in $\operatorname{span} \mathcal{C}^K_{\mathbb{1}}$ such that $\|f_n - f\| \leqslant \frac{1}{n}$ and, therefore, $|f_n - f| \leqslant \frac{1}{n}\mathbb{1}$. It follows that $\bigl|(f_n - f)x\bigr| \leqslant \frac{1}{n}|x|$. Using Theorem 14.4.4(ii), we conclude that $\bigl|Tf_n x - Tfx\bigr| \leqslant \frac{1}{n}\bigl|Tx\bigr|$. It follows that $(Tf_n x)$ converges uniformly to Tfx in Y. On the other hand, $Tf_n x = (Sf_n)(Tx)$ for every n. It follows from $Sf_n \to Sf$ in the sup-norm that $(Sf_n)(Tx) \to (Sf)(Tx)$ uniformly in Y. We conclude that $Tfx = (Sf)(Tx)$. □

14.4.27 Remark. It was observed in the proof that $S\mathbb{1} = \mathbb{1}_Q$, where Q is the closure of the support of $\operatorname{Range} T$, that is, $Q = \overline{\bigcup_{x \in X} \operatorname{supp} Tx}$. It now follows from Theorem 6.11.6 that there exists function $\varphi\colon L \to K$ such that φ is continuous on Q and $Sf = \mathbb{1}_Q(f \circ \varphi)$ for every $f \in C(K)$ (the values of φ on $L \setminus Q$ are irrelevant).

We now pass to weighted composition representations. Theorem 14.4.22 asserts that every order bounded disjointness preserving operator $T\colon C(K) \to C(L)$ is a weighted composition operator. That is, there exist $w \in C(L)$ and $\varphi\colon L \to K$ continuous on $\{w \neq 0\}$ such that $Tf = w \cdot (f \circ \varphi)$ for every $f \in C(K)$. Let's tentatively denote such an operator by $W_{w,\varphi}$. If φ is continuous on all of L then $W_{w,\varphi} = M_w C_\varphi$, the product of a multiplication operator and a composition operator. However, if we do not assume that φ is continuous on all of L then C_φ is not even defined. Note also that if $v \in C(K)$ then

$$W_{w,\varphi} M_v f = w \cdot \bigl((vf) \circ \varphi\bigr) = w \cdot \bigl(v \circ \varphi\bigr) \cdot \bigl(f \circ \varphi\bigr) = W_{w\cdot(v\circ\varphi),\varphi} f, \tag{14.2}$$

so that $W_{w,\varphi} M_v = W_{w\cdot(v\circ\varphi),\varphi}$. That is, a multiplication operator on the right is "absorbed" into the weighted composition operator.

Suppose now that $T\colon X \to Y$ be an operator between vector lattices; represent X^u and Y^u as $C^\infty(K)$ and $C^\infty(L)$ as before. In Theorem 14.2.14 we showed that T is an orthomorphism iff it can be represented as a multiplication operator. It is now natural to conjecture that maybe T is order bounded disjointness preserving iff it can be represented as a weighted composition operator?

We start with some good news. First, by Polar Decomposition Theorem 14.4.24, it suffices to find a weighted composition representation for $|T|$, so we may often assume WLOG that T is a lattice homomorphism. Second, we can usually replace Y with $C^\infty(L)$. This does not interfere with the previous step, because $|T|$ is computed pointwise and, therefore, is not affected by replacing Y with $C^\infty(L)$ as the co-domain.

But there are also some bad news. We have to overcome some subtle pitfalls in order to even define a "weighted composition operator" in this setting! What exactly do we mean by an expression of the form $w \cdot (f \circ \varphi)$? Let's first look at just composition operators. Let $\varphi\colon L \to K$ be a continuous function and $f \in C^\infty(K)$. Then $f \circ \varphi$ may happen to be infinite on a non-negligible set! For example, if $\operatorname{Range}\varphi$ consists of a single point, and f takes the value ∞ at this point, then $f\circ\varphi$ is identically infinity. So the map $f\mapsto f\circ\varphi$ need not define an operator on $C^\infty(K)$. However, it may happen that $x\circ\varphi$ is in $C^\infty(K)$ for every $x\in X$. In this case, it clearly defines a lattice homomorphism from X to $C^\infty(L)$.

14.4.28. Suppose now that $w\in C^\infty(L)$ and $\varphi\colon L\to K$ is continuous on $\{w\neq 0\}$. Let $f\in C^\infty(K)$. Is $w\cdot(f\circ\varphi)$ defined in $C^\infty(L)$? Generally no, because $f\circ\varphi$ may be infinite on a non-negligible set, so the product $w\cdot(f\circ\varphi)$ is not defined in $C^\infty(L)$. Even requiring that the set $\{x\circ\varphi=\pm\infty\}$ is contained in $\{w=0\}$ and setting the product to zero there does not resolve the issue because the resulting function may fail to be continuous. Put $Q=\overline{\{w\neq 0\}}$. Then Q is clopen and $Q^C\subseteq\{w=0\}$. Suppose that φ is continuous on Q and $x\circ\varphi$ is finite almost everywhere on Q. Then we can define $w\cdot(f\circ\varphi)$ to be zero on Q^C, while on Q we define it as a product in $C^\infty(Q)$. This yields a function in $C^\infty(L)$, which we will denote by $w\cdot(f\circ\varphi)$.

Suppose, that $w\cdot(x\circ\varphi)$ is in $C^\infty(L)$ for every $x\in X$, with the product defined as in the previous paragraph. The map $T\colon x\mapsto w\cdot(x\circ\varphi)$ may be viewed as a ***weighted composition operator*** on X; we then again write $T=W_{w,\varphi}$. This operator clearly preserves disjointness. Can every order bounded disjointness preserving operator represented in this way? The following example shows that the answer is "no".

14.4.29 Example ([DvR17])**.** Let $K=\beta\mathbb{N}$, the Stone-Čhech compactification of $\mathbb{N}$; see page 35. Define $u\in C(\beta\mathbb{N})$ via $u(n)=\frac{1}{n}$ when $n\in\mathbb{N}$ and $u(n)=0$ otherwise. Define $v\in C^\infty(\beta\mathbb{N})$ via $v(n)=n$ when $n\in\mathbb{N}$ and $v(n)=\infty$ otherwise. It follows from $(uv)(n)=1$ for all $n\in\mathbb{N}$ that $uv=\mathbb{1}$ in $C^\infty(\beta\mathbb{N})$.

Fix $k\in\beta\mathbb{N}\setminus\mathbb{N}$. Let X be set of $x\in C^\infty(\beta\mathbb{N})$ such that $(vx)(k)$ is finite. That is, $x\in X$ iff the sequence $\bigl(nx(n)\bigr)_{n\in\mathbb{N}}$ is bounded. It is easy to see that X is an ideal in $C^\infty(\beta\mathbb{N})$. It is order dense because $\mathbb{1}_{\{n\}}\in X$ for every $n\in\mathbb{N}$. Note that $u\in X$ and $x(k)=0$ for all $x\in X$.

Let $Y=\mathbb{R}=C^\infty\bigl(\{0\}\bigr)$. Define $T\colon X\to Y$ via $Tx=(vx)(k)$. It is a straightforward verification that T is a lattice homomorphism. It is non-zero as $Tu=1$. We will show that T can not be represented as a weighted composition operator. Suppose, for the sake of contradiction, that there exist $w\in\mathbb{R}$ and $\varphi\colon\{0\}\to\beta\mathbb{N}$ such that $Tx=w(x\circ\varphi)$ for every $x\in X$. We claim that $\varphi(0)=k$. Indeed, suppose not. Then there exists a clopen set $U\subset\beta\mathbb{N}$ such that $k\in U\not\ni\varphi(0)$. It follows from $vu\mathbb{1}_U=\mathbb{1}_U$ that $u\mathbb{1}_U\in X$ and $T(u\mathbb{1}_U)=(vu\mathbb{1}_U)(k)=\mathbb{1}_U(k)=1$. On the other hand, $w\bigl((u\mathbb{1}_U)\circ\varphi\bigr)=0$; a contradiction. This proves that $\varphi(0)=k$. It follows that $Tx=wx(k)=0$ for every $x\in X$, hence $T=0$; this is again a contradiction.

We now proceed with some positive results. By Corollary 6.11.7, every lattice homomorphism from $C(K)$ to $C(L)$ that maps $\mathbb{1}$ to $\mathbb{1}$ (more precisely, $\mathbb{1}_K$ to $\mathbb{1}_L$) is a composition operator. We now prove an analogous result in the new setting.

14.4.30 Theorem. *Let $T\colon X \to Y$ be lattice homomorphism, and suppose that $\mathbb{1} \in X$ and $T\mathbb{1} = \mathbb{1}$. Then there exist a continuous $\varphi\colon L \to K$ such that $Tx = x \circ \varphi$ for all $x \in X$.*

Proof. By Theorem 6.9.7, T extends to a lattice homomorphism from X^δ to Y^δ. Replacing X and Y with their order completions, we may assume that X and Y are order complete. The lattice homomorphism $S\colon C(K) \to C(L)$ in Hart's Lemma 14.4.26 satisfies $\mathbb{1} = T\mathbb{1} = T(\mathbb{1}\cdot\mathbb{1}) = (S\mathbb{1})(T\mathbb{1}) = S\mathbb{1}$. As in Remark 14.4.27, there exists a continuous map $\varphi\colon L \to K$ such that $Sf = f \circ \varphi$ for every $f \in C(K)$.

Let $x \in X$. Assume first that $x \geqslant \mathbb{1}$, and put $f = \frac{1}{x}$. Then $f \in C(K)_+$, so that $\mathbb{1} = T\mathbb{1} = Tfx = (Sf)(Tx) = (f\circ\varphi)(Tx)$. It follows that $Tx = (f \circ \varphi)^{-1} = x \circ \varphi$.

Suppose now that $x \geqslant 0$. Applying the preceding paragraph to $\mathbb{1} + x$, we get $\mathbb{1} + Tx = T(\mathbb{1} + x) = (\mathbb{1} + x) \circ \varphi = \mathbb{1} + (x \circ \varphi)$; it follows that $Tx = x \circ \varphi$. Finally, for a general $x \in X$, we have $Tx = Tx^+ - Tx^- = x^+ \circ \varphi - x^- \circ \varphi = x \circ \varphi$. $\square$

Recall that if u is a weak unit in X then it is also a weak unit in $C^\infty(K)$, the function $1/u$ is well defined, is also a weak unit, and the multiplication operator M_u is a lattice isomorphism on $C^\infty(K)$ with $M_u^{-1} = M_{1/u}$; see Exercise 14.2.12.

14.4.31 Theorem ([AAK92])**.** *Let $T\colon X \to Y$ be an order bounded disjointness preserving operator, and suppose that* $\operatorname{Range} T \subseteq B_{Tu}$ *for some $u \in X$. Then there exist $w \in B_{Tu}$, a weak unit e in $C^\infty(K)$, and a continuous function $\varphi\colon L \to K$ such that $Tx = w \cdot \big((ex) \circ \varphi\big)$ for all $x \in X$.*

Proof. As before, we may assume WLOG that $Y = C^\infty(L)$ and, replacing T with $\lvert T\rvert$ we may assume that T is a lattice homomorphism. Using $\lvert Tu\rvert = \lvert T\rvert\lvert u\rvert$, we may assume that $u \geqslant 0$ be replacing u with $\lvert u\rvert$. Put $v = Tu$.

Suppose first u and v are weak units in $C^\infty(K)$ and $C^\infty(L)$, respectively. Define $X' = M_{u^{-1}}(X)$, $Y' = M_{v^{-1}}(Y)$, and $T\colon X' \to Y'$ via $T' = M_{v^{-1}}TM_u$, so that the following diagram is commutative:

$$\begin{tikzcd}
X \arrow[r, "T"] & Y \\
X' \arrow[u, "M_u"] \arrow[r, "T'"] & Y' \arrow[u, "M_v"']
\end{tikzcd}$$

It is east to see that $T'\mathbb{1} = \mathbb{1}$. By Theorem 14.4.30, there exists a continuous function $\varphi\colon K \to L$ such that $Tx = x \circ \varphi$ for every $x \in X'$. It follows that for every $x \in X$ we have $Tx = v \cdot \big((u^{-1}x) \circ \varphi\big)$, so we are done.

The case when u and v are no longer weak units can be reduced to the preceding special case as follows. Extending T to X^δ, by Theorem 6.9.7, if necessary, we may assume that X is order complete. Then $X = B_u \oplus B_u^d$. If $x \perp u$ for some $x \in X$ then $Tx \perp Tu$ implies $Tx \perp (\operatorname{Range} T)$, hence $x = 0$; it follows that T vanishes on B_u^d. Let the clopen sets K_0 and L_0 be the supports of B_u and B_v in $C^\infty(K)$ and $C^\infty(L)$, respectively. We can view B_u and B_v as order dense ideals in $C^\infty(K_0)$ and $C^\infty(L_0)$. Then T induces a lattice homomorphism $T_0\colon B_u \to B_v$. By the preceding, there exists a continuous function $\varphi_0\colon L_0 \to K_0$ such that $T_0x = v_0 \cdot \big((u_0^{-1}x) \circ \varphi_0\big)$, for every $x \in B_u$, where we view x as an element of $C^\infty(K_0)$, and u_0 and v_0 are the restrictions of u and v to K_0 and L_0, respectively.

Let e be the function in $C^\infty(L)$ that agrees with u_0^{-1} on K_0 and is constant one on $K \setminus K_0$; then e is a weak unit in $C^\infty(K)$. Extend φ to a continuous function from L to K by setting it to some constant value on $L \setminus L_0$. We claim that $Tx = v \cdot \big((ex) \circ \varphi\big)$ for every $x \in X$. Observe that Tx vanishes on $L \setminus L_0$ by assumption, while $v \cdot \big((ex) \circ \varphi\big)$ vanishes on $L \setminus L_0$ because so does v; see 14.4.28. It is left to verify that the two functions agree on L_0. If $x \in B_u$, then on L_0 we have

$$Tx = T_0x = v_0 \cdot \big((u_0^{-1}x) \circ \varphi_0\big) = v \cdot \big((ex) \circ \varphi\big).$$

Finally, let $x \in B_u^d$. Then $Tx = 0$. For every $s \in L_0$ where $v(s)$ is finite, we get $\varphi(s) \in K_0$, so $e\big(\varphi(s)\big) = u^{-1}\big(\varphi(s)\big)$ and $x\big(\varphi(s)\big) = 0$ as x vanishes on K_0. Hence $v \cdot \big((ex) \circ \varphi\big)$ vanishes on L_0. □

In Theorem 14.4.31, we require that $\operatorname{Range} T \subseteq B_{Tu}$ for some u. This assumption is rather mild. It is obviously satisfied when $\operatorname{Range} T$ contain a weak unit of Y. It is also satisfied when X and Y are Banach lattices and X has a weak unit. Indeed, as before, we may assume WLOG that T is a lattice homomorphism. In particular, T is norm bounded. Let $u \in X_+$ be a quasi-interior point. Since I_u is dense in X, $T(I_u)$ is dense in $\operatorname{Range} T$. It is easy to see that $T(I_u) \subseteq B_{Tu}$, hence $\operatorname{Range} T \subseteq B_{Tu}$.

With a minor abuse of notation, Theorem 14.4.31 yields a representation of the form $T = M_uC_\varphi M_e$. Example 14.4.29 shows that, in general, we cannot get rid of the second multiplication operator M_e. Unlike in the case of $C(K)$-spaces in (14.2), we no can no longer absorb M_e into the weighted composition operator because while $x \circ \varphi$ and $(ex) \circ \varphi$ may be finite almost everywhere in $\overline{\{u \neq 0\}}$, this may no longer true for $e \circ \varphi$, hence the product $u \cdot (e \circ \varphi) \cdot (x \circ \varphi)$ may not be defined.

In Theorem 14.4.31, we fixed Maeda-Ogasawara representations of X and Y in $C^\infty(K)$ and $C^\infty(L)$, where K and L is the Stone spaces of X and Y, respectively. One can remove multiplication by e in the theorem if we allow changing a representation for X. While we cannot change K because K is the Stone space of X, hence is determined by X, we can change the way X is embedded into $C^\infty(K)$. As before, we may always assume WLOG that $Y = C^\infty(L)$.

14.4.32 Corollary. *Let $T\colon X \to C^\infty(L)$ be an order bounded disjointness preserving operator, and suppose that* $\operatorname{Range} T \subseteq B_{Tu}$ *for some $u \in X$. There exist a representation of X as an order dense sublattice of $C^\infty(K)$, where K is the Stone space of X, $w \in B_{Tu}$, and a continuous function $\varphi\colon L \to K$ such that $Tx = w \cdot \big(x \circ \varphi\big)$ for all $x \in X$. That is, $T = W_{w,\varphi}$.*

Proof. Fix *some* representation of X as an order dense sublattice of $C^\infty(K)$. Let w, e, and φ be as in Theorem 14.4.31 for this representation. Let $X' = M_e(X)$. Since M_e is a lattice isomorphism on $C^\infty(K)$, X' is lattice isomorphic to X and is an order dense sublattice of $C^\infty(K)$. We can now view T as a composition of $M_{e|X}\colon X \to X'$ and $W_{w,\varphi}\colon X' \to C^\infty(L)$. Thus, if we view the $M_{e|X}\colon X \to C^\infty(K)$ as a new representation of X as an order dense sublattice in $C^\infty(K)$; then T is $M_{w,\varphi}$ with respect to this representation. □

14.4.33 Theorem ([AAK92])**.** *Every order continuous disjointness preserving operator $T\colon X \to Y$ can be represented as a weighted composition operator as in Theorem 14.4.31 and Corollary 14.4.32.*

Proof. By Theorem 6.5.5, T is order bounded. It follows from $\lvert Tx\rvert = \lvert T\rvert\lvert x\rvert$ that $\lvert T\rvert$ is order continuous. Therefore, replacing T with $\lvert T\rvert$ and using the polar decomposition in Theorem 14.4.24, we may assume that T is a lattice homomorphism. By Fremlin's Theorem 7.4.36, we may assume WLOG that $X = C^\infty(K)$ and $Y = C^\infty(L)$. Recall that $\mathbb{1}$ is a weak unit in $C^\infty(K)$. For every $f \in C^\infty(K)_+$, it follows from $(n\mathbb{1} \wedge f) \uparrow f$ that $T(n\mathbb{1} \wedge f) \uparrow Tf$ and, therefore, $Tf \in B_{T\mathbb{1}}$. Thus, $\operatorname{Range} T \subseteq B_{T\mathbb{1}}$. Now apply Theorem 14.4.31. □

In $C(K)$ spaces, real-valued lattice homomorphisms are exactly the positive scalar multiples of point evaluation functionals. This is no longer true for $C^\infty(K)$ (or a sublattice of it) as point evaluations take values in $\overline{\mathbb{R}}$ rather than $\mathbb{R}$. In general, a vector lattice need not have any non-zero real-valued lattice homomorphisms. However, if they do exist, they can be characterized as follows.

14.4.34 Exercise. Every non-zero real-valued lattice homomorphism φ on X is of the form $\varphi(x) = \lambda(ex)(t_0)$ for some $\lambda \in \mathbb{R}_+$, $t_0 \in K$, and a weak unit $e \in C^\infty(K)$.

It is a natural conjecture that every order bounded disjointness preserving operator (or even every lattice homomorphism) admits a representation as in Theorem 14.4.31 and Corollary 14.4.32. There have been several papers claiming to contain counterexamples to this conjecture, however we do not find them convincing.

Chapter 15

Unbounded convergences

This chapter is incomplete.

15.1 Uo-convergence

In this section, we will discuss the so called ***unbounded order convergence*** or ***uo-convergence***, which was introduced in [Nak48, Wick77b]. Just as uniform convergence is naturally related to $C(K)$-representations of vector lattices, uo-convergence is naturally related to L_1-representations.

Let (x_α) be a net in a vector lattice X. We say that (x_α) ***uo-converges*** to x if $|x_\alpha - x| \wedge u \xrightarrow{\text{o}} 0$ for every $u \in X_+$. We write $x_\alpha \xrightarrow{\text{uo}} x$.

15.1.1 Exercise. $x_\alpha \xrightarrow{\text{uo}} x$ iff $x - x_\alpha \xrightarrow{\text{uo}} 0$ iff $|x - x_\alpha| \xrightarrow{\text{uo}} 0$.

15.1.2 Exercise. If $x_\alpha \xrightarrow{\text{o}} x$ then $x_\alpha \xrightarrow{\text{uo}} x$. If (x_α) is order bounded then $x_\alpha \xrightarrow{\text{o}} x$ iff $x_\alpha \xrightarrow{\text{uo}} x$.

In the following proofs, we will often use the fact that $x \wedge (a+b) \leqslant x \wedge a + x \wedge b$ for any $x, a, b \in X_+$; see Exercise 1.3.11(iv).

15.1.3 Exercise. Uo-limits are unique. We will occasionally write $\text{uo-lim}_\alpha x_\alpha$ for the uo-limit of (x_α).

15.1.4 Exercise. Suppose that $x_\alpha \xrightarrow{\text{uo}} x$, $y_\alpha \xrightarrow{\text{uo}} y$, and $\lambda \in \mathbb{R}$. Then

(i) $x_\alpha + y_\alpha \xrightarrow{\text{uo}} x + y$ and $\lambda x_\alpha \xrightarrow{\text{uo}} \lambda x$, that is, the uo-convergence is linear;

(ii) $|x_\alpha| \xrightarrow{\text{uo}} |x|$, so that the lattice operations are uo-continuous;

(iii) If $x_\alpha \geqslant 0$ for every α then $x \geqslant 0$, so that X_+ is uo-closed.

15.1.5 Exercise. Lattice operations are uo-continuous.

15.1.6 Exercise. For monotone nets, uo-convergence agrees with order convergence. That is, if $x_\alpha\uparrow$ and $x_\alpha \xrightarrow{\text{uo}} x$ then $x=\sup x_\alpha$.

15.1.7 Proposition. *(i) $x_\alpha \xrightarrow{\text{uo}} x$ iff $a\vee(x_\alpha\wedge b)\xrightarrow{\text{o}} a\vee(x\wedge b)$ for all $a,b\in X$.*
(ii) If $x_\alpha\geqslant 0$ for all α then $x_\alpha \xrightarrow{\text{uo}} x$ iff $x_\alpha\wedge b\xrightarrow{\text{o}} x\wedge b$ for all $b\in X$.

Proof. (i) If $x_\alpha \xrightarrow{\text{uo}} x$ and $a,b\in X$, uo-continuity of lattice operations yields $a\vee(x_\alpha\wedge b)\xrightarrow{\text{uo}} a\vee(x\wedge b)$. Since the net $\bigl(a\vee(x_\alpha\wedge b)\bigr)$ is order bounded, uo and order convergences agree for it. The converse follows from the fact that for every $u\geqslant 0$ we have

$$\begin{aligned}|x_\alpha-x|\wedge u&=\Bigl|(x-u)\vee\bigl(x_\alpha\wedge(x+u)\bigr)-x\Bigr|\\&=\Bigl|(x-u)\vee\bigl(x_\alpha\wedge(x+u)\bigr)-(x-u)\vee\bigl(x\wedge(x+u)\bigr)\Bigr|\xrightarrow{\text{o}}0.\end{aligned}$$

(ii) Again, if $x_\alpha \xrightarrow{\text{uo}} x$ then uo-continuity of lattice operations yields that $x_\alpha\wedge b\xrightarrow{\text{uo}} x\wedge b$. Since this net is in $[0,b]$, we have $x_\alpha\wedge b\xrightarrow{\text{o}} x\wedge b$. Conversely, suppose that $x_\alpha\wedge b\xrightarrow{\text{o}} x\wedge b$. For every $a\in X$, order continuity of lattice operations implies $a\vee(x_\alpha\wedge b)\xrightarrow{\text{o}} a\vee(x\wedge b)$. Now apply (i). $\square$

15.1.8 Exercise. On $L_p(\mu)$ spaces (with $0\leqslant p\leqslant\infty$), the uo-convergence of sequences agrees with the almost everywhere (a.e.) convergence. Hence, one may think of uo-convergence as a generalization of a.e. convergence to abstract vector lattices.

15.1.9 Exercise. The uo-convergence agrees with the coordinate-wise convergence on $\mathbb{R}^{\mathbb{N}}$, on ℓ_p with $0<p\leqslant\infty$, and on c_0.

15.1.10 Example. The sequence (e_n) in c_0 fails to converge in order, but uo-converges to zero.

We proved in Exercise 3.2.1 that a sublattice Y of X is regular iff $x_\alpha\xrightarrow{\text{o}}0$ in Y implies $x_\alpha\xrightarrow{\text{o}}0$ in X. We will see that things are even better for uo-convergence. We write $x_\alpha\xrightarrow[Y]{\text{uo}}x$ to indicate that the net uo-converges to x in Y. In general, $x_\alpha\xrightarrow[Y]{\text{uo}}x$ does not imply $x_\alpha\xrightarrow[X]{\text{uo}}x$; the converse is also false in general. We will show that both implications hold exactly when Y is regular. Of course, WLOG, we may assume that $x=0$; otherwise replace x_α with $x_\alpha-x$.

15.1.11 Theorem. *Let Y be a sublattice of an Archimedean vector lattice X. TFAE:*

(i) *Y is regular;*
(ii) $x_\alpha\xrightarrow[Y]{\text{uo}}0 \quad\Rightarrow\quad x_\alpha\xrightarrow[X]{\text{uo}}0$ *for every net (x_α) in Y;*
(iii) $x_\alpha\xrightarrow[Y]{\text{uo}}0 \quad\Leftrightarrow\quad x_\alpha\xrightarrow[X]{\text{uo}}0$ *for every net (x_α) in Y.*

Proof. The implication (iii)⇒(ii) is trivial.

(ii)⇒(i) Suppose $x_\alpha \downarrow 0$ in Y. Passing to a tail, we may assume that the net is order bounded (in both Y and X). Using the fact that for order bounded nets the uo-convergence agrees with the order convergence, we have $x_\alpha \xrightarrow[Y]{\mathrm{uo}} 0$, which yields $x_\alpha \xrightarrow[X]{\mathrm{uo}} 0$, hence $x_\alpha \xrightarrow[X]{\mathrm{o}} 0$ and, therefore, $x_\alpha \downarrow 0$ in X.

(i)⇒(iii) Let Y be a regular sublattice of X and (x_α) a net in Y.

Suppose that $x_\alpha \xrightarrow{\mathrm{uo}} 0$ in X. Fix $u \in Y_+$. Then $|x_\alpha| \wedge u \xrightarrow{\mathrm{o}} 0$ in X; since this net is order bounded in Y, Theorem 3.2.5 yields $|x_\alpha| \wedge u \xrightarrow{\mathrm{o}} 0$ in Y. Therefore, $x_\alpha \xrightarrow{\mathrm{uo}} 0$ in Y.

Suppose now that $x_\alpha \xrightarrow[Y]{\mathrm{uo}} 0$. We want to prove that $x_\alpha \xrightarrow[X]{\mathrm{uo}} 0$. Replacing x_α with $|x_\alpha|$, we may assume WLOG that $x_\alpha \geqslant 0$ for every α.

Special case: we first prove the implication in the case when X is order complete and Y is an ideal in X. Recall that, in this case, $Y \oplus Y^d$ is an order dense ideal; see Exercise 5.1.11.

Let $y \in Y_+$ and $z \in Y^d_+$. By assumption, $x_\alpha \wedge y \xrightarrow{\mathrm{o}} 0$ in Y and, therefore in X, because Y is regular. It follows that

$$x_\alpha \wedge (y+z) \leqslant x_\alpha \wedge y + x_\alpha \wedge z = x_\alpha \wedge y \xrightarrow{\mathrm{o}} 0$$

in X. Thus, $x_\alpha \wedge v \xrightarrow[X]{o} 0$ for every positive v in $Y \oplus Y^d$.

Fix $u \in X_+$. For every positive $v \in Y \oplus Y^d$, we have $u \wedge v \in Y \oplus Y^d$, so that $x_\alpha \wedge u \wedge v \xrightarrow{\mathrm{o}} 0$ in X. Since X is order complete, by Proposition 2.4.21 we have

$$0 = \inf_\alpha \sup_{\beta \geqslant \alpha} x_\beta \wedge u \wedge v = v \wedge \inf_\alpha \sup_{\beta \geqslant \alpha} x_\beta \wedge u.$$

It follows that $\inf_\alpha \sup_{\beta \geqslant \alpha} x_\beta \wedge u \in (Y \oplus Y^d)^d = \{0\}$ by Exercise 5.1.10. This yields $\inf_\alpha \sup_{\beta \geqslant \alpha} x_\beta \wedge u = 0$, so that $x_\alpha \wedge u \xrightarrow{\mathrm{o}} 0$ and, therefore, $x_\alpha \xrightarrow[X]{\mathrm{uo}} 0$. This proves the *Special case.*

Now we will prove the general case. Let Y be a regular sublattice of X and (x_α) a positive net in Y such that $x_\alpha \xrightarrow[Y]{\mathrm{uo}} 0$. Since X is a regular sublattice of X^δ, we may view Y as a regular sublattice of X^δ. Let J be the ideal generated by Y in X^δ.

We claim that $x_\alpha \xrightarrow[J]{\mathrm{uo}} 0$. Indeed, take any $u \in J_+$. There exists $y \in Y$ such that $u \leqslant y$. It follows that $x_\alpha \wedge u \leqslant x_\alpha \wedge y$. By assumption, $x_\alpha \wedge y \xrightarrow{\mathrm{o}} 0$ in Y. Since Y is regular in X^δ, we have $x_\alpha \wedge y \xrightarrow{\mathrm{o}} 0$ in X^δ. Therefore, the net $(x_\alpha \wedge y)$ is dominated by a net (v_β) in X^δ with $v_\beta \downarrow 0$ in X^δ. Replacing v_β with $v_\beta \wedge y$, we may assume WLOG that $v_\beta \in J$. Since J is a sublattice of X^δ, $v_\beta \downarrow 0$ in X^δ yields $v_\beta \downarrow 0$ in J. Thus, the net $x_\alpha \wedge u$ in J is dominated by the net (v_β) in J and $v_\beta \downarrow 0$ in J. Therefore, $x_\alpha \wedge u \xrightarrow{\mathrm{o}} 0$ in J. This proves the claim.

It now follows from the *Special case* that $x_\alpha \xrightarrow[X^\delta]{\text{uo}} 0$. In particular, given any $u \in X_+$, we have $x_\alpha \wedge u \xrightarrow{\text{o}} 0$ in X^δ. It follows from Corollary 3.2.32 that $x_\alpha \wedge u \xrightarrow{\text{o}} 0$ in X. Therefore, $x_\alpha \xrightarrow[X]{\text{uo}} 0$.

□

15.1.12 Corollary. *Let $L_1(\mu)$ be an L_1-representation for a vector lattice X and an order continuous strictly positive functional h. Let (x_n) be a sequence in X. Then $x_n \xrightarrow{\text{uo}} 0$ in X iff $x_n \xrightarrow{\text{a.e.}} 0$ in $L_1(\mu)$.*

Proof. By Theorem 9.2.7(i), X is regular in $\widetilde{X}$. Therefore, by Theorem 15.1.11, $x_n \xrightarrow{\text{uo}} 0$ in X iff $x_n \xrightarrow{\text{uo}} 0$ in $L_1(\mu)$. Now apply Exercise 15.1.8. □

15.1.13 Theorem. *Every disjoint sequence is uo-null.*

15.1.14 Example. Let $f_n \in C[0,1]$ such that $f(0) = n$, $(\frac{1}{n})$ vanishes on $[\frac{1}{n}, 1]$, and f is linear on $[0, \frac{1}{n}]$. Then $f_n \xrightarrow{\text{uo}} 0$. Since $f_n(0) = n$. However, even after passing to a subsequence, one cannot approximate (f_n) with a disjoint sequence (in order). That is, assume, for the sake of the contradiction, that there is a subsequence (f_{n_k}) and a disjoint sequence (d_k) such that $f_{n_k} - d_k \xrightarrow{\text{o}} 0$. But since all but one of the d_k's (or none at all) must vanish at 0, we have $(f_{n_k} - d_k)(0) \to \infty$, hence the sequence $(f_{n_k} - d_k)$ is not even order bounded; a contradiction.

15.1.15 Corollary. *Let X be an Archimedean vector lattice. Uo-convergence in X agrees with order convergence iff X is finite-dimensional.*

Proof. It is easy to see that if X is finite-dimensional then the two convergences agree. Suppose that X is infinite-dimensional. Let (x_n) be a disjoint sequence of non-zero vectors. Define a double sequence $y_{n,m} = mx_n$. Since X is Archimedean, no tail of this net is order bounded, hence it fails to converge in order. We claim that it is uo-null.

WLOG, replacing X with X^δ, we may assume that X is order complete. Fix $u \in X_+$. For every n, consider the band projection of u to onto x_n; it is given by $P_{x_n}u = \sup_m(u \wedge mx_n)$. The sequence $(P_{x_n}u)$ is disjoint and order bounded by u, hence $P_{x_n}u \xrightarrow{\text{o}} 0$. On the other hand, $y_{n,m} \wedge u \leqslant P_{x_n}u$ for all n and m; it follows that $y_{n,m} \wedge u \xrightarrow{\text{o}} 0$ and, therefore, $y_{n,m} \xrightarrow{\text{uo}} 0$. □

15.1.16 Theorem. *Let (x_α) be a net in a vector lattice with a weak unit w. Then $x_\alpha \xrightarrow{\text{uo}} x$ iff $|x_\alpha - x| \wedge w \xrightarrow{\text{o}} 0$.*

15.1.17 Proposition. *A solid set is uo-closed iff it is order closed.*

Proof. The forward implication follows from the fact that every order convergent net is uo-convergent. Suppose that A is order closed, (x_α) a net in A, and $x_\alpha \xrightarrow{\mathrm{uo}} x$. Then $|x_\alpha| \xrightarrow{\mathrm{uo}} |x|$, so that $|x| \wedge |x_\alpha| \xrightarrow{\mathrm{o}} |x|$. Since A is solid, $|x| \wedge |x_\alpha| \in A$ for every α. We conclude that $|x| \in A$ and, therefore, $x \in A$. □

In Theorem 2.4.30, we characterized order convergence of order bounded nets in $C(K)$. The proof extends easily to uo-convergence:

15.1.18 Theorem. *For an net (f_α) in $C(K)_+$, $f_\alpha \xrightarrow{\mathrm{uo}} 0$ iff for every non-empty open set U and every $\varepsilon > 0$ there exists an open non-empty $V \subseteq U$ and an index α_0 such that f_α is less than ε on V whenever $\alpha \geqslant \alpha_0$.*

Proof. By Theorem 15.1.16, $f_\alpha \xrightarrow{\mathrm{uo}} 0$ iff $f_\alpha \wedge \mathbb{1} \xrightarrow{\mathrm{o}} 0$. By Theorem 2.4.30, the latter is equivalent to the following: for every non-empty open U and every $0 < \varepsilon < 1$ there exists an open non-empty $V \subseteq U$ and an α_0 such that $f_\alpha(t) \wedge 1 < \varepsilon$ for all $\alpha \geqslant \alpha_0$ and all $t \in V$. Now observe that $f_\alpha(t) \wedge 1 < \varepsilon$ iff $f_\alpha(t) < \varepsilon$. □

Recall that if K is an extremally disconnected compact space then $C^\infty(K)$ is a universally complete vector lattice. The preceding result extends to $C^\infty(K)$ spaces:

15.1.19 Theorem. *Let $X = C_\infty(K)$ and K is extremally disconnected. For an net (f_α) in $C(K)_+$, $f_\alpha \xrightarrow{\mathrm{uo}} 0$ iff for every non-empty open set U and every $\varepsilon > 0$ there exists an open non-empty $V \subseteq U$ and an index α_0 such that f_α is less than ε on V whenever $\alpha \geqslant \alpha_0$.*

Proof. Here is a sketch of the proof: just observe that the proofs of Lemma 2.4.29 and Theorem 2.4.30 remain valid in $C_\infty(K)$. □

15.1.20 Proposition. *If $f_\alpha \xrightarrow{\mathrm{uo}} f$ in $C(K)$ then f_α converges to f pointwise on a co-meagre set. The converse is true for sequences.*

Proof. The result follows from Theorem 2.4.32 and the observation that (f_α) converges to zero pointwise on some set iff $\bigl(|f_\alpha| \wedge \mathbb{1}\bigr)$ converges to zero pointwise on the set. □

15.1.21 Theorem. *[Grobler] For a sequence (x_n) in universally complete vector lattice,*

$$(x_n) \text{ uo-converges} \quad\Leftrightarrow\quad (x_n) \text{ is uo-Cauchy} \quad\Leftrightarrow\quad (x_n) \text{ o-converges} \quad\Leftrightarrow\quad (x_n) \text{ is o-Cauchy.}$$

Proof. Let X be a universally complete vector lattice. We first show that uo-convergence and order convergence agree for sequences in X. We know that order convergence implies uo-convergence, so we only need to prove that if $x_n \xrightarrow{\mathrm{uo}} x$ then $x_n \xrightarrow{\mathrm{o}} x$. WLOG, $x = 0$ and $x_n \geqslant 0$ for every n. It suffices to show that (x_n) is order bounded.

By Corollary 7.4.31, we may identify X with a $C^\infty(K)$ space for some extremally disconnected compact K. We will use Lemma 7.4.12. Fix an open non-empty set $U \subseteq K$. It follows from Theorem 15.1.19 that there exists an open non-empty $V \subseteq U$ and $m \in \mathbb{N}$ such that x_n is bounded by 1 on V for all $n \geqslant m$. Since every function in $C^\infty(K)$ is continuous and finite except on a nowhere dense set, we can find an open non-empty $W \subseteq V$ such that $x_1, \dots, x_{m-1}$ are bounded by some r on W. It follows from Lemma 7.4.12 that (x_n) is order bounded.

We have proved that uo-convergence and order convergence agree for sequences in X. A quick glance at the preceding proof shows that it remains valid for nets with finite heads; in particular, it remains valid for double sequences. We conclude that (x_n) is uo-Cauchy iff the double sequence $(x_n - x_m)$ is uo-null iff it is order null iff (x_n) is order Cauchy.

It is clear that every order convergent sequence is order Cauchy. Since X or order complete, the converse is also true by Theorem 6.10.9. □

Combining Theorems 15.1.11 and 15.1.21, we get

15.1.22 Corollary. *A sequence (x_n) in an Archimedean vector lattice X is uo-null in X iff it is order null in X^u; it is uo-Cauchy in X iff it is order convergent in X^u.*

15.1.23 Theorem. *In a discrete vector lattice, uo-convergence agrees with coordinate-wise convergence.*

Proof. Let X be a discrete vector lattice. By Proposition 5.4.19, we may view X as an order dense (hence regular) sublattice of $\mathbb{R}^A$, where A is a maximal disjoint collection of atoms in X. By Theorem 15.1.11, it suffices to show that uo-convergence agrees with point-wise convergence in $\mathbb{R}^A$. Let (x_α) be a net in $\mathbb{R}^A$; it suffices to show that $x_\alpha \xrightarrow{\text{uo}} x$ in $\mathbb{R}^A$ iff (x_α) converges to x point-wise. WLOG, $x = 0$ and $x_\alpha \geqslant 0$ for all α.

Suppose that $x_\alpha \xrightarrow{\text{uo}} 0$. Fix $a \in A$. Then $x_\alpha \wedge \mathbb{1}_{\{a\}} \xrightarrow{\text{o}} 0$. However, $x_\alpha \wedge \mathbb{1}_{\{a\}}$ is a scalar multiple of $\mathbb{1}_{\{a\}}$; namely, $x_\alpha \wedge \mathbb{1}_{\{a\}} = \big(x_\alpha(a) \wedge 1\big)\mathbb{1}_{\{a\}}$. It follows that $x_\alpha(a) \wedge 1 \to 0$, hence $x_\alpha(a) \to 0$.

Suppose now that (x_α) converges to 0 point-wise. Fix $0 \leqslant u \in \mathbb{R}^A$. We need to show that $x_\alpha \wedge u \xrightarrow{\text{o}} 0$. It suffices to show that $v_\alpha \downarrow 0$, where $v_\alpha = \sup_{\beta \geqslant \alpha}(x_\beta \wedge u)$; see Proposition 2.4.21. This follows immediately from the observation that $v_\alpha(a) \downarrow 0$ for every $a \in A$. □

The following theorem provides an equivalent definition of uo-convergence, which bypasses going through order convergence. It is somewhat similar to Theorem 15.1.18 and captures the idea of "almost everywhere convergence".

15.1.24 Theorem ([Bil23]). *For a net (x_α) in X_+, $x_\alpha \xrightarrow{\text{uo}} 0$ iff for every $u > 0$ there exists $0 < v \leqslant u$ and α_0 such that $(x_\alpha - u)^+ \perp v$ for all $\alpha \geqslant \alpha_0$.*

Proof. Suppose that $x_\alpha \xrightarrow{\text{uo}} 0$. Let $u > 0$. Then $u \wedge x_\alpha \xrightarrow{\text{o}} 0$. Let (a_γ) be a dominating net for $u \wedge x_\alpha \xrightarrow{\text{o}} 0$. It follows from $a_\gamma \downarrow 0$ then $u \nleqslant a_{\gamma_0}$ for some γ_0. We can then find α_0 such that $u \wedge x_\alpha \leqslant a_{\gamma_0}$ for all $\alpha \geqslant \alpha_0$. Then $u \wedge x_\alpha \leqslant u \wedge a_{\gamma_0} < u$. Then $v = u - u \wedge a_{\gamma_0}$. Then $0 < v \leqslant u$ and

$$v \leqslant u - u \wedge x_\alpha = (u - x_\alpha)^+ \perp (x_\alpha - u)^+ \quad \text{whenever} \quad \alpha \geqslant \alpha_0.$$

To prove the converse, take $a \in X_+$; we need to show that $a \wedge x_\alpha \xrightarrow{\text{o}} 0$. It suffices to prove that $a \wedge x_\alpha \xrightarrow{\text{o}} 0$ in X^δ by Corollary 3.2.32. By Proposition 2.4.21, it suffices to show that $b := \inf_\alpha \sup_{\beta \geqslant \alpha}(a \wedge x_\beta) = 0$ in X^δ. Suppose not; suppose that $b > 0$. Since X is order dense in X^δ, we can find $u \in X$ such that $0 < 2u \leqslant b$. By the assumption, we can find a non-zero $v \leqslant u$ and an index α_0 such that $(x_\alpha - u)^+ \perp v$ whenever $\alpha \geqslant \alpha_0$. It follows that $(a \wedge x_\alpha - u)^+ \perp v$. Passing to $\limsup$, we conclude that $(b - u)^+ \perp v$. This contradicts $b - u \geqslant u \geqslant v$. $\square$

15.1.25 Corollary. *Suppose that X has PPP; let (x_α) be a net in X_+. Then $x_\alpha \xrightarrow{\text{uo}} 0$ iff for every $u > 0$ there exists a component w of u and α_0 such that $P_w x_\alpha \leqslant w$ for all $\alpha \geqslant \alpha_0$.*

Proof. Suppose that $x_\alpha \xrightarrow{\text{uo}} 0$, and fix $u > 0$. Let v and α_0 be as in Theorem 15.1.24. Take $w = P_v u$. It is easy to see that $P_w = P_v$, so that

$$P_w x_\alpha - w = P_v x_\alpha - w = P_v(x_\alpha - u) = -P_v\big((x_\alpha - u)^-\big) \leqslant 0$$

and, therefore, $P_w x_\alpha \leqslant w$ for all $\alpha \geqslant \alpha_0$.

To prove the converse, let w and α_0 be as in the condition. Then

$$P_w\big((x_\alpha - u)^+\big) = (P_w x_\alpha - P_w u)^+ = (P_w x_\alpha - w)^+ = 0,$$

so that $(x - u)^+ \perp w$. $\square$

While we know that uo-convergence agrees with a.e. convergence for *sequences* in $L_p(\mu)$ spaces, this fails for nets; moreover, a.e. convergence makes no sense for nets (as the index set of a net need not be countable). The following is a characterization of uo-convergent nets in $L_p(\mu)$ spaces, analogous to Theorem 15.1.18:

15.1.26 Proposition. *For a positive net (f_α) in $L_p(\mu)$, $f_\alpha \xrightarrow{\text{uo}} 0$ iff for every set U with $\mu(U) > 0$ and every $\varepsilon > 0$ there exists $V \subseteq U$ with $\mu(V) > 0$ and an index α_0 such that f_α is less than or equal to ε on V for all $\alpha \geqslant \alpha_0$.*

Proof. Suppose that $f_\alpha \xrightarrow{\mathrm{uo}} 0$. Let $\mu(U) > 0$ and $\varepsilon > 0$. Put $u = \varepsilon \mathbb{1}_U$. Let w and α_0 be as in Corollary 15.1.25. Then $w = \varepsilon \mathbb{1}_V$ for some subset V of U with $\mu(V) > 0$. For every $\alpha \geqslant \alpha_0$ we have $P_w x_\alpha \leqslant w$, so that f_α is less than or equal to ε on V.

We will use Theorem 15.1.24 to prove the converse. Let $u > 0$. Find $\varepsilon > 0$ and a set U with $\mu(U) > 0$ such that $\varepsilon \mathbb{1}_U \leqslant u$. Let V and α_0 be as in the assumption. Take $v = \varepsilon \mathbb{1}_V$. Then $0 < v \leqslant u$ and it is easy to see that $(f_\alpha - u)^+$ vanishes on V, hence $(f_\alpha - u)^+ \perp v$ for all $\alpha \geqslant \alpha_0$. □

15.2 Uo-completeness

A net $(x_\alpha)_{\alpha \in \Lambda}$ in a vector lattice is ***uo-Cauchy*** if the double net $(x_\alpha - x_\beta)_{(\alpha,\beta) \in \Lambda^2}$ is uo-null. A vector lattice is ***uo-complete*** if every uo-Cauchy net is uo-convergent.

15.2.1 Theorem. *An Archimedean vector lattice is uo-complete iff it is universally complete.*

Proof. Suppose that X is uo-complete. Every order Cauchy net is eventually order bounded by Exercise 6.10.8, hence uo-Cauchy, hence uo-convergent and, therefore, by order boundedness of the tail, order convergent. Therefore, X is order complete.

Let D be a disjoint set in X_+; it suffices to show that $\sup D$ exists. Let Λ be the collection of all finite subsets of D; put $x_\alpha = \sup \alpha$. Then $x_\alpha \uparrow$. Fix $u \in X_+$. Since X is order complete, $v := \sup(D \wedge u)$ exists. Fix $\gamma \in \Lambda$; let $\alpha, \beta \geqslant \gamma$. It is easy to see that $|x_\alpha - x_\beta| \perp x_\gamma$ and $|x_\alpha - x_\beta| + x_\gamma \leqslant x_{\alpha\cup\beta}$. It follows that $|x_\alpha - x_\beta| \wedge u + x_\gamma \wedge u \leqslant x_{\alpha\cup\beta} \wedge u \leqslant v$, so that $|x_\alpha - x_\beta| \wedge u \leqslant v - x_\gamma \wedge u$. It follows from the definition of v that $(v - -x_\gamma \wedge u) \downarrow 0$. Hence, $|x_\alpha - x_\beta| \wedge u \xrightarrow{\mathrm{o}} 0$ and, therefore, the net (x_α) is uo-Cauchy. By assumption, it uo-converges to some x. Since the net is increasing, we have $x = \sup x_\alpha = \sup D$ by Exercise 15.1.6. This completes the proof that X is universally complete.

Suppose now that X is universally complete. By Maeda-Ogasawara's Theorem, we can represent it as $C^\infty(K)$ for some extremally disconnected K. Let (x_α) be a uo-Cauchy net in X; we need to show that is uo-converges. WLOG, $x_\alpha \geqslant 0$ for all α; otherwise, consider (x_α^+) and (x_α^-). Let $a \in X_+$. Then $|x_\alpha \wedge a - x_\beta \wedge a| \leqslant |x_\alpha - x_\beta|$ that the net $(x_\alpha \wedge a)$ is uo-Cauchy. Since this net is order bounded, it is order Cauchy. Hence, by Theorem 6.10.9, it converges in order to some v_a in X_+. We may view (v_a) as a net indexed by X_+.

It is clear that $a \geqslant v_a \uparrow$. Also, if $a \leqslant b$ then $a \wedge v_b = v_a$ because

$$a \wedge v_b = a \wedge \operatorname*{o-lim}_\alpha (x_\alpha \wedge b) = \operatorname*{o-lim}_\alpha (a \wedge x_\alpha \wedge b) = \operatorname*{o-lim}_\alpha (x_\alpha \wedge a) = v_a.$$

Let F be the subset of K given by $F = \bigcap_{a \in X_+} \{v_a = a\}$. Clearly, F is closed. We claim that $\operatorname{Int} F = \varnothing$. Suppose not; suppose that there exists a non-empty open set $U \subseteq F$. By Theorem 15.1.19, there exists an index α_0 and an open non-empty set $V \subseteq U$ such that $|x_\alpha - x_\beta|$ is less than 1 on V for all $\alpha, \beta \geqslant \alpha_0$. Since $x_{\alpha_0} \in C^\infty(K)$, we can find a non-empty clopen set $W \subseteq V$ such that x_{α_0} is bounded above on W by some $m \in \mathbb{N}$. It follows that x_α is bounded above by $m+1$ on W for all $\alpha \geqslant \alpha_0$. Take $a = (m+2)\mathbb{1}$, then $x_\alpha \wedge a$ equals x_α on W, so v_a is bounded by $m+1$ there. On the other hand, since $W \subseteq F$, v_a must equal a on W, which is $m+2$. This contradiction proves the claim.

Let $t \notin F$. Then there exists $a \in X_+$ such that $v_a(t) < a(t)$. For every $b \geqslant a$, it follows from $v_a = a \wedge v_b$ that $v_a(t) = a(t) \wedge v_b(t)$, hence $v_a(t) = v_b(t)$. This means that the net $\big(v_a(t)\big)$ stabilizes. Let $w(t)$ be the stabilized value: $w(t) \coloneqq v_a(t)$. It is clear that w is well defined on F^C.

Again, let $t \in F^C$ and find $a \in X_+$ such that $v_a(t) < a(t)$. There exists a neighborhood U of t such that $v_a(s) < a(s)$ for all $s \in U$ and, therefore, $w(s) = v_a(s)$. Hence, w and v_a agree on U. It follows that w is continuous on F^C. By Lemma 7.4.8, w extends uniquely to an element v of $C^\infty(K)$.

We claim that $v = \sup v_a$. Indeed, for each $t \notin F$, we can, as before, find $a \in X_+$ such that $v_a(t) < a(t)$ and, therefore, $v(t) = w(t) = v_a(t) = v_b(t)$ for all $b \geqslant a$; it follows that v agrees with the point-wise supremum of the net (v_a) outside of F. This yields that $v = \sup v_a$.

Observe that for every fixed $b \in X_+$, we have

$$b \wedge v = b \wedge \sup_{a \in X_+} v_a = b \wedge \sup_{a \geqslant b} v_a = \sup_{a \geqslant b} b \wedge v_a = \sup_{a \geqslant b} v_b = v_b$$

This yields $x_\alpha \wedge b \to v \wedge b$. It now follows from Exercise 15.1.7(ii) that $x_\alpha \xrightarrow{\text{uo}} v$. $\square$

The following corollary complements Proposition 15.1.7(ii):

15.2.2 Corollary. *Let e be a weak unit in X, and (x_α) a net in X_+. Then $x_\alpha \xrightarrow{\text{uo}} x$ iff $x_\alpha \wedge ne \xrightarrow{\text{o}} x \wedge ne$ for every $n \in \mathbb{N}$.*

Proof. The forward implication follows from Proposition 15.1.7(ii). Suppose that $x_\alpha \wedge ne \xrightarrow{\text{o}} x \wedge ne$ for every $n \in \mathbb{N}$. Since the net is order bounded, we have $x_\alpha \wedge ne \xrightarrow{\text{uo}} x \wedge ne$.

Observe that WLOG, $X = C^\infty(K)$ and $e = \mathbb{1}$. Indeed, we can identify X^u with $C^\infty(K)$, so that e corresponds to $\mathbb{1}$. Since X is order dense and, therefore, regular in X^u, uo-convergence in X agrees with that in X^u. So we can use Theorem 15.1.19 to characterize uo-convergrence.

Fix an open non-empty subset U of K and $\varepsilon \in (0,1)$. Find an open non-empty $V \subseteq U$ such that x is bounded on V. We can find $n \in \mathbb{N}$ such that $x(t) < n-1$ for all $t \in V$. Then $x \wedge n\mathbb{1}$ agrees with x on V. Since $x_\alpha \wedge ne \xrightarrow{\text{uo}} x \wedge ne$, we can find an open non-empty $W \subseteq V$ such that the difference between $x_\alpha \wedge n\mathbb{1}$ and $x \wedge n\mathbb{1}$ is less than ε on W; in particular, it is less than one. Since $x \wedge n\mathbb{1}$ agrees with x on W, $x_\alpha \wedge n\mathbb{1}$ is less than n on W and, therefore, agrees with x_α. It follows that the difference between x_α and x is less than ε on W. $\square$

For a Banach lattice X, we say that its ball B_X is uo-complete if it is complete with respect to the uo-convergence of X. That is, every net (x_α) in B_X, which is uo-Cauchy in X, is uo-convergent in X to an element of B_X. The following two theorems relate this property with monotonic completeness (Levi Property); see Definition 10.6.2. The theorem are based on [Tay18]; the proofs presented below is due to E. Bilokopytov.

15.2.3 Theorem. *For a Banach lattice X, the ball B_X is uo-complete iff X is monotonically complete and has the Fatou Property.*

Proof. Recall that since X is order dense and, therefore regular in X^u, uo-convergences in X and in X^u agree.

Suppose that B_X is uo-complete. It follows that B_X is uo-closed in X. Let (x_α) be an increasing net in B_X. We use Exercise 7.4.23 to show that (x_α) is dominable. Fix $u > 0$. WLOG, by scaling, we may assume that $\|u\| > 1$. Since B_X is uo-closed, we cannot have $u \wedge x_\alpha \uparrow u$. So there exists $0 < v \leqslant u$ such that $u \wedge x_\alpha \leqslant u - v$ for all α. It follows from

$$(x_\alpha - u)^+ \perp (u - x_\alpha)^+ = u - u \wedge x_\alpha \geqslant v$$

that $(x_\alpha - u)^+ \perp v$. This proves that (x_α) is dominable.

Therefore, (x_α) is order bounded in X^u. Being increasing and order bounded, it is order Cauchy in X^u by Proposition 6.10.5. It follows that it is uo-Cauchy in X^u and, therefore, in X. Since B_X is uo-complete, we conclude that (x_α) uo-converges in X to some x in B_X. Again, since (x_α) is increasing, we get $x_\alpha \uparrow x$.

To prove the converse, suppose that every increasing positive net in B_X has supremum in B_X. This implies that X is order complete and, therefore, is an ideal in X^u and B_X is a solid subset of X^u. Let (x_α) be a net in B_X which is uo-Cauchy in X. Then it is still uo-Cauchy in X^u. Since X^u is uo-complete, $x_\alpha \xrightarrow{\text{uo}} a$ in X^u for some $a \in X^u$. Since lattice operations are uo-continuous, we have $|x_\alpha| \xrightarrow{\text{uo}} |a|$ and, therefore, $|a| \wedge |x_\alpha| \xrightarrow{\text{o}} |a|$ in X^u. Using the definition of order convergence, we find a net (y_γ) in X^u such that $0 \leqslant y_\gamma \uparrow |a|$ in X^u and for every γ there exists an α such that $y_\gamma \leqslant |a| \wedge x_\alpha$. This implies that $y_\gamma \in B_X$. The assumption now yields that $\sup y_\alpha$

exists in X and is contained in B_X. Regularity of X in X^u implies that this supremum is $|a|$, hence $|a|$ in B_X and, therefore, $a \in X$. We clearly then have $x_\alpha \xrightarrow{\text{uo}} a$ in X. □

The next theorem shows that monotonic completeness may be interpreted as a "bounded uo-completenes":

15.2.4 Theorem. *A Banach lattice X monitonically complete iff every norm bounded uo-Cauchy sequence is uo-convergent.*

Proof. Suppose X is monotonically complete. Then X is order complete, hence X is an ideal in X^u. Let (x_α) be a norm bounded uo-Cauchy net in X. WLOG, (x_α) is contained in B_X. Since (x_α) is uo-Cauchy in X^u and X^u is uo-complete, (x_α) uo-converges in X^u to some $a \in X^u$. Then $|a| \wedge |x_\alpha| \xrightarrow{\text{o}} |a|$. We can find an increasing net (y_γ) in X^u such that for every γ there exists α such that $0 \leqslant y_\gamma \leqslant |a| \wedge |x_\alpha|$. It follows that $y_\gamma \in B_X$. Since X is monotonically complete, there exists $x \in X$ such that $y_\gamma \uparrow x$ in X and, therefore, in X^u (by regularity of X in X^u). It follows that $x = |a|$, hence $|a| \in X$. Since X is an ideal in X^u, we conclude that $a \in X$ and $x_\alpha \xrightarrow{\text{uo}} a$ in X.

Converse: TBC

□

15.3 Un-convergence

Let X be a Banach lattice. For a net (x_α) in X, we day that (x_α) ***unboundedly norm converges (un-converges)*** to x if for every $u > 0$ the net $|x_\alpha - x| \wedge u$ converges to zero in norm.

We established some properties of un-convergence in Theorems 9.6.4 and 9.6.5. In particular, for "nice" Banach Function Spaces, un-convergence agrees with convergence in measure.

15.4 Uaw-convergence

To be completed.

Here X is a Banach lattice. A net (x_α) in X is uaw-converges to x if $|x_\alpha - x| \wedge u \xrightarrow{\text{w}} 0$. Here "uaw" stands for "unbounded absolute weak convergence": for an order bounded sequence (x_k), $x_k \xrightarrow{\text{uaw}} 0$ iff $|x_k| \xrightarrow{\text{w}} 0$.

15.4.1 Exercise. If $A \subseteq X$ is weakly almost order bounded then $x_\alpha \xrightarrow{\text{uaw}} x$ implies $|x_\alpha| \xrightarrow{\text{w}} 0$ for every net (x_α) in A.

15.4.2 Proposition. *Every disjoint sequence is uaw-null.*

Proof. Follows immediately from Exercise 6.8.8(i). □

We show that in the definition of a confined set, one may replace disjoint sequences with uaw-null ones.

15.4.3 Theorem. *Let A be a bounded solid subset of X and ρ a continuous seminorm on X. TFAE:*

(i) *A is ρ-confined;*

(ii) *every uaw-null sequence in A is ρ-null;*

(iii) *every uaw-null net in A is ρ-null.*

Proof. (iii)⇒(ii) it trivial. (ii)⇒(i) because every disjoint sequence is uaw-null.

(i)⇒(iii) Suppose that A is ρ-confined. By Exercise 11.1.5, we have $\rho = \rho_B$ for some bounded set $B \subseteq X^*$. In view of Exercise 11.1.13, replacing B with $\mathrm{Sol}(B)$, we may assume that B is solid and, therefore, ρ_B is a lattice seminorm. Let (x_α) be a uaw-null sequence in A. We need to show that $x_\alpha \to 0$ uniformly on B. Fix $\varepsilon > 0$.

Since A is ρ_B-confined, by Burkinshaw-Dodds Theorem 11.3.6, A is ρ_B-almost order bounded and B is ρ_A-almost order bounded. Hence, we can find $u \in X_+$ and $g \in X^*_+$ such that $A \subseteq [-u,u] + \varepsilon B_{\rho_B}$ and $B \subseteq [-g,g] + \varepsilon B_{\rho_A}$. It follows from $x_\alpha \xrightarrow{\text{uaw}} 0$ that $g\bigl(\lvert x_\alpha\rvert \wedge u\bigr) \to 0$. Find α_0 such that $\Bigl\lvert g\bigl(\lvert x_\alpha\rvert \wedge u\bigr)\Bigr\rvert < \varepsilon$ whenever $\alpha \geqslant \alpha_0$. Let $\alpha \geqslant \alpha_0$ and $f \in B$. Then

$$\bigl\lvert f(x_\alpha)\bigr\rvert \leqslant \lvert f\rvert\bigl(\lvert x_\alpha\rvert\bigr) \leqslant \lvert f\rvert\Bigl(\bigl(\lvert x_\alpha\rvert - u\bigr)^+\Bigr) + \lvert f\rvert\bigl(\lvert x_\alpha\rvert \wedge u\bigr)$$

It follows from $\lvert f\rvert \in B$ and $\bigl(\lvert x_\alpha\rvert - u\bigr)^+ \in \varepsilon B_{\rho_B}$ that $\lvert f\rvert\Bigl(\bigl(\lvert x_\alpha\rvert - u\bigr)^+\Bigr) \leqslant \varepsilon$. Therefore,

$$\begin{aligned}\bigl\lvert f(x_\alpha)\bigr\rvert \leqslant \varepsilon + \bigl(\lvert f\rvert \wedge g\bigr)\bigl(\lvert x_\alpha\rvert \wedge u\bigr) + \Bigl(\bigl(\lvert f\rvert - g\bigr)^+\Bigr)\bigl(\lvert \alpha_n\rvert \wedge u\bigr)&\\ \leqslant \varepsilon + g\bigl(\lvert x_\alpha\rvert \wedge u\bigr) + \Bigl(\bigl(\lvert f\rvert - g\bigr)^+\Bigr)\bigl(\lvert x_\alpha\rvert\bigr) < 3\varepsilon&\end{aligned}$$

because $\lvert x_\alpha\rvert \in A$ and $\bigl(\lvert f\rvert - g\bigr)^+ \in \varepsilon B_{\rho_A}$. It follows that $\rho_B(x_\alpha) < 3\varepsilon$. □

15.4.4 Corollary. *For a bounded set A in X and a continuous seminorm ρ on X, A is ρ-confined iff ρ is uaw-continuous on* $\mathrm{Sol}(A)$.

Using a similar argument, we can prove the following dual statement.

15.4.5 Theorem. *Let B be a bounded solid subset of X^* and A a bounded subset of X. That B is ρ_A-confined iff every uaw*-null sequence (or net) in B is ρ_A-null.*

Applying the preceding results with ρ being the norm, we get the following:

15.4.6 Corollary. (i) *A bounded solid subset A of X is confined iff for every (x_n) in A, if $x_n \xrightarrow{\text{uaw}} 0$ then $x_n \to 0$;*

(ii) *A bounded solid subset B of X is confined iff for every (f_n) in B, if $f_n \xrightarrow{\text{uaw}^*} 0$ then $f_n \to 0$.*

Again, sequences may be replaced with nets in the preceding corollary. Recall that an operator $T\colon X \to Y$ is M-weakly compact if it maps disjoint sequences in B_X to null sequences in Y. That is, if B_X is r_T-confined. Applying Theorem 15.4.3, we immediately get the following:

15.4.7 Corollary. *Let $T\colon X \to Y$ be a bounded operator from a Banach lattice X to a Banach space Y. TFAE:*

(i) *T is M-weakly compact;*

(ii) *if (x_n) is bounded and $x_n \xrightarrow{\text{uaw}} 0$ then $Tx_n \to 0$;*

(iii) *if (x_α) is bounded and $x_\alpha \xrightarrow{\text{uaw}} 0$ then $Tx_\alpha \to 0$.*

Thus we may replace "disjoint" with "uaw-null" in the definition of an M-weakly compact operator. Furthermore, it show that M-weak compactness of an operator may be viewed as a kind of continuity: T is M-weakly compact if it is (sequentially) uaw-to-norm continuous on B_X.

It was shown in Exercise 11.3.7 that a set A is confined iff every disjoint bounded sequence in X^* converges to zero uniformly on A. We show next that "disjoint" may be replaced with uaw-null or uaw*-null:

15.4.8 Theorem. *For a bounded set A in X, TFAE*

(i) *A is confined;*

(ii) *if (f_n) is bounded in X^* and $f_n \xrightarrow{\text{uaw}} 0$ then $f_n \to 0$ uniformly on A;*

(iii) *if (f_n) is bounded in X^* and $f_n \xrightarrow{\text{uaw}^*} 0$ then $f_n \to 0$ uniformly on A.*

Proof. (i)⇒(iii) WLOG, A is solid. Applying Burkinshaw-Dodds Theorem 11.3.6 with $B = B_{X^*}$, we conclude that B_{X^*} is ρ_A-confined. By Theorem 15.4.5, this implies (iii).

(iii)⇒(ii) is trivial. To show that (ii)⇒(i), observe that every bounded disjoint sequence (f_n) in X^* is uaw-null and, therefore, $f_n \to 0$ uniformly on A. Now apply Exercise 11.3.7. □

One can replace sequences with nets in the preceding proposition: just modify the proof using Exercise 11.1.14. In a similar fashion, we can prove the dual version:

15.4.9 Proposition. *For a bounded set B in X^*, B is confined iff every bounded uaw-null sequence (or net) in X converges to zero on B.*

Chapter 16

(p, q)-convexity/concavity

Expressions of the form $\left(\sum_{i=1}^{n}|t_i|^p\right)^{\frac{1}{p}}$ with $t_1, \dots, t_n \in \mathbb{R}$, $n \in \mathbb{N}$, and $0 < p < \infty$ are important and common in modern analysis. In particular, they are involved in the definition of ℓ_p^n spaces and in various classical inequalities, including Hölder, Cauchy-Schwart, and the Khinchin's inequality. Let X be a Banach lattice and $x_1, \dots, x_n \in X$. We may define the expression $\left(\sum_{i=1}^{n}|x_i|^p\right)^{\frac{1}{p}}$ as an element of X using function calculus. In this section, we will use such expressions to study ℓ_p-type structures in Banach lattices, as well as related classes of operators.

16.1 Expressions of the form $\left(\sum_{i=1}^{n}|x_i|^p\right)^{\frac{1}{p}}$

Throughout this section X is a uniformly complete Archimedean vector lattice, though, most interesting results will happen when X is a Banach lattice (recall that every Banach lattice is Archimedean and uniformly complete). For $x_1, \dots, x_n \in X$ and $0 < p < \infty$, the expression $\left(\sum_{i=1}^{n}|x_i|^p\right)^{\frac{1}{p}}$ is defined via (positively homogeneous) function calculus as in Section 4.4, because this expression is positively homogeneous. Let us briefly recall the construction, as it will be helpful later. Take any $e \in X_+$ such that $x_1, \dots, x_n$ are in the principal ideal I_e (for example, one may take $e = |x_1| \vee \dots \vee |x_n|$). Since X in uniformly complete, I_e is lattice isomorphic to $C(K)$ for some compact space K. This way, we may view $x_1, \dots, x_n$ as functions in $C(K)$; the expression $\left(\sum_{i=1}^{n}|x_i|^p\right)^{\frac{1}{p}}$ is now defined point-wise as an element of $C(K)$ and, therefore, of X. As shown in Section 4.4, the resulting vector does not depend on the particular choice of e.

We will be mostly interested in the case when $1 \leqslant p \leqslant \infty$; in this case, p^* is defined via $\frac{1}{p} + \frac{1}{p^*} = 1$. When $p = 1$, we have $\left(\sum_{i=1}^{n}|x_i|^p\right)^{\frac{1}{p}} = \sum_{i=1}^{n}|x_i|$; when $p = \infty$, we put

$\left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}} = \bigvee_{i=1}^n |x_i|$.

Given every identity or inequality which only involves only positively homogeneous continuous functions and finitely many scalars, if it is valid in $\mathbb{R}$, it remains valid when the scalars (or just some of the scalars) are replaced with elements of X. This is because function calculus is a lattice homomorphism; see Theorem 4.4.3. This allows us to carry over many results about expressions of the form $\left(\sum_{i=1}^n |t_i|^p\right)^{\frac{1}{p}}$ from real numbers to X. For example, it is a standard fact that if $0 < p \leqslant q \leqslant \infty$ and $(t_i)_{i=1}^n \in \mathbb{R}^n$ then $\left\|(t_i)_{i=1}^n\right\|_{\ell_p^n} \geqslant \left\|(t_i)_{i=1}^n\right\|_{\ell_q^n}$. It follows immediately that

$$\left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}} \geqslant \left(\sum_{i=1}^n |x_i|^q\right)^{\frac{1}{q}} \text{ whenever } 0 < p \leqslant q \leqslant \infty \tag{16.1}$$

for any $x_1 \dots, x_n \in X$.

16.1.1 Proposition. *If $x_1 \dots, x_n \in X$ are disjoint then $\left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}} = \sum_{i=1}^n |x_i|$ for any $0 < p \leqslant \infty$.*

Proof. If $1 \leqslant p \leqslant \infty$ then $\|\cdot\|_{\ell_1} \geqslant \|\cdot\|_{\ell_p} \geqslant \|\cdot\|_{\ell_\infty}$, so that

$$\sum_{i=1}^n |x_i| \geqslant \left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}} \geqslant \bigvee_{i=1}^n |x_i| = \sum_{i=1}^n |x_i|;$$

the last identity follows from the fact that the vectors are disjoint. It follows that $\left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}} \leqslant \sum_{i=1}^n |x_i|$.

Here is another proof, which works for an arbitrary p. Find $e \in X_+$ such that $x_1, \dots, x_n \in I_e$. Identifying I_e with $C(K)$ for some compact K, we may view $x_1, \dots, x_n$ as functions on K. Disjointness of x_i's implies that for every $t \in K$ there is at most one j such that $x_j(t) \neq 0$; it follows that

$$\left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}}(t) = \left(\sum_{i=1}^n |x_i(t)|^p\right)^{\frac{1}{p}} = |x_j(t)| = \left(\sum_{i=1}^n |x_i|\right)(t).$$

Hence, the two expressions agree as functions in $C(K)$ and, therefore, as vectors in X. $\square$

16.1.2 Exercise. Let $0 < p \leqslant \infty$ and $x_1, \dots, x_n, y_1, \dots, y_n \in X$ such that $|x_i| \leqslant |y_i|$ as $i = 1, \dots, n$. Then $\left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}} \leqslant \left(\sum_{i=1}^n |y_i|^p\right)^{\frac{1}{p}}$.

The next result may be viewed as an alternative definition of $\left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}}$ for $p \geqslant 1$, which does not require function calculus. Note that for scalars $\alpha_1, \dots, \alpha_n$, the condition $(\alpha_i) \in B_{\ell_p^n}$ means $\sum_{i=1}^n |\alpha_i|^p \leqslant 1$. Recall that if $1 \leqslant p \leqslant \infty$ then $(\ell_p^n)^* = \ell_q^n$, where $q = p^*$, so that

$$\Bigl(\sum_{i=1}^n |t_i|^p\Bigr)^{\frac{1}{p}} = \sup\Bigl\{\sum_{i=1}^n \alpha_i t_i \ :\ (\alpha_i) \in B_{\ell_q^n}\Bigr\} \quad \text{for any} \quad t_1, \dots, t_n \in \mathbb{R}. \tag{16.2}$$

16.1.3 Proposition. *Let $x_1, \dots, x_n \in X$, $1 \leqslant p \leqslant \infty$, and $q = p^*$. Then*

$$\Bigl(\sum_{i=1}^n |x_i|^p\Bigr)^{\frac{1}{p}} = \sup\Bigl\{\sum_{i=1}^n \alpha_i x_i \ :\ (\alpha_i) \in B_{\ell_q^n}\Bigr\}.$$

Proof. Note that we cannot just use function calculus to transfer (16.2) into X because it involves a supremum over an infinite set. Put

$$e = \Bigl(\sum_{i=1}^n |x_i|^p\Bigr)^{\frac{1}{p}} \qquad \text{and} \qquad A = \Bigl\{\sum_{i=1}^n \alpha_i x_i \ :\ (\alpha_i) \in B_{\ell_q^n}\Bigr\}.$$

We need to prove that $\sup A = e$. Recall that I_e is lattice isomorphic to $C(K)$ for some compact K, with e corresponding to function $\mathbb{1}$. Since $|x_i| \leqslant e$ for every i, $x_i \in I_e$, so that we may view x_i as an element of $C(K)$. For every $t \in K$, (16.2) yields

$$1 = \Bigl(\sum_{i=1}^n |x_i(t)|^p\Bigr)^{\frac{1}{p}} = \sup\Bigl\{\sum_{i=1}^n \alpha_i x_i(t) \ :\ (\alpha_i) \in B_{\ell_q^n}\Bigr\}.$$

This implies that the required identity holds in $C(K)$, because if a point-wise supremum of a subset of $C(K)$ happens to be a continuous function then it is the supremum of the set in $C(K)$. Since the lattice isomorphism between $C(K)$ and I_e preserves suprema of arbitrary sets by Theorem 1.6.13, it follows that $\sup A = e$ in I_e and, therefore, in X. $\square$

16.1.4 Exercise. Under the asssumptions of Proposition 16.1.3,

$$\Bigl(\sum_{i=1}^n |x_i|^p\Bigr)^{\frac{1}{p}} = \sup\Bigl\{\Bigl|\sum_{i=1}^n \alpha_i x_i\Bigr| \ :\ (\alpha_i) \in B_{\ell_q^n}\Bigr\}$$

Hölder's inequality, as well as its numerous corollaries, will be a major tool throughout this section. The usual Hölder's inequality for real numbers implies that

$$\Bigl|\sum_{i=1}^n \alpha_i x_i\Bigr| \leqslant \Bigl(\sum_{i=1}^n |\alpha_i|^q\Bigr)^{\frac{1}{q}} \Bigl(\sum_{i=1}^n |x_i|^p\Bigr)^{\frac{1}{p}},$$

when $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, $x_1, \dots, x_n \in X$, $1 \leqslant p \leqslant \infty$, and $q = p^*$.

16.1.5. *Khinchin's inequality.* Applying function calculus to Khinchin's Inequality in the form presented in 6.4.4(i), we conclude that

$$A_1\Bigl(\sum_{i=1}^n |x_i|^2\Bigr)^{\frac{1}{2}} \leqslant \operatorname*{Ave}_{\varepsilon_i=\pm 1}\Bigl|\sum_{i=1}^n \varepsilon_i x_i\Bigr| \leqslant B_1\Bigl(\sum_{i=1}^n |x_i|^2\Bigr)^{\frac{1}{2}} \tag{16.3}$$

for any $x_1,\dots,x_n \in X$. For a general p, a similar argument yields

$$A_p\Bigl(\sum_{i=1}^n |x_i|^2\Bigr)^{\frac{1}{2}} \leqslant \Bigl(\frac{1}{2^n}\sum_{\varepsilon_i=\pm 1}\Bigl|\sum_{i=1}^n \varepsilon_i x_i\Bigr|^p\Bigr)^{\frac{1}{p}} \leqslant B_p\Bigl(\sum_{i=1}^n |x_i|^2\Bigr)^{\frac{1}{2}}, \tag{16.4}$$

These inequalities will be referred to as Khinchin's inequality for vector lattices.

16.1.6 Lemma. *Let X be a Banach lattice, $x,y \in X$, and $\lambda,\mu \in \mathbb{R}_+$ with $\lambda+\mu=1$. Then $\bigl\||x|^\lambda|y|^\mu\bigr\| \leqslant \|x\|^\lambda\|y\|^\mu$.*

Proof. Note that the expression $|x|^\lambda|y|^\mu$ is defined via function calculus because the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ given by $f(s,t)=|s|^\lambda|t|^\mu$ is continuous and positively homogeneous. WLOG, x and y are non-zero.

Concavity of the natural log yields that $\lambda\ln s+\mu\ln s \leqslant \ln(\lambda s+\mu t)$ for any $s,t\in\mathbb{R}_+$. Exponentiating, we get $s^\lambda t^\mu \leqslant \lambda s+\mu t$. Let $\alpha>0$, then $s^\lambda t^\mu = \bigl(\alpha^{\frac{1}{\lambda}}s\bigr)^\lambda\bigl(\alpha^{-\frac{1}{\mu}}t\bigr)^\mu \leqslant \lambda\alpha^{\frac{1}{\lambda}}s+\mu\alpha^{-\frac{1}{\mu}}t$. It follows that $|s|^\lambda|t|^\mu \leqslant \lambda\alpha^{\frac{1}{\lambda}}|s|+\mu\alpha^{-\frac{1}{\mu}}|t|$ for any $s,t\in\mathbb{R}$. By function calculus, $|x|^\lambda|y|^\mu \leqslant \lambda\alpha^{\frac{1}{\lambda}}|x|+\mu\alpha^{-\frac{1}{\mu}}|y|$. Now Triangle Inequality yields $\bigl\||x|^\lambda|y|^\mu\bigr\| \leqslant \lambda\alpha^{\frac{1}{\lambda}}\|x\|+\mu\alpha^{-\frac{1}{\mu}}\|y\|$. Plugging in $\alpha=\Bigl(\frac{\|y\|}{\|x\|}\Bigr)^{\lambda\mu}$ yields the required inequality. ☐

16.1.7. Let $1\leqslant p<r<q\leqslant\infty$. Then $\frac{1}{q}<\frac{1}{r}<\frac{1}{p}$, hence $\frac{1}{r}$ may be written as a convex combination of $\frac{1}{p}$ and $\frac{1}{q}$, that is, there exist $\lambda,\mu\geqslant 0$ such that $\lambda+\mu=1$ and $\frac{1}{r}=\frac{\lambda}{p}+\frac{\mu}{q}$. It is follows easily from Hölder's inequality that

$$\Bigl(\sum_{i=1}^n \alpha_i|t_i|^r\Bigr)^{\frac{1}{r}} \leqslant \Bigl(\sum_{i=1}^n \alpha_i|t_i|^p\Bigr)^{\frac{\lambda}{p}}\Bigl(\sum_{i=1}^n \alpha_i|t_i|^q\Bigr)^{\frac{\mu}{q}},$$

for any real numbers $t_1,\dots,t_n$ and any positive real numbers $\alpha_1,\dots,\alpha_n$. It now follows by function calculus that

$$\Bigl(\sum_{i=1}^n \alpha_i|x_i|^r\Bigr)^{\frac{1}{r}} \leqslant \Bigl(\sum_{i=1}^n \alpha_i|x_i|^p\Bigr)^{\frac{\lambda}{p}}\Bigl(\sum_{i=1}^n \alpha_i|x_i|^q\Bigr)^{\frac{\mu}{q}},$$

for any $\alpha_1,\dots,\alpha_n\in\mathbb{R}_+$ and $x_1\dots,x_n\in X$. In a Banach lattice setting, we can further apply Lemma 16.1.6 and get

$$\Bigl\|\Bigl(\sum_{i=1}^n \alpha_i|x_i|^r\Bigr)^{\frac{1}{r}}\Bigr\| \leqslant \Bigl\|\Bigl(\sum_{i=1}^n \alpha_i|x_i|^p\Bigr)^{\frac{1}{p}}\Bigr\|^\lambda\cdot\Bigl\|\Bigl(\sum_{i=1}^n \alpha_i|x_i|^q\Bigr)^{\frac{1}{q}}\Bigr\|^\mu.$$

Next, we discuss the relationship of p-sums with operators.

16.1.8 Proposition. *Let X and Y be two uniformly complete Archimedean vector lattices, $x_1,\dots,x_n\in X$, $1\leqslant p\leqslant\infty$, and $T\colon X\to Y$ a linear operator.*

(i) *If T is a lattice homomorphism then $\Bigl(\sum_{i=1}^n\lvert Tx_i\rvert^p\Bigr)^{\frac{1}{p}}=T\Bigl(\sum_{i=1}^n\lvert x_i\rvert^p\Bigr)^{\frac{1}{p}}$;*

(ii) *If $T\geqslant 0$ then $\Bigl(\sum_{i=1}^n\lvert Tx_i\rvert^p\Bigr)^{\frac{1}{p}}\leqslant T\Bigl(\sum_{i=1}^n\lvert x_i\rvert^p\Bigr)^{\frac{1}{p}}$.*

Proof. (i) follows immediately from the fact that lattice-homomorphisms preserve function calculus; see Exercise 4.4.6.

(ii) By Proposition 16.1.3, we have $\Bigl(\sum_{i=1}^n\lvert x_i\rvert^p\Bigr)^{\frac{1}{p}}=\sup A$, where $A=\Bigl\{\sum_{i=1}^n\alpha_ix_i\,:\,(\alpha_i)\in B_{\ell_q^n}\Bigr\}$ and $q=p^*$. Also,

$$\Bigl(\sum_{i=1}^n\lvert Tx_i\rvert^p\Bigr)^{\frac{1}{p}}=\sup\Bigl\{\sum_{i=1}^n\alpha_iTx_i\,:\,(\alpha_i)\in B_{\ell_q^n}\Bigr\}=\sup T(A).$$

It is left to note that $\sup T(A)\leqslant T(\sup A)$. □

16.1.9 Exercise. Under the assumptions of Proposition 16.1.8, suppose that T is an operator such that $\lvert T\rvert$ exists. Then $\Bigl(\sum_{i=1}^n\lvert Tx_i\rvert^p\Bigr)^{\frac{1}{p}}\leqslant\lvert T\rvert\Bigl(\sum_{i=1}^n\lvert x_i\rvert^p\Bigr)^{\frac{1}{p}}$.

16.1.10 Exercise. Suppose that $1\leqslant p\leqslant\infty$, $X^\sim$ separates points of X, and $x_1,\dots,x_n,y_1,\dots,y_m\in X$. If $\Bigl(\sum_{i=1}^n\bigl\lvert f(x_i)\bigr\rvert^p\Bigr)^{\frac{1}{p}}\leqslant\Bigl(\sum_{j=1}^m\bigl\lvert f(y_j)\bigr\rvert^p\Bigr)^{\frac{1}{p}}$ for every $f\in X^\sim_+$ then $\Bigl(\sum_{i=1}^n\lvert x_i\rvert^p\Bigr)^{\frac{1}{p}}\leqslant\Bigl(\sum_{j=1}^m\lvert y_j\rvert^p\Bigr)^{\frac{1}{p}}$.

16.2 p-sums of functionals

Throughout this section, X is a Banach lattice, $1\leqslant p\leqslant\infty$, and $q=p^*$. We have the following version of Hölder's inequality.

16.2.1 Theorem. *For $x_1,\dots,x_n\in X$ and $f_1,\dots,f_n\in X^*$, one has*

$$\Bigl\lvert\sum_{i=1}^n\langle f_i,x_i\rangle\Bigr\rvert\leqslant\Bigl\langle\Bigl(\sum_{i=1}^n\lvert f_i\rvert^q\Bigr)^{\frac{1}{q}},\Bigl(\sum_{i=1}^n\lvert x_i\rvert^p\Bigr)^{\frac{1}{p}}\Bigr\rangle. \tag{16.5}$$

Proof. The idea of the proof is simple: we will interpret x_i's and f_i's as measurable functions and then apply the usual Hölder's inequality. Find $e\in X_+$ such that $x_1,\dots,x_n\in I_e$. For every $f\in X^*$ and $x\in I_e$, we have $\bigl\lvert f(x)\bigr\rvert\leqslant\lVert f\rVert_{X^*}\lVert x\rVert_X\leqslant\lVert f\rVert_{X^*}\lVert x\rVert_e\lVert e\rVert_X$, it follows that the restriction of f to I_e induces a bounded functional in $\bigl(I_e,\lVert\cdot\rVert_e\bigr)^*$.

Let $J\colon\bigl(I_e,\lVert\cdot\rVert_e\bigr)\to X$ be the formal inclusion map. Then $J^*\colon X^*\to\bigl(I_e,\lVert\cdot\rVert_e\bigr)^*$ is the restriction map. Both J and J^* are norm bounded lattice homomorphisms (see

Exercise 6.6.8). Let $\mu_i = J^* f_i$; let $y_i \in I_e$ such that $x_i = Jy_i$ as $i = 1,\dots,n$ (formally, $y_i = x_i$). Clearly, the left hand side of (16.5) does not change if we replace x_i's and f_i's with y_i's and μ_i's, respectively. The right hand side does not change either because by Proposition 16.1.8(i) we have

$$\Big\langle \Big(\sum_{i=1}^n |f_i|^q\Big)^{\frac{1}{q}}, \Big(\sum_{i=1}^n |x_i|^p\Big)^{\frac{1}{p}} \Big\rangle = \Big\langle \Big(\sum_{i=1}^n |f_i|^q\Big)^{\frac{1}{q}}, J\Big[\Big(\sum_{i=1}^n |y_i|^p\Big)^{\frac{1}{p}}\Big] \Big\rangle$$
$$= \Big\langle J^*\Big[\Big(\sum_{i=1}^n |f_i|^q\Big)^{\frac{1}{q}}\Big], \Big(\sum_{i=1}^n |y_i|^p\Big)^{\frac{1}{p}} \Big\rangle = \Big\langle \Big(\sum_{i=1}^n |\mu_i|^q\Big)^{\frac{1}{q}}, \Big(\sum_{i=1}^n |y_i|^p\Big)^{\frac{1}{p}} \Big\rangle.$$

By Krein-Kakutani's Representation Theorem, we may now assume that $X = C(K)$ for some compact space K, and by Riesz Representation Theorem 8.4.2, we may identify X^* with $\mathcal{M}(K)$. Hence, we view y_i's and μ_i's as elements of $C(K)$ and $\mathcal{M}(K)$, respectively. Let $\mu \in \mathcal{M}(K)_+$ such that $|\mu_i| \leqslant \mu$ for every i. Then $\mu_i \in B_\mu$, the principal band of μ in $\mathcal{M}(K)$. Radon-Nikodym Theorem 8.3.14 and Proposition 8.3.13 tell us that B_μ is lattice isometric to $L_1(\mu)$ via the Radon-Nikodym derivative. For each i, put $h_i = \frac{d\mu_i}{d\mu}$. It follows that $h_i \in L_1(\mu)$ and the function $\big(\sum_{i=1}^n |h_i|^q\big)^{\frac{1}{q}}$ in $L_1(\mu)$ is the Radon-Nikodym derivative of the measure $\big(\sum_{i=1}^n |\mu_i|^q\big)^{\frac{1}{q}}$ with respect to μ. Applying Hölder's inequality pointwise, we get

$$\Big|\sum_{i=1}^n \langle \mu_i, y_i\rangle\Big| = \Big|\sum_{i=1}^n \int y_i h_i \, d\mu\Big| \leqslant \int \Big|\sum_{i=1}^n y_i h_i\Big| d\mu$$
$$\leqslant \int \Big(\sum_{i=1}^n |y_i|^p\Big)^{\frac{1}{p}} \Big(\sum_{i=1}^n |h_i|^q\Big)^{\frac{1}{q}} d\mu = \Big\langle \Big(\sum_{i=1}^n |\mu_i|^q\Big)^{\frac{1}{q}}, \Big(\sum_{i=1}^n |y_i|^p\Big)^{\frac{1}{p}} \Big\rangle.$$

□

As a corollary, we get the following generalization of Riesz-Kantorovich formulas. When $p = 1$, it agrees with a Riesz-Kantorovich formula.

16.2.2 Theorem. *Let $f_1,\dots,f_n \in X^*$ and $x \in X_+$. Then*

$$\Big(\sum_{i=1}^n |f_i|^q\Big)^{\frac{1}{q}}(x) = \sup\Big\{\sum_{i=1}^n \langle f_i, x_i\rangle \;:\; x_1,\dots,x_n \in X,\ \Big(\sum_{i=1}^n |x_i|^p\Big)^{\frac{1}{p}} \leqslant x\Big\}.$$

Proof. Suppose that $x_1,\dots,x_n \in X$ and $\big(\sum_{i=1}^n |x_i|^p\big)^{\frac{1}{p}} \leqslant x$. It follows from Theorem 16.2.1 that $\sum_{i=1}^n \langle f_i, x_i\rangle \leqslant \big(\sum_{i=1}^n |f_i|^q\big)^{\frac{1}{q}}(x)$.

On the other hand, Proposition 16.1.3 yields $\big(\sum_{i=1}^n |f_i|^q\big)^{\frac{1}{q}} = \sup A$ where $A = \big\{\sum_{i=1}^n \alpha_i f_i \;:\; (\alpha_i)_{i=1}^n \in B_{\ell_p^n}\big\}$. Since $\sup A = \sup A^\vee$ and $A^\vee$ may be viewed as an increasing net, Riesz-Kantorovich Theorem 6.3.2 yields

$$\Big(\sum_{i=1}^{n}|f_i|^q\Big)^{\frac{1}{q}}(x)=\sup\big\{g(x)\ :\ g\in A^{\vee}\big\}.$$

Let $g\in A^{\vee}$. Then $g=\bigvee_{j=1}^m g_j$ for some $g_1,\dots,g_m\in A$. For every $j=1,\dots,m$ one has $g_j=\sum_{i=1}^n\alpha_{ji}f_i$ for some sequence $(\alpha_{ji})_{i=1}^n\in B_{\ell_p^n}$. By Riesz-Kantorovich formulas,

$$g(x)=\sup\Big\{\sum_{j=1}^{m}g_j(y_j)\ :\ y_1,\dots,y_m\geqslant 0,\ x=y_1+\dots+y_m\Big\}.$$

Let $x=y_1+\dots+y_m$ for some $y_1,\dots,y_m\in X_+$. Then

$$\sum_{j=1}^{m}g_j(y_j)=\sum_{j=1}^{m}\sum_{i=1}^{n}\alpha_{ji}f_i(y_j)=\sum_{i=1}^{n}f_i\Big(\sum_{j=1}^{m}\alpha_{ji}y_j\Big)=\sum_{i=1}^{n}\langle f_i,x_i\rangle$$

where $x_i=\sum_{j=1}^m\alpha_{ji}y_j$. It is left to show that $\Big(\sum_{i=1}^n|x_i|^p\Big)^{\frac{1}{p}}\leqslant x$.

Let (v_{ji}) be an $m\times n$ real matrix; let v_j be the j-th row of this matrix as $j=1,\dots,m$. Expanding the triangle inequality in ℓ_p^n, $\big\|\sum_{j=1}^m v_j\big\|_{\ell_p^n}\leqslant\sum_{j=1}^m\|v_j\|_{\ell_p^n}$, we get

$$\Big(\sum_{i=1}^{n}\Big|\sum_{j=1}^{m}v_{ji}\Big|^p\Big)^{\frac{1}{p}}\leqslant\sum_{j=1}^{m}\Big(\sum_{i=1}^{n}|v_{ji}|^p\Big)^{\frac{1}{p}}.$$

By function calculus, this inequality remains valid when v_{ji}'s are elements of X. Hence,

$$\begin{aligned}\Big(\sum_{i=1}^{n}|x_i|^p\Big)^{\frac{1}{p}}=\Big(\sum_{i=1}^{n}\Big|\sum_{j=1}^{m}\alpha_{ji}y_j\Big|^p\Big)^{\frac{1}{p}}&\leqslant\sum_{j=1}^{m}\Big(\sum_{i=1}^{n}|\alpha_{ji}y_j|^p\Big)^{\frac{1}{p}}\\&=\sum_{j=1}^{m}\Big(\sum_{i=1}^{n}|\alpha_{ji}|^p\Big)^{\frac{1}{p}}y_j=\sum_{j=1}^{m}y_j=x.\end{aligned}$$

□

We also have a dual version of Proposition 16.2.2

16.2.3 Proposition. *Let $x_1,\dots,x_n\in X$ and $f\in X_+^*$. Then*

$$\Big\langle f,\Big(\sum_{i=1}^{n}|x_i|^q\Big)^{\frac{1}{q}}\Big\rangle=\sup\Big\{\sum_{i=1}^{n}\langle f_i,x_i\rangle\ :\ f_1,\dots,f_n\in X^*,\ \Big(\sum_{i=1}^{n}|f_i|^p\Big)^{\frac{1}{p}}\leqslant f\Big\}.$$

Proof. Let $x=\Big(\sum_{i=1}^n|x_i|^q\Big)^{\frac{1}{q}}$. Since the canonical embedding $x\in X\mapsto\hat{x}\in X^{**}$ is a lattice homomorphism, it preserves function calculus. Applying Theorem 16.2.2 to

$\hat{x}_1,\dots,\hat{x}_n$, we get

$$\langle f,x\rangle=\langle\hat{x},f\rangle=\Big\langle\Big(\sum_{i=1}^n|\hat{x}_i|^q\Big)^{\frac{1}{q}},f\Big\rangle$$
$$=\sup\Big\{\sum_{i=1}^n\langle\hat{x}_i,f_i\rangle\ :\ f_1,\dots,f_n\in X^*,\ \Big(\sum_{i=1}^n|f_i|^p\Big)^{\frac{1}{p}}\leqslant f\Big\},$$

which yields the required result. □

16.3 Power norms: $\|\bar{x}\|_{\text{sum-}p}$, $\|\bar{x}\|_{\text{weak-}p}$, and $\|\bar{x}\|_{\text{lat-}p}$.

The role of this section is to provide motivation for the rest of the chapter, — to supply a "pig picture". A norm on a vector space E measures the "size" of a single vector in E. In modern Analysis, one often needs to measure the "size" of a (finite) sequence of vectors. A finite sequence of vectors $(x_1,\dots,x_n)$ in E may be viewed as an element of E^n; we sometimes call it a ***multivector*** and denote by $\bar{x}$. It is often convenient to identify $\bar{x}$ with an infinite sequence with only finitely many terms $(x_1,\dots,x_n,0,0,\dots)$; the set of all such sequences will be denoted by $c_{00}(E)$. That is, we may identify the vector space $c_{00}(E)$ with the set of all multivectors in E or with $\bigcup_{i=1}^n E^n$.

16.3.1 Definition. Let E be a vector space. Following [DLOT17], we say that a norm on $c_{00}(E)$ is a ***power-norm*** if it is invariant under permutations as well as under and padding the multivector with zeros, and is "subhomogeneous":

(i) $\big\|(x_1,\dots,x_n)\big\|=\big\|(x_{\sigma(1)},\dots,x_{\sigma(n)})\big\|$ for every permutation σ;
(ii) $\big\|(x_1,\dots,x_n)\big\|=\big\|(x_1,\dots,x_n,0)\big\|$, and
(iii) $\big\|(\alpha_1x_1,\dots,\alpha_nx_n)\big\|_n\leqslant\big(\max|\alpha_i|\big)\big\|(x_1,\dots,x_n)\big\|$.

Here $n\in\mathbb{N}$, $x_1,\dots,x_n\in E$, and $\alpha_1,\dots,\alpha_n\in\mathbb{R}$ are arbitrary.

In particular, taking $n=1$, every power-norm induces a norm on E via $\|x\|=\big\|(x)\big\|$. Usually, power-norms are defined on a Banach space E rather than just a vector space, in such a way that the norm on E induced by the power-norm agrees with the original norm of E. So one can think of a power-norm as an extension of the original norm to multivectors.

16.3.2 Exercise. Condition (iii) in the definition of a power norm may be replaced with $\big\|(x_1,\dots,x_{n-1},\lambda x_n)\big\|_n\leqslant\big\|(x_1,\dots,x_{n-1},x_n)\big\|_n$ whenever $|\lambda|\leqslant 1$.

16.3.3 Definition. Let E be a Banach space and $\bar{x}=(x_1,\dots,x_n)$ a multivector in E. The expression $\|\bar{x}\|_{\text{sum}}=\sum_{i=1}^n\|x_i\|$ defines a power-norm on E. More generally, for $1 \leqslant p \leqslant \infty$, the expression

$$\|\bar{x}\|_{\text{sum-}p} \coloneqq \Bigl(\sum_{i=1}^{n} \|x_i\|^p\Bigr)^{\frac{1}{p}}$$

defines a power-norm on E; we will call it the ***p-sum power-norm***. This is just the norm in ℓ_p^n of the vector $\bigl(\|x_i\|\bigr)_{i=1}^n$. When $p=\infty$, we put $\|\bar{x}\|_{\text{sum-}\infty}$ to be $\max_i \|x_i\|$.

16.3.4 Definition. Again, let E be a Banach space, $1 \leqslant p \leqslant \infty$, and $\bar{x} = (x_1, \dots, x_n)$ a multivector in E. It is easy to see that the expression

$$\|\bar{x}\|_{\text{weak-}p} \coloneqq \sup\Bigl\{ \Bigl(\sum_{i=1}^{n} |f(x_i)|^p\Bigr)^{\frac{1}{p}} \;:\; f \in B_{E^*} \Bigr\}$$

defines a a power-norm on E; it is called the ***weak p-summing norm***. Note that $\Bigl(\sum_{i=1}^n |f(x_i)|^p\Bigr)^{\frac{1}{p}} = \Bigl\| \bigl(f(x_i)\bigr)_{i=1}^n \Bigr\|_{\ell_p^n}$.

It is easy to see that the weak p-summing norm is less than or equal to the p-sum norm.

16.3.5 Definition. Let X be a Banach lattice, $1 \leqslant p \leqslant \infty$, and $\bar{x} = (x_1, \dots, x_n)$ a multivector in X. The expression

$$\|\bar{x}\|_{\text{lat-}p} \coloneqq \Bigl\| \Bigl(\sum_{i=1}^{n} |x_i|^p\Bigr)^{\frac{1}{p}} \Bigr\|$$

defines a a power-norm on X. It is called the ***(canonical) lattice p-multinorm*** of X. In the case $p = \infty$, we have $\|\bar{x}\|_{\text{lat-}\infty} = \Bigl\| \bigvee_{i=1}^n |x_i| \Bigr\|$.

Let $T \colon E \to F$ be a linear operator between two vectors spaces equipped with power-norms. Let $\overline{T} \colon c_{00}(E) \to c_{00}(F)$ be defined via

$$\overline{T}(x_1, \dots, x_n) = (Tx_1, \dots, Tx_n).$$

We say that T is ***power-bounded*** or ***multibounded*** with respect to the power-norms if $\overline{T}$ is bounded. That is, there exists a constant $K > 0$ such that $\|\overline{T}\bar{x}\| \leqslant K\|\bar{x}\|$ or, in expanded form,

$$\bigl\|(Tx_1, \dots, Tx_n)\bigr\| \leqslant K \bigl\|(x_1, \dots, x_n)\bigr\|$$

for any finite sequesce $x_1, \dots, x_n \in E$. If the precise value of the constant is not important, we will write $\|\overline{T}\bar{x}\| \lesssim \|\bar{x}\|$. Clearly, $\|\overline{T}\|$ is the least value of K for which this inequality holds; it is referred as the ***power-norm*** or ***multinorm*** of T and denoted $\|T\|_{\text{mb}}$. It is clear that every power-bounded operator is bounded and $\|T\| \leqslant \|T\|_{\text{mb}}$.

Hence, the set of all multibounded operators forms a subspace of $L(E,F)$, which is a normed space under $\|\cdot\|_{\mathrm{mb}}$.

In particular, this language allow us to compare two power-norms on the same space: given two power norms $\|\cdot\|$ and $|||\cdot|||$ on a vector space E, we write $\|\cdot\| \lesssim |||\cdot|||$ if the identity operator on E is power-bounded from $|||\cdot|||$ to $\|\cdot\|$. That is, there exists a constant $K>0$ such that $\|\bar{x}\| \leqslant K\,|||\bar{x}|||$ for every multivector $\bar{x}$.

16.3.6. Let $1 \leqslant p \leqslant q \leqslant \infty$. It follows easily from $\|\cdot\|_{\ell_1} \leqslant \|\cdot\|_{\ell_p}$ that

$$\|\cdot\|_{\text{sum-}q} \leqslant \|\cdot\|_{\text{sum-}p}, \qquad \|\cdot\|_{\text{weak-}q} \leqslant \|\cdot\|_{\text{weak-}p}, \qquad \text{and} \qquad \|\cdot\|_{\text{lat-}q} \leqslant \|\cdot\|_{\text{lat-}p}.$$

16.3.7 Exercise. Consider an $L_p(\mu)$ space for some $1 \leqslant p \leqslant \infty$ and some measure μ. Then $\|\bar{x}\|_{\text{lat-}p} = \|\bar{x}\|_{\text{sum-}p}$ for any multivector $\bar{x}$ in $L_p(\mu)$.

16.3.8 Exercise. Let $1 \leqslant p \leqslant \infty$. Every bounded operator between Banach spaces is multibounded with respect the p-sum power-norms.

16.3.9 Exercise. Let $T\colon X \to Y$ be a regular operator between Banach lattices, $1 \leqslant p \leqslant \infty$. Then T is power-bounded with respect to the lattice p-multinorms. More precisely, $\|\overline{T}\bar{x}\|_{\text{lat-}p} \leqslant \|T\|_r \|\bar{x}\|_{\text{lat-}p}$ for every multivector $\bar{x} = (x_1,\dots,x_n)$. That is

$$\Big\|\Big(\sum_{i=1}^n |Tx_i|^p\Big)^{\frac{1}{p}}\Big\| \leqslant \|T\|_r \Big\|\Big(\sum_{i=1}^n |x_i|^p\Big)^{\frac{1}{p}}\Big\|.$$

Since every functional f on a Banach lattice is regular, with $\|f\|_r = \|f\|$, this yields the following:

16.3.10 Corollary. *On every Banach lattice,* $\|\cdot\|_{\text{weak-}p} \leqslant \|\cdot\|_{\text{lat-}p}$.

16.3.11 Proposition. *In* $C(K)$,

$$\|\bar{f}\|_{\text{weak-}p} = \|\bar{f}\|_{\text{lat-}p} = \sup\Big\{\Big\|\sum_{i=1}^n \alpha_i f_i\Big\| \;:\; (\alpha_i) \in B_{\ell^n_{p^*}}\Big\}$$

whenever $1 \leqslant p \leqslant \infty$ *and* $\bar{f}$ *is a multivector in* $C(K)$.

Proof. For $f_1,\dots,f_n$ in $C(K)$, we have

$$\begin{aligned}\sup\Big\{\Big\|\sum_{i=1}^n \alpha_i f_i\Big\| \;:\; (\alpha_i) \in B_{\ell^n_{p^*}}\Big\} &= \sup\Big\{\Big|\sum_{i=1}^n \alpha_i f_i(t)\Big| \;:\; (\alpha_i) \in B_{\ell^n_{p^*}},\ t \in K\Big\} \\ &= \sup\Big\{\Big(\sum_{i=1}^n |f_i(t)|^p\Big)^{\frac{1}{p}} \;:\; t \in K\Big\} = \|\bar{f}\|_{\text{lat-}p}.\end{aligned}$$

We already know that $\|\bar f\|_{\text{weak-}p} \leqslant \|\bar f\|_{\text{lat-}p}$. For the converse inequality, observe that for every $t \in K$ we have $\delta_t \in B_{\mathcal{M}(K)}$, so that

$$\sup\Big\{\Big(\sum_{i=1}^n |f_i(t)|^p\Big)^{\frac{1}{p}} \,:\, t\in K\Big\} \leqslant \sup\Big\{\Big(\sum_{i=1}^n \big|\mu(f_i)\big|^p\Big)^{\frac{1}{p}} \,:\, \mu \in B_{\mathcal{M}(K)}\Big\}.$$

□

16.3.12 Definition. An operator $T\colon E \to F$ between two Banach spaces is said to be (p,q)***-summing*** if it is power-bounded from $(E, \|\cdot\|_{\text{weak-}q})$ to $(F, \|\cdot\|_{\text{sum-}p})$. That is, $\|\overline{T}\bar x\|_{\text{sum-}p} \lesssim \|\bar x\|_{\text{weak-}q}$.

We will focus on the interplay between p-summing power norms and lattice p-multinorms. While weak p-summing norms and (p,q)-summing operators will only appear occasionally as auxiliary tools in this chapter, they are important concepts in the modern theory of Banach spaces; we refer the reader to detailed studies of them in [DJT95, AK06].

It is a deep fact that in the case $p = 2$, Exercise 16.3.9 extends to all operators, not just the regular ones; this was deduced by Krivine from Grothendieck's inequality:

16.3.13 Theorem (Grothendieck-Krivine)**.** *Let $T\colon X \to Y$ be a bounded operator between Banach lattices X and Y then*

$$\Big\|\Big(\sum_{i=1}^n |Tx_i|^2\Big)^{\frac{1}{2}}\Big\| \leqslant K_G\|T\|\Big\|\Big(\sum_{i=1}^n |x_i|^2\Big)^{\frac{1}{2}}\Big\| \tag{16.6}$$

for any $x_1, \dots, x_n$ in X, where K_G is Grothendieck's universal constant.

Proof. We will use representation theorems for AM and AL spaces to reduce to the special case $T\colon C(K) \to L_1(\mu)$ and then further to the case $T\colon \ell_\infty^m \to \ell_1^m$. First, we show that WLOG we may assume that $X = C(K)$ for some compact space K. Indeed, suppose we know that (16.6) holds operators defined on $C(K)$ spaces. Now put $e = \Big(\sum_{i=1}^n |x_i|^2\Big)^{\frac{1}{2}}$ in X. Consider the diagram $I_e \xrightarrow{J} X \xrightarrow{T} Y$, where I_e is the principal ideal of e in X equipped with $\|\cdot\|_e$ norm and J is the formal inclusion. Since $\|x\|_X \leqslant \|x\|_e\|e\|_X$ for every $x \in I_e$, we have $\|J\| \leqslant \|e\|_X$. Since $\big(I_e, \|\cdot\|_e\big)$ is lattice isometric to some $C(K)$ space by Theorem 4.1.10, the inequality holds for TJ, so that

$$\Big\|\Big(\sum_{i=1}^n |Tx_i|^2\Big)^{\frac{1}{2}}\Big\|_Y \leqslant K_G\|TJ\|\Big\|\Big(\sum_{i=1}^n |x_i|^2\Big)^{\frac{1}{2}}\Big\|_e \leqslant K_G\|T\|\|J\|\|e\|_e \leqslant K_G\|T\|\Big\|\Big(\sum_{i=1}^n |x_i|^2\Big)^{\frac{1}{2}}\Big\|_X,$$

and we are done.

Second, we show that WLOG we may assume that $Y = L_1(\mu)$ for some measure μ. Suppose that the result is known to be true for operators acting from a $C(K)$ space into an L_1-space. Since $C(K)^{**}$ is again a unital AM-space and, therefore, is lattice isometric to some $C(K)$-space, replacing T with T^{**} we may assume that Y is a dual Banach lattice. In particular, Y is order continuous. Let $y=\left(\sum_{i=1}^n|Tx_i|^2\right)^{\frac12}$. Using Hahn-Banach Theorem, find $h\in Y_+^*$ such that $\|h\|=1$ and $\|y\|_Y=h(y)$. Since Y is order continuous, we can decompose Y into a band decomposition $Y=C_h\oplus N_h$, where C_h and N_h are the carrier and the null ideal if h, respectively, as in 6.5.12. Let $P\colon Y\to C_h$ be the corresponding band projection. Note that $0\leqslant y-Py\in N_h$ implies $h(Py)=h(y)$. Clearly, h is strictly positive on C_h. As in Section 9.2, $\|y\|_h=h\bigl(|y|\bigr)$ defines an AL-norm on C_h, and the completion of $\bigl(C_h,\|\cdot\|_h\bigr)$ is an AL-space and is, therefore, lattice isometric to $L_1(\mu)$ for some μ. Thus, we have a lattice isomorphic embedding $R\colon C_h\to L_1(\mu)$. It follows from $\|Ry\|_{L_1}=h\bigl(|y|\bigr)\leqslant\|h\|_{Y^*}\|y\|_Y=\|y\|_Y$ that $\|R\|\leqslant 1$. Consider the diagram $C(K)\xrightarrow{T}Y\xrightarrow{P}C_h\xrightarrow{R}L_1(\mu)$. By the assumption, the theorem is valid for the operator RPT, hence

$$\Bigl\|\Bigl(\sum_{i=1}^n|RPTx_i|^2\Bigr)^{\frac12}\Bigr\|_{L_1(\mu)}\leqslant K_G\|RPT\|\Bigl\|\Bigl(\sum_{i=1}^n|x_i|^2\Bigr)^{\frac12}\Bigr\|_{C(K)}\leqslant K_G\|T\|\Bigl\|\Bigl(\sum_{i=1}^n|x_i|^2\Bigr)^{\frac12}\Bigr\|_{C(K)}.$$

On the other hand, since RP is a lattice homomorphism, we have

$$\Bigl\|\Bigl(\sum_{i=1}^n|RPTx_i|^2\Bigr)^{\frac12}\Bigr\|_{L_1(\mu)}=\|RPy\|_{L_1(\mu)}=h(Py)=h(y)=\|y\|_Y=\Bigl\|\Bigl(\sum_{i=1}^n|Tx_i|^2\Bigr)^{\frac12}\Bigr\|_Y.$$

We have reduced the theorem to the case $X=C(K)$ and $Y=L_1(\mu)$. Again, replacing T with T^{**} if necessary, we may assume that $C(K)$ is a dual space; in particular, is has PPP. Freudenthal's Theorem 7.2.14 implies that simple functions are norm dense in $C(K)$. Since both $\left(\sum_{i=1}^n|x_i|^2\right)^{\frac12}$ and $\left(\sum_{i=1}^n|Tx_i|^2\right)^{\frac12}$ depend (norm-to-norm) continuously on $x_1,\dots,x_n$, it suffices to prove the theorem under the assumption that each x_i is a step function. We can then find a disjoint collection $A_1,\dots,A_m$ of clopen subsets of K such that each x_i is in $\operatorname{span}\{\mathbb{1}_{A_1},\dots,\mathbb{1}_{A_m}\}$. This span is a sublattice of $C(K)$ and is clearly lattice isometric to ℓ_∞^m. Hence, restricting T to this span, we may assume that $X=\ell_\infty^m$.

Let $e_1,\dots,e_m$ be the standard unit vectors in ℓ_∞^m. Then $\operatorname{Range}T=\operatorname{span}\{Te_1,\dots,Te_m\}$. Again, as simple functions are dense in $L_1(\mu)$, we may approximate T with an operator that sends $e_1,\dots,e_m$ into a simple function, so, WLOG, Te_i is a simple function as $i=1$. Then $\operatorname{Range}T$ is contained in a sublattice of $L_1(\mu)$ spanned by a disjoint set of characteristic functions. This sublattice forms a lattice copy of ℓ_1^k for some k. Hence, we may assume WLOG that $Y=\ell_1^k$. Padding with zeros, if necessary, we may assume WLOG that $m=k$.

Hence, it suffices to prove the theorem for $T\colon\ell_\infty^m\to\ell_1^m$. We can now treat T as an $m\times m$ matrix, so the theorem becomes a statement about matrices, and the expressions $\left(\sum_{i=1}^n|x_i|^2\right)^{\frac12}$ and $\left(\sum_{i=1}^n|Tx_i|^2\right)^{\frac12}$ are computed component-wise. Let $y_i=Tx_i$ for every $i=1,\dots,n$; let $x_{i,j}$ and $y_{i,j}$ be the j-th components of x_i and y_i, respectively, as $j=1,\dots,m$. In this notation, (16.6) becomes

$$\sum_{j=1}^m\Bigl(\sum_{i=1}^n|y_{i,j}|^2\Bigr)^{\frac12}\leqslant K_G\|T\|\max_{j=1,\dots,m}\Bigl(\sum_{i=1}^n|x_{i,j}|^2\Bigr)^{\frac12}.$$

This is an equivalent form of Grothendieck's inequality; see, e.g., Theorem 1.f.14 in [LT79] or [Pis12]. □

16.4 (p,q)-convex/concave operators and spaces

Throughout this section, let X and Y be Banach lattices, E and F Banach spaces, and $p,q \in [1,\infty]$. A linear operator $T\colon E \to X$ is said to be (p,q)***-convex*** if it is power-bounded from the p-sum power-norm on E to the lattice q-multinorm on X, that is, $\|\overline{T}\bar{x}\|_{\text{lat-}q} \lesssim \|\bar{x}\|_{\text{sum-}p}$. This means that there exists a constant $K > 0$ such that

$$\Big\|\Big(\sum_{i=1}^{n}|Tx_i|^q\Big)^{\frac{1}{q}}\Big\| \leqslant K\Big(\sum_{i=1}^{n}\|x_i\|^p\Big)^{\frac{1}{p}}$$

for any $x_1,\dots,x_n \in E$. We then say that T is (p,q)-convex with constant K. Similarly, a linear operator $T\colon X \to E$ said to be (p,q)***-concave*** with constant K if $\|\overline{T}\bar{x}\|_{\text{sum-}p} \leqslant K\|\bar{x}\|_{\text{lat-}q}$ for every multivector $\bar{x}$ in X

Clearly, the constants above are just the appropriate power-norms of T. It follows immediately that T itself is bounded, and $\|T\| \leqslant K$. It is clear that restricting $T\colon E \to X$ to a closed subspace preserves (p,q)-convexity, while restricting $T\colon X \to E$ to a closed sublattice preserves (p,q)-concavity.

The reader may have an impression at this point that we have introduced a huge number of classes of operators, as (p,q) could be any pair in $[1,\infty]$. However, we will see later that only pairs of the form (p,p) and (p,∞) are interesting for convexity, because on one hand, there are no non-trivial (p,q)-convex operators when $q < p$ and, on the other hand, we will show in Theorem 17.3.6 that for a fixed p, the classes of (p,q)-convex operators for all $q \in (p,\infty]$ coincide. The situation with concavity is similar: only (p,p)-concave and $(p,1)$-concave operators are of interest.

16.4.1 Exercise. If $p > q$ then the only (p,q)-convex operator is zero. If $p < q$ then the only (p,q)-concave operator is zero.

Using the fact that $\|\cdot\|_{\ell_p} \leqslant \|\cdot\|_{\ell_r}$ when $r \leqslant p$, it is easy to show that convexity and concavity are preserved if we increase the spread between p and q:

16.4.2 Exercise. (i) If T is (p,q)-convex then T is (r,s)-convex whenever $r \leqslant p$ and $q \leqslant s$, with the same constant.

(ii) If T is (p,q)-concave then T is (r,s)-concave for any $r \geqslant p$ and $s \leqslant q$, with the same constant.

The following exercise shows that the classes of (p,q)-convex and of (p,q)-concave operators are stable under composition with a bounded operator on one side and with a regular operator on the other side:

16.4.3 Exercise. (i) Let $T\colon E \to X$ be (p,q)-convex with constant K.

(a) If $S\colon F\to E$ is a bounded operator then TS is (p,q)-convex with constant $K\|S\|$;

(b) If $S\colon X\to Y$ is a regular operator then ST is (p,q)-convex with constant $K\|S\|_r$.

(ii) Let $T\colon X\to E$ be (p,q)-concave with constant K.

(a) If $S\colon E\to F$ is a bounded operator then ST is (p,q)-concave with constant $K\|S\|$;

(b) If $S\colon Y\to X$ is a regular operator then TS is (p,q)-concave with constant $K\|S\|_r$.

In the special case when $p=q$, we say that T is ***p*-convex** instead of (p,p)-convex; same with concavity.

16.4.4 Exercise. Every (bounded) operator $T\colon E\to X$ is 1-convex, with $\|T\|$ being the 1-convexity constant. Every operator $T\colon X\to E$ is ∞-concave, with $\|T\|$ being the ∞-concavity constant.

A Banach lattice X is **(p,q)*-convex*** or **(p,q)*-concave*** with constant K if the identity operator on it has the corresponding property. That is, X is (p,q)-convex if $\|\cdot\|_{\text{lat-}q}\lesssim\|\cdot\|_{\text{sum-}p}$ and (p,q)-concave if $\|\cdot\|_{\text{sum-}p}\lesssim\|\cdot\|_{\text{lat-}q}$. We say that X is p-convex if it is (p,p)-convex; same for concavity. That is, X is ***p*-convex** with constant K if

$$\Big\|\Big(\sum_{i=1}^n|x_i|^p\Big)^{\frac{1}{p}}\Big\|\leqslant K\Big(\sum_{i=1}^n\|x_i\|^p\Big)^{\frac{1}{p}}\tag{16.7}$$

for any $x_1,\dots,x_n\in X$; X is ***p*-concave** with constant K if

$$\Big(\sum_{i=1}^n\|x_i\|^p\Big)^{\frac{1}{p}}\leqslant K\Big\|\Big(\sum_{i=1}^n|x_i|^p\Big)^{\frac{1}{p}}\Big\|\tag{16.8}$$

for any $x_1,\dots,x_n\in X$.

As observed in Exercise 16.3.7, $\|\cdot\|_{\text{lat-}p}$ and $\|\cdot\|_{\text{sum-}p}$ agree on $L_p(\mu)$. Hence, $L_p(\mu)$ is p-convex and p-concave with constant 1. On the other hand, suppose that X is a Banach lattice which is p-convex and p-concave with constant 1. Then, in particular, $\|x+y\|^p=\|x\|^p+\|y\|^p$ whenever $x\perp y$. By Theorem 10.7.1, X is an abstract L_p-space and, therefore, is lattice isometric to $L_p(\mu)$ for some measure μ. Thus, we can view p-convex and p-concave Banach lattices as one-sided isomorphic variants of abstract L_p-spaces.

16.4.5 Exercise. (i) Every Banach lattice X is 1-convex and ∞-concave with constant 1;

(ii) $L_p(\mu)$ is q-convex iff $q \leqslant p$ and q-concave iff $q \geqslant p$ (provided $\dim L_p(\mu) = \infty$).
(iii) X is 1-concave with constant 1 iff X is an AL-space;
(iv) X is ∞-convex with constant 1 iff X is an AM-space;
(v) If X is p-convex and q-concave then $p \leqslant q$.

We will see later that the constant K in the definitions of (p,q)-convex and (p,q)-concave spaces may be removed by renorming the space. In practice, this often allows one to say that "WLOG, up to a renorming, we may assume that the constant is 1". In the case when the constant is 1, there are easier equivalent definitions of these p-convexity and p-concavity. In this case, it clearly suffices to state the definition for just two vectors.

The following proposition shows that the concepts of convexity and concavity are dual. As usual, we write p^* for the conjugate of p.

16.4.6 Theorem.

(i) $T\colon E \to X$ *is* (p,q)*-convex iff* T^* *is* (p^*,q^*)*-concave;*
(ii) $T\colon X \to E$ *is* (p,q)*-concave iff* T^* *is* (p^*,q^*)*-convex.*

In both cases, the constant of convexity/concavity remains the same.

Proof. Suppose that $T\colon E \to X$ is (p,q)-convex with constant K. Let $f_1,\dots,f_n \in X^*$. We may view $(T^*f_1,\dots,T^*f_n)$ as an element of the ℓ_{p^*}-sum of n copies of E^*:

$$(E^* \oplus \dots \oplus E^*)_{\ell_{p^*}} = \big((E \oplus \dots \oplus E)_{\ell_p}\big)^*,$$

where the duality is given by

$$\big\langle (g_1,\dots,g_n),(x_1,\dots,x_n)\big\rangle = \sum_{i=1}^n g_i(x_i),$$

where $x_1,\dots,x_n \in E$ and $g_1,\dots,g_n \in E^*$. It follows that

$$\|\bar g\|_{\text{sum-}p^*} = \sup\Big\{\sum_{i=1}^n g_i(x_i) \;:\; \|\bar x\|_{\text{sum-}p} \leqslant 1\Big\}. \tag{16.9}$$

Suppose that $\|\bar x\|_{\text{sum-}p} \leqslant 1$ in E. with By Proposition 16.2.1,

$$\sum_{i=1}^n (T^*f_i)(x_i) = \sum_{i=1}^n f_i(Tx_i) \;\leqslant \Big\langle \Big(\sum_{i=1}^n |f_i|^{q^*}\Big)^{\frac{1}{q^*}}, \Big(\sum_{i=1}^n |Tx_i|^q\Big)^{\frac{1}{q}} \Big\rangle$$
$$\leqslant \|\bar f\|_{\text{lat-}q^*}\|\overline{T}\bar x\|_{\text{lat-}q} \leqslant K\|\bar f\|_{\text{lat-}q^*}.$$

It follows from (16.9) that $\|\overline{T^*}\bar f\|_{\text{sum-}p^*} \leqslant K\|\bar f\|_{\text{lat-}q^*}$. Hence, T^* is (p^*,q^*)-concave with constant K.

Suppose now that $T\colon X\to E$ is (p,q)-concave with constant K. Let $f_1,\dots,f_n\in E^*$. Put $g=\Bigl(\sum_{i=1}^n\lvert T^*f_i\rvert^{q^*}\Bigr)^{\frac{1}{q^*}}$. Clearly, $\lVert g\rVert=\sup_{x\in B_X^+}g(x)$. Fix $x\in B_X^+$. We will use Theorem 16.2.2 to estimate $g(x)$. Let $x_1,\dots,x_n\in X$ such that $\Bigl(\sum_{i=1}^n\lvert x_i\rvert^q\Bigr)^{\frac{1}{q}}\leqslant x$. Then

$$\sum_{i=1}^n\langle T^*f_i,x_i\rangle=\sum_{i=1}^n\langle f_i,Tx_i\rangle\leqslant\sum_{i=1}^n\lVert f_i\rVert\lVert Tx_i\rVert\leqslant\lVert\bar f\rVert_{\text{sum-}p^*}\lVert\overline{T}\bar x\rVert_{\text{sum-}p}\leqslant K\lVert\bar f\rVert_{\text{sum-}p^*}.$$

It follows from Theorem 16.2.2 that $g(x)\leqslant K\lVert\bar f\rVert_{\text{sum-}p^*}$ and, therefore, $\lVert\overline{T^*}\bar f\rVert_{\text{lat-}q^*}=\lVert g\rVert\leqslant K\lVert\bar f\rVert_{\text{sum-}p^*}$. Hence, T^* is (p^*,q^*)-convex.

Suppose that T^* is (p^*,q^*)-concave. Then T^{**} is (p,q)-convex. It follows that T is (p,q)-convex as a restriction of T^{**} to a sublattice. The last implication is proved in a similar manner. □

16.4.7 Corollary.

(i) *X is p-convex iff X^* is p^*-concave;*
(ii) *X is p-concave iff X^* is p^*-convex.*

In both cases, the constant of convexity/concavity remains the same.

16.4.8 Exercise. If X is (p,q)-convex or (p,q)-convex then the same is true for every closed sublattice of X and for every quotient of X over a closed ideal; with the same constants.

16.4.9 Exercise. (p,q)-convexity is stable under ℓ_∞-sums: if $\{X_\gamma\}_{\gamma\in\Gamma}$ is a family of Banach lattices such that X_γ is (p,q)-convex with constant 1 for every $\gamma\in\Gamma$, then $\Bigl(\bigoplus_{\gamma\in\Gamma}X_\gamma\Bigr)_\infty$ is (p,q)-convex with constant 1. Similarly, (p,q)-concavity is stable under ℓ_1-sums.

We will show next that p-convexity is preserved if we lower the exponent, both for operators and for spaces. This is non-trivial because both sides of the inequality in the definition of convexity increase as we lower the exponent. Similarly, concavity is preserved when the exponent goes up.

16.4.10 Theorem.

(i) *If $T\colon E\to X$ is p-convex then it is s-convex for every $s\in[1,p]$;*
(ii) *If $T\colon X\to E$ is p-concave then it is s-concave for every $s\in[p,\infty]$.*

Proof. (i) Suppose that $T\colon E\to X$ is p-convex with constant K, and $s\in[1,p]$. The cases when $s=1$ or $s=p$ are trivial, so we assume that $1<s<p$. The idea is the

following: we use 16.1.7 with appropriate coefficients to express the s-sum in question as an interpolation between the 1-sum and the p-sum; we then estimate the 1-sum using the triangle inequality and the p-sum using the assumption.

Let $x_1,\dots,x_n\in E$. WLOG, $x_i\neq 0$ for every i. For each i, put $y_i=\frac{x_i}{\|x_i\|}$ and $\alpha_i=\|x_i\|^s$. Since $\frac{1}{p}<\frac{1}{s}<1$, we can write $\frac{1}{s}$ as a convex combination $\frac{1}{s}=\lambda+\frac{1-\lambda}{p}$ for some $\lambda\in(0,1)$. Using 16.1.7, we get

$$\begin{aligned}
\Bigl\|\Bigl(\sum_{i=1}^n|Tx_i|^s\Bigr)^{\frac{1}{s}}\Bigr\| &= \Bigl\|\Bigl(\sum_{i=1}^n\alpha_i|Ty_i|^s\Bigr)^{\frac{1}{s}}\Bigr\| \\
&\leqslant \Bigl\|\sum_{i=1}^n\alpha_i|Ty_i|\Bigr\|^{\lambda}\cdot\Bigl\|\Bigl(\sum_{i=1}^n|\alpha_i^{\frac{1}{p}}Ty_i|^p\Bigr)^{\frac{1}{p}}\Bigr\|^{1-\lambda} \\
&\leqslant \|T\|^{\lambda}\Bigl(\sum_{i=1}^n\alpha_i\|y_i\|\Bigr)^{\lambda}\cdot K\Bigl(\sum_{i=1}^n\|\alpha_i^{\frac{1}{p}}y\|^p\Bigr)^{\frac{1-\lambda}{p}} \\
&= K\|T\|^{\lambda}\Bigl(\sum_{i=1}^n\alpha_i\Bigr)^{\lambda}\Bigl(\sum_{i=1}^n\alpha_i\Bigr)^{\frac{1-\lambda}{p}} = K\|T\|^{\lambda}\Bigl(\sum_{i=1}^n\|x_i\|^s\Bigr)^{\frac{1}{s}}.
\end{aligned}$$

(ii) follows by duality. Indeed, suppose that T is p-concave and $s\geqslant p$. By Theorem 16.4.6, T^* is p^*-convex. Since $s^*\leqslant p^*$, it follows from (i) that T^* is s^*-convex. Applying Theorem 16.4.6 again, we conclude that T is s-concave. □

16.4.11 Remark. With some more work, one can modify the preceding proof so that it also shows that if X is p-convex then the convexity constant is a continuous non-decreasing function on $[1,p]$; see, e.g., Proposition 1.d.5 in [LT79].

16.4.12 Corollary.

(i) *If X is p-convex then it is s-convex for every $s\in[1,p]$;*
(ii) *If X is p-concave then it is s-concave for every $s\in[p,\infty]$.*

16.4.13 Exercise. Let $T\colon E\to X$. TFAE:

(i) T is (p,q)-convex with constant 1
(ii) $\Bigl(\sum_{i=1}^n|\alpha_iTx_i|^q\Bigr)^{\frac{1}{q}}\in B_X$ whenever $x_1,\dots,x_n\in B_E$ and $(\alpha_i)_{i=1}^n\in B_{\ell_p^n}$;
(iii) $\Bigl(\sum_{i=1}^n|\alpha_iTx_i|^q\Bigr)^{\frac{1}{q}}\in B_X$ whenever $x_1,\dots,x_n\in S_E$ and $(\alpha_i)_{i=1}^n\in S_{\ell_p^n}$.

We proceed with the dual version of the preceding exercise; B_X° stands for the open unit ball of X:

16.4.14 Exercise. Let $T\colon X\to E$. TFAE:

(i) T is (p,q)-concave with constant 1

(ii) $\left(\sum_{i=1}^n |\alpha_i x_i|^q\right)^{\frac{1}{q}} \notin B_X^\circ$ whenever $Tx_1,\dots,Tx_n \notin B_E^\circ$ and $(\alpha_i)_{i=1}^n \notin B_{\ell_p^n}^\circ$;

(iii) $\left(\sum_{i=1}^n |\alpha_i x_i|^q\right)^{\frac{1}{q}} \notin B_X^\circ$ whenever $Tx_1,\dots,Tx_n \in S_E$ and $(\alpha_i)_{i=1}^n \in S_{\ell_p^n}$.

In particular, this exercise provides an equivalent characterization of (p,q)-convex and (p,q)-concave Banach lattices:

16.4.15 Corollary.

(i) *X is (p,q)-convex with constant 1 iff $\left(\sum_{i=1}^n |\alpha_i x_i|^q\right)^{\frac{1}{q}} \in B_X$ whenever $x_1,\dots,x_n \in B_X$ and $(\alpha_i)_{i=1}^n \in B_{\ell_p^n}$.*

(ii) *X is (p,q)-concave with constant 1 iff $\left(\sum_{i=1}^n |\alpha_i x_i|^q\right)^{\frac{1}{q}} \notin B_X^\circ$ whenever $x_1,\dots,x_n \notin B_X^\circ$ and $(\alpha_i)_{i=1}^n \notin B_{\ell_p^n}^\circ$.*

In both (i) *and* (ii)*, it suffices to take $x_1,\dots,x_n$ in S_X and $(\alpha_i)_{i=1}^n$ in $S_{\ell_p^n}$.*

The preceding corollary has a geometric interpretation. Condition (i) says that (p,q)-convexity of X may be expressed as closedness of the ball with respect to certain expressions, similar to convexity; this somewhat justifies the name "(p,q)-convex". Similarly, condition (ii) says that $X^+ \setminus B_X^\circ$ is closed under certain expressions. For comparison, recall that if $X = \ell_1^2$ then both B_X and $X^+ \setminus B_X^\circ$ are convex sets.

Recall that if X is order continuous and has a weak unit, then X and X^* may be represented as ideals in some $L_1(\mu)$ space; see Theorem 9.3.2.

16.4.16 Theorem ([JMST79])**.** *Let X be as in Theorem 9.3.2 and $1 < p,q < \infty$.*

(i) *If X is p-convex with constant 1 then $X \subseteq L_p(\mu)$.*

(ii) *If X is p-concave with constant 1 then $L_p(\mu) \subseteq X$.*

In both cases, the inclusion is continuous.

We first prove the following lemma:

16.4.17 Lemma. *Let X be a Banach lattice, contractively embedded as an ideal in $L_1(\mu)$ for some probability measure μ. Let $1 < p < \infty$. Suppose that X is p-convex with constant 1. Then $\|x\|_{L_p} \leqslant \|x\|_X$ for every simple function $x \in X$. Moreover, if X has the Fatou Property then $\|x\|_{L_p} \leqslant \|x\|_X$ for all $x \in X$, hence $X \subseteq L_p(\mu)$.*

Proof. Fix a simple function x in X_+ and $t \geqslant 0$. Using p-convexity, we have

$$\int \left(tx^p(\omega)+1\right)^{\frac{1}{p}} d\mu(\omega) = \left\|(tx^p+\mathbb{1})^{\frac{1}{p}}\right\|_{L_1} \leqslant \left\|\left((t^{\frac{1}{p}}x)^p + \mathbb{1}^p\right)^{\frac{1}{p}}\right\|_X$$
$$\leqslant \left(\|t^{\frac{1}{p}}x\|_X^p + \|\mathbb{1}\|_X^p\right)^{\frac{1}{p}} = \left(t\|x\|_X^p + 1\right)^{\frac{1}{p}}.$$

Define $F(t) = \int\Bigl(tx^p(\omega)+1\Bigr)^{\frac{1}{p}}d\mu(\omega)$ and $G(t) = \bigl(t\lVert x\rVert_X^p + 1\bigr)^{\frac{1}{p}}$, then $F(t) \leqslant G(t)$ for all $t \geqslant 0$, and $F(0) = 1 = G(0)$. It follows that $F'(0) \leqslant G'(0)$. Observe that $F'(0) = \frac{1}{p}\int x^p(\omega)\,d\mu(\omega) = \frac{1}{p}\lVert x\rVert_{L_p}^p$. We do not have to worry about differentiability of F or interchanging integration with differentiation because x is a simple function, hence the integral is actually a finite sum. Furthermore, $G'(0) = \frac{1}{p}\lVert x\rVert_X^p$. It follows that $\lVert x\rVert_{L_p} \leqslant \lVert x\rVert_X$.

Now suppose that X has the Fautou Property and $x \in X_+$. Let (x_n) be a sequence of simple functions such that $x_n \uparrow x$. By the preceding argument, $\lVert x_n\rVert_{L_p} \leqslant \lVert x_n\rVert_X$ for every n. Since both X and $L_p(\mu)$ have the Fatou Property, passing to the limit, we conclude that $\lVert x\rVert_{L_p} \leqslant \lVert x\rVert_X$. It follows that $X \subseteq L_p(\mu)$. □

Proof of Theorem 16.4.16. (i) follows by the lemma because every order continuous Banach lattice has the Fatou Property. Note that we may assume WLOG that the embedding $X \hookrightarrow L_1(\mu)$ is a contraction.

(ii) Suppose that X is p-concave. Then X^* is q-convex by Corollary 16.4.7, where $q = p^*$, and X^* has the Fatou Property by Exercise 10.6.8. Scaling the norm of X, if necessary, we may assume WLOG that the embedding $X^* \hookrightarrow L_1(\mu)$ is a contraction. By the lemma, $\lVert y\rVert_{L_q} \leqslant \lVert y\rVert_{X^*}$ for all $y \in X^*$.

Let $x \in X_+$ be a simple function, and $\varepsilon > 0$. Find $y \in X_+^*$ such that $\lVert y\rVert_{X^*} = 1$ and

$$\lVert x\rVert_X - \varepsilon \leqslant \langle x, y\rangle = \int xy\,d\mu \leqslant \lVert x\rVert_{L_p}\lVert y\rVert_{L_q} \leqslant \lVert x\rVert_{L_p}\lVert y\rVert_{X^*} \leqslant \lVert x\rVert_{L_p}.$$

We conclude that $\lVert x\rVert_X \leqslant \lVert x\rVert_{L_p}$ for every simple x in X_+.

Now let $0 \leqslant x \in L_p(\mu)$. Find a sequence of simple functions (x_n) such that $0 \leqslant x_n \uparrow x$. Since $L_p(\mu)$ is order continuous, $\lVert x_n - x\rVert_{L_p} \to 0$. It follows that the sequence (x_n) is Cauchy in $L_p(\mu)$. By the preceding, $\lVert x_n - x_m\rVert_X \leqslant \lVert x_n - x_m\rVert_{L_p}$ for every m, n, so that sequence (x_n) is Cauchy in X, hence converges in X to some y. It follows that $y = \sup x_n = x$. (Note that since both $L_p(\mu)$ and X are ideals in $L_1(\mu)$, suprema of sequences are the same in them.) It follows that $x \in X$ and that $\lVert x\rVert_X = \lim_n\lVert x_n\rVert_X \leqslant \lim_n\lVert x_n\rVert_{L_p} = \lVert x\rVert_{L_p}$. □

In Definition 16.3.12, we introduced (p,q)-summing operators. Recall that by Proposition 16.3.11, the weak q-summing norm on $C(K)$ agrees with the canonical q-multinorm. This immediately yields the following:

16.4.18 Proposition. *An operator $T\colon C(K) \to E$ is (p,q)-summing iff it is (p,q)-concave (with the same constants).*

It is a natural question whether there are any interesting relationships between (p,q)-convexity/concavity and various vector lattice properties of operators and spaces. Say, could it be that every (p,q)-convex operator is order bounded or regular, or vice versa? In general, the answer is "no". For example, every bounded operator on ℓ_2 is 2-convex and 2-concave, while there are bounded operators on ℓ_2 that are not order bounded; see, e.g., Krengel's example 6.4.1. On the other hand, the identity operator on a Banach lattice is certainly order bounded and regular, but it need not be p-convex or p-concave in general. The space $C[0,1]$ is an AM-space, so it is p-convex for every p, yet is fails to be even order complete. On the bright side, we will show in Section 16.6 that every $(p,1)$-concave operator is order weakly compact and that every $(p,1)$-concave Banach lattice is a KB-space.

16.5 p-convexification and p-concavication

16.5.1 Vector lattice case

Throughout this subsection, X is a uniformly complete Archimedean vector lattice and $p\in(0,\infty)$. In previous sections, we considered the expression $\left(\sum_{i=1}^n|x_i|^p\right)^{\frac{1}{p}}$. In this section, we will need the expression $\left(\sum_{i=1}^n x_i^p\right)^{\frac{1}{p}}$. Formally, we cannot define this expression via homogeneous function calculus because the function $t\mapsto t^p$ is defined on $[0,\infty)$, but, generally, not for $t<0$. It will be convenient for us to extend this function to $(-\infty,0)$ via $(-t)^p=-t^p$ for $t>0$. Note that this notation conflicts with the common one because, for example, $(-5)^2$ is now -25 rather than 25! Under this convention, however, the expression $\left(\sum_{i=1}^n t_i^p\right)^{\frac{1}{p}}$ defines a continuous positively homogeneous function on $\mathbb{R}^n$. Hence, the expression $\left(\sum_{i=1}^n x_i^p\right)^{\frac{1}{p}}$ is now defined for $x_1,\dots,x_n\in X$ via the function calculus. We use the non-standard notation for t^p for this definition only.

We start with some motivation. Let μ be a measure. The map $f\mapsto f^p$ is a non-linear isometry between $L_p(\mu)$ and $L_1(\mu)$. Similarly, there is a non-linear isometry between $L_p(\mu)$ and $L_q(\mu)$ for any $p,q\in(0,\infty)$, given be $f\mapsto f^{\frac{p}{q}}$.

By Maeda-Ogasawara theory, specifically, by Theorem 7.4.15, X may be represented as an order dense sublattice of $C^\infty(K)$ for some extremally disconnected compact space K. Fix such a representation. Recall that, by Proposition 7.4.21, the (continuous positively homogeneous) function calculus in X agrees with the "almost everywhere pointwise" functional calculus on $C^\infty(K)$. For $f\in C^\infty(K)$, we define f^p pointwise: $f^p(t)=\big(f(t)\big)^p$; it is clear that f^p is again an element of $C^\infty(K)$. Let

$X^p := \{f^p : f \in X\}$, viewed as a subset of $C^\infty(K)$. We call X^p the ***p-concavification*** of X (the motivation for this name will become clear later).

16.5.1 Proposition. *X^p is an order dense sublattice of $C^\infty(K)$; hence is itself an Archimedean vector lattice under the operations induced from $C^\infty(K)$.*

Proof. X^p is closed under scalar multiplication because if $x \in X$ and $\alpha \in \mathbb{R}$ then $\alpha(x^p) = \big(\alpha^{\frac{1}{p}} x)^p \in X^p$. To show that X^p is closed under addition, let $x, y \in X$ and define $z \in X$ via $z = (x^p + y^p)^{\frac{1}{p}}$ in the sense of the function calculus of X and, therefore, of $C^\infty(K)$, so that $x^p + y^p = z^p$ in $C^\infty(K)$. It follows that $x^p + y^p \in X^p$. Finally, if $x \in X$ then $|x^p| = |x|^p$ (pointwise), hence $|x^p| \in X^p$. We conclude that X^p is a sublattice of $C^\infty(K)$. To see that it is order dense, let $0 < f \in C^\infty(K)$. Then $0 < f^{\frac{1}{p}} \in C^\infty(K)$, so order density of X implies that there exists $x \in X$ such that $0 < x \leqslant f^{\frac{1}{p}}$. This yields $0 < x^p \leqslant f$. $\square$

Let $\Phi\colon X \to X^p$ via $\Phi(x) = x^p$. Clearly, Φ is an order preserving (non-linear) bijection. We can use Φ to define new vector lattice operations on X by transfering the vector lattice structure from X^p onto X: for $x, y \in X$ and $\lambda \in \mathbb{R}$, we define (in X)

$$\lambda \odot x = \Phi^{-1}\big(\lambda\Phi(x)\big) = \lambda^{\frac{1}{p}} x \quad \text{and} \quad x \oplus y = \Phi^{-1}\big(\Phi(x) + \Phi(y)\big) = \big(x^p + y^p\big)^{\frac{1}{p}},$$

where the latter expression may be interpreted either in $C^\infty(K)$ or, equivalently, using the function calculus of X. (Note that $\ominus x$ agrees with $-x$.) Equipped with these new vector operation and with the original order and lattice operations, $\big(X, \oplus, \odot, \leqslant\big)$ is a vector lattice which is lattice isomorphic to X^p. This provides an equivalent definition of X^p that does not involve a $C^\infty(K)$ representation. In some literature, it is $\big(X, \oplus, \odot, \leqslant\big)$ rather than X^p that is called the p-concavification of X. It follows, in particular, that X^p is well-defined in the sense that it does not depend on the choice of a $C^\infty(K)$ representation, up to a lattice isomorphism. Depending on circumstances, it may be more convenient to use either $\big(X, \oplus, \odot, \leqslant\big)$ or X^p. One can view $\big(X, \oplus, \odot, \leqslant\big)$ as an "abstract" p-concavification, and X^p as its functional representation. In the former approach, the p-convexification agrees with X as a set, but is equipped with different vector space operations. In the latter approach, X^p is a different set, but the operations are more "natural".

16.5.2 Proposition. *X^p is uniformly complete.*

Proof. It suffices to show that $\big(X, \oplus, \odot, \leqslant\big)$ is uniformly complete. Fix $e > 0$ in X. By Krein-Kakutani Representation Theorem, we can find a lattice isomorphism $T\colon I_e \to C(L)$ for some compact Hausdorff space L. By uniqueness of function calculus, the function calculus on I_e agrees with the pointwise function calculus on $C(L)$ via T. It is easy to see that as a set, I_e agrees with the principal ideal of e in $\bigl(X,\oplus,\odot,\leqslant\bigr)$, hence the I_e notation is unambiguous (though, vector operations are different). Consider the map $\Phi\colon C(L)\to C(L)$ given by $\Phi(f)=f^p$. Clearly, Φ is a non-linear bijection from $C(L)$ onto itself.

$$\begin{tikzcd} I_e \arrow[r, "T"] \arrow[dr, "S"'] & C(L) \arrow[d, "\Phi"] \\ & C(L) \end{tikzcd}$$

Let $S=\Phi\circ T$. Observe that S is linear with respect to $\oplus$ and $\odot$. In particular, $x\oplus y=S^{-1}\bigl(Sx+Sy\bigr)$ and $\alpha\odot x=S^{-1}\bigl(\alpha Sx\bigr)$ for $x,y\in I_e$. It follows that S is a lattice isomorphism between I_e and $C(L)$, where the former is viewed as an ideal in $\bigl(X,\oplus,\odot,\leqslant\bigr)$. Therefore, $\bigl(X,\oplus,\odot,\leqslant\bigr)$ is uniformly complete. $\square$

Since X^p is uniformly complete, it has a function calculus of its own. Since $C^\infty(K)$ is the universal completion of X^p, this function calculus again agrees with the "almost everywhere pointwise" functional calculus of $C^\infty(K)$. In particular, if $0<q<0$, we can now apply the q-concavification procedure to X^p. It is now easy to see that

$$(X^p)^q=X^{pq}=\bigl\{x^{pq}\ :\ x\in X\bigr\}\subseteq C^\infty(K).$$

Since $\frac{1}{p}$ is also in $(0,\infty)$, one can also perform the preceding construction with p replaced with $\frac{1}{p}$. We call $X^{\frac{1}{p}}$ the ***p-convexification*** of X. (Again, the motivation for this term will become clear later.) Clearly, $(X^p)^{\frac{1}{p}}=X$.

16.5.3. Let Y be a uniformly complete sublattice of a X. By uniqueness of function calculus, the function calculus in Y agrees with the restriction to Y of the function calculus on X. It follows that $\bigl(Y,\oplus,\odot,\leqslant\bigr)$ may be viewed as a sublattice of $\bigl(Y,\oplus,\odot,\leqslant\bigr)$ and, therefore, Y^p may be identified with the sublattice of X^p consisting of the elements of the form y^p for $y\in Y$.

Let $T\colon X\to Y$ be a linear operator between uniformly complete Archimedean vector lattices. It need not be linear with respect to $\oplus$! However, if T is a lattice homomorphisms then it preserve function calculus by Exercise 4.4.6; it follows that $T(x\oplus y)=(Tx)\oplus(Ty)$. Hence, we have the following:

16.5.4 Proposition. *Let $T\colon X\to Y$ be a lattice homomorphism between two uniformly complete Archimedean vector lattices. Viewed as a map from $\bigl(X,\oplus,\odot,\leqslant\bigr)$ to $\bigl(Y,\oplus,\odot,\leqslant\bigr)$, T is still a linear operator and a lattice homomorphism. We may also* *view it as a (linear) lattice homomorphism from X^p to Y^p defined vie $x^p \mapsto (Tx)^p$, where $x \in X$.*

The following examples show that in function spaces, one can usually realize X^p as a space of functions on the same domain.

16.5.5 Example. Consider the map $\Phi\colon C(K) \to C(K)$ defined via $\Phi(f) = f^p$. Clearly, Φ is non-linear. However, Φ is linear lattice isomorphism when viewed as a map from $C(K)$ to $\bigl(C(K), \oplus, \odot, \leqslant\bigr)$. It follows that $C(K)^{(p)}$ is lattice isomorphic to $C(K)$.

In our definition of X^p, we used $C^\infty(K)$ as an ambient space. The following example shows that in case of spaces of measurable functions, one may usually use $L_0(\mu)$ as an ambient space instead of $C^\infty(K)$.

16.5.6 Example. Let X be a uniformly complete sublattice of $L_0(\mu)$ for some measure space $(\Omega, \mathcal{F}, \mu)$. Let $\Psi\colon L_0(\mu) \to L_0(\mu)$ be defined by $\Psi(f) = f^p$ in the sense that $\bigl(\Psi(f)\bigr)(t) = \bigl(f(t)\bigr)^p$ for all $t \in \Omega$. As in Proposition 16.5.1, one can show that $Y \coloneqq \operatorname{Range}\Psi$ is a sublattice of $L_0(\mu)$. It is also a straightforward verification that Ψ is a (linear) lattice isomorphism between Y and $\bigl(X, \oplus, \odot, \leqslant\bigr)$ and, therefore, Y is lattice isomorphic to X^p. So for most practical purposes, we may just identify X^p with Y.

In particular, under this identification, we have $\bigl(L_p(\mu)\bigr)^p = L_1(\mu)$ and $\bigl(L_p(\mu)\bigr)^{\frac{1}{p}} = L_p(\mu)$.

16.5.2 Banach lattice case

For the rest of the section, we will assume that X is a Banach lattice and $0 < p < \infty$. Does X^p have a natural Banach lattice norms? We will see that the answer is always affirmative when $p \leqslant 1$, while in the case when $p > 1$ we need X to be p-convex.

Let's start with some motivation and examples. Given $1 \leqslant r < \infty$, we have $\bigl(L_r(\mu)\bigr)^p = L_{\frac{r}{p}}(\mu)$. This is a Banach lattice precisely when $p \leqslant r$, i.e., either when $p \leqslant 1$ or when $p > 1$ and $L_r(\mu)$ is p-convex.

As before, we view X and X^p as sublattices of $C^\infty(K)$. Motivated by the $L_r(\mu)$ example, it seems natural to define a norm on X^p via $\|x^p\|_{X^p} = \|x\|_X^p$, where $x \in X$. This means that the unit ball of this norm on X^p is the set $\bigl\{x^p : x \in B_X\bigr\}$. We will denote this set by $(B_X)^p$. It is easy to see that this set is solid in X^p. But is it convex?

16.5.7 Example. Let $X = \ell_r^n$, where $1 \leqslant r < \infty$ and $n \in \mathbb{N}$. The set $(B_X)^p$ is the unit ball of $\ell^n_{r/p}$. If $p < 1$, it is an inflation of the original ball, it is "closer" to the ℓ^n_∞, and it is still convex, because $\ell^n_{r/p}$ is a Banach space. On the other hand, if $p > 1$ then

$(B_X)^p$ is a deflation of the original ball. At p goes up from 1 to r, this ball morphs from $B_{\ell_r^n}$ into $B_{\ell_1^n}$ when $p=r$. If we keep increasing p beyond r, $(B_X)^p$ is no longer convex, so it is not a unit ball of a norm any more.

To proceed, we will need the following a technical calculus lemma, which, essentially, says that the expressions $t-s$ and t^p-s^p are, in some sense, equivalent on bounded sets in $\mathbb{R}$.

16.5.8 Exercise. Let $1<q<\infty$. There exist positive constants A and B such that if $s<t$ in $\mathbb{R}$ then

$$(t^{\frac{1}{q}}-s^{\frac{1}{q}})^q \leqslant A(t-s) \quad \text{and} \quad (t^q-s^q)^{\frac{1}{q}} \leqslant B\big(|s|\vee|t|\big)^{\frac{1}{q^*}}(t-s)^{\frac{1}{q}}.$$

16.5.9 Proposition. *Let X be a Banach lattice and $0<p<1$. The formula $|||x^p|||=\|x\|^p$ defines a Banach lattice norm on X^p. Moreover, $\big(X^p,|||\cdot|||\big)$ is $\frac{1}{p}$-convex with constant 1.*

Proof. Observe that $|||x^p|||=0$ iff $\|x\|=0$ iff $x=0$. Homogenuity follows from $|||\alpha x^p|||=\big\|\alpha^{\frac{1}{p}}x\big\|^p=|\alpha|\,|||x^p|||$. It is clear that the unit ball of this norm is $(B_X)^p$. In order to verify the triangle inequality, it suffices that $(B_X)^p$ is a convex set in X^p. Indeed, let $x,y\in B_X$ and $0\leqslant\lambda\leqslant 1$, then, using the facts that the function $t\mapsto t^p$ is concave on $[0,\infty)$ and that B_X is convex, we have

$$\Big|\lambda x^p+(1-\lambda)y^p\Big| \leqslant \lambda|x|^p+(1-\lambda)|y|^p \leqslant \Big(\lambda|x|+(1-\lambda)|y|\Big)^p \in (B_X)^p.$$

Since $(B_X)^p$ is solid in X^p, we conclude that $\lambda x^p+(1-\lambda)y^p\in(B_X)^p$. Therefore, $|||\cdot|||$ is a lattice norm.

We will now show completeness. Put $q=\frac{1}{p}$. By the second inequality in Exercise 16.5.8, we have

$$t-s=\big(t^{pq}-s^{pq}\big)^{\frac{1}{q}q} \leqslant B^q\big(|t|^p\vee|s|^p\big)^{\frac{1}{q^*}q}(t^p-s^p)^{\frac{1}{q}q}$$

Let $x,y\in B_X$. Applying Function Calculus in X, we get

$$|x-y| \leqslant B^q\big(|x|^p\vee|y|^p\big)^{q\frac{1}{q^*}}\big|x^p-y^p\big|^{q\frac{1}{q}}.$$

Lemma 16.1.6 yileds

$$\|x-y\| \leqslant B^q\Big\|\Big(|x|^p\vee|y|^p\Big)^q\Big\|^{\frac{1}{q^*}}\big\|(x^p-y^p)^q\big\|^{\frac{1}{q}}$$

Note that

$$\Big\|\Big(|x|^p\vee|y|^p\Big)^q\Big\|^{\frac{1}{q^*}} \leqslant \Big|\Big|\Big|\,|x|^p\vee|y|^p\,\Big|\Big|\Big|^{\frac{q}{q^*}} \leqslant \big(|||x^p|||+|||y^p|||\big)^{\frac{q}{q^*}} \leqslant 2^{\frac{q}{q^*}}.$$

and $\left\|(x^p-y^p)^q\right\|^{\frac{1}{q}} = |||x^p-y^p|||$. It follows that $\|x-y\| \leqslant 2^{\frac{q}{q^*}}B^q\,|||x^p-y^p|||$.

Let (x_n) be a sequence in X such that the sequence (x_n^p) is Cauchy in $\left(X^p, |||\cdot|||\right)$. It follows that (x_n^p) is bounded; WLOG, $|||x_n^p||| \leqslant 1$ and, therefore, $\|x_n\| \leqslant 1$ for every n. By the preceding argument, we have $\|x_n - x_m\| \leqslant 2^{\frac{q}{q^*}}B^q\,|||x_n^p - x_m^p||| \to 0$. It follows that the sequence (x_n) is Cauchy in X, hence it converges to some $x \in X$. Using the first inequality in Exercise 16.5.8, we get

$$|||x_n^p - x^p||| = \left\|\left(x_n^{\frac{1}{q}} - x^{\frac{1}{q}}\right)^q\right\|^p \leqslant A^p\|x_n - x\|^p \to 0.$$

To see that $\left(X^p, |||\cdot|||\right)$ is $\frac{1}{p}$-convex with constant 1, observe that if $x_1,\dots,x_n \in X$ then $x_1^p,\dots,x_n^p \in X^p$ and

$$\left|\left|\left|\left(\sum_{i=1}^n |x_i^p|^{\frac{1}{p}}\right)^p\right|\right|\right| = \left\|\sum_{i=1}^n |x_i|\right\|^p \leqslant \left(\sum_{i=1}^n \|x_i\|\right)^p = \left(\sum_{i=1}^n |||x_i^p|||^{\frac{1}{p}}\right)^p$$

□

We now move on to the case when $p > 1$. The following proposition may be viewed as an equivalent characterization of p-convexity with constant 1:

16.5.10 Proposition. *Let $1 < p < \infty$. X is p-convex with constant 1 iff $(B_X)^p$ is convex.*

Proof. Suppose that X is p-convex with constant 1; let $x, y \in B_X$ and $\lambda, \mu \geqslant 0$ with $\lambda + \mu = 1$. It suffices to show that $\left(\lambda x^p + \mu y^p\right)^{\frac{1}{p}} \in B_X$. Indeed,

$$\left\|\left(\lambda x^p + \mu y^p\right)^{\frac{1}{p}}\right\| \leqslant \left\|\left(|\lambda^{\frac{1}{p}}x|^p + |\mu^{\frac{1}{p}}y|^p\right)^{\frac{1}{p}}\right\| \leqslant \left(\left\|\lambda^{\frac{1}{p}}x\right\|^p + \left\|\mu^{\frac{1}{p}}y\right\|^p\right)^{\frac{1}{p}} \leqslant (\lambda+\mu)^{\frac{1}{p}} = 1.$$

Conversely, suppose that $(B_X)^p$ is convex. Let $x, y \in B_X$ and suppose that $\alpha^p + \beta^p \leqslant 1$. Convexity of $(B_X)^p$ yields $\alpha^p x^p + \beta^p y^p \in (B_X)^p$, so that $\left(\alpha^p x^p + \beta^p y^p\right)^{\frac{1}{p}} \in B_X$. It follows from Exercise 16.4.15(i) that X is p-convex with constant 1. □

Suppose that X is p-convex for some $1 < p < \infty$ with constant 1, and define $|||x^p||| = \|x\|^p$ for $x \in X$. By Proposition 16.5.10, the corresponding unit ball is $(B_X)^p$, which is a convex subset of X^p. So it is easy to verify that $|||\cdot|||$ is a lattice norm on X^p; the p-convexity assumption becomes the triangle inequality. As we observed earlier,

However, in case when the constant of convexity is greater than 1, setting $|||x^p||| = \|x\|^p$ no longer defined a norm on X^p because it now satisfies the triangle inequality with a constant. In this case, the following standard tool, which may be viewed as variant of Exercise 11.1.6, comes to rescue. It says essentially that if we replace the triangle inequality in the definition of a lattice norm with a version with a constant which does not depend on the number of vectors, the function is equivalent to an actual lattice norm:

16.5.11 Proposition. *Let $\|\cdot\|$ be a function from a vector lattice Z to $\mathbb{R}_+$ such that $\|x\|=0$ iff $x=0$, $\|\lambda x\|=|\lambda|\|x\|$, $|x|\leqslant|y|$ implies $\|x\|\leqslant\|y\|$, and there is a constant $C\geqslant 1$ such that $\bigl\|\sum_{i=1}^n x_i\bigr\|\leqslant C\sum_{i=1}^n\|x_i\|$ for any $x_1,\dots,x_n$. For $x\in Z$, put*

$$|||x|||=\inf\Bigl\{\sum_{i=1}^n\|v_i\|\ :\ v_1,\dots,v_n\geqslant 0,\ |x|\leqslant\sum_{i=1}^n v_i\Bigr\}.$$

Then $|||\cdot|||$ is a lattice norm on Z, and $\frac{1}{C}\|x\|\leqslant|||x|||\leqslant\|x\|$ for every $x\in Z$. One may replace $|x|\leqslant\sum_{i=1}^n v_i$ with $|x|=\sum_{i=1}^n v_i$ in the definition of $|||x|||$.

Proof. It is clear from the definition that $|||\lambda x|||=|\lambda|\,|||x|||$ and that if $|x|\leqslant|y|$ then $|||x|||\leqslant|||y|||$. Riesz Decomposition Theorem allows us to replace $|x|\leqslant\sum_{i=1}^n v_i$ with $|x|=\sum_{i=1}^n v_i$ in the definition of $|||x|||$. We clearly have $|||x|||\leqslant\|x\|$. For every decomposition $|x|=\sum_{i=1}^n v_i$ with $v_1,\dots,v_n\geqslant 0$, we have $\|x\|\leqslant C\sum_{i=1}^n\|v_i\|$; it follows that $\frac{1}{C}\|x\|\leqslant|||x|||$.

To verify the triangle inequality for $|||\cdot|||$, suppose that $|x|\leqslant\sum_{i=1}^n v_i$ and $|y|\leqslant\sum_{j=1}^m w_j$ for some $v_1,\dots,v_n,\ w_1,\dots,w_m\in Z_+$. It follows from $|x+y|\leqslant\sum_{i=1}^n v_i+\sum_{j=1}^m w_j$ that $|||x+y|||\leqslant\sum_{i=1}^n\|v_i\|+\sum_{j=1}^m\|w_j\|$. Taking infima over all decompositions of x and of y, we get $|||x+y|||\leqslant|||x|||+|||y|||$. □

16.5.12 Proposition. *Let X be a p-convex Banach lattice, for some $1<p<\infty$, with constant K. Then X^p is a Banach lattice under the norm*

$$|||x^p|||=\inf\Bigl\{\sum_{i=1}^n\|u_i\|^p\ :\ u_1,\dots,u_n\geqslant 0,\ |x^p|\leqslant\sum_{i=1}^n u_i^p\Bigr\},$$

and $\frac{1}{K^p}\|x\|^p\leqslant|||x^p|||\leqslant\|x\|^p$ for all $x\in X$. One may replace $|x^p|\leqslant\sum_{i=1}^n u_i^p$ with $|x^p|=\sum_{i=1}^n u_i^p$ in the definition of $|||x^p|||$.

Proof. It is easy to verify that the function $|||x^p|||_0:=\|x\|^p$ satisfies the assumptions of Proposition 16.5.11. In particular,

$$\Bigl|\Bigl|\Bigl|\sum_{i=1}^n x_i^p\Bigr|\Bigr|\Bigr|_0=\Bigl\|\Bigl(\sum_{i=1}^n x_i^p\Bigr)^{\frac1p}\Bigr\|^p\leqslant\Bigl\|\Bigl(\sum_{i=1}^n|x_i|^p\Bigr)^{\frac1p}\Bigr\|^p\leqslant K^p\sum_{i=1}^n\|x_i\|^p=K^p\sum_{i=1}^n|||x_i^p|||_0$$

for every $x_1,\dots,x_n\in X$. Now Proposition 16.5.11 yields all the required conclusions except completeness.

The proof of completeness is similar to that in Proposition 16.5.9. Let (x_n) be a sequence in X such that (x_n^p) is Cauchy in $\bigl(X^p,|||\cdot|||\bigr)$. Using the first inequality in Exercise 16.5.8 with s and t replaces with s^p and t^p, we get $\bigl|t-s\bigr|\leqslant A^{\frac1p}\bigl|\bigl(t^p-s^p\bigr)^{\frac1p}\bigr|$ for any real s and t. We conclude that $|x_n-x_m|\leqslant A^{\frac1p}\bigl|\bigl(x_n^p-x_m^p\bigr)^{\frac1p}\bigr|$, and, therefore,

$$\|x_n-x_m\|\leqslant A^{\frac1p}\bigl\|\bigl(x_n^p-x_m^p\bigr)^{\frac1p}\bigr\|\leqslant A^{\frac1p}K\,|||x_n^p-x_m^p|||^{\frac1p}\to 0.$$

This yields $\|x_n - x_m\| \to 0$, hence (x_n) is Cauchy in X, and, therefore, $\|x_n - x\| \to 0$ for some $x \in X$. It also follows that there exists $M > 0$ such that $\|x_n\| \leqslant M$ for all n. Using Exercise 16.5.8 and Lemma 16.1.6, we get

$$|||x_n^p - x^p||| \leqslant \left\|\left(x_n^p - x^p\right)^{\frac{1}{p}}\right\|^p \leqslant B\left\||x_n| \vee |x|\right\|^{\frac{p}{p^*}} \|x_n - x\| \leqslant B(2M)^{\frac{p}{p^*}} \|x_n - x\| \to 0$$

□

As we have seen, the cases $p < 1$ and $p > 1$ are quite different. Instead of taking $p \in (0, \infty)$, it is often convenient to restrict p to the range $[1, \infty)$, and consider the p-concavification X^p and the p-convexification $X^{\frac{1}{p}}$. Summarizing Propositions 16.5.9 and 16.5.12, the p-convexification $X^{\frac{1}{p}}$ of a Banach lattice X is always a Banach lattice, which is, moreover, p-convex with constant 1, while the p-concavification X^p is a Banach lattice when X is p-convex. In the latter case, $(X^p)^{\frac{1}{p}} = X$ as vector lattices; however, $(X^p)^{\frac{1}{p}}$ and X may have different norms.

16.5.13 Proposition. *If X is p-convex for some $1 \leqslant p < \infty$ then X is lattice isomorphic to $(X^p)^{\frac{1}{p}}$. If, in addition, a Banach lattice Y is lattice isomorphic to X^p then $Y^{\frac{1}{p}}$ is lattice isomorphic to X.*

Proof. The case $p = 1$ is trivial, so suppose that $1 < p < \infty$. By Proposition 16.5.12, X^p is a Banach lattice, so that $(X^p)^{\frac{1}{p}}$ is defined by Proposition 16.5.9. We know that $(X^p)^{\frac{1}{p}}$ is lattice isomorphic to X as a vector lattice and, therefore, as a Banach lattice by Corollary 6.1.10. Alternatively, from the formulae for the norms in Propositions 16.5.9 and 16.5.12, it follows that the norm on $(X^p)^{\frac{1}{p}}$ is equivalent to the original norm of X.

If Y is lattice isomorphic to X^p then by Proposition 16.5.4, $Y^{\frac{1}{p}}$ is isomorphic to $(X^p)^{\frac{1}{p}}$ as a vector lattice, hence as a Banach lattice. It follows that $Y^{\frac{1}{p}}$ is (Banach) lattice isomorphic to X. □

As the names suggest, the procedures of p-convexification and p-concavification affect the convexity and the concavity of the space in the natural way (again, one can use $L_p(\mu)$ spaces for motivation).

16.5.14 Exercise. Let $1 \leqslant r, s \leqslant \infty$ and $1 \leqslant p < \infty$. Suppose that X is (r,s)-convex (or concave) with constant K. Then

(i) $X^{\frac{1}{p}}$ is (rp, sp)-convex (respectively, concave) with constant $K^{\frac{1}{p}}$;

(ii) If $p \leqslant \min\{r, s\}$ and X is p-convex with constant M then X^p is $(\frac{r}{p}, \frac{s}{p})$-convex (respectively, concave) with constant $M^p K^p$,

The following result complements Proposition 16.5.10 and Corollary 16.4.15(ii). For motivation, observe that if $Z = \ell_q$ with $0 < q \leqslant 1$ then $Z_+ \setminus B_Z^\circ$ is convex.

16.5.15 Exercise. Let $1 \leqslant p < \infty$ and X is p-concave with constant 1 then the set $(X_+ \setminus B_X^\circ)^p$ is convex in X^p.

16.6 Upper and lower p-estimates

Throughout this section, X and Y are Banach lattices, E and F are Banach spaces, and $p, q \in [1, \infty]$.

16.6.1 (p,∞)-convex and $(p,1)$-concave operators

Recall that an operator $T\colon E \to X$ is (p,∞)-convex when $\left\|\bigvee_{i=1}^n |Tx_i|\right\| \lesssim \|\bar{x}\|_{\text{sum-}p}$ for any $x_1, \dots, x_n \in E$; an operator $T\colon X \to E$ is $(p,1)$-concave if $\|\overline{T}\bar{x}\|_{\text{sum-}p} \lesssim \left\|\sum_{i=1}^n |x_i|\right\|$ for any $x_1, \dots, x_n \in X$. By Theorem 16.4.6, there two concepts are in duality. We will show that these definitions may be reformulated in terms of infinite sequences.

16.6.1 Proposition. *An operator $T\colon X \to E$ is $(p,1)$-concave iff the sequence $\big(\|Tx_i\|\big)$ is in ℓ_p whenever $\sum_{i=1}^\infty |x_i|$ converges.*

Proof. Suppose that T is $(p,1)$-concave and $\sum_{i=1}^\infty |x_i|$ converges. For each n, we have

$$\Big(\sum_{i=1}^n \|Tx_i\|^p\Big)^{\frac{1}{p}} \lesssim \Big\|\sum_{i=1}^n |x_i|\Big\| \leqslant \Big\|\sum_{i=1}^\infty |x_i|\Big\|.$$

It follows that $\big(\|Tx_i\|\big)$ is in ℓ_p.

To prove the converse, suppose that T fails to be $(p,1)$-concave. Then for every $m \in \mathbb{N}$, we can find a finite set $z_1^{(m)}, \dots, z_{n_m}^{(m)}$ such that

$$\Big\|\sum_{i=1}^{n_m} \big|z_i^{(m)}\big|\Big\| \leqslant \tfrac{1}{2^m} \quad \text{and} \quad \Big(\sum_{i=1}^{n_m} \|Tz_i^{(m)}\|^p\Big)^{\frac{1}{p}} \geqslant 1.$$

Let (x_i) be the concatenation of all these finite sequences. It is easy to see that $\sum_{i=1}^\infty |x_i|$ converges, but $\big(\|Tx_i\|\big)$ is not in ℓ_p. □

We leave the dual version as an exercise; the proof is similar:

16.6.2 Exercise. Let $1 \leqslant p < \infty$. An operator $T\colon E \to X$ is (p,∞)-convex iff the sequence $\big(\bigvee_{i=1}^n |Tx_i|\big)_n$ is norm bounded whenever the sequence $\big(\|x_i\|\big)$ is in ℓ_p. (The case $p = \infty$ will be considered later in Theorem 16.9.6.)

Our next goal is to characterize (p,∞)-convex and $(p,1)$-concave operators in terms of disjoint vectors.

16.6.3 Proposition. *Let $1 \leqslant p \leqslant \infty$. An operator $T\colon E \to X$ is (p,∞)-convex with constant K iff $\left\|\bigvee_{i=1}^n z_i\right\| \leqslant K\|\bar{x}\|_{\text{sum-}p}$ whenever $x_1,\dots,x_n$ in E, $z_1,\dots,z_n$ are disjoint in X, and $0 \leqslant z_i \leqslant |Tx_i|$ for every $i=1,\dots,n$.*

Proof. The forward implication is immediate. To prove the converse, let $x_1,\dots,x_n \in E$. Put $u = \bigvee_{i=1}^n |Tx_i|$. Fix $\varepsilon > 0$. Find $x^* \in S_{X^*}^+$ such that $\|u\| \leqslant x^*(u) + \varepsilon$. Using Theorem 11.3.4, find disjoint $z_1,\dots,z_n$ in X such that $0 \leqslant z_i \leqslant |Tx_i|$ for every i and $x^*(u) \leqslant x^*\left(\bigvee_{i=1}^n z_i\right) + \varepsilon$. It follows that

$$\left\|\bigvee_{i=1}^n |Tx_i|\right\| = \|u\| \leqslant x^*\left(\bigvee_{i=1}^n z_i\right) + 2\varepsilon \leqslant \left\|\bigvee_{i=1}^n z_i\right\| + 2\varepsilon \leqslant K\|\bar{x}\|_{\text{sum-}p} + 2\varepsilon.$$

We conclude that T is (p,∞)-convex. □

For $(p,1)$-concavity, the situation is even better, — it suffices to verify the definition for disjoint vectors only:

16.6.4 Proposition. *Let $1 \leqslant p \leqslant \infty$ and $T\colon X \to E$. T is $(p,1)$-concave with constant K iff*

$$\|\overline{T}\bar{x}\|_{\text{sum-}p} \leqslant K\left\|\sum_{i=1}^n x_i\right\| \tag{16.10}$$

for any disjoint $x_1,\dots,x_n$ in X.

Proof. The forward implication is immediate. To prove the converse, by Theorem 16.4.6, it suffices to prove that T^* is (p^*,∞)-convex. We will show this using Proposition 16.6.3. Suppose that $x_1^*,\dots,x_n^* \in E^*$, $z_1^*,\dots,z_n^*$ are disjoint in X^*, and $z_i^* \leqslant |T^*x_i^*|$ for every $i=1,\dots,n$. Fix $\varepsilon > 0$. Find $x \in S_X^+$ such that $\left\|\bigvee_{i=1}^n z_i^*\right\| \leqslant \left\langle\bigvee_{i=1}^n z_i^*, x\right\rangle + \varepsilon = \sum_{i=1}^n z_i^*(x) + \varepsilon$. By Proposition 11.3.2, there exist disjoint $v_1,\dots,v_n$ in $[0,x]$ such that $z_i^*(v_i) > z_i^*(x) - \frac{\varepsilon}{n}$ for every $i=1,\dots,n$. It follows that $\sum_{i=1}^n z_i^*(x) \leqslant \sum_{i=1}^n z_i^*(v_i) + \varepsilon$. For each i, we have $0 \leqslant z_i^*(v_i) \leqslant |T^*x_i^*|(v_i)$. By Riesz-Kantorovich Formulae, we can find $w_i \in [-v_i, v_i]$ such that $|T^*x_i^*|(v_i) \leqslant \left|(T^*x_i^*)(w_i)\right| + \frac{\varepsilon}{n}$. This yields $\sum_{i=1}^n z_i^*(v_i) \leqslant \sum_{i=1}^n \left|(T^*x_i^*)(w_i)\right| + \varepsilon$. Note that $\left|(T^*x_i^*)(w_i)\right| = \left|x_i^*(Tw_i)\right| \leqslant \|x_i^*\|\|Tw_i\|$. Combining the estimates and using the fact that $\left|\sum_{i=1}^n w_i\right| \leqslant x$, we get

$$\begin{aligned}\left\|\bigvee_{i=1}^n z_i^*\right\| &\leqslant \sum_{i=1}^n \|x_i^*\|\|Tw_i\| + 3\varepsilon \leqslant \|\overline{x^*}\|_{\text{sum-}p^*}\|\overline{T}\bar{w}\|_{\text{sum-}p} + 3\varepsilon \\ &\leqslant \|\overline{x^*}\|_{\text{sum-}p^*} \cdot K\left\|\sum_{i=1}^n w_i\right\| + 3\varepsilon \leqslant K\|\overline{x^*}\|_{\text{sum-}p^*} + 3\varepsilon.\end{aligned}$$

It follows that $\left\|\bigvee_{i=1}^n z_i^*\right\| \leqslant K\|\overline{x^*}\|_{\text{sum-}p^*}$ so that T^* is (p^*,∞)-convex by Proposition 16.6.3. □

16.6.5 Remark. In the converse implication of Theorem 16.6.4, it suffices to consider disjoint *positive* $x_1,\dots,x_n$ at the price of replacing K with $\frac{K}{2}$ in (16.10). Indeed, suppose that

$$\|\overline{T}\bar{z}\|_{\text{sum-}p} \leqslant \frac{K}{2}\Big\|\sum_{i=1}^n z_i\Big\|$$

for any disjoint positive $z_1,\dots,z_n$ in X. If $x_1,\dots,x_n$ are disjoint in X, using the fact that ℓ_p^n-norm satisfies the triangle inequality, we get

$$\|\overline{T}\bar{x}\|_{\text{sum-}p} \leqslant \|\overline{T}\bar{x}^+\|_{\text{sum-}p} + \|\overline{T}\bar{x}^-\|_{\text{sum-}p} \leqslant \frac{K}{2}\Big\|\sum_{i=1}^n x_i^+\Big\| + \frac{K}{2}\Big\|\sum_{i=1}^n x_i^-\Big\| \leqslant K\Big\|\sum_{i=1}^n x_i\Big\|.$$

Our next goal is to prove a variant of Proposition 16.6.1 for disjoint sequences. We will use the following:

16.6.6 Lemma. *Let $T\colon X\to E$ and $1\leqslant p<\infty$. For each $x\in X$, put*

$$\rho(x) = \sup\Big\{\|\overline{T}\bar{u}\|_{\text{sum-}p} \;:\; u_1,\dots,u_n \textit{ positive disjoint, and } \sum_{i=1}^n u_i \leqslant |x|\Big\}.$$

Then ρ is positively homogeneous, subadditive, and $|x|\leqslant|y|$ implies $\rho(x)\leqslant\rho(y)$.

(That is, ρ satisfies the definition of a lattice seminorm, except that it may be infinite.)

Proof. It is clear that ρ is positively homogeneous, and that $|x|\leqslant|y|$ implies $\rho(x)\leqslant\rho(y)$. To prove subadditivity, fix $x,y\in X$. It follows from $|x+y|\leqslant\big||x|+|y|\big|$ that $\rho(x+y)\leqslant\rho\big(|x|+|y|\big)$. Take any disjoint $u_1,\dots,u_n$ such that $\sum_{i=1}^n u_i\leqslant|x|+|y|$. For every i, we have $u_i\leqslant|x|+|y|$, so, by Riesz Decomposition Theorem, we can find positive v_i and w_i such that $u_i=v_i+w_i$, $v_i\leqslant|x|$, and $w_i\leqslant|y|$. It follows that each of the two collections, $v_1,\dots,v_n$ and $w_1,\dots,w_n$, is disjoint, $\sum_{i=1}^n v_i\leqslant|x|$ and $\sum_{i=1}^n w_i\leqslant|y|$. Then

$$\|\overline{T}\bar{u}\|_{\text{sum-}p} \leqslant \|\overline{T}\bar{v}\|_{\text{sum-}p} + \|\overline{T}\bar{w}\|_{\text{sum-}p} \leqslant \rho(x)+\rho(y).$$

It follows that $\rho\big(|x|+|y|\big)\leqslant\rho(x)+\rho(y)$. $\square$

In our proof of Proposition 16.6.1, we found a sequence of finite sets with the "right" properties, and we built our sequence as a concatenation of those sets. This idea will not work for disjoint sequences because the union of disjoint sets need not be disjoint. Instead, we will use an inductive "splitting" argument: we will find a disjoint set with the appropriate property, then we will split one of its members into a further disjoint set, etc. In deciding which member to split, we will use ρ from the preceding lemma; we will always split a member whose ρ is infinite.

16.6.7 Theorem. *An operator $T\colon X \to E$ is $(p,1)$-concave for some $1 \leqslant p < \infty$ iff $\sum_{i=1}^{\infty}\|Tx_i\|^p$ converges for every disjoint order bounded sequence (x_n) in X.*

Proof. Suppose that T is $(p,1)$-concave. Let (x_n) be a disjoint sequence contained in $[-u,u]$ for some $u \in X_+$. For each n, we have

$$\Bigl(\sum_{i=1}^{n}\|Tx_i\|^p\Bigr)^{\frac{1}{p}} \lesssim \Bigl\|\sum_{i=1}^{n}|x_i|\Bigr\| \leqslant \|u\|.$$

It follows that $\sum_{i=1}^{\infty}\|Tx_i\|^p$ converges.

To prove the converse, suppose that T fails to be $(p,1)$-concave. Let ρ be as in Lemma 16.6.6. We claim that there exists $v \in X_+$ with $\rho(v) = \infty$. Indeed, for every $m \in \mathbb{N}$, by Proposition 16.6.4, we can find a disjoint finite set $z_1^{(m)},\dots,z_{n_m}^{(m)}$ such that

$$\Bigl\|\sum_{i=1}^{n_m} z_i^{(m)}\Bigr\| \leqslant \tfrac{1}{2^m} \quad\text{and}\quad \Bigl(\sum_{i=1}^{n_m}\|Tz_i^{(m)}\|^p\Bigr)^{\frac{1}{p}} \geqslant m.$$

By Remark 16.6.5, we may choose $z_1^{(m)},\dots,z_{n_m}^{(m)}$ to be positive. Put $y_m = \sum_{i=1}^{n_m} z_i^{(m)}$; then $\|y_m\| \leqslant \frac{1}{2^m}$. Put $v = \sum_{m=1}^{\infty} y_m$. For each m, we have

$$\rho(v) \geqslant \rho(y_m) \geqslant \Bigl(\sum_{i=1}^{n_m}\|Tz_i^{(m)}\|^p\Bigr)^{\frac{1}{p}} \geqslant m;$$

it follows that $\rho(v) = \infty$.

Without loss of generality, $\|T\| \leqslant 1$ and $\|v\| \leqslant 1$. We will inductively construct a positive sequence $v_0 \geqslant v_1 \geqslant v_2 \geqslant \dots$ such that $\rho(v_n) = \infty$ for all n, and so that for every n there is a finite set $x_1^{(n)},\dots,x_{k_n}^{(n)}$ in $[0,v_{n-1}]$ such that the vectors $x_1^{(n)},\dots,x_{k_n}^{(n)}$, and v_n are disjoint and $\sum_{i=1}^{k_n}\|Tx_i^{(n)}\|^p \geqslant 1$. Once this is achieved, we will complete the proof by taking the concatenation of all the finite sequences $x_1^{(n)},\dots,x_{k_n}^{(n)}$: the resulting sequence, say, (z_i), will be disjoint and $\sum_{i=1}^{\infty}\|Tz_i\|^p = \infty$, so we will be done.

Take $v_0 = v$. Suppose we have already constructed v_j, k_j, and $x_i^{(j)}$ for all $j < n$. Since $\rho(v_{n-1}) = \infty$, we can find a finite disjoint set $w_1,\dots,w_m$ in $[0,v_{n-1}]$ with $\sum_{i=1}^{m}\|Tw_i\|^p \geqslant 2$. We now consider two cases.

Case 1: $\rho(w_j) = \infty$ for some $j \leqslant m$. Take this w_j to be v_n, while the rest of w_i's become $x_1^{(n)},\dots,x_{k_n}^{(n)}$. We get $k_n = m-1$. Since $\|Tw_j\| \leqslant 1$, we have $\sum_{i=1}^{k_n}\|Tx_i^{(n)}\|^p \geqslant 1$.

Case 2: $\rho(w_i)$ is finite for all $i = 1,\dots,m$. Take $k_n = m$. Find a sufficiently small $\varepsilon > 0$ such that $\sum_{i=1}^{m}\bigl\|T(w_i - \varepsilon v_{n-1})^+\bigr\|^p \geqslant 1$. For each $i = 1,\dots,k_n$, put $x_i^{(n)} = (w_i - \varepsilon v_{n-1})^+$. So we have $\sum_{i=1}^{k_n}\|Tx_i^{(n)}\|^p \geqslant 1$. Also, since $0 \leqslant x_i^{(n)} \leqslant w_i$ for every i, $x_i^{(n)}$'s are disjoint.

Put $w=\sum_{i=1}^m w_i$ and $v_n=(\varepsilon v_{n-1}-w)^+$. It follows from $x_i^{(n)}\leqslant(w-\varepsilon v_{n-1})^+$ that $x_i^{(n)}\perp v_n$ for every i. Further, $v_n=(\varepsilon v_{n-1})\vee w-w$ yields that $v_n+w\geqslant\varepsilon v_{n-1}$. Then $\rho(\varepsilon v_{n-1})\leqslant\rho(v_n)+\rho(w)\leqslant\rho(v_n)+\sum_{i=1}^m\rho(w_i)$. Since $\rho(\varepsilon v_{n-1})=\infty$ but $\rho(w_i)$ is finite for all $i=1,\dots,m$, we conclude that $\rho(v_n)=\infty$. $\square$

Recall that in Section 11.8 we defined an operator $T\colon X\to E$ to be order weakly compact if $T[0,e]$ is relatively weakly compact for every $e>0$. We observed in Exercise 11.8.1 that this is equivalent to T mapping order bounded disjoint sequences to convergent sequences. Theorem 16.6.7 now immediately yields the following:

16.6.8 Corollary. *If $T\colon X\to E$ is $(p,1)$-concave for some $1\leqslant p<\infty$ then T is order weakly compact.*

16.6.2 (p,∞)-convex and $(p,1)$-concave spaces

Recall that X is (p,∞)-convex when the identity operator on X is (p,∞)-convex, that is, $\left\|\bigvee_{i=1}^n|x_i|\right\|\lesssim\|\bar{x}\|_{\text{sum-}p}$ for any $x_1,\dots,x_n\in X$. Further, X is $(p,1)$-concave when the identity operator on X is $(p,1)$-concave, that is $\|\bar{x}\|_{\text{sum-}p}\lesssim\left\|\sum_{i=1}^n|x_i|\right\|$ for any $x_1,\dots,x_n\in X$. Propositions 16.6.3 and 16.6.4 allow us to restrict these definitions to disjoint vectors. Recall that in case of disjoint vectors we have $\sum_{i=1}^n|x_i|=\left|\sum_{i=1}^n x_i\right|=\bigvee_{i=1}^n|x_i|=\left|\bigvee_{i=1}^n x_i\right|$.

16.6.9 Theorem.

(i) *X is (p,∞)-convex with constant K iff $\left\|\sum_{i=1}^n x_i\right\|\leqslant K\|\bar{x}\|_{\text{sum-}p}$ for any disjoint $x_1,\dots,x_n\in X$;*

(ii) *X is $(p,1)$-concave with constant K iff $\|\bar{x}\|_{\text{sum-}p}\leqslant K\left\|\sum_{i=1}^n x_i\right\|$ for any disjoint $x_1,\dots,x_n\in X$.*

Proof. The forward implications in both statements are trivial. The converse implication in (ii) follows immediately from Proposition 16.6.4. For the converse implication in (i), use Proposition 16.6.3: observe that if $x_1,\dots,x_n$ in X and $z_1,\dots,z_n$ are disjoint in X with $z_i\leqslant|Tx_i|$ for every $i=1,\dots,n$, then $\left\|\bigvee_{i=1}^n z_i\right\|\leqslant K\|\bar{z}\|_{\text{sum-}p}\leqslant K\|\bar{x}\|_{\text{sum-}p}$. $\square$

Recall that in case of disjoint vectors, Proposition 16.1.1 yields $\left(\sum_{i=1}^n|x_i|^p\right)^{\frac{1}{p}}=\left|\sum_{i=1}^n x_i\right|$ In view of this, Theorem 16.6.9 says that (p,∞)-convex and $(p,1)$-concave Banach lattices are exactly those that satisfy the definitions of p-convexity and p-concavity for disjoint vectors only! This earned this concepts special names: a (p,∞)-convex Banach lattice is said to satisfy an ***upper p-estimate***, while a $(p,1)$-concave

Banach lattices said to satisfy an ***lower p-estimate***, So upper and lower p-estimates may be viewed as "disjoint" variants of p-convexity and p-concavity.

It is clear that

$$\begin{array}{lllll} p\text{-convexity} & \Rightarrow & \text{upper } p\text{-estimate} & \Rightarrow & \text{upper } q\text{-estimate for all } q \leqslant p, \text{ and} \\ p\text{-concavity} & \Rightarrow & \text{lower } p\text{-estimate} & \Rightarrow & \text{lower } q\text{-estimate for all } q \geqslant p, \end{array}$$

with the same constants. Theorem 16.4.6 immediately yields

16.6.10 Corollary.

(i) *X has an upper p-estimate iff X^* has a lower p^*-estimate;*

(ii) *X has an lower p-estimate iff X^* has an upper p^*-estimate.*

In both cases, the constants remain the same.

16.6.11. Suppose that X is satisfies an upper p-estimate and a lower p-estimate, both with constant one. Then $\|x+y\| = \big(\|x\|^p + \|y\|^p\big)^{\frac{1}{p}}$ whenever $x \perp y$. This means that X is an abstract L_p-space when $p < \infty$ or an AM-space when $p = \infty$.

16.6.12 Proposition. *If X satisfies a lower p-estimate (in particular, if X is p-concave) for some $p < \infty$ then X is a KB-space (in particular, it is order continuous).*

Proof. It is easy to see that c_0 fails a lower p-estimate. Therefore, X has no subspace isomorphic to c_0. It follows that X is a KB-space by Theorem 10.4.4. □

The preceding proposition fails for upper p-estimates or p-convexity. Indeed, being an AM-space, c is p-convex for every p. Yet, c is not even order complete.

The next result is an analogue of Exercise 16.5.14.

16.6.13 Exercise. Suppose that X is p-convex with constant M. If X satisfies a lower r-estimate with constant K then its p-concavification X^p satisfies a lower $\frac{r}{p}$-estimate with constant M^pK^p, where $1 \leqslant p \leqslant r < \infty$.

Recall that by Meyer-Nieberg Theorem a Banach lattice is order continuous iff every order bounded disjoint sequence converges to zero. The following corollary of Theorems 16.6.7 may be viewed as a refinement of this fact:

16.6.14 Corollary. *X satisfies a lower p-estimate for some $1 \leqslant p < \infty$ iff the series $\sum_{i=1}^\infty \|x_i\|^p$ converges for every order bounded disjoint sequence (x_i).*

16.7 Renormings with constant 1.

Throughout this section, X stands for a Banach lattice and $p,q \in [1,\infty]$. It is clear that p-convexity is an isomorphic property: if X is p-convex then every Banach lattice isomorphic to X is also p-convex, though, the constant of p-convexity may be different. Same for p-concavity, as well as for upper and lower p-estimates. Hence, these properties are stable under renormings, though the constants may change. We will show in this section that it is often possible to renorm X in such a way that constants become equal to one.

16.7.1 Theorem. *Suppose that X satisfies the lower p-estimate for some $1 \leqslant p < \infty$, with constant K. For $x \in X$, put*

$$|||x||| = \sup\Big\{\|\bar{u}\|_{\text{sum-}p} \,:\, \sum_{i=1}^{n} u_i \leqslant |x|,\ u_1,\dots,u_n \text{ positive disjoint}\Big\}.$$

Then $|||\cdot|||$ is an lattice norm on X with $\|\cdot\| \leqslant |||\cdot||| \leqslant K\|\cdot\|$, and $|||\cdot|||$ satisfies lower p-estimate with constant 1.

Proof. It is easy to see that $|||x||| \geqslant \|x\|$. It follows from the assumption that $|||x||| \leqslant K\|x\|$. We conclude that $|||x|||$ is finite and $|||x||| = 0$ implies $x = 0$. It follows from Lemma 16.6.6 that $|||\cdot|||$ is a lattice norm.

To show that $|||\cdot|||$ satisfies a lower p-estimate with constant 1, let $x,y \in X_+$ with $x \perp y$. Fix $\varepsilon > 0$. Find disjoint positive collections $v_1,\dots,v_n$ and $w_1,\dots,w_m$ such that $\sum_{i=1}^n v_i \leqslant x$, $\sum_{j=1}^m w_j \leqslant y$ such that $|||x|||^p \leqslant \sum_{i=1}^n \|v_i\|^p + \varepsilon$ and $|||y|||^p \leqslant \sum_{j=1}^m \|w_j\|^p + \varepsilon$. It follows from $\sum_{i=1}^n v_i + \sum_{j=1}^m w_j \leqslant x+y$ that

$$|||x+y|||^p \geqslant \sum_{i=1}^{n}\|v_i\|^p + \sum_{j=1}^{n}\|w_j\|^p \geqslant |||x|||^p + |||y|||^p - 2\varepsilon.$$

We conclude that $|||x+y||| \geqslant \Big(|||x|||^p + |||y|||^p\Big)^{\frac{1}{p}}$. ☐

Recall that p-concavity implies lower p-estimate. Since every Banach lattice satisfies an upper 1-estimate with constant 1, in the special case when $p=1$, Theorem 16.7.1 combined with 16.6.11 yield the following:

16.7.2 Proposition. *TFAE:*

(i) *X is 1-concave;*
(ii) *X satisfies a lower 1-estimate;*
(iii) *X is lattice isomorphic to an AL-space.*

By duality, we also have:

16.7.3 Corollary. *TFAE:*

(i) *X is ∞-convex;*
(ii) *X satisfies an upper ∞-estimate;*
(iii) *X is lattice isomorphic to an AM-space.*

16.7.4 Proposition. *Under the assumptions of Theorem 16.7.1, suppose that, in addition, X satisfies an upper p-estimate with constant K. Then $|||\cdot|||$ satisfies an upper p-estimate with the same constant K.*

Proof. Let $x_1,\dots,x_n\in X$ be disjoint vectors. Put $x=\sum_{i=1}^n x_i$. Take any positive disjoint $u_1,\dots,u_n$ with $\sum_{i=1}^n u_i\leqslant|x|$. Put $v_{i,j}=|x_i|\wedge u_j$. Then $|x_i|\geqslant\sum_{j=1}^m v_{i,j}$ for every i and $u_j=\sum_{i=1}^n v_{i,j}$ for every j. These imply that $|||x_i|||^p\geqslant\sum_{j=1}^m\|v_{i,j}\|^p$ for every i and $\|u_j\|\leqslant K\Bigl(\sum_{i=1}^n\|v_{i,j}\|^p\Bigr)^{\frac{1}{p}}$ for every j. It follows that

$$\sum_{j=1}^n\|u_j\|^p\leqslant K^p\sum_{j=1}^m\sum_{i=1}^n\|v_{i,j}\|^p\leqslant K^p\sum_{i=1}^n|||x_i|||^p.$$

We conclude that

$$\Bigl|\Bigl|\Bigl|\sum_{i=1}^n x_i\Bigr|\Bigr|\Bigr|\leqslant K\Bigl(\sum_{i=1}^n|||x_i|||^p\Bigr)^{\frac{1}{p}}.$$

□

16.7.5 Theorem. *If X satisfies both an upper and a lower p-estimate for some $1\leqslant p<\infty$ then X is lattice isomorphic to $L_p(\mu)$ for some measure μ.*

Proof. It suffices to show that X can be equivalently renormed so that both constants become 1 because this would imply that the renormed space is an AL$_p$-space, hence is lattice isometric to some $L_p(\mu)$ by Theorem 10.7.1. In case $p=1$, this is Proposition 16.7.2.

Suppose that $1<p<\infty$; suppose that X satisfies an upper p-estimate with constant K and a lower p-estimate with constant M. By Theorem 16.7.1 and Proposition 16.7.4, X is lattice isomorphic to some Y which satisfies an upper p-estimate with constant K and a lower p-estimate with constant 1. By Corollary 16.6.10, Y^* satisfies a lower p^*-estimate with constant K and an upper p^*-estimate with constant 1. Applying Theorem 16.7.1 and Proposition 16.7.4 again, we conclude that Y^* is lattice isomorphic to some Z which satisfies a lower p^*-estimate with constant 1 and an upper p^*-estimate with constant 1. Then Z^* satisfies an upper p-estimate with constant 1 and a lower p-estimate with constant 1. Finally, observe that X is lattice isomorphic to a closed sublattice of Z^* because $X\hookrightarrow X^{**}\simeq Y^{**}\simeq Z^*$. □

As a corollary, we get the following:

16.7.6 Theorem. *If X is p-convex and p-concave for some $1 \leqslant p < \infty$ then X is lattice isomorphic to $L_p(\mu)$ for some measure μ.*

We will focus on renorming p-convex and q-concave spaces.

16.7.7 Lemma. *Every p-convex Banach lattice X with $1 < p < \infty$ can be renormed so that the constant of p-concavity becomes 1.*

Proof. By Proposition 16.5.13, X is lattice isomorphic to $\big(X^p\big)^{\frac{1}{p}}$. Being a p-convexification of a Banach lattice, $\big(X^p\big)^{\frac{1}{p}}$ is p-convex by Proposition 16.5.9. □

(Analyzing the preceding proof, one can see that the distortion of this renorming is at most the p-convexity constant of X.)

16.7.8 Lemma. *Every q-concave Banach lattice X with $1 \leqslant q \leqslant \infty$ can be renormed so that the constant of q-concavity becomes 1.*

Proof. If $q = \infty$, the statement is trivial, as every Banach lattice is ∞-concave with constant 1. If $q = 1$ then, by Proposition 16.7.2, X is lattice isomorphic to an AL-space, which is 1-concave with constant 1. Suppose now that $1 < q < \infty$. By Corollary 16.4.7, X^* is q^*-convex. By Lemma 16.7.7, X^* is lattice isomorphic to a Banach lattice Y which is q^*-convex with constant 1. Applying Corollary 16.4.7 again, we conclude that Y^* is q-concave with constant 1. It follows that X is lattice isomorphic to a closed sublattice of Y^*, and the latter is q-concave with constant 1. □

We can, actually, replace both constants with ones simultaneously for arbitrary values of p and q:

16.7.9 Theorem. *Suppose that X is p-convex and q-concave for some $p, q \in [1, \infty]$. There is a Banach lattice Y, which is lattice isomorphic to X, and which is p-convex and q-concave with both constants equal to one.*

Proof. By Exercise 16.4.5(v), we have $p \leqslant q$. If $p = \infty$ then $q = \infty$, and X^* is 1-concave. By Lemma 16.7.8, X^* is lattice isomorphic to a Banach lattice whose 1-concavity constant is 1. It follows that X is lattice isomorphic to a Banach lattice whose 1-convexity constant is 1. Since $q = \infty$, the concavity constant is 1.

Suppose now that $p < \infty$. Then the p-concavification X^p is a Banach lattice. By Exercise 16.5.14, X^p is $\frac{q}{p}$-concave. By Lemma 16.7.8, we find a Banach lattice Z which is lattice isomorphic to X^p and is $\frac{q}{p}$-concave with constant 1. Being a Banach lattice, Z is 1-convex with constant 1. By Exercise 16.5.14, $Z^{\frac{1}{p}}$ is p-convex and q-concave, both with constants one. By Proposition 16.5.13, $Z^{\frac{1}{p}}$ is lattice isomorphic to X. □

Finally, we provide an analogue of Theorem 16.7.9 where we simultaneously renorm convexity with lower estimate:

16.7.10 Theorem. *Let X be a Banach lattice.*

(i) *If X is p-convex and satisfies a lower q-estimate for some $1 \leqslant p < q < \infty$, then X can be renormed so that both constants become 1.*

(ii) *If X is q-concave and satisfies an upper p-estimate for some $1 < p < q \leqslant \infty$, then X can be renormed so that both constants become 1.*

Proof. (i) If $p = 1$, then the result follows from Theorem 16.7.1 because every Banach lattice is 1-convex with constant 1. Suppose that $p > 1$. Since X is p-convex, its p-concavification X^p is a Banach lattice. By Exercise 16.6.13, X^p satisfies a lower $\frac{q}{p}$-estimate. Applying Theorem 16.7.1, we find a Banach lattice Y which is lattice isomorphic to X^p and satisfies a lower $\frac{q}{p}$-estimate or, equivalently, is $(\frac{q}{p}, 1)$-concave, with constant 1. Also, being a Banach lattice, Y is 1-convex with constant 1. It follows from Exercise 16.5.14 that $Y^{\frac{1}{p}}$ is p-convex with constant 1 and (q,p)-concave with constant 1. The latter implies that $Y^{\frac{1}{p}}$ is $(q,1)$-concave and, therefore satisfies a lower q-estimate with constant 1. Finally, observe that $Y^{\frac{1}{p}}$ is lattice isomorphic to X by Proposition 16.5.13.

(i) By duality, X^* is q^*-convex and satisfies a lower p^*-estimate by Corollaries 16.4.7 and 16.6.10. Now apply (i), and then duality again. □

16.8 Weak L_p-spaces

This section is incomplete and misplaced. Most of this section will be moved into a chapter on Function Spaces earlier, except the facts that $L_{p,\infty}$ has upper p-estimate but not p-convex will be moved into the section on upper/lower estimates.

There are two main reasons why weak L_p-spaces are important for us. First, they provide examples of spaces that satisfy upper p-estimates without being p-convex, thus distinguishing the two properties. Second, they will be instrumental in proofs of factorization and representation theorems for spaces with upper p-estimates later.

Fix a measure space $(\Omega, \mathcal{F}, \mu)$ and $1 \leqslant p < \infty$. Let $f \in L_0(\mu)$. For simplicity, throughout this section, we will write $\|f\|_p$ for $\|f\|_{L_p(\mu)}$. Put

$$\|f\|_{p,\infty} = \sup_{t>0} t \cdot \mu\big(|f| > t\big)^{\frac{1}{p}}.$$

It is easy to see that $\|\lambda f\|_{p,\infty} = |\lambda| \|f\|_{p,\infty}$, $\|f\|_{p,\infty} = 0$ iff $f = 0$, and $\|f\|_{p,\infty} = \big\||f|\big\|_{p,\infty} = 0$. The latter often allows one to restrict attention to positive functions.

Let's provide some motivation for this definition. For simplicity, let $f \geqslant 0$. First, it is easy to see that $\mu(f > t)$ may be replaced with $\mu\bigl(f \geqslant t\bigr)$. Indeed, if $\varepsilon > 0$ and $0 < s < t$ then $\{f > t\} \subseteq \{f \geqslant t\} \subseteq \{f > s\}$, so that

$$t \cdot \mu(f > t)^{\frac{1}{p}} \leqslant t \cdot \mu(f \geqslant t)^{\frac{1}{p}} \leqslant t \cdot \mu(f > s)^{\frac{1}{p}} \leqslant s \cdot \mu(f > s)^{\frac{1}{p}} + \varepsilon$$

when s is sufficiently close to t.

Second, note that $t \cdot \mu(f \geqslant t)^{\frac{1}{p}}$ represents the L_p-norm of the function $t\mathbb{1}_A$ where $A = \{f \geqslant t\}$, that is, $t\mathbb{1}_A$ is the biggest "box" of height t that fits under the graph of f. So we have

$$\|f\|_{p,\infty} = \sup\Bigl\{\|t\mathbb{1}_A\|_p \;:\; t > 0,\ A \in \mathcal{F},\ t\mathbb{1}_A \leqslant |f|\Bigr\}.$$

16.8.1 Example. $\|\cdot\|_{p,\infty}$ may fail the triangle inequality. Consider $f, g \in L_0[0,1]$ given by $f(t) = t$ and $g(t) = 1 - t$. Then $\|f + g\|_{p,\infty} = \|\mathbb{1}\|_{p,\infty} = 1$. However, it is easy to see that $\|f\|_{1,\infty} = \|g\|_{1,\infty} = \frac{1}{4}$, so that $\|f\|_{1,\infty} + \|g\|_{1,\infty} = \frac{1}{2}$. It is now easy to see that $\|f\|_{p,\infty} + \|g\|_{p,\infty} < 1$ for all p sufficiently close to 1.

16.8.2. We put $L_{p,\infty}(\mu) = \bigl\{f \in L_0(\mu) \;:\; \|f\|_{p,\infty} < \infty\bigr\}$, we call this a weak L_p space on $(\Omega, \mathcal{F}, \mu)$. It is clear from the definition that $L_p(\mu) \subseteq L_{p,\infty}(\mu)$. This inclusion may be proper: for example, on $[0,1]$, the function $f(t) = t^{-\frac{1}{p}}$ is in $L_{p,\infty}[0,1]$ but not in $L_p[0,1]$. We will now show that while $\|\cdot\|_{p,\infty}$ is not a norm, if $p > 1$ then $L_{p,\infty}(\mu)$ admits an equivalent norm that makes it into a Banach lattice. In fact, we will produce a whole range of such norms. Suppose that $1 \leqslant r < p$; for $f \in L_0(\mu)$ we define

$$\|f\|_{p,\infty,r} = \sup_{0<\mu(A)<\infty} \mu(A)^{\frac{1}{p}-\frac{1}{r}} \|f\mathbb{1}_A\|_r.$$

The expression under the supremum may be written as $\mu(A)^{\frac{1}{p}}\Bigl(\frac{1}{\mu(A)}\int_A |f|^r\Bigr)^{\frac{1}{r}}$, where the second factor may be viewed as a kind of an average of f over A. The following fact asserts that the quantities $\|f\|_{p,\infty}$ and $\|f\|_{p,\infty,r}$ are equivalent; the proof is essentially a Measure Theory exercise:

16.8.3 Exercise. Let $1 \leqslant r < p$ and $f \in L_{p,\infty}(\mu)$. Then

$$\|f\|_{p,\infty} \leqslant \|f\|_{p,\infty,r} \leqslant \Bigl(\frac{p}{p-r}\Bigr)^{\frac{1}{r}} \|f\|_{p,\infty}.$$

16.8.4 Exercise. Suppose that $1 \leqslant r < p < \infty$. If μ is finite then $L_p(\mu) \subseteq L_{p,\infty}(\mu) \subseteq L_r(\mu)$. If μ is a probability measure and $h \in L_{p,\infty}(\mu)$ then $\|h\|_{L_r(\mu)} \leqslant \|h\|_{L_{p,\infty,r}(\mu)}$.

16.8.5 Theorem. *Let $1 \leqslant r < p$. Then $\|f\|_{p,\infty,r}$ is a complete lattice norm on $L_{p,\infty}(\mu)$. Moreover, it satisfies the upper p-estimate with constant 1.*

Proof. It is straightforward that $\|\lambda f\|_{p,\infty,r} = |\lambda|\|f\|_{p,\infty,r}$ and that $\|f\|_{p,\infty,r} = 0$ iff $f = 0$. To verify the triangle inequality, take $f, g \in L_{p,\infty}(\mu)$ and a set A with $0 < \mu(A) < \infty$, and observe that

$$\mu(A)^{\frac{1}{p}-\frac{1}{r}}\big\|(f+g)\mathbb{1}_A\big\|_r \leqslant \mu(A)^{\frac{1}{p}-\frac{1}{r}}\|f\mathbb{1}_A\|_r + \mu(A)^{\frac{1}{p}-\frac{1}{r}}\|g\mathbb{1}_A\|_r \leqslant \|f\|_{p,\infty,r} + \|g\|_{p,\infty,r}.$$

Taking the supremum over all A, we get the $\|f+g\|_{p,\infty,r} \leqslant \|f\|_{p,\infty,r} + \|g\|_{p,\infty,r}$.

It is straightforward that $\|f\|_{p,\infty,r}$ is a lattice norm. It is left to prove completeness. By Exercise 16.8.3, we may assume WLOG that $r = 1$. Let (f_n) be a Cauchy sequence in $L_{p,\infty}(\mu)$. In particular, it is bounded: there exists $M > 0$ such that $\|f_n\|_{p,\infty,1} \leqslant M$ for all n. For every set A with $0 < \mu(A) < \infty$, it follows from the definition of $\|\cdot\|_{p,\infty,1}$ that $\big\|f_n\mathbb{1}_A - f_m\mathbb{1}_A\big\|_1 \leqslant \mu(A)^{\frac{1}{p^*}}\|f_n - f_m\|_{p,\infty,1} \to 0$, so that the sequence $(f_n\mathbb{1}_A)$ is Cauchy in $L_1(A,\mu)$ and, therefore, converges to some g_A there. It is easy to see that if B is another measurable set with $0 < \mu(B) < \infty$ then g_A and g_B agree on $A \cap B$. It follows that there exists $g \in L_0(\mu)$ such that g_A is the restriction of g to A.

We will show next that $g \in L_{p,\infty}(\mu)$. Again, fix A with $0 < \mu(A) < \infty$. Observe that

$$\begin{aligned}\mu(A)^{\frac{1}{p}-1}\|g\mathbb{1}_A\|_1 &\leqslant \mu(A)^{\frac{1}{p}-1}\big\|(g-f_n)\mathbb{1}_A\big\|_1 + \mu(A)^{\frac{1}{p}-1}\|f_n\mathbb{1}_A\|_1 \\ &\leqslant \mu(A)^{\frac{1}{p}-1}\big\|(g-f_n)\mathbb{1}_A\big\|_1 + \|f_n\|_{p,\infty,1} < 2M\end{aligned}$$

for all sufficiently large n, because $\|(g-f_n)\mathbb{1}_A\|_1 \to 0$. Taking supremum over A we conclude that $\|g\|_{p,\infty,1} \leqslant 2M$, so that $g \in L_{p,\infty}(\mu)$.

Finally, we will show that $\|g - f_n\|_{p,\infty,1} \to 0$. Fix $\varepsilon > 0$. Fix m such that $\|f_n - f_m\|_{p,\infty,1} < \varepsilon$ for all $n \geqslant m$. Using the definition of $\|g - f_m\|_{p,\infty,1}$, we find a set A with $0 < \mu(A) < \infty$ such that

$$\big\|g - f_m\big\|_{p,\infty,1} < \mu(A)^{\frac{1}{p}-1}\big\|(g-f_m)\mathbb{1}_A\big\|_1 + \varepsilon.$$

It follows that

$$\big\|g - f_m\big\|_{p,\infty,1} < \mu(A)^{\frac{1}{p}-1}\big\|(g-f_n)\mathbb{1}_A\big\|_1 + \mu(A)^{\frac{1}{p}-1}\big\|(f_n-f_m)\mathbb{1}_A\big\|_1 + \varepsilon$$

for every $n \geqslant m$. Note that

$$\mu(A)^{\frac{1}{p}-1}\big\|(f_n-f_m)\mathbb{1}_A\big\|_1 \leqslant \|f_n - f_m\|_{p,\infty,1} < \varepsilon,$$

while the definition of g implies that $\big\|(g-f_n)\mathbb{1}_A\big\|_1 \to 0$, so that $\mu(A)^{\frac{1}{p}-1}\big\|(g-f_n)\mathbb{1}_A\big\|_1 < \varepsilon$ for sufficiently large n. It follows that $\big\|g - f_m\big\|_{p,\infty,1} < 3\varepsilon$.

Finally, let's verify that $\|\cdot\|_{p,\infty,r}$ satisfies the upper p-estimate with constant 1. It follows from the definition of the norm that $\int_A f^r \leqslant \mu(A)^{1-\frac{p}{r}}\|f\|^r_{p,\infty,r}$ whenever

$0 \leqslant f \in L_{p,\infty}(\mu)$ and $0 < \mu(A) < \infty$. Let $f_1, \dots, f_n$ be a disjoint positive collection in $L_{p,\infty}(\mu)$ and $0 < \mu(A) < \infty$. Put $A_i = A \cap \operatorname{supp} f_i$. Then, using disjointness and Hölder's inequality with the exponent $q = \frac{p}{r}$, we get

$$\begin{aligned}
\mu(A)^{\frac{1}{p}-\frac{1}{r}}\Bigl(\int_A\Bigl(\sum_{i=1}^n f_i\Bigr)^r\Bigr)^{\frac{1}{r}} &= \Bigl(\frac{1}{\mu(A)^{1-\frac{r}{p}}}\sum_{i=1}^n\int_{A_i} f_i^r\Bigr)^{\frac{1}{r}} \\
&\leqslant \Bigl(\sum_{i=1}^n\Bigl(\frac{\mu(A_i)}{\mu(A)}\Bigr)^{1-\frac{r}{p}}\|f_i\|_{p,\infty,r}^r\Bigr)^{\frac{1}{r}} \\
&\leqslant \Bigl(\sum_{i=1}^n\frac{\mu(A_i)}{\mu(A)}\Bigr)^{\frac{1}{q^*r}}\Bigl(\sum_{i=1}^n\|f_i\|_{p,\infty,r}^p\Bigr)^{\frac{1}{p}} \leqslant \Bigl(\sum_{i=1}^n\|f_i\|_{p,\infty,r}^p\Bigr)^{\frac{1}{p}}
\end{aligned}$$

because $\sum_{i=1}^n \frac{\mu(A_i)}{\mu(A)} = 1$. Taking the supremum over A, we get

$$\Bigl\|\sum_{i=1}^n f_i\Bigr\|_{p,\infty,r} \leqslant \Bigl(\sum_{i=1}^n\|f_i\|_{p,\infty,r}^p\Bigr)^{\frac{1}{p}}.$$

□

16.8.6 Example. $L_{p,\infty}[0,1]$ *fails to be p-convex for every* $1 < p < \infty$. Indeed, suppose that $L_{p,\infty}[0,1]$ is p-convex for some $1 < p < \infty$. WLOG, we may assume that $L_{p,\infty}[0,1]$ is equipped with $\|\cdot\|_{p,\infty,1}$. For $f \in L_{p,\infty}[0,1]$, taking $A = [0,1]$ in the definition of $\|\cdot\|_{p,\infty,1}$, we conclude that $\|f\|_{p,\infty,1} \geqslant \|f\|_1$. It follows that $L_{p,\infty}[0,1]$ is a subset (even an ideal) in $L_1[0,1]$ and the inclusion map is continuous. By Lemma 16.7.7, there is an equivalent lattice norm $|||\cdot|||$ on $L_{p,\infty}[0,1]$ which is p-convex with constant 1. Clearly, the embedding of $L_{p,\infty}[0,1]$ into $L_1[0,1]$ remains continuous under $|||\cdot|||$. Scaling $|||\cdot|||$, if necessary, we may assume WLOG that the embedding is contractive. Now by Lemma 16.4.17, $\|f\|_p \leqslant |||f|||$ for every simple function $f \in L_{p,\infty}[0,1]$. Take $f(t) = t^{-\frac{1}{p}}$, and let (f_n) be an increasing sequence of positive simple functions which converges to f a.e.. Then $\|f_n\|_p \to \infty$. On the other hand, $\|f_n\|_p \leqslant |||f_n|||$ for every n, and it follows from $0 \leqslant f_n \leqslant f \in L_{p,\infty}[0,1]$ that the sequence (f_n) is bounded in $|||\cdot|||$, which is a contradiction.

We will now briefly introduce $L_{p,1}(\mu)$; we refer the reader to [Graf14] for an excellent exposition on spaces $L_{p,1}(\mu)$ and $L_{p,\infty}(\mu)$. Let $f \in L_0(\mu)$. There exists a non-decreasing function $f^*\colon [0,\infty) \to [0,\infty]$ which has the same distribution as $|f|$, that is $\mu(f^* > t) = \mu\bigl(|f| > t\bigr)$ for every t. One can define f^* via

$$f^*(t) = \inf\bigl\{\lambda > 0 \,:\, \mu\bigl(|f| > \lambda\bigr) \leqslant t\bigr\}.$$

We say that f^* as the ***decreasing rearrangement*** of f. It follows from A.3.14 that

$$\int |f| = \int_0^\infty \mu\big(|f|>t\big) = \int_0^\infty \mu\big(f^*>t\big) = \int_0^\infty f^*.$$

For $f \in L_0(\mu)$ and $1<p<\infty$, we define

$$\|f\|_{p,1} = \int_0^\infty t^{\frac{1}{p}-1} f^*(t)\,dt. \tag{16.11}$$

Let $L_{p,1}(\mu)$ be the set of all functions $f \in L_0(\mu)$ such that $\|f\|_{p,1}$ is finite. It is shown in [Graf14] that while $\|f\|_{p,1}$ is, generally, not a norm, there is an equivalent lattice norm, which makes $L_{p,1}(\mu)$ into an order continuous Banach lattice. Moreover, $L_{p,1}(\mu)^* = L_{p^*,\infty}(\mu)$.

We have observed that that $L_{p,\infty}(\mu)$ satisfies an upper p-estimate with constant 1, but fails to be p-convex. It follows, by duality, $L_{p,1}(\mu)$ satisfies a lower p-estimate with constant 1, but fails to be p-concave. It follows from Proposition 16.6.12 that $L_{p,1}(\mu)$ is a KB-space; in particular, it is order continuous.

16.8.7 Example. For every $f \in L_{p,1}(\mu)$ where μ is a finite measure, we have $f\wedge n\mathbb{1} \xrightarrow{\text{a.e.}} f$, hence $f\wedge n\mathbb{1} \xrightarrow{\text{o}} f$; it follows from order continuity that $(f\wedge n\mathbb{1})$ converges to f in norm. We conclude that $L_\infty(\mu)$ is dense in $L_{p,1}(\mu)$.

The situation is different for $L_{p,\infty}$. We claim that $L_\infty[0,1]$ is not dense in $L_{p,\infty}(\mu)$ for every $1<p<\infty$. Indeed, suppose that $L_\infty[0,1]$ is dense in $L_{p,\infty}(\mu)$. Take $f(t)=t^{-\frac{1}{p}}$, then $f\in L_{p,\infty}(\mu)$. Fix any $\varepsilon>0$; we can then find $g\in L_\infty(\mu)$ such that $\|f-g\|_{p,\infty}<\varepsilon$. Put $s=\|g\|_\infty$. It follows from $\big|f-f\wedge s\mathbb{1}\big| \leqslant |f-g|$ that $\big\|f-f\wedge s\mathbb{1}\big\|_{p,\infty}<\varepsilon$. On the other hand, taking $t=\frac{s}{2}$ in the formula for $\big\|f-f\wedge s\mathbb{1}\big\|_{p,\infty}$. yields $\big\|f-f\wedge s\mathbb{1}\big\|_{p,\infty} \geqslant 1-\frac{1}{2^p}$. So taking $\varepsilon<1-\frac{1}{2^p}$ yields a contradiction.

This example provides a negative answer to the question in 9.3.6: take $X = L_{p,1}[0,1]$, then X satisfies the assumptions of Theorem 9.3.3, yet $L_\infty[0,1]$ is not dense in X^*.

The following exercise shows that, in some sense, we may view $L_{p,1}(\mu)$ as an intermediate space between $L_q(\mu)$ and $L_\infty(\mu)$:

16.8.8 Exercise. Suppose that $1\leqslant q<p<\infty$. Then $\|f\|_{p,1}\leqslant C\|f\|_q^{\frac{q}{p}}\|f\|_\infty^{1-\frac{q}{p}}$, for every $f\in L_\infty(\mu)\cap L_1(\mu)$, where $C=p+\big(\frac{1}{q}-\frac{1}{p}\big)^{\frac{1}{q^*}}$.

16.8.9 Proposition. *If μ is σ-finite then $L_{p,1}(\mu)^*=L_{p^*,\infty}(\mu)$.*

16.8.10 Proposition. *If μ is a probability measure, $1<p<\infty$, and $1\leqslant q<r<\infty$ then $\|\cdot\|_{p,q}\geqslant\|\cdot\|_{p,r}$.*

The following complements Exercise 16.8.4

16.8.11 Proposition. *If μ is a finite measure and $1 < p < r \leqslant \infty$ then $L_r(\mu) \subseteq L_{p,1}(\mu) \subseteq L_p(\mu)$, and the inclusion maps are continuous.*

Proof. The second inclusion (and its continuity) follows from Proposition 16.8.10 applied with $q = 1$ and $r = p$.

For the first inclusion, observe that every $f \in L_r(\mu)$, applying Hölder's Inequality to the definition of $\|f\|_{p,1}$, we get

$$\|f\|_{p,1} = \int_0^{\mu(K)} t^{-\frac{1}{p^*}} f^*(t)\,dt \leqslant \Big(\int_0^{\mu(K)} t^{-\frac{r^*}{p^*}} dt\Big)^{\frac{1}{r^*}} \Big(\int_0^{\mu(K)} f^*(t)^r dt\Big)^{\frac{1}{r}} = C\|f\|_r.$$

□

16.8.12 Proposition. *Simple functions are dense in $L_{p,1}(\mu)$ when $p > 1$ and μ is a probability measure.*

16.9 1-concave and ∞-convex operators.

Just as in the scale of L_p-spaces, the extreme cases when $p = 1$ and $p = \infty$ have various special properties, 1-concave and ∞-convex operators stand out among all (p,q)-convex/concave operators. Proofs of certain properties are easier for these classes, and they have quite a few extra properties.

16.9.1 1-concave operators

Recall that an operator $T\colon X \to E$ is 1-concave if $\sum_{i=1}^n \|Tx_i\| \lesssim \Big\|\sum_{i=1}^n |x_i|\Big\|$ for all $x_1, \dots, x_n$ in X. Note also that T is 1-concave iff it satisfies a lower 1-estimate.

16.9.1 Exercise. In the definition of an 1-concave operator, it suffices to consider positive disjoint vectors.

Similarly, it suffices to consider positive vectors in Proposition 16.6.1:

16.9.2 Exercise. An operator $T\colon X \to E$ is 1-concave iff $\sum_{i=1}^\infty \|Tx_i\|$ converges whenever $\sum_{i=1}^\infty x_i$ converges for every positive sequence (x_i).

Justified by this fact, 1-concave operators are sometimes called ***cone absolutely summing***. The 1-concavity constant of T is denoted by $\|T\|_\ell$. Since this is the multinorm of T with respect to the appropriate power-norms, $\|T\|_\ell$ is a norm on the space of all 1-concave operators.

16.9.3 Proposition. *An operator $T\colon X\to E$ is 1-concave with constant K iff there exists $x^*\in B^+_{X^*}$ such $\|Tx\|\leqslant Kx^*\big(|x|\big)$ for every $x\in X$.*

Proof. Suppose that T is 1-concave. Scaling T, if necessary, we may assume that $K=1$. For $x\in X_+$, put

$$\tau(x)=\sup\Big\{\sum_{i=1}^n\|Tu_i\|\ :\ u_1,\dots,u_n\geqslant 0,\sum_{i=1}^n u_i\leqslant x\Big\}.$$

It follows from 1-concavity that the supremum is finite with $\|Tx\|\leqslant\tau(x)\leqslant\|x\|$. As in 11.1.7, we see τ is additive on X_+ and, therefore, there exists $x^*\in X^*_+$ such that $\tau(x)=x^*(x)$ for every $x\in X_+$. For every $x\in X$, we have

$$\|Tx\|\leqslant\|Tx^+\|+\|Tx^-\|\leqslant x^*(x^+)+x^*(x^-)=x^*\big(|x|\big)$$

It follows from $|x^*(x)|\leqslant x^*\big(|x|\big)=\tau\big(|x|\big)\leqslant\|x\|$ that $\|x^*\|\leqslant 1$.

Conversely, suppose such an x^* exists. Then

$$\sum_{i=1}^n\|Tx_i\|\leqslant\sum_{i=1}^n Kx^*\big(|x_i|\big)=Kx^*\Big(\sum_{i=1}^n|x_i|\Big)\leqslant K\Big\|\sum_{i=1}^n|x_i|\Big\|$$

for any $x_1,\dots,x_n\in X$. □

16.9.4 Theorem. *An operator $T\colon X\to E$ is 1-concave with constant K iff it factors as $T=SR$ through an $L_1(\mu)$-space such that $\|S\|\|R\|\leqslant K$ and $R\geqslant 0$. Moreover, one can chose R to be a lattice homomorphism with dense range.*

Proof. Suppose that T is 1-concave with constant K. Let x^* be as in Proposition 16.9.3. We now argue as in 9.5.1. Consider the quotient map $Q\colon X\to X/N_{x^*}$, where N_{x^*} is the null ideal of X^*. The map $\tilde{x}\to x^*\big(|x|\big)$ induces an AL-norm on X/N_{x^*} (where $\tilde{x}=Qx$), so that the completion of X/N_{x^*} is lattice isometric to $L_1(\mu)$ for some measure μ. Let $J\colon X/N_{x^*}\hookrightarrow L_1(\mu)$ be the inclusion map. Put $R=JQ$. Clearly, R is a lattice homomorphism with dense range, and $\|R\|=\|x^*\|\leqslant 1$.

$$\begin{tikzcd} X \arrow[r, "T"] \arrow[d, "Q"'] & E \\ X/N_{x^*} \arrow[r, hook, "J"] & L_1(\mu) \arrow[u, "S"'] \end{tikzcd}$$

It follows from $\|Tx\|\leqslant Kx^*\big(|x|\big)$ that T vanishes on N_{x^*}. Hence, T induces an operator $\widetilde{T}\colon X/N_{x^*}\to E$ via $\widetilde{T}\tilde{x}=Tx$. Furthermore, $\big\|\widetilde{T}\tilde{x}\big\|=\|Tx\|\leqslant Kx^*\big(|x|\big)=K\|\tilde{x}\|$. It follows that $\|\widetilde{T}\|\leqslant K$, so that $\widetilde{T}$ extends to an operator $S\colon L_1(\mu)\to E$ with $\|S\|\leqslant K$.

Conversely, suppose that T admits a factorization $X\xrightarrow{R}L_1(\mu)\xrightarrow{S}E$ with $\|S\|\|R\|\leqslant K$ and $R\geqslant 0$. Since the identity operator on $L_1(\mu)$ is 1-concave with constant 1, the conclusion follows from Exercise 16.4.3. □

16.9.2 ∞-convex operators.

Recall that $T\colon E\to X$ is ∞-convex if $\bigl\|\bigvee_{i=1}^n|Tx_i|\bigr\|\lesssim\max_i\|x_i\|$ for all $x_1,\dots,x_n$ in E. By Theorem 16.4.6, the classes of 1-concave and of ∞-convex operators are in duality.

We will show that the concept of an ∞-convex operator may be extended to a more general context. An operator $T\colon E\to V$ from a Banach space E to an Archimedean vector lattice V is said to be ***majorizing*** if $z_n\xrightarrow{\|\cdot\|}0$ in E implies that the sequence (Tz_n) is order bounded in H. The following result is somewhat parallel to characterizations of sequentially uniformly continuous operators in Proposition 13.4.1:

16.9.5 Proposition. *Let $T\colon E\to V$ be an operator from a Banach space E to an Archimedean vector lattice V. TFAE:*

(i) *T is majorizing;*
(ii) *T is sequentially norm-to-uniform continuous[1];*
(iii) *T is sequentially norm-to-order continuous.*

Proof. (ii)$\Rightarrow$(iii)$\Rightarrow$(i) is trivial. To show that (i)$\Rightarrow$(ii), suppose that $z_n\xrightarrow{\|\cdot\|}0$ in E. Find an increasing sequence (λ_n) in $\mathbb{R}_+$ such that $\lambda_n\nearrow\infty$ but $\lambda_nz_n\xrightarrow{\|\cdot\|}0$. Then (i) yields that (λ_nTz_n) is order bounded, hence $Tz_n\xrightarrow{\mathrm{u}}0$. $\square$

The next theorem is somewhat parallel to Theorem 13.4.3 and Exercise 16.6.2:

16.9.6 Theorem. *Let $T\colon E\to X$. TFAE:*

(i) *T is majorizing;*
(ii) *if $z_n\xrightarrow{\|\cdot\|}0$ in E then the sequence $\bigl(\bigvee_{k=1}^n|Tz_k|\bigr)_n$ is norm bounded;*
(iii) *if (z_n) is norm bounded in E then $\bigl(\bigvee_{k=1}^n|Tz_k|\bigr)_n$ is norm bounded;*
(iv) *T is ∞-convex.*

Proof. (i)$\Rightarrow$(ii) is trivial.

(ii)$\Rightarrow$(i) Suppose that $z_n\xrightarrow{\|\cdot\|}0$. Then the sequence $\bigl(\bigvee_{k=1}^n|Tz_k|\bigr)_n$ is norm bounded in X and, therefore, in X^{**} (we identify X with its canonical image in X^{**}). Since X^{**} is monotonically bounded by Exercise 10.6.8, this sequence is also order bounded in X^{**}. Hence, viewed as an operator from E to X^{**}, T is majorizing. Proposition 16.9.5, it follows that $Tz_n\xrightarrow{\mathrm{u}}0$ in X^{**} and, therefore, in X, by Theorem 2.3.17. Now apply Proposition 16.9.5 again.

(iv)$\Rightarrow$(iii)$\Rightarrow$(ii) is straightforward

(ii)$\Rightarrow$(iv) Suppose that T fails to be ∞-convex. Then for every m, one can find vectors $z_1^{(m)},\dots,z_{n_m}^{(m)}$ such that $\bigvee_{k=1}^n\|z_k\|<\frac{1}{m}$ and $\Bigl\|\bigvee_{k=1}^n|Tz_k|\Bigr\|>m$. Let (z_n) be the

[1] That is, for every sequence (x_n) in E, if (x_n) converges in norm to some x, then (Tx_n) converges (relatively) uniformly to Tx.

concatination of these finite sequences. It is easy to see that $z_n \xrightarrow{\|\cdot\|} 0$, yet the sequence $\left(\bigvee_{k=1}^n |Tz_k|\right)_n$ fails to be norm bounded. $\square$

The ∞-convexity constant of T is denoted by $\|T\|_m$.

Combining Proposition 16.9.5 with Theorems 16.9.6 and 13.4.3, we get the following:

16.9.7 Corollary. *Every ∞-convex operator is pre-regular.*

16.9.8 Proposition. *An operator $T\colon E \to X$ is ∞-convex iff TB_E is order bounded in X^{**}. Moreover, $T\colon E \to X$ is ∞-convex with constant K iff $TB_E \subseteq [-\xi,\xi]$ for some $\xi \in X^{**}$ with $\|\xi\| \leqslant K$.*

Proof. Suppose that T is ∞-convex with constant K. WLOG, we may assume that $K = 1$. Consider the collection of all the expressions of the form $\bigvee_{i=1}^n |Tx_i|$, where $x_1, \dots, x_n \in B_E$. We may view this collection as an increasing net in B_X indexed by finite subsets of B_E, ordered by inclusion. By Exercise 10.6.8, X^{**} is monotonically bounded, hence this net is order bounded in X^{**}. Moreover, Exercise 10.6.8 asserts that this net is contained in $[-\xi,\xi]$ for some $\xi \in X^{**}$ with $\|\xi\| \leqslant 1$. The required conclusion follows trivially.

The converse is straightforward. $\square$

The following result is a counterpart of Theorem 16.9.4:

16.9.9 Theorem. *An operator $T\colon E \to X$ is ∞-convex with constant K iff T factors as $T = RS$ via an AM-space such that $\|R\|\|S\| \leqslant K$ and $R \geqslant 0$. Moreover, one may choose R to be a lattice homomorphism*

Proof. Throughout the proof, we identify X with its canonical image in X^{**}. Suppose that T is ∞-convex with constant K. By Proposition 16.9.8, $TB_E \subseteq [-\xi,\xi]$ for some $0 \leqslant \xi \in X^{**}$ with $\|\xi\| \leqslant K$. Since X^{**} is uniformly complete, the principal ideal I_ξ is complete under the norm $\|\cdot\|_\xi$ induced by ξ. For $\zeta \in I_\xi$, we have $|\zeta| \leqslant \|\zeta\|_\xi \xi$, so that $\|\zeta\|_{X^{**}} \leqslant \|\zeta\|_\xi \|\xi\|_{X^{**}} \leqslant K\|\zeta\|_\xi$. Therefore, the inclusion map $J\colon \left(I_\xi, \|\cdot\|_\xi\right) \to X^{**}$ is continuous with $\|J\| \leqslant K$.

It follows from $TB_E \subseteq [-\xi,\xi]$ that $\operatorname{Range} T \subseteq I_\xi$. Let Y be the closed sublattice of $\left(I_\xi, \|\cdot\|_\xi\right)$ generated by of $\operatorname{Range} T$. Since $\left(I_\xi, \|\cdot\|_\xi\right)$ is an AM-space, so is Y. Define $S\colon E \to Y$ via $Sx = Tx$. It follows from $SB_E = TB_E \subseteq [-\xi,\xi]$ that $\|S\| \leqslant 1$.

Observe that $J(\operatorname{Range} T) = \operatorname{Range} T \subseteq X$. Since J is a lattice homomorphism, J maps $\operatorname{Lat}(\operatorname{Range} T)$ into X, and the restriction of J to $\operatorname{Lat}(\operatorname{Range} T)$ is still $\|\cdot\|_\xi$-to-$\|\cdot\|_{X^{**}}$ continuous, hence $\|\cdot\|_\xi$-to-$\|\cdot\|_X$ continuous, with norm $\leqslant K$. It follows that J extends to a lattice homomorphism $R\colon Y \to X$, with $\|R\| \leqslant K$. Clearly, $T = RS$.

To prove the converse, note that the identity map on an AM-space is ∞-convex with constant 1; the result now follows from Exercise 16.4.3. $\square$

Every compact set in a Banach space is contained in the closure of the convex hull of a null sequence; see, e.g., Theorem 5.3 in [AB06]. Combined with the definition of a majorizing operator, it immediately yields the following:

16.9.10 Proposition. *An operator $T\colon E\to X$ is majorizing iff it maps compact sets to order bounded sets.*

16.9.11 Theorem. *If $T\colon E\to X$ is finite-rank then T is ∞-convex with constant K, where $K=\bigl\lVert\sup TB_E\bigr\rVert$.*

Proof. Applying Theorem 11.5.2 with $A=TB_E$, we conclude that $u:=\sup A$ exists and equals the limit of $A^\vee$ viewed as an increasing net. Note that if $x\in B_E$ then $\pm Tx\in A$, so that $\lvert Tx\rvert\in A^\vee$.

On one hand, for every $x_1,\dots,x_n\in B_E$ we have $\bigvee_{i=1}^n\lvert Tx_i\rvert\in A^\vee$ and, therefore, $\Bigl\lVert\bigvee_{i=1}^n\lvert Tx_i\rvert\Bigr\rVert\leqslant\lVert u\rVert$; this implies that T is ∞-convex with constant $\leqslant\lVert u\rVert$. To prove the other inequality, fix $\varepsilon>0$. It follows from $u=\lim A^\vee$ that there exists $z\in A^\vee$ with $\lVert u-z\rVert<\varepsilon$. Observe that $z=\bigvee_{i=1}^n Tx_i$ for some $x_1,\dots,x_n\in B_E$. It follows that $z\leqslant\bigvee_{i=1}^n\lvert Tx_i\rvert\leqslant u$. We conclude that $\Bigl\lVert\bigvee_{i=1}^n\lvert Tx_i\rvert\Bigr\rVert\geqslant\lVert u\rVert-\varepsilon$. $\square$

Note that many results of this section remain valid if E is just a normed space. We refer the reader to [Sch74, MN91] for further properties of majorizing and cone absolutely summing operators.

Chapter 17

Factorization of (p,q)-convex and (p,q)-concave operators

This chapter is a continuation of the preceding one. We observed in Theorems 16.9.4 and 16.9.9 that 1-concave and ∞-convex operators factor through AL- and AM-spaces, respectively. In this chapter, we will prove similar results for other classes of (p,q)-convex and (p,q)-concave operators. This will help us establish further properties of these operators. Even though this chapter is focused on operators, as byproducts we will also establish some facts about spaces. most of the results in this chapter go back to Krivine, Maurey, and Pisier.

Throughout this section, X, Y, and Z are a Banach lattice, E and F are Banach spaces, $p,q\in[1,\infty]$, and $(\Omega,\mathcal{F},\mu)$ is a measure space.

17.1 Factorization through (p,q)-convex spaces

We proved in Theorems 16.9.4 that every 1-concave operator factors through an AL-space. In this section, we extend this result to (p,q)-concave operators. The idea of the proof is similar.

Suppose that an operator $T\colon X\to E$ factors through a (p,q)-concave Banach lattice Z, that is, $T\colon X\xrightarrow{R}Z\xrightarrow{S}E$ such that R is regular. Then T factors through the identity operator on Z; it now follows from Exercise 16.4.3 that T is itself (p,q)-concave. The following theorem provides a kind of converse:

17.1.1 Theorem. *Let $1\leqslant q\leqslant p\leqslant\infty$ and $T\colon X\to E$ be a (p,q)-concave operator with constant K. Then T factors through a Banach lattice Z such that Z is (p,q)-concave with constant 1, Q is a densely interval preserving*[1] *lattice homomorphism,*

[1] That is, $Q[0,x]$ is dense in $[0,Qx]$ for every $x\geqslant 0$; see Subsection 6.11.1.

and $\|Q\|\|S\| \leqslant K$*:*

$$\begin{tikzcd} X \arrow[rr, "T"] \arrow[dr, dotted, "Q"'] & & E \\ & Z \arrow[ur, dotted, "S"'] & \end{tikzcd}$$

Proof. The proof is long but straightforward. For $x \in X$, we define[2]

$$\rho(x) = \sup\Big\{ \|\overline{T}\bar{x}\|_{\text{sum-}p} \ :\ \Big(\sum_{i=1}^{n} |x_i|^q\Big)^{\frac{1}{q}} \leqslant |x| \Big\}.$$

Clearly, $\|Tx\| \leqslant \rho(x) \leqslant K\|x\|$. We claim that ρ is a lattice seminorm. The only non-trivial part of this claim is the triangle inequality. Let $x, y \in X$. Put $e = |x|+|y|$. Using Krein-Kakutani's Representation Theorem, we can identify I_e with $C(K)$ for some compact Hausdorff K so that e corresponds to $\mathbb{1}$. Suppose that $\big(\sum_{i=1}^n |z_i|^q\big)^{\frac{1}{q}} \leqslant |x+y|$. Clearly, $z_1, \dots, z_n \in I_e$, so we may view them as functions in $C(K)$. For $i = 1, \dots, n$, put $x_i = z_i|x|$ and $y_i = z_i|y|$. Then $x_i + y_i = z_i$. Note that

$$\Big(\sum_{i=1}^{n} |x_i|^q\Big)^{\frac{1}{q}} = \Big(\sum_{i=1}^{n} |z_i|^q\Big)^{\frac{1}{q}} |x| \leqslant |x|, \quad \text{so that} \quad \|\overline{T}\bar{x}\|_{\text{sum-}p} \leqslant \rho(x).$$

Similarly, $\|\overline{T}\bar{y}\|_{\text{sum-}p} \leqslant \rho(y)$. Since $\|Tz_i\| \leqslant \|Tx_i\| + \|Ty_i\|$, we get

$$\|\overline{T}\bar{z}\|_{\text{sum-}p} \leqslant \|\overline{T}\bar{x}\|_{\text{sum-}p} + \|\overline{T}\bar{y}\|_{\text{sum-}p} \leqslant \rho(x) + \rho(y).$$

We conclude that $\rho(x+y) \leqslant \rho(x)+\rho(y)$ and, therefore, ρ is indeed a lattice seminorm.

It follows that $\ker\rho$ is an ideal in X. Consider the quotient map $Q\colon X \to X/\ker\rho$. Being a quotient map, it is an interval preserving lattice homomorphism. Note that ρ induces a lattice norm on $X/\ker\rho$. Let Z be the completion of $X/\ker\rho$ under this norm. Clearly, Z is a Banach lattice. Viewed as an operator from X to Z, Q as a densely interval preserving lattice homomorphism. It follows from $\rho(x) \leqslant K\|x\|$ that $\|Q\| \leqslant K$.

Since $\|Tx\| \leqslant \rho(x)$ for every x, we conclude that T vanishes on $\ker\rho$, hence T induces a linear operator $\widetilde{T}\colon X/\ker\rho \to E$ with $\|\widetilde{T}\| \leqslant 1$. Then $\widetilde{T}$ extends to some $S\colon Z \to E$ with $\|S\| \leqslant 1$. Clearly, $T = SQ$.

It is left to show that Z is (p,q)-concave with constant 1. It suffices to verify the condition for just two vectors. First, we show that ρ has the property. Fix $x, y \in X$, let $\varepsilon > 0$, and find $x_1, \dots, x_n$, $y_1, \dots, y_m$ in X such that

$$\Big(\sum_{i=1}^{n} |x_i|^q\Big)^{\frac{1}{q}} \leqslant |x|,\ \Big(\sum_{j=1}^{m} |y_j|^q\Big)^{\frac{1}{q}} \leqslant |y|,\ \|\overline{T}\bar{x}\|_{\text{sum-}p} \geqslant (1-\varepsilon)\rho(x),\ \text{and}\ \|\overline{T}\bar{y}\|_{\text{sum-}p} \geqslant (1-\varepsilon)\rho(y).$$

[2]Cf. the definition of τ in the proof of Proposition 16.9.3, which was then used in the proof of Theorems 16.9.4.

It follows from
$$\Bigl(\sum_{i=1}^n |x_i|^q + \sum_{j=1}^m |y_j|^q\Bigr)^{\frac{1}{q}} \leqslant \bigl(|x|^q + |y|^q\bigr)^{\frac{1}{q}}$$
that
$$\rho\Bigl(\bigl(|x|^q+|y|^q\bigr)^{\frac{1}{q}}\Bigr) \geqslant \Bigl(\sum_{i=1}^n \|Tx_i\|^p + \sum_{j=1}^m \|Ty_j\|^p\Bigr)^{\frac{1}{p}} \geqslant (1-\varepsilon)\Bigl(\rho(x)^p+\rho(y)^p\Bigr)^{\frac{1}{p}}.$$

Hence, $\rho\Bigl(\bigl(|x|^q+|y|^q\bigr)^{\frac{1}{q}}\Bigr) \geqslant \Bigl(\rho(x)^p+\rho(y)^p\Bigr)^{\frac{1}{p}}$. It follows that
$$\begin{aligned}\Bigl\|\Bigl(|Qx|^q+|Qy|^q\Bigr)^{\frac{1}{q}}\Bigr\| = \Bigl\|Q\Bigl(|x|^q+|y|^q\Bigr)^{\frac{1}{q}}\Bigr\| &= \rho\Bigl(\bigl(|x|^q+|y|^q\bigr)^{\frac{1}{q}}\Bigr)\\ &\geqslant \Bigl(\rho(x)^p+\rho(y)^p\Bigr)^{\frac{1}{p}} = \Bigl(\|Qx\|^p+\|Qy\|^p\Bigr)^{\frac{1}{p}}.\end{aligned}$$

Finally, since $\operatorname{Range} Q = X/\ker\rho$ is dense in Z and the function calculus is norm continuous by 4.4.10, we get $\bigl\|\bigl(|u|^q+|v|^q\bigr)^{\frac{1}{q}}\bigr\| \geqslant \bigl(\|u\|^p+\|v\|^p\bigr)^{\frac{1}{p}}$ for every $u,v\in Y$. $\square$

A few remarks about Theorem 17.1.1 are due at this point:

17.1.2. In Section 16.4, we first defined (p,q)-concave operators; we then defined a Banach lattice to be (p,q)-concave if the identity operator is (p,q)-concave. In view of Theorem 17.1.1 and the remark preceding it, one can do the opposite: first, define (p,q)-concave Banach lattices, and then define an operator to be (p,q)-concave if it factors through a (p,q)-concave Banach lattices with appropriate factors.

17.1.3. If the operator T in the theorem is positive then the operator S will also be positive. Indeed, let $0\leqslant a\in X/\ker\rho$. We can find $0\leqslant x\in a$; we then have $Sa=Tx\geqslant 0$. Since $X/\ker\rho$ is dense in Z, we get $S\geqslant 0$.

Similarly, if T is a lattice homomorphism then S may be chosen to be a lattice homomorphism. Indeed, for any $x\in X$, we have
$$|SQx| = |Tx| = T|x| = SQ|x| = s|Qx|.$$

It follows that S is a lattice homomorphism on $X/\ker\rho$, and, therefore, (by density), on Z.

17.1.4. Theorem 17.1.1 yields a simple alternative proof of Lemma 16.7.8 that every q-concave Banach lattice X may be renormed so that the constant of q-concavity becomes 1. Indeed, the identity operator $I\colon X\to X$ is q-concave, hence factors via Banach lattice Z which is q-concave with constant 1, and both factors are lattice homomorohisms by 17.1.3. This implies that Z is lattice isomorphic to X.

We now prove the dual version of Theorem 17.1.1:

17.1.5 Theorem. *Let $1 \leqslant p \leqslant q \leqslant \infty$ and $T\colon E \to X$ be a (p,q)-convex operator with constant K. Then T factors through a Banach lattice Y such that Y is (p,q)-convex with constant 1, Q is a lattice homomorphism, and $\|S\|\|Q\| \leqslant K$:*

$$\begin{tikzcd} E \arrow[rr, "T"] \arrow[dr, dotted, "S"'] & & X . \\ & Y \arrow[ur, dotted, "Q"'] & \end{tikzcd}$$

Proof. By Theorem 16.4.6, $T^*\colon X^* \to E^*$ is (p^*,q^*)-concave with constant K. We factor it as in Theorem 17.1.1, then dualize:

$$\begin{tikzcd} X^* \arrow[rr, "T^*"] \arrow[dr, dotted, "Q_0"'] & & E^* \\ & Z \arrow[ur, dotted, "S_0"'] & \end{tikzcd} \qquad \begin{tikzcd} E^{**} \arrow[rr, "T^{**}"] \arrow[dr, "S_0^*"'] & & X^{**} \\ & Z^* \arrow[ur, "Q_0^*"'] & \end{tikzcd}$$

Here Z is (p,q)-concave with constant 1, $\|S_0\| \leqslant \|T^*\| = \|T\|$, and Q_0 is a densely interval preserving lattice homomorphism with $\|Q_0\| \leqslant K$. By Lotz's Theorem 6.11.12, Q_0^* is a lattice homomorphism. We view E and X as subspaces of E^{**} and X^{**}, respectively.

Let Y be the closed sublattice of Z^* generated by $S_0^*(E)$; let $S\colon E \to Y$ be the restriction of S_0^* to E. By Theorem 16.4.6, Z^* is (p,q)-convex with constant 1, hence so is Y. For every $x \in E$, $Q_0^* S_0^* x = T^{**}x = Tx \in X$. It follows that $Q_0^*\big(S_0^*(E)\big) \subseteq X$. Since Q_0^* is a lattice homomorphism, it maps $\operatorname{Lat} S_0^*(E)$ into X and, therefore, extends to a lattice homomorphism from Y to X; we denote this extension by Q. It is now clear that $T = QS$. $\square$

17.2 Factorization through $L_p(\mu)$

.

Theorem 16.7.6 asserts that a Banach lattice that is both p-convex and p-concave is lattice isomorphic to an $L_p(\mu)$-space. In this section, we provide variants of this result where the properties of p-convexity and of p-concavity are separated amongst two spaces or even two operators.

17.2.1 Theorem ([Kriv74])**.** *Suppose that $1 \leqslant p < \infty$, X is p-convex and Y is p-concave and $T\colon X \to Y$ is a lattice homomorphism. Then T factors via $L_p(\mu)$ for some measure μ, with the factors being lattice homomorphisms.*

Proof. Without loss of generality, we may make a few additional assumptions. By Lemmas 16.7.7 and 16.7.8, we may renorm X and Y so that the constants of p-covexity and of p-concavity are both 1. Scaling T, we may assume that $\|T\|=1$. We may also assume that T is one-to-one. Indeed, T factors as $T\colon X \xrightarrow{A} (X/\ker T) \xrightarrow{B} Y$ with A being the quotient map and B an injection. Since $\ker T$ is a closed ideal, $X/\ker T$ is a Banach lattice; it is p-convex with constant 1 by Exercise 16.4.8. Being a quotient map, A is a lattice homomorphism. Since T is a lattice homomorphism, so is B. Note also that $\|A\|=\|B\|=1$. It now suffices to prove the theorem for B. Thus, replacing T with B, we may assume that T is one-to-one.

Consider the p-concavifications X^p and Y^p; we use the notation from Section 16.5. Note that by Proposition 16.5.12, X^p is a Banach lattice under the norm $\|x^p\|=\|x\|^p$, while Y^p is just a vector lattice. By Proposition 16.5.4, T induces a (linear) lattice homomorphism $T^p\colon X^p \to Y^p$ via $T^p x^p = (Tx)^p$.

By Exercise 16.5.15, the set $(Y_+ \setminus B_Y^\circ)^p$ is convex in Y^p. It follows that it's pre-image under T^p is a convex subset of X^p; denote it by C. Since T is one-to-one, we have $C \subseteq Y_+^p$. If $x\in X$ with $\|x\|<1$ then $\|T^p x^p\| = \|Tx\|^p < 1$, so that $T^p x^p \in (B_Y^\circ)^p$ and, therefore, $x \notin C$. We conclude that $B_{X^p}^\circ = (B_X^\circ)^p$ and C are disjoint convex sets. By Hahn-Banach Theorem, there exists a functional $f \in (X^p)^*$ such that $f \leqslant 1$ on $B_{X^p}^\circ$ and $f \geqslant 1$ on C. The former condition implies that $\|f\|\leqslant 1$. Since $C \subseteq Y_+^p$, replacing f with $|f|$, we may assume that $f\geqslant 0$.

We claim that $f(x^p) \geqslant \|Tx\|^p$ for every $x \in X_+$. Indeed, put $y=\frac{x}{\|Tx\|}$, then $Ty \in S_Y^+$, so that $y^p \in C$. It follows that $f(y^p)\geqslant 1$, which yields $f(x^p)\geqslant \|Tx\|^p$. Since T is one-to-one, we conclude that f is strictly positive. As in Section 9.2, $f\big(|x|^p\big)$ defines an AL-norm on X^p (here $x\in X$). It follows that the completion of X^p under this norm is lattice isometric to $L_1(\nu)$ for some measure ν; we may view X^p as a sublattice of $L_1(\nu)$. Since $(X^p)^{\frac{1}{p}} = X$ and $L_1(\nu)^{\frac{1}{p}} = L_p(\nu)$ (as vector lattices), by 16.5.3 we may view X as a sublattice of $L_p(\nu)$; and the norm of $L_p(\nu)$ corresponds to $f\big(|x|^p\big)^{\frac{1}{p}}$ on X; denote the latter by $|||x|||$. It follows that $|||\cdot|||$ is an AL$_p$-norm on X, hence the completion of $\big(X, |||\cdot|||\big)$ is lattice isometric to $L_p(\mu)$ for some measure μ by Theorem 10.7.1. (In fact, $\mu=\nu$; but this is not needed for the proof.)

Let $J\colon X \to L_p(\mu)$ be the inclusion map. For every $x\in X$ we have $|||x||| = f\big(|x|^p\big)^{\frac{1}{p}} \leqslant \big(\|f\|\|x^p\|\big)^{\frac{1}{p}} = \|x\|$. It follows that $\|J\|\leqslant 1$.

Finally, for every $x \in X$, we have $|||x||| = f\big(|x|^p\big)^{\frac{1}{p}} \geqslant \|Tx\|$, so that T is a contraction as an operator from $\big(X, |||\cdot|||\big)$ to Y and, therefore, extends to a contractive lattice homomorphism R from $L_p(\mu)$ to Y. Clearly, $T=RJ$. $\square$

17.2.2. Recall that, by Theorem 16.7.6, if X is both p-convex and p-concave then X

is lattice isomorphic to some $L_p(\mu)$. This may actually be easily deduced from Theorem 17.2.1 by applying it to the identity operator. Thus, one may view Theorem 17.2.1 as a generalization of Theorem 16.7.6.

17.2.3. One can actually control the norms of the operators in Theorem 17.2.1. The proof yields that if X is p-convex with constant 1, Y is p-concave with constant 1, and T is a contraction then the factors may also be chosen to be contractions.

In the general case, it can be verified that the product of the norms of the factors is less than or equal to $K^p M^p \|T\|$, where K is the p-convexity constant of X and M is the p-convexity constant of Y. Indeed, K^p and M^p come from proofs of Lemmas 16.7.7 and 16.7.8 when we renorm X and Y, while $\|T\|$ comes from scaling T to make it into a contraction.

17.2.4 Theorem (Krivine's Factorization Theorem, [Kriv74])**.** *Let $1 \leqslant p < \infty$, suppose that $T\colon E \to X$ and $S\colon X \to F$ are two operators such that T is p-convex with constant K and S is p-concave with constant M. Then ST factors through $L_p(\mu)$ for some measure μ such that $\|U\|\|V\| \leqslant KM$:*

$$\begin{tikzcd}
E \arrow[r, "T"] \arrow[dr, dotted, "U"'] & X \arrow[r, "S"] & F \\
& L_p(\mu) \arrow[ur, dotted, "V"'] &
\end{tikzcd}$$

Proof. The proof is illustrated by the following diagram:

$$\begin{tikzcd}
E \arrow[rr, "T"] \arrow[dr, "U_1"'] & & X \arrow[rr, "S"] \arrow[dr, "R_2"] & & F \\
& Y \arrow[ur, "R_1"] \arrow[rr, "R_2R_1"'] \arrow[dr, "U_2"'] & & Z \arrow[ur, "V_1"'] & \\
& & L_p(\mu) \arrow[ur, "V_2"'] & &
\end{tikzcd}$$

By Theorem 17.1.5, T factors through some Banach lattice Y with factors U_1 and R_1, such that Y is p-convex with constant 1, R_1 is a lattice homomorphism, $\|U_1\| \leqslant K$, and $\|R_1\| \leqslant 1$. Similarly, by Theorem 17.1.1, S factors through some Banach lattice Z with factors R_2 and V_1, such that Z is p-concave with constant 1, R_2 is a lattice homomorphism, $\|V_1\| \leqslant K$ and $\|R_2\| \leqslant 1$. Applying Theorem 17.2.1, we can factor R_2R_1 via $L_1(\mu)$ for some measure μ with factors U_2 and V_2. Moreover, it follows from 17.2.3 that we may chose U_2 and V_2 to be contractions. Now taking $U = U_2U_1$ and $V = V_1V_2$ completes the proof. $\square$

17.3 Factorization through $L_{p,1}(\mu)$

In this section, we will prove that for $q<p$, an operator $T\colon C(K)\to E$ is (p,q)-concave iff it factors through a Lorentz space $L_{p,1}(\mu)$. As a corollary, we will show that $T\colon X\to E$ is (p,q)-concave iff T is $(p,1)$-concave. Using duality, we will show that $T\colon E\to X$ is (p,q)-convex for some $q>p$ iff it is (p,∞)-convex. As usual, for $f\in L_0(\mu)$ we write $\|f\|_p=\|f\|_{L_p(\mu)}$, while $\|f\|_{p,1}$ is interpreted as in 16.11. For $f\in C(K)$, we write $\|f\|_\infty$ for its norm in $C(K)$, or just $\|f\|$, when it causes no confusion.

The following proposition is somewhat complementary to the fact that the class of (p,q)-concave operators is stable under multiplication on the left; see Exercise 16.4.3.

17.3.1 Proposition. *An operator $T\colon X\to E$ is (p,q)-concave with constant M iff TJ is (p,q)-concave with constant $M\|J\|$ for every injective lattice homomorphism $J\colon C(K)\to X$, where K is a compact Hausdorff space.*

Proof. The forward implication follows from Exercise 16.4.3. To prove the converse, let $x_1,\dots,x_n\in X$, and put $e=\Bigl(\sum_{i=1}^n|x_i|^q\Bigr)^{\frac1q}$. WLOG, $\|e\|=1$. By Krein-Kakutani Representation theorem, there is a compact Hausdorff space K and a surjective lattice isometry $J\colon C(K)\to\bigl(I_e,\|\cdot\|_e\bigr)$ such that $J\mathbb{1}=e$. Let $f_1,\dots,f_n\in C(K)$ such that $Jf_i=x_i$. Then $\Bigl(\sum_{i=1}^n|f_i|^q\Bigr)^{\frac1q}=\mathbb{1}$.

Since $\|x\|\leqslant\|x\|_e$ for every $x\in I_e$, we may view J as a contractive operator from $C(K)$ into X. By assumption, TJ is (p,q)-concave with constant M. It follows that

$$\|\overline{T}\bar{x}\|_{\text{sum-}p}=\|\overline{TJ}\bar{f}\|_{\text{sum-}p}\leqslant M\|\bar{f}\|_{\text{lat-}q}=M\|\mathbb{1}\|=M.$$

□

The following is a variant of Pietch Factorization Theorem:

17.3.2 Theorem. *Let $T\colon C(K)\to E$ be p-concave with constant 1. Then there exists a probability measure $\mu\in(K)$ such that $\|Tf\|\leqslant\|f\|_p$ for every $f\in C(K)$.*

The conclusion of the theorem may be interpreted as follows: T factors through the formal inclusion of $C(K)$ into $L_p(K,\mu)$,

$$\begin{tikzcd} C(K) \arrow[r, "T"] \arrow[d, hook] & E \\ L_p(K,\mu) \arrow[ur, dotted, "S"'] & \end{tikzcd}$$

such that $\|S\|\leqslant 1$.

Proof. Let

$$W = \Bigl\{ f \in C(K) \,:\, \exists f_1,\dots,f_n \in C(K),\ f = \sum_{i=1}^n |f_i|^p \text{ and } \sum_{i=1}^n \|Tf_i\|^p = 1 \Bigr\}.$$

Observe that W is convex. Indeed, let $f,g \in W$ and $\alpha,\beta \geqslant 0$ with $\alpha+\beta = 1$. Decompose $f = \sum_{i=1}^n |f_i|^p$ and $g = \sum_{j=1}^m |g_j|^p$ as in the definition of W. Now observe that the functions $\alpha^{\frac{1}{p}} f_i$ and $\beta^{\frac{1}{p}} g_j$ for all i and j do the job for $\alpha f + \beta g$.

Let $f \in W$; let $f = \sum_{i=1}^n |f_i|^p$ as in the definition of W. Then

$$1 = \Bigl(\sum_{i=1}^n \|Tf_i\|^p\Bigr)^{\frac{1}{p}} \leqslant \Bigl\| \Bigl(\sum_{i=1}^n |f_i|^p\Bigr)^{\frac{1}{p}} \Bigr\|_\infty \leqslant \bigl\| f^{\frac{1}{p}} \bigr\|_\infty = \|f\|_\infty^{\frac{1}{p}}.$$

It follows that $\|f\|_\infty \geqslant 1$ for every $f \in W$. We can, therefore, use Hahn-Banach Theorem to separate W from $B^\circ_{C(K)}$: there exists $\mu \in C(K)^* = \mathcal{M}(K)$ such that $\|\mu\| = 1$ and $\mu(f) \geqslant 1$ for all $f \in W$. Since $W \subseteq C(K)_+$, replacing μ with $|\mu|$, we may assume that $\mu \geqslant 0$, that is, μ is a probability measure.

Let $g \in C(K)$ such that $Tg \neq 0$. Put $f = \bigl(\frac{|g|}{\|Tg\|}\bigr)^p$. It is easy to see that $f \in W$ (with a decomposition consisting of the single function $\frac{g}{\|Tg\|}$). It follows that $\mu(f) \geqslant 1$, which yields $\|Tg\| \leqslant \mu\bigl(|g|^p\bigr)^{\frac{1}{p}} = \|g\|_p$. □

Recall that every Banach space is a closed subspace of some $C(K)$ space. Since an operator $T\colon C(K) \to E$ is p-concave iff it is p-summing by Proposition 16.4.18, the following result may be viewed as a generalization of Theorem 17.3.2.

17.3.3 Theorem. *Let F be a closed subspace of a $C(K)$-space and $T\colon X \to E$ is p-summing with constant 1. Then there exists a probability measure $\mu \in M(K)$ such that $\|Tf\| \leqslant \|f\|_p$ for every $f \in F$.*

Proof. The proof is analogous to that of Theorem 17.3.2 with two modifications. First, in the definition of W we take f_i's in F. Second, by the assumption, $\Bigl(\sum_{i=1}^n \|Tf_i\|^p\Bigr)^{\frac{1}{p}}$ is dominated by the weak p-summing norm of $(f_1,\dots,f_i)$ in F, hence in $C(K)$ (because the weak p-summing norm is stable under passing to subspaces), and by Proposition 16.3.11, the latter equals $\Bigl\|\Bigl(\sum_{i=1}^n |f_i|^p\Bigr)^{\frac{1}{p}}\Bigr\|_\infty$. □

17.3.4 Lemma. *Let $1 \leqslant q \leqslant p < \infty$, K a compact space, and $0 \leqslant \mu \in \mathcal{M}(K)$.*

(i) *The formal inclusion of $C(K)$ into $L_p(\mu)$ is p-concave;*

(ii) *If $q < p$ then the formal inclusion of $C(K)$ into $L_{p,1}(\mu)$ is (p,q)-concave.*

Proof. Let $f_1,\dots,f_n \in C(K)_+$. For (i), observe that

$$\Bigl(\sum_{i=1}^n \|f_i\|_p^p\Bigr)^{\frac{1}{p}} = \Bigl(\sum_{i=1}^n \int f_i^p\Bigr)^{\frac{1}{p}} \leqslant \Bigl\|\Bigl(\sum_{i=1}^n f_i^p\Bigr)^{\frac{1}{p}}\Bigr\|_\infty \mu(K)^{\frac{1}{p}}.$$

For (ii), using Exercise 16.8.8 and then (i), we get

$$\Big(\sum_{i=1}^n \|f_i\|_{p,1}^p\Big)^{\frac{1}{p}} \leqslant C\Big(\sum_{i=1}^n \|f_i\|_q^q \|f_i\|_\infty^{q-p}\Big)^{\frac{1}{p}} \leqslant C\max_i \|f_i\|_\infty^{1-\frac{q}{p}}\Big(\sum_{i=1}^n \|f_i\|_q^q\Big)^{\frac{1}{p}}$$
$$\leqslant C\Big\|\bigvee_{i=1}^n f_i\Big\|_\infty^{1-\frac{q}{p}} \cdot \Big\|\Big(\sum_{i=1}^n f_i^q\Big)^{\frac{1}{q}}\Big\|_\infty^{\frac{q}{p}} \mu(K)^{\frac{1}{p}} = C\mu(K)^{\frac{1}{p}}\Big\|\Big(\sum_{i=1}^n f_i^q\Big)^{\frac{1}{q}}\Big\|_\infty.$$

□

17.3.5 Theorem (Pisier's Factorization Theorem)**.** *Let $T\colon C(K) \to E$ and $1 \leqslant q < p < \infty$. TFAE:*

(i) *T is (p,q)-concave;*

(ii) *There exists a probability measure $\mu \in M(K)$ and a constant C_1 such that $\|Tf\| \leqslant C_1\|f\|_q^{\frac{q}{p}}\|f\|_\infty^{1-\frac{q}{p}}$ for every $f \in C(K)$;*

(iii) *There exists a probability measure $\mu \in M(K)$ and a constant C_2 such that $\|Tf\| \leqslant C_2\|f\|_{p,1}$ for every $f \in C(K)$;*

(iv) *There exists a probability measure $\mu \in M(K)$ and an operator $S\colon L_{p,1}(\mu) \to E$ such that $Tf = Sf$ for every $f \in C(K)$.*

A few remarks are due before we proceed to the proof. Condition (iv) essentially means that T factors through the formal inclusion $J\colon C(K) \hookrightarrow L_{p,1}(\mu)$:

$$\begin{tikzcd} C(K) \arrow[r, "T"] \arrow[d, hook, "J"] & E . \\ L_{p,1}(K,\mu) \arrow[ur, dotted, "S"'] & \end{tikzcd}$$

(Note that J need not be one-to-one as $Jf = 0$ when $\operatorname{supp} f \cap \operatorname{supp} \mu = \varnothing$.) Hence, this theorem may be viewed as an analogue of Pietch Factorization Theorem 17.3.2. Note also that q is missing in (iii) and (iv), which means the condition is, actually, independent of q.

Proof. (i)⇒(ii) Scaling T, we may assume WLOG that the (p,q)-concavity constant is 1. Fix $m \in \mathbb{N}$. Find $f_1, \dots, f_n \in C(K)$ such that

$$\Big\|\Big(\sum_{i=1}^n |f_i|^q\Big)^{\frac{1}{q}}\Big\|_\infty \leqslant 1 \quad \text{and} \quad 1 - \frac{1}{m} < \Big(\sum_{i=1}^n \|Tf_i\|^p\Big)^{\frac{1}{p}} \leqslant 1.$$

The former implies $\sum_{i=1}^n |f_i|^q \leqslant \mathbb{1}$; we may assume that $\sum_{i=1}^n |f_i|^q = \mathbb{1}$ by adding an extra function.

There exists a finite sequence $(\alpha_i)_{i=1}^n$ in the unit ball of $\ell_{p^*}^n$ such that $1 - \frac{1}{m} < \sum_{i=1}^n \alpha_i\|Tf_i\|$. For every i, we pick $z_i^* \in S_{E^*}$ such that $\|Tf_i\| = z_i^*(f_i)$. Define a functional μ_m on $C(K)$ via $\mu_m(f)=\sum_{i=1}^n \alpha_i z_i^*(Tff_i)$, where ff_i is the point-wise product. It is clear that μ_m is linear and $\mu_m(\mathbb{1})>1-\frac{1}{m}$. For every $f\in C(K)$, we have

$$\bigl|\mu_m(f)\bigr|\leqslant\Bigl(\sum_{i=1}^n|\alpha_i|^{p^*}\Bigr)^{\frac{1}{p^*}}\Bigl(\sum_{i=1}^n\bigl|z_i^*(Tff_i)\bigr|^p\Bigr)^{\frac{1}{p}}\leqslant\Bigl(\sum_{i=1}^n\|Tff_i\|^p\Bigr)^{\frac{1}{p}}$$
$$\leqslant\Bigl\|\Bigl(\sum_{i=1}^n|ff_i|^q\Bigr)^{\frac{1}{q}}\Bigr\|_\infty=\Bigl\||f|\Bigl(\sum_{i=1}^n|f_i|^q\Bigr)^{\frac{1}{q}}\Bigr\|_\infty=\|f\|_\infty,$$

so that $\|\mu_m\|\leqslant 1$.

Let $f\in C(K)$ with $\|f\|_\infty=1$. Put $g=\bigl(\mathbb{1}-|f|^q\bigr)^{\frac{1}{q}}$. Using $\sum_{i=1}^n|f_i|^q=\mathbb{1}$, we get

$$\mathbb{1}=\bigl(g^q+|f|^q\bigr)^{\frac{1}{q}}=\Bigl(\sum_{i=1}^n|gf_i|^q+|f|^q\Bigr)^{\frac{1}{q}}.$$

Further, (p,q)-concavity of T yields

$$1\geqslant\Bigl(\sum_{i=1}^n\bigl\|Tgf_i\bigr\|^p+\|Tf\|^p\Bigr)^{\frac{1}{p}}.$$

As above, we have $\bigl|\mu_m(g)\bigr|\leqslant\Bigl(\sum_{i=1}^n\|Tgf_i\|^p\Bigr)^{\frac{1}{p}}$, hence

$$\|Tf\|^p\leqslant 1-\sum_{i=1}^n\bigl\|Tgf_i\bigr\|^p\leqslant 1-\bigl|\mu_m(g)\bigr|^p.$$

Since the unit ball of $C(K)^*$ is weak*-compact, there is a weak*-limit point μ of the sequence (μ_m). Then $\|\mu\|\leqslant 1$ and it follows from $\mu_m(\mathbb{1})\geqslant 1-\frac{1}{m}$ that $\mu(\mathbb{1})=1$. Then μ is a probability measure because $1=\mu(\mathbb{1})\leqslant\mu^+(\mathbb{1})\leqslant\|\mu^+\|\leqslant\|\mu\|=1$, so that $\mu^-(\mathbb{1})=0$ and, therefore, $\mu\geqslant 0$.

This also yields that $\|Tf\|^p\leqslant 1-\bigl|\mu(g)\bigr|^p$. Since $\|f\|=1$, we have $g\geqslant\mathbb{1}-|f|^q\geqslant 0$, so that $\mu(g)\geqslant 1-\mu\bigl(|f|^q\bigr)\geqslant 0$. It is an easy exercise that $1-(1-t)^p\leqslant pt$ when $t\in[0,1]$. We now get

$$\|Tf\|^p\leqslant 1-\mu(g)^p\leqslant 1-\Bigl(1-\mu\bigl(|f|^q\bigr)\Bigr)^p\leqslant p\mu\bigl(|f|^q\bigr)=p\|f\|_q^q=p\|f\|_q^q\|f\|_\infty^{p-q}$$

because $\|f\|_\infty=1$. This yields the required inequality.

(ii)⇒(iii) This implication is a well known fact in Interpolation Theory; we will, however, present a direct proof. We will show first that T extends to $L_\infty(\mu)$. Take $f\in L_\infty(\mu)$. By Luzin's Theorem, we find a sequence (f_n) in $C(K)$ such that $f_n\xrightarrow{\text{a.e.}}f$ and $\|f_n\|_\infty\leqslant\|f\|_\infty$ for every n. This yields $f_n\xrightarrow{\|\cdot\|_q}f$. It follows from

$$\bigl\|Tf_n-Tf_m\bigr\|\leqslant C_1\|f_n-f_m\|_q^{\frac{q}{p}}\|f_n-f_m\|_\infty^{1-\frac{q}{p}}\leqslant C_1\|f_n-f_m\|_q^{\frac{q}{p}}\bigl(2\|f\|_\infty\bigr)^{1-\frac{q}{p}}\to 0$$

that the sequence (Tf_n) is Cauchy, hence convergent. Put $\widetilde{T}f = \lim_n Tf_n$. It is straightforward that $\widetilde{T}f$ is well-defined and the resulting operator $\widetilde{T}\colon L_\infty(\mu) \to E$ is linear. Since $\|f_n\|_\infty \leqslant \|f\|_\infty$ for every n and $\|f_n\|_q \to \|f\|_q$, passing to the limit in $\big\|Tf_n\big\| \leqslant C_1\|f_n\|_q^{\frac{q}{p}}\|f_n\|_\infty^{1-\frac{q}{p}}$ we get $\big\|\widetilde{T}f\big\| \leqslant C_1\|f\|_q^{\frac{q}{p}}\|f\|_\infty^{1-\frac{q}{p}}$.

Next, we will prove the required inequality for a simple function $f = \sum_{i=1}^m a_i \mathbb{1}_{A_i}$. WLOG, A_i's are disjoint and $|a_1| > |a_2| > \cdots > |a_m| > 0$. Then f^* equals $|a_i|$ on $[t_{i-1}, t_i)$ for some $0 = t_0 < t_1 < \cdots < t_m$. It follows that

$$\|f\|_{p,1} = \sum_{i=1}^m \int_{t_{i-1}}^{t_i} t^{\frac{1}{p}-1}|a_i|\,dt = p\sum_{i=1}^m |a_i|\big(t_i^{\frac{1}{p}} - t_{i-1}^{\frac{1}{p}}\big).$$

To compute $\|f\|_q^{\frac{q}{p}}\|f\|_\infty^{1-\frac{q}{p}}$, we will "slice the function horizontally", adjusting for signs. For every $k = 1,\dots,m$, put $h_k = \sum_{i=1}^k (\operatorname{sign} a_i)\mathbb{1}_{A_i}$. Note that $\|h_k\|_\infty = 1$ and $\|h_k\|_q = t_k^{\frac{1}{q}}$, so that $\|\widetilde{T}h_k\| \leqslant C_1 t_k^{\frac{1}{p}}$. Observe that $f = \sum_{k=1}^m \big(|a_k| - |a_{k+1}|\big)h_k$ (we put $a_{m+1} = 0$). It follows that

$$\|\widetilde{T}f\| \leqslant C_1\sum_{k=1}^m \big(|a_k| - |a_{k+1}|\big)t_k^{\frac{1}{p}} = C_1\sum_{i=1}^m |a_i|\big(t_i^{\frac{1}{p}} - t_{i-1}^{\frac{1}{p}}\big) \leqslant \frac{C_1}{p}\|f\|_{p,1}.$$

We now extend the estimate to all of $L_\infty(\mu)$ (and, therefore, to $C(K)$, where we need it). Let $f \in L_\infty(\mu)$. Find a sequence (f_n) of simple functions such that $f_n \xrightarrow{\text{a.e.}} f$ and $\|f_n\|_\infty \leqslant \|f\|_\infty$. It follows that (f_n) converges to f in order and, therefore, (f_n) converges to f in $\|\cdot\|_q$ and in $\|\cdot\|_{p,1}$. It follows from

$$\big\|\widetilde{T}(f_n - f)\big\| \leqslant C_1\|f_n - f\|_q^{\frac{q}{p}}\|f_n - f\|_\infty^{1-\frac{q}{p}} \to 0$$

that

$$\|\widetilde{T}f\| = \lim_n \|\widetilde{T}f_n\| \leqslant \lim_n \frac{C_1}{p}\|f_n\|_{p,1} = \frac{C_1}{p}\|f\|_{p,1}.$$

(iii)⇒(iv) Observe that $C(K)$ is dense in $L_{p,1}(\mu)$. Indeed, as before, for every $f \in L_\infty(\mu)$, we use Luzin's Theorem to find a sequence (f_n) in $C(K)$ such that $f_n \xrightarrow{\text{a.e.}} f$ and $\|f_n\|_\infty \leqslant \|f\|_\infty$; it follows that (f_n) converges to f in $\|\cdot\|_{p,1}$. We conclude that $C(K)$ is $\|\cdot\|_{p,1}$-dense in $L_\infty(\mu)$. We then observe that $L_\infty(\mu)$ is dense in $L_{p,1}(\mu)$ arguing as in Example 16.8.7. It follows from (iii) that T extends to a bounded operator $\widetilde{T}\colon L_{p,1}(\mu) \to E$, and $T = \widetilde{T}J$, where $J\colon C(K) \to L_{p,1}(\mu)$ is the formal inclusion.

(iv)⇒(i) follows from the fact that the formal inclusion of $C(K)$ into $L_{p,1}(\mu)$ is (p,q)-concave by Lemma 17.3.4(ii), hence T is (p,q)-concave by Exercise 16.4.3. □

Theorem 17.3.5 leads to some important consequences.

17.3.6 Theorem. *Let $1<p<\infty$. The classes of (p,q)-concave operators $T\colon X\to E$ coincide for all $q\in[1,p)$. In particular, the classes of (p,q)-concave Banach lattices agree for all $q\in[1,p)$. Similarly, the classes of (p,q)-convex operators or Banach lattices agree for all $q\in(p,\infty]$.*

Proof. For concavity, Proposition 17.3.1 allows us to reduce the problem to the special case $X=C(K)$. In this case, the result follows from the equivalence (i)⇔(iii) in Theorem 17.3.5. The statement for convexity follows by duality. □

This means that among all possible pairs (p,q), it suffices to consider the pairs (p,p) and $(p,1)$ for concavity and the pairs (p,p) and (p,∞) for convexity. These correspond to p-concavity, p-lower estimate, p-convexity, and p-upper estimate, respectively.

17.3.7 Theorem. *Let $1<p<\infty$. If $T\colon X\to E$ is $(p,1)$-concave then T is q-concave for every $q\in(p,\infty]$. If $T\colon E\to X$ is (p,∞)-convex then T is q-convex for every $q\in[1,p)$.*

Proof. For concavity, by Proposition 17.3.1 we may assume WLOG that $X=C(K)$. By Theorem 17.3.5, T factors through the canonical inclusion $C(K)\hookrightarrow L_{p,1}(K,\mu)$. Recall that $L_q(\mu)\subseteq L_{p,1}(\mu)$, and the inclusion map is continuous by Proposition 16.8.11. Also, it is easy to see that the inclusion map from $C(K)$ to $L_q(\mu)$ is continuous. Hence, we have a factorization as follows:

$$\begin{tikzcd} C(K) \arrow[r, "T"] \arrow[d] & E \\ L_q(\mu) \arrow[r] & L_{p,1}(K,\mu) \arrow[u] \end{tikzcd}$$

Since $L_q(\mu)$ is q-concave, T is q-concave by Exercise 16.4.3. The statement for convexity now follows by duality. □

17.3.8. By Theorems 17.3.6, we get the following hierarchies of properties for Banach lattices:

$$p\text{-convex} \quad\Rightarrow\quad \text{upper } p\text{-estimate} \quad\Rightarrow\quad q\text{-convex for all } q<p,$$

and

$$p\text{-concave} \quad\Rightarrow\quad \text{lower } p\text{-estimate} \quad\Rightarrow\quad q\text{-concave for all } q>p.$$

Compare this with Exercise 16.8.4. We have observed that L_p spaces are prototypical examples of p-convex and p-concave spaces. We can now view $L_{p,\infty}(\mu)$ and $L_{p,1}(\mu)$ as prototypical examples of spaces with upper or lower p-estimates. Exercises 16.8.4 and 16.8.11 naturally correlate with 17.3.8.

17.4 Factorization through $L_p(g\mu)$

The following result is an easy application of Krivine's Factorization Theorem 17.2.4 with S being the identity operator on $L_r(\mu)$:

17.4.1 Theorem. *Suppose that $1 \leqslant r < p < \infty$ and $T\colon E \to L_r(\mu)$ is a p-convex operator for some measure μ. Then T factors through $L_p(\nu)$ for some measure ν:*

$$\begin{tikzcd} E \arrow[rr, "T"] \arrow[dr, dotted, "U"'] & & L_r(\mu) \\ & L_p(\nu) \arrow[ur, dotted, "V"'] & \end{tikzcd}$$

In his section, we will prove Maurey-Nisishin's Theorem, which strengthens the preceding result in the following way: ν can be chosen to be of the form $g\mu$, and the operator V is a multiplication operator. The function g here is a ***density*** in the sense that $g \geqslant 0$ and $\|g\|_{L_1(\mu)} = 1$. For a fixed density g, we will often consider functions f such that $\ker g \subseteq \ker f$ (up to a set of measure zero). This is equivalent to saying that f is contained in the band B_g generated by g. This will allow us to consider the function $g^{-s}f$ for some $s > 0$; this function is well-defined on $\{g \neq 0\}$; we extend it by zero on $\{g = 0\}$.

Maurey-Nikishin's theorem says that if $T\colon E \to L_r(\mu)$ is a p-convex operator with constant 1 for some $p > r$ then we can find a density g such that $g^{-1}TB_E$ is contained in the unit ball of $L_p(g\mu)$. This allows us to factor T via $L_p(g\mu)$, with the second factor being the operator of multiplication by g. The "heavy lifting" part of the proof of Maurey-Nisishin's Theorem is the following lemma:

17.4.2 Lemma. *Suppose that $1 \leqslant r < p < \infty$ and $A \subseteq L_r(\mu)_+$. TFAE:*

(i) $\Bigl\|\Bigl(\sum_{i=1}^n (\alpha_i f_i)^p\Bigr)^{\frac{1}{p}}\Bigr\|_{L_r(\mu)} \leqslant \bigl\|(\alpha_i)_{i=1}^n\bigr\|_{\ell_p^n}$ *for all $f_1,\dots,f_n \in A$ and $\alpha_1,\dots,\alpha_n \in \mathbb{R}_+$;*

(ii) *There exists a density g such that $A \subseteq B_g$ and $\bigl\|g^{-\frac{1}{r}}f\bigr\|_{L_p(g\mu)} \leqslant 1$ for every $f \in A$.*

(Note that $\bigl\|g^{-\frac{1}{r}}f\bigr\|_{L_p(g\mu)} = \bigl\|hf\bigr\|_{L_p(\mu)}$ where $h = g^{\frac{1}{p}-\frac{1}{r}}$. Thus, (ii) says that hA is contained in $B_{L_p(\mu)}$.)

Proof. Let $q = \frac{p}{r}$. Throughout the proof, we will write $\|f\|_p$ for $\|f\|_{L_p(\mu)}$.

(ii)$\Rightarrow$(i) Let $f_1,\dots,f_n \in A$ and $\alpha_1,\dots,\alpha_n \in \mathbb{R}_+$. Put $h = \sum_{i=1}^n \bigl(\alpha_i f_i g^{-\frac{1}{r}}\bigr)^p$. Then

$$\begin{aligned}\Bigl\|\Bigl(\sum_{i=1}^n (\alpha_i f_i)^p\Bigr)^{\frac{1}{p}}\Bigr\|_r &= \Bigl(\int \Bigl(\sum_{i=1}^n (\alpha_i f_i)^p\Bigr)^{\frac{r}{p}} d\mu\Bigr)^{\frac{1}{r}} \\ &= \Bigl(\int \Bigl(\sum_{i=1}^n \bigl(\alpha_i f_i g^{-\frac{1}{r}}\bigr)^p\Bigr)^{\frac{r}{p}} g\, d\mu\Bigr)^{\frac{1}{r}} = \Bigl(\int h^{\frac{r}{p}} g\, d\mu\Bigr)^{\frac{1}{r}} = \Bigl(\int (hg)^{\frac{1}{q}} g^{\frac{1}{q^*}}\, d\mu\Bigr)^{\frac{1}{r}}\end{aligned}$$

(using Hölder inequality with q and q^* and the fact that g is a density)

$$\leqslant \Big(\int hg\,d\mu\Big)^{\frac{1}{rq}} = \Big(\int \sum_{i=1}^n \alpha_i^p \big(f_i g^{-\frac{1}{r}}\big)^p g\,d\mu\Big)^{\frac{1}{p}}$$
$$= \Big(\sum_{i=1}^n \alpha_i^p \int \big(f_i g^{-\frac{1}{r}}\big)^p g\,d\mu\Big)^{\frac{1}{p}} = \Big(\sum_{i=1}^n \alpha_i^p \big\|f_i g^{-\frac{1}{r}}\big\|^p_{L_p(g\mu)}\Big)^{\frac{1}{p}} \leqslant \Big(\sum_{i=1}^n \alpha_i^p\Big)^{\frac{1}{p}}.$$

(i)⇒(ii) Let

$$K = \sup\Big\{\Big\|\Big(\sum_{i=1}^n (\alpha_i f_i)^p\Big)^{\frac{1}{p}}\Big\|_r \;:\; f_1,\dots,f_n \in A,\ (\alpha_i)_{i=1}^n \in B^+_{\ell_p^n}\Big\}.$$

Then $K \leqslant 1$. Replacing A with $\frac{1}{K}A$, we may assume WLOG that $K=1$. Put

$$W = \Big\{g \in L_1(\mu) \;:\; 0 \leqslant g \leqslant \Big(\sum_{i=1}^n (\alpha_i f_i)^p\Big)^{\frac{1}{q}}, f_1,\dots,f_n \in A, (\alpha_i)_{i=1}^n \in B^+_{\ell_p^n}\Big\}.$$

Clearly, $\sup_{g\in W}\|g\|_1 = 1$,

Let $W^q = \{g^q \;:\; g \in W\}$. It is easy to see that W^q is the set of all positive functions in $L_{\frac{1}{q}(\mu)}$ dominated by convex combinations of elements of A^p. Hence, W^q is convex. It follows that W is *q-convex* in the following sense: if $g_1,\dots,g_n \in W$ and $\lambda_1,\dots,\lambda_n \in \mathbb{R}_+$ with $\sum_{i=1}^n \lambda_i = 1$ then $\Big(\sum_{i=1}^n \lambda_i g_i^q\Big)^{\frac{1}{q}} \in W$. Indeed, for each $i=1,\dots,n$, find $h_i \in \operatorname{conv} A^p$ with $0 \leqslant g_i^q \leqslant h_i$. Then $\sum_{i=1}^n \lambda h_i \in \operatorname{conv} A^p$, and

$$\Big(\sum_{i=1}^n \lambda_i g_i^q\Big)^{\frac{1}{q}} \leqslant \Big(\sum_{i=1}^n \lambda h_i\Big)^{\frac{1}{q}} \in (W^q)^{\frac{1}{q}} = W, \quad \text{so that} \quad \Big(\sum_{i=1}^n \lambda_i g_i^q\Big)^{\frac{1}{q}} \in W.$$

It follows that W is convex. Indeed, $\sum_{i=1}^n \lambda_i g_i \leqslant \Big(\sum_{i=1}^n \lambda_i g_i^q\Big)^{\frac{1}{q}}$ because the map $t \mapsto t^q$ is convex on $\mathbb{R}_+$; this yields $\sum_{i=1}^n \lambda_i g_i \in W$. Convexity of W yields $\overline{W} = \overline{W}^{\mathrm{w}}$. Since function calculus is norm continuous by 4.4.10, $\overline{W}$ is still q-convex.

We claim that W is relatively weakly compact. Suppose not. Then by by part (vi) of Dunford-Pettis Theorem 9.8.6, we can find $\delta > 0$, a sequence (g_m) in W, and a disjoint sequence (B_m) of measurable sets such that $\int_{B_m} g_m > \delta$. For every $k \in \mathbb{N}$, using additivity of $\|\cdot\|_1$ on disjoint vectors, we have

$$\Big\|\bigvee_{m=1}^k g_m\Big\|_1 \geqslant \Big\|\bigvee_{m=1}^k g_m \mathbb{1}_{B_m}\Big\|_1 = \sum_{m=1}^k \|g_m \mathbb{1}_{B_m}\|_1 \geqslant k\delta.$$

On the other hand, using $\|\cdot\|_{\ell_\infty^n} \leqslant \|\cdot\|_{\ell_q^n}$ and the q-convexity of W, we get

$$\Big\|\bigvee_{m=1}^k g_m\Big\|_1 \leqslant \Big\|\Big(\sum_{m=1}^k g_m^q\Big)^{\frac{1}{q}}\Big\|_1 = k^{\frac{1}{q}}\Big\|\Big(\sum_{m=1}^k \frac{1}{k} g_m^q\Big)^{\frac{1}{q}}\Big\|_1 \leqslant k^{\frac{1}{q}}.$$

This yields $k\delta \leqslant k^{\frac{1}{q}}$ for every k, which is a contradiction.

We conclude that $\overline{W}$ is weakly compact. It follows from $1 = \sup_{g\in W}\|g\|_1 = \sup_{g\in W}\langle g, \mathbb{1}\rangle$ that $\overline{W} \subseteq B^+_{L_1(\mu)}$ and that we can then find (and fix for the rest of the proof) some $g \in \overline{W}$ such that $\|g\|_1 = 1$, i.e., g is a density.

Fix $f \in A$. Let $s > 0$. Since $f^r \in W$ and $\overline{W}$ is q-convex, we have

$$\Bigl(\frac{1}{1+s}g^q + \frac{s}{1+s}(f^r)^q\Bigr)^{\frac{1}{q}} \in \overline{W}.$$

Taking the L_1-norm, we conclude that

$$\int \bigl(g^q + sf^p\bigr)^{\frac{1}{q}}\, d\mu \leqslant (1+s)^{\frac{1}{q}}. \tag{17.1}$$

Splitting this integral over the sets $\{g > 0\}$ and $\{g = 0\}$, we get

$$(1+s)^{\frac{1}{q}} \geqslant \int_{\{g=0\}} s^{\frac{1}{q}} f^r\, d\mu + \int_{\{g>0\}} g\, d\mu = s^{\frac{1}{q}} \int_{\{g=0\}} f^r\, d\mu + 1$$

It follows that $\int_{\{g=0\}} f^r\, d\mu \leqslant \frac{(1+s)^{\frac{1}{q}}-1}{s^{\frac{1}{q}}}$. By L'Hôpital's rule, the right hand side converges to zero as $s \to 0$, hence $\int_{\{g=0\}} f^r\, d\mu = 0$. Therefore, f vanishes on $\ker g$. We conclude that $A \subseteq B_g$.

By L'Hôpital's rule again, $\lim_{s\to 0^+} \frac{(1+\lambda s)^{\frac{1}{q}}-1}{s} = \frac{\lambda}{q}$ for every $\lambda > 0$. Put

$$h_s = \frac{\bigl(g^q + sf^p\bigr)^{\frac{1}{q}} - g}{s} = g \cdot \frac{\bigl(\mathbb{1} + sf^pg^{-q}\bigr)^{\frac{1}{q}} - \mathbb{1}}{s}.$$

Then h_s converges point-wise to $h = g \cdot \frac{1}{q} f^p g^{-q}$. Further, (17.1) yields

$$0 \leqslant \int h_s\, d\mu = \frac{1}{s}\Bigl[\int \Bigl(g^q + sf^p\Bigr)^{\frac{1}{q}} d\mu - \int g\, d\mu\Bigr] \leqslant \frac{(1+s)^{\frac{1}{q}}-1}{s} \to \frac{1}{q}$$

as $s \to 0$. By Fatou's Lemma, we have

$$\bigl\|g^{-\frac{1}{r}} f\bigr\|^p_{L_p(g\mu)} = \int (g^{-\frac{1}{r}} f)^p \cdot g\, d\mu = q \int h\, d\mu \leqslant q \liminf_{s\to 0^+} \int h_s\, d\mu = 1.$$

□

17.4.3 Theorem (Maurey-Nikishin). *Let* $1 \leqslant r < p < \infty$, $C > 0$, *and* $T\colon E \to L_r(\mu)$. *TFAE:*

(i) T *is* p*-convex with constant* C*;*

(ii) *There exists a density* g *in* $L_1(\mu)$ *such that* $\operatorname{Range} T$ *is contained in the band* B_g *and* $\|g^{-\frac{1}{r}} Tx\|_{L_p(g\mu)} \leqslant C\|x\|$ *for all* $x \in E$*;*

(iii) *T factors via $L_p(g\mu)$, where g is a density, $\|R\| \leqslant C$, $\|M\| \leqslant 1$, and $Mh = g^{\frac{1}{r}}h$ for all $h \in L_p(g\mu)$:*

$$\begin{tikzcd} E \arrow[rr, "T"] \arrow[dr, dotted, "R"'] & & L_r(\mu) \\ & L_p(g\mu) \arrow[ur, dotted, "M"'] & \end{tikzcd}$$

Proof. WLOG, $C = 1$. The statements (i) and (ii) are equivalent because each is equivalent to the corresponding statement in Lemma 17.4.2 with $A = \{|Tx| : x \in B_E\}$. For (ii) this trivial. For (i), this follows from Exercise 16.4.13. Thus, (i)$\Leftrightarrow$(ii).

(iii)$\Rightarrow$(ii) as $\operatorname{Range} T \subseteq \operatorname{Range} M \subseteq B_g$ and $\|g^{-\frac{1}{r}}Tx\|_{L_p(g\mu)} = \|Rx\|_{L_p(g\mu)} \leqslant \|x\|$ for all $x \in E$.

(ii)$\Rightarrow$(iii) For $x \in E$, define $Rx = g^{-\frac{1}{r}}Tx$. By (ii), R is a well-defined contractive linear operator from E to $L_p(g\mu)$. For $h \in L_p(g\mu)$, put $Mh = g^{\frac{1}{r}}h$. Observe that

$$\|Mh\|_r = \Big(\int g|h|^r \, d\mu\Big)^{\frac{1}{r}} = \|h\|_{L_r(g\mu)} \leqslant \|h\|_{L_p(g\mu)},$$

because $r < p$ and $g\mu$ is a probability measure. It follows that M is a contractive linear operator from $L_p(g\mu)$ to $L_r(\mu)$. Finally, we clearly have $MRx = Tx$ for every $x \in E$. $\square$

17.4.4 Remark. It is clear that the operator R in (iii) is given by $Rx = g^{-\frac{1}{r}}Tx$. In particular, if T is a lattice homomorphism then so are R and M.

We next present the dual version of Maurey-Nikishin's Theorem 17.4.3:

17.4.5 Corollary. *Let $1 < q < s < \infty$, $C > 0$, and $S\colon L_s(\mu) \to F$. TFAE:*

(i) *S is q-concave with constant C;*

(ii) *There exists a density g in $L_1(\mu)$ such that $\|Sf\| \leqslant C\|g^{-\frac{1}{s}}f\|_{L_q(g\mu)}$ for all $f \in L_s(\mu)$;*

(iii) *T factors via $L_q(g\mu)$, where g is a density, $\|V\| \leqslant C$, $\|N\| \leqslant 1$, and $Nf = g^{-\frac{1}{s}}f$ for all $f \in L_s(\mu)$:*

$$\begin{tikzcd} L_s(\mu) \arrow[rr, "S"] \arrow[dr, dotted, "N"'] & & F \\ & L_q(g\mu) \arrow[ur, dotted, "V"'] & \end{tikzcd}$$

(As before, we define $g^{-\frac{1}{s}}f$ to be zero on $\{g = 0\}$; this causes no problems because we view $g^{-\frac{1}{s}}f$ as an element of $L_q(g\mu)$, where the set $\{g = 0\}$ has measure zero.)

Proof. Again, WLOG, $C=1$. Put $r=s^*$, $p=q^*$, $E=F^*$, and $T=S^*$. We will first show that (i) and (iii) are equivalent because each of them is equivalent to the corresponding condition in Maurey-Nikishin's Theorem 17.4.3. We will then complete the proof by showing that (ii)$\Leftrightarrow$(iii).

(i) is equivalent to Theorem 17.4.3(i) by Theorem 16.4.6.

Suppose S satisfies (iii). Then T factors as in the diagram in Theorem 17.4.3(iii) with $R=V^*$ and $M=N^*$. Put $\nu=g\mu$. For every $h\in L_p(\nu)$ and $f\in L_s(\mu)$, we have

$$\langle Mh,f\rangle=\langle h,Nf\rangle=\int h\cdot g^{-\frac{1}{s}}f\cdot g\,d\mu=\int hg^{\frac{1}{r}}\cdot f\,d\mu=\langle hg^{\frac{1}{r}},f\rangle.$$

This yields $Mh=hg^{\frac{1}{r}}$. It follows from $T=MR$ that $Rx=g^{-\frac{1}{r}}Tx$ for every $x\in E$. Hence, T satisfies Theorem 17.4.3(iii).

Suppose T satisfies Theorem 17.4.3(iii). Dualizing, we get

$$\begin{tikzcd} L_s(\mu) \arrow[rr, "S^{**}"] \arrow[dr, "N=M^*"'] & & F^{**} \\ & L_q(g\mu) \arrow[ur, "V=R^*"'] & \end{tikzcd}$$

Note that since $L_s(\mu)$ is reflexive, S^{**} agrees with S and actually acts into F rather that F^{**} (we identify F with its canonical image in F^{**}). Take $N=M^*$ and $V=R^*$. For $f\in L_s(\mu)$ and $h\in L_q(g\mu)$, we have

$$\langle Nf,h\rangle=\langle f,Mh\rangle=\int f\cdot g^{\frac{1}{r}}h\,d\mu=\int g^{-\frac{1}{s}}f\cdot h\cdot g\,d\mu=\langle g^{-\frac{1}{s}}f,h\rangle,$$

where the last expression corresponds to the dual pair $\big\langle L_q(g\mu),L_q(g\mu)\big\rangle$. It follows that $Nf=g^{-\frac{1}{s}}f$.

Note that M is one-to-one: if $Mh=0$ for some $h\in L_p(g\mu)$ then h vanishes on $\{g\neq 0\}$, so $h=0$ in $L_p(g\mu)$. It follows that $\operatorname{Range}N$ is weak*-dense in $L_q(g\mu)$. Since the latter space is reflexive, $\operatorname{Range}N$ is weakly dense and, therefore, norm dense. It follows that

$$\operatorname{Range}V=V\big(\overline{\operatorname{Range}N}\big)\subseteq\overline{\operatorname{Range}VN}=\overline{\operatorname{Range}S^{**}}\subseteq F.$$

Hence, we may view V as an operator from $L_q(g\mu)$ to F.

(iii)$\Rightarrow$(ii): $\|Sf\|=\|VNf\|\leqslant\|Nf\|=\|g^{-\frac{1}{s}}f\|$.

(ii)$\Rightarrow$(iii) Put $\nu=g\mu$. For $f\in L_s(\mu)$, put $Nf=g^{-\frac{1}{s}}f$. Since $g\mu$ is a probability measure and $q<s$, we have

$$\|Nf\|_{L_q(\nu)}\leqslant\|Nf\|_{L_s(\nu)}=\Big(\int f^sg^{-1}\cdot g\,d\mu\Big)^{\frac{1}{s}}=\|f\|_{L_s(\mu)},$$

so that N is a contractive operator from $L_s(\mu)$ to $L_q(\nu)$. Further, we define V on $\operatorname{Range} N$ via $V(Nf) = Sf$; this is well-defined because N is one-to-one. We then clearly have $S = VN$. Furthermore, it follows from the assumption that $\|V(Nf)\| \leqslant \|g^{-\frac{1}{s}}f\|_{L_q(\nu)} = \|Nf\|_{L_q(\nu)}$, so that V is a contraction (on $\operatorname{Range} N$). It extends to a contraction on $L_q(\nu)$ because $\operatorname{Range} N$ contains $L_s(\nu)$ and, therefore, is dense in $L_q(\nu)$. Indeed, if $h \in L_s(\nu)$ then $f := hg^{\frac{1}{s}} \in L_s(\mu)$ and $h = Nf$. $\square$

The next result is an application of Krivine's, Maurey's and Maurey-Nikishin's Theorems.

17.4.6 Theorem (Reisner)**.** *The product of a p-convex and a q-concave operators, where $1 \leqslant q < p < \infty$, factors through a multiplication operator from $L_p(\mu)$ to $L_q(\mu)$ for some measure μ:*

$$\begin{tikzcd}
E \arrow[r, "U", "p\text{-convex}"'] \arrow[d, "R"'] & X \arrow[r, "V", "q\text{-concave}"'] & F \\
L_p(\mu) \arrow[rr, "M_h", "\text{where } 0\leqslant h\in L_r(\mu),\ \frac{1}{r}=\frac{1}{q}-\frac{1}{p}"'] & & L_q(\mu) \arrow[u, "S"']
\end{tikzcd}$$

Proof. The proof is illustrated by the following diagram:

$$\begin{tikzcd}
E \arrow[rr, "U"] \arrow[dr, "U_1"'] & & X \arrow[rr, "V"] \arrow[dr, "Q_2"] & & F \\
& Y \arrow[ur, "Q_1"] \arrow[rr, "Q_2Q_1"'] \arrow[dl, dotted, "R"'] \arrow[d, dashed, "P"'] \arrow[drr, "T"'] & & Z \arrow[ur, "V_1"'] & \\
L_p(\mu) \arrow[rrr, dotted, bend right, "M"'] & L_p(g\mu) \arrow[rr, dashed, "N"'] & & L_q(\mu) \arrow[u, "S"'] &
\end{tikzcd}$$

We start as in the proof of Theorem 17.2.4. By Theorem 17.1.5, U factors through some p-convex Banach lattice Y, $U = Q_1U_1$, such that Q_1 is a lattice homomorphism. Similarly, by Theorem 17.1.1, V factors as $V = V_1Q_2$ through some q-concave Banach lattice Z, such that Q_2 is a lattice homomorphism. Now Q_2Q_1 is a lattice homomorphism from Y to Z. Since $q < p$, Y is q-convex. By Theorem 17.2.1, Q_2Q_1 factors as $Q_2Q_1 = ST$ via $L_q(\mu)$ for some measure μ, such that S and T are lattice homomorphisms. Since Y is p-convex, so is T.

By Maurey-Nikishin's Theorem 17.4.3, T factors as $T=NP$ through $L_p(g\mu)$, where g is a density in $L_1(\mu)$, $Py=g^{-\frac{1}{q}}Ty$ and $Nf=g^{\frac{1}{q}}f$ for $y\in Y$ and $f\in L_p(g\mu)$. Take r so that $\frac{1}{q}=\frac{1}{p}+\frac{1}{r}$ and put $h=g^{\frac{1}{r}}$. Then $0\leqslant h\in L_r(\mu)$. Define $R\colon Y\to L_p(\mu)$ and $M\colon L_p(\mu)\to L_q(\mu)$ via $Ry=h^{-1}Ty$ and $Mf=hf$. They are well-defined and bounded because

$$\|Ry\|^p_{L_p(\mu)}=\int h^{-p}|Ty|^p\,d\mu=\int\bigl|g^{-\frac{1}{q}}Ty\bigr|^p\,g\,d\mu=\|Py\|^p_{L_p(g\mu)}\leqslant\|P\|\|y\|,\text{ and}$$

$$\|Mf\|_{L_q(\mu)}=\Bigl(\int|f|^qg^{\frac{q}{r}}\,d\mu\Bigr)^{\frac{1}{q}}\leqslant\Bigl(\int|f|^p\,d\mu\Bigr)^{\frac{1}{p}}\Bigl(\int g\,d\mu\Bigr)^{\frac{1}{r}}=\|f\|_{L_p(\mu)}$$

by Hölder's inequality. Finally, it is clear that $MR=T$. □

The following result is another application of Maurey-Nikishin's Theorem. However, unlike in the original Maurey-Nikishin's Theorem, we factor though L_2 on the same measure space, and the multiplier comes from $L_2(\mu)$ rather than $L_1(\mu)$.

17.4.7 Corollary (Maurey-Rosenthal)**.** *Let $T\colon E\to L_1(\mu)$ be 2-convex with constant 1. Then T factors as $E\xrightarrow{R}L_2(\mu)\xrightarrow{M_h}L_1(\mu)$ such $\|R\|\leqslant 1$ and M_h is a multiplication operator with $0\leqslant h\in L_2(\mu)$ and $\|h\|_{L_2(\mu)}=1$.*

Proof. One way to prove this is to apply Reisner's Theorem with $p=2$, $q=1$, $X=F=L_1(\mu)$, and $U=T$ and V being the identity operator on $L_1(\mu)$. Alternatively, one can apply the Maurey-Nikishin's Theorem 17.4.3 directly as follows. Take $r=1$ and $p=2$. Find a density g using the implication (i)⇒(ii) in the theorem. Put $h=g^{\frac{1}{2}}$. Then $0\leqslant h\in L_2(\mu)$ and $\|h\|_{L_2(\mu)}=1$. For $x\in E$, define $Rx=h^{-1}Tx$. By (ii), we have $\|x\|\geqslant\|g^{-1}Tx\|_{L_2(g\mu)}=\|Rx\|_{L_2(\mu)}$, so that R is a contraction from E to $L_2(\mu)$. It is easy to see that $\|M_h\|=1$ and $T=M_hR$. □

17.5 Factorization through $L_{p,1}(g\mu)$

This section is somewhat parallel to the preceding one. In the preceding section we proved Maurey-Nikishin's Theorem that an operator $T\colon E\to L_r(\mu)$ is p-convex iff it factors through $L_p(g\mu)$ for some density g. Recall that an operator is (p,q)-convex for some $q>p$ iff it is (p,∞)-convex by Theorem 17.3.6. In this section, we will present Theorem 17.5.2, due to Pisier, that asserts that T is (p,∞)-convex iff it factors through $L_{p,\infty}(g\mu)$. We will equip $L_{p,\infty}(\mu)$ with the norm $\|\cdot\|_{p,\infty,r}$; see 16.8.2 and Theorem 16.8.5. We denote the resulting Banach lattice by $L_{p,\infty,r}(\mu)$. The proof combines ideas of the proof of Maurey-Nikishin's Theorem and of Theorem 17.3.5. Again, the "heavy lifting" is delegated to the following lemma:

17.5.1 Lemma. *Let $1 \leqslant r < p < \infty$ and $F \subseteq L_r(\mu)_+$. TFAE:*

(i) *There exists $C > 0$ such that $\bigl\|\bigvee_{i=1}^n \alpha_i f_i\bigr\|_{L_r(\mu)} \leqslant C\bigl\|(\alpha_i)_{i=1}^n\bigr\|_{\ell_p^n}$ for all $f_1,\dots,f_n \in F$ and $\alpha_1,\dots,\alpha_n \in \mathbb{R}_+$;*

(ii) *There exists $C' > 0$ and a density $g \in L_1(\mu)$ such that $\bigl\|f\mathbb{1}_B\bigr\|_{L_r(\mu)} \leqslant C'\Bigl(\int_B g\,d\mu\Bigr)^{\frac{1}{r}-\frac{1}{p}}$ for every $f \in F$ and every measurable set B;*

(iii) *There exists $C'' > 0$ and a density $g \in L_1(\mu)$ such that $F \subseteq B_g$ and $\bigl\|g^{-\frac{1}{r}}f\bigr\|_{L_{p,\infty,r}(g\mu)} \leqslant C''$.*

Proof. Throughout the proof, we write $\|\cdot\|_r$, $\|\cdot\|_p$, and $\|\cdot\|_1$, for $\|\cdot\|_{L_r(\mu)}$, $\|\cdot\|_{L_p(\mu)}$, and $\|\cdot\|_{L_1(\mu)}$, respectively. We will also write $\nu = g\mu$.

(ii)⇒(iii) Fix $f \in F$. Applying the assumption with $B = \{g = 0\}$, we get $\bigl\|f\mathbb{1}_B\bigr\|_{L_r(\mu)} = 0$, so that f vanishes on B and, therefore, $f \in B_g$. Let B be an arbitrary measurable set with $\nu(B) > 0$. Then

$$\nu(B)^{\frac{1}{p}-\frac{1}{r}}\|g^{-\frac{1}{r}}f\mathbb{1}_B\|_{L_r(\nu)} = \Bigl(\int_B g\,d\mu\Bigr)^{\frac{1}{p}-\frac{1}{q}}\Bigl(\int_B f^r\,d\mu\Bigr)^{\frac{1}{r}} \leqslant C'$$

by the assumption. It follows from the definition of $\|\cdot\|_{p,\infty,r}$ in 16.8.2 that $\bigl\|g^{-\frac{1}{r}}f\bigr\|_{L_{p,\infty,r}(\nu)} \leqslant C'$.

(iii)⇒(i) Let $f_1,\dots,f_n \in F$ and $\alpha_1,\dots,\alpha_n \in \mathbb{R}_+$. Since $L_{p,\infty,r}(\mu)$ satisfies an upper p-estimate with constant 1 by Theorem 16.8.5, it is (p,∞)-convex. Also, by Exercise 16.8.4, we have $\|\cdot\|_{L_r(\nu)} \leqslant \|\cdot\|_{L_{p,\infty,r}(\nu)}$ This yields

$$\begin{aligned}\Bigl\|\bigvee_{i=1}^n \alpha_i f_i\Bigr\|_r = \Bigl\|\bigvee_{i=1}^n \alpha_i f_i g^{-\frac{1}{r}}\Bigr\|_{L_r(\nu)} &\leqslant \Bigl\|\bigvee_{i=1}^n \alpha_i f_i g^{-\frac{1}{r}}\Bigr\|_{L_{p,\infty,r}(\nu)}\\ &\leqslant \Bigl(\sum_{i=1}^n \|\alpha_i f_i g^{-\frac{1}{r}}\|^p_{L_{p,\infty,r}(\nu)}\Bigr)^{\frac{1}{p}} \leqslant C''\Bigl(\sum_{i=1}^n \alpha_i^p\Bigr)^{\frac{1}{p}}.\end{aligned}$$

(i)⇒(ii) WLOG, we assume that $C = 1$. Scaling (inflating) F, if necessary, we may also assume that

$$\sup\Bigl\{\bigl\|\bigvee_{i=1}^n \alpha_i f_i\bigr\|_r \,:\, n \in \mathbb{N},\ f_1,\dots,f_n \in F,\ (\alpha_i)_{i=1}^n \in B_{\ell_p^n}\Bigr\} = 1.$$

Hence, for every m we can find $\alpha_{m,1},\dots,\alpha_{m,n_m}$ in $\mathbb{R}_+$ and $f_{m,1},\dots,f_{m,n_m}$ in F such that

$$\sum_{\imath=1}^{n_m} \alpha_{m,i}^p = 1 \quad\text{and}\quad 1 - \frac{1}{m} < \bigl\|\bigvee_{i=1}^n \alpha_{m,i} f_{m,i}\bigr\|_r \leqslant 1.$$

For every $m \in \mathbb{N}$, put $g_m = \Bigl(\bigvee_{i=1}^n \alpha_{m,i} f_{m,i}\Bigr)^r$. Then $g_m \geqslant 0$ and $(1-\frac{1}{m})^r \leqslant \|g_m\|_1 \leqslant 1$.

We claim that the sequence (g_m) is relatively weakly compact in $L_1(\mu)$. Suppose not. Then, by part (vi) of Dunford-Pettis Theorem 9.8.6, we can find $\delta > 0$ and a disjoint sequence (B_m) of measurable sets such that $\int_{B_m} g_m > \delta$. For every $k \in \mathbb{N}$, using additivity of $\|\cdot\|_1$ on disjoint vectors, we have

$$\Big\|\bigvee_{m=1}^{k} g_m\Big\|_1 \geqslant \Big\|\bigvee_{m=1}^{k} g_m \mathbb{1}_{B_m}\Big\|_1 = \sum_{m=1}^{k} \|g_m \mathbb{1}_{B_m}\|_1 \geqslant k\delta.$$

On the other hand, using the assumption, we have

$$\begin{aligned}\Big\|\bigvee_{m=1}^{k} g_m\Big\|_1 = \Big\|\bigvee_{m=1}^{k}\Big(\bigvee_{i=1}^{n_m}\alpha_{m,i}f_{m,i}\Big)^r\Big\|_1 &= \Big\|\Big(\bigvee_{m=1}^{k}\bigvee_{i=1}^{n_m}\alpha_{m,i}f_{m,i}\Big)^r\Big\|_1 \\ &= \Big\|\bigvee_{m=1}^{k}\bigvee_{i=1}^{n_m}\alpha_{m,i}f_{m,i}\Big\|_r^r \leqslant \Big(\sum_{m=1}^{k}\sum_{i=1}^{n_m}\alpha_{m,i}^p\Big)^{\frac{r}{p}} \leqslant k^{\frac{r}{p}}.\end{aligned}$$

This yields $k\delta \leqslant k^{\frac{r}{p}}$ for every k, which is a contradiction. This proves that (g_m) is relatively weakly compact.

Therefore, it has a weak accumulation point, say g. Clearly, $g \geqslant 0$. It follows from $(1-\frac{1}{m})^r \leqslant \langle g_m, \mathbb{1}\rangle \leqslant 1$ and that $\|g\| = \langle g, \mathbb{1}\rangle = 1$, hence g is a density.

Fix $f \in F$ and a measurable set B; let $s > 0$ and $m \in \mathbb{N}$. We have

$$\begin{aligned}s^{\frac{r}{p}}\int_B f^r\,d\mu + \int_{\Omega\setminus B} g_m\,d\mu &\leqslant \int \big(s^{\frac{1}{p}}f\big)^r \vee g_m\,d\mu = \int\Big(\big(s^{\frac{1}{p}}f\big)\vee\bigvee_{i=1}^{n_m}\alpha_{m,i}f_{m,i}\Big)^r d\mu \\ &= \Big\|\big(s^{\frac{1}{p}}f\big)\vee\bigvee_{i=1}^{n_m}\alpha_{m,i}f_{m,i}\Big\|_r^r \leqslant \Big(s+\sum_{i=1}^{n_m}\alpha_{m,i}^p\Big)^{\frac{r}{p}} \leqslant (s+1)^{\frac{r}{p}} \leqslant \tfrac{r}{p}s+1,\end{aligned}$$

where the last inequality follows from Bernoulli's Inequality. This yields

$$s^{\frac{r}{p}}\int_B f^r\,d\mu \leqslant \tfrac{r}{p}s + \int_B g_m\,d\mu$$

because $\int g\,d\mu = 1$. Since $\int_B g_m\,d\mu = \langle g_m, \mathbb{1}_B\rangle$, the last inequality remains valid if we replace g_m with g. Therefore,

$$\int_B f^r\,d\mu \leqslant s^{-\frac{r}{p}}\Big(\tfrac{r}{p}s + \int_B g\,d\mu\Big).$$

We interpret the expression in the right hand side as a function of s. It is easy to see that it attains minimum when $s = \frac{p}{p-r}\int g\,d\mu$. Hence $\int_B f^r\,d\mu$ is less than or equal to the value of the function at this point, which is $\big(\frac{p}{p-r}\big)^{1-\frac{r}{p}}\Big(\int g\,d\mu\Big)^{1-\frac{r}{p}}$. This yields the required inequality. □

17.5.2 Theorem (Pisier)**.** *For $1 \leqslant r < p < \infty$ and $T\colon E \to L_r(\mu)$, TFAE:*

(i) *T is (p,∞)-convex;*

(ii) *There exists $C'>0$ and a density g such that $\big\|(Tx)\cdot\mathbb{1}_B\big\|_r \leqslant C'\|x\|\Big(\int_B g\,d\mu\Big)^{\frac{1}{r}-\frac{1}{p}}$ for every $x\in E$ and every measurable set B;*

(iii) *There exists $C''>0$ and a density g such that $Tx\in B_g$ and $\big\|g^{-\frac{1}{r}}Tx\big\|_{L_{p,\infty,r}(g\mu)} \leqslant C''\|x\|$ for all $x\in E$;*

(iv) *There exists a density g such that T factors through $L_{p,\infty}(g\mu)$, $T=MR$, with $Mh=g^{\frac{1}{r}}h$ for all $h\in L_{p,\infty}(g\mu)$.*

Proof. As in the proof of Maurey-Nikishin's Theorem 17.4.3, it can be easily verified that each of (i), (ii), and (iii) is equivalent to the corresponding statement in Lemma 17.5.1 with $F=\big\{|Tx| : x\in B_E\big\}$. It follows that (i)⇔(ii)⇔(iii).

(iv)⇒(iii) because $\operatorname{Range}T\subseteq \operatorname{Range}M\subseteq B_g$ and

$$\big\|g^{-\frac{1}{r}}Tx\big\|_{L_{p,\infty,r}(g\mu)} = \big\|Rx\big\|_{L_{p,\infty,r}(g\mu)} \leqslant \|R\|\|x\|.$$

(iii)⇒(iv) Let $h\in L_{p,\infty(g\mu)}$. Put $Mh=g^{\frac{1}{r}}h$. Taking $B=\Omega$ in the definition of $\|\cdot\|_{p,\infty,r}$ in 16.8.2 and using the fact that $\int g\,d\mu=1$, we get

$$\|h\|_{L_{p,\infty,r}(\nu)} \geqslant \Big(\int g\,d\mu\Big)^{\frac{1}{p}-\frac{1}{r}}\Big(\int |h|^r g\,d\mu\Big)^{\frac{1}{r}} = \|g^{\frac{1}{r}}h\|_r = \|Mh\|_r,$$

so that M is a contractive operator from $L_{p,\infty,r}(g\mu)$ to $L_r(\mu)$. For $x\in E$, put $Rx=g^{-\frac{1}{r}}Tx$. It follows from (iii) that R is a bounded operator from E to $L_{p,\infty,r}(g\mu)$. Clearly, $MR=T$. □

Unlike in Maurey-Nikishin's Theorem 17.4.3, the constants in Theorem 17.5.2 may be different. A careful analysis of the proof yields $C\leqslant C'\leqslant C''\leqslant(1-\frac{r}{p})^{\frac{1}{p}-\frac{1}{r}}C$, where C is the (p,∞)-convexity constant of T. Also, $\|M\|\|R\|\leqslant C''$.

Next, we present the dual version of Theorem 17.5.2:

17.5.3 Theorem. *Suppose that $1<q<s<\infty$ and $S\colon L_s(\mu)\to F$. TFAE:*

(i) *S is $(q,1)$-concave;*

(ii) *There exists $C'>0$ and a density g such that $\big\|S(f\mathbb{1}_B)\big\| \leqslant C'\|f\|_{L_s(\mu)}\Big(\int_B g\,d\mu\Big)^{\frac{1}{q}-\frac{1}{s}}$ for every $f\in L_s(\mu)$ and every measurable set B;*

(iii) *There exists $C''>0$ and a density g such that $\|Sf\|\leqslant C''\|g^{-\frac{1}{s}}f\|_{L_{q,1}}(g\mu)$ for every $f\in L_s(\mu)$;*

(iv) *There exists a density g such that S factors through $L_{q,1}(g\mu)$, $S=VN$, where $Nf=g^{-\frac{1}{s}}f$.*

Proof. Let $r=s^*$, $p=q^*$, $E=F^*$, and $T=S^*$. For a density g we will write $\nu=g\mu$. Note that this is a probability measure. By Proposition 16.8.9, $L_{q,1}(\nu)^*=L_{p,\infty}(\nu)$.

Throughout the proof, we will assume that $L_{q,1}(\nu)$ is equipped with the norm coming from duality with $L_{p,\infty,r}(\nu)$. We will prove that (i) and (ii) are equivalent by showing that each of them is equivalent to the corresponding part of Theorem 17.5.2. We will then show that (ii)⇒(iii)⇒(iv)⇒(i).

For (i), this follows by duality as per Theorem 16.4.6.

For (ii), observe that for every measurable set B, every $f \in L_s(\mu)$ and $x \in E$ we have

$$\bigl\langle x, S(f\mathbb{1}_B)\bigr\rangle = \bigl\langle Tx, f\mathbb{1}_B\bigr\rangle = \bigl\langle (Tx)\mathbb{1}_B, f\bigr\rangle. \tag{17.2}$$

Hence, if S satisfies (ii) then

$$\Bigl|\bigl\langle (Tx)\mathbb{1}_B, f\bigr\rangle\Bigr| \leqslant \|x\|\bigl\|S(f\mathbb{1}_B)\bigr\| \leqslant \|x\| \cdot C'\|f\|_{L_s(\mu)}\Bigl(\int_B g\,d\mu\Bigr)^{\frac{1}{q}-\frac{1}{s}}.$$

Taking supremum over all $f \in B_{L_s(\mu)}$, and using the fact that $\frac{1}{r} - \frac{1}{p} = \frac{1}{q} - \frac{1}{s}$, we get $\bigl\|(Tx)\mathbb{1}_B\bigr\|_r \leqslant C'\|x\|\Bigl(\int_B g\,d\mu\Bigr)^{\frac{1}{r}-\frac{1}{p}}$, so that T satisfies Theorem 17.5.2(ii). Conversely, if T satisfies Theorem 17.5.2(ii) then (17.2) yields

$$\bigl|\bigl\langle x, S(f\mathbb{1}_B)\bigr\rangle\bigr| \leqslant \bigl\|(Tx)\mathbb{1}_B\bigr\|\|f\|_{L_s(\mu)} \leqslant C'\|x\|\|f\|_{L_s(\mu)}\Bigl(\int_B g\,d\mu\Bigr)^{\frac{1}{r}-\frac{1}{p}}.$$

Taking supremum over $x \in B_E$, we get $\bigl\|S(f\mathbb{1}_B)\bigr\| \leqslant C'\|f\|_{L_s(\mu)}\Bigl(\int_B g\,d\mu\Bigr)^{\frac{1}{q}-\frac{1}{s}}$, so that S satisfies (ii).

(ii)⇒(iii) Take $f \in L_s(\mu)$. Then $g^{-\frac{1}{s}}f \in L_s(\nu) \subseteq L_{q,1}(\nu)$ by Proposition 16.8.11. For every $x \in B_E$ we have

$$\begin{aligned}\langle Sf, x\rangle = \langle f, Tx\rangle = \int f \cdot (Tx)\,d\mu &= \int (g^{-\frac{1}{s}}f)\cdot(g^{-\frac{1}{r}}Tx)\cdot g\,d\mu \\ &\leqslant \|g^{-\frac{1}{s}}f\|_{L_{q,1}(\nu)}\|g^{-\frac{1}{r}}Tx\|_{L_{p,\infty,r}(\nu)} \leqslant \|g^{-\frac{1}{s}}f\|_{L_{q,1}(\nu)} \cdot C''\|x\|\end{aligned}$$

because T satisfies condition (iii) of Theorem 17.5.2. Taking supremum over $x \in B_E$, we get $\|Sf\| \leqslant C''\|g^{-\frac{1}{s}}f\|_{L_{q,1}(\nu)}$.

(iii)⇒(iv) Let $f \in L_s(\mu)$. For every $h \in L_{p,\infty}(\nu)$, in the duality of $L_{q,1}(\nu)$ and $L_{p,\infty,r}(\nu)$, we have

$$\begin{aligned}\Bigl|\langle g^{-\frac{1}{s}}f, h\rangle\Bigr| = \Bigl|\int g^{-\frac{1}{s}}f\cdot h\cdot g\,d\mu\Bigr| &\leqslant \int |f|\cdot|h|\cdot g^{\frac{1}{r}}\,d\mu \\ &\leqslant \Bigl(\int |f|^s\,d\mu\Bigr)^{\frac{1}{s}}\Bigl(\int |h|^r\cdot g\,d\mu\Bigr)^{\frac{1}{r}} \leqslant \|f\|_{L_s(\mu)}\|h\|_{L_{p,\infty,r}(\nu)}\end{aligned}$$

by taking $A = \Omega$ in the definition of $\|\cdot\|_{p,\infty,r}$ in 16.8.2. It follows that $\|g^{-\frac{1}{s}}f\|_{L_{q,1}(\nu)} \leqslant \|f\|_{L_s(\mu)}$. Hence, $Nf = g^{-\frac{1}{s}}f$ defines a contractive operator from $L_s(\mu)$ to $L_{q,1}(\nu)$.

Furthermore, Range $N = L_s(\nu)$ and is, therefore, dense in $L_{q,1}(\nu)$ as it contain the simple functions; see Proposition 16.8.12. Define V on the range on N via $V(Nf) = Sf$. Since N is one-to-one, this is a well-defined linear operator from Range N to F. By (iii), it is bounded and, therefore, extends to a bounded operator from $L_{q,1}(\nu)$ to F.

(iv)⇒(i) Note that $L_{q,1}(\nu)$ is $(q,1)$-concave by Theorem 16.4.6. Since $N \geqslant 0$, S is $(q,1)$-concave by Exercise 16.4.3. □

17.6 Representations for p- and (p,∞)-convex spaces

Recall that Theorem 9.5.2 asserts that every Banach lattices embeds isometrically into an ℓ_∞-sum of L_1-spaces. We will now adjust its proof and get analogous results for p- and (p,∞)-convex Banach latices.

17.6.1 Theorem. *If X is p-convex with constant 1, then there exists a family Γ of probability measures and a lattice isometric embedding of X into $\Big(\bigoplus_{\mu\in\Gamma} L_p(\mu)\Big)_\infty$.*

Proof. Fix $x \in S_X^+$. Find $x^* \in S_{X^*}^+$ such that $x^*(x) = 1$, and define a lattice seminorm ρ_x on X via $\rho_x(z) = x^*\big(|z|\big)$. As in Section 9.2 and in 9.5.1, ρ_x induces an AL norm on $X/\ker\rho_x$, which we will denote $\tilde\rho_x$. The completion of $\big(X/\ker\rho_x, \tilde\rho_x\big)$ can be represented as $L_1(\mu_x)$ for some measure μ_x. Let $Q_x\colon X \to X/\ker\rho_x$ be the canonical quotient map and $J_x\colon X/\ker\rho_x \hookrightarrow L_1(\mu_x)$ the inclusion map.

$$\begin{tikzcd} X \arrow[r, "Q_x"] \arrow[dr, "R_x"'] & X/\ker\rho_x \arrow[r, "J_x"] & L_1(\mu_x) \\ & L_p(g_x\mu_x) \arrow[ur, "M_x"'] & \end{tikzcd}$$

Since X is p-convex with constant 1 and J_xQ_x is a lattice homomorphism, J_xQ_x is p-convex with constant 1. By Maurey-Nikishin Factorization Theorem 17.4.3 and Remark 17.4.4, we can find a density $g_x \in L_1(\mu_x)$ such that J_xQ_x factors through $L_p(g_x\mu_x)$ as on the diagram, with R_x and M_x being contractive lattice homomorphisms. Define

$$T\colon X \to \Big(\bigoplus_{x\in S_X^+} L_p(g_x\mu_x)\Big)_\infty \quad \text{via} \quad Jz = (R_xz)_{x\in S_X^+}.$$

Since each R_x is a contractive lattice homomorphism, so is T. Since M_x is a contraction, for every $x \in S_X^+$ we have

$$\|Tx\| \geqslant \|R_xx\| \geqslant \|J_xQ_xx\| = \rho_x(x) = 1.$$

Since T is a lattice homomorphism, we conclude that $\|Tx\| = \big\|T|x|\big\| = 1$ for all $x \in S_X$, hence T is an isometry. $\square$

If X is p-convex with constant K, then we can first renorm it as in Lemma 16.7.7 so that the p-convexity constant becomes 1 and then apply the theorem. In this case, viewed as an operator on X, the resulting embedding T is no longer be an isometry, but we have $\|x\| \leqslant \|Tx\| \leqslant K\|x\|$.

Replacing Maurey-Nikishin Factorization Theorem 17.4.3 with Theorem 17.5.2 with $r = 1$ in the preceding proof, we get a similar result for (p,∞)-convex spaces, that is, Banach lattices satisfying an upper p-estimate, except that the resulting map is now only an isomorphism rather than an isometry:

17.6.2 Theorem. *If X is (p,∞)-convex then there exists a family Γ of measures and a lattice isomorphic embedding T of X into $\Big(\bigoplus_{\mu\in\Gamma} L_{p,\infty}(\mu)\Big)_\infty$, with $\|x\| \leqslant \|Tx\| \leqslant K\big(1-\frac{1}{p}\big)^{\frac{1}{p}-1}\|x\|$ for every x, where K is the (p,∞)-convexity constant of T.*

Recall that ℓ_∞-sums preserve (p,q)-convexity by Exercise 16.4.9. This implies that the two theorems above provide equivalent characterizations of p-convex and (p,∞)-convex Banach lattices. Namely, p-convex Banach lattices are just closed sublattices of ℓ_∞-sums of $L_p(\mu)$-spaces, while (p,∞)-convex Banach lattices are precisely closed sublattices of ∞-sums of $L_{p,\infty}(\mu)$-spaces (up to an isomorphism)

17.7 Type and cotype

The concepts of p-convexity and p-concavity in Banach lattices are related to the concepts of type and cotype in Banach spaces. We start by reviewing some important facts of the latter theory; for detailed expositions of type and cotype in Banach spaces we refer the reader to [LT79, DJT95, AK06].

The concepts of type and cotype naturally fit into the framework of power norms, described in Section 16.3. Let E be a Banach space. For a multivector $\bar{x}$ in E, we define $\|\bar{x}\|_{\mathrm{rad}} = \operatorname*{Ave}_{\varepsilon_i=\pm1}\big\|\sum_{i=1}^n \varepsilon_i x_i\big\|$, where the average is taken over all choices of signs. This expression may be viewed as the average distance from the vertices of the "parallelepiped" generated by $x_1,\dots,x_n$ to the origin. We observed in 6.4.2 that for $\alpha_1,\dots,\alpha_n \in \mathbb{R}$, we have

$$\operatorname*{Ave}_{\varepsilon_i=\pm1}\Big|\sum_{i=1}^n \varepsilon_i\alpha_i\Big| = \int_0^1\Big|\sum_{i=1}^n \alpha_i r_i(t)\Big|dt.$$

In view of this, one often writes

$$\|\bar{x}\|_{\mathrm{rad}} = \int_0^1 \Big\|\sum_{i=1}^n r_i(t)x_i\Big\|dt$$

for $x_1,\dots,x_n \in E$; the integral is interpreted as a finite sum. If we treat the Rademacher functions as random variables, this expression may be viewed as the expectation of the random variable $\Big\|\sum_{i=1}^n r_i x_i\Big\|$.

More generally, for a multivector $\bar{x}$ in E and $1 \leqslant p < \infty$, we define

$$\|\bar{x}\|_{\mathrm{rad}\text{-}p} = \Big(\operatorname*{Ave}_{\varepsilon_i=\pm1}\Big\|\sum_{i=1}^n \varepsilon_i x_i\Big\|^p\Big)^{\frac{1}{p}}.$$

17.7.1 Exercise. $\|\cdot\|_{\mathrm{rad}\text{-}p}$ a power norm on E.

Naturally, it is interesting to compare this power norm to the power-norms introduced in Section 16.3. First, it turns out that the definition with p is equivalent to the original one:

17.7.2 Theorem (Kahane)**.** *Let $1 \leqslant p < \infty$. Then $\|\cdot\|_{\mathrm{rad}\text{-}p} \sim \|\cdot\|_{\mathrm{rad}}$.*

We refer the reader to [AK06] for a proof of Kahane's Theorem. Khinchin's Inequality may be viewed as a special case of Kahane's Theorem with $E = \mathbb{R}$.

Let $1 \leqslant p \leqslant \infty$. We say that E has ***(Rademacher) type*** p if $\|\cdot\|_{\mathrm{rad}} \lesssim \|\cdot\|_{\mathrm{sum}\text{-}p}$ on E. Similarly, E has ***(Rademacher) cotype*** p if $\|\cdot\|_{\mathrm{rad}} \gtrsim \|\cdot\|_{\mathrm{sum}\text{-}p}$. It is clear that these definitions may be interpreted as the identity operator being power-bounded with respect to the appropriate norms on $c_{00}(E)$. It is easy to see that if E has type p then it also has type r for every $r \in [1,p]$. Similarly, if E has cotype p then it has cotype r for every $r \in [p,\infty]$. Clearly, type and cotype are inherited by subspaces and are preserved by isomorphisms. By Kahane's Theorem, the power norm $\|\cdot\|_{\mathrm{rad}}$ in the definitions of type and cotype may be replaced with $\|\cdot\|_{\mathrm{rad}\text{-}p}$.

It follows from the triangle inequality that every Banach space has type 1. The following argument shows that every Banach space E has cotype ∞. For every $x_1, x_2 \in E$, we have

$$\|x_1\| = \Big\|\frac{x_1+x_2}{2} + \frac{x_1-x_2}{2}\Big\| \leqslant \frac{1}{2}\Big(\|x_1+x_2\| + \|x_1-x_2\|\Big) = \operatorname*{Ave}_{\varepsilon_i=\pm1}\Big\|\sum_{i=1}^2 \varepsilon_i x_i\Big\|.$$

Similarly, if $x_1,\dots,x_n \in E$ then $\|x_k\| \leqslant \operatorname*{Ave}_{\varepsilon_i=\pm1}\big\|\sum_{i=1}^n \varepsilon_i x_i\big\|$ for every k.

Khinchin's inequality says that $\mathbb{R}$ has type 2 and cotype 2. It follows that every one-dimensional subspace has type 2 and cotype 2. Since type and cotype are inherited by subspaces, it follows that no Banach space has type greater than 2 or cotype less than 2. Thai is, the effective range for type is the interval $[1,2]$, while the effective range for cotype is the interval $[2,\infty]$.

There is a partial duality between type and cotype.

17.7.3 Theorem. *If E has type p then E^* has cotype p^*.*

We refer the reader to [LT79] for a proof. It follows that if X^* has type p then X has cotype p^* (because X^{**} has cotype p^*). We will show in Example 17.7.11 that the converse of Theorem 17.7.3 is false.

We will now consider type and cotype in Banach lattices. For the rest of the section, X will be a Banach lattice. We observed in 16.1.5 that Khinchin's Inequality remains valid in Banach lattices, as long as we use moduli rather than norms. The first part of the inequality extends to norms by the triangle inequality:

$$A_1\Big\|\Big(\sum_{i=1}^n |x_i|^2\Big)^{\frac{1}{2}}\Big\| \leqslant \Big\|\operatorname*{Ave}_{\varepsilon_i=\pm1}\Big|\sum_{i=1}^n \varepsilon_i x_i\Big|\Big\| \leqslant \operatorname*{Ave}_{\varepsilon_i=\pm1}\Big\|\sum_{i=1}^n \varepsilon_i x_i\Big\|,$$

which yields $\|\cdot\|_{\text{lat-2}} \lesssim \|\cdot\|_{\text{rad}}$. The second part of the inequality may fail for norms in general. However, we will show that it is satisfies in a large class of Banach lattices. We say that X has ***nontrivial concavity*** if X is p-concave for some $p<\infty$. In view of 17.3.8, this is equivalent to X having a lower p-estimate for some $p<\infty$. WLOG, $p>2$ because concavity is preserved when p increases. By Proposition 16.6.12, every Banach lattice with nontrivial concavity is a KB space.

17.7.4 Lemma. *If $T\colon X\to E$ is p-concave for some $p<\infty$ then $\|\overline{T}\bar{x}\|_{\text{rad}} \lesssim \|\bar{x}\|_{\text{lat-2}}$ for every multivector $\bar{x}$ in X.*

Proof. Using Kahane's Theorem 17.7.2 and p-concavity of T, we get

$$\|\overline{T}\bar{x}\|_{\text{rad}} = \|\overline{T}\bar{x}\|_{\text{rad-}p} = \Big(\sum_{\varepsilon_i=\pm1}\Big\|\frac{1}{2^{np}}\sum_{i=1}^n \varepsilon_i Tx_i\Big\|^p\Big)^{\frac{1}{p}} \lesssim \Big\|\Big(\sum_{\varepsilon_i=\pm1}\Big|\frac{1}{2^{np}}\sum_{i=1}^n \varepsilon_i x_i\Big|^p\Big)^{\frac{1}{p}}\Big\|.$$

The result now follows from the second part of Khinchin's Inequality (16.4). $\square$

17.7.5 Theorem (Maurey-Krivine)**.** *If X has nontrivial concavity then $\|\cdot\|_{\text{rad}} \sim \|\cdot\|_{\text{lat-2}}$.*

Proof. We have already observed that $\|\cdot\|_{\text{lat-2}} \lesssim \|\cdot\|_{\text{rad}}$. For the converse, apply the lemma to the identity operator. $\square$

This theorem is most helpful as it will allow us to reduce type and cotype to convexity and concavity when X has nontrivial concavity.

17.7.6 Proposition. *Let $2 < p < \infty$. Then X has cotype p iff X satisfies a lower p-estimate.*

Proof. Suppose X has cotype p. Take any disjoint $x_1, \dots, x_n$. Then $\left\|\sum_{i=1}^n \pm x_i\right\| = \left\|\sum_{i=1}^n x_i\right\|$ for any choice of signs. It follows that $\left\|\sum_{i=1}^n x_i\right\| = \|\bar{x}\|_{\mathrm{rad}} \gtrsim \|\bar{x}\|_{\text{sum-}p}$. We conclude that X satisfies a lower p-estimate.

Conversely, suppose that X satisfies a lower p-estimate. It follows that X has nontrivial concavity. Furthermore, X is $(p,1)$-concave, and, therefore, $(p,2)$-concave by Theorem 17.3.6. Combining Theorem 17.7.5 with $(p,2)$-concavity, we get $\|\bar{x}\|_{\mathrm{rad}} \sim \|\bar{x}\|_{\text{lat-}2} \gtrsim \|\bar{x}\|_{\text{sum-}p}$. It follows that X has cotype p. ☐

In the extreme case $p = 2$, a lower estimate is replaced with concavity:

17.7.7 Proposition. *X has cotype 2 iff it is 2-concave.*

Proof. Suppose that X is 2-concave. Let $x_1, \dots, x_n \in X$. By Theorem 17.7.5, $\|\bar{x}\|_{\mathrm{rad}} \sim \|\bar{x}\|_{\text{lat-}2} \gtrsim \|\bar{x}\|_{\text{sum-}2}$, so X has cotype 2.

Suppose now that X has cotype 2. Observe that X has a nontrivial concavity because for every $r > 2$, X has cotype r, hence a lower r-estimate by Proposition 17.7.6. Let $x_1, \dots, x_n \in X$. By Theorem 17.7.5, $\|\bar{x}\|_{\text{lat-}2} \sim \|\bar{x}\|_{\mathrm{rad}} \gtrsim \|\bar{x}\|_{\text{sum-}}$, so that X is 2-concave. ☐

Using similar arguments we will now prove dual versions of Propositions 17.7.6 and 17.7.7 for type. However, as the proofs utilize Theorem 17.7.5, we have to assume additionally that X has non-trivial concavity:

17.7.8 Proposition. *Let $1 < p < 2$. If X has type 2 then X has an upper p-estimate. The converse it true provided X has a nontrivial concavity.*

17.7.9 Proposition. *Suppose that X has nontrivial concavity. Then X has cotype 2 iff it is 2-concave.*

Since we know convexity/concavity and upper/lower estimates of L_p-spaces, we can now determine their types and cotypes:

17.7.10 Corollary. *Let $X = L_p(\mu)$ for some $1 \leqslant p < \infty$. If $p \geqslant 2$ then X has type 2 and cotype p, but no cotype less than p. If $p \leqslant 2$ then X has cotype 2 and type p, but no type greater than p.*

Theorem 17.7.3 asserts that if E has type p then E^* has cotype p^*. The following example shows that the converse is false:

17.7.11 Example. Let $X = C[0,1]$. Since X has a subspace isomorphic to ℓ_1, and ℓ_1 has no type greater than 1, we conclude that X has no type greater than 1. However, X^* is an AL-space, so it 2-concave and, therefore, has cotype 2.

However, for Banach lattices with non-trivial concavity, we have a complete duality between type and cotype. This follows immediately from Propositions 17.7.6–17.7.9 and from dualities between convexity and concavity and between upper and lower estimates in Corollaries 16.4.7 and 16.6.10:

17.7.12 Proposition. *Let X be a Banach lattice with nontrivial concavity; let $1 < p \leqslant 2$. Let $q = p^*$.*

(i) *X has type p iff X^* has cotype q;*

(ii) *X has cotype q iff X^* has type p.*

A deep result of Kwapień asserts that if a Banach space E has type 2 and cotype 2, it is isomorphic to a Hilbert space. In case of Banach lattices, this follows easily from the fact that such a space is 2-convex and 2-concave, hence is lattice isomorphic to $L_2(\mu)$ for some measure μ by Theorem 16.7.6.

Chapter 18

Orlicz spaces

Until now, our primary examples of Banach lattices were $L_p(\mu)$ and $C(K)$ spaces. An Orlicz space $L_\varphi(\mu)$ can be thought of as a generalization of an $L_p(\mu)$ space with the function $t \mapsto |t|^p$ used in the definition of $L_p(\mu)$ replaced with a more general function φ. This procedure gives rise to a class of spaces which is sufficiently rich to provide models for various applications. Nevertheless, this class is not "too large". Firstly, Orlicz spaces are Banach lattices, so all the theory of Banach lattices applies to them. Secondly, $L_\varphi(\mu)$ is a sublattice of the space of all measurable functions $L_0(\mu)$, so that it is a function space and one can use techniques of measurable functions. In particular, Lebesgue Dominated Convergence Theorem and Monotone Convergence Theorem are indispensable tools for studies of Orlicz spaces.

Throughout this chapter, $(\Omega, \mathcal{F}, \mu)$ will be a measure space. Since the measure μ will be fixed, we write $\int f$ instead of $\int f\,d\mu$ and L_φ instead of $L_\varphi(\mu)$. Unless specified otherwise, we do not assume the measure be finite or σ-finite. We assume, though, that every set of infinite measure contains a subset of positive finite measure. A measurable function is ***finitely supported*** if its support has finite measure.

As we mentioned before, the idea is to replace the function $t \mapsto |t|^p$ in the definition of $L_p(\mu)$ with a more general function φ. A naive idea could be to define $\|f\|_\varphi = \varphi^{-1}\Bigl(\int \varphi\bigl(|f|\bigr)\Bigr)$. However, this expression fails the axioms of a norm even for very "nice" functions φ; it need not even be positively homogeneous. We will take a somewhat different route: we will still use the expression $\int \varphi\bigl(|f|\bigr)$ as a "measure of the size of the function", called the ***modular*** of f, but we will define $\|f\|_\varphi$ in a more subtle way. Namely, we first define the "unit ball" of the modular by $f \in B_\varphi$ if $\int \varphi\bigl(|f|\bigr) \leqslant 1$; we then define $\|\cdot\|_\varphi$ as the Minkowski functional of B_φ, so that B_φ becomes the unit ball of this norm.

18.1 Orlicz function and its modular

An ***Orlicz function*** is a convex function $\varphi\colon \mathbb{R}_+ \to \mathbb{R}_+$ such that $\varphi(t) = 0$ iff $t = 0$.

18.1.1 Exercise. Let φ be an Orlicz function. Verify the following:

(i) φ is continuous and strictly increasing.

(ii) Let $s, t \geqslant 0$. Then $\varphi(st) \leqslant s\varphi(t)$ if $s \leqslant 1$ and $\varphi(st) \geqslant s\varphi(t)$ if $s \geqslant 1$.

(iii) Let $t_0 > 0$. There exists $\alpha > 0$ such that $\varphi(t) \leqslant \alpha t$ whenever $0 \leqslant t \leqslant t_0$ and $\varphi(t) \geqslant \alpha t$ whenever $t \geqslant t_0$.

(iv) $\lim_{t\to\infty} \varphi(t) = \infty$.

With a minor abuse of notation, for $f \in L_0(\mu)$, we write $\varphi(f)$ for $\varphi\circ f$. In particular, we write $\varphi\bigl(|f|\bigr)$ for the function $t \mapsto \varphi\bigl(|f(t)|\bigr)$. We call the expression $I_\varphi(f) = \int \varphi\bigl(|f|\bigr)$ the ***modular*** of f. Note that I_φ is a non-linear functional, and $I_\varphi(f) = \infty$ is possible. Recall that for sequences in $L_0(\mu)$, a.e. convergence agrees with order convergence; see Exercise 2.4.28.

18.1.2 Exercise. Prove that

(i) $I_\varphi(0) = 0$ and $I_\varphi\bigl(|f|\bigr) = I_\varphi(f)$.

(ii) I_φ is strictly monotone: if $0 \leqslant f < g$ in $L_0(\mu)$ then $I_\varphi(f) < I_\varphi(g)$.

(iii) I_φ is absolutely convex: if $|\alpha|+|\beta| = 1$ in $\mathbb{R}$ and $f, g \in L_0(\mu)$ then $I_\varphi\bigl(\alpha f+\beta g\bigr) \leqslant |\alpha|I_\varphi(f) + |\beta|I_\varphi(g)$.

(iv) If $0 \leqslant f_n \uparrow f$ in $L_0(\mu)$ then $I_\varphi(f_n) \uparrow I_\varphi(f)$.

(v) If $(f_n) \subseteq [0, g]$ for some g with $I_\varphi(g) < \infty$ and $f_n \xrightarrow{\text{a.e.}} f$ then $I_\varphi(f_n) \to I_\varphi(f)$.

(vi) If $I_\varphi(f) < \infty$ and $\varepsilon > 0$ then $I_\varphi(\lambda f) < \varepsilon$ for all sufficiently small $\lambda > 0$.

(iv) and (v) essentially say that I_φ is σ-order continuous. This extends to nets as follows:

18.1.3 Proposition. *If $u \geqslant f_\alpha \downarrow 0$ and $I_\varphi(u) < \infty$ then $I_\varphi(f_\alpha) \to 0$.*

Proof. Observe that $\varphi(u) \geqslant \varphi(f_\alpha) \downarrow 0$, and all these functions are in $L_1(\mu)$. By Exercise 2.1.18, there is an increasing sequence of indices (α_n) such that $\varphi(f_{\alpha_n}) \downarrow 0$. Lebesgue Dominated Convergence Theorem yields $I_\varphi(f_{\alpha_n}) = \int \varphi(f_{\alpha_n}) \to 0$. By monotonicity, $I_\varphi(f_\alpha) \to 0$. $\square$

We will also use the following variant of the triangle inequality for I_φ.

18.1.4 Lemma. $I_\varphi(f + g) \leqslant I_\varphi(2f) + I_\varphi(2g)$ *for every $f, g \in L_0(\mu)$.*

Proof. Since φ is increasing, we have $\varphi(a \vee b) = \varphi(a) \vee \varphi(b)$ for every $a, b \in \mathbb{R}_+$. It follows from $|f+g| \leqslant (2|f|) \vee (2|g|)$ that

$$\varphi\big(|f+g|\big) \leqslant \varphi\big(2|f|\big) \vee \varphi\big(2|g|\big) \leqslant \varphi\big(2|f|\big) + \varphi\big(2|g|\big).$$

Integrating, we get the required inequality. □

The following exercise shows that functions of finite modular have "relatively small" support.

18.1.5 Exercise. Suppose that $I_\varphi(f)$ is finite.

(i) The restriction of μ to the support of f is σ-finite.

(ii) For every $\varepsilon > 0$ there exists a set A of finite measure such that the restriction of f to A is bounded and $I_\varphi(f \cdot \mathbb{1}_A) \geqslant (1-\varepsilon) I_\varphi(f)$.

18.2 L_φ as a vector lattice

Again, by analogy with L_p spaces, it is natural to consider all the measurable functions whose modular is finite:

$$L_\varphi^0 = \Big\{ f \in L_0(\mu) \ : \ I_\varphi(f) < \infty \Big\}.$$

We call this set the ***Orlicz class*** of φ (but this is not the Orlicz space yet). This set is obviously solid. Furthermore, Exercise 18.1.2(iii) implies that L_φ^0 is absolutely convex.

18.2.1 Exercise. Let μ be the Lebesgue measure on $[0,1]$ and $\varphi(t) = e^t - 1$ (observe that this is an Orlicz function). Find $f \in L_0(\mu)$ such that $I_\varphi(f) = 1$ but $I_\varphi(2f) = \infty$. It follows that $f \in L_\varphi^0$ while $2f \notin L_\varphi^0$.

This example shows that L_φ^0 need not be closed under scaling; in particular, it is not a vector subspace of $L_0(\mu)$. The following observation is a standard exercise in linear algebra. Let A be an absolutely convex subset of a vector space V. Put $L = \bigcup_{\lambda > 0} \lambda A$ and $H = \bigcap_{\lambda > 0} \lambda A$. Then $H \subseteq A \subseteq L$, H and L are linear subspaces of V and $L = \operatorname{span} A$. Hence, L is the least linear subspace of V which contains A, while H is the greatest subspace of V contained in A. Applying this construction to L_φ^0, we get two subspaces of $L_0(\mu)$:

$$L_\varphi = \bigcup_{\lambda > 0} \lambda L_\varphi^0 \quad \text{and} \quad H_\varphi = \bigcap_{\lambda > 0} \lambda L_\varphi^0.$$

That is, $f \in L_\varphi$ if $\lambda f \in L^0_\varphi$ for *some* $\lambda > 0$, while $f \in H_\varphi$ if $\lambda f \in L^0_\varphi$ for *every* $\lambda > 0$. Then $H_\varphi \subseteq L^0_\varphi \subseteq L_\varphi$, $L_\varphi = \operatorname{span} L^0_\varphi$, L_φ is the least linear subspace of $L_0(\mu)$ which contains L^0_φ, and H_φ is the greatest linear subspace of $L_0(\mu)$ which is contained in L^0_φ. L_φ is called the ***Orlicz space*** corresponding to φ and μ; H_φ is called the ***heart*** of L_φ.

In Exercise 18.2.1, $H_\varphi \neq L_\varphi$. On the other hand, if $\varphi(t) = \lvert t\rvert^p$ for $1 \leqslant p < \infty$ then $H_\varphi = L^0_\varphi = L_\varphi = L_p(\mu)$. We will later discuss characterizations of those φ for which $H_\varphi = L^0_\varphi = L_\varphi$.

18.2.2 Proposition. *L_φ and H_φ are ideals in $L_0(\mu)$,*

$$L_\varphi \;=\; \Bigl\{f \in L_0(\mu) \;:\; \exists\lambda > 0 \quad I_\varphi(\lambda f) \leqslant 1\Bigr\}, \text{ and} \tag{18.1}$$

$$H_\varphi \;=\; \Bigl\{f \in L_0(\mu) \;:\; \forall\lambda > 0 \quad I_\varphi(\lambda f) < \infty\Bigr\}. \tag{18.2}$$

Proof. (18.2) is trivial. To prove the identity for L_φ, suppose first that $I_\varphi(\lambda f) \leqslant 1$ for some f and $\lambda > 0$; then $\lambda f \in L^0_\varphi$, so that $f \in L_\varphi$. Conversely, suppose that $f \in L_\varphi$. Then $I_\varphi(\lambda f) < \infty$ for some $\lambda > 0$. By Exercise 18.1.2(vi), $I_\varphi(\lambda_1 \lambda f) \leqslant 1$ for some positive λ_1. Finally, (18.1) and (18.2) together with properties of I_φ imply that L_φ and H_φ are ideals. $\square$

18.2.3 Exercise. For a measurable set A, $\mathbb{1}_A \in L_\varphi$ iff $\mathbb{1}_A \in H_\varphi$ iff $\mu(A)$ is finite.

18.2.4 Exercise. The restriction of μ to the support of every $f \in L_\varphi$ is σ-finite.

18.2.5 Proposition. *L_φ is order complete.*

Proof. Let $0 \leqslant u \in L_\varphi$. Consider I_u, the principal ideal of u in L_φ. Let $A = \operatorname{supp} u$. We may view u as an element of $L_0(\mu_A) = L_0(A, \mathcal{F}_A, \mu_A)$, where μ_A is the restriction of μ to A. Clearly, I_u is also an ideal in $L_0(\mu_A)$. Note that μ_A is σ-finite by Exercise 18.2.4. It follows that $L_0(\mu_A)$ is order complete by Theorem 2.1.16. Being an ideal in $L_0(\mu_A)$, I_u is order complete by Proposition 3.3.18. Since every principal ideal in L_φ is order complete, we conclude that L_φ is complete by, again, Proposition 3.3.18. $\square$

18.2.6. There is another way to see that L_φ is order complete (and other things). Let $(L^0_\varphi)_+$ be the set of all positive functions in L^0_φ. The map $\tau\colon f \mapsto \varphi(f)$ is a bijection from $(L^0_\varphi)_+$ onto $L_1(\mu)_+$. It extends to a bijection between L^0_φ and $L_1(\mu)$ via $\tau(f) = \varphi\bigl(\lvert f\rvert\bigr) \cdot \operatorname{sign} f$. Note that τ is order-preserving in the sense that $f \leqslant g$ iff $\tau(f) \leqslant \tau(g)$. Thus, as ordered sets, $L_1(\mu)$ and L^0_φ are isomorphic. In particular, order intervals in L^0_φ are order isomorphic to order intervals in $L_1(\mu)$, with $[f, g]$ in L^0_φ order isomorphic to $\bigl[\tau(f), \tau(g)\bigr]$ in $L_1(\mu)$. Furthermore, let $f \leqslant g$ in L_φ. There exists $\lambda > 0$ such that $\lambda f, \lambda g \in L^0_\varphi$. Clearly, the map $h \mapsto \lambda h$ is an order-preserving map from the order interval $[f, g]$ in L_φ onto the order interval $[\lambda f, \lambda g]$ in L^0_φ. Thus, order intervals in L_φ are order isomorphic to order intervals in $L_1(\mu)$. Therefore, every property that may be stated in terms of order on order intervals is valid in L_φ iff it is valid in $L_1(\mu)$ (note that this property may not use the linear structure, only the order). Order completeness is an example of such a property. Since $L_1(\mu)$ is order complete by Theorem 2.1.16, this yields that L_φ is also order complete.

Here is another example of this technique.

18.2.7 Proposition. *L_φ satisfies the countable sup property.*

Proof. The countable sup property may be stated in terms of order on intervals of the form $[0, u]$ in L_φ. □

Recall that in $L_p(\mu)$, $0 \leqslant p < \infty$, $f_n \xrightarrow{\text{o}} f$ iff (f_n) is order bounded and $f_n \xrightarrow{\text{a.e.}} f$; see Proposition 2.4.27. We observe that this remains valid in L_φ and H_φ.

18.2.8 Proposition. *For an order bounded sequence (f_n) in L_φ and $f \in L_\varphi$, $f_n \xrightarrow{\text{o}} f$ in L_φ iff $f_n \xrightarrow{\text{a.e.}} f$. The same is true for H_φ.*

Proof. We only prove it for L_φ; the proof for H_φ is similar. One can prove this analogously to the proof of Proposition 2.4.27. Alternatively, since L_φ is an ideal in $L_0(\mu)$ and, therefore, a regular sublattice, the statement follows immediately from Exercise 3.2.1(iv) and Lemma 3.2.4.

One may instead prove the proposition using the trick described in 18.2.6. Indeed, replacing f_n with $|f_n - f|$, we may assume WLOG that $f = 0$ and $f_n \geqslant 0$ for every n. It is given that (f_n) is order bounded, say, (f_n) is in $[0, u]$ for some $u \in L_\varphi$. As in 18.2.6, we find $\lambda > 0$ such that $\lambda u \in L^0_\varphi$; then $f_n \xrightarrow{\text{o}} 0$ in L_φ iff $\tau(\lambda f_n) \xrightarrow{\text{o}} 0$ in $L_1(\mu)$. The latter means that $\varphi(\lambda f_n) \xrightarrow{\text{a.e.}} 0$, which is equivalent to $f_n \xrightarrow{\text{a.e.}} 0$. □

18.3 L_φ as a Banach lattice

Let $B_\varphi = \{f \in L_0(\mu) : I_\varphi(f) \leqslant 1\}$. It follows from properties of I_φ that B_φ is absolutely convex; (18.1) yields that $L_\varphi = \bigcup_{\lambda>0} \lambda B_\varphi$. Therefore, the Minkowski functional of B_φ defines a seminorm on L_φ:

$$\|f\|_\varphi = \inf\{\lambda > 0 : \tfrac{f}{\lambda} \in B_\varphi\} = \inf\Big\{\lambda > 0 : I_\varphi\big(\tfrac{f}{\lambda}\big) \leqslant 1\Big\} \quad \text{for} \quad f \in L_\varphi.$$

18.3.1 Exercise. For $f \in L_\varphi$, $\|f\|_\varphi = 1$ iff $I_\varphi(f) = 1$.

18.3.2 Theorem. $(L_\varphi, \|\cdot\|_\varphi)$ *is a Banach lattice.*

Proof. First, we show that $\|\cdot\|_\varphi$ is a norm and not just a seminorm. Let $0 \neq f \in L_\varphi$; it suffices to show that $\|f\|_\varphi \neq 0$. WLOG, $f \geqslant 0$; otherwise, replace f with $|f|$. Since $f \neq 0$, we have $f \geqslant \delta \mathbb{1}_A$ with $\delta > 0$ and $\mu(A) > 0$. Let $\lambda > 0$, we then have

$$I_\varphi(\lambda f) \geqslant I_\varphi(\lambda\delta\mathbb{1}_A) = \varphi(\lambda\delta)\mu(A).$$

Let α be as in Exercise 18.1.1(iii). If λ is sufficiently large, we have

$$\varphi(\lambda\delta)\mu(A) \geqslant \alpha\lambda\delta\mu(A) > 1.$$

It follows that $I_\varphi(\lambda f) > 1$, so that $\|f\|_\varphi > 0$.

It is easy to see that $\|\cdot\|_\varphi$ is a lattice norm. Next, we show that it is complete. The proof is similar to the proof of the completeness of $L_p(\mu)$. Let (f_n) be a Cauchy sequence; we need to show that it converges. It suffices to find a convergent subsequence. Therefore, passing to a subsequence, we may assume that $\|f_{n+1} - f_n\|_\varphi < \frac{1}{2^n}$. For every m, put $g_m = \sum_{n=1}^m |f_{n+1} - f_n|$. We have

$$\Big\|\sum_{n=1}^m |f_{n+1} - f_n|\Big\|_\varphi \leqslant \sum_{n=1}^m \|f_{n+1} - f_n\|_\varphi \leqslant \sum_{n=1}^m \frac{1}{2^n} < 1.$$

It follows that $g_m \in B_\varphi$ and, therefore, $\int \varphi(g_m) = I_\varphi(g_m) \leqslant 1$. Note that the sequences (g_m) and $\big(\varphi(g_m)\big)$ are increasing. Let $g\colon \Omega \to \mathbb{R} \cup \{\infty\}$ be the point-wise limit of (g_m), that is, $g(t) = \sum_{n=1}^\infty \big|f_{n+1}(t) - f_n(t)\big|$ for every $t \in \Omega$. Then $\varphi(g_m) \to \varphi(g)$ point-wise. Monotone Convergence Theorem yields that $\int \varphi(g) \leqslant 1$. It follows, in particular, that g is a.e. finite and $g \in B_\varphi$.

For each m, put $h_m = g - g_m$, so that $h_m = \sum_{n=m+1}^\infty |f_{n+1} - f_n|$ a.e. Let $k > m$. Then

$$\Big\|\sum_{n=m+1}^k |f_{n+1} - f_n|\Big\|_\varphi \leqslant \frac{1}{2^m}, \text{ so that } \Big\|2^m \sum_{n=m+1}^k |f_{n+1} - f_n|\Big\|_\varphi \leqslant 1$$

$$\text{and, therefore, } \int \varphi\Big(2^m \sum_{n=m+1}^k |f_{n+1} - f_n|\Big) \leqslant 1.$$

Again, the sequence

$$\varphi\Big(2^m \sum_{n=m+1}^k |f_{n+1} - f_n|\Big) \text{ is increasing and converges point-wise to } \varphi(2^m h_m)$$

as $k \to \infty$. By Monotone Convergence Theorem, we have $\int \varphi(2^m h_m) \leqslant 1$, so that $\|h_m\|_\varphi \leqslant \frac{1}{2^m}$.

Since $\sum_{n=1}^{m}\lvert f_{n+1}-f_n\rvert$ converges a.e. to g, it follows that $f_{m+1}-f_1=\sum_{n=1}^{m}(f_{n+1}-f_n)$ converges a.e. to some function, which is dominated by g and, therefore, is in L_φ. Adding f_1, we conclude that (f_m) converges a.e. to some $f\in L_\varphi$.

For each $m\geqslant 2$, the following is satisfied a.e.:

$$\lvert f-f_m\rvert=\Bigl\lvert\sum_{n=m}^{\infty}(f_{n+1}-f_n)\Bigr\rvert\leqslant h_{m-1}.$$

It follows that $\lVert f-f_m\rVert_\varphi\leqslant\lVert h_{m-1}\rVert_\varphi\to 0$. □

The norm $\lVert\cdot\rVert_\varphi$ is called the ***Luxemburg norm*** on L_φ.

18.3.3 Proposition. *L_φ is monotonically complete and satisfies the Fatou Property.*

Proof. We will first show that L_φ has the Fatou property. Suppose that $0\leqslant f_\alpha\uparrow f$ in L_φ. We need to show that $\lVert f_\alpha\rVert_\varphi\to\lVert f\rVert_\varphi$. WLOG, $\lVert f\rVert_\varphi=1$. Since L_φ has the countable sup property by Proposition 18.2.7, $f_{\alpha_n}\uparrow f$ for some increasing sequence of indices (α_n). Fix $\lambda\in(0,1)$. Then $I_\varphi\bigl(\frac{f}{\lambda}\bigr)>1$. It follows from $\frac{f_{\alpha_n}}{\lambda}\uparrow\frac{f}{\lambda}$ and the Monotone Convergence Theorem that $I_\varphi\bigl(\frac{f_{\alpha_n}}{\lambda}\bigr)>1$ for all sufficiently large n, and, therefore, $\lVert f_{\alpha_n}\rVert_\varphi>\lambda$. This yields $\lVert f_{\alpha_n}\rVert_\varphi\to 1$ and, consequently, $\lVert f_\alpha\rVert_\varphi\to 1$.

Next, we show that L_φ is monotonically complete. Since L_φ is order complete, it suffices to show that it is monotonically bounded. Suppose that $0\leqslant f_\alpha\uparrow$ and $\lVert f_\alpha\rVert_\varphi\leqslant 1$ for every α. It follows that $\varphi(f_\alpha)\uparrow$ and $\bigl\lVert\varphi(f_\alpha)\bigr\rVert_{L_1(\mu)}\leqslant 1$. Being a KB-space, $L_1(\mu)$ is monotonically bounded, hence there exists $h\in L_1(\mu)$ such that $\varphi(f_\alpha)\leqslant h$ for every α. WLOG, $h(t)\geqslant 0$ for every t. Define $f(t)=\varphi^{-1}\bigl(h(t)\bigr)$. Then $f_\alpha\leqslant f$ for every α. It follows from $\varphi(f)=h\in L_1(\mu)$ that $f\in L_\varphi$.

□

Recall that a function of the form $f=\sum_{i=1}^{n}\lambda_i\mathbb{1}_{A_i}$, where $A_1,\dots,A_n$ are disjoint sets of positive measure, is said to be ***simple***. An easy calculation yields $I_\varphi(f)=\sum_{i=1}^{n}\varphi\bigl(\lvert\lambda_i\rvert\bigr)\mu(A_i)$. Suppose that $f\geqslant 0$. Then $\lambda_i\geqslant 0$ for all i. Put $h=\sum_{i=1}^{n}\varphi^{-1}(\lambda_i)\mathbb{1}_{A_i}$ then h is again a simple function and $\varphi(h)=f$. In this case, we write $h=\varphi^{-1}(f)$.

18.3.4 Lemma. *Let $0\leqslant f\in L_\varphi$. Then there exists a sequence (h_n) of finitely supported simple functions such that $0\leqslant h_n\uparrow f$ and $\lVert h_n\rVert_\varphi\to\lVert f\rVert_\varphi$.*

Proof. WLOG, $\lVert f\rVert_\varphi=1$, so that $\int\varphi(f)=1$. It follows from $\varphi(f)\in L_1(\mu)$ that there is a sequence of finitely supported simple functions (g_n) such that $0\leqslant g_n\uparrow\varphi(f)$. Put $h_n=\varphi^{-1}(g_n)$. Then h_n is again a finitely supported simple function, and $0\leqslant h_n\uparrow f$. It follows from Proposition 18.3.3 that $\lVert h_n\rVert_\varphi\to\lVert f\rVert_\varphi$. □

18.3.5 Corollary. *The set of all finitely supported simple functions is an order dense sublattice of L_φ.*

The following is an "infinite" version of Lemma 18.3.4:

18.3.6 Exercise. Suppose that μ is σ-finite and $0 \leqslant f \in L_0(\mu)$ such that $\|f\|_\varphi = \infty$. Then there exists a sequence (h_n) of finitely supported simple functions such that $0 \leqslant h_n \uparrow f$ and $\|h_n\|_\varphi \to \infty$.

18.4 Properties of H_φ

18.4.1 Proposition. *H_φ is the order continuous part of L_φ, i.e., $H_\varphi = (L_\varphi)_{\mathrm{oc}}$.*

Proof. Let $0 \leqslant u \in H_\varphi$ and $u \geqslant f_\alpha \downarrow 0$. Fix $\varepsilon > 0$. Then $\frac{1}{\varepsilon}u \geqslant \frac{1}{\varepsilon}f_\alpha \downarrow 0$. It follows from $u \in H_\varphi$ that $I_\varphi\bigl(\frac{1}{\varepsilon}u\bigr)$ is finite. Proposition 18.1.3 yields that $I_\varphi(\frac{1}{\varepsilon}f_\alpha) \to 0$. In particular, there exists α_0 such that for every $\alpha \geqslant \alpha_0$ we have $I_\varphi(\frac{1}{\varepsilon}f_\alpha) \leqslant 1$ and, therefore, $\|f_\alpha\|_\varphi \leqslant \varepsilon$.

To prove the converse, let $0 \leqslant f \in L_\varphi \setminus H_\varphi$. WLOG, $\int \varphi(f) = \infty$. For every n, put $A_n = \{\frac{1}{n} \leqslant f \leqslant n\}$. It follows from $f \geqslant \frac{1}{n}\mathbb{1}_{A_n}$ that $\mu(A_n) < \infty$; see Exercise 18.2.3. It follows that $\int \varphi\bigl(f \cdot \mathbb{1}_{A_n}\bigr) \leqslant \varphi(n)\mu(A_n) < \infty$, so that $\int \varphi\bigl(f \cdot \mathbb{1}_{A_n^C}\bigr) = \infty$. Put $f_n = f \cdot \mathbb{1}_{A_n^C}$. Then $\|f_n\|_\varphi \geqslant 1$ for every n. On the other hand, $f \geqslant f_n \downarrow 0$. Hence, $f \notin (L_\varphi)_{\mathrm{oc}}$. □

18.4.2 Theorem. *H_φ is an order continuous Banach lattice and a closed order dense ideal in L_φ.*

Proof. Combining Proposition 18.4.1 with Theorem 10.2.3, we conclude that H_φ is an order continuous Banach lattice and a closed ideal in L_φ. To see that H_φ is order dense, observe that it contains all finitely supported simple functions by Exercise 18.2.3; now apply Corollary 18.3.5. □

Combining the theorem with Exercise 18.2.3, we immediately get the following.

18.4.3 Corollary. *H_φ contains every bounded measurable function supported on a set of finite measure.*

18.4.4 Theorem. *H_φ is the closure of the set of all finitely supported simple functions in L_φ.*

Proof. Since H_φ is closed, by Exercise 18.2.3, it contains the closure of the set of all finitely supported simple functions.

Conversely, let $f \in H_\varphi$. It suffices to show that f can be approximated in norm by finitely supported simple functions. WLOG, $f \geqslant 0$; otherwise, consider f^+ and f^-. The result now follows from Lemma 18.3.5 and the fact that H_φ is order continuous. □

Since H_φ is a closed ideal in L_φ, the quotient L_φ/H_φ is a Banach lattice by Theorem 3.4.6.

18.4.5 Theorem. *L_φ/H_φ is an AM-space.*

Proof. Let $x, y \in L_\varphi/H_\varphi$ with $x \perp y$; we need to show that $\|x+y\| = \|x\| \vee \|y\|$. It is easy to see that $\|x+y\| \geqslant \|x\| \vee \|y\|$; it is left to verify the reverse inequality. WLOG, $x, y \geqslant 0$. It suffices to show that for every λ with $\|x\| \vee \|y\| < \lambda$ we have $\|x+y\| \leqslant \lambda$. Scaling x and y, we may assume that $\lambda = 1$. Thus, suppose that $\|x\| \vee \|y\| < 1$; we need to show that $\|x+y\| \leqslant 1$.

By Exercise 3.4.9, we find positive representatives f and g of x and y, respectively, in L_φ, so that $\|f\|_\varphi < 1$, $\|g\|_\varphi < 1$, and $f \perp g$. Applying Exercise 18.1.5(ii) with $\varepsilon = \frac{1}{2}$, we find a set A of finite measure such that f is bounded on A and $I_\varphi(f \cdot \mathbb{1}_A) \geqslant \frac{1}{2} I_\varphi(f)$. Since $\mathbb{1}_A \in H_\varphi$ and H_φ is an ideal, we have $f \cdot \mathbb{1}_A \in H_\varphi$. Put $u = f - f \cdot \mathbb{1}_A = f \cdot \mathbb{1}_{A^C}$, then $\tilde{u} = \tilde{f} = x$ and $I_\varphi(u) \leqslant \frac{1}{2} I_\varphi(f) \leqslant \frac{1}{2}$. Similarly, we find $v \in [0, g]$ such that $\tilde{v} = \tilde{g} = y$ and $I_\varphi(v) \leqslant \frac{1}{2}$. It follows from $0 \leqslant u \leqslant f$ and $0 \leqslant v \leqslant g$ that $u \perp v$. It follows that

$$I_\varphi(u+v) = \int \varphi(u+v) = \int \varphi(u) + \int \varphi(v) = I_\varphi(u) + I_\varphi(v) \leqslant \tfrac{1}{2} + \tfrac{1}{2} = 1.$$

This yields that $\|u+v\|_\varphi \leqslant 1$ and, therefore, $\|x+y\| \leqslant 1$. □

18.4.6 Exercise. H_φ has the Fatou Property but need not be sequentially monotonically complete.

18.5 Order continuity and bands in L_φ

The following theorem provides several equivalent characterizations of order continuity of L_φ.

18.5.1 Theorem. *TFAE:*

(i) *$H_\varphi = L_\varphi$;*
(ii) *If $f \in L_0(\mu)_+$ with $I_\varphi(f) \leqslant 1$ then $I_\varphi(\lambda f) < \infty$ for every $\lambda > 1$;*
(iii) *L_φ is order continuous;*
(iv) *L_φ is σ-order continuous;*

(v) *L_φ contains no lattice copy of ℓ_∞;*

(vi) *L_φ is a KB-space;*

(vii) *H_φ is a KB-space.*

Proof. (i)$\Leftrightarrow$(ii) follows from Proposition 18.2.2. (i)$\Leftrightarrow$(iii) by Proposition 18.4.1. Since L_φ is order complete by Proposition 18.2.5, we have (iii)$\Leftrightarrow$(iv)$\Leftrightarrow$(v) by Theorems 2.5.24 and 10.3.1.

(iii)$\Rightarrow$(vi) Suppose that $0 \leqslant f_n \uparrow$ and $\|f_n\|_\varphi \leqslant 1$ for every n; it suffices to show that (f_n) converges. WLOG, $f_n(t)\uparrow$ for every $t \in \Omega$; put $f(t) = \lim_n f_n(t)$. Note that $f(t) = \infty$ is possible. Then $\varphi(f_n(t)) \uparrow \varphi(f(t))$ where, again, we put $\varphi(\infty) = \infty$. It follows from $\|f_n\|_\varphi \leqslant 1$ that $\int \varphi(f_n) \leqslant 1$. Monotone Convergence Theorem yields that $\varphi(f)$ is a.e. finite, and so f is a.e. finite, and $\int \varphi(f) \leqslant 1$, so that $f \in L_\varphi$. It follows from $f_n \xrightarrow{\text{a.e.}} f$ that $f_n \xrightarrow{\text{o}} f$. Order continuity of L_φ yields that $f_n \to f$.

(vi)$\Rightarrow$(vii) because H_φ is a closed ideal in L_φ.

(vii)$\Rightarrow$(ii) Suppose that (ii) fails. Then there exists $f \in L_0(\mu)_+$ such that $I_\varphi(f) = \infty$, and $I_\varphi(\lambda f) \leqslant 1$ for some $\lambda \in (0,1)$. We will show that H_φ fails condition of Corollary 10.4.5 and, therefore, fails to be a KB-space.

We claim that there exists a set A of finite measure such that f is bounded on A and $\int_A \varphi(f) \geqslant 1$. Indeed, since $I_\varphi(\lambda f) \leqslant 1$, it follows from Exercise 18.1.5(i) that the restriction of μ to $\operatorname{supp} f$ is σ-finite and, therefore, $\operatorname{supp} f = \bigcup_{n=1}^\infty D_n$ for some nested increasing sequence (D_n) of sets of finite measure. Put $C_n = D_n \cap \{f \leqslant n\}$. Then $f \cdot \mathbb{1}_{C_n} \uparrow f$, hence $\varphi\big(f \cdot \mathbb{1}_{C_n}\big) \uparrow \varphi(f)$. It follows from Fatou's Lemma that $\lim_n \int \varphi\big(f \cdot \mathbb{1}_{C_n}\big) = \infty$, hence $\int_{C_n} \varphi(f) = \int \varphi\big(f \cdot \mathbb{1}_{C_n}\big) \geqslant 1$ for some n. Note that $f \leqslant n$ on C_n. Take $A = C_n$. This proves the claim.

Let $A_1 = A$. Since $\int_A \varphi(f)$ is finite, $\int_{A^C} \varphi(f) = \infty$. Applying the claim to the restriction of f to A^C and proceeding inductively, we obtain a disjoint sequence (A_n) of sets of finite measure such that f is bounded on A_n and $\int_{A_n} \varphi(f) \geqslant 1$. Put $g_n = f \cdot \mathbb{1}_{A_n}$. Note that $g_n \in H_\varphi$ by Corollary 18.4.3. It follows from $\int \varphi(g_n) = \int_{A_n} \varphi(f) \geqslant 1$ that $\|g_n\|_\varphi \geqslant 1$. On the other hand, since g_n's are disjoint and $0 \leqslant g_n \leqslant f$ for every n, we have $\lambda \sum_{n=1}^m g_n \leqslant \lambda f$ for every m. This yields $I_\varphi\big(\lambda \sum_{n=1}^m g_n\big) \leqslant I_\varphi(\lambda f) \leqslant 1$ and, consequently, $\big\|\sum_{n=1}^m g_n\big\|_\varphi \leqslant \frac{1}{\lambda}$. This contradicts Corollary 10.4.5. $\square$

If $\varphi(t) = t^p$ with $1 \leqslant p < \infty$ then $H_\varphi = L_\varphi = L_p(\mu)$, and this space is a KB-space. On the other hand, if μ and φ is as in Exercise 18.2.1, then $H_\varphi \neq L_\varphi$ and, therefore, L_φ is not order continuous.

Next, we are going to show that bands in L_φ and H_φ have the same structure as in $L_p(\mu)$. Let φ be an Orlicz function. For a set $A \in \mathcal{F}$, we write $L_\varphi(A) = \big\{f \in L_\varphi :$

$\operatorname{supp} f \subseteq A\big\}$, where the inclusion $\operatorname{supp} f \subseteq A$ is up to a set of measure zero. It is easy to see that $L_\varphi(A)$ may be identified with the Orlicz space corresponding to φ and the restriction of μ to A. The following theorem is analogous to Theorem 5.1.26 for $L_p(\mu)$-spaces.

18.5.2 Theorem. *If μ is σ-finite then bands in L_φ are exactly the sets of the form $L_\varphi(A)$ where $A \in \mathcal{F}$.*

Proof. It is easy to see that $L_\varphi(A)$ is a band for every $A \in \mathcal{F}$. It is also clear that the corresponding band projection is defined by $P_A f = f \cdot \mathbb{1}_A$. To prove the converse, let B be a band in L_φ; we will show that $B = L_\varphi(A)$ for some $A \in \mathcal{F}$.

First, we assume that μ is finite. Let D be the set of all characteristic functions in B. Since L_φ is order complete and $D \leqslant \mathbb{1} \in L_\varphi$, we conclude that $\sup D$ exists in L_φ. Since L_φ satisfies the countable sup property, $\sup D = \sup \mathbb{1}_{A_n}$ for some increasing sequence of characteristic functions $(\mathbb{1}_{A_n})$ in D. It follows that $\sup D = \mathbb{1}_A$ where $A = \bigcup_{n=1}^\infty A_n$. In particular, $\mathbb{1}_A \in B$. Moreover, $\mathbb{1}_C \in B$ for every measurable subset C of A. We claim that $B = L_\varphi(A)$.

Suppose that $f \in L_\varphi(A)_+$. For every n, put $C_n = A \cap \{f \leqslant n\}$. Then $\mathbb{1}_{C_n} \in B$. Furthermore, $f \cdot \mathbb{1}_{C_n} \leqslant n\mathbb{1}_{C_n} \in B$ yields $f \cdot \mathbb{1}_{C_n} \in B$. It follows from $f \cdot \mathbb{1}_{C_n} \uparrow f$ that $f \in B$. On the other hand, suppose that $f \in B_+$. For every n, we have $\frac{1}{n}\mathbb{1}_{\{f \geqslant \frac{1}{n}\}} \leqslant f$, so that $\mathbb{1}_{\{f \geqslant \frac{1}{n}\}} \in B$. It follows from $\mathbb{1}_{\{f \geqslant \frac{1}{n}\}} \uparrow \mathbb{1}_{\operatorname{supp} f}$ that $\mathbb{1}_{\operatorname{supp} f} \in B$. Consequently, $\mathbb{1}_{\operatorname{supp} f} \in D$ and, therefore, $\mathbb{1}_{\operatorname{supp} f} \leqslant \mathbb{1}_A$. It follows that $f \in L_\varphi(A)$.

Suppose now that μ is σ-finite. Let $\Omega = \bigcup \Omega_n$ for some disjoint sequence (Ω_n) of sets of finite measure. Clearly, $L_\varphi(\Omega_n)$ is a band in L_φ for every n. It follows that $B \cap L_\varphi(\Omega_n)$ is a band in $L_\varphi(\Omega_n)$. By the first part of the proof, $B \cap L_\varphi(\Omega_n) = L_\varphi(A_n)$ for some measurable $A_n \subseteq \Omega_n$. This yields the identity $P_B P_{\Omega_n} = P_{A_n}$ for the corresponding band projections; note that $P_{\Omega_n} f = f \cdot \mathbb{1}_{\Omega_n}$ and $P_{A_n} f = f \cdot \mathbb{1}_{A_n}$. Put $A = \bigcup_{n=1}^\infty A_n$. We claim that $B = L_\varphi(A)$. Let $f \in L_\varphi(A)_+$. Observe that $f \cdot \mathbb{1}_{A_n} \in L_\varphi(A_n) \subseteq B$. It follows from $B \ni \sum_{n=1}^m f \cdot \mathbb{1}_{A_n} \uparrow f$ that $f \in B$. Conversely, let $f \in B_+$. Clearly, $\sum_{n=1}^m P_{\Omega_n} f \uparrow f$. Since band projections are order continuous by Exercise 6.5.4, we have

$$\sum_{n=1}^m P_{A_n} f = \sum_{n=1}^m P_B P_{\Omega_n} f \ \uparrow\ P_B f = f.$$

Note that $P_{A_n} f \in L_\varphi(A_n) \subseteq L_\varphi(A)$, so that $\sum_{n=1}^m P_{A_n} f \in L_\varphi(A)$. Since $L_\varphi(A)$ is a band, this yields $f \in L_\varphi(A)$. $\square$

Exercises 5.2.20 and 5.1.17 immediately yield the following:

18.5.3 Corollary. *Let μ be a σ-finite measure. Closed ideals in H_φ agree with bands and are exactly the sets of the form $\{f \in H_\varphi : \operatorname{supp} f \subseteq A\}$ for some measurable set A.*

We will denote such a band by $H_\varphi(A)$. Clearly, it can be identified with the H_φ space corresponding to φ and the restriction of μ to A.

Note that bands and closed ideals in L_φ need not agree. For example, H_φ is a closed ideal in L_φ, but it is not of the form $L_\varphi(A)$, hence it is not a band (unless $H_\varphi = L_\varphi$).

18.6 Equivalence of Orlicz functions and the Δ_2-condition

It is well known that if $p \neq q$ then $L_p \neq L_q$. It may, however, happen that $L_\varphi = L_\psi$ for two different Orlicz functions φ and ψ. This leads to the concept of equivalence of Orlicz functions: φ and ψ are equivalent if they have "the same growth rates"; in this case, $L_\varphi = L_\psi$.

18.6.1 Definition. Given two Orlicz functions φ and ψ, we write $\varphi \preceq \psi$ if there are two positive scalars c_1 and c_2 such that $\varphi(t) \leqslant c_1\psi(c_2 t)$ for all $t > 0$. We say that $\varphi \sim \psi$ if $\varphi \preceq \psi$ and $\psi \preceq \varphi$.

To visualize this condition, recall that the graph of the function $t \mapsto c_1\psi(c_2 t)$ is obtained by stretching the graph of ψ vertically by a factor of c_1 and horizontally by the factor of $\frac{1}{c_2}$. So $\varphi \preceq \psi$ means that φ is dominated by the graph of ψ after appropriate stretching of one of them. It is clear that the "$\preceq$" relationship is transitive, while "$\sim$" is an equivalence relation.

18.6.2 Lemma. *$\varphi \preceq \psi$ iff there exists a constant $c > 0$ such that $\varphi(t) \leqslant \psi(ct)$ for every $t > 0$.*

Proof. Suppose $\varphi \preceq \psi$ with c_1 and c_2 as in Definition 18.6.1. WLOG, $c_1 \geqslant 1$; otherwise, replace it with 1. It follows from Exercise 18.1.1(ii) that $\varphi(t) \leqslant \psi(c_1 c_2 t)$ for every $t > 0$.

The converse implication is trivial. □

18.6.3 Proposition. *If $\varphi \preceq \psi$ then $L_\psi \subseteq L_\varphi$ and the inclusion map is continuous. If $\varphi \sim \psi$ then $L_\varphi = L_\psi$, $H_\varphi = H_\psi$, and the norms $\|\cdot\|_\varphi$ and $\|\cdot\|_\psi$ are equivalent.*

Proof. Suppose that $\varphi(t) \leqslant \psi(ct)$ for all $t > 0$. Let $f \in L_\psi$ with $\|f\|_\psi \leqslant 1$. WLOG, $f \geqslant 0$. Then $\int \varphi\bigl(\frac{1}{c}f\bigr) \leqslant \int \psi(f) \leqslant 1$. It follows that $f \in L_\varphi$ and $\|f\|_\varphi \leqslant c$. Hence, the embedding of L_ψ into L_φ has norm at most c.

It follows immediately that if $\varphi \sim \psi$ then $L_\varphi = L_\psi$ and the norms are equivalent. Proposition 18.4.1 yields that $H_\varphi = H_\psi$. □

This implies that, when dealing with L_φ, we may often assume by passing to an equivalent Orlicz function and, therefore, to an equivalent norm on the space, that φ has certain additional properties.

18.6.4 Exercise. Let φ and ψ be Orlicz functions which agree outside of an interval $[s,t]$, where $0 < s < t$. Then $\varphi \sim \psi$. This exercise implies that the properties of L_φ are determined by the rates at which φ approaches zero and infinity, and do not depend on the behaviour of φ "in the middle".

18.6.5 Exercise. Let φ be an Orlicz function and ψ the function that agrees with φ at all points 2^n as $n \in \mathbb{Z}$ and is linear in between. Then ψ is an Orlicz function and $\psi \sim \varphi$.

This exercise implies that every equivalence class of Orlicz function contains a piece-wise[1] linear function. It is often convenient, when dealing with L_φ, to assume that φ is (infinitely) piece-wise linear. Note also that the function ψ in this exercise is completely determined by its slopes on the intervals $[2^n, 2^{n+1}]$ as $n \in \mathbb{Z}$.

Conversely, let $(a_n)_{n\in\mathbb{Z}}$ be a bi-directional increasing sequence of positive reals. Let u be the piece-wise constant function on $(0,\infty)$ such that $u(t) = a_n$ whenever $2^n < t \leqslant 2^{n+1}$. For each $t \geqslant 0$, put $\varphi(t) = \int_0^t u(s)\,ds$. Then φ is a piece-wise linear Orlicz function.

18.6.6 Exercise. For every Orlicz function φ there exists a smooth Orlicz function equivalent to it.

18.6.7 Remark. It is a standard calculus exercise that an Orlicz function φ is differentiable except on a countable set. Moreover, φ has right and left derivatives at every point (except that it obviously has no left derivative at zero). Let u be the right derivative of φ. Then u is right continuous, increasing, and $u(t) > 0$ whenever $t > 0$. Observe that $\varphi(t) = \int_0^t u(s)\,ds$ for every t. Thus, φ and u completely determine each other.

An Orlicz function φ is said to satisfy the Δ_2***-condition*** if there exists $C > 1$ such that $\varphi(2t) \leqslant C\varphi(t)$ for all $t \geqslant 0$.

[1]There is a minor sloppiness of notation here as normally a piece-wise linear function is determined by *finitely many* linear functions.

18.6.8 Exercise. An Orlicz function satisfies the Δ_2-condition iff the function $\frac{\varphi(2t)}{\varphi(t)}$ is bounded on $(0,+\infty)$.

18.6.9 Exercise. Verify that the function $\varphi(t) = Kt^q$, where $1 \leqslant q < \infty$, satisfies the Δ_2-condition.

The following exercise shows that the number 2 in the definition of the Δ_2-condition is not important and may be replaced with any other number $\lambda > 1$ at the price of adjusting the constant C.

18.6.10 Exercise. Suppose that an Orlicz function φ satisfies the Δ_2-condition with constant C. Put $q = \log_2 C$. Then

$$\varphi(\lambda t) \leqslant C\lambda^q \varphi(t) \text{ whenever } \lambda \geqslant 1 \text{ and } t \geqslant 0. \tag{18.3}$$

We will see that the number q from this exercise is a more important parameter than C. The following exercise shows that the growth of φ as $t \to \infty$ as well as the decay when $t \to 0^+$ is slower than polynomial.

18.6.11 Exercise. Let φ be an Orlicz function satisfying the Δ_2-condition; let q be as in Exercise 18.6.10. There is a constant K such that $\varphi(t) \leqslant Kt^q$ when $t \geqslant 1$ and $\varphi(t) \geqslant Kt^q$ when $t \leqslant 1$.

18.6.12 Example ([KR58])**.** The converse to Exercise 18.6.11 is false: we are going to construct an Orlicz function whose growth at zero and infinity is polynomial, yet it fails the Δ_2-condition. For $t \geqslant 0$, put

$$p(t) = \begin{cases} t & \text{if } 0 \leqslant t < 1, \\ n! & \text{if } n! \leqslant t < (n+1)! \text{ for some } n \in \mathbb{N}, \end{cases}$$

and define $\varphi(t) = \int_0^t p$. It is easy to see that φ is an Orlicz function. Observe that $p(t) \leqslant t$ for every t and, therefore, $\varphi(t) \leqslant \frac{t^2}{2}$. For every n, we have

$$\varphi(n!) < (n-1)!n! \quad \text{and} \quad \varphi(2n!) > \int_{n!}^{2n!} p = (n!)^2.$$

It follows that $\frac{\varphi(2n!)}{\varphi(n!)} > n$ and, therefore, φ fails the Δ_2-condition by Exercise 18.6.8.

18.6.13 Exercise. Let φ and ψ be Orlicz functions.

(i) If φ satisfies the Δ_2-condition and $\psi \sim \varphi$ then ψ also satisfies the Δ_2-condition.
(ii) If both φ and ψ satisfy the Δ_2-condition then $\varphi \sim \psi$ iff there exist positive K_1 and K_2 such that $K_1 \leqslant \frac{\psi(t)}{\varphi(t)} \leqslant K_2$ for all $t > 0$.

18.6.14 Proposition. *If an Orlicz function φ satisfies the Δ_2-condition then L_φ is order continuous.*

Proof. We will apply Theorem 18.5.1(ii). Suppose that φ satisfies the Δ_2-condition. Let C and q be as in Exercise 18.6.10. Let $f \in L_0(\mu)_+$ such that $I_\varphi(f) \leqslant 1$. Then $\int \varphi(\lambda f) \leqslant C\lambda^q \int \varphi(f) < \infty$. □

We can improve the preceding result as follows. Recall that if a Banach lattice X satisfies a lower q-estimate for some $q < \infty$ then is order continuous; see Proposition 16.6.12.

18.6.15 Proposition. *Let φ be an Orlicz function satisfying the Δ_2-condition. Then L_φ satisfies a lower q-estimate with q as in Exercise 18.6.10.*

Proof. It follows from Exercise 18.6.10 that $\varphi(\lambda t) \geqslant \frac{\lambda^q}{C}\varphi(t)$ whenever $\lambda \in (0,1)$ and $t > 0$.

Let $f_1,\dots,f_n$ be a disjoint positive normalized sequence in L_φ, and $a_1,\dots,a_n \in \mathbb{R}_+$. In particular, $I_\varphi(f_i) = 1$ for every $i = 1,\dots,n$. Put $M = \Bigl(\sum_{i=1}^n a_i^q\Bigr)^{\frac{1}{q}}$. Using disjointness and Exercise 18.1.1(ii), we get

$$\int \varphi\Bigl(\frac{C}{M}\sum_{i=1}^n a_i f_i\Bigr) = \int \sum_{i=1}^n \varphi\Bigl(\frac{a_i}{M}Cf_i\Bigr) \geqslant \int \sum_{i=1}^n \Bigl(\frac{a_i}{M}\Bigr)^q \frac{1}{C}\varphi(Cf_i)$$
$$\geqslant \frac{1}{M^q}\sum_{i=1}^n a_i^q \int \varphi(f_i) = \frac{1}{M^q}\sum_{i=1}^n a_i^q = 1.$$

It follows that $\bigl\|\sum_{i=1}^n a_i f_i\bigr\|_\varphi \geqslant \frac{M}{C}$. □

18.7 Orlicz spaces on a finite measure

In the theory of $L_p(\mu)$ spaces, there are two important special cases: first, $L_p[0,1]$ or, more generally, when μ is a finite measure, and, second, ℓ_p, which corresponds to μ being the counting measure on $\mathbb{N}$. In some respects, these two cases are opposite to each other. For example, if $p < q$ then $\ell_p \subset \ell_q$ while $L_p[0,1] \supset L_q[0,1]$. We will see that the situation is similar for Orlicz spaces. We will see that in cases of a finite or a counting measure, properties of L_φ are determined by the asymptotic behaviour of φ as $t \to +\infty$ or as $t \to 0^+$, respectively. More precisely, given two Orlicz functions, φ and ψ, we say that $\varphi \preceq \psi$ at infinity if Definition 18.6.1 is satisfied for all $t > t_0$ for some positive t_0. Similarly, we say that $\varphi \preceq \psi$ at zero if Definition 18.6.1 is satisfied for all $t \in (0, t_0)$ for some positive t_0. We define equivalence of φ and ψ at infinity or at zero analogously. Finally, we also define the Δ_2-condition at infinity in the similar way.

Throughout this section, we assume that μ is a finite measure (we will consider counting measures in the next section). Let φ be an Orlicz function, $f \in L_0(\mu)$, and $t_0 > 0$. Then

$$I_\varphi(f) = \int \varphi\bigl(\lvert f\rvert\bigr) = \int_{\{\lvert f\rvert<t_0\}} \varphi\bigl(\lvert f\rvert\bigr) + \int_{\{\lvert f\rvert\geqslant t_0\}} \varphi\bigl(\lvert f\rvert\bigr) \leqslant \varphi(t_0)\mu(\Omega) + \int_{\{\lvert f\rvert\geqslant t_0\}} \varphi\bigl(\lvert f\rvert\bigr).$$

It follows that $I_\varphi(f)$ is finite iff $\int_{\{\lvert f\rvert\geqslant t_0\}} \varphi\bigl(\lvert f\rvert\bigr)$ is finite. The convergence of the latter integral does not depend on the behaviour of φ on $[0, t_0)$. Therefore, L_φ is determined by the asymptotic behaviour of φ as $t \to +\infty$.

It is easy to see that Lemma 18.6.2 remains valid if we replace $t > 0$ with $t > t_0 > 0$. We proceed with the new version of Proposition 18.6.3.

18.7.1 Proposition. *Let μ be a finite measure, and φ and ψ two Orlicz functions. If $\varphi \preceq \psi$ at infinity then $L_\psi \subseteq L_\varphi$ and the inclusion is continuous. If $\varphi \sim \psi$ at infinity then $L_\varphi = L_\psi$, $H_\varphi = H_\psi$, and the norms $\lVert\cdot\rVert_\varphi$ and $\lVert\cdot\rVert_\psi$ are equivalent.*

Proof. The proof is analogous to that of Proposition 18.6.3 with a minor modification. Suppose that $\varphi(t) \leqslant \psi(ct)$ for every $t \geqslant t_0 > 0$. Let $f \in L_\psi$; we will show that $f \in L_\varphi$. WLOG, $f \geqslant 0$ and $I_\psi(f) < \infty$. It follows that

$$\int \varphi\bigl(\tfrac{1}{c}f\bigr) = \int_{\{f<ct_0\}} \varphi\bigl(\tfrac{1}{c}f\bigr) + \int_{\{f\geqslant ct_0\}} \varphi\bigl(\tfrac{1}{c}f\bigr) \leqslant \varphi(t_0)\mu(\Omega) + \int \psi(f) < \infty.$$

It follows that $f \in L_\varphi$ and, therefore, $L_\psi \subseteq L_\varphi$. Since L_φ and L_ψ are both ideals of $L_0(\mu)$, L_ψ is an ideal in L_φ, so that the inclusion of L_ψ into L_φ is continuous by Exercise 6.1.13. $\square$

Many of the results of Section 18.6 remain valid if we only make assumptions on φ on (t_0, ∞) instead of $\mathbb{R}_+$. In particular, every Orlicz function is equivalent at infinity to an Orlicz function which is linear on intervals $[0, 1]$, $[1, 2]$, $[2, 4]$, $[4, 8]$, etc, which is determined by the sequence of slopes on these intervals.

Suppose now that φ satisfies the Δ_2-condition at infinity. Then there is an Orlicz function ψ which is equivalent to φ at infinity (and, therefore, produces the same Orlicz space), but ψ also satisfies the Δ_2-condition on all of $(0, \infty)$. This implies that L_φ is order continuous and satisfies a lower q-estimate. Also, as in Exercise 18.6.11, there exist $K, q, t_0 \geqslant 0$ such that $\varphi(t) \leqslant Kt^q$ for all $t > t_0$, i.e., φ has at most polynomial growth as $t \to \infty$.

Let φ be an Orlicz function. By Exercise 18.1.1(iii), there exist positive t_0 and α such that $\varphi(t) \geqslant \alpha t$ whenever $t \geqslant t_0$; this means that the growth rate of φ as $t \to \infty$ is at least linear. It follows that $L_\varphi \subseteq L_1(\mu)$. Indeed, let $f \in L_\varphi$. By scaling and taking the modulus, we may assume $f \geqslant 0$ and $I_\varphi(f)$ is finite. Then

$$\int f = \int_{\{f<t_0\}} f + \int_{\{f\geqslant t_0\}} f \leqslant t_0\mu(\Omega) + \tfrac{1}{\alpha}\int_{\{f\geqslant t_0\}} \varphi(f) \leqslant t_0\mu(\Omega) + \tfrac{1}{\alpha} I_\varphi(f) < \infty.$$

It follows that $f \in L_1$. On the other hand, Corollary 18.4.3 yields $L_\infty(\mu) \subseteq H_\varphi$. Thus, we have proved the following:

18.7.2 Proposition. *If μ is finite and φ is an Orlicz function then $L_\infty(\mu) \subseteq H_\varphi \subseteq L_\varphi \subseteq L_1(\mu)$.*

An Orlicz function φ grows linearly at ∞ if there exists $\beta, t_0 > 0$ such that $\varphi(t) \leqslant \beta t$ whenever $t > t_0$ or, equivalently, if $\lim\limits_{t\to\infty} \frac{\varphi(t)}{t}$ is finite. The following proposition shows that linear growth corresponds to $L_1(\mu)$.

18.7.3 Proposition. *Let μ be finite and φ an Orlicz function. If $\lim\limits_{t\to\infty} \frac{\varphi(t)}{t}$ is finite then $L_\varphi = L_1(\mu)$. If, in addition, μ is non-atomic, then the converse is also true.*

Proof. If $\lim\limits_{t\to\infty} \frac{\varphi(t)}{t}$ is finite then there exist $0 < \alpha < \beta$ and $t_0 > 0$ such that $\alpha t \leqslant \varphi(t) \leqslant \beta t$ whenever $t > t_0$. This means that $\varphi \sim \psi$ at infinity where $\psi(t) = t$. It follows that $L_\varphi = L_\psi = L_1(\mu)$.

Suppose now that μ is non-atomic and $\lim\limits_{t\to\infty} \frac{\varphi(t)}{t} = \infty$, but $L_\varphi = L_1(\mu)$. Since $L_1(\mu)$ is order continuous, Theorem 18.5.1 yields $H_\varphi = L_1(\mu)$. By assumption, for every $t_0 > 0$ and every $n \in \mathbb{N}$ one can find $t > t_0$ such that $\varphi(t) > nt$. Find a sequence (t_n) such that $\varphi(t_n) > nt_n$ and $t_n > 2^n$. Since μ is non-atomic, we can find a disjoint sequence (A_n) of measurable sets with $\mu(A_n) = \frac{\mu(\Omega)}{n^2 t_n}$; this is possible because $\sum_{n=1}^\infty \frac{\mu(\Omega)}{n^2 t_n} < \mu(\Omega)$. Define $f = \sum_{n=1}^\infty t_n \mathbb{1}_{A_n}$. Then

$$\int f = \sum_{n=1}^\infty t_n \mu(A_n) \leqslant \mu(\Omega) \sum_{n=1}^\infty \frac{1}{n^2} < \infty,$$

so that $f \in L_1(\mu)$. However,

$$\int \varphi(f) = \sum_{n=1}^\infty \varphi(t_n)\mu(A_n) \geqslant \sum_{n=1}^\infty n t_n \frac{\mu(\Omega)}{n^2 t_n} = \mu(\Omega) \sum_{n=1}^\infty \frac{1}{n} = \infty.$$

It follows that $f \notin H_\varphi$; a contradiction. $\square$

18.7.4 Proposition. *Suppose that μ is a finite measure. Then $L_1(\mu) = \bigcup L_\varphi$ where the union is over all Orlicz functions φ with non-linear growth at infinity (i.e., with $\lim\limits_{t\to\infty} \frac{\varphi(t)}{t} = \infty$).*

Proof. Let $f \in L_1(\mu)$. We will show that $f \in L_\varphi$ for an appropriate φ. WLOG, $f \geqslant 0$ and $\|f\|_{L_1} = 1$. For each n, put $G_n = \{n - 1 \leqslant f < n\}$. It can be easily verified that

$$1 = \int f \geqslant \sum_{n=1}^{m} (n-1)\mu(G_n) \geqslant \sum_{n=1}^{m} n\mu(G_n) - \mu(\Omega)$$

for every m. It follows that the series $\sum_{n=1}^{\infty} n\mu(G_n)$ converges. It is a calculus exercise that there exists an increasing sequence α_n in $[1, +\infty)$ such that $\lim_n \alpha_n = \infty$ and the series $\sum_{n=1}^{\infty} \alpha_n n\mu(G_n)$ converges. Put $\alpha_0 = 1$ and let φ be the infinitely piece-wise linear function such that the slope of φ on $[n, n+1]$ is α_n as $n = 0, 1, 2, \dots$. It follows from $\alpha_n \uparrow \infty$ that φ has non-linear growth. However, it is easy to see that $\varphi(n) \leqslant \alpha_n n$ for every n and, therefore,

$$\int \varphi(f) = \sum_{n=1}^{\infty} \int_{G_n} \varphi(f) \leqslant \sum_{n=1}^{\infty} \alpha_n n\mu(G_n) < \infty.$$

It follows that $f \in L_\varphi$. $\square$

Note that the analogous statement for L_p-spaces fails; in fact, $L_1 \neq \bigcup_{1<q<\infty} L_q$.

Recall that B_φ stands for the unit ball of L_φ.

18.7.5 Proposition. *If μ is finite and $\lim\limits_{t\to\infty} \frac{\varphi(t)}{t} = \infty$ then B_φ is equi-integrable.*

Proof. Find a sequence (t_n) such that $t_n \to \infty$ and $\varphi(t_n) \geqslant nt_n$. Convexity of φ yields that $\varphi(s) \geqslant ns$ whenever $s \geqslant t_n$. For each $0 \leqslant f \in B_\varphi$, we have

$$\lim_{M\to\infty} \int_{\{f \geqslant M\}} f = \lim_{n\to\infty} \int_{\{f \geqslant t_n\}} f \leqslant \lim_{n\to\infty} \frac{1}{n} \int_{\{f \geqslant t_n\}} \varphi(f) \leqslant \lim_{n\to\infty} \frac{1}{n} \to 0$$

uniformly on f. The result now follows from Exercise 9.7.9 and 9.7.6. $\square$

We already know that if μ is finite and φ satisfies the Δ_2-condition at infinity then L_φ is order continuous. We will show next that if, in addition, μ is non-atomic, then the converse is also true.

18.7.6 Proposition. *Let μ be a finite non-atomic measure and φ an Orlicz function. L_φ is order continuous iff φ satisfies the Δ_2-condition at infinity.*

Proof. We only need to prove the forward implication. Suppose that φ fails the Δ_2-condition at infinity. We can then find a sequence $t_n \to \infty$ such that $\varphi(t_n) > 2^n \varphi\big(\frac{t_n}{2}\big)$. Clearly, $\varphi(t_n) \to \infty$. Passing to a subsequence, we may assume that $\sum_{n=1}^{\infty} \frac{1}{\varphi(t_n)} \leqslant \mu(\Omega)$. Since μ is non-atomic, there exists a disjoint sequence of measurable sets (A_n)

with $\mu(A_n)=\frac{1}{\varphi(t_n)}$. Put $f=\sum_{n=1}^\infty t_n\mathbb{1}_{A_n}$. Then $I_\varphi(f)=\sum_{n=1}^\infty \varphi(t_n)\mu(A_n)=\infty$. However,

$$I_\varphi\bigl(\tfrac{f}{2}\bigr)=\sum_{n=1}^\infty \varphi\bigl(\tfrac{t_n}{2}\bigr)\mu(A_n)<\sum_{n=1}^\infty \tfrac{1}{2^n}\varphi(t_n)\frac{1}{\varphi(t_n)}<1.$$

Theorem 18.5.1(ii) yields that L_φ is not order continuous. □

18.8 Discrete Orlicz spaces

In this section, we consider the case when μ is a counting measure on some infinite set Γ[2]. In this case, it is common to write $\ell_\varphi(\Gamma)$ and $h_\varphi(\Gamma)$ for L_φ and H_φ, respectively. An $x\in\ell_\varphi(\Gamma)$ may be viewed as a family $x=(x_\gamma)_{\gamma\in\Gamma}$, and $I_\varphi(x)=\sum_{\gamma\in\Gamma}\varphi\bigl(\lvert x_\gamma\rvert\bigr)$. We will see that, in some sense, the case of a counting measure is "dual" to that of a finite non-atomic measure.

18.8.1 Exercise. Show that $\ell_1(\Gamma)\subseteq h_\varphi(\Gamma)\subseteq\ell_\varphi(\Gamma)\subseteq\ell_\infty(\Gamma)$. Furthermore, if Γ is infinite then $\ell_\varphi(\Gamma)\neq\ell_\infty(\Gamma)$.

Suppose that $x\in\ell_\infty(\Gamma)$ and fix $t_0>0$. After appropriately scaling x, we may assume that $\lvert x_\gamma\rvert<t_0$ for every γ, so that the convergence of $I_\varphi(x)=\sum_{\gamma\in\Gamma}\varphi\bigl(\lvert x_\gamma\rvert\bigr)$ is determined by the behaviour of φ on $[0,t_0]$. Therefore, most of the (isomorphic) properties of $\ell_\varphi(\Gamma)$ are determined by the behaviour of φ near zero.

Let φ and ψ be Orlicz functions. In Section 18.7, we defined what it means that $\varphi\preceq\psi$ at infinity, $\varphi\sim\psi$ at infinity, or φ satisfies the Δ_2-condition at infinity. We say that $\varphi\preceq\psi$ near zero, $\varphi\sim\psi$ near zero, or φ satisfies the Δ_2-condition near zero if $t\in(t_0,\infty)$ is replaced with $t\in(0,t_0)$ in the appropriate definitions.

18.8.2 Exercise. Let $1\leqslant p<q<\infty$, and put $\varphi(t)=t^p$ and $\psi(t)=t^q$. Then $\varphi\preceq\psi$ at infinity but $\varphi\succeq\psi$ near zero.

Next, we present a variant of Propositions 18.6.3 and 18.7.1 for counting measure. The proof is similar so we leave it is as an exercise.

18.8.3 Exercise. Let φ and ψ be two Orlicz functions and Γ a set. If $\varphi\preceq\psi$ near zero then $\ell_\psi(\Gamma)\subseteq\ell_\varphi(\Gamma)$ and the inclusion is continuous. If $\varphi\sim\psi$ near zero then $\ell_\varphi(\Gamma)=\ell_\psi(\Gamma)$, $h_\varphi(\Gamma)=h_\psi(\Gamma)$ and the norms are equivalent.

Suppose that an Orlicz function φ satisfies the Δ_2-condition near zero. Then there is an Orlicz function ψ which is equivalent to φ near zero, but ψ also satisfies the

[2]Some of the results of this section easily extend to a slightly more general situation when μ is purely atomic and the measure of atoms is bounded away from zero.

Δ_2-condition on all of $(0,\infty)$. This implies that $\ell_\varphi(\Gamma)=\ell_\psi(\Gamma)$ and, therefore, $\ell_\varphi(\Gamma)$ is order continuous and satisfies a lower q-estimate. Also, as in Exercise 18.6.11, there exist $K,q,t_0\geqslant 0$ such that $\varphi(t)\geqslant Kt^q$ for all $t\in[0,t_0)$, i.e., φ has at most polynomial decay as $t\to 0^+$.

We write $\ell_\varphi=\ell_\varphi(\mathbb{N})$ and $h_\varphi=h_\varphi(\mathbb{N})$. This notation is consistent with the ℓ_p notation. We call ℓ_φ an ***Orlicz sequence space***. As before, we write e_i for the i-th unit vector. It is clear that (e_i) is a disjoint sequence in ℓ_φ, hence an unconditional basic sequence. By Theorem 18.4.4, we have $[e_i]=h_\varphi$. This yields the following:

18.8.4 Proposition. *The sequence (e_i) is an unconditional basis of h_φ.*

Theorem 18.5.1 applies to ℓ_φ and provides equivalent characterizations of order continuity of ℓ_φ. In addition, we have the following:

18.8.5 Theorem. *Let φ be an Orlicz function. TFAE:*

(i) *ℓ_φ is order continuous;*
(ii) *(e_i) is a basis of ℓ_φ;*
(iii) *ℓ_φ is separable;*
(iv) *φ satisfies the Δ_2-condition near zero.*

Proof. (i)$\Leftrightarrow$(ii) By Theorem 18.5.1, (i) is equivalent to $\ell_\varphi=h_\varphi$. Now apply Proposition 18.8.4.

(ii)$\Rightarrow$(iii) This is a standard fact: if (e_i) is a basis then the set $\bigl\{\sum_{i=1}^n r_ie_i\ :\ r_1,\dots,r_n\in\mathbb{Q}\bigr\}$ is a countable dense subset of ℓ_φ.

(iii)$\Rightarrow$(i) Suppose (i) fails. Then, by Theorem 18.5.1, ℓ_φ contains a subspace which is isomorphic to ℓ_∞. It follows that ℓ_φ is not separable.

(iv)$\Rightarrow$(i) By a preceding remark, we may assume WLOG that φ satisfies the Δ_2-condition on $[0,\infty)$. Now apply Proposition 18.6.14.

(i)$\Rightarrow$(iv) Assume that φ fails the Δ_2-condition near zero. It suffices to show that condition (ii) in Theorem 18.5.1 fails. We will construct $x\in\ell_\varphi$ such that $I_\varphi(x)\leqslant 1$ but $I_\varphi(2x)=\infty$.

By assumption, we can find a sequence (t_n) in $(0,+\infty)$ such that $t_n\to 0$ and $\varphi(2t_n)\geqslant 2^n\varphi(t_n)$. Continuity of φ yields $\varphi(t_n)\to 0$. Passing to a subsequence, we may assume that $\varphi(t_n)\leqslant\frac{1}{2^{n+1}}$ for every n. For each n, let $m_n\in\mathbb{N}$ be the integer part of $\frac{1}{2^n\varphi(t_n)}$, i.e.,

$$m_n\leqslant\frac{1}{2^n\varphi(t_n)}<m_n+1.$$

It follows that $m_n\varphi(t_n) \leqslant \frac{1}{2^n}$. On the other hand,

$$\frac{1}{2^n} < m_n\varphi(t_n) + \varphi(t_n) \leqslant m_n\varphi(t_n) + \frac{1}{2^{n+1}},$$

so that $m_n\varphi(t_n) \geqslant \frac{1}{2^{n+1}}$.

Define x as follows:

$$x = (\underbrace{t_1, \dots, t_1}_{m_1}, \underbrace{t_2, \dots, t_2}_{m_2}, \dots).$$

Then

$$I_\varphi(x) = \sum_{i=1}^\infty \varphi(x_i) = \sum_{n=1}^\infty m_n\varphi(t_n) \leqslant \sum_{n=1}^\infty \frac{1}{2^n} \leqslant 1.$$

On the other hand,

$$I_\varphi(2x) = \sum_{i=1}^\infty \varphi(2x_i) = \sum_{n=1}^\infty m_n\varphi(2t_n) \geqslant \sum_{n=1}^\infty m_n 2^n\varphi(t_n) \geqslant \sum_{n=1}^\infty 2^n \frac{1}{2^{n+1}} = \infty.$$

□

The following exercise shows that the case of linear decay of φ at zero corresponds to the case $\ell_\varphi(\Gamma) = \ell_1(\Gamma)$.

18.8.6 Exercise. Let Γ be an infinite set. Then $\lim\limits_{t\to 0^+} \frac{\varphi(t)}{t} > 0$ iff $\ell_\varphi(\Gamma) = \ell_1(\Gamma)$.

18.8.7 Example. Consider the function $e^{-\frac{1}{t}}$. It is positive and convex on $(0, \frac{1}{2})$ and its right limit at zero is zero, so we may find an Orlicz function φ which agrees with $e^{-\frac{1}{t}}$ on $(0, \frac{1}{3})$. It is easy to see that φ fails the Δ_2-condition near zero. Let $x = (t_1, t_2, \dots)$ where $t_n = 1/\ln(n + e^3)$. This sequence is contained in $(0, \frac{1}{3})$ and $I_\varphi(x) = \sum_{n=1}^\infty \frac{1}{n+e^3} = \infty$, while $I_\varphi(x/2) = \sum_{n=1}^\infty \frac{1}{(n+e^3)^2} < \infty$, so that $x \in \ell_\varphi \setminus h_\varphi$.

Let $x_n = t_n e_n$. Then $\|x_n\| = t_n\|e_n\| = t_n\|e_1\| \to 0$, hence (x_n) is a norm null sequence in h_φ. Clearly, $x = \sup x_n$ in ℓ_φ; in particular, (x_n) is order bounded in ℓ_φ. However, since h_φ is an ideal in ℓ_φ and $x \notin h_\varphi$, (x_n) is not order bounded in h_φ. Thus, (x_n) is a norm null order bounded sequence in a Banach lattice, which is not order bounded in a norm closed ideal (even order dense); this answers the question in 1.4.18.

Furthermore, put $u_n = (0, \dots, 0, t_n, t_{n+1}, \dots)$. Then $0 \leqslant u_n \leqslant x$ implies that (u_n) is a sequence in ℓ_φ. Clearly, $0 \leqslant x_n \leqslant u_n \downarrow 0$, so that $x_n \xrightarrow{\sigma o} 0$ and, in particular, $x_n \xrightarrow{o} 0$ in ℓ_φ. However, (x_n) is not order or σ-order null in h_φ because it is not order bounded in h_φ. Thus, (x_n) is a norm and order null sequence in a Banach lattice which fails to be order null in a closed ideal; this answers the question in 2.5.8.

Alternative definition of Orlicz functions. In the literature, there are considerable variations in the definition of an Orlicz function. However, they do not lead to major changes in the theory. Let us comment on some of the variations.

While we assume that φ only vanishes at zero, some authors allow φ to vanish on $[0, t_0]$ for some $t_0 > 0$. For finite measure, this makes no difference as only the behaviour of $\varphi(t)$ as $t \to \infty$ matters. For counting measures, this condition yields $\ell_\varphi(\Gamma) = \ell_\infty(\Gamma)$.

While we assume that φ takes values in $\mathbb{R}_+$, one may allow φ to take the value infinity on $[t_0, \infty)$ for some $t_0 > 0$. In this case, for every non-zero simple function g we have $I_\varphi(\lambda g) = \infty$ for some $\lambda > 0$, hence $g \notin H_\varphi$. For finite measure, this yields that $L_\varphi = L_\infty(\mu)$. For counting measure, this condition is irrelevant because only the behaviour of $\varphi(t)$ as $t \to 0^+$ matters.

One may replace the assumption that φ is convex with some weaker assumptions. For example, one can instead require that φ is increasing, continuous, and satisfies $\lim_{t\to\infty} \varphi(t) = \infty$. Under these assumptions, one can still define H_φ and L_φ and they are still nested ideals in $L_0(\mu)$. If, in addition, one requires that there exists a positive constant C such that $\varphi(Ct) \geqslant 2\varphi(t)$ for every $t > 0$, then $\|\cdot\|_\varphi$ is defined; however, it will only be a lattice quasi-norm on L_φ.

18.9 Duality

In this section, we identify the dual space of L_φ. We will make several important assumptions which will streamline the exposition; we will see that these assumptions are not too restrictive so that our results still apply in the most important special cases. First, we will assume throughout this section that the measure μ is σ-finite. This includes the cases when μ is finite and when μ is the counting measure on $\mathbb{N}$. Second, we assume that

$$\lim_{t\to 0^+} \frac{\varphi(t)}{t} = 0 \quad \text{and} \quad \lim_{t\to\infty} \frac{\varphi(t)}{t} = \infty. \tag{18.4}$$

For finite measure, the first condition in (18.4) has no effect because only the behaviour of φ as $t \to \infty$ matters, while the second condition just means, by Proposition 18.7.3, that $L_\varphi \neq L_1(\mu)$. This is not very restrictive because we know the dual of $L_1(\mu)$ already. In the case of $\ell_\varphi(\Gamma)$, the second condition has no effect, while the first condition means that $\ell_\varphi(\Gamma) \neq \ell_1(\Gamma)$ by Exercise 18.8.6; so, again, the conditions are not restrictive.

For $s \geqslant 0$, define $\varphi^*(s) = \sup_{t\geqslant 0}\big(st - \varphi(t)\big)$. We say that φ^* is the ***conjugate*** of φ.

18.9.1 Exercise. (i) The sup in the definition of φ^* is attained, so that $\varphi^*(s) = \max_{t\geqslant 0}\big(st - \varphi(t)\big)$.

(ii) φ^* is again an Orlicz function.

(iii) If $\varphi \preceq \psi$ then $\psi^* \preceq \varphi^*$. In particular, if $\varphi \sim \psi$ then $\varphi^* \sim \psi^*$.

(iv) If $\varphi(t) = \frac{t^p}{p}$ for $1 < p < \infty$ then $\varphi^*(s) = \frac{s^q}{q}$, where $q = p^*$.

(v) (***Young's inequality***) $\varphi(t) + \varphi^*(s) \geqslant st$ for all $s, t \geqslant 0$. Moreover, for every $s > 0$ there exists $t > 0$ such that $\varphi(t) + \varphi^*(s) = st$.

18.9.2. There is an alternative more visual definition of the conjugate of an Orlicz function. Let φ be an Orlicz function satisfying (18.4). As in Remark 18.6.7, let u be the right derivative of φ; then u is right continuous, increasing, and $u(t) > 0$ whenever $t > 0$. Furthermore, $\varphi(t) = \int_0^t u$ for every $t \geqslant 0$. The conditions in (18.4) yield that $u(0) = 0$ and $\lim_{t\to\infty} u(t) = \infty$. This means that u has a generalized right continuous inverse, i.e., there is a unique right continuous function whose graph is obtained by reflecting the graph of u with respect to the diagonal and then re-defining it at the jump discontinuities. Denote this function by u^*. It is now easy to see that u^* is defined on all of $\mathbb{R}_+$ and is increasing. Furthermore, $u^*(s) > 0$ whenever $s > 0$, $u^*(0) = 0$ and $\lim_{s\to\infty} u^*(s) = \infty$.

We claim that $\varphi^*(s) = \int_0^s u^*$. Indeed, let $\psi(s) = \int_0^s u^*$. It is easy to see that $\varphi(t) + \psi(s) \geqslant st$ for every $s, t \geqslant 0$. Fix $s > 0$. It follows from $\psi(s) \geqslant st - \varphi(t)$ for all $t \geqslant 0$ that $\psi(s) \geqslant \varphi^*(s)$. On the other hand, choose t_0 so that $s = u(t_0)$, (more precisely, so that $\lim\limits_{t\to t_0^-} u(t) \leqslant s \leqslant u(t_0)$); then $\varphi(t_0) + \psi(s) = st_0$, so that $\psi(s) = st_0 - \varphi(t_0) \leqslant \varphi^*(s)$. Thus, φ^* is the function whose right derivative is the inverse of the right derivative of φ which vanishes at zero.

It follows, in particular, that $\varphi^{**} = \varphi$. Thus, instead of saying that φ^* is the conjugate of φ, it is more appropriate to say that φ and φ^* form a dual pair of Orlicz functions.

For the rest of this section, φ and ψ are a dual pair of Orlicz functions.

18.9.3 Proposition. *If $f \in L_\varphi$ and $g \in L_\psi$ then* $\left|\int fg\right| \leqslant 2\|f\|_\varphi \|g\|_\psi$.

Proof. WLOG, $f, g \geqslant 0$. Suppose first that $\|f\|_\varphi \leqslant 1$ and $\|g\|_\psi \leqslant 1$. By Young's inequality, $fg \leqslant \varphi(f) + \psi(g)$. Integrating, we get $\int fg \leqslant I_\varphi(f) + I_\psi(g) \leqslant 2$. The general case now follows easily by scaling. □

18.9.4 Example. The following example shows that the constant 2 in the proposition cannot be removed. View $L_2[0,1]$ as an Orlicz space L_φ with $\varphi(t) = t^2$. Then $\psi(s) \coloneqq \varphi^*(s) = \frac{s^2}{4}$. For every $f \in L_2[0,1]$, $\|f\|_\varphi = \|f\|_2$, while $\|f\|_\psi = \frac{1}{2}\|f\|_2$. Take $f = g = \mathbb{1}$, then $\int fg = 1$ but $\|f\|_\varphi \|g\|_\psi = \frac{1}{2}$.

For $f \in L_\varphi$ and $g \in L_\psi$, we define $\langle f, g\rangle = \int fg$. Proposition 18.9.3 guarantees that this is defined and $\left|\langle f, g\rangle\right| \leqslant 2\|f\|_\varphi \|g\|_\psi$. In particular, L_φ and L_ψ form a dual pair. It also implies that every function g in L_ψ defines a bounded functional on L_φ via $f \mapsto \langle f, g\rangle$. Hence L_ψ may be viewed as a subset of L_φ^*. Similarly, L_φ may be viewed as a subset of L_ψ^* as the situation is completely symmetrical. We will see that, unlike L_p-spaces for $1 < p < \infty$, these inclusion may be proper.

For $f \in L_0(\mu)$, define

$$|||f|||_\varphi = \sup\Big\{\big|\int fg\big| \;:\; g \in L_\psi,\ \|g\|_\psi \leqslant 1\Big\} = \sup\Big\{\big|\int fg\big| \;:\; I_\psi(g) \leqslant 1\Big\}.$$

18.9.5 Exercise. Verify that $|||\cdot|||_\varphi$ is a lattice norm on L_φ.

We will refer to it as the ***Orlicz norm*** and show that it is equivalent to the Luxemburg norm.

18.9.6 Proposition. *Let $f \in L_0(\mu)$. Its Orlicz norm $|||f|||_\varphi$ is finite iff $f \in L_\varphi$; in this case, $\|f\|_\varphi \leqslant |||f|||_\varphi \leqslant 2\|f\|_\varphi$.*

Proof. WLOG, $f \geqslant 0$. The second inequality follows immediately from Proposition 18.9.3. It follows that if $f \in L_\varphi$ then $|||f|||_\varphi$ is finite. To complete the proof, it suffices to show that if $|||f|||_\varphi$ is finite then $\|f\|_\varphi \leqslant |||f|||_\varphi$ (and, therefore, $f \in L_\varphi$). WLOG, $|||f|||_\varphi = 1$. For the sake of contradiction, suppose that $\|f\|_\varphi > 1$. Applying Lemma 18.3.4 in the case when $\|f\|_\varphi$ is finite or Exercise 18.3.6 if it is infinite, we find a finitely supported non-zero simple function $h = \sum_{i=1}^n t_i \mathbb{1}_{A_i}$ such that $0 \leqslant h \leqslant f$ and $\|h\|_\varphi \geqslant 1$. WLOG the sets A_i are disjoint. For each i, find s_i such that $\varphi(t_i) + \psi(s_i) = t_i s_i$. Let $g = \sum_{i=1}^n s_i \mathbb{1}_{A_i}$. It follows that $hg = \varphi(h) + \psi(g)$. Integrating, we get

$$I_\varphi(h) + I_\psi(g) = \langle h, g\rangle. \tag{18.5}$$

We claim that $I_\psi(g) \leqslant 1$. Suppose not. Then, by Exercise 18.1.1(ii), we have $I_\psi\big(\frac{g}{I_\psi(g)}\big) \leqslant 1$. It follows that $\big\|\frac{g}{I_\psi(g)}\big\|_\psi \leqslant 1$. Consequently, $\big\langle h, \frac{g}{I_\psi(g)}\big\rangle \leqslant |||h|||_\varphi \leqslant |||f|||_\varphi = 1$. Therefore, $\langle h, g\rangle \leqslant I_\psi(g)$. Combining this with (18.5), we get $I_\varphi(h) = 0$, so that $h = 0$; a contradiction.

Finally, combining (18.5) with $I_\psi(g) \leqslant 1$, we get

$$I_\varphi(h) + I_\psi(g) = \langle h, g\rangle \leqslant |||h|||_\varphi \leqslant |||f|||_\varphi = 1.$$

This yields $I_\varphi(h) \leqslant 1$ and, therefore, $\|h\|_\varphi \leqslant 1$; a contradiction. □

The following lemma asserts that in the definition of the Orlicz norm, it suffices to take supremum over simple functions.

18.9.7 Lemma. *Let $f \in L_\varphi(\mu)$. Then*

$$\begin{aligned} |||f|||_\varphi &= \sup\Big\{\big|\int fg\big| \;:\; g \text{ is simple and } I_\psi(g) \leqslant 1\Big\} && (18.6)\\ &= \sup\Big\{\big|\int fg\big| \;:\; g \in H_\psi \text{ and } \|g\|_\psi \leqslant 1\Big\}. && (18.7)\end{aligned}$$

Proof. We prove (18.6) first. WLOG, $f \geqslant 0$. Fix $\varepsilon > 0$ and find $0 \leqslant g \in L_\psi$ with $I_\psi(g) \leqslant 1$ and $\int fg \geqslant |||f|||_\varphi - \varepsilon$. By Lemma 18.3.4, there is a sequence (h_n) of finitely supported simple functions such that $0 \leqslant h_n \uparrow g$. It follows that $fh_n \uparrow fg$ and, therefore, $\int fh_n \to \int fg$. Find n such that $\int fh_n \geqslant \int fg - \varepsilon$. It follows that $\int fh_n \geqslant |||f|||_\varphi - 2\varepsilon$. Clearly, $\|h_n\|_\psi \leqslant \|g\|_\psi \leqslant 1$.

To prove (18.7), simply observe that the set in (18.7) contains the set in (18.6) and is contained in the set used in the definition of $|||f|||_\varphi$. ☐

18.9.8 Theorem. *$H_\varphi^* \simeq L_\psi$. More precisely, the map $T\colon \big(L_\psi, |||\cdot|||_\psi\big) \to \big(H_\varphi, \|\cdot\|_\varphi\big)^*$ defined by $(Tg)(f) = \langle f, g\rangle$ is a lattice isometry.*

Proof. Fix $g \in L_\psi$ and put $\xi = Tg$, i.e., $\xi(f) = \langle f, g\rangle$ for $f \in H_\varphi$. By Lemma 18.9.7(18.7), we have

$$|||g|||_\psi = \sup\Big\{\big|\langle f, g\rangle\big| \;:\; f \in H_\varphi \text{ and } \|f\|_\varphi \leqslant 1\Big\} = \|\xi\|,$$

where $\|\xi\|$ is computed with respect to $\|\cdot\|_\varphi$ on H_φ. It follows that T is an isometry.

It is clear that T is one-to-one. We will show next that T is onto; hence, a bijection. Fix $\xi \in H_\varphi^*$; we will find $g \in L_\psi$ such that $\xi = Tg$. WLOG, $\xi \geqslant 0$. Since H_φ is order continuous, so is ξ.

First, we assume that μ is finite. In this case, $\mathbb{1}_A \in H_\varphi$ for every $A \in \mathcal{F}$. Put $\nu(A) = \xi(\mathbb{1}_A)$. It is easy to see that ν is a finitely additive measure on $\mathcal{F}$. It is countably additive by Exercise 8.3.1. Indeed, suppose that $A_n \downarrow \varnothing$ in $\mathcal{F}$. It follows that $\mathbb{1}_{A_n} \downarrow 0$ and, therefore, $\xi(\mathbb{1}_{A_n}) \to 0$. Hence, $\nu(A_n) \to 0$, so that ν is countably additive. We have $\nu \ll \mu$ because if $\mu(A) = 0$ then $\mathbb{1}_A = 0$ in H_φ, so that $\xi(\mathbb{1}_A) = 0$. By Radon-Nikodym's Theorem 8.3.14, there exists $g \in L_1(\mu)$ such that $\nu = g \cdot \mu$.

It is easy to verify that $\xi(h) = \int gh\,d\mu$ for every simple function h. It follows from Lemma 18.9.7 that $|||g|||_\psi = \|\xi\|$, so that $g \in L_\psi$. We now have $\xi(h) = \langle h, g\rangle$ for every simple h; since simple functions are dense in H_φ, we have $\xi(f) = \langle f, g\rangle$ for every $f \in H_\varphi$. It follows that $\xi = Tg$. Also, $\|g\|_\psi \leqslant |||g|||_\psi = \|\xi\|$.

Suppose now that μ is σ-finite. Let $\Omega = \bigcup_{n=1}^\infty \Omega_n$ where Ω_n is a nested increasing sequence of sets of finite measure. Clearly, the restriction of ξ to $H_\varphi(\Omega_n)$ is a bounded linear functional. By the finite measure case, we find $g_n \in L_\psi(\Omega_n)$ such that $|||g_n|||_\psi \leqslant \|\xi\|$ and $\xi(f) = \langle f, g_n\rangle$ whenever $f \in H_\varphi(\Omega_n)$.

In particular, for every simple function h with $\operatorname{supp} h \subseteq \Omega_n$ we can view h as an element of both $H_\varphi(\Omega_n)$ and $H_\varphi(\Omega_{n+1})$ and, therefore, $\int hg_n\,d\mu = \xi(h) = \int hg_{n+1}\,d\mu$. It follows that g_n is the restriction of g_{n+1} to Ω_n. Hence, the sequence (g_n) is increasing and norm bounded in L_ψ. By Proposition 18.3.3, $g_n \uparrow g$ for some $g \in L_\psi$. Fix a positive f in H_φ. Then $f \cdot \mathbb{1}_{\Omega_n} \uparrow f$. It follows that $\xi(f \cdot \mathbb{1}_{\Omega_n}) \to \xi(f)$. On the other hand,

$\xi(f\cdot\mathbb{1}_{\Omega_n})=\int fg_n\to\int fg$ because $fg_n\uparrow fg$. It follows that $\xi(f)=\int fg=\langle f,g\rangle$. Therefore, $\xi=Tg$. This proves that T is a bijection.

It is trivial that T is positive. On the other hand, if a non-zero $g\in L_\psi$ is not positive then there exists a characteristic function h such that $(Tg)(h)=\int gh<0$ and, therefore, Tg is not positive. By Theorem 1.6.13(iv), T is a lattice isomorphism. $\square$

18.9.9 Corollary. *If φ satisfies the Δ_2-condition then $L_\varphi^*=L_\psi$. More precisely, $\big(L_\varphi,\|\cdot\|_\varphi\big)^*$ is lattice isometric to $\big(L_\psi,|||\cdot|||_\psi\big)$.*

We now proceed to characterizing L_φ^*. Define a map $S\colon L_\psi\to L_\varphi^*$ via $(Sg)(f)=\langle f,g\rangle=\int fg$, where $f\in L_\varphi$ and $g\in L_\psi$. Note that the operator T in Theorem 18.9.8 is related to this operator S as follows: for $g\in L_\psi$, Tg is the restriction of Sg to H_φ. Observe also that it follows immediately from the definition of $|||\cdot|||_\psi$ that S is an isometry (non-surjective) from $\big(L_\psi,|||\cdot|||_\psi\big)$ to $\big(L_\varphi,\|\cdot\|_\varphi\big)^*$. As usual, by $H_\varphi^\perp$ we mean the annihilator of H_φ in L_φ^*, i.e., the set of all functionals in L_φ^* which vanish on H_φ.

18.9.10 Theorem. *$S(L_\psi)$ and $H_\varphi^\perp$ are complementary bands in L_φ^*. $H_\varphi^\perp$ is lattice isometric to $L_1(\nu)$ for some measure ν. Hence, after appropriate identifications, L_φ^* admits a band decomposition: $L_\varphi^*=L_\psi\oplus L_1(\nu)$. Furthermore, L_ψ is the order continuous dual of L_φ, i.e., $L_\psi=(L_\varphi)^\sim_{\mathrm{oc}}$.*

Proof. First, we prove that L_ψ or, more precisely, $S(L_\psi)$ is the order continuous dual of L_φ. Let $g\in L_\psi$; we will show that Sg is order continuous. WLOG, $g\geqslant 0$. Let $f_n\downarrow 0$ in L_φ. Since L_φ satisfies the countable sup property, it suffices to verify that $\langle f_n,g\rangle\to 0$. But this is trivial by Lebesgue Dominated Convergence Theorem because $f_ng\downarrow 0$ in $L_1(\mu)$.

Conversely, suppose that ξ is an order continuous functional in L_φ^*; we need to show that ξ is in $S(L_\psi)$. WLOG, $\xi\geqslant 0$. Notice that the restriction of ξ to H_φ is a positive linear functional on H_φ so, by Theorem 18.9.8, there exists a positive $g\in L_\psi$ such that $\xi(f)=(Tg)(f)=(Sg)(f)$ for every $f\in H_\varphi$. It follows that ξ and Sg agree on H_φ. By the first part of the proof, Sg is order continuous. So we have two order continuous functionals, ξ and Sg, which agree on H_φ. Since H_φ is order dense in L_φ by Theorem 18.4.2, it follows that $\xi=Sg$.

By Ogasawara's Theorem 6.6.12, $S(L_\psi)$ is a projection band in L_φ^*.

It is easy to see that $H_\varphi^\perp$ is an ideal in L_φ^*. Indeed, if $0\leqslant\zeta\leqslant\xi\in H_\varphi^\perp$ then trivially $\zeta\in H_\varphi^\perp$. If $\xi\in H_\varphi^\perp$ then $|\xi|\in H_\varphi^\perp$ by the Riesz-Kantorovich formula.

Observe that $S(L_\psi)\cap H_\varphi^\perp=\{0\}$. Indeed, let $0\leqslant\xi\in S(L_\psi)\cap H_\varphi^\perp$. Then $\xi=Sg$ for some $0\leqslant g\in L_\psi$ and $\int fg=0$ for every $f\in H_\varphi$ and, in particular, for every f of the form $\mathbb{1}_A$ where $\mu(A)$ is finite. It follows that $g=0$. Thus, $S(L_\psi)\cap H_\varphi^\perp=\{0\}$.

Next, we show that $S(L_\psi)+H_\varphi^\perp=L_\varphi^*$. Fix $\xi\in L_\varphi$ and let ξ_0 be the restriction of ξ to H_φ. By Theorem 18.9.8, $\xi_0=Tg$ for some $g\in L_\psi$. For every $f\in H_\varphi$, we have $(\xi-Sg)(f)=\xi_0(f)-(Tg)(f)=0$, so that $\xi-Sg\in H_\varphi^\perp$. It follows that $\xi\in S(L_\psi)+H_\varphi$.

By Proposition 5.2.7, $H_\varphi^\perp$ is a band and $L_\varphi^*=S(L_\psi)\oplus H_\varphi^\perp$ is a band decomposition. It is a standard fact that $H_\varphi^\perp$ is lattice isometric to $(L_\varphi/H_\varphi)^*$. Since L_φ/H_φ is an AM-space by Theorem 18.4.5, it follows by Theorem 8.1.7 that $H_\varphi^\perp$ is an AL-space. It then follows from Kakutani's Representation Theorem 8.5.5 that $H_\varphi^\perp$ is lattice isometric to $L_1(\nu)$ for some measure ν. $\square$

Next, we characterize how H_ψ sits inside L_ψ in terms of duality with L_φ.

18.9.11 Lemma. *Suppose that μ is finite, (f_n) is a norm bounded sequence in L_φ and $f_n\xrightarrow{\text{a.e.}}0$. Then $\int f_n\to 0$.*

Proof. WLOG, $\|f_n\|_\varphi\leqslant 1$ for every n. Fix $\varepsilon>0$. Since B_φ is equi-integrable by Proposition 18.7.5, there exists $\delta>0$ such that if $\mu(A)<\delta$ then $\int_A|f|<\varepsilon$ for every $f\in B_\varphi$. Using Egorov's Theorem, find a measurable set A such that $\mu(A)<\delta$ and (f_n) converges to zero uniformly on A^C. It follows that $\int_{A^C}f_n\to 0$ and $\left|\int_A f_n\right|<\varepsilon$ for each n. $\square$

18.9.12 Proposition. *Let $\xi\in L_\varphi^*$. TFAE:*

(i) *$\xi\in H_\psi$ or, more precisely, $\xi=Sg$ for some $g\in H_\psi$;*
(ii) *$\xi(f_n)\to 0$ whenever (f_n) is norm bounded in L_φ and $f_n\xrightarrow{\text{a.e.}}0$.*

Proof. (i)⇒(ii) Suppose that (f_n) is norm bounded and $f_n\xrightarrow{\text{a.e.}}0$. Note that $\xi(f)=\langle f,g\rangle=\int fg$ for every $f\in L_\varphi$. We first consider the special case when $g=\mathbb{1}_A$ with $\mu(A)<\infty$. Since the restriction of μ to A is finite, applying Lemma 18.9.11 to the restrictions $f_{n|A}$ viewed as elements of $L_\varphi(A)$, we get $\langle f_n,\mathbb{1}_A\rangle=\int_A f_n\to 0$. It follows that if g is a finitely supported simple function then $\langle f_n,g\rangle\to 0$. Since finitely supported simple functions are dense in H_ψ by Theorem 18.4.4, we have $\langle f_n,g\rangle\to 0$ for every $g\in H_\psi$.

(ii)⇒(i) Suppose that ξ satisfies (ii). WLOG, $\xi\geqslant 0$. First, we show that ξ is order continuous. Since L_φ has the countable sup property, it suffices to verify that if $f_n\downarrow 0$ in L_φ then $\xi(f_n)\to 0$, which easily follows from (ii). Thus, $\xi\in(L_\varphi)^\sim_{\mathrm{oc}}$, so that ξ may be viewed as an element of L_ψ, that is $\xi=Sg$ for some $g\in L_\psi$. It is left to prove that $g\in H_\psi$.

By Proposition 18.4.1, it suffices to prove that $g\in(L_\psi)_{\mathrm{oc}}$. We will use Theorem 10.2.3(v). Let (h_n) be a disjoint sequence in $[0,g]$. For every n, from the definition of $|||\cdot|||_\psi$, we find $0\leqslant f_n\in L_\varphi$ such that $\|f_n\|_\varphi\leqslant 1$ and $\langle f_n,h_n\rangle\geqslant\frac{1}{2}|||h_n|||_\psi$. WLOG,

$\operatorname{supp} f_n \subseteq \operatorname{supp} h_n$. It follows that (f_n) is disjoint and, therefore, $f_n \xrightarrow{\text{a.e.}} 0$. By (ii), we have $\langle f_n, g\rangle \to 0$, so that $\langle f_n, h_n\rangle \to 0$ and, therefore, $|||h_n|||_\psi \to 0$. $\square$

18.9.13 Exercise. In this exercise, we consider the dual pair e^t and $s\ln s$. Well, not exactly; these functions are not even Orlicz functions. Find two Orlicz functions φ and ψ such that $\varphi(t) = e^t$ and $\psi(s) = s\ln s$ for all sufficiently large t and s, and such that both φ and ψ satisfy (18.4). Show that $\psi \sim \varphi^*$ at infinity. Show that ψ satisfies the Δ_2-condition while φ fails it.

Let μ be the Lebesgue on $[0,1]$. Show that L_ψ is order continuous while L_φ is not. Show that L_φ contains a lattice copy of c_0 while L_ψ contains a lattice copy of ℓ_1.

Chapter 19

Tensor products

We start this chapter with a quick review of the algebraic tensor product of two vector spaces. We then construct the "vector lattice tensor product" $X \bar{\otimes} Y$ of two Archimedean vector lattices. We define it via a universal property for lattice bimorphism. Our construction is along the lines of on [Frem72, Sch80, WicTP]. We prove that $X \otimes Y$ sits "densely" in $X \bar{\otimes} Y$. We then show that $X \bar{\otimes} Y$ satisfies a universal property of positive bimorphisms.

We then pass to a normed case: we construct a "positive projective tensor product" $X \otimes_{|\pi|} Y$ of two Banach lattices. This object is somewhat analogous to the projective tensor product in the category of Banach spaces. Our construction is based on [Frem74].

All vector lattices in this chapter are assumed to be Archimedean.

19.1 Algebraic tensor product

We start this section with a quick review of the classical algebraic tensor products in linear algebra. For details, we refer the reader to Chapter 1 in [Ryan02] or Section IV(5) of [Hung80].

Let X, Y, and Z be three vector spaces. We write $X^{\#}$ for the linear dual of X, i.e., the vector space of all linear maps from X to $\mathbb{R}$. A map $\varphi\colon X \times Y \to Z$ is said to be ***bilinear*** if it is linear in each of the variables. We write $\mathcal{B}(X,Y;Z)$ for the vector space of all bilinear maps from $X \times Y$ to Z. We write $\mathcal{B}(X,Y) = \mathcal{B}(X,Y;\mathbb{R})$.

For every pair X and Y of vector spaces, there exist a vector space $X \otimes Y$ and a bilinear map $j\colon X \times Y \to X \otimes Y$ that satisfy the following *universal property*: for every vector space Z and every bilinear map $\varphi\colon X \times Y \to Z$ there exists a unique linear map $\varphi^\otimes$ such that $\varphi = \varphi^\otimes \circ j$.

$$\begin{tikzcd} X \times Y \arrow[r, "\varphi"] \arrow[d, "j"'] & Z \\ X \otimes Y \arrow[ur, dotted, "\varphi^\otimes"'] & \end{tikzcd}$$

The space $X \otimes Y$ is unique up to a linear isomorphism that preserves the image of $X \times Y$; it is called the ***algebraic tensor product*** of X and Y. The map $\varphi^\otimes$ is called the ***linearization*** of φ. The map j is called the ***canonical tensor map***. We write $x \otimes y$ for $j(x, y)$ and call it the ***elementary tensor*** of x and y. In this notation, the universal property states that $\varphi(x, y) = \varphi^\otimes(x \otimes y)$. The tensor product $X \otimes Y$ is the linear span of the range of j, hence, every element of $X \otimes Y$ may be written as a linear combination of elementary tensors. However, this expansion is not unique: for example, $(x_1 + x_2) \otimes y = x_1 \otimes y + x_2 \otimes y$ and $(\lambda x) \otimes y = x \otimes (\lambda y) = \lambda(x \otimes y)$. It follows that every $a \in X \otimes Y$ may be expressed as

$$a = \sum_{i=1}^{n} x_i \otimes y_i \tag{19.1}$$

for some $x_1, \dots, x_n \in X$ and $y_1, \dots, y_n \in Y$ (again, this decomposition is not unique).

If X_0 and Y_0 are subspaces of X and Y, respectively, then $X_0 \otimes Y_0$ may be viewed as the subspace of $X \otimes Y$ spanned by $\{x \otimes y : x \in X_0,\ y \in Y_0\}$. If A and B are linearly independent subsets of X and Y, respectively, then the set $C \coloneqq \{x \otimes y : x \in A, y \in B\}$ is linearly independent in $X \otimes Y$. Furthermore, if A and B are (Hamel) bases then so is C. It follows that $\mathbb{R}^n \otimes \mathbb{R}^m$ may be identified with the space $M_{n,m}(\mathbb{R})$ of all $n \times m$ real matrices, with elementary tensors corresponding to matrices of rank one.

There are several ways to construct $X \otimes Y$ (and, by doing so, prove that it actually exists). We will briefly mention three approaches. In the first one, we define a bilinear map j from $X \times Y$ to the linear dual $\mathcal{B}(X, Y)^\#$ via $\big(j(x, y)\big)(\psi) = \psi(x, y)$, where $x \in X$, $y \in Y$, and $\psi \in \mathcal{B}(X, Y)$. We then define $X \otimes Y$ as the subspace of $\mathcal{B}(X, Y)^\#$ generated by the range of j.

In the second approach, we consider the vector space of all formal linear combinations of elements of $X \times Y$ (which may be thought as a ***free vector space*** over $X \times Y$). We then quotient it over the subspace generated by differences of those expressions that have to be equal in $X \otimes Y$, e.g., $(\lambda x, y) - (x, \lambda y)$, etc. In the third approach, we pick bases A and B of X and Y, respectively, and define $X \otimes Y$ as the vector space of all formal linear combinations of elements of $A \times B$.

The universal property of $X \otimes Y$ implies that the map $\varphi \mapsto \varphi^\otimes$ is a linear isomorphism between $\mathcal{B}(X, Y; Z)$ and $\mathcal{L}(X \otimes Y, Z)$. In the case when $Z = \mathbb{R}$, this yields $\mathcal{B}(X,Y) \simeq (X\otimes Y)^{\#}$. One can identify $\mathcal{B}(X,Y)$ with $\mathcal{L}(X,Y^{\#})$ as follows: for $\varphi\in\mathcal{B}(X,Y)$, the map $x\mapsto\varphi(x,\cdot)$ is in $\mathcal{L}(X,Y^{\#})$ and, conversely, for every $S\in\mathcal{L}(X,Y^{\#})$ the map $(x,y)\mapsto(Sx)(y)$ is in $\mathcal{B}(X,Y)$. Furthermore, one can identify $X^{\#}\otimes Y$ with the subspace of $\mathcal{L}(X,Y)$ consisting of all operators of finite rank: for $f\in X^{\#}$ and $y\in Y$, we view the elementary tensor $f\otimes y$ as an operator from X to Y via $(f\otimes y)(x)=f(x)y$, where $x\in X$.

19.1.1. The non-uniqueness of an expansion in (19.1) leads to considerable difficulties. One often needs to verify that two expansions actually correspond to the same element of $X\otimes Y$. Or, equivalently, that an expansion corresponds to zero.

Let E be a subset of $X^{\#}$ such that E separates points of X, i.e., if $f(x)=0$ for all $f\in E$ then $x=0$. Similarly, let F be a subset of $Y^{\#}$ such that F separates points of Y. Let $a\in X\otimes Y$, with $a=\sum_{i=1}^n x_i\otimes y_i$. Then $a=0$ iff $\sum_{i=1}^n f(x_i)g(y_i)=0$ for all $f\in E$ and $g\in F$. See Section 1.1 of [Ryan02] for a proof of this fact.

Recall that by $\mathbb{R}^A$ we denote the vector space of all real-valued functions on a set A.

19.1.2 Proposition. *Let A and B be two sets. We define $\varphi\colon\mathbb{R}^A\times\mathbb{R}^B\to\mathbb{R}^{A\times B}$ via $\big(\varphi(f,g)\big)(s,t)=f(s)g(t)$. Then $\mathbb{R}^A\otimes\mathbb{R}^B$ may be identified with the span of* $\operatorname{Range}\varphi$ *in $\mathbb{R}^{A\times B}$, with φ acting as the canonical tensor map. Moreover, $\mathbb{R}^A\otimes\mathbb{R}^B$ is a subalgebra of $\mathbb{R}^{A\times B}$.*

Proof. Clearly, φ is bilinear. Let $j\colon\mathbb{R}^A\times\mathbb{R}^B\to\mathbb{R}^A\otimes\mathbb{R}^B$ be the canonical tensor map.

$$\begin{tikzcd} \mathbb{R}^A\times\mathbb{R}^B \arrow[r, "\varphi"] \arrow[d, "j"'] & \mathbb{R}^{A\times B} \\ \mathbb{R}^A\otimes\mathbb{R}^B \arrow[ur, dashed, "\varphi^{\otimes}"'] & \end{tikzcd}$$

It suffices to show that $\varphi^{\otimes}$ is one-to-one. We will use 19.1.1 to show this. Let $a=\sum_{i=1}^n f_i\otimes g_i$ in $\mathbb{R}^A\otimes\mathbb{R}^B$ such that $\varphi^{\otimes}(a)=0$. Let E and F be the sets of all point evaluation functionals on $\mathbb{R}^A$ and $\mathbb{R}^B$, respectively. It is clear that they separate points. For every $s\in A$ and $t\in B$, we have

$$\begin{aligned}\sum_{i=1}^n\delta_s(f_i)\delta_t(g_i)=\sum_{i=1}^n f_i(s)g_i(t)=\sum_{i=1}^n\varphi(f_i,g_i)(s,t)\\ =\sum_{i=1}^n\varphi^{\otimes}(f_i\otimes g_i)(s,t)=\varphi^{\otimes}(a)(s,t)=0.\end{aligned}$$

By 19.1.1, we conclude that $a=0$ and, therefore, $\varphi^{\otimes}$ is injective. We can now identify (through $\varphi^{\otimes}$) the tensor product $\mathbb{R}^A\otimes\mathbb{R}^B$ with $\operatorname{Range}\varphi^{\otimes}=\operatorname{span}(\operatorname{Range}\varphi)$.

Given $f \in \mathbb{R}^A$ and $g \in \mathbb{R}^B$, we identify the elementary tensor $f \otimes g$ with the function in $\mathbb{R}^{A\times B}$ defined via $(f \otimes g)(s,t) = f(s)g(t)$. Note that $(f_1 \otimes g_1) \cdot (f_2 \otimes g_2) = (f_1 f_2) \otimes (g_1 g_2)$; it follows that $\mathbb{R}^A \otimes \mathbb{R}^B$ is a subalgebra of $\mathbb{R}^{A\times B}$. □

In the following, given $f \in \mathbb{R}^A$ and $g \in \mathbb{R}^B$, we will identify the elementary tensor $f \otimes g$ with the function in $\mathbb{R}^{A\times B}$ defined via $(f \otimes g)(s,t) = f(s)g(t)$. In this notation, Proposition 19.1.2 says that $\mathbb{R}^A \otimes \mathbb{R}^B$ is the subspace of $\mathbb{R}^{A\times B}$ formed by all sums of such elementary tensors.

19.1.3 Proposition. *Let Ω and Σ be two Hausdorff topological spaces. We can identify $C(\Omega) \otimes C(\Sigma)$ with the subspace of $C(\Omega \times \Sigma)$ consisting of all sums of elementary tensors $f \otimes g$ where $f \in C(\Omega)$ and $g \in C(\Sigma)$. Under this identification, $C(\Omega) \otimes C(\Sigma)$ is a unital subalgebra of $C(\Omega \times \Sigma)$.*

Proof. Since $C(\Omega)$ and $C(\Sigma)$ are subspaces of $\mathbb{R}^\Omega$ and $\mathbb{R}^\Sigma$, by the preceding argument we may identify $C(\Omega) \otimes C(\Sigma)$ with the subalgebra of $\mathbb{R}^{\Omega\times\Sigma}$ that consists of all the sums of elementary tensors $f \otimes g$ where $f \in C(\Omega)$ and $g \in C(\Sigma)$. Viewed as a function on $\Omega \times \Sigma$, $f \otimes g$ is a continuous function; we conclude that $C(\Omega) \otimes C(\Sigma)$ is contained in $C(\Omega \times \Sigma)$. Finally, $\mathbb{1}_{\Omega\times\Sigma} = \mathbb{1}_\Omega \otimes \mathbb{1}_\Sigma \in C(\Omega) \otimes C(\Sigma)$. □

In view of the preceding proposition, we will always consider $C(\Omega) \otimes C(\Sigma)$ as a subset of $C(\Omega \times \Sigma)$.

19.1.4 Proposition. *Let K and L be compact spaces. Then $C(K) \otimes C(L)$ is norm dense in $C(K \times L)$.*

Proof. From the preceding proposition, we already know that $C(K) \otimes C(L)$ is a unital subalgebra of $C(K \times L)$. To show that it is dense, we use Stone-Weierstrass Theorem 3.1.31; it is straightforward that functions of the form $f \otimes \mathbb{1}$ and $\mathbb{1} \otimes g$ in $C(K) \otimes C(L)$ separate points of $K \times L$. □

Recall that in Section 7.6 we introduced $S(\Omega)$, — the vector lattice of all real-valued continuous functions defined on open dense subsets of a Hausdorff topological space Ω. We will now prove an analogue of Proposition 19.1.3 for $S(\Omega)$ spaces.

19.1.5 Proposition. *Let Ω and Σ be two Hausdorff topological spaces. We define $j\colon S(\Omega) \times S(\Sigma) \to S(\Omega \times \Sigma)$ via $\big(j(f,g)\big)(s,t) = f(s)g(t)$, where $f \in S(\Omega)$, $g \in S(\Sigma)$, $s \in \operatorname{dom} f$, and $t \in \operatorname{dom} g$. Then $S(\Omega) \otimes S(\Sigma)$ may be identified with the span of $\operatorname{Range} j$ in $S(\Omega \times \Sigma)$, with j being the canonical tensor map.*

Proof. It is easy to see that $j(f,g)$ is indeed an element of $S(\Omega\times\Sigma)$, and that j is bilinear. We will verify the universal property. Let $\varphi\colon S(\Omega)\times S(\Sigma)\to Z$ be a bilinear map to some vector space Z.

$$\begin{tikzcd}
C(\Omega_0)\times C(\Sigma_0) \arrow[r, hook] \arrow[d, "j_0"'] & S(\Omega)\times S(\Sigma) \arrow[r, "\varphi"] \arrow[d, "j"'] & Z \\
C(\Omega_0)\otimes C(\Sigma_0) \arrow[r, hook] \arrow[d, hook] & \operatorname{span}\operatorname{Range} j \arrow[ur, dashed, "\varphi^{\otimes}"'] \arrow[d, hook] & \\
C(\Omega_0\times\Sigma_0) \arrow[r, hook] & S(\Omega\times\Sigma) &
\end{tikzcd}$$

We define $\varphi^{\otimes}$ from $\operatorname{span}\operatorname{Range} j$ to Z via $\varphi^{\otimes}\Bigl(\sum_{i=1}^n j(f_i,g_j)\Bigr)=\sum_{i=1}^n\varphi(f_i,g_i)$. We need to verify that this map is well defined; it will then follow immediately that it is linear and $\varphi^{\otimes}j=\varphi$, and that $\varphi^{\otimes}$ is uniquely determined by these properties; so the proof will then be complete.

Suppose that $\sum_{i=1}^n j(f_i,g_j)=0$; it suffices to show that $\sum_{i=1}^n\varphi(f_i,g_i)=0$. Let $\Omega_0=\bigcup_{i=1}^n\operatorname{dom} f_i$ and $\Sigma_0=\bigcup_{i=1}^n\operatorname{dom} g_i$. Then $f_1,\dots,f_n\in C(\Omega_0)$, $g_1,\dots,g_n\in C(\Sigma_0)$. We may view $C(\Omega_0)$ and $C(\Sigma_0)$ as subspaces of $S(\Omega)$ and $S(\Sigma)$, respectively. Let j_0 and φ_0 be the restrictions of j and φ to $C(\Omega_0)\times C(\Sigma_0)$, respectively. By Proposition 19.1.3, j_0 is the canonical tensor map for the tensor product $C(\Omega_0)\otimes C(\Sigma_0)$, where the latter may be viewed as a subspace of $C(\Omega_0\times\Sigma_0)$ spanned by $\operatorname{Range} j_0$. We conclude that there exists a linearization $\varphi_0^{\otimes}$ of φ_0 such that $\varphi_0^{\otimes}\colon C(\Omega_0)\otimes C(\Sigma_0)\to Z$ with $\varphi_0^{\otimes}j_0=\varphi_0$. It follows from $\sum_{i=1}^n j_0(f_i,g_j)=\sum_{i=1}^n j(f_i,g_j)=0$ that

$$0=\varphi_0^{\otimes}\Bigl(\sum_{i=1}^n j_0(f_i,g_j)\Bigr)=\sum_{i=1}^n\varphi_0(f_i,g_j)=\sum_{i=1}^n\varphi(f_i,g_j).$$

$\square$

19.1.6. Similar to the case of $C(\Omega)$ spaces, given $f\in S(\Omega)$ and $g\in S(\Sigma)$, we may view the elementary tensor $f\otimes g$ as the function in $S(\Omega\times\Sigma)$ given by $(f\otimes g)(s,t)=f(s)g(t)$, where $f\in S(\Omega)$, $g\in S(\Sigma)$, $s\in\operatorname{dom} f$, and $t\in\operatorname{dom} g$. In this notation, the preceding proposition says that $S(\Omega)\otimes S(\Sigma)$ may be identified with the subspace of $S(\Omega\times\Sigma)$ spanned by $\bigl\{f\otimes g\,:\,f\in S(\Omega),\ g\in S(\Sigma)\bigr\}$.

19.1.7 Example. Let $0<p\leqslant\infty$. Since ℓ_p is a (linear) subspace of $\mathbb{R}^{\mathbb{N}}$, we can use Proposition 19.1.2 to identify $\ell_p\otimes\ell_p$ with a subspace of $\mathbb{R}^{\mathbb{N}^2}$. We may view $\mathbb{R}^{\mathbb{N}^2}$ as a

space of infinite matrices. For $x, y \in \ell_p$, the elementary tensor $x \otimes y$ corresponds to the matrix $(x_i y_j)$. When $p < \infty$, it follows from

$$\sum_{i,j=1}^{\infty} |x_i y_j|^p = \Bigl(\sum_{i=1}^{\infty} |x_i|^p\Bigr)\Bigl(\sum_{j=1}^{\infty} |y_j|^p\Bigr) < \infty$$

that $x \otimes y$ is in $\ell_p(\mathbb{N}^2)$. Similarly, if $p = \infty$ then we clearly have $x \otimes y \in \ell_\infty(\mathbb{N}^2)$. In either case, $\|x \otimes y\| = \|x\|\|y\|$. Thus, we may identify $\ell_p \otimes \ell_p$ with the subspace of $\ell_p(\mathbb{N}^2)$ spanned by elementary tensors. Similarly, we may view $c_0 \otimes c_0$ as a subspace of $C_0(\mathbb{N}^2)$, spanned by elementary tensors.

We could not use the same trick to represent the tensor product of $L_p(\Omega, \mathcal{F}, \mu)$ with itself as a subspace of $\mathbb{R}^{\Omega^2}$ because its elements are equivalence classes rather then individual functions, so that $L_p(\Omega, \mathcal{F}, \mu)$ is not a subset of $\mathbb{R}^\Omega$. We will revisit this issue in Example 19.3.4.

Let φ be a bilinear map. Note that φ is never injective because $\varphi(x, 0) = 0 = \varphi(0, y)$ for all $x \in X$ and $Y \in Y$. We say that φ is ***bi-injective*** if $\varphi(x, y) = 0$ implies $x = 0$ or $y = 0$. The canonical tensor map $j\colon X \times Y \to X \otimes Y$ is bi-injective.

19.1.8 Exercise. Let φ be a bilinear map. Suppose that φ is bi-injective, does this imply that its linearization $\varphi^\otimes$ is injective?

19.1.9. The spaces $X \otimes Y$ and $X^\# \otimes Y^\#$ form a dual pair. We first define this action on elementary tensors via $\bigl\langle f \otimes g, x \otimes y\bigr\rangle = f(x)g(y)$, and then extend it by additivity:

$$\Bigl\langle \sum_{i=1}^{n} f_i \otimes g_i, \sum_{j=1}^{m} x_j \otimes y_j \Bigr\rangle = \sum_{i,j} f_i(x_j) g_i(y_j).$$

By 19.1.1, this action separates points of both spaces. In particular, we may view $X^\# \otimes Y^\#$ as a subspace of $(X \otimes Y)^\#$.

19.2 Vector lattice tensor product $X \bar{\otimes} Y$

The symbols X, Y, and Z will stand for (Archimedean) vector lattices. A bilinear map $\varphi\colon X \times Y \to Z$ is said to be ***positive*** if $\varphi(x, y) \geqslant 0$ whenever $x, y \geqslant 0$.

19.2.1 Exercise. Let $\varphi\colon X \times Y \to Z$ be a positive bilinear map. Then $\bigl|\varphi(x, y)\bigr| \leqslant \varphi\bigl(|x|, |y|\bigr)$ for all $x \in X$ and $y \in Y$.

A bilinear map $\varphi\colon X \times Y \to Z$ is a ***lattice bimorphism*** if $\bigl|\varphi(x, y)\bigr| = \varphi\bigl(|x|, |y|\bigr)$ for all $x \in X$ and $y \in Y$. Every lattice bimorphism is positive because if $x, y \geqslant 0$ then $0 \leqslant \bigl|\varphi(x, y)\bigr| = \varphi\bigl(|x|, |y|\bigr) = \varphi(x, y)$.

19.2.2 Exercise. A bilinear map $\varphi\colon X\times Y\to Z$ is a lattice bimorphism iff the maps $\varphi(x,\cdot)$ and $\varphi(\cdot,y)$ are lattice homomorphism for all $x,y>0$.

The definition of a vector lattice tensor product of vector lattices is similar to that of vector spaces, we just replace (bi)linear maps with lattice (bi)morphisms:

19.2.3 Theorem (Fremlin)**.** *Given two vector lattices X and Y, there exists a vector lattice $X\mathbin{\bar{\otimes}}Y$ and a lattice bimorphism $j\colon X\times Y\to X\mathbin{\bar{\otimes}}Y$ such that for every vector lattice Z and every lattice bimorphism $\varphi\colon X\times Y\to Z$ there exists a unique (linear) lattice homomorphism $\bar\varphi$ such that $\varphi=\bar\varphi\circ j$.*

$$\begin{tikzcd}
X\times Y \arrow[r, "\varphi"] \arrow[d, "j"'] & Z \\
X\mathbin{\bar{\otimes}}Y \arrow[ur, dotted, "\bar\varphi"'] &
\end{tikzcd}$$

The space $X\mathbin{\bar{\otimes}}Y$ is unique up to a lattice homomorphism that preserves the image of $X\times Y$. Furthermore, $X\mathbin{\bar{\otimes}}Y$ contains $X\otimes Y$ as a subspace, $\operatorname{Range} j\subseteq X\otimes Y$*, and j is the canonical tensor map when viewed as a map from $X\times Y$ to $X\otimes Y$.*

We call $X\mathbin{\bar{\otimes}}Y$ the ***vector lattice tensor product***of X and Y. It is sometimes also called the ***Riesz tensor product*** or the ***Fremlin tensor product***. The rest of the section will be a proof of the theorem (and more). Here is the plan of attack. First, we will prove the uniqueness clause in the theorem. Second, we will prove an analogue of the theorem with vector lattices replaced with $C(K)$-spaces. Next, we will prove the theorem in the "unital" case when X and Y have strong unit by reducing it to $C(K)$-spaces using Krein-Kakutani Representation. Next, we will prove that every bi-injective lattice bimorphism on $X\times Y$ yields a copy of $X\mathbin{\bar{\otimes}}Y$. This reduces the theorem to proving existence bi-injective lattice bimorphism. We will complete the proof by explicitly constructing such a bimorphism.

Uniqueness in the theorem. We employ a standard categorical argument. Suppose that there are vector lattices E and F and lattice bimorphisms $i\colon X\times Y\to E$ and $j\colon X\times Y\to F$ such that both pairs (E,i) and (F,j) satisfy the conditions of the theorem. Then there exist lattice homomorphisms $\bar\imath$ and $\bar\jmath$ such that the diagram on the left is commutative:

$$\begin{tikzcd}
X\times Y \arrow[r, "j"] \arrow[d, "i"'] & F \arrow[dl, shift left, "\bar\imath"] \\
E \arrow[ur, shift left, "\bar\jmath"] &
\end{tikzcd}
\qquad
\begin{tikzcd}
X\times Y \arrow[r, "i"] \arrow[d, "i"'] & E \\
E \arrow[ur, dotted, "id"'] &
\end{tikzcd}$$

Then $\bar\imath\bar\jmath\colon E\to E$ is a lattice homomorphism which preserves the range of i. On the other hand, applying the conditions of the theorem to the diagram on the right, we conclude that the identity map is the unique lattice homomorphism from E to itself that preserves the range of i. It follows that $\overline{i}\overline{j} = id_{|E}$. Similarly, $\overline{j}\overline{i} = id_{|F}$. Hence, $\overline{j}$ is a lattice isomorphism of E onto F that preserves the image of $X \times Y$.

19.2.1 Unital case

Throughout this subsection, K, L, and M stand for compact spaces. Recall that, by Proposition 4.1.1, real-valued lattice homomorphisms on $C(K)$ are precisely the non-negative scalar multiples of point evaluation functionals.

19.2.4 Proposition. *Let $\varphi\colon C(K) \times C(L) \to \mathbb{R}$ be a lattice bimorphism. Then there exist $s \in K$, $t \in L$, and $\lambda \in \mathbb{R}_+$ such that $\varphi(f,g) = \lambda f(s)g(t)$ for all $f \in C(K)$ and $g \in C(L)$.*

Proof. The map $\varphi(\cdot,\mathbb{1})$ is a real-valued lattice homomorphism on $C(K)$. Hence, there exist $s \in K$ and $\lambda \in \mathbb{R}_+$ such that $\varphi(f,\mathbb{1}) = \lambda f(s)$ for every $f \in C(K)$. Similarly, there exist $t \in L$ and $\mu \in \mathbb{R}_+$ such that $\varphi(\mathbb{1},g) = \mu g(t)$ for every $g \in C(L)$. Note that $\lambda = \varphi(\mathbb{1},\mathbb{1}) = \mu$. For every $f \in C(K)$ and $g \in C(L)$, we have

$$\begin{aligned}
&\Bigl|\varphi(f,g) - \lambda f(s)g(t)\Bigr| \\
&\quad = \Bigl|\varphi(f,g) - \varphi(f,\mathbb{1})g(t)\Bigr| = \Bigl|\varphi(f,g) - \varphi\bigl(f,g(t)\mathbb{1}\bigr)\Bigr| = \Bigl|\varphi\bigl(f,g-g(t)\mathbb{1}\bigr)\Bigr| \\
&\quad = \varphi\Bigl(\lvert f\rvert, \bigl\lvert g-g(t)\mathbb{1}\bigr\rvert\Bigr) \leqslant \varphi\Bigl(\lVert f\rVert\mathbb{1}, \bigl\lvert g-g(t)\mathbb{1}\bigr\rvert\Bigr) = \lVert f\rVert\varphi\Bigl(\mathbb{1}, \bigl\lvert g-g(t)\mathbb{1}\bigr\rvert\Bigr) \\
&\qquad\qquad = \lVert f\rVert \cdot \mu \cdot \bigl\lvert g-g(t)\mathbb{1}\bigr\rvert(t) = 0.
\end{aligned}$$

□

As in Proposition 19.1.3, we identify the algebraic tensor product $C(K) \otimes C(L)$ with a subspace of $C(K \times L)$.

19.2.5 Theorem. *Let $j\colon C(K) \times C(L) \to C(K) \otimes C(L)$ be the canonical tensor map. For every lattice bimorphism $\varphi\colon C(K) \times C(L) \to C(M)$ there exists a lattice homomorphism $T\colon C(K \times L) \to C(M)$ such that $\varphi = Tj$:*

$$\begin{tikzcd}
C(K) \times C(L) \arrow[r, "\varphi"] \arrow[d, "j"'] & C(M) \\
C(K \times L) \arrow[ur, dotted, "T"'] &
\end{tikzcd}$$

Proof. Fix $w \in M$. The map $\delta_w \circ \varphi$ is a real-valued lattice bimorphism. By the preceding proposition, there exist $s_w \in K$, $t_w \in L$, and $\lambda_w \in \mathbb{R}_+$ such that

$$\lambda_w f(s_w)g(t_w) = (\delta_w \circ \varphi)(f,g) = \bigl(\varphi(f,g)\bigr)(w).$$

Let $u = \varphi(\mathbb{1}, \mathbb{1})$. Then $u(w) = \lambda_w$.

Define $T\colon C(K \times L) \to \mathbb{R}^M$ via $(Th)(w) = u(w)h(s_w, t_w)$. It is straightforward that T is a lattice homomorphism. If $f \in C(K)$, $g \in C(L)$, and $w \in M$ then

$$\bigl(T(f \otimes g)\bigr)(w) = u(w)f(s_w)g(t_w) = \bigl(\varphi(f, g)\bigr)(w),$$

hence $\varphi = Tj$.

It is left to show that $\operatorname{Range} T \subseteq C(M)$. It follows from the definition of T that $\|Th\| \leqslant \|u\|\|h\|$ for every $h \in C(K \times L)$, hence T is bounded (with respect to the supremum norm). It follows from $T(f \otimes g) = \varphi(f, g) \in C(M)$ that $T\bigl(C(K) \otimes C(L)\bigr) \subseteq C(M)$. Since $C(K) \otimes C(L)$ is dense in $C(K \times L)$ by Proposition 19.1.4, we conclude that $\operatorname{Range} T \subseteq C(M)$. $\square$

Recall that by Krein-Kakutani's Representation Theorem 4.1.8, every vector lattice X with a strong unit u may be represented as a norm dense sublattice of $C(K)$ for some K, with u corresponding to $\mathbb{1}_K$.

19.2.6 Theorem. *Let X and Y be norm dense sublattices of $C(K)$ and $C(L)$ respectively, containing constant functions. Then $X \bar{\otimes} Y$ exists and may be represented as the sublattice of $C(K \times L)$ generated by $j(X, Y)$, where $j\colon C(K) \times C(L) \to C(K \times L)$ is the canonical tensor map; $j(f, g) = f \otimes g$.*

Proof. Let Z be a vector lattice and $\varphi\colon X \times Y \to Z$ a lattice bimorphism. Put $u = \varphi(\mathbb{1}, \mathbb{1})$. For every $f \in X$ and $g \in Y$, we have

$$\bigl|\varphi(f, g)\bigr| = \varphi\bigl(|f|, |g|\bigr) \leqslant \varphi\bigl(\|f\|\mathbb{1}, \|g\|\mathbb{1}\bigr) = \|f\|\|g\|u.$$

It follows that $\varphi(f, g) \in I_u$, hence $\operatorname{Range} \varphi \subseteq I_u$. Using Krein-Kakutani's Representation Theorem, we may represent I_u as a norm dense sublattice of $C(M)$ for some compact space M with u corresponding to $\mathbb{1}_M$. The preceding computation then shows that $\bigl\|\varphi(f, g)\bigr\| \leqslant \|f\|\|g\|$. It follows that φ is jointly continuous. Therefore, it extends to a lattice bimorphism from $C(K) \times C(L)$ to $C(M)$:

$$\begin{tikzcd}
X \times Y \arrow[r, "\varphi"] \arrow[d, hook] & I_u \arrow[r, hook] \arrow[d, hook] & Z \\
C(K) \times C(L) \arrow[r] \arrow[d] & C(M) & \\
C(K \times L) \arrow[ur, dotted, "T"'] & &
\end{tikzcd}$$

Let T be as in Theorem 19.2.5. Let E be the sublattice of $C(K \times L)$ generated by $j(X,Y)$. Then the restriction of T to E is a lattice homomorphism from E to $C(M)$ satisfying $Tj(x,y) = \varphi(x,y)$ for all $x \in X$ and $y \in Y$. Furthermore,

$$T(E) = T\big(\operatorname{Lat} j(X,Y)\big) = \operatorname{Lat} Tj(X,Y) = \operatorname{Lat}\varphi(X,Y) \subseteq I_u \subseteq Z.$$

So we may view $T_{|E}$ as a lattice homomorphism from E to Z.

It is left to show uniqueness. Let $S\colon E \to Z$ be a lattice homomorphism such that $Sj(x,y) = \varphi(x,y)$ for all $x \in X$ and $y \in Y$. In particular, $Sj(x,y) = Tj(x,y)$. Hence, S and T agree on $j(X,Y)$. Since both S and T are lattice homomorphisms, they agree on E. □

It follows that, under the setting of the theorem, $X \bar{\otimes} Y$ is the sublattice of $C(K \times L)$ generated by $X \otimes Y$, when the latter is viewed as a subspace of $C(K \times L)$.

Note that Theorem 19.2.5 is *not* a special case of Theorem 19.2.6 because the former yields a lattice homomorphism defined on all of $C(K \times L)$.

19.2.2 General case

Throughout this section, symbols X, Y, Z, and E stand for (Archimedean) vector latices. We will show that the sublattice spanned by the range of every bi-injective lattice bimorphism on $X \times Y$ yields a copy of $X \bar{\otimes} Y$. We will then show that such a bimorphism always exists. Together, these two facts will complete the proof of Fremlin's Theorem 19.2.3 on existence of vector lattice tensor products.

19.2.7 Lemma. *Suppose that X and Y have strong units; let $\varphi\colon X \times Y \to Z$ be a lattice bimorphism. Let $\bar{\varphi}\colon X \bar{\otimes} Y \to Z$ be the induced lattice homomorphism ($X \bar{\otimes} Y$ and $\bar{\varphi}$ exist by Theorem 19.2.6). If φ is bi-injective then $\bar{\varphi}$ is injective.*

Proof. As in Theorem 19.2.6, we represent $X \bar{\otimes} Y$ as a sublattice of $C(K \times L)$. Take a non-zero $a \in X \bar{\otimes} Y$. Find non-zero $f \in C(X)_+$ and $g \in C(Y)_+$ such that $f \otimes g \leqslant |a|$. By Exercise 3.2.12, X and Y are order dense in $C(K)$ and $C(L)$, respectively. Hence, there exist non-zero $x \in E_+$ and $y \in F_+$ such that $x \leqslant f$ and $y \leqslant g$. It follows from $0 \leqslant x \otimes y \leqslant f \otimes g \leqslant a$ that

$$\big|\bar{\varphi}(a)\big| = \bar{\varphi}\big(|a|\big) \geqslant \bar{\varphi}(x \otimes y) = \varphi(x,y) \neq 0.$$

□

19.2.8 Theorem. *Let $\psi\colon X \times Y \to E$ be a bi-injective lattice bimorphism. Then* $\operatorname{Lat}(\operatorname{Range}\psi)$ *is $X \bar{\otimes} Y$.*

Proof. Let $E_0 = \mathrm{Lat}(\mathrm{Range}\,\psi)$. We need to verify that E_0 satisfies the universal property. Let $\varphi\colon X \times Y \to Z$ be a lattice bimorphism. Fix positive $u \in X$ and $v \in Y$. Let $\varphi_{u,v}$ and $\psi_{u,v}$ be the restrictions of φ and ψ to $I_u \times I_v$, respectively. Let $E_{u,v} = \mathrm{Lat}(\mathrm{Range}\,\psi_{u,v})$.

$$\begin{tikzcd}
I_u \times I_v \arrow[r, "\varphi_{u,v}"] \arrow[d, "\psi_{u,v}"'] \arrow[dr, "j"] & Z \\
E_{u,v} \arrow[ur, dotted, "S_{u,v}"'] & I_u \bar{\otimes} I_v \arrow[l, dashed, "\bar{\psi}_{u,v}"] \arrow[u, dashed, "\bar{\varphi}_{u,v}"']
\end{tikzcd}$$

By Theorem 19.2.6, $I_u \bar{\otimes} I_v$ exists. Let $j\colon I_u \times I_v \to I_u \bar{\otimes} I_v$ be the canonical lattice bimorphism. Let $\bar{\varphi}_{u,v}$ and $\bar{\psi}_{u,v}$ be the induced lattice homomorphisms as in the diagram, so that $\varphi_{u,v} = \bar{\varphi}_{u,v} j$ and $\psi_{u,v} = \bar{\psi}_{u,v} j$. By Lemma 19.2.7, $\bar{\psi}_{u,v}$ is injective. It is also surjective. Indeed, since $\bar{\psi}_{u,v}$ is a lattice homomorphisms, $\mathrm{Range}\,\bar{\psi}_{u,v}$ is a sublattice. Since $\mathrm{Range}\,\bar{\psi}_{u,v}$ contains $\mathrm{Range}\,\psi_{u,v}$ it also contains $\mathrm{Lat}(\mathrm{Range}\,\psi_{u,v}) = E_{u,v}$. Hence, $\bar{\psi}_{u,v}$ is a lattice isomorphism.

Define $S_{u,v}\colon E_{u,v} \to Z$ via $S_{u,v} = \bar{\varphi}_{u,v}\bar{\psi}_{u,v}^{-1}$. By construction, S is a lattice homomorphism. Clearly, $S_{u,v}\psi_{u,v}(x,y) = \bar{\varphi}_{u,v}\bar{\psi}_{u,v}^{-1}\bar{\psi}_{u,v} j(x,y) = \varphi_{u,v}(x,y)$, hence $S_{u,v}\psi(x,y) = \varphi(x,y)$ for all $x \in I_u$ and $y \in I_v$. Recall that $E_{u,v} = \mathrm{Lat}(\mathrm{Range}\,\psi_{u,v})$; we conclude that $S_{u,v}$ is a unique lattice homomorphism from $E_{u,v}$ to Z that satisfies $S_{u,v}\psi(x,y) = \varphi(x,y)$ for all $x \in I_u$ and $y \in I_v$.

Suppose now that $0 \leqslant u' \leqslant u$ and $0 \leqslant v' \leqslant v$. Then $I_{u'} \subseteq I_u$, $I_{v'} \subseteq I_v$, $E_{u',v'}$ is a sublattice of $E_{u,v}$, and $S_{u',v'}$ is a unique lattice homomorphism from $E_{u',v'}$ to Z such that $S_{u',v'}\psi(x,y) = \varphi(x,y)$ for all $x \in I_{u'}$ and $y \in I_{v'}$. It follows that $S_{u',v'}$ equals the restriction of $S_{u,v}$ to $E_{u',v'}$.

This yields a family of lattice homomorphisms $(S_{u,v})$ indexed by $(u,v) \in X_+ \times Y_+$, with $\mathrm{dom}\, S_{u,v} = E_{u,v}$. The preceding paragraph implies that these operators agree on intersections of their domains. It is easy to see that union of all $E_{u,v}$'s is E_0. This yields a lattice homomorphism S from E_0 to Z such that $S\psi(x,y) = \varphi(x,y)$ for all $x \in X$ and $y \in Y$. The latter identity implies that S in uniquely determined on $\mathrm{Range}\,\psi$ and, therefore, on E_0. $\square$

We are now ready to complete the proof of Fremlin's Theorem 19.2.3. Let X and Y be two vector lattices. By Corollary 7.6.2, we may realize X and Y as sublattices of $S(K)$ and $S(L)$ for some K and L, respectively. Let $j\colon S(K) \times S(L) \to S(K \times L)$ be as in Proposition 19.1.5. Then j is a bi-injective lattice bimorphism. So the restriction of j to $X \times Y$ yields a bi-injective lattice bimorphism from $X \times Y$ to $S(K \times L)$. Theorem 19.2.8 tells us that $X \bar{\otimes} Y$ exists, and equals the sublattice of $S(K \times L)$

generated by $j(X \times Y)$. In particular, it contains the subspace of $S(K \times L)$ generated by $j(X \times Y)$, which we may identify with $X \otimes Y$, and the restriction of j to $X \times Y$ may also be viewed as the canonical tensor map from $X \times Y$ to $X \otimes Y$. Since $X \bar{\otimes} Y$ may be identified with a sublattice of $S(K \times L)$, and the latter is Archimedean, we conclude that $X \bar{\otimes} Y$ is Archimedean.

19.2.9 Remark. We use $S(K)$ and $S(L)$ rather than $C^\infty(K)$ and $C^\infty(L)$ in the proof because $K \times L$ need not be extremally disconnected and, therefore, $C^\infty(K \times L)$ may fail to be a vector lattice.

There are several alternative constructions of $X \bar{\otimes} Y$ in the literature. We mentioned in Section 19.1 that the algebraic tensor product of two vector spaces may be constructed as the quotient of the free vector space over $X \times Y$ over the subspace generated by differences of elements that "should be equal" in $X \otimes Y$. This is a special case of a more general construction in Universal Algebra. This method also provides an alternative construction of vector lattice tensor product via free vector lattices. Here is the outline: consider all formal lattice-linear expressions of $X \times Y$. This is itself a vector lattice, let's call it Z. Take the ideal J of it, generated by differences of expressions that should be equal in $X \bar{\otimes} Y$. E.g., $|x \times y| - |x| \times |x|$ should be there. Define $X \bar{\otimes} Y$ to be the quotient of Z over the uniform closure of J. We take the uniform closure of J instead of J itself to ensure that the resulting space is Archimedean; without the closure, this construction yields the tensor product for the category of all (not necessarily Archimedean) vector lattices.

While this approach provides a shorter construction of $X \bar{\otimes} Y$, it also has a major disadvantage. In the next section, we will use our "concrete" construction to deduce several important properties of $X \bar{\otimes} Y$. Deducing these properties from the "universal algebra" construction of $X \bar{\otimes} Y$ is much more difficult.

There is also a construction of $X \bar{\otimes} Y$ due to Grobler and Labuschagne that avoids functional representations; see [GL88].

19.3 Properties of $X \bar{\otimes} Y$

As before, symbols X, Y, and Z stand for Archimedean vector latices. We write $j\colon X \times Y \to X \bar{\otimes} Y$ for the canonical lattice bimorphism in Fremlin's Theorem 19.2.3. As before, we write $x \otimes y$ for $j(x, y)$. By Fremlin's Theorem 19.2.3,j agrees with the canonical tensor map from $X \times Y$ to $X \otimes Y$, so the notation $x \otimes y$ is unambiguous: $x \otimes y$ means the same object in $X \otimes Y$ and in $X \bar{\otimes} Y$. Since j is a lattice bimorphism, we have $\left|x \otimes y\right| = |x| \otimes |y|$; Furthermore, j is positive, hence $x \otimes y \geqslant 0$ whenever $x, y \geqslant 0$. Vector lattice tensor product "respects" sublattices:

19.3.1 Proposition. *Let X_0 and Y_0 be sublattices of X and Y, respectively. Then $X_0 \bar{\otimes} Y_0$ equals the sublattice of $X \bar{\otimes} Y$ generated by all $x \otimes y$ with $x \in X$ and $y \in Y$.*

Proof. Let $j\colon X \times Y \to X \bar{\otimes} Y$ be the canonical (bi-injective) lattice bi-morphism. The restriction j_0 of j to $X_0 \times Y_0$ is a bi-injective lattice bi-morphism. Hence, by Theorems 19.2.8,

$$X_0 \bar{\otimes} Y_0 = \mathrm{Lat}(\mathrm{Range}\, j_0) = \mathrm{Lat}\bigl\{x \otimes y \ :\ x \in X, y \in Y\bigr\}.$$

□

19.3.2 Example. As in Example 19.1.2, let $X = \mathbb{R}^A$ and $Y = \mathbb{R}^B$ for some sets A and B. Define $\varphi\colon \mathbb{R}^A \times \mathbb{R}^B \to \mathbb{R}^{A \times B}$ via $\bigl(\varphi(f, g)\bigr)(s, t) = f(s)g(t)$. Then φ is a bi-injective linear bimorphism. It follows from Proposition 19.2.8 that $\mathbb{R}^A \bar{\otimes} \mathbb{R}^B$ may be represented as the sublattice of $\mathbb{R}^{A \times B}$ generated by elementary tensors.

19.3.3 Example. Similarly, building upon Example 19.1.7, we can identify $\ell_p \bar{\otimes} \ell_p$ with the sublattice of $\ell_p(\mathbb{N}^2)$ generated by elementary tensors (here $0 < p \leqslant \infty$). So we may view $\ell_p \bar{\otimes} \ell_p$ as a space of infinite matrices.

19.3.4 Example. Consider spaces $L_p(\mu)$ and $L_p(\nu)$ and some $p \in [0, \infty]$. We will identify $L_p(\mu) \bar{\otimes} L_p(\nu)$ is a sublattice of $L_p(\mu \times \nu)$.

Let $f \in L_p(\mu)$ and $g \in L_p(\nu)$. Define $h(s, t) = f(s)g(t)$. It is an exercise in measure theory that this function is measurable in the product measure space. Indeed, WLOG, $f, g \geqslant 0$. Fix $z \in (0, \infty)$. Let $H = \bigl\{(x, y) \in \mathbb{R}^2_+ \ :\ xy > z\bigr\}$, and $H_0 = H \cap \mathbb{Q}^2$. It can be easily verified that $h(s, t) > z$ iff there exists $(x, y) \in H_0$ with $f(s) > x$ and $g(t) > y$. That is,

$$\{h > z\} = \bigcup_{(x,y) \in H_0} \{f > x\} \times \{g > y\}.$$

Being a countable union of measurable sets, this set is itself measurable. It follows that h is measurable. Furthermore, when $p > 0$, one can argue that h is in $L_p(\mu \times \nu)$; see, e.g., Exercise 18 on p. 272 of [Roy88].

Define $\varphi\colon L_p(\mu) \times L_p(\nu) \to L_p(\mu \times \nu)$ via $\varphi(f, g)(s, t) = f(s)g(t)$. It can be easily verified that φ is a bi-injective lattice bimorphism. By Theorem 19.2.8, $L_p(\mu) \bar{\otimes} L_p(\nu)$ may be identified with the sublattice of $L_p(\mu \times \nu)$ generated by the range of φ. As before, we write $f \otimes g = \varphi(f, g)$. Hence, $L_p(\mu) \bar{\otimes} L_p(\nu)$ is the sublattice of $L_p(\mu \times \nu)$ generated by the elementary tensors.

As a byproduct, this construction yields that $L_p(\mu) \otimes L_p(\nu)$ may be identified with the subspace of $L_p(\mu \times \nu)$ spanned by the elementary tensors.

19.3.5 Proposition. *If u and v are strong units in X and Y, respectively, then $u\otimes v$ is a strong unit in $X \mathbin{\bar{\otimes}} Y$.*

Proof. We represent X and Y as dense sublattices of $C(K)$ and $C(L)$, with u and v corresponding to $\mathbb{1}_K$ and $\mathbb{1}_L$, respectively. Then represent $X \mathbin{\bar{\otimes}} Y$ as a sublattice of $C(K\times L)$ as in Theorem 19.2.6. Then $u\otimes v$ becomes $\mathbb{1}_K\otimes\mathbb{1}_L=\mathbb{1}_{K\times L}$, which is a strong unit in $C(K\times L)$ and, therefore, in $X \mathbin{\bar{\otimes}} Y$. □

$X\otimes Y$ is order dense and majorizing in $X \mathbin{\bar{\otimes}} Y$:

19.3.6 Proposition. *Given $0<a\in X \mathbin{\bar{\otimes}} Y$, we have $0<x\otimes y\leqslant a\leqslant u\otimes v$ for some positive x, y, u, and v.*

Proof. Since elementary tensors are generating in $X \mathbin{\bar{\otimes}} Y$, we can express a as a lattice linear expression of $x_1\otimes y_1,\dots,x_n\otimes y_n$ for some $x_1,\dots,x_n\in X$ and $y_1,\dots,y_n\in Y$. Find $u\in X_+$ and $v\in Y_+$ such that $x_1,\dots,x_n\in I_u$ and $y_1,\dots,y_n\in I_v$. Then $x_i\otimes y_i\in I_u \mathbin{\bar{\otimes}} I_v$ for every i and, therefore, $a\in I_u \mathbin{\bar{\otimes}} I_v$. Since $u\otimes v$ is a strong unit in $I_u \mathbin{\bar{\otimes}} I_v$, we have $a\leqslant\lambda(u\otimes v)=(\lambda u)\otimes v$ for some $\lambda\geqslant 0$. This yields an upper estimate.

The proof of the lower estimate is similar to that of Lemma 19.2.7. Again, represent $I_u \mathbin{\bar{\otimes}} I_v$ as a sublattice of $C(K\times L)$ as in Theorem 19.2.6. Then a is a non-zero positive function in $C(K\times L)$. We can then find $f\in C(K)$ and $g\in C(L)$ with $0<f\otimes g\leqslant a$. Since I_u and I_v are order dense in $C(K)$ and $C(L)$, respectively, we find $x\in X$ and $y\in Y$ with $0<x\leqslant f$ and $0<y\leqslant g$. It follows that $0<x\otimes y\leqslant f\otimes g\leqslant a$. □

Next, we show that $X\otimes Y$ is "uniformly dense" in $X \mathbin{\bar{\otimes}} Y$, with an elementary tensor as a regulator:

19.3.7 Proposition. *For every $a\in X \mathbin{\bar{\otimes}} Y$ there exist $u\in X_+$ and $v\in Y_+$ such that for every $\varepsilon>0$ there exists $b\in X\otimes Y$ such that $\lvert a-b\rvert\leqslant\varepsilon(u\otimes v)$.*

Proof. Fix $a\in X \mathbin{\bar{\otimes}} Y$. As in the previous propositions, we may assume WLOG that X and Y have strong units u and v and are represented in $C(K)$ and $C(L)$ as in Theorem 19.2.6, with $u=\mathbb{1}_K$ and $v=\mathbb{1}_L$. Since $C(K)\otimes C(L)$ is norm dense in $C(K\times L)$ by Proposition 19.1.4, we can find $c\in C(K)\otimes C(L)$ such that $\lVert a-c\rVert<\frac{\varepsilon}{2}$. Then $c=\sum_{i=1}^n f_i\otimes g_i$ for some $f_1,\dots,f_n\in C(K)$ and $g_1,\dots,g_n\in C(L)$. It can be easily verified that the tensor map $(f,g)\to f\otimes g$ is jointly continuous on $C(K)\times C(L)$, hence the expression $\sum_{i=1}^n f_i\otimes g_i$ depends continuously on the parameters. Since X and Y are norm dense in $C(K)$ and $C(L)$, respectively, we can, for every i, approximate f_i and g_i with some x_i in X and y_i in Y, respectively, so that $\lVert b-c\rVert<\frac{\varepsilon}{2}$, where

$b=\sum_{i=1}^n x_i\otimes y_i$. It follows that $b\in X\otimes Y$ and $\|a-b\|<\varepsilon$, so that $|a-b|\leqslant\varepsilon\mathbb{1}_{K\times L}=\varepsilon(u\otimes v)$. ☐

19.3.8 Corollary. *Let $\varphi\colon X\times Y\to Z$ be a lattice bimorphism. Then φ is bi-injective then $\bar\varphi$ is injective.*

Proof. If φ is not injective then there exist non-zero x and y such that $0=\varphi(x,y)=\bar\varphi(x\otimes y)$, so that $\bar\varphi$ fails to be injective. Suppose now that φ is bi-injective. Let $0\neq a\in X\bar\otimes Y$. By Proposition 19.3.6, we have $0<x\otimes y\leqslant|a|$ for some $x\in X$ and $y\in Y$. It follows that x and y are non-zero and

$$\bigl|\bar\varphi(a)\bigr|=\bar\varphi\bigl(|a|\bigr)\geqslant\bar\varphi(x\otimes y)=\varphi(x,y)>0.$$

☐

Vector lattice tensor product "respects" disjointness:

19.3.9 Proposition. *Let $x_1,x_2\in X$ and $y_1,y_2\in Y$. If $x_1\perp x_2$ then $(x_1\otimes y_1)\perp(x_2\otimes y_2)$.*

Proof. Since the canonical tensor map j is a lattice bimorphism, we may assume that $x_1,x_2,y_1,y_2\geqslant 0$. Further, $j(\cdot,y_1+y_2)$ is a lattice homomorphism, so $x_1\otimes(y_1+y_2)\perp x_2\otimes(y_1+y_2)$. Finally, observe that $x_1\otimes y_1\leqslant x_1\otimes(y_1+y_2)$ and $x_2\otimes y_2\leqslant x_2\otimes(y_1+y_2)$. ☐

We now move on to some bad news. $X\bar\otimes Y$ does not inherit many nice properties of X and Y.

19.3.10 Example ([Th23])**.** *Even when X and Y are order complete, $X\bar\otimes Y$ need not be order complete.* Let K and L be two compact Hausdorff spaces such that both K and L are extremally disconnected, but $K\times L$ is not. Then $C(K)$ and $C(L)$ are order complete, while $C(K\times L)$ is not. We claim that $C(K)\bar\otimes C(L)$ is not order complete. Suppose, for the sake of contradiction, that $C(K)\bar\otimes C(L)$ is order complete. It follows that it is uniformly complete. As before, we may view $C(K)\bar\otimes C(L)$ as a sublattice of $C(K\times L)$. Since $\mathbb{1}_{K\times L}$ is a strong unit in $C(K)\bar\otimes C(L)$, we conclude that $C(K)\bar\otimes C(L)$ is complete with respect to the norm induced by $\mathbb{1}_{K\times L}$, which is exactly the supremum norm. It follows that $C(K)\bar\otimes C(L)$ is norm closed in $C(K\times L)$. Since $C(K)\otimes C(L)$ and, therefore, $C(K)\bar\otimes C(L)$ is norm dense in $C(K\times L)$, we conclude that $C(K)\bar\otimes C(L)=C(K\times L)$. But this is a contradiction because $C(K)\bar\otimes C(L)$ is order complete while $C(K\times L)$ is not.

19.4 Tensor cones

Being a subspace of $X \bar{\otimes} Y$, the vector space $X \otimes Y$ becomes an ordered vector space with respect to the induced order. This order is given by the cone $(X \otimes Y)_+ \coloneqq (X \bar{\otimes} Y)_+ \cap (X \otimes Y)$. On the other hand, there is another natural cone in $X \otimes Y$, namely, the one generated by positive elementary tensors:

$$X_+ \otimes Y_+ \coloneqq \Big\{\sum_{i=1}^n x_i \otimes y_i \;:\; x_1, \dots, x_n \in X_+,\; y_1, \dots, y_n \in Y_+\Big\}.$$

Since $x \otimes y$ is positive in $X \bar{\otimes} Y$ when $x, y \geqslant 0$, we have $X_+ \otimes Y_+ \subseteq (X \otimes Y)_+$. The following example, due to Fremlin, shows that these two cones need not be equal.

19.4.1 Example. Put $X = Y = C[0,1]$. Let $f(t) = e^t$ and $g(t) = e^{-t}$, and put $a = f \otimes g + g \otimes f - 2\mathbb{1}$. It is easy to see that $a \geqslant 0$ in $C[0,1]^2$, hence $a \in (X \otimes Y)_+$. We claim that $a \notin X_+ \otimes Y_+$. Suppose that $a = \sum_{i=1}^n f_i \otimes g_i$ for some $0 \leqslant f_1, \dots, f_n, g_1, \dots, g_n \in C[0,1]$. Let D be the diagonal of $[0,1]^2$. For every t, we have $a(t,t) = 0$, hence $f_i(t)g_i(t) = 0$ for every i. It follows that f_i and g_i have disjoint supports. Hence, if $(t,t) \in \partial \operatorname{supp} f_i \otimes g_i$ then $t \in \partial \operatorname{supp} f_i$. We conclude that the set $D \cap \partial \operatorname{supp} f_i \otimes g_i$ is nowhere dense in D for every i and, therefore, $D \cap \partial \operatorname{supp} u$ is nowhere dense in D. This is a contradiction as $\operatorname{supp} a = [0,1]^2 \setminus D$.

We will show that, nevertheless, $X_+ \otimes Y_+$ is dense in $(X \otimes Y)_+$ in a very strong sense. As before, we start with the $C(K)$ case. Recall that a function f in $C(K)$ is *strictly positive* if $f(t) > 0$ for all $t \in K$. Compactness of K implies that $f \geqslant \delta \mathbb{1}_K$ for some $\delta > 0$.

19.4.2 Theorem (Fremlin-Talagrand)**.** *Let K and L be compact Hausdorff spaces; X and Y norm dense sublattices of $C(K)$ and $C(L)$ containing constant functions, and $a \in X \otimes Y$. If a is strictly positive as function in $C(K \times L)$ then $a \in X_+ \otimes Y_+$.*

Proof. Let $a = \sum_{i=1}^n x_i \otimes y_i$ for some $x_1, \dots, x_n \in X$ and $y_1, \dots, y_n \in Y$. It follows from $a = \sum_{i=1}^n x_i^+ \otimes y_i + \sum_{i=1}^n x_i^- \otimes (-y_i)$ that we may assume WLOG that $x_i \geqslant 0$ for all i. Put $x = \sum_{i=1}^n x_i$. WLOG, scaling, we may assume that $\|x\| = 1$ and $\|y_i\| \leqslant 1$ for all i. Find $\delta > 0$ such that $a \geqslant \delta \mathbb{1}_{K \times L}$.

For each i, cover $\operatorname{Range} y_i$ with finitely many intervals of length less than δ. Using pre-images of these intervals under y_i, and taking all possible intersections if these pre-images as i runs from 1 to n yields an open cover $V_1, \dots, V_m$ of L such that for every $i = 1, \dots, n$ and every $j = 1, \dots, m$, the oscillation of y_i on V_j is less that δ, that is, $\sup_{V_j} y_i - \inf_{V_j} y_i < \delta$. WLOG, by dropping redundant V_j's, we may assume that the cover is minimal in the sense that no V_j is covered by the rest of V_j's.

Using Theorem 5.6.15, we find $z_1, \dots, z_m \in Y_+$ such that $\sum_{j=1}^m z_j = \mathbb{1}_L$ an $\overline{\operatorname{supp}}\, z_j \subseteq V_j$ for every j. For every i and j, put $\alpha_{i,j} = \inf_{V_j} y_i$ (note that $\alpha_{i,j}$ may be negative). We can re-write a as follows:

$$a = \sum_{i=1}^n x_i \otimes \Big(y_i - \sum_{j=1}^m \alpha_{i,j} z_j\Big) + \sum_{i=1}^n x_i \otimes \Big(\sum_{j=1}^m \alpha_{i,j} z_j\Big). \tag{19.2}$$

It suffices to show that both expressions in the right hand side are in $X_+ \otimes Y_+$.

Fix $t \in L$. For every j, if $z_j(t) > 0$ then $t \in V_j$, so that $\alpha_{i,j} \leqslant y_i(t) \leqslant \alpha_{i,j} + \delta$ for all i. It follows that for every i we have

$$\sum_{j=1}^m \alpha_{i,j} z_j(t) = \sum_{j=1}^m y_i(t) z_j(t) \leqslant y_i(t)$$

and

$$\sum_{j=1}^m \alpha_{i,j} z_j(t) \geqslant \sum_{j=1}^m \big(y_i(t) - \delta\big) z_j(t) = y_i(t) - \delta.$$

This yields $0 \leqslant v_i \leqslant \delta \mathbb{1}_L$, where $v_i := y_i - \sum_{j=1}^m \alpha_{i,j} z_j$. Put $b = \sum_{i=1}^n x_i \otimes v_i$. Clearly, $b \in X_+ \otimes Y_+$, and $b \leqslant \sum_{i=1}^n x_i \otimes (\delta \mathbb{1}_L) = \delta(x \otimes \mathbb{1}_L)$. Recall that $\|x\| = 1$. We conclude that $0 \leqslant b \leqslant \delta \mathbb{1}_{K \times L} \leqslant a$. Put $c := a - b$. Then $a = b + c$ is the decomposition in (19.2). Further,

$$0 \leqslant c = \sum_{i=1}^n x_i \otimes \Big(\sum_{j=1}^m \alpha_{i,j} z_j\Big) = \sum_{j=1}^m \Big(\sum_{i=1}^n \alpha_{i,j} x_i\Big) \otimes z_j = \sum_{j=1}^m u_j \otimes z_j,$$

where $u_j = \sum_{i=1}^n \alpha_{i,j} x_i$. It suffices to show that $u_j \geqslant 0$ for every j.

Fix j_0. By the minimality of the cover $V_1, \dots, V_m$, there exists $t_0 \in V_{j_0}$ such that $t_0 \notin V_j$ and, therefore, $z_j(t_0) = 0$ for all $j \neq j_0$. It follows that $z_{j_0}(t_0) = 1$. For every $s \in K$, we have

$$0 \leqslant c(s, t_0) = \sum_{j=1}^m u_j(s) z_j(t_0) = u_{j_0}(s).$$

We conclude that $u_{j_0} \geqslant 0$. □

The following theorem, which is due to Fremlin and Talagrand, shows that for $a \in (X \otimes Y)_+$ iff it is the end-point of a ray in $X_+ \otimes Y_+$. So $(X \otimes Y)_+$ may be viewed as the closure of $X_+ \otimes Y_+$ along (certain) rays.

19.4.3 Theorem. *Let $a \in X \otimes Y$. Then $a \in (X \bar{\otimes} Y)_+$ iff there exist $u, v \geqslant 0$ such that $a + r(u \otimes v) \in X_+ \otimes Y_+$ for all $r > 0$.*

Proof. Suppose that $a \in (X \mathbin{\bar{\otimes}} Y)_+$. As in the proof of Proposition 19.3.6, we find $u \in X_+$ and $v \in Y_+$ such that $a \in I_u \mathbin{\bar{\otimes}} I_v$ and $a \leqslant u \otimes v$. Here $I_u \mathbin{\bar{\otimes}} I_v$ is viewed as a sublattice of $X \mathbin{\bar{\otimes}} Y$. So a is positive in $I_u \mathbin{\bar{\otimes}} I_v$. Represent I_u and I_v as norm dense sublattices of some $C(K)$ and $C(L)$, such that u and v become $\mathbb{1}_K$ and $\mathbb{1}_L$, respectively. Then a may be viewed as a positive function in $C(K \times L)$, and $u \otimes v$ corresponds to $\mathbb{1}_{K\times L}$. For every $r > 0$, $a + r(u \otimes v) = a + r\mathbb{1}_{K\times L}$ is strictly positive, hence, by Theorem 19.4.2, it belongs to $(I_u)_+ \otimes (I_v)_+$ and, therefore, to $X_+ \otimes Y_+$.

To prove the converse, observe that $a + r(u \otimes v) \geqslant 0$ in $X \mathbin{\bar{\otimes}} Y$ for every $r > 0$. Since $X \mathbin{\bar{\otimes}} Y$ is Archimedean, passing to the limit as $r \to 0$ we get $a \geqslant 0$ in $X \mathbin{\bar{\otimes}} Y$. □

19.4.4 Corollary. *$X_+ \otimes Y_+$ is uniformly dense in $(X \mathbin{\bar{\otimes}} Y)_+$.*

19.4.5 Proposition. *$C(K)_+ \otimes C(L)_+$ is norm dense in $C(K \times L)_+$.*

Proof. Let $a \in C(K \times L)_+$ and $\varepsilon > 0$. Since $C(K) \mathbin{\bar{\otimes}} C(L)$ is norm dense in $C(K \times L)$, there exists $b \in C(K) \mathbin{\bar{\otimes}} C(L)$ such that $\|a - b\| < \varepsilon$. Replacing b with b^+, if necessary, we may assume that $b \in \big(C(K) \mathbin{\bar{\otimes}} C(L)\big)_+$. Since uniform convergence implies norm convergence, by Corollary 19.4.4 we can find $c \in C(K)_+ \otimes C(L)_+$ such that $\|b - c\| < \varepsilon$ and, therefore, $\|a - c\| < 2\varepsilon$. □

19.5 Positive bilinear maps

We will show in this section that the universal property in Fremlin's Theorem 19.2.3 remain valid if lattice bi/homo-morphisms are replaced with positive maps, as long as the co-domain is uniformly complete. The results are based on work by Fremlin. As before, we first tackle the $C(K)$ case.

Recall from Section 19.4 that $a \in C(K) \otimes C(L)$ and $a \in C(K \times L)_+$ need not imply $a \in C(K)_+ \otimes C(L)_+$. The following result is an analogue of Theorem 19.2.5, but with lattice bi/homomorphisms replaced with positive maps.

19.5.1 Theorem. *For every positive bilinear map $\varphi\colon C(K) \times C(L) \to C(M)$, its linearization $\varphi^{\otimes}$ extends uniquely to a positive linear operator $T\colon C(K \times L) \to C(M)$.*

Proof. It is clear that $\varphi^{\otimes}(a) \geqslant 0$ for every $a \in C(K)_+ \otimes C(L)_+$. We claim that $\varphi^{\otimes}(a) \geqslant 0$ for every $a \in (C(K) \otimes C(L))_+$. Indeed, a is positive as a function in $C(K \times L)$. For every $r > 0$, the function $a + r\mathbb{1}$ is strictly positive, where $\mathbb{1} = \mathbb{1}_{K\times L}$. By Theorem 19.4.2, $a + r\mathbb{1} \in C(K)_+ \otimes C(L)_+$. It follows that $0 \leqslant \varphi^{\otimes}(a + r\mathbb{1}) = \varphi^{\otimes}(a) + r\varphi(\mathbb{1}_K, \mathbb{1}_L)$. Taking the limit as $r \to 0$, we conclude that $\varphi^{\otimes}(a) \geqslant 0$. This proves the claim.

Let now $a \in C(K)\otimes C(L)$, viewed as an element of $C(K\times L)$. If $\|a\| \leqslant 1$ then $-\mathbb{1} \leqslant a \leqslant \mathbb{1}$ with respect to the order in $C(K\times L)$. It follows that $-\varphi^{\otimes}(\mathbb{1}) \leqslant \varphi^{\otimes}(a) \leqslant \varphi^{\otimes}(\mathbb{1})$. We conclude that $\varphi^{\otimes}$ is bounded with respect to the supremum norms of $C(K\times L)$ and $C(M)$, with $\|\varphi^{\otimes}\| \leqslant \|\varphi^{\otimes}(\mathbb{1})\|$. Since $C(K)\otimes C(L)$ is dense in $C(K\times L)$, $\varphi^{\otimes}$ extends to a continuous operator $T\colon C(K\times L)\to C(M)$.

Since T is clearly positive on $C(K)_+\otimes C(L)_+$, it follows from Proposition 19.4.5 that $T \geqslant 0$. The uniqueness clause follows from the fact that T is continuous and agrees with $\varphi^{\otimes}$ on $C(K)\otimes C(L)$, and the latter is dense in $C(K\times L)$. □

By Riesz-Markov Theorem 8.4.1, positive linear functionals on $C(K)$ correspond to finite Baire measures on K. We now extend this result to bilinear functionals: applying Theorem 19.5.1 with $C(M)=\mathbb{R}$, we see that positive bilinear functionals on $C(K)\times C(L)$ correspond to positive linear functional on $C(K\times L)$ and, therefore, by Riesz-Markov Theorem 8.4.1, to finite Baire measure on $K\times L$:

19.5.2 Corollary. *A map $\varphi\colon C(K)\times C(L)\to\mathbb{R}$ is positive bilinear iff there exists a finite Baire measure μ on $K\times L$ such that $\varphi(f,g)=\int f(s)g(t)\,d\mu(s,t)$ for all $f\in C(K)$ and $g\in C(L)$.*

We are now ready to prove that $X\bar{\otimes}Y$ satisfies the universal property of Fremlin's Theorem 19.2.3 even when bi/homomorphisms are replaced with positive (bi)linear maps, as long as the destination space is uniformly complete.

19.5.3 Theorem. *Suppose that Z is uniformly complete and $\varphi\colon X\times Y\to Z$ is a positive bilinear map. Then there exists a unique positive linear operator $\bar{\varphi}\colon X\bar{\otimes}Y\to Z$ such that $S(x\otimes y)=\varphi(x,y)$ for all $x\in X$ and $y\in Y$.*

Proof. Fix $u\in X_+$ and $v\in Y_+$. Put $w=\varphi(u,v)$. Let $\varphi_{u,v}$ be the restriction of φ to $I_u\times I_v$. For every $x\in I_u$ and $y\in I_v$, Exercise 19.2.1 yields

$$\bigl|\varphi(x,y)\bigr| \leqslant \varphi\bigl(|x|,|y|\bigr) \leqslant \varphi\bigl(\|x\|_u u,\|y\|_v V\bigr) = \|x\|_u\|y\|_v w. \tag{19.3}$$

It follows that $\varphi_{u,v}$ acts into I_w. As before, we can find compact spaces K, L, and M such that I_u, I_v, and I_v are norm dense sublattices of $C(K)$, $C(L)$, and $C(M)$, respectively, with u, v, and w corresponding to $\mathbb{1}_K$, $\mathbb{1}_L$, and $\mathbb{1}_M$, respectively. Since Z is uniformly complete, we actually have $I_w=C(M)$. It follows from (19.3) that $\bigl\|\varphi_{u,v}(x,y)\bigr\| \leqslant \|x\|\|y\|$. In particular, $\varphi_{u,v}$ is norm continuous. It can be easily verified that $\varphi_{u,v}$ extends uniquely to a positive bilinear map $\psi\colon C(K)\times C(L)\to C(M)$. By Corollary 19.5.1, there exists a unique positive operator $T\colon C(K\times L)\to C(M)$ such that $T(f \otimes g) = \psi(f,g)$ for all $f \in C(K)$ and $g \in C(L)$. Let $S_{u,v}$ be the restriction of T to $I_u \bar{\otimes} I_v$.

Thus, for every $u \in X_+$ and $v \in Y_+$ there is a unique positive linear operator $S_{u,v}\colon I_u \bar{\otimes} I_v \to Z$ such that $S_{u,v}(x \otimes y) = \varphi(x,y)$ for all $x \in I_u$ and $y \in I_v$. The uniqueness clause ensures that as u and v run over X_+ and Y_+, these operators $S_{u,v}$ agree on intersections of their domains. Therefore, "gluing" them together, we get an operator $\bar{\varphi}\colon X \bar{\otimes} Y \to Z$ with the required properties. □

It is clear that in the special case when φ is a lattice bimorphism the operator $\bar{\varphi}$ constructed in Theorem 19.5.3 is exactly $\bar{\varphi}$ from Fremlin's Theorem 19.2.3, this follows from the uniqueness clauses in both theorems. So our $\bar{\varphi}$ notation is consistent.

19.5.4 Example. The following example, due to Buskes, shows that the condition that Z is uniformly complete in Theorem 19.5.3 cannot be removed. Define $S\colon C\bigl([0,1]^2\bigr) \to C[0,1]$ via $(Sa)(t) = \int_0^1 a(s,t)\,ds$. It is clear that S is a positive linear operator. Define $\varphi\colon C[0,1]\times C[0,1] \to C[0,1]$ via $\varphi(f,g) = S(f\otimes g) = \bigl(\int_0^1 f(s)\,ds\bigr)g$. Then φ is positive and bilinear. Let $\bar{\varphi}\colon C[0,1]\bar{\otimes}C[0,1] \to C[0,1]$ be as in Theorem 19.5.3. By uniqueness, $\bar{\varphi}$ agrees to the restriction of S to $C[0,1] \bar{\otimes} C[0,1]$, where the latter space is viewed as a sublattice of $C\bigl([0,1]^2\bigr)$.

Let X be the sublattice of $C[0,1]$ consisting of all piece-wise affine functions in $C[0,1]$. It is easy to see that X fails to be uniformly complete. Then $X \bar{\otimes} X$ is a sublattice of $C[0,1] \bar{\otimes} C[0,1]$ and, therefore, of $C\bigl([0,1]^2\bigr)$. Note that if $f,g \in X$ then $\varphi(f,g) \in X$. Let $\varphi_0\colon X \times X \to X$ be the restriction of φ to $X \times X$. Suppose, for the sake of contradiction, that there exists $\bar{\varphi}_0\colon X \bar{\otimes} X \to X$ as in Theorem 19.5.3. Then, by uniqueness, $\bar{\varphi}_0$ must be the restriction of S to $X \bar{\otimes} X$.

Take $f,g \in X$ given by $f(s) = s$ and $g(t) = t$. Put $a \in X\bar{\otimes}X$ by $a = \bigl(f\otimes g - \frac{1}{2}\mathbb{1}\bigr)^+$, i.e., $a(s,t) = \bigl(st - \frac{1}{2}\bigr)^+$. It is easy to verify that Sa is not piece-wise affine, so $\bar{\varphi}_0(a) = Sa \notin X$; a contradiction.

Recall that $X^\sim$ separates points if for every $0 < x \in X$ there exists $0 < f \in X^\sim$ such that $f(x) \neq 0$; cf. Exercise 6.6.16.

19.5.5 Corollary. *If $X^\sim$ and $Y^\sim$ separate points then $(X \bar{\otimes} Y)^\sim$ separates points.*

Proof. Suppose that $0 < a \in X \bar{\otimes} Y$. By Proposition 19.3.6, we find $0 < x_0 \in X$ and $0 < y_0 \in Y$ such that $x_0 \otimes y_0 \leqslant a$. Find $0 < f \in X^\sim$ and $0 < g \in Y^\sim$ such that $f(x_0) > 0$ and $g(y_0) > 0$. Define $\varphi\colon X \times Y \to \mathbb{R}$ via $\varphi(x,y) = f(x)g(y)$. Clearly, φ is a positive bilinear map. It follows that $\bar{\varphi}$ is a positive linear functional on $X \bar{\otimes} Y$, and

$$\bar{\varphi}(a) \geqslant \bar{\varphi}(x_0 \otimes y_0) = \varphi(x_0, y_0) = f(x_0)g(y_0) > 0.$$

□

Recall that $(X \otimes Y)_+ = (X \bar{\otimes} Y)_+ \cap (X \otimes Y)$. We now get another characterization of this cone:

19.5.6 Theorem. *Suppose that $X^\sim$ and $Y^\sim$ separate points. For $a \in X \times Y$, we have $a \in (X \bar{\otimes} Y)_+$ iff $(f \otimes y)(a) \geqslant 0$ for all $f \in X^\sim_+$ and $g \in Y^\sim_+$.*

Proof. Suppose that $a \in (X \bar{\otimes} Y)_+$ and $f \in X^\sim_+$ and $g \in Y^\sim_+$. For $x \in X$ and $y \in Y$, define $\varphi(x, y) = f(x)g(y)$. Then $\varphi^\otimes = f \otimes g$. By Theorem 19.5.3, $\bar{\varphi}$ is a positive linear functional on $X \bar{\otimes} Y$. It follows that $0 \leqslant \bar{\otimes}\varphi(a) = \varphi^\otimes(a) = (f \otimes g)(a)$.

Suppose now that $(f \otimes y)(a) \geqslant 0$ for all $f \in X^\sim_+$ and $g \in Y^\sim_+$. Write $a = \sum_{i=1}^n x_i \otimes y_i$. Find $u \in X_+$ and $v \in Y_+$ such that $x_1, \dots, x_n \in I_u$ and $y_1, \dots, y_n \in I_v$. By Krein-Kakutani's Representation Theorem, we can find compact spaces K and L such that I_u and I_v are sublattices of $C(K)$ and $C(L)$, respectively.

For all $f \in X^\sim_+$ and $g \in Y^\sim_+$, we have

$$0 \leqslant (f \otimes g)(a) = \sum_{i=1}^n f(x_i)g(y_i) = f\Bigl(\sum_{i=1}^n g(y_i)x_i\Bigr).$$

Since $X^\sim$ separates points, it follows that for every $g \in Y^\sim_+$ the expression $\sum_{i=1}^n g(y_i)x_i$ is positive in X, hence in I_u, hence in $C(K)$. Therefore, for every $s \in K$ we have $0 \leqslant \sum_{i=1}^n g(y_i)x_i(s) = g\Bigl(\sum_{i=1}^n x_i(s)y_i\Bigr)$. Since this holds for and $g \in Y^\sim_+$, we conclude that the expression $\sum_{i=1}^n x_i(s)y_i$ is positive in Y_+, hence in I_v, hence in $C(L)$. Therefore, for every $t \in L$, we have $0 \leqslant \sum_{i=1}^n x_i(s)y_i(t) = a(s, t)$. We conclude that a is positive in $C(K) \bar{\otimes} C(L)$, hence in $I_u \bar{\otimes} I_v$, hence in $X \bar{\otimes} Y$. □

We saw that the tensor product of vector spaces has the property that $(X \otimes Y)^\#$ is linearly isomorphic to $\mathcal{L}(X, Y^\#)$. We now prove a somewhat similar result for vector lattices. Since $X \bar{\otimes} Y$ is a vector lattice, so is its order dual $(X \bar{\otimes} Y)^\sim$. On the other hand, since $Y^\sim$ is order complete, the space $\mathcal{L}_r(X, Y^\sim)$ is also a vector lattice.

19.5.7 Proposition. *$(X \bar{\otimes} Y)^\sim$ is lattice isomorphic to $\mathcal{L}_r(X, Y^\sim)$.*

Proof. First, we construct a map τ between the positive cones. Given $\xi \in (X \bar{\otimes} Y)^\sim_+$, we define a positive operator $T \colon X \to Y^\sim$ via $(Tx)(y) = \xi(x \otimes y)$. Denote this T by $\tau\xi$. Conversely, given a positive operator $T \in \mathcal{L}(X, Y^\sim)$, we define $\varphi(x, y) \coloneqq \langle Tx, y\rangle$; this yields a positive bilinear functional on $X \otimes Y$. Put $\xi = \bar{\varphi}$ as in Theorem 19.5.3; then $\xi \in (X \bar{\otimes} Y)^\sim_+$, and it is clear that $T = \tau\xi$. It follows that τ is a bijection between $\xi \in (X \otimes Y)^\sim_+$ and $\mathcal{L}(X, Y^\sim)_+$. It is easy to see that τ is additive. By Corollary 6.2.3, τ extends to a lattice isomorphism between $(X \bar{\otimes} Y)^\sim$ and $\mathcal{L}_r(X, Y^\sim)$ via $\tau\xi = \tau(\xi^+) - \tau(\xi^-)$. □

19.5.8 Example (Kusraev). *A lattice bimorphism φ such that $\varphi(x,\cdot)$ and $\varphi(\cdot,y)$ are order continuous for all $x,y \geqslant 0$, yet $\bar{\varphi}$ fails to be order continuous.* Take $X = Y = L_\infty[0,1]$. As in Example 19.3.4, we view $L_\infty[0,1] \bar{\otimes} L_\infty[0,1]$ as a sublattice of $L_\infty[0,1]^2$, generated by elementary tensors $(f\otimes g)(s,t) = f(s)g(t)$. Define $\varphi\colon L_\infty[0,1]\times L_\infty[0,1] \to L_\infty[0,1]$ via $\varphi(f,g)(s) = f(s)g(s)$. It can be easily verified that φ is a lattice bimorphism and $\varphi(f,\cdot)$ and $\varphi(\cdot,g)$ are order continuous for all $f,g \geqslant 0$. It is easy to see that for $h \in L_\infty[0,1] \bar{\otimes} L_\infty[0,1]$, $\bar{\varphi}(h)$ is the restriction of h to the diagonal. Note that while this operation is not well-defined on $L_\infty[0,1]^2$ because the diagonal has measure zero, it is well defined for elementary tensors and, therefore, for every $h \in L_\infty[0,1] \bar{\otimes} L_\infty[0,1]$.

For $n \in \mathbb{N}$, let h_n be the characteristic function of the set of n squares of size $\frac{1}{n} \times \frac{1}{n}$ covering the diagonal. That is, $h = \mathbb{1}_A$ where $A = \bigcup_{k=1}^n [\frac{k-1}{n}, \frac{k}{n}] \times [\frac{k-1}{n}, \frac{k}{n}]$. Clearly, $h_n \in L_\infty[0,1] \otimes L_\infty[0,1]$. Furthermore, $h_n \downarrow 0$ in $L_\infty[0,1]^2$ and, therefore, in $L_\infty[0,1] \bar{\otimes} L_\infty[0,1]$. Nevertheless $\bar{\varphi}(h_n) = \mathbb{1}$ for every n. This shows that $\bar{\varphi}$ fails to be order continuous.

19.6 Positive projective tensor product

We start with an quick overview of the projective tensor product of Banach spaces. For further details, we refer the reader to [Ryan02]. Let X, Y, and Z be Banach spaces. The ***projective tensor product*** $X \otimes_\pi Y$ is defined as the completion of $X \otimes Y$ with respect to the ***projective tensor norm***:

$$\|a\|_\pi = \inf\Big\{\sum_{i=1}^n \|x_i\|\|y_i\| \ :\ a = \sum_{i=1}^n x_i \otimes y_i,\ x_1,\dots,x_n \in X,\ y_1,\dots,y_n \in Y\Big\}$$

for $a \in X \otimes Y$. This is a ***cross-norm*** in the sense that $\|x \otimes y\|_\pi = \|x\|\|y\|$ for all $x \in X$, $y \in Y$. For every $a \in X \otimes_\pi Y$, we have

$$\|a\|_\pi = \inf\Big\{\sum_{i=1}^\infty \|x_i\|\|y_i\| \ :\ a = \sum_{i=1}^\infty x_i \otimes y_i,\ (x_i) \text{ in } X,\ (y_i) \text{ in } Y\Big\}.$$

The unit ball of $X \otimes_\pi Y$ is the closed convex hull of the set $B_X \otimes B_Y$.

A bilinear map φ from $X \times Y$ to a Banach space Z is said to be bounded if there exists $C > 0$ such that $\big\|\varphi(x,y)\big\| \leqslant C\|x\|\|y\|$. The infimum of such C's is denoted $\|\varphi\|$. It is easy to see that

$$\|\varphi\| = \sup\Big\{\big|\varphi(x,y)\big| \ :\ x \in B_X,\ y \in B_Y\Big\}.$$

This is a norm on the space $B(X,Y;Z)$ of all bounded bilinear maps from $X \times Y$ to Z. Naturally, we write $B(X,Y) = B(X,Y;\mathbb{R})$.

19.6.1. $B(X,Y)$ is isometric to $L(X,Y^*)$. Indeed, for $\varphi \in B(X,Y)$, the map $T\colon x \mapsto \varphi(x,\cdot)$ is in $L(X,Y^*)$. Conversely, for $T \in L(X,Y^*)$, the map $(x,y) \mapsto (Tx)(y)$ is in $B(X,Y)$. Furthermore,

$$\|T\| = \sup_{x\in B_X} \|Tx\| = \sup\Big\{ \big\|\varphi(x,\cdot)\big\| \;:\; x \in B_X \Big\}$$
$$= \sup\Big\{ \big\|\varphi(x,y)\big\| \;:\; x \in B_X,\ y \in B_Y \Big\} = \|\varphi\|.$$

The space $X \otimes_\pi Y$ is characterized by the following universal property: for every Banach space Z and every bounded bilinear map $\varphi\colon X \times Y \to Z$, the linearization $\varphi^\otimes\colon X \otimes Y \to Z$ is bounded with respect to $\|\cdot\|_\pi$ and, therefore, extends uniquely to a bounded operator φ^π from $X \otimes_\pi Y$ to Z. The map $\varphi \mapsto \varphi^\pi$ is an isometry between $B(X,Y;Z)$ and $L(X\otimes_\pi Y, Z)$. Taking $Z = \mathbb{R}$, we conclude that $B(X,Y)$ is isometric to $(X\otimes_\pi Y)^*$. Combining this isometry with the isometry between $B(X,Y)$ and $L(X,Y^*)$ in the preceding paragraph, we get an isometry between $(X \otimes_\pi Y)^*$ and $L(X,Y^*)$. It follows that $X \otimes_\pi Y$ may be viewed as a closed subspace of $L(X,Y^*)^*$.

For every measure μ, $X \otimes_\pi L_1(\mu)$ can be identified with $L_1(\mu, X)$, the space of all X-valued Bochner integrable functions.

For the rest of the section, let X, Y and Z be Banach lattices. Recall that every positive operator between Banach lattices is bounded.

19.6.2 Proposition. *Every positive bilinear map $\varphi\colon X \times Y \to Z$ is bounded.*

Proof. For every $x \in X$, put $\varphi_x = \varphi(x,\cdot)$. If $x \geqslant 0$ then φ_x is a positive linear operator from Y to Z; hence it is bounded. It follows that for every $x \in X$, $\varphi_x \in L(Y,Z)$. Consider the map $R\colon X \to L(Y,Z)$ defined via $Rx = \varphi_x$. It suffices to show that R is continuous because then $\big\|\varphi(x,y)\big\| = \big\|(Rx)(y)\big\| \leqslant \|R\|\|x\|\|y\|$ for all $x \in X$ and $y \in Y$.

Suppose that $x_n \to 0$ in X; put $T_n = Rx_n$; we need to show that $T_n \to 0$ in the operator norm. It suffices to show that (T_n) has a subsequence that converges to zero (by Proposition A.2.1). So, passing to a subsequence, we may assume that $x_n \xrightarrow{\mathrm{u}} 0$ with some regulator a. Put $S = Ra$. Clearly, S is a positive operator. Fix $\varepsilon > 0$. Find n_0 such that $-\varepsilon a \leqslant x_n \leqslant \varepsilon a$ for all $n \geqslant n_0$. Then $-\varepsilon S \leqslant T_n \leqslant \varepsilon S$, hence $-\varepsilon Sy \leqslant T_n y \leqslant \varepsilon Sy$ for every $y \in Y_+$ and, therefore, $\|T_n y\| \leqslant \varepsilon\|S\|\|y\|$; it follows that $\|T_n\| \leqslant \varepsilon\|S\|$. We conclude that $T_n \to 0$. $\square$

We define ***Fremlin projective tensor norm*** on $X \bar{\otimes} Y$ via

$$\|a\|_{|\pi|} = \inf\Big\{ \sum_{i=1}^n \|x_i\|\|y_i\| \;:\; |a| \leqslant \sum_{i=1}^n x_i \otimes y_i,\ x_1,\dots,x_n \in X_+,\ y_1,\dots,y_n \in Y_+ \Big\}.$$

19.6.3 Proposition. *$\|\cdot\|_{|\pi|}$ is a cross-norm and a lattice norm on $X \bar{\otimes} Y$.*

Proof. By Proposition 19.3.6, for every $a \in X \bar{\otimes} Y$ there exist $x \in X_+$ and $y \in Y_+$ with $|a| \leqslant x \otimes y$; it follows that $\|a\|_{|\pi|}$ is finite. It also follows immediately from the definition of $\|\cdot\|_{|\pi|}$ that if $|a| \leqslant |b|$ in $X \bar{\otimes} Y$ then $\|a\|_{|\pi|} \leqslant \|b\|_{|\pi|}$.

We will show next that $\|x_0 \otimes y_0\|_{|\pi|} = \|x_0\|\|y_0\|$ for all $x_0 \in X$ and $y_0 \in Y$. First, suppose that $x_0, y_0 \geqslant 0$. It follows immediately from the definition of $\|\cdot\|_{|\pi|}$ that $\|x_0 \otimes y_0\|_{|\pi|} \leqslant \|x_0\|\|y_0\|$. To prove the opposite inequality, fix $f \in S_X^+$ and $g \in S_Y^+$ with $f(x_0) = \|x_0\|$ and $g(y_0) = \|y_0\|$. Define $\varphi \colon X \times Y \to \mathbb{R}$ via $\varphi(x,y) = f(x)g(y)$. Clearly, φ is bilinear and positive. Let $\bar{\varphi} \colon X \bar{\otimes} Y \to \mathbb{R}$ be as in Theorem 19.5.3. If $x_0 \otimes y_0 \leqslant \sum_{i=1}^n x_i \otimes y_i$ for some $x_1, \dots, x_n \in X_+$ and $y_1, \dots, y_n \in Y_+$ then

$$\begin{aligned}\|x_0\|\|y_0\| &= f(x_0)g(y_0) = \varphi(x_0,y_0) = \bar{\varphi}(x_0 \otimes y_0)\\ &\leqslant \bar{\varphi}\Big(\sum_{i=1}^n x_i \otimes y_i\Big) = \sum_{i=1}^n \varphi(x_i,y_i) = \sum_{i=1}^n f(x_i)g(y_i) \leqslant \sum_{i=1}^n \|x_i\|\|y_i\|.\end{aligned}$$

It follows that $\|x_0\|\|y_0\| \leqslant \|x_0 \otimes y_0\|_{|\pi|}$. Hence, $\|x_0\|\|y_0\| = \|x_0 \otimes y_0\|_{|\pi|}$.

Now for all $x_0 \in X$ and $y_0 \in Y$, it follows from $|x_0 \otimes y_0| = |x_0| \otimes |y_0|$ that

$$\|x_0 \otimes y_0\|_{|\pi|} = \big\||x_0 \otimes y_0|\big\|_{|\pi|} = \big\||x_0| \otimes |y_0|\big\|_{|\pi|} = \|x_0\|\|y_0\|.$$

It is left to verify that $\|\cdot\|_{|\pi|}$ is a norm. One can easily verify that $\|\lambda a\|_{|\pi|} = |\lambda|\|a\|_{|\pi|}$. If $a \neq 0$ then, by Theorem 19.5.3, there exist non-zero $x \in X_+$ and $y \in Y_+$ such that $x \otimes y \leqslant |a|$. It follows that $\|a\|_{|\pi|} \geqslant \|x \otimes y\|_{|\pi|} = \|x\|\|y\| > 0$. We leave the triangle inequality as an exercise. □

19.6.4 Exercise. Verify that $\|\cdot\|_{|\pi|}$ satisfies the triangle inequality.

It follows that the completion of $X \bar{\otimes} Y$ under $\|\cdot\|_{|\pi|}$ is a Banach lattice. It is called the ***positive projective tensor product*** or the ***Fremlin projective tensor product*** of X and Y and denoted $X \otimes_{|\pi|} Y$. By construction, $X \otimes_{|\pi|} Y$ contains $X \bar{\otimes} Y$ as a dense sublattice, and, therefore, $X \otimes_{|\pi|} Y$ contains $X \otimes Y$ as a subspace. Since $X \otimes Y$ is uniformly dense in $X \bar{\otimes} Y$ by Proposition 19.3.7, it is dense in $X \bar{\otimes} Y$ with respect to $\|\cdot\|_{|\pi|}$. It follows that $X \otimes Y$ is dense in $X \otimes_{|\pi|} Y$. Similarly, by Corollary 19.4.4, $X_+ \otimes Y_+$ is uniformly dense in $(X \bar{\otimes} Y)_+$, hence it is $\|\cdot\|_{|\pi|}$-dense there and, therefore, in $(X \otimes_{|\pi|} Y)_+$.

The space $X \otimes_{|\pi|} Y$ satisfies a universal property for bounded positive bilinear maps:

19.6.5 Theorem. *For every positive bilinear map $\varphi \colon X \times Y \to Z$, there exists a unique positive linear operator $\varphi^{|\pi|} \colon X \otimes_{|\pi|} Y \to Z$ such that $\varphi^{|\pi|}(x \otimes y) = \varphi(x,y)$ for all $x \in X$ and $y \in Y$. Moreover, $\|\varphi^{|\pi|}\| = \|\varphi\|$.*

(By uniqueness, $\varphi^{|\pi|}$ has to be an extension of $\varphi^{\otimes}$, as well as of $\bar{\varphi}$ from Theorem 19.5.3.)

Proof. Being a Banach lattice, Z is uniformly complete. Let $\bar{\varphi}\colon X \bar{\otimes} Y \to Z$ be as in Theorem 19.5.3. It suffices to show that $\bar{\varphi}$ is bounded with respect to $\|\cdot\|_{|\pi|}$, and $\|\bar{\varphi}\| = \|\varphi\|$. WLOG, $\|\varphi\| = 1$. Fix $a \in X \bar{\otimes} Y$ with $\|a\|_{|\pi|} < 1$. We can then find $x_1, \dots, x_n \in X_+$ and $y_1, \dots, y_n \in Y_+$ with $|a| \leqslant \sum_{i=1}^n x_i \otimes y_i$ and $\sum_{i=1}^n \|x_i\|\|y_i\| < 1$ Then

$$|\bar{\varphi}(a)| \leqslant \bar{\varphi}\big(|a|\big) \leqslant \sum_{i=1}^n \bar{\varphi}(x_i \otimes y_i) = \sum_{i=1}^n \varphi(x_i, y_i),$$

and, therefore, $\big\|\bar{\varphi}(a)\big\| \leqslant \sum_{i=1}^n \|\varphi\|\|x_i\|\|y_i\| < 1$. We conclude that $\|\bar{\varphi}\| \leqslant 1$.

It is left prove the converse inequality. Again, fix $\varepsilon > 0$. It follows from $\|\varphi\| = 1$ that there exist $x \in B_X$ and $y \in B_Y$ with

$$1 - \varepsilon \leqslant \big\|\varphi(x, y)\big\| = \big\|\bar{\varphi}(x \otimes y)\big\| \leqslant \|\bar{\varphi}\|\|x \otimes y\|_{|\pi|} = \|\bar{\varphi}\|\|x\|\|y\| \leqslant \|\bar{\varphi}\|.$$

This yields $\|\bar{\varphi}\| \geqslant 1$. $\square$

19.6.6. Conversely, every positive operator $T\colon X \otimes_{|\pi|} Y \to Z$ yields a positive bilinear map $\varphi\colon X \times Y \to Z$ via $\varphi(x, y) = T(x \otimes y)$. Thus, Theorem 19.6.5 establishes a bijection between $B(X, Y; Z)_+$ and $L(X \otimes_{|\pi|} Y, Z)_+$, which preserves order and norm.

19.6.7. Theorem 19.6.5 also works for homomorphisms. Indeed, if the map φ in the proposition is a lattice bimorphism then the map $\bar{\varphi}\colon X \bar{\otimes} Y \to Z$ in Theorem 19.2.3 is a lattice homomorphism. Since $X \bar{\otimes} Y$ is dense in $X \otimes_{|\pi|} Y$, $\varphi^{|\pi|}$ extends $\bar{\varphi}$ and, therefore, $\varphi^{|\pi|}$ is a lattice homomorphism. Conversely, every lattice homomorphism $T\colon X \otimes_{|\pi|} Y \to Z$ yields a lattice bimorphism $\varphi\colon X \times Y \to Z$ via $\varphi(x, y) = T(x \otimes y)$. Thus, a restriction of the bijection from the preceding paragraph yields a bijection between lattice bimorphisms in $B(X, Y; Z)$ and lattice homomorphisms in $L(X \otimes_{|\pi|} Y, Z)$.

19.6.8. In 19.6.1, we constructed an isometry between $B(X, Y)$ and $L(X, Y^*)$. It is easy to see that this isometry preserves order. Hence, restricting it to the positive cones, we get an additive bijection between $B(X, Y)_+$ and $L(X, Y^*)_+$ which preserves order and norm. On the other hand, applying 19.6.6 with $Z = \mathbb{R}$, we get an additive bijection between $B(X, Y)_+$ and $(X \otimes_{|\pi|} Y)^*_+$ which preserves order and norms. Composing the two maps, we get an additive bijection between $(X \otimes_{|\pi|} Y)^*_+$ and $L(X, Y^*)_+$, which preserves order and norm. Notice that $(X \otimes_{|\pi|} Y)^*_+$ is the positive cone of a Banach lattice $(X \otimes_{|\pi|} Y)^*$. Also, since Y^* is order complete, $L_r(X, Y^*)$ is a Banach lattice; see 13.2; and its positive cone is $L(X, Y^*)_+$. It follows from Corollary 6.2.3 that the map constructed above extends to a lattice isometry between $(X \otimes_{|\pi|} Y)^*$ and $L_r(X, Y^*)$. Summarizing:

19.6.9 Corollary. *$(X\otimes_{|\pi|}Y)^*$ is lattice isometric to $L_r(X,Y^*)$. Consequently, $X\otimes_{|\pi|}Y$ may be identified with a closed sublattice of $L_r(X,Y^*)^*$.*

Using the construction in 19.6.8, we can somewhat explicitly describe the two isometries in the proposition. For $T\in L_r(X,Y^*)$ we associate the functional in $(X\otimes_{|\pi|}Y)^*$ such that $x\otimes y\mapsto\langle Tx,y\rangle$. The second isometry is the restriction of the adjoint of the first one to $X\otimes_{|\pi|}Y$, so $x\otimes y$ corresponds to the functional on $L_r(X,Y^*)$ that sends T to $\langle Tx,y\rangle$.

The argument in 19.6.8 also shows that $(X\otimes_{|\pi|}Y)^*$ is lattice isomorphic (as a vector lattice) to $B_r(X,Y)$, the space of all differences of positive bilinear functionals.

Since $X\otimes Y$ is uniformly dense in $X\mathbin{\bar\otimes}Y$, it is norm dense in $X\otimes_{|\pi|}Y$. Thus, in view of Corollary 19.6.9, we may view $X\otimes_{|\pi|}Y$ as the closure of $X\otimes Y$ in $L_r(X,Y^*)^*$. This provides an alternative way of introducing $X\otimes_{|\pi|}Y$.

Fix $a\in X\mathbin{\bar\otimes}Y$. For every positive bilinear map $\varphi\colon X\times Y\to Z$ with $\lVert\varphi\rVert\leqslant 1$, Theorem 19.6.5 implies that $\bigl\lVert\bar\varphi(a)\bigr\rVert\leqslant\lVert\bar\varphi\rVert\lVert a\rVert_{|\pi|}=\lVert\varphi\rVert\lVert a\rVert_{|\pi|}=\lVert a\rVert_{|\pi|}$. On the other hand, if we take φ to be the canonical tensor map $\varphi\colon X\times Y\to X\otimes_{|\pi|}Y$ given by $\varphi(x,y)=x\otimes y$ then φ is positive bilinear, and $\lVert\varphi\rVert=1$. Observe that $\bar\varphi$ is just the embedding of $X\mathbin{\bar\otimes}Y$ into $X\otimes_{|\pi|}Y$. It follows that $\bigl\lVert\bar\varphi(a)\bigr\rVert=\lVert a\rVert_{|\pi|}$. Thus, one could say that $\lVert a\rVert_{|\pi|}$ is the supremum (even the maximum) of $\bigl\lVert\bar\varphi(a)\bigr\rVert$, where the supremum is taken over all positive bilinear maps $\varphi\colon X\times Y\to Z$ with $\lVert\varphi\rVert\leqslant 1$, where Z is an arbitrary Banach lattice. In fact, it suffices to take $Z=\mathbb{R}$:

19.6.10 Proposition. *For $a\in X\otimes_{|\pi|}Y$, we have*

$$\lVert a\rVert_{|\pi|}=\max\bigl\{\bigl\lvert\varphi^{|\pi|}(a)\bigr\rvert\ :\ \varphi\colon X\times Y\to\mathbb{R}\text{ is positive and bilinear, and }\lVert\varphi\rVert\leqslant 1\bigr\}.$$

Proof. On one hand, for every such φ we have $\bigl\lvert\varphi^{|\pi|}(a)\bigr\rvert\leqslant\bigl\lVert\varphi^{|\pi|}\bigr\rVert\lVert a\rVert_{|\pi|}=\lVert\varphi\rVert\lVert a\rVert_{|\pi|}\leqslant\lVert a\rVert_{|\pi|}$. Using Hahn-Banach Theorem, find $\xi\in(X\otimes_{|\pi|}Y)^*$ such that $\lVert\xi\rVert=1$ and $\lVert\xi(a)\rVert=\lVert a\rVert_{|\pi|}$. Replacing ξ with $\lvert\xi\rvert$, if necessary, we may assume that $\xi\geqslant 0$. As before, ξ induces a positive bilinear functional $\varphi\colon X\times Y\to\mathbb{R}$ via $\varphi(x,y)=\xi(x\otimes y)$. Then $\xi=\varphi^{|\pi|}$ and so $\lVert\varphi\rVert=\lVert\xi\rVert\leqslant 1$. Finally, $\bigl\lvert\varphi^{|\pi|}(a)\bigr\rvert=\lvert\xi(a)\rvert=\lVert a\rVert_{|\pi|}$. □

19.6.11 Proposition. *For every $a\in X\otimes_{|\pi|}Y$ we have*

$$\lVert a\rVert_{|\pi|}=\inf\Bigl\{\sum_{i=1}^{\infty}\lVert x_i\rVert\lVert y_i\rVert\ :\ \lvert a\rvert\leqslant\sum_{i=1}^{\infty}x_i\otimes y_i,\ x_1,\dots,x_n\in X_+,\ y_1,\dots,y_n\in Y_+\Bigr\}.$$

Proof. To reduce clutter, we will write $\lVert a\rVert$ instead of $\lVert a\rVert_{|\pi|}$. On one hand, since it is a lattice norm and a cross-norm, we conclude that if $\lvert a\rvert\leqslant\sum_{i=1}^{\infty}x_i\otimes y_i$ then

$\|a\| \leqslant \sum_{i=1}^{\infty}\|x_i \otimes y_i\| = \sum_{i=1}^{\infty}\|x_i\|\|y_i\|$. This proves that $\|a\|$ is less than or equal to the desired infimum.

To prove the converse inequality, fix $\varepsilon > 0$ and find a sequence $(a_n)_{n=0}^{\infty}$ in $X \bar{\otimes} Y$ such that $\|a_n - a\| < \frac{\varepsilon}{2^n}$. In particular, $\|a - a_0\| < \varepsilon$, so that $\|a_0\| < \|a\| + \varepsilon$. Also, $\|a_{n+1} - a_n\| \leqslant \frac{\varepsilon}{2^{n-1}}$ for every n. Using the definition of the norm, find $x_1^{(0)}, \dots, x_{k_0}^{(0)}$ in X_+ and $y_1^{(0)}, \dots, y_{k_0}^{(0)}$ in Y_+ such that $|a_0| \leqslant \sum_{i=1}^{k_0} x_i^{(0)} \otimes y_i^{(0)}$ and $\sum_{i=1}^{k_0}\|x_i^{(0)}\|\|y_i^{(0)}\| \leqslant \|a\| + \varepsilon$. Similarly, for each $n = 1, 2 \dots$, find $x_1^{(n)}, \dots, x_{k_n}^{(n)}$ in X_+ and $y_1^{(n)}, \dots, y_{k_n}^{(n)}$ in Y_+ such that $|a_{n+1} - a_n| \leqslant \sum_{i=1}^{k_n} x_i^{(n)} \otimes y_i^{(n)}$ and

$$\sum_{i=1}^{k_n}\|x_i^{(n)}\|\|y_i^{(n)}\| \leqslant \|a_{n+1} - a_n\| + \frac{\varepsilon}{2^{n+1}}.$$

It follows from $a = a_0 + \sum_{n=0}^{\infty}(a_{n+1} - a_n)$ that

$$|a| \leqslant |a_0| + \sum_{n=0}^{\infty}|a_{n+1} - a_n| \leqslant \sum_{n=0}^{\infty}\sum_{i=1}^{k_n} x_i^{(n)} \otimes y_i^{(n)}.$$

On the other hand,

$$\sum_{n=0}^{\infty}\sum_{i=1}^{k_n}\|x_i^{(n)}\|\|y_i^{(n)}\| \leqslant \|a_0\| + \varepsilon + \sum_{n=0}^{\infty}\|a_{n+1} - a_n\| + \varepsilon \leqslant \|a\| + 5\varepsilon.$$

□

It follows easily from the definitions that $\|a\|_{|\pi|} \leqslant \|a\|_{\pi}$ for every $a \in X \otimes Y$. The canonical tensor map $\varphi\colon X \times Y \to X \otimes_{|\pi|} Y$ is bilinear with $\|\varphi\| = 1$. Using the universal property for the projective tensor product, we find a bounded linear operator $\varphi^{\pi}\colon X \otimes_{\pi} Y \to X \otimes_{|\pi|} Y$.

19.6.1 Examples of positive projective tensor products

This section is based on results of Fremlin.

19.6.12 Theorem. *$X \otimes_{\pi} L_1(\mu)$ and $X \otimes_{|\pi|} L_1(\mu)$ are isomorphic for every measure μ.*

Proof. Fix $a \in X \otimes L_1(\mu)$. We already know that $\|a\|_{|\pi|} \leqslant \|a\|_{\pi}$. Recall that $X \otimes_{\pi} L_1(\mu) = L_1(\mu, X)$, and the latter is a Banach lattice. Consider the canonical tensor map $\varphi\colon X \times L_1(\mu) \to X \otimes_{\pi} L_1(\mu)$ given by $\varphi(x, y) = x \otimes y$. In $L_1(\mu, X)$, $x \otimes y$ corresponds to the function xy, given by $t \mapsto y(t)x$. It is clear that φ is positive. It follows that $\varphi^{\otimes}$ extends to a positive operator $\varphi^{|\pi|}\colon X \otimes_{|\pi|} L_1(\mu) \to X \otimes_{\pi} L_1(\mu)$. Note that $\varphi^{|\pi|}$ is the identity on $X \otimes L_1(\mu)$. It follows that $\|a\|_{\pi} \leqslant \|\varphi^{|\pi|}\|\|a\|_{|\pi|}$. Hence, the two norms are equivalent on $X \otimes L_1(\mu)$. Since $X \otimes L_1(\mu)$ is dense both in $X \otimes_{\pi} L_1(\mu)$ and in $X \otimes_{|\pi|} L_1(\mu)$, the two spaces are isomorphic. □

19.6.13 Corollary. *$L_1(\mu)\otimes_{|\pi|}L_1(\nu)$ is lattice isometric to $L_1(\mu\times\nu)$, with $f\otimes g$ corresponding to the function $(s,t)\mapsto f(s)g(t)$ (cf. Example 19.3.4).*

Proof. By Theorem 19.6.12, $L_1(\mu)\otimes_{|\pi|}L_1(\nu)$ is lattice isometric to the Bochner space $L_1\bigl(\nu,L_1(\mu)\bigr)$. It is easy to see that the latter is lattice isometric to $L_1(\mu\times\nu)$. □

Let X and Y be two Banach lattices; for $a\in X\otimes_{|\pi|}Y$, define

$$\rho(a)=\inf\bigl\{\|x\|\|y\| \,:\, x,y\geqslant 0,\ |a|\leqslant x\otimes y\bigr\}. \tag{19.4}$$

It is clear that $\|a\|_{|\pi|}\leqslant\rho(a)$ for every $a\in X\otimes_{|\pi|}Y$. If $|a|\leqslant|b|$ in $X\otimes_{|\pi|}Y$ then $\rho(a)\leqslant\rho(b)$. Given any $x\in X$ and $y\in Y$, it follows from $\|x\|\|y\|=\|x\otimes y\|_{|\pi|}\leqslant\rho(x\otimes y)\leqslant\|x\|\|y\|$ that $\rho(x\otimes y)=\|x\|\|y\|$. We will show that for many pairs spaces, (19.4) actually yields an alternative (and much easier) formula for the positive projective tensor norm.

19.6.14 Theorem. *If ρ is subadditive on $X\otimes Y$ then $\|\cdot\|_{|\pi|}$ agrees with ρ on $X\otimes_{|\pi|}Y$. Moreover, in this case, $X\bar{\otimes}Y$ is order dense in $X\otimes_{|\pi|}Y$.*

Proof. To reduce clutter, we will write $\|\cdot\|$ instead of $\|\cdot\|_{|\pi|}$. We will first show that it agrees with ρ on $X\bar{\otimes}Y$. Let $a\in X\bar{\otimes}Y$. Find $x_1,\dots,x_n\in X_+$ and $y_1,\dots,y_n\in Y_+$ such that $|a|\leqslant\sum_{i=1}^n x_i\otimes y_i$ and $\|a\|\geqslant\sum_{i=1}^n\|x_i\|\|y_i\|-\varepsilon$. By monotonicity and subadditivity of ρ, we have

$$\rho(a)\leqslant\rho\Bigl(\sum_{i=1}^n x_i\otimes y_i\Bigr)\leqslant\sum_{i=1}^n\rho(x_i\otimes y_i)=\sum_{i=1}^n\|x_i\|\|y_i\|\leqslant\|a\|+\varepsilon.$$

It follows that $\rho(a)\leqslant\|a\|$ and, therefore, $\rho(a)=\|a\|$.

Now let $a\in X\otimes_{|\pi|}Y$. We claim that there is a sequence (a_n) in $X\bar{\otimes}Y$ which converges to a uniformly with an elementary tensor as a regulator; we will use ideas of the proof of Theorem 2.3.7 to show this. Since $X\otimes_{|\pi|}Y$ is the completion of $X\bar{\otimes}Y$, there is a sequence (a_n) in $X\bar{\otimes}Y$ such that $\|a_n-a\|\to 0$. Passing to a subsequence, we may assume that $\|a_n-a\|<\frac{1}{2^{4n+1}}$ for every n. It follows that $\|a_n-a_{n+1}\|<\frac{1}{2^{4n}}$ for every n. By the first part of the proof, we can find positive $x_n\in X$ and $y_n\in Y$ such that $|a_n-a_{n+1}|\leqslant x_n\otimes y_n$ and $\|x_n\|\|y_n\|<\frac{1}{2^{4n}}$. WLOG, $\|x_n\|=\|y_n\|<\frac{1}{2^{2n}}$. Put $x_0=\sum_{n=1}^\infty 2^n x_n$ and $y_0=\sum_{n=1}^\infty 2^n y_n$; it is easy to see that the two series converge. For $m>n$, we have

$$|a_m-a_n|\leqslant\sum_{k=1}^{m-1}|a_k-a_{k+1}|\leqslant\sum_{k=1}^{m-1}x_k\otimes y_k\leqslant\frac{1}{2^{2n}}\sum_{k=1}^{m-1}(2^kx_k)\otimes(2^ky_k)\leqslant\frac{1}{2^{2n}}x_0\otimes y_0.$$

Passing to the limit on m, we get $|a-a_n| \leqslant \frac{1}{2^{2n}} x_0 \otimes y_0$. This proves the claim. Scaling x_0 and y_0, if necessary, we may also assume that $\|x_0\| = \|y_0\| = 1$.

Next, we will prove that $\|a\| = \rho(a)$. Fix $\varepsilon > 0$. Find n such that $|a_n - a| < \varepsilon x_0 \otimes y_0$. It follows that $\big|\|a_n\| - \|a\|\big| \leqslant \|a_n - a\| \leqslant \varepsilon$. Since $a_n \in X \bar{\otimes} Y$, we have $\|a_n\| = \rho(a)$, so we can find $u, v \geqslant 0$ such that $|a_n| \leqslant u \otimes v$ and $\|a_n\| \geqslant \|u\|\|v\| - \varepsilon$. WLOG, $\|u\| = \|v\|$. It follows that $\|u\|^2 = \|u\|\|v\| \leqslant \|a_n\| + \varepsilon \leqslant \|a\| + 2\varepsilon$. Observe that $|a| \leqslant |a_n| + \varepsilon x_0 \otimes y_0 \leqslant u \otimes v + \varepsilon x_0 \otimes y_0 \leqslant x \otimes y$, where $x = u + \varepsilon x_0$ and $y = v + \varepsilon y_0$. Observe also that

$$\begin{aligned} \|x\|\|y\| \leqslant \big(\|u\| + \varepsilon\big)\big(\|v\| + \varepsilon\big) = \|u\|\|v\| + 2\varepsilon\|u\| + \varepsilon^2 \\ \leqslant \|a\| + 2\varepsilon + 2\varepsilon\sqrt{\|a\| + 2\varepsilon} + \varepsilon^2. \end{aligned}$$

Since ε is arbitrary, we conclude that $\rho(a) \leqslant \|a\|$ and, therefore, $\rho(a) = \|a\|$.

Finally, we will show that $X \bar{\otimes} Y$ is order dense in $X \otimes_{|\pi|} Y$. Let $a \in X \otimes_{|\pi|} Y$. By the claim, there exists a sequence (a_n) in $X \bar{\otimes} Y$ which converges to a uniformly with $x_0 \otimes y_0$ as a regulator. By the Archimedean property, there exists $\varepsilon > 0$ such that $a - \varepsilon x_0 \otimes y_0 \not\leqslant 0$. That is, $(a - \varepsilon x_0 \otimes y_0)^+ > 0$. Find n such that $|a_n - a| < \frac{\varepsilon}{2} x_0 \otimes y_0$. This yields

$$a + \tfrac{\varepsilon}{2} x_0 \otimes y_0 \geqslant a_n \geqslant a - \tfrac{\varepsilon}{2} x_0 \otimes y_0.$$

Subtracting $\frac{\varepsilon}{2} x_0 \otimes y_0$, we get

$$a \geqslant a_n - \tfrac{\varepsilon}{2} x_0 \otimes y_0 \geqslant a - \varepsilon x_0 \otimes y_0.$$

It follows that

$$a \geqslant \big(a_n - \tfrac{\varepsilon}{2} x_0 \otimes y_0\big)^+ \geqslant (a - \varepsilon x_0 \otimes y_0)^+ > 0.$$

Finally, observe that $\big(a_n - \frac{\varepsilon}{2} x_0 \otimes y_0\big)^+ \in X \bar{\otimes} Y$. □

19.6.15 Remark. We proved in Proposition 19.3.6 that $X \otimes Y$ is order dense and majorizing in $X \bar{\otimes} Y$. Under the assumptions of Theorem 19.6.14, we can extend this to $X \otimes_{|\pi|} Y$: given any $0 < a \in X \otimes_{|\pi|} Y$, we have $0 < x \otimes y \leqslant a \leqslant u \otimes v$ for some positive x, y, u, and v. Indeed, the lower estimate follows from the facts that $X \otimes Y$ is order dense in $X \bar{\otimes} Y$ and the latter is order dense in $X \otimes_{|\pi|} Y$. The upper estimate follows from the the definition of $\rho(a)$ and the fact that $\rho(a)$ equals $\|a\|_{|\pi|}$ and is, therefore, finite.

19.6.16 Corollary. *If X is an AM-space then $\|a\|_{|\pi|}$ is given by* (19.4)*.*

Proof. By Theorem 19.6.14, it suffices to show that ρ is subadditive on $X \otimes Y$. Take $a_1, a_2 \in X \otimes Y$ and fix $\varepsilon > 0$. Find $x_1, x_2 \in X_+$ and $y_1, y_2 \in Y_+$ such that

$$|a_1| \leqslant x_1 \otimes y_1, \ |a_2| \leqslant x_2 \otimes y_2, \ \rho(a_1) \geqslant \|x_1\|\|y_1\| - \varepsilon, \text{ and } \rho(a_2) \geqslant \|x_2\|\|y_2\| - \varepsilon.$$

WLOG, we may assume that $\|x_1\| = \|x_2\| = 1$. Note that

$$|a_1 + a_2| \leqslant |a_1| + |a_2| \leqslant x_1 \otimes y_1 + x_2 \otimes y_2$$
$$\leqslant (x_1 \vee x_2) \otimes y_1 + (x_1 \vee x_2) \otimes y_2 = (x_1 \vee x_2) \otimes (y_1 + y_2).$$

It follows that

$$\rho(a_1 + a_2) \leqslant \|x_1 \vee x_2\| \|y_1 + y_2\| \leqslant \|y_1\| + \|y_2\| \leqslant \rho(a_1) + \rho(a_2) + 2\varepsilon.$$

We conclude that $\rho(a_1 + a_2) \leqslant \rho(a_1) + \rho(a_2)$. □

19.6.17 Theorem. *$C(K) \otimes_{|\pi|} C(L)$ is lattice isometric to $C(K \times L)$.*

Proof. Let $a \in C(K) \otimes C(L)$. As before, we may view a as an element of $C(K \times L)$. By Corollary 19.6.16, $\|a\|_{|\pi|}$ is given by y (19.4). Suppose that $|a| \leqslant x \otimes y$ with $x, y \geqslant 0$. Note that

$$x \leqslant \|x\| \mathbb{1}_K,\ y \leqslant \|y\| \mathbb{1}_L, \|x\| = \big\| \|x\| \mathbb{1}_K \big\|, \text{ and } \|y\| = \big\| \|y\| \mathbb{1}_L \big\|.$$

WLOG, replacing x and y with $\|x\|\mathbb{1}_K$ and $\|y\|\mathbb{1}_L$, respectively, we may assume that x and y to be scalar multiples of $\mathbb{1}_K$ and $\mathbb{1}_L$, respectively. It follows from $\mathbb{1}_K \otimes \mathbb{1}_L = \mathbb{1}_{K\times L}$ that

$$\|a\|_{|\pi|} = \inf\big\{\lambda \,:\, |a| \leqslant \lambda \mathbb{1}_{K\times L}\big\} = \|a\|_\infty.$$

We conclude that $\|\cdot\|_{|\pi|}$ and $\|\cdot\|_\infty$ agree on $C(K) \otimes C(L)$. Since $C(K) \otimes C(L)$ is uniformly dense in $C(K \times L)$, it is dense with respect to $\|\cdot\|_\infty$. Hence, $C(K \times L)$ equals the completion of $C(K) \otimes C(L)$ with respect to $\|\cdot\|_{|\pi|}$, which is $C(K) \otimes_{|\pi|} C(L)$. □

We will show next that $L_2(\mu) \otimes_{|\pi|} L_2(\nu)$ may be viewed a sublattice of $L_2(\mu \times \nu)$.

19.6.18 Proposition. *The norm $\|\cdot\|_{|\pi|}$ on $L_2(\mu) \otimes_{|\pi|} L_2(\nu)$ is given by* (19.4)*. Furthermore the lattice embedding $L_2(\mu) \bar{\otimes} L_2(\nu) \hookrightarrow L_2(\mu \times \nu)$ from Example 19.3.4 extends to an injective lattice homomorphism $L_2(\mu) \otimes_{|\pi|} L_2(\nu) \hookrightarrow L_2(\mu \times \nu)$.*

Proof. As in Example 19.3.4, let $\varphi \colon L_2(\mu) \times L_2(\nu) \to L_2(\mu \times \nu)$ given by $\varphi(x, y)(s, t) = x(s)y(t)$; then $\bar{\varphi} \colon L_2(\mu) \bar{\otimes} L_2(\nu) \hookrightarrow L_2(\mu \times \nu)$ is a lattice isomorphic embedding, so we may view $L_2(\mu) \bar{\otimes} L_2(\nu)$ as a sublattice of $L_2(\mu \times \nu)$.

Let ρ be as in (19.4). We will show that ρ is subadditive on $L_2(\mu) \otimes L_2(\nu)$, so that we can apply Theorem 19.6.14. Fix $a_1, a_2 \in L_2(\mu) \otimes L_2(\nu)$; put $a = a_1 + a_2$. Fix $\varepsilon > 0$. Find positive $x_1, x_2 \in L_2(\mu)$ and $y_1, y_2 \in L_2(\nu)$ such that

$$|a_1| \leqslant x_1 \otimes y_1,\ |a_2| \leqslant x_2 \otimes y_2,\ \rho(a_1) \geqslant \|x_1\| \|y_1\| - \varepsilon, \text{ and } \rho(a_2) \geqslant \|x_2\| \|y_2\| - \varepsilon.$$

WLOG, $\|x_1\| = \|y_1\|$ and $\|x_2\| = \|y_2\|$. Note that $|a| \leqslant |a_1| + |a_2| \leqslant x_1 \otimes y_1 + x_2 \otimes y_2$. Put $x = (x_1^2 + x_2^2)^{\frac{1}{2}}$ and $y = (y_1^2 + y_2^2)^{\frac{1}{2}}$. By Hölder's inequality, we have

$$\bigl|a(s,t)\bigr| \leqslant x_1(s)y_1(t) + x_2(s)y_2(t) \leqslant x(s)y(t)$$

a.e., so that $|a| \leqslant x \otimes y$. Note also that

$$\|x\|^2 = \|x_1\|^2 + \|x_2\|^2 = \|y_1\|^2 + \|y_2\|^2 = \|y\|^2.$$

It follows that

$$\|x\|\|y\| = \|x\|^2 = \|x_1\|^2 + \|x_2\|^2 = \|x_1\|\|y_1\| + \|x_2\|\|y_2\| \leqslant \rho(a_1) + \rho(a_2) + 2\varepsilon.$$

This yields $\rho(a) \leqslant \rho(a_1) + \rho(a_2)$.

By Theorem 19.6.14, $\|\cdot\|_{|\pi|}$ agrees with ρ on $L_2(\mu) \otimes_{|\pi|} L_2(\nu)$ and, furthermore, $L_2(\mu) \bar{\otimes} L_2(\nu)$ is order dense in $L_2(\mu) \otimes_{|\pi|} L_2(\nu)$.

By 19.6.7, $\varphi^{|\pi|} \colon L_2(\mu) \otimes_{|\pi|} L_2(\nu) \to L_2(\mu \times \nu)$ is a lattice homomorphism. It is one-to-one because if $0 \neq a \in L_2(\mu) \otimes_{|\pi|} L_2(\nu)$ then, by order density, we have $|a| \geqslant x \otimes y > 0$ for some positive x and y; we then have $\bigl|\varphi^{|\pi|}(a)\bigr| = \varphi^{|\pi|}\bigl(|a|\bigr) \geqslant \varphi^{|\pi|}(x \otimes y) = \varphi(x,y) > 0$. □

19.6.19 Example (Fremlin)**.** Let $X = Y = L_2[0,1]$. We claim that $X \otimes_{|\pi|} Y$ is not order complete. For every $\varepsilon \in (0,1)$, define two positive bilinear functionals φ_ε and ψ_ε on $X \times Y$ via

$$\varphi_\varepsilon(x,y) = \int_0^{1-\varepsilon} x(t)y(t+\varepsilon)\,dt \quad \text{and} \quad \psi_\varepsilon(x,y) = \int_\varepsilon^1 x(t)y(t-\varepsilon)\,dt.$$

(the integrals exist by Hölder's inequality). It can be verified that $\lim_{\varepsilon\to 0} \varphi_\varepsilon(x,y) - \psi_\varepsilon(x,y) = 0$: first, show this when y is continuous (and, therefore, uniformly continuous); then use density of $C[0,1]$ in $L_2[0,1]$ to extend it to any $g \in L_2[0,1]$. It follows that $\lim_{\varepsilon\to 0} \varphi_\varepsilon^{\otimes}(a) - \psi_\varepsilon^{\otimes}(a) = 0$ for every $a \in X \otimes Y$. Again, it follows from Hölder's inequality that $\|\varphi_\varepsilon^{|\pi|}\| = \|\varphi_\varepsilon\| = 1$ and $\|\psi_\varepsilon^{|\pi|}\| = \|\psi_\varepsilon\| = 1$, so that the family $\bigl\{\varphi_\varepsilon^{|\pi|} - \psi_\varepsilon^{|\pi|} \,:\, \varepsilon \in (0,1)\bigr\}$ is uniformly bounded. Since $X \otimes Y$ is dense in $X \otimes_{|\pi|} Y$, we conclude that $\lim_{\varepsilon\to 0} \varphi_\varepsilon^{|\pi|}(a) - \psi_\varepsilon^{|\pi|}(a) = 0$ for every $a \in X \otimes_{|\pi|} Y$.

Let $n > 1$. Partition $[0,1]^2$ into n^2 squares of size $\frac{1}{n} \times \frac{1}{n}$. Let a_n and b_n be the characteristic functions of the unions of all the little squares above (respectively, below) the diagonal. Assume for the sake of contradiction that $X \otimes_{|\pi|} Y$ is order complete. It follows from $0 \leqslant a_n \uparrow\leqslant \mathbb{1}$ that $a \coloneqq \sup a_n$ exists. If $\varepsilon > \frac{1}{n}$ then $\varphi_\varepsilon^{|\pi|}(a_n) = 1 - \varepsilon$, this implies that $\varphi_\varepsilon^{|\pi|}(a) \geqslant \sup \varphi_\varepsilon^{|\pi|}(a_n) \geqslant 1 - \varepsilon$. On the other hand, we have $a_n + b_m \leqslant \mathbb{1}$ for all $n, m > 1$, so that $a \leqslant \mathbb{1} - b_m$ and, therefore, $\psi_\varepsilon^{|\pi|}(a) \leqslant \psi_\varepsilon^{|\pi|}(\mathbb{1} - b_m) = 0$. This contradicts $\lim_{\varepsilon\to 0} \varphi_\varepsilon^{|\pi|}(a) - \psi_\varepsilon^{|\pi|}(a) = 0$.

Projective tensor product of Banach spaces generally does not respect unconditional bases: it was shown in [KP70] that $\ell_p \otimes_\pi \ell_p$ has no unconditional basis. The situation is better for positive projective tensor product. Recall that Banach sequence space is a Banach lattice whose lattice structure is defined by a Schauder basis (in which case, the basis is 1-unconditional); see 1.5.11.

19.6.20 Proposition ([BBPTT13])**.** *Let X and Y be Banach sequence spaces, with bases (e_i) and (f_i), respectively. Then any enumeration of the set $B \coloneqq \{e_i \otimes f_j : i, j \in \mathbb{N}\}$ is a 1-unconditional basis of $X \otimes_{|\pi|} Y$.*

Proof. Note that the sequences (e_i) and (f_i) are disjoint in X and in Y, respectively. By Proposition 19.3.9, the set B is disjoint in $X \otimes_{|\pi|} Y$. It follows that any enumeration of it is a 1-unconditional basic sequence. It is left to prove that $\operatorname{span} B$ is dense in $X \otimes_{|\pi|} Y$.

Fix $x \in X$ and $y \in Y$. Given any $\varepsilon \in (0, 1)$, find $x_0 \in \operatorname{span}\{e_i\}$ and $y_0 \in \operatorname{span}\{f_i\}$ such that $\|x - x_0\| < \varepsilon$ and $\|y - y_0\| < \varepsilon$. It follows that

$$\begin{aligned} \|x \otimes y - x_0 \otimes y_0\|_{|\pi|} &= \|x_0 \otimes (y - y_0) + (x - x_0) \otimes y_0 + (x - x_0) \otimes (y - y_0)\|_{|\pi|} \\ &\leqslant \|x_0\|\|y - y_0\| + \|x - x_0\|\|y_0\| + \|x - x_0\|\|y - y_0\| \leqslant 3\varepsilon. \end{aligned}$$

It follows that $\overline{\operatorname{span}}\, B$ contains $X \otimes Y$. Since the latter is dense in $X \otimes_{|\pi|} Y$, we conclude that $\overline{\operatorname{span}}\, B = X \otimes_{|\pi|} Y$. $\square$

19.6.21 Corollary. *If X and Y are two Banach sequence spaces then $X \otimes_{|\pi|} Y$ is again a Banach sequence space. In particular, it is order continuous by Proposition 2.5.14.*

19.6.22 Example. Let $1 \leqslant p < \infty$. It is known that $\ell_p \otimes_\pi \ell_p$ contains a copy of ℓ_q where $q = \max\{\frac{p}{2}, 1\}$. Things works even better for $\ell_p \otimes_{|\pi|} \ell_p$. By Proposition 19.6.20, any enumeration of the set $\{e_i \otimes e_j : i, j \in \mathbb{N}\}$ is a 1-unconditional basis of $\ell_p \otimes_{|\pi|} \ell_p$. Here (e_i) stands for the standard basis of ℓ_p. So we may view $\ell_p \otimes_{|\pi|} \ell_p$ as a space of infinite matrices. Let D be the subspace of all the diagonal matrices in $\ell_p \otimes_{|\pi|} \ell_p$, that is, $D = [e_i \otimes e_i]_{i \in \mathbb{N}}$. We claim that D is lattice isometric to $\ell_{\frac{p}{2}}$ if $p \geqslant 2$ and to ℓ_1 if $1 \leqslant p \leqslant 2$. Note that if $x, y \in \ell_p$ then Hölder's inequality implies that their component-wise product xy is in $\ell_{\frac{p}{2}}$, and $\|xy\|_{\frac{p}{2}} \leqslant \|x\|_p \|y\|_p$. Fix $u = \sum_{i=1}^m \alpha_i e_i \otimes e_i$ for some $\alpha_1, \ldots, \alpha_m \geqslant 0$.

Suppose first that $p \geqslant 2$. We write (f_i) for the standard basis of $\ell_{\frac{p}{2}}$. Take $x = \sum_{i=1}^m \alpha_i^{\frac{1}{2}} e_i$ in ℓ_p. Then $u \leqslant x \otimes x$, so that $\|u\|_{|\pi|} \leqslant \|x\|_p^2 = \big\|(\alpha_i)_{i=1}^m\big\|_{\frac{p}{2}}$. By the preceding paragraph, the map $\varphi \colon \ell_p \times \ell_p \to \ell_{\frac{p}{2}}$ defined via $\varphi(x, y) = xy$ is positive, bilinear, and $\|\varphi\| \leqslant 1$. We conclude that $\varphi^{|\pi|} \colon \ell_p \otimes_{|\pi|} \ell_p \to \ell_{\frac{p}{2}}$ is a contraction. It follows from

$\varphi^{|\pi|}(u) = \sum_{i=1}^m \alpha_i \varphi(e_i, e_i) = \sum_{i=1}^m \alpha_i f_i$ that $\|u\|_{|\pi|} \geqslant \left\|\sum_{i=1}^m \alpha_i f_i\right\|_{\frac{p}{2}} = \left\|(\alpha_i)_{i=1}^m\right\|_{\frac{p}{2}}$. This proves that $\|u\|_{|\pi|} = \left\|(\alpha_i)_{i=1}^m\right\|_{\frac{p}{2}}$. It is now easy to deduce that D is lattice isometric to $\ell_{\frac{p}{2}}$.

Suppose now that $1 \leqslant p \leqslant 2$. By the triangle inequality, $\|u\|_{|\pi|} \leqslant \sum_{i=1}^m \alpha_i \|e_i \otimes e_i\|_{|\pi|} = \sum_{i=1}^m \alpha_i = \left\|(\alpha_i)_{i=1}^m\right\|_1$. To prove the converse inequality, observe that since $\ell_{\frac{p}{2}} \subseteq \ell_1$ and $\|\cdot\|_{\frac{p}{2}} \geqslant \|\cdot\|_1$, the positive bilinear map from $\ell_p \times \ell_p$ to ℓ_1 defined via $\varphi(x, y) = xy$ satisfies $\|\varphi\| \leqslant 1$. Arguing as in the preceding paragraph, we get $\|u\|_{|\pi|} = \left\|(\alpha_i)_{i=1}^m\right\|_1$, and, therefore, D is lattice isometric to ℓ_1.

In particular, if $1 \leqslant p \leqslant 2$ then $\ell_p \otimes_{|\pi|} \ell_p$ contains a lattice copy of ℓ_1. It follows that $\ell_p \otimes_{|\pi|} \ell_p$ is not reflexive and $(\ell_p \otimes_{|\pi|} \ell_p)^*$ fails to be order continuous by Theorem 10.5.3. Note that by Corollary 19.6.9, $(\ell_p \otimes_{|\pi|} \ell_p)^* = L_r(\ell_p, \ell_{p'})$.

19.7 Positive injective tensor product

We start by recalling the definition of the injective tensor product of Banach spaces X and Y. As in 19.1.9, we may view $X \otimes Y$ as a subspace of $B(X^*, Y^*)$ via $(x \otimes y)(f, g) = f(x)g(y)$. We then define the injective tensor product $X \otimes_\varepsilon Y$ as the closure of $X \otimes Y$ in $B(X^*, Y^*)$. Since one can identify $B(X^*, Y^*)$ with $L(X^*, Y^{**})$ as in 19.6.1, one may replace $B(X^*, Y^*)$ with $L(X^*, Y^{**})$ in the preceding construction. This way, $X \otimes_\varepsilon Y$ may be viewed as a space of operators. In this case, we view $x \otimes y$ as an operator $x^* \mapsto x^*(x)\hat{y}$, where $\hat{y}$ is the canonical image of y in Y^{**}. In fact, one may replace $L(X^*, Y^{**})$ with $L(X^*, Y)$. Indeed, we may identify $X \otimes Y$ with a subspace $L(X^*, Y)$ by sending $x \otimes y$ into the operator $x^* \mapsto x^*(x)y$. Hence, we can identify $X \otimes_\varepsilon Y$ with the closure of $X \otimes Y$ in $L(X^*, Y)$.

For the rest of this section, X and Y will be Banach lattices. As in Section 13.2, $L_r(X^*, Y)$ is an ordered Banach space, while $L_r(X^*, Y^{**})$ is a Banach lattice.

19.7.1 Proposition. *The map $\varphi\colon X \times Y \to L_r(X^*, Y^{**})$ defined via $\varphi(x, y)(x^*) = x^*(x)\hat{y}$ is a bi-injective lattice bimorphism.*

Proof. It is straightforward that φ is bilinear and bi-injective. Recall that the canonical embedding of Y into Y^{**} is a lattice homomorphism, so that $|\hat{y}| = \widehat{|y|}$. For $x \in X$, $y \in Y$, and $x^* \in X^*_+$, Riesz-Kantorovich formulae yield

$$\begin{aligned}
\big|\varphi(x, y)\big|(x^*) &= \sup\big\{\big|\varphi(x, y)(z^*)\big| \ : \ z^* \in X^*, \ |z^*| \leqslant x^*\big\} \\
&= \sup\big\{\big|z^*(x)\hat{y}\big| \ : \ z^* \in X^*, \ |z^*| \leqslant x^*\big\} \\
&= \sup\big\{\big|z^*(x)\big| \ : \ z^* \in X^*, \ |z^*| \leqslant x^*\big\} \cdot |\hat{y}| \\
&= x^*\big(|x|\big)\widehat{|y|} = \varphi\big(|x|, |y|\big)(x^*).
\end{aligned}$$

We conclude that $\bigl|\varphi(x,y)\bigr| = \varphi\bigl(|x|,|y|\bigr)$. $\square$

It follows from Corollary 19.3.8 that $\bar{\varphi}\colon X\bar{\otimes}Y \to L_r(X^*,Y^{**})$ is an injective lattice homomorphism. Thus, we may identify $X\bar{\otimes}Y$ with a sublattice of $L_r(X^*,Y^{**})$. We now define the ***positive injective tensor product*** $X\otimes_{|\varepsilon|}Y$ as the closure of this sublattice. It follows immediately that $X\otimes_{|\varepsilon|}Y$ is a Banach lattice and $X\bar{\otimes}Y$ is a dense sublattice in it (up to the identification above). Since $X\otimes Y$ is uniformly dense in $X\bar{\otimes}Y$, it is norm dense in $X\otimes_{|\varepsilon|}Y$. We will write $\|\cdot\|_{|\varepsilon|}$ for the norm of $X\otimes_{|\varepsilon|}Y$, and call it the positive injective tensor norm. As we see, this is just the restriction of the (regular) norm of $L_r(X^*,Y^{**})$ to $X\otimes_{|\varepsilon|}Y$. In particular, $\|a\|_{|\varepsilon|}=\|\bar{\varphi}(a)\|_r$ for every $a\in X\bar{\otimes}Y$. Since $X\bar{\otimes}Y$ is a sublattice of $X\otimes_{|\varepsilon|}Y$, an element $a\in X\otimes Y$ is positive in $X\otimes_{|\varepsilon|}Y$ iff it is positive in $X\bar{\otimes}Y$. That is, $(X\otimes_{|\varepsilon|}Y)\cap(X\otimes Y)=(X\otimes Y)_+$. It is also easy to see from this definition that $\|\cdot\|_{|\varepsilon|}$ is a cross-norm in the sense that $\|x\otimes y\|_{|\varepsilon|}=\|x\|\|y\|$ for all $x\in X$ and $y\in Y$.

Can one replace Y^{**} with Y in the preceding construction, similarly to what we did with $X\otimes_\varepsilon Y$? One can define $\psi\colon X\times Y\to L_r(X^*,Y)$ via $\psi(x,y)(x^*)=x^*(x)y$; this is clearly bi-linear and bi-injective. The obvious obstacle is that $L_r(X^*,Y)$ generally fails to be a vector lattice. However, there is a way to fix this issue. By Proposition 13.5.2, the set $G(X^*,Y)=F(X^*,Y)^\vee-F(X^*,Y)^\vee$ is a vector lattice inside $L_r(X^*,Y)$, containing all operators of finite rank. We may view ψ as a function from $X\times Y$ to $G(X^*,Y)$. Reproducing the preceding argument, we show that ψ is a bi-injective lattice bimorphism and, therefore, $\bar{\psi}$ is a lattice isomorphic embedding from $X\bar{\otimes}Y$ into $G(X^*,Y)$. Hence, we may view $X\bar{\otimes}Y$ as a sublattice of $G(X^*,Y)$, and a vector lattice inside $L(X^*,Y)$. In particular, lattice operations for elements of $X\bar{\otimes}Y$ are defined in $L_r(X^*,Y)$. We've got the following diagram:

$$X\otimes Y\subseteq X\bar{\otimes}Y\overset{\bar{\psi}}{\hookrightarrow}G(X^*,Y)\subseteq L_r(X^*,Y)\overset{J}{\hookrightarrow}L_r(X^*,Y^{**}),$$

where $JT=jT$, the composition of T with the canonical embedding $j\colon Y\hookrightarrow Y^{**}$. Since the restriction of J to $G(X^*,Y)$ is a lattice isometry by Proposition 13.5.4, by uniqueness in Fremlin's Theorem 19.5.3, $J\bar{\psi}=\bar{\varphi}$. That is, the two preceding constructions are "compatible", they yield the same "copy" of $X\bar{\otimes}Y$ in $L_r(X^*,Y^{**})$. Moreover, since J is an isometry on $G(X^*,Y)$ by Proposition 13.5.4, we have $\|\bar{\psi}(a)\|_r=\|\bar{\varphi}(a)\|_r$ for all $a\in X\bar{\otimes}Y$. Hence, the both constructions yield the same norm on $X\bar{\otimes}Y$. We may now identify $X\otimes_{|\varepsilon|}Y$ with the closure of $X\bar{\otimes}Y$ in $L_r(X^*,Y)$ in the regular norm. For $a\in X\bar{\otimes}Y$, we write $T_a=\bar{\psi}(a)\in G(X^*,Y)$.

Next, we discuss the relationship between the norms $\|\cdot\|_{|\varepsilon|}$ and $\|\cdot\|_\varepsilon$ on $X\otimes Y$. For every $a\in X\otimes Y$ we have $\|a\|_{|\varepsilon|}=\|T_a\|_r\geqslant\|T_a\|=\|a\|_\varepsilon$. Observe that if $a\in(X\otimes Y)_+$

(in particular, when $a \in X_+ \oplus Y_+$) then $T_a \geqslant 0$, so that $\|T_a\|_r = \|T_a\|$ and, therefore, $\|a\|_{|\varepsilon|} = \|a\|_\varepsilon$.

19.7.2 Proposition. *Let $a \in X \bar{\otimes} Y$. Then*

$$\|a\|_{|\varepsilon|} = \inf\bigl\{\|c\|_\varepsilon \,:\, c \in X_+ \otimes Y_+, |a| \leqslant c\bigr\}.$$

Proof. On one hand, if $c \in X_+ \otimes Y_+$ and $|a| \leqslant c$ then $\|a\|_{|\varepsilon|} \leqslant \|c\|_{|\varepsilon|} = \|c\|_\varepsilon$.

To prove the other direction, fix $a \in X \bar{\otimes} Y$ and $\delta > 0$. It suffices to show that there exists $c \in X_+ \otimes Y_+$ such that $|a| \leqslant c$ and $\|c\|_\varepsilon \leqslant \|a\|_{|\varepsilon|} + \delta$.

By Proposition 19.3.7, we can find $u \in X_+$, $v \in Y_+$, and $b \in X \otimes Y$ such that $\bigl||a| - b\bigr| \leqslant \varepsilon u \otimes v$, where $\varepsilon = \frac{\delta}{4\|u\|\|v\|}$. It follows that $|a| \leqslant b + \varepsilon u \otimes v$. It also follows that

$$\bigl||a| - (b + \varepsilon u \otimes v)\bigr| \leqslant \bigl||a| - b\bigr| + \varepsilon u \otimes v \leqslant 2\varepsilon u \otimes v.$$

Since $\bigl\|2\varepsilon u \otimes v\bigr\|_{|\varepsilon|} = \frac{\delta}{2}$, this yields $\|a\|_{|\varepsilon|} \geqslant \|b + \varepsilon u \otimes v\|_{|\varepsilon|} - \frac{\delta}{2}$.

It follows from $|a| \leqslant b+\varepsilon u\otimes v$ that $b+\varepsilon u\otimes v \in (X\otimes Y)_+$. By Theorem 19.4.3, there exist $x \in X_+$ and $y \in Y_+$ such that $b+\varepsilon u\otimes v+rx\otimes y$ is in $X_+\otimes Y_+$ for every $r > 0$. Take $r = \frac{\delta}{2\|x\|\|y\|}$ and put $c = b+\varepsilon u\otimes v+rx\otimes y$. Then $c \in X_+\otimes Y_+$, $c \geqslant |a|$, and it follows from $\bigl\|c-(b+\varepsilon u\otimes v)\bigr\|_{|\varepsilon|} = \|rx\otimes y\|_\varepsilon = \frac{\delta}{2}$ that $\|c\|_\varepsilon = \|c\|_{|\varepsilon|} \leqslant \|b+\varepsilon u\otimes v\|_{|\varepsilon|}+\frac{\delta}{2} \leqslant \|a\|_{|\varepsilon|}+\delta$. $\square$

While the definition of the positive projective tensor norm is clearly symmetric, yielding that $X \otimes_{|\pi|} Y$ is lattice isometric to $Y \otimes_{|\pi|} X$, our definition of $X \otimes_{|\varepsilon|} Y$ was not symmetric. Yet $X \otimes_{|\varepsilon|} Y$ is lattice isometric to $Y \otimes_{|\varepsilon|} X$. One way to see this is to use Proposition 19.7.2 and the (well known) fact that the injective tensor product is symmetric. Alternatively, one can argue as follows. Let $a \in X \bar{\otimes} Y$. We can express a as a lattice linear combination of elementary tensors: $a = \Phi(x_1 \otimes y_1, \dots, x_n \otimes y_n)$. Let $b \in Y \bar{\otimes} X$ given by $b = \Phi(y_1 \otimes x_1, \dots, y_n \otimes x_n)$. As before, let $\bar{\psi}\colon X \bar{\otimes} Y \to G(X^*, Y)$ such that $\bar{\psi}(x \otimes y)(x^*) = x^*(x)y$; then $\|a\|_{|\varepsilon|} = \|\bar{\psi}(a)\|_r$. Similarly, let $\bar{\phi}\colon Y \bar{\otimes} X \to G(Y^*, X^{**})$ given by $\bar{\phi}(y \otimes x)(y^*) = y^*(y)\hat{x}$; then $\|b\|_{|\varepsilon|} = \|\bar{\phi}(b)\|_r$. With a minor abuse of notation, we may write $\bar{\psi}(a) = \Phi(x_1 \otimes y_1, \dots, x_n \otimes y_n)$, where we view $x_i \otimes y_i$ as an operator in $F(X, Y)$. Since the adjoint map is a lattice homomorphism on $G(X^*, Y)$ by Proposition 13.5.5, we have

$$\bigl(\bar{\psi}(a)\bigr)^* = \Phi\bigl((x_1 \otimes y_1)^*, \dots, (x_n \otimes y_n)^*\bigr) = \Phi\bigl((y_1 \otimes \hat{x}_1)^*, \dots, (y_n \otimes \hat{x}_n)^*\bigr) = \bar{\phi}(b).$$

It follows from Corollary 13.5.6 that

$$\|a\|_{|\varepsilon|} = \bigl\|\bar{\psi}(a)\bigr\|_r = \bigl\|\bar{\psi}(a)^*\bigr\|_r = \bigl\|\bar{\phi}(b)\bigr\| = \|b\|_{|\varepsilon|}.$$

This means that the canonical lattice isomorphism between $X \bar{\otimes} Y$ and $Y \bar{\otimes} X$ is an isometry with respect to the lattice injective tensor norms. By density, it extends to a surjective lattice isometry between $X \otimes_{|\varepsilon|} Y$ and $Y \otimes_{|\varepsilon|} X$. To summarize:

19.7.3 Corollary. *The map* $\sum_{i=1}^n x_i \otimes y_i \mapsto \sum_{i=1}^n y_i \otimes x_i$ *between* $X \otimes Y$ *and* $Y \otimes X$ *extends to a surjective lattice isometry between* $X \otimes_{|\varepsilon|} Y$ *and* $Y \otimes_{|\varepsilon|} X$.

We used properties of regular operators to define and investigate positive injective tensor products. Now we will go the other way: we will apply positive injective tensor products to study regular operators. Recall that $G(X,Y)$ is the vector lattice in $L_r(X,Y)$ generated by operators of finite rank; cf. Proposition 13.5.2.

19.7.4 Proposition. *There is a surjective lattice isomorphism* $V\colon X^* \bar{\otimes} Y \to G(X,Y)$ *such that* $\big(V(x^* \otimes y)\big)(x) = x^*(x)y$ *for all* $x \in X$, $y \in Y$, *and* $x^* \in X^*$. *Furthermore,* $\big\|Va\big\|_r = \|a\|_{|\varepsilon|}$ *for every* $a \in X^* \bar{\otimes} Y$. *It follows that* V *extends to a lattice isometry between* $X^* \otimes_{|\varepsilon|} Y$ *and the closure of* $G(X,Y)$ *in* $L_r(X,Y)$.

Proof. Define $\varphi\colon X^* \times Y \to G(X,Y)$ via $\big(\varphi(x^*,y)\big)(x) = x^*(x)y$. Clearly, $\operatorname{Range}\varphi^{\otimes}$ consists of all operators of rank one in $L(X,Y)$. As in the proof of Proposition 19.7.1, we show that φ is a bi-injective lattice bimorphism. It follows that $\bar{\varphi}\colon X \bar{\otimes} Y \to G(X,Y)$ is a lattice isomorphic embedding. It is onto because $\operatorname{Range}\bar{\varphi}$ contains all operators of finite rank, hence is a generating set in $G(X,Y)$. Put $V = \bar{\varphi}$.

Let $a \in X^* \otimes Y$, with $a = \sum_{i=1}^n x_i^* \otimes y_i$. Then $\|a\|_{|\varepsilon|} = \|T\|_r$, where $T \in L_r(X^{**},Y)$ is given by $Tx^{**} = \sum_{i=1}^n x^{**}(x_i^*)y_i$. Let $S = Va$. It is easy to see that S is the operator in $L(X,Y)$ given by $Sx = \sum_{i=1}^n x_i^*(x)y_i$. By Proposition 13.5.7, we have $\|T\|_r = \|S\|_r$ and, therefore, $\big\|Va\big\|_r = \|a\|_{|\varepsilon|}$. It follows that the restriction of V to $X^* \otimes Y$ is a $\|\cdot\|_{|\varepsilon|}$-to-$\|\cdot\|_r$ isometry.

Since $X^* \otimes Y$ is dense in $X^* \otimes_{|\varepsilon|} Y$, V extends to a lattice isometry between $X^* \otimes_{|\varepsilon|} Y$ and $\overline{G(X,Y)}$. □

19.7.5 Corollary. $X^* \otimes_{|\varepsilon|} Y^*$ *embeds lattice isometrically into* $(X \otimes_{|\pi|} Y)^*$.

Proof. $X^* \otimes_{|\varepsilon|} Y^* \simeq \overline{G(X,Y^*)} \subseteq L_r(X,Y^*) = (X \otimes_{|\pi|} Y)^*$. □

It is easy to see that the duality in Corollary 19.7.5 is the "trace duality" in the sense that $\langle f \otimes g, x \otimes y\rangle = f(x)g(y)$ whenever $f \in X^*$, $g \in Y^*$, $x \in X$, and $y \in Y$.

We know that every operator T in $G(X,Y)$ is compact. Combining the Proposition 19.7.4, with Theorem 19.3.7, we can now show more: T is a uniform limit of finite-rank operators (cf. [Sch84, Proposition 3.16]):

19.7.6 Corollary. *For every* $T \in G(X,Y)$ *there exists a positive operator* U *of rank one such that for every* $\varepsilon > 0$ *there exists* $S \in F(X,Y)$ *such that* $|T - S| \leqslant \varepsilon U$.

Note that in the setting of Corollary 19.7.6, we have $\|T - S\| \leqslant \|T - S\|_r \leqslant \varepsilon \|U\|_r$. We conclude that $F(X,Y)$ is dense in $G(X,Y)$ in the regular and in the operator norm. It follows that the closure of $G(X,Y)$ in the regular or the operator norm agrees with the closure of $F(X,Y)$ is the same norm. Combining this with 13.5.3 and with Propositions 13.5.4 and 13.5.5, we get the following result of Schlotterbeck and Arendt (see [Ar81]):

19.7.7 Corollary. *The closure of $F(X,Y)$ in the regular norm is a Banach lattice. The adjoint map and the map $T \mapsto j_Y T$ are lattice isometries on it (with respect to the regular norms).*

We finish this section with a couple of special cases.

19.7.8 Proposition. $X \otimes_{|\varepsilon|} Y = X \otimes_\varepsilon Y$ *if* $Y = C(K)$ *for some compact space* K.

Proof. Since $X \otimes Y$ is dense in both spaces, it suffices to show that the norms $\|\cdot\|_{|\varepsilon|}$ and $\|\cdot\|_\varepsilon$ agree on $X \otimes Y$. Let $a \in X \otimes C(K)$. By the definitions of the two norms, $\|a\|_{|\varepsilon|}$ and $\|a\|_\varepsilon$ equal $\|T\|_r$ and $\|T\|$, respectively, for some $T \in F(X^*, Y^{**})$. So it suffices to show that $\|T\|_r = \|T\|$.

We always have $\|T\| \leqslant \|T\|_r = \big\||T|\big\|$. Let $x \in B_X^+$. By Riesz-Kantorovich Theorem, $|T|x = \sup\big\{|Tz| \ :\ |z| \leqslant x\big\}$. Since $Y^{**} = C(Q)$ for some compact space Q, the norm and the supremum are interchangeable for sets in Y^{**}, so $\big\||T|x\big\| = \sup\big\{\|Tz\| \ :\ |z| \leqslant x\big\} \leqslant \|T\|\|x\|$. It follows that $\|T\|_r \leqslant \|T\|$. □

Since it is known that $C(K) \otimes_\varepsilon C(L) = C(K \times L)$, we immediately get the following:

19.7.9 Corollary. $C(K) \otimes_{|\varepsilon|} C(L) = C(K \times L)$.

19.7.10 Proposition. $L_1(\mu) \otimes_{|\varepsilon|} L_1(\nu) = L_1(\mu \times \nu)$ *for any two* σ*-finite measures* μ *and* ν.

Proof. By Theorem 19.6.12 and Corollary 19.6.13, $L_1(\mu) \otimes_{|\pi|} L_1(\nu) = L_1(\mu) \otimes_\pi L_1(\nu) = L_1(\mu \times \nu)$. In particular, $\|\cdot\|_{|\pi|}$ and $\|\cdot\|_\pi$ agree on $L_1(\mu) \otimes L_1(\nu)$.

Let $a \in L_1(\mu)_+ \otimes L_1(\nu)_+$. That is, $a = \sum_{i=1}^n f_i \otimes g_i$ for some positive f_i's and g_i's. Then $\|a\|_{|\varepsilon|} = \|T\|_r$ for $T \in L\big(L_\infty(\mu), L_1(\nu)\big)$ given by $Th = \sum_{i=1}^n \langle f_i, h\rangle g_i$ for every $h \in L_\infty(\mu)$. Since $T \geqslant 0$, we get

$$\begin{aligned}\|a\|_{|\varepsilon|} = \|T\|_r = \|T\| = \|T\mathbb{1}\| = \Big\|\sum_{i=1}^n \langle f_i, \mathbb{1}\rangle g_i\Big\| \\ = \Big\|\sum_{i=1}^n \|f_i\| g_i\Big\| = \sum_{i=1}^n \|f_i\|\|g_i\| = \|a\|_\pi = \|a\|_{|\pi|}.\end{aligned}$$

It follows that $\|\cdot\|_{|\varepsilon|}$ and $\|\cdot\|_{|\pi|}$ agree on $L_1(\mu)_+ \otimes L_1(\nu)_+$. Since this cone is uniformly dense in $\big(L_1(\mu)\bar{\otimes}L_1(\nu)\big)_+$ by Corollary 19.4.4, the two norms agree on $\big(L_1(\mu)\bar{\otimes}L_1(\nu)\big)_+$. Since they are both lattice norms, they agree on $L_1(\mu) \bar{\otimes} L_1(\nu)$. We conclude that $L_1(\mu) \otimes_{|\varepsilon|} L_1(\nu) = L_1(\mu) \otimes_{|\pi|} L_1(\nu)$, because the two spaces are the completions of $L_1(\mu) \bar{\otimes} L_1(\nu)$ under $\|\cdot\|_{|\varepsilon|}$ and $\|\cdot\|_{|\pi|}$, respectively. $\square$

Chapter 20

Multinormed spaces

This chapter is based on [DLOT17].

20.1 Definition and basic properties

20.1.1 Multinorms

Let X be a vector space over $\mathbb{R}$. For each $n \in \mathbb{N}$, let $\|\cdot\|_n$ be a norm on X^n. Hence, for each n-tuple $x_1, \dots, x_n \in X$, the norm $\big\|(x_1, \dots, x_n)\big\|_n$ is defined. We can view $\bar{x} = (x_1, \dots, x_n) \in X^n$ as a ***multivector*** and write $\|\bar{x}\|_n$ instead of $\big\|(x_1, \dots, x_n)\big\|_n$. We will often write $\|\bar{x}\|$ instead of $\|\bar{x}\|_n$ when the value of n is clear from the context. Such a sequence of norms is said to be a ***multinorm*** on X if the following axioms are satisfied for every n and every $x_1, \dots, x_n \in X$:

(A1) $\big\|(x_{\sigma(1)}, \dots, x_{\sigma(n)})\big\|_n = \big\|(x_1, \dots, x_n)\big\|_n$ for every permutation σ of $\{1, \dots, n\}$. That is, the norms are invariant under permutations of vectors;

(A2) $\big\|(x_1, \dots, x_n, 0)\big\|_{n+1} = \big\|(x_1, \dots, x_n)\big\|_n$, that is, padding an n-tuple with zeros does not affect its norm;

(A3) $\big\|(\alpha_1 x_1, \dots, \alpha_n x_n)\big\|_n \leqslant \big(\max|\alpha_i|\big)\big\|(x_1, \dots, x_n)\big\|_n$ for any scalars $\alpha_1, \dots, \alpha_n$; and

(A4) $\big\|(x_1, \dots, x_{n-1}, x_n, x_n)\big\|_{n+1} = \big\|(x_1, \dots, x_{n-1}, x_n)\big\|_n$. That is, repeating an entry does not affect the norm.

While the first three axioms are very natural, the fourth one is quite restrictive as it essentially enforces ℓ_∞-behaviour.

20.1.1 Exercise. If $X = \mathbb{R}$, the only multinorm on $\mathbb{R}$ is the ℓ_∞-norm, i.e., $\|(t_1, \dots, t_n)\|_n = \max|t_i|$.

20.1.2 Exercise. Let X be a normed space. For $x_1, \dots, x_n \in X$, define $\big\|(x_1, \dots, x_n)\big\| = \max\|x_i\|$. Verify that this is a multinorm.

20.1.3 Example. Let X be a normed lattice. For $x_1, \dots, x_n \in X$, define

$$\big\|(x_1, \dots, x_n)\big\| = \big\||x_1| \vee \dots \vee |x_n|\big\|$$

It is easy to see that this is a multinorm. We will call it the ***canonical multinorm*** on X.

Given a scalar matrix $A \in M_{m,n}$, $A = (a_{ij})$, we write $\big\|A\colon \ell_p^n \to \ell_q^m\big\|$ for the norm of A viewed as an operator from ℓ_p^n to ℓ_q^m. In this setting, the transpose matrix A^t may be identified with the adjoint of A, hence

$$\big\|A^t\colon \ell_{q^*}^m \to \ell_{p^*}^n\big\| = \big\|A\colon \ell_p^n \to \ell_q^m\big\|.$$

Given a multivector $\bar{x} = (x_1, \dots, x_n) \in X^n$, we write $A\bar{x}$ for the multivector in X^m whose i-th component is $\sum_{j=1}^n a_{ij}x_j$ (we view $\bar{x}$ as a *column* multivector). Let f be a linear functional on X. We write $f(\bar{x})$ for the vector $\big(f(x_1), \dots, f(x_n)\big)$ in $\mathbb{R}^n$. It can be easily verified that $f(A\bar{x}) = Af(\bar{x})$.

20.1.4 Proposition. *Let X be a vector space, and for every $n \in \mathbb{N}$ let $\|\cdot\|_n$ be a norm on X^n. This sequence of norms is a multinorm iff*

$$\|A\bar{x}\|_m \leqslant \big\|A\colon \ell_\infty^n \to \ell_\infty^m\big\| \cdot \|\bar{x}\|_n.$$

for every $\bar{x} \in X^n$ and every scalar matrix $A \in M_{m,n}$.

Proof. For an $m \times n$ scalar matrix A, we write $\|A\| := \|A\colon \ell_\infty^n \to \ell_\infty^m\|$. It is easy to see that it is equal to the maximum of the ℓ_1-norms of the rows of A.

The converse implication is easy as each axiom of a multinorm can we written in the form $\|A\bar{x}\| \leqslant \|\bar{x}\|$ for an appropriate matrix A with $\|A\| \leqslant 1$.

Suppose now that X is a multinormed space. Let $\bar{x} \in X^n$ and A be an $m \times n$ matrix. We first prove the required inequality in the special case when A has at most one non-zero entry in each row, say, $a_{1,j_1}, \dots, a_{m,j_m}$ (some of them may be zeros). Applying (A3) followed by a combination of (A1) and (A4), we get

$$\begin{aligned}\|A\bar{x}\|_m &= \big\|\big(a_{1,j_1}x_{j_1}, \dots, a_{m,j_m}x_{j_m}\big)\big\|_m \\ &\leqslant \max_i |a_{i,j_i}| \cdot \big\|(x_{j_1}, \dots, x_{j_m})\big\|_m \leqslant \|A\|\|\bar{x}\|_n.\end{aligned}$$

Now we proceed to the general case. We will find a matrix B in which every row has at most one non-zero entry, such that $\|A - B\| = \|A\| - \|B\|$. First, in each row $i = 1, \dots, m$, take an entry with the greatest absolute value: let j_i be such that $|a_{i,j_i}| = \max_j |a_{i,j}|$. Among these, take the smallest non-zero one:

$$\delta = \min\big\{|a_{i,j_i}|, a_{i,j_i} \neq 0, i = 1, \dots, m\big\}$$

We now define a matrix B as follows: if $a_{i,j_i} \neq 0$ then we define b_{i,j_i} so that $\operatorname{sign} b_{i,j_i} = \operatorname{sign} a_{i,j_i}$ and $|b_{i,j_i}| = \delta$. Otherwise, put $b_{i,j} = 0$. Then B has at most one non-zero entry in each row, $\|B\| = \delta$ and $\|A - B\| = \|A\| - \delta$. Note also that $A - B$ has more zero entries than A because the entry where δ was attained in A will become zero in $A - B$.

It follows that if we iterate this process, it will terminate in finitely many steps. Therefore, we can write $A = \sum_{k=1}^{r} B_k$ such that each B_k has at most one non-zero entry in every row and $\|A\| = \sum_{k=1}^{r} \|B_k\|$. It now follows that

$$\|A\bar{x}\| \leqslant \sum_{k=1}^{r} \|B_k \bar{x}\| \leqslant \sum_{k=1}^{r} \|B_k\| \|\bar{x}\| = \|A\| \|\bar{x}\|.$$

□

This proposition shows once again that the concept of a multinorm is naturally related to ℓ_∞ (or rather c_0).

20.1.2 1-multinorms

A sequence of norms on X^n is said to be a ***1-multinorm*** if it satisfies the axioms of a multinorm with (A4) replaced with

(A4') $\big\|(x_1, \dots, x_{n-1}, x_n, x_n)\big\|_{n+1} = \big\|(x_1, \dots, x_{n-1}, 2x_n)\big\|_n$.

20.1.5 Exercise. A sequence of norms is a 1-multinorm iff

$$\|A\bar{x}\|_m \leqslant \big\|A\colon \ell_1^n \to \ell_1^m\big\| \cdot \|\bar{x}\|_n.$$

for every multivector $\bar{x} \in X^n$ and every scalar matrix $A \in M_{m,n}$.

20.1.3 p-multinorms

Proposition 20.1.4 and Exercise 20.1.5 motivate the following definition. Let X be a vector space X and $1 \leqslant p \leqslant \infty$. For every $n \in \mathbb{N}$, let $\|\cdot\|_n$ be a norm on X^n. This sequence of norms is a ***p-multinorm*** if

$$\|A\bar{x}\|_m \leqslant \big\|A\colon \ell_p^n \to \ell_p^m\big\| \cdot \|\bar{x}\|_n.$$

for every multivector $\bar{x} \in X^n$ and every scalar matrix $A \in M_{m,n}$. In this case, we say that X is a p-multinormed space.

Under this definition, a *multinorm* is an ∞-*multinorm*.

Note that every p-multinormed space is, in particular, a normed space with respect to $\|\cdot\|_1$. However, there may be many different p-multinorms on X which agree on the "ground level".

20.1.6 Exercise. Show that every p-multinorm satisfies Axioms (A1)–(A3) in the definition of a multinorm.

20.1.7 Exercise. Show that the only p-multinorm on $\mathbb{R}$ is given by $\big\|(t_1,\dots,t_n)\big\| = \Big(\sum_{i=1}^n |t_i|^p\Big)^{\frac{1}{p}}$.

20.1.8 Exercise. Show that for every p-multinormed space X and every $x_1,\dots,x_n$ in X one has

$$\max_{1\leqslant i\leqslant n}\|x_i\| \leqslant \big\|(x_1,\dots,x_n)\big\| \leqslant \sum_{i=1}^n \|x_i\|.$$

20.1.9 Exercise. Let X be a p-multinormed space. If $\|\cdot\|_1$ is complete then $\|\cdot\|_n$ is complete for every n.

Let X be a p-multinormed space for some $1\leqslant p\leqslant\infty$. We say that the p-multinorm is ***p-convex*** if

$$\big\|(x_1,\dots,x_n)\big\|^p \leqslant \big\|(x_1,\dots,x_k)\big\|^p + \big\|(x_{k+1},\dots,x_n)\big\|^p$$

for any $x_1,\dots,x_n\in X$ and any $k<n$. Clearly, this property is inherited by subspaces.

20.2 Subspaces, quotients, duality, and operators

Let X be a p-multinormed space and Y a vector subspace of X. The restriction of the p-multinorm to Y makes Y into a p-multinormed space. It is also easy to see that the quotient space X/Y can be naturally equipped with a p-multinorm via

$$\Big\|\big(x_1+Y,\dots,x_n+Y\big)\Big\| = \inf_{y_1,\dots,y_n\in Y}\Big\|\big(x_1+y_1,\dots,x_n+y_n\big)\Big\|.$$

20.2.1 Exercise. Verify that this formula indeed defines a p-multinorm on X/Y.

One can say that the category of p-multinormed spaces is closed under taking subspaces and quotients.

20.2.1 Duality

Let X be a p-multinormed space. In particular, it is a normed space. For every $n\in\mathbb{N}$, one can identify $(X^*)^n$ with $(X^n)^*$ as follows: for $\bar f=(f_1,\dots,f_n)\in(X^*)^n$ and $\bar x=(x_1,\dots,x_n)\in X^n$, put

$$\langle \bar f,\bar x\rangle = \sum_{i=1}^n \langle f_i,x_i\rangle.$$

Since $\|\cdot\|_n$ is a norm on X^n, we have the dual norm on $(X^n)^*$ and, therefore, on $(X^*)^n$. Specifically,

$$\|\bar{f}\| = \sup\bigl\{\bigl|\langle \bar{f}, \bar{x}\rangle\bigr| \;:\; \bar{x} \in X^n,\ \|\bar{x}\| \leqslant 1\bigr\}$$

for every $\bar{f} \in (X^*)^n$. This procedure produces a sequence of norms on the powers of X^*. We claim that this is a q-multinorm, where q is the conjugate of p. Indeed, take any $\bar{f} = (f_1, \dots, f_m) \in (X^*)^m$ and $A \in M_{n,m}$. Let $\bar{x} = (x_1, \dots, x_n) \in X^n$ with $\|\bar{x}\| \leqslant 1$. A straightforward calculation shows that

$$\begin{aligned}\bigl|\langle A\bar{f}, \bar{x}\rangle\bigr| = \bigl|\langle \bar{f}, A^t\bar{x}\rangle\bigr| \leqslant \|\bar{f}\|\|A^t\bar{x}\| \\ \leqslant \|\bar{f}\|\|A^t\colon \ell_p^n \to \ell_p^m\|\|\bar{x}\| \leqslant \|\bar{f}\|\|A\colon \ell_q^m \to \ell_q^n\|,\end{aligned}$$

so that $\|A\bar{f}\| \leqslant \|A\colon \ell_q^m \to \ell_q^n\|\|\bar{f}\|$. Thus, the dual of a p-multinormed space is a q-multinormed space.

20.2.2 Exercise. Use duality to solve Exercise 20.1.5.

20.2.2 Operators

Let X and Y be two p-multinormed spaces and $T\colon X \to Y$ a linear operator. We say that T is ***multibounded*** if there exists a constant C such that

$$\bigl\|(Tx_1, \dots, Tx_n)\bigr\| \leqslant C\|(x_1, \dots, x_n)\|$$

for any $x_1, \dots, x_n \in X$. There is another way to look at it. For every n, consider the n-th ***amplification*** of T:

$$T^{(n)}\colon X^n \to Y^n, \qquad T^{(n)}(x_1, \dots, x_n) = (Tx_1, \dots, Tx_n).$$

So T is multibounded if all its amplifications are uniformly bounded by some constant. The infimum of all such constants is denoted $\|T\|_{\mathrm{mb}}$. Clearly, $\|T\| \leqslant \|T\|_{\mathrm{mb}}$. We say that T is a ***multiisometry*** if

$$\bigl\|(Tx_1, \dots, Tx_n)\bigr\| = \|(x_1, \dots, x_n)\|$$

for any $x_1, \dots, x_n \in X$ or, equivalently, if all the amplifications of T are isometries. We say that X and Y are ***multiisometric*** if there exists a surjective multiisometry from X to Y.

20.2.3 Exercise. Suppose that $T\colon X \to Y$ is multibounded. Then $T^*\colon Y^* \to X^*$ is multibounded with respect to the dual multinorms and $\|T^*\|_{\mathrm{mb}} = \|T\|_{\mathrm{mb}}$.

20.2.4 Exercise. Let $T\colon X \to Y$ be a linear operator between two p-multinormed spaces. Then the induced operator $\widetilde{T}\colon X/\ker T \to Y$ is a surjective multiisometry iff $T^{(n)}(B^o_{X^n}) = B^o_{Y^n}$ for every n. (Recall that B^o_X stands for the open unit ball of X.)

20.3 p-multinorms and tensor norms on $\ell_p \otimes X$

Let X be a p-multinormed space. We may view the p-multinorm as a norm on finite sequences of elements of X. More precisely, we write $c_{00}(X)$ for the vector space of all infinite sequences of elements of X which have only finitely many non-zero terms. Due to (A2), a p-multinorm on X induces a norm on $c_{00}(X)$. We will show in this section that this norm can be extended further to $\ell_p \otimes X$.

As usual, for two vector spaces X and Y, we write $X \otimes Y$ for the algebraic tensor product of the spaces; its elements are of the form $\sum_{i=1}^n x_i \otimes y_i$. Given linear operators $S\colon X \to X$ and $T\colon Y \to Y$, they induce an operator $S \otimes T\colon X \otimes Y \to X \otimes Y$ via

$$(S \otimes T)\Big(\sum_{i=1}^n x_i \otimes y_i\Big) = \sum_{i=1}^n Sx_i \otimes Ty_i.$$

In particular,

$$(S \otimes I_Y)\Big(\sum_{i=1}^n x_i \otimes y_i\Big) = \sum_{i=1}^n Sx_i \otimes y_i.$$

Recall that if X and Y are normed spaces, a norm on $X \otimes Y$ is a ***cross-norm*** if $\|x \otimes y\| \leqslant \|x\|\|y\|$ for all $x \in X$ and $y \in Y$.

Throughout this section, X is a vector space and $1 \leqslant p < \infty$. However all the content of this section remains valid for $p = \infty$ when ℓ_∞ is replaced with c_0. For an $m \times n$ scalar matrix A we write $\|A\|$ for the norm of A as an operator from ℓ_p to ℓ_p. We denote by (e_i) the standard unit basis of ℓ_p, c_0, or c_{00}.

20.3.1 Lemma. *There is a linear bijection between $c_{00}(X)$ and $c_{00} \otimes X$ given by $(x_i) \mapsto \sum e_i \otimes x_i$. In particular, every element of $c_{00} \otimes X$ may be written in the form $\sum_{j=1}^n e_j \otimes x_j$ for some $x_1, \dots, x_n \in X$.*

Proof. It is clear that the map is linear and one-to-one. Hence, it suffices to prove the second part of the statement. Let $\xi \in c_{00} \otimes X$, then $\xi = \sum_{i=1}^k u_i \otimes z_i$ for some $u_1, \dots, u_k \in c_{00}$ and $z_1, \dots, z_k \in X$. Choose n sufficiently large so that $\operatorname{supp} u_i \subseteq \{1, \dots, n\}$ for every $i = 1, \dots, k$. So we can write $u_i = \sum_{j=1}^n \alpha_{ij} e_j$. It follows that

$$\xi = \sum_{i=1}^k \Big(\sum_{j=1}^n \alpha_{ij} e_j\Big) \otimes z_i = \sum_{j=1}^n e_j \otimes \Big(\sum_{i=1}^k \alpha_{ij} z_i\Big) = \sum_{j=1}^n e_j \otimes x_j$$

where $x_j = \sum_{i=1}^k \alpha_{ij} z_i$. □

20.3.2 Lemma. *Let A be an $m \times n$ matrix and $\bar{x} \in X^n$. Put $\bar{y} = A\bar{x}$. Then $(A \otimes I_X)\Big(\sum_{j=1}^n e_j \otimes x_j\Big) = \sum_{i=1}^m e_i \otimes y_i$.*

Proof.

$$(A \otimes I_X)\Big(\sum_{j=1}^{n} e_j \otimes x_j\Big) = \sum_{j=1}^{n} Ae_j \otimes x_j = \sum_{j=1}^{n}\Big(\sum_{i=1}^{m} a_{ij}e_i\Big) \otimes x_j$$
$$= \sum_{i=1}^{m} e_i \otimes \Big(\sum_{j=1}^{n} a_{ij}x_j\Big) = \sum_{i=1}^{m} e_i \otimes y_i.$$

□

20.3.3 Theorem. *Let X be a vector space. There is a one-to-one correspondence between the following classes of objects:*

(i) *the p-multinorms on X;*

(ii) *the cross-norms on $c_{00} \otimes X$ such that $\|A \otimes I_X\| \leqslant \|A\|$ for every matrix A; we view c_{00} as a subspace of ℓ_p here;*

(iii) *the cross-norms on $\ell_p \otimes X$ such that $\|T \otimes I_X\| \leqslant \|T\|$ for every operator $T\colon \ell_p \to \ell_p$, i.e., $\big\|\sum_{i=1}^n Tu_i \otimes x_i\big\| \leqslant \|T\|\big\|\sum_{i=1}^n u_i \otimes x_i\big\|$.*

The correspondence is given by $\big\|(x_1,\dots,x_n)\big\| = \big\|\sum_{j=1}^n e_j \otimes x_j\big\|$.

Proof. (i)⇒(ii) Suppose that X is a p-multinormed space. For $x_1,\dots,x_n \in X$, define $\Big\|\sum_{j=1}^n e_j \otimes x_j\Big\| \coloneqq \big\|(x_1,\dots,x_n)\big\|$. In view of Lemma 20.3.1, this defines a norm on $c_{00} \otimes X$. Let A be an $m \times n$ matrix and $\xi \in c_{00} \otimes X$. We can write $\xi = \sum_{j=1}^k e_j \otimes x_j$ for some $x_1,\dots,x_k \in X$. "Padding" A or $\bar{x}$ with zeros, WLOG $n = k$. Put $\bar{y} = A\bar{x}$. By Lemma 20.3.2,

$$\big\|(A \otimes I_X)\xi\big\| = \Big\|\sum_{i=1}^{m} e_i \otimes y_i\Big\| = \|\bar{y}\| \leqslant \|A\|\|\bar{x}\| = \|A\|\|\xi\|.$$

To show that this norm is a cross-norm, consider an elementary tensor $u \otimes z$ with $z \in X$ and $u = \sum_{j=1}^n \alpha_j e_j$ in c_{00}. Then

$$\big\|u \otimes z\big\| = \Big\|\sum_{j=1}^{n} e_j \otimes (\alpha_j z)\Big\| = \big\|(\alpha_1 z,\dots,\alpha_n z)\big\| = \|Az\|,$$

where A is a column matrix, $A = (\alpha_1,\dots,\alpha_n)^T$. It follows that $\big\|u \otimes z\big\| \leqslant \|A\|\|z\| = \|u\|\|z\|$.

(ii)⇒(i) Suppose we start with a norm on $c_{00} \otimes X$ such that $\|A \otimes I_X\| \leqslant \|A\|$ for every matrix A. For $x_1,\dots,x_n \in X$, put $\big\|(x_1,\dots,x_n)\big\| \coloneqq \Big\|\sum_{i=1}^n e_i \otimes x_i\Big\|$. Let's verify that this is a p-multinorm. Given $\bar{x} \in X^n$ and $A \in M_{m,n}$, put $\bar{y} = A\bar{x}$. By Lemma 20.3.2,

$$\|\bar{y}\| = \Big\|\sum_{i=1}^{m} e_i \otimes y_i\Big\| = \Big\|(A \otimes I_X)\Big(\sum_{j=1}^{n} e_j \otimes x_j\Big)\Big\| \leqslant \|A\|\Big\|\sum_{j=1}^{n} e_j \otimes x_j\Big\| = \|A\|\|\bar{x}\|.$$

(iii)⇒(ii) Clearly, the restriction of any norm in (iii) to $c_{00}\otimes X$ gives a norm in (ii).

(ii)⇒(iii) Suppose we start with a norm as in (ii). We will extend it to a norm on $\ell_p\otimes X$ as in (iii) as follows. For every n, let $P_n\colon\ell_p\to\ell_p$ be the n-th basis projection, i.e., $P_n\Bigl(\sum_{j=1}^\infty\alpha_je_j\Bigr)=\sum_{j=1}^n\alpha_je_j$. Let $\xi\in\ell_p\otimes X$, so that $\xi=\sum_{i=1}^k u_i\otimes z_i$ for some $u_1,\dots,u_k\in\ell_p$ and $z_1,\dots,z_k\in X$. WLOG, the two k-tuples are linearly independent. For every n we have $(P_n\otimes I_X)\xi\in c_{00}\otimes X$, so that $\lambda_n:=\bigl\lVert(P_n\otimes I_X)\xi\bigr\rVert$ is defined. Note that

$$\begin{aligned}\lambda_n=\bigl\lVert(P_nP_{n+1}\otimes I_X)\xi\bigr\rVert=\bigl\lVert(P_n\otimes I_X)(P_{n+1}\otimes I_X)\xi\bigr\rVert\\ \leqslant\lVert P_n\rVert\bigl\lVert(P_{n+1}\otimes I_X)\xi\bigr\rVert\leqslant\lambda_{n+1}.\end{aligned}$$

On the other hand, $\lambda_n=\bigl\lVert\sum_{i=1}^kP_nu_i\otimes z_i\bigr\rVert\leqslant\sum_{i=1}^k\lVert u_i\rVert\lVert z_i\rVert$. Hence, the sequence (λ_n) converges. We now define

$$\lVert\xi\rVert:=\lim_{n\to\infty}\bigl\lVert(P_n\otimes I_X)\xi\bigr\rVert=\sup_n\bigl\lVert(P_n\otimes I_X)\xi\bigr\rVert.$$

Check that this is a norm on $\ell_p\otimes X$. It is easy to see that it is positively homogeneous and satisfies the triangle inequality. Let $\xi\neq0$. Since $u_1,\dots,u_k$ are linearly independent, there exists n such that $P_nu_1,\dots,P_nu_k$ are linearly independent. It follows that $\sum_{i=1}^kP_nu_i\otimes z_i\neq0$, so that $\lVert\xi\rVert\geqslant\Bigl\lVert\sum_{i=1}^kP_nu_i\otimes z_i\Bigr\rVert>0$.

Let's check that this is a cross-norm. Let $u\in\ell_p$ and $z\in X$. Since we know that the norm on $c_{00}\otimes X$ is a cross-norm, we have

$$\lVert u\otimes z\rVert=\sup_n\bigl\lVert P_nu\otimes z\bigr\rVert\leqslant\sup_n\lVert P_nu\rVert\lVert z\rVert\leqslant\lVert u\rVert\lVert z\rVert.$$

Let T be an operator on ℓ_p and $\xi\in\ell_p\otimes X$. We need to show that $\bigl\lVert(T\otimes I_X)\xi\bigr\rVert\leqslant\lVert T\rVert\lVert\xi\rVert$. WLOG, $\lVert T\rVert=1$. As before, let $\xi=\sum_{i=1}^ku_i\otimes z_i$. Since

$$\bigl\lVert(T\otimes I_X)\xi\bigr\rVert=\sup_n\bigl\lVert(P_n\otimes I_X)(T\otimes I_X)\xi\bigr\rVert=\sup_n\bigl\lVert(P_nT\otimes I_X)\xi\bigr\rVert,$$

it suffices to prove that $\bigl\lVert(P_nT\otimes I_X)\xi\bigr\rVert\leqslant\lVert\xi\rVert$ for every n. Fix n. For each m, put $\xi_m:=(P_m\otimes I_X)\xi\in c_{00}\otimes X$. Then

$$\begin{aligned}\bigl\lVert(P_nT\otimes I_X)\xi\bigr\rVert\leqslant\bigl\lVert(P_nT\otimes I_X)(\xi-\xi_m)\bigr\rVert+\bigl\lVert(P_nT\otimes I_X)\xi_m\bigr\rVert\\ \leqslant\sum_{i=1}^k\lVert P_nT\rVert\lVert(I-P_m)u_i\rVert\lVert z_i\rVert+\bigl\lVert(P_nTP_m\otimes I_X)\xi_m\bigr\rVert.\end{aligned}$$

As $m\to\infty$, the first expression converges to zero because $\lVert(I-P_m)u\rVert\to0$ for every $u\in\ell_p$. On the other hand,

$$\bigl\lVert(P_nTP_m\otimes I_X)\xi_m\bigr\rVert\leqslant\lVert P_nTP_m\rVert\lVert\xi_m\rVert\leqslant\lVert\xi\rVert.$$

Therefore, $\bigl\lVert(P_nT\otimes I_X)\xi\bigr\rVert\leqslant\lVert\xi\rVert$. □

20.3.4 Remark. It follows from the proof that in (iii), it suffices to only consider compact or finite-rank operators.

20.4 Strong p-multinorms

Again, let X be a vector space, and for every $n \in \mathbb{N}$ let $\|\cdot\|_n$ be a norm on X^n. In particular, X itself is a normed space. We say that this sequence of norms is a ***strong p-multinorm*** if the following condition is true: given any $x_1, \dots, x_n, y_1, \dots, y_m \in X$;

$$\text{if } \Big(\sum_{i=1}^{n} \big|f(x_i)\big|^p\Big)^{\frac{1}{p}} \leqslant \Big(\sum_{j=1}^{m} \big|f(y_j)\big|^p\Big)^{\frac{1}{p}} \text{ for every } f \in X^*$$
$$\text{then } \big\|(x_1, \dots, x_n)\big\| \leqslant \big\|(y_1, \dots, y_m)\big\|.$$

That is,

$$\text{if } \big\|f(\bar{x})\big\|_{\ell_p^n} \leqslant \big\|f(\bar{y})\big\|_{\ell_p^m} \text{ for every } f \in X^* \text{ then } \|\bar{x}\|_n \leqslant \|\bar{y}\|_m.$$

One can think of this property as a kind of monotonicity. Our motivation for introducing it will become clear in Section 20.7.

20.4.1 Proposition. *Every strong p-multinorm is a p-multinorm.*

Proof. Let $\bar{x} \in X^n$ and $A \in M_{m,n}$ with $\|A \colon \ell_p^n \to \ell_p^m\| \leqslant 1$. For every $f \in X^*$ one has

$$\big\|f(A\bar{x})\big\|_{\ell_p^m} = \big\|Af(\bar{x})\big\|_{\ell_p^m} \leqslant \big\|f(\bar{x})\big\|_{\ell_p^n}.$$

It follows that $\|A\bar{x}\| \leqslant \|\bar{x}\|$. □

20.4.2 Proposition. *If $p = 2$ or $p = \infty$ then every p-multinorm is strong.*

Proof. Let X be a p-multinormed space. Suppose that $x_1, \dots, x_n, y_1, \dots, y_m \in X$ and $\big\|f(\bar{x})\big\|_{\ell_p^n} \leqslant \big\|f(\bar{y})\big\|_{\ell_p^m}$ for every $f \in X^*$. Let $Z = \big\{f(\bar{y}) \;:\; f \in X^*\big\}$. We can view Z as a subspace of $\mathbb{R}^m$. Define $T \colon Z \to \mathbb{R}^n$ via $T \colon f(\bar{y}) \mapsto f(\bar{x})$. Note that T is well defined because if $f(\bar{y}) = 0$ then, by assumption, $f(\bar{x}) = 0$. The assumption $\big\|f(\bar{x})\big\|_{\ell_p^n} \leqslant \big\|f(\bar{y})\big\|_{\ell_p^m}$ means that T is a contraction when we view it as an operator $T \colon Z \subseteq \ell_p^m \to \ell_p^n$. When $p = 2$ or $p = \infty$, we can extend T to a contraction $A \colon \ell_p^m \to \ell_p^n$. Indeed, in the case $p = \infty$ this follows from the injectivity of ℓ_∞^n, while in the case $p = 2$ we extend T be setting it to zero on the orthogonal complement of Z in ℓ_2^m. In particular, for every $f \in X^*$ we have $f(\bar{x}) = Tf(\bar{y}) = Af(\bar{y}) = f(A\bar{y})$, so that $\bar{x} = A\bar{y}$. Since A is a contraction, it follows from the definition of p-multinorm that $\|\bar{x}\| \leqslant \|\bar{y}\|$. □

20.4.3 Example. Let X be a normed space. For $x_1,\dots,x_n\in X$, define

$$\big\|(x_1,\dots,x_n)\big\| = \sup\big\{\big\|f(\bar{x})\big\|_{\ell_p^n} \,:\, f\in B_{X^*}\big\}.$$

It is easy to see that this is a strong p-multinorm on X.

20.4.4 Example. *A 1-multinorm which is not strong.* Let $X=\ell_\infty^N$. For $x_1,\dots,x_n\in X$, put $\big\|(x_1,\dots,x_n)\big\| = \sum_{i=1}^n\|x_i\|$. It follows from Exercise 20.1.5 that this is a 1-multinorm. We will now show that it is not strong when $N=2^m$ and m is sufficiently large.

Let $x_i=\frac{2m}{N}e_i$ as $i=1,\dots,N$, where e_i is the i-th unit vector of ℓ_∞^N. Let $y_j=r_j$ as $j=1,\dots,m$, where r_j is the j-th discrete Rademacher vector in ℓ_∞^N, that is,

$$\begin{aligned}
r_1 &= (1,\dots, && 1,-1, && \dots,-1)\\
r_2 &= (1,\dots,1,-1,\dots, && -1,1, && \dots,1,-1,\dots,-1)\\
&\ \ \vdots && \ \ \vdots && \\
r_m &= (1,-1,\dots && && \dots,1,-1)
\end{aligned}$$

Let $f\in X^*=\ell_1^N$ be any functional of norm one; $f=(f_1,\dots,f_N)$. Then

$$\sum_{i=1}^N\big|f(x_i)\big| = \frac{2m}{N}\sum_{i=1}^N|f_i| = \frac{2m}{N}.$$

On the other hand, $\sum_{j=1}^m\big|f(y_j)\big| = m\operatorname*{Ave}_{\varepsilon_i=\pm1}\big|\sum_{i=1}^N\varepsilon_i f_i\big|$, where the average is taken over all possible choices of pluses and minuses. By Khinchin's inequality (16.3) in 16.1.5, $\operatorname*{Ave}_{\varepsilon_i=\pm1}\Big|\sum_{i=1}^N\varepsilon_i f_i\Big| \geqslant A_1\big(\sum_{i=1}^N|f_i|^2\big)^{\frac12}$. Hölder's inequality yields $1=\sum_{1=1}^N|f_i|\leqslant\big(\sum_{i=1}^N|f_i|^2\big)^{\frac12}N^{\frac12}$, so that $\sum_{j=1}^m\big|f(y_j)\big|\geqslant\frac{mA_1}{\sqrt{N}}$. It follows that $\sum_{i=1}^N\big|f(x_i)\big|\leqslant\sum_{j=1}^m\big|f(y_j)\big|$ provided that m is sufficiently large. On the other hand, $\big\|(x_1,\dots,x_N)\big\| = \frac{2m}{N}N = 2m$, while $\big\|(y_1,\dots,y_m)\big\|=m$, so that $\big\|(x_1,\dots,x_N)\big\|>\big\|(y_1,\dots,y_m)\big\|$.

20.5 Canonical p-multinorms in Banach lattices

Let X be a Banach lattice. For $x_1,\dots,x_n\in X$, put $\big\|(x_1,\dots,x_n)\big\| \coloneqq \big\|\bigvee_{i=1}^n|x_i|\big\|$. It is easy to see that this defines a multinorm as the axioms of a multinorm are satisfied. Thus, every Banach lattice has a "natural" multinorm. We will call it the ***canonical*** multinorm on X.

In a similar fashion, one can define a canonical 1-multinorm on X via $\big\|(x_1,\dots,x_n)\big\| \coloneqq \big\|\sum_{i=1}^n|x_i|\big\|$. It follows from Exercise 20.1.5 that this is indeed a 1-multinorm.

We are now going to define a canonical p-multinorm for every $1 \leqslant p \leqslant \infty$ in a similar fashion. Let $x_1, \dots, x_n \in X$. We define the expression $\left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}}$ as in Section 16.1, and put $\left\|(x_1, \dots, x_n)\right\| := \left\|\left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}}\right\|$. By Function Calculus, it satisfies the triangle inequality, hence defines a lattice norm on X^n. It follows immediately from Exercise 16.1.10 that this sequence of norms is a strong p-multinorm; in particular, it is a p-multinorm. We will refer to it as the ***canonical** p**-multinorm*** on X. Moreover, every linear subspace and every quotient space of X inherits this p-multinorm.

20.5.1 Theorem. *Let $1 \leqslant p \leqslant \infty$ and X a Banach lattice with the canonical p-multinorm. Then the dual multinorm on X^* is exactly the canonical q-multinorm on X^*, where q is the conjugate of p.*

Proof. Consider X^* with the dual multinorm. Let $\bar{f} = (f_1, \dots, f_n) \in (X^*)^n$. We need to show that $\|\bar{f}\| = \left\|\left(\sum_{i=1}^n |f_i|^q\right)^{\frac{1}{q}}\right\|$. For every $\bar{x} = (x_1, \dots, x_n) \in X^n$, we have

$$\langle \bar{f}, \bar{x} \rangle = \sum_{i=1}^n \langle f_i, x_i \rangle \leqslant \left\langle \left(\sum_{i=1}^n |f_i|^q\right)^{\frac{1}{q}}, \left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}} \right\rangle$$

by Proposition 16.2.1. It follows that

$$\left|\langle \bar{f}, \bar{x} \rangle\right| \leqslant \left\|\left(\sum_{i=1}^n |f_i|^q\right)^{\frac{1}{q}}\right\| \|\bar{x}\|,$$

so that $\|\bar{f}\| \leqslant \left\|\left(\sum_{i=1}^n |f_i|^q\right)^{\frac{1}{q}}\right\|$. For the converse inequality, we will use Theorem 16.2.2. Fix $x \geqslant 0$. Let $\bar{x} = (x_1, \dots, x_n) \in X^n$ with $\left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}} \leqslant x$. Then $\|\bar{x}\| \leqslant \|x\|$, so that

$$\sum_{i=1}^n \langle f_i, x_i \rangle = \langle \bar{f}, \bar{x} \rangle \leqslant \|\bar{f}\| \|\bar{x}\| \leqslant \|\bar{f}\| \|x\|.$$

It follows from Theorem 16.2.2 that $\left(\sum_{i=1}^n |f_i|^q\right)^{\frac{1}{q}}(x) \leqslant \|\bar{f}\| \|x\|$, hence $\left\|\left(\sum_{i=1}^n |f_i|^q\right)^{\frac{1}{q}}\right\| \leqslant \|\bar{f}\|$. □

20.5.2 Exercise. The canonical p-multinorm on a Banach lattice X is p-convex iff X itself is p-convex with constant 1.

20.6 p-multibounded operators between Banach lattices

Throughout this section $T\colon X \to Y$ will be a bounded operator between two Banach lattices. We say that T is ***p-multibounded*** if it is multibounded with respect to the canonical p-multinorms on X and Y. We will write $\|T\|_{p-\mathrm{mb}}$ for the multinorm of T. We will show that p-multibounded operators are closely related to pre-regular operators, which were discussed in Sections 13.3 and 13.4.

It follows immediately from Proposition 16.1.8 that every positive operator is p-multibounded for every p. Therefore, every regular operator is p-multibounded. Actually, a stronger statement is true.

20.6.1 Proposition. *If T is pre-regular then T is p-multibounded. In this case, $\|T\|_{p-\mathrm{mb}} \leqslant \|T^*\|_r$.*

Proof. Let $x_1, \dots, x_n \in X$. Then, using Proposition 16.1.8, we get

$$\Big\|\Big(\sum_{i=1}^n |Tx_i|^p\Big)^{\frac{1}{p}}\Big\|_Y = \Big\|\Big(\sum_{i=1}^n |jTx_i|^p\Big)^{\frac{1}{p}}\Big\|_{Y^{**}} \leqslant \Big\|\Big(\sum_{i=1}^n \big(|jT||x_i|\big)^p\Big)^{\frac{1}{p}}\Big\|_{Y^{**}}$$
$$\leqslant \Big\||jT|\Big(\sum_{i=1}^n \big(|x_i|\big)^p\Big)^{\frac{1}{p}}\Big\|_{Y^{**}} \leqslant \big\||jT|\big\|\Big\|\Big(\sum_{i=1}^n \big(|x_i|\big)^p\Big)^{\frac{1}{p}}\Big\|$$

Re-writing this in terms of the canonical p-multinorms, we get

$$\big\|(Tx_1, \dots, Tx_n)\big\| \leqslant \|T\colon X \to Y^{**}\|_r \big\|(x_1, \dots, x_n)\big\|.$$

By the remark before the proposition, $\|T\colon X \to Y^{**}\|_r = \|T^*\|_r$. □

Is the converse true? That is, suppose T is p-multibounded, does this imply that T is pre-regular? We will see that the answer is “yes” when $p = 1$ and $p = \infty$, but “no” in the case $p = 2$.

20.6.2 Proposition. *If T is order bounded then it is ∞-multibounded, with $\|T\|_{\infty-\mathrm{mb}} \leqslant \|T\|_{\mathrm{ob}}$.*

Proof. Let $x_1, \dots, x_n \in X$ with $\big\|(x_1, \dots, x_n)\big\| \leqslant 1$. Put $x = \bigvee_{i=1}^n |x_i|$; then $\|x\| \leqslant 1$. By Proposition 13.1.2, there exists $y \in Y_+$ such that $T[-x, x] \subseteq [-y, y]$ and $\|y\| \leqslant \|T\|_{\mathrm{ob}}$. It follows that $|Tx_i| \leqslant y$ for every i, so that

$$\big\|(Tx_1, \dots, Tx_n)\big\| = \Big\|\bigvee_{i=1}^n |Tx_i|\Big\| \leqslant \|y\| \leqslant \|T\|_{\mathrm{ob}}.$$

□

Recall that X has the Fatou Property if $0 \leqslant x_\alpha \uparrow x$ implies $\|x_\alpha\| \uparrow \|x\|$. Recall that for every Banach lattice X, the dual lattice X^* is monotonically bounded and has the Fatou Property; see Exercises 10.6.8.

20.6.3 Theorem. *Let T be an operator $T\colon X \to Y$ between two Banach lattices. TFAE*

(i) *T is ∞-multibounded;*
(ii) *T is 1-multibounded;*
(iii) *T is pre-regular (i.e., T^* is regular).*

In this case, $\|T\|_{1-\mathrm{mb}} = \|T\|_{\infty-\mathrm{mb}} = \|T^*\|_r$.

Proof. Theorem 13.4.3 asserts that T is pre-regular iff it is ∞-multibounded. By Proposition 20.6.1, we know that if T^* is regular that T is ∞-multibounded and 1-multibounded; in this case $\|T\|_{1-\mathrm{mb}}$ and $\|T\|_{\infty-\mathrm{mb}}$ are dominated by $\|T^*\|_r$.

Suppose that T is 1-multibounded. Then $T^*\colon Y^* \to X^*$ is ∞-multibounded by Exercise 20.2.3 and Theorem 20.5.1. Then T^* is pre-regular, hence T is pre-regular by Theorem 13.3.1.

To prove the equality of the three norms, fix $x > 0$ and let

$$A = \Bigl\{\bigvee_{i=1}^{n}|Tx_i| \;:\; x_1,\dots,x_n \in [0,x]\Bigr\}.$$

We may view A is an increasing net indexed by finite subsets of $[0,x]$ ordered by inclusion. Let $x_1,\dots,x_n \in [0,x]$. Then

$$\Bigl\|\bigvee_{i=1}^{n}|Tx_i|\Bigr\| \leqslant \|T\|_{\infty-\mathrm{mb}}\Bigl\|\bigvee_{i=1}^{n}|x_i|\Bigr\| \leqslant \|T\|_{\infty-\mathrm{mb}}\|x\|.$$

Hence, A is norm bounded by $\|T\|_{\infty-\mathrm{mb}}\|x\|$. Viewed as a subset of Y^{**}, A is still norm bounded. Since Y^{**} is monotonically complete, $u \coloneqq \sup A$ exists in Y^{**}. By the Fatou Property, $\|u\| = \sup_{y\in A}\|y\| \leqslant \|T\|_{\infty-\mathrm{mb}}\|x\|$. It follows that $|jT|x = \sup_{|y|\leqslant x}|jTy| \leqslant u$, so that $\bigl\||jT|x\bigr\| \leqslant \|u\| \leqslant \|T\|_{\infty-\mathrm{mb}}\|x\|$. Therefore,

$$\|T^*\|_r = \|jT\|_r = \bigl\||jT|\bigr\| \leqslant \|T\|_{\infty-\mathrm{mb}}.$$

Hence, $\|T\|_{\infty-\mathrm{mb}} = \|T^*\|_r$. Finally, we have

$$\|T\|_{1-\mathrm{mb}} = \|T^*\|_{\infty-\mathrm{mb}} = \|T^{**}\|_r = \|T^*\|_r.$$

□

On the other hand, in the case $p = 2$, every bounded operator T is 2-multibounded by Theorem 16.3.13 and $\|T\|_{2-\mathrm{mb}} \leqslant K_G\|T\|$.

20.7 Representing p-multinormed spaces as subspaces of Banach lattices

The goal of this section is to show that multinorms are naturally related to Banach lattices. We will show, in particular, that every multinormed space is multiisometric to a subspace of a Banach lattice with the canonical multinorm. That is, for every multinormed space X there exists a Banach lattice E and a linear operator $J\colon X \to E$ such that

$$\left\|(x_1,\dots,x_n)\right\| = \Bigl\|\bigvee_{i=1}^{n}\lvert Jx_i\rvert\Bigr\|$$

for any $x_1,\dots,x_n \in X$. That is, multinormed spaces are exactly subspaces of Banach lattices. In other words, given a vector space X, every multinorm on X corresponds to a linear embedding of X into a Banach lattice.

Furthermore, we will show that for a p-multinormed space X satisfying certain assumptions there exists a Banach lattice E and a linear embedding $J\colon X \to E$ which is a multiisometry with respect to the canonical p-multinorm on E.

Let X be a strong p-multinormed space for some $1 \leqslant p \leqslant \infty$. In particular, X is a normed space. Let $K = B_{X^*}$ with the weak* topology; then K is a compact space and $C(K)$ is a Banach lattice. Recall that the lattice operations on $C(K)$ are point-wise. As usual, for each $x \in X$ we write $\hat{x}$ for its canonical image in X^{**}. The restriction of $\hat{x}$ to K belongs to $C(K)$. With a slight abuse of notation, we will still denote this restriction by $\hat{x}$, so we can consider $\hat{x}$ as an element of $C(K)$. We will be especially interested in functions of the form $\bigl(\sum_{i=1}^{n}\lvert\hat{x}_i\rvert^p\bigr)^{\frac{1}{p}}$ for $x_1,\dots,x_n \in X$. Note that since the lattice operations in $C(K)$ are point-wise, the same is true for Function Calculus, so we have

$$\Bigl(\sum_{i=1}^{n}\lvert\hat{x}_i\rvert^p\Bigr)^{\frac{1}{p}}(f) = \Bigl(\sum_{i=1}^{n}\lvert f(x_i)\rvert^p\Bigr)^{\frac{1}{p}} \tag{20.1}$$

for every $f \in K$. It follows from the definition of a strong p-multinorm that

$$\text{if } \Bigl(\sum_{i=1}^{n}\lvert\hat{x}_i\rvert^p\Bigr)^{\frac{1}{p}} \leqslant \Bigl(\sum_{j=1}^{m}\lvert\hat{y}_j\rvert^p\Bigr)^{\frac{1}{p}} \text{ then } \left\|(x_1,\dots,x_n)\right\| \leqslant \left\|(y_1,\dots,y_m)\right\| \tag{20.2}$$

for any $x_1,\dots,x_n,y_1,\dots,y_m \in X$. (This was, actually, the motivation for the definition).

Let V be the (order) ideal of $C(K)$ generated by $\{\hat{x} \,:\, x \in X\}$. That is, for $\varphi \in C(K)$, $\varphi \in V$ iff $\lvert\varphi\rvert \leqslant \sum_{i=1}^{n}\lvert\hat{x}_i\rvert$ for some $x_1,\dots,x_n \in X$. Since the $\|\cdot\|_{\ell_p}$ and $\|\cdot\|_{\ell_1}$ norms on $\mathbb{R}^n$ are equivalent, it follows from Function Calculus that

$$C_1\sum_{i=1}^{n}\lvert\hat{x}_i\rvert \leqslant \Bigl(\sum_{i=1}^{n}\lvert\hat{x}_i\rvert^p\Bigr)^{\frac{1}{p}} \leqslant C_2\sum_{i=1}^{n}\lvert\hat{x}_i\rvert$$

for some constants C_1 and C_2. Therefore, $\varphi \in V$ iff $|\varphi| \leqslant \left(\sum_{i=1}^n |\hat{x}_i|^p\right)^{\frac{1}{p}}$ for some $x_1, \ldots, x_n \in X$.

20.7.1 Lemma. *Let X be a strong p-multinormed space, $K = (B_{X^*}, w^*)$, and V the order ideal in $C(K)$ generated by $\{\hat{x} : x \in X\}$. Define $\rho\colon V \to \mathbb{R}_+$ via*

$$\rho(\varphi) = \inf\Big\{\big\|(x_1, \ldots, x_n)\big\| \,:\, |\varphi| \leqslant \Big(\sum_{i=1}^n |\hat{x}_i|^p\Big)^{\frac{1}{p}} \text{ for some } x_1, \ldots, x_n \in X\Big\}.$$

Then

(i) $\rho(\lambda\varphi) = |\lambda|\rho(\varphi)$ *for any scalar* λ,
(ii) $\rho\big(|\varphi|\big) = \rho(\varphi)$,
(iii) *if* $|\varphi| \leqslant |\psi|$ *then* $\rho(\varphi) \leqslant \rho(\psi)$, *and*
(iv) $\rho\Big(\big(\sum_{i=1}^n |\hat{x}_i|^p\big)^{\frac{1}{p}}\Big) = \big\|(x_1, \ldots, x_n)\big\|$ *for any* $x_1, \ldots, x_n \in X$.

Proof. (i), (ii), and (iii) are straightforward; (iv) follows from (20.2). □

Suppose ρ satisfies the triangle inequality. Then it is a lattice seminorm, hence $\ker\rho$ is an ideal, so that $V/\ker\rho$ is a normed lattice under the norm induced by ρ. Let E be its completion; hence E is a Banach lattice. Note that the quotient map $q\colon V \to E$ preserves Function Calculus by Corollary 4.4.7. Define $J\colon X \to E$ via $Jx = \hat{x} + \ker\rho$. We claim that J is a multiisometry with respect to the canonical p-multinorm on E. Indeed,

$$\begin{aligned}\Big\|\Big(\sum_{i=1}^n |Jx_i|^p\Big)^{\frac{1}{p}}\Big\| = \Big\|\Big(\sum_{i=1}^n |q(\hat{x}_i)|^p\Big)^{\frac{1}{p}}\Big\| &= \Big\|q\Big(\Big(\sum_{i=1}^n |\hat{x}_i|^p\Big)^{\frac{1}{p}}\Big)\Big\| \\ &= \rho\Big(\Big(\sum_{i=1}^n |\hat{x}_i|^p\Big)^{\frac{1}{p}}\Big) = \big\|(x_1, \ldots, x_n)\big\|\end{aligned}$$

for any $x_1, \ldots, x_n \in X$ by Lemma 20.7.1(iv). The following lemma summarizes this computation.

20.7.2 Lemma. *Let X be a strong p-multinormed space. Suppose that ρ in Lemma 20.7.1 satisfies the triangle inequality. Let E be the completion of $V/\ker\rho$. Then E is a Banach lattice and the map $J\colon X \to E$ given by $Jx = \hat{x} + \ker\rho$ is a multiisometry with respect to the canonical p-multinorm on E.*

20.7.3 Theorem. *Every multinormed space is multiisometric to a subspace of a Banach lattice (with the canonical multinorm).*

Proof. Let X be a multinormed space. By Proposition 20.4.2, the multinorm is strong. Let K, V, and ρ be as in Lemma 20.7.1. In view of Lemma 20.7.2, it suffices to show that ρ satisfies the triangle inequality. Take $\varphi, \psi \in V$ and fix $\varepsilon > 0$. By the definition of ρ, one can find multivectors $\bar{x} = (x_1, \dots, x_n) \in X^n$ and $\bar{y} = (y_1, \dots, y_m) \in X^m$ such that

$$|\varphi| \leqslant \bigvee_{i=1}^{n} |\hat{x}_i|, \quad |\psi| \leqslant \bigvee_{i=1}^{m} |\hat{y}_i|, \quad \|\bar{x}\| \leqslant \rho(\varphi) + \varepsilon, \quad \text{and} \quad \|\bar{y}\| \leqslant \rho(\psi) + \varepsilon.$$

WLOG, $n = m$ as, otherwise, we can "pad" the shorter multivector with zeros.

Recall that every separable Banach space embeds isometrically into ℓ_∞, see, e.g., [AK06, Theorem 2.5.7]. In particular, $\ell_\infty^n \oplus_1 \ell_\infty^n$ embeds isometrically into ℓ_∞. Since this space is finite-dimensional one can easily check (or confer with [Jam87, Proposition 0.13]) that there exists a sufficiently large k and map $S\colon \ell_\infty^n \oplus_1 \ell_\infty^n \to \ell_\infty^k$ which is a $(1+\varepsilon)$-isometry, that is, $\|w\| \leqslant \|Sw\| \leqslant (1+\varepsilon)\|w\|$ for every $w \in \ell_\infty^n \oplus_1 \ell_\infty^n$. We can write $w = \left[\begin{smallmatrix} u \\ v \end{smallmatrix}\right]$ for $u, v \in \ell_\infty^n$. It is easy to see that there are two $k \times n$ matrices A and B such that $Sw = Au + Bv$. Note that $\|A\|, \|B\| \leqslant \|S\| \leqslant (1+\varepsilon)$.

Put $\bar{z} = A\bar{x} + B\bar{y}$. Take any $f \in K$. Then

$$f(\bar{z}) = f\big(A\bar{x} + B\bar{y}\big) = Af(\bar{x}) + Bf(\bar{y}) = S\left[\begin{smallmatrix} f(\bar{x}) \\ f(\bar{y}) \end{smallmatrix}\right].$$

It follows that

$$\big\|f(\bar{z})\big\|_{\ell_\infty^k} \geqslant \left\|\left[\begin{smallmatrix} f(\bar{x}) \\ f(\bar{y}) \end{smallmatrix}\right]\right\|_{\ell_\infty^n \oplus_1 \ell_\infty^n} = \big\|f(\bar{x})\big\|_{\ell_\infty^n} + \big\|f(\bar{y})\big\|_{\ell_\infty^n},$$

hence $\bigvee_{i=1}^n |\hat{x}_i| + \bigvee_{i=1}^n |\hat{y}_i| \leqslant \bigvee_{i=1}^k |\hat{z}_i|$. This yields $|\varphi + \psi| \leqslant \bigvee_{i=1}^k |\hat{z}_i|$, so that

$$\begin{aligned}\rho(\varphi+\psi) \leqslant \|\bar{z}\| \leqslant \|A\bar{x}\| + \|B\bar{y}\| \leqslant (1+\varepsilon)\big(\|\bar{x}\| + \|\bar{y}\|\big) \\ \leqslant (1+\varepsilon)\big(\rho(\varphi) + \rho(\psi) + 2\varepsilon\big).\end{aligned}$$

It follows that $\rho(\varphi + \psi) \leqslant \rho(\varphi) + \rho(\psi)$. Thus, ρ is a seminorm. □

20.7.4 Theorem. *Every strong 1-multinormed space is multiisometric to a subspace of a Banach lattice with the canonical 1-multinorm.*

Proof. Again, by Lemma 20.7.2, it suffices to verify that ρ satisfies the triangle inequality. Take $\varphi, \psi \in V$, fix $\varepsilon > 0$; one can find $\bar{x}, \bar{y} \in X^n$ such that

$$|\varphi| \leqslant \sum_{i=1}^{n} |\hat{x}_i|, \ |\psi| \leqslant \sum_{i=1}^{n} |\hat{y}_i|, \ \|\bar{x}\| \leqslant \rho(\varphi) + \varepsilon, \text{ and } \|\bar{y}\| \leqslant \rho(\psi) + \varepsilon.$$

Let $\bar{z}$ be the concatenation of $\bar{x}$ and $\bar{y}$. Then $|\varphi+\psi| \leqslant \sum_{i=1}^{2n}|\hat{z}_i|$. This yields

$$\begin{aligned}\rho(\varphi+\psi) &\leqslant \big\|(z_1,\dots,z_{2n})\big\| \\ &= \big\|(x_1,\dots,x_n,0,\dots,0)+(0,\dots,0,y_1,\dots,y_n)\big\| \\ &\leqslant \big\|(x_1,\dots,x_n,0,\dots,0)\big\| + \big\|(0,\dots,0,y_1,\dots,y_n)\big\| \\ &= \big\|(x_1,\dots,x_n)\big\| + \big\|(y_1,\dots,y_n)\big\| \leqslant \rho(\varphi)+\rho(\psi)+2\varepsilon.\end{aligned}$$

□

The latter theorem can be generalized to an arbitrary p as follows.

20.7.5 Theorem. *Let X be a strong p-multinormed p-convex space. Then X is multiisometric to a subspace of a Banach lattice E with the canonical p-multinorm, where E is p-convex with constant 1.*

Proof. We already know that the statement is true for $p=1$ and $p=\infty$, so we assume now that $1<p<\infty$. Let q be the conjugate of p. Again, by Lemma 20.7.2, it suffices to verify that ρ satisfies the triangle inequality. Take $\varphi,\psi\in V$, fix $\varepsilon>0$; one can find $\bar{x},\bar{y}\in X^n$ such that

$$|\varphi| \leqslant \Big(\sum_{i=1}^n|\hat{x}_i|^p\Big)^{\frac1p},\ |\psi| \leqslant \Big(\sum_{i=1}^n|\hat{y}_i|^p\Big)^{\frac1p},\ \|\bar{x}\|\leqslant\rho(\varphi)+\varepsilon,\ \text{and}\ \|\bar{y}\|\leqslant\rho(\psi)+\varepsilon.$$

Put

$$\alpha=\Big(\frac{\|\bar{x}\|}{\|\bar{x}\|+\|\bar{y}\|}\Big)^{\frac1q}\quad\text{and}\quad\beta=\Big(\frac{\|\bar{y}\|}{\|\bar{x}\|+\|\bar{y}\|}\Big)^{\frac1q}.$$

Note that $\alpha^q+\beta^q=1$. By Hölder's inequality, $|s|+|t|\leqslant\Big(\frac{|s|^p}{\alpha^p}+\frac{|t|^p}{\beta^p}\Big)^{\frac1p}$ for any real numbers s and t. It follows that $|\varphi|+|\psi|\leqslant\Big(\frac{|\varphi|^p}{\alpha^p}+\frac{|\psi|^p}{\beta^p}\Big)^{\frac1p}$ and, therefore,

$$\rho(\varphi+\psi)\leqslant\rho\bigg(\Big(\frac{|\varphi|^p}{\alpha^p}+\frac{|\psi|^p}{\beta^p}\Big)^{\frac1p}\bigg).$$

Note that

$$\frac{|\varphi|^p}{\alpha^p}\leqslant\sum_{i=1}^n\Big|\frac{\hat{x}_i}{\alpha}\Big|^p\quad\text{and}\quad\frac{|\psi|^p}{\beta^p}\leqslant\sum_{i=1}^n\Big|\frac{\hat{y}_i}{\beta}\Big|^p.$$

Let $\bar{z}$ be the concatenation of $\frac1\alpha\bar{x}$ and $\frac1\beta\bar{y}$. Then

$$\Big(\frac{|\varphi|^p}{\alpha^p}+\frac{|\psi|^p}{\beta^p}\Big)^{\frac1p}\leqslant\Big(\sum_{i=1}^{2n}|\hat{z}_i|^p\Big)^{\frac1p}.$$

It follows that

$$\rho\bigg(\Big(\frac{|\varphi|^p}{\alpha^p}+\frac{|\psi|^p}{\beta^p}\Big)^{\frac1p}\bigg)\leqslant\|\bar{z}\|.$$

By p-convexity, we have

$$\|\bar{z}\| \leqslant \Bigl(\bigl\|\tfrac{1}{\alpha}\bar{x}\bigr\|^p + \bigl\|\tfrac{1}{\beta}\bar{y}\bigr\|^p\Bigr)^{\frac{1}{p}} = \left(\frac{\|\bar{x}\|^p}{\alpha^p} + \frac{\|\bar{y}\|^p}{\beta^p}\right)^{\frac{1}{p}} = \|\bar{x}\| + \|\bar{y}\|$$

after we plug in α and β. Therefore,

$$\rho(\varphi+\psi) \leqslant \|\bar{x}\| + \|\bar{y}\| \leqslant \rho(\varphi) + \rho(\psi) + 2\varepsilon.$$

It follows that $\rho(\varphi+\psi) \leqslant \rho(\varphi) + \rho(\psi)$.

Now let $\bar{u}$ be the concatenation of $\bar{x}$ and $\bar{y}$. It follows that

$$\bigl(|\varphi|^p + |\psi|^p\bigr)^{\frac{1}{p}} \leqslant \Bigl(\sum_{i=1}^{2n} |\hat{u}_i|^p\Bigr)^{\frac{1}{p}},$$

so that

$$\rho\Bigl(\bigl(|\varphi|^p + |\psi|^p\bigr)^{\frac{1}{p}}\Bigr) \leqslant \|\bar{u}\| \leqslant \bigl(\|\bar{x}\|^p + \|\bar{y}\|^p\bigr)^{\frac{1}{p}} \leqslant \Bigl(\bigl(\rho(\varphi)+\varepsilon\bigr)^p + \bigl(\rho(\psi)+\varepsilon\bigr)^p\Bigr)^{\frac{1}{p}}.$$

It follows that

$$\rho\Bigl(\bigl(|\varphi|^p + |\psi|^p\bigr)^{\frac{1}{p}}\Bigr) \leqslant \bigl(\rho(\varphi)^p + \rho(\psi)^p\bigr)^{\frac{1}{p}}.$$

Hence, ρ and, therefore, the resulting Banach lattice, are p-convex with constant 1. □

20.7.6 Remark. Note that the condition in the preceeding theorem that the p-multinorm be strong is necessary because every canonical p-multinorm is strong, and this property is inherited by subspaces. In particular, the 1-multinorm in Example 20.4.4 is not strong, hence it cannot be represented as a subspace of a Banach lattice.

The following example shows that the preceding theorem fails without the convexity assumption.

20.7.7 Example. For every $1 < p < \infty$, we will construct a p-multinormed space X which is not multiisometric to a subspace of a Banach lattice. In particular, if $p = 2$ then this p-multinorm is strong by Proposition 20.4.2 (hence it cannot be convex).

Fix n, put $X = \ell_q^n$. Equip X with the p-multinorm coming from the projective tensor norm on $\ell_p \otimes X$ in the sense of Theorem 20.3.3. Suppose that T is an isomorphism from X into a Banach lattice E, with E equipped with the canonical p-multinorm. Suppose that $\|T\| \leqslant 1$. Let M be the multibounded norm of the inverse of T defined on the range of T. We will show that is is "large" for a sufficiently large n, so that T cannot be a multiisometry.

Let $(e_i)_{i=1}^{\infty}$ and $(d_i)_{i=1}^{n}$ be the unit vector bases of ℓ_p and ℓ_q^n, respectively. Let $\bar{d} = (d_1, \ldots, d_n)$. Then $\|\bar{d}\| = \bigl\|\sum_{i=1}^n e_i \otimes d_i\bigr\|$ in $\ell_p \otimes \ell_q^n$.

Recall that for any two Banach spaces Y and Z, the projective tensor product $Z\widehat{\otimes}Y^*$ is isometric to the space of nuclear operators $N(Y,Z)$, with $z \otimes y^*$ in $Z\widehat{\otimes}Y^*$ corresponding to the rank-one operator from Y to Z sending y to $y^*(y)z$. We apply this with $Y = \ell_p$ and $X = \ell_p^n$. The sum $\sum_{i=1}^n e_i \otimes d_i$ corresponds to the formal inclusion of ℓ_p^n into ℓ_p, whose nuclear norm is n. It follows that $\|\bar{d}\| = n$.

For each $i = 1, \ldots, n$, put $x_i = Td_i$. It follows that

$$\Bigl\|\Bigl(\sum_{i=1}^n |x_i|^p\Bigr)^{\frac{1}{p}}\Bigr\| = \|\bar{x}\| \geqslant \frac{\|\bar{d}\|}{M} = \frac{n}{M}. \tag{20.3}$$

By Khinchin's Inequality, $\operatorname*{Ave}_{\pm}\Bigl|\sum_{i=1}^n \pm x_i\Bigr| \geqslant A_1\Bigl(\sum_{i=1}^n |x_i|^2\Bigr)^{\frac{1}{2}}$, where the average is taken over all combinations of signs. Also,

$$\Bigl\|\sum_{i=1}^n \pm x_i\Bigr\| = \Bigl\|T\sum_{i=1}^n \pm d_i\Bigr\| \leqslant \Bigl\|\sum_{i=1}^n \pm d_i\Bigr\| = n^{\frac{1}{q}}$$

for any combination of signs. Therefore,

$$\Bigl\|\Bigl(\sum_{i=1}^n |x_i|^2\Bigr)^{\frac{1}{2}}\Bigr\| \leqslant \frac{1}{A_1}\Bigl\|\operatorname{Ave}\Bigl|\sum_{i=1}^n \pm x_i\Bigr|\Bigr\| \leqslant \frac{1}{A_1}\operatorname{Ave}\Bigl\|\sum_{i=1}^n \pm x_i\Bigr\| \leqslant \frac{n^{\frac{1}{q}}}{A_1}.$$

If $p \geqslant 2$ then $\Bigl(\sum_{i=1}^n |x_i|^p\Bigr)^{\frac{1}{p}} \leqslant \Bigl(\sum_{i=1}^n |x_i|^2\Bigr)^{\frac{1}{2}}$, so that $\Bigl\|\Bigl(\sum_{i=1}^n |x_i|^p\Bigr)^{\frac{1}{p}}\Bigr\| \leqslant \frac{n^{\frac{1}{q}}}{A_1}$. It follows from (20.3) that $M \geqslant A_1 n^{\frac{1}{p}}$.

If $p \leqslant 2$ then, by Hölder's Inequality, $\Bigl(\sum_{i=1}^n |x_i|^p\Bigr)^{\frac{1}{p}} \leqslant n^{\frac{1}{p}-\frac{1}{2}}\Bigl(\sum_{i=1}^n |x_i|^2\Bigr)^{\frac{1}{2}}$, so that $\Bigl\|\Bigl(\sum_{i=1}^n |x_i|^p\Bigr)^{\frac{1}{p}}\Bigr\| \leqslant n^{\frac{1}{p}-\frac{1}{2}} \cdot \frac{n^{\frac{1}{q}}}{A_1} = \frac{n^{\frac{1}{2}}}{A_1}$. It follows that $M \geqslant A_1 n^{\frac{1}{2}}$. In either case, $M > 1$ provided that n is sufficiently large.

Now take $X = \ell_q$ and, again, equip X with the p-multinorm coming from the projective tensor norm on $\ell_p \otimes X$. For every n, one may view ℓ_q^n as a subspace of X, hence there is no multiisometry T from X to a Banach lattice.

We finish with a few observations for the case of a general $1 \leqslant p < \infty$. Again, let X be a strong p-multinormed space and let K, V, and ρ be as in Lemma 20.7.1. Let q be the conjugate of p.

20.7.8 Lemma. *Under these assumptions,*

(i) $\rho\Bigl(\bigl(|\varphi|^p + |\psi|^p\bigr)^{\frac{1}{p}}\Bigr) \leqslant \rho(\varphi) + \rho(\psi)$ *and*

(ii) $\rho(\varphi + \psi) \leqslant 2^{\frac{1}{q}}\bigl(\rho(\varphi) + \rho(\psi)\bigr)$

for any $\varphi, \psi \in V$.

Proof. Let $\varphi, \psi \in V$ and fix $\varepsilon > 0$. Again, find $n \in \mathbb{N}$ and $\bar{x}, \bar{y} \in X^n$ such that

$$|\varphi| \leqslant \Big(\sum_{i=1}^{n} |\hat{x}_i|^p\Big)^{\frac{1}{p}}, |\psi| \leqslant \Big(\sum_{i=1}^{n} |\hat{y}_i|^p\Big)^{\frac{1}{p}}, \|\bar{x}\| \leqslant \rho(\varphi) + \varepsilon, \text{ and } \|\bar{y}\| \leqslant \rho(\psi) + \varepsilon.$$

Let $\bar{z}$ be the concatenation of $\bar{x}$ and $\bar{y}$. Then $\big(|\varphi|^p + |\psi|^p\big)^{\frac{1}{p}} \leqslant \Big(\sum_{i=1}^{2n} |\hat{z}_i|^p\Big)^{\frac{1}{p}}$. It follows that

$$\rho\Big(\big(|\varphi|^p + |\psi|^p\big)^{\frac{1}{p}}\Big) \leqslant \|\bar{z}\| \leqslant \|\bar{x}\| + \|\bar{y}\| \leqslant \rho(\varphi) + \rho(\psi) + 2\varepsilon.$$

This proves (i).

By Hölder's inequality, we have $|s + t| \leqslant 2^{\frac{1}{q}}\big(|s|^p + |t|^p\big)^{\frac{1}{p}}$ for any $s, t \in \mathbb{R}$. It follows that $|\varphi + \psi| \leqslant 2^{\frac{1}{q}}\big(|\varphi|^p + |\psi|^p\big)^{\frac{1}{p}}$. Therefore,

$$\rho(\varphi + \psi) \leqslant 2^{\frac{1}{q}} \rho\Big(\big(|\varphi|^p + |\psi|^p\big)^{\frac{1}{p}}\Big) \leqslant 2^{\frac{1}{q}}\big(\rho(\varphi) + \rho(\psi)\big).$$

□

Recall some background: Suppose that X is a vector lattice. For every $e \in X_+$, let I_e be the ideal generated by e. For $x \in I_e$, define

$$\|x\|_e = \inf\big\{\lambda > 0 \,:\, |x| \leqslant \lambda e\big\}.$$

This defines a lattice norm on I_e. One has $|x| \leqslant \|x\|_e e$.

If $\big(I_e, \|\cdot\|_e\big)$ is complete for every $e \in X_+$, we say that X is uniformly complete. Every Banach lattice is uniformly complete. Every Archimedean uniformly complete vector lattice has a well defined Function Calculus for positively homogeneous functions. It is easy to see that every ideal in a uniformly complete vector lattice is itself uniformly complete.

If $T\colon X \to Y$ is a lattice homomorphism between two vector lattices and X is uniformly complete then so is Y, see Exercise 2(b) on p. 110 of [AB06]. Moreover, in this case, T preserves Function Calculus. That is, $Tf(x_1, \dots, x_n) = f(Tx_1, \dots, Tx_n)$ for any $x_1, \dots, x_n \in X$ and any positively homogeneous function $f\colon \mathbb{R}^n \to \mathbb{R}$.

A sequence (x_n) in X uniformly converges to some $x \in X$ if $\|x_n - x\|_e \to 0$ for some $e \in X_+$. Same for nets. A subset of X is uniformly closed if the limit of every uniformly convergent net in A belongs to A. Actually, it suffices to consider sequences. For a vector lattice X and an ideal J in X, the quotient vector lattice X/J is Archimedean iff J is uniformly closed; see Proposition 3.4.7.

20.7.9 Lemma. $\ker\rho$ *is an ideal and* $V/\ker\rho$ *is an Archimedean uniformly complete vector lattice.*

Proof. Being a Banach lattice, $C(K)$ is Archimedean and uniformly complete. Since V is an ideal in $C(K)$, it is also Archimedean and uniformly complete. It follows from Lemmas 20.7.1 and 20.7.8 that $\ker\rho$ is an ideal. It is easy to see that it is uniformly closed in V. Indeed, let (φ_n) be a sequence in $\ker\rho$ with $\|\varphi_n-\varphi\|_\psi\to 0$ for some $\varphi\in V$ and $\psi\in V_+$. Then

$$\rho(\varphi)\leqslant 2^{\frac{1}{q}}\bigl(\rho(\varphi_n)+\rho(\varphi_n-\varphi)\bigr)\leqslant 2^{\frac{1}{q}}\rho\bigl(\|\varphi_n-\varphi\|_\psi\psi\bigr)=2^{\frac{1}{q}}\|\varphi_n-\varphi\|_\psi\rho(\psi)\to 0.$$

Hence, $\varphi\in\ker\rho$.

It follows that $V/\ker\rho$ is Archimedean. Since the quotient map is a lattice homomorphism, $V/\ker\rho$ is uniformly complete by Proposition 2.2.16. □

20.8 Representing 1-multinormed spaces as quotients of Banach lattices

Theorem 20.7.3 asserts that every multinormed space is multiisometric to a subspace of a Banach lattice with the canonical multinorm. We are going to prove the dual result.

20.8.1 Theorem. *Every 1-multinormed space is multiisometric to a quotient of a Banach lattice (with the canonical 1-multinorm) over a closed subspace.*

Proof. Recall that if X and Y are normed spaces and $T\colon X\to Y$ is an isometry then $T^*(B^o_{Y^*})=B^o_{X^*}$. Let X be a 1-multinormed space. In view of Exercise 20.2.4, it suffices to find a Banach lattice E (with the canonical 1-multinorm) and an operator $T\colon E\to X$ such that

$$T^{(n)}(B^o_{E^n})=B^o_{X^n}\text{ for every }n.\tag{20.4}$$

A naive approach would be to argue that X^* is multinormed, hence there exists a multiisometry $S\colon X^*\to F$ for some Banach lattice F with the canonical multinorm and, therefore, the adjoint operator $S^*\colon F^*\to X^{**}$ satisfies $(S^*)^{(n)}(B^o_{(F^*)^n})=B^o_{(X^{**})^n}$ for every n. Unfortunately, S^* acts F^* onto X^{**} instead of X. So we will refine this argument.

Let $\mathcal{I}=\bigcup_{n=1}^\infty X^n$. For each $\bar{x}\in\mathcal{I}$, $\bar{x}=(x_1,\dots,x_n)$, put $Y_{\bar{x}}=\operatorname{span}\{x_1,\dots,x_n\}$. As $Y_{\bar{x}}$ is a finite-dimensional subspace of X, we may identify $Y_{\bar{x}}^{**}$ with $Y_{\bar{x}}$. Since $Y_{\bar{x}}^*$ is a multinormed space, there exists a Banach lattice $F_{\bar{x}}$ and a multiisometry $S_{\bar{x}}\colon Y_{\bar{x}}^*\to F_{\bar{x}}$ (w.r.t. the canonical multinorm of $F_{\bar{x}}$). It follows that the adjoint operator $S_{\bar{x}}^*\colon F_{\bar{x}}^*\to Y_{\bar{x}}$ satisfies $(S_{\bar{x}}^*)^{(n)}\bigl(B^o_{(F_{\bar{x}}^*)^n}\bigr)=B^o_{Y_{\bar{x}}^n}$ for every n. In particular, $\bigl\|(S_{\bar{x}}^*)^{(n)}\bigr\|=1$.

Put

$$E=\bigoplus_{\bar{x}\in\mathcal{I}}^{\ell_1}F_{\bar{x}}^*.$$

That is, E consists of all families $u = (u_{\bar{x}})_{\bar{x}\in\mathcal{I}}$ such that $u_{\bar{x}} \in F^*_{\bar{x}}$ for every $\bar{x}$, and $\|u\| = \sum_{\bar{x}\in\mathcal{I}}\|u_{\bar{x}}\| < \infty$. This means, in particular, that at most countably many components of u are non-zero. Being an ℓ_1-sum of Banach lattices, E itself is a Banach lattice.

Define $T\colon E \to X$ via $Tu = \sum_{\bar{x}\in\mathcal{I}} S^*_{\bar{x}}u_{\bar{x}}$. The sum is well-defined because it has at most countably many non-zero terms, $\left\|S^*_{\bar{x}}u_{\bar{x}}\right\| \leqslant \|u_{\bar{x}}\|$ for each $\bar{x}$, and the sum $\sum_{\bar{x}\in\mathcal{I}}\|u_{\bar{x}}\|$ converges.

It is left to verify that (20.4) is satisfied. This verification is straightforward; we leave it as an exercise. $\square$

20.8.2 Exercise. Complete the proof of the theorem.

20.9 Quotient of a canonical p-multinorm over an ideal

Let X be a Banach lattice equipped with the canonical p-multinorm. Let Y be a closed ideal in X. Then X/Y is again a Banach lattice. Write q for the quotient map; recall that since q is a lattice homomorphism, we have

$$q\Big(\Big(\sum_{i=1}^n |x_i|^p\Big)^{\frac{1}{p}}\Big) = \Big(\sum_{i=1}^n |q(x_i)|^p\Big)^{\frac{1}{p}}$$

for any $x_1, \dots, x_n \in X$ by Corollary 4.4.7.

On one hand, X/Y has the quotient p-multinorm defined as on page 552:

$$\begin{aligned}\big\|(x_1 + Y, \dots, x_n + Y)\big\|_{\text{quotient}} &= \inf_{y_1,\dots,y_n\in Y}\big\|(x_1 - y_1, \dots, x_n - y_n)\big\| \\ &= \inf_{y_1,\dots,y_n\in Y}\Big\|\Big(\sum_{i=1}^n |x_i - y_i|^p\Big)^{\frac{1}{p}}\Big\|.\end{aligned}$$

On the other hand, being a Banach lattice itself, X/Y has the canonical p-multinorm:

$$\begin{aligned}\big\|(x_1 + Y, \dots, x_n + Y)\big\|_{\text{canonical}} &= \Big\|\Big(\sum_{i=1}^n |x_i + Y|^p\Big)^{\frac{1}{p}}\Big\| = \Big\|\Big(\sum_{i=1}^n |q(x_i)|^p\Big)^{\frac{1}{p}}\Big\| \\ &= \Big\|q\Big(\Big(\sum_{i=1}^n |x_i|^p\Big)^{\frac{1}{p}}\Big)\Big\| = \inf_{y\in Y}\Big\|\Big(\sum_{i=1}^n |x_i|^p\Big)^{\frac{1}{p}} - y\Big\|.\end{aligned}$$

We claim that these two p-multinorms coincide. On one hand,

$$q\Big(\Big(\sum_{i=1}^n |x_i - y_i|^p\Big)^{\frac{1}{p}}\Big) = q\Big(\Big(\sum_{i=1}^n |x_i|^p\Big)^{\frac{1}{p}}\Big),$$

for any $y_1,\dots,y_n\in Y$, so that

$$\Bigl\|\Bigl(\sum_{i=1}^n|x_i-y_i|^p\Bigr)^{\frac{1}{p}}\Bigr\|\geqslant\Bigl\|q\Bigl(\Bigl(\sum_{i=1}^n|x_i|^p\Bigr)^{\frac{1}{p}}\Bigr)\Bigr\|=\bigl\|(x_1+Y,\dots,x_n+Y)\bigr\|_{\mathrm{canonical}},$$

hence $\|\cdot\|_{\mathrm{quotient}}\geqslant\|\cdot\|_{\mathrm{canonical}}$. To prove the converse inequality, it suffices to show that for every $y\in Y$ there exist $y_1,\dots,y_n\in Y$ such that

$$\Bigl(\sum_{i=1}^n|x_i-y_i|^p\Bigr)^{\frac{1}{p}}\leqslant\Bigl|\Bigl(\sum_{i=1}^n|x_i|^p\Bigr)^{\frac{1}{p}}-y\Bigr|. \tag{20.5}$$

Fix $y\in Y$. WLOG, $0\leqslant y\leqslant\bigl(\sum_{i=1}^n|x_i|^p\bigr)^{\frac{1}{p}}$ as, otherwise, replacing y with $y^+\wedge\bigl(\sum_{i=1}^n|x_i|^p\bigr)^{\frac{1}{p}}$ will make the right hand side in (20.5) even smaller. It suffices to prove that for every $0\leqslant y\leqslant\bigl(\sum_{i=1}^n|x_i|^p\bigr)^{\frac{1}{p}}$ there exist $y_1,\dots,y_n$ such that (20.5) holds and $|y_i|\leqslant y$ for each i; the last condition guarantees that $y_i\in Y$.

By Yudin's Representation Theorem, it suffices to prove this for continuous functions. Put $y_i=\bigl(|x_i|\wedge y\bigr)\operatorname{sign}x_i$, then y_i is a continuous function and $|y_i|\leqslant y$. Also, $|x_i-y_i|=|x_i|-|y_i|$. Replacing $|x_i-y_i|$ with $|x_i|-|y_i|$ in the left hand side of (20.5), we may assume that $x_i\geqslant 0$ for each i. Hence $y_i=x_i\wedge y$. Thus, we need to prove that for any positive continuous functions $x_1,\dots,x_n$ and y, if $y\leqslant\bigl(\sum_{i=1}^n|x_i|^p\bigr)^{\frac{1}{p}}$ then

$$\Bigl(\sum_{i=1}^n(x_i-x_i\wedge y)^p\Bigr)^{\frac{1}{p}}\leqslant\Bigl(\sum_{i=1}^n x_i^p\Bigr)^{\frac{1}{p}}-y. \tag{20.6}$$

Since the order is point-wise, it suffices to prove (20.6) for positive real numbers. View $x_1,\dots,x_n$ and y as positive real numbers, put $\bar{x}=(x_1,\dots,x_n)$. WLOG, by scaling, $\|\bar{x}\|_{\ell_p^n}=1$. Thus, we need to prove that $\bigl\|(\bar{x}-y\mathbb{1})^+\bigr\|_{\ell_p^n}\leqslant 1-y$.

The following lemma asserts that this is indeed true, and not only for ℓ_p^n but for a much larger class of spaces. Recall that every Banach space with a 1-unconditional basis is a Banach lattice in the order determined by the basis, so we can view X as a sequence space. We write $\mathbb{1}$ for the vector whose all coordinates equal 1, provided such a vector is in the space.

20.9.1 Lemma. *Let X be a finite-dimensional Banach space with a 1-unconditional normalized basis and $0\leqslant x\in S_X$. Then $\|(x-\delta\mathbb{1})^+\|\leqslant 1-\delta$ for every $0<\delta<1$.*

Proof. Let $n=\dim X$, $(e_i)_{i=1}^n$ the basis of X, and $(e_i^*)_{i=1}^n$ the bi-orthogonal basis of X^*. Put $y=(x-\delta\mathbb{1})^+$. Assume for the sake of contradiction that $\|y\|>1-\delta$. Suppose that $x=(x_1,\dots,x_n)$. WLOG, (up to renumbering the coordinates of x)

$y = (y_1, \dots, y_k, 0, \dots, 0)$ for some $1 \leqslant k \leqslant n$, $x_i = y_i + \delta$ for each $i = 1, \dots, k$, and $x_i \leqslant \delta$ for all $i = k+1, \dots, n$. By 1-unconditionality,

$$\|x\| = \big\|(x_1, \dots, x_k, x_{k+1}, \dots, x_n)\big\| \geqslant \big\|(y_1 + \delta, \dots, y_k + \delta, 0, \dots, 0)\big\|.$$

Since $\|y\| > 1 - \delta$, there exists $0 \leqslant f \in S_{X^*}$ such that $f(y) > 1 - \delta$. Moreover, we can choose f of the form $f = (f_1, \dots, f_k, 0, \dots, 0)$. Since $\|e_i^*\| = 1$ for every i, we have

$$1 = \|f\| = \big\|(f_1, \dots, f_k, 0, \dots, 0)\big\| \leqslant \sum_{i=1}^{k} f_i.$$

Therefore,

$$\|x\| \geqslant f(x) = \sum_{i=1}^{k} f_i x_i = \sum_{i=1}^{k} f_i y_i + \delta \sum_{i=1}^{k} f_i > 1 - \delta + \delta = 1.$$

□

Chapter 21

Order convergences and order topology

This chapter is not complete!

The concept of order convergence in vector lattices is subtle. There are two main issue with this concept. First, the very definition of order convergence is controversial. The order limit of an increasing (or decreasing) net is universally defined as its supremum (respectively, infimum). Hence, for monotone nets we have an easy and commonly accepted definition of order convergence. However, in the case of a non-monotone sequence or net, there have been several non-equivalent definitions of order convergence in the literature! In this chapter, we will discuss and compare several common definitions.

The second issue with order convergence is the fact that it is generally not topological. Even though we use a lot of "topological" terminology like "order closed set", "order continuous operator", and "order dense sublattice", order convergence on a vector lattice is not actually given by a topology. However, the collection of all order closed sets does form a topology. There are some goon news. First, this topology is the same for all common definitions of order convergence. Second, some common terms become proper in this setting, for example, a linear functional is order continuous iff it is continuous with respect to order topology. But there are also some bad news. First, order convergence does *not* agree with convergence with respect to the order topology. Second, the order topology need not be Hausdorff. Furthermore, it need not be linear: addition need not be jointly continuous. Yet, in order continuous Banach lattices, order topology agrees with norm topology; this is one reason that order continuous Banach lattices are so "nice".

Because of these two issues, for a long time, a general trend was to avoid order convergence whenever possible and reduce things to monotone nets. For example, order complete vector lattices, order continuous operators, and order continuous normed lattices may be defined in terms of order convergent monotone nets. Yet, with some recent advances, including uo-convergence, it has become clear that general order convergence cannot be avoided. This is why we decided to include this chapter.

21.1 Definitions of order convergence

In this section, we will compare the common definitions of order convergence. To distinguish them, we will refer to them as o1, o2, and o3. Suppose that $(x_\alpha)_{\alpha\in\Lambda}$ is a net in a vector lattice X and $x \in X$. We say that

$$x_\alpha \xrightarrow{\text{o1}} x \text{ if there exists a net } (u_\alpha)_{\alpha\in\Lambda} \text{ such that } u_\alpha \downarrow 0 \quad \text{and } \forall\alpha\in\Lambda\ |x_\alpha - x| \leqslant u_\alpha.$$

We believe that historically this is the earliest definition. In this definition the dominating net (u_α) is indexed over the same index set as the original net (x_α). The following example illustrated a major shortcoming of this definition.

21.1.1 Example. *A net with a tail of zeros which is not o_1-convergent.* It is natural to expect that the limit must be a "tail property" in the sense that the limit of a net should coincide with the limit of every tail. That is, adding more terms at the head should not affect convergence. However, consider the net (x_α) in $\mathbb{R}$ whose index set consists of two consecutive copies of $\mathbb{N}$ and

$$(x_\alpha) = (1, 2, 3, \dots, 0, 0, 0 \dots).$$

Note that (x_α) is identically zero on the second copy of $\mathbb{N}$. This net has a tail of zeros, so it is natural to expect it to converge to zero. However, it is easy to see that there is no decreasing net on the same index set which dominates (x_α).

The next definition partially compensated for this: we say that $x_\alpha \xrightarrow{\text{o2}} x$ if some tail of (x_α) o1-converges to x. That is,

$$x_\alpha \xrightarrow{\text{o2}} x \text{ if there exists a net } (u_\alpha)_{\alpha\in\Lambda} \text{ such that } u_\alpha \downarrow 0 \quad \text{and } \exists\alpha_0\ \forall\alpha\geqslant\alpha_0\ |x_\alpha - x| \leqslant u_\alpha.$$

Finally, let o3 be the ε-δ style definition used throughout this book. That is,

$$x_\alpha \xrightarrow{\text{o3}} x \text{ if there exists a net } (u_\gamma)_{\gamma\in\Gamma} \text{ such that } u_\gamma \downarrow 0 \quad \text{and } \forall\gamma\in\Gamma\ \exists\alpha_0\in\Lambda\ \forall\alpha\geqslant\alpha_0\ |x_\alpha - x| \leqslant u_\gamma. \tag{21.1}$$

It is easy to see that $x_\alpha \xrightarrow{\text{o3}} x$ iff there exist nets $a_\gamma \uparrow x$ and $b_\gamma \downarrow x$ such that for every γ, the order interval $[a_\gamma, b_\gamma]$ contains a tail of (x_α). In a similar fashion, one can equivalently reformulate the definitions of o1- and o2-convergence replacing a single net (u_α) with a pair of nets (a_α) and (b_α). The advantage of these "two-net" versions of the definitions is that it allows to generalize most of the results in this chapter to ordered vector spaces. For simplicity, we will stick to vector lattices, where the "one-net" and the "two-net" approaches are clearly equivalent.

It is easy to see that

$$x_\alpha \xrightarrow{\text{o1}} x \quad \Rightarrow \quad x_\alpha \xrightarrow{\text{o2}} x \quad \Rightarrow \quad x_\alpha \xrightarrow{\text{o3}} x \tag{21.2}$$

21.1.2 Proposition. *Let X be an order complete vector lattice. Then*

(i) *o1, o2, and o3 convergences agree for order bounded nets;*
(ii) *o1, o2, and o3 convergences agree for sequences;*
(iii) *o2 and o3 convergences agree for arbitrary nets;*

Proof. (i) follows immediately from 2.4.22. (ii) follows from (i) and (21.2) because every o3-convergent sequence is order bounded. (iii) follows from (i) and (21.2) because every o3-convergent net has an order bounded tail. □

In general, however, these three convergences are different. The net in Example 21.1.1 is o2- and o3-convergent to zero, yet it is not o1-convergent. The following example, due to Fremlin, shows that o2 and o3 are different.

21.1.3 Example. *A net which is o3-convergent but not o2-convergent.* Let Γ be an uncountable set endowed with the discrete topology. Let K be the one-point compactification of Γ. That is, $K = \Gamma \cup \{\infty\}$ such that a set U is a neighbourhood of ∞ whenever $\infty \in U$ and $K \setminus U$ is finite. It follows that if A is an infinite subset of K then A meets every neighbourhood of ∞, so that $\infty \in \bar{A}$.

Let $X = C(K)$. Fix a distinct sequence (γ_n) in Γ and put $x_n = \mathbb{1}_{\{\gamma_n\}}$. Clearly, $x_n \in X$. We claim that $x_n \xrightarrow{\text{o3}} 0$ in X. Indeed, consider the collection Λ of all finite subsets of Γ. Then Λ is directed by inclusion. For each $F \in \Lambda$, put $u_F \in C(K)$ to be the characteristic function of $K \setminus F$. Viewed as a net over Λ, $u_F \downarrow 0$. For every finite $F \subseteq \Gamma$, there exists n_0 such that $\gamma_n \notin F$ for all $n \geqslant n_0$, hence $u_F(\gamma_n) = 1$ and, therefore, $0 \leqslant x_n \leqslant u_F$. It follows that $x_n \xrightarrow{\text{o3}} 0$.

Finally, we will show that (x_n) is not o2-convergent. Suppose that there exits a sequence (u_n) in X such that $u_n \downarrow 0$ and $n_0 \in \mathbb{N}$ such that $x_n \leqslant u_n$ for all $n \geqslant n_0$. Fix $n \geqslant n_0$. For every $k \geqslant n$, we have $u_n \geqslant u_k \geqslant x_k$, which yields $u_n(\gamma_k) \geqslant 1$. Since ∞ is in the closure of $\{\gamma_k : k \geqslant n\}$, it follows that $u_n(\infty) \geqslant 1$. Then $u_n \geqslant \frac{1}{2}$ on some neighbourhood of ∞. That is, there exists a finite subset F_n of Γ such that $u_n \geqslant \frac{1}{2}$ on $K \setminus F_n$. Put $F = \bigcup_{n=n_0}^{\infty} F_n$. Then F is countable, so that there exists $\gamma_0 \in \Gamma \setminus F$. It follows that $u_n(\gamma_0) \geqslant \frac{1}{2}$ and, therefore, $u_n \geqslant \frac{1}{2}\mathbb{1}_{\{\gamma_0\}}$ for all $n \geqslant n_0$. This contradicts $u_n \downarrow 0$.

Example 21.1.1 shows that o1 is a poor convergence. There are several reasons why in this book we prefer o3 over o2. One reason is that o3 is consistent with the order completion. Indeed, let X be a vector lattice and X^δ its order completion. By Proposition 21.1.2, o2 and o3 convergences agree on X^δ, so there is no ambiguity there. However, by Corollary 3.2.32, o3 convergence is the convergence in X which agrees with *the* order convergence on X^δ.

Another reason why we prefer o3 is that it comes out naturally as a *convergent structure*, which will be discussed later.

From now on, we will often refer to o3-convergence as the order convergence or o-convergence.

21.2 Order convergences and order topology

Clearly, each of the three convergences o1, o2, and o3, is translation-invariant in the sense that if $x_\alpha \to x$ and y is a fixed vector in X then $x_\alpha + y \to x + y$.

Surprisingly, these three convergences are not topological, i.e., they are not given by a topology. Indeed, for order bounded sequences in $L_p(\mu)$, the three convergences agree by Proposition 21.1.2; furthermore, they all agree with the a.e. convergence by Proposition 2.4.27. Now arguing as in Example A.3.8 we conclude that order convergences are not topological.

Even though order convergences are not topological, they, nevertheless, give rise to a certain topology, called the order topology, as follows. A subset A of X is said to be o1-closed if for every net (x_α) in A which o1-converges to some $x \in X$ we have $x \in A$. We define o2- and o3-closed sets in the similar fashion. (Note that o3-closed sets are exactly the order closed sets in other parts of the book). Let τ_{o1}, τ_{o2}, and τ_{o3} be the collections of all o1-, o2-, and o3-closed sets, respectively. It is easy to see that there collections are topologies on X. Clearly, these topologies are translation invariant. It follows from (21.2) that $\tau_{o1} \subseteq \tau_{o2} \subseteq \tau_{o3}$. We are going to show that actually $\tau_{o1} = \tau_{o2} = \tau_{o3}$.

We will make use of the following class of sets: a set U is said to be ***net-catching*** for x if for every net $u_\alpha \downarrow 0$ there exists α such that $[x - u_\alpha, x + u_\alpha] \subseteq U$. Equivalently, if $a_\alpha \uparrow x$ and $b_\alpha \downarrow x$ then $[a_\alpha, b_\alpha] \subseteq U$ for some α (and, therefore, for all sufficently large α).

21.2.1 Lemma. *Every τ_{o1}-open neighbourhood of x is net-catching for x.*

Proof. WLOG, $x = 0$. Let U be a τ_{o1}-open neighbourhood of 0 which is not net-catching. Then there exists a net (u_α) such that $u_\alpha \downarrow 0$ but $[-u_\alpha, u_\alpha]$ is not entirely contained in U for every α, so that there exists some $x_\alpha \in [-u_\alpha, u_\alpha] \cap U^C$. Then $x_\alpha \xrightarrow{\text{o1}} 0$. Since U^C is τ_{o1}-closed, we have $0 \in U^C$, which is a contradiction. $\square$

21.2.2 Lemma. *If $x_\alpha \xrightarrow{\text{o3}} 0$ then $x_\alpha \xrightarrow{\tau_{o1}} 0$.*

Proof. Let $(x_\alpha)_{\alpha\in\Lambda}$ be a net such that $x_\alpha \xrightarrow{\text{o3}} 0$; let $(u_\gamma)_{\gamma\in\Gamma}$ be a dominating net as in the definition of o3. Let U be a τ_{o1}-open neighborhood of zero. By Lemma 21.2.1, U is net-catching, so that there exists γ such that U contains $[-u_\gamma, u_\gamma]$, which, in tern, contains a tail of (x_α). $\square$

21.2.3 Theorem. $\tau_{o1} = \tau_{o2} = \tau_{o3}$.

Proof. We already have $\tau_{o1} \subseteq \tau_{o2} \subseteq \tau_{o3}$, so it suffices to show that every o3-closed set is o1-closed. Let A be an o3-closed set, (x_α) a net in A, and $x_\alpha \xrightarrow{\text{o1}} x$. By Lemma 21.2.2, $x_\alpha \xrightarrow{\tau_{o1}} x$. Since A is o3-closed, this yields $x \in A$. Therefore, A is o1-closed. $\square$

From now on, we refer to this topology as ***order topology*** and denote it τ_o. Combining Lemmas21.2.1 and 21.2.2, Theorem 21.2.3, and (21.2), we get the following.

21.2.4 Corollary. *Each of o1-, o2-, and o3-convergences implies τ_o convergence. In particular, $x_\alpha \xrightarrow{\text{o}} x$ implies $x_\alpha \xrightarrow{\tau_o} x$. Furthermore, every open order neighbourhood of x is net-catching for x.*

In general, $x_\alpha \xrightarrow{\tau_o} x$ does not imply $x_\alpha \xrightarrow{\text{o}} x$ because, as we already know, order convergence need not be topological.

21.2.5 Exercise. A set A in a vector lattice is order open iff $x_\alpha \xrightarrow{\text{o}} x \in A$ implies that A contains a tail of (x_α). Here o-convergence may be replaced with o1 or o2-convergence.

21.3 Properties of order topology

Throughout this section, order convergence may be replaced with o1 or o2 convergence.

We have observed earlier that order topology is translation invariant. It follows immediately from the definition that points are closed. The following example shows, however, that order topology need not be Hausdorff or linear!

21.3.1 Example. *Order topology on $C[0,1]$ is not Hausdorff and not linear.* Suppose it is Hausdorff. Let U and V be open neighbourhoods of 0 and of $\mathbb{1}$, respectively with $U \cap V = \varnothing$. Let (a_k) be an enumeration of the rational numbers in $(0,1)$. For each $k, n \in \mathbb{N}$, let $x_{k,n}$ be the continuous function such that $x_{k,n}(a_k) = 1$, $x_{n,k}$ vanishes outside of $\left[a_k - \frac{1}{n}, a_k + \frac{1}{n}\right]$, and is linear on $\left[a_k - \frac{1}{n}, a_k\right]$ and on $\left[a_k, a_k + \frac{1}{n}\right]$. For each fixed k, we clearly have $x_{k,n} \downarrow 0$ as $n \to \infty$. It follows that $x_{k,n} \xrightarrow{\mathrm{o}} 0$ and, therefore, $x_{k,n} \xrightarrow{\tau_o} x$ as $n \to \infty$. Choose n_1 such that $y_1 \coloneqq x_{1,n_1} \in U$. Observe that $y_1 \vee x_{2,n} \downarrow y_1$. It follows from $y_1 \in U$ that $y_2 \coloneqq y_1 \vee x_{2,n_2} \in U$ for some n_2. Iterating this process, we construct a sequences (n_k) in $\mathbb{N}$ and (y_k) in U such that $y_k = y_{k-1} \vee x_{k,n_k}$ for every $k > 1$. It follows that $y_k \uparrow$ and $y_k(a_i) = 1$ as $i = 1, \dots, k$, which yields $y_k \uparrow \mathbb{1}$. Therefore, there exists k_0 such that $y_{k_0} \in V$. This contradicts U and V being disjoint.

In particular, the order topology on $C[0,1]$ is not linear, i.e., it is not a topological vector space because a topological vector space is Hausdorff iff points are closed; see Section A.2.

Recall that a linear functional $f \colon X \to \mathbb{R}$ is said to be order continuous if $f(x_\alpha) \to f(x)$ whenever $x_\alpha \xrightarrow{\mathrm{o}} x$. We will show next that this is equivalent to f being continuous with respect to τ_o.

21.3.2 Proposition. *Let (Y, σ) be a topological space. Then a function $f \colon (X, \tau_o) \to (Y, \sigma)$ is continuous iff $x_\alpha \xrightarrow{\mathrm{o}} x$ implies $f(x_\alpha) \xrightarrow{\sigma} f(x)$.*

Proof. Suppose that f is τ_o-to-σ continuous. If $x_\alpha \xrightarrow{\mathrm{o}} x$ then $x_\alpha \xrightarrow{\tau_o} x$ by Corollary 21.2.4, hence $f(x_\alpha) \xrightarrow{\sigma} f(x)$.

Conversely, suppose that $x_\alpha \xrightarrow{\mathrm{o}} x$ implies $f(x_\alpha) \xrightarrow{\sigma} f(x)$. Take a closed set F in Y. If (x_α) is a net in $f^{-1}(F)$ and $x_\alpha \to x$ then $f(x_\alpha) \xrightarrow{\sigma} f(x)$, so that $f(x) \in F$, hence $x \in f^{-1}(F)$. It follows that $f^{-1}(F)$ is closed. Hence, f is τ_o-to-σ continuous. □

21.3.3 Remark. Note that in Proposition 21.3.2, order convergence may be replaced with o1 or o2-convergence.

21.3.4 Corollary. *The order topology is the finest topology on X that makes order convergent nets topologically convergent.*

Proof. Corollary 21.2.4 asserts that every order convergent net is τ_o-convergent. Let σ be a topology on X such that if $x_\alpha \xrightarrow{\mathrm{o}} x$ then $x_\alpha \xrightarrow{\sigma} x$. It follows from Proposition 21.3.2 that the identity map on X is τ_o-to-σ-continuous, hence $\sigma \leqslant \tau_o$. □

Combining Proposition 21.3.2 with the fact that $x_\alpha \to x$ implies $x_\alpha \xrightarrow{\tau_o} x$, we immediately obtain the following.

21.3.5 Corollary. *Let $f\colon X \to Y$ be a function between vector lattices. If $f(x_\alpha) \xrightarrow{\mathrm{o}} f(x)$ whenever $x_\alpha \xrightarrow{\mathrm{o}} x$ then f is continuous with respect to the order topologies of X and Y.*

We observed in Example 21.3.1 that order topology need not be linear, i.e., addition and scalar multiplication need not be jointly continuous. However Corollary 21.3.5 yields the following.

21.3.6 Proposition. *Vector and lattice operations are separately continuous with respect to the order topology.*

21.3.7 Proposition. *A Banach lattice X is order continuous iff its order topology agrees with its norm topology.*

Proof. If X is order continuous, then norm closed sets and order closed sets agree by Exercise 2.5.15, hence the two topologies coincide.

Conversely, suppose that the two topologies agree. Then $x_\alpha \xrightarrow{\mathrm{o}} x$ implies $x_\alpha \xrightarrow{\tau_o}$ by Corollary 21.2.4, hence $x_\alpha \xrightarrow{\|\cdot\|} x$ and, therefore, X is order continuous. □

In particular, the order topology on an order continuous Banach lattice is Hausdorff and linear.

21.4 When is order convergence topological?

We observed in the preceding section that, in general, order convergence is not topological, i.e., not given by a topology. In that section, we will show that order convergence is topological precisely when the space is finite-dimensional. This section is based on [DEM].

21.4.1 Theorem. *Let τ be a linear topology on a vector lattice X. If $x_\alpha \xrightarrow{\tau} 0$ implies $x_\alpha \xrightarrow{\mathrm{o}} 0$ for every net (x_α) in X then X is Archimedean, has a strong unit and $x_\alpha \xrightarrow{\tau} 0$ implies $x_\alpha \xrightarrow{\mathrm{u}} 0$.*

Proof. To show that X is Archimedean, suppose that $0 \leqslant x \leqslant \frac{1}{n}u$ for all $n \in \mathbb{N}$. Since scalar multiplication is τ-continuous, we have $\frac{1}{n}u \xrightarrow{\tau} 0$, which yields $\frac{1}{n}u \xrightarrow{\mathrm{o}} 0$ and, therefore, $x = 0$.

Let $\mathcal{N}_0$ be a base of neighborhoods of zero for τ. As in Section A.1, we can construct a net in X such that elements of $\mathcal{N}_0$ are tails of this net. Namely, consider the set $\Lambda = \bigl\{(U,x)\ :\ x \in U \in \mathcal{N}_0\bigr\}$ ordered by $(U,x) \leqslant (V,y)$ whenever $V \subseteq U$. This is a directed set. For $\alpha \in \Lambda$ with $\alpha = (U,x)$, put $x_\alpha = x$. Then $(x_\alpha)_{\alpha\in\Lambda}$ is a net in X. Note that for every $\alpha = (U,x)$ in λ we have $\{x_\beta\}_{\beta\geqslant\alpha}$.

Observe that $x_\alpha \xrightarrow{\tau} 0$. Indeed, fix $U_0 \in \mathcal{N}_0$, take any $x_0 \in U$ and put $\alpha = (U_0, x_0)$. For every $\beta = (V, x)$ with $\beta \geqslant \alpha$, we have $x_\beta = x \in V \subseteq U$. It follows that $x_\alpha \xrightarrow{\tau} 0$.

By assumption, we have $x_\alpha \xrightarrow{\mathrm{o}} 0$ and, therefore, the net has an order bounded tail. That is, there exist $\alpha = (U, x)$ in Λ and $e \in X_+$ such that $x_\beta \in [-e, e]$ whenever $\beta \geqslant \alpha$. It follows that $U \subseteq [-e, e]$. For every $x \in X$, we have $\lambda x \in U$ for some $\lambda > 0$, and, therefore, $\lambda x \in [-e, e]$. It follows that e is a strong unit.

To prove the last claim, suppose that $x_\alpha \xrightarrow{\tau} 0$ for some net (x_α). Fix $\varepsilon > 0$. Let $U \in \mathcal{N}_0$ such that $U \subseteq [-e, e]$. Then εU is a τ-neighborhood of zero, so there exists α_0 such that $x_\alpha \in \varepsilon U$ whenever $\alpha \geqslant \alpha_0$. It follows that $|x_\alpha| \leqslant \varepsilon e$. It follows that $\|x_\alpha\|_e \to 0$. □

21.4.2 Theorem. *Order convergence on a vector lattice X arises from a linear topology iff X is finite-dimensional and Archimedean.*

Proof. Let X be an Archimedean finite-dimensional vector lattice. Then X is lattice-isomorphic to $\mathbb{R}^n$ for some n by Theorem 5.4.11, so WLOG, $X = \mathbb{R}^n$. Then order convergence clearly agrees with the coordinate-wise convergence, which is topological.

Suppose that there is a linear topology τ on X such that $x_\alpha \xrightarrow{\tau} 0$ iff $x_\alpha \xrightarrow{\mathrm{o}} 0$ for every net (x_α) in X. By Theorem 21.4.1, X has a strong unit e and $x_\alpha \xrightarrow{\tau} 0$ implies $\|x_\alpha\|_e \to 0$. For the sake of contradiction, assume that X is infinite-dimensional. By Theorem 5.4.11, X contains an (infinite) disjoint sequence of non-zero vectors, say, (x_n). WLOG, $\|x_n\|_e = 1$. Then $x_n \xrightarrow{\mathrm{o}} 0$ by Proposition 3.2.33, so that, $x_n \xrightarrow{\tau} 0$ and, therefore, $\|x_n\|_e \to 0$; a contradiction. □

21.4.3 Corollary. *Order convergence on an Archimedean vector lattice X arises from a topology iff X is finite-dimensional.*

Proof. Suppose that order convergence on X comes from a topology τ. Since addition and scalar multiplication are jointly continuous with respect to order convergence by Proposition 2.4.7(ii) and Exercise 2.4.12, it follows that τ is linear. Now apply Theorem 21.4.2. □

21.4.4 Example. Let $X = \mathbb{R}^2$ with lexicographic order. Note that X fails the Archimedean Property. The sequence $(\frac{1}{n}, 0)$ converges to zero in every Hausdorff linear topology on X, yet it does not converge to zero in order.

21.4.5 Proposition. *Uniform convergence on an Archimedean vector lattice X is topological iff X has a strong unit.*

Proof. If uniform convergence arises from a topology, then X has a strong unit by Theorem 21.4.1. Conversely, if X has a strong unit, say, e, then uniform convergence agrees with convergence in $\|\cdot\|_e$-norm. □

21.5 Order convergence structure

In this section we will show that the order convergence (that is, the o3-convergence) may be realized as a natural *convergence structure*. We start with the definition of a convergence structure; we refer the reader to [BB02] for further details on convergence structures. The definitions a filter, a tail filter, a filter base, etc can be found in Section A.1. Let Ω be an arbitrary set. For $x \in \Omega$, we write $[x]$ for the *principal filter* of x, i.e., the collection of all subsets of Ω that contain x.

Fix a map λ from Ω to the set of all filters on Ω. We will write $\mathcal{F} \xrightarrow{\lambda} x$ instead of $\mathcal{F} \in \lambda(x)$. We say that λ is a a ***convergence structure*** on Ω if for every $x \in \Omega$ and every two filters $\mathcal{F}$ and $\mathcal{G}$ on Ω the following are satisfied:

(i) $[x] \xrightarrow{\lambda} x$;

(ii) if $\mathcal{F} \xrightarrow{\lambda}$ and $\mathcal{F} \subseteq \mathcal{G}$ then $\mathcal{G} \xrightarrow{\lambda} x$;

(iii) if $\mathcal{F} \xrightarrow{\lambda} x$ and $\mathcal{G} \xrightarrow{\lambda} x$ then $\mathcal{F} \cap \mathcal{G} \xrightarrow{\lambda} x$.

Given a net (x_α) in Ω, we say that (x_α) converges to x relative to the convergent structure λ if the tail filter of (x_α) is in $\lambda(x)$. Thus, every convergent structure induces a convergence.

Let τ be a topology on Ω. For each $x \in \Omega$, let $\lambda(x)$ be the filter of all neighbourhoods of x. It is easy to see that this defines a convergence structure on Ω, and that a net converges relative to this convergent structure iff it converges in τ. In this sense, the concept of a convergence structure extends the concept of topology.

Let X be a vector lattice and $x \in X$. For a filter $\mathcal{F}$ on X, we write $\mathcal{F} \xrightarrow{\mathrm{o}} x$ and say that $\mathcal{F}$ ***order converges to*** x if $\mathcal{F}$ contains a filter base $\mathcal{B}$ consisting of order intervals such that $\bigcap \mathcal{B} = \{x\}$. We are going to show that this forms a convergence structure. First, we will present an equivalent definition of $\mathcal{F} \xrightarrow{\mathrm{o}} x$.

21.5.1. Suppose that $x \in X$ and there are two nets (a_α) and (b_α) in X such that $a_\alpha \uparrow x$ and $b_\alpha \downarrow x$. Let $\mathcal{B}$ be the set of all order intervals $[a_\alpha, b_\alpha]$. It is easy to see that $\mathcal{B}$ is a filter base and $\bigcap \mathcal{B} = \{0\}$. Conversely, let $\mathcal{B}$ be a filter base consisting of order intervals with $\bigcap \mathcal{B} = \{x\}$. Since $\mathcal{B}$ is directed by reverse inclusion, the "left ends" and the "right ends" of the intervals that comprise $\mathcal{B}$ form two nets, say, (a_α) and (b_α), indexed by $\mathcal{B}$. It is easy to see that $a_\alpha \uparrow x$ and $b_\alpha \downarrow x$.

21.5.2 Proposition. *Let X be a vector lattice, $x \in X$ and $\mathcal{F}$ a filter on X. TFAE:*

(i) $\mathcal{F} \xrightarrow{\mathrm{o}} x$*;*

(ii) *There exist two nets (a_α) and (b_α) such that $a_\alpha \uparrow x$, $b_\alpha \downarrow x$, and $[a_\alpha, b_\alpha] \in \mathcal{F}$ for every α;*

(iii) *There exists a net (u_α) such that $u_\alpha\downarrow 0$ and $[x-u_\alpha,x+u_\alpha]\in\mathcal{F}$ for every α.*

Proof. The equivalence of (i) and (ii) follows immediately from 21.5.1. (iii)⇒(ii) trivially. Suppose (ii) is satisfied. Put $u_\alpha=(-a_\alpha)\vee b_\alpha$. Then $u_\alpha\downarrow 0$ and $[a_\alpha,b_\alpha]\subseteq[-u_\alpha,u_\alpha]$, so that $[-u_\alpha,u_\alpha]\in\mathcal{F}$. □

21.5.3 Proposition. *Order convergent filters form a convergence structure.*

Proof. WLOG, $x=0$. Taking $\mathcal{B}=\bigl\{\{0\}\bigr\}$, we conclude that $[0]\xrightarrow{\mathrm{o}}0$. It is immediate from the definition that if $\mathcal{F}\xrightarrow{\mathrm{o}}0$ and $\mathcal{F}\subseteq\mathcal{G}$ then $\mathcal{G}\xrightarrow{\mathrm{o}}0$. Finally, suppose that $\mathcal{F}\xrightarrow{\mathrm{o}}0$ and $\mathcal{G}\xrightarrow{\mathrm{o}}0$. Then there exist nets (u_α) and (v_β) such that $u_\alpha\downarrow 0$, $v_\beta\downarrow 0$, $[-u_\alpha,u_\alpha]\in\mathcal{F}$ and $[-v_\beta,v_\beta]\in\mathcal{G}$ for all α and β. Consider a double net $(w_{(\alpha,\beta)})$ where $w_{(\alpha,\beta)}=u_\alpha\vee v_\beta$. Then $w_{(\alpha,\beta)}\downarrow 0$. For every α and β, we have $[-u_\alpha,u_\alpha]\subseteq[-w_{(\alpha,\beta)},w_{(\alpha,\beta)}]$, so that $[-w_{(\alpha,\beta)},w_{(\alpha,\beta)}]\in\mathcal{F}$. Similarly, $[-w_{(\alpha,\beta)},w_{(\alpha,\beta)}]\in\mathcal{G}$, hence $[-w_{(\alpha,\beta)},w_{(\alpha,\beta)}]\in\mathcal{F}\cap\mathcal{G}$. It follows that $\mathcal{F}\cap\mathcal{G}\xrightarrow{\mathrm{o}}0$. □

Naturally, we call the resulting convergent structure the ***order convergence structure***.

The following example shows that “contains a filter base $\mathcal{B}$” in the definition of $\mathcal{F}\xrightarrow{\mathrm{o}}x$ cannot be replaced with “generated by a filter base $\mathcal{B}$”.

21.5.4 Example. Let $X=\mathbb{R}^2$ with the coordinate-wise order. Let $\mathcal{B}_1$ be the sets of all intervals in the x-axis centred at zero. Let $\mathcal{B}_2$ be the sets of all intervals in the y-axis centred at zero. Let $\mathcal{F}_1$ and $\mathcal{F}_2$ be the filters on X generated by $\mathcal{B}_1$ and $\mathcal{B}_2$, respectively. By the preceding proposition, we have $\mathcal{F}_1\xrightarrow{\mathrm{o}}0$ and $\mathcal{F}_2\xrightarrow{\mathrm{o}}0$. It follows that $\mathcal{F}\xrightarrow{\mathrm{o}}0$, where $\mathcal{F}=\mathcal{F}_1\cap\mathcal{F}_2$. Note that $\mathcal{F}$ is generated by the filter base $\mathcal{B}$ of all “crosses” centred at zero, i.e., unions of intervals in $\mathcal{B}_1$ and $\mathcal{B}_2$. One can also see directly that $\mathcal{F}\xrightarrow{\mathrm{o}}0$ because it contains all squares centred at the origin. However, the elements of filter base $\mathcal{B}$ are not order intervals. Moreover, it is easy to see that no filter base consisting of order intervals generates $\mathcal{F}$.

Recall that every convergence structure induces convergence of nets. We now claim that the order convergence structure induces precisely our usual order convergence of nets, i.e., the o3-convergence. This is another reason to believe that o3 is the “right” order convergence.

21.5.5 Proposition. *A net in a vector lattice converges with respect to the order convergence structure iff it converges in order.*

Proof. Let (x_α) be a net in a vector lattice X and $x\in X$. Suppose that $x_\alpha\xrightarrow{\mathrm{o}}x$. Then there exists a net (u_γ) as in (21.1). WLOG, $x=0$. Then every interval of the form

$[-u_\gamma, u_\gamma]$ contains a tail of (x_α), hence $[-u_\gamma, u_\gamma]$ is contained in the tail filter of (x_α). It follows that the filter converges in order to 0.

Conversely, suppose that the tail filter of (x_α) converges in order to zero. Then there exists a net (u_γ) such that $u_\gamma \downarrow 0$ and for every γ the order interval $[-u_\gamma, u_\gamma]$ is in the tail filter of (x_α). This means that $[-u_\gamma, u_\gamma]$ contains a tail of (x_α). Therefore, $x_\alpha \xrightarrow{\mathrm{o}} 0$. □

Given a convergence structure λ on some set Ω, one can "modify" it into a genuine topology as follows. For every $x \in \Omega$, put $\mathcal{U}_x$ be the intersection of all filters which converge to x. We now declare a subset V of Ω to be open if $V \in \mathcal{U}_x$ whenever $x \in V$. That is,

$$V \quad \text{is open iff} \quad V \in \mathcal{F} \text{ whenever } \mathcal{F} \xrightarrow{\lambda} x \in V$$

It can be easily verified that the collection of all open sets forms a topology on Ω. This topology is called the ***topological modification of*** λ.

We are now going to apply this to the order convergence structure. First, we observe that $\mathcal{U}_x$ consists precisely of the net-catching sets for x:

21.5.6 Proposition. *Let $x \in V \subseteq X$. Then V is net-catching for x iff $V \in \mathcal{U}_x$, that is, $V \in \mathcal{F}$ whenever $\mathcal{F} \xrightarrow{\mathrm{o}} x$.*

Proof. Suppose that V is net-catching for x and $\mathcal{F} \xrightarrow{\mathrm{o}} x$. (a_α) and (b_α) be as in Proposition 21.5.2(ii). Since $[a_\alpha, b_\alpha] \subseteq V$ for some α, we conclude that $V \in \mathcal{F}$.

To prove the converse, suppose that $a_\alpha \uparrow x$ and $b_\alpha \downarrow x$. Let $\mathcal{F}$ be the filter generated by the order intervals $[a_\alpha, b_\alpha]$ over all α. Then $\mathcal{F} \xrightarrow{\mathrm{o}} x$ by Proposition 21.5.2. By assumption, $V \in \mathcal{F}$. It follows that $[a_\alpha, b_\alpha] \subseteq V$ for some α. □

21.5.7 Theorem. *The topological modification of the order convergence structure is the order topology.*

Proof. Let τ stand for the topological modification of the order convergence structure; let $V \subseteq X$. Suppose that V is order open. Then, for every $x \in V$, Corollary 21.2.4 yields that V is net-catching for x and, therefore, $V \in \mathcal{F}$ whenever $\mathcal{F} \xrightarrow{\mathrm{o}} x$ by Proposition 21.5.6. It follows that V is τ-open.

Conversely, suppose that V is τ-open. Let $x_\alpha \xrightarrow{\mathrm{o}} x \in V$. Then there exist nets (a_γ) and (b_γ) such that $a_\gamma \uparrow x$, $b_\gamma \downarrow x$, and for every γ the order interval $[a_\gamma, b_\gamma]$ contains a tail of (x_α). Let $\mathcal{F}$ be the filter generated by the intervals $[a_\gamma, b_\gamma]$ over all γ. Then $\mathcal{F} \xrightarrow{\mathrm{o}} x$ by Proposition 21.5.2. It follows that $V \in \mathcal{F}$, so that V contains $[a_\gamma, b_\gamma]$ for some γ, which, in turn, contains a tail of (x_α). This proves that V is order open by Exercise 21.2.5. □

Chapter 22

Free vector and Banach lattices

22.1 Free vector lattice over a set: $\mathrm{FVL}(A)$

We start with a bit of motivation. "Free vector lattice" is a categorical concept. Free objects appear in various branches of mathematics. For example, free groups are prominent in Algebra. Given a set A, a free group over A is defined as the set of all formal words with letters being elements of A or their formal inverses. We identify two words if they agree after adding or removing subwords of the form aa^{-1} or $a^{-1}a$; for example, we identify $a_1a_2a_2^{-1}a_3$ and $a_1a_3a_4^{-1}a_4$. One defines the product of two words as their concatenation. This leads to a group, called the ***free group*** over A. By interpreting every element of A as a single letter word, we may identify A with a subset of its free group; the elements of A are called ***generators*** of the group. It is easy to see that every map φ from the set of the generators A to an arbitrary group G extends uniquely to a group homomorphism $\hat{\varphi}$ from the free group to G. One may actually take it as a defining property of free groups. More generally, an object X in a category $\mathcal{C}$ is free over a subset $A \subseteq X$ when every map φ from A to any object Y in $\mathcal{C}$ extends uniquely to a morphism $\hat{\varphi}\colon X \to Y$. Yet another familiar example is that of a (Hamel) basis of a vector space. A subset A of a vector space X is a basis iff every map φ from A to an arbitrary vector space Y extends uniquely to a linear map $\hat{\varphi}\colon X \to Y$. Given an arbitrary set A, let X be the set of all formal linear combinations of elements of A, i.e., the set of all formal expressions of the form $\lambda_1a_1 + \dots + \lambda_na_n$, where $n \in \mathbb{N}$, $a_1, \dots, a_n \in A$, and $\lambda_1, \dots, \lambda_n$ are scalars. If we define addition and scalar multiplication on X in the natural way then X becomes a vector space with basis A. This shows that every set A may be "embedded" as a basis into a vector space X. So we may view X as a ***free vector space*** over A.

There are several common features in these two examples. First, since we do not assume the set of generators to have any a priori structure, the construction really depends only the cardinality of A. We may think of A being a subset of the free object X, or we may instead consider a set A and an injective map $i\colon A \hookrightarrow X$; the two approaches are equivalent as one may just identify A and $i(A)$. Second, both in the free group and in the free vector space construction, the free object may be build as a set of equivalence classes of certain formal expressions of generators with appropriately defined operations on these formal expressions.

Let A be a subset of a vector lattice X. We say that X is a ***free vector lattice*** over A if every map φ from A to an arbitrary vector lattice Y extends uniquely to a (linear) lattice homomorphism $\hat{\varphi}\colon X \to Y$. The elements of A are said to be ***generators*** of X. In the next two sections, we will prove that free vector lattices exist, that is, for every set A one may find a vector lattice X such that A is a subset of X (or, equivalently, A may be identified with a subset of X via an injective map $i\colon A \to X$) and X is a free vector lattice over A.

$$\begin{tikzcd} X \arrow[dr, dotted, "\hat{\varphi}"] & \\ A \arrow[u, hook, "i"] \arrow[r, "\varphi"'] & Y \end{tikzcd}$$

22.1.1 Exercise. A free vector lattice over A is unique in the following sense: let X and Y be two free vector lattices over A, then there is a lattice isomorphism T from X onto Y which fixes A, i.e., $Ta = a$ for all $a \in A$.

We write $\mathrm{FVL}(A)$ for the free vector lattice over A. It is straightforward that $\mathrm{FVL}(A)$ only depends on the cardinality of A. It is not uncommon in the literature to write $\mathrm{FVL}(\mathfrak{a})$ instead of $\mathrm{FVL}(A)$, where $\mathfrak{a}$ is the cardinality of A. In particular, if $n \in \mathbb{N}$, we write $\mathrm{FVL}(n)$ for the free vector lattice over a set of n elements.

22.2 Existence of $\mathrm{FVL}(A)$: concrete approach

In this section, we will prove the existence of $\mathrm{FVL}(A)$ for every set A. Our construction will be based on lattice-linear function calculus, which was developed in Section 4.3.

We start with the special case when A is finite, say, $A = \{a_1, \dots, a_n\}$. We claim that $\mathrm{FVL}(n) = L_n$. Recall that L_n is the set of all lattice-linear functions $f\colon \mathbb{R}^n \to \mathbb{R}$; one may view L_n as the sublattice of $\mathbb{R}^{\mathbb{R}^n}$ generated by the coordinate functionals $\delta_1, \dots, \delta_n$. Lattice operations in L_n are computed point-wise.

The map $i\colon A \to L_n$ given by $i(a_j) = \delta_j$ is clearly one-to-one and allows us to identify A with the subset $\{\delta_1, \dots, \delta_n\}$ of L_n. Let Y be an arbitrary vector lattice; let $\varphi\colon A \to Y$ be any map. For $f \in L_n$, use lattice-linear function calculus to define

$\hat{\varphi}(f) \coloneqq f\big(\varphi(a_1), \dots, \varphi(a_n)\big)$.

$$\begin{tikzcd} L_n \arrow[dr, dotted, "\hat{\varphi}"] & \\ A \arrow[u, hook, "i"] \arrow[r, "\varphi"'] & Y \end{tikzcd}$$

That is, we take any formal lattice-linear expression $\Phi[t_1, \dots, t_n]$ representing f, and define $\hat{\varphi}(f) = \Phi\big(\varphi(a_1), \dots, \varphi(a_n)\big)$. It follows from Exercise 4.3.6, $\hat{\varphi}(f)$ is well-defined, $\hat{\varphi}$ is a lattice homomorphism, $\hat{\varphi}$ extends φ in the sense that $\hat{\varphi}(\delta_j) = \varphi(a_j)$ as $j = 1, \dots, n$, and $\hat{\varphi}$ is a unique lattice homomorphism with this property.

We now generalize the preceding construction to the case when A is an arbitrary set by replacing $\mathbb{R}^n$ with $\mathbb{R}^A$. For each $a \in A$, we define a "coordinate functional" $\delta_a \colon \mathbb{R}^A \to \mathbb{R}$ via $\delta_a(x) = x(a)$. Let L be the sublattice of $\mathbb{R}^{\mathbb{R}^A}$ generated by all these coordinate functionals. Clearly, L consists of all lattice-linear combinations of finite sets of coordinate functionals. The map $i \colon A \to L$ defined by $i(a) = \delta_a$ is an injection. We will show that that $L = \mathrm{FVL}(A)$.

The following lemma is a partial inverse to Yudin's Theorem 4.3.3.

22.2.1 Lemma. *Let $\Phi[t_1, \dots, t_n]$ be a lattice-linear expression; let $a_1, \dots, a_n$ be distinct elements of A. If $\Phi(\delta_{a_1}, \dots, \delta_{a_n}) = 0$ in L then $\Phi(t_1, \dots, t_n) = 0$ for any $t_1, \dots, t_n \in \mathbb{R}$.*

Proof. For every $x \in \mathbb{R}^A$, we have

$$0 = \big[\Phi(\delta_{a_1}, \dots, \delta_{a_n})\big](x) = \Phi\big(\delta_{a_1}(x), \dots, \delta_{a_n}(x)\big) = \Phi\big(x(a_1), \dots, x(a_n)\big).$$

Since x is arbitrary, the vector $\big(x(a_1), \dots, x(a_n)\big)$ runs over all of $\mathbb{R}^n$. It follows that $\Phi(t_1, \dots, t_n) = 0$ for all $t_1, \dots, t_n \in \mathbb{R}$. $\square$

22.2.2 Corollary. *The set $\{\delta_a : a \in A\}$ is linearly independent in L.*

This justifies the following definition. Vectors $x_1, \dots, x_n$ in a vector lattice X are said to be ***lattice-linearly independent*** if $f(x_1, \dots, x_n) = 0$ implies $f = 0$ for every $f \in L_n$. A subset of X is ***lattice-linearly independent*** if every finite subset of it has this property. Clearly, lattice-linear independence implies linear independence. We know that n vectors are linearly independent iff their linear span is linearly isomorphic to $\mathbb{R}^n$. We claim that vectors $x_1, \dots, x_n$ in a vector lattice are lattice-linear independent iff the sublattice $\mathrm{Lat}(x_1, \dots, x_n)$ they generate is lattice isomorphic to L_n:

22.2.3 Proposition. *Vectors $x_1, \dots, x_n$ in a vector lattice X are lattice-linear independent iff the map $T \colon L_n \to X$ defined by $Tf = f(x_1, \dots, x_n)$ is a lattice isomorphism between L_n and* $\mathrm{Lat}(x_1, \dots, x_n)$.

Proof. It is easy to see that if T is a lattice isomorphism then $x_1, \dots, x_n$ are lattice-linear independent. Suppose now that $x_1, \dots, x_n$ are lattice-linear independent. By Exercise 4.3.6, T is a lattice homomorphism. Since Tf is a lattice-linear combination of x_i's, it is contained in $\mathrm{Lat}(x_1, \dots, x_n)$. On the other hand, since $T\delta_i = x_i$ as $i = 1, \dots, n$, we have $\operatorname{Range} T = \mathrm{Lat}(x_{,} \dots, x_n)$. Finally, lattice-linear independence of $x_1, \dots, x_n$ guarantees that T is one-to-one.

□

Lemma 22.2.1 assets that $\delta_{a_1}, \dots, \delta_{a_n}$ are lattice-linearly independent in L for any distinct $a_1, \dots, a_n$ in A.

22.2.4 Corollary. *Let $a_1, \dots, a_n$ be distinct elements of A. The map $T\colon L_n \to L$ defined by $Tf = f(\delta_{a_1}, \dots, \delta_{a_n})$ is a lattice isomorphism between L_n and $\mathrm{Lat}(\delta_{a_1}, \dots, \delta_{a_n})$.*

22.2.5 Example. Let $n = 3$; take $f \in L_3$ given by $f(t_1, t_2, t_3) = 2t_1 + t_2 \wedge t_3$. Then $f = 2\delta_1 + \delta_2 \wedge \delta_3$ and $Tf = 2\delta_{a_1} + \delta_{a_2} \wedge \delta_{a_3}$. For every function $x\colon A \to \mathbb{R}$, we have $(Tf)(x) = 2x(a_1) + x(a_2) \wedge x(a_3)$.

Let Y be a vector lattice; let $\varphi\colon A \to Y$ be any map. For $f \in L$, we can write f as a lattice-linear combination of some $\delta_{a_1}, \dots, \delta_{a_n}$. That is, $f = \Phi(\delta_{a_1}, \dots, \delta_{a_n})$ for some lattice-linear expression $\Phi[t_1, \dots, t_n]$ and some distinct $a_1, \dots, a_n$ in A. We now define $\hat{\varphi}(f) = \Phi\big(\varphi(a_1), \dots, \varphi(a_n)\big)$.

$$\begin{tikzcd} L \arrow[dr, dotted, "\hat{\varphi}"] & \\ A \arrow[u, hook, "i"] \arrow[r, "\varphi"'] & Y \end{tikzcd}$$

22.2.6 Exercise. Show that $\hat{\varphi}(f)$ is well-defined, i.e., does not depend on the choice Φ and of $a_1, \dots, a_n$. (Hint: use Lemma 22.2.1.)

It follows easily from the definition of $\hat{\varphi}$ that it is a lattice homomorphism on L. Clearly, $\hat{\varphi}$ extends φ in the sense that $\hat{\varphi}(\delta_a) = \varphi(a)$ for every $a \in A$. Since L is generated by δ_a's, every lattice homomorphism from L to Y that extends φ has to agree with $\hat{\varphi}$. This completes the proof that $L = \mathrm{FVL}(A)$.

22.2.7. There is an alternative way to view $\mathrm{FVL}(A)$. Let $n \in \mathbb{N}$, $a_1, \dots, a_n$ distinct points of A, and $\Phi[t_1, \dots, t_n]$ and $\Psi[t_1, \dots, t_n]$ two lattice-linear expressions. Combining Yudin's Theorem 4.3.3 with Lemma 22.2.1, we conclude that Φ and Ψ correspond to the same lattice-linear function in L_n, that is, $\Phi(t_1, \dots, t_n) = \Psi(t_1, \dots, t_n)$ for all $t_1, \dots, t_n \in \mathbb{R}$ iff $\Phi(\delta_{a_1}, \dots, \delta_{a_n}) = \Psi(\delta_{a_1}, \dots, \delta_{a_n})$ in L. Thus, we may identify L with the set of all formal lattice-linear expressions $\Phi[a_1, \dots, a_n]$ of finite sets of elements of

A, where we identify two expressions when they correspond to the same lattice-linear function on $\mathbb{R}^n$. This is somewhat analogous the the fact that the free vector space over A may be constructed as the set of all formal finite linear combinations of elements of A or that the free group over A may be constructed as the set of all formal words where we identify two words if one may be obtained from each the other by cancellations.

22.2.8. We realized $\mathrm{FVL}(A)$ as a sublattice L of the function space $\mathbb{R}^{\mathbb{R}^A}$ generated by $\{\delta_a : a \in A\}$. It is sometimes convenient to represent $\mathrm{FVL}(A)$ inside a "smaller" function space. First, observe that every point evaluation functional δ_a is continuous with respect to the topology of point-wise convergence on $\mathbb{R}^A$ (which is the same as the product topology when we view $\mathbb{R}^A$ as the product of A copies of $\mathbb{R}$). It follows that L is a sublattice of $C(\mathbb{R}^A)$.

Let $\Delta = [-1,1]^A$, viewed as a subset of $\mathbb{R}^A$. By Tikhonov's Theorem, Δ is compact with respect to the product topology. The restriction map $R\colon L \to C(\Delta)$ which maps $f \in L$ to its restriction to Δ is a clearly a lattice homomorphism. The following exercise implies that we may represent $\mathrm{FVL}(A)$ as a sublattice of $C(\Delta)$.

22.2.9 Exercise. The restriction map $R\colon L \to C(\Delta)$ which maps $f \in L$ to its restriction to Δ is a lattice isomorphic embedding.

22.3 * Existence of $\mathrm{FVL}(A)$: universal algebra approach

We observed in 22.2.7 that $\mathrm{FVL}(A)$ may be identified with equivalence classes of formal lattice-linear expressions. This is related to an alternative construction of $\mathrm{FVL}(A)$, based on a technique of *universal algebras*. Here is the idea. Consider the set of all formal lattice-linear expressions of elements of A. Define lattice and linear operations on this set in a natural way. Identify two expressions if their equality may be deduced from the axioms of vector lattices. For example, we identify $(a_1 \vee a_2) + a_3$ and $(a_3 + a_1) \vee (a_3 + a_2)$ because their equality follows from the distributive law and the commutativity of addition. Verify that the lattice and the linear operations are well defined for the equivalence classes of this relation.

An advantage of this approach is its universality: it works not only for vector lattices but also for groups, rings, vector spaces, etc.; for every category that can be axiomatized in a certain way. Its disadvantage is that it does not yield a functional model for it like the one we constructed earlier. We will briefly sketch this approach here; we refer the reader to [Bir67, pp. 143-144] or [Jac89, Ber12] for further details; see also [Ble73].

Ω-*algebras.* We start by fixing a set Ω of formal operations. For example, if one is interested in groups, one would take Ω to consist of one binary operation (which would usually be call a product), one unary operation (inversion), and one 0-nary operation, i.e., a constant, called 1. Since we are interested in vector lattices, we take Ω to consist of two binary operations $+$ and $\vee$, one 0-nary operation 0 (which will stand for the zero vector); also, for every $\lambda \in \mathbb{R}$ we include a unary operation to represent the scalar multiplication by λ: $x \mapsto \lambda x$. By an ***Ω-algebra*** (or a ***universal algebra*** with operations Ω) we mean any set equipped with operations in Ω. At this stage, we impose no restrictions or compatibility assumptions.

Ω-*homomorphisms and congruences.* An ***Ω-homomorphism*** is a function $f\colon X \to Y$ between two Ω-algebras that respects the operations, i.e., $f(x+y) = f(x) + f(x)$, $f(x \vee y) = f(x) \vee f(y)$, $f(0) = 0$, and $f(\lambda x) = \lambda f(x)$ for every $\lambda \in \mathbb{R}$. An equivalence relation "$\cong$" on an Ω-algebra X is said to be a ***congruence*** if it respects the operations, i.e., if $x_1 \cong y_1$ and $x_2 \cong y_2$ then $x_1 + x_2 \cong y_1 + y_2$, $x_1 \vee x_2 \cong y_1 \vee y_2$, and $\lambda x_1 \cong \lambda y_1$ for every $\lambda \in \mathbb{R}$. We view "$\cong$" as a subset of $X \times X$. It may be easily verified that the intersection of any family of congruences is again a congruence. Given any subset $C \subseteq X \times X$, the intersection of all congruences containing C is clearly the least congruence containing C; we call it the congruence generated by C. There is a correspondence between homomorphisms and congruences. Given an Ω-homomorphism $f\colon X \to Y$ between two Ω-algebras, one can define a congruence on X by setting $x \cong y$ whenever $f(x) = f(y)$. We call this congruence the kernel of f. On the other hand, given a congruence "$\cong$" on an Ω-algebra X, the operations of Ω induce operations on the congruence classes, making set of equivalence classes a Ω-algebra. Naturally, we can call it the quotient space of X with respect to $\cong$. The quotient map that sends $x \in X$ to its congruence class $[x]$ is an Ω-homomorphism.

Expressions. Let A be an arbitrary set. We define formal lattice-linear expressions or, just expressions on A. The definition is inductive on the complexity of the expression. We declare every element of A an expression of complexity 1; we also declare the symbol 0 (which is the symbol of our only 0-nary operation) to be an expression of complexity 1. If Φ and Ψ are expressions of complexity n, we declare that the following are expressions of complexity $n+1$: $\Phi + \Psi$, $\Phi \vee \Psi$, and $\lambda\Phi$ for every $\lambda \in \mathbb{R}$. With some extra effort, one can include parentheses in expressions. Alternatively, one may use postfix notation which does not require parentheses: for example, one writes $\Phi\Psi+$ instead of $\Phi + \Psi$ and $a_1 a_2 a_3 \vee + a_4 \vee$ instead of $\bigl(a_1 + (a_2 \vee a_3)\bigr) \vee a_4$. We write $\mathcal{E}(A, \Omega)$ for the set of all expressions. Note that this is indeed a set because

$\mathcal{E}(A,\Omega)=\bigcup_{n=1}^{n}\mathcal{E}_n(A,\Omega)$, where $\mathcal{E}_n(A,\Omega)$ consists of all expressions of complexity n; $\mathcal{E}_n(A,\Omega)$ is a subset of the set of all words in the alphabet $A\cup\mathbb{R}\cup\{+,\vee,0\}$. Clearly, $A\subset\mathcal{E}(A,\Omega)$. For $\Phi,\Psi\in\mathcal{E}(A,\Omega)$, we can define $\Phi+\Psi$, $\Phi\vee\Psi$, and $\lambda\Phi$ in the natural way; thus, $\mathcal{E}(A,\Omega)$ is an Ω-algebra.

Substitutions. Let A be any set and X an Ω-algebra. Given a map $\varphi\colon A\to X$, it extends in a natural way (by induction on complexity) to a map $\tilde{\varphi}\colon\mathcal{E}(A,\Omega)\to X$. For example, if Φ is $2a_1+a_2\vee a_3$ then $\tilde{\varphi}(\Phi)$ is $2\varphi(a_1)+\varphi(a_2)\vee\varphi(a_3)$. That is, $\tilde{\varphi}(\Phi)$ is obtained by substituting elements of A in Φ with their images under φ and then evaluating the resulting expression in X. It is easy to see that $\hat{\varphi}$ is an Ω-homomorphism.

Conversely, let $f\colon\mathcal{E}(A,\Omega)\to X$ be an Ω-homomorphism. Put $\varphi=f_{|A}$. It is easy to see that $f=\tilde{\varphi}$. Thus, $\mathcal{E}(A,\Omega)$ is the free object over A in the category of Ω-algebras: every map from A to any Ω-algebra X extends uniquely to an Ω-homomorhism from $\mathcal{E}(A,\Omega)$ to X.

Axioms. Fix a countable set $T=\{t_1,t_2,\dots\}$; we will view elements of T as "formal variables". The reader may verify that the axioms of a vector lattice may be written in the form

$$\forall t_1,\dots,t_n\quad\Phi(t_1,\dots,t_n)=\Psi(t_1,\dots,t_n),$$

where $\Phi,\Psi\in\mathcal{E}(T,\Omega)$ with n variables. Let $\mathcal{A}$ be the set of all such pairs (Φ,Ψ); we think of $\mathcal{A}$ as the set of axioms of vector lattices.

22.3.1 Exercise. Verify that the axioms of vector lattices may indeed be reformulated in this form. That is, vector lattices are ***equationally definable***.

One can actually define a vector lattice as an Ω-algebra X such that for every (substitution) map $\varphi\colon T\to X$ and every pair (Φ,Ψ) in $\mathcal{A}$ we have $\tilde{\varphi}(\Phi)=\tilde{\varphi}(\Psi)$. That is, after replacing $t_1,t_2,\dots$ in an axiom with arbitrary elements of X we get an equality in X. This exactly means that every vector lattice axiom is satisfied in X.

Construction of FVL(A). First, we define a relation $\sim$ on $\mathcal{E}(A,\Omega)$ as follows: let $(\Phi,\Psi)\in\mathcal{A}$ and $f\colon\mathcal{E}(T,\Omega)\to\mathcal{E}(A,\Omega)$ an Ω-homomorphism; we then declare $f(\Phi)\sim f(\Psi)$. In other words, if there is an axiom of vector lattices of the form $\forall t_1,\dots,t_n\quad\Phi(t_1,\dots,t_n)=\Psi(t_1,\dots,t_n)$ and $\Phi_1,\dots,\Phi_n\in\mathcal{E}(A,\Omega)$ then we declare that

$$\Phi(\Phi_1,\dots,\Phi_n)\sim\Psi(\Phi_1,\dots,\Phi_n).$$

Let "$\cong$" be the congruence on $\mathcal{E}(A,\Omega)$ generated by "$\sim$'. Let L_A be the quotient of $\mathcal{E}(A,\Omega)$ with respect to this congruence.

It follows immediately that L_A is an Ω-algebra. We claim that L_A is a vector lattice. For example, let's verify that addition on L_A is commutative. Pick any $\Phi,\Psi\in\mathcal{E}(A,\Omega)$.

Since there is an axiom that states that $\forall t_1, t_2\ \ t_1+t_2 = t_2+t_1$, we have $\Phi+\Psi \sim \Psi+\Phi$, hence $\Phi + \Psi \cong \Psi + \Phi$, so that $[\Phi + \Psi] = [\Psi + \Phi]$ and, therefore, $[\Phi] + [\Psi] = [\Psi] + [\Phi]$. Other axioms may be verified in a similar fashion.

Finally, we will show that L_A is $\mathrm{FVL}(A)$. Let $\varphi\colon A \to Y$ be a map from A to an arbitrary vector lattice Y. Then φ extends to a unique Ω-homomorphism $\tilde{\varphi}\colon \mathcal{E}(A, \Omega) \to Y$. Suppose that $\Phi \sim \Psi$ in $\mathcal{E}(A, \Omega)$. Then

$$\Phi = \Phi_0(\Phi_1, \dots, \Phi_n) \text{ and } \Psi = \Psi_0(\Phi_1, \dots, \Phi_n)$$

for some pair (Φ_0, Ψ_0) in $\mathcal{A}$ and $n \in \mathbb{N}$. It follows that $\Phi_0(x_1, \dots, x_n) = \Psi_0(x_1, \dots, x_n)$ for any $x_1, \dots, x_n$ in any vector lattice X. Therefore,

$$\tilde{\varphi}(\Phi) = \Phi_0\big(\tilde{\varphi}(\Phi_1), \dots, \tilde{\varphi}(\Phi_n)\big) = \Psi_0\big(\tilde{\varphi}(\Phi_1), \dots, \tilde{\varphi}(\Phi_n)\big) = \tilde{\varphi}(\Psi).$$

It follows that the kernel of $\tilde{\varphi}$ contains "$\sim$" and, therefore, it contains "$\cong$". That is, $\tilde{\varphi}(\Phi) = \tilde{\varphi}(\Psi)$ whenever $\Phi \cong \Psi$. It follows that $\tilde{\varphi}$ induces a map on equivalence classes: $\hat{\varphi}\colon L_A \to Y$. Being an Ω-homomorphism between two vector lattices, it is a lattice homomorphism.

To check uniqueness, let $f\colon L_A \to Y$ be a lattice homomorphism extending φ. Define a map $F\colon \mathcal{E}(A, \Omega) \to Y$ via $F(\Phi) = f\big([\Phi]\big)$. Then F is an Ω-homomorphism which agrees with φ on A. It follows that $F = \tilde{\varphi}$, so that $f = \hat{\varphi}$.

22.4 Properties of $\mathrm{FVL}(A)$

Throughout this section, A is an arbitrary set. If A is a subset of a vector lattice X, we say that A generates X when $X = \mathrm{Lat}(A)$, i.e., the sublattice generated by A is all of X. Recall that in the preceding section, we constructed $\mathrm{FVL}(A)$ as the sublattice of $\mathbb{R}^{\mathbb{R}^A}$ generated by the canonical image of A there, namely, by $\{\delta_a : a \in A\}$.

22.4.1 Exercise. (i) A generates $\mathrm{FVL}(A)$.

(ii) $\mathrm{FVL}(A)$ is Archimedean.

(iii) If $A \subseteq B$ then $\mathrm{FVL}(A)$ is a sublattice of $\mathrm{FVL}(B)$.

22.4.2 Exercise. Let A be a subset of a vector lattice X. TFAE:

(i) X is a free vector lattice over A;

(ii) A is lattice-linear independent and A generates X;

(iii) A generates X and every map φ from A to any vector lattice Y extends to a lattice homomorphism $\hat{\varphi}$ from X to Y.

Note that (ii) suggests that the concept of $\mathrm{FVL}(A)$ is indeed analogous to the concept of a Hamel basis of a vector space, while (iii) essentially says that the uniqueness of extension in the definition of $\mathrm{FVL}(A)$ may be replaced with $X = \mathrm{Lat}(A)$.

Recall that for any distinct $a_1, \dots, a_n$ in A, the corresponding elements $\delta_{a_1}, \dots, \delta_{a_n}$ are lattice-linear independent in $\mathrm{FVL}(A)$ and generate a lattice copy of L_n.

22.4.3 Proposition. *If A is infinite then every finitely generated vector lattice is a quotient of* $\mathrm{FVL}(A)$.

Proof. Let X be a vector lattice generated by a finite subset $\{x_1, \dots, x_n\}$. Pick any n distinct elements of A, say $a_1, \dots, a_n$. Define $\varphi\colon A \to X$ via $\varphi(a_i) = x_i$ if $i \leqslant n$ and $\varphi(a_i) = 0$ otherwise. Then φ extends to a lattice homomorphism $\hat{\varphi}\colon \mathrm{FVL}(A) \to X$. Since its range contains $x_1, \dots, x_n$, $\hat{\varphi}$ is onto. Let $J = \ker\hat{\varphi}$. Then J is an ideal in $\mathrm{FVL}(A)$ and $\mathrm{FVL}(A)/J$ is lattice isomorphic to $\mathrm{Range}\,\hat{\varphi} = X$. $\square$

This proposition suggests that one cannot expect $\mathrm{FVL}(A)$ to be a "nice" space.

Let's look more carefully at the structure of $\mathrm{FVL}(n)$ when $n \in \mathbb{N}$. Recall that $\mathrm{FVL}(n) = L_n$. Clearly, L_n is a sublattice of $C(\mathbb{R}^n)$. We observed in Section 4.4 that the map $R\colon L_n \to C(S^n_\infty)$ that sends $f \in L_n$ to its restriction onto the unit sphere of ℓ^n_∞ is a one-to-one lattice homomorphism. It allows us to identify L_n with a sublattice of $C(S^n_\infty)$.

22.4.4 Exercise. L_2 is lattice isomorphic to the set of all piece-wise affine functions on S^2_∞.

22.4.5 Exercise. $\mathrm{FVL}(1) = \mathbb{R}^2$.

Recall that δ_i stands for the i-th coordinate functional in L_n. Let $e \in L_n$ defined by $e = |\delta_1| + \dots + |\delta_n|$. It is easy to see that e is identically one on S^n_∞. This yields the following.

22.4.6 Proposition. *The element $|\delta_1| + \dots + |\delta_n|$ is a strong unit in* $\mathrm{FVL}(n)$.

Let now A be arbitrary. We observed in the previous section that every f in $\mathrm{FVL}(A)$ may be written as a lattice-linear combination of some finite set of δ_a's, hence f is in the sublattice $\mathrm{Lat}(\delta_{a_1}, \dots, \delta_{a_n})$ generated by some distinct $a_1, \dots, a_n$ in A. We may identify this sublattice with $\mathrm{FVL}\bigl(\{a_1, \dots, a_n\}\bigr)$. We also observed that this sublattice may be identified with L_n so that δ_{a_i} corresponds to δ_i. Now Proposition 22.4.6 yields the following.

22.4.7 Lemma. *For every $f \in \mathrm{FVL}(A)$ there exists a finite subset $\{a_1, \dots, a_n\}$ of A and $\lambda \in \mathbb{R}_+$ such that $|f| \leqslant \lambda\bigl(|\delta_{a_1}| + \dots + |\delta_{a_n}|\bigr)$.*

22.4.8 Proposition. *For every $a \in A$, δ_a is a weak unit in* $\mathrm{FVL}(A)$.

Proof. Suppose that $f \perp \delta_a$ in $\mathrm{FVL}(A)$; we need to show that $f = 0$. Find a finite subset B of A such that $f \in \mathrm{FVL}(B)$. WLOG, $a \in B$. Thus, WLOG, A is finite, $\mathrm{FVL}(A) = L_n$, and $\delta_a = \delta_1$. Now $f \perp \delta_1$ yields that $\operatorname{supp} f$ is contained in $\{\delta_1 = 0\}$. Since the latter set is a proper subspace of $\mathbb{R}^n$ and f is continuous, this yields $f = 0$. $\square$

22.4.9 Exercise. If A is infinite, $\mathrm{FVL}(A)$ has no strong units.

22.4.10 Proposition. *Real-valued lattice homomorphisms on* $\mathrm{FVL}(A)$ *are precisely the maps of the form $f \mapsto f(x)$ for some $x \in \mathbb{R}^A$.*

Proof. Maps of this form are lattice-homomorphisms because lattice operations on $\mathrm{FVL}(A)$ are pointwise, so that $|f|(x) = |f(x)|$ for every $f \in \mathrm{FVL}(A)$ and $x \in \mathbb{R}^A$. The converse follows from the definition of $\mathrm{FVL}(A)$: every lattice homomorphism on $\mathrm{FVL}(A)$ is the unique extension of its restriction to A. $\square$

We have observed earlier that if $B \subseteq A$ then $\mathrm{FVL}(B)$ may be viewed as a sublattice of $\mathrm{FVL}(A)$.

22.4.11 Example. If $B \subsetneq A$ then $\mathrm{FVL}(B)$ is not an ideal in $\mathrm{FVL}(A)$. Indeed, if $a \in A \setminus B$ and $b \in B$ then $|\delta_a| \wedge |\delta_b| \leqslant |\delta_b|$ but $|\delta_a| \wedge |\delta_b| \notin \mathrm{FVL}(B)$. In the language of lattice-linear functions, the function $|t_1| \wedge |t_2|$ is dominated by $|t_1|$ in L_2, but is not contained in the sublattice generated by t_1 in L_2.

Let $B \subseteq A$. Let $\varphi\colon A \to \mathrm{FVL}(A)$ be defined by $\varphi(a) = a$ (or, rather, δ_a) when $a \in B$ and $\varphi(a) = 0$ otherwise. Then φ extends to a lattice homomorphism $P\colon \mathrm{FVL}(A) \to \mathrm{FVL}(A)$. If $f \in \mathrm{FVL}(A)$, $f = \Phi(\delta_{a_1}, \dots, \delta_{a_n})$ for some lattice-linear expression Φ and some $a_1, \dots, a_n \in A$, then Pf is obtained by replacing all the δ_{a_i}'s in this expression for which $a_i \notin B$ with zeros. It is easy to see that P is a projection with $\operatorname{Range} P = \mathrm{FVL}(B)$. This yields the following.

22.4.12 Proposition. *Let $B \subseteq A$. The lattice homomorphism $P\colon \mathrm{FVL}(A) \to \mathrm{FVL}(A)$ defined via $P\delta_a = \delta_a$ when $a \in B$ and $P\delta_a = 0$ otherwise is a projection onto* $\mathrm{FVL}(B)$.

Equip $\mathbb{R}^A$ with the topology of pointwise convergence. This is exactly the topology that makes every evaluation functional δ_a continuous. Since every $f \in \mathrm{FVL}(A)$ is a lattice-linear combination of evaluation functionals, f is continuous with respect to this topology. It follows that its support $\operatorname{supp} f = \bigl\{x \in \mathbb{R}^A \;:\; f(x) \neq 0\bigr\}$ is an open set. For a subset $S \subseteq \mathrm{FVL}(A)$, put $\operatorname{supp} S = \bigcup_{f \in S} \operatorname{supp} f$. Being a union of open sets, $\operatorname{supp} S$ is open.

22.4.13 Proposition. *Suppose that* $\operatorname{card}(A) > 1$. *Then* $\mathrm{FVL}(A)$ *contains no proper non-trivial projection bands.*

(By Propositon 5.2.7, this means that one cannot decompose $\mathrm{FVL}(A)$ into a direct sum of two disjoint non-trivial ideals.)

Proof. Suppose that $\mathrm{FVL}(A) = J_1 \oplus J_2$, where J_1 and J_2 are two disjoint non-trivial ideals.

Claim: $\operatorname{supp} J_1 \cup \operatorname{supp} J_2 = \mathbb{R}^A \setminus \{0\}$. Indeed, let $0 \neq x \in \mathbb{R}^A$. Then there exists $a \in A$ such that $0 \neq |x(a)| = \bigl|\delta_a(x)\bigr| = |\delta_a|(x)$. Find positive $f \in J_1$ and $g \in J_2$ such that $|\delta_a| = f + g$ (see Exercise 3.3.14(iv)). Then $f(x) > 0$ or $g(x) > 0$, hence $x \in \operatorname{supp} J_1$ or $x \in \operatorname{supp} J_2$.

Claim: $\operatorname{supp} J_1 \cap \operatorname{supp} J_2 = \varnothing$. Indeed, let $x \in \operatorname{supp} J_1 \cap \operatorname{supp} J_2$. Then there exist positive $f \in J_1$ and $g \in J_2$ with $f(x) > 0$ and $g(x) > 0$. Then $(f \wedge g)(x) = f(x) \wedge g(x) > 0$, hence $f \wedge g \neq 0$. But $f \wedge g \in J_1 \cap J_2$; a contradiction.

The two claims together yield a disconnection of $\mathbb{R}^A \setminus \{0\}$, i.e., a partitioning of $\mathbb{R}^A \setminus \{0\}$ into two disjoint open sets. This contradict the fact that the set $\mathbb{R}^A \setminus \{0\}$ is connected. $\square$

Since the principal ideal generated by an atom is a projection band by Lemma 5.4.9, we have the following.

22.4.14 Corollary. *If* $\operatorname{card}(A) > 1$ *then* $\mathrm{FVL}(A)$ *contains no atoms.*

22.4.15 Corollary. *If* $\operatorname{card}(A) > 1$ *then* $\mathrm{FVL}(A)$ *fails PPP. In particular,* $\mathrm{FVL}(A)$ *is not σ-order complete.*

Proof. Suppose X has PPP. Since $\dim \mathrm{FVL}(A) > 1$, it contains a pair of non-zero disjoint vectors, say, x and y (see 1.5.2). Then B_x is a projection band. It is non-trivial because $x \in B_x$ and it is proper because $y \perp B_x$. This contradicts Proposition 22.4.13.

Hence, $\mathrm{FVL}(A)$ fails PPP. It follows from Corollary 5.2.23 that X is not σ-order continuous. $\square$

22.4.16 Theorem. *A disjoint collection in* $\mathrm{FVL}(A)$ *is at most countable.*

Proof. Consider a disjoint collection of functions in $\mathrm{FVL}(A)$. Their supports form a disjoint collection of open subsets of $\mathbb{R}^A$ (again, we consider the topology of pointwise convergence on $\mathbb{R}^A$.) The result now follows from the following fact of topology. $\square$

22.4.17 Theorem (Šanin)**.** *A disjoint collection of non-empty open sets in* $\mathbb{R}^A$ *is at most countable.*

Proof. Let $\mathcal{G}$ be an uncountable disjoint collection of non-empty open subsets of $\mathbb{R}^A$.

Since the topology of pointwise convergence on $\mathbb{R}^A$ is exactly the product topology, it has a base consisting of open sets of the form $U = \prod_{a\in A} O_a$ where each O_a is an open subset of $\mathbb{R}$ and only finitely many of them are different from $\mathbb{R}$. Note that $O_a = \delta_a(U)$.

Since every set in $\mathcal{G}$ contains a base open set, we may assume WLOG that $\mathcal{G}$ consists of base open sets. For every $U \in \mathcal{G}$, U is of the form $U = \prod_{a\in A} O_a$; let $\alpha(U) = \{a \in A : O_a \neq \mathbb{R}\}$. Then α is a function from $\mathcal{G}$ to finite subsets of A. For any distinct $U, V \in \mathcal{G}$, it follows from $U \cap V = \varnothing$ that $\alpha(U)\cap\alpha(V)$ is non-empty. Since $\mathcal{G}$ is uncountable, there exists $m \in \mathbb{N}$ such that $\bigl\{U \in \mathcal{G} : \operatorname{card}\bigl(\alpha(U)\bigr) = m\bigr\}$ is uncountable. WLOG, $\operatorname{card}\bigl(\alpha(U)\bigr) = m$ for all $U \in \mathcal{G}$.

We will now prove the result by induction on m. If $m = 1$ then there exists $a \in A$ such that $\alpha(U) = \{a\}$ for all $U \in \mathcal{G}$. Then $\{\delta_a(U) : U \in \mathcal{G}\}$ is an uncountable disjoint family of open subsets of $\mathbb{R}$. This is impossible because each of these sets contains a rational number.

Suppose now that $m > 1$. Fix any $V \in \mathcal{G}$, let $\alpha(V) = \{a_1, \dots, a_m\}$. Since $\alpha(U)$ meets $\alpha(V)$ for all $U \in \mathcal{G}$, some of the a_i's (say, a_1) is contained in $\alpha(U)$ for uncountably many $U \in \mathcal{G}$. Then $\{\delta_{a_1}(U) : U \in \mathcal{G},\ a_1 \in \alpha(U)\}$ is an uncountable collection of open subsets of $\mathbb{R}$. Since each of them contains a rational number, there is a rational number that is contained in uncountably many members of this collection. Let $\mathcal{G}'$ be the subcollection of $\mathcal{G}$ formed by these sets. Let $T\colon \mathbb{R}^A \to \mathbb{R}^{A\setminus a_0}$ be the restriction map. Then $\{T(U) : U \in \mathcal{G}'\}$ is an uncountable disjoint collection of base open subsets of $\mathbb{R}^{A\setminus a_0}$ with $\alpha\bigl(T(U)\bigr) = m - 1$ for every $U \in \mathcal{G}'$. Now apply the induction hypothesis. □

22.5 Free Banach lattice over a set: FBL(A)

The concept of a free Banach lattice over a set was introduced and investigated in [dPW15]. Let A be a subset of a Banach lattice X. We say that X is a free Banach lattice over A if every bounded map φ from A to an arbitrary Banach lattice Y extends uniquely to a lattice homomorphism $\hat{\varphi}\colon X \to Y$ with $\|\hat{\varphi}\| = \sup_{a\in A}\bigl\|\varphi(a)\bigr\|$. By φ being bounded here we mean that $\sup_{a\in A}\bigl\|\varphi(a)\bigr\|$ is finite.

22.5.1 Exercise. Let X be a free Banach lattice over a subset A. Then $A \subseteq S_X$.

As before, instead of assuming that A is a subset of X, one may assume that there is an injective map from A to X (or, rather, to S_X). Similarly to Exercise 22.1.1, X is uniquely defined by A:

22.5.2 Exercise. Let X and Y be two Banach lattices which are both free over a common subset A. Then there is a lattice isometry between X and Y that fixes A.

We will write FBL(A) for the free Banach lattice over A. The following theorem proves the existence of FBL(A). As before, we identify FVL(A) with the sublattice of $\mathbb{R}^{\mathbb{R}^A}$ generated by $\{\delta_a : a \in A\}$; by identifying $a \in A$ with $\delta_a \in \mathrm{FVL}(A)$, we may view A as a subset of FVL(A).

22.5.3 Theorem. *There exists the greatest lattice seminorm* ν *on* $\mathrm{FVL}(A)$ *with* $\nu(a) \leqslant 1$ *for all* $a \in A$. *It is a lattice norm, and the completion of* $\mathrm{FVL}(A)$ *with respect to it is* $\mathrm{FBL}(A)$.

Proof. Let $\mathcal{N}$ be the set of all lattice seminorms ν on $\mathrm{FVL}(A)$ such that $\nu(\delta_a) \leqslant 1$ for every $a \in A$.

Let $x \in \mathbb{R}^A$ such that $|x(a)| \leqslant 1$ for all $a \in A$. For $f \in \mathrm{FVL}(A)$, put $\nu_x(f) = \bigl|f(x)\bigr|$. It can be easily verified that $\nu_x \in \mathcal{N}$.

For $f \in \mathrm{FVL}(A)$, put $\|f\| = \sup\limits_{\nu \in \mathcal{N}} \nu(f)$. We claim that this is a lattice norm on $\mathrm{FVL}(A)$.

First, observe that $\|f\|$ is finite. By Lemma 22.4.7, we can find $a_1, \dots, a_n \in A$ and $\lambda \in \mathbb{R}_+$ such that

$$|f| \leqslant \lambda\bigl(|\delta_{a_1}| + \dots + |\delta_{a_n}|\bigr).$$

It follows that $\nu(f) \leqslant \lambda n$ for every $\nu \in \mathcal{N}$, hence $\|f\| \leqslant \lambda n < \infty$.

It is straightforward that $\|\cdot\|$ is positively homogeneous. To verify the triangle inequality, let $f, g \in \mathrm{FVL}(A)$ and fix $\varepsilon > 0$. There exists $\nu \in \mathcal{N}$ such that

$$\|f + g\| - \varepsilon < \nu(f + g) \leqslant \nu(f) + \nu(g) \leqslant \|f\| + \|g\|.$$

It follows that $\|f + g\| \leqslant \|f\| + \|g\|$.

Suppose that $f \neq 0$. Find $a_1, \dots, a_n \in A$ such that f is a lattice-linear combination of $\delta_{a_1}, \dots, \delta_{a_n}$, that is, $f \in \mathrm{FVL}\bigl(\{a_1, \dots, a_n\}\bigr)$. It follows that there is $x \in \mathbb{R}^A$ such that $f(x) \neq 0$ and $\operatorname{supp} x \subseteq \{a_1, \dots, a_n\}$. Without loss of generality, scaling x if necessary, $|x(a)| \leqslant 1$ for all $a \in A$. Then $\nu_x(f) = \bigl|f(x)\bigr| > 0$, hence $\|f\| \neq 0$.

If $|f| \leqslant |g|$ in $\mathrm{FVL}(A)$ then $\nu(f) \leqslant \nu(g)$ for every $\nu \in \mathcal{N}$, hence $\|f\| \leqslant \|g\|$. Thus, $\|\cdot\|$ is a lattice norm on $\mathrm{FVL}(A)$.

Let X be the completion of $\bigl(\mathrm{FVL}(A), \|\cdot\|\bigr)$. We claim that X is a free Banach lattice over A.

Let $\varphi\colon A \to Y$, where Y is a Banach lattice and $\sup_{a \in A}\|\varphi(a)\| \leqslant 1$. Then φ extends to a lattice homomorphism $\hat{\varphi}\colon \mathrm{FVL}(A) \to Y$. For $f \in \mathrm{FVL}(A)$, put $\nu(f) = \bigl\|\hat{\varphi}(f)\bigr\|$. It is easy to see that $\nu \in \mathcal{N}$, hence $\bigl\|\hat{\varphi}(f)\bigr\| = \nu(f) \leqslant \|f\|$. It follows that $\|\hat{\varphi}\| \leqslant 1$ and, therefore, $\hat{\varphi}$ extends to a lattice homomorphism $\tilde{\varphi}$ from X to Y with $\|\tilde{\varphi}\| \leqslant 1$.

Uniqueness of the extension follows from the fact that any lattice homomorphism extension of φ to X has to agree with $\hat{\varphi}$ on $\mathrm{FVL}(A)$, hence with $\tilde{\varphi}$ on X as $\mathrm{FVL}(A)$ is dense in X. $\square$

It follows that the FBL norm is the greatest norm on $\mathrm{FVL}(A)$ such that $\|a\| \leqslant 1$ for every $a \in A$. When there is a risk confusion, we will denote this norm by

$\|\cdot\|_{\mathrm{FBL}}$. We will see that for many applications, it suffices to consider the normed lattice $\big(FVL(A), \|\cdot\|_{\mathrm{FBL}}\big)$ instead of the full $\mathrm{FBL}(A)$.

22.5.4 Exercise. On $\mathrm{FVL}(A)$, the norm $\|\cdot\|_{\mathrm{FBL}}$ agrees with the Minskowski functional of $\mathrm{conv}\big(\mathrm{Sol}(A)\big)$.

22.5.5 Exercise. For any distinct $a_1, \dots, a_n \in A$ and scalars $\lambda_1, \dots, \lambda_n$,

(i) $\big\|\sum_{i=1}^n \lambda_i a_i\big\|_{\mathrm{FBL}} = \sum_{i=1}^n |\lambda_i|$;
(ii) $\big\||a_1| \vee \dots \vee |a_n|\big\|_{\mathrm{FBL}} = \big\||a_1| + \dots + |a_n|\big\|_{\mathrm{FBL}} = n$.
(iii) $\big\||a_1| \wedge \dots \wedge |a_n|\big\|_{\mathrm{FBL}} = 1$.

It follows, in particular, that the closed subspace generated by A in $\mathrm{FBL}(A)$ is isometric to $\ell_1(A)$. This is not surprising given that $\ell_1(A)$ is the free Banach space generated by A. One may think of it as an intermediate step on the way from A to $\mathrm{FBL}(A)$. We will see, however, that the FBL-norm is very far from being an AL-norm.

We observed in Proposition 22.4.12 that if $B \subseteq A$ then $\mathrm{FVL}(B)$ is a sublattice of $\mathrm{FVL}(A)$ and there is a natural projection $P_B\colon \mathrm{FVL}(A) \to \mathrm{FVL}(A)$ such that P_B is a lattice homomorphism and $\mathrm{Range}\, P_B = \mathrm{FVL}(B)$. It follows that for $b \in B$ the notation δ_b in unambiguous; it means the same thing in $\mathrm{FVL}(B)$, $\mathrm{FVL}(A)$, $\mathrm{FBL}(B)$, and $\mathrm{FBL}(A)$.

22.5.6 Proposition. *If $B \subseteq A$ then the inclusion map extends to a lattice isometry between $\mathrm{FBL}(B)$ and the closed sublattice of $\mathrm{FBL}(A)$ generated by $\{\delta_b : b \in B\}$. Furthermore, $P_B\colon \mathrm{FVL}(A) \to \mathrm{FVL}(A)$ extends to a lattice homomorphic projection $\widetilde{P}_B\colon \mathrm{FBL}(A) \to \mathrm{FBL}(A)$ with $\mathrm{Range}\, \widetilde{P}_B = \mathrm{FBL}(B)$ and $\|\widetilde{P}_B\| = 1$.*

Proof. Recall that generators of a free Banach lattice have norm one. The map $b \in B \mapsto \delta_b \in \mathrm{FBL}(A)$ extends to a lattice homomorphism $S\colon \mathrm{FBL}(B) \to \mathrm{FBL}(A)$ with $\|S\| = 1$. Similarly, the map $\varphi\colon A \to \mathrm{FBL}(B)$ defined by $\varphi(b) = \delta_b$ if $b \in B$ and $\varphi(a) = 0$ when $a \notin B$ extends to a lattice homomorphism $T\colon \mathrm{FBL}(A) \to \mathrm{FBL}(B)$ with $\|T\| = 1$.

$$\begin{tikzcd} \mathrm{FBL}(B) \arrow[dr, dotted, "S"] & \\ B \arrow[u, hook] \arrow[r, "b\mapsto\delta_b"'] & \mathrm{FBL}(A) \end{tikzcd} \qquad \begin{tikzcd} \mathrm{FBL}(A) \arrow[dr, dotted, "T"] & \\ A \arrow[u, hook] \arrow[r, "\varphi"'] & \mathrm{FBL}(B) \end{tikzcd}$$

Note that $TS\colon \mathrm{FBL}(B) \to \mathrm{FBL}(B)$ is a lattice homomorphism such that $TS\delta_b = \delta_b$ for every $b \in B$; it follows that TS is the identity operator on $\mathrm{FBL}(B)$, i.e., $TS = I_{\mathrm{FBL}(B)}$. It is now straightforward that S is one-to-one, T is onto, and $(ST)^2 = ST$, so that

$ST\colon \mathrm{FBL}(A)\to\mathrm{FBL}(A)$ is a projection. Put $\widetilde{P}_B = ST$. For every $f\in\mathrm{FBL}(B)$, we have $\|f\|_{\mathrm{FBL}(B)} = \|TSf\|_{\mathrm{FBL}(B)} \leqslant \|Sf\|_{\mathrm{FBL}(A)} \leqslant \|f\|_{\mathrm{FBL}(B)}$, so that S is an isometry. It follows that $\operatorname{Range}\widetilde{P}_B$ is a lattice isometric copy of $\mathrm{FBL}(B)$ in $\mathrm{FBL}(A)$. Finally, it follows from $\widetilde{P}_B\delta_b=\delta_b$ when $b\in B$ and $\widetilde{P}_B\delta_a=0$ when $a\in A\setminus B$ that the restriction of $\widetilde{P}_B$ to $\mathrm{FVL}(A)$ agrees with P_B, and that $\operatorname{Range}\widetilde{P}_B$ is the closed sublattice of $\mathrm{FBL}(A)$ generated by $\{\delta_b : b\in B\}$. $\square$

Is it possible to realize $\mathrm{FBL}(A)$ as a "nice" function space? Recall that we constructed $\mathrm{FVL}(A)$ as the sublattice L of $\mathbb{R}^{\mathbb{R}^A}$ generated by $\{\delta_a : a\in A\}$. We observed in 22.2.8 that L may be identified with a sublattice of $C(\Delta)$, where $\Delta=[-1,1]^A$ equipped with the product topology, through the restriction operator $R\colon L\to C(\Delta)$, where $Rf=f_{|\Delta}$ is a lattice isomorphic embedding. Define $\|f\|_\infty=\|Rf\|_{C(\Delta)}=\sup_{x\in\Delta}|f(x)|$. Since $C(\Delta)$ is a Banach lattice, this is a lattice norm on L. Since $|\delta_a(x)|=\big|x(a)\big|\leqslant 1$ for all $x\in\Delta$, we have $\|\delta_a\|_\infty\leqslant 1$. Since the FBL norm is the greatest norm on FVL with this property, we conclude that $\|f\|_\infty\leqslant\|f\|_{\mathrm{FBL}}$ for every $f\in L$.

Consider the special case when A is finite; we then write $\mathrm{FBL}(n)$ instead of $\mathrm{FBL}(A)$. In this case, $\Delta=B^n_\infty$; we identify $\mathrm{FVL}(n)$ with a sublattice of $C(B^n_\infty)$ or of $C(S^n_\infty)$ via the restriction map. For $f\in\mathrm{FVL}(n)$ we have $\|f\|_\infty\leqslant\|f\|_{\mathrm{FBL}}$ as before. On the other hand, $e=|\delta_1|+\dots+|\delta_n|$ is a strong unit in this sublattice; viewed as an element of $C(S^n_\infty)$, $e=\mathbb{1}$. Hence, for $f\in\mathrm{FBL}(n)$ we have $|f|\leqslant\|f\|_\infty e$ and, therefore, $\|f\|_{\mathrm{FBL}}\leqslant\|f\|_\infty\|e\|_{\mathrm{FBL}}=n\|f\|_\infty$. It follows that the FBL norm is equivalent to the $\|\cdot\|_\infty$ on $\mathrm{FVL}(n)$, hence $\mathrm{FBL}(n)$ agrees with the closure of $\mathrm{FVL}(n)$ in $C(B^n_\infty)$ or in $C(S^n_\infty)$ in $\|\cdot\|_\infty$. Thus, by the results of Section 4.4, $\mathrm{FBL}(n)$ is nothing but the space H_n of all continuous positively homogeneous functions of $\mathbb{R}^n$. It also follows that $\mathrm{FBL}(n)$ is lattice isomorphic to an AM-space. Note that even though the norms $\|\cdot\|_{\mathrm{FBL}}$ and $\|\cdot\|_\infty$ are equivalent, they are different, and the constant of equivalence tends to infinity as $n\to\infty$. We summarize this as follows:

22.5.7 Proposition. $\mathrm{FBL}(n)$ *equals the space* H_n *of all continuous positively homogeneous functions* $\mathbb{R}^n\to\mathbb{R}$, *with* $\|f_{|B^n_\infty}\|_\infty\leqslant\|f\|_{\mathrm{FBL}}\leqslant n\|f_{|B^n_\infty}\|_\infty$.

For an general set A, we are now going to show that $\mathrm{FBL}(A)$ may be represented inside a $C(K)$ space. As before, let $\Delta=[-1,1]^A$ equipped with the product topology and let $R\colon \mathrm{FVL}(A)\to C(\Delta)$ be the restriction map $Rf=f_{|\Delta}$.

22.5.8 Theorem (de Pagtor, Wickstead)**.** *R extends to a lattice homomorphic injection* $\widetilde{R}\colon \mathrm{FBL}(A)\hookrightarrow C(\Delta)$ *with* $\|\widetilde{R}\|\leqslant 1$.

Proof. It was observed in 22.2.8 that $R\colon \mathrm{FVL}(A)\to C(\Delta)$ is a lattice isomorphic embedding. We observed earlier that $\|Rf\|_{C(\Delta)}\leqslant\|f\|$ for every $f\in\mathrm{FVL}(A)$, where $\|f\|$ stands for $\|f\|_{\mathrm{FBL}}$. It follows that R extends to a lattice homomorphism $\widetilde{R}\colon \mathrm{FBL}(A)\to C(\Delta)$ with $\|\widetilde{R}\|\leqslant 1$. It is left to verify that $\widetilde{R}$ is one-to-one. We need two auxiliary claims.

Claim 1: every lattice homomorphism $\omega\colon \mathrm{FBL}(A)\to\mathbb{R}$ *vanishes on* $\ker\widetilde{R}$. WLOG, $\|\omega\|=1$. As in Proposition 22.4.10, let x be the restriction of ω to A, then by the definition of $\mathrm{FVL}(A)$ we have $\omega(f)=f(x)$ for all $f\in\mathrm{FVL}(A)$. Note also that $|x(a)|=|\omega(\delta_a)|\leqslant\|\omega\|\|\delta_a\|\leqslant 1$, so that $x\in\Delta$. Now let $h\in\ker\widetilde{R}$. It follows from $h\in\mathrm{FBL}(A)$ that there is a sequence (f_n) in $\mathrm{FVL}(A)$ such that $\|f_n-h\|\to 0$. It follows that $Rf_n\to\widetilde{R}h=0$ in $C(\Delta)$, hence $\omega(f_n)=f_n(x)\to 0$. It follows that $\omega(h)=0$.

Claim 2: for every finite subset B of A, the canonical projection $\widetilde{P}_B$ vanishes on $\ker\widetilde{R}$. By Proposition 22.5.7 and the preceding discussion, we may identify $\mathrm{FBL}(B)$ with a sublattice of $C\big([-1,1]^B\big)$. Fix $x\in[-1,1]^B$. The map that sends $f\in\mathrm{FBL}(A)$ to $(\widetilde{P}_Bf)(x)$ is clearly a lattice homomorphism. By Claim 1, it is zero on $\ker\widetilde{R}$. Let $h\in\ker\widetilde{R}$, then $(\widetilde{P}_Bh)(x)=0$ for every $x\in[-1,1]^B$, so that $\widetilde{P}_Bh=0$.

We are now ready to show that $\widetilde{R}$ is one-to-one. Let $h\in\ker\widetilde{R}$. Fix $\varepsilon>0$. Find $f\in\mathrm{FVL}(A)$ with $\|f-h\|<\varepsilon$. Find a finite subset B of A such that $f\in\mathrm{FVL}(B)$. It follows from $\widetilde{P}_Bf=P_Bf=f$ and $\widetilde{P}_Bh=0$ that

$$\|f\|=\big\|\widetilde{P}_Bf-\widetilde{P}_Bh\big\|\leqslant\|\widetilde{P}_B\|\|f-h\|<\varepsilon.$$

This yields $\|h\|\leqslant\|f\|+\|f-h\|<2\varepsilon$. It follows that $h=0$. □

We observe next that vector lattice properties of $\mathrm{FBL}(A)$ are as bad (or as rich?) as those of $\mathrm{FVL}(A)$:

22.5.9 Proposition. *Suppose that $|A|>1$. Then* $\mathrm{FBL}(A)$ *has no projection bands other than* $\{0\}$ *and the entire space.* $\mathrm{FBL}(A)$ *fails PPP; it is not Dedekind complete; it has no atoms, and every disjoint family in it is at most countable.*

Proof. Let $X=\mathrm{FBL}(A)$. By the preceding theorem, we may view X as a sublattice of $C(\Delta)$. First, show that X has no projection bands other than $\{0\}$ and X. By Exercise 5.2.6, it suffices to show that $\operatorname{supp}X$ is a connected subset of Δ. For every non-zero $x\in\Delta$, there exists $a\in A$ such that $0\neq x(a)=\delta_a(x)$. It follows that $x\in\operatorname{supp}\delta_a$ and, therefore, in $\operatorname{supp}X$. Hence, $\operatorname{supp}X=\Delta\setminus\{0\}$, which is a connected set.

As is in Section 22.4, we conclude that X fails PPP, is not Dedekind complete, and has no atoms. The proof that every disjoint family in it is at most countable is similar to that of Theorem 22.4.16, except that now we consider supports in Δ. □

22.6 Free vector lattice over a vector space: $\mathrm{FVL}[E]$

Given a vector space E, we now want to find the "largest" vector lattice X such that E is a subspace of X and E generates X in the sense that the sublattice generated by E in X is all of X. Here is a formal definition. Let E be a subspace of a vector lattice X; we say that X is a free vector lattice over a vector space E if every linear map T from E to an arbitrary vector lattice Y extends uniquely to a lattice homomorphism $\widehat{T}\colon X \to Y$. As in Exercise 22.1.1, it is easy to see that X is uniquely determined by E in the sense that if E is also a subspace of a vector lattice Y and Y is also free over E than there is a lattice isomorphism between X and Y that fixes E. So we will write $X = \mathrm{FVL}[E]$; we use square brackets to distinguish $\mathrm{FVL}[E]$ from $\mathrm{FVL}(E)$. As in the previous sections, it is clear that instead of assuming that E is a subspace of X one may assume that E is a vector space which is linearly isomorphic to a subspace of X.

$$\begin{tikzcd}
\mathrm{FVL}[E] \arrow[dr, dotted, "\widehat{T}"] & \\
E \arrow[u, hook, "i"] \arrow[r, "T"'] & Y
\end{tikzcd}$$

We will now present an explicit construction of $\mathrm{FVL}[E]$. Let E be a vector space; let $E^{\#}$ stand for the linear dual of E, that is, the vector space of all linear functionals from E to $\mathbb{R}$. The space $\mathbb{R}^{E^{\#}}$ of all functions $f\colon E^{\#} \to \mathbb{R}$ is a vector lattice under pointwise order and lattice operations. For every $x \in E$, we define the evaluation functional $\hat{x}\colon E^{\#} \to \mathbb{R}$ in the usual way: $\hat{x}(x^*) = x^*(x)$. We may view $\hat{x}$ as an element of $\mathbb{R}^{E^{\#}}$. Let L be the sublattice of $\mathbb{R}^{E^{\#}}$ generated by $\{\hat{x} : x \in E\}$. Since linear functionals separate points of E, the map $i\colon E \to L$ defined by $i(x) = \hat{x}$ is a linear injection, so we may identify E with a subspace of L. We claim that $L = \mathrm{FVL}[E]$.

Let $T\colon E \to Y$ be a linear map from E to a vector lattice Y. Take any $f \in L$. Then f is a lattice-linear combination of $\hat{x}_1, \dots, \hat{x}_n$ for some $x_1, \dots, x_n \in E$. That is, there exists a formal lattice-linear expression $\Phi[t_1, \dots, t_n]$ such that $f = \Phi(\hat{x}_1, \dots, \hat{x}_n)$, where Φ is evaluated in L. We define $\widehat{T}f$ to be the same lattice-linear expression of $Tx_1, \dots, Tx_n$, i.e., we put $\widehat{T}f = \Phi(Tx_1, \dots, Tx_n)$, where the right hand side is evaluated in Y.

22.6.1 Exercise. Verify that $\widehat{T}f$ is well-defined.

It can be easily verifies that $\widehat{T}$ is a lattice homomorphism. Clearly, $\widehat{T}\hat{x} = Tx$. This completes the proof that $L = \mathrm{FVL}[E]$.

22.6.2 Exercise. The sublattice generated by E in $\mathrm{FVL}[E]$ is all of $\mathrm{FVL}[E]$.

22.6.3 Exercise. Let $x_1,\dots,x_n$ be linearly independent vectors in a vector lattice E. Then their images $\hat{x}_1,\dots,\hat{x}_n$ in $\mathrm{FVL}[E]$ are lattice-linearly independent.

Combining this with Propositions 22.2.3 and 22.4.6, we get the following.

22.6.4 Corollary. *Let $x_1,\dots,x_n$ be linearly independent vectors in E. Then the sublattice generated by $\hat{x}_1,\dots,\hat{x}_n$ in $\mathrm{FVL}[E]$ is lattice isomorphic to L_n, and $\lvert\hat{x}_1\rvert+\dots+\lvert\hat{x}_n\rvert$ is a strong unit in it.*

22.6.5 Corollary. *For every $f\in\mathrm{FVL}[E]$ there exist $x_1\dots,x_n\in E$ and $\lambda>0$ such that $\lvert f\rvert\leqslant\lambda\bigl(\lvert\hat{x}_1\rvert+\dots+\lvert\hat{x}_n\rvert\bigr)$. If E is a normed space, one may chose $x_1\dots,x_n\in B_E$.*

Proof. Every $f\in\mathrm{FVL}[E]$ may be written as a lattice-linear combination of some generators, say, $\hat{x}_1,\dots,\hat{x}_n$. Iterating the inequalities $\lvert x\pm y\rvert\leqslant\lvert x\rvert+\lvert y\rvert$, $\lvert x\wedge y\rvert\leqslant\lvert x\rvert+\lvert y\rvert$, and $\lvert x\vee y\rvert\leqslant\lvert x\rvert+\lvert y\rvert$, one gets the required result. Alternatively, by a linear substitution, we may assume WLOG that $x_1,\dots,x_n$ are linearly independent in E. Now apply Corollary 22.6.4.

In the case when E is a normed space, we scale x_i's and λ so that $\lVert x_i\rVert\leqslant 1$ for all i. $\square$

22.6.6 Exercise. Let A be a Hamel basis of a vector space E. Then $\mathrm{FVL}(A)=\mathrm{FVL}[E]$. That is, for every set A we have $\mathrm{FVL}(A)=\mathrm{FVL}\bigl[\mathrm{FVS}(A)\bigr]$, where $\mathrm{FVS}(A)$ stands for the free vector space over A.

22.6.7. The proof of the existence of $\mathrm{FVL}[E]$ presented above was based on Exercise 22.6.1, which is direct but rather technical. We now outline a shorter and more abstract proof that avoids Exercise 22.6.1. As before, let E be a vector space and L the sublattice of $\mathbb{R}^{E^\#}$ generated by $\{\hat{x}:x\in E\}$. Let $T\colon E\to Y$ be a linear map from E to a vector lattice Y. We need to show that there exists a unique lattice homomorphism $\widehat{T}\colon L\to Y$ which extends T is the sense that $\widehat{T}\hat{x}=Tx$ for all $x\in E$. Since the set $\{\hat{x}:x\in E\}$ generates L, it s clear that such a lattice homomorphism would have to be unique, so only its existence is an issue.

First, as in Exercise 22.6.3, show that $x_1,\dots,x_n$ are linearly independent in E iff $\hat{x}_1,\dots,\hat{x}_n$ are lattice-linear independent in L. Fix a Hamel basis[1] A of E. As in Exercise 22.6.6, we identify L with $FVL(A)$ in such a way that for every $a\in A$, δ_a in $\mathrm{FVL}(A)$ corresponds to $\hat{a}$ in L.

Let $\varphi\colon A\to Y$ be the restriction of T to A. Let $\hat{\varphi}\colon FVL(A)\to Y$ be its canonical lattice homomorphic extension as in the definition of $\mathrm{FVL}(A)$. We can view $\hat{\varphi}$ as a

[1]For those who prefer to avoid Hamel bases: in the construction, we only use finite sets of vectors, so with a minor tweaking of the argument, one may replace use of a Hamel basis with restricting to finite-dimensional subspaces of E.

map $\widehat{T}$ from L to Y. Since it is linear and agrees with T on A, it also agrees with T on E. More precisely, it follows from $\widehat{T}\hat{a} = \hat{\varphi}(\delta_a) = \varphi(a) = Ta$ for every $a \in A$ that $\widehat{T}\hat{x} = Tx$ for every $x \in E$. Thus, $\widehat{T}$ is a lattice homomorphism from L to Y extending T. This proves that L is $\mathrm{FVL}[E]$.

Let $f \in L$; we can write $f = \Phi(\hat{x}_1, \dots, \hat{x}_n)$, where Φ is a lattice-linear expression. Since $\widehat{T}$ is a lattice homomorphism, we have

$$\widehat{T}f = \Phi\big(\widehat{T}\hat{x}_1, \dots, \widehat{T}\hat{x}_n\big) = \Phi(Tx_1, \dots, Tx_n).$$

This implies that the resulting operator agrees with that in our previous construction and does not depend on the choice of A.

22.6.8. In the special case when E is a normed space, the preceding construction may be somewhat improved. First, it is easy to see that one may replace $E^{\#}$ with E^*, i.e., we may identify $\mathrm{FVL}[E]$ with the of $\mathbb{R}^{E^*}$ generated by all the evaluation functionals. With a minor abuse of notation, we will still denote it by L. Observe that each $\hat{x}$ is weak*-continuous, so L is actually a sublattice of $C(E^*, w^*)$. Second, let T be the restriction map that sends $f \in L$ to its restriction to the unit ball B_{E^*}. Since every f in L is positively homogeneous, T is a lattice isomorphic embedding. It is easy to see that if $f \in L$ is a lattice-linear combination of $\hat{x}_1, \dots, \hat{x}_n$ for some $x_1, \dots, x_n$ in E then Tf is the same lattice-linear combination of their restrictions to B_{E^*}. Thus, we may view L as the sublattice of $C(B_{E^*}, w^*)$ generated by the restrictions of the evaluaion functionals to B_{E^*}. Recall that (B_{E^*}, w^*) is a compact space, so we have represented $\mathrm{FVL}[E]$ as a sublattice of a $C(K)$-space.

22.7 Free Banach lattice over a Banach space: $\mathrm{FBL}[E]$

In this section, for a given Banach space E we want to find the "largest" Banach lattice X which contains E as a closed sublattice and is generated by it. More precisely, let E be a closed subspace of a Banach lattice X; we say that X is a free Banach lattice over the subspace E if every bounded operator $T\colon E \to Y$ from E to an arbitrary Banach lattice Y extends uniquely to a lattice homomorphism $\widehat{T}\colon X \to Y$ with $\|\widehat{T}\| = \|T\|$. This concept was first introduced and investigated in [ART18, ATV19].

$$\begin{tikzcd} \mathrm{FBL}[E] \arrow[dr, dotted, "\widehat{T}"] & \\ E \arrow[u, hook] \arrow[r, "T"'] & Y \end{tikzcd}$$

As before, it is a straightforward verification that X is determined by E up to a lattice isometry preserving E, so we may write $X = \mathrm{FBL}[E]$. Next, we are going to prove that $\mathrm{FBL}[E]$ may be defined for every Banach lattice E as the completion of $\mathrm{FVL}[E]$ in the appropriate norm. For every $x \in E$, let $\delta_x = \hat{x}_{|B_{E^*}}$, the restriction of the evaluation functional $\hat{x} \in E^{**}$ to the unit ball B_{E^*}.[2] As in 22.6.8, we identify $\mathrm{FVL}[E]$ with the sublattice L of $C(B_{E^*})$ generated by $\{\delta_x : x \in E\}$. Note that order and lattice operations on L are defined pointwise. The map $x \in E \mapsto \delta_x \in L$ is a linear embedding, so that we view E as a linear subspace of L. The following theorem is analogous to Theorem 22.5.3.

22.7.1 Theorem. *Let E be a Banach space. There is a greatest lattice seminorm ν on $\mathrm{FVL}[E]$ satisfying $\nu(x) \leqslant \|x\|$ for all $x \in E$. It is a lattice norm; the completion of $\mathrm{FVL}[E]$ with respect to it is $\mathrm{FBL}[E]$.*

Proof. Let L be as above; let $\mathcal{M}$ be the set of all lattice seminorms ν on L such that $\nu(\delta_x) \leqslant \|x\|$ for all $x \in E$.

For every $x^* \in B_{E^*}$, define $\nu_{x^*}(f) = \bigl|f(x^*)\bigr|$ for $f \in L$. It is easy to see that ν_{x^*} is a seminorm on L. Note that ν_{x^*} is a lattice seminorm because $|f|(x^*) = \bigl|f(x^*)\bigr|$. Furthermore, if $x \in E$ then $\nu_{x^*}(\delta_x) = \bigl|\delta_x(x^*)\bigr| = \bigl|x^*(x)\bigr| \leqslant \|x\|$, so that $\nu_{x^*} \in \mathcal{M}$.

For $f \in L$, define $|||f||| = \sup_{\nu \in \mathcal{M}} \nu(f)$. We claim that $|||\cdot|||$ is a lattice norm on L.

Observe that it is finite. Indeed, let $f \in L$, then Corollary 22.6.5 guarantees that $|f| \leqslant \lambda\bigl(|\delta_{x_1}| + \cdots + |\delta_{x_n}|\bigr)$ for some $x_1, \dots, x_n \in B_E$ and $\lambda > 0$. This yields $\nu(f) \leqslant \lambda n$ for every $\nu \in \mathcal{M}$, hence $|||f||| \leqslant \lambda n < \infty$.

It is straightforward that $|||\cdot|||$ is a lattice seminorm on L. Suppose that $0 \neq f \in L$. Then $f(x^*) \neq 0$ for some $x^* \in B_{E^*}$. Then $|||f||| \geqslant \nu_{x^*}(f) = \bigl|f(x^*)\bigr| > 0$. Thus, $|||\cdot|||$ is a lattice norm on L.

Note also that $|||\delta_x||| = \|x\|$ for all $x \in E$. Indeed, $\nu(\delta_x) \leqslant \|x\|$ for every $\nu \in \mathcal{M}$, hence $|||\delta_x||| \leqslant \|x\|$. On the other hand, let $x^* \in B_{E^*}$ such that $x^*(x) = \|x\|$; then $|||\delta_x||| \geqslant \nu_{x^*}(\delta_x) = \bigl|\delta_x(x^*)\bigr| = \|x\|$. Thus, the canonical embedding of E into L is an isometry.

Let X be the completion of $\bigl(L, |||\cdot|||\bigr)$. We claim that $X = \mathrm{FBL}[E]$. Let $T\colon E \to Y$ be a linear operator from E to an arbitrary Banach lattice Y. WLOG, $\|T\| = 1$. Since $L = \mathrm{FVL}[E]$, T extends to a lattice homomorphism $\widehat{T}\colon L \to Y$. Since $\widehat{T}$ extends T, we have $\|\widehat{T}\| \geqslant 1$. For $f \in L$, define $\nu(f) = \|\widehat{T}f\|$ in Y. It is easy to see that ν is a lattice seminorm on L. For every $x \in X$, one has $\nu(\delta_x) = \|\widehat{T}\delta_x\| = \|Tx\| \leqslant \|x\|$. It follows that $\nu \in \mathcal{M}$, so that $\|\widehat{T}f\| = \nu(f) \leqslant |||f|||$, so that $\|\widehat{T}\| \leqslant 1$. We conclude that $\widehat{T}$ extends to a contractive lattice homomorphism $\widetilde{T}\colon X \to Y$.

Uniqueness of the extension follows from the fact that any contractive lattice homomorphism that extends T to X has to agree with $\widehat{T}$ on L and, therefore, with $\widetilde{T}$

[2] Caution: this is not to be confused with δ_a from sections of $\mathrm{FVL}(A)$ and $\mathrm{FBL}(A)$!

on X. $\square$

Let T be a bounded operator from a Banach space E to a Banach lattice Y. By the definition of $\mathrm{FBL}[E]$, T extends to a lattice homomorphism $\widehat{T}\colon \mathrm{FBL}[E]\to Y$ with $\|\widehat{T}\|=\|T\|$. Since $\mathrm{FVL}[E]$ is a sublattice of $\mathrm{FBL}[E]$ and the restriction of $\widehat{T}$ to $\mathrm{FVL}[E]$ is a lattice isomorphism, it has to agree with the extension of T in the definition of $\mathrm{FVL}[E]$.

22.7.2 Corollary. *Let E be a Banach lattice and $f\in\mathrm{FVL}(E)$. Then $\|f\|_{\mathrm{FBL}}=\sup\|\widehat{T}f\|$, where the supremum is taken over all operators T from E to a Banach lattice Y with $\|T\|\leqslant 1$.*

Proof. Denote the supremum in the statement above by $|||f|||$. Let T be an operator from E to a Banach lattice Y with $\|T\|\leqslant 1$. Since $\widehat{T}\colon \mathrm{FBL}[E]\to Y$ is a lattice homomorphism, the map $f\mapsto\|\widehat{T}f\|$ is a lattice seminorm on $\mathrm{FVL}[E]$. Since $\|\widehat{T}\delta_x\|=\|Tx\|\leqslant\|x\|$ for every $x\in E$, it follows from Theorem 22.7.1 that this seminorm is dominated by the FBL-norm, i.e., $\|\widehat{T}f\|\leqslant\|f\|_{\mathrm{FBL}}$. Therefore, $|||f|||\leqslant\|f\|_{\mathrm{FBL}}$.

On the other hand, take $Y=\mathrm{FBL}[E]$ and $T\colon E\hookrightarrow\mathrm{FBL}[E]$ the canonical embedding. Then $\widehat{T}$ is the identity map on $\mathrm{FBL}[E]$ by the definition of $\mathrm{FBL}[E]$, hence $|||f|||\geqslant\|\widehat{T}f\|_{\mathrm{FBL}}=\|f\|_{\mathrm{FBL}}$. $\square$

Theorem 22.7.1 guarantees thee existence of $\mathrm{FBL}[E]$. Its proof can be used to prove existence of free objects in several related categories, e.g., free p-convex Banach lattices or free Banach lattices with upper p-estimates. A major disadvantage is that this theorem does not provide a functional representation of $\mathrm{FBL}[E]$. While $\mathrm{FVL}[E]$ may be viewed as a sublattice of $C(B_{E^*})$ by 22.6.8, it is not clear from Theorem 22.7.1 whether $\mathrm{FBL}[E]$ is still a subset of $C(B_{E^*})$. We are going to show that it is, but we need some additional tools. The fact that $\mathrm{FBL}[E]$ has a functional representatio turns out to me most useful.

We start by showing that in Corollary 22.7.2, instead of *all* Banach lattices, it suffices to consider only ℓ_1^n's for all $n\in\mathbb{N}$. This result is essentially due to Avilés, Rodríguez, and Tradacete [ART18].

22.7.3 Theorem. *For $f\in\mathrm{FBL}[E]$, $\|f\|_{\mathrm{FBL}}=\sup\|\widehat{T}f\|$, where the supremum is taken over all operators T from E to ℓ_1^n with $n\in\mathbb{N}$ and $\|T\|\leqslant 1$.*

Proof. Let $0\leqslant f\in\mathrm{FVL}[E]$. Find $\xi\in\mathrm{FBL}^*_+$ such that $\|\xi\|=1$ and $\xi(f)=\|f\|$. We will now use the construction from 9.5.1: let $Q\colon \mathrm{FBL}[E]\to\mathrm{FBL}[E]/N_\xi$ be the quotient map. Then ξ induces a strictly positive functional on $\mathrm{FBL}[E]/N_\xi$ which leads to a lattice isomorphic contractive embedding $J\colon \mathrm{FBL}[E]/N_\xi\hookrightarrow L_1(\mu)$ for some measure μ, such that $\|JQf\| = \xi(f) = \|f\|$. Let R be the restriction of JQ to E. Since JQ is a lattice homomorphism, we have $JQ = \widehat{R}$, and, therefore, $\|\widehat{R}f\| = \|f\|$.

Being an element of $\mathrm{FVL}[E]$, f can be written as a lattice-linear combination $f = \Phi(\delta_{x_1}, \dots, \delta_{x_m})$ for some $x_1, \dots, x_m \in E$. Then $\widehat{R}f = \Phi(Rx_1, \dots, Rx_m)$ in $L_1(\mu)$. WLOG, $L_1(\mu)$ has a weak unit; otherwise, replace $L_1(\mu)$ with the band generated by $Rx_1, \dots, Rx_m$, and replace R with its composition with the restriction operator from $L_1(\mu)$ onto this band. Therefore, up to a lattice isometry, we may assume that μ is a probability measure.

Fix $\delta > 0$ (to be determined later). For each $i = 1, \dots, m$, find a simple function y_i such that $\|Rx_i - y_i\| < \delta$. Let $\mathcal{G}$ be the (finite) sub-σ-algebra generated by $y_1, \dots, y_m$. Let $P\colon L_1(\mu) \to L_1(\mathcal{G}, \mu)$ be the conditional expectation. Since $\|P\| = 1$, the operator $PR\colon E \to L_1(\mathcal{G}, \mu)$ is a contraction, so that $\widehat{PR}\colon \mathrm{FBL}[E] \to L_1(\mathcal{G}, \mu)$ is a contractive lattice homomorphism. It follows from $Py_i = y_i$ that

$$\|PRx_i - Rx_i\| \leqslant \|PRx_i - Py_i\| + \|y_i - Rx_i\| < 2\delta$$

for every $i = 1, \dots, m$. Since function calculus in norm continuous,

$$\big\|\widehat{R}f - \widehat{PR}f\big\| = \Big\|\Phi(Rx_1, \dots, Rx_m) - \Phi(PRx_1, \dots, PRx_m)\Big\| < \varepsilon$$

provided that δ is sufficiently small. It follows that $\|\widehat{PR}f\| > \|\widehat{R}f\| - \varepsilon = \|f\| - \varepsilon$. Note $L_1(\mathcal{G}, \mu)$ is lattice isometric to ℓ_1^n for some n; let $U\colon L_1(\mathcal{G}, \mu)$ be a lattice isometry. Take $T = UPR$; it is now easy to see that $\|T\| = 1$ and $\|\widehat{T}f\| = \|U\widehat{PR}f\| = \|\widehat{PR}f\| > \|f\| - \varepsilon$.

Finally, let $f \in \mathrm{FBL}[E]$. Using density of $\mathrm{FVL}[E]$, we find $g \in \mathrm{FVL}[E]_+$ with $\big\||f| - g\big\| < \frac{\varepsilon}{3}$. Using the previous argument, we find $T\colon E \to \ell_1^n$ such that $\|T\| \leqslant 1$ and $\|\widehat{T}g\| > \|g\| - \frac{\varepsilon}{3}$. It follows from the triangle inequality that $\|\widehat{T}f\| > \|f\| - \varepsilon$. $\square$

Recall that we identify $\mathrm{FVL}[E]$ with a sublattice of $C(B_{E^*})$.

22.7.4 Corollary (Avilés, Rodríguez, Tradacete)**.** *For $f \in \mathrm{FBL}[E]$,*

$$\|f\|_{\mathrm{FBL}} = \sup\Big\{\sum_{k=1}^{n}\big|f(x_k^*)\big| \;:\; x_1^*, \dots, x_n^* \in B_{E^*},\ \sup_{x \in B_E}\sum_{k=1}^{n}|x_k^*(x)| \leqslant 1\Big\}. \tag{22.1}$$

Proof. By the preceding theorem,

$$\|f\|_{\mathrm{FBL}} = \sup\Big\{\|\widehat{T}f\| \;:\; T\colon E \to \ell_1^n,\ \|T\| \leqslant 1\Big\}.$$

Let $T\colon E \to \ell_1^n$. For every $x \in E$, we have $Tx = \sum_{i=1}^n x_i^*(x)e_i$, where $x_i^* = T^*e_i^*$, and $\|Tx\| = \sum_{i=1}^n\big|x_i^*(x)\big|$. Hence, the condition $\|T\| \leqslant 1$ is equivalent to $\sup_{x \in B_E}\sum_{k=1}^n|x_k^*(x)| \leqslant$

1. Furthermore, $\widehat{T}f = \sum_{k=1}^n f(x_k^*)e_k$ because the map $f \mapsto \sum_{k=1}^n f(x_k^*)e_k$ is a lattice homomorphism which agrees with T on E. It follows that $\|\widehat{T}f\| = \sum_{k=1}^n \bigl|f(x_k^*)\bigr|$. □

The following proposition shows that we may view $\mathrm{FBL}[E]$ as a sublattice of $C(B_{E^*})$ (but with a different norm).

22.7.5 Proposition. $\mathrm{FBL}[E]$ *is a sublattice of* $C(B_{E^*})$. *More precisely, the inclusion map* $\mathrm{FVL}[E] \hookrightarrow C(B_{E^*})$ *extends to a lattice (but not norm) isomorphic embedding* $\mathrm{FBL}[E] \to C(B_{E^*})$ *of norm one.*

Proof. For $f \in C(B_{E^*})$, define $|||f|||$ as in (22.1). Let Z be the set of all those $f \in C(B_{E^*})$, for which $|||f||| < \infty$. One can easily verify that $\bigl(Z, |||\cdot|||\bigr)$ is a normed lattice. For every $x^* \in B_{E^*}$, we have $|||f||| \geqslant \bigl|f(x^*)\bigr|$, so that $|||f||| \geqslant \|f\|_{C(B_{E^*})}$.

We will show that $\bigl(Z, |||\cdot|||\bigr)$ is a Banach lattice. Indeed, let (f_i) be a Cauchy sequence in it. Then it is also Cauchy in $C(B_{E^*})$, hence $\|f_i - f\|_{C(B_{E^*})} \to 0$ for some $f \in C(B_{E^*})$. In particular, $f_i(x^*) \to f(x^*)$ for every $x^* \in B_{E^*}$. We claim that $f \in Z$ and $|||f_i - f||| \to 0$. Indeed, fix $\varepsilon > 0$, find m_0 such that $|||f_m - f_i||| < \varepsilon$ whenever $i, m \geqslant m_0$. Let $x_1^*, \dots, x_n^* \in B_{E^*}$ such that $\sup_{x \in B_E} \sum_{k=1}^n \bigl|x_k^*(x)\bigr| \leqslant 1$. Then for every $i, m \geqslant m_0$ we have

$$\varepsilon > \Bigl|\sum_{k=1}^n (f_m - f_i)(x_k^*)\Bigr| = \Bigl|\sum_{k=1}^n f_m(x_k^*) - \sum_{k=1}^n f(x_k^*)\Bigr|.$$

Passing to the limit as $i \to \infty$, we get

$$\Bigl|\sum_{k=1}^n (f_m - f)(x_k^*)\Bigr| = \Bigl|\sum_{k=1}^n f_m(x_k^*) - \sum_{k=1}^n f(x_k^*)\Bigr| \leqslant \varepsilon.$$

It follows that $|||f_m - f||| \leqslant \varepsilon$; hence $f \in Z$ and $|||f_i - f||| \to 0$. Thus, $\bigl(Z, |||\cdot|||\bigr)$ is a Banach lattice.

By Corollary 22.7.4, $\mathrm{FVL}[E]$ is a sublattice of Z. Being the completion of $\mathrm{FVL}[E]$ under $|||\cdot|||$, $\mathrm{FBL}[E]$ is a closed sublattice of Z and, therefore, a sublattice of $C(B_{E^*})$. It follows from $|||f||| \geqslant \|f\|_{C(B_{E^*})}$ that the inclusion map is a contraction. □

Recall that when we view $\mathrm{FVL}[E]$ as a lattice of functions on E^* (or on B_{E^*}), the lattice operations are computed point-wise. By the preceding proposition, this remains valid for $\mathrm{FBL}[E]$.

One could define the concept of a free Banach space over a set, $\mathrm{FBS}(A)$, as follows: a Banach space X is free over a subset A if every bounded map φ from A to an arbitrary Banach space Y extends to a bounded operator $\hat{\varphi}$ with $\|\hat{\varphi}\| = \sup_{a \in A}\|\varphi(a)\|$. However, this concept is of little value as it is an easy exercise that $\mathrm{FBS}(A)$ is nothing but $\ell_1(A)$. In view of this, the following exercise essentially asserts that $\mathrm{FBL}(A) = \mathrm{FBL}\bigl[\mathrm{FBS}(A)\bigr]$:

22.7.6 Exercise. Show that $\mathrm{FBL}(A) = \mathrm{FBL}\bigl[\ell_1(A)\bigr]$ for every set A.

22.8 Properties of $\mathrm{FBL}[E]$

Throughout this section, we will again assume that $\mathrm{FBL}[E]$ is represented as a space of positively homogeneous w*-continuous functions on B_{E^*}. Of course, such functions extend uniquely to functions on E^*. We write δ_E for the natural embedding of E into $\mathrm{FBL}[E]$. i.e., $\delta_E x = \delta_x$. We view $\mathrm{FVL}[E]$ as the sublattice of $\mathrm{FBL}[E]$ generated by $\{\delta_x : x \in E\}$. We equip B_{E^*} with the weak* topology.

22.8.1 Theorem. *Let $H(E)$ be the vector lattice of all continuous positively homogeneous functions in $C(B_{E^*})$. Then* $\mathrm{FVL}[E]$ *is order dense in $H(E)$.*

22.8.2 Exercise. Let $x^* \in E^*$. Viewed as a map from $E \to \mathbb{R}$, it has the canonical extension to a lattice homomorphism $\widehat{x^*}\colon \mathrm{FBL}[E] \to \mathbb{R}$. Then $\widehat{x^*} f = f(x^*)$ for every $f \in \mathrm{FBL}[E]$.

22.8.3. Let $T\colon E \to F$ be a bounded operator between two Banach spaces. Let $\delta_E\colon E \hookrightarrow \mathrm{FBL}[E]$ and $\delta_F\colon F \hookrightarrow \mathrm{FBL}[F]$ be the canonical embedding. Consider the operator $S = \delta_F T$. The definition of $\mathrm{FBL}[E]$ guarantees that there exists a lattice homomorphism $\overline{T}\colon \mathrm{FBL}[E] \to \mathrm{FBL}[F]$ such that $\overline{T} = \widehat{S}$, thus, making the following diagram commutative:

$$\begin{tikzcd}
\mathrm{FBL}[E] \arrow[r, dotted, "\overline{T}"] & \mathrm{FBL}[F] \\
E \arrow[u, hook, "\delta_E"] \arrow[ur, "S"] \arrow[r, "T"'] & F \arrow[u, hook', "\delta_F"']
\end{tikzcd}$$

Thus, T "lifts" to a lattice homomorphism $\overline{T}$ between the free lattices. Clearly, $\|\overline{T}\| \leqslant \|S\| = \|T\|$. On the other hand, since $\overline{T}$ extends T, we have $\|\overline{T}\| \geqslant \|T\|$. Thus, $\|\overline{T}\| = \|T\|$.

22.8.4 Exercise. Let T and $\overline{T}$ be as above. Then $\overline{T} f = f \circ T^*$ for every $f \in \mathrm{FBL}[E]$.

22.8.5 Proposition. *[OTTT] Let $T\colon E \to F$ be as above.*

(i) *T is one-to-one iff $\overline{T}$ is one-to-one;*
(ii) *T has dense range iff $\overline{T}$ has dense range;*
(iii) *T is onto iff $\overline{T}$ is onto;*

This proposition provides a useful tool for establishing various properties of free Banach lattices.

22.8.6 Theorem. $\mathrm{FBL}[E]$ *has a strong unit iff E is finite-dimensional.*

Proof. Let $n = \dim E$ be finite. Then E is isomorphic to ℓ_1^n. By Proposition 22.8.5, $\mathrm{FBL}[E]$ is lattice isomorphic to $\mathrm{FBL}[\ell_1^n]$. The latter equals H_n by Exercise 22.7.6 and Proposition 22.5.7 and, therefore, has a strong unit.

Suppose now that $\mathrm{FBL}[E]$ has a strong unit, say e, but $\dim E = \infty$. By Exercise 2.2.11, we may assume that e dominates the unit ball of $\mathrm{FBL}[E]$. In particular, $|\delta_x| \leqslant e$ for every $x \in B_E$. Since $\mathrm{FVL}[E]$ is dense in $\mathrm{FBL}[E]$, we can find $f \in \mathrm{FVL}[E]$ such that $\|e - f\| < \frac{1}{2}$. By Proposition 22.7.5, we have $\|e - f\|_{C(B_{E^*})} < \frac{1}{2}$. Since $f \in \mathrm{FVL}[E]$, we can write it as a lattice-linear combination of some $\delta_{x_1}, \dots, \delta_{x_n}$. Find $x^* \in S_{E^*}$ which vanishes on $x_1, \dots, x_n$. It follows that $f(x^*) = 0$. Then for every $x \in B_X$ we have

$$\bigl|x^*(x)\bigr| = |\delta_x|(x^*) \leqslant e(x^*) = \bigl|e(x^*) - f(x^*)\bigr| \leqslant \|e - f\|_{C(B_{E^*})} < \tfrac{1}{2},$$

which contradicts $\|x^*\| = 1$. $\square$

$\mathrm{FBL}[E]$ always has weak units: for every $x \in X$, $|\delta_x|$ is a weak unit in $\mathrm{FBL}[E]$. Indeed, it is a weak unit in $\mathrm{FVL}[E]$ and the latter is order dense in $\mathrm{FBL}[E]$.

22.8.7 Theorem. $\mathrm{FBL}[E]$ *has a quasi-interior point iff* $\mathrm{FBL}[E]$ *is separable iff* E *is separable.*

Proof. Suppose that E is separable. View it as a separable subspace of $\mathrm{FBL}[E]$. Since the closed sublattice generated by E in $\mathrm{FBL}[E]$ is all of $\mathrm{FBL}[E]$, the latter is separable by Exercise 3.1.16. If $\mathrm{FBL}[E]$ is separable , it contains a quasi-interior point by Exercise 5.5.5.

Suppose now that $e \in \mathrm{FBL}[E]_+$ is a quasi-interior point. If $x^* \in B_{E^*}$ satisfies $e(x^*) = 0$, that is, $\widehat{x^*}(e) = 0$, then $\widehat{x^*}$ vanishes on I_e and, therefore, $\widehat{x^*} = 0$, so that $x^* = 0$. Thus, e only vanishes at 0.

For every n, let $U_n = \{x^* \in B_{E^*} : e(x^*) < \frac{1}{n}\}$. Then $\bigcap_{n=1}^\infty U_n = \{0\}$. For every n, U_n is a weak*-open subset of B_{E^*} , hence there exists a finite set $A_n \subseteq E$ such that U_n contains

$$V_n \coloneqq \bigl\{x^* \in B_{E^*} : \bigl|x^*(x)\bigr| < 1 \text{ for all } x \in A_n\bigr\}.$$

Let F be the closed span of $\bigcup_{n=1}^\infty A_n$. We claim that $F = E$ and, therefore, E is separable. Indeed, otherwise, there exists $x^* \in S_{E^*}$ that vanishes on F. On the other hand, $x^* \notin V_n$ for some n; it follows that $\bigl|x^*(x)\bigr| \geqslant 1$ for all $x \in A_n$; which is a contradiction as $A_n \subseteq F$. $\square$

Proposition 22.8.5 leaves out the important case when T is an embedding (isomorphic or isometric). In other words, suppose that E is a closed subspace of F, does this induce an isometric or isomorphic embedding of FBL[E] into FBL[F]? In the case when E is complemented, things work pretty well: then FBL[E] may be viewed as a complemented sublattice of FBL[F] and the corresponding projection is a lattice homomorphism:

22.8.8 Proposition ([ART18]). *Suppose that E is a complemented subspace of F. Let $\iota\colon E \hookrightarrow F$ be the inclusion map and $P\colon F \to F$ the corresponding projection. Then $\overline{P}$ is a projection onto $\bar{\iota}\big(\mathrm{FBL}[E]\big)$.*

In general, we can say that if E is a subspace of F then FBL[E] is a regular sublattice of FBL[F]:

22.8.9 Theorem ([OTTT]). *Let E be a closed subspace of F, with $\iota\colon E \hookrightarrow F$ being the inclusion map. The map $\bar{\iota}\colon \mathrm{FBL}[E] \to \mathrm{FBL}[F]$ is order continuous.*

Next, we characterize pairs E and F, where E is a closed subspace of F, such that FBL[E] is a closed subspace of FBL[F]:

22.8.10 Theorem ([OTTT]). *Let E be a closed subspace of F, with $\iota\colon E \hookrightarrow F$ being the inclusion map. The map $\bar{\iota}\colon \mathrm{FBL}[E] \to \mathrm{FBL}[F]$ is an isomorphic embedding iff there exists $C > 0$ such that every operator T from E to an $L_1(\mu)$ space extends to $\widetilde{T}\colon F \to L_1(\mu)$ with $\|\widetilde{T}\| \leqslant C\|T\|$.*

Let (x_k) be a basic sequence in a Banach lattice E. We may consider its image (δ_{x_k}) in FBL[E]. Since the embedding of E into FBL[E] is an isometry, (δ_{x_k}) is equivalent to (x_k). What can be said about the sequence $\big(|\delta_{e_k}|\big)$ in FBL[E]?

22.8.11 Proposition ([ATV19]). *If (x_k) is a basis of E then $\big(|\delta_{x_k}|\big)$ is a basic sequence. If, in addition, (x_k) is unconditional then so is $\big(|\delta_{x_k}|\big)$.*

Proof. Let M be the basis constant of (x_k). Let $n \in \mathbb{N}$; let $P_n\colon E \to E$ be the n-th basis projection of (x_n). Let $\overline{P_n}\colon \mathrm{FBL}[E] \to \mathrm{FBL}[E]$ be its lifting as in 22.8.3. We have $\|\overline{P_n}\| = \|P_n\| \leqslant M$. For every $m \geqslant n$ and coefficients $\alpha_1, \dots, \alpha_m$, we have

$$\overline{P_n}\sum_{k=1}^{m} \alpha_k|\delta_{x_k}| = \sum_{k=1}^{m} \alpha_k|\overline{P_n}\delta_{x_k}| = \sum_{k=1}^{m} \alpha_k|\delta_{P_n x_k}| = \sum_{k=1}^{n} \alpha_k|\delta_{x_k}|.$$

It follows that

$$\Big\|\sum_{k=1}^{n} \alpha_k|\delta_{x_k}|\Big\| \leqslant M\Big\|\sum_{k=1}^{m} \alpha_k|\delta_{x_k}|\Big\|.$$

Hence, $\big(|\delta_{x_k}|\big)$ is a basic sequence by Theorem A.4.26.

The proof that unconditionality is preserved is similar; just replace P_n with $T\colon E \to E$ defined by $T\Big(\sum_{k=1}^{\infty} \alpha_k x_k\Big) \sum_{k\in A}^{\infty} \alpha_k x_k$ where $A \subseteq \mathbb{N}$. ☐

The following theorem asserts that if (e_k) is the standard basis of ℓ_p or c_0 then $\big(|\delta_{e_k}|\big)$ is equivalent to the standard basis of some ℓ_r:

22.8.12 Theorem ([ATV19, OTTT]). *Let $E = \ell_p$ with $1 \leqslant p < \infty$ or $E = c_0$; let (e_k) be the standard unit basis of E. The sequence $\big(|\delta_{e_k}|\big)$ in $\mathrm{FBL}[E]$ is equivalent to the unit basis of*

(i) *ℓ_1 if $E = \ell_p$ with $p \leqslant 2$*
(ii) *ℓ_r, where $\frac{1}{r} = \frac{1}{2} + \frac{1}{p}$ if $E = \ell_p$ with $p \geqslant 2$;*
(iii) *ℓ_2 if $E = c_0$.*

The preceding theorem yields an application of free Banch lattices to the general theory of Banach lattices:

22.8.13 Theorem ([OTTT]). *Let X be a Banach lattice and (x_k) a basic sequence in X that is equivalent to the unit vector basis of c_0 or ℓ_p with $p > 2$. Then $|x_k| \xrightarrow{\mathrm{w}} 0$.*

Proof. Let $E = c_0$ or $E = \ell_p$ with $p > 2$, respectively; let $T\colon E \to X$ be the isomorphic embedding defined via $Te_k = x_k$. Let $\widehat{T}\colon \mathrm{FBL}[E] \to X$ be the canonical extension of T. By the preceding theorem, the sequence$\big(|\delta_{e_k}|\big)$ in $\mathrm{FBL}[E]$ is equivalent to the unit vector basis of ℓ_r for some $r > 1$; it follows that it is weakly null in its closed span and, therefore, in $\mathrm{FBL}[E]$. Since $\widehat{T}$ is continuous, it is weak-to-weak continuous; it follows that $\widehat{T}|\delta_{e_k}| \xrightarrow{\mathrm{w}} 0$ in X. Finally, since $\widehat{T}$ is a lattice homomorphism, we have $\widehat{T}|\delta_{e_k}| = \big|\widehat{T}\delta_{e_k}\big| = \big|Te_k\big| = |x_k|$. □

The following observation is due to D. de Hevia and P. Tradacete:

22.8.14 Proposition. *A Banach space E is isomorphic (as a Banach space) to a Banach lattice iff $\mathrm{FBL}[E] = (\mathrm{Range}\,\delta_E) \oplus J$, where the sum is a Banach space sum, for some closed ideal J in $\mathrm{FBL}[E]$.*

Proof. Suppose that $T\colon E \to X$ is an isomorphism from E onto some Banach space X. Let $\widehat{T}\colon \mathrm{FBL}[E] \to X$ be the canonical extension of T. It is easy to see that $P \coloneqq \delta_E T^{-1}\widehat{T}$ is a projection onto $\mathrm{Range}\,\delta_E$, and $\ker P = \ker \widehat{T} =: J$ is a closed ideal, because it is the kernel of a lattice homomorphism. We then have $\mathrm{FBL}[E] = (\mathrm{Range}\,\delta_E) \oplus J$.

To prove the converse, observe that E is isomorphic to $\mathrm{FBL}[E]/J$, and the latter is a Banach lattice. □

Appendix A

Background material

A.1 Nets

A relation "$\leqslant$" on a set $\mathcal{S}$ is called a ***(partial) order*** if

(i) $\forall x \in \mathcal{S} \quad x \leqslant x \quad$ (reflexivity);

(ii) $\forall x, y \in \mathcal{S} \quad x \leqslant y$ and $y \leqslant x$ imply $x = y \quad$ (antisymmetry);

(iii) $\forall x, y, z \in \mathcal{S} \quad x \leqslant y$ and $y \leqslant z$ imply $x \leqslant z \quad$ (transitivity).

In this case, we say that $\mathcal{S}$ is ***(partially) ordered***. We write $x \geqslant y$ if $y \leqslant x$, $x < y$ if $x \leqslant y$ and $x \neq y$, and $x > y$ when $y < x$.

A relation which is reflexive and transitive is called a ***pre-order***. In this case, we say that $x \sim y$ if $x \leqslant y$ and $y \leqslant x$. This is, clearly, an equivalence relation. Every pre-order can be made into an order in two ways. One way is to replace each equivalence class with a single point; this would produce a "quotient order". Another approach is to "straighten up" each equivalence class: using the Well Orderdness Principle, redefine the relation on each equivalence class to make it into a linear order.

A set Λ is ***directed*** if it is pre-ordered and every two elements have a common descendant, i.e.,

$$\forall \alpha, \beta \in \Lambda \quad \exists \gamma \in \Lambda \text{ such that } \alpha \leqslant \gamma \text{ and } \beta \leqslant \gamma.$$

A function from a directed set Λ to any set A is referred to as a ***net*** in A. In this case, instead of writing $x\colon \Lambda \to A$ we write $(x_\alpha)_{\alpha \in \Lambda}$ or just (x_α) if the index set is clear from the context. Note that every directed set can be viewed as a net over itself via the identity map. A ***sequence*** is a net over $\mathbb{N}$.

Given a sequence, one may consider its tails and subsequences. Similar concepts may be defined for nets. Let $(x_\alpha)_{\alpha \in \Lambda}$ be a net in A. Fix $\alpha_0 \in \Lambda$; put $\Lambda_0 = \{\alpha \in \Lambda : \alpha \geqslant \alpha_0\}$. The restriction of x to Λ_0 is called a ***tail*** of (x_α); it is sometimes denoted by $(x_\alpha)_{\alpha\geqslant\alpha_0}$. Next, we define a ***subnet*** of $(x_\alpha)_{\alpha\in\Lambda}$. Let Γ be another directed set and $\varphi\colon\Gamma\to\Lambda$ satisfies

(i) φ is monotone, i.e., $\gamma_1\leqslant\gamma_2$ implies $\varphi(\gamma_1)\leqslant\varphi(\gamma_2)$;
(ii) $\operatorname{Range}\varphi$ is *co-final* in Λ, i.e., for every $\alpha\in\Lambda$ there is $\gamma\in\Gamma$ with $\alpha\leqslant\varphi(\gamma)$.

The composition $x\circ\varphi\colon\Gamma\to A$ is a net in A indexed by Γ, it is called a subnet of $(x_\alpha)_{\alpha\in\Lambda}$. The co-finality condition means that the subnet "goes as far as the original net goes".

There are a few subtleties in the definition of a net. First, in some literature, an index set of a net is required to be a partially ordered, not just pre-ordered. This does not cause any difficulties though because, as we mentioned before, every pre-ordered set can be made into an ordered set.

The second subtlety is related to Set Theory. In Functional Analysis, it is quite common to have statements of the form "for every net in X something is true". Such statements are "illegal" from the point of view of Set Theory because they implicitly involve quantifying over the collection of all directed sets, which is not a set! Here is a way to circumvent this problem. Let X be a set. A collection $\mathcal{B}$ of non-empty subsets of X is called a ***filter base*** if for all $A,B\in\mathcal{B}$ there exists $C\in\mathcal{B}$ such that $C\subseteq A\cap B$. A collection $\mathcal{F}$ of non-empty subsets of X is a ***filter*** if it is closed under taking supersets and finite intersections. It is easy to see that every filter is a filter base, and if $\mathcal{B}$ is a filter base then the collection of all supersets of sets in $\mathcal{B}$ forms a filter. Given a net $(x_\alpha)_{\alpha\in\Lambda}$ in X, consider the collection of sets $\{T_\alpha\,:\,\alpha\in\Lambda\}$, where $T_\alpha=\{x_\beta\,:\,\beta\geqslant\alpha\}$. Note that these sets are the ranges of tails of $(x_\alpha)_{\alpha\in\Lambda}$. These sets form a filter base in X. We call it the ***tail filter base*** of (x_α). The filter generated by this base is called the ***tail filter*** of (x_α). Two nets in X are equivalent if they have the same tail filter bases. In this book, we identify equivalent nets. That is, all statements involving nets in this book should be understood as statements about equivalence classes of nets. On the other hand, let $\mathcal{B}$ be a filter base on X. Let $\Lambda=\big\{(A,x)\,:\,x\in A\in\mathcal{B}\big\}$, ordered via $(x,A)\leqslant(y,B)$ whenever $B\subseteq A$. Clearly, this is a directed set. Consider the net in X indexed by Λ given by $\varphi_{(A,x)}=x$. It is easy to see that the tail filter base of this net is exactly $\mathcal{B}$. Thus, there is a one-to-one correspondence between all equivalence classes of nets on X and all filter bases in X. The latter collection is clearly a set.

The third subtlety is that our definition of a net allows the index set to have a greatest element. Hence, a net may have a "terminal node" (in particular, a finite sequence would be a net). This is inconvenient in some of applications of net. In such a case, we will "extend" our net by a constant sequence "at the end", to avoid having terminal nodes. Observe that this operation does not affect the tail filter base of the net, so it produces a net which is equivalent to the original one.

A.2 Topology

We refer the reader to [Kel55, KN76] for a detailed exposition of topology.

Let Ω be a set. By a ***topology*** on Ω we mean a collection τ of subsets of Ω which includes $\varnothing$ and Ω and is closed under finite intersections and arbitrary unions. The pair (Ω, τ) is called a ***topological space***. Members of τ are called ***open***. The complements of open sets are termed ***closed***. The collection of all closed sets is closed under finite unions and arbitrary intersections. Given $x \in \Omega$ and $A \subseteq \Omega$, if there is an open set U such that $x \in U \subseteq A$ then we say that A is a ***neighborhood*** of x and that x is an ***interior point*** of A. The set of all interior points of A is called the ***interior*** of A and denoted $\operatorname{Int} A$ or A°. Observe that $\operatorname{Int} A$ is the union of all open sets contained in A. We define the closure of A as the intersection of all closed sets containing A; we denote it by $\operatorname{Cl} A$ or $\overline{A}$. When dealing with several topologies simultaneously, we sometimes write $\overline{A}^\tau$ to emphasize that the closure is taken with respect to τ. Note that $\operatorname{Int} A$ is open and $\overline{A}$ is a closed. We write ∂A for the ***boundary*** of A, $\partial A = \overline{A} \setminus \operatorname{Int} A$.

Fix $x \in \Omega$. A collection $\mathcal{N}$ of neighborhoods of x is said to be a ***base*** of the topology at x if every neighborhood of x contains a member of $\mathcal{N}$. A set U is open iff with every point x it also contains a neighborhood (equivalently, a base neighborhood) of x. Thus, a topology is determined by its neighborhoods.

Let (x_α) be a net in Ω and $x \in \Omega$. We say that (x_α) converges to x with respect to τ and write $x_\alpha \xrightarrow{\tau} x$ if every neighborhood of x contains a tail of the net. We write $x_\alpha \to x$ when there is no risk of confusion.

A topological spaces is said to be ***Hausdorff*** if for any two distinct x and y in Ω there exist neighborhoods U and V of x and y, respectively, such that $U \cap V = \varnothing$. A topological space is Hausdorff iff every net has at most one limit. All topological spaces in this book are assumed to be Hausdorff, unless specified otherwise. A topological space Ω is ***normal*** if every two disjoint closed sets F and G may be separated by open neighborhoods, i.e., there exist disjoint open sets U and V such that $F \subseteq U$ and $G \subseteq V$.

We would like to point out that a topology may be "recovered" from convergence. A set F is closed iff it contains limits of all convergent nets in F. Given $x \in \Omega$ and $U \subseteq \Omega$, U is a neighborhood of x if every net which converges to x has a tail in U.

A.2.1 Proposition. *Let (x_n) be a sequence in a topological space. Then $x_n \to x$ iff every subsequence (x_{n_k}) of (x_n) has a further subsequence $(x_{n_{k_i}})$ such that $x_{n_{k_i}} \to x$.*

A function f between two topological spaces is said to be ***continuous*** at a point x if the pre-image of every neighborhood of $f(x)$ is a neighborhood of x or, equivalently, $f(x_\alpha) \to f(x)$ whenever $x_\alpha \to x$. We say that f is continuous if it is continuous at every point or, equivalently, if the pre-image of every open set is open. We write $C(\Omega)$ for $C(\Omega, \mathbb{R})$, where $\mathbb{R}$ is equipped with the usual topology.

Let A be a subset of a topological space. We that that A is F_σ if it is the union of a sequence of closed sets. A is said to be a G_δ set if it is the intersection of a sequence of open sets. We say that A is ***clopen*** if it is both closed and open; equivalently, the characteristic function of A is continuous.

Let A be a subset of a topological space Ω. The sets of the form $A \cap U$ where U is an open subset of Ω forms a topology on A. Note that a net in A converges to a point in A in this ***induced*** topology iff it converges in the original topology of Ω. We say that Ω is ***disconnected*** if it may be written as a union of two non-empty disjoint open subsets; such a pair of subsets is called a ***disconnection*** of Ω. We say that Ω is ***connected*** if it is not disconnected.

A topological space K is said to be ***compact*** if every ***open cover*** of K admits a finite subcover. That is, if $\mathcal{O}$ is a family of open subsets of K such that its union is all of K then there exists a finite subcollection of $\mathcal{O}$ whose union is K. K is compact iff it has the ***finite intersection property***, i.e., if $\mathcal{C}$ is a collection of closed subsets of K such that every finite subcollection of $\mathcal{C}$ has a non-empty intersection then the the intersection of the entire collection is non-empty. A subset K of a topological space Ω is said to be compact if it is compact in the induced topology. A closed subset of a compact set is again compact. Every compact set in a topological space is closed. A set K in a topological space is compact iff every net in K has a subnet which converges to a point of K. A set A in a topological space is said to be ***relatively compact*** if it is contained in a compact set or, equivalently, if its closure is compact. We say that K is ***sequentially compact*** if every sequence in K admits a subsequence which converges to an element of K. The image of a compact set under a continuous function is compact. In particular, if K is compact that ever function in $C(K)$ attains its maximum and minimum.

A.2.2 Theorem (Urysohn's Lemma)**.** *Let A and B be two disjoint closed subsets of a normal (Hausdorff) topological space Ω. Then there exists $f \in C(\Omega)$ such that $0 \leqslant f(x) \leqslant 1$ for all x, f vanishes on A and equals one on B.*

A compact (Hausdorff) topological space is normal. In particular, Urysohn's lemma is satisfied in such spaces.

A.2.3 Proposition. *Let K be a compact space and $F \subseteq U \subseteq K$ such that F closed and U open. Then there exists an open set V such that $F \subseteq V \subseteq \overline{V} \subseteq U$.*

A.2.4 Theorem (Baire Category Theorem)**.** *Let Ω be a complete metric space or a compact space. The intersection of a sequence of open dense subsets of Ω is dense.*

A.2.5. A subset A of a topological space is ***nowhere dense*** if $\operatorname{Int}\overline{A} = \varnothing$. It is easy to see that the boundary of an open set is nowhere dense. We say that A ***meagre*** or ***of first category*** if it is a union of a sequence of nowhere dense sets. The class of meagre sets is closed under taking subsets and countable unions. A set is ***co-meagre*** if its complement is meagre.

A.2.6 Proposition. *A is co-meagre iff A contains an intersection of a sequence of open dense sets.*

Proof. If A is co-meagre then $A^C = \bigcup_{n=1}^{\infty} D_n$, where each D_n is nowhere dense. It follows that $(\overline{D_n})^C$ is open and dense and $A \supseteq \bigcap_{n=1}^{\infty}(\overline{D_n})^C$. Conversely, if A contains the intersection of a sequence (U_n) of open dense sets then $A^C \subseteq \bigcup_{n=1}^{\infty} U_n^C$ is meagre. □

Combining this with Baire Category Theorem, we get the following.

A.2.7 Corollary. *Let K be a compact space. Every co-meagre set is dense. No non-empty open subset of K is meagre.*

Let $(\Omega_\gamma)_{\gamma\in\Gamma}$ be a family of topological spaces. Let $\Omega = \prod_{\gamma\in\Gamma} \Omega_\gamma$ be the Cartesian product. For every γ, let $P_\gamma \colon \Omega \to \Omega_\gamma$ be the coordinate projection map, i.e., $Px = x(\gamma)$ for $x \in \Omega$. One defines the ***product topology*** on Ω as the weakest topology that makes all the coordinate projections continuous. It is easy to see that this is exactly the topology of coordinate-wise (or point-wise) convergence: for a net (x_α) in Ω, we have $x_\alpha \to x$ in Ω iff $P_\gamma(x_\alpha) \to P_\gamma(x)$ in Ω_γ for every γ. If Ω_γ is Hausdorff for every γ then Ω is also Hausdorff. A function from Ω to some topological space Γ is ***jointly continuous*** if it is continuous with respect to the product topology on Ω.

A.2.8 Theorem (Tychonoff Product Theorem)**.** *The product of a family of topological spaces is compact in the product topology iff each factor is compact.*

A topology on a vector space X is said to be ***linear*** if addition and scalar multiplication are jointly continuous. In this case, we say that X is a ***topological vector space***. We refer the reader to [KN76] for details on topological vector spaces. In particular, Theorem 2.5.1 [KN76] asserts that a topology is linear if there is a base on neighbourhoods of zero satisfying certain natural properties, and that a linear topology is Hausdorff iff the intersection of all open neighbourhoods of 0 is $\{0\}$, which is equivalent to singletons being closed sets.

A.3 Measure and Integration

In this section, we provide a brief overview of Measure Theory and Integration. We refer the reader to [Roy88] for further details.

A.3.1 Measures and measurable functions

Let Ω be a set. A collection $\mathcal{F}$ of subsets of Ω is called an ***algebra*** of sets if it contains $\varnothing$ and is closed under taking complements and unions, i.e., $A, B \in \mathcal{F}$ implies $\Omega \setminus A$ and $A \cup B$ are in $\mathcal{F}$. If, in addition, $\mathcal{F}$ is closed under countable unions, it is a ***σ-algebra***.

A.3.1 Proposition. *Let $\mathcal{F}$ be an algebra of sets. Then $\mathcal{F}$ is a σ-algebra iff the union of every nested increasing sequence of sets in $\mathcal{F}$ is again in $\mathcal{F}$.*

Let $\mathcal{F}$ be a σ-algebra. Members of $\mathcal{F}$ are referred to as ***measurable*** sets. A ***measure*** on $\mathcal{F}$ is a function $\mu\colon \mathcal{F} \to [0, \infty]$ such that $\mu(\varnothing) = 0$ and μ is countably additive, i.e., if (A_n) is a disjoint sequence of measurable sets then $\mu\bigl(\bigcup_{n=1}^{\infty} A_n\bigr) = \sum_{n=1}^{\infty} \mu(A_n)$. The triple $(\Omega, \mathcal{F}, \mu)$ is called a ***measure space***.

Let μ be a measure. We say that μ is ***finite*** if $\mu(\Omega)$ is finite; μ is a ***probability measure*** if $\mu(\Omega) = 1$. A measure μ is ***σ-finite*** if Ω is the union of a sequence of measurable sets (Ω_n) such that $\mu(\Omega_n)$ is finite for every n. It is easy to see that one can choose the sequence (Ω_n) to be disjoint or nested increasing. A measurable set A is said to be an ***atom*** for μ if $\mu(A) > 0$ and for every measurable subset B of A one has either $\mu(B) = 0$ or $\mu(B) = \mu(A)$. We say that a measure space is ***non-atomic*** if it contains no atoms.

Throughout this book, we only consider measures which have no atoms of infinite measure. Such measures are sometimes called ***semifinite***. This condition is not very restrictive as one may always "remove" these atoms without losing any essential information about the measure space.

Let Ω be an arbitrary set. The ***counting measure*** on Ω is the measure μ defined on $\mathcal{P}(\Omega)$ such that $\mu(A)$ is the number of elements in A.

There is a unique measure space $(\Omega, \mathcal{F}, \mu)$ where $\Omega = \mathbb{R}$, $\mathcal{F}$ contains all intervals, the measure of every interval equals its length, and μ is complete in the sense that if $A \in \mathcal{F}$ with $\mu(A) = 0$ and $B \subseteq A$ then $B \in \mathcal{F}$ and $\mu(B) = 0$. We call μ the ***Lebesgue***

measure. The sets in $\mathcal{F}$ are called ***Lebesgue measurable***. Given an interval I in $\mathbb{R}$, the restriction of the Lebesgue measure to the Lebesgue subsets of I is called the Lebesgue measure on I.

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. A measurable set A is said to have ***full*** measure if $\mu(\Omega \setminus A) = 0$. A logical statement $P(\omega)$ depending on a parameter $\omega \in \Omega$ is said to be satisfies ***almost everywhere*** or ***a.e.*** if there exists a measurable set A of full measure such that $P(\omega)$ is satisfied for all $\omega \in A$.

Let $f, g \colon \Omega \to \mathbb{R}$ and $\lambda \in \mathbb{R}$. We write $\{f > \lambda\}$ for the set $\bigl\{\omega \in \Omega : f(\omega) > \lambda\bigr\}$, that is, $\{f > \lambda\}$ is the pre-image of (λ, ∞). One defies notations $\{f \geqslant \lambda\}$, $\{f \leqslant g\}$, etc., in a similar way. A function $f \colon \Omega \to \mathbb{R}$ is said to be ***measurable*** if the pre-image of every interval under f is a measurable set. In a similar fashion, one can define measurable extended real valued functions, i.e., functions with values in $[-\infty, \infty]$. Two measurable functions f and g are said to be equivalent if they agree almost everywhere, i.e., the set $\{f = g\}$ has full measure. This is an equivalence relations on the set of measurable functions. It is common to identify measurable functions with their equivalence classes. We write $L_0(\mu)$ for the set of all equivalence classes of measurable functions which are a.e. finite. For every real number a, we define $a^+ = \max\{a, 0\}$ and $a^- = \max\{-a, 0\}$. For a measurable function f, the functions f^+ and f^- defined by $f^+(\omega) = \bigl(f(\omega)\bigr)^+$ and $f^-(\omega) = \bigl(f(\omega)\bigr)^-$ are measurable. Note that $f = f^+ - f^-$ and $|f| = f^+ + f^-$.

A.3.2 Theorem. *If f and g are measurable functions and $\lambda \in \mathbb{R}$ then $f + g$ and λf are measurable.*

A.3.3 Theorem. *Let (f_n) be a sequence of measurable functions. For every $t \in \Omega$, put $g(t) = \sup f_n(t)$ and $h(t) = \inf f_n(t)$. Then g and h are measurable extended real valued functions.*

A sequence (f_n) be a sequence of measurable functions converges a.e. to a function f if there exists a set A of full measure such that $f_n(\omega) \to f(\omega)$ for all $\omega \in A$. We write $f_n \xrightarrow{\text{a.e.}} f$. The following proposition says that, in this case, f is “almost” measurable.

A.3.4 Proposition. *Let (f_n) be a sequence of measurable functions which converges a.e. to a function f. Then there exists a measurable function g such that the set $\{f \neq g\}$ is contained in a set of zero measure.*

A net (f_α) of measurable functions is said to ***converge in measure*** to a measurable function f if for any positive δ, the measure of the set $\bigl\{|f_\alpha - f| > \delta\bigr\}$ converges to zero in α. We write $f_\alpha \xrightarrow{\mu} f$.

A.3.5 Proposition. *Let μ be a finite measure and let (f_n) and f be in $L_0(\mu)$. If $f_n \xrightarrow{\text{a.e.}} f$ then $f_n \xrightarrow{\mu} f$.*

A.3.6 Proposition. *Let (f_n) and f be in $L_0(\mu)$. If $f_n \xrightarrow{\mu} f$ then $f_{n_k} \xrightarrow{\text{a.e.}} f$ for some subsequence (f_{n_k}) of (f_n).*

We say that f is positive if $f(\omega) \geqslant 0$ a.e. A ***simple function*** is a linear combination of characteristic functions of measurable set. It is easy to see that these sets may be chosen to be disjoint.

A.3.7 Proposition. *Let f be a positive measurable function. Then there is an increasing sequence (h_n) of simple functions such that h_n converges to f a.e. If the measure is σ-finite then we may choose h_n so that it vanishes outside of a set of finite measure.*

A.3.8 Example. *A.e. convergence is not topological.* Consider the standard sequence in $L_0[0,1]$: let $s_0 = 0$ and $s_n = \sum_{k=1}^n \frac{1}{k}$ for $n \in \mathbb{N}$, so that $s_n \uparrow \infty$. Put $A_n = [s_{n-1}, s_n]$ mod 1. Then $A_n \subseteq [0,1]$ and the Lebesgue measure of A_n is $\frac{1}{n}$. Let $x_n = \mathbb{1}_{A_n}$. We will call (x_n) the ***typewriter sequence***. Clearly, (x_n) does not converge to zero a.e. However, it is easy to see that every subsequence of (x_n) has a further subsequence which converges to zero a.e. Proposition A.2.1 yields that a.e. convergence is not given by a topology.

A.3.2 Integration

Let h be a non-negative simple function, $h = \sum_{i=1}^n \alpha_i \mathbb{1}_{A_i}$. We write $\int h\,d\mu = \sum_{i=1}^n \alpha_i \mu(A_i)$. For a function $f\colon \Omega \to [0,\infty]$, we define $\int f\,d\mu$ to be the supremum of the integrals of simple functions h satisfying $0 \leqslant h \leqslant f$. We say that f is ***integrable*** if it is measurable and $\int f\,d\mu < \infty$. An arbitrary function f is integrable if f^+ and f^- are integrable; in this case, we define $\int f\,d\mu = \int f^+\,d\mu - \int f^-\,d\mu$. We write $\int f$ instead of $\int f\,d\mu$ when there is no risk of confusion.

A.3.9 Proposition. *Let f and g be integrable functions and $\lambda \in \mathbb{R}$.*

(i) *$f+g$ and λf are integrable and $\int(f+g) = \int f + \int g$ and $\int(\lambda f) = \lambda \int f$;*
(ii) *If $f \leqslant g$ then $\int f \leqslant \int g$;*
(iii) *If $|h| \leqslant |f|$ for some measurable h then h is integrable.*

A.3.10 Theorem (Fatou's Lemma)**.** *Let (f_n) be a sequence of positive measurable functions that converge a.e. to a function f. Then $\int f \leqslant \liminf \int f_n$.*

A.3.11 Theorem. *Let (f_n) be a sequence of positive measurable functions which converge a.e. to a function f and suppose that $f_n \leqslant f$ for all n. Then $\int f = \lim \int f_n$.*

Proof. By Fatou's lemma, we have

$$\int f \leqslant \liminf \int f_n \leqslant \limsup \int f_n \leqslant \int f.$$

□

A.3.12 Corollary (Monotone Convergence Theorem)**.** *Let (f_n) be an increasing sequence of positive measurable functions which converge a.e. to a function f. Then $\int f = \lim \int f_n$.*

A.3.13 Theorem (Lebesgue Dominated Convergence Theorem)**.** *Let g be an integrable function and (f_n) a sequence of measurable functions such that $|f_n| \leqslant g$ and $f_n \xrightarrow{\text{a.e.}} f$. Then $\int f = \lim \int f_n$.*

Proof. It follows from $|f| \leqslant g$ that f is integrable. The sequences $(g + f_n)$ and $(g - f_n)$ are positive and converge a.e. to $g + f$ and $g - f$, respectively. Now apply Fatou's Lemma to $(g + f_n)$ and $(g - f_n)$. □

For a measurable set A and a function f, we define $\int_A f = \int (f \cdot \mathbb{1}_A)$.

A.3.14. Let $0 \leqslant f \in L_0(\mu)$. The function $t \mapsto \mu(f > t)$ is called the ***distribution function*** of f; here $t \in [0, \infty)$. We have

$$\int f = \int_0^\infty \mu(f \geqslant t)\, dt = \int_0^\infty \mu(f > t)\, dt.$$

It follows easily (via a change of variable) that $\int f^p = p \int_0^\infty s^{p-1} \mu(f > s)\, ds$ when $p > 1$.

A.3.3 Baire and Borel measure

The following is based on Sections 13.1 and 13.2 of [Roy88] and Sections 7.1 and 7.3 of [Dud02]. While both [Roy88] and [Dud02] develop their theory for locally compact spaces, we only consider the case of compact spaces. Let K be a compact (Hausdorff) space. The smallest σ-algebra $\mathcal{B}$ of subsets of K which makes every function in $C(K)$ measurable is called the ***Baire*** σ-algebra. The σ-algebra $\mathcal{BO}$ generated by all closed (equivalently, all open) subsets of K is called the ***Borel*** σ-algebra. Members of $\mathcal{B}$ and $\mathcal{BO}$ are called Baire and Borel sets, respectively. It is easy to see that $\mathcal{B} \subseteq \mathcal{BO}$. A measure defined on $\mathcal{B}$ or on $\mathcal{BO}$ is called a Baire or a Borel measure, respectively.

Let μ be a measure defined on a σ-algebra $\mathcal{F}$ which contain $\mathcal{B}$. We say that μ is ***regular*** if for every measurable set A we have

$$\mu(A) = \inf\bigl\{\mu(U) \,:\, A \subseteq U,\ U \in \mathcal{F},\ U \text{ open}\bigr\}$$

or, equivalently

$$\mu(A) = \sup\bigl\{\mu(F) \,:\, F \subseteq A,\ F \in \mathcal{F},\ F \text{ closed}\bigr\}.$$

Every finite Baire measure on K is regular and extends to a unique regular Borel measure (though not every Borel measure needs to be regular). Thus, there is a one-to-one correspondence between Baire and regular Borel finite measures.

The following is a special case of Proposition 13.14 in [Roy88]:

A.3.15 Proposition. *A Borel measure μ on a compact space is regular iff for every open set U and every $\varepsilon > 0$ there exists a closed set $F \subseteq U$ with $\mu(U \setminus F) < \varepsilon$.*

A.4 Banach spaces

Throughout this section, X stands for a normed space. We write B_X for the closed unit ball of X and S_X for the unit sphere of X. Given normed spaces X and Y, we write $L(X,Y)$ for the space of continuous linear operators from X to Y. We write X^* for $L(X,\mathbb{R})$, the ***dual space*** of X. A ***Banach space*** is a complete normed space.

A function $p\colon V \to \mathbb{R}$, where V is a vector space, is called ***sublinear*** if

(i) $p(x+y) \leqslant p(x) + p(y)$ for all $x, y \in V$, and
(ii) $p(\lambda x) = \lambda p(x)$ for all $x \in V$ and $0 \leqslant \lambda \in \mathbb{R}$.

A.4.1 Theorem (Hahn-Banach)**.** *Let W be a subspace of a vector space V over $\mathbb{R}$, $p\colon V \to \mathbb{R}$ a sublinear map, and $f\colon W \to \mathbb{R}$ a linear functional such that $f(x) \leqslant p(x)$ for all $x \in W$. Then f extends to a linear functional $\tilde{f}\colon V \to \mathbb{R}$ such that $\tilde{f}(x) \leqslant p(x)$ for all $x \in V$.*

The following two results are corollaries of Hahn-Banach Theorem.

A.4.2 Theorem. *Let Y be a subspace of X and $f \in Y^*$. Then f extends to $\tilde{f} \in X^*$ with $\|\tilde{f}\| = \|f\|$.*

A.4.3 Theorem. *Let A be a non-empty convex closed set in a Banach space X and $x \notin A$. Then they can be strictly separated by a hyperplane, i.e., there exists $f \in X^*$ such that $f(x) > \sup f(A)$.*

A.4.4 Theorem (Banach)**.** *A continuous bijection between Banach spaces is an isomorphism.*

A.4.5 Theorem (Uniform Boundedness Principle)**.** *Let $\mathcal{A}$ be a subset of $L(X,Y)$, where X is a Banach space and Y is a normed space. If the set $\{Tx : T \in \mathcal{A}\}$ is bounded in Y for every $x \in X$ then $\mathcal{A}$ is bounded in $L(X,Y)$.*

A.4.6 Proposition. *Let $A \subseteq X$. A is relatively compact iff for every $\varepsilon > 0$ there exists a finite set F such that $A \subseteq F + \varepsilon B_X$.*

A.4.7 Corollary. *Let $A \subseteq X$. A is relatively compact if for every $\varepsilon > 0$ there exists a relatively compact set C such that $A \subseteq C + \varepsilon B_X$.*

Let X and Y be normed spaces and $T \in L(X, Y)$. We say that T is an ***isometric embedding*** if T if $\|Tx\| = \|x\|$ for every x; if, in addition, T is onto, we say that T is an ***isometry*** and that X and Y are ***isometric***. We say that T is an ***isomorphic embedding*** if there are positive constants c_1 and c_2 such that $c_1\|x\| \leqslant \|Tx\| \leqslant c_2\|x\|$ for all $x \in X$. If, in addition, T is onto, we say that T is an ***isomorphism*** and that X and Y are ***isomorphic***. For $T \in L(X, Y)$, its ***adjoint*** is the operator from Y^* to X^* defined by $T^*g = g \circ T$. For an operator T between Banach spaces, we have $\|T^*\| = \|T\|$, furthermore, T^* is surjective iff T is an isomorphic embedding; T is surjective iff T^* is an isomorphic embedding. If Y is a closed subspace of X and $Q \colon X \to X/Y$ is the quotient map then Q^* is an isometric embedding with $\operatorname{Range} Q^* = Y^\perp$, the set of all functionals in X^* that vanish on Y.

Given $x \in X$, we write $\hat{x}$ for the element of $X^{**} = (X^*)^*$ defined by $\hat{x}(f) = f(x)$ for every $f \in X^*$. The map $j(x) = \hat{x}$ defines an isometric embedding $j \colon X \to X^{**}$. In particular, X can be identified with a closed subspace of X^{**}. It is a common convention (though still an abuse of notation) to view X as a subset of X^{**}.

X is said to be reflexive when j is onto. X is reflexive iff X^* is reflexive. Every closed subspace of a reflexive Banach space is itself reflexive.

A net (x_α) in a normed space X converges ***weakly*** to x if $f(x_\alpha) \to f(x)$ for every $f \in X^*$; we write $x_\alpha \xrightarrow{\mathrm{w}} x$. We say that (x_α) is ***weakly null*** if $x_\alpha \xrightarrow{\mathrm{w}} 0$. Weak convergence defines ***weak topology*** on X. Symmetrically, a net (f_α) in X^* converges ***weak**** to f if $f_\alpha(x) \to f(x)$ for all $x \in X$; denoted $f_\alpha \xrightarrow{\mathrm{w}^*} f$. This defines the ***weak* topology*** on X^*. For a net (x_α) in X and $x \in X$, $x_\alpha \xrightarrow{\mathrm{w}} x$ in X iff $\hat{x}_\alpha \xrightarrow{\mathrm{w}^*} \hat{x}$ in X^{**}. That is, the map j is a homeomorphism between the weak topology on X and the restriction of the weak* topology of X^{**} to $j(X)$.

A.4.8 Proposition. *If $x_\alpha \xrightarrow{\mathrm{w}} x$ in X then $\|x\| \leqslant \limsup\|x_\alpha\|$. If $f_\alpha \xrightarrow{\mathrm{w}^*} f$ in X^* then $\|f\| \leqslant \limsup\|f_\alpha\|$.*

A.4.9 Proposition. *If $T \colon X \to Y$ is a continuous operator between normed spaces then T is weak-to-weak continuous.*

Proof. Suppose that $x_\alpha \xrightarrow{\mathrm{w}} x$ in X and $g \in Y^*$. Then $g(Tx_\alpha) = (T^*g)x_\alpha \to (T^*g)x = g(Tx)$. It follows that $Tx_\alpha \xrightarrow{\mathrm{w}} Tx$. □

A.4.10 Proposition. *A subset of X is weakly bounded iff it is bounded.*

Proof. It is easy to see that every bounded set is weakly bounded. Suppose that A is weakly bounded. Applying Uniform Boundedness Principle to the set $j(A) = \{\hat{x} : x \in A\}$ in X^{**}, we conclude that $j(A)$ and, therefore, A, is bounded. $\square$

A.4.11 Proposition. *Every convex closed subset of a normed space is weakly closed.*

A.4.12 Proposition. *Suppose that $x_n \xrightarrow{w} x$. Then there exists a sequence (y_n) such that y_n is a convex combination of $x_1 \dots, x_n$ for every n, and $y_n \to x$.*

A.4.13 Theorem. *In ℓ_1, weak and norm convergences agree on sequences.*

A.4.14 Theorem (Alaoglu)**.** *The unit ball of X^* is weak* compact.*

A.4.15 Proposition. *The unit ball of X is weakly closed. It is weakly compact iff X is reflexive.*

A.4.16 Theorem. *Let X be a Banach space and $\xi \in X^{**}$. Then ξ is weak* continuous as a functional on X^* iff $\xi = \hat{x}$ for some $x \in X$.*

A.4.17 Theorem. *Let A be a non-empty weak*-closed convex closed set in X^{**} and $\xi \notin A$. Then ξ may be strictly separated from A by an element of X^*. That is, there exists $f \in X^*$ such that $\langle f, \xi\rangle > \sup\bigl\{\langle f, \zeta\rangle : \zeta \in A\bigr\}$.*

(Theorems A.4.3 and A.4.17 are special cases of the fact that a closed convex set in a locally convex space may be strictly separated from a point by a continuous linear functional; see, e.g., Theorem 5.59 in [AB99].)

Let A be a subset of X. We write $\overline{A}^w$ for the ***weak closure*** of A, i.e., the closure of A in the weak topology. A is said to be ***weakly compact*** if it is compact with respect to the weak topology of X. Clearly, A is weakly compact iff every net in A has an accumulation point in A iff every net in A has a subnet which converges weakly to a point of A. A set A is ***relatively weakly compact*** if it is contained in a weakly compact set or, equivalently, if $\overline{A}^w$ is compact. A is relatively weakly compact iff every net in A has an accumulation point (which may not be in A) iff every net in A has a weakly convergent subnet (whose limit need not be in A). Every relatively weakly compact set is weakly bounded, and, therefore, is bounded by Proposition A.4.10. It follows from Proposition A.4.15 that a subset of a reflexive Banach space is relatively weakly compact iff it is bounded. Given a subset A of X^*, we write $\overline{A}^{w^*}$ for its weak* closure; we define (relatively) weak* compact sets in X^* in a similar fashion.

A.4.18 Proposition. *A subset A of X is relatively weakly compact iff A is bounded and the weak* closure of A in X^{**} is contained in X (that is, $\overline{j(A)}^{w^*} \subseteq j(X)$).*

Proof. Suppose that A is relatively weakly compact. Then $A \subseteq C$ for some weakly compact set C. It follows that $f(A) \subseteq f(C)$ and, therefore, $f(A)$ is bounded for every $f \in X^*$. Hence, A is weakly bounded and, therefore, bounded by Proposition A.4.10.

Suppose that $\xi \in \overline{A}^{w^*}$ in X^{**}. Then there is a net (x_α) in A such that $\hat{x}_\alpha \xrightarrow{\text{w}^*} \xi$ in X^{**}. Passing to a subnet, we may also assume that $x_\alpha \xrightarrow{\text{w}} x$ for some $x \in X$. It follows that $\hat{x}_\alpha \xrightarrow{\text{w}^*} \hat{x}$ in X^{**}, hence $\xi = \hat{x} \in j(X)$.

To prove the converse, note first that boundedness of A implies that $j(A)$ is relatively weak* compact by Alaoglu's Theorem A.4.14. Let (x_α) be a net in A. Then $(\hat{x}_\alpha)$ has a weak* accumulation point in X^{**}. By assumption, this accumulation point is, in fact, in X; hence it is a weak accumulation point of (x_α). $\square$

A.4.19 Proposition (Grothendieck)**.** *[FHH11, Lemma 13.32] Let $A \subseteq X$. If for every $\varepsilon > 0$ there exists a relatively weakly compact set C such that $A \subseteq C + \varepsilon B_X$ then A itself is relatively weakly compact.*

Proof. We use Proposition A.4.18. It is easy to see that A is bounded. Let $\xi \in \overline{A}^{w^*}$ in X^{**}. Then there exists a net (x_α) in A such that $\hat{x}_\alpha \xrightarrow{\text{w}^*} \xi$ in X^{**}. Fix $\varepsilon > 0$. Find a relatively weakly compact set C in X such that $A \subseteq C + \varepsilon B_X$. For each α, find $y_\alpha \in C$ with $\|x_\alpha - y_\alpha\| \leqslant \varepsilon$. Since C is relatively weakly compact, there exists a weak accumulation point y of (y_α). It follows from Proposition A.4.8 that $\|\hat{y} - \xi\| \leqslant \varepsilon$, hence $\operatorname{dist}(\xi, X) \leqslant \varepsilon$. Therefore, $\xi \in X$. $\square$

A.4.20 Theorem (Eberlein-Šmulian)**.** *A subset A of a Banach space is relatively weakly compact iff every sequence in A has a weakly convergent subsequence.*

A.4.21. It follows that relative weak compactness is a "sequential" property in the sense that one can determine whether a set A is relatively weakly compact by analyzing sequences in A; one does not have to consider nets. It follows, in particular, that A is relatively weakly compact iff every countable subset of A is relatively weakly compact.

Let X and Y be Banach space. We say that X ***contains a copy*** of Y if X has a closed subspace which is isomorphic to Y. Let $T \in L(X, Y)$. We say that T is ***compact*** if $T(B_X)$ is relatively compact in Y. Similarly, T is ***weakly compact*** if $T(B_X)$ is relatively weakly compact.

A.4.22 Theorem (Schauder)**.** *[FHH11, Theorem 15.3] Let $T\colon X \to Y$ be a bounded operator between Banach spaces. Then T is compact iff T^* is compact;*

A.4.23 Theorem (Gantmacher)**.** *[FHH11, Theorem 13.34] Let $T\colon X \to Y$ be a bounded operator between Banach spaces. Then T is weakly compact iff T^* is weakly compact.*

A Banach space is ***weakly sequentially complete*** if every weakly Cauchy sequence is weakly convergent. We prove in Corollary 9.8.11 that $L_1(\mu)$ is weakly sequentially complete.

A.4.24 Proposition. *c_0 is not weakly sequentially complete.*

Proof. Put $u_n = (\overbrace{1,\dots,1}^{n},0,\dots)$. For every positive $f \in \ell_1$, $f(u_n) \to \|f\|$. It follows that $f(u_n)$ converges for every $f \in \ell_1$, so that (u_n) is weakly sequentially Cauchy. However, (u_n) is not weakly convergent. Suppose $u_n \xrightarrow{\mathrm{w}} u$. For each k, let f_k be the the k-th unit vector basis of ℓ_1, then $f_k(u) = \lim_n f_k(u_n) = 1$. Hence, $u = (1,1,\dots)$; a contradiction to $u \in c_0$. $\square$

A.4.25 Proposition. *A Banach space X is weakly sequentially complete iff it is sequentially weak* closed in X^{**}.*

Proof. Let $j\colon X \to X^{**}$ be the canonical inclusion, $j(x) = \hat{x}$. We need to prove that X is weakly sequentially complete iff $j(X)$ is sequentially weak* closed in X. This is an easy corollary of the following observation: a sequence (x_n) in X is weakly Cauchy iff $(\hat{x}_n)$ converges weak* in X^{**}. Indeed, suppose that (x_n) is weakly Cauchy. By Uniform Boundedness Principle, it is bounded. Say, $\|x_n\| \leqslant M$ for all n. For every $f \in X^*$, put $\xi(f) = \lim_n f(x_n)$. Clearly, $\xi\colon X^* \to \mathbb{R}$ is linear. It follows from $\bigl|f(x_n)\bigr| \leqslant M\|f\|$ that $\|\xi\| \leqslant M$, hence $\xi \in X^{**}$. Clearly, $\hat{x}_n \xrightarrow{\mathrm{w}^*} \xi$. Conversely, it is easy to see that if $\hat{x}_n \xrightarrow{\mathrm{w}^*} \xi$ for some $\xi \in X^{**}$ then (x_n) is weakly Cauchy in X. $\square$

A.4.1 Bases and basic sequences

A sequence (x_i) in a Banach space X is a ***(Schauder) basis*** of X if every vector x in X admits a unique expansion of the form $x = \sum_{i=1}^\infty \alpha_i x_i$, where α_i's are scalars and the series converges in norm. We say that (x_i) is a ***basic sequence*** if it is a basis of its closed linear span $[x_i]$.

A.4.26 Theorem. *A sequence (x_i) of non-zero vectors in a Banach space X is basic iff there exists a constant $M \geqslant 1$ such that for every $n \leqslant m$ and every choice of coefficients $\alpha_1,\dots,\alpha_m$, one has*

$$\Bigl\|\sum_{i=1}^n \alpha_i x_i\Bigr\| \leqslant M \Bigl\|\sum_{i=1}^m \alpha_i x_i\Bigr\|. \tag{A.1}$$

If, in addition, $[x_i] = X$ then (x_i) is a basis of X.

The sequence shown on Figure ??? is a basis of $C[0,1]$; it is called the Schauder basis of $C[0,1]$.

A basic sequence is said to be ***monotone*** if one can take $M = 1$ in (A.1).

Let (x_i) be a basis of X. For every $n \in \mathbb{N}$, define maps $x_n^*\colon X \to \mathbb{R}$ and $P_n\colon X \to X$ via

$$x_n^*\Big(\sum_{i=1}^{\infty} \alpha_i x_i\Big) = \alpha_n \text{ and } P_n\Big(\sum_{i=1}^{\infty} \alpha_i x_i\Big) = \sum_{i=1}^{n} \alpha_i x_i.$$

Clearly, x_n^* and P_n are linear; x_n^* is called the ***n-th biorthogonal functional*** and P_n is called the ***n-th basis projection*** for the basis (x_n). It follows from Theorem A.4.26 that they are bounded and, moreover, uniformly bounded: $\|P_n\| \leqslant M$ and $\|x_n^*\| \leqslant 2M$ for every n.

Given basic sequences (x_n) and (y_n) in Banach spaces X and Y, respectively, we say that they are ***equivalent*** and write $(x_n) \sim (y_n)$ if for every sequence of scalars (α_n), the series $\sum_{n=1}^{\infty} \alpha_n x_n$ converges in X iff the series $\sum_{n=1}^{\infty} \alpha_n y_n$ converges in Y.

A.4.27 Proposition. *Let (x_n) and (y_n) be basic sequences in Banach spaces X and Y, respectively. TFAE:*

(i) $(x_n) \sim (y_n)$*;*

(ii) *There is an isomorphism from $[x_n]$ onto $[y_n]$ which maps x_n to y_n for every n;*

(iii) *There are constants $C_1, C_2 > 0$ such that*

$$C_1\Big\|\sum_{n=1}^{m} \alpha_n x_n\Big\| \leqslant \Big\|\sum_{n=1}^{m} \alpha_n y_n\Big\| \leqslant C_2\Big\|\sum_{n=1}^{m} \alpha_n x_n\Big\|$$

for any $m \in \mathbb{N}$ and any scalars $\alpha_1, \dots, \alpha_n$.

For the condition (iii) (and in similar statements where the precise value of the constants is irrelevant), one can use a shorthand notation

$$\Big\|\sum_{n=1}^{m} \alpha_n x_n\Big\| \lesssim \Big\|\sum_{n=1}^{m} \alpha_n y_n\Big\| \lesssim \Big\|\sum_{n=1}^{m} \alpha_n x_n\Big\|$$

or simply

$$\Big\|\sum_{n=1}^{m} \alpha_n x_n\Big\| \sim \Big\|\sum_{n=1}^{m} \alpha_n y_n\Big\|.$$

A basic sequence (x_i) is ***unconditional*** if the convergence of $\sum_{i=1}^{\infty} \alpha_i x_i$ implies the convergence of $\sum_{i=1}^{\infty} \pm\alpha_i x_i$ for any choice of signs or, equivalently, if the convergence of $\sum_{i=1}^{\infty} \alpha_i x_i$ implies the convergence of $\sum_{i=1}^{\infty} \beta_i x_i$ whenever $|\beta_i| \leqslant |\alpha_i|$ for every i.

A.4.28 Theorem. *A sequence (x_i) of non-zero vectors is an unconditional basic sequence iff there exists a constant $K > 0$ such that for every for every n, every scalars $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n$, if $|\alpha_i| \leqslant |\beta_i|$ for $i = 1, \dots, n$ then*

$$\Bigl\|\sum_{i=1}^{n} \alpha_i x_i\Bigr\| \leqslant K \Bigl\|\sum_{i=1}^{n} \beta_i x_i\Bigr\|.$$

We then say that (x_i) is K-***unconditional***.

A.4.29 Theorem (Principle of Small Perturbations)**.** *Let (x_n) be a seminormalized basic sequence. There exists a constant C such that every sequence (y_n) satisfying $\sum_{n=1}^{\infty} \|x_n - y_n\| < C$ is basic. Moreover, if $[x_n]$ is complemented then so is $[y_n]$.*

Here is how this principle is often used: suppose that (x_n) is a seminormalized basic sequence and (y_n) is another sequence such that $x_n - y_n \to 0$. Then (y_n) has a subsequence (y_{n_k}) which is basic and equivalent to (x_{n_k}). If $[x_{n_k}]$ is complemented then so is $[y_{n_k}]$.

For more details on bases and basic sequences, we refer the reader to [LT77].

A.4.2 $C(K)$, $L_p(\mu)$, and ℓ_p spaces

Let Ω be a topological space. We write $C(\Omega)$ for the vector space of all real-valued continuous functions on Ω. We will usually assume that Ω is Hausdorff. The space $C_b(\Omega)$ of all bounded continuous functions on Ω is a Banach space under the sup-norm $\|f\| = \sup\bigl\{|f(t)| \ : \ t \in \Omega\bigr\}$. In particular, if K is a compact topological space, then $C(K) = C_b(K)$, hence $C(K)$ is a Banach space under the sup-norm. In Theorem 8.4.2, we identify the dual space of $C(K)$.

A.4.30 Theorem (Arzelá–Ascoli)**.** *Let K be a compact Hausdorff space. A subset A of $C(K)$ is relatively compact iff it is equicontinuous and pointwise bounded.*

Suppose that Ω is ***locally compact***, i.e., every point has a compact neighborhood or, equivalently, there exists a compact space K such that Ω is a topological subspace of K and $K \setminus \Omega$ is a singleton. For $f \in C(\Omega)$, we say that f ***vanishes at infinity*** if for every $\varepsilon > 0$ there exists a compact set K such that $|f(t)| < \varepsilon$ whenever $t \notin K$. We write $C_0(\Omega)$ for the set of all such functions. Then $C_0(\Omega)$ is a Banach space under the sup-norm. Again, if K is compact then $C_0(K) = C(K)$.

A.4.31 Theorem (Banach-Mazur)**.** *Every separable Banach space embeds isometrically into $C[0,1]$.*

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, and consider the space $L_0(\mu)$ of all real-valued measurable functions on Ω. As usual, we identify functions that are equal almost everywhere. Let $0 < p < \infty$. $L_p(\mu)$ stands for the linear subspace of $L_0(\mu)$ consisting of all $f \in L_0(\mu)$ such that $\int |f|^p$ is finite. For $f \in L_p(\mu)$, we put $\|f\| = \bigl(\int |f|^p\bigr)^{\frac{1}{p}}$. If $1 \leqslant p < \infty$, this norm makes $L_p(\mu)$ into a Banach space.

We write $L_\infty(\mu)$ for the set of all $f \in L_0(\mu)$ such that $\|f\|_\infty < \infty$, where $\|f\|_\infty$ is the infimum of all $\lambda \in \mathbb{R}_+$ such that the $\mu\bigl(\{|f| > \lambda\}\bigr) = 0$. Under this norm, $L_\infty(\mu)$ is a Banach space.

We write $L_p[a, b]$ or $L_p(\mathbb{R})$ for the L_p-space on $[a, b]$ or $\mathbb{R}$, respectively, equipped with the Lebesgue measure. For $1 < p < \infty$, we write p^* for the ***conjugate*** of p defined by $\frac{1}{p} + \frac{1}{p^*} = 1$. Clearly, $p^{**} = p$. In addition, we define that 1 and ∞ are conjugates of each other.

If μ is σ-finite and $1 \leqslant p < \infty$ then the dual space of $L_p(\mu)$ may be identified with $L_q(\mu)$, where $q = p^*$, under the duality

$$\langle f, g\rangle = \int fg, \quad f \in L_p(\mu),\ g \in L_q(\mu).$$

This remains valid for non-σ-finite measures provided $1 < p < \infty$.

As we mentioned earlier, in this book we restrict our attention to measures which have no atoms of infinite measure. Here is another motivation for this: if a set A is an atom of a measure μ with $\mu(A) = \infty$ and $0 < p < \infty$ then every function f in $L_p(\mu)$ has to vanish on A. Thus, the "contribution" of A to $L_p(\mu)$ is trivial and replacing Ω with $\Omega \setminus A$ does not really affect $L_p(\mu)$.

Let Ω be an arbitrary set and $0 < p \leqslant \infty$. We write $\ell_p(\Omega)$ for $L_p(\mu)$ where μ the counting measure on Ω. In particular, if $\Omega = \{1, \dots, n\}$, $\ell_p(\Omega)$ is just $\mathbb{R}^n$ equipped with the norm $\|x\| = \Bigl(\sum_{i=1}^n |x_i|^p\Bigr)^{\frac{1}{p}}$ when $p < \infty$ and $\|x\| = \max|x_i|$ when $p = \infty$. We denote this space by ℓ_p^n. We write ℓ_p for $\ell_p(\mathbb{N})$. For $p < \infty$, it consists of all sequences (x_i) such that the series $\sum_{i=1}^\infty |x_i|^p$ converges, while ℓ_∞ is the space of all bounded sequences. Again, if $1 \leqslant p < \infty$ we have $(\ell_p)^* = \ell_q$ where $q = p^*$ and the duality is given by

$$\langle x, y\rangle = \sum_{i=1}^\infty x_i y_i, \quad x \in \ell_p,\ y \in \ell_q.$$

We write c for the space of all convergent real sequences and c_0 for the space of all real sequences which converge to zero. We equip c and c_0 with $\|\cdot\|_\infty$ and view them as closed subspaces of ℓ_∞. Note that $(c_0)^* = \ell_1$ under the same duality as above. As a Banach space, c is isomorphic to c_0. We write c_{00} for the vector space of all real-valued infinite sequences having finitely many non-zero terms.

It may be deduced from Hahn-Banach Theorem that there exists a bounded linear functional φ on ℓ_∞ which is invariant under translations and satisfies $\liminf_n x_n \leqslant \varphi(x) \leqslant \limsup_n x_n$ for every $x = (x_n) \in \ell_\infty$; in particular, $\varphi(x) = \lim_n x_n$ whenever $x \in c$. Such a functional is called a ***Banach limit***.

A.4.32. *ℓ_p-sums.* Let (X_n) be a sequence of Banach spaces and $1 \leqslant p \leqslant \infty$. By $\bigoplus_{\ell_p} X_n$ we mean the subspace of the Cartesian product $\prod_{n=1}^\infty X_n$ consisting of the sequences $x = (x_n)$, where $x_n \in X_n$ for every n, for which the sequence $\bigl(\|x_n\|_{X_n}\bigr)$ is in ℓ_p-summable; we then put the $\|x\|$ to be the ℓ_p-norm of this sequence. In a similar fashion, one can define $\bigoplus_{c_0} X_n$. The resulting spaces are again Banach spaces.

More generally, let $(X_\alpha)_{\alpha\in\Lambda}$ be a family of Banach spaces and $1 \leqslant p < \infty$; let X be the subspace of $\prod_{\alpha\in\Lambda} X_\alpha$ which consists of those families $x = (x_\alpha)$ for which the family of norm $\bigl(\|x_\alpha\|\bigr)$ belongs to $\ell_p(\Lambda)$. We put $\|x\|$ to be the norm of the family of norms in $\ell_p(\Lambda)$. It is easy to see that this may only happen when x has at most countable non-zero components, say, $(x_{\alpha_i})_{i=1}^\infty$, and $\|x\| = \Bigl(\sum_{i=1}^\infty \|x_{\alpha_i}\|^p\Bigr)^{\frac{1}{p}}$.

A.4.33 Theorem (Rosenthal's ℓ_1 Theorem)**.** *Every bounded sequence in a Banach space contains a subsequence that is either weakly Cauchy or is equivalent to the unit vector basis of ℓ_1.*

Appendix B

Solutions to exercises

1.1.1 The sets $\inf \bigcup \mathcal{C}$ and $\{\inf A \,:\, A \in \mathcal{C}\}$ have the same lower bounds.

1.1.2 $x_1 \leqslant y_1 \leqslant y_1 \vee y_2$. Similarly, $x_2 \leqslant y_1 \vee y_2$. It follows that $x_1 \vee x_2 \leqslant y_1 \vee y_2$. The proof for infima is similar.

1.1.5 The forward implication is trivial. To prove the converse, suppose that $A \leqslant z$. Then $A \leqslant x \wedge z \leqslant x$ implies $x \wedge z \leqslant x$, hence $x \leqslant z$ and, therefore, $x = \sup A$.

1.1.7 Obviously, (ii)⇒(i). To show that (i)⇒(iii), note that

$$a \leqslant b \quad \Rightarrow \quad a = a \wedge b \quad \Rightarrow \quad \varphi(a) = \varphi(a) \wedge \varphi(b) \quad \Rightarrow \quad \varphi(a) \leqslant \varphi(b).$$

On the other hand, if $\varphi(a) \leqslant \varphi(b)$ then $\varphi(a \vee b) = \varphi(a) \vee \varphi(b) = \varphi(b)$. Since φ is a bijection, it follows that $a \vee b = b$, so that $a \leqslant b$.

(iii)⇒(ii) Let $A \subseteq L$ with $x = \sup A$; we need to show that $\sup \varphi(A) = \varphi(x)$. Since $a \leqslant x$ for each $a \in A$, we have $\varphi(a) \leqslant \varphi(x)$, hence $\varphi(A) \leqslant \varphi(x)$. On the other hand, suppose that $\varphi(A) \leqslant u$ for some $u \in M$. Since φ is a bijection, there exists $y \in L$ such that $u = \varphi(y)$. Then $\varphi(a) \leqslant u$ for each $a \in A$ implies $a \leqslant y$, so that $x \leqslant y$ and, therefore, $\varphi(x) \leqslant u$. The proof that φ preserves infima is similar.

1.2.1(i) $x_1 + x_2 \leqslant y_1 + x_2 \leqslant y_2 + x_2$.

1.2.1(ii) It follows from $-x = y$ that $-x \geqslant 0$. Adding x to both sides, we get $0 \geqslant x$. Since also $0 \leqslant x$, we conclude that $x = 0$. Similarly, $y = 0$.

1.2.5 (i)⇒(ii) trivially. (ii)⇒(i): take $y = x - z$. (iv)⇔(iii)⇒(ii) trivially. (i)⇒(iii) because $-(x+y) \leqslant z \leqslant (x+y)$.

1.2.7 Suppose that X is directed and $x \in X$. Since X is directed there exists $u \in X$ such that $x \leqslant u$ and $0 \leqslant u$. Hence x is regular. Conversely, suppose that X_+ is generating and let $x, y \in X$. Since x and y are regular, there exist $u, v \in X_+$ such that $x \leqslant u$ and $y \leqslant v$. Then both x and y are less than or equal to $u + v$.

1.2.9 The sequence $(\frac{1}{n}u)$ is decreasing because $\frac{1}{n}u - \frac{1}{n+1}u = \frac{1}{n(n+1)}u \geqslant 0$ for every n. Suppose that X is Archimedean and $u \in X_+$. Clearly, $0 \leqslant \frac{1}{n}u$ for every n. Suppose that for all $n \in \mathbb{N}$ we have $x \leqslant \frac{1}{n}u$ and, therefore, $nx \leqslant u$. It follows that $x \leqslant 0$. Hence $0 = \inf \frac{1}{n}u$. Conversely, suppose that $\frac{1}{n}u \downarrow 0$ for every $u \in X_+$. Suppose that $x \in X$ and $u \in X_+$ are such that $nx \leqslant u$ for all $n \in \mathbb{N}$. It follows that $x \leqslant \frac{1}{n}u$ for all n, hence $x \leqslant \inf \frac{1}{n}u = 0$.

1.2.12 Suppose not. Since $\dim X > 1$, there exist two vectors x and y such that they are not scalar multiples of each other. Negating x or y, if necessary, we may assume that both x and y are positive. By the Archimedean property, there exists $n \in \mathbb{N}$ such that $x - ny$ is not positive. Let $\lambda = \sup\{t \geqslant 0 \,:\, x - ty \geqslant 0\}$. It follows that $x - ty$ is positive whenever $t < \lambda$ and negative whenever $t > \lambda$. In particular, for every $n \in \mathbb{N}$, we have $x - (\lambda - \frac{1}{n})y > 0$ and $x - (\lambda + \frac{1}{n})y < 0$. It follows that $n(\lambda y - x) < y$ and $n(x - \lambda y) < y$. By the Archimedean property, $\lambda y - x \leqslant 0$ and $x - \lambda y \leqslant 0$, so that $x = \lambda y$; a contradiction.

1.3.3(i) It follows from $x \wedge y \leqslant x, y$ that $-(x \wedge y) \geqslant -x, -y$. To show that $-(x \wedge y)$ is the least upper bound of $-x$ and $-y$, let $z \geqslant -x, -y$, then $-z \leqslant x, y$, so that $-z \leqslant (x \wedge y)$ and, therefore, $z \geqslant -(x \wedge y)$.

1.3.3(ii) It follows from $y \wedge z \leqslant y, z$ that $x + y \wedge z \leqslant x + y, x + z$, so that $x + y \wedge z \leqslant (x + y) \wedge (x + z)$. To show that $x + y \wedge z$ is the *greatest* lower bound of $x + y$ and $x + z$, suppose that $v \leqslant x + y$ and $v \leqslant x + z$ for some v, then $v - x \leqslant y, z$, hence $v - x \leqslant y \wedge z$ and, therefore, $v \leqslant x + y \wedge z$. The proof of the other identity is similar.

1.3.3(iii) Using (ii), we get $x \vee y - y = (x - y) \vee (y - y) = (x - y)^+$. Further, $x - x \wedge y = (-x) \vee (-y) - (-x) = \big((-y) - (-x)\big)^+ = (x - y)^+$.

1.3.3(iv) Follows immediately from (iii).

1.3.3(vi) Let $a = x \wedge (y \vee z)$ and $b = (x \wedge y) \vee (x \wedge z)$. It follows from $y \vee z \geqslant y$ that $a \geqslant x \wedge y$. Similarly, $a \geqslant x \wedge z$. Therefore, $a \geqslant (x \wedge y) \vee (x \wedge z)$, so that $a \geqslant b$.

To show that $a \leqslant b$, observe that by (iv) we have

$$y = x \wedge y + x \vee y - x \leqslant b + x \vee y \vee z - x.$$

Similarly, $z \leqslant b + x \vee y \vee z - x$. Hence, $y \vee z \leqslant b + x \vee y \vee z - x$, so that $b \geqslant x + y \vee z - x \vee y \vee z$. On the other hand, applying (iv) to x and $y \vee z$, we have

$$a = x \wedge (y \vee z) = x + y \vee z - x \vee (y \vee z) \leqslant b.$$

The proof of the other identity is similar.

1.3.5(i) Let $a_0 = \inf A$. By Exercise 1.3.4, we have

$$\inf A + \inf B = \inf(a_0 + B) = \inf_{b \in B}(b + \inf A) = \inf_{b \in B} \inf(b + A) = \inf_{b \in B} \inf_{a \in A}(a + b) = \inf(A + B).$$

1.3.5(ii) Let A and B be the sets of the terms of the nets. By (i), we have $x + y = \inf(A + B) = \inf\{a_\alpha + b_\beta \,:\, \alpha, \beta\}$. Since the nets are monotone, the latter set has the same lower bounds as the net $a_\alpha + b_\alpha$.

1.3.6 x^+ and x^- are positive by definition. It follows from $|x| \geqslant \pm x$ that $0 = x + (-x) \leqslant 2|x|$, hence $|x| \geqslant 0$. Using Exercise 1.3.3, we get $x^+ - x^- = x \vee 0 - (-x) \vee 0 = x \vee 0 + x \wedge 0 = x + 0 = x$. Further, $x + |x| = x + x \vee (-x) = (2x) \vee (x - x) = 2(x \vee 0) = 2x^+$. It now follows from $x + |x| = 2x^+$ and $x = x^+ - x^-$ that $|x| = x^+ + x^-$. Observe that

$$x^+ \vee x^- = (x \vee 0) \vee \big((-x) \vee 0\big) = x \vee (-x) \vee 0 = |x| \vee 0 = |x| \geqslant x.$$

It follows that $0 \leqslant x^+ \wedge x^- = x - x^+ \vee x^- \leqslant 0$ and, therefore, $x^+ \wedge x^- = 0$. The identity $|\lambda x| = |\lambda||x|$ follows from Exercise 1.3.3(v).

1.3.7 Exercise 1.3.3(iv) yields $x^+ = (u - v)^+ = u - u \wedge v = u$. Similarly, $x^- = v$.

1.3.11(iii) It follows from $x \leqslant x^+$ and $y \leqslant y^+$ that $x+y \leqslant x^+ + y^+$, so that $(x+y)^+ \leqslant (x^+ + y^+)^+ = x^+ + y^+$;

1.3.11(iv) Note that $(a+b) \wedge a = a$. Applying Exercise 1.3.3(iii), we get

$$x \wedge (a+b) - x \wedge a = x \wedge (a+b) - x \wedge (a+b) \wedge a = \big[x \wedge (a+b) - a\big]^+ \leqslant \big[(a+b) - a\big]^+ = b,$$

Also, we clearly have $x \wedge (a+b) - x \wedge a \leqslant x \wedge (a+b) \leqslant x$. Combining the two inequalities, we get $x \wedge (a+b) - x \wedge a \leqslant x \wedge b$.

1.3.13 Let $x, y \in Z$; put $z = \frac{1}{2}(x+y) + \frac{1}{2}|x-y|$. We claim that $z = x \vee y$. Note first that

$$z \geqslant \tfrac{1}{2}(x+y) + \tfrac{1}{2}(x-y) = x \quad \text{and} \quad z \geqslant \tfrac{1}{2}(x+y) - \tfrac{1}{2}(x-y) = y.$$

Suppose that $u \geqslant x, y$. Note that $u \geqslant x$ yields $u - \frac{1}{2}(x+y) \geqslant \frac{1}{2}(x-y)$ while $u \geqslant y$ yields $u - \frac{1}{2}(x+y) \geqslant \frac{1}{2}(y-x)$. It follows that $u - \frac{1}{2}(x+y) \geqslant \frac{1}{2}|x-y|$, so that $u \geqslant z$. The proof of the other identity is similar.

1.3.14 The forward implication is immediate. Suppose that Z_+ is generating and is a lattice; let $z \in Z$. By Exercise 1.3.13, it suffices to show that $z \vee (-z)$ exists in Z. Since Z_+ is generating, z is regular and, by Exercise 1.2.5(iii) we can find x such that $x \pm z \geqslant 0$. Since Z_+ is a lattice, $(x+z) \vee (x-z)$ exists. Using a distributive law, we conclude that $(x+z) \vee (x-z) - x = z \vee (-z)$ exists.

1.3.15 Those of the form $y = \alpha x$ with $\alpha \geqslant 0$, and the vertical axis.

1.3.16 The forward implication is trivial. To prove the converse, let $x, u \in X$ such that $nx \leqslant u$ for all n. Then $nx^+ \leqslant u^+$ for every n, hence $x^+ = 0$ and $x = -x^- \leqslant 0$.

1.3.18 Note that $x \geqslant 0$ iff $x \in C$ iff $m(x) = x$. To show that X is a vector lattice, it suffices to prove that $x^+ = x \vee 0$ exists for every $x \in X$. Let $x \in X$. Since C is generating, $x = y - z$ for some $y, z \geqslant 0$. It follows from $m(y) \leqslant m(y-z) + m(z)$ that $x = y - z = m(y) - m(z) \leqslant m(y-z) = m(x)$. Similarly, $-x \leqslant m(-x) = m(x)$.

Put $v = \frac{m(x)+x}{2}$. We claim that $v = x^+$. It follows from $\pm x \leqslant m(x)$ that $v \geqslant \frac{x+x}{2} = x$ and $v \geqslant \frac{-x+x}{2} = 0$. Hence, v is an upper bound of $\{x, 0\}$. Suppose that $u \geqslant x$ and $u \geqslant 0$. Then

$$m(x) \leqslant m(u) + m(x-u) = m(u) + m(u-x) = u + (u-x) = 2u - x.$$

It follows that $u \geqslant \frac{m(x)+x}{2} = v$.

This proves that X is a vector lattice. Under the same notation as above, $\frac{m(x)+x}{2} = v = x^+ = \frac{|x|+x}{2}$; it follows that $m(x) = |x|$.

1.3.22 Suppose that there exist $x \in X$ and $u \in X_+$ such that $nx \leqslant u$ for all $n \in \mathbb{N}$. Taking the positive parts, we get $nx^+ \leqslant u$, hence $n\|x^+\| = \|nx^+\| \leqslant \|u\|$ for all n. It follows that $x^+ = 0$, so that $x \leqslant 0$.

1.4.4 WLOG, $f \geqslant 0$. For every $\lambda > 0$ we have $\infty > \int f^p \geqslant \int_{\{f \geqslant \lambda\}} f^p \geqslant \lambda^p \mu\{f \geqslant \lambda\}$, so that $\mu\{f \geqslant \lambda\} < \infty$. The result now follows from $\operatorname{supp} f = \bigcup_{n=1}^\infty \{f \geqslant \frac{1}{n}\}$.

1.4.10 Being a subspace of $\mathbb{R}^{[0,1]}$, $C^1[0,1]$ is an Archimedean ordered vector space. Observe that $g \in C^1[0,1]$ defined via $g(\omega) = \omega - \frac{1}{2}$ does not have a modulus. Indeed, suppose that $|g|$ exists. One can easily find a decreasing sequence (f_n) in $C^1[0,1]$ such that $f_n(\omega) \downarrow \big|g(\omega)\big|$ for every $\omega \in [0,1]$. It follows from $\pm g \leqslant f_n$ that $|g| \leqslant f_n$ for all n, hence $|g|(\omega) = \big|g(\omega)\big|$ for every $\omega \in [0,1]$. But then $|g| \notin C^1[0,1]$, contradiction.

1.4.11 It follows from Exercise 1.3.22 that this space admits no lattice norms.

1.4.15 In the case $p = \infty$, the order interval $[-\mathbb{1}, \mathbb{1}]$ is the unit ball of the space, hence not compact. Let $1 \leqslant p < \infty$; consider an order interval $[a, b]$ in ℓ_p. Let (x_n) be a sequence in $[a, b]$. First, using a diagonal process we find a subsequence of (x_n) that converges coordinate-wise. That is, find first a subsequence $(x_n^{(1)})$ of (x_n) which converges in the first coordinate, then a a subsequence $(x_n^{(2)})$ of $(x_n^{(1)})$ which converges in the second coordinate, etc; now consider $y_n = x_n^{(n)}$; it is easy to see that (y_n) converges coordinate-wise to some sequence y. Clearly, $a_n \leqslant y_n \leqslant b_n$ for each n; it follows that $y \in \ell_p$. It is left to show that $y_n \to y$ (in norm). This could be done directly; alternatively, one can view ℓ_p as $L_p(\mathbb{N}, \mu)$, where μ is the counting measure, and apply Lebesgue Dominated Convergence Theorem A.3.13 to show that the sequence $|y_n - y|^p$ converges in $L_1(\mu)$-norm to zero, which implies norm convergence $y_n \to y$ in ℓ_p.

1.4.16 Let $\|\cdot\|$ be a lattice norm on $\mathbb{R}^{\mathbb{N}}$. For every $n \in \mathbb{N}$, put $r_n = \|e_n\|$. Let $x \in \mathbb{R}^{\mathbb{N}}$ such that its n-th component $x_n = \frac{n}{r_n}$. Then $0 \leqslant \frac{n}{r_n} e_n \leqslant x$ implies $\|x\| \geqslant \|\frac{n}{r_n} e_n\| = n$ for every n, which is a contradiction.

1.5.3(ii) Using the triangle inequality and Exercise 1.3.8(iii), we have

$$0 \leqslant |x| + |y| - |x + y| \leqslant |x| + |y| - \big||x| - |y|\big| = 2\big(|x| \wedge |y|\big) = 0.$$

It follows that $|x + y| = |x| + |y|$. It follows from Exercise 1.3.3(iv) that $|x| + |y| = |x| \vee |y|$.

1.5.3(iii) $0 \leqslant |x| \wedge |z| \leqslant |x| \wedge |y| = 0$, so that $|x| \wedge |z| = 0$.

1.5.3(iv) Using Exercise 1.3.11(iv), we get

$$0 \leqslant |x| \wedge |y + z| \leqslant |x| \wedge \big(|y| + |z|\big) \leqslant |x| \wedge |y| + |x| \wedge |z| = 0,$$

so that $x \perp (y + z)$. Similarly, $x \perp \big(|y| + |z|\big)$. Finally, observe that $|y \vee z| \leqslant |y| \vee |z| \leqslant |y| + |z|$ and apply (iii).

1.5.4 Let $z = x - y$. By Exercise 1.3.3(iii), we have

$$x - x \wedge y = (x - y)^+ = z^+ \quad \text{and} \quad y - x \wedge y = (y - x)^+ = (-z)^+ = z^-.$$

The result now follows from Exercise 1.3.6.

1.5.6 Applying Exercise 1.5.3(i) inductively, we conclude that the vectors $\alpha_1 x_1, \dots, \alpha_n x_n$ are (pairwise) disjoint. Applying Exercise 1.5.3(iv) inductively, we conclude that if $x \perp x_i$ for some x and for each $i = 1, \dots, n$ then $x \perp \big(\sum_{i=1}^n x_i\big)$. Now we are ready to prove the required statement; the proof is by induction on n. Suppose the statement is true for n, prove it now for $n + 1$. Suppose $x_1, \dots, x_{n+1}$ be a disjoint collection and $\alpha_1, \dots, \alpha_{n+1}$ be scalars. We already know that the vectors $\alpha_1 x_1, \dots, \alpha_{n+1} x_{n+1}$ are disjoint. It follows that $\Big(\sum_{i=1}^n \alpha_i x_i\Big) \perp (\alpha_{n+1} x_{n+1})$. Applying Exercise 1.5.3(ii) and the induction hypothesis, we get

$$\Big|\sum_{i=1}^{n+1} \alpha_i x_i\Big| = \Big|\sum_{i=1}^{n} \alpha_i x_i\Big| + |\alpha_{n+1} x_{n+1}| = \sum_{i=1}^{n} |\alpha_i||x_i| + |\alpha_{n+1}||x_{n+1}| = \sum_{i=1}^{n+1} |\alpha_i||x_i|.$$

Similarly,

$$\Big|\sum_{i=1}^{n+1} \alpha_i x_i\Big| = \Big|\sum_{i=1}^{n} \alpha_i x_i\Big| \vee |\alpha_{n+1} x_{n+1}| = \Big(\bigvee_{i=1}^{n} |\alpha_i||x_i|\Big) \vee |\alpha_{n+1}||x_{n+1}| = \bigvee_{i=1}^{n+1} |\alpha_i||x_i|.$$

1.5.7 Let $x_1,\dots,x_n$ be a disjoint collection and $\alpha_1,\dots,\alpha_n$ scalars such that $\sum_{i=1}^n \alpha_i x_i = 0$. Then

$$0 = \Big|\sum_{i=1}^n \alpha_i x_i\Big| = \bigvee_{i=1}^n |\alpha_i||x_i|.$$

It follows that for each i one has $|\alpha_i||x_i| = 0$, so that $\alpha_i = 0$.

1.5.8 Suppose that (x_n) is a disjoint sequence and $x_n \to x$. Since $|x_n| \wedge |x_m| = 0$ when $n \neq m$, passing to the limit on n we get $|x| \wedge |x_m| = 0$ for every m. Now passing to the limit on m we get $|x| \wedge |x_| = 0$, hence $x = 0$.

1.5.10 Consider the map that sends an element $x \in X$ with $x = \sum_{i=1}^\infty \alpha_i e_i$ to the sequence (α_i) in $\mathbb{R}^{\mathbb{N}}$. This allows us to view X as a subspace of $\mathbb{R}^{\mathbb{N}}$. The order we defined on X agrees with the one inherited from $\mathbb{R}^{\mathbb{N}}$. In order to show that X is a vector lattice, it suffices to show that it is a sublattice of $\mathbb{R}^{\mathbb{N}}$.

Let $x \in X$ with $x = \sum_{i=1}^\infty \alpha_i e_i$. It follows from the 1-unconditionality that $\Big\|\sum_{i=n}^m |\alpha_i| e_i\Big\| = \Big\|\sum_{i=n}^m \alpha_i e_i\Big\|$ whenever $n \leqslant m$. This implies that the series $\sum_{i=n}^\infty |\alpha_i| e_i$ converges in X. The latter series corresponds to $|x|$ in $\mathbb{R}^{\mathbb{N}}$. Hence, X is a sublattice of $\mathbb{R}^{\mathbb{N}}$ and, therefore, a vector lattice.

To show that X is a Banach lattice, suppose that $\Big|\sum_{i=1}^\infty \alpha_i e_i\Big| \leqslant \Big|\sum_{i=1}^\infty \beta_i e_i\Big|$. It follows that $\sum_{i=1}^\infty |\alpha_i| e_i \leqslant \sum_{i=1}^\infty |\beta_i| e_i$. We then conclude that $|\alpha_i| \leqslant |\beta_i|$ for every i. 1-unconditionality yields $\Big\|\sum_{i=1}^n \alpha_i e_i\Big\| \leqslant \Big\|\sum_{i=1}^n \beta_i e_i\Big\|$ for every n. Passing to the limit on n, we get $\Big\|\sum_{i=1}^\infty \alpha_i e_i\Big\| \leqslant \Big\|\sum_{i=1}^\infty \beta_i e_i\Big\|$.

1.5.13(i) $\mathbb{1}$ is a strong unit because every function in $C(K)$ is bounded. Let $0 \leqslant f \in C(K)$. Observe that f is a strong unit iff $f \geqslant \lambda\mathbb{1}$ for some $\lambda > 0$ iff $f(t) > 0$ for every $t \in K$.

1.5.13(iii) If $f \in C_b(\Omega)$ then there exists $\lambda > 0$ such that $|f(t)| \leqslant \lambda$ for all $t \in \Omega$; it follows that $|f| \leqslant \lambda\mathbb{1}$. On the other hand, $\mathbb{1}$ need not be a strong unit in $C(\Omega)$. Take, for example, $\Omega = \mathbb{R}$ and $f(t) = t$. Nevertheless, $\mathbb{1}$ is a weak unit in $C(\Omega)$ as its support is all of Ω.

1.5.13(v) It is easy to see that every function f in $C_0(\mathbb{R})$ which vanishes nowhere is a weak unit. We claim that $C_0(\mathbb{R})$ has no strong units. Indeed, assume that u is a strong unit in $C_0(\mathbb{R})$. Put $v = \sqrt{u}$, then $v \in C_0(\mathbb{R})$, hence $v \leqslant \lambda u$ for some positive λ. This yields $u \geqslant \frac{1}{\lambda^2}\mathbb{1}$, which contradicts $u \in C_0(\mathbb{R})$.

1.5.13(vii) Let $0 < p < \infty$, and suppose that $e > 0$ is a strong unit in $L_p[0,1]$. Fix an arbitrary $M \in \mathbb{R}_+$. Pick $a \in [0,1]$ and define $f_a(t) = |t-a|^{-\frac{1}{2p}}$, then $f \in L_p[0,1]$. Then there exists $\lambda_a \in \mathbb{R}_+$ such that $f_a \leqslant \lambda_a e$. Put $I_a = \big\{t \,:\, f_a(t) > \lambda_a M\big\}$. Then I_a is a non-empty open interval centred at a, and for almost all $t \in I_a$ we have $\lambda_a e(t) \geqslant f_a(t) \geqslant \lambda_a M$, so that $e(t) \geqslant M$. As a runs over $[0,1]$, the intervals I_a form an open cover. Hence, there is a finite sub-cover. It follows that $e(t) \geqslant M$ for almost all $t \in [0,1]$, hence $e \geqslant M\mathbb{1}$. Since M was chosen arbitrarily, this is a contradiction. Finally, the same proof works for $p = 0$ if one takes $f_a(t) = \frac{1}{|t-a|}$.

1.5.13(x) Suppose $f \in X$ is a weak unit. Then $\operatorname{supp} f$ has full measure. Since the support of every function is σ-finite by Exercise 1.4.4, it follows that μ is σ-finite.

Conversely, suppose that μ is σ-finite. Let (Ω_n) be a disjoint sequence of sets in $\mathcal{F}$ of finite measure such that $\bigcup_{n=1}^\infty \Omega_n = \Omega$. Let $f = \sum_{n=1}^\infty \lambda_n \mathbb{1}_{\Omega_n}$, where λ_n is a sequence of positive reals which converges to zero sufficiently fast that $\sum_{n=1}^\infty \lambda^p \mu(\Omega_n)$ converges, so that $f \in L_p(\mu)$. Clearly, $\operatorname{supp} f = \Omega$, so that f is a weak unit.

1.5.13(xi) It is a classical fact of analysis that there exists a closed subset F of $[0,1]$ such that F has positive Lebesgue measure yet empty interior; see, e.g., Exercise 14a in [Roy88, Section 3.3]. Let $f \in C[0,1]$ such that $\ker f = F$; for example, one can define $f(t)$ to be the distance from t to F. No non-zero continuous function is disjoint to f because F has empty interior; thus, f is a weak unit in $C[0,1]$. However, f is not a weak unit in L_p because it vanishes on a set of positive measure.

1.5.17 Take $u_{k+1} = x+y-\sum_{i=1}^{k} u_i$. Then $u_{k+1} \geqslant 0$ and $\sum_{i=1}^{k+1} u_i = x+y$. Now apply Theorem 1.5.16

1.5.18 Applying Theorem 1.5.16 to the equality $x^+ + x^- = u+v$, we find four positive vectors vectors a, b, c, and d such that $u = a+b$, $v = c+d$, $x^+ = a+c$, and $x^- = b+d$. Put $y = a-b$ and $z = c-d$. Then $y+z = x^+ - x^- = x$. It follows from $0 \leqslant a \leqslant x^+$ and $0 \leqslant b \leqslant x^-$ that $a \perp b$ and, therefore, $|y| = |a-b| = a+b = u$. Similarly, $c \perp d$, and, therefore, $|z| = v$.

1.5.20 Apply Riesz Decomposition Theorem.

1.6.3(i) It follows from $x \vee y \geqslant x, y$ that $T(x \vee y) \geqslant Tx, Ty$, hence $T(x \vee y) \geqslant (Tx) \vee (Ty)$.

1.6.3(iii) $T|x| = T\big(x \vee (-x)\big) \geqslant (Tx) \vee (-Tx) = |Tx|$.

1.6.8 Let $T\colon X \to Y$ be a lattice homomorphism between vector lattices. Then for every $y \in \operatorname{Range} T$ we have $y = Tx$ for some $x \in X$, so that $|y| = |Tx| = T|x| \in \operatorname{Range} T$. Hence $\operatorname{Range} T$ is a sublattice.

Next, let Z be a sublattice of Y and $x \in T^{-1}(Z)$. Then $Tx \in Z$, so that $T|x| = |Tx| \in Z$, hence $|x| \in T^{-1}(Z)$. It follows that $T^{-1}(Z)$ is a sublattice.

1.6.9 If $Sx = Tx$ then $|Sx| = |Tx|$ implies $S|x| = T|x|$.

1.6.10 Suppose that T is a lattice homomorphism and $a \perp b$ for some $a, b \in X$. Then $|Ta| \wedge |Tb| = T\big(|a| \wedge |b|\big) = 0$.

Suppose now that $a \perp b$ implies $Ta \perp Tb = 0$ for all $a, b \in X_+$. Let $x \in X$. It follows from $x^+ \wedge x^- = 0$ that $Tx^+ \wedge Tx^- = 0$. Applying Exercise 1.5.3(ii), we get

$$|Tx| = |Tx^+ - Tx^-| = |Tx^+| + |Tx^-| = Tx^+ + Tx^- = T|x|.$$

1.6.11 Let $a \geqslant 0$. Since $T \geqslant 0$, we have $T[0,a] \subseteq [0, Ta]$. To prove the converse inclusion, let $y \in [0, Ta]$. Since T is surjective, $y = Tx$ for some x. Put $z = x^+ \wedge a$. Then $z \in [0,a]$ and $Tz = (Tx)^+ \wedge Ta = y^+ \wedge a = y$.

1.6.16 (i)⇒(ii) Whenever $i \neq j$ we have $0 = T(e_i \wedge e_j) = Te_i \wedge Te_j$. Hence, the columns of T are pairwise disjoint. Since they are non-zero, each column has exactly one non-zero entry. Also, by Theorem 1.6.13(iv), $T \geqslant 0$ and $T^{-1} \geqslant 0$, so that $T^t \geqslant 0$ and $(T^t)^{-1} \geqslant 0$, where T^t stands for the transpose matrix of T, hence T^t is a lattice isomorphism. By the preceding argument, T^t has exactly one non-zero entry in each column, so that T has exactly one non-zero entry in each row.

Implications (ii)⇒(iii) and (iii)⇒(i) are straightforward.

2.1.3 The implications (i)⇒(ii)⇒(iii) are trivial. It is left to prove (iii)⇒(i). Assume (iii) and suppose that $x_\alpha \uparrow\leqslant u$. Consider a tail $(x_\alpha)_{\alpha \geqslant \alpha_0}$ for a fixed α_0; it is clear that $\sup x_\alpha$ exists iff $\sup_{\alpha \geqslant \alpha_0} x_\alpha$ exists. The net $(x_\alpha - x_{\alpha_0})_{\alpha \geqslant \alpha_0}$ is increasing, positive, and bounded above by $u - x_{\alpha_0}$, hence $z \coloneqq \sup_{\alpha \geqslant \alpha_0} (x_\alpha - x_{\alpha_0})$ exists. It follows from Exercise 1.3.4(iii) that $\sup_{\alpha \geqslant \alpha_0} x_\alpha = x_{\alpha_0} + z$.

2.1.4 The equivalences (i)⇔(ii) and (iii)⇔(iv) follow easily from exercise 1.3.4(i). The implications (i)⇒(iii)⇒(v) are trivial. To show that (iii)⇒(i), note that given any sequence (x_n) which is bounded above by some u, one can define a new sequence $y_n = x_1 \vee \cdots \vee x_n$; then $y_n \uparrow\leqslant u$ and $\sup y_n = \sup x_n$. To show that (v)⇒(iii), observe that given $x_n \uparrow\leqslant u$, the sequence $(x_n - x_1)$ is positive, increasing, and bounded above by $u - x_1$, hence $\sup_n (x_n - x_1)$ exists by (v). It now follows from Exercise 1.3.4(iii) that $\sup x_n$ exists.

2.1.7(i) Let $A \subseteq \mathbb{R}^\Omega_+$ with A non-empty and $A \leqslant u$ for some $u \in \mathbb{R}^{\not\leqslant}$. Consider the point-wise supremum of A: for every $t \in \Omega$, define $s(t) = \sup\big\{a(t) \,:\, a \in A\big\}$. Note that the supremum exists because the set $\big\{a(t) \,:\, a \in A\big\}$ is bounded above by $u(t)$. It is easy to see that $s = \sup A$.

2.1.7(ii) We will prove the statement for ℓ_p; the proofs for c_0 is similar. Let $A \in (\ell_p)_+$ be non-empty and $A \leqslant u$ for some $u \in \ell_p$. Then $u \geqslant 0$. For every $i \in \mathbb{N}$, the set $\{a_i : a \in A\}$ is bounded above by u_i, where a_i is the i-th component of a. Then $s_i \coloneqq \sup\{a_i : a \in A\}$ exists and $0 \leqslant s_i \leqslant u_i$. It follows that $s \in \ell_p$. It is now clear that $s = \sup A$.

2.1.10 X is a sublattice of $\mathbb{R}^\Omega$; hence X is a vector lattice. To check that X is σ-order complete, suppose that $0 \leqslant f_n\!\uparrow \leqslant f_0$ as $n = 1, 2, \dots$ in X. Let f be the supremum of $(f_n)_{n=1}^\infty$ in $\mathbb{R}^\Omega$, i.e., the point-wise supremum. Then $f_n \leqslant f \leqslant f_0$ for every $n \geqslant 1$. It suffices to show that $f \in X$, then $f = \sup_{n\geqslant 1} f_n$ in X. Suppose $f_n = \lambda_n \mathbb{1} + g_n$ such that $\operatorname{supp} g_n$ is countable as $n = 0, 1, \dots$. Let $A = \bigcup_{n=0}^\infty \operatorname{supp} g_n$; then A is countable, hence its complement A^C is non-empty. It follows that each f_n equals $\lambda_n \mathbb{1}$ on A^C, hence $0 \leqslant \lambda_n\!\uparrow \leqslant \lambda_0$, so that $\lambda = \sup_{n\geqslant 1} \lambda_n$ exists and f agrees with $\lambda\mathbb{1}$ on A^C. It follows that $f \in X$.

To show that X is not order complete, take $A \subset \Omega$ such that both A and A^C are uncountable. Put $F = \{\mathbb{1}_{\{\omega\}} : \omega \in A\}$. Then $F \leqslant \mathbb{1}$. However, we claim that $\sup F$ does not exist. Suppose $f = \sup F$. It follows from $F \leqslant \mathbb{1}$ that $f \leqslant \mathbb{1}$. Then $f = \lambda\mathbb{1} + g$ with $\operatorname{supp} g$ countable. We claim that $\lambda = 1$. Indeed, take any $\omega \in A \setminus \operatorname{supp} g$; such an ω exists because $\operatorname{supp} g$ is countable while A is not; then $\lambda = f(\omega)$. It follows from $\mathbb{1}_{\{\omega\}} \leqslant f \leqslant \mathbb{1}$ that $\lambda = 1$. On the other hand, pick any $\omega \in A^C \setminus \operatorname{supp} g$ and put $h = f - \mathbb{1}_{\{\omega\}}$. Then $F \leqslant h < f$; a contradiction.

2.1.18 We may view $D^\vee$ as an increasing net. As in Exercise 2.1.3, we may assume that this net is positive. The statement follows from a careful analysis of the proof of Theorem 2.1.16. In *Special case*, (f_n) is the required sequence. In *General case*, we have $\sup D = \sup h_m = \sup_m \sup D_m$. By the preceding, for each m we can find a sequence $(f_n^{(m)})$ in D_m so that $\sup D_m = \sup_n f_n^{(m)}$. It follows that $\sup D$ is the supremum of the double sequence $(f_n^{(m)})_{m,n}$. Rearrange this double sequence into a sequence (y_k) and then take $x_k = y_1 \vee \dots \vee y_k$, then $x_k \uparrow$ and $\sup D = \sup_{m,n} f_n^{(m)} = \sup y_k = \sup x_k$.

2.1.21 One way to see this is to observe that $\mathbb{R}^\mathbb{N} = L_0(\mathbb{N}, \mu)$, where μ is the counting measure on $\mathbb{N}$, and apply Remark 2.1.19. It can also be shown directly as follows. Let $D \subseteq \mathbb{R}^\mathbb{N}$ with $\inf D = 0$; we need to find a countable subset C of D such that $\inf C = 0$. For every $n, m \in \mathbb{N}$, $\frac{1}{m} e_n$ is not a lower bound for D, hence there exists $x_{n,m} \in D$ such that $\frac{1}{m} e_n \not\leqslant x_{n,m}$, that is, the n-th component of $x_{n,m}$ is less than $\frac{1}{m}$. We can now take $C = \{x_{n,m} : n, m \in \mathbb{N}\}$.

2.1.24 (i)$\Rightarrow$(ii) Let $U = \bigcup_{n=1}^\infty F_n$, where each F_n is closed. Find an open set V_1 such that $F_1 \subseteq V_1 \subseteq \overline{V_1} \subseteq U$ (as in Proposition A.2.3). Since $F_2 \cup \overline{V_1}$ is a closed subset of U, we can find an open set V_2 such that $F_2 \cup \overline{V_1} \subseteq V_2 \subseteq \overline{V_2} \subseteq U$. Proceeding inductively, we get a required sequence.

(ii)$\Rightarrow$(iii) By Urysohn's Lemma, for every n we find $f_n \in C(K)$ such that $0 \leqslant f_n \leqslant \mathbb{1}$, f_n equals 1 on $\overline{V_n}$, and vanishes outside of V_{n+1}. Put $f = \sum_{n=1}^\infty \frac{1}{2^n} f_n$; since $C(K)$ is a Banach space and this series converges absolutely, it also converges in norm. It is easy to see that $U = \{f \neq 0\}$.

(iii)$\Rightarrow$(i) Observe that $U = \bigcup_{n=1}^\infty \{|f| \geqslant \frac{1}{n}\}$.

2.1.25 The first implication is trivial. Suppose that K is basically disconnected. Let $t \in K$ and V an open neighborhood of t. It suffices to show that V contains a clopen neighborhood of t. Find an open set W with $t \in W \subseteq \overline{W} \subseteq V$. By Urysohn's Lemma, there exists $f \in C(K)$ such that $0 \leqslant f \leqslant \mathbb{1}$, $f(t) = 1$, and f vanishes outside of W. Put $U = \{f > 0\}$. Then U is an open F_σ set, so that $\overline{U}$ is clopen. Clearly, $t \in \overline{U}$. It follows from $U \subseteq W$ that $\overline{U} \subseteq \overline{W} \subseteq V$.

2.1.26 Clearly, singletons are always connected. Suppose that K is totally disconnected and let $A \subseteq K$, $s, t \in A$, and $s \neq t$. Since K is totally disconnected, we can find a clopen set U in K such that $s \in U$ and $t \notin U$. Then $A \cap U$ and $A \cap U^C$ are two non-empty clopen subsets of A (clopen in the induced topology of A), so A is disconnected.

Conversely, suppose that every point of K has a base of clopen neighbourhoods. Let $s \in K$; let V be an open neighborhood of s. For each $t \in V^C$, since the set $\{s, t\}$ is not connected, we can find a clopen U_t such that $t \in U_t$ and $s \notin U_t$. The collection $\{U_t : t \in V^C\}$ forms an open cover of V^C. The latter set is closed and, therefore, compact. Hence, we can find a finite subcover $U_{t_1}, \dots, U_{t_n}$. Put $U = \bigcup_{i=1}^n U_{t_i}$; then U is clopen, $x \notin U$, and $V^C \subseteq U$. It follows that U^C is a clopen set satisfying $x \in U^C \subseteq V$.

2.1.27 Recall that Δ consists of those reals in $[0,1]$ which admit a ternary expansion that does not include any 1's. Here is an alternative construction of Δ: let $I_{1,1}=[0,\frac{1}{3}]$, $I_{1,2}=[\frac{2}{3},1]$, $I_{2,1}=[0,\frac{1}{9}]$, $I_{2,2}=[\frac{2}{9},\frac{1}{3}]$, $I_{2,3}=[\frac{2}{3},\frac{7}{9}]$, $I_{2,4}=[\frac{8}{9},1]$, $I_{3,1}=[0,\frac{1}{27}]$, etc. For each n, put $C_n=\bigcup_{m=1}^{2^n}I_{n,m}$, the union of all intervals "at level n". Then $\Delta=\bigcap_{n=1}^{\infty}C_n$. Being an intersection of compact sets, Δ is compact in the topology induced from $[0,1]$, and, hence Δ is a compact space.

Put $J_{n,m}=I_{n,m}\cap\Delta$. Clearly, $J_{n,m}$ is closed. Since its complement in Δ is a finite union of sets of the same kind, the complement is also closed, hence $J_{n,m}$ is clopen. For every $t\in[0,1]$, let $\mathcal{N}$ be the collection of all $J_{n,m}$'s that contain t. Then $\mathcal{N}$ is a base of clopen neighborhoods of t in Δ. Hence, Δ is totally disconnected.

To show that Δ is not basically disconnected, it suffices to prove that $C(\Delta)$ is not σ-order complete. Fix t_0 in $(0,1)$ such that the ternary expansion of t_0 never stabilizes. Equivalently, t_0 is not an endpoint of any $I_{n,m}$. Then there exist two sequences in Δ converging to t_0, one strictly increasing and the other strictly decreasing. For every n, let $f_n\in C[0,1]$ such that f_n equals 1 on $[0,t_0]$, vanishes on $[t_0+\frac{1}{n},1]$, and is linear in between. Let g_n be the restriction of f_n to Δ. Clearly, $g_n\downarrow\geqslant 0$ in $C(\Delta)$. It is straightforward that $\inf g_n$ does not exist.

2.1.28 Let U be an open set in $\beta\Gamma$; we need to show that $\overline{U}$ is clopen. Let $f\colon\Gamma\to\mathbb{R}$ be the characteristic function of $U\cap\Gamma$, and let g be the unique continuous extension of f to $\beta\Gamma$. Since Γ is dense in $\beta\Gamma$, g is again a characteristic function. It suffices to verify that $g=\mathbb{1}_{\overline{U}}$; this implies that $\mathbb{1}_{\overline{U}}$ is continuous and, therefore $\overline{U}$ is clopen. Since U is open and Γ is dense in $\beta\Gamma$, we have $\overline{U\cap\Gamma}=\overline{U}$. Since f takes value 1 on $U\cap\Gamma$, we conclude that g takes value 1 on $\overline{U}$. It is a straightforward verification that g takes value zero on $\beta\Gamma\setminus\overline{U}$.

2.2.1(ii) I_e is closed under scalar multiplication: if $x\in I_e$ then $|x|\leqslant\lambda e$ for some $\lambda\in\mathbb{R}_+$, then for every $r\in\mathbb{R}$ we have $|rx|\leqslant|r|\lambda e$, hence $rx\in I_e$. I_e is closed under addition: if $x,y\in I_e$ then $|x|\leqslant\lambda e$ and $|y|\leqslant\mu e$ for some $\lambda,\mu\in\mathbb{R}_+$; then $|x+y|\leqslant|x|+|y|\leqslant(\lambda+\mu)e$, hence $x+y\in I_e$. Finally, if $x\in I_e$ then $|x|\in I_e$.

2.2.1(viii) Without loss of generality, we may scale each x_n so that the series $\sum_{n=1}^{\infty}\|x_n\|$ converges and, therefore, $e:=\sum_{n=1}^{\infty}|x_n|$ converges. Alternatively, we may just put $e=\sum_{n=1}^{\infty}\frac{|x_n|}{2^n\|x_n\|}$. In either case, each $|x_n|$ is dominated by a scalar multiple of e.

2.2.3(i) For every $n\in\mathbb{N}$, we have $|x|\leqslant\big(\|x\|_e+\frac{1}{n}\big)e$, so that $|x|-\|x\|_e e\leqslant\frac{1}{n}e$. It follows from the Archimedean Property that $|x|-\|x\|_e e\leqslant 0$.

2.2.3(ii) It is clear that $\|\alpha x\|_e=|\alpha|\|x\|_e$ for all $x\in X$ and $\alpha\in\mathbb{R}$. It follows from (i) that $\|x\|_e=0$ iff $x=0$.

To prove the triangle inequality, take any $x,y\in I_e$. Then

$$|x+y|\leqslant|x|+|y|\leqslant\|x\|_e e+\|y\|_e e=\big(\|x\|_e+\|y\|_e\big)e.$$

It follows that $\|x+y\|_e\leqslant\|x\|_e+\|y\|_e$. Hence, $\|\cdot\|_e$ is a norm on I_e. It is easy to see that $|x|\leqslant|y|$ implies $\|x\|_e\leqslant\|y\|_e$, so that $\big(I_e,\|\cdot\|_e\big)$ is a normed lattice.

2.2.3(iii) Follows from (i).

2.2.3(v) Taking the norms of the both sides in i we conclude that $\|x\|\leqslant C\|x\|_e$ for every $x\in I_e$, where $C=\|e\|$.

2.2.5(i) It follows from $x\vee y\geqslant x$ and $x\vee y\geqslant y$ that $\|x\vee y\|_e\geqslant\|x\|_e$ and $\|x\vee y\|_e\geqslant\|y\|_e$, so that $\|x\vee y\|_e\geqslant\|x\|_e\vee\|y\|_e$. On the other hand, for every $\varepsilon>0$, we have $0\leqslant x\leqslant\big(\|x\|_e+\varepsilon\big)e$ and $0\leqslant y\leqslant\big(\|y\|_e+\varepsilon\big)e$. Hence,

$$0\leqslant x\vee y\leqslant\big(\|x\|_e+\varepsilon\big)e\vee\big(\|y\|_e+\varepsilon\big)e=\big(\|x\|_e\vee\|y\|_e+\varepsilon\big)e.$$

It follows that $\|x\vee y\|_e\leqslant\|x\|_e\vee\|y\|_e+\varepsilon$. Since ε is arbitrary, we get $\|x\vee y\|_e\leqslant\|x\|_e\vee\|y\|_e$.

2.2.5(ii) We have $|x+y| = |x| \vee |y|$ by Exercise 1.5.3(ii). It follows that

$$\|x+y\|_e = \big\||x+y|\big\|_e = \big\||x| \vee |y|\big\|_e = \big\||x|\big\|_e \vee \big\||y|\big\|_e = \|x\|_e \vee \|y\|_e.$$

2.2.6(i) Suppose that $g \in I_u$, so $|g| \leqslant \lambda u$ for some $\lambda \geqslant 0$. If $t \notin \Omega$ then $|g(t)| \leqslant \lambda u(t) = 0$, hence g vanishes on Ω^C. For every $t \in \Omega$, define $h(t) = \frac{g(t)}{u(t)}$. It follows that $|h(t)| \leqslant \lambda$. Let $t \in \Omega$. Since $u(t) > 0$, there exists $\varepsilon > 0$ and a neighborhood V of t such that u is greater than ε on V; it follows that h is continuous at t; hence h is continuous on Ω.

Conversely, suppose that there exists $h \in C_b(\Omega)$ such that $g(t) = u(t)h(t)$ for every $t \in \Omega$ and g vanishes on Ω^C. Let $\lambda = \|h\|_{\sup}$. Clearly, $|g| \leqslant \lambda u$. It is left to verify that g is continuous. On the open set Ω, g is continuous being the product of two continuous functions. Let $t \in \Omega^C$; then $g(t) = 0$. Fix $\varepsilon > 0$. Since $u(t) = 0$, one can find a neighborhood V of t such that $u(s) < \varepsilon/\lambda$ for all $s \in V$. Let $s \in V$. If $s \in \Omega$ then $|g(s)| = u(s)|h(s)| \leqslant \frac{\varepsilon}{\lambda} \cdot \lambda = \varepsilon$, and if $s \in \Omega^C$ then $g(s) = 0$. Therefore, $g(t) = 0 = \lim_{s \to t} g(s)$.

Let T the map that sends g to h. The preceding argument shows that T is a bijection between I_u and $C_b(\Omega)$. It is easy to see that T is linear and $Tg_1 \leqslant Tg_2$ iff $g_1 \leqslant g_2$; it follows from Theorem 1.6.13 that T is a lattice isomorphism.

2.2.6(ii) Fix $\varepsilon > 0$. Put $g = (\frac{1}{\varepsilon}u) \wedge \sqrt{u}$. It follows from $0 \leqslant g \leqslant \frac{1}{\varepsilon}u$ that $g \in I_u$. For $t \in K$, $g(t) \neq \sqrt{u(t)}$ iff $\frac{1}{\varepsilon}u(t) < \sqrt{u(t)}$ iff $\sqrt{u(t)} < \varepsilon$. Therefore, g agrees with $\sqrt{u}$ on the set $\{\sqrt{u} \geqslant \varepsilon\}$. Since $0 \leqslant g \leqslant \sqrt{u}$, we have $\|\sqrt{u} - g\| < \varepsilon$, so that $\sqrt{u} \in \overline{I_u}$.

2.2.6(iii) By (ii), $\sqrt{u} \in \overline{I_u}$. Since the ratio of $\sqrt{u}$ and u is unbounded on $(0,1]$, $\sqrt{u} \notin I_u$ by (i).

2.2.6(iv) WLOG, $0 \leqslant u \leqslant \mathbb{1}$. Let $\Omega = \{u \neq 0\}$, Clearly, Ω is open. Suppose that Ω is clopen. Then Ω and $\Omega^c = \ker u$ form a disconnection of K. Also, Ω is compact, hence u is bounded below by some $\varepsilon > 0$ on Ω and, therefore, $\varepsilon \mathbb{1}_\Omega \leqslant u \leqslant \mathbb{1}_\Omega$. It follows that $I_u = I_{\mathbb{1}_\Omega}$, which consists of all the functions supported on Ω. This set is clearly closed in $C(K)$.

Suppose now that I_u is closed. Then $\sqrt{u} \in I_u$ by (ii). Therefore, $\sqrt{u} \leqslant \lambda u$ for some $\lambda > 0$. It follows that $u(t) \geqslant \frac{1}{\lambda^2}$ for every $t \in \Omega$, hence $\Omega = u^{-1}\big[\frac{1}{\lambda^2}, \infty\big)$ and, therefore, Ω is closed.

2.2.7 The argument is similar to that in Exercise 2.2.6(i), except that we have to take care of sets of measure zero. As usual, u stands for an equivalence class of functions that are equal a.e. Fix a specific representative u_0. WLOG, we may assume that $u_0(t) \geqslant 0$ for all t. Put $A = \{u_0 > 0\}$. Clearly, A is measurable. We are going to show that for every $g \in I_u$ there exists a unique $h \in L_\infty(A, \mu)$ such that g equals $u \cdot h$ on A and vanishes on A^C. The converse implication is straightforward because $|g| \leqslant \|h\|_\infty u$. Let $g \in I_u$. There exists $\lambda \geqslant 0$ such that $|t| \leqslant \lambda u$. Choose a representative g_0 of g. We may assume WLOG that $\big|g_0(t)\big| \leqslant \lambda u_0(t)$ for every t; otherwise replace $g_0(t)$ with 0 for every t for which $|g_0(t)| > \lambda u_0(t)$ and note that this set has measure zero. It follows that g_0 vanishes outside of A. For $t \in A$, put $h_0(t) = g_0(t)/u_0(t)$. It is clear that h_0 is measurable on A and $\big|h_0(t)\big| \leqslant \lambda$ for all $t \in A$, so the corresponding equivalence class h lives in $L_\infty(A, \mu)$. It follows that $g = u \cdot h$ on A.

The proof that the resulting map is a lattice isomorphism is the same as in Exercise 2.2.6(i).

2.2.11 (i)$\Rightarrow$(iii) By Corollary 2.2.10, the norm of X is equivalent to $\|\cdot\|_e$, hence there exists $\lambda > 0$ such that λB_X is contained in the unit ball of $\|\cdot\|_e$. The latter ball equals $[-e, e]$ by Exercise 2.2.3(iii).

(iii)$\Rightarrow$(i) For every non-zero $x \in X$, we have $\frac{x}{\|x\|} \in B_x$, hence $\lambda \frac{x}{\|x\|} \in [-e, e]$. It follows that $|x| \leqslant \frac{\|x\|}{\lambda} e$.

(ii)$\Rightarrow$(iii) Suppose that $e \in \operatorname{Int} X_+$, so that $e + \lambda B_X \subset X_+$ for some positive real λ. For every $x \in B_X$ we have $e \pm \lambda x \geqslant 0$, so that $-e \leqslant \lambda x \leqslant e$. This yields $\lambda B_X \subseteq [-e, e]$.

(iii)$\Rightarrow$(ii) $e + \lambda B_X \subseteq e + [-e, e] = [0, 2e] \subseteq X_+$.

2.3.1(i) Suppose that $x_\alpha \xrightarrow{\mathrm{u}} x$ and $x_\alpha \xrightarrow{\mathrm{u}} y$. Then there exist $e_1, e_2 \in X_+$ such that $\|x_\alpha - x\|_{e_1} \to 0$ and $\|x_\alpha - y\|_{e_2} \to 0$. Put $e = e_1 \vee e_2$, then $\|x_\alpha - x\|_e \to 0$ and $\|x_\alpha - y\|_e \to 0$ by Exercise 2.2.3(iv). It follows that $\|x - y\|_e \leqslant \|x_\alpha - x\|_e + \|x_\alpha - y\|_e \to 0$, so that $x = y$.

2.3.1(iv) Find $e \in X_+$ such that $\|x_\alpha - x\|_e \to 0$. WLOG, $x \in I_e$; otherwise, replace e with $e \vee |x|$. We have

$$\|\lambda_\alpha x_\alpha - \lambda x\|_e \leqslant \|\lambda_\alpha x_\alpha - \lambda_\alpha x\|_e + \|\lambda_\alpha x - \lambda x\|_e = |\lambda_\alpha| \|x_\alpha - x\|_e + |\lambda_\alpha - \lambda| \|x\|_e \to 0$$

because (λ_α) has a bounded tail and $\|x\|_e$ is finite.

2.3.1(v) By assumption, there exist $e_1, e_2 \in X_+$ such that $\|x_\alpha - x\|_{e_1} \to 0$ and $\|y_\alpha - y\|_{e_2} \to 0$. Put $e = e_1 \vee e_2$, then $\|x_\alpha - x\|_e \to 0$ and $\|y_\alpha - y\|_e \to 0$ by Exercise 2.2.3(iv). Since $\|\cdot\|_e$ is a lattice norm, we have $\big\|(x_\alpha + y_\alpha) - (x + y)\big\|_e \to 0$, $\big\|(x_\alpha \wedge y_\alpha) - (x \wedge y)\big\|_e \to 0$, and $\big\|(x_\alpha \vee y_\alpha) - (x \vee y)\big\|_e \to 0$,

2.3.1(vii) Suppose $x_\alpha \xrightarrow{\mathrm{u}} x$. Let e be a regulator; taking $\varepsilon = 1$ in (2.3) we conclude that $x_\alpha \in [x - e, x + e]$ for all $\alpha \geqslant \alpha_0$.

2.3.2 $\|x_\alpha - x\|_e \to 0$ clearly implies $x_\alpha \xrightarrow{\mathrm{u}} x$. Suppose $x_\alpha \xrightarrow{\mathrm{u}} x$. Find $e_1 \in X_+$ such that $\|x_\alpha - x\|_{e_1} \to 0$. Since e is a strong unit, there exists $\lambda > 0$ such that $e_1 \leqslant \lambda e$. Fix $\varepsilon > 0$. There exists α_0 such that $\|x_\alpha - x\|_{e_1} < \frac{\varepsilon}{\lambda}$ whenever $\alpha \geqslant \alpha_0$. This implies $|x_\alpha - x|_{e_1} \leqslant \frac{\varepsilon}{\lambda} e_1 \leqslant \varepsilon e$, hence $\|x_\alpha - x\|_e \leqslant \varepsilon$. Therefore, $x_\alpha \xrightarrow{\mathrm{u}} x$.

2.3.4 The forward implication is trivial. To prove the converse, suppose that $x_n \xrightarrow{\mathrm{u}} x$ for some sequence (x_n) in A and some $x \in X$. Find $e \in X_+$ such that $\|x_n - x\|_e \to 0$. Put $u = e \vee |x|$, then $x \in I_u$ and $\|x_n - x\|_u \leqslant \|x_n - x\|_e \to 0$. It follows that, after passing to a tail, (x_n) is in I_u and $x_n \to x$ in $\big(I_u, \|\cdot\|_u\big)$. By assumption, we have $x \in A$.

2.3.15 The first claim is straightforward. Let Y be a subspace of c_{00} and $0 < e \in c_{00}$. It is easy to see that I_e is finite-dimensional, hence $Y \cap I_e$ is closed in $\big(I_e, \|\cdot\|_e\big)$. Now apply Exercise 2.3.4.

2.3.18 It is a Calculus exercise that there exists a sequence (λ_n) in $(0, +\infty)$ such that $\lambda_n \to \infty$ and the series $\sum_{n=1}^{\infty} \lambda_n \|x_n\|$ converges. Now proceed as in the proof of Proposition 2.3.6.

2.4.2 WLOG, $x = 0$. The first implication is trivial. To show that $x_\alpha \xrightarrow{\mathrm{u}} x$ implies $x_\alpha \xrightarrow{\sigma o} x$, take $u_n = \frac{1}{n} e$ for the dominating sequence, where e is a regulator for the uniform convergence.

2.4.3 WLOG, $x = 0$. Suppose that $x_n \xrightarrow{\sigma o} x$. That is, there exists a sequence (u_m) such that $u_m \downarrow 0$ and for every m there exists $n_m \in \mathbb{N}$ such that $|x_n| \leqslant u_m$ whenever $n \geqslant n_m$. WLOG, (n_m) is an increasing sequence in X. Also, just as in the preceding exercise, (x_n) is order bounded by some, say, u_0. Replacing u_0 with $u_0 \vee u_1$, we may assume WLOG that $u_0 \geqslant u_1$. Now we define a new sequence (v_n) as follows: put $v_n = u_m$ whenever $n_m \leqslant n < n_{m+1}$ (we take $n_0 = 1$). It is easy to see that $v_n \downarrow 0$ and $|x_n| \leqslant v_n$ for every n.

The converse implication is trivial.

2.4.8 WLOG, $a = 0$; otherwise replace x_α with $x_\alpha - a$. By Proposition 2.4.7, $0 = x_\alpha^- \xrightarrow{\mathrm{uo}} x^-$, hence $x^- = 0$, so that $x \geqslant 0$.

2.4.9 Suppose $x_\alpha \xrightarrow{\mathrm{o}} y$ and $x_\alpha \xrightarrow{\mathrm{o}} z$. Taking differences, we conclude that a constant zero net converges in order to both $y - z$ and $z - y$. By Exercise 2.4.8, $y - z \geqslant 0$ and $z - y \geqslant 0$; hence $y = z$.

2.4.11 If $x_\alpha \uparrow x$ then $x_\alpha \xrightarrow{\mathrm{o}} x$; just take $u_\alpha = x - x_\alpha$ as a dominating net. Suppose that $x_\alpha \uparrow$ and $x_\alpha \xrightarrow{\mathrm{o}} x$. Fix α; the tail $(x_\beta)_{\beta \geqslant \alpha}$ converges in order to x by Exercise 2.4.10; it follows that $x_\alpha \leqslant x$ by Exercise 2.4.8. Thus, x is an upper bound of the net (x_α). On the other hand, if u is an upper bound of the net then $x \leqslant u$ by Exercise 2.4.8. Thus, $x = \sup x_\alpha$.

The proofs for decreasing nets and for σ-order convergence are similar.

2.4.12 First, observe that for every $a \in X_+$, we have $\lambda_\alpha a \xrightarrow{\sigma o} \lambda a$. Indeed, for every k there exists α_k such that $-\frac{1}{k} \leqslant \lambda - \lambda_\alpha \leqslant \frac{1}{k}$ whenever $\alpha \geqslant \alpha_k$. It follows that $-\frac{1}{k}a \leqslant \lambda a - \lambda_\alpha a \leqslant \frac{1}{k}a$. Now $\frac{1}{k}a \downarrow 0$ yields $\lambda_\alpha a \xrightarrow{\sigma o} \lambda a$.

Passing to a tail, we may assume that (λ_α) is bounded, so that there exists $M > 0$ such that $|\lambda_\alpha| \leqslant M$ for all α.

$$\bigl|\lambda_\alpha x_\alpha - \lambda x\bigr| \leqslant |\lambda_\alpha||x_\alpha - x| + |\lambda_\alpha - \lambda||x| \leqslant M|x_\alpha - x| + |\lambda_\alpha - \lambda||x| \xrightarrow{\mathrm{o}} 0.$$

2.4.15 Subtracting x from everything, we may assume WLOG that $x = 0$.

(i)$\Rightarrow$(ii) Suppose $x_\alpha \xrightarrow{\mathrm{o}} 0$. Let (u_γ) be a dominating net for this convergence. Put $a_\gamma = -u_\gamma$ and $b_\gamma = u_\gamma$.

(ii)$\Rightarrow$(i) Put $u_\gamma = b_\gamma \wedge (-a_\gamma)$.

(ii)$\Rightarrow$(iii) Put $A = \{a_\gamma\}_{\gamma\in\Gamma}$ and $B = \{b_\gamma\}_{\gamma\in\Gamma}$. If $a \in A$ and $b \in B$ then $a = a_{\gamma_1}$ and $b = b_{\gamma_2}$ for some $\gamma_1, \gamma_2 \in \Gamma$. Find $\gamma_0 \in \Gamma$ such that $\gamma_0 \geqslant \gamma_1, \gamma_2$. Then $[a, b] \supseteq [a_{\gamma_0}, b_{\gamma_0}]$, and the latter order interval contains a tail of (x_α).

(iii)$\Rightarrow$(ii) Take the nets $A^\wedge$ and $B^\vee$.

(i)$\Rightarrow$(iv) Let $(u_\gamma)_{\gamma\in\Gamma}$ be a dominating net; take $C = \{u_\gamma\}_{\gamma\in\Gamma}$.

(iv)$\Rightarrow$(i) Take $C^\wedge$ for the dominating net.

(iii)$\Rightarrow$(v) Take $\mathcal{F} = \bigl\{[a, b] \,:\, a \in A, b \in B\bigr\}$.

(v)$\Rightarrow$(iii) Let A and B be the sets of all left (respectively, right) endpoints of intervals in $\mathcal{F}$.

2.4.16 (ii)$\Rightarrow$(i) follows from Exercise 2.4.15(iii).

(i)$\Rightarrow$(ii) By Exercise 2.4.15(iii), there exist non-empty sets A_0 and B_0 with $\sup A_0 = 0 = \inf B_0$ and for every $a \in A_0$ and $b \in B_0$, the order interval $[a, b]$ contains a tail of (x_α). It follows that $A_0 \subseteq A$ and $B_0 \subseteq B$. It follows that A and B are non-empty and $\sup A = 0 = \inf B$.

(i)$\Leftrightarrow$(iii) is similar.

2.4.17 Replacing x_α with $|x_\alpha - x|$, we may assume WLOG that $x = 0$ and $x_\alpha \geqslant 0$ for every α. Let A be a dominating set as Exercise 2.4.16. Replacing A with $A^\wedge$, we may assume that A is closed under taking infima of finite sets. Since X has the countable sup property, we can find a sequence (w_n) in A so that $\inf w_n = 0$. Replacing each w_n with $w_1 \wedge \dots \wedge w_n$, we may assume that $w_n \downarrow 0$. Since (x_α) is order bounded, we may assume that $x_\alpha \leqslant w_1$ for all α (by enlarging w_1 if necessary).

For each α, let n_α be the least natural number n such that the tail $(x_\beta)_{\beta\geqslant\alpha}$ is dominated by w_n. Put $u_\alpha = w_{n_\alpha}$. It follows that $x_\alpha \leqslant u_\alpha$. If $\alpha \geqslant \beta$ then $n_\alpha \leqslant n_\beta$ and, therefore, $u_\alpha \geqslant u_\beta$; hence $u_\alpha\downarrow$. Finally, for every n, it follows from $w_n \in A$ that there exists α such that the tail $(x_\beta)_{\beta\geqslant\alpha}$ is dominated by w_n, hence $n \geqslant n_\alpha$ and, therefore, $u_\alpha = w_{n_\alpha} \leqslant w_n$. It now follows from $\inf w_n = 0$ that $\inf u_\alpha = 0$.

2.4.20 Suppose that σ-order convergence on an Archimedean vector lattice X is stable. Uniform convergence always implies σ-order convergence. Suppose $x_n \xrightarrow{\sigma o} 0$. Find (λ_n) such that $0 < \lambda_n \uparrow \infty$ and $\lambda_n x_n \xrightarrow{\sigma o} 0$. In particular, $(\lambda_n x_n)$ is order bounded, say, by some e. It follows that $|x_n| \leqslant \frac{1}{\lambda_n}e$ and, therefore, $x_n \xrightarrow{\mathrm{u}} 0$.

The converse is trivial because uniform convergence is stable.

2.4.26 Let (x_n) be a disjoint sequence in $[-u, u]$ for some $u \in X_+$. Replacing x_n with $|x_n|$, we may assume that $x_n \geqslant 0$ for every n. For every n, put $v_n = x_1 \vee \dots \vee x_n = x_1 + \dots x_n$. Then $v_n\uparrow\leqslant u$; since X is σ-order complete, $v = \sup_n v_n$ exists. For every $m \in \mathbb{N}$, we have

$$x_m \leqslant \sup_{n\geqslant m} x_n = \sup_{n\geqslant m} (x_m + \dots + x_n) = \sup_{n\geqslant m} (x_1 + \dots + x_n) - (x_1 + \dots + x_{m-1}) = v - v_{m-1} \downarrow 0.$$

2.4.28 In view of Proposition 2.4.27, it suffices to prove that if $f_n \xrightarrow{\text{a.e.}} f$ then (f_n) is order bounded. Suppose that $f_n \xrightarrow{\text{a.e.}} f$. WLOG (modifying f_n's on a set of measure zero), we may assume that $f_n(t) \to f(t)$ for every t. In particular, the sequence $\bigl(f_n(t)\bigr)$ is bounded. Let $u(t) = \sup_n\bigl|f_n(t)\bigr|$. Then $u \in L_0(\mu)$ by Theorem A.3.3 and $|f_n| \leqslant u$ for every n.

2.5.1 Suppose that $x_n \uparrow$ and $x_{n_k} \to x$. Fix n; for every sufficiently large k, we have $x_{n_k} \geqslant x_n$. Passing to the limit on k, we conclude that $x \geqslant x_n$. It follows that the sequence $\big(\|x - x_n\|\big)$ is decreasing. It has a subsequence that converges to zero, namely, $\big(\|x - x_{n_k}\|\big)$. It follows that $\|x - x_n\| \to 0$. The proof for decreasing sequences is similar.

2.5.2 The forward implication is trivial. To prove the converse, suppose that every increasing Cauchy sequence in X_+ is norm convergent. Let (x_n) be a Cauchy sequence in X; we need to show that (x_n) is norm convergent. It suffices to show that (x_n) has a convergent subsequence.

We claim that, passing to a subsequence, we may assume that the series $\sum_{n=1}^{\infty}\|x_{n+1} - x_n\|$ converges. Indeed, find n_1 such that $\|x_n - x_{n_1}\| < \frac{1}{2}$ for all $n \geqslant n_1$, then find $n_2 > n_1$ such that $\|x_n - x_{n_2}\| < \frac{1}{4}$ for all $n \geqslant n_2$, etc.

Put

$$y_n = \sum_{i=1}^{n}(x_i - x_{i-1})^+ \quad \text{and} \quad z_n = \sum_{i=1}^{n}(x_i - x_{i-1})^-$$

taking $x_0 = 0$. Then (y_n) and (z_n) are increasing sequences in X_+ and $y_n - z_n = x_n$. Note that

$$\|y_m - y_n\| = \Big\| \sum_{i=n+1}^{m} (x_i - x_{i-1})^+ \Big\| \leqslant \sum_{i=n+1}^{m} \|x_i - x_{i-1}\| \to 0$$

as $n, m \to \infty$ with $n < m$. It follows that (y_n) is Cauchy, hence converges by assumption. Similarly, (z_n) converges. It follows from $y_n - z_n = x_n$ that (x_n) converges.

2.5.5 Let A be an order closed set and $x_n \to x$ for some sequence (x_n) in A. By Theorem 2.5.4, $x_{n_k} \xrightarrow{\text{o}} x$ for some subsequence (x_{n_k}) of (x_n); hence $x \in A$. It follows that A is closed.

2.5.6(i) No (to both questions); see Example 2.3.20.

2.5.6(ii) No (to both questions). Here is a counterexample. Fix an uncountable set Ω. Put $X = \ell_1(\Omega)$. Let Γ be the set of all finite non-empty subsets of Ω, ordered by inclusion. For $\gamma \in \Gamma$, put $x_\gamma = \frac{1}{|\gamma|^2}\mathbb{1}_\gamma$, where $|\gamma|$ is the cardinality of γ. Clearly, $\|x_\gamma\| = \frac{1}{|\gamma|} \to 0$.

Suppose that there is a subnet $(x_{\gamma_\alpha})_{\alpha\in A}$ of $(x_\gamma)_{\gamma\in\Gamma}$ which is order bounded by some $u \in \ell_1(\Omega)_+$. Since A is co-final in Γ, for every $t \in \Omega$, $\{t\} \subseteq \gamma_\alpha$ for some $\alpha \in A$; we conclude that $u(t) \geqslant x_{\gamma_\alpha}(t) > 0$. This means that u has uncountable support, which is impossible in $\ell_1(\Omega)$.

2.5.7 Consider the space c of all convergent real sequences, but instead of the usual $\|\cdot\|_\infty$ norm, equip it with the norm $\|x\| = \|x\|_\infty + \lim|x|$. It follows from $\|x\|_\infty \leqslant \|x\| \leqslant 2\|x\|_\infty$ that this norm is equivalent to $\|\cdot\|_\infty$, hence makes c into a Banach space. It is easy to verify that this norm is a lattice norm, so we get a Banach lattice. Consider the sequence

$$x_n = (\underbrace{1, \dots, 1,}_{n \text{ times}} 0, 0, \dots).$$

Then (x_n) is in the unit ball of this norm. However, $x_n \uparrow \mathbb{1}$, and $\|\mathbb{1}\| = 2$.

2.5.12 Observe that the sequence $\mathbb{1}_{[0,\frac{1}{n}]}$ converges in order to zero, but fails to converge in norm.

2.5.19 Since uniform convergence always implies order convergence, we only need to show the converse implication. Suppose that $f_n \xrightarrow{\text{o}} 0$ in $L_p(\mu)$. Replacing f_n with $|f_n|$, we may assume WLOG that $f_n \geqslant 0$ for every n. By Proposition 2.4.27, $f_n \xrightarrow{\text{a.e.}} 0$ and there exists $u \in L_p(\mu)$ such that $f_n \leqslant u$.

Suppose that $0 < p < \infty$. We have $f_n^p \xrightarrow{\text{a.e.}} 0$ and $f_n^p \leqslant u^p$ in $L_1(\mu)$, hence $f_n^p \xrightarrow{\text{o}} 0$ and, therefore, $f_n^p \xrightarrow{\text{u}} 0$ in $L_1(\mu)$ by Theorem 2.5.17. It follows that there exists $0 \leqslant v \in L_1(\mu)$ such that for every $\varepsilon > 0$ there exists n_0 such that $f_n^p \leqslant \varepsilon^{\frac{1}{p}} v$ and, therefore, $f_n \leqslant \varepsilon v^{\frac{1}{p}}$ whenever $n \geqslant n_0$. Note that $v^{\frac{1}{p}} \in L_p(\mu)$. It follows that $f_n \xrightarrow{\text{u}} 0$ in $L_p(\mu)$.

Suppose that $p = 0$ and μ is σ-finite. We claim that there exists $w \in L_0(\mu)$ such that $w \geqslant \mathbb{1}$ and $\frac{u}{w} \in L_1(\mu)$. Indeed, write $\Omega = \bigcup_{k=1}^{\infty} \Omega_k$ such that (Ω_k) is a disjoint sequence and $\mu(\Omega_k)$ is finite for every k. On Ω_k, put w to agree with $\mathbb{1} \vee 2^k \mu(\Omega_k) u$. Then w never vanishes, so that $\frac{u}{w}$ is defined, and $\int \frac{u}{w} \leqslant \sum_{k=1}^{\infty} \int_{\Omega_k} \frac{1}{2^k \mu(\Omega_k)} = 1$.

Let $h_n \coloneqq \frac{f_n}{w} \leqslant \frac{u}{w}$. Viewed as a sequence in $L_1(\mu)$, (h_n) is order bounded and $h_n \xrightarrow{\text{a.e.}} 0$, hence that $h_n \xrightarrow{\text{o}} 0$ and, therefore, $h_n \xrightarrow{\text{u}} 0$ in $L_1(\mu)$ by Theorem 2.5.17. It is now easy to see that $f_n \xrightarrow{\text{u}} 0$ in $L_0(\mu)$.

2.5.21 We already know that in every vector lattice, uniform convergence implies order convergence, and order convergence implies order boundedness. It can be easily verified that order convergence in $\mathbb{R}^{\mathbb{N}}$ implies coordinate-wise convergence (prove it first for monotone nets decreasing to zero).

Suppose that a net $(x^{(\alpha)})$ converges to zero coordinate-wise and is order bounded, i.e., there exists $u \geqslant 0$ such that $|x^{(\alpha)}| \leqslant u$ for all α. WLOG $u_n \geqslant n$ for every n; otherwise, replace u with $u \vee v$ where $v_n = n$ for all n. We will show that $(x^{(\alpha)})$ converges to zero uniformly, with u^2 being a regulator. Fix $n \in \mathbb{N}$. Using coordinate-wise convergence of the net, find α_0 such that for every $\alpha \geqslant \alpha_0$ and every $k = 1, \dots, n$, we have $|x_k^{(\alpha)}| \leqslant \frac{1}{n} u_k^2$. On the other hand, for $k > n$, we have $|x_k^{(\alpha)}| \leqslant u_k \leqslant \frac{1}{k} u_k^2 \leqslant \frac{1}{n} u_k^2$. Hence, $|x^{(\alpha)}| \leqslant \frac{1}{n} u^2$ for all $\alpha \geqslant \alpha_0$.

The order boundedness condition may be dropped for sequences because every sequence in $\mathbb{R}^{\mathbb{N}}$ that converges point-wise is automatically order bounded. However, this may fail for nets. For example, consider the double-sequence $e_{n,m} = n e_m$. It converges to zero coordinate-wise as $n, m \to \infty$, but no tail of it is order bounded.

2.5.23 Let Y be a closed sublattice of an order continuous Banach lattice X and $y_\alpha \downarrow 0$ in Y. It suffices to show that $y_\alpha \downarrow 0$ in X as well. Clearly, we still have $y_\alpha \downarrow\geqslant 0$ in X. Since X is order complete by Theorem 2.5.22, we have $y_\alpha \downarrow x$ for some $x \in X_+$. Also, $y_\alpha \to x$ because X is order continuous, so that $x \in Y$ because Y is closed. Applying the Monotone Convergence Lemma 2.5.3 in Y we get $x = 0$.

3.0.1(ii) Suppose that Z is a regular sublattice in X and $A \subseteq Z$ such that $z \coloneqq \inf_Z A$ exists. WLOG, replacing A with $A - z$, we assume that $z = 0$. By assumption, $\inf_X A = 0$. Hence no positive non-zero element of X (and, therefore, of Y) is a lower bound of A. It follows that $\inf_Y A = 0$.

3.1.2 The forward implication is trivial. The converse follows from the fact that $z = x \wedge y$ iff $0 = x \wedge y - z = (x - z) \wedge (y - z)$.

3.1.7 No. Take $X = \mathbb{R}^3$, pick two positive vectors u and v in X, and put $Y = \operatorname{span} u$ and $Z = \operatorname{span} v$. Clearly, Y and Z are sublattices of X. However, $Y + Z$ need not be a sublattice. Take, for example, $u = (1, 1, 0)$ and $v = (0, 1, 1)$, then $u - v \in Y + Z$ but $|u - v| \notin Y + Z$.

3.1.8 $\operatorname{Lat}(A)$ is the intersection of all the sublattices of X containing A. Now note that every sublattice of X containing A also contains $\operatorname{span} A$.

3.1.9 It is easy to see that Y is closed under lattice operations in $C[0, 1]$, hence Y is a sublattice of $C[0, 1]$. Let Z be the sublattice generated by $\mathbb{1}$ and u; we need to show that $Y = Z$. Since $\mathbb{1}$ and u belong to Y and Z is the intersection of all sublattices of $C[0, 1]$ containing $\mathbb{1}$ and u, it follows that $Z \subseteq Y$. We will prove that every $f \in Y$ is contained in Z by induction on the number of pieces. For the base of induction, note that every linear function is clearly in Z. If f has two pieces, it may be written as a supremum or an infimum of two linear functions, so that $f \in Z$. Assume that for some $n > 2$ all functions in Y which have less than n pieces are in Z, and suppose that f has n pieces. Then there exist $0 = t_0 < t_1 < \cdots < t_n = 1$ such that f is linear on each $[t_{i-1}, t_i]$ as $i = 1, \dots, n$. Let h be the piece-wise affine function which agrees with f on $[0, t_{n-1}]$ and is linear on $[t_{n-2}, 1]$. Then h has $n - 1$ pieces, so that $h \in Z$, and either $h \geqslant f$ or $h \leqslant f$. Suppose $h \geqslant f$. Find a function g such that g agrees with f on $[t_{n-1}, 1]$, g is linear on $[0, t_{n-1}]$, and $g \geqslant f$. Since g has at most two pieces, $g \in Z$. It now follows from $f = g \wedge h$ that $f \in Z$. The case $h \leqslant f$ is similar.

3.1.10 Let L be the set of all lattice linear combinations of elements of A. Then $A \subseteq L$. Clearly, L is closed under vector and lattice operations, hence L is a sublattice. Furthermore, every sublattice of X containing A also contains L.

3.1.11 It is straightforward that $W^\vee$ and $W^\wedge$ are closed under positive scalar multiplication. It follows from Exercise 1.3.5(i) that they are closed under addition.

3.1.13 It is easy to see that $W - W$ is a subspace of X as in Section 1.2. It suffices to show that if a is in $W - W$ then so is a^+. Indeed, let $a \in W - W$, then $a = y - x$ for some $x, y \in W$. Applying Exercise 1.3.3(iii), we get $a^+ = x \vee y - x \in W - W$.

3.1.16 Let D be a dense countable subset of Y. Since lattice operations are continuous, $D^\vee$ is dense in $Y^\vee$. Note that $D^\vee$ is still countable. Similarly, $D^{\vee\wedge}$ is a countable dense subset of $Y^{\vee\wedge} = \mathrm{Lat}(Y)$.

Since $\operatorname{span} A$ is separable and $\mathrm{Lat}(A) = \mathrm{Lat}(\operatorname{span} A)$, by the preceding argument we conclude that $\mathrm{Lat}(A)$ is separable and, therefore, its closure is separable.

3.1.24 Indeed, if it were, then, since $\mathbb{1} \in c$, we could approximate $\mathbb{1}$ by a linear combination of finitely many vectors in the sequence. It follows that at least one of these vectors must have infinite support. It follows that its support contains a tail of $\mathbb{N}$. Hence, the sequence has finitely many terms, so that c is finite-dimensional; a contradiction.

This exercise does not contradict Theorem 3.1.20 as ℓ_∞ is not a Banach function space.

3.1.26(i) $\mathcal{G} = \bigl\{\varnothing, [-1,0], [0,1], [-1,1]\bigr\}$.

3.1.26(ii) $\mathcal{G}$ consists of $\mathcal{F}$-measurable sets A that are almost symmetric with respect to the origin, i.e., $\mu\bigl(A \triangle (-A)\bigr) = 0$.

3.2.1 (i)$\Leftrightarrow$(ii) and (ii)$\Rightarrow$(iii) are trivial. To show that (iii)$\Rightarrow$(ii), let $A \subseteq Y$ with $u = \inf_Y A$; show that $u = \inf A$. WLOG, replacing A with $A - u$, we may assume that $u = 0$. Then $0 = \inf A = \inf A^\wedge$. Since $A^\wedge$ is a decreasing net, we are done.

(iv)$\Leftrightarrow$(v) is trivial; just replace (y_α) with $(y_\alpha - y)$. (v)$\Rightarrow$(iii) trivially. (iii)$\Rightarrow$(v) because the dominating net in the definition of order convergence goes down to zero in Y and, therefore, in X. (iv)$\Leftrightarrow$(vi) is trivial.

3.2.3 Let $y_\alpha \downarrow 0$ in Y. Then $y_\alpha \xrightarrow{\|\cdot\|} 0$ in Y and, therefore, in X. Monotone Convergence Lemma 2.5.3 yields $y_\alpha \downarrow 0$ in X.

3.2.4 Let
$$a = \sup_\alpha \inf_{\beta \geqslant \alpha} y_\beta \quad \text{and} \quad b = \inf_\alpha \sup_{\beta \geqslant \alpha} y_\beta;$$
these expressions are defined in Y and, by regularity, in X. It follows easily that $x = a = b$. Indeed, let A be the set of all eventual lower bounds of (y_α) in X. Then $x = \sup A$. For each α, put $v_\alpha = \inf\limits_{\beta \geqslant \alpha} y_\beta$. Then $v_\alpha \in A$ and for every $z \in A$ there exists α with $z \leqslant v_\alpha$. It follows that $a = \sup v_\alpha = x$. Similarly, $b = x$.

3.2.8 Suppose that Y is majorizing and $x \in X$. Then there exists $y \in Y$ such that $x \leqslant |x| \leqslant y$.

3.2.11 It is clear that if e is a weak unit in X then it remains a weak unit in Y. Suppose e is a weak unit in Y. If $0 < x \in X_+$ then there exists $y \in Y$ with $0 < y \leqslant x$. It follows that $e \wedge x \geqslant e \wedge y > 0$, hence e is a weak unit in X.

3.2.12 Let $0 < f \in C(K)$; we will find $w \in Y$ such that $0 < w \leqslant f$. WLOG, $\|f\| = 1$. Using density of Y, find $u, v \in Y$ such that and $\|u - \frac{1}{2}\mathbb{1}\| < \frac{1}{6}$ and $\|v - f\| < \frac{1}{3}$. Then $\frac{1}{3} < u(t) < \frac{2}{3}$ and $f(t) - \frac{1}{3} < v(t) < f(t) + \frac{1}{3}$ for every $t \in K$. It follows that $v - u \leqslant f$, so that $w \coloneqq (v-u)^+ \leqslant f$. Clearly, $w \in Y$. It is left to show that $w \neq 0$. Since $\|f\| = 1$, we have $f(s) = 1$ for some $s \in K$. Then $w(s) \geqslant v(s) - u(s) > f(s) - \frac{1}{2} - \frac{2}{3} = 0$.

Now for counterexamples. The space of all functions in $C[0,1]$ that vanish at 0 is an ideal. The space of all functions f in $C[0,1]$ satisfying $f(0) = f(1)$ is a sublattice containing $\mathbb{1}$. Both are order dense but not norm dense.

3.2.24(iv) $y \in x^{\downarrow}$ iff $y \leqslant x$ iff $x \leqslant x^{\uparrow}$ iff $y \in x^{\uparrow\downarrow}$.

3.2.24(v) By (ii), we have $A^{\uparrow} \subseteq A^{\uparrow\downarrow\uparrow}$. To prove the converse inclusion, suppose that $x \in A^{\uparrow\downarrow\uparrow}$. Then $x \geqslant A^{\uparrow\downarrow}$. Since $A \subseteq A^{\uparrow\downarrow}$ by (ii), we conclude that $x \geqslant A$ and, therefore, $x \in A^{\uparrow}$. The other identity can be proved in a similar fashion.

3.2.24(vii) It follows from $B \subseteq B^{\uparrow\downarrow}$ that $A + B \subseteq A + B^{\uparrow\downarrow}$ and, therefore, $(A+B)^{\uparrow\downarrow} \subseteq (A + B^{\uparrow\downarrow})^{\uparrow\downarrow}$. For the converse inclusion, observe that for every $x \in A$, we have $x + B^{\uparrow\downarrow} = (x+B)^{\uparrow\downarrow} \subseteq (A+B)^{\uparrow\downarrow}$. It follows that $A + B^{\uparrow\downarrow} \subseteq (A+B)^{\uparrow\downarrow}$, and, therefore, $(A + B^{\uparrow\downarrow})^{\uparrow\downarrow} \subseteq (A+B)^{\uparrow\downarrow}$.

3.2.25(i) Use Exercise 3.2.24(v).

3.2.25(iii) Let a be a down-cone. Since $a = a^{\uparrow\downarrow}$ and $a \neq X$, we have $a^{\uparrow} \neq \varnothing$, hence a is bounded above. To show that a is convex, observe that $a = a^{\uparrow\downarrow} = \bigcap_{x \in a^{\uparrow}} x^{\downarrow}$, and note that $x^{\downarrow}$ is convex for every x, and the intersection of convex sets is convex.

3.2.25(iv) Let $\mathcal{F} \subseteq X^{\delta}$ such that $\bigcap \mathcal{F} \neq \varnothing$. We have $\bigcap \mathcal{F} \subseteq (\bigcap \mathcal{F})^{\uparrow\downarrow}$ by Exercise 3.2.24(ii). For every $a \in \mathcal{F}$, we have $\bigcap \mathcal{F} \subseteq a$, hence $(\bigcap \mathcal{F})^{\uparrow\downarrow} \subseteq a$, hence $(\bigcap \mathcal{F})^{\uparrow\downarrow} \subseteq \bigcap \mathcal{F}$.

3.2.25(v) Let $\mathcal{F}$ be a subset of X^{δ} bounded below by some $a \in X^{\delta}$. Then $a \subseteq \bigcap \mathcal{F}$, hence $\bigcap \mathcal{F}$ is non-empty. By (iv), it is a down-cone. It follows that $\bigcap \mathcal{F} = \inf \mathcal{F}$. By Proposition 2.1.1, every subset of X^{δ} that is bounded above has supremum.

It is left to show that every two point set $\{a, b\}$ in X^{δ} is bounded both above and below, as then $a \vee b$ and $a \wedge b$ exist by the preceding paragraph. Since X_+ is generating, X is directed both upwards and downwards; see Exercise 1.2.7. By (iii), there exists $z \in X$ such that $a, b \leqslant x$ in X. It follows that $a, b \leqslant z^{\downarrow}$ in X^{δ}. On the other hand, take any $x \in a$ and $y \in b$, and find $v \in X$ such that $v \leqslant x, y$. Then $v^{\downarrow} \leqslant a, b$.

3.2.26 It is clear that $a \oplus b = b \oplus a$. We will now show associativity. By the definition of $\oplus$, Exercise 3.2.24(vii), and associativity of Minkowski sum, we have

$$a \oplus (b \oplus c) = \big(a + (b+c)^{\uparrow\downarrow}\big)^{\uparrow\downarrow} = (a + (b+c))^{\uparrow\downarrow} = ((a+b)+c))^{\uparrow\downarrow} = (a \oplus b) \oplus c.$$

It follows from $a + 0^{\downarrow} = a - X_+ = a$ that $a \oplus 0^{\downarrow} = a^{\uparrow\downarrow} = a$.

Let $b \coloneqq -(a^{\uparrow})$. Then $a + b = a - a^{\uparrow}$. By Proposition 1.2.13, $\sup(a - a^{\uparrow}) = 0$ (it is at this point that we use the assumption that X is Archimedean). It follows from the definition of $a \oplus b$ and Exercise 3.2.24(vi) and (iv) that $a \oplus b = (a+b)^{\uparrow\downarrow} = 0^{\uparrow\downarrow} = 0^{\downarrow}$.

3.2.28 To show that X^{δ} is a vector space, in view of Exercises 3.2.26 and 3.2.27, it suffices to verify the axioms mentioned in Exercise 3.2.27. Let $a, b \in X^{\delta}$ and $r, s > 0$. One can easily verify that $r \odot (s \odot a) = (rs) \odot a$ and $1 \odot a = a$. We also have

$$r \odot (a \oplus b) = r(a \oplus b) = r(a+b)^{\uparrow\downarrow} = (ra + rb)^{\uparrow\downarrow} = ra \oplus rb = r \odot a \oplus r \odot b.$$

We will next show that $(r+s) \odot a = r \odot a \oplus s \odot a$. First, we claim that $(r+s)a = ra + sa$. Indeed, we trivially have $(r+s)a \subseteq ra + sa$. To show the converse inclusion, we use convexity of a from Exercise 3.2.25(iii):

$$ra + sa = (r+s)\Big(\frac{r}{r+s}a + \frac{s}{r+s}a\Big) \subseteq (r+s)a.$$

Furthermore, by definition of $\odot$, the set $(r+s)\odot a$ is a down-cone. It follows that

$$(r+s)\odot a=(r+s)a=ra+sa=r\odot a+s\odot a=(r\odot a+s\odot a)^{\uparrow\downarrow}=r\odot a\oplus s\odot a.$$

Let's now verify that X^δ is a vector lattice. Suppose that $a,b,c\in X^\delta$ with $a\leqslant b$ and $r\geqslant 0$. It follows from $a\subseteq b$ that $ra\subseteq rb$ and, therefore, $r\odot a\leqslant r\odot b$. Furthermore, $a+c\subseteq b+c$, so that $(a+c)^{\uparrow\downarrow}\subseteq(b+c)^{\uparrow\downarrow}$, which yields $a\oplus c\leqslant b\oplus c$. It follows that X^δ is an ordered vector space. Now Exercise 3.2.25(v) completes the proof.

3.2.29(i) This follows from $x=\sup x^\downarrow$.

3.2.29(ii) $x\geqslant y \quad\Leftrightarrow\quad x^\downarrow\supseteq y^\downarrow \quad\Leftrightarrow\quad Tx\geqslant Ty.$

3.2.29(iii) Clearly, $T0=\not{\vdash}$. To show that T is additive, let $x,y\in X$ and observe that by Exercise 1.3.5(i), $\sup(x^\downarrow+y^\downarrow)=\sup x^\downarrow+\sup y^\downarrow=x+y$. It now follows from Exercise 3.2.24(vi) that

$$Tx\oplus Ty=x^\downarrow\oplus y^\downarrow=(x^\downarrow+y^\downarrow)^{\uparrow\downarrow}=(x+y)^\downarrow=T(x+y).$$

For $r>0$, we have $T(rx)=(rx)^\downarrow=r(x^\downarrow)=r\odot Tx$. It follows that

$$\not{\vdash}=T0=T\bigl(rx+(-rx)\bigr)=r\odot Tx\oplus T(-rx),$$

and, therefore, $T(-rx)=\ominus r\odot Tx$. Finally $T(0x)=T0=\not{\vdash}=0\odot Tx$.

3.2.29(iv) By Exercise 3.2.25(ii), we have $a=\bigcup_{x\in a}x^\downarrow$. It follows that $a=\sup_{x\in a}x^\downarrow=\sup_{x\in a}Tx$.

3.2.30 c is order dense and majorizing in ℓ_∞ by Exercise 3.2.6(iii). Recall that ℓ_∞ is order complete.

3.2.35 Verifying that $|||\cdot|||$ is a norm on X^δ which extends $\|\cdot\|$ is straightforward. Suppose X is norm complete. Let (z_n) be a Cauchy sequence in X^δ; we need to show that it converges in $|||\cdot|||$. By Exercise 2.5.2, we may assume WLOG that (z_n) is increasing and positive. Passing to a subsequence, we may assume WLOG that the series $\sum_{n=1}^\infty|||z_{n+1}-z_n|||$ converges. Put $z_0=0$.

For each n, use the definition of $|||\cdot|||$ to find $x_n\in X_+$ such that $0\leqslant z_n-z_{n-1}\leqslant x_n$ and $\|x_n\|\leqslant|||z_n-z_{n-1}|||+\frac{1}{2^n}$. It follows that $\sum_{n=1}^\infty\|x_n\|$ converges. Since X is complete, $\sum_{n=1}^\infty x_n$ converges. For every n, we have $z_n=\sum_{i=1}^n(z_i-z_{i-1})\leqslant\sum_{i=1}^n x_i\leqslant\sum_{i=1}^\infty x_i$ It follows that (z_n) is bounded above. Since X^δ is order complete, $z_n\uparrow z$ for some $z\in X^\delta$.

Let $n<m$, then $0\leqslant z_m-z_n=\sum_{i=n+1}^m(z_i-z_{i-1})\leqslant\sum_{i=n+1}^m x_i\leqslant\sum_{i=n+1}^\infty x_i$. Taking supremum over m, we get $0\leqslant z-z_n\leqslant\sum_{i=n+1}^\infty x_i$, Since $\sum_{i=n+1}^\infty x_i$ converges in norm to zero as $n\to\infty$, we conclude that $z_n\to z$.

3.3.1(i) It follows from $-|x|\leqslant x\leqslant|x|$ that $0\leqslant x+|x|\leqslant 2|x|$, so that $x+|x|\in J$ and, therefore, $x\in J$.

3.3.1(ii) Suppose that J is an ideal, $x\in J$ and $|y|\leqslant|x|$. Since J is a sublattice, we have $|x|\in J$. It follows that $|y|\in J$ and, therefore, $y\in J$.

To prove the converse, observe that if $x\in J$ then it follows from $\bigl||x|\bigr|\leqslant|x|$ that $|x|\in J$, hence J is a sublattice. It now follows from $0\leqslant y\leqslant x\in J$ that $|y|\leqslant|x|$ and, therefore, $y\in J$.

3.3.3(ii) Let Y be a sublattice and J an ideal. It suffices to show that $|y+z|\in Y+J$ whenever $y\in Y$ and $z\in J$. Observe that $|y+z|=|y|+|y+z|-|y|$. Note that $|y|\in Y$. The triangle inequality yields $\bigl||y+z|-|y|\bigr|\leqslant|y+z-y|=|z|$, so that $|y+z|-|y|\in J$.

3.3.3(iii) Suppose that $T\colon X\to Y$ is interval preserving, let J be an ideal in X, $u\in T(J)$ and $|v|\leqslant|u|$ for some $v\in Y$. Find $x\in J$ with $Tx=u$. Then $|v|\leqslant|Tx|\leqslant T|x|$ implies $v\in\bigl[-T|x|,T|x|\bigr]=T\bigl[-|x|,|x|\bigr]\subseteq T(J)$.

3.3.3(vi) Let J be an ideal. Since J is a sublattice, $\overline{J}$ is a sublattice by Exercise 3.1.6. Suppose $x \in \overline{J}$ and $0 \leqslant y \leqslant x$. Find a sequence (x_n) in J such that $x_n \to x$. Then $x_n^+ \in J$ and $x_n^+ \to x^+ = x$, so that, replacing x_n with x_n^+, we may assume that $x_n \geqslant 0$. Then $J \ni y \wedge x_n \to y \wedge x = y$, so that $y \in \overline{J}$.

3.3.4 If X is Archimedean then it contains two non-zero disjoint vectors x and y as in 1.5.2. The principal ideal I_x is non-trivial as it contains x. It follows from Exercise 1.5.3 that $y \perp I_x$, hence $y \notin I_x$, so that I_x is proper.

If X fails to be Archimedean then there exist non-zero $x, y \in X_+$ such that $nx \leqslant y$ for all $n \in \mathbb{N}$. This implies that $y \notin I_x$, hence I_x is a proper non-trivial ideal.

3.3.5 Suppose $x \in I_a$. Then $|x| \leqslant \lambda a$ for some $\lambda > 0$. It follows that $|Tx| \leqslant T|x| \leqslant \lambda Ta = 0$, so that $Tx = 0$.

3.3.8 It is clear that sets of this form are ideals. Conversely, let J be an ideal in $\mathbb{R}^n$. Let A be the set of all indices i such that the i-th component is non-zero for some vector in J; denote such a vector by x_i. Put $J_A = \operatorname{span}\{e_i : i \in A\}$; clearly, $J \subseteq J_A$. Put $u = \sum_{i \in A} |x_i|$. Then $u \in J$ and $A = \operatorname{supp} u$. It is now easy to see that $J_A = I_u \subseteq J$. Hence, $J = J_A$.

3.3.9 Clearly, it suffices to prove the statement for $C_0(\Omega)$. Let J be an order ideal on $C_0(\Omega)$, and take $f \in C_0(\Omega)$ and $g \in J$. Then $|fg| \leqslant \|f\|_\infty |g|$, so that $fg \in J$.

3.3.12 By the preceding theorem, the map is onto. It is left to verify that $F = \bigcap_{f \in J_F} \ker f$. The implication $F \subseteq \bigcap_{f \in J_F} \ker f$ is straightforward. On the other hand, if $t \notin F$ then, by Urysohn's Lemma, we can find $f \in C(K)$ such that $f(t) = 1$ and f vanishes on F. It follows that $f \in J_F$ and, therefore, $t \notin \bigcap_{f \in J_F} \ker f$.

3.3.14 (i) If $J_1 \not\perp J_2$ then there exists $x_1 \in J_1$ and $x_2 \in J_2$ such that $x_1 \not\perp x_2$ or, equivalently, $|x_1| \wedge |x_2| \neq 0$. Now note that $|x_1| \wedge |x_2| \in J_1 \cap J_2$. Conversely, if $0 \neq x \in J_1 \cap J_2$ then $x \not\perp x$ yields $J_1 \not\perp J_2$.

(ii) Suppose $0 < x \in X$. There exists $y_1 \in J_1$ with $0 < y_1 \leqslant x$. There exists $y_2 \in J_2$ with $0 < y_2 \leqslant y_1$. Then $y_2 \in J_1 \cap J_2$ and $0 < y_2 \leqslant x$.

For the rest of the exercise, we will use the following observation. Suppose that $x \in J_1 + J_2$ and $0 \leqslant y \leqslant |x|$. We have $x = x_1 + x_2$ for some $x_1 \in J_1$ and $x_2 \in J_2$. Therefore, $0 \leqslant y \leqslant |x| \leqslant |x_1| + |x_2|$. It follows from the Riesz Decomposition Theorem that $y = y_1 + y_2$ where $0 \leqslant y_1 \leqslant |x_1|$ and $0 \leqslant y_2 \leqslant |x_2|$. We clearly have $y_1 \in J_1$ and $y_2 \in J_2$ because J_1 and J_2 are ideals, hence $y \in J_1 + J_2$.

To prove that $J_1 + J_2$ is an ideal, suppose that $x \in J_1 + J_2$ and $|y| \leqslant |x|$. Applying the preceding paragraph of the proof to y^+ and y^-, we conclude that $y^+, y^- \in J_1 + J_2$, so that $y \in J_1 + J_2$. Applying the argument in the preceding paragraph with $y = x \geqslant 0$, we get (iv).

To prove (v), let $y \in J_1 + J_2$, and apply (iv) to y^+ and y^-. We can write $y^+ = u_1 + u_2$ and $y^- = v_1 + v_2$ where $0 \leqslant u_1, v_1 \in J_1$ and $0 \leqslant u_2, v_2 \in J_2$. Then $y = y^+ - y^- = (u_1 - v_1) + (u_2 - v_2) = y_1 + y_2$ where $y_1 \coloneqq u_1 - v_1 \in J_1$ and $y_2 \coloneqq u_2 - v_2 \in J_2$. Also,

$$|y_1| \leqslant u_1 + v_1 \leqslant (u_1 + u_2) + (v_1 + v_2) = y^+ + y^- = |y|.$$

Similarly, $|y_2| \leqslant |y|$.

3.3.16 $J_1, J_2 \subseteq J_1 + J_2$ implies $\overline{J_1}, \overline{J_2} \subseteq \overline{J_1 + J_2}$, hence $\overline{J_1} + \overline{J_2} \subseteq \overline{J_1 + J_2}$. To prove the converse inclusion, observe that by Theorem 3.3.15, $\overline{J_1} + \overline{J_2}$ is a closed ideal containing J_1 and J_2, hence it contains $J_1 + J_2$ and, therefore, $\overline{J_1 + J_2}$.

3.3.17 It follows from $J \subseteq J_0$ that $\overline{J}^X \subseteq \overline{J_0}^X$. To prove the converse inclusion, it suffices to show that $J_0 \subseteq \overline{J}^X$. Let $0 < x \in J_0$. Fix $\varepsilon > 0$. There exists $y \in J$ such that $x \leqslant y$. Since Y is dense, there exists $z \in Y$ such that $\|x - z\| < \varepsilon$. WLOG, $z \geqslant 0$; otherwise, replace z with z^+ and use the fact that $|x - z^+| = |x^+ - z^+| \leqslant |x - z|$. Observe that $y \wedge z \in J$. It follows from

$$|x - y \wedge z| = |x \wedge y - y \wedge z| \leqslant |x - z|$$

that $\|x - y \wedge z\| < \varepsilon$. This concludes the proof that $\overline{J}^X = \overline{J_0}^X$. This set is an ideal in X by Exercise 3.3.3(vi). The fact that $\overline{J}^Y = \overline{J}^X \cap Y$ now follows from the remark before the exercise.

To show that the statement may fail without the density assumption, take

$$X = C[-1,1],\ Y = \{f \in X \,:\, f(-1) = f(1)\},\ \text{and}\ J = \{f \in Y \,:\, f(0) = 0\}.$$

Then Y and J are already closed. J is an ideal in Y, but not in X.

3.3.20(ii) Let J be a solid subspace. Then $x \in J$ implies $|x| \in J$, hence J is a sublattice. Clearly, if $x \in J$ and $0 \leqslant y \leqslant x$ then $y \in J$.

On the other hand, suppose that J is an ideal, $x \in J$ and $|y| \leqslant |x|$. Since J is a sublattice, we have $|x| \in J$. It follows from $-|x| \leqslant y \leqslant |x|$ that $0 \leqslant y + |x| \leqslant 2|x|$ and, therefore, $y + |x| \in J$. Since J is a subspace, this yields $y \in J$.

3.3.20(iii) Let $\|\cdot\|$ be a norm on a vector lattice X. If the norm is a lattice norm then B_X is solid because if $x \in B_X$ and $|y| \leqslant |x|$ then $\|y\| \leqslant \|x\| \leqslant 1$, so that $y \in B_X$. Conversely, suppose that B_X is solid; let's verify that this forces $\|\cdot\|$ to be lattice norm. Suppose that $|y| \leqslant |x|$. WLOG, $y \neq 0$. Then $\frac{|y|}{\|x\|} \leqslant \frac{|x|}{\|x\|} \in B_X$, implies $\frac{y}{\|x\|} \in B_X$, which yields $\|y\| \leqslant \|x\|$.

3.3.21(i) Follows immediately from the fact that the intersection of any collection of solid sets is again solid.

3.3.21(ii) Put $B = \{y \in X \,:\, |y| \leqslant |x| \text{ for some } x \in A\}$. Then B is solid and $A \subseteq B$, hence $\operatorname{Sol}(A) \subseteq B$. On the other hand, every solid set containing A also contains B, so that $B \subseteq \operatorname{Sol}(A)$.

3.3.21(iii) Follows from (ii).

3.3.22 Let A be a solid set. Clearly, if A is order closed then it is closed under taking sups of increasing positive nets, so we only need to prove the converse. Suppose that (x_α) is a net in A such that $x_\alpha \xrightarrow{\mathrm{o}} x$ for some x; we need to show that $x \in A$. Since A is solid, it suffices to prove that $|x| \in A$.

By order continuity of lattice operations, we have $|x_\alpha| \xrightarrow{\mathrm{o}} |x|$. By definition of order convergence, there exists a net (a_γ) such that $a_\gamma \uparrow x$ and for every γ there exists an α such that $a_\gamma \leqslant |x_\alpha|$. Replacing a_γ with a_γ^+, we may assume WLOG that $a_\gamma \geqslant 0$. It now follows that $a_\gamma \in A$ and, therefore, $|x| \in A$.

3.3.26 Let $B = \bigcup_{a \in A}[0,a]$; let $x, y \in B$, and $\lambda, \mu \in \mathbb{R}_+$ with $\lambda + \mu = 1$. Find $a, b \in A$ with $x \in [0,a]$ and $y \in [0,b]$. Then $0 \leqslant \lambda x + \mu y \leqslant \lambda a + \mu b \in A$. It follows that $\lambda x + \mu y \in B$.

3.3.28 Here is a counterexample. Let $X = C[-1,1]$ and $h \in X$ where $h(t) = t$. Let $A = \operatorname{Sol}(h + \varepsilon B_X)$ where ε is a small positive real, say, $\varepsilon = \frac{1}{4}$. By definition, A is solid and $h \in \operatorname{Int} A$. Note that every f in the ball $h + \varepsilon B_X$ has a root, and, therefore, every $f \in A$ has a root. It is easy to see that $|h|$ may be approximated by functions with no roots, hence $|h| \in \partial A$ and, therefore, $|h| \notin \operatorname{Int} A$. It follows that $\operatorname{Int} A$ is not solid.

3.4.1 Clearly, $\ker T$ is a subspace. Being a lattice homomorphism, T is positive. Suppose that $x \in \ker T$ and $|y| \leqslant |x|$. Then $|Ty| = T|y| \leqslant T|x| = |Tx| = 0$, hence $Ty = 0$.

In fact, this exercise is a special case of the fact that the pre-image of an ideal under a lattice homomorphism is again an ideal; see Exercise 3.3.3(iv).

3.4.4(i) Follows from our definition of the quotient order.

3.4.4(ii) See the proof of Theorem 3.4.2.

3.4.4(iii) Being a surjective lattice homomorphism, the quotient map Q is interval preserving by Exercise 1.6.11.

3.4.4(iv) Take any $x_0 \in \varphi$ (think of it as an initial guess). Then $|x_0| \in |\varphi|$, so that $|x_0| = y + a$ for some $a \in J$. Put $b = |a| \in J$, then $|x_0| \leqslant y + b$, so that $\bigl(|x_0| - b\bigr)^+ \leqslant y$. Put

$$x = (x_0^+ - x_0^+ \wedge b) - (x_0^- - x_0^- \wedge b) = x_0 - (x_0^+ \wedge b - x_0^- \wedge b)$$

It follows from $x_0^+ \perp x_0^-$ that $(x_0^+ - x_0^+ \wedge b) \perp (x_0^- - x_0^- \wedge b)$, hence Exercise 1.3.7 yields $x^+ = x_0^+ - x_0^+ \wedge b$ and $x^- = x_0^- - x_0^- \wedge b$. Since

$$|x - x_0| \leqslant |x_0^+ \wedge b| + |x_0^- \wedge b| \leqslant 2b,$$

we have $x - x_0 \in J$, hence $x \in \varphi$. Finally,

$$|x| = x^+ + x^- = |x_0| - (x_0^+ \wedge b + x_0^- \wedge b) = |x_0| - |x_0| \wedge b = \bigl(|x_0| - b\bigr)^+ \leqslant y.$$

3.4.4(v) Since $\varphi_1 \geqslant 0$, there exists $0 \leqslant x_1 \in \varphi_1$. Choose the rest of the sequence inductively. Suppose $x_1, \dots, x_{n-1}$ have been selected. It follows from $\varphi_n - \varphi_{n-1} \geqslant 0$ that there exists $0 \leqslant z \in \varphi_n - \varphi_{n-1}$. Put $x_n = x_{n-1} + z$. Then $x_n \in \varphi_n$ and $x_n \geqslant x_{n-1}$.

3.4.5(i) Let $0 \leqslant \varphi \in X/J$. Find $0 \leqslant x \in \varphi$. Then $\widetilde{T}\varphi = Tx \geqslant 0$.

3.4.5(ii) Let $\varphi \in X/J$. Pick any $x \in \varphi$. Then $|x| \in |\varphi|$, so that $\bigl|\widetilde{T}\varphi\bigr| = |Tx| = T|x| = \widetilde{T}|\varphi|$.

3.4.9 Putting $x_0 = x - x \wedge y$ and $y_0 = y - x \wedge y$ solves the first part. Suppose now that X is a Banach lattice and $\varepsilon > 0$. Find x_1 and y_1 with $\tilde{x}_1 = \tilde{x}$, $\tilde{y}_1 = \tilde{y}$, $\|x_1\| \leqslant (1+\varepsilon)\|\tilde{x}\|$ and $\|y_1\| \leqslant (1+\varepsilon)\|\tilde{y}\|$. Put $x_2 = x_0 \wedge |x_1|$ and $y_2 = y_0 \wedge |y_1|$. Since the quotient map is a lattice homomorphism, we have $\tilde{x}_2 = \tilde{x}$ and $\tilde{y}_2 = \tilde{y}$. It follows from $0 \leqslant x_2 \leqslant x_0$ and $0 \leqslant y_2 \leqslant y_0$ that $x_2 \perp y_2$. So x_2 and y_2 satisfy the requirements.

3.4.11 Using the definition of the quotient norm, find $w \in X$ with $Qw = Qu$ and $\|w\| \leqslant \|Qu\| + \varepsilon$. WLOG, $w \geqslant 0$; otherwise, replace it with $|w|$. Put $v = u \wedge w$. Then $0 \leqslant v \leqslant u$, $Qv = Qu$, and it follows from $0 \leqslant v \leqslant w$ that $\|v\| \leqslant \|Qu\| + \varepsilon$.

3.4.12(ii) Suppose that X is order complete. Since each X_α may be viewed as an ideal in X, it is order complete by Proposition 3.3.18. Suppose now that X_α is order complete for every α. Let $A \leqslant u$ where $A \subseteq X_+$ and $u \in X_+$. Then $\pi_\alpha(A) \leqslant \pi_\alpha(u) = u_\alpha$ in X_α for every α. Since X_α is order complete, $v_\alpha := \sup \pi_\alpha(A)$ exists in X_α. Clearly, $v_\alpha \leqslant u_\alpha$. Put $v = (v_\alpha)$ in X. It is straightforward that $A \leqslant v \leqslant u$. Since u was an arbitrary upper bound of A, it follows that $v = \sup A$.

4.1.4 Let $q\colon X \to X/M$ be the quotient map. Suppose that $\dim X/M > 1$. By Exercise 3.3.4, X/M contains a proper non-trivial ideal, say J. It is easy to see that $q^{-1}(J)$ is a proper ideal in X containing M. Maximality of M yields $q^{-1}(J) = M$, which contradicts J being non-trivial.

4.1.5 Let M be a proper ideal in a vector lattice X. If M is maximal then $\dim X/M = 1$, so that X/M is lattice isomorphic to $\mathbb{R}$. Since the quotient map $q\colon X \to X/M$ is a lattice homomorphism, the diagram $X \xrightarrow{q} X/M \cong \mathbb{R}$ yields a required lattice homomorphism.

Conversely, let $\varphi\colon X \to \mathbb{R}$ be a non-zero lattice homomorphism, and put $M = \ker\varphi$. Them M is a non-trivial ideal. It is maximal because it has linear co-dimension 1.

4.2.1 Let X be an Archimedean vector lattice with $\dim X > 1$. By Exercise 1.2.12, there is a vector $x \in X$ which is neither positive nor negative. It follows that x^+ and x^- are both non-zero. It now follows from $x^+ \wedge x^- = 0$ that they are not comparable.

4.2.6 (i) Let $f, g \in C(K)$ with $f \wedge g = 0$. Then either $f(t_0) = 0$, in which case $f \in J_{\{t_0\}}$ or $g(t_0) = 0$, in which case $g \in J_{\{t_0\}}$. Now use Exercise 4.2.5(iv).

(ii) By 4.1.2 and 4.1.3, there exists a maximal proper ideal M containing J. By Exercise 4.1.5, $M = \ker \varphi$ for some lattice homomorphism $\varphi \colon C(K) \to \mathbb{R}$. By Proposition 4.1.1, φ may be chosen to be a point evaluation at some $t \in K$. This yields $M = J_{\{t\}}$. Hence, $J \subseteq J_{\{t\}}$.

To prove uniqueness, suppose that $J \subseteq J_{\{t\}}$ and $J \subseteq J_{\{s\}}$ for some $s \neq t$. Since J is prime, it follows that J equals either $J_{\{t\}}$ or $J_{\{s\}}$, hence one of them contains the other, which is a contradiction.

4.2.9 It is a straightforward verification that $a > 0$ in ${}^n\mathbb{R}$ iff $Ta > 0$ in X. It follows that $Ta^+ = (Ta)^+$, so that T is a lattice homomorphism. If $a \neq 0$ then either $a > 0$ or $a < 0$, so that $Ta > 0$ or $Ta < 0$, hence $Ta \neq 0$; it follows that T is one-to-one.

4.2.10 Either $na < b$ for every $n \in \mathbb{N}$, which is equivalent to $a \ll b$, or $b \leqslant na$ for some $n \in \mathbb{N}$, which is equivalent to $I_b \subseteq I_a$.

4.2.12 Note that $I_e = X$. Take a non-zero $x \in X$ such that x is not a scalar multiple of e. Replacing x with $-x$, if necessary, we may assume that $x > 0$. If $x \ll e$, we are done. Otherwise, Exercise 4.2.10 yields $X = I_e = I_x$. Now Lemma 4.2.11 says that there exists $\lambda > 0$ such that $|x - \lambda e| \ll e$. Finally, observe that $|x - \lambda e| > 0$ because x is not a scalar multiple of e.

4.2.16(i) The forward implication is trivial. To prove the converse, suppose hat $x \wedge y \in Y$. It follows from $\bigl(x - x \wedge y\bigr) \wedge \bigl(y - x \wedge y\bigr) = 0$ that either $x - w \wedge y$ or $y - x \wedge y$ is in Y, hence either x or y is in Y.

4.2.16(ii) Let Y be a prime subspace and $x \in Y$. It follows from $x^+ \wedge x^-$ that either x^+ or x^- is in Y; hence both are in Y.

4.3.1(ii) Let $e = |b| \vee |c|$, then $a, b, c \in I_e$. By Theorem 4.1.7, we may view I_e as a sublattice of $C(K)$. Thus, we may view a, b, and c as continuous function, and it suffices to verify the statement in question pointwise; hence it is enough to verify it for real numbers, which is straightforward.

4.3.1(iii) Again, it suffices to prove the statement for real numbers. Let $d = (a - b - c)^+$. It is clear that $d \leqslant a$, so that $a - d \geqslant 0$. On the other hand, $d \geqslant (a-b)^+ - c$, so that $c \geqslant (a-b)^+ - d = a - a \wedge b - d$.

4.3.6 In order to verify that T is a lattice homomorphism, fix $f, g \in L_n$. Choose formal expressions Φ and Ψ representing f and g, respectively. Then

$$f(x_1, \dots, x_n) = \Phi(x_1, \dots, x_n) \text{ and } g(x_1, \dots, x_n) = \Psi(x_1, \dots, x_n).$$

Note that $\Phi + \Psi$, $\lambda(\Phi)$, and $|\Phi|$ represent $f + g$, λf, and $|f|$ respectively (here λ is a real scalar), so that

$$(f+g)(x_1, \dots, x_n) = \Phi(x_1, \dots, x_n) + \Psi(x_1, \dots, x_n) = f(x_1, \dots, x_n) + g(x_1, \dots, x_n),$$

$$(\lambda f)(x_1, \dots, x_n) = \lambda\bigl(\Phi(x_1, \dots, x_n)\bigr) = \lambda f(x_1, \dots, x_n), \text{ and}$$

$$|f|(x_1, \dots, x_n) = \bigl|\Phi(x_1, \dots, x_n)\bigr| = \bigl|f(x_1, \dots, x_n)\bigr|.$$

This shows that T is a lattice homomorphism.

Observe that $T\delta_i = x_i$ as $i = 1, \dots, n$. Moreover, T is a unique lattice homomorphism from L_n to X with this property because L_n is generated by $\delta_1, \dots, \delta_n$, so that every element of L_n is a lattice-linear combination of δ_i's,

4.4.6 Let V be as in Theorem 4.4.3; $V\colon H_n \to I_e$ where $e = |x_1| \vee \cdots \vee |x_n|$. By Remark 4.4.4, V is bounded. Similarly, let W be the positively homogeneous function calculus for $Tx_1, \dots, Tx_n$. Note that $W\colon H_n \to I_u$ where $u = Te$. It is easily verified that T viewed as an operator from $\big(I_e, \|\cdot\|_e\big)$ to $\big(I_u, \|\cdot\|_u\big)$ is bounded. We need to prove that $W = TV$. Since T is a lattice homomorphism, W and TV agree on L_n. Finally, recall that L_n is dense in H_n.

5.1.2 Proposition 5.1.1 yields that J is order dense in $B(J)$. In particular, if $X = B(J)$ then J is order dense in X. If X is Archimedean and J is order dense in X then $x = \sup J \cap [0, x]$ by Proposition 3.2.15 for every $x \in X_+$; now Theorem 5.1.1 yields that $x \in B(J)$; hence $B(J) = X$.

5.1.10 Combine Exercise 5.1.2 with Corollary 5.1.9.

5.1.11 By Exercise 3.3.14(iii), $J + J^d$ is an ideal. It suffices to show that $(J + J^d)^d = \{0\}$. Indeed, let $x \in X$ such that $x \perp (J + J^d)$. Then $x \perp J$ and $x \perp J^d$, hence $x \perp x$ and, therefore, $x = 0$.

To show that $J + J^d$ may be proper, let $X = C[0,1]$ and J be the ideal of all functions in X vanishing on $[0, \frac{1}{2}]$. Then J^d is the set of all functions in X vanishing on $[\frac{1}{2}, 1]$, and, therefore, $J + J^d = J_{\{\frac{1}{2}\}} \neq X$.

5.1.12 First, show that $C_0(\mathbb{R})$ is an ideal (and, therefore, a sublattice) of $C(\mathbb{R})$. Suppose that $f \in C_0(\mathbb{R})$ and $|g| \leqslant |f|$. It follows from $\lim\limits_{|t|\to\infty} f(t) = 0$ that $\lim\limits_{|t|\to\infty} g(t) = 0$, hence $g \in C_0(\mathbb{R})$.

$C_0(\mathbb{R})$ is not a band. If it were, we would have $C_0(\mathbb{R})^{dd} = C_0(\mathbb{R})$. However, since $C_0(\mathbb{R})$ contains functions that vanish nowhere, $C_0(\mathbb{R})^d = \{0\}$ and, therefore, $C_0(\mathbb{R})^{dd} = C(\mathbb{R})$.

5.1.13 If e is a weak unit then $\{e\}^d = \{0\}$ by Exercise 5.1.5(vii), so that $B_e = \{e\}^{dd} = X$. If $B_e = X$ then Proposition 5.1.3 yields that $x \wedge ne \uparrow x$ for every positive x.

On the other hand, if e is not a weak unit then there exists $x > 0$ such that $x \perp e$. In this case, $x \wedge ne = 0$ for every n.

5.1.14 We use Exercise 5.1.13(iii). Let $0 \leqslant x \in B_e$. By Proposition 5.1.3, $x \wedge ne \uparrow x$ in X and, therefore in B_e. Hence, e is a weak unit in B_e by Exercise 5.1.13.

5.1.15 Follows from continuity of lattice operations.

5.1.16 Let B be a band in a normed lattice X. Suppose that $x_n \to x$ for some sequence (x_n) in B and $x \in X$; we need to show that $x \in B$. If X is a Banach lattice, this follows immediately from Theorem 2.5.4. If X is only a normed lattice, one may argue as follows. By Exercise 1.3.22, X is Archimedean, and, therefore, $B = A^d$ for some A by Corollary 5.1.8. We have $x_n \perp a$ for every $a \in A$ and $n \in \mathbb{N}$, which is equivalent to $|x_n| \wedge |a| = 0$. Continuity of lattice operations yield $|x| \wedge |a| = 0$, so that $x \in A^d$.

5.1.17 A band is an order closed ideal. In an order continuous Banach lattice, order closed and norm closed sets agree; see Exercise 2.5.15.

5.1.27(i) It is trivial that c_0 is a closed ideal and it is not of the form $\mathcal{J}_A$. To see that c_0 is not order closed in ℓ_∞, let $x_n = (\underbrace{1, \dots, 1}_{n\text{ times}}, 0, \dots)$, then $x_n \in c_0$, $x_n \uparrow \mathbb{1}$ in ℓ_∞, but $\mathbb{1} \notin c_0$.

5.1.27(ii) It is easy to see that J_0 is an ideal in X, hence J is a closed ideal. Note that the distance from J_0 to $\mathbb{1}$ is 1, so that $\mathbb{1} \notin J$. It is easy to see that it is not of the form $\mathcal{J}_A$. To see that it is not order closed, let $f_n = \mathbb{1}_{(\frac{1}{n}, 1]}$. Then $f_n \in J$ and $f_n \uparrow \mathbb{1}$, yet $\mathbb{1} \notin J$.

5.1.28 We have $B_f = \{f\}^{dd}$ by Corollary 5.1.9. It is easy to see that $\{f\}^{dd} = L_p(A)$.

5.1.30 Let Y be a closed sublattice of $L_p(\mu)$. We will show that there exists a measurable set A, a function $h \in L_p(\mu)$ with $\operatorname{supp} h = A$, and a sub-$\sigma$-algebra $\mathcal{G}$ of the restriction of $\mathcal{F}$ to A such that the support of every function in Y is contained in A and $Y = h \cdot L_p(A, \mathcal{G}, \mu|_{\mathcal{G}}) = \{hf : f \in L_p(A, \mathcal{G}, \mu|_{\mathcal{G}}\}$.

Let B be the band generated by Y in $L_p(\Omega, \mathcal{F}, \mu)$. By Theorem 5.1.26, $B = L_p(A)$ for some $A \in \mathcal{F}$. It is clear that the support of every function in Y is contained in A. We claim that there exists $h \in Y_+$ with $\operatorname{supp} h = A$. To prove the claim, note that B is a norm closed ideal by Proposition 5.1.25, hence B is the norm closure of the ideal $I(Y)$. It follows that there is a sequence (f_n) in $I(Y)$ such that $f_n \xrightarrow{\|\cdot\|} \mathbb{1}_A$. Passing to a subsequence, we may assume that $f_n \xrightarrow{\mathrm{o}} \mathbb{1}_A$ in $L_p(A)$ by Theorem 2.5.4. It follows from Proposition 2.4.27 that $f_n \xrightarrow{\text{a.e.}} \mathbb{1}_A$. For each n, find $g \in Y$ with $|f_n| \leqslant g_n$. For almost every $t \in A$ there exists n such that $f_n(t)$ and, therefore, $g_n(t)$ is positive. Put $h = \sum_{n=1}^{\infty} \frac{g_n}{2^n \|g_n\|}$. The series converges in norm because it converges absolutely. Then $h \in Y$ and $h(t) > 0$ a.e. on A. This proves the claim.

Proceeding as in 1.6.17, the map $Tf = \frac{f}{h}$ defines a lattice isometry from $L_p(A, \mathcal{F}|_A, \mu|_A)$ to $L_p(A, \mathcal{F}|_A, \nu)$ for some finite measure ν, with $Th = \mathbb{1}$. If we view Y as closed sublattice of $L_p(A, \mathcal{F}|_A, \mu|_A)$ then $T(Y)$ is a closed sublattice of $L_p(A, \mathcal{F}|_A, \nu)$. By Theorem 3.1.25, there exists a sub-σ-algebra $\mathcal{G}$ of $\mathcal{F}|_A$ such that $T(Y) = L_p(A, \mathcal{G}, \nu)$. Applying T^{-1}, we get the desired result.

5.2.6 The following solution is borrowed from Proposition 6.1 of [dPW15]. Suppose that $X = B \oplus B^d$ for some band B in X. We have $\operatorname{supp} f \cap \operatorname{supp} g = \varnothing$ whenever $f \in B$ and $g \in B^d$; it follows that $\operatorname{supp} B \cap \operatorname{supp} B^d = \varnothing$. We claim that $\operatorname{supp} B \cup \operatorname{supp} B^d = \operatorname{supp} X$. Indeed, we trivially have $\operatorname{supp} B \cup \operatorname{supp} B^d \subseteq \operatorname{supp} X$. Let $t \in \operatorname{supp} X$, then $h(t) \neq 0$ for some $h \in X$. Find $f \in B$ and $g \in B^d$ with $h = f + g$; then $f(t) \neq 0$ or $g(t) \neq 0$, so that $t \in \operatorname{supp} B \cup \operatorname{supp} B^d$. Thus, $X = \operatorname{supp} B \cup \operatorname{supp} B^d$ is a disconnection of $\operatorname{supp} X$; a contradiction.

5.2.9 We clearly have $I(A) + I(B) = X$ and $I(A) \perp I(B)$. By Proposition 5.2.7, $I(A)$ and $I(B)$ are complementary projection bands. Let P be the band projection onto $I(A)$. Let $x \in I(A)$. By assumption, $x = a + b$ for some $a \in A$ and $b \in B$. It follows that $x = Px = a$, hence $I(A) \subseteq A$ and, therefore, $I(A) = A$. Similarly, $I(B) = B$.

5.2.10 Let B be a projection band in X, P the corresponding band projection, and $x \in B$. Then we can write $x = y + z$ for some $y \in B$ and $z \in B^d$. Then $|x| = |y| + |z|$ Exercise 1.5.3(ii). Since B and B^d are bands, we have $|y| \in B$ and $|z| \in B^d$. Therefore, $P|x| = |y| = |Px|$.

5.2.13
$$B_1 \subseteq B_2 \quad \Leftrightarrow \quad \operatorname{Range} P_1 \subseteq \operatorname{Range} P_2 \quad \Leftrightarrow \quad P_1 = P_1 P_2 \quad \Rightarrow \quad P_1 \leqslant P_2.$$
On the other hand, if $P_1 \leqslant P_2$ then $P_1 = P_1^2 \leqslant P_1 P_2 \leqslant P_1$, so that $P_1 P_2 = P_1$.

5.2.16 By Theorem 5.2.15, it suffices to show that $\sup B_a \cap [0, x]$ exists iff $\sup_n x \wedge na$ exists for every $x \in X_+$,and the two suprema are equal. Take any $x \in X_+$ and let $D = \{x \wedge na : n \in \mathbb{N}\}$. It suffices to show that $B_a \cap [0, x]$ and D have the same upper bounds. Since $D \subseteq B_a \cap [0, x]$, every upper bound of $B_a \cap [0, x]$ is also an upper bound of D. Conversely, suppose that $D \leqslant u$ and $y \in B_a \cap [0, x]$. By Proposition 5.1.3, $y = \sup\{y \wedge na : n \in \mathbb{N}\}$. Since $0 \leqslant y \wedge na \leqslant x \wedge na \leqslant u$, it follows that $y \leqslant u$. Hence, $B_a \cap [0, x] \leqslant u$.

5.2.17 It follows from $[0, x] \cap J \subseteq [0, x] \cap B(J)$ that $\big([0, x] \cap B(J)\big)^{\uparrow} \subseteq \big([0, x] \cap J\big)^{\uparrow}$. To prove the other inclusion, suppose that $[0, x] \cap J \leqslant v$. Take $z \in [0, x] \cap B(J)$. By Theorem 5.1.1(iv), $z = \sup[0, z] \cap J$. It follow from $0 \leqslant z \leqslant x$ that $[0, z] \cap J \subseteq [0, x] \cap J \leqslant v$, so that $z \leqslant v$. Therefore, $\big([0, x] \cap J\big)^{\uparrow} = \big([0, x] \cap B(J)\big)^{\uparrow}$.

The second claim now follows from Theorem 5.2.15.

5.2.18 The forward implication is obvious. To prove the converse, let $B = \operatorname{Range} P$, the projection band corresponding to P. It follows from $a = Pa$ that $a \in B$ and, therefore, $B_a \subseteq B$. On the other hand, since P vanishes on $\{a\}^d = B_a^d$, we have $B_a^d \subseteq \ker P$ and, therefore, $B = \operatorname{Range} P \subseteq B_a^{dd} = B_a$. It follows that $B = B_a$.

5.2.19 Recall also that P is a lattice homomorphism by Exercise 5.2.10 and $0 \leqslant P \leqslant I$ implies that P is order continuous. Let $0 \leqslant x \in B_a$. Then $(x \wedge na) \uparrow x$, hence $(Px) \wedge (nPa) \uparrow Px$ and, therefore, $Px \in B_{Pa}$. It follows that $P(B_a) \subseteq B_{Pa}$.

To prove the reverse inclusion, let $0 \leqslant x \in B_{Pa}$. It follows from $Pa \leqslant a$ that $B_{Pa} \subseteq B_a$, so that $x \in B_a$. Let $B = \operatorname{Range} P$, then B is a band, and $Pa \in B$ implies $B_{Pa} \subseteq B$, hence $x \in B$; it follows that $x = Px \in P(B_a)$.

5.2.20(i) We clearly have $B \subseteq Y \cap B^{dd}$. Let's write B^d_X and B^d_Y to denote the disjoint complement of B in X and in Y, respectively; same for B^{dd}_X and B^{dd}_Y. Clearly, $B^d_Y \subseteq B^d_X$, so that

$$B^{dd} \cap Y = B^{dd}_X \cap Y = \big\{x \in Y \;:\; x \perp B^d_X\big\} \subseteq \big\{x \in Y \;:\; x \perp B^d_Y\big\} = B^{dd}_Y = B,$$

because B is a band in Y.

5.2.20(ii) It is clear that $B \cap Y$ is an ideal in Y. Suppose that $x_\alpha \uparrow x$ in Y for some net (x_α) in $B \cap Y$ and some $x \in Y$. Then $x_\alpha \uparrow x$ in X by, hence $x \in B$.

5.2.20(iii) We write $\mathcal{B}(X)$ and $\mathcal{B}(Y)$ for the sets of all bands in X and in Y, respectively. By (ii), the map $\varphi(B) = B \cap Y$ is a map from $\mathcal{B}(X)$ to $\mathcal{B}(Y)$. On the other hand, the map $\psi(B) = B^{dd}$ is a map from $\mathcal{B}(Y)$ to $\mathcal{B}(X)$. It is left to show that $\varphi \circ \psi$ and $\psi \circ \varphi$ are the identities. In one direction, if $B \in \mathcal{B}(Y)$ then $B = B^{dd} \cap Y$ by (i). To prove the converse, let $B \in \mathcal{B}(X)$, and put $B_0 = B \cap Y$; we need to show that $B = B_0^{dd}$. It follows from $B_0 \subseteq B$ and B being a band in X that $B_0^{dd} \subseteq B$. To show the converse inclusion, fix $0 \leqslant x \in B$. By Proposition 3.2.15, $x = \sup [0, x] \cap Y$, where the supremum is computed in X. Since $[0, x] \cap Y \subseteq B_0$, by Proposition 5.1.1(iii), x belongs to the band generated by B_0 in X, i.e., to B_0^{dd}.

5.2.20(iv) By (ii), $B \cap Y$ is a band in Y. It follows from $0 \leqslant P \leqslant I$ that P leaves Y invariant. The restriction $P_{|Y}$ still satisfies $0 \leqslant P_{|Y} \leqslant I$ and $P^2_{|Y} = P_{|Y}$, hence it is a band projection on Y. It is easy to see that $\operatorname{Range} P_{|Y} = B \cap Y$, hence this set is a projection band.

5.2.20(v) We use Theorem 5.2.15 and the fact that ideals are regular. Suppose that B is a projection band and $0 \leqslant x \in I_a$. Then the set $[0, x] \cap B$ is contained in I_a and $y \coloneqq \sup [0, x] \cap B$ exists in X. It follows from $0 \leqslant y \leqslant x$ that $y \in I_a$, hence $y \coloneqq \sup_{I_a} [0, x] \cap B = \sup_{I_a} [0, x] \cap (B \cap I_a)$. It follows that $B \cap I_a$ is a projection band in I_a.

To prove the converse, fix an arbitrary $a \in X_+$. Then $[0, a] \cap B \subseteq I_a$ and, by assumption, $\sup [0, a] \cap B = \sup_{I_a} [0, a] \cap B = \sup_{I_a} [0, a] \cap (B \cap I_a)$ exits.

5.2.20(vi) Recall that for a net (x_α) in Y and $x \in Y$, we have $x_\alpha \uparrow x$ in X iff $x_\alpha \uparrow x$ in Y. Let $0 \leqslant x \in B^Y(Y \cap J)$. Since $Y \cap J$ is an ideal in Y, there exists a net (x_α) in $Y \cap J$ such that $0 \leqslant x_\alpha \uparrow x$ in Y and, therefore, in X. It follows that $x \in B(J)$, hence $x \in Y \cap B(J)$. Conversely, let $0 \leqslant x \in Y \cap B(J)$. Then there exists a net (x_α) in J such that $0 \leqslant x_\alpha \uparrow x$ in X. Then $x \wedge x_\alpha \in Y \cap J$ for every α and $0 \leqslant x \wedge x_\alpha \uparrow x$ in X and, therefore, in Y. It follows that $x \in B^Y(Y \cap J)$.

5.2.20(vii) By (ii), $B^X_a \cap Y$ is a band in Y. Since it contains a, it also contains B^Y_a, hence $B^Y_a \subseteq B^X_a \cap Y$. By (i), $B^Y_a = Y \cap (B^Y_a)^{dd} \supseteq Y \cap \{a\}^{dd} = Y \cap B^X_a$, where the disjoint complements are computed in X. It follows that $B^Y_a = B^X_a \cap Y$. The last claim now follows from (iv).

5.2.20(viii) No, use Exercise 1.5.13(xi). Let $X = L_p[0, 1]$, $Y = C[0, 1]$, let f be as in the solution of Exercise 1.5.13(xi). Since f is a weak unit in $C[0, 1]$, we have $B^Y_f = C[0, 1]$. In particular, $\mathbb{1} \in B^Y_f$. On the other hand, B^X_f is the set of all functions in $L_p[0, 1]$ whose support is contained in $\operatorname{supp} f$; hence $\mathbb{1} \notin B^X_f$.

5.2.20(ix) For $y \in Y_+$, we have $P^X_a y = \sup_X (y \wedge na) = \sup_Y (y \wedge na) = P^X_a y$ by Exercise 5.2.16.

5.2.22 Since every projection band is a band, and every band is order and, therefore, uniformly closed, we only need to prove that if I_e is uniformly closed then it is a projection band. Suppose that I_e is uniformly closed. Let $x \in X_+$. Find $u \in X_+$ such that $x, e \in I_u$. Since X is uniformly complete, we may identify I_u with $C(K)$ for some compact K. Since I_e is uniformly closed in X, it is closed in $\big(I_u, \|\cdot\|_u\big)$. Viewing e as an element of $C(K)$ and applying Exercise 2.2.6(iv), we conclude that the set $V := \{e \neq 0\}$ is clopen, and, therefore, compact. In particular, the characteristic function $\mathbb{1}_V$ is continuous. It follows that $e \geqslant r\mathbb{1}_V$ for some $r > 0$. If we view x as an element of $C(K)$ then its restriction to V is bounded above by a scalar multiple of $\mathbb{1}_V$ and, therefore, of e. We can then write x as a disjoint sum $x = x \cdot \mathbb{1}_V + x \cdot \mathbb{1}_{V^C}$, where $x \cdot \mathbb{1}_V \in I_e$ and $x \cdot \mathbb{1}_{V^C} \perp e$. Hence $X = I_e \oplus I_e^d$. Now apply Corollary 5.2.8.

5.2.27 Since PP implies PPP, it suffices to prove the claim for PPP. Suppose that there exist non-zero positive x and a such that $x \geqslant na$ for all $n \in N$, hence $x \wedge na = na$. It follows from Exercise 5.2.16 that $h := \sup\{na \,:\, n \in \mathbb{N}\}$ exists. This is a contradiction because

$$2h = \sup\{2na \,:\, n \in \mathbb{N}\} \leqslant \sup\{na \,:\, n \in \mathbb{N}\} = h,$$

so that $h = 0$; but $h \geqslant a$ yields $a = 0$.

5.4.1 (i)$\Leftrightarrow$(ii)$\Rightarrow$(iii) is straightforward. To show that (iii)$\Rightarrow$(ii), suppose that $\dim I_a > 1$, then I_a contains two non-zero disjoint elements by 1.5.2. Scaling them if necessary, we may assume that they are in $[0, a]$.

5.4.2 Suppose a and b are two atoms such that $x := a \wedge b \neq 0$. Then $0 < x \leqslant a$ and $0 < x \leqslant b$ yields that x is a scalar multiple of both a and b, hence a and b are scalar multiples of each other.

5.4.3 Let a be an atom in X. There exists $y \in Y$ such that $0 \leqslant y \leqslant a$, which implies that a is a scalar multiple of y, hence $a \in Y$. Conversely, suppose that $a \in Y$ in an atom in Y, but not in X. Then we find non-zero disjoint $x, y \in X_+$ in $[0, a]$ and then non-zero disjoint $b, c \in Y$ such that $b \in [0, x]$ and $c \in [0, y]$; a contradiction.

5.4.6 Let a be an atom in X; we will show that $a \oplus 0$ is an atom in $X \oplus Y$. Suppose that $0 \leqslant x \oplus y \leqslant a \oplus 0$. Then $0 \leqslant y \leqslant 0$ yields that $y = 0$, while $0 \leqslant x \leqslant a$ implies that x is a scalar multiple of a. Therefore, $x \oplus y$ is a scalar multiple of $a \oplus 0$ and, hence, $a \oplus 0$ is an atom in $X \oplus Y$. In a similar fashion, one can show that $0 \oplus b$ is an atom in $X \oplus Y$ whenever b is an atom in Y.

Conversely, suppose that $a \oplus b$ is an atom in $X \oplus Y$. WLOG, $a \neq 0$. It follows from $0 \leqslant 0 \oplus b \leqslant a \oplus b$ that $0 \oplus b$ is a scalar multiple of $a \oplus b$, which yields $b = 0$. It is now easy to see that a has to be an atom in X.

5.4.7 By Lemma 5.4.1(iii), we find disjoint non-zero x and y in $[0, e]$. Take $u = P_x e$ and $v = x - u = (I - P_x)e$. Then $x \leqslant e$ implies $u \geqslant P_x x = x > 0$ and $v \geqslant y > 0$.

5.4.15 Let Y be a finite-dimensional sublattice of an Archimedean vector lattice X. Then $Y = \operatorname{span}\{e_1, \dots, e_m\}$ for some positive disjoint $e_1, \dots, e_n$; put $e = e_1 + \dots + e_m$. Identifying Y with $\mathbb{R}^m$, we mat view e as $(1, \dots, 1)$. Suppose that $y_\alpha \downarrow 0$ in Y; we need to show that $y_\alpha \downarrow 0$ in X. Let $x \in X_+$ such that $x \leqslant y_\alpha$ for all α. Since order convergence in $\mathbb{R}^m$ is coordinate-wise, for every $n \in \mathbb{N}$ there exists α such that $\frac{1}{n} e \geqslant y_\alpha \geqslant x$. The Archimedean property now yields $x = 0$.

5.4.20 Let x be a positive vector in a discrete order continuous Banach lattice X. Write $x = \sup\big\{\lambda_a(x)a \,:\, a \in A\big\}$ as in Lemma 5.4.18. For each $a \in A$, the vector $\lambda_a a$ equals $P_a x$, and is itself an atom if it is non-zero. So we have expressed x as a supremum of a collection of atoms. By Decomposition Lemma 5.3.5, this collection is countable.

Here is an alternative proof that avoids Decomposition Lemma 5.3.5. Again, write $x = \sup B = \sup B^\vee$ for *some* collection of atoms B. We interpret $B^\vee$ as an increasing net; denote it by (v_α). It follows from $v_\alpha \uparrow x$ that $v_\alpha \to x$. Hence there exist an increasing sequence of indices (α_n) such that $v_{\alpha_n} \to x$. By Monotone Convergence Lemma 2.5.3, $x = \sup v_{\alpha_n} = \sup C$, where C is the set of those members of B that occur in v_{α_n}'s. Clearly, C is countable.

5.4.21 Write Ω as a union of a nested increasing sequence (Ω_n) of sets of finite measure. Note that each atom of $L_\infty(\mu)$ of norm one is of the form $\mathbb{1}_A$, where A is an atom on μ. Let $\mathcal{A}$ be the set of all atoms of $L_\infty(\mu)$ of norm one; this is clearly a maximal collection of atoms in X; let $\mathcal{A}_0$ be the corresponding collection of atoms of μ. For every n and every $A \in \mathcal{A}_0$, either A is almost contained in Ω_n or almost disjoint with it. That is, either $\mu(A \setminus \Omega_n) = 0$ or $\mu(A \cap \Omega_n) = 0$. Since $\mu(\Omega_n) < \infty$, there are at most countable A's in $\mathcal{A}_0$ that are almost contained in Ω_n. Since every member of $\mathcal{A}_0$ is almost contained in some Ω_n, the set $\mathcal{A}_0$ is at most countable. Hense so is $\mathcal{A}$. Enumerate $\mathcal{A}_0$ as (A_i). Let $C = \bigcup_{i=1}^\infty A_i$. Then no atom dominated by $\mathbb{1}_{\omega \setminus C}$, hence $\mu(\Omega \setminus C) = 0$. WLOG, $C = \Omega$. For $f \in L_\infty(\mu)$, its restriction to each A_i is constant, say, m_i, and the sequence (m_i) belongs to ℓ_∞. It is clear that this defines a lattice isometry between $L_\infty(\mu)$ and ℓ_∞.

5.4.22 It is clear that every function of this form is an atom. Suppose now that $a \in C(K)_+$ is an atom and $s, t \in \operatorname{supp} a$ with $s \neq t$. Using Urysohn's Lemma, we find $f \in [0, \mathbb{1}]$ such that $f(t) = 1$ and $f(s) = 0$; it follows that $0 < af < a$, and af is not a scalar multiple of a; a contradiction.

5.4.26(i) Observe that X and X_a have the same atoms. It follows that the band in X_a generated by its atoms is X_a.

5.4.26(ii) Suppose that $0 \leqslant x \in X_a$. By Proposition 5.1.1(iii), $x = \sup D$ for some $D \subseteq I(A)_+$, where A is the set of all atoms of X. Hence, every element of D may be written as $v = \lambda_1 a_1 + \dots \lambda_n a_n$ for some $a_1, \dots, a_n \in A$. Since any two atoms are either proportional or disjoint, we may assume WLOG that $a_1, \dots, a_n$ are disjoint. It follows from $v \in D$ that $v \geqslant 0$ and, therefore, $\lambda_i \geqslant 0$ for all i and, furthermore, $\lambda_i a_i$ is again an atom. Thus, $v = (\lambda_1 a_n) \vee \dots \vee (\lambda_n a_n)$ is the supremum of a set of atoms, and so is x.

5.4.26(iii) Every atom of X_a^d would also be an atom in X, and, therefore, must be in X_a.

5.5.1 If $x \wedge ne \to x$ then $x \in \overline{I_e}$ because $x \wedge nx \in I_e$ for every n.

Suppose $x \in \overline{I_e}$. Find a sequence (y_n) in I_e such that $y_n \to x$. Replacing y_n with $|y_n|$, we may assume WLOG that $y_n \geqslant 0$ for all n. For every n we have $y_n \leqslant me$ for some $m \in \mathbb{N}$. WLOG, we may assume that $y_n \leqslant ne$ (otherwise, repeat y_{n-1} m times; put $y_0 = 0$). It follows from $y_n \to x$ that $x \wedge y_n \to x$ and, therefore, $x - x \wedge y_n \to 0$. Now $0 \leqslant x - x \wedge ne \leqslant x - x \wedge y_n$ yields $x - x \wedge ne \to 0$.

5.5.5 Follow the proof of Corollary 5.1.18.

5.6.8 Suppose that $F \cap G = \varnothing$ and let $h \in X$. By Urysohn's Lemma for dense sublattices 5.6.1, there exists $f \in X$ such that f vanishes on F and f agrees with h on G. It follows that $h = f + (h - f)$, where $f \in J_F^X$ and $h - f \in J_G^X$.

Now suppose that $X = J_F^X + J_G^X$. There exists a strictly positive $e \in X$ (because $\mathbb{1} \in \overline{X}$). It follows from $X = J_F^X + J_G^X$ that e vanishes on $F \cap G$, hence $F \cap G = \varnothing$.

5.6.10 By Proposition 5.6.9, it suffices to verify that $\operatorname{supp} D_U^X = U$. It is clear that $\operatorname{supp} D_U^X \subseteq U$. To prove the converse inclusion, fix $t \in U$. Find an open set V such that $t \in V \subseteq \overline{V} \subseteq U$. Applying Urysohn's Lemma for dense sublattices 5.6.1 with $F = \{t\}$ and $G = V^C$, and with f being any function in X with $f(t) \neq 0$, we find $g \in X_+$ such that $g(t) \neq 0$ and $\operatorname{supp} g \subseteq V$. It follows that $\overline{\operatorname{supp}}\, g \subseteq U$, so that $g \in D_U^X$.

5.6.17(ii)

$$\begin{aligned} g \in A^d \quad &\Leftrightarrow \quad \forall f \in A \quad g \in \{f\}^d \quad \Leftrightarrow \quad \forall f \in A \quad g \text{ vanishes on } (\ker f)^C \\ &\Leftrightarrow \quad g \text{ vanishes on } \bigcup_{f \in A} (\ker f)^C \quad \Leftrightarrow \quad g \text{ vanishes on } F. \end{aligned}$$

5.6.17(iii) Applying the previous part with $A = J_F^X$, we get $(J_F^X)^d = J_G^X$, where $G = \overline{\left(\bigcap_{f \in J_F^X} \ker f\right)^C} = \overline{F^C}$. Since F is closed, we have $\overline{F^C} = (\operatorname{Int} F)^C$.

5.6.17(iv) By the preceding parts, $J^d = J_F^X$ where $F = \overline{U}$, where $U = \left(\bigcap_{f\in J}\ker f\right)^C = \bigcup_{f\in J}\operatorname{supp} f = \operatorname{supp} J$.

5.6.19 $A \subseteq J_{U^C} = J_{\overline{K\setminus\overline{U}}}$. The latter set is a band because $K\setminus\overline{U}$ is open, hence f is in this band, so that f vanishes outside of $\overline{U}$.

5.6.24 Suppose that X is disjointly order complete, $e \in X_+$, and A a disjoint set in $(I_e)_+$ which is bounded above by some $u \in I_e$. Then $z \coloneqq \sup_X A$ exists and $z \leqslant u$; it follows that $z \in I_e$, hence $z = \sup_{I_e} A$ and, therefore, $\sup_{I_e} A$ exists.

Conversely, suppose that every principal ideal in X is disjointly order complete, and let A be a disjoint subset of X_+ bounded above by some e. Then $A \subseteq I_e$, hence $\sup A$ exists in I_e and, therefore, in X because ideals are regular.

The proof for the σ- version is analogous.

6.1.1 Suppose that $\pm T \leqslant S$. Then for every $x \in X$ we have $Tx = Tx^+ - Tx^- \leqslant Sx^+ + Sx^- = S|x|$. It follows that $\pm Tx = T(\pm x) \leqslant S|\pm x| = Sx$, so that $|Tx| \leqslant S|x|$.

Conversely, if $|Tx| \leqslant S|x|$ for every $x \in X$ then for every $x \geqslant 0$ we have $(\pm T)x = T(\pm x) \leqslant Sx$, so that $\pm T \leqslant S$.

6.1.5 It follows from $\pm T \leqslant |T|$ that T is regular and

$$Tx = Tx^+ - Tx^- \leqslant |T|x^+ + |T|x^- = |T||x|.$$

6.1.13 The inclusion map $T\colon Y \to X$ is positive; hence it is norm continuous by Theorem 6.1.9.

6.1.14 It is clear that $\|T\| \geqslant \sup\limits_{x\in S_X^+}\|Tx\|$. Fix $\varepsilon > 0$; find $x \in X_X$ with $\|Tx\| \geqslant \|T\| - \varepsilon$. Then $|x| \in S_X^+$ and it follows from $|Tx| \leqslant T|x|$ that $\big\|T|x|\big\| \geqslant \|Tx\| \geqslant \|T\| - \varepsilon$.

6.1.15 It is easy to see that if the infinite matrix $\big(|t_{ij}|\big)$ represents a linear operator, say S, then $S = T \vee (-T)$, hence $S = |T|$. Being positive, $|T|$ is norm bounded. Conversely, suppose that $S \coloneqq |T|$ exists. Being positive, S is norm bounded. Let (s_{ij}) be the matrix of S. It follows from $\pm T \leqslant S$ that $\pm t_{ij} \leqslant s_{ij}$ and, therefore, $|t_{ij}| \leqslant s_{ij}$ for all $i, j \in \mathbb{N}$. Therefore, the series $\sum_{j=1}^\infty |t_{ij}|x_j$ converges for every $x \in X$ and every $i \in \mathbb{N}$ and

$$\Big|\sum_{j=1}^\infty |t_{ij}|x_j\Big| \leqslant \sum_{j=1}^\infty s_{ij}|x_j| = \big(S|x|\big)_i.$$

Define $Rx \in \mathbb{R}^{\mathbb{N}}$ via $(Rx)_i = \sum_{j=1}^\infty |t_{ij}|x_j$. Suppose $Y = \ell_p$. For $x \in X$, we have

$$\|Rx\|_{\ell_p}^p = \sum_{i=1}^\infty \Big|\sum_{j=1}^\infty |t_{ij}|x_j\Big|^p \leqslant \sum_{i=1}^\infty \Big|\sum_{j=1}^\infty s_{ij}|x_j|\Big|^p = \big\|S|x|\big\|_{\ell_p}^p.$$

It follows that $\|Rx\|_{\ell_p} \leqslant \|S\|\|x\|_{\ell_p}$, and, therefore, $R \in L(\ell_p)$. Clearly, the matrix of R is $\big(|t_{ij}|\big)$. If $Y = c_0$, the proof is similar.

6.3.5 On one hand, if $x_1, \dots, x_m \geqslant 0$ with $\sum_{j=1}^m x_j = x$ then

$$\sum_{j=1}^m \bigvee_{i=1}^n T_i x_j \leqslant \sum_{j=1}^m \Big(\bigvee_{i=1}^n T_i\Big)x_j = \Big(\bigvee_{i=1}^n T_i\Big)\sum_{j=1}^m x_j = \Big(\bigvee_{i=1}^n T_i\Big)x.$$

On the other hand, if we take $m = n$ and $x_1, \dots, x_n \geqslant 0$ with $\sum_{j=1}^n x_j = x$ then

$$\sum_{j=1}^n \bigvee_{i=1}^n T_i x_j \geqslant \sum_{i=1}^n T_i x_i.$$

Now apply 6.3.3.

One can argue as in 6.3.4 that the set of all positive partitions of X is directed with respect to refinement, hence the set under the supremum may be viewed as a net. It is left to show that this net is increasing. Observe that if we refine our partition by replacing, say, x_m with $y+z$, this results in replacing the term $\bigvee_{i=1}^n T_i x_m$ in the sum with $\bigvee_{i=1}^n T_i y + \bigvee_{i=1}^n T_i z$. For every i, we have $T_i x_m = T_i y + T_i z \leqslant \bigvee_{k=1}^n T_k y + \bigvee_{k=1}^n T_k z$. It follows that $\bigvee_{i=1}^n T_i x_m \leqslant \bigvee_{i=1}^n T_i y + \bigvee_{i=1}^n T_i z$.

6.3.13 Let $T \geqslant 0$. Since $\mathcal{R}_2 T$ vanishes on J_1, we have

$$(\mathcal{R}_1\mathcal{R}_2 T)(x) = \sup\bigl\{(\mathcal{R}_2 T)v \,:\, v \in J_1 \text{ and } v \leqslant x\bigr\} = 0$$

for every $x \geqslant 0$. It follows that $\mathcal{R}_1\mathcal{R}_2 = 0$ and, therefore, $(\operatorname{Range}\mathcal{R}_1) \cap (\operatorname{Range}\mathcal{R}_2) = 0$.

6.3.16 Since X is order complete by Theorem 2.1.16, $\mathcal{L}_r(X)$ is a vector lattice, so that $S \coloneqq T \wedge I$ exists and $S \geqslant 0$. It suffices to show that $Sf = 0$ for every $f \in X_+$. First, observe that if $f = \mathbb{1}_A$ for some measurable set A then $Sf \leqslant f$ and $Sf \leqslant Tf = \lambda(A)\mathbb{1}$ so that $Sf \leqslant \lambda(A)\mathbb{1}_A$. Next, consider $S\mathbb{1}$. Fix $n \in \mathbb{N}$, then $\mathbb{1} = \sum_{i=1}^n \mathbb{1}_{[\frac{i-1}{n},\frac{i}{n}]}$, so that

$$0 \leqslant S\mathbb{1} = \sum_{i=1}^n S\mathbb{1}_{[\frac{i-1}{n},\frac{i}{n}]} \leqslant \tfrac{1}{n}\sum_{i=1}^n \mathbb{1}_{[\frac{i-1}{n},\frac{i}{n}]} = \tfrac{1}{n}\mathbb{1}.$$

Since n is arbitrary, $S\mathbb{1} = 0$. Now let $f \in X_+$ be arbitrary. It is a standard (and easy) fact that $\int_{\{f>M\}} f \to 0$ as $M \to \infty$. Fix $\varepsilon > 0$ and find $M > 0$ such that $\int_{\{f>M\}} f < \varepsilon$. Then

$$0 \leqslant Sf = Sf \cdot \mathbb{1}_{\{f \leqslant M\}} + Sf \cdot \mathbb{1}_{\{f>M\}} \leqslant S(M\mathbb{1}) + Tf\dot{\mathbb{1}}_{\{f>M\}} \leqslant \varepsilon\mathbb{1}.$$

Since ε is arbitrary, it follows that $Sf = 0$.

6.3.17(i) Since order intervals are norm bounded, the image of every order interval in X is norm bounded in $C(K)$, hence order bounded.

6.3.17(ii) As in (i), the operator is order bounded. Since ℓ_∞ is order complete, the operator is regular by Riesz-Kantorovich Theorem.

6.5.4 Let T be central, that is, $\pm T \leqslant \lambda I$ for some positive λ. WLOG, $\lambda = 1$. Then $-I \leqslant T \leqslant I$, so that $0 \leqslant I + T \leqslant 2I$. If $x_\alpha \xrightarrow{\mathrm{o}} 0$ then

$$\Bigl|(I+T)x_\alpha\Bigr| \leqslant (I+T)\lvert x_\alpha\rvert \leqslant 2\lvert x_\alpha\rvert \xrightarrow{\mathrm{o}} 0.$$

It follows that $(I+T)x_\alpha \xrightarrow{\mathrm{o}} 0$, hence $x_\alpha + Tx_\alpha \xrightarrow{\mathrm{o}} 0$ and, therefore, $Tx_\alpha \xrightarrow{\mathrm{o}} 0$.

6.5.13 Clearly, $\ker f = c_0$. Note that $f \geqslant 0$; note also that $x \in c_0$ iff $\lvert x\rvert \in c_0$. We now conclude that $N_f = \ker f = c_0$. Then $N_f = (c_0)^d = \{0\}$.

6.5.15 Suppose that T is order continuous. Let $u = \sup A$. Then $u = \sup A^\vee$. Since T is a lattice homomorphism, we have $T(A^\vee) = (TA)^\vee$. Recall that we may view $A^\vee$ as an increasing net, say, (x_α). Then $x_\alpha \uparrow u$ implies $Tx_\alpha \uparrow Tu$, hence

$$Tu = \sup Tx_\alpha = \sup T(A^\vee) = \sup (TA)^\vee = \sup TA.$$

The converse implication is straightforward.

6.5.16 Combine the preceding exercise with Theorem 1.6.13(ii).

6.6.3 Suppose $f, g \in X_+^\sim$ such that $nf \leqslant g$ for all $n \in \mathbb{N}$. Then for every $x \in X_+$ we have $0 \leqslant nf(x) \leqslant g(x)$ for all $n \in \mathbb{N}$, which yields $f(x) = 0$. It follows that $f = 0$.

Alternatively, one may argue that $X^\sim$ is order complete and, therefore, Archimedean by Proposition 2.1.6.

6.6.4 Being positive, f and g are order bounded. The claim that $f \wedge g = 0$ follows from Riesz-Kantorovich formulas 6.6.2. Indeed, let $x \geqslant 0$ and $\varepsilon > 0$. Let (f_n) be as in Example 1.4.2. Then $\int f_n x \searrow 0$, hence $f(u) < \varepsilon$ where $u = f_n x$ for a sufficiently large n. Put $v = x - u$, then $g(v) = v(0) = 0$. It follows that $f(u) + g(v) < \varepsilon$. Now Riesz-Kantorovich formulae 6.6.2 yield $(f \wedge g)(x) = 0$.

6.6.8 Let $h = f_{|Y}$. Clearly, $h \in J^\sim$. For every $x \in J_+$ Riesz-Kantorovich formula (6.3) yields

$$|h|(x) = \sup\bigl\{h(y) \;:\; y \in J,\; |y| \leqslant x\bigr\} = \sup\bigl\{f(y) \;:\; y \in X,\; |y| \leqslant x\bigr\} = |f|(x)$$

because $|y| \leqslant x$ implies $y \in J$. It follows that $|h|$ agrees with $|f|$ on J.

6.6.14 Suppose that $T\colon C[0,1] \to X$ is a non-trivial order continuous operator. Then $Tx \neq 0$ for some $x \in C[0,1]$. By Hahn-Banach Theorem, there exists $f \in X^*$ such that $f(Tx) \neq 0$. Since f is order continuous, the composition $f \circ T$ is order continuous and non-trivial. This contradicts Theorem 6.6.13. The proof of the σ-version is similar.

6.6.15 To see that f is not order continuous, let $x_n = (\overbrace{1, \dots, 1}^{n}, 0, \dots)$. Then $x_n \uparrow \mathbb{1}$ and $f(x_n) = 0$ for every n, but $f(\mathbb{1}) = 1$. Similarly for g: since a free ultrafilter contains no finite sets, $g(x_n) = 0$ for every n, yet $g(\mathbb{1}) = 1$.

Let $x = (1, -1, 1, -1, \dots)$. Then $f(x) = 0$, while $f\bigl(|x|\bigr) = 1$. Hence, f is not a lattice homomorphism. The fact that g is a lattice homomorphism is straightforward from the definition of g.

6.6.16 (i)$\Rightarrow$(ii) Let $x > 0$. Then $f(x) \neq 0$ for some $f \in Z$. Since $f = f^+ - f^-$ it follows that either $f^+(x) > 0$ or $f^-(x) > 0$.

(ii)$\Rightarrow$(i) Let $x \neq y$ in X. Then $|x - y| \neq 0$, so that $f\bigl(|x - y|\bigr) > 0$ for some positive $f \in Z$. It follows from 6.6.2 that $g(x - y) > 0$ for some $g \in [-f, f]$; hence $g(x) \neq g(y)$. Note that as Z is an ideal, $g \in Z$.

(ii)$\Rightarrow$(iii) It is trivial that if $x \geqslant 0$ then $f(x) \geqslant 0$ for every $f \in Z_+$. Suppose now that $x \in X$ and $f(x) \geqslant 0$ for every $f \in Z_+$. It follows from 6.6.2 that $f(x^-) = 0$ (again, we are using the fact that Z is an ideal). Now (ii) yields $x^- = 0$, hence $x \geqslant 0$.

(iii)$\Rightarrow$(ii) Suppose (ii) fails. Then there exists $x > 0$ such that $f(x) = 0$ for all $f \in Z_+$. Applying (iii) to $-x$ yields that $-x \geqslant 0$; a contradiction.

6.7.2 Suppose that f is an order bounded functional on $L_p[0.1]$ for some $0 \leqslant p < 1$; show that $f = 0$. WLOG, replacing f with $|f|$, we may assume that $f \geqslant 0$. Since $L_1[0,1] \subseteq L_p[0,1]$, f is also a positive functional on $L_1[0,1]$, hence $f \in L_\infty[0,1]$. Then f dominates a scalar multiple of the characteristic function of some measurable set A; cf. Example 3.2.6(viii). WLOG, $f \geqslant \mathbb{1}_A$. Take any $0 \leqslant h \in L_p[0,1]$ such that $\int_A h = \infty$, then $\langle f, h\rangle = \infty$; a contradiction.

6.7.6(i) WLOG, $\|x\| = 1$. Use Hahn-Banach Theorem to find $g \in S_{X^*}$ such that $g(x) = 1$. Since X^* is a Banach lattice, $|g|$ exists and $\bigl\||g|\bigr\| = \|g\| = 1$. It follows from

$$1 = g(x) \leqslant |g|(x) \leqslant \bigl\||g|\bigr\|\|x\| = 1$$

that $|g|(x) = 1$. Take $f = |g|$.

6.7.6(ii) Find $x \in S_X$ such that $f(x) \geqslant \|f\| - \varepsilon$. It follows from $|x| \geqslant x$ that $f\bigl(|x|\bigr) \geqslant f(x) \geqslant \|f\| - \varepsilon$, so that $|x|$ does the job.

6.7.9 Let $g \in Y_+^*$. For every $x \in X_+$, we have $(T^*g)(x) = g(Tx) \geqslant 0$, hence $T^*g \geqslant 0$.

6.7.11 By Proposition 6.7.10, X is lattice isometric to the sublattice $j(X)$ in X^{**}. Then $\overline{X}$ is isometric to the norm closure $\overline{j(X)}$ in X^{**}. Being the closure of a sublattice, $\overline{j(X)}$ is a closed sublattice of X^{**} by 3.1.6, hence a Banach lattice. This induces an order on $\overline{X}$ which makes it into a Banach lattice. Uniqueness follows from the fact that the positive cone of $\overline{X}$ is the closure of X_+ in $\overline{X}$

6.7.15 Note that $0 \leqslant \varphi \in \ell_\infty^* = X^{**}$. We will show that $\varphi \perp X$ in X^{**}; this would imply that $\varphi \in X^d$ and, therefore, $\varphi \notin X^{dd}$. It suffices to show that $\varphi \wedge \hat{e}_i = 0$ for every i. We will use the Riesz-Kantorovich formula. Fix i. Take an arbitrary $0 \leqslant f \in \ell_\infty$. Define $g, h \in \ell_\infty$ as follows: $g_i = f_i$ and $g_j = 0$ if $j \neq i$; put $h = f - g$. Then $f = g + h$, $g, h \geqslant 0$, and $\varphi(g) = \hat{e}_i(h) = 0$. By Riesz-Kantorovich formula, $(\varphi \wedge \hat{e}_i)(f) = 0$. Therefore, $\varphi \perp \hat{e}_i$.

6.8.1(i) Suppose that $x_\alpha \xrightarrow{\mathrm{w}} x$ for some net (x_α) in X_+. For every $f \in X_+^*$, it follows from $0 \leqslant f(x_\alpha) \to f(x)$ that $f(x) \geqslant 0$. It follows that $x \geqslant 0$ by Remark 6.7.5. Alternatively, one can simply argue that every convex norm closed set is weakly closed.

6.8.1(ii) Suppose that $f_\alpha \xrightarrow{\mathrm{w}^*} f$ for some net (f_α) in X_+^*. For every $x \in X_+$, it follows from $0 \leqslant f_\alpha(x) \to f(x)$ that $f(x) \geqslant 0$ and, therefore, $f \geqslant 0$.

6.8.1(iii) Let $a \leqslant b$ in X. It follows from $[a, b] = (a + X_+) \cap (b - X^+)$ that $[a, b]$ is an intersection of two weakly closed sets, hence $[a, b]$ is weakly closed. The second statement is similar.

6.8.7 The forward implication is obvious. Suppose that $x_n \xrightarrow{\mathrm{w}} 0$ implies $|x_n| \xrightarrow{\mathrm{w}} 0$. Assume now that $x_n \xrightarrow{\mathrm{w}} x$. Then $|x_n - x| \xrightarrow{\mathrm{w}} 0$. For every $f \in X_+^*$, we have

$$\Bigl|f\bigl(|x_n| - |x|\bigr)\Bigr| \leqslant f\Bigl(\bigl||x_n| - |x|\bigr|\Bigr) \leqslant f\bigl(|x_n - x|\bigr) \to 0,$$

hence $|x_n| - |x| \xrightarrow{\mathrm{w}} 0$ and, therefore, the modulus is weakly sequentially continuous. Since lattice operations may be expressed via each other, this completes the proof.

6.8.8(i) Suppose that $|x_n| \leqslant u$ for all n. Fix $f \in X^*$; we need to prove that $f(x_n) \to 0$. Since $f = f^+ - f^-$, we may assume WLOG that $f \geqslant 0$. Note that

$$\sum_{k=1}^{n} \bigl|f(x_k)\bigr| \leqslant \sum_{k=1}^{n} f\bigl(|x_k|\bigr) = f\Bigl(\sum_{k=1}^{n} |x_k|\Bigr) \leqslant f(u)$$

for every n. It follows that the series $\sum_{k=1}^\infty \bigl|f(x_k)\bigr|$ converges and, therefore, $f(x_n) \to 0$.

6.8.8(ii) Let $x_n \xrightarrow{\mathrm{w}} x$. Let $Y = [x_n]$. Since Y is a closed subspace of X, it is weakly closed; see Proposition A.4.11. It follows that $x \in Y$ and $x_n \xrightarrow{\mathrm{w}} x$ in Y. By Proposition 1.5.9, (x_n) is a basic sequence, hence a basis of Y. Let $x_k^* \in Y^*$ be the k-th biorthogonal functional for (x_n). It follows that $x_k^*(x_n) \to x_k^*(x)$ as $n \to \infty$. But $x_k^*(x_n) = \delta_{nk} \to 0$ as $n \to \infty$, so that $x_k^*(x) = 0$ for every k. It follows that $x = 0$.

6.8.8(iii) WLOG, (x_n) is contained in B_X. Since X is reflexive, B_X is weakly compact; see Proposition A.4.15. By Eberlein-Šmulian's Theorem A.4.20, (x_n) has a weakly convergent subsequence, say, (x_{n_k}). By (ii), $x_{n_k} \xrightarrow{\mathrm{w}} 0$. Using the same argument, we can show that every subsequence of (x_n) has a weakly null further subsequence. It follows that (x_n) is weakly null by Proposition A.2.1.

6.9.4 Extend f_1 to $\tilde{f}_1$ in X_+^* as in Theorem 6.9.3. In particular, $\|\tilde{f}_1\| \leqslant 1$. Then inductively use Theorem 6.9.2 to extend $f_2, f_3, \dots$ to positive functionals $\tilde{f}_2, \tilde{f}_3, \dots$, respectively, so that $\tilde{f}_1 \geqslant \tilde{f}_2 \geqslant \tilde{f}_3 \dots$. In particular, $\|\tilde{f}_n\| \leqslant \|\tilde{f}_1\| \leqslant 1$ for all $n \geqslant 2$.

6.9.8 For the sake of contradiction, suppose that $Tx = 0$ for some non-zero $x \in X$. Find $y \in Y$ with $0 < y \leqslant |x|$. Then $Ty \leqslant T|x| = |Tx| = 0$, hence $Ty = 0$, which is a contradiction.

6.9.9 Let $x \in X$. By Proposition 3.2.15, we have $x = \sup[0,x] \cap Y$. We may view the set $\sup[0,x] \cap Y$ as an increasing net. It follows that $Tx = \sup T\bigl([0,x] \cap Y\bigr)$.

6.9.13 Let $x \perp y$ in X_+. By Exercise 1.6.10, it suffices to show that $\widetilde{T}x \perp \widetilde{T}y$. Using (6.11) for $\widetilde{T}x$ and $\widetilde{T}y$ followed by the distributive law, we get

$$\widetilde{T}x \wedge \widetilde{T}y = \sup\bigl\{Tv \wedge Tu \;:\; u \in [0,x] \cap Y,\; v \in [0,y] \cap Y\bigr\} = 0$$

because, under the assumptions, we have $u \wedge v = 0$ and, therefore, $Tv \wedge Tu = 0$.

Suppose now that $T \colon X \to Y$ is an order continuous lattice homomorphism between two Archimedean vector. Since Y is regular in Y^δ, we may view T as an order continuous lattice homomorphism from X to Y^δ. By the preceding argument, it can be extended uniquely to an order continuous lattice homomorphism $T^\delta \colon X^\delta \to Y^\delta$.

6.10.6 Since $x_n \uparrow\leqslant \mathbb{1}$, Proposition 6.10.5 guarantees that (x_n) is order Cauchy. One can also see this directly as follows. For $m,n \in \mathbb{N}$, put $u_{m,n} = (\overbrace{0,\dots,0}^{2k},1,1,\dots)$, where $k = \min\{m,n\}$. Then $u_{m,n} \downarrow 0$ and $|x_m - x_n| \leqslant u_{m,n}$, so that $(x_m - x_n)_{m,n} \xrightarrow{\text{o}} 0$.

For the sake of contradiction, assume that $x_n \xrightarrow{\text{o}} x$ for some $x \in c$. Since $\ell_\infty = c^\delta$, we conclude that $x_n \xrightarrow{\text{o}} x$ in ℓ_∞, so that $x = \sup x_n = (0,1,0,1,\dots)$ in ℓ_∞. However, this vector is not in c.

6.10.8 Let (x_α) be a Cauchy net. Then $(x_\alpha - x_\beta)_{(\alpha,\beta)}$ is order null and, therefore, has an order bounded tail. That is, there exists $u \geqslant 0$ and a pair of indices (α_0,β_0) such that $|x_\alpha - x_\beta| \leqslant u$ whenever $(\alpha,\beta) \geqslant (\alpha_0,\beta_0)$. In particular, for every $\alpha \geqslant \alpha_0$ we have $|x_\alpha| \leqslant |x_\alpha - x_{\beta_0}| + |x_{\beta_0}| \leqslant u + |x_{\beta_0}|$.

6.10.10 (iii)⇒(ii) is trivial. To show that (ii)⇒(i), suppose that $0 \leqslant x_n \uparrow\leqslant u$. Then (x_n) is order Cauchy by Proposition 6.10.5, hence order convergent. It follows that $\sup x_n$ exists.

(i)⇒(iii) Suppose that X is σ-order complete and (x_n) is order Cauchy. We need to prove that (x_n) is σ-order convergent. It is easy to see that (x_n) is order bounded. Put

$$x = \sup a_n \text{ where } a_n = \inf_{m \geqslant n} x_m \text{ and } y = \inf b_n \text{ where } b_n = \sup_{m \geqslant n} x_m.$$

By Proposition 2.4.24, it suffices to show that $x = y$. As before, we have $a_n \leqslant x \leqslant y \leqslant b_n$ for every n. Let (u_γ) be a net such that $u_\gamma \downarrow 0$ and for every γ there exist n_0 such that $|x_m - x_n| \leqslant u_\gamma$ whenever $m,n \geqslant n_0$. Fix γ. Let n_0 be as before. Let $m \geqslant n \geqslant n_0$, then $|x_m - x_n| \leqslant u_\gamma$, so that $x_m \in [x_n - u_\gamma, x_n + u_\gamma]$. It follows that $a_n, b_n \in [x_n - u_\gamma, x_n + u_\gamma]$, so that $0 \leqslant y - x \leqslant b_n - a_n \leqslant 2u_\gamma$. Since γ was arbitrary, it follows that $y - x = 0$.

6.10.12 Being order dense, Y is regular in X, hence Y^δ may be viewed as a sublattice of X^δ by Theorem 6.10.11. Since X is order dense in X^δ, we conclude that Y and, therefore, Y^δ are both order dense in X^δ. Now use Exercise 3.3.19.

6.10.13 To forward implication is Corollary 2.3.14. To prove the converse, suppose that $x_n \xrightarrow{\text{u}} x$ in X for some (x_n) in Y and $x \in X$. Then (x_n) is uniformly Cauchy in X. That is, $x_n - x_m \xrightarrow{\text{u}} 0$ in X. Since Y is majorizing, $x_n - x_m \xrightarrow{\text{u}} 0$ in Y. Hence, (x_n) is uniformly Cauchy in Y. Then $x_n \xrightarrow{\text{u}} y$ in Y for some $y \in Y$. It follows that $x_n \xrightarrow{\text{u}} y$ in X; hence $x = y$ and, therefore, $x \in Y$.

6.11.1 Suppose f is a lattice homomorphism and $0 \leqslant g \leqslant f$. For every $x \in \ker f$ we have

$$\bigl|g(x)\bigr| \leqslant g\bigl(|x|\bigr) \leqslant f\bigl(|x|\bigr) = \bigl|f(x)\bigr| = 0.$$

It follows that $\ker f \subseteq \ker g$, so that g is a scalar multiple of f. Conversely, suppose that f is an atom. Let $x \in X$. We use Riesz-Kantorovich formulas 6.6.2 to find $f(x^+)$. Since $0 \leqslant g \leqslant f$ iff $g = \lambda f$ for some $0 \leqslant \lambda \leqslant 1$; it follows that $f(x^+) = \max\bigl\{0, f(x)\bigr\} = f(x)^+$. Hence, f is a lattice homomorphism.

6.11.2 Follows from Exercise 6.11.1.

6.11.3 Since a lattice isomorphism preserves all vector lattice structures, it maps atoms to atoms. Combining this with Exercise 5.4.4(i), we conclude that for every i we have $Te_i = w_i e_{\sigma(i)}$ for some $w_i > 0$ and some index $\sigma(i)$. Since T is a linear isomorphism, it follows that σ is a bijection on $\mathbb{N}$.

6.11.4 Let $T\colon L_p[0,1] \to L_q(\mu)$ be a non-trivial lattice homomorphism and $p < q < \infty$. Then $Tf \neq 0$ for some f. Since $T|f| = |Tf|$, replacing f with $|f|$, we may assume that $f > 0$. Being positive, T is norm bounded. Scaling f and T, we may assume WLOG that $\|f\|_{L_p} = 1$ and $\|Tf\|_{L_q} = 1$. Fix $n \in \mathbb{N}$. Partitioning the support of f appropriately and using the fact that the Lebesgue measure is non-atomic, we can decomposition f into a sum of n disjoint positive functions, $f = f_1 + \dots + f_n$, having the same norms. It follows from

$$1 = \|f\|_{L_p}^p = \|f_1\|_{L_p}^p + \dots + \|f_n\|_{L_p}^p$$

that $\|f_k\|_{L_p} = n^{-\frac{1}{p}}$ for every $k = 1, \dots, n$. Since $Tf_1, \dots, Tf_n$ are disjoint, we have

$$1 = \|Tf\|_{L_q}^q = \|Tf_1\|_{L_q}^q + \dots + \|Tf_n\|_{L_q}^q.$$

Therefore, there exists k such that $\|Tf_k\|_{L_q}^q \geqslant \frac{1}{n}$, so that $\|Tf_k\|_{L_q} \geqslant n^{-\frac{1}{q}}$. It follows that $\frac{\|Tf_k\|_{L_q}}{\|f_k\|_{L_p}} \geqslant n^{\frac{1}{p}-\frac{1}{q}}$. Since n is arbitrary, this contradicts T being bounded.

The proof for operators to $C(K)$ and $L_\infty(\mu)$ is similar.

6.11.5 Let $x \in X$ such that $\varphi(x) = 0$ for every lattice homomorphisms in $X^\sim$. By $C(K)$ Representation Theorem, we identify X with a dense sublattice of some $C(K)$ space; we view x as an element of $C(K)$. For every $t \in K$, the restriction of δ_t to X is a lattice homomorphism, so that $0 = \delta_t(x) = x(t)$. It follows that $x = 0$.

Similarly, if $\varphi(x) \leqslant \varphi(y)$ for every lattice homomorphism in $\varphi \in X^\sim$ then $x(t) \leqslant y(t)$ for every $t \in K$, hence $x \leqslant y$.

6.11.8 If K and L are homeomorphic, it is easy to see that the resulting composition operator is a surjective lattice isomorphism (even isometry). Conversely, suppose that $T\colon C(K) \to C(L)$ be a surjective lattice isomorphism.

Let $w \coloneqq T\mathbb{1}_K$. Since $\mathbb{1}_K$ is a strong unit in $C(K)$, w is a strong unit in $C(L)$. It follows that w is bounded both above and below by positive constants, and, therefore, $w^{-1} \in C(L)$. It is easy to see that the multiplication by w^{-1} operator $M_{w^{-1}}$ is a surjective lattice isomorphism on $C(L)$, hence $M_{w^{-1}}T\colon C(K) \to C(L)$ is a surjective lattice isomorphism taking $\mathbb{1}_K$ to $\mathbb{1}_L$. Relabelling $M_{w^{-1}}T$ as T, we may just assume WLOG that $T\mathbb{1}_K = \mathbb{1}_L$.

Applying Corollary 6.11.7 to T and T^{-1}, we find continuous functions $\varphi\colon L \to K$ and $\psi\colon K \to L$ such that $Tf = f \circ \varphi$ and $T^{-1}g = g \circ \psi$ for all $f \in C(K)$ and $g \in C(L)$. It suffices to show that $\psi = \varphi^{-1}$. Suppose not. Then there exists $t \in K$ such that $s \coloneqq \varphi\bigl(\psi(t)\bigr) \neq t$. Find $f \in C(K)$ such that $f(s) \neq f(t)$. Then

$$f(s) = f\Bigl(\varphi\bigl(\psi(t)\bigr)\Bigr) = (T^{-1}Tf)(t) = f(t),$$

which is a contradiction.

6.11.14 We already know that Q^* is an isometric embedding; it is left to show that it is a lattice homomorphism. Being a surjective lattice homomorphism, Q is interval preserving by Exercise 1.6.11. Let $\varphi \in (X/J)^*$ and $x \in X_+$. By Riesz-Kantorovich formulae,

$$\begin{aligned}\bigl|Q^*\varphi\bigr|(x) &= \sup\bigl\{(Q^*\varphi)(y) \,:\, -x \leqslant y \leqslant x\bigr\} = \sup\Bigl\{\varphi(Qy) \,:\, -x \leqslant y \leqslant x\Bigr\}\\ &= \sup\varphi\bigl(Q[-x,x]\bigr) = \sup\varphi\bigl[-Qx, Qx]\bigr) = |\varphi|(Qx) = \bigl(Q^*|\varphi|\bigr)(x).\end{aligned}$$

It follows that $\bigl|Q^*\varphi\bigr| = Q^*|\varphi|$.

7.1.1(i)

$$y = y \wedge \mathbb{1} = y \wedge (x \vee \bar{x}) = (y \wedge x) \vee (y \wedge \bar{x}) = \mathbb{0} \vee (y \wedge \bar{x}) = y \wedge \bar{x}.$$

It follows that $y \leqslant \bar{x}$. Similarly, $\bar{x} \leqslant y$. Hence, $y = \bar{x}$.

7.1.1(ii) Follows from (i).

7.1.1(iii) Let $z = \bar{x} \wedge \bar{y}$; we need to show that $z = \overline{x \vee y}$. By (i), it suffices to show that $z \wedge (x \vee y) = \mathbb{0}$ and $z \vee (x \vee y) = \mathbb{1}$. These follow easily from the distributive laws. The second identity may be proved in a similar fashion.

7.1.1(iv) If $x \wedge y = \mathbb{0}$ then $\bar{y} = \mathbb{0} \vee \bar{y} = (x \wedge y) \vee \bar{y} = (x \vee \bar{y}) \wedge (y \vee \bar{y}) = x \vee \bar{y}$, so that $\bar{y} \geqslant x$. If $x \leqslant \bar{y}$ then $x \wedge y \leqslant \bar{y} \wedge y = \mathbb{0}$, hence $x \wedge y = \mathbb{0}$.

7.1.8 If for every $a \in \mathcal{B}$ we have $a \in F$ or $\bar{a} \in F$ then F is maximal because, by definition, a filter cannot contain disjoint elements. Conversely, suppose that F is maximal. Assume that there exists $a \in \mathcal{B}$ such that neither a nor $\bar{a}$ is in F. Note that if $b \leqslant a$ and $c \leqslant \bar{a}$ then $b \wedge c = \mathbb{0}$, hence b and c cannot both be in F. Suppose F contains no elements less than $\bar{a}$ (the proof for the other case is similar, just interchange a with $\bar{a}$). To reach a contradiction, we will construct a filter that contains F and a; this will contradict the maximality of F. From the definition of a filter, this new filter must also contain all elements of the form $a \wedge b$ with $b \in F$ as well as everything that dominates them. But this set is already a filter. More precisely, let

$$G = \bigl\{x \in \mathcal{B} \,:\, x \geqslant b \wedge a \text{ for some } b \in F\bigr\}.$$

Check that G is a filter. It is easy to see that if $x \in G$ and $x \leqslant y$ then $y \in G$. If $x, y \in G$ then $x \geqslant b \wedge a$ and $y \geqslant c \wedge a$ for some $b, c \in F$, so that $x \wedge y \geqslant (b \wedge c) \wedge a$, hence $x \wedge y \in G$. To show that $\mathbb{0} \notin G$, assume otherwise; then $b \wedge a = \mathbb{0}$ for some $b \in F$. Then $b \leqslant \bar{a}$ by Exercise 7.1.1(iv); this contradicts our assumption. Therefore, G is a filter. Clearly, G contains F and a; this proves the claim.

7.1.9 If $a \in p$ then $a \leqslant a \vee b$ implies $a \vee b \in p$. Similarly, if $b \in p$ then $a \vee b \in p$. Suppose now that $a \vee b \in p$ but $a \notin p$; we will show that $b \in p$. Indeed, by Exercise 7.1.8, we have $\bar{a} \in p$. It follows that p contains $\bar{a} \wedge (a \vee b) = \bar{a} \wedge b$. Since $\bar{a} \wedge b \leqslant b$, it follows that $b \in p$.

7.2.6 By Theorem 7.2.5, it suffices to prove the exercise in $\mathcal{C}_{\mathbb{1}}$, which is straightforward.

7.2.8 First, verify that T is well-defined. This may be done by a direct verification, or by using 7.1.12.

Let $x \in \operatorname{span}\mathcal{C}_e$; let $x = \sum_{i=1}^n \lambda_i x_i$ be a disjoint representation of x. Then $\varphi(x_1), \dots, \varphi(x_n)$ are pairwise disjoint in $\mathcal{C}_u$ so that

$$\lvert Tx\rvert = \Bigl\lvert\sum_{i=1}^n \lambda_i \varphi(x_i)\Bigr\rvert = \sum_{i=1}^n \lvert\lambda_i\rvert \varphi(x_i) = T\Bigl(\sum_{i=1}^n \lvert\lambda_i\rvert x_i\Bigr) = T\lvert x\rvert.$$

It follows that T is a lattice homomorphism.

7.3.5 Let $P_1, P_2 \in \mathcal{P}(X)$; let B_1 and B_2 be the corresponding projection bands. We will show that $P_1P_2 = \inf\{P_1, P_2\}$ in $\mathcal{L}(X)$. It follows from $0 \leqslant P_1, P_2 \leqslant I$ that $P_1P_2 \leqslant P_1$ and $P_1P_2 \leqslant P_2$. Suppose that $T \in \mathcal{L}(X)_+$ satisfies $T \leqslant P_1$ and $T \leqslant P_2$. Let $x \in X_+$. It follows from $0 \leqslant Tx \leqslant P_1x$ that $Tx \in B_1$. Similarly, $Tx \in B_2$, hence $Tx \in B_1 \cap B_2$. Since P_1P_2 is the band projection for $B_1 \cap B_2$, we have $Tx = P_1P_2Tx \leqslant P_1P_2P_2x = P_1P_2x$. It follows that $T \leqslant P_1P_2$. The result for supremum follows from the fact that $P_1 \vee P_2 = P_1 + P_2 - P_1 \wedge P_2$ both in $\mathcal{P}(X)$ and in $\mathcal{L}(X)$.

7.3.6 Let $x = Pe$. It follows from $0 \leqslant P \leqslant I$ that $0 \leqslant x \leqslant e$. Since $I - P$ is a complementary projection, we have $x \wedge (e - x) = (Pe) \wedge (I - P)e = 0$.

7.3.8 Implications (iv)⇒(iii)⇒(ii) are trivial. (ii)⇒(i) by Exercise 7.3.6. To show (i)⇒(iv), observe that $e=(e-x)+x$ and $x\perp(e-x)$ yield $P_xe=P_xx+P_x(e-x)=x$.

7.3.9 $Px=PP_xe\in\mathcal{C}_e$ by Exercise 7.3.6 because PP_x is a band projection by Theorem 5.2.14.

7.3.13 Let $P\in\mathcal{P}(X)$. Fix a weak unit $e\in X_+$. Put $x=Pe$. By Exercise 7.3.8, $x=P_xe$. It follows from Theorem 7.3.11 that $P=P_x$.

7.3.15(i) WLOG, $a\geqslant 0$. Let $W=\overline{\{a\neq 0\}}$. Since K is totlly disconnected, W is clopen. Let $x=P_ae$. Then $P_a=P_x$ and, therefore, $a=P_aa=P_xa$, and $x=\mathbb{1}_U$, where U is the generalized support of a, so $a=a\cdot\mathbb{1}_U$ in $C(K)$. It follows that $W\subseteq U$.

On the other hand, let $y=\mathbb{1}_{W^C}$. Then $y\in\mathcal{C}_{\mathbb{1}}$, hence in I_e by Theorem 7.2.5. Clearly, $a\perp y$ in $C(K)$ and, therefore, in I_e, and, hence in X. It follow that $P_xy=P_ay=0$. In $C(K)$, this becomes $\mathbb{1}_U\mathbb{1}_{W^C}=0$, hence $U\subseteq W$.

7.3.15(ii) WLOG, $a\geqslant 0$. By the first part, it suffices to prove that $P_a=P_{a\wedge e}$ or, equivalently, $B_a=B_{a\wedge e}$. Since $a\wedge e\leqslant a$, we trivially have $B_{a\wedge e}\subseteq B_a$. On the other hand, we use Proposition 5.1.3 and the fact that e is a weak unit to show that $a\in B_{a\wedge a}$ because

$$\sup\bigl\{a\wedge n(a\wedge e)\bigr\}=\sup\bigl\{a\wedge ne\bigr\}=a.$$

It follows that $B_a\subseteq B_{a\wedge e}$.

7.3.16 For every $x\in X_+$, we have $P_ax=\sup_X(x\wedge na)=\sup_{X^\delta}(x\wedge na)=\widetilde{P}_ax$ by Corollary 3.2.32.

7.4.4 $L_0(\mu)$ is order complete by Theorem 2.1.16.

Let D be a disjoint subset of $L_0(\Omega,\mathcal{F},\mu)$; we need to find an upper bound for D. If D is countable, the solution is easy: for each member of D, we pick a representative in such a way that we get a sequence (f_n) of measurable functions that are disjoint point-wise, i.e., for every $t\in\Omega$ there is at most one n with $f_n(t)>0$. We then define $h=\sup f_n$ point-wise. This function is measurable by Theorem A.3.3 and for every t, $h(t)$ equals $f_n(t)$ for some t, hence is finite. If follows that h (or rather its equivalence class) is in $L_0(\mu)$, and it is clear that $h\geqslant D$.

We will now reduce the case of a general D to the countable case using σ-additivity of μ. Suppose first that μ is finite. Since $L_1(\mu)$ is order complete, $h_n\coloneqq\sup\{f\wedge n\mathbb{1}\ :\ f\in D\}$ exists for every $n\in\mathbb{N}$. Note that $L_1(\mu)$ is order dense and, therefore, regular in $L_0(\mu)$, hence the latter supremum is the same in $L_1(\mu)$ and in $L_0(\mu)$. Since $L_1(\mu)$ has countable sup property, as in Exercise 2.1.18, we can find a countable subset A_n of D such that $h_n=\sup\{f\wedge n\mathbb{1}\ :\ f\in A_n\}$. Let $A=\bigcup_{n=1}^\infty A_n$. Then A is a countable subset of D. By the first part of the argument, find $h\in L_0(\mu)$ such that $h\geqslant A$. It follows that $h\geqslant\sup A_n\geqslant h_n\geqslant f\wedge n\mathbb{1}$ for every $f\in D$ and $n\in\mathbb{N}$. It follows from $f=\sup_n f\wedge n\mathbb{1}$ that $h\geqslant f$. Hence, $h\geqslant D$.

In case of a σ-finite measure μ, find a strictly positive u in $L_0(\mu)$ and replace $\mathbb{1}$ with u in the preceding argument.

7.4.5 Suppose that X is an infinite-dimensional universally complete Banach lattice. By Theorem 5.4.11, X contains a disjoint sequence of non-zero vectors, say, (x_n). WLOG, we may assume that $x_n\geqslant 0$ and $\|x_n\|=n$ (by replacing x_n with $\frac{n|x_n|}{\|x_n\|}$). Since X is universally complete, $u\coloneqq\sup x_n$ exists. For every n, it follows from $u\geqslant x_n$ that $\|u\|\geqslant n$, which is impossible.

7.4.6 Let X be a universally complete vector lattice. Let A be a maximal (with respect to inclusion) disjoint collection in X_+. Since X is universally complete, $w\coloneqq\sup A$ exists in X. It is clear that w is a weak unit in X because otherwise we can find $v>0$ with $v\perp w$, hence $v\in A^d$, which would contradict maximality of A.

7.4.9 It is straightforward that $C(K)$ is an ideal. Let $0<f\in C^\infty(K)$. Put $g=\mathbb{1}\wedge f$. Then $g\in C(K)$ and $0<g\leqslant f$.

7.4.20 This is corollary of Theorem 7.4.18. We need to show that $B_a = B_V$. By Theorem 7.4.18, $B_a = B_U$ for the clopen set $U = \overline{\bigcup_{f\in B} \operatorname{supp} f}$. It follows from $a \in B_a$ that $V \subseteq U$ and, therefore, $B_V \subseteq B_U = B_a$. To prove the opposite inclusion, suppose that $f \in B_V^d$. It follows from $B_V^d = B_{K\setminus V}$ that $(\operatorname{supp} f) \cap (\operatorname{supp} a) = \varnothing$, so that $f \perp a$ and, therefore, $f \in B_a^d$. It follows that $B_V^d \subseteq B_a^d$; we conclude that $B_V \supseteq B_a$.

The statement about the band projection follows immediately from Theorem 7.4.18.

7.4.24 It is immediate from the definition that a subset of a dominable set is dominable. Suppose A is order bounded in X_+, i.e., $A \leqslant h$ for some $h \in X_+$. Let $0 < u \in X$. By the Archimedean property, we can find $n \in \mathbb{N}$ such that $nu \nleqslant h$. Put $v := (nu - h)^+ > 0$. For every $a \in A$, $nu \wedge a \leqslant nu \wedge h = nu - v$, which yields $(nu - a)^+ = nu - nu \wedge a \geqslant v$.

7.4.25 Suppose that A is dominable in X. Let $0 < u \in Y$. Since X is order dense, find $w \in X$ such that $0 < w \leqslant u$. Since A is dominable in X, find $0 < v \in X$ and $r > 0$ such that $v \leqslant w$ and $(rw - a)^+ \geqslant v$. It follows from $w \leqslant u$ that $v \leqslant u$ and $(ru - a)^+ \geqslant v$.

Suppose that A is dominable in Y. Fix $0 < u \in X$. Find $0 < w \in Y$ and $r > 0$ such that $w \leqslant u$ and $(ru - a)^+ \geqslant w$ for all $a \in A$. Since X is order dense, find $v \in X$ such that $0 < v \leqslant w$. Then $v \leqslant u$ and $(ru - a)^+ \geqslant v$ for all $a \in A$.

7.4.28 Since X is order dense in X^u, the claim follows from Propositions 3.3.18 and 3.3.19.

7.4.32 Clearly, $L_p(\mu)$ is an order dense ideal in $L_0(\mu)$. By Exercise 7.4.4, the latter is universally complete.

7.5.4(i) If e is a weak unit in X^u and $x^* = 0$ then $x \wedge e = 0$, hence $x = 0$. Conversely, suppose that e fails to be a weak unit. Then there exists $u \in X_+^u$ such that $u \wedge e = 0$. Since X is order dense in X^u, we find $x \in X_+$ such that $0 < x \leqslant u$. However, it follows from $x^* = x \wedge e \leqslant u \wedge e = 0$.

7.5.4(ii) If $X \subseteq I_e$ and $x \in X_+$ then there exists $\lambda > 0$ with $\lambda x \leqslant e$; it follows that $(\lambda x)^* = (\lambda x) \wedge e = \lambda x$. To prove the converse, let $x \in X_+$; find $\lambda > 0$ such that $(\lambda x)^* = \lambda x$; it follows from $(\lambda x)^* = (\lambda x) \wedge e$ that $\lambda x \leqslant e$ and, therefore, $x \in I_e$.

7.6.3 To show that $S(\Omega)$ is laterally complete, take a disjoint family A in $S(\Omega)_+$. For each $f \in A$, put $U_f = \{f \neq 0\}$. This yields a disjoint family of open sets; let U be the union of this family; put $U_0 := (\overline{U})^C$. Clearly, U and U_0 are open; it follows that $U \cup U_0$ is open and dense. We define $g \colon U \cup U_0 \to \mathbb{R}$ as follows: for each $f \in A$, g agrees with f on U_f; we define g to be zero on U_0. It is now easy to see that $g \in S(\Omega)$ and $g = \sup A$.

8.0.2 Let $\overline{X}$ stand for the completion of X. Let $x, y \in \overline{X}$ with $x \wedge y = 0$. Find sequences (x_n) and (y_n) in X such that $x_n \to x$ and $y_n \to y$ in $\overline{X}$. Since $|x_n| \to |x| = x$ and $|y_n| \to |y| = y$, we may assume, WLOG, that $x_n, y_n \geqslant 0$ for all n. Put $u_n = x_n - x_n \wedge y_n$ and $v_n = y_n - x_n \wedge y_n$. Then $u_n \to x - x \wedge y = x$ and $v_n \to y - x \wedge y = y$. It follows from $u_n \wedge v_n = 0$ that $\|u_n + v_n\| = \|u_n\| + \|v_n\|$. Passing to the limit, we conclude that $\|x + y\| = \|x\| + \|y\|$. A similar argument works for AM.

8.1.6 By Theorem 6.1.9 we have $\mathcal{L}_{\mathrm{ob}}(X, Y) \subseteq L(X, Y)$. To prove the converse, note that by Corollary 4.1.11, we may assume WLOG that $Y = C(K)$ for some compact K. Let $T \colon X \to C(K)$ be a norm bounded operator. An order bounded set in X is norm bounded, hence its image under T is norm bounded in $C(K)$. Finally, observe that every norm bounded set in $C(K)$ is order bounded.

8.1.13 For every $x \in X$, it follows from $x^+ \perp x^-$ that $Tx^+ \perp Tx^-$, so that

$$\|Tx\| = \|Tx^+\| + \|Tx^-\| = \|x^+\| + \|x^-\| = \|x\|.$$

8.2.2 Let $\mu \in \mathrm{ba}(\mathcal{F})$ and $\varphi = T\mu$. It is clear that φ is linear. Let's verify that φ is order bounded. Since μ is bounded, there exists $K > 0$ such that $\lvert\mu(A)\rvert \leqslant K$ for all $A \in \mathcal{F}$. Since for every $f \in \mathrm{s}(\mathcal{F})$ we have $[0, f] \subseteq \bigl[0, \|f\|_\infty \mathbb{1}\bigr]$, it suffices to verify that $\varphi\bigl([0, \mathbb{1}]\bigr)$ is order bounded. Let $f \in [0, \mathbb{1}]$ in $\mathrm{s}(\mathcal{F})$. Find a representation $f = \sum_{i=1}^n r_i \mathbb{1}_{A_i}$ where the sets $A_1, \dots, A_n$ are pairwise disjoint. Then $0 \leqslant r_i \leqslant 1$ for every i. Let $I_+ = \bigl\{i : \mu(A_i) \geqslant 0\bigr\}$ and $I_- = \bigl\{i : \mu(A_i) < 0\bigr\}$. Put $g = \sum_{i \in I_+} r_i \mathbb{1}_{A_i}$ and $h = \sum_{i \in I_-} r_i \mathbb{1}_{A_i}$. Then $f = g + h$, so that

$$\begin{aligned}\bigl\lvert\varphi(f)\bigr\rvert \leqslant \bigl\lvert\varphi(g)\bigr\rvert + \bigl\lvert\varphi(h)\bigr\rvert &= \sum_{i \in I_+} r_i \mu(A_i) - \sum_{i \in I_-} r_i \mu(A_i) \\ &\leqslant \sum_{i \in I_+} \mu(A_i) - \sum_{i \in I_-} \mu(A_i) = \mu\Bigl(\bigcup_{i \in I_+} A_i\Bigr) - \mu\Bigl(\bigcup_{i \in I_-} A_i\Bigr) \leqslant 2K.\end{aligned}$$

This proves that φ is order bounded. Hence, T is an operator from $\mathrm{ba}(\mathcal{F})$ to $\bigl(\mathrm{s}(\mathcal{F})\bigr)^\sim$. It is clear that T is linear. To show that T is a bijection, we construct its inverse. Let $\varphi \in \bigl(\mathrm{s}(\mathcal{F})\bigr)^\sim$. Define $\mu \colon \mathcal{F} \to \mathbb{R}$ via $\mu(A) = \varphi(\mathbb{1}_A)$. It is clear that μ is finitely-additive; we only need to verify that μ is bounded. Since φ is order bounded, there exists $C > 0$ such that $\varphi\bigl([0, \mathbb{1}]\bigr) \subseteq [-C, C]$. It follows that $\mu(A) = \varphi(\mathbb{1}_A) \in [-C, C]$. Therefore, μ is bounded, hence $\mu \in \mathrm{ba}(\mathcal{F})$. It is clear that $T\mu = \varphi$.

The fact that $\mu \geqslant 0$ iff $T\mu \geqslant 0$ is straightforward.

8.2.5 If $\mu_\alpha \uparrow \mu$ and $A \in \mathcal{F}$, Theorem 6.3.2 yields $\mu_\alpha(A) = \mu_\alpha(\mathbb{1}_A) \uparrow \mu(\mathbb{1}_A) = \mu(A)$.

Conversely, suppose that $\mu_\alpha(A) \to \mu(A)$ for every $A \in \mathcal{F}$. It follows from $\mu_\alpha(A) \uparrow$ that $\mu_\alpha(A) \leqslant \mu(A)$, so that $\mu_\alpha \leqslant \mu$. Suppose that $\nu \in \mathrm{ba}(\mathcal{F})$ satisfies $\mu_\alpha \leqslant \nu$ for every α. Then $\mu(A) = \lim_\alpha \mu_\alpha(A) \leqslant \nu(A)$ yields $\mu \leqslant \nu$. Therefore, $\mu = \sup_\alpha \mu_\alpha$.

8.2.6 One implication is trivial because $\bigl\lvert\nu(A)\bigr\rvert \leqslant \lvert\nu\rvert(A)$. To prove the converse, let $\varepsilon > 0$ and find $\delta > 0$ such that $\lvert\mu\rvert(A) < \delta$ implies $\bigl\lvert\nu(A)\bigr\rvert < \frac{\varepsilon}{3}$. Take A with $\lvert\mu\rvert(A) < \delta$. By Theorem 8.2.3, there exists a measurable subset B of A such that $\lvert\nu\rvert(A) \leqslant \nu(B) - \nu(A \setminus B) + \frac{\varepsilon}{3}$. Note that $\lvert\mu\rvert(B) \leqslant \lvert\mu\rvert(A) < \delta$, so that $\bigl\lvert\nu(B)\bigr\rvert < \frac{\varepsilon}{3}$. Similarly, $\bigl\lvert\nu(A \setminus B)\bigr\rvert < \frac{\varepsilon}{3}$. It follows that $\lvert\nu\rvert(A) \leqslant \bigl\lvert\nu(B)\bigr\rvert + \bigl\lvert\nu(A \setminus B)\bigr\rvert + \frac{\varepsilon}{3} < \varepsilon$.

8.3.1 Suppose μ is countably additive and (A_n) is a nested decreasing sequence in $\mathcal{F}$ with $\bigcap_{n=1}^\infty A_n = \varnothing$. Let $B_n = A_n \setminus A_{n+1}$. Then (B_n) is a disjoint sequence with $A_1 = \bigcup_{n=1}^\infty B_n$. This yields $\mu(A_1) = \sum_{k=1}^\infty \mu(B_k)$. In particular, the series converges. It follows from $A_n = \bigcup_{k=n}^\infty B_k$ that $\mu(A_n) = \sum_{k=n}^\infty \mu(B_k) \to 0$ as $n \to \infty$.

Conversely, suppose that $\mu(A_n) \to 0$ for every nested decreasing sequence (A_n) in $\mathcal{F}$ with $\bigcap_{n=1}^\infty A_n = \varnothing$. Let (B_k) be a disjoint sequence in $\mathcal{F}$ and let $B = \bigcup_{k=1}^\infty B_k$. Put $A_n = \bigcup_{k=n+1}^\infty B_k = B \setminus \bigcup_{k=1}^n B_k$. The sequence (A_n) is nested decreasing and $\bigcap_{n=1}^\infty A_n = \varnothing$, hence $\mu(A_n) \to 0$. This yields $\mu(B) - \sum_{k=1}^n \mu(B_k) \to 0$, hence $\mu(B) = \sum_{k=1}^\infty \mu(B_k)$.

8.3.4 Replacing μ with $\lvert\mu\rvert$, we may assume WLOG that $\mu \geqslant 0$. This is because μ is σ-order continuous iff $\lvert\mu\rvert$ is σ-order continuous by Theorem 6.5.10, and μ is countably additive iff $\lvert\mu\rvert$ is countably additive by Theorem 8.3.3.

Suppose that μ is countably additive and $f_n \downarrow 0$ in $\mathrm{s}(\mathcal{F})$. (Note that f_n's are actual functions, not equivalence classes.). Let f be the point-wise infimum of (f_n). Then f is measurable (by Theorem A.3.3). We claim that f equals zero μ-a.e. Indeed, otherwise there exists $\varepsilon > 0$ such that $\mu(A) > 0$, where $A = \{f \geqslant \varepsilon\}$. It follows that $0 < \varepsilon \mathbb{1}_A \leqslant f \leqslant f_n$ for every n, which contradicts $f_n \downarrow 0$.

Let $\tilde{f}_n$ be the equivalence class of f_n in $L_1(\mu)$. Then $\tilde{f}_n \downarrow$ and, by the preceding paragraph, $\tilde{f}_n \xrightarrow{\text{a.e.}} 0$. By Monotone Convergence Theorem, $\mu(f_n) = \int \tilde{f}_n \, d\mu \to 0$.

Conversely, suppose that μ is σ-order continuous. Suppose that $A_n \downarrow \varnothing$ in $\mathcal{F}$. Then $\mathbb{1}_{A_n} \downarrow 0$ in $\mathrm{s}(\mathcal{F})$, so that $\mu(A_n) = \mu(\mathbb{1}_{A_n}) \to 0$. Hence, μ is countably additive by Exercise 8.3.1.

8.3.8 Suppose that $\mu \perp \nu$. Let $s = \sup\bigl\{\mu(A) : \nu(A) = 0\bigr\}$. Find a sequence (A_n) such that $\mu(A_n) \uparrow s$ and $\nu(A_n) = 0$ for all n. Put $B = \bigcup_{n=1}^\infty A_n$, then $\nu(B) = 0$ and $\mu(B) = s$. It suffices to show that $\mu(B^C) = 0$.

Suppose not. Find a sequence (α_n) of positive reals such that $\sum_{n=1}^{\infty}\alpha_n < \mu(B^C)$. Since $\mu\wedge\nu = 0$, by Theorem 8.2.3, for every n we can find $D_n \subset B^C$ such that $\mu(D_n) < \alpha_n$ and $\nu(B^C \setminus D_n) < \alpha_n$. Put $D = \bigcup_{n=1}^{\infty} D_n$. Then $\mu(D) \leqslant \sum_{n=1}^{\infty}\mu(D_n) < \sum_{n=1}^{\infty}\alpha_n < \mu(B^C)$, so that $\mu(B^C \setminus D) > 0$. On the other hand, $\nu(B^C \setminus D) \leqslant \nu(B^C \setminus D_n) < \alpha_n$ for all n, hence $\nu(B^C \setminus D) = 0$. It follows that $\mu\big(B \cup (B^C \setminus D)\big) > s$ while $\nu\big(B \cup (B^C \setminus D)\big) = 0$; this contradicts the definition of s.

To prove the converse, suppose that there exists $B \in \mathcal{F}$ such that $\mu(B) = 0$ while $\nu(B^C) = 0$. Let $\lambda = \mu\wedge\nu$. Then $\lambda(\Omega) = \lambda(B) + \lambda(B^C) \leqslant \mu(B) + \nu(B^C) = 0$, hence $\lambda = 0$.

8.3.9 Since μ^+ and μ^- are disjoint measures, by Exercise 8.3.8 we can find $D \in \mathcal{F}$ such that $\mu^-(D) = 0$ and $\mu^+(D^C) = 0$. Put $B = A \cap D$. It follows from $\mu^-(B) = 0$ that $\mu(B) = \mu^+(B)$, hence the supremum in the formula for μ^+ is attained. Furthermore,

$$\begin{aligned}|\mu|(A) = \mu^+(A) + \mu^-(A) = \mu^+(B) + \mu^+(A \setminus B) + \mu^-(B) + \mu^-(A \setminus B)\\ = \mu^+(B) + \mu^-(A \setminus B) = \mu(B) - \mu(A \setminus B),\end{aligned}$$

hence the supremum in the formula for $|\mu|$ is attained. The formulae for $\mu\vee\nu$ and $\mu\wedge\nu$ follow from Exercise 1.3.3(iii).

8.4.5 WLOG, $\mu \geqslant 0$; otherwise, consider μ^+ and μ^-. Being an AL-space, $\mathcal{M}(K)$ is order continuous, so it follows from Exercise 5.4.20 that $\mu = \sup_n \nu_n$ for a sequence (ν_n) of atoms in $\mathcal{M}(K)$. By the preceding discussion, $\nu_n = r_n\delta_{t_n}$ for some positive real r_n and some $t_n \in K$. Let $C = \{t_n\}$. Put $\mu_m = \bigvee_{n=1}^{m}\nu_n$. Then $\mu_m \uparrow \mu$, so if $A \in \mathcal{F}$ with $A \cap C = \varnothing$ then, by Exercise 8.2.5, we have $\mu(A) = \lim_n \mu_n(A) = 0$.

8.4.6 Every weakly null sequence in a Banach space is norm bounded. If $f_n \xrightarrow{\text{w}} 0$ then $f_n(t) = \delta_t(f_n) \to 0$ for every $t \in K$.

Conversely, suppose that (f_n) is norm bounded in $C(K)$ (and, therefore, order bounded), and f_n converges to zero pointwise. Let $\mu \in C(K)^*$; we need to show that $\mu(f_n) \to 0$. WLOG, $\mu \geqslant 0$; otherwise, consider μ^+ and μ^-. By Riesz-Markov Theorem 8.4.1, we may then view μ as a measure. By Lebesgue Dominated Convergence Theorem A.3.13, $\mu(f_n) = \int f_n\,d\mu \to 0$.

8.4.7 Let X be a Banach lattice and $x_n \xrightarrow{\text{w}} 0$ in X. By Exercise 6.8.7, it suffices to prove that $|x_n| \xrightarrow{\text{w}} 0$. By Corollary 8.1.10, we may identify X with a closed sublattice of some $C(K)$ space; WLOG, $X = C(K)$. The conclusion now follows from Exercise 8.4.6.

8.4.8 Suppose that X is an order continuous Banach lattice such that $X^* = \mathcal{M}[0,1]$. Since $\mathcal{M}[0,1]$ is an AL-space, its dual is an AM-space. Being a sublattice of $X^{**} = \mathcal{M}[0,1]^*$, X itself is an AM-space. By Theorem 8.1.16, $X = c_0(\Gamma)$ for some Γ. It follows that $\mathcal{M}[0,1] = X^* = \ell_1(\Gamma)$. This is impossible: $\ell_1(\Gamma)$ is atomic, but the Lebesgue measure in $\mathcal{M}[0,1]$ dominates no atoms.

8.6.3 We will show that if any of the three spaces has atoms then the other two have atoms. Suppose that μ has an atom, say, A. We claim that $[A]$ is an atom of $\bar{\mu}$. Indeed, suppose that $[B] \leqslant [A]$ for some $B \in \mathcal{F}$. Then $\mu(B \setminus A) = 0$. Since $B \cap A \subseteq A$ and A is an atom, we either have $\mu(B \cap A) = 0$, in which case $\mu(B) = \mu(B \cap A) + \mu(B \setminus A) = 0$ and, therefore, $[B] = \not{\mathbb{0}}$, or $\mu(B \cap A) = \mu(A)$, in which case $\mu(A \setminus B) = 0$, so that $[A \setminus B] = \not{\mathbb{0}}$ and, therefore, $[A] = [B] \vee [A \setminus B] = [B]$.

Suppose that $\bar{\mu}$ has an atom, say $[A]$. Then $\mathbb{1}_A$ is an atom in $\mathcal{C}_{\mathbb{1}}$. We claim that $\mathbb{1}_A$ is also an atom in $L_p(\mu)$. Suppose not, that $0 < f < \mathbb{1}_A$ for some $f \in L_p(\mu)$. Then there exists $\lambda \in (0,1)$ such that $\mu(B)$ and $\mu(A \setminus B)$ are both positive, where $B = \{f > \lambda\}$. It follows that $0 < \mathbb{1}_B < \mathbb{1}_A$; a contradiction.

Finally, suppose f is a (positive) atom of $L_p(\mu)$. Since $f > 0$, we have $f \geqslant \lambda\mathbb{1}_A$ for some $A \in \mathcal{F}$ with $\mu(A) > 0$ and some $\lambda > 0$. Since f is an atom, it follows that $f = \lambda\mathbb{1}_A$. Hence, $\mathbb{1}_A$ is an atom of $L_p(\mu)$. We claim that A is an atom of μ. Suppose that $B \in \mathcal{F}$ and $B \subseteq A$. Then $\mathbb{1}_B \leqslant \mathbb{1}_A$. It follows that $\mathbb{1}_B$ is a scalar multiple of $\mathbb{1}_A$, which is only possible if either $\mu(B) = 0$ or $\mu(B) = \mu(A)$.

8.6.4 Suppose that $L_p(\mu)$ is separable. Since $L_p(\mu)$ is dense in $L_1(\mu)$, the latter is separable as well. As we observed earlier, $(\mathcal{F}/\mathcal{N},\bar{\mu})$ is isometric to $(\mathcal{C}_{\mathbb{1}},\|\cdot\|_{L_1})$, which may be viewed as a metric subspace of $L_1(\mu)$, hence is separable.

Conversely, suppose that $(\mathcal{C}_{\mathbb{1}},\|\cdot\|_{L_1})$ is separable. Let D be a countable subset of $\mathcal{C}_{\mathbb{1}}$ such that D is dense in $\mathcal{C}_{\mathbb{1}}$ with respect to $\|\cdot\|_{L_1}$. This means that for every $A\in\mathcal{F}$ and every $\varepsilon>0$ there exists $B\in\mathcal{F}$ such that $\|\mathbb{1}_A-\mathbb{1}_B\|_{L_1}<\varepsilon$ or, equivalently, $\mu(A\triangle B)<\varepsilon$. It follows that every simple function may be approximated in $\|\cdot\|_{L_p}$ by simple functions in $\operatorname{span} D$, hence $\operatorname{span} D$ is dense in $L_p(\mu)$.

9.1.3 Suppose not; suppose h is a strictly positive functional on X. For every $\omega\in\Omega$, let e_ω be the characteristic function of the singleton $\{\omega\}$. For every finite set $F\subseteq\Omega$ we have $\sum_{\omega\in F}e_\omega=\mathbb{1}_F\leqslant\mathbb{1}$, so that $\sum_{\omega\in F}h(e_\omega)\leqslant h(\mathbb{1})$. Thus, the uncountable family $\{h(e_\omega)\,:\,\omega\in\Omega\}$ of positive real numbers has uniformly bounded finite sums, which is impossible.

9.1.4 Let h be a strictly positive functional on X. If $0\leqslant x\leqslant\frac{1}{n}u$ for every n then $0\leqslant h(x)\leqslant\frac{1}{n}h(u)$. It follows that $h(x)=0$ and, therefore, $x=0$.

9.1.6 Suppose that h is a strictly positive functional on X, $A\subseteq X$, and $u=\sup A$. Note that $u=\sup A^\vee$. Suppose we can find a countable subset B of $A^\vee$ with $\sup B=u$. Every element of B is the supremum of a finite subset of A. Let $C\subseteq A$ be the union of all these subsets; then C is countable and $B\subseteq C^\vee$, hence $u=\sup C$, and we are done. Therefore, we assume that $A=A^\vee$.

It follows from $h(A)\leqslant h(u)$ that $\lambda:=\sup h(A)$ exists. There exists (x_n) in A such that $h(x_n)\to\lambda$. It suffices to show that $u=\sup x_n$. Clearly, $u\geqslant x_n$ for every n. Suppose that $v\geqslant x_n$ for every n. Fix $x\in A$. For every n, we have $\lambda\geqslant h(x\vee x_n)\geqslant h(x_n)\to\lambda$. This implies that $h(x\vee x_n)\to\lambda$. Furthermore, $x_n\leqslant v$ yields

$$0\leqslant h\big((x-v)^+\big)\leqslant h\big((x-x_n)^+\big)=h\big(x\vee x_n-x_n\big)=h(x\vee x_n)-h(x_n)\to\lambda-\lambda=0.$$

It follows that $h\big((x-v)^+\big)=0$, so that $(x-v)^+=0$ and, consecutively, $x\leqslant v$. Therefore, $u\leqslant v$.

To show that the converse is false, observe that $\mathbb{R}^{\mathbb{N}}$ has the countable sup property by Exercise 2.1.21, yet it does not admit a strictly positive functional; see Exercise 9.1.2.

9.2.1 First, verify that $\|\cdot\|$ is a norm. Let $x,y\in X$. For every $\lambda\in\mathbb{R}$. $\|\lambda x\|_h=h\big(|\lambda x|\big)=|\lambda|h\big(|x|\big)=|\lambda|\|x\|_h$. Furthermore, it follows from $|x+y|\leqslant|x|+|y|$ that

$$\|x+y\|_h=h\big(|x+y|\big)\leqslant h\big(|x|+|y|\big)=h\big(|x|\big)+h\big(|y|\big)=\|x\|_h+\|y\|_h.$$

If $\|x\|_h=0$ then strict positivity of h yields $|x|=0$ and, therefore, $x=0$. This proves that $\|\cdot\|_h$ is a norm on X.

Next, verify that it is a lattice norm: if $|x|\leqslant|y|$ then $h\big(|x|\big)\leqslant h\big(|y|\big)$, so that $\|x\|_h\leqslant\|y\|_h$.

Finally, suppose that $x\perp y$. Then $|x+y|=|x|+|y|$, so that $\|x+y\|_h=h\big(|x+y|\big)=h\big(|x|\big)+h\big(|y|\big)=\|x\|_h+\|y\|_h$.

9.2.5 $\|x_n\|_{L_1(\mu)}=h(x_n)\to 0$.

The assumption that (x_n) is positive is necessary. For example, the Rademacher sequence (r_n) in L_p with $1<p<\infty$ is weakly null, but $\|r_n\|_{L_1}=1$ for all n.

9.2.8 Observe that $\hat{e}$ is strictly positive and order continuous by Proposition 6.6.19 and Theorem 9.1.10. Therefore, we can consider the L_1-representation $\widetilde{X^*}$ for X^* and $\hat{e}$. We can identify $\widetilde{X^*}$ with $L_1(\mu)$ for some measure μ. We have $\int f\,d\mu=\hat{e}(f)=f(e)$ for every $f\in X^*$.

X^* is order complete by Corollary 6.7.12. It follows from Theorem 9.2.7 that X^* is an ideal in $L_1(\mu)$.

9.6.1(i) Let (x_n), (d_n), and (r_n) be as in the definition, and suppose that $x_n\geqslant 0$ for all n. For each n, put $d_n'=d_n^+\wedge x_n$ (we truncate d_n both above and below). It follows from $0\leqslant d_n'\leqslant d_n^+$ that the sequence (d_n') is still disjoint. Observe that $0\leqslant x_n-d_n'\leqslant|x_n-d_n|=|r_n|$. It follows that $x_n-d_n'\to 0$.

9.6.1(ii) Let (x_n), (d_n), and (r_n) be as in the definition, For every n, put $d_n' = P_{d_n}x_n$ and $r_n' = P_{d_n^d}x_n$. Then $d_n' \perp r_n'$ and $d_n' + r_n' = x_n$. The sequence (d_n') is clearly still disjoint. Observe that $r_n' = P_{d_n^d}x_n = P_{d_n^d}(d_n + r_n) = P_{d_n^d}r_n$. Since $r_n \to 0$, we conclude that $r_n' \to 0$.

9.6.1(iii) Suppose that (x_n) is asymptotically disjoint; let (d_n) be a disjoint sequence with $x_n - d_n \to 0$. Then $\bigl(|d_n|\bigr)$ is also a disjoint sequence; it follows from $\bigl||x_n| - |d_n|\bigr| \leqslant |x_n - d_n| \to 0$ that $|x_n| - |d_n| \to 0$.

Conversely, suppose that $\bigl(|x_n|\bigr)$ is asymptotically disjoint. Then we can decompose $|x_n| = d_n + r_n$ such that (d_n) is disjoint, $r_n \to 0$, and $d_n, r_n \geqslant 0$ for all n. By Exercise 1.5.18, we can now split $x_n = w_n + s_n$ such that $|w_n| = d_n$ and $|s_n| = r_n$. It follows that (w_n) is disjoint and $s_n \to 0$.

9.6.8 (i)$\Rightarrow$(ii) Suppose that (ii) fails. Then there exists a sequence (x_n) in Y_+ such that $\|x_n\|_X = 1$ and $\mu(x_n \geqslant \frac{1}{n}) < \frac{1}{n}$. It follows that (x_n) converges to zero in measure. By Theorem 9.6.6, (x_n) has a subsequence which is asymptotically disjoint (as a sequence in X).

(ii)$\Rightarrow$(i) Suppose that Y contains a sequence (x_n) which, when viewed as a sequence in X, is normalized, and $x_n = d_n + r_n$, where (d_n) is disjoint and $\|r_n\|_X \to 0$. Since μ is finite, $\mu(\operatorname{supp} d_n) \to 0$; it follows that (d_n) converges to zero in measure. Furthermore, we have $\|r_n\|_{L_1(\mu)} \to 0$ and, therefore, (r_n) converges to zero in measure. It follows that (x_n) converges to zero in measure. This contradicts (ii).

(ii)$\Rightarrow$(iii) It follows from (ii) that $\|x\|_{L_1(\mu)} > \delta^2$ whenever $x \in Y_+$ with $\|x\|_X = 1$. Therefore, $\|x\|_{L_1(\mu)} \geqslant \delta^2\|x\|_X$ for every $x \in X$.

To see that (iii) need not imply (i) or (ii), take $Y = X = L_1(\mu)$.

9.6.9 There exists a disjoint sequence (d_n) in $L_p(\mu)$ such that $\|x_n - d_n\|_{L_p(\mu)} \to 0$. It follows that, after passing to a tail, (d_n) is seminormalized in $L_p(\mu)$. Being a disjoint seminormalized sequence in $L_p(\mu)$, (d_n) is equivalent to the unit vector basis of ℓ_p. Since $\|x_n - d_n\|_{L_p(\mu)} \to 0$, after passing to a further subsequence and applying the Principle of Small Perturbations A.4.29, we conclude that (x_n) itself is equivalent to the unit vector basis of ℓ_p.

9.7.1 (i) Suppose that $x \in [-u, u] + \varepsilon B_X$. Then $x = y + z$ with $|y| \leqslant u$ and $\|z\| \leqslant \varepsilon$. The Triangle Inequality yields that $|x| - |y| \leqslant |z|$. It follows that

$$\bigl(|x| - u\bigr)^+ \leqslant \bigl(|x| - |y|\bigr)^+ \leqslant |z|, \text{ so that } \bigl\|\bigl(|x| - u\bigr)^+\bigr\| \leqslant \|z\| \leqslant \varepsilon.$$

Conversely, suppose that $\|b\| \leqslant \varepsilon$ where $b = \bigl(|x| - u\bigr)^+$. By Exercise 1.3.3(iii), $|x| = a + b$ where $a = |x| \wedge u$. By Exercise 1.5.18, there exist y and z satisfying $x = y + z$, $|y| = a$, and $|z| = b$. It follows that $|y| \leqslant u$, so that $y \in [-u, u]$, while $\|z\| = \|b\| \leqslant \varepsilon$, so that $z \in \varepsilon B_X$.

Now observe that (i)$\Rightarrow$(ii)$\Rightarrow$(iii).

9.7.3 Let A be an almost order bounded subset of Y and (x_n) a sequence in A. If $x_n \xrightarrow{\|\cdot\|_Y} x$ in Y then $x_n \xrightarrow{\|\cdot\|_X} x$ because the inclusion is continuous. Suppose that $x_n \xrightarrow{\|\cdot\|_X} x \in Y$. WLOG, $x = 0$ and $x_n \geqslant 0$ for all n. Fix $\varepsilon > 0$. There exists $u \in Y_+$ with $A \subseteq [-u, u] + \varepsilon B_Y$. It follows from $0 \leqslant x_n \wedge u \leqslant x_n$ that $x_n \wedge u \xrightarrow{\|\cdot\|_X} 0$. By the original Amemiya's Theorem 9.3.9, $x_n \wedge u \xrightarrow{\|\cdot\|_Y} 0$, hence $\|x_n \wedge u\|_Y \leqslant \varepsilon$ for all sufficiently large values of n. It follows that $\|x_n\|_Y \leqslant \|x_n \wedge u\|_Y + \bigl\|(x_n - u)^+\bigr\|_Y \leqslant 2\varepsilon$. Hence, $x_n \xrightarrow{\|\cdot\|_Y} 0$.

9.7.4 The forward implication is straightforward with $\delta = \frac{1}{n}$. To prove the converse, let ε, (A_n) and (f_n) as in the statement. For every $\delta > 0$ there exists n such that $\mu(A_n) < \delta$, yet $\int_{A_n} |f_n|\,d\mu > \varepsilon$, hence F is not equi-integrable.

9.7.9 Suppose that F is equi-integrable. Fix $\varepsilon > 0$ and let $\delta > 0$ be as in the definition of equi-integrability. Since F is bounded, there is a $K > 0$ such that $\|f\| \leqslant K$ for every $f \in F$. Choose

$M > K\delta^{-1}$. For every $f \in F$, Chebyshev's Inequality yields $\mu(f \geqslant M) \leqslant \frac{K}{M} < \delta$. Consequently, $\int_{\{f \geqslant M\}} f < \varepsilon$.

Conversely, suppose that for every $\varepsilon > 0$ there exists $M > 0$ such that $\int_{\{f \geqslant M\}} f < \varepsilon$ for all $f \in F$. Let $\varepsilon > 0$, fix M as before, and put $\delta = \frac{\varepsilon}{M}$. Then for all $f \in F$ and $A \in \Sigma$ with $\mu(A) < \delta$ we have

$$\int_A f = \int_{A \cap \{f < M\}} f + \int_{A \cap \{f \geqslant M\}} f \leqslant M\mu(A) + \int_{\{f \geqslant M\}} f \leqslant M\frac{\varepsilon}{M} + \varepsilon = 2\varepsilon.$$

9.7.11 Suppose $f_n \to 0$. Clearly, (f_n) is almost order bounded. It now follows from Proposition 9.7.10 that (f_n) is equi-integrable.

9.7.13 Fix $\varepsilon > 0$. Since g is integrable, there exists $M > 0$ such that $\int_{g \geqslant M} g < \varepsilon$. It follows from (9.2) that $\int_{f \geqslant M} f \leqslant \int_{g \geqslant M} g < \varepsilon$ for every $f \in F$, Hence F is equi-integrable by Exercise 9.7.9.

9.8.1 Let A be a relatively compact set. Fix $\varepsilon > 0$. By Proposition A.4.6, there exists a finite set F such that $A \subseteq F + \varepsilon B_X$. Clearly, there exists $u > 0$ such that $F \subseteq [-u, u]$, hence $A \subseteq [-u, u] + \varepsilon B_X$.

9.8.2 (ii)$\Leftrightarrow$(i)$\Rightarrow$(iii) is trivial. Suppose that (f_n) fails to converge to zero uniformly on A. Then there exists $\varepsilon > 0$ such that $\forall n_0\, \exists n > n_0\, \exists x \in A\, \lvert f_n(x) \rvert \geqslant \varepsilon$. Fix such an ε. Inductively, construct $n_1 < n_2 < n_3 < \dots$ and $x_1, x_2, x_3, \dots$ in A with $\lvert f_{n_k}(x_k) \rvert \geqslant \varepsilon$. For each $m \in \mathbb{N}$, if $m = n_k$ for some k then put $a_m = x_{n_k}$; otherwise, let a_m be an arbitrary element of A. Then $f_m(a_m) \not\to 0$.

9.8.5 Define $S\colon \ell_1 \to X$ via $Se_n = x_n$. It can be easily verified that S is bounded. The operator $TS\colon \ell_1 \to \ell_1$ satisfies $\lVert TSe_n - e_n \rVert \leqslant \lambda$ for every n. It follows that $\lVert TS - I \rVert \leqslant \lambda < 1$. By a standard fact of Spectral Theory, TS is invertible. It follows that S is an isomorphism onto its range, and, therefore, (x_n) is a basic sequence equivalent to (e_n). Furthermore, put $U = (TS)^{-1}$. Then $SUT\colon X \to X$ is a projection onto $\operatorname{Range} S = [x_n]$.

9.8.8 Observe that each of the three conditions implies that F is bounded by Proposition A.4.18. By Dunford-Pettis Theorem, weak compactness of F is equivalent to almost order boundedness. It is clear that F, $\mathrm{Sol}(F)$, and $\lvert F \rvert$ are either all almost order bounded or all fail to be almost order bounded.

9.8.9 If $1 < p < \infty$ then $L_p(\mu)$ is reflexive, order intervals are weakly compact because they are bounded and weakly closed. In case $p = 1$, this follows from Dunford-Pettis Theorem.

9.8.12 Let (f_n) be a bounded sequence in $L_1(\mu)$. Let (g_k) and (h_k) be as in Subsequence Splitting Lemma 9.7.14. Since the set $\{h_k\}$ is almost order bounded, it is relatively weakly compact by Dunford-Pettis Theorem 9.8.6. By Eberlein-Šmulian Theorem A.4.20, (h_k) has a weakly convergent subsequence.

10.1.3 Let X be a reflexive Banach lattice. Order intervals in X are norm bounded, hence they are relatively weak* compact by Alaoglu's Theorem. Since the space is reflexive, they are relatively weakly compact. Since order intervals are weakly closed by Exercise 6.8.1(iii), they are weakly compact. Now apply Theorem 10.1.2(iii).

10.2.1 Let $y_\alpha \downarrow 0$ in Y. By regularity, $y_\alpha \downarrow 0$ in X, hence $y_\alpha \xrightarrow{\lVert\cdot\rVert} 0$.

10.2.5 Since c_0 is order continuous, we have $c_0 \subseteq (\ell_\infty)_{\mathrm{oc}}$. On the other hand, let $0 \leqslant u \in \ell_\infty \setminus c_0$ and put $x_n = P_n u$, the projection of u onto the first n coordinates. Then $0 \leqslant x_n \uparrow u$, but (x_n) is not convergent, so that $u \notin (\ell_\infty)_{\mathrm{oc}}$ by Theorem 10.2.3(iv).

10.3.2 (i)$\Rightarrow$(iii) Since X is separable, it cannot contain a copy of ℓ_∞. Hence, if X is σ-order complete then it is order continuous by Theorem 10.3.1. Implications (iii)$\Rightarrow$(ii)$\Rightarrow$(i) have been proved earlier.

10.4.1(i) By Nakano's Theorem 2.5.24.

10.4.1(iii) Suppose that $B_X^+ \ni x_n \uparrow$. Since B_X^+ is weakly compact, By Eberlein-Šmulian's Theorem A.4.20 we conclude that (x_n) has a weakly convergent subsequence, say, (x_{n_k}). By Dini's Theorem 6.8.2, (x_{n_k}) is norm convergent. Since (x_n) is monotone, (x_n) itself is norm convergent.

10.4.1(iv) Let $X = L_1(\mu)$ and $B_X^+ \ni x_n \uparrow$. The sequence $\bigl(\lVert x_n\rVert\bigr)$ is increasing and bounded by 1, hence is convergent. If $k \leqslant m$, we have $\lVert x_m - x_k\rVert = \lVert x_m\rVert - \lVert x_k\rVert$; it follows that (x_n) is Cauchy and, therefore, convergent.

10.4.1(v) Let X be an infinite-dimensional AM-space. Let (e_n) be a disjoint positive normalized sequence; such a sequence exists by Theorem 5.4.11. Put $u_n = \sum_{i=1}^n e_i$. Then $B_X^+ \ni x_n \uparrow$, yet $\lVert u_n - u_{n-1}\rVert = \lVert e_n\rVert = 1$, so (u_n) is not convergent.

10.4.2 Clearly, if every increasing bounded net in X is convergent then X is a KB-space. Conversely, suppose that X is a KB-space; let (x_α) be an increasing bounded net in X. Fix any index α_0; then the positive net $(x_\alpha - x_{\alpha_0})_{\alpha\geqslant\alpha_0}$ is still increasing and bounded; it converges iff the original net converges. Thus, WLOG, we may assume that (x_α) is positive. Suppose that (x_α) is not convergent. Then it is not Cauchy, hence one can find a positive real ε and indices $\alpha_1 < \alpha_2 < \dots$ such that $\lVert x_{\alpha_{n+1}} - x_{\alpha_n}\rVert > \varepsilon$ for all n. But (x_{α_n}) is an increasing norm bounded sequence, hence is convergent; a contradiction.

10.4.9 Suppose that $\lVert f_n\rVert \leqslant M$ for all n. For every $x \in X_+$, we have $f_n(x) \uparrow\leqslant M\lVert x\rVert$, so that $g(x) \coloneqq \lim_n f_n(x)$ exists. By Extension Lemma 6.2.1, g extends to a positive functional on X via $g(x) = g(x^+) - g(x^-)$ for $x \in X$. Therefore, $f_n(x) \to g(x)$ for every x. It follows from $\bigl\lvert f_n(x)\bigr\rvert \leqslant M\lVert x\rVert$ that $\lvert g(x)\rvert \leqslant M\lVert x\rVert$ and, therefore, $g \in X^*$. It is clear that $f_n \uparrow g$.

10.6.3 Let (x_α) be any increasing norm bounded net. Fix any index α_0 and consider the net $(x_\alpha - x_{\alpha_0})_{\alpha\geqslant\alpha_0}$. This net is positive, it is still increasing and norm bounded, and it is order bounded or has supremum iff the original net does.

10.6.8 We first show that X^* is monotonically bounded. Let (f_α) be a norm bounded increasing net in X^*. By Banach-Alaoglu's theorem, it has a subnet (g_γ) which weak* converges to some $f \in X^*$. For every $x \in X$ we have $g_\gamma(x) \uparrow$ and $g_\gamma(x) \to f(x)$, hence $g_\gamma \leqslant f$ for every γ. It follows that $f_\alpha \leqslant f$ for every α. Therefore, X^* is monotonically bounded. Since X^* is order complete by Corollary 6.7.12, it is also monotonically complete.

Suppose that (f_α) is a net in X^* and $0 \leqslant f_\alpha \uparrow f$. Clearly, $\lVert f_\alpha\rVert \uparrow\leqslant \lVert f\rVert$. Fix $\varepsilon > 0$. There exists $x \in B_X$ such that $\lVert f\rVert \leqslant \lvert f(x)\rvert + \varepsilon$. WLOG, $x \geqslant 0$, otherwise replace x with $\lvert x\rvert$. By Riesz-Kantorovich Theorem 6.3.2, $f_\alpha(x) \uparrow f(x)$, so that there exists α_0 such that for all $\alpha \geqslant \alpha_0$ one has $f_\alpha(x) \geqslant f(x) - \varepsilon \geqslant \lVert f\rVert - 2\varepsilon$. It follows that $\lVert f_\alpha\rVert \geqslant \lVert f\rVert - 2\varepsilon$.

10.6.9(ii) WLOG, A is solid. Observe that this set equals $A^\vee$ by Proposition 5.2.26.

10.6.10 Suppose that X is a KB-space. Clearly, X is order continuous. Suppose A is order bounded above in X^{**}. Replacing A with $A^\vee$, we may view it as an increasing net, say (x_α). Then $(\hat{x}_\alpha)$ is order and, therefore, norm bounded in X^{**}, hence (x_α) is norm bounded in X. By the KB-property, $x_\alpha \to x$ for some $x \in X$ and, therefore, $x_\alpha \uparrow x$ by Monotone Convergence Lemma 2.5.3. It follows that $A \leqslant x$.

To prove the converse, note that by Theorem 10.4.10, it suffices to show that X is a band in X^{**}. By Theorem 10.1.8, we know that X is an ideal and is regular in X^{**}. It suffices to show that if $\hat{x}_\alpha \uparrow \xi$ in X^{**} for some increasing net (x_α) in X_+ then $\xi \in X$. Since $(\hat{x}_\alpha)$ is order bounded in X^{**}, by the assumption, it is order bounded in X. Since X is order continuous and, therefore, order complete, $x \coloneqq \sup x_\alpha$ exists in X. By regularity, we have $\hat{x}_\alpha \uparrow \hat{x}$ in X^{**}; it follows that $\xi = \hat{x}$ is in X.

10.6.12 For every $x \in X$, we have $\lvert Tx\rvert \leqslant T\lvert x\rvert$ and $\bigl\lVert T\lvert x\rvert\bigr\rVert = \lVert x\rVert = \lVert Tx\rVert = \bigl\lVert\lvert Tx\rvert\bigr\rVert$. It follows that $T\lvert x\rvert = \lvert Tx\rvert$.

10.8.10 Suppose that $a \in X_+$ is an atom. Then $B_a = \operatorname{span}\{a\}$ is a projection band in X by Lemma 5.4.9. It follows that $X = B_a \oplus B_a^d$, hence X^* may be identified with $(B_a)^* \oplus (B_a^d)^*$. Viewed as a one-dimensional band in X^*, $(B_a)^*$ is spanned by an atom of X^*

11.1.1 Suppose that ρ is a lattice seminorm, $|x| \leqslant |y|$ and $y \in B_\rho$. Then $\rho(x) \leqslant \rho(y) \leqslant 1$, hence $x \in B_\rho$. Conversely, suppose that ρ is a seminorm such that B_ρ is a solid set, and $|x| \leqslant |y|$. We consider two cases. If $\rho(y) \neq 0$ then $\frac{1}{\rho(y)} y \in B_\rho$; it follows from $\bigl|\frac{1}{\rho(y)}x\bigr| \leqslant \bigl|\frac{1}{\rho(y)}y\bigr|$ that $\frac{1}{\rho(y)}x \in B_\rho$, which implies $\rho(x) \leqslant \rho(y)$. If $\rho(y) = 0$ then for every $\lambda > 0$ we have $\rho(\lambda y) = 0$, so that $\lambda y \in B_\rho$. It follows from $|\lambda x| \leqslant |\lambda y|$ that $\lambda x \in B_\rho$, hence $\rho(x) \leqslant \frac{1}{\lambda}$. Since λ was arbitrary, this yields $\rho(x) = 0$.

11.1.3 Suppose that A is solid and $|f| \leqslant |g|$ in X^*. Let $x \in A$. Using Riesz-Kantorovich formula, we have

$$\bigl|f(x)\bigr| \leqslant |f|\bigl(|x|\bigr) \leqslant |g|\bigl(|x|\bigr) = \sup\bigl\{\bigl|g(y)\bigr| \;:\; |y| \leqslant |x|\bigr\} \leqslant \rho_A(g)$$

so that $\rho_A(f) \leqslant \rho_A(g)$. The proof for ρ_B is similar.

11.1.4 See the solution of Exercise 11.1.5

11.1.5 We will solve Exercises 11.1.4 and 11.1.5 simultaneously.

(i)⇒(ii) Let ρ be a continuous seminorm on a Banach space X. Then $N \coloneqq \ker\rho$ is closed and ρ induces a norm on X/N via $|||x + N||| = \rho(x)$. Let Y be the completion of $\bigl(X/N, |||\cdot|||\bigr)$. Let $Q\colon X \to X/N$ be the quotient map and $J\colon X/N \to Y$ be the inclusion map. Put $T = JQ$. It is easy to see that $\rho = r_T$.

If, in addition, X is a Banach lattice and ρ is a lattice seminorm, then N is an ideal, so that X/N is a vector lattice, Q is a lattice homomorphism, $|||\cdot|||$ is a lattice norm, so Y is a Banach lattice. It follows that T is a lattice homomorphism.

(ii)⇒(iii) Let $T\colon X \to Y$ be a bounded operator between Banach spaces. Put $B = T^* B_{Y^*}$. For $x \in X$,

$$r_T(x) \leqslant 1 \quad\Leftrightarrow\quad \|Tx\| \leqslant 1 \quad\Leftrightarrow\quad \forall g \in B_{Y^*}\ \bigl|g(Tx)\bigr| \leqslant 1 \quad\Leftrightarrow\quad \forall f \in B\ \bigl|f(x)\bigr| \leqslant 1 \quad\Leftrightarrow\quad \rho_B(x) \leqslant 1.$$

Hence, $r_T = \rho_B$. If, in addition, X and Y are Banach lattices and T is a lattice homomorphism, then T^* is interval preserving by Kim-Andô Theorem 6.11.11, so that B is solid.

(iii)⇒(i) is straightforward.

11.1.6 It is clear that $|x| \leqslant |y|$ implies $\tau(x) \leqslant \tau(y)$. Take $x, y \in X$. Let $v_1, \dots, v_n, w_1, \dots, w_m \in X_+$ such that $|x| \leqslant \sum_{i=1}^n v_i$ and $|y| \leqslant \sum_{j=1}^m w_j$. It follows from $|x + y| \leqslant \sum_{i=1}^n v_i + \sum_{j=1}^m w_j$ that $\tau(x + y) \leqslant \sum_{i=1}^n \rho(v_i) + \sum_{j=1}^m \rho(w_j)$. Therefore, $\tau(x + y) \leqslant \tau(x) + \tau(y)$.

Let τ' be a lattice seminorm on X such that $\tau'(x) \leqslant \rho(x)$ for every $x \geqslant 0$. Suppose that $x \in X$ and $u_1, \dots, u_n \geqslant 0$ such that $|x| \leqslant \sum_{i=1}^n u_n$. Then $\tau'(x) \leqslant \sum_{i=1}^n \tau'(u_i) \leqslant \sum_{i=1}^n \rho(u_i)$. Therefore, $\tau'(x) \leqslant \tau(x)$.

11.1.8 Suppose not. Fix $\varepsilon > 0$ such that for every $u \geqslant 0$ we have $A \not\subseteq [-u, u] + \varepsilon B_\rho$. Let $u \in X_+$. Then $A \not\subseteq [-2u, 2u] + \varepsilon B_\rho$. Find $a \in A$ such that $a \notin [-2u, 2u] + \varepsilon B_\rho$. Taking x to be either a^+ or a^-, we get $0 \leqslant x \in A$ and $x \notin [-u, u] + \frac{\varepsilon}{2} B_\rho$. It follows from $x = x \wedge u + (x - u)^+$ and $x \wedge u \in [-u, u]$ that $(x - u)^+ \notin \frac{\varepsilon}{2} B_\rho$, hence $\rho\bigl((x - u)^+\bigr) > \frac{\varepsilon}{2}$.

To see that the converse is false, take $X = \mathbb{R}^2$, A the line $y = -x$, and ρ the distance from a point to A.

11.1.10(i) Suppose that A is weakly confined. Let $f \in X_+^*$ and (x_n) a disjoint sequence in $\operatorname{Sol}(A)$. Then $\rho_f(x_n) = f\bigl(|x_n|\bigr) = f(x_n^+) + f(x_n^-) \to 0$ because (x_n^+) and (x_n^-) are disjoint sequences in $\operatorname{Sol}(A)$. For the converse, suppose that A is ρ_f-confined for every $f \in X_+^*$. Let (x_n) be a disjoint sequence in $\operatorname{Sol}(A)$ and $f \in X^*$. Then

$$f(x_n) = f^+(x_n^+) - f^+(x_n^-) - f^-(x_n^+) + f^-(x_n^-) = \rho_{f^+}(x_n^+) - \rho_{f^+}(x_n^-) - \rho_{f^-}(x_n^+) + \rho_{f^-}(x_n^-) \to 0.$$

11.1.11 Use Exercise 1.5.8.

11.1.12 Since ρ is a lattice seminorm, if every positive disjoint sequence in X is ρ-null then every disjoint sequence (x_n) satisfies $\rho(x_n) = \rho(|x_n|) \to 0$.

Suppose that every positive disjoint sequence in A is weakly null. Let (x_n) be a disjoint sequence in A and $f \in X_+^*$. It follows from $|x_n| \xrightarrow{\mathrm{w}} 0$ that $|f(x_n)| \leqslant f(|x_n|) \to 0$, which yields $x_n \xrightarrow{\mathrm{w}} 0$.

The proof for weak*-confinement is similar.

11.1.13(i) WLOG, A is solid. It is clear that if A is $\rho_{\mathrm{Sol}(B)}$-confined then then A is ρ_B-confined. Suppose that A is ρ_B-confined but not $\rho_{\mathrm{Sol}(B)}$-confined. Then there exists a disjoint sequence (x_n) is A and $\varepsilon > 0$ such that $\rho_{\mathrm{Sol}(B)}(x_n) > \varepsilon$ for all n. WLOG, replacing x_n with $|x_n|$, we may assume that $x_n \geqslant 0$. For every n, find $g_n \in \mathrm{Sol}(B)$ such that $|g_n(x_n)| > \varepsilon$. Find $f_n \in B$ such that $|g_n| \leqslant |f_n|$. It follows that $|f_n|(x_n) \geqslant |g_n|(x_n) \geqslant |g_n(x_n)| > \varepsilon$. By Riesz-Kantorovich formula, there exists $y_n \in [-x_n, x_n]$ (and, therefore, $y_n \in A$) such that $|f_n(y_n)| > \varepsilon$. But A being ρ_B-confined implies $f_n(y_n) \to 0$; a contradiction.

11.1.13(ii) The proof is similar.

11.2.3 In the last paragraph, we now argue that $0 \leqslant w_n - v_n \leqslant \frac{1}{2^n} e$ implies $\|w_n - v_n\| \leqslant \frac{1}{2^n}\|e\|$, hence $\|w_n - v_n\| \to 0$ and, therefore, $\rho(w_n - v_n) \to 0$.

11.2.5 (i)⇒(ii) is trivial. Let A be a ρ-almost order bounded set and (x_n) a disjoint sequence in A. Fix $\varepsilon > 0$. Find $u \in X_+$ such that $\rho\Big[\big(|x| - u\big)^+\Big] < \varepsilon$ for all $x \in A$. The sequence $\big(|x_n| \wedge u\big)$ is disjoint and order bounded, hence ρ-null. It follows that

$$\rho(x_n) = \rho\big(|x_n|\big) = \rho\big(|x_n| \wedge u\big) + \rho\Big[\big(|x_n| - u\big)^+\Big] < 2\varepsilon$$

for all sufficiently large n.

11.2.8 WLOG, A is solid. Suppose A is confined in X. Then for every $x \in A$ we have $\big[0, |x|\big] \subseteq A$, hence $\big[0, |x|\big]$ is confined. It follows that $x \in X_{\mathrm{oc}}$ and, therefore, $A \subseteq X_{\mathrm{oc}}$. Being confined in X_{oc}, it is almost order bounded there by Corollary 11.2.4. The converse implication follows directly from Proposition 11.2.6.

11.3.7 By Burkinshaw-Dodds Theorem, $\mathrm{Sol}(A)$ is confined, that is, $\rho_{B_{X^*}}$-confined, iff B_{X^*} is $\rho_{\mathrm{Sol}(A)}$-confined. Now use Exercise 11.1.13.

11.4.2 If B_X were almost order bounded then we could find $u > 0$ such that $\|(e_n - u)^+\| < \frac{1}{2}$ for every n. It is easy to see that $e_n \wedge u \to 0$. It follows from $e_n = e_n \wedge u + (e_n - u)^+$ that $\|e_n\| < 1$ for all sufficiently large n; a contradiction.

11.4.6 Since the norm on X agrees with $\rho_{B_{X^*}}$, the result follows from Burkinshaw-Dodds Theorem 11.3.6.

11.5.3 Suppose that A is solid and relatively compact; let (x_n) be a disjoint sequence in A. Every subsequence of (x_n) has a further subsequence which is norm convergent and, therefore, norm null by Exercise 1.5.8. Hence (x_n) is norm null.

11.6.2 Suppose that X is order continuous and A is relatively compact; in particular, A is bounded. By Exercise 9.8.1, A is almost order bounded. It now follows from Proposition 11.2.6 that A is confined.

To prove the converse, note that for every $x \in X_+$, the singleton $\{x\}$ is compact, hence its solid hull $[-x, x]$ is confined. By Meyer-Nieberg's Theorem 10.1.2, X is order continuous.

11.7.1 Suppose X fails to be a KB-space. By Theorem 10.4.4, there is a positive disjoint sequence (x_n) in X which is equivalent to the unit vector basis of c_0. Then $x_n \xrightarrow{\mathrm{w}} 0$ (because this holds true in c_0), but (x_n) does not converge to zero in norm. Therefore, X fails the PSP.

11.7.4(i) Every disjoint seminormalized sequence in $L_1(\mu)$ is equivalent to the unit vector basis of ℓ_1.

11.7.4(ii) No reflexive space contains a copy of ℓ_1.

11.8.1 Let $Y = \big(I_e, \|\cdot\|_e\big)$. By $C(K)$ Representation Theorem, we identify (up to a lattice isometry) Y with $C(K)$ for some K. Since $[0,e]$ is norm bounded in X, the restriction of T to Y is a bounded operator; denote it by S. Then $T[0,e]$ is relatively weakly compact iff S is weakly compact iff S^* is weakly compact iff $S^*(B_X)$ is relatively weakly compact in Y iff (by Grothendieck Theorem) every disjoint sequence in $[0,e]$ converges uniformly to zero on $S^*(B_X)$ iff $Sx_n \to 0$ for every disjoint sequence in $[0,e]$.

11.8.9 Being rank-one, T is compact. Observe that $\mathrm{Sol}(TB_{\ell_1}) = [-\mathbb{1},\mathbb{1}] = B_{\ell_\infty}$, which is not confined. On the other hand, for $x \in \ell_1$, we have $r_T(x) = \|x\|_{\ell_1}$, and B_{ℓ_1} is not confined, hence not r_T-confined.

12.1.3 Suppose that $x \in X$ and $u \in X_+$ satisfy $nx \leqslant u$ for every $n \in \mathbb{N}$. It follows that $\frac{u}{n} - x \geqslant 0$. Passing to the limit, we get $-x \geqslant 0$.

13.1.1 Let $T\colon X \to Y$ be an operator between ordered vector spaces. Suppose T is order bounded and consider an order interval $[a,b]$ in X. Since $b-a \geqslant 0$ and T is order bounded, there exists $c \in Y_+$ such that $T\big[-(b-a), b-a\big] \subseteq [-c,c]$. For every $x \in [a,b]$ we have

$$Tx = Ta + T(x-a) \in Ta + [-c,c] = [Ta-c, Ta+c].$$

Conversely, suppose that T maps order intervals into order intervals. Let $a \in X_+$. Then $T[-a,a] \subseteq [b,c]$ for some $b,c \in Y$. It follows from $0 \in [-a,a]$ that $b \leqslant 0 \leqslant c$. Put $d = c-b$. Then $d \geqslant 0$ and $[b,c] \subseteq [-d,d]$.

13.2.1 Let $R \geqslant 0$ with $-R \leqslant T \leqslant R$. By Exercise 6.1.1, for every $x \in X$ we have $|Tx| \leqslant R|x|$, so that $\|Tx\| \leqslant \|R\|\|x\|$. It follows that $\|T\| \leqslant \|R\|$. Taking the infimum over all such R, we conclude that $\|T\| \leqslant \|T\|_r$.

13.2.3 Fix $\varepsilon > 0$. There exists $0 \leqslant R\colon X \to Y$ such that $-R \leqslant T \leqslant R$ and $\|T\|_r + \varepsilon \geqslant \|R\|$. Then $-R_{|Z} \leqslant T_{|Z} \leqslant R_{|Z}$, so that $T_{|Z}$ is regular and $\big\|T_{|Z}\big\|_r \leqslant \|R_{|Z}\| \leqslant \|R\| \leqslant \|T\|_r + \varepsilon$. Since $\varepsilon > 0$ is arbitrary, it follows that $\big\|T_{|Z}\big\|_r \leqslant \|T\|_r$.

13.2.9 Suppose that T is regular. Fix $\varepsilon > 0$ and find $0 \leqslant R\colon X \to Y$ such that $-R \leqslant T \leqslant R$ and $\|T\|_r \geqslant \|R\| - \varepsilon$. Then $-R^* \leqslant T^* \leqslant R^*$, so that T^* is regular and $\|T^*\|_r \leqslant \|R^*\| = \|R\| \leqslant \|T\|_r + \varepsilon$.

13.2.12 Fix $\varepsilon > 0$. Find $0 \leqslant R \in L(X,Y)$ such that $\pm T \leqslant R$ and $\|R\| \leqslant \|T\|_r + \varepsilon$. It follows from $\pm jT \leqslant jR$ that $\|jT\|_r \leqslant \|jR\| = \|R\| \leqslant \|T\|_r + \varepsilon$. We conclude that $\|jT\|_r \leqslant \|T\|_r$.

Suppose that $|T|$ exists. It follows from $\pm T \leqslant |T|$ that $\pm jT \leqslant j|T|$ and, therefore, $|jT| \leqslant j|T|$.

13.4.2 Similar to the proof of Theorem 6.1.9.

14.1.7 It is clear that $S \geqslant 0$. Since $\pm Tx \leqslant Sx$ for all $x \in X_+$, we have $\pm T \leqslant S$. Furthermore, S is the least operator with this property: if $R \geqslant \pm T$ then for all $x \in X_+$ we have $\pm Tx \leqslant Rx$ and, therefore, $Sx = |Tx| \leqslant Rx$, so that $S \leqslant R$. It follows that $S = T \vee (-T) = |T|$.

14.1.8 For $x \in X_+$, define $\tau(x) = |Tx|$. Then τ is additive. It follows from Extension Lemma 6.2.1 that τ extends to an operator S on all of X. It follows that $S = |T|$ by Exercise 14.1.7.

14.1.10 Let $T \in \mathcal{Z}(X)$. WLOG, $\pm T \leqslant I$. View X as a linear subspace of $C[0,1]$ and T as an operator from X to $C[0,1]$. It follows from $|Tx| \leqslant |x|$ for every $x \in X$ that T is bounded and, therefore, extends to an operator $\widetilde{T}\colon C[0,1] \to C[0,1]$, and the extension still satisfies $\pm\widetilde{T} \leqslant I$, hence is central. By Proposition 14.1.3, $\widetilde{T} = M_f$ for some $f \in C[0,1]$. Note that $f = \widetilde{T}\mathbb{1} = T\mathbb{1}$, hence $f \in X$. Also, $f^2 = Tf$ has to be in X. It follows that f is constant, so that T is a scalar multiple of the identity.

14.1.11 The forward implication is trivial as $\pm T \leqslant \lambda I$ implies $\pm T^\sim \leqslant \lambda I^\sim$. Suppose that $X^\sim$ separates the points of X and $T^\sim$ is central. Then $\pm T^\sim \leqslant \lambda I^\sim$ for some $\lambda \geqslant 0$. For every $x \in X_+$ and $f \in X_+^\sim$, we have $\pm T^\sim f \leqslant \lambda f$, so that $\pm(T^\sim f)(x) \leqslant \lambda f(x)$ and, therefore, $f(\pm Tx) \leqslant f(\lambda x)$. Since this is true for every $f \in X_+^\sim$, we conclude that $\pm Tx \leqslant \lambda x$ and, therefore, T is central.

14.1.12 It is clear that if $f \in L_\infty(\mu)$ then M_f is a central operator on X. Conversely, suppose that $T \in \mathcal{Z}(X)$. By Exercise 14.1.1, we may assume WLOG that $0 \leqslant T \leqslant I$. Let $f = T\mathbb{1}$. Then $0 \leqslant f \leqslant \mathbb{1}$, hence $f \in L_\infty(\mu)$. We will show that $T = M_f$.

Fix a measurable set A. It follows from $0 \leqslant T\mathbb{1}_A \leqslant \mathbb{1}_A$ that $T\mathbb{1}_A$ vanishes on A^C. On the other hand, it follows from $0 \leqslant T\mathbb{1} - T\mathbb{1}_A = T\mathbb{1}_{A^C} \leqslant \mathbb{1}_{A^C}$ that $T\mathbb{1} - T\mathbb{1}_A$ vanishes on A and, therefore, $T\mathbb{1}_A$ agrees with $T\mathbb{1} = f$ on A. We conclude that $T\mathbb{1}_A = M_f\mathbb{1}_A$. It follows that T and M_f agree on simple functions. Since simple functions are dense in $L_p(\mu)$ and the two operators are bounded, we have $T = M_f$.

It is straightforward that the map $f \mapsto M_f$ is a lattice homomorphism

14.1.14 The argument is somewhat similar to that in our solution of Exercise 14.1.10. On one hand, if $f \in X$ then the multiplication operator $g \mapsto fg$ is clearly a central operator on X. Conversely, if T is a central operator on X, T is bounded with respect to the supremum norm, hence extends to an bounded operator $\widetilde{T}$ on $C[0,1]$. It is clear that $\widetilde{T}$ is still central. Hence, $\widetilde{T} = M_f$ where $f = \widetilde{T}\mathbb{1} = T\mathbb{1}$, hence $f \in X$. This yields a lattice isomorphism between X and $\mathcal{Z}(X)$.

14.1.16 Let $X = C(K)$. By Proposition 14.1.3, $\mathcal{Z}(X) \simeq C(K)$. It is easy to see that the map is an isometry.

14.1.22 As in the proof of Proposition 14.1.18, we may identify I_e with a $C(K)$ space, with e corresponding to $\mathbb{1}$ and v corresponding to some $f \in [0,\mathbb{1}]$. Let T be the central operator on I_e corresponding to M_f. It follows from $f = M_f\mathbb{1}$ that $Te = v$. It follows from $0 \leqslant f \leqslant \mathbb{1}$ that $0 \leqslant T \leqslant I$ on I_e. The last inequality remains valid for the extension of T to X.

14.1.24 Let T be central, that is, $\pm T \leqslant \lambda I$ for some $\lambda > 0$. Let J be an ideal and $x \in J_+$. Then $\pm Tx \leqslant \lambda x$, so that $|Tx| \leqslant \lambda x$ and, therefore, $Tx \in J$. It follows that $T(J) \subseteq J$.

14.2.1 (i)$\Rightarrow$(ii) If $x \perp y$ then $y \perp B_x$, but $Tx \in T(B_x) \subseteq B_x$, hence $y \perp Tx$.

(ii)$\Rightarrow$(iii) For each $y \in B_x^d$, we have $y \perp x$, so that $y \perp Tx$. It follows that $Tx \in B_x^{dd} = B_x$.

(iii)$\Rightarrow$(i) Let B be a band and $x \in B$; then $Tx \in B_x \subseteq B$.

14.2.2 If $x \wedge y = 0$ then $Tx \wedge y = 0$ and, further, $Tx \wedge Ty = 0$. Now apply Exercise 1.6.10.

14.2.3 Indeed, every central operator is ideal preserving by Exercise 14.1.24. Every orthomorphism is band preserving by definition. Every ideal preserving operator is clearly band preserving; it is order bounded by Corollary 14.1.27; it follows that it is an orthomorphism.

14.2.6(ii) Being order closed, B is uniformly closed, hence X/B is an Archimedean vector lattice. $\widetilde{T}$ is well defined because B is invariant under T. To verify that $\widetilde{T}$ is band preserving, suppose that $a \wedge b = 0$ in X/B. By Exercise 3.4.9, there exist $0 \leqslant x \in a$ and $0 \leqslant y \in b$ with $x \perp y$. It follows that $Tx \perp y$ and, therefore, $\widetilde{T}a \perp b$.

14.2.7 There exists $w \in C^\infty(K)$ such that $Tf = wf$ for every $f \in X$. It follows that if $f \perp g$ for some $f, g \in X_+$ then $Tf \perp g$, hence T is band preserving. To show that T is order bounded, fix $f \in X_+$ and let $g \in X$ with $|g| \leqslant f$. Then $|Tg| = |wg| \leqslant |wf| = |Tf|$.

14.2.9 WLOG, f is bounded on U; otherwise, replace U with an open subset on which f is bounded. Say, $f \leqslant M$ on U. Since $\overline{U}$ is clopen, $\mathbb{1}_{\overline{U}} \in C^\infty(K)$. Since X is order dense, there exists $v \in X_+$ such that $0 < v \leqslant \mathbb{1}_{\overline{U}}$. Since $v > 0$, we can find $\delta > 0$ and an open subset V of U such that $v > \delta$ on V. It follows that on V we have $\frac{M}{\delta}v > M \geqslant f$. Take $g = \left(\frac{M}{\delta}v\right) \wedge f$; it is easy to see that g satisfies the requirements.

14.2.11 Let $t \in U \in \mathcal{U}$. Define $w(t) = w_U(t)$. By assumption, $w(t)$ is well-defined. This yields a continuous $\overline{\mathbb{R}}$-valued function on $\bigcup\mathcal{U}$. For each $U \in \mathcal{U}$, the set $\{w_U \text{ is finite}\}$ is open and dense in U; it follows that the set $\{w \text{ is finite}\}$ is open and dense in $\bigcup\mathcal{U}$ and, therefore, in K. By Lemma 7.4.8, w extends to a function in $C^\infty(K)$.

14.2.12 The closed set $\{e = 0\}$ has empty interior because if U is a non-empty open subset of $\{e \neq 0\}$ then $\overline{U}$ is clopen, $\mathbb{1}_{\overline{U}} \in C^\infty(K)$, and $e \perp \mathbb{1}_{\overline{U}}$. Hence the function $1/e$ is continuous and finite except on a nowhere dense set, so we may extend it to an element of $C^\infty(K)$ by Lemma 7.4.8. Note that the set $\{1/e = 0\} = \{e = \pm\infty\}$ is nowhere dense, hence $1/e$ is a weak unit. It is clear that M_e is a lattice homomorphism and $M_e^{-1} = M_{1/e}$.

14.2.13 The set $\left(\overline{U} \cap \overline{V}\right) \setminus \overline{U \cap V}$ is open and contained in $\partial U \cup \partial V$, hence nowhere dense. Therefore, this set is empty.

14.2.23 Find a $C^\infty(K)$ representation for X^u such that the weak unit becomes $\mathbb{1}$. Let w be as in Theorem 14.2.14. Then $w = w\mathbb{1} = T\mathbb{1} = 0$, which implies that $T = 0$.

14.2.25 If $x \in (\operatorname{Range} T)^d$ then $x \perp Tx$ and, therefore, $Tx \perp Tx$, so that $Tx = 0$, hence $x \in \ker T$. To prove the converse, we represent X^u as $C^\infty(K)$ and T as M_w for some $w \in C^\infty(K)_+$. If $x \in \ker T$ then $wx = 0$, so that the support of x is disjoint from that of w and, therefore, from that of $wy = Ty$ for every $y \in X$, hence $x \in (\operatorname{Range} T)^d$.

14.2.26 (i) $\Rightarrow$ (ii) trivially. (ii) $\Rightarrow$ (iii) $\operatorname{Range} T = X$, hence is a band. By the preceding exercise, $\ker T = (\operatorname{Range} T)^d = \{0\}$. (iii) $\Rightarrow$ (i) We only need to prove that T is surjective. Note that $\operatorname{Range} T = (\operatorname{Range} T)^{dd} = (\ker T)^d = \{0\}^d = X$.

14.2.28 Suppose that T is injective and $S \perp T$ for some $S \in \operatorname{Orth}(X)$. Then $TS = 0$ by Corollary 14.2.19, hence $S = 0$.

Suppose now that X is order complete and T is a weak unit in $\operatorname{Orth}(X)$. Then $|T|$ is a weak unit as well, so we have $n|T| \wedge I \uparrow I$ by Proposition 5.1.3 and then $n|T|x \wedge x \uparrow x$ for every $x \geqslant 0$ by Theorem 6.3.2. If $x \in X$ and $Tx = 0$ then $|T||x| = 0$, hence $n|T||x| \wedge |x| \uparrow |x|$ implies $x = 0$.

14.2.29 Suppose that $T \in \operatorname{Orth}(X)$ and T^{-1} exists. Represent X^u as $C^\infty(K)$ as before; let w be as in Theorem 14.2.14. We claim that the set $\{w = 0\}$ is nowhere dense. Indeed, otherwise, one could find a clopen subset U of it; by order density of X in $C^\infty(K)$ we can find a non-zero $f \in X_+$ with $f \leqslant \mathbb{1}_U$ and, therefore, $Tf = wf = 0$; a contradiction.

It follows that $1/w$ is again in $C^\infty(K)$. For $f \in X$, we have $(1/w)Tf = (1/w)wf = f$, hence T^{-1} agrees with $M_{1/w}$ on X. It follows that T^{-1} is an orthomorphism by Theorem 14.2.14.

14.2.36 Since B_e is a projection band in X, replacing X with B_e we may assume WLOG that e is a weak unit in X. Represent X^u as $C^\infty(K)$ with e corresponding to $\mathbb{1}$. Since X is order complete, it is an ideal in $C^\infty(K)$.

Consider the operator M_w on $C^\infty(K)$. For every $x \in X_+$ we have $0 \leqslant M_w x = wx \leqslant x$ as $w \leqslant \mathbb{1}$. It follows that M_w leaves X invariant and $0 \leqslant T \leqslant I$, where T is the restriction of M_w to X. Clearly, $Te = w\mathbb{1} = w$.

14.2.37 The answer is "no". Let $X = L_1[0,1]$ and $e = \mathbb{1}$. Then $B_e = X$. Let $w \in B_e$ be any unbounded function. Let $T \in \mathrm{Orth}(X)$. By Theorem 14.2.33, T is central. By Example 14.1.20, $T = M_v$ for some $v \in L_\infty[0,1]$. Hence $Te = v \in L_\infty[0,1]$. Thus, $Te = w$ is impossible.

14.2.38 Suppose that T is an orthomorphism. Let $x \wedge y$. By Exercise 1.6.10, it suffices to show that $(x+Tx) \wedge (y+Ty) = 0$. Indeed, we have $Tx \wedge y = 0$, so that $(x+Tx) \wedge y = 0$. Furthermore, $(x+Tx) \wedge Ty = 0$, so that $(x+Tx) \wedge (y+Ty) = 0$.

Conversely, suppose that $I+T$ is a lattice homomorphism and $x \perp y$. Applying $I+T$ to $|x| \wedge |y| = 0$ we get $\big(|x| + T|x|\big) \wedge \big(|y| + T|y|\big) = 0$ and, therefore, $0 = |x| \wedge T|y| = |x| \wedge |Ty|$, hence $x \perp Ty$.

14.3.2 Suppose that $R, S, T \in \mathrm{Orth}(X)_+$ with $S \perp T$. Then for every $x \in X_+$, Theorem 14.2.16 yields $Sx \perp Tx$. It follows that $(RSx) \perp Tx$ and, therefore, $RS \perp T$. Since orthomorphisms commute, we also have $SR \perp T$.

14.3.3 Clearly, L_v is linear. If follows from the definition of f-algebra that L_v is band preserving. It follows from $L_v = L_{v^+} - L_{v^-}$ and the fact that X is a vector lattice algebra that L_v is regular, hence order bounded.

14.3.6 Fix $v \in X$ and define $S, T\colon X \to X$ via $Sx = (xv)v$ and $Tx = x(vv)$. As in the proof of Theorem 14.3.5, S and T agree on B_v and on B_v^d, hence $S = T$. So $(xv)v = (xv)v$ for all $x, v \in X$.

Now also fix $w \in X$ and define $R, V\colon X \to X$ via $Rx = (wx)v$ and $Vx = w(xv)$. Again, since $Rv = Vv$ and R and V both vanish on $\{v\}^d$, we conclude that $R = V$ and, therefore, $(wx)v = w(xv)$.

14.3.7 We have $e = e^2 \geqslant 0$ by Proposition 14.3.4(iii). If $x \perp e$ then Proposition 14.3.4(i) yields $x = xe = 0$.

14.4.6 (i)⇒(ii) and (i)⇒(ii) by Theorem 14.4.4. To show that (ii)⇒(i), note that $S = |T|$ by Exercise 14.1.7 and apply Theorem 14.4.4(iii). Finally, let R be as in (iii). By Exercise 14.1.7, $R = |T|$. If $|x| \leqslant |y|$ then $R|x| \leqslant R|y|$, so that $|Tx| \leqslant |Ty|$. Now apply Theorem 14.4.4(ii).

14.4.7 No. Every positive operator T satisfies the condition.

14.4.8 The forward implication follows from Theorem 14.4.4(ii). For the converse, it follows from $\big||x|\big| = |x|$ that $|Tx| = \big|T|x|\big|$; now use Theorem 14.4.2.

14.4.9(i) Suppose that $x, y \geqslant 0$. Then

$$0 \leqslant T^+x \leqslant T^+(x \vee y) = \big(T(x \vee y)\big)^+ \text{ and } 0 \leqslant T^-y \leqslant T^-(x \vee y) = \big(T(x \vee y)\big)^-.$$

It follows that $T^+x \wedge T^-y = 0$. Therefore, for any $x, y \in X$, we have $\big|T^+x\big| \wedge \big|T^-y\big| = T^+|x| \wedge T^-|y| = 0$.

14.4.9(ii) Let $y \in Y_+$. Then $y = Tx$ for some $x \in X$. It follows that $y = |Tx| = |T||x|$, hence $y \in \mathrm{Range}|T|$. Therefore, $\mathrm{Range}|T| = Y$.

14.4.9(iii) $\ker T = \ker|T|$ follows from $\big||T|x\big| = |T||x| = |Tx|$ for every $x \in X$. The fact that it is an ideal follows from Theorem 14.4.4 or from the fact that $|T|$ is a lattice homomorphism.

14.4.9(iv)

$$Tx \in J \quad\Leftrightarrow\quad |Tx| \in J \quad\Leftrightarrow\quad |T||x| \in J \quad\Leftrightarrow\quad \big||T|x\big| \in J \quad\Leftrightarrow\quad |T|x \in J.$$

14.4.15 Let $a, b \in A_T$. Then $T\big[-|a|, |a|\big] \subseteq [-u, u]$ and $T\big[-|b|, |b|\big] \subseteq [-v, v]$ for some $u, v \geqslant 0$. Suppose that $|x| \leqslant |a+b|$. Then $|x| \leqslant |a| + |b|$. By Riesz Decomposition Theorem 1.5.19, we can write $x = y + z$, where $|y| \leqslant |a|$ and $|z| \leqslant |b|$. It follows that $|Tx| \leqslant |Ty| + |Tz| \leqslant u + v$, hence $a + b \in A_T$.

It is easy to see that $\lambda a \in A_T$ for every $\lambda \in \mathbb{R}$. It is also clear from the definition that if $a \in A_T$ and $|b| \leqslant |a|$ then $b \in A_T$; hence A_T is an ideal.

Let $a > 0$ be an atom. Then $[-a, a] = \big\{\lambda a : -1 \leqslant \lambda \leqslant 1\big\}$ and, therefore, $T[-a, a] \subseteq \big[-|Ta|, |Ta|\big]$, so that $a \in A_T$.

14.4.34 We represent $\mathbb{R}^u$ as $C^\infty\bigl(\{q\}\bigr)$. Clearly, φ satisfies the assumptions of Theorem 14.4.31. Let e, w, and φ be as the theorem. Let $t_0=\varphi(q)$ and $\lambda=w(q)$. Then φ has the required form.

15.1.2 Suppose that $x_\alpha\xrightarrow{\text{o}}x$ and $u\geqslant 0$. It follows from $0\leqslant|x_\alpha-x|\wedge u\leqslant|x_\alpha-x|\xrightarrow{\text{o}}0$ that $x_\alpha\xrightarrow{\text{uo}}x$. Suppose now that $x_\alpha\xrightarrow{\text{uo}}x$ and (x_α) is order bounded. Then the net $(x_\alpha-x)$ is order bounded, so there exists $u\geqslant 0$ such that $|x_\alpha-x|\leqslant u$ for all α. It follows that $|x_\alpha-x|=|x_\alpha-x|\wedge u\xrightarrow{\text{o}}0$, hence $x_\alpha\xrightarrow{\text{uo}}x$.

15.1.3 Suppose that $x_\alpha\xrightarrow{\text{uo}}x$ and $x_\alpha\xrightarrow{\text{uo}}y$. Let $u\in X_+$. It follows from

$$|x-y|\leqslant|x_\alpha-x|+|x_\alpha-y|\qquad\text{that}\qquad|x-y|\wedge u\leqslant|x_\alpha-x|\wedge u+|x_\alpha-y|\wedge u\xrightarrow{\text{o}}0.$$

Therefore, $|x-y|\wedge u=0$. Taking $u=|x-y|$, we get $|x-y|=0$, hence $x=y$.

15.1.4(i) The case $\lambda=0$ is trivial, so assume that $\lambda\neq 0$. For every $u\in X_+$ we have

$$|\lambda x_\alpha-\lambda x|\wedge u=|\lambda|\Bigl[|x-x_\alpha|\wedge(|\lambda|^{-1}u)\Bigr]\xrightarrow{\text{o}}0,\qquad\text{and}$$

$$\Bigl|(x_\alpha+y_\alpha)-(x+y)\Bigr|\wedge u\leqslant|x_\alpha-x|\wedge u+|y_\alpha-y|\wedge u\xrightarrow{\text{o}}0.$$

15.1.4(ii) For every $u\in X_+$, we have $\Bigl||x_\alpha|-|x|\Bigr|\wedge u\leqslant|x_\alpha-x|\wedge u\xrightarrow{\text{o}}0$.

15.1.4(iii) Since lattice operations are uo-continuous, we have $x_\alpha=x_\alpha^+\xrightarrow{\text{uo}}x^+$, so that $x=x^+$ and, therefore, $x\geqslant 0$.

15.1.8 Suppose that $f_n\xrightarrow{\text{a.e.}}0$ and $u\geqslant 0$ in $L_p(\mu)$. The sequence $\bigl(|f_n|\wedge u\bigr)$ is order bounded and converges to zero a.e., hence in order by Proposition 2.4.27. Conversely, suppose that $f_n\xrightarrow{\text{uo}}0$ in $L_p(\mu)$. Fix any measurable set A of finite measure, put $u=\mathbb{1}_A$. Then $|f_n|\wedge u\xrightarrow{\text{o}}0$. By Proposition 2.4.27, $|f_n|\wedge u\xrightarrow{\text{a.e.}}0$. Thus, (f_n) converges to zero a.e. on A. Since A was arbitrary, we have $f_n\xrightarrow{\text{a.e.}}0$ on the entire space.

15.1.9 Given a net (x_α) in X, we write $x_{\alpha,i}$ for the i-th coordinate of x_α. Suppose that $x_\alpha\xrightarrow{\text{uo}}0$. Fix i, and let e_i be the i-th unit vector. Then $|x_\alpha|\wedge e_i\xrightarrow{\text{o}}0$ in α. This implies that $x_{\alpha,i}\to 0$ as $\alpha\to\infty$. Conversely, suppose that (x_α) converges to zero coordinate-wise and let $u\in X_+$; $u=(u_i)$. For every m, let $a_m\in X$ be defined as follows: $a_{m,i}=\frac{1}{m}\wedge u_i$ if $i\leqslant m$ and $a_{m,i}=u_i$ otherwise. It is easy to see that $a_m\downarrow 0$. Since (x_α) converges to zero coordinate-wise, for every m there exists α_0 such that $|x_{\alpha,i}|<\frac{1}{m}$ for all $\alpha\geqslant\alpha_0$ and $i=1,\dots,m$. It follows that $|x_\alpha|\wedge u\leqslant a_m$ for all $\alpha\geqslant\alpha_0$. Therefore, $|x_\alpha|\wedge u\xrightarrow{\text{o}}0$.

15.4.1 WLOG A is solid; otherwise, replace A with $\operatorname{Sol}A$. Fix $f\in X_+^*$ and $\varepsilon>0$. There exists $u\in X_+$ such that $A\subseteq[-u,u]+\varepsilon B_{\rho_f}$. Then

$$f\bigl(|x_\alpha|\bigr)\leqslant f\bigl(|x_\alpha|\wedge u\bigr)+f\bigl((|x_\alpha|-u)^+\bigr).$$

It follows from $x_\alpha\xrightarrow{\text{uaw}}0$ that $f\bigl(|x_\alpha|\wedge u\bigr)<\varepsilon$ for all sufficiently large α. It follows from $(|x_\alpha|-u)^+\in\varepsilon B_{\rho_f}$ that $f\bigl((|x_\alpha|-u)^+\bigr)<\varepsilon$.

16.1.2 Choose $e\in X_+$ such that $x_1,\dots,x_n,y_1,\dots,y_n\in I_e$. We may identify I_e with $C(K)$ for some compact space K. Then $\bigl|x_i(t)\bigr|\leqslant\bigl|y_i(t)\bigr|$ for every $i=1,\dots,n$ and every $t\in K$, so that $\Bigl(\sum_{i=1}^n|x_i(t)|^p\Bigr)^{\frac1p}\leqslant\Bigl(\sum_{i=1}^n|y_i(t)|^p\Bigr)^{\frac1p}$. It follows that $\Bigl(\sum_{i=1}^n|x_i|^p\Bigr)^{\frac1p}\leqslant\Bigl(\sum_{i=1}^n|y_i|^p\Bigr)^{\frac1p}$ in $C(K)$ and, therefore, in X.

Alternatively, for any $x_1,\dots,x_n,y_1,\dots,y_n\in X$, function calculus yields

$$\Bigl(\sum_{i=1}^n\bigl(|x_i|\wedge|y_i|\bigr)^p\Bigr)^{\frac1p}\leqslant\Bigl(\sum_{i=1}^n|y_i|^p\Bigr)^{\frac1p}.$$

In the case when $|x_i|\leqslant|y_i|$ as $i=1,\dots,n$, the required inequality follows immediately.

16.1.9 Since $|Tx_i| \leqslant |T||x_i| = \big||T||x_i|\big|$ for every i, Exercise 16.1.2 and Proposition 16.1.8(ii) yield

$$\Big(\sum_{i=1}^{n}|Tx_i|^p\Big)^{\frac{1}{p}} \leqslant \Big(\sum_{i=1}^{n}\big||T||x_i|\big|^p\Big)^{\frac{1}{p}} \leqslant |T|\Big(\sum_{i=1}^{n}|x_i|^p\Big)^{\frac{1}{p}}.$$

16.1.10 We will use Proposition 16.1.3 and Hölder's Inequality. Let $(a_i) \in B_{\ell_q^n}$. For every $f \in X_+^\sim$ we have

$$\begin{aligned} f\Big(\sum_{i=1}^{n} a_i x_i\Big) &\leqslant \Big|f\Big(\sum_{i=1}^{n} a_i x_i\Big)\Big| \leqslant \sum_{i=1}^{n}|a_i|\big|f(x_i)\big| \\ &\leqslant \Big(\sum_{i=1}^{n}|a_i|^q\Big)^{\frac{1}{q}}\Big(\sum_{i=1}^{n}\big|f(x_i)\big|^p\Big)^{\frac{1}{p}} \leqslant \Big(\sum_{j=1}^{m}\big|f(y_j)\big|^p\Big)^{\frac{1}{p}} \leqslant f\bigg(\Big(\sum_{j=1}^{m}|y_j|^p\Big)^{\frac{1}{p}}\bigg), \end{aligned}$$

where the last inequality follows from Proposition 16.1.8. This yields $\sum_{i=1}^n a_i x_i \leqslant \Big(\sum_{j=1}^m |y_j|^p\Big)^{\frac{1}{p}}$. The conclusion now follows from Proposition 16.1.3.

16.3.2 It is clear that this condition is implied by (iii). To prove the converse, let $x_1, \dots, x_n \in E$, and $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. For each $k = 1, \dots, n$ let $\lambda_k = \frac{\alpha_k}{\max|\alpha_i|}$. Using positive homogenuity of $\|\cdot\|_n$, and applying the condition n times, we get

$$\begin{aligned} \big\|(\alpha_1 x_1, \dots, \alpha_{n-1}x_{n-1}, \alpha_n x_n)\big\|_n &= \max|\alpha_i|\big\|(\lambda_1 x_1, \dots, \lambda_{n-1}x_{n-1}, \lambda_n x_n)\big\|_n \\ &\leqslant \max|\alpha_i|\big\|(\lambda_1 x_1, \dots, \lambda_{n-1}x_{n-1}, x_n)\big\|_n \leqslant \dots \leqslant \max|\alpha_i|\big\|(x_1, \dots, x_n)\big\|. \end{aligned}$$

Note that we can "remove" $\lambda_{n-1}, \dots, \lambda_1$ using the assumption and condition (i).

16.3.9 Suppose that $R\colon X \to Y$ be a positive operator satisfying $-R \leqslant T \leqslant R$. Then $|Tx_i| \leqslant R|x_i| = \big|R|x_i|\big|$ for every i, hence

$$\Big(\sum_{i=1}^{n}|Tx_i|^p\Big)^{\frac{1}{p}} \leqslant \Big(\sum_{i=1}^{n}\big|R|x_i|\big|^p\Big)^{\frac{1}{p}} \leqslant R\Big(\sum_{i=1}^{n}|x_i|^p\Big)^{\frac{1}{p}}$$

by Proposition 16.1.8(ii). It follows that $\|\overline{T}\bar{x}\|_{\text{lat-}p} \leqslant \|R\|\|\bar{x}\|_{\text{lat-}p}$. Taking infimum over all such R, we get the required inequality.

16.4.1 Suppose T is (p,q)-convex for some $p > q$. Take any non-zero vector $x \in E$, and let $x_i = x$ as $i = 1, \dots, n$. Then the definition of (p,q)-convexity yields $n^{\frac{1}{q}}\|Tx\| \leqslant Kn^{\frac{1}{p}}\|x\|$, so that $\|Tx\| \leqslant Kn^{\frac{1}{p}-\frac{1}{q}}\|x\|$. Passing to the limit on n, we conclude that $\|Tx\| = 0$. The proof for concavity is similar.

16.4.2(i) Using (16.1), we get $\|\overline{T}\bar{x}\|_{\text{lat-}s} \leqslant \|\overline{T}\bar{x}\|_{\text{lat-}q} \lesssim \|\bar{x}\|_{\text{sum-}p} \leqslant \|\bar{x}\|_{\text{sum-}r}$.

16.4.3(ia) $\|\overline{TS}\bar{x}\|_{\text{lat-}q} \leqslant K\|\overline{S}\bar{x}\|_{\text{sum-}p} \leqslant K\|S\|\|\bar{x}\|_{\text{sum-}p}$.

16.4.3(ib) By Exercise 16.3.9, $\|\overline{ST}\bar{x}\|_{\text{lat-}q} \leqslant \|S\|_r\|\overline{T}\bar{x}\|_{\text{lat-}q} \leqslant K\|S\|_r\|\bar{x}\|_{\text{sum-}p}$.

16.4.3 One direction is straightforward; for the other, use Exercise 16.3.9.

16.4.4 For 1-convexity, use the triangle inequality:

$$\Big\|\sum_{i=1}^{n}|Tx_i|\Big\| \leqslant \sum_{i=1}^{n}\|Tx_i\| \leqslant \|T\|\sum_{i=1}^{n}\|x_i\|.$$

For ∞-concavity, note that for each $k \leqslant n$ we have $|Tx_k| \leqslant \bigvee_{i=1}^n|Tx_i|$. It follows that $\|Tx_k\| \leqslant \Big\|\bigvee_{i=1}^n|Tx_i|\Big\|$ and, therefore, $\max_i\|Tx_i\| \leqslant \Big\|\bigvee_{i=1}^n|Tx_i|\Big\|$.

16.4.5(ii) By the remark before the exercise, $L_p(\mu)$ is p-convex. By Exercise 16.4.2, $L_p(\mu)$ is q-convex for all $q \leqslant p$. Suppose that $L_p(\mu)$ is q-convex. Fix a disjoint normalized sequence (x_n) in X. For every n, we have $\|\bar{x}\|_{\text{sum-}q} = Kn^{\frac{1}{q}}$, while $\left(\sum_{i=1}^n |x_i|^q\right)^{\frac{1}{q}} = \sum_{i=1}^n |x_i| = \left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}}$ because of disjointness, so that $\|\bar{x}\|_{\text{lat-}q} = n^{\frac{1}{p}}$. Now q-convexity implies that $n^{\frac{1}{p}} \leqslant Kn^{\frac{1}{q}}$ for every n. This yields $q \leqslant p$.

16.4.5(iii) If $x \perp y$ in X, 1-concavity and the triangle inequality yield $\|x\| + \|y\| \leqslant \big\||x| + |y|\big\| = \|x + y\| \leqslant \|x\| + \|y\|$. Hence, $\|x + y\| = \|x\| + \|y\|$. It follows that X is an AL-space.

16.4.5(iv) Since $\|x \vee y\| \geqslant \|x\| \vee \|y\|$ is always true when $x, y \geqslant 0$, ∞-convexity with constant 1 means $\|x \vee y\| = \|x\| \vee \|y\|$. This is equivalent to X being an AM-space by Proposition 8.1.8.

16.4.5(v) Suppose X is p-convex with constant K and q-concave with constant M. Let $n \in \mathbb{N}$ and $x_1, \dots, x_n$ be disjoint and normalized. By Proposition 16.1.1,

$$\Big\|\sum_{i=1}^n |x_i|\Big\| = \|\bar{x}\|_{\text{lat-}p} \leqslant K\|\bar{x}\|_{\text{sum-}p} = Kn^{\frac{1}{p}},$$

and

$$\Big\|\sum_{i=1}^n |x_i|\Big\| = \|\bar{x}\|_{\text{lat-}q} \geqslant \frac{1}{M}\|\bar{x}\|_{\text{sum-}q} = \frac{1}{M}n^{\frac{1}{q}}.$$

It follows that $Kn^{\frac{1}{p}} \geqslant \frac{1}{M}n^{\frac{1}{q}}$ for every n, hence $p \leqslant q$.

16.4.8 The statement for sublattices is straightforward. Let J be a closed ideal in X. Suppose that X is (p, q)-convex. By Theorem 16.4.6, X^* is (p^*, q^*)-concave. By Exercise 6.11.14, $(X/J)^*$ is lattice isometric to a closed sublattice of X^*. It follows that $(X/J)^*$ is (p^*, q^*)-concave. Theorem 16.4.6 now yields that X/J is (p, q)-convex. The proof for concavity is similar.

16.4.9 Suppose that X_γ is (p, q)-convex with constant 1 for every $\gamma \in \Gamma$. Let $X = \left(\bigoplus_{\gamma \in \Gamma} X_\gamma\right)_\infty$. For each $\gamma \in \Gamma$, let $P_\gamma \colon X \to X_\gamma$ be the coordinate projection. Take $x_1, \dots, x_n$ in X. Put $x_{i,\gamma} = P_\gamma x_i$. Since P_γ is a lattice homomorphism, we have $P_\gamma\left(\sum_{i=1}^n |x_i|^q\right)^{\frac{1}{q}} = \left(\sum_{i=1}^n |x_{i,\gamma}|^q\right)^{\frac{1}{q}}$. It follows that

$$\Big\|\Big(\sum_{i=1}^n |x_i|^q\Big)^{\frac{1}{q}}\Big\| = \sup_{\gamma \in \Gamma}\Big\|\Big(\sum_{i=1}^n |x_{i,\gamma}|^q\Big)^{\frac{1}{q}}\Big\| \leqslant \sup_{\gamma \in \Gamma}\Big(\sum_{i=1}^n \|x_{i,\gamma}\|^p\Big)^{\frac{1}{p}} \leqslant \Big(\sum_{i=1}^n \|x_i\|^p\Big)^{\frac{1}{p}}.$$

For concavity, the argument is similar; we use the fact that the ℓ_p-norm satisfies the triangle inequality:

$$\begin{aligned}\Big(\sum_{i=1}^n \|x_i\|^p\Big)^{\frac{1}{p}} = \Big(\sum_{i=1}^n \Big[\sum_{\gamma \in \Gamma} \|x_{i,\gamma}\|\Big]^p\Big)^{\frac{1}{p}} &\leqslant \sum_{\gamma \in \Gamma}\Big(\sum_{i=1}^n \|x_{i,\gamma}\|^p\Big)^{\frac{1}{p}} \\ &\leqslant \sum_{\gamma \in \Gamma}\Big\|\Big(\sum_{i=1}^n |x_{i,\gamma}|^q\Big)^{\frac{1}{q}}\Big\| = \sum_{\gamma \in \Gamma}\Big\|P_\gamma\Big(\sum_{i=1}^n |x_i|^q\Big)^{\frac{1}{q}}\Big\| = \Big\|\Big(\sum_{i=1}^n |x_i|^q\Big)^{\frac{1}{q}}\Big\|.\end{aligned}$$

16.4.13 (i)$\Rightarrow$(ii) Suppose X is (p, q)-convex, $x_1, \dots, x_n \in B_E$ and $(\alpha_i)_{i=1}^n \in B_{\ell_p^n}$. Then

$$\Big\|\Big(\sum_{i=1}^n |\alpha_i T x_i|^q\Big)^{\frac{1}{q}}\Big\| \leqslant \Big(\sum_{i=1}^n \|\alpha_i x_i\|^p\Big)^{\frac{1}{p}} \leqslant \Big(\sum_{i=1}^n |\alpha_i|^p\Big)^{\frac{1}{p}} \leqslant 1.$$

(ii)$\Rightarrow$(iii) is trivial.

(iii)⇒(i) Let $x_1,\dots,x_n\in E$. WLOG, $x_i\neq 0$ for all i. Put

$$r=\Bigl(\sum_{i=1}^n\|x_i\|^p\Bigr)^{\frac{1}{p}},\qquad y_i=\frac{x_i}{\|x_i\|},\quad\text{and}\quad \alpha_i=\frac{\|x_i\|}{r}\quad\text{as}\quad i=1,\dots,n.$$

We have $y_i\in S_E$ and $x_i=r\alpha_i y_i$ for every i, and $(\alpha_i)_{i=1}^n\in S_{\ell_p^n}$. Then

$$\Bigl\|\Bigl(\sum_{i=1}^n|Tx_i|^q\Bigr)^{\frac{1}{q}}\Bigr\|=r\Bigl\|\Bigl(\sum_{i=1}^n|\alpha_i Ty_i|^q\Bigr)^{\frac{1}{q}}\Bigr\|\leqslant r.$$

16.4.14 (i)⇒(ii) Suppose X is (p,q)-concave, $Tx_1,\dots,Tx_n\notin B_E^\circ$ and $(\alpha_i)_{i=1}^n\notin B_{\ell_p^n}^\circ$. Then

$$\Bigl\|\Bigl(\sum_{i=1}^n|\alpha_i x_i|^q\Bigr)^{\frac{1}{q}}\Bigr\|\geqslant\Bigl(\sum_{i=1}^n\|\alpha_i Tx_i\|^p\Bigr)^{\frac{1}{p}}\geqslant\Bigl(\sum_{i=1}^n|\alpha_i|^p\Bigr)^{\frac{1}{p}}\geqslant 1.$$

(ii)⇒(iii) is trivial.

(iii)⇒(i) Let $x_1,\dots,x_n\in X$. WLOG, $Tx_i\neq 0$ for all i. Put

$$r=\Bigl(\sum_{i=1}^n\|Tx_i\|^p\Bigr)^{\frac{1}{p}},\qquad y_i=\frac{x_i}{\|Tx_i\|},\quad\text{and}\quad \alpha_i=\frac{\|Tx_i\|}{r}\quad\text{as}\quad i=1,\dots,n.$$

We have $Ty_i\in S_E$ and $x_i=r\alpha_i y_i$ for every i, and $(\alpha_i)_{i=1}^n\in S_{\ell_p^n}$. Then

$$\Bigl\|\Bigl(\sum_{i=1}^n|x_i|^q\Bigr)^{\frac{1}{q}}\Bigr\|=r\Bigl\|\Bigl(\sum_{i=1}^n|\alpha_i y_i|^q\Bigr)^{\frac{1}{q}}\Bigr\|\geqslant r.$$

16.5.8 Since the function $f(r)=r^q$ is differentiable, by Mean Value Theorem there exists $r\in[s,t]$ such that

$$t^q-s^q=f'(r)(t-s)=q|r|^{q-1}(t-s)\leqslant q\bigl(|s|\vee|t|\bigr)^{q-1}(t-s).$$

The second required inequality now follows easily.

We now move on to the first inequality. If $s<0<t$ then, using Hölder's inequality, we get

$$t^{\frac{1}{q}}-s^{\frac{1}{q}}=t^{\frac{1}{q}}+(-s)^{\frac{1}{q}}\leqslant\bigl(t+(-s)\bigr)^{\frac{1}{q}}(1+1)^{\frac{1}{q^*}}=2^{\frac{1}{q^*}}(t-s)^{\frac{1}{q}}.$$

The case $0\leqslant s<t$ splits into two subcases. If $0\leqslant s\leqslant\frac{t}{2}$ then

$$t^{\frac{1}{q}}-s^{\frac{1}{q}}\leqslant t^{\frac{1}{q}}=2^{\frac{1}{q}}\bigl(\tfrac{t}{2}\bigr)^{\frac{1}{q}}=2^{\frac{1}{q}}\bigl(t-\tfrac{t}{2}\bigr)^{\frac{1}{q}}\leqslant 2^{\frac{1}{q}}(t-s)^{\frac{1}{q}}.$$

Suppose that $0<\frac{t}{2}<s<t$. Applying the Mean Value Therem to the function $g(r)=r^{\frac{1}{q}}$ on $[s,t]$ we find $r\in[s,t]$ such that

$$t^{\frac{1}{q}}-s^{\frac{1}{q}}=\tfrac{1}{q}r^{\frac{1}{q}-1}(t-s)\leqslant\tfrac{1}{q}(t-s)^{\frac{1}{q}-1}(t-s)=\tfrac{1}{q}(t-s)^{\frac{1}{q}}$$

because $r>\frac{t}{2}>t-s$. Finally, if $s<t\leqslant 0$ then

$$\bigl(t^{\frac{1}{q}}-s^{\frac{1}{q}}\bigr)^q=\Bigl((-s)^{\frac{1}{q}}-(-t)^{\frac{1}{q}}\Bigr)^q\leqslant A\bigl((-s)-(-t)\bigr)=A(t-s).$$

16.5.14 (i) If X is (r,s)-convex with constant K, and $x_1,\dots,x_n\in X$, then

$$\Bigl|\!\Bigl|\!\Bigl|\Bigl(\sum_{i=1}^n|x_i^{1/p}|^{sp}\Bigr)^{\frac{1}{sp}}\Bigr|\!\Bigr|\!\Bigr|=\Bigl\|\Bigl(\sum_{i=1}^n|x_i|^s\Bigr)^{\frac{1}{s}}\Bigr\|^{\frac{1}{p}}$$
$$\leqslant\Bigl(K\Bigl(\sum_{i=1}^n\|x_i\|^r\Bigr)^{\frac{1}{r}}\Bigr)^{\frac{1}{p}}=K^{\frac{1}{p}}\Bigl(\sum_{i=1}^n\bigl|\!\bigl|\!\bigl|x_i^{1/p}\bigr|\!\bigr|\!\bigr|^{rp}\Bigr)^{\frac{1}{rp}}.$$

If X is (r,s)-concave with constant K, then

$$\Big(\sum_{i=1}^n \Big\|\Big|x_i^{1/p}\Big\|\Big|^{rp}\Big)^{\frac{1}{rp}} = \Big(\Big(\sum_{i=1}^n \|x_i\|^r\Big)^{\frac{1}{r}}\Big)^{\frac{1}{p}} \leqslant \Big(K\Big\|\Big(\sum_{i=1}^n |x_i|^s\Big)^{\frac{1}{s}}\Big\|\Big)^{\frac{1}{p}} = K^{\frac{1}{p}}\Big\|\Big|\Big(\sum_{i=1}^n |x_i^{1/p}|^{sp}\Big)^{\frac{1}{sp}}\Big\|\Big|$$

(ii) If X is (r,s)-convex with constant K then Theorem 16.5.12 yields

$$\Big\|\Big|\Big(\sum_{i=1}^n |x_i^p|^{\frac{r}{p}}\Big)^{\frac{p}{r}}\Big\|\Big| \leqslant \Big\|\Big(\sum_{i=1}^n |x_i|^r\Big)^{\frac{1}{r}}\Big\|^p \leqslant \Big(M\Big(\sum_{i=1}^n \|x_i\|^s\Big)^{\frac{1}{s}}\Big)^p = M^p\Big(\sum_{i=1}^n \big(\|x_i\|^p\big)^{\frac{s}{p}}\Big)^{\frac{p}{s}} \leqslant M^pK^p\Big(\sum_{i=1}^n \||x_i^p\||^{\frac{s}{p}}\Big)^{\frac{p}{s}}.$$

The proof for concavity is similar.

16.5.15 Let $x,y \in X_+ \setminus B_X^\circ$ and $\alpha,\beta \geqslant 0$ with $\alpha+\beta=1$. Put $z = \big(\alpha x^p + \beta y^p\big)^{\frac{1}{p}}$. Clearly, $z \geqslant 0$. It follows from $\|z\| \geqslant \big(\|\alpha^{\frac{1}{p}}x\| + \|\beta^{\frac{1}{p}}y\|\big)^{\frac{1}{p}} \geqslant 1$ that $z \notin B_X^\circ$.

16.6.2 The forward implication is straightforward. Suppose that T fails to be (p,∞)-convex. Then for every m, one can find vectors $x_1^{(m)},\dots,x_{n_m}^{(m)}$ such that

$$\sum_{k=1}^n \|x_k\|^p \leqslant \frac{1}{2^m} \quad\text{and}\quad \Big\|\bigvee_{k=1}^n |Tx_k|\Big\| > m$$

Let (x_n) be the concatenation of these finite sequences. It is easy to see that the sequence $\big(\|x_i\|\big)$ is in ℓ_p, yet the sequence $\big(\bigvee_{k=1}^n |Tz_k|\big)_n$ fails to be norm bounded.

16.6.13 We use Proposition 16.5.12 and Theorem 16.6.9. Let $x_1,\dots,x_n$ be disjoint vectors in X. Then $x_1^p,\dots,x_n^p$ are disjoint in X^p, so by Proposition 16.5.12, we have

$$\Big\|\Big|\sum_{i=1}^n x_i^p\Big\|\Big| = \Big\|\Big|\Big(\sum_{i=1}^n x_i\Big)^p\Big\|\Big| \geqslant \frac{1}{M^p}\Big\|\sum_{i=1}^n x_i\Big\|^p \geqslant \frac{1}{M^p}\Big(\Big(\frac{1}{K}\sum_{i=1}^n \|x\|^r\Big)^{\frac{1}{r}}\Big)^p = \frac{1}{M^pK^p}\Big(\sum_{i=1}^n \||x_i^p\||^{\frac{r}{p}}\Big)^{\frac{p}{r}}.$$

16.8.3 WLOG, $f \geqslant 0$. To show the first inequality, fix $t > 0$. Take $A = \{f > t\}$. Then $\|f\mathbb{1}_A\|_r \geqslant t\mu(A)^{\frac{1}{r}}$. It follows that $\mu(A)^{\frac{1}{p}-\frac{1}{r}}\|f\mathbb{1}_A\|_r \geqslant t\mu(A)^{\frac{1}{p}}$ and, therefore, $\|f\|_{p,\infty,r} \geqslant t\mu(f > t)^{\frac{1}{p}}$. Taking supremum over $t > 0$, we get $\|f\|_{p,\infty,r} \geqslant \|f\|_{p,\infty}$.

For the second inequality, observe that

$$\|f\|_{p,\infty} = \sup_{s>0} s\cdot\mu(f^r > s^r)^{\frac{1}{p}} = \sup_{t>0} t^{\frac{1}{r}}\cdot\mu(f^r > t)^{\frac{1}{p}}.$$

It follows that $\mu(f^r > t) \leqslant t^{-\frac{p}{r}}\|f\|_{p,\infty}^p$ for every $t > 0$.

Fix a measurable set A with $0 < \mu(A) < \infty$. Applying (A.3.14) to $f^r\mathbb{1}_A$, we get

$$\int_A f^r = \int_0^\infty \mu\big(A \cap \{f^r > t\}\big)\,dt. \tag{B.1}$$

We will estimate the integral in the right hand side of (B.1) by splitting it into two integrals. Put $\alpha = \mu(A)^{1-\frac{r}{p}}\|f\|_{p,\infty}^r$. It is clear that

$$\int_0^\alpha \mu\big(A \cap \{f^r > t\}\big)\,dt \leqslant \int_0^\alpha \mu(A)\,dt = \alpha\mu(A) = \mu(A)^{1-\frac{r}{p}}\|f\|_{p,\infty}^r.$$

On the other hand, it follows from

$$\mu\big(A \cap \{f^r > t\}\big) \leqslant \mu(f^r > t) \leqslant t^{-\frac{p}{r}}\|f\|_{p,\infty}^p$$

that

$$\int_\alpha^\infty \mu\big(A \cap \{f^r > t\}\big)\,dt \leqslant \|f\|_{p,\infty}^p \int_\alpha^\infty t^{-\frac{p}{r}}\,dt$$
$$= \frac{r}{p-r}\alpha^{1-\frac{p}{r}}\|f\|_{p,\infty}^p = \frac{r}{p-r}\mu(A)^{1-\frac{p}{r}}\|f\|_{p,\infty}^r.$$

It now follows from (B.1) that

$$\int_A f^r \leqslant \mu(A)^{1-\frac{r}{p}}\|f\|_{p,\infty}^r + \frac{r}{p-r}\mu(A)^{1-\frac{p}{r}}\|f\|_{p,\infty}^r = \frac{p}{p-r}\mu(A)^{1-\frac{p}{r}}\|f\|_{p,\infty}^r.$$

We now conclude that

$$\mu(A)^{\frac{1}{p}}\Big(\int_A f^r\Big)^{\frac{1}{r}} \leqslant \Big(\frac{p}{p-r}\Big)^{\frac{1}{r}}\|f\|_{p,\infty}.$$

Taking supremum over A yields the required inequality.

16.8.4 We have already observed that $L_p(\mu) \subseteq L_{p,\infty}(\mu)$. Now let $f \in L_{p,\infty}(\mu)$. It follows that $\|f\|_{p,\infty}$ is finite, so that $\|f\|_{p,\infty,r}$ is finite by Exercise 16.8.3. Taking $A = \Omega$ in the definition of $\|f\|_{p,\infty,r}$, we conclude that $\|f\|_r$ is finite.

16.8.8 WLOG, $f > 0$. Put $\alpha = \Big(\frac{\|f\|_q}{\|f\|_\infty}\Big)^q$. We have

$$\|f\|_{p,1} = \int_0^\alpha t^{\frac{1}{p}-1}f^*(t)\,dt + \int_\alpha^\infty t^{\frac{1}{p}-1}f^*(t)\,dt.$$

We will estimate the two integrals separately. Observe that

$$\int_0^\alpha t^{\frac{1}{p}-1}f^*(t)\,dt \leqslant \|f\|_\infty \int_0^\alpha t^{\frac{1}{p}-1}\,dt = p\|f\|_\infty \alpha^{\frac{1}{p}} = p\|f\|_q^{\frac{q}{p}}\|f\|_\infty^{1-\frac{q}{p}}.$$

For the second integral, we using Hölder's inequality

$$\int_\alpha^\infty t^{\frac{1}{p}-1}f^*(t)\,dt \leqslant \Big(\int_\alpha^\infty f^*(t)^q\,dt\Big)^{\frac{1}{q}}\Big(\int_\alpha^\infty \big(t^{\frac{1}{p}-1}\big)^{q^*}\,dt\Big)^{\frac{1}{q^*}}$$
$$\leqslant \|f\|_q \cdot \alpha^{\frac{1}{q^*}-\frac{1}{p^*}} \cdot \Big(\frac{p(q-1)}{q-p}\Big)^{\frac{1}{q^*}} = \Big(\frac{p(q-1)}{q-p}\Big)^{\frac{1}{q^*}}\|f\|_q^{\frac{q}{p}}\|f\|_\infty^{1-\frac{q}{p}}.$$

Adding together the two estimates yields the required conclusion.

16.9.1 By Theorem 16.6.9, it suffices to consider disjoint vectors. Suppose that T satisfies the definition for positive disjoint vectors. Let $x_1, \dots, x_n$ be disjoint vectors in X. Then

$$\sum_{i=1}^n \|Tx_i\| \leqslant \sum_{i=1}^n \big(\|Tx_i^+\| + \|Tx_i^-\|\big) \leqslant K\Big\|\sum_{i=1}^n (x_i^+ + x_i^-)\Big\| = K\Big\|\sum_{i=1}^n |x_i|\Big\|.$$

16.9.2 Adjust the proof of Proposition 16.6.1 using Exercise 16.9.1.

17.7.1 Define $\|x_1,\dots,x_n\|_n=\Bigl(\operatorname*{Ave}_{\varepsilon_i=\pm1}\bigl\|\sum_{i=1}^n\varepsilon_i x_i\bigr\|^p\Bigr)^{\frac{1}{p}}$. It is easy to verify that this is a norm on E^n. Conditions (i) and (ii) in Definition 16.3.1 are straightforward. Instead of (iii), we will verify the condition in Exercise 16.3.2. We need an auxiliary statement.

Claim: $\|x+\lambda y\|^p+\|x-\lambda y\|^p\leqslant\|x+y\|^p+\|x-y\|^p$ for all $x,y\in E$ whenever $|\lambda|\leqslant 1$. Indeed, take $\alpha=\frac{1+\lambda}{2}$. Then $0\leqslant\alpha\leqslant 1$ and

$$x+\lambda y=\alpha(x+y)+(1-\alpha)(x-y)\quad\text{and}\quad x-\lambda y=(1-\alpha)(x+y)+\alpha(x-y).$$

Using the triangle inequality for ℓ_p^2, we get

$$\begin{aligned}&\Bigl[\|x+\lambda y\|^p+\|x-\lambda y\|^p\Bigr]^{\frac{1}{p}}\\&\qquad\leqslant\Bigl[\Bigl(\alpha\|x+y\|+(1-\alpha)\|x-y\|\Bigr)^p+\Bigl((1-\alpha)\|x+y\|+\alpha\|x-y\|\Bigr)^p\Bigr]^{\frac{1}{p}}\\&\qquad\leqslant\Bigl[\Bigl(\alpha\|x+y\|\Bigr)^p+\Bigl(\alpha\|x-y\|\Bigr)^p\Bigr]^{\frac{1}{p}}+\Bigl[\Bigl((1-\alpha)\|x+y\|\Bigr)^p+\Bigl((1-\alpha)\|x-y\|\Bigr)^p\Bigr]^{\frac{1}{p}}\\&\qquad=\Bigl[\|x+y\|^p+\|x-y\|^p\Bigr]^{\frac{1}{p}}.\end{aligned}$$

Suppose that $x_1,\dots,x_n\in E$ and $|\lambda|\leqslant 1$. Using the claim, we get

$$\begin{aligned}&\|x_1,\dots,x_{n-1},\lambda x_n\|_n\\&\qquad=\Bigl(\frac{1}{2^n}\sum_{\varepsilon_1,\dots,\varepsilon_{n-1}=\pm1}\Bigl\|\sum_{i=1}^{n-1}\varepsilon_i x_i+\lambda x_n\Bigr\|^p+\Bigl\|\sum_{i=1}^{n-1}\varepsilon_i x_i-\lambda x_n\Bigr\|^p\Bigr)^{\frac{1}{p}}\\&\qquad\leqslant\Bigl(\frac{1}{2^n}\sum_{\varepsilon_1,\dots,\varepsilon_{n-1}=\pm1}\Bigl\|\sum_{i=1}^{n-1}\varepsilon_i x_i+x_n\Bigr\|^p+\Bigl\|\sum_{i=1}^{n-1}\varepsilon_i x_i-x_n\Bigr\|^p\Bigr)^{\frac{1}{p}}\\&\qquad=\|x_1,\dots,x_{n-1},x_n\|_n.\end{aligned}$$

Exercise 16.3.2 completes the proof.

18.1.2(iii) Since φ is convex, we have

$$\begin{aligned}I_\varphi(\alpha f+\beta g)&=\int\varphi\bigl(|\alpha f+\beta g|\bigr)\leqslant\int\varphi\bigl(|\alpha||f|+|\beta||g|\bigr)\\&\leqslant\int|\alpha|\varphi\bigl(|f|\bigr)+|\beta|\varphi\bigl(|g|\bigr)=|\alpha|\int\varphi\bigl(|f|\bigr)+|\beta|\int\varphi\bigl(|g|\bigr)=|\alpha|I_\varphi(f)+|\beta|I_\varphi(g).\end{aligned}$$

18.1.2(iv) This follows immediately from the Monotone Convergence Theorem.

18.1.2(v) This follows immediately from the Lebesgue Dominated Convergence Theorem.

18.1.2(vi) WLOG, $f\geqslant 0$; otherwise, replace f with $|f|$. Clearly, it suffices to find just one such λ; any number in $(0,\lambda)$ will then also work because I_φ is monotone. It follows from $f\geqslant\frac{1}{n}f\downarrow 0$ a.e. that $I_\varphi\bigl(\frac{1}{n}f\bigr)\downarrow 0$ by (v). Therefore $I_\varphi\bigl(\frac{1}{n}f\bigr)<\varepsilon$ for some n.

18.1.5(i) Let $A_n=\bigl\{|f|>\frac{1}{n}\bigr\}$. Then $\operatorname{supp}f=\bigcup A_n$. For each n, $I_\varphi(f)=\int\varphi\bigl(|f|\bigr)\geqslant\varphi(\frac{1}{n})\mu(A_n)$, so that $\mu(A_n)$ is finite.

18.1.5(ii) WLOG, $f\geqslant 0$. For each n, put $A_n=\bigl\{\frac{1}{n}\leqslant|f|\leqslant n\bigr\}$. As in (i), $\mu(A_n)$ is finite. Clearly, the restriction of f to A_n is bounded by n. Note that $f\cdot\mathbb{1}_{A_n}\to f$ a.e., so that $I_\varphi(f\cdot\mathbb{1}_{A_n})\to I_\varphi(f)$ by Exercise 18.1.2(v).

18.2.1 This is a standard calculus exercise. Put $f(t)=\ln\bigl(1+\frac{1}{2\sqrt{t}}\bigr)$. Then $I_\varphi(f)=\int_0^1\frac{dt}{2\sqrt{t}}=1$, while

$$I_\varphi(2f)=\int_0^1\biggl(\Bigl(1+\frac{1}{2\sqrt{t}}\Bigr)^2-1\biggr)dt=\int_0^1\Bigl(\frac{1}{\sqrt{t}}+\frac{1}{4t}\Bigr)dt=\infty.$$

18.2.3 If $\mu(A)$ is finite and $\lambda>0$ then $I_\varphi(\lambda\mathbb{1}_A)=\varphi(\lambda)\mu(A)<\infty$, so that $\mathbb{1}_A\in H_\varphi$. If $\mathbb{1}_A\in H_\varphi$ then $\mathbb{1}_A\in L_\varphi$ because $H_\varphi\subseteq L_\varphi$. If $\mathbb{1}_A\in L_\varphi$ then there exists $\lambda>0$ such that $\infty>I_\varphi(\lambda\mathbb{1}_A)=\varphi(\lambda)\mu(A)$. It follows that $\mu(A)<\infty$.

18.2.4 There exists $\lambda>0$ such that $I_\varphi(\lambda f)$ is finite. Now apply Exercise 18.1.5(i) to λf.

18.3.1 Recall that $\|f\|_\varphi=\inf A$, where $A=\bigl\{\lambda>0\ :\ I_\varphi\bigl(\frac{f}{\lambda}\bigr)\leqslant 1\bigr\}$. It follows from Exercise 18.1.2 that the map $\lambda\mapsto I_\varphi\bigl(\frac{f}{\lambda}\bigr)$ is continuous and strictly decreasing on $(0,\infty)$. It follows that A is closed. It also follows that $I_\varphi(f)=1$ iff $A=[1,\infty)$, which is equivalent to $\|f\|_\varphi=1$.

18.3.6 By Proposition A.3.7, that there is a sequence of finitely supported simple functions (g_n) such that $0\leqslant g_n\uparrow\varphi(f)$ in $L_0(\mu)$. Put $h_n=\varphi^{-1}(g_n)$. Then h_n is again a finitely supported simple function and $h_n\uparrow f$.

Fix $M>0$. It follows from $f\notin L_\varphi$ that $\int\varphi\bigl(\frac{f}{M}\bigr)=\infty$. Observe that $\varphi\bigl(\frac{h_n}{M}\bigr)\uparrow\varphi\bigl(\frac{f}{M}\bigr)$. By Fatou's Lemma, $\int\varphi\bigl(\frac{h_n}{M}\bigr)\to\infty$. In particular, there exists n_n such that $I_\varphi\bigl(\frac{h_n}{M}\bigr)\geqslant 1$, so that $\|h_n\|_\varphi\geqslant M$.

18.4.6 Being order continuous, H_φ has the Fatou Property. Take μ and φ such that μ is finite and $H_\varphi\subsetneq L_\varphi$, e.g., as in Exercise 18.2.1. Take any positive f in $L_\varphi\setminus H_\varphi$. Put $f_n=f\cdot\mathbb{1}_{\{f\leqslant n\}}$. Then (f_n) is an increasing norm bounded sequence in H_φ. However, $\sup f_n$ in H_φ does not exist.

18.6.5 It is straightforward that ψ is an Orlicz function. We claim that $\varphi(t)\leqslant\psi(t)\leqslant\varphi(2t)$ for every t. To see this, observe that the portion of the graph of ψ on $[2^n,2^{n+1}]$ is to the left of the graph of φ on the same interval and to the right of the graph of $\varphi(2t)$ as $t\in[2^{n-1},2^n]$.

18.6.10 Let n be the non-negative integer such that $2^n\leqslant\lambda<2^{n+1}$. Then

$$\varphi(\lambda t)<\varphi(2^{n+1}t)\leqslant C^{n+1}\varphi(t)=C2^{qn}\varphi(t)\leqslant C\lambda^q\varphi(t).$$

18.6.11 Let $t\geqslant 1$. Plugging t for λ and 1 for t into (18.3), we get $\varphi(t)\leqslant Ct^q\varphi(1)$. When $t<1$, using $\lambda=\frac{1}{t}$ in (18.3), we get $\varphi(1)=\varphi\bigl(\frac{1}{t}t\bigr)\leqslant C\bigl(\frac{1}{t}\bigr)^q\varphi(t)$, so that $\varphi(t)\geqslant\frac{1}{C}\varphi(1)t^q$. Now put $K=\varphi(1)\cdot\max\{C,\frac{1}{C}\}$.

18.6.13(i) Let C and q be as in Exercise 18.6.10. Since $\psi\sim\varphi$, there exist constants $c_1,c_2>0$ with $\varphi(c_1t)\leqslant\psi(t)\leqslant\varphi(c_2t)$ for all $t\geqslant 0$. WLOG, $c_2\geqslant\frac{c_1}{2}$; otherwise replace c_2 with $\frac{c_1}{2}$. Then

$$\psi(2t)\leqslant\varphi(2c_2t)\leqslant C\Bigl(\frac{2c_2}{c_1}\Bigr)^q\varphi(c_1t)\leqslant C\Bigl(\frac{2c_2}{c_1}\Bigr)^q\psi(t).$$

18.6.13(ii) Suppose $\varphi\sim\psi$. It follows that there exist $c_1,c_2>0$ such that $\varphi(c_1t)\leqslant\psi(t)\leqslant\varphi(c_2t)$ for every $t>0$. WLOG, $c_1\leqslant 1\leqslant c_2$. Let C and q be as in Exercise 18.6.10, then $\psi(t)\leqslant\varphi(c_2t)\leqslant Cc_2^q\varphi(t)$, so that $\frac{\psi(t)}{\varphi(t)}\leqslant Cc_2^q$. The estimate on the other side is similar. The converse implication is trivial.

18.8.1 To show that $\ell_\varphi(\Gamma)\subseteq\ell_\infty(\Gamma)$, suppose that $x=(x_\gamma)_{\gamma\in\Gamma}$ is unbounded. Then $\varphi\bigl(|x|\bigr)=\bigl(\varphi\bigl(|x_\gamma|\bigr)\bigr)_{\gamma\in\Gamma}$ is also unbounded and, therefore, $I_\varphi(x)=\sum_{\gamma\in\Gamma}\varphi\bigl(|x_\gamma|\bigr)=\infty$. It follows that $x\notin\ell_\varphi(\Gamma)$.

Now suppose that $x\in\ell_1(\Gamma)_+$. It follows from Exercise 18.1.1(iii) that there exist positive α and t_0 such that $\varphi(t)\leqslant\alpha t$ on $[0,t_0]$. Put $A=\bigl\{\gamma\in\Gamma\ :\ x_\gamma\geqslant t_0\bigr\}$. It follows from $x\in\ell_1(\Gamma)_+$ that A is finite. Note that $I_\varphi(x)=\sum_{\gamma\in A}\varphi(x_\gamma)+\sum_{\gamma\notin A}\varphi(x_\gamma)$. The first sum is finite because A is finite, while the second sum is dominated by $\sum_{\gamma\notin A}\alpha x_\gamma\leqslant\alpha\|x\|_{\ell_1}$. It follows that $I_\varphi(x)$ is finite. Similarly, $I_\varphi(\lambda x)$ is finite for every $\lambda>0$, hence $x\in h_\varphi(\Gamma)$.

If Γ is infinite then $\mathbb{1}\in\ell_\infty(\Gamma)$, but $I_\varphi(\lambda\mathbb{1})=\infty$ for every $\lambda>0$, so that $\mathbb{1}\notin\ell_\varphi(\Gamma)$. Hence, $\ell_\varphi(\Gamma)\neq\ell_\infty(\Gamma)$.

18.8.3 The proof is similar to that of Propositions 18.6.3 and 18.7.1. Suppose that there exist positive t_0 and c such that $\varphi(t) \leqslant \psi(ct)$ whenever $t < t_0$. Let $x \in \ell_\psi(\Gamma)$; we will show that $x \in \ell_\varphi(\Gamma)$. WLOG, $x \geqslant 0$ and, after scaling, $\|x\|_\infty < t_0$ and $\|cx\|_\psi \leqslant 1$. It follows that

$$I_\varphi(x) = \sum_{\gamma\in\Gamma} \varphi(x_\gamma) \leqslant \sum_{\gamma\in\Gamma} \psi(cx_\gamma) = I_\psi(cx) \leqslant 1$$

and, therefore, $x \in \ell_\varphi$. It follows that $\ell_\psi(\Gamma) \subseteq \ell_\varphi(\Gamma)$. The inclusion is continuous by Exercise 6.1.13. The rest of the proof is analogous to those of Propositions 18.6.3 and 18.7.1.

18.8.6 We know by Exercise 18.8.1 that $\ell_1(\Gamma) \subseteq \ell_\varphi(\Gamma)$. Suppose that $\alpha := \lim\limits_{t\to 0_+} \frac{\varphi(t)}{t} > 0$. It follows that $\varphi(t) \geqslant \alpha t$ for every t. Let $x \in \ell_\varphi(\Gamma)$ with $\|x\|_\varphi \leqslant 1$. We have $1 \geqslant \sum_{\gamma\in\Gamma} \varphi(|x_\gamma|) \geqslant \alpha \sum_{\gamma\in\Gamma} |x_\gamma|$, so that $x \in \ell_1(\Gamma)$. It follows that $\ell_\varphi(\Gamma) \subseteq \ell_1(\Gamma)$.

Suppose now that $\lim\limits_{t\to 0^+} \frac{\varphi(t)}{t} = 0$. We will show that there exists $x \in \ell_\varphi(\Gamma) \setminus \ell_1(\Gamma)$. Since ℓ_φ naturally embeds into $\ell_1(\Gamma)$, it suffices to prove the claim for the case $\Gamma = \mathbb{N}$, so that $\ell_\varphi(\Gamma) = \ell_\varphi$. The proof is analogous to that of Theorem 18.8.5. It follows from the assumption that there exists a sequence (t_n) in $[0, \frac{1}{2}]$ such that $t_n \downarrow 0$ and $\varphi(t_n) \leqslant \frac{t_n}{n^2}$. For each m, choose $m_n \in \mathbb{N}$ so that $\frac{1}{m_n+1} \leqslant t_n < \frac{1}{m_n}$. It follows that $m_n t_n < 1$ and $m_n t_n + t_n \geqslant 1$, so that $m_n t_n \geqslant \frac{1}{2}$. Put

$$x = (\underbrace{t_1, \dots, t_1}_{m_1}, \underbrace{t_2, \dots, t_2}_{m_2}, \dots).$$

Then $\|x\|_{\ell_1} = \sum_{n=1}^\infty m_n t_n = \infty$, so that $x \notin \ell_1$. On the other hand,

$$I_\varphi(x) = \sum_{i=1}^\infty \varphi\big(|x_i|\big) = \sum_{n=1}^\infty m_n \varphi(t_n) \leqslant \sum_{n=1}^\infty m_n \frac{t_n}{n^2} < \sum_{n=1}^\infty \frac{1}{n^2} < \infty,$$

so that $x \in \ell_\varphi$.

18.9.1(i) Fix $s > 0$ and consider the function $f(t) = st - \varphi(t)$ for $t \geqslant 0$. Note that $f(t) = t\big(s - \frac{\varphi(t)}{t}\big)$ when $t > 0$. Observe that f is continuous, $f(0) = 0$, the right derivative of f as zero is s and, therefore, is positive, and $f(t) < 0$ for all sufficiently large t. It follows that f attains its maximum on $(0, \infty)$, and it is a positive real number.

18.9.1(ii) It is clear that $\varphi^*(0) = 0$. The solution of (i) yields $\varphi^*(s) > 0$ whenever $s > 0$. It is left to verify that φ is convex. Fix non-negative real numbers s_1, s_2, λ_1, and λ_2 such that $\lambda_1 + \lambda_2 = 1$. By definition of φ^*, there exists $t \geqslant 0$ such that

$$\varphi^*(\lambda_1 s_1 + \lambda_2 s_2) = (\lambda_1 s_1 + \lambda_2 s_2)t - \varphi(t) = \lambda_1\big(s_1 t - \varphi(t)\big) + \lambda_2\big(s_2 t - \varphi(t)\big) \leqslant \lambda_1 \varphi^*(s_1) + \lambda_2 \varphi^*(s_2).$$

18.9.1(iii) Suppose that $\varphi(t) \leqslant \psi(ct)$ for every $t \geqslant 0$. It follows that

$$\psi^*(s) = \sup_{t\geqslant 0}\big(st - \psi(t)\big) \leqslant \sup_{t\geqslant 0}\Big(st - \varphi\big(\tfrac{t}{c}\big)\Big) = \varphi^*(cs).$$

18.9.1(iv) Fix s, put $f(t) = st - \varphi(t) = st - \frac{t^p}{p}$. Then $\frac{df}{dt} = s - t^{p-1}$, which vanishes when $t = s^{\frac{1}{p-1}}$. It follows that $\varphi^*(s) = f\big(s^{\frac{1}{p-1}}\big) = \frac{s^q}{q}$.

18.9.1(v) This follows exactly from (i).

18.9.5 It is easy to see that it is positively homogeneous and satisfies the triangle inequality. It follows from $\langle f, g\rangle = \big\langle |f|, (\operatorname{sign} f)g\big\rangle$ that $|||f|||_\varphi = \big|\big|\big|\,|f|\,\big|\big|\big|_\varphi$. It is easy to see that if $f \geqslant 0$ then it suffices to only consider positive g's in the definition of $|||f|||_\varphi$; it follows that if $0 \leqslant f_1 \leqslant f_2$ then $|||f_1|||_\varphi \leqslant |||f_2|||_\varphi$.

18.9.13 Take

$$\varphi(t)=\begin{cases}\frac{e^2}{4}t^2 & \text{if } 0\leqslant t\leqslant 2;\\ e^t & \text{if } t\geqslant 2,\end{cases}\qquad\text{and}\qquad \psi(s)=\begin{cases}\frac{s^2}{e} & \text{if } 0\leqslant s\leqslant e;\\ s\ln s & \text{if } s\geqslant e.\end{cases}$$

It is easy to verify that these are Orlicz functions (their derivatives are increasing) and that they satisfy (18.4).

We compute φ^* as described in 18.9.2. Observe that

$$u(t)=\varphi'(t)=\begin{cases}\frac{e^2}{2}t & \text{when } 0\leqslant t\leqslant 2;\\ e^t & \text{when } t\geqslant 2.\end{cases}$$

It follows that

$$u^*(t)=\begin{cases}\frac{2}{e^2}s & \text{when } 0\leqslant s\leqslant e^2,\\ \ln s & \text{when } s\geqslant e^2,\end{cases}\qquad \varphi^*(t)=\begin{cases}\frac{s^2}{e^2} & \text{when } 0\leqslant s\leqslant e^2,\\ s\ln s-s+1-e^2 & \text{when } s\geqslant e^2.\end{cases}$$

To show that $\varphi^*\sim\psi$ at infinity, observe that $s\ln s\geqslant\varphi^*(s)\geqslant\frac{1}{2}s\ln s$ for all sufficiently large s. It is easy to verify that ψ satisfies the Δ_2-condition, while φ fails it. By Proposition 18.7.6, L_ψ is order continuous, while L_φ is not. By Theorem 18.5.1, $H_\psi=L_\psi$ while both H_φ and L_φ fail to be KB-spaces and, therefore, contain copies of c_0. Observe that $L_\psi^*=H_\psi^*\simeq L_\varphi$. By Theorem 10.5.3, L_ψ contains a lattice copy of ℓ_1.

19.1.8 No, here is a counterexample. Define $\varphi\colon\mathbb{R}^2\times\mathbb{R}^2\to\mathbb{R}^3$ via

$$\varphi\Big(\big(\begin{smallmatrix}x\\y\end{smallmatrix}\big),\big(\begin{smallmatrix}s\\t\end{smallmatrix}\big)\Big)=\Big(\begin{smallmatrix}xs+yt\\xt+yt\\ys+yt\end{smallmatrix}\Big).$$

It can be easily verified that φ is bilinear. To see that φ is bi-injective, suppose that $\big(\begin{smallmatrix}x\\y\end{smallmatrix}\big)$ and $\big(\begin{smallmatrix}s\\t\end{smallmatrix}\big)$ are both non-zero, yet $\varphi\Big(\big(\begin{smallmatrix}x\\y\end{smallmatrix}\big),\big(\begin{smallmatrix}s\\t\end{smallmatrix}\big)\Big)=0$. It follows that $xs=xt=ys=-yt$. Since at least one of the four products is non-zero, all the four are non-zero, hence x, y, s, and t are all non-zero. It now follows from $xs=xt$ that $s=t$ and from $ys=-yt$ that $s=-t$; a contradiction.

However, its linearization $\varphi^\otimes\colon\mathbb{R}^2\otimes\mathbb{R}^2\to\mathbb{R}^3$ cannot be injective as $\dim\mathbb{R}^3<\dim(\mathbb{R}^2\otimes\mathbb{R}^2)$.

19.2.1 Note that

$$\varphi(x,y)=\varphi(x^+,y^+)-\varphi(x^+,y^-)-\varphi(x^-,y^+)+\varphi(x^+,y^+).$$

Since the four terms in the right hand side are positive, we have

$$\big|\varphi(x,y)\big|\leqslant\varphi(x^+,y^+)+\varphi(x^+,y^-)+\varphi(x^-,y^+)+\varphi(x^+,y^+)=\varphi\big(|x|,|y|\big)$$

19.2.2 The forward implication is straightforward: if $x>0$ then

$$\big|\varphi(x,y)\big|=\varphi\big(|x|,|y|\big)=\varphi\big(x,|y|\big).$$

To prove the converse, fix $x\in X$ and $y\in Y$, and observe that

$$\varphi(x,y)=\big[\varphi(x^+,y^+)+\varphi(x^-,y^-)\big]-\big[\varphi(x^+,y^-)+\varphi(x^-,y^+)\big].$$

Since lattice homomorphisms preserve disjointness, it follows from the assumption that every term is the first sum is disjoint to every term in the second sum; hence the two sums are disjoint. Also, all the four terms are positive. It follows that

$$\big|\varphi(x,y)\big|=\big[\varphi(x^+,y^+)+\varphi(x^-,y^-)\big]+\big[\varphi(x^+,y^-)+\varphi(x^-,y^+)\big]=\varphi(x^++x^-,y^++y^-)=\varphi\big(|x|,|y|\big).$$

19.6.4 Let $a,b \in X \bar{\otimes} Y$. Fix $\varepsilon > 0$. From the definition of $\|\cdot\|_{|\pi|}$, find

$$x_1,\dots,x_n,x_1',\dots,x_m' \text{ in } X_+ \text{ and } y_1,\dots,y_n,y_1',\dots,y_m' \text{ in } X_+ \text{ such that}$$

$$|a| \leqslant \sum_{i=1}^n x_i \otimes y_i,\ |b| \leqslant \sum_{j=1}^m x_j' \otimes y_j',\ \|a\|_{|\pi|} \geqslant \sum_{i=1}^n \|x_i\|\|y_i\| - \varepsilon \text{ and } \|b\|_{|\pi|} \geqslant \sum_{j=1}^m \|x_j'\|\|y_j'\| - \varepsilon.$$

It follows that $|a+b| \leqslant |a|+|b| \leqslant \sum_{i=1}^n x_i \otimes y_i + \sum_{j=1}^m x_j' \otimes y_j'$, and, therefore,

$$\|a+b\|_{|\pi|} \leqslant \sum_{i=1}^n \|x_i\|\|y_i\| + \sum_{j=1}^m \|x_j'\|\|y_j'\| \leqslant \|a\|_{|\pi|} + \|b\|_{|\pi|} + 2\varepsilon.$$

20.1.5 Follow the proof of Proposition 20.1.4, but use matrices with at most one non-zero entry in every *column*.

20.1.9 Follows from Exercise 20.1.8.

20.2.1 It is easy to see that this formula defines a norm on $(X/Y)^n$. Given any $\bar{x} \in X^n$ and an $m \times n$ matrix A, we have

$$\left\| A \begin{bmatrix} x_1 + Y \\ \vdots \\ x_n + Y \end{bmatrix} \right\| = \left\| \begin{bmatrix} a_{11}x_1 + \dots + a_{1n}x_n + Y \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n + Y \end{bmatrix} \right\|$$

$$= \inf_{\bar{y} \in Y^m} \|A\bar{x} + \bar{y}\| \leqslant \inf_{\bar{z} \in Y^n} \|A\bar{x} + A\bar{z}\| \leqslant \|A\colon \ell_p^n \to \ell_p^m\| \inf_{\bar{z} \in Y^n} \|\bar{x} + \bar{z}\| = \|A\colon \ell_p^n \to \ell_p^m\| \left\| \begin{bmatrix} x_1 + Y \\ \vdots \\ x_n + Y \end{bmatrix} \right\|.$$

20.2.2 Fix a 1-multinorm. Verify that the dual sequence of norms satisfies (A1)-(A4), hence is an ∞-multinorm, so that the second dual sequence of norms is a 1-multinorm by duality; it follows that the original sequence of norms is a 1-multinorm as well.

20.2.3 Note that $\|T\|_{\mathrm{mb}} = \sup_n \|T^{(n)}\|$, where $T^{(n)}\colon X^n \to Y^n$ is defined by $T^{(n)}(x_1,\dots,x_n) = (Tx_1,\dots,Tx_n)$. Note that $(T^*)^{(n)} = (T^{(n)})^*$ because

$$\langle (T^*)^{(n)} \bar{f}, \bar{x} \rangle = \sum_{i=1}^n \langle T^* f_i, x_i \rangle = \sum_{i=1}^n \langle f_i, Tx_i \rangle = \langle \bar{f}, (T^{(n)})\bar{x} \rangle$$

for all $\bar{f} \in (Y^*)^n$ and $\bar{x} \in X^n$. It follows that

$$\|T^*\|_{\mathrm{mb}} = \sup_n \|(T^*)^{(n)}\| = \sup_n \|(T^{(n)})^*\| = \sup_n \|T^{(n)}\| = \|T\|_{\mathrm{mb}}.$$

20.2.4 Note that $\ker T^{(n)} = (\ker T)^n$. By the definition of quotient p-multinorm, $(X/\ker T)^n$ is isometric to $X^n/(\ker T)^n$. Under this isometry, $\widetilde{T}^{(n)}$ may be identified with $\widetilde{T^{(n)}}$. Therefore,
$\widetilde{T}$ is a multiisometry $\Leftrightarrow (\widetilde{T})^{(n)}$ is an isometry for each n
$\Leftrightarrow\ \widetilde{T^{(n)}}$ is an isometry for each n $\Leftrightarrow$ $T^{(n)}(B^o_{X^n}) = B^o_{Y^n}$ for every n.

20.8.2 We will first show that $\|T^{(n)}\| \leqslant 1$ for every n, so that $T^{(n)}(B^o_{E^n}) \subseteq B^o_{X^n}$. Take any $\bar{u} \in E^m$, $\bar{u} = (u_1,\dots,u_m)$, $u_i = (u_{i,\bar{x}})_{\bar{x} \in \mathcal{I}}$ for every $i = 1,\dots,m$. Note that at most countably many of the $u_{i,\bar{x}}$'s are non-zero. Then

$$T^{(n)}\bar{u} = \Big(\sum_{\bar{x} \in \mathcal{I}} S^*_{\bar{x}} u_{1,\bar{x}}, \dots, \sum_{\bar{x} \in \mathcal{I}} S^*_{\bar{x}} u_{m,\bar{x}} \Big) = \sum_{\bar{x} \in \mathcal{I}} \big(S^*_{\bar{x}} u_{1,\bar{x}}, \dots, S^*_{\bar{x}} u_{m,\bar{x}} \big).$$

The last identity follows from the fact that the sum converges absolutely. Indeed, for each $\bar{x} \in \mathcal{I}$ we have

$$\Big\|\big(S^*_{\bar{x}}u_{1,\bar{x}}, \dots, S^*_{\bar{x}}u_{m,\bar{x}}\big)\Big\| \leqslant \sum_{i=1}^m \big\|S^*_{\bar{x}}u_{i,\bar{x}}\big\| \leqslant \sum_{i=1}^m \big\|u_{i,\bar{x}}\big\|.$$

It follows that

$$\sum_{\bar{x}\in\mathcal{I}} \Big\|\big(S^*_{\bar{x}}u_{1,\bar{x}}, \dots, S^*_{\bar{x}}u_{m,\bar{x}}\big)\Big\| \leqslant \sum_{i=1}^m \sum_{\bar{x}\in\mathcal{I}} \|u_{i,\bar{x}}\| = \sum_{i=1}^m \|u_i\| < \infty.$$

Therefore,

$$T^{(n)}\bar{u} = \sum_{\bar{x}\in\mathcal{I}} \big(S^*_{\bar{x}}\big)^{(n)}(u_{1,\bar{x}}, \dots, u_{m,\bar{x}}).$$

It follows from $\big\|(S^*_{\bar{x}})^{(n)}\big\| = 1$ that

$$\big\|T^{(n)}\bar{u}\big\| \leqslant \sum_{\bar{x}\in\mathcal{I}} \big\|(u_{1,\bar{x}}, \dots, u_{m,\bar{x}})\big\| \leqslant \sum_{\bar{x}\in\mathcal{I}} \Big\|\sum_{i=1}^m |u_{i,\bar{x}}|\Big\| = \Big\|\sum_{i=1}^m |u_i|\Big\| = \|\bar{u}\|.$$

Second, show that $T^{(n)}$ maps $B^o_{E^n}$ *onto* $B^o_{X^n}$. Fix $\bar{x} \in B^o_{X^n}$. Then $\bar{x} \in B^o_{Y^n_{\bar{x}}}$. It follows from $(S^*_{\bar{x}})^{(n)}\big(B^o_{(F^*_{\bar{x}})^n}\big) = B^o_{Y^n_{\bar{x}}}$ that $\bar{x} = (S^*_{\bar{x}})^{(n)}\bar{f}$ for some $\bar{f} \in B^o_{(F^*_{\bar{x}})^n}$. Define $\bar{u} \in E^n$ as follows: $u_{i,\bar{x}} = f_i$ and $u_{i,\bar{y}} = 0$ whenever $\bar{y} \neq \bar{x}$. Then

$$\|\bar{u}\| = \big\|\sum_{i=1}^n |u_i|\big\| = \sum_{\bar{y}\in\mathcal{I}} \big\|\sum_{i=1}^n |u_{i,\bar{y}}|\big\| = \big\|\sum_{i=1}^n |f_i|\big\| = \|\bar{f}\| < 1,$$

so that $\bar{u} \in B^o_{E^n}$. On the other hand, $Tu_i = \sum_{\bar{y}\in\mathcal{I}} S^*_{\bar{y}}u_{i,\bar{y}} = S^*_{\bar{x}}f_i$, so that $T^{(n)}\bar{u} = \big(S^*_{\bar{x}}\big)^{(n)}\bar{f} = \bar{x}$.

21.2.5 Suppose that A is order open, a net $(x_\alpha)_{\alpha\in\Lambda}$ order converges to some $x \in A$, yet no tail of the net is entirely contained in A. Let $\Lambda_0 = \big\{\alpha \in \Lambda \ : \ x_\alpha \notin A\big\}$. Then Λ_0 is co-final in Λ, i.e., $\forall\alpha \in \Lambda\ \exists\beta \in \Lambda_0\ \alpha \leqslant \beta$. It follows that Λ_0 is a directed set, hence $(x_\alpha)_{\alpha\in\Lambda_0}$ is a net. Since $(x_\alpha)_{\alpha\in\Lambda}$ order converges to x, there exists a net (u_γ) such that $u_\gamma \downarrow 0$ and for every gamma, the order interval $[x - u_\gamma, x + u_\gamma]$ contains a tail of $(x_\alpha)_{\alpha\in\Lambda}$. It follows that $[x - u_\gamma, x + u_\gamma]$ contains a tail of $(x_\alpha)_{\alpha\in\Lambda_0}$, hence $(x_\alpha)_{\alpha\in\Lambda_0}$ converges in order to x. Since A^C is order closed, it follows that $x \in A^C$, which is a contradiction.

The converse implication is straightforward.

22.1.1 First, observe that the identity map from X to itself is the unique lattice homomorphism extending the identity map from A to itself. Now consider the inclusion maps $i\colon A \to X$ and $j\colon A \to Y$. They extend uniquely to lattice homomorphisms $T\colon X \to Y$ and $S\colon Y \to X$. The composition $ST\colon X \to X$ is a lattice homomorphism from X to X which fixes A, so, by the preceding observation, ST is the identity map on X. Similarly, TS is the identity map of Y. It follows that $S = T^{-1}$, hence T is a lattice isomorphism.

22.2.6 Suppose that $f = \Phi_1(\delta_{a_1}, \dots, \delta_{a_n})$ for some lattice-linear expression $\Phi[t_1, \dots, t_n]$ and some $a_1, \dots, a_n \in A$. Suppose that also $f = \Phi'(\delta_{b_1}, \dots, \delta_{b_k})$ for some lattice-linear expression $\Phi'[t_1, \dots, t_k]$ and some $b_1, \dots, b_k \in A$. Choose distinct $c_1, \dots, c_m \in A$ so that $\{c_1, \dots, c_m\} = \{a_1, \dots, a_n\} \cup \{b_1, \dots, b_k\}$. Consider the formal lattice-linear expression $\Phi[t_1, \dots, t_n] - \Phi'[s_1, \dots, s_k]$. By appropriately replacing the variables $t_1, \dots, t_n, s_1, \dots, s_k$ in this expression with new variables $r_1, \dots, r_m$, we produce a lattice-linear expression $\Psi[r_1, \dots, r_m]$ such that

$$\Psi(\delta_{c_1}, \dots, \delta_{c_m}) = \Phi(\delta_{a_1}, \dots, \delta_{a_n}) - \Phi'(\delta_{b_1}, \dots, \delta_{b_k}) = 0.$$

Lemma 22.2.1 now yields that $\Psi(r_1, \dots, r_m) = 0$ for any $r_1, \dots, r_m$ in $\mathbb{R}$. It follows by lattice-linear calculus that $\Psi\big(\varphi(c_1), \dots, \varphi(c_m)\big) = 0$. This yields

$$\Phi\big(\varphi(a_1), \dots, \varphi(a_n)\big) - \Phi'\big(\varphi(b_1), \dots, \varphi(b_k)\big) = 0,$$

so that the two expressions are equal.

22.2.9 Since the lattice operations on L are defined pointwise, T is a lattice homomorphism. To show that T is one-to-one, assume that $Rf = 0$ for some $f \in L$. Then $f = \Phi(\delta_{a_1}, \dots, \delta_{a_n})$ for some lattice-linear expression Φ and some $a_1, \dots, a_n$ in A. For every function $x\colon A \to [-1, 1]$, we have

$$0 = \big[\Phi(\delta_{a_1}, \dots, \delta_{a_n})\big](x) = \Phi\big(x(a_1), \dots, x(a_n)\big).$$

Since x is arbitrary, it follows that $\Phi(t_1, \dots, t_n) = 0$ for all $t_1, \dots, t_n \in [-1, 1]$. Since Φ is positively homogeneous, it follows that Φ vanishes on $\mathbb{R}$, hence $f = 0$.

22.3.1 We are only allowed to use binary operations $+$ and $\vee$, the symbol 0 for the zero vector, and for every $\lambda \in \mathbb{R}$ we have a unary operation to represent the scalar multiplication: $x \mapsto \lambda x$. We start with

$$\begin{aligned}
&\forall t_1, t_2, t_3 && (t_1 + t_2) + t_3 = t_1 + (t_2 + t_3)\\
&\forall t_1, t_2, t_3 && t_1 + t_2 = t_2 + t_1\\
&\forall t && t + 0 = t\\
&\forall t && 0t = 0\\
&\forall t && 1t = t.
\end{aligned}$$

Regarding the other axioms about scalar multiplication, there is a subtlety: we do not allow quantifies over scalars. Here is a way to get around this. Instead of the axiom

$$\forall \lambda \in \mathbb{R}\ \forall t_1, t_2 \quad \lambda(t_1 + t_2) = \lambda t_1 + \lambda t_2,$$

we include for every $\lambda \in \mathbb{R}$ the following axiom:

$$\forall t_1, t_2 \quad \lambda(t_1 + t_2) = \lambda t_1 + \lambda t_2.$$

This yields an continuum collection of axioms. It follows, in particular, that $(-1)t$ is the additive inverse of t because $(-1)t + t = (-1)t + 1t = (-1 + 1)t = 0t = 0$. Similarly, for every $\lambda_1, \lambda_2 \in \mathbb{R}$ put $\lambda = \lambda_1 + \lambda_2$ and $\mu = \lambda_1 \lambda_2$ and include the following two axioms.

$$\begin{aligned}
&\forall t && \lambda t = \lambda_1 t + \lambda_2 t\\
&\forall t && \mu t = \lambda_1(\lambda_2 t).
\end{aligned}$$

By now, we have axiomitized a vector space; we now proceed to the order structure. Our Ω contains no symbols for $\leqslant$. One can get around this replacing $x \leqslant y$ with $x \vee y = y$. Here we go:

$$\begin{aligned}
&\forall t && t \vee t = t\\
&\forall t_1, t_2 && t_1 \vee t_2 = t_2 \vee t_1\\
&\forall t_1, t_2, t_3 && (t_1 \vee t_2) \vee t_3 = t_1 \vee (t_2 \vee t_3)\\
&\forall t_1, t_2, t_3 && (t_1 \vee t_2) \vee t_3 = (t_1 \vee t_3) \vee (t_2 \vee t_3)\\
&\forall t_1, t_2, t_3 && (t_1 \vee t_2) + t_3 = (t_1 + t_3) \vee (t_2 + t_3)
\end{aligned}$$

In addition, for every $\lambda \in \mathbb{R}_+$, we include

$$\forall t_1, t_1 \quad \lambda(t_1 \vee t_2) = (\lambda t_1) \vee (\lambda t_2).$$

It is clear that every vector lattice satisfies all these axioms. To show the converse, suppose that X is an Ω-algebra satisfying all these axioms. It is clear that X is a vector space. For $x, y \in X$, define $x \leqslant y$ if $x \vee y = y$. First, observe that this is a partial order. Indeed, $x \vee x = x$ yields $x \leqslant x$. If $x \leqslant y$ and $y \leqslant x$ then $y = x \vee y = y \vee x = x$, so that $x = y$. If $x \leqslant y$ and $y \leqslant z$ then $x \vee y = y$ and $y \vee z = z$, so that

$$x \vee z = x \vee (y \vee z) = (x \vee y) \vee z = y \vee z = z,$$

so that $x \leqslant z$. Next, we observe that the order is linear. If $x, y, z \in X$ satisfy $x \leqslant y$, then $(x + z) \vee (y + z) = (x \vee y) + z = y + z$, so that $x + z \leqslant y + z$. Similarly, if $x \leqslant y$ and $\lambda \in \mathbb{R}_+$ then $(\lambda x) \vee (\lambda y) = \lambda(x \vee y) = \lambda y$, so that $\lambda x \leqslant \lambda y$. Finally, we show that $\sup\{x, y\}$ exists for every $x, y \in X$ and, actually, equals $x \vee y$. To show that $x \vee y$ is an upper bound of the set $\{x, y\}$, observe that $x \vee (x \vee y) = (x \vee x) \vee y = x \vee y$, so that $x \leqslant x \vee y$. Similarly, $y \leqslant x \vee y$. Now suppose that $z \in X$ such that $x \leqslant z$ and $y \leqslant z$. Then $(x \vee y) \vee z = (x \vee z) \vee (y \vee z) = z \vee z = z$, so that $x \vee y \leqslant z$. It is easy to see that $\inf\{x, y\}$ exists as well.

22.4.1(ii) Since $\mathbb{R}^{\mathbb{R}^A}$ is Archimedean, so is every its sublattice, including $\mathrm{FVL}(A)$.

22.4.1(iii) As in 22.2.7, we may view $\mathrm{FVL}(B)$ as the set of all formal lattice-linear expressions of elements of B, where we identify two expressions whenever they generate the same lattice-linear real function. Clearly, those expressions that only involve elements of A form a sublattice which may be identified with $\mathrm{FVL}(A)$.

22.4.2 (i)⇒(ii) It is clear that A generates X. A is lattice-linear independent by Lemma 22.2.1 and the discussion after it.

(ii)⇒(iii) Let $x \in X$. Since $X = \mathrm{Lat}(A)$, we can write $x = \Phi(a_1, \dots, a_n)$ for some lattice-linear expression Φ. We define $\hat{\varphi}(x)$ to be $\Phi\big(\varphi(a_1), \dots, \varphi(a_n)\big)$. Using the argument from page 587, we conclude that $\hat{\varphi}(x)$ is well-defined, is a lattice homomorphism, and extends φ.

(iii)⇒(i) Let φ be a map from A to an arbitrary vector lattice Y. By (iii), it extends to a lattice-homomorphism $\hat{\varphi}\colon X \to Y$. It is left to show that this extension is unique. Let $x \in X$. It follows from $X = \mathrm{Lat}(A)$ that $x = \Phi(a_1, \dots, a_n)$ for some lattice-linear expression Φ ans some $a_1, \dots, a_n \in A$. Then $\hat{\varphi}(a) = \Phi\big(\varphi(a_1), \dots, \varphi(a_n)\big)$, so that $\hat{\varphi}(a)$ is uniquely defined.

22.4.5 L_1 may be identified with the set of all piece-wise affine functions on S^1_∞. Since this sphere consists of just two points, the resulting set may be identified with $\mathbb{R}^2$.

Alternatively, L_1 may be viewed as the set of all lattice-linear combinations of a single symbol a. Since a^+ and a^- are disjoint, the span of a^+ and a^- is closed under lattice operations, hence $L_1 = \mathrm{span}\{a^+, a^-\}$ and, therefore, is lattice isometric to $\mathbb{R}^2$.

22.4.9 Let $e \in \mathrm{FVL}(A)_+$ be a strong unit. Then $e \in \mathrm{FVL}(B)$ for some finite subset B of A. Pick $a \in A \setminus B$. Let $x \in \mathbb{R}^A$ defined by $x = \mathbb{1}_{\{a\}}$. Then $\delta_a(x) = 1$, while $\delta_b(x) = 0$ for all $b \in B$; it follows that $e(x) = 0$ because e is a lattice-linear combination of $\{\delta_b : b \in B\}$. Clearly, no scalar multiple of e dominates δ_a.

22.5.1 Let $\phi\colon A \to X$ be the inclusion map, then $\hat{\varphi}$ has to agree with the identity operator on X, which yields that $\|a\| = \big\|\varphi(a)\big\| \leqslant \|I\| = 1$ for every $a \in A$.

To prove the converse inequality, consider the constant map $\psi\colon A \to \mathbb{R}$ such that $\psi(a) = 1$ for all a. Let $\hat{\psi}\colon X \to \mathbb{R}$ be the extension; then $\|\hat{\psi}\| = 1$, so that $1 = \big|\psi(a)\big| = \big|\hat{\psi}(a)\big| \leqslant \|a\|$ for every $a \in A$.

22.5.2 Let $i\colon A \hookrightarrow X$ and $j\colon A \hookrightarrow Y$ be the inclusion maps; let $\hat{i}\colon Y \to X$ and $\hat{j}\colon X \to Y$ be the corresponding extensions. It follows from Exercise 22.5.1 that $\|\hat{i}\| = 1 = \|\hat{j}\|$. The composition $\hat{i}\hat{j}\colon X \to X$ is a lattice homomorphism which fixes A. Since $\hat{i}\hat{j}(a) = a$ for every $a \in A$, we have $\|\hat{i}\hat{j}\| \geqslant 1$. On the other hand, $\|\hat{i}\hat{j}\| \leqslant \|\hat{i}\|\|\hat{j}\| = 1$. It follows that $\|\hat{i}\hat{j}\| = 1$, hence $\hat{i}\hat{j}$ is the unique extension of i to a norm one lattice homomorphism from X to X. By uniqueness, we conclude that $\hat{i}\hat{j} = I_X$. Similarly, $\hat{j}\hat{i} = I_Y$, so that $\hat{j} = \hat{i}^{-1}$.

22.5.4 Put $B = \mathrm{conv}\big(\mathrm{Sol}(A)\big)$. By Nakano's Theorem 3.3.24, B is a convex solid set. Let ρ be the Minkowski functional of B. It follows from Lemma 22.4.7 that $\rho(f)$ is finite for every $f \in \mathrm{FVL}(A)$, hence ρ is a seminorm on $\mathrm{FVL}(A)$.

Since B is solid, ρ is a lattice seminorm. Clearly, $A \subseteq B$, so that $\rho(a) \leqslant 1$ for all $a \in A$. It follows from Theorem 22.5.3 that ρ is dominated by $\|\cdot\|_{\mathrm{FBL}}$.

To prove the converse inequality, let ν be a lattice seminorm on $\mathrm{FVL}(A)$ satisfying $\nu(a) \leqslant 1$ for all $a \in A$. Then the ball B_ν is a convex solid set containing A, hence it contains B. It follows that ν is dominated by ρ. It follows from Theorem 22.5.3 that $\|\cdot\|_{\mathrm{FBL}}$ is dominated by ρ.

22.5.5 We already know that $\|a\|_{\mathrm{FBL}} = 1$ for every $a \in A$. Let $\varphi\colon A \to \ell_1^n$ given by $\varphi(a_i) = e_i$ when $i = 1, \dots, n$ and zero elsewhere. Let $\hat{\varphi}\colon \mathrm{FBL}(A) \to \ell_1^n$ be the canonical lattice homomorphism extending φ. Then $\|\hat{\varphi}\| = 1$. We have

$$\sum_{i=1}^n |\lambda_i| = \Bigl\|\sum_{i=1}^n \lambda_i e_i\Bigr\| = \Bigl\|\hat{\varphi}\Bigl(\sum_{i=1}^n \lambda_i a_i\Bigr)\Bigr\| \leqslant \Bigl\|\sum_{i=1}^n \lambda_i a_i\Bigr\|_{\mathrm{FBL}} \leqslant \sum_{i=1}^n |\lambda_i|$$

Since $\|\cdot\|_{\mathrm{FBL}}$ is a lattice norm, we have

$$\bigl\||a_1| \vee \dots \vee |a_n|\bigr\|_{\mathrm{FBL}} \leqslant \bigl\||a_1| + \dots + |a_n|\bigr\|_{\mathrm{FBL}} \leqslant n.$$

On the other hand, it follows from $\hat{\varphi}\bigl(\bigvee_{i=1}^n |a_i|\bigr) = \bigvee_{i=1}^n |e_i| = \sum_{i=1}^n e_i$, that $\Bigl\|\bigvee_{i=1}^n |a_i|\Bigr\|_{\mathrm{FBL}} \geqslant n$.

For the last identity, consider the constant one map $\varphi\colon A \to \mathbb{R}$.

22.6.1 To show that $\widehat{T}f$ is well-defined, suppose that f may be also written as $f = \Psi(\hat{y}_1, \dots, \hat{y}_m)$ for some $y_1, \dots, y_m \in E$ and some lattice-linear expression $\Psi[t_1, \dots, t_m]$. We need to show that

$$\Phi(Tx_1, \dots, Tx_n) = \Psi(Ty_1, \dots, Ty_m)$$

in Y. Let $z_1, \dots, z_k$ be a basis of the linear span of $x_1, \dots, x_n, y_1, \dots, y_m$ in E. Then z_j's are linearly independent, and each x_i and y_i may be expressed as a linear combination of x_j's:

$$x_i = \sum_{j=1}^k \alpha_{ij} z_j,\ (i = 1, \dots, n) \quad \text{and } y_i = \sum_{j=1}^k \beta_{ij} z_j,\ (i = 1, \dots, m).$$

Since the hat operation is linear, we have

$$\hat{x}_i = \sum_{j=1}^k \alpha_{ij} \hat{z}_j,\ (i = 1, \dots, n) \quad \text{and } \hat{y}_i = \sum_{j=1}^k \beta_{ij} \hat{z}_j,\ (i = 1, \dots, m).$$

It follows that

$$f = \Phi(\hat{x}_1, \dots, \hat{x}_n) = \Phi'(\hat{z}_1, \dots, \hat{z}_k)$$

and

$$f = \Psi(\hat{y}_1, \dots, \hat{y}_m) = \Psi'(\hat{z}_1, \dots, \hat{z}_k),$$

where Φ' and Ψ' are formal lattice-linear expressions defined by

$$\begin{aligned} \Phi'[t_1, \dots, t_k] &= \Phi\Bigl[\sum_{j=1}^k \alpha_{1j} t_j, \dots, \sum_{j=1}^k \alpha_{nj} t_j\Bigr] \quad \text{and} \\ \Psi'[t_1, \dots, t_k] &= \Psi\Bigl[\sum_{j=1}^k \beta_{1j} t_j, \dots, \sum_{j=1}^k \beta_{mj} t_j\Bigr]. \end{aligned}$$

Let $x^* \in E^\#$. Then

$$\begin{aligned} \Phi'\bigl(x^*(z_1), \dots, x^*(z_k)\bigr) = \Phi'\bigl(\hat{z}_1(x^*), \dots, \hat{z}_k(x^*)\bigr) &= f(x^*) \\ &= \Psi'\bigl(\hat{z}_1(x^*), \dots, \hat{z}_k(x^*)\bigr) = \Psi'\bigl(x^*(z_1), \dots, x^*(z_k)\bigr). \end{aligned}$$

Since $z_1, \dots, z_k$ are linearly independent, the k-tuple $\bigl(x^*(z_1), \dots, x^*(z_k)\bigr)$ runs over all of $\mathbb{R}^k$ as x^* runs over $E^\#$. It follows that

$$\Phi'(t_1, \dots, t_k) = \Psi'(t_1, \dots, t_k)$$

for every k-tuple $(t_1, \dots, t_k)$ in $\mathbb{R}^k$, so that

$$\Phi'(Tz_1, \dots, Tz_k) = \Psi'(Tz_1, \dots, Tz_k)$$

in Y. Since T is linear, we have

$$\Phi(Tx_1,\dots,Tx_n)=\Phi'(Tz_1,\dots,Tz_k) \text{ and } \Psi'(Tz_1,\dots,Tz_k)=\Psi(Ty_1,\dots,Ty_m),$$

which yields

$$\Phi(Tx_1,\dots,Tx_n)=\Psi(Ty_1,\dots,Ty_m).$$

22.6.2 We constructed $\mathrm{FVL}[E]$ as the sublattice of $\mathbb{R}^{E^\#}$ generated by (the canonical image of) E.

22.6.3 Let $\Phi[t_1,\dots,t_n]$ be a lattice-linear expression such that $\Phi(\hat{x}_1,\dots,\hat{x}_n)=0$. It follows that for every $x^*\in E^\#$ we have

$$0=\big(\Phi(\hat{x}_1,\dots,\hat{x}_n)\big)(x^*)=\Phi\big(x^*(x_1),\dots,x^*(x_n)\big).$$

As x^* runs over $E^\#$, the vector $\big(x^*(x_1),\dots,x^*(x_n)\big)$ runs over $\mathbb{R}^n$. It follows that $\Phi(t_1,\dots,t_n)=0$ for any $t_1,\dots,t_n\in\mathbb{R}$.

22.6.6 Consider the map $\varphi\colon A\to\mathrm{FVL}[E]$ defined by $\varphi(a)=\hat{a}$, where $a\in A$ (and, therefore, $a\in E$). By the definition of $\mathrm{FVL}(A)$, φ extends to a lattice homomorphism $\hat{\varphi}\colon \mathrm{FVL}(A)\to\mathrm{FVL}[E]$. As before, we identify $\mathrm{FVL}(A)$ with the sublattice of $\mathbb{R}^{\mathbb{R}^A}$ generated by $\{\delta_a : a\in A\}$. By definition of $\hat{\varphi}$, we have $\hat{\varphi}(\delta_a)=\varphi(a)=\hat{a}$ for every $a\in A$. It follows that $\mathrm{Range}\,\hat{\varphi}$ contains $\hat{a}$ for every $a\in A$ and, hence it contains $\hat{x}$ for every $x\in E$. Since this set generates $\mathrm{FVL}[E]$, $\hat{\varphi}$ is onto.

To see that $\hat{\varphi}$ is one-to-one, suppose that $\hat{\varphi}(f)=0$ for some $f\in\mathrm{FVL}(A)$. We can write $f=\Phi(\delta_{a_1},\dots,\delta_{a_n})$ for some distinct $a_1,\dots,a_n$ in A and some lattice-linear expression Φ. Since $\hat{\varphi}$ is a lattice homomorphism, we have $0=\hat{\varphi}(f)=\Phi(\hat{a}_1,\dots,\hat{a}_n)$. Now Exercise 22.6.3 yields that $f=0$.

22.7.6 Let $X=\mathrm{FBL}\big[\ell_1(A)\big]$; we need to show that X is a free Banach lattice over A. Let φ be a bounded map from A to an arbitrary Banach lattice Y. Then φ extends to a bounded operator $\tilde{\varphi}\colon \ell_1(A)\to Y$ with $\lVert\tilde{\varphi}\rVert=\sup_{a\in A}\lVert\varphi(a)\rVert$ We can then extend $\tilde{\varphi}$ to a lattice homomorphism $\hat{\varphi}\colon X\to Y$ with $\lVert\hat{\varphi}\rVert=\lVert\tilde{\varphi}\rVert=\sup_{a\in A}\lVert\varphi(a)\rVert$. Uniqueness follows from the observation that X is generated by the image of A in X.

22.8.2 WLOG, scaling if necessary, we may assume that $x^*\in B_{E^*}$. Consider the map $\theta\colon \mathrm{FBL}[E]\to\mathbb{R}$ given by $\theta f=f(x^*)$. Since the lattice operations $\mathrm{FBL}[E]$ are point-wise, this map is a lattice homomorphism. For every $x\in E$, $(\theta\delta_E)(x)=\theta\delta_x=\delta_x(x^*)=x^*(x)$. Hence, $\theta\delta_E=x^*$. It follows that $\theta=\widehat{x^*}$.

22.8.4 WLOG, $\lVert T\rVert\leqslant 1$. Fix $y^*\in B_{F^*}$. It suffices to show that $(\overline{T}f)(y^*)=f(T^*y^*)$. Define $\theta\colon \mathrm{FBL}[E]\to\mathbb{R}$ via $\theta(f)=f(T^*y^*)$. Clearly, θ is a lattice homomorphism. Fix $x\in E$. We have

$$\theta\delta_x=\delta_x(T^*y^*)=(T^*y^*)(x)=y^*(Tx)=\widehat{y^*}(\delta_{Tx})=(\widehat{y^*}\overline{T})\delta_x.$$

It follows that the two lattice homomorphisms θ and $\widehat{y^*}\overline{T}$ agree on E (or, rather, on its image in $\mathrm{FBL}[E]$), hence they are equal. Applying Exercise 22.8.2, we get $f(T^*y^*)=\theta(f)=\widehat{y^*}\overline{T}f=(\overline{T}f)(y^*)$

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